GeeksforGeeks

A computer science portal for geeks

CHAPTER WISE

GATE

PREVIOUS YEAR QUESTIONS

WITH DETAILED SOLUTIONS & VIDEO SOLUTIONS

22 YEARS
OF QUESTIONS
2005 – 2026

CHAPTER-WISE
ARRANGEMENT

DETAILED
EXPLANATIONS

QR CODE
VIDEO SOLUTIONS

COMPUTER SCIENCE & INFORMATION TECHNOLOGY

2005 – 2026 | **UPDATED TILL GATE 2026**

LinkedIn Youtube Telegram GeeksforGeeks

GeeksforGeeks

Head Office: 7th Floor, Sovereign Corporate Tower,
A-143, Sector 136, Noida, Uttar Pradesh 201305

Hyderabad: 3rd Floor, SRR COMPLEX, House No 9-12,
Gaddiannaram, Vijeta Complex, Lalitha Nagar,
Dilsukhnagar, Hyderabad, Telangana 500060

Ph.: 08069289001
E-mail: support@geeksforgeeks.org
Website: www.geeksforgeeks.org

© All rights reserved by GeeksforGeeks. No part of this book may be reproduced or distributed in any form or any means, electronic, mechanical, photocopying or otherwise without the prior permission of the @GeeksforGeeks, Sanchhaya Education Private Limited

SYLLABUS

CS Computer Science and Information Technology

01 Section 1: Engineering Mathematics

Discrete Mathematics: Propositional and first order logic. Sets, relations, functions, partial orders and lattices. Monoids, Groups. **Graphs:** connectivity, matching, colouring. Combinatorics: counting, recurrence relations, generating functions.

Linear Algebra: Matrices, determinants, system of linear equations, eigenvalues and eigenvectors, LU decomposition.

Calculus: Limits, continuity and differentiability, Maxima and minima, Mean value theorem, Integration.

Probability and Statistics: Random variables, Uniform, normal, exponential, Poisson and binomial distributions. Mean, median, mode and standard deviation. Conditional probability and Bayestheorem.

02 Section 2: Digital Logic

Boolean algebra and minimization – algebraic technique, Karnaugh map, tabular method. Design of combinational and sequential circuits. Number representation and arithmetic (fixed and floating point).

03 Section 3: Computer Organization and Architecture

Instruction set and addressing modes. Design of arithmetic and logic unit (ALU). Design of control unit – hardwired and microprogrammed. Memory interfacing and hierarchy: performance, cache memory mapping. I/O interface (interrupt and DMA). Instruction pipelining, pipeline hazards.

04 Section 4: Programming and Data Structures

Programming in C. Recursion. Arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs.

05 Section 5: Algorithms

Searching, sorting, hashing. Asymptotic worst case time and space complexity. Algorithm design techniques: greedy, dynamic programming and divide-and-conquer. Graph traversals, minimum spanning trees, shortest paths.

06 Section 6: Theory of Computation

Regular expressions and finite automata. Context-free grammars and push-down automata. Regular and context-free languages, pumping lemma. Turing machines and undecidability.

07 Section 7: Compiler Design

Lexical analysis, parsing, syntax-directed translation. Runtime environments. Intermediate code generation. Local optimisation, Data flow analyses: constant propagation, liveness analysis, common sub expression elimination.

08 Section 8: Operating System

System calls, processes, threads, inter-process communication, concurrency and synchronization. Deadlock. CPU and I/O scheduling. Memory management and virtual memory. File systems.

CS Computer Science and Information Technology

09 Section 9: Databases

ER-model. Relational model: relational algebra, tuple calculus, SQL. Integrity constraints, normal forms. File organization, indexing (e.g., B and B+ trees). Transactions and concurrency control.

10 Section 10: Computer Networks

Principles of Layering; Basics of switching (circuit, packet and virtual circuit) and performance metrics; Data link layer: error detection, Medium Access Control, Ethernet; Distance vector and link state routing; IPv4 - Fragmentation, CIDR Notation, Network Address Translation; TCP- flow control and congestion control, socket API; DNS and HTTP.

GENERAL APTITUDE

Verbal Aptitude

Basic English grammar: Tenses, articles, adjectives, prepositions, conjunctions, verb-noun agreement, and other parts of speech.

Basic vocabulary

Words, idioms, and phrases in context.
Reading and comprehension, Narrative sequencing.

Quantitative Aptitude

Data interpretation: Data graphs (bar graphs, pie charts, and other graphs representing data), 2- and 3-dimensional plots, maps, and tables.

Numerical computation and estimation

Ratios, percentages, powers, exponents and logarithms, permutations and combinations, and series Mensuration and geometry Elementary statistics and probability.

Analytical Aptitude

Logic: Deduction and induction, Analogy, Numerical relations and reasoning.

Spatial Aptitude

Transformation of shapes: Translation, rotation, scaling, mirroring, assembling, and grouping paper folding, cutting, and patterns in 2 and 3 dimensions.

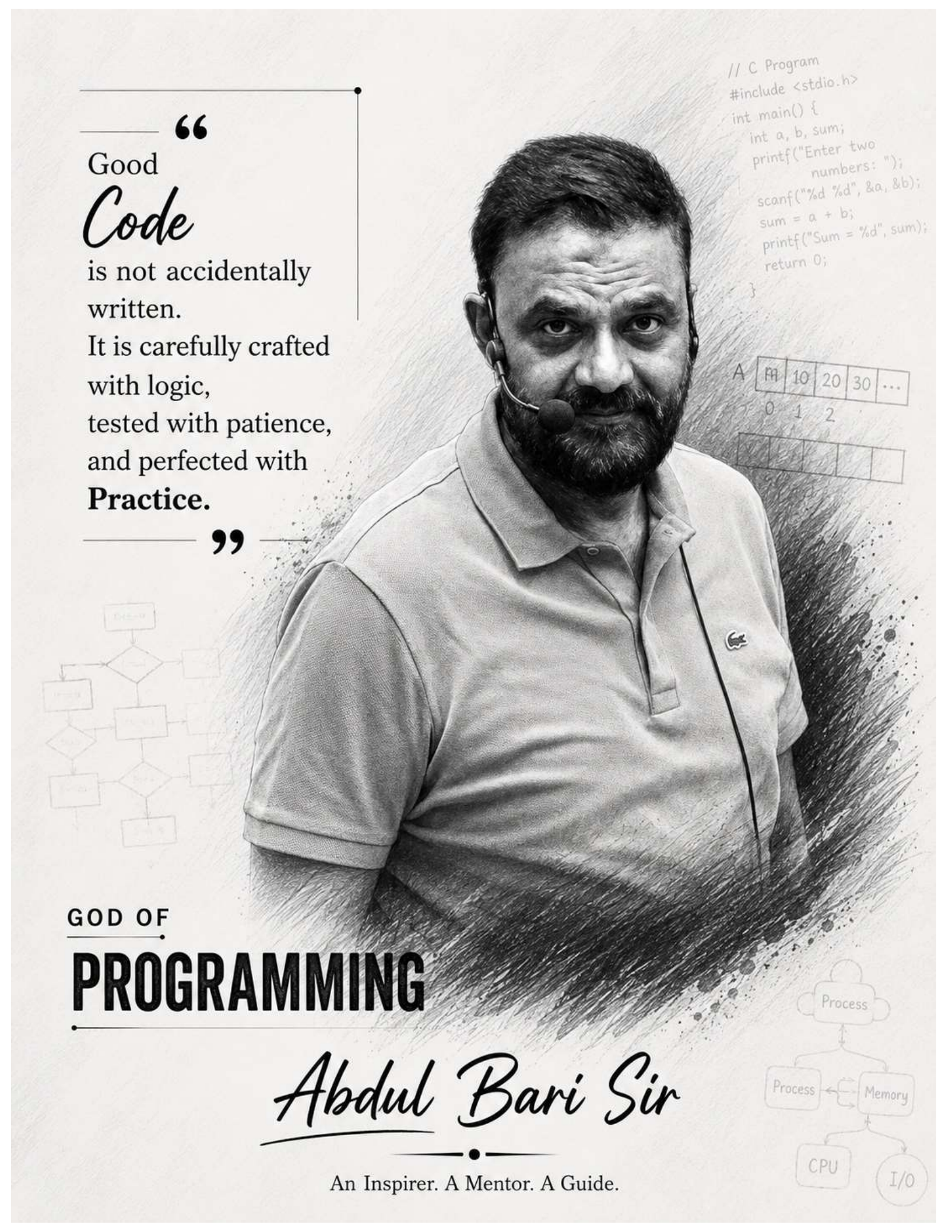
>>> GATE CS/IT LAST 5 YEARS WEIGHTAGE : <<<

Subject	2026-SET 2	2026-SET 1	2025-SET 2	2025-SET 1	2024-SET 2	2024-SET 1	2023	2022
C Programming	6	3	6	5	4	4	3	5
Data Structure	6	9	6	8	5	4	7	5
Algorithms	11	9	8	5	7	10	7	9
Digital Logic	5	5	7	6	5	7	6	5
TOC	6	6	9	10	7	7	9	8
Compiler Design	5	5	4	6	8	5	7	4
Operating System	9	6	7	8	9	8	11	9
DBMS	6	6	9	6	8	7	5	7
COA	11	11	9	8	9	11	8	8
Computer Networks	9	10	6	8	9	9	9	10
Discrete Mathematics	2	6	3	8	7	6	8	9
Engineering Mathematics	9	9	11	7	6	7	6	6
General Aptitude	15	15	15	15	15	15	15	15

C PROGRAMMING

“Scan Here for Video Solution”

Link to GfG corresponding Short Notes: GATE FARRAY


“Good Code is not accidentally written. It is carefully crafted with logic, tested with patience, and perfected with Practice.”
GOD OF
PROGRAMMING
Abdul Bari Sir
An Inspirer. A Mentor. A Guide.

1 DATA TYPE AND OPERATOR

GATE CSE – 2022 MCQ (2-MARKS)

Q1. What is printed by the following ANSI C program?

```
#include<stdio.h>
int main (int argc, char *argv[])
{
char a = 'P';
char b = ' x ';
char c = (a & b)+' * ';
char d = (b) -  ' - ';
char e = (a ∧ b) + ' + ';
printf("%c %c %c \n", c, d, e);
return 0;
}
```

ASCII encoding for relevant characters is given below.

A	B	C	.....	Z
65	66	67	.....	90

a	b	c	... ...	z
97	98	99	... ...	122

*	+	-
42	43	45

(a) z K S
(b) 122 75 83
(c) * - +
(d) P x +

GATE CSE – 2017 SET 2 NAT (2-MARKS)

Q2. Consider the following C program

```
#include<stdio.h>
int main ( ) {
int m = 10
int n, nl;
n=++m;

nl=m++;
n--;
--n1;
n−= nl;
printf("%d", n);
return 0;
}
```

The output of the program is ___.

GATE CSE – 2014 MCQ (1-MARKS)

Q3. Suppose n and p are unsigned int variables in a C program. We wish to set p to n_{C_3}. If n is large, which one of the following statements is most likely to set p correctly?

(a) $p = n^*(n-1)^*(n-2)/6;$
(b) $p = n^*(n-1)/2^*(n-2)/3;$
(c) $p = n^*(n-1)/3^*(n-2)/2;$
(d) $p = n^*(n-1)^*(n-2)/6.0;$

GATE CSE – 2008 MCQ (2-MARKS)

Q4. Which combination of the integer variables x, y,and z makes the variable a get the value 4 in the following expression?

$a = (x > y)?((x > z)?x:z):((y > z)?y:z)$

(a) x = 3, y = 4, z = 2
(b) x = 6, y = 5, z = 3
(c) x = 6, y = 3, z = 5
(d) x = 5, y = 4, z = 5

Link to GfG corresponding Short Notes: GATE FARRAY

ANSWER KEY

1. (a)	2. (0 to 0)	3. (b)	4. (a)

HINTS & SOLUTIONS

1. Answer: (A)
Solution:
char a = 'P'
char b = 'x'
As per precedence of operators, () evaluated prior to + /-
Note that &,^ and | are bit wise operators and Addition/Subtraction of characters applied on their ASCII values.
a = 'P' ===> a = Ascii value of ('P') = 65+15 = 80 ==> a = $(0101\ 0000)_2$
b = 'x' ===> b = Ascii value of ('x') = 97+23 = 120 ==> b = $(0111\ 1000)_2$

a & b = $(0101\ 0000)_2$ [apply & logic on corresponding bits] = $(80)_{10}$
'*' = 42 = $(0010\ 1010)_2$
a&b + '*' = $(0111\ 1010)_2$ = 64+32+16+8+2 = 122 [add corresponding bits], [simply 80+42 = 122]
print character (122) = small 'z'

a | b = $(0111\ 1000)_2$ [apply | logic on corresponding bits] = $(120)_{10}$ '-' = 45
a|b - '-' = 120-45 = 75 = 65 + 10
print character (75) = Capital 'K'

a ^ b = $(0010\ 1000)_2$ [apply ^ logic on corresponding bits] = $(40)_{10}$
'+' = 43
a^b + '+' = 40+43 = 83 = 65 + 18
print character (83) = Capital 'S'

2. Answer:(0 to 0)
Solution:

1. int m = 10;
   o $m = 10$
2. $n = ++m$;
- Pre-increment (++m): m is incremented first, changing from 10 to 11 .
- The updated value (11) is then assigned to n.
- Current State: $m = 11, n = 11$
3. $n1 = m++$;
- Post-increment ($m++$): The current value of $m(11)$ is assigned to n 1 first.
- After assignment, m is incremented from 11 to 12 .
- Current State: $m = 12, n = 11, n1 = 11$
4. n--;
- n is decremented by 1 .
- Current State: $n = 10$
5. --n1;
- n1 is decremented by 1 .
- Current State: n1 = 10
6. n -= n1;
- This is equivalent to $n = n - n1$.
- $n = 10 - 10 = 0$
- Current State: $n = 0$
7. printf("%d", n);
- Prints the final value of n, which is **0**.

3. Answer:**(b)**
Solution:
We need to set $p = nC3$ where $nC3 = n(n-1)(n-2)/6$.
n and p are unsigned int. If n is large, overflow can happen during multiplication before division.
Check options:
(a) $p = n^*(n-1)(n-2)/6$;
Here multiplication happens first. For large n, $n(n-1)^*(n-2)$ becomes very large and may overflow before dividing by 6 . So result may be wrong.
(b) $p = n^*(n-1)/2^*(n-2)/3$;
Here division is done early. Since $6 = 2 \times 3$, we divide step by step. This reduces intermediate values and avoids overflow. This is most safe and correct.
(c) $p = n^*(n-1)/3^*(n-2)/2$;
Division by 3 first may not divide exactly at that stage, so integer truncation can happen. This may give wrong result.
(d) $p = n^*(n-1)^*(n-2)/6.0$;
6.0 is double. Expression becomes floating point, then converted back to unsigned int. This is not safe and may lose precision.

4. Answer: (A)

Solution:

Given expression:

a = (x > y) ? ((x > z) ? x : z) : ((y > z) ? y : z)

This nested ternary operator essentially finds the maximum of x, y, and z.

Case 1: If x > y → a = max(x, z)

Case 2: Otherwise → a = max(y, z)

So overall, a = max(x, y, z)

Checking options:

A. x = 3, y = 4, z = 2

max = 4 (Correct)

B. x = 6, y = 5, z = 3

max = 6 (Incorrect)

C. x = 6, y = 3, z = 5

max = 6(Incorrect)

D. x = 5, y = 4, z = 5

max = 5 (Incorrect)

2 CONTROL FLOW STATEMENT

GATE CSE – 2025 SET- 1 NAT (2-MARKS)

Q1

```
#include < stdio.h >
int gate(int n) {
   int d, t, newnum, turn;
   newnum = turn = 0; t=1;
   while(n >= t) t *= 10;
                 t /= 10;
  while(t > 0) {
     d = n / t;
     n = n % t;
     t /= 10;
     if(turn) newnum = 10 * newnum + d;
     turn = (turn + 1) % 2;
   }
  return newnum;
}
int main () {
   printf("%d", gate(14362));
   return 0;
}
```

The value printed by the given C program is _____ .
(Answer in integer)

GATE CSE – 2025 SET- 1 NAT (2-MARKS)

Q2

```
int x=126,y=105;
  do {
    if(x>y) x=x-y;
 else y=y-x;
    } while(x!=y);
printf("%d",x);
```

The output of the given C code segment is_____ .
(Answer in integer)

GATE CSE – 2021 SET - 2 MCQ (2-MARKS)

Q3. Consider the following ANSI C code segment:

```
z = x + 3 + y->f1 + y→f2;
 for (i = 0; i< 200; i = i + 2)
{
   if (z >i) {
     p = p + x + 3:
     q = q + y→f2;
   } else {
     p = p + y→f2:
     q = q + x + 3:
   }
}
```

Assume that the variable y points to a struct (allocated on the heap) containing two fields f1 and f2, and the local variables x, y, z, p, q, and i are allotted registers. Common sub-expression elimination (CSE) optimization is applied on the code. The number of addition and dereference operations (of the form y->f1 or y->f2) in the optimized code, respectively, are:

(a) 303 and 102
(b) 303 and 2
(c) 403 and 102
(d) 203 and 2

GATE CSE – 2021 SET- 1 MCQ (2-MARKS)

Q4. Consider the following ANSI C program.

```
#include <stdio.h>
 int main()
        {
              int i, j, count;
              count = 0;
              i = 0;
              for (j = –3, j <= 3; j++)
                    {
                     if ((j >= 0) && (i++))
                     count = count + j;
                    }
              count = count + i;
              printf("%d", count);
              return 0;
        }
```

Check Out Our GATE Courses & Secure Your Dream Rank!

Which of one of the following options is correct?

(a) The program will compile successfully and output 10 when executed.
(b) The program will not compile successfully.
(b) The program will compile successfully and output 8 when executed.
(d) The program will compile successfully and output 13 when executed.

GATE CSE - 2019 MCQ (2-MARKS)

Q5. Consider the following C program:

```
#include <stdio.h>
int r ( )
{
  static int num=7;
  return num--;
}
int main ( )
{
for (r ( ); r ( ); r ( ))
  printf ("%d", r ( ));
return 0;
}
```

Which one of the following values will be displayed on execution of the programs?

(a) 41
(b) 63
(c) 52
(d) 630

GATE CSE - 2019 MCQ (2-MARKS)

Q6. Consider the following C function.

```
void convert (int n)
{
  if (n < 0)
  printf ("%d", n);
  else {
          convert (n/2);
          printf ("%d", n%2);
       }
}
```

Which one of the following will happen when the function convert is called with any positive integer n as argument?

(a) It will print the binary representation of n in the reverse order and terminate
(b) It will not print anything and will not terminate
(c) It will print the binary representation of n and terminate
(d) It will print the binary representation of n but will not terminate

GATE CSE - 2019 NAT (2-MARKS)

Q7. Consider the following C program:

```
#include <stdio.h>
int main ()  {
   float sum = 0.0, j = 1.0, i = 2.0;
   while (i/j > 0.0625) {
        j = j + j;
        sum = sum + i/j;
        printf ("%f \n", sum);
    }
    return 0;
}
```

The number of times the variable sum will be printed, when the above program is executed, is _____.

GATE CSE – 2017 SET - 2 MCQ (2-MARKS)

Q8. Consider the C program fragment below which is meant to divide x by y using repeated subtractions. The variables x, y, q and r are all unsigned int.

```
while (r >= y) {
   r = r - y;
   q = q + 1;
}
```

Which of the following conditions on the variables x, y, q and r before the execution of the fragment will ensure that the loop terminates in a state satisfying the condition x == (y*q + r)?

(a) (q = = r) && (r = = 0)
(b) (x > 0) && (r = = x) && (y > 0)
(c) (q = = 0) && (r = = x) && (y > 0)
(d) (q = = 0) && (y > 0)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2016 SET - 2 MCQ (2-MARKS)

Q9. The following function computes X^Y for positive integers X and Y.

```
int exp (int X, int Y) {
   int res = 1;
   int a = X;
   int b = Y;

  while ( b != 0 ){
  if ( b%2 == 0)
                    {
                  a = a*a;
                  b = b/2;
                    }
            else{
                    res = res*a;
                    b = b-1;
            }
   }
   return res;
 }
```

Which one of the following conditions is TRUE before every iteration of the loop?

(a) $X^Y = a^b$
(b) $(res * a)^Y = (res * X)^b$
(c) $X^Y = res * a^b$
(d) $X^Y = (res * a)^b$

GATE CSE – 2015 SET – 3 NAT (2-MARKS)

Q10. Consider the following C program:

```
# include <stdio.h>
int main( )
{
int i, j, k = 0;
j = 2 * 3 / 4 + 2.0 / 5 + 8 / 5;
k  - = --j;
for (i = 0; i< 5; i++)
{
switch(i + k)
            {
            case 1:
            case 2: printf("\n%d", i + k);
            case 3: printf("\n%d", i + k);
            default: printf("\n%d", i + k);
             }
 }
return 0;
}
```

Total number of times printf() executed ?

GATE CSE – 2014 SET - 3 MCQ (2-MARKS)

Q11. Consider the following pseudo code. What is the total number of multiplications to be performed?

```
D = 2
for i = 1 to n do
  for j = i to n do
     for k = j + 1 to n do
        D = D * 3
```

(a) Half of the product of the 3 consecutive integers.
(b) One-third of the product of the 3 consecutive integers.
(c) One-sixth of the product of the 3 consecutive integers
(d) None of the above.

GATE CSE - 2014 SET – 2 NAT (2-MARKS)

Q12. Consider the following function

```
#include <math.h>
double f(double x){
                    if (abs(x*x - 3) < 0.01) return x;
  `                 else return f(x/2 + 1.5/x);
                    }
```

Give a value q (to 2 decimals) such that f(q) will return q: _____.

GATE CSE - 2013 MCQ (2-MARKS)

Q13. Consider the following function:

```
int unknown(int n) {
   int i, j, k = 0;
   for (i  = n/2; i<= n; i++)
      for (j = 2; j <= n; j = j * 2)
k = k + n/2;
   return k;
}
```

The return value of the function is__

(a) $\Theta(n^2)$
(b) $\Theta(n^2 \log n)$
(c) $\Theta(n^3)$
(d) $\Theta(n^3 \log n)$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2012 MCQ (2-MARKS)

Q14. What will be the output of the following C program segment?

```
char inchar = 'A';
 switch (inchar)
        {
                case 'A' :
                printf ("choice A ") ;
                case 'B' :
                printf ("choice B ") ;
                case 'C' :
                case 'D' :
                case 'E' :
                default:
                printf ("No Choice") ;
        }
```

(a) No Choice
(b) Choice A
(c) Choice A Choice B No Choice
(d) Program gives no output as it is erroneou

GATE CSE – 2007 SET – 1 MCQ (1-MARKS)

Q15. Consider the following segment of C-code:

```
int j, n;
j = 1;
while (j <= n)
j = j * 2;
```

The number of comparisons made in the execution of the loop for any n>0 is:

(a) $\lceil \log_2 n \rceil + 1$
(b) n
(c) $\lceil \log_2 n \rceil$
(d) $\lfloor \log_2 n \rfloor + 1$

Answer Key

1.46	**2.21**	**3.(b)**
4.(a)	**5.(c)**	**6.(b)**
7.(5)	**8.(c)**	**9.(c)**
10.(10)	**11.(c)**	**12.(1.73)**
13.(b)	**14.(c)**	**15.(d)**

HINTS AND SOLUTIONS

1. Answer:46

The code peels off the digits one by one from left to right: 1, then 4, then 3, then 6, and finally 2.

The Switch (turn): The variable turn acts exactly like a light switch alternating between 0 (OFF) and 1 (ON).

- It starts at 0 (OFF).
- Every time a digit is processed, the switch flips: $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0$.

By crossing out the skipped positions, the process looks like this:

$$\not{1} \rightarrow 4 \rightarrow \not{3} \rightarrow 6 \rightarrow \not{2} \implies 46$$

2. Answer:21

Initially, x = 126 and y = 105. The loop continues as long as $x \neq y$, and in each iteration, it subtracts the smaller value from the larger one.

In the first iteration: Since x > y, x = x - y = 126 - 105 = 21.

Now x = 21 and y = 105. In the next iteration: y = y - x = 105 - 21 = 84.

The process continues until both values become equal, and at that point, the loop ends. After the loop, both x and y will be 21, which is the GCD of 126 and 105

3. ANSWER:- (B)

Solution:

```
t1 = x + 3      || 1 addition
t2 = y → f1; || 1 dereference
t3 = y → f2; || 1 dereference
z = t1 + t2 + t3|| 2 additions
for (i = 0;  i< 200; i+ = 2) {
if (z >i) {
p = p + t1 ;||  1 addition
q = q + t2 ;|| 1 addition
}
else{
        p = p + t3;   || 1 addition
        q = q + t1;   || 1 addition
   }
}
```

loop will run 100 times and both if and else are performing 2 addition operations, so from loop we get 2 × 100 = 200 additions and 100 times i is incrementing, so, 200 + 100 = 300 addition from the

Check Out Our GATE Courses & Secure Your Dream Rank!

loop. Before loop, 3 additions and 2 dereference operations.
So, 303 addition, 2 dereference.

4. ANSWER:- (A)
Solution:
Initial values
i = 0
count = 0
Loop runs from j = −3 to 3.
Condition:
(j >= 0) && (i++)
Important point:
&& is short-circuit operator.
If first condition is false, second part (i++) is NOT executed.
Now check each j:

j = -3 → (j >= 0) is false → i++ not executed
j = -2 → false → i still 0
j = -1 → false → i still 0

j = 0 → true && (i++)
i++ returns 0 (old value), then i becomes 1
So condition = true && 0 → false
count not updated.
j = 1 → true && (i++)
i++ returns 1, then i becomes 2
Condition = true && 1 → true
count = count + 1 → count = 1

j = 2 → true && (i++)
i++ returns 2, then i becomes 3
Condition true
count = 1 + 2 = 3

j = 3 → true && (i++)
i++ returns 3, then i becomes 4
Condition true
count = 3 + 3 = 6

After loop:
count = count + i
count = 6 + 4 = 10

Final output:
10

5. ANSWER:- (C)
Solution:
r() has a static variable num = 7.
Each call returns num--.
Execution flow:
Init: r() → returns 7 → num = 6
Condition: r() → returns 6 → num = 5 → true
printf: r() → returns 5 → num = 4 → prints 5
Increment: r() → returns 4 → num = 3
Condition: r() → returns 3 → num = 2 → true
printf: r() → returns 2 → num = 1 → prints 2
Increment: r() → returns 1 → num = 0
Condition: r() → returns 0 → loop ends

Output: 52

6. ANSWER:- (B)
Solution:
a = {2, 4, 6, 8, 10}
Sequence of function calls
Convert(6)
Convert(3)
Convert(1)
Convert(0)
:
Convert(0)
:
:
It will not terminate and never produce any output.
Note:
There is no instruction which stops the loop.

7. ANSWER:- (5)
Solution:

i = 2.0
j = 1.0

Condition:
while (i/j > 0.0625)

Since i = 2,

Condition becomes:
2/j > 0.0625

0.0625 = 1/16

So:
2/j > 1/16

Multiply both sides by j:

2 > j/16

Check Out Our GATE Courses & Secure Your Dream Rank!

Multiply by 16:

32 > j

So loop runs while:
j < 32

Now track j values.
Inside loop:

j = j + j → j doubles every time.

Start:
j = 1

Iteration 1:
Check 2/1 = 2 > 0.0625 → true
j becomes 2 → print

Iteration 2:
Check 2/2 = 1 > 0.0625 → true
j becomes 4 → print

Iteration 3:
Check 2/4 = 0.5 > 0.0625 → true
j becomes 8 → print

Iteration 4:
Check 2/8 = 0.25 > 0.0625 → true
j becomes 16 → print

Iteration 5:
Check 2/16 = 0.125 > 0.0625 → true
j becomes 32 → print

Next check:
2/32 = 0.0625

Now:
0.0625 > 0.0625 → false

Loop stops.

So sum is printed 5 times.

Final Answer: 5

8. ANSWER:- (C)
Solution:
Divide x by y.

x, y, q, r are unsigned integers.
while (r >= y)
{
r = r – y;
q = q + 1;
}
Loop terminates in a state satisfying the condition
x == (y * q + r)
y ⇒ Dividend = Divisor * Quotient + Remainder
So, to divide a number with repeated subtractions, the Quotient should be initialized to 0 and it must be incremented for every subtraction.
So initially q=0 which represents
x = 0 + r ⇒ x = r
and y must be a positive value (>0).

9. ANSWER:- (C)
Solution:
Consider X=2, y=3,
After 1 iteration, res=2, a=2,b=2
Only C satisfies the values
X^Y=res* a^b
2^3=2*2

10. ANSWER : (10)
Solution:
Arithmetic Evaluation
In C, integer division truncates decimals, and prefix operators update variables immediately.
j calculation: 2 * 3 / 4 (integer) is 1. 2.0 / 5 (float) is 0.4. 8 / 5 (integer) is 1. The sum is 2.4, which truncates to 2 for the integer j.
k calculation: k -= --j first decrements j to 1. Then k = 0 - 1, resulting in k = -1.
2. Switch Trace with Fall-through
The loop runs for i = 0 to 4. Because there are no break statements, once a case matches, every subsequent printf in the block executes.
i = 0 (i+k = -1): Matches default. (1 execution)
i = 1 (i+k = 0): Matches default. (1 execution)
i = 2 (i+k = 1): Matches case 1, falls through case 2, case 3, and default. (3 executions)
i = 3 (i+k = 2): Matches case 2, falls through case 3, and default. (3 executions)
i = 4 (i+k = 3): Matches case 3, falls through default. (2 executions)
Final Total
Adding the executions per iteration: 1 + 1 + 3 + 3 + 2 = 10.
The total number of times printf is executed is 10.

Check Out Our GATE Courses & Secure Your Dream Rank!

11. ANSWER: (C)

Solution:

We count how many times

D = D * 3 is executed.

Conditions from loops:

$1 \le i \le j < k \le n$

This means we are choosing 3 numbers such that

$i \le j < k$.

The total number of such triples is:

$(n - 1)\, n\, (n + 1) / 6$
This equals one-sixth of the product of three consecutive integers.

12. ANSWER:- (1.73)
Solution:
The stopping condition is:
$|x^2 - 3| < 0.01$
We need a value q (to 2 decimal places) that satisfies:

$|q^2 - 3| < 0.01$
This means q2 must be between 2.99 and 3.01:
$2.99 < q^2 < 3.01$
Taking the square root gives the range:
$\sqrt{2.99} < q < \sqrt{3.01}$
$1.72916... < q < 1.73493...$

The value 3
≈ 1.73205 is in this range. Rounding this to two decimal places gives 1.73.
Verification: $1.732 = 2.9929$. The absolute difference is $|2.9929 - 3| = 0.0071$, which is less than 0.01.
The value q is 1.73.

13. ANSWER:- (B)
Solution:
Outer loop runs for (n/2) times and inner loop runs for (logn) times.
So, the total number of times loop runs is (n/2 logn).
So, the final k value will be n/2*(n/2 logn) =
$O(n^2 \log n)$
$= (n/2+1).n/2 \cdot \log n$
$= (n^2 \log n)$

14. ANSWER:- (C)
Solution:
The switch case has no break statements.
inchar = 'A' → matches case 'A', prints "choice A ".
Then it falls through to case 'B', prints "choice B ".
Then it falls through to case 'C', case 'D', case 'E' (no code, so no print), then default, prints "No Choice".

15. ANSWER:- (D)
Solution:
Initially j = 1.

Each iteration doubles j:

1, 2, 4, 8, 16, ...

The loop continues while $j \le n$.

Suppose after k multiplications:

$j = 2^k$

The loop stops when $2^k > n$.

So the smallest k such that $2^k > n$ is:

$k = \lfloor \log_2 n \rfloor + 1$. The condition ($j \le n$) is checked

one extra time when it becomes false.

So total number

3 Function, Storage Class and Recursion

GATE CSE – 2026 SET – 2 MCQ (1-MARKS)

Q1. Consider the following three ANSI – C programs. P1, P2, and P3

P1	P2	P3

#include <stdio.h>
 	
int a = 5;
int main(){
int a = 7;
return(0);
}
	#include <stdio.h>
	
	int main(){
	int a=5;
	int a=7;
	return(0);
	}
		#include <stdio.h>
		
		int main(){
		int a=5;
		float a=7;
		return(0);
		}

Which one of the following statement is true?

(a) Only P1 will compile without any error
(b) Only P2 will compile without any error
(c) Only P3 will compile without any error
(d) All three programs P1, P2, and P3 will compile without any error

GATE CSE – 2026 SET – 1 NAT (1-MARKS)

Q2. Consider the recursive functions represented by the following code segment:

```
int bar(int n)
        {
            if (n == 1)  return 0;
            else
            return 1 + bar(n/2);
        }

int foo(int n) {
        if (n == 1)
            return 1;
        else   return 1 + foo(bar(n));
        }
```

The smallest positive integer n for which foo(n) returns 5 is _____. (answer in integer)
Note: Assume that the program compiles and runs successfully.

GATE CSE – 2026 SET – 2 NAT (1-MARKS)

Q3. Consider the following ANSI-C function.

```
int func(int start, int end) {
        int length = end + 1 - start;
if ((length < 1) || (start < 0) || (end < 0)) {
        return 0;
}
if (length % 3 == 0) {
        return (func(start + 1, end));
}
else if (length % 3 == 1) {
        return 1 + func(start, end - 1);}

else {
        return func(start + 2, end);
        }
}
```

The maximum possible value that can be returned from this function is _____. (answer in integer)
Note: Ignore syntax errors (if any) in the function.

GATE CSE – 2025 SET – 2 NAT (2 MARKS)

Q4 Consider the following C program:

```
#include <stdio.h>
int g(int n) {
   return (n+10);
}
int f(int n) {
   return g(n*2);
}
int main() {
   int sum, n;
   sum=0;

   for (n=1; n<=3; n++)
      sum += f(n);

   printf("%d", sum);
   return 0;
}
```

The output of the given C program is_____ (Answer in integer)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2025 SET – 2 NAT (2 MARKS)

Q5

```
#include < stdio.h >
int foo(int S[],int size){
   if(size == 0) return 0;
   if(size == 1) return 1;
if(S[0] != S[1]) return      1+foo(S+1,size-1);
   return foo(S+1,size-1);
}
int main(){
   int A[]={0,1,2,2,2,0,0,1,1};
   printf("%d",foo(A,9));
   return 0;
}
```

The value printed by the given C program is_____ . (Answer in integer).

GATE CSE – 2024 SET – 2 MCQ (2 MARKS)

Q6. Consider the following C program. Assume parameters to a function are evaluated from right to left.

```
#include <stdio.h>
int g(int p) {
   printf("%d", p);
   return p;
}
int h(int q) {
   printf("%d", q);
   return q;
}
void f(int x, int y) {
   g(x);
   h(y);
}
int main() {
   f(g(10), h(20));
}
```

Which one of the following options is the CORRECT output of the above C program?

(a) 20101020
(b) 10202010
(c) 20102010
(d) 10201020

GATE CSE – 2023 SET – 1 NAT (1-MARKS)

Q7. Consider the following program in C:

```
#include <stdio.h>
void func(int i, int j) {
if (i < j) {
                int i = 0;
                while (i < 10) {
                   j += 2;

                      i++
                   }
                }
                    printf("%d", i);
        }
                int main() {
                  int i = 9, j = 10;
                func(i, j);
                return 0;
        }
```

The output of the program is _____. (answer in integer)

Note: Assume that the program compiles and runs successfully.

GATE CSE - 2023 MCQ (2-MARKS)

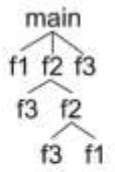
Q8. Consider the following C program:

```
int main()
{
   f1();
   f2(2);
   f3();
   return(0);
}
int f1()
{
   return(1);
}

int f2(int x)
{
   f3();
   if (x == 1)
      return f1();
   else
      return (x * f2(x - 1));
}
int f3()
{
   return(5);
}
```

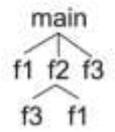
Check Out Our GATE Courses & Secure Your Dream Rank!

Which of the following options represent the activation function corresponding to the main function?

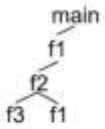
a)



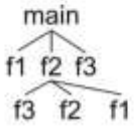
b)



C)



D)



GATE CSE - 2021 NAT (2-MARKS)

Q9. Consider the following ANSI C program.

```
#include <stdio.h>
int foo(int x, int y, int q)
{
   if ((x <= 0) && (y <= 0))
        return q;
   if (x <= 0)
        return foo(x, y-q, q);
   if (y <= 0)
        return foo(x-q, y, q);
        return foo(x, y–q, q) + foo(x-q, y, q);
}
int main()
{
   int r = foo(15,15,10);
printf("%d", r);
   return 0;
}
```

The output of the program upon execution is _____ .

GATE CSE – 2021 SET - 1 NAT (2-MARKS)

Q10. Consider the following ANSI C function :

```
int SimpleFunction(int Y[], int n, int x)
{
  int total = Y[0], loopIndex;
  for (loopIndex=1; loopIndex<= n –1; loopIndex++)

  total = x * total + y[loopIndex];
  return total;
}
```

Let Z be an array of 10 elements with Z[i]=1, for all i such that 0 <= i <=9

The value returned by SimpleFunction(Z, 10, 2) is _____

GATE CSE - 2020 NAT (2-MARKS)

Q11. Consider the following ANSI C function:

```
int SomeFunction(int x, int y)
{
   if ((x == 1) || (y == 1)) return 1;
   if (x == y) return x;
   if (x > y) return SomeFunction(x - y, y);
   if (y > x) return SomeFunction (x, y - x);
}
```

The value returned by Some Function(15, 255) is _____

GATE CSE - 2020 NAT (2-MARKS)

Q12. Consider the following C functions.

```
int fun1(int n) {
   static int i = 0;
     if (n > 0) {
        ++i;
        fun1(n-1);
     }
   return (i);
}
```

```
int fun2(int n) {
   static int i = 0;
     if (n > 0) {
     i = i + fun1(n);
     fun2(n-1);
   }
   return (i);
}
```

The return value of fun2(5) is ____

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2020 NAT (2-MARKS)

Q13. Consider the following C Functions.

```
int tob (int b, int* arr) {
   int i;
   for (i = 0; b >0; i++) {
      if (b % 2)
         arr[i] = 1;
      else arr[i]=0;
         b= b /2;
      }
   return (i);
}
```

```
int pp(int a, int b) {
   int arr[20];
   int i, tot = 1, ex, len;
   ex= a;
   len = tob(b, arr);
for (i=0; i < len; i++ ) {
      if (arr[i] == 1
         tot = tot * ex;
          ex= ex * ex;
   }
} return (tot);
```

The value returned by pp (3,4) is _____.

GATE CSE - 2019 NAT (1-MARKS)

Q14. Consider the following C program:

```
#include <stdio.h>
int jumble (int x, int y) {
   x = 2 * x + y;
   return x;
}
int main ()  {
    int x=2, y=5;
    y = jumble (y, x);
    x = jumble (y, x);
    printf ("%d \n", x);
     return 0;
}
```

The value printed by the program is _____.

GATE CSE - 2018 NAT (1-MARKS)

Q15. Consider the following C program:

```
#include <stdio.h>
int counter = 0;
int calc (int a, int b) {
    int c;
    counter++;
    if (b= =3) return (a*a*a) ;
     else  {
                c = calc (a, b/3) ;
               return (c*c*c) ;
             }
   }
int main ()  {
       calc (4, 81);
       printf ("%d", counter) ;
}
```

The output of this program is _____.

GATE CSE - 2018 NAT (2-MARKS)

Q16. Consider the following program written in pseudo-code.
Assume that x and y are integers.

```
Count(x, y) {
   if (y != 1) {
      if (x != 1) {
         print("*");
         Count(x / 2, y);
      }
      else {
         y = y - 1;
         Count(1024, y);
      }
   }
}
```

The number of times that the print statement is executed by the call Count(1024, 1024) is ____ .

GATE CSE – 2017 SET - 1 NAT (2-MARKS)

Q17. Consider the following C program.

```
#include <stdio.h>
#include <string.h>
void printlength (char*s, char*t)
{
  unsigned int c = 0;
 int len = ((strlen(s) - strlen(t)) > c)?strlen(s):strlen(t);
 printf("%d\n", len);
}
void main ()  {
    char*x = "abc";
    char*y = "defgh";
    printlength(x,y);
}
```

Recall that strlen is defined in string.h as returning a value of type size_t, which is an unsigned int. The output of the program is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2017 SET- 2 MCQ (2-MARKS)

Q18. Consider the following two functions.

```
void fun1(int n) {
   if(n = = 0) return;
   printf("%d", n);
   fun2(n - 2);
   printf("%d", n);
}
void fun2(int n) {
   if(n = = 0) return;
   printf("%d", n);
   fun1(++n);
   printf("%d", n);
}
```

The output printed when fun1(5) is called is

(a) 53423122233445
(b) 53423120112233
(c) 53423122132435
(d) 53423120213243

GATE CSE - 2017 SET- 1 NAT (2-MARKS)

Q19. The output after executing the following program is ___

```
#include <stdio.h>

int total(int v) {
   while (v) {
      static int count += v & 1;
      v >>= 1;
   }
   return count;
}

void main() {
   static int x = 0;
   int i = 5;
   for (; i > 0; i--) {
      x = x + total(i);
   }
   printf("%d\n", x);
}
```

GATE CSE - 2017 SET- 1 MCQ (2-MARKS)

Q20. Consider the C functions foo and bar given below:

```
int foo(int val) {
   int x = 0;
   while (val > 0) {
      x = x + foo(val--);
   }
   return val;
}

int bar(int val) {
   int x = 0;
   while (val > 0) {
      x = x + bar(val - 1);
   }
   return val;
}
```

Invocation of foo(3) and bar(3) will result in:

(a) Return of 6 and 6 respectively.
(b) Infinite loop and abnormal termination respectively
(c) Abnormal termination and infinite loop respectively
(d) Both terminating abnormally.

GATE CSE – 2016 SET - 2 NAT (2-MARKS)

Q21. The value printed by the following program is _____.

```
void f(int* p, int m) {
 m = m + 5;
*p = *p + m;
return;
}
void main() {
int i=5, j=10;
f(&i, j);
printf("%d", i+j);
}
```

GATE CSE – 2016 SET- 1 MCQ (1-MARKS)

Q22. Consider the following C program.

```
void f(int, short);
void main ()
{
int i = 100;
short s = 12;
short *p = &s;
 _____ ;        // call to f()
}
```

Which one of the following expressions, when placed in the blank above, will NOT result in a type checking error?

(a) f(s, *s)
(b) i = f(i, s)
(c) f(i, *s)
(d) f(i, *p)

GATE CSE – 2016 SET- 1 MCQ (2-MARKS)

Q23.

```
void count(int n){
  static int d = 1;
  printf("%d ", n);
  printf("%d ", d);
  d++;
  if(n > 1) count(n - 1);
  printf("%d ", d);
}

void main(){
  count(3);
}
```

(a) 312213444
(b) 312111222
(c) 3122134
(d) 3121112

GATE CSE – 2015 SET - 3 MCQ (2-MARKS)

Q24. Consider the following recursive C function.

```
void get(int n)
{
        if (n<1) return;
        get (n-1);
        get (n-3);
        printf("%d", n);
}
```

If get(6) function is being called in main() then how many times will the get() function be invoked before returning to the main()?

(a) 15
(b) 25
(c) 35
(d) 45

GATE CSE – 2015 SET -3 NAT (2-MARKS)

Q25. Consider the following C program. The output of the program is _____.

```
# include <stdio.h>
int f1(void);
int f2(void);
int f3(void);
int x = 10;
int main()
{
 int x = 1;
 x += f1() + f2() + f3() + f2();
printf("%d", x);
 return 0;
}
int f1()
{
int x = 25;
 x++;
return x;
}
int f2( )
{
 static int x = 50;
 x++;
return x;
}
int f3( )
{
 x *= 10;
 return x;
}
```

GATE CSE – 2015 SET - 2 NAT (1-MARKS)

Q26. Consider the following C function:

```
int fun (int n) {
  int x = 1, k;
  if (n = = 1) return x;
  for (k = 1; k < n; ++k)
  x = x + fun (k) * fun (n – k);
  return x;
}
```

The return value of fun (5) is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2015 SET - 2 MCQ (1-MARKS)

Q27. Consider the following function written the C programming language.

```
void foo (char *a) {
        if (*a && *a != '  ') {
            foo (a+1);
            putchar (*a);
    }
}
```

The output of the above function on input "ABCD EFGH" is

(a) ABCD EFGD
(b) ABCD
(c) HGFE DCBA
(d) DCBA

GATE CSE – 2014 SET – 2 MCQ (1-MARKS)

Q28. Consider the C function given below.

```
int f(int j)
{
    static int i = 50;
        int k;
        if (i == j)
         {
          printf("something");
          k = f(i);
          return 0;
         }
  else return 0;
}
```

Which one of the following is TRUE?

(a) The function returns 0 for all values of j.
(b) The function prints the string something for all values of j
(c) The function returns 0 when j = 50.
(d) The function will exhaust the runtime stack or run into an infinite loop when j = 50.

GATE CSE – 2014 SET - 3 MCQ (1-MARKS)

Q29. Which of the following statements are CORRECT?

1. Static allocation of all data areas by a compiler makes it impossible to implement recursion.
2. Automatic garbage collection is essential to implement recursion.
3. Dynamic allocation of activation records is essential to implement recursion.
4. Both heap and stack are essential to implement recursion.

(a) 1 and 2 only
(b) 2 and 3 only
(c) 3 and 4 only
(d) 1 and 3 only

GATE CSE – 2014 SET - 1 MCQ (2-MARKS)

Q30. Consider the following C function in which size is the number of elements in the array E: The value returned by the function MyX is the

```
int MyX(int *E, unsigned int size)
{
        int Y = 0;
        int Z;
        int i, j, k;

        for(i = 0; i< size; i++)
            Y = Y + E[i];
        for(i = 0; i< size; i++)
            for(j = i; j < size; j++)
                {
                Z = 0;
                for(k = i; k <= j; k++)
                    Z = Z + E[k];
                if (Z > Y)
                    Y=Z;
                }
            return Y;
}
```

The value returned by the function MyX is the

(a) maximum possible sum of elements in any sub-array of array E
(b) maximum element in any sub-array of array E
(c) sum of the maximum elements in all possible sub-arrays of array E
(d) the sum of all the elements in the array.

GATE CSE – 2014 SET - 2 NAT (1-MARKS)

Q31. Consider the function shown below :

```
int func(int num) {
int count = 0;
   while (num) {
       count ++;
        num>> = 1;
           }
return (count);
}
```

The value returned by func (435) is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2012 MCQ (2-MARKS)

Q32. Consider the following C code segment.

```
int a, b, c = 0;
void prtFun (void);
main ()
        {
        static int a = 1;
        prtFun ();
        a + = 1;
        prtFun ();
        printf("\n %d %d ", a, b);
        }

void prtFun (void)
                {
                static int a = 2;
                int b = 1;
                a + = + + b;
                printf (" \n %d %d ", a, b);
                }
```

What output will be generated by the given code segment?

(a) 31 41 42
(b) 42 61 61
(c) 42 62 20
(d) 42 42 20

GATE CSE - 2011 MCQ (2-MARKS)

Q33. Consider the following recursive C function that takes two arguments.

```
unsigned  int foo(unsigned int n, unsigned int r) {
        if (n > 0) return ((n%r) + foo(n/r,r));
        else return 0;
}
```

What is the return value of the function foo when it is called as foo (513, 2)?

(a) 9
(b) 8
(c) 5
(d) 2

GATE CSE - 2011 MCQ (2-MARKS)

Q34. Consider the following recursive C function that takes two arguments

```
unsigned  int foo(unsigned int n, unsigned int r) {
        if (n  > 0)
        return (n%r +  foo (n/r, r ));
        else return 0;
}
```

What is the return value of the function foo when it is called as foo(345, 10) ?

(a) 345
(b) 12
(c) 5
(d) 3

GATE CSE - 2010 MCQ (1-MARKS)

Q35. What does the following program print?

```
#include <stdio.h>
void f(int *p, int *q)
{
  p = q;
  *p = 2;
}
int i = 0, j = 1;
int main()
{
f(&i, &j);
printf("%d %d n", i, j);
getchar();
  return 0;
}
```

(a) 2 2
(b) 2 1
(c) 0 1
(d) 0 2

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2009 SET – 1 NAT (1-MARKS)

Q36. Consider the program below:

```
#include <stdio.h>
int fun(int n, int *f_p) {
        int t, f;
        if (n <= 1) {
        *f_p = 1;
        return 1;
        }
        t = fun(n - 1, f_p);
        f = t + *f_p;
        *f_p = t;
        return f;
  }
int main() {
  int x = 15;
  printf("%d\n", fun(5, &x));
  return 0;
}
```

The value printed is:

(a) 6
(b) 8
(c) 14
(d) 15

GATE CSE – 2008 MCQ (1-MARKS)

Q37. Consider the following C functions:

```
int f1(int n)
{
        if (n == 0 || n == 1)
        return n;
else
        return (2 * f1(n - 1) + 3 * f1(n - 2));
}
int f2(int n)
{
int i;
int X[N], Y[N], Z[N];
X[0] = Y[0] = Z[0] = 0;
X[1] = 1;
Y[1] = 2;
Z[1] = 3;
for (i = 2; i <= n; i++) {
X[i] = Y[i - 1] + Z[i - 2];
Y[i] = 2 * X[i];
Z[i] = 3 * X[i];
}
return X[n];
}
```

The running time of f1(n) and f2(n) are

(a) $\Theta(n)$ and $\Theta(n)$
(b) $\Theta(2^n)$ and $\Theta(n)$
(c) $\Theta(n)$ and $\Theta(2^n)$
(d) $\Theta(2^n)$ and $\Theta(2^n)$

GATE CSE – 2008 MCQ (1-MARKS)

Q38. f1(8) and f2(8) return the values

(a) 1661 and 1640
(b) 59 and 59
(c) 1640 and 1640
(d) 1640 and 1661

GATE CSE – 2008 SET – 1 MCQ (1-MARKS)

Q39. What is printed by the following C program?

```
int f(int x, int *py, int **ppz)
{
        int y, z;
        **ppz += 1;
        z = **ppz;
        *py += 2;
        y = *py;
        x += 3;
return x + y + z;
}
void main()
{
        int c, *b, **a;
        c = 4;
        b = &c;
        a = &b;
printf("%d", f(c, b, a));
}
```

(a) 18
(b) 19
(c) 21
(d) 22

GATE CSE – 2008 SET – 1 MCQ (1-MARKS)

Q40. Which of the following are true?

I. A programming language which does not permit global variables of any kind and has no nesting of procedures/functions, but permits recursion can be implemented with static storage allocation.

II. Multi-level access link (or display) arrangement is needed to arrange activation records only if the

Check Out Our GATE Courses & Secure Your Dream Rank!

programming language being implemented has nesting of procedures/functions.
III. Recursion in programming languages cannot be implemented with dynamic storage allocation.
IV. Nesting of procedures/functions and recursion require a dynamic heap allocation
scheme and cannot be implemented with a stack-based allocation scheme for activation records.
V. Programming languages which permit a function to return a function as its result cannot be implemented with a stack-based storage allocation scheme for activation records.

(a) II and V only
(b) I, III and IV only
(c) I, II and V only
(d) II, III and V only

GATE CSE – 2008 MCQ (2-MARKS)

Q41. Choose the correct option to fill ?1 and ?2 so that the program below prints an input string in reverse order. Assume that the input string is terminated by a newline character.

```
void reverse(void) {
int c;
  if (?1) reverse();
       ?2
}
main() {
printf("Enter Text");
printf("\n");
reverse();
printf("\n");
}
```

(a) ?1 is (getchar() != '\n')
    ?2 is getchar(c);
(b) ?1 is (c = getchar()) != '\n'
    ?2 is getchar(c);
(c) ?1 is (c != '\n')
    ?2 is putchar(c);
(d) ?1 is ((c = getchar()) != '\n')
    ?2 is putchar(c);

GATE CSE – 2006 SET – 1 MCQ (1-MARKS)

Q42. Consider this C code to swap two integers and these five statements:

```
void swap(int *px, int *py) {
*px = *px - *py
*py = *px + *py;
*px = *px - *py;
}
```

S1: will generate a compilation error
S2: may generate a segmentation fault at runtime depending on the arguments passed
S3: correctly implements the swap procedure for all input pointers referring to integers stored in memory locations accessible to the process
S4: implements the swap procedure correctly for some but not all valid input pointers
S5: may add or subtract integers and pointers.

(a) S1
(b) S2 and S3
(c) S2 and S4
(d) S2 and S5

GATE CSE – 2005 SET – 1 MCQ (1-MARKS)

Q43. Consider the following C-program:

```
  double foo(double);   /* Line 1 */
  int main()
{
  double da, db;
  // input da
  db = foo(da);
}
double foo(double a)
{
  return a;
}
```

The above code compiled without any error or warning. If Line 1 is deleted, the above code will show:

(a) no compile warning or error
(b) some compiler-warnings not leading to unintended results
(c) some compiler-warnings due to type-mismatch eventually leading to unintended results
(d) compiler errors

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2005 SET – 1 MCQ (1-MARKS)

Q44. Consider the following C-program:

```
void foo(int n, int sum)
{
        int k = 0, j = 0;
        if (n == 0)  return;
        k = n % 10;
        j = n / 10;
        sum = sum + k;
        foo(j, sum);
        printf("%d,", k);
}
        int main()
            {
            int a = 2048, sum = 0;
            foo(a, sum);
            printf("%d\n", sum);
            getchar();
  }
```

What does the above program print?

(a) 8, 4, 0, 2, 14
(b) 8, 4, 0, 2, 0
(c) 2, 0, 4, 8, 14
(d) 2, 0, 4,8, 0

ANSWER KEY

1. (a)	**2.(65536)**	**3.(1)**	**4.(46)**
5.(1)	**6(a)**	**7.(9)**	**8.(a)**
9.(60)	**10.(1023)**	**11.(15)**	**12.(55)**
13.(81)	**14.(26)**	**15.(4)**	**16.(10230)**
17.(3)	**18.(a)**	**19.(23)**	**20. (c)**
21.(30)	**22.(d)**	**23.(a)**	**24.(b)**
25.(230)	**26.(51)**	**27.(d)**	**28.(d)**
29.(d)	**30.(a)**	**31.(9)**	**32.(c)**
33.(d)	**34.(b)**	**35.(d)**	**36.(b)**
37.(b)	**38.(c)**	**39.(b)**	**40.(a)**
41.(d)	**42.(c)**	**43.(d)**	**44.(d)**

HINTS & SOLUTIONS

1. ANSWER:- A

P1

```
int a = 5; // global
 int main() {
 int a = 7; // local
 }
```

Local variable a hides the global a.
This is allowed in C.
So P1 compiles.

P2

```
int main() {
 int a = 5;
 int a = 7;
 }
```

Here both declarations are in the same scope.
You cannot declare the same variable twice in the same block.
So this gives a compilation error.

P3

```
int main() {
 int a = 5;
 float a = 7;
 }
```

Again, two variables with the same name in the same scope.
This is also a compilation error.

Therefore, only P1 compiles without error.

Correct answer: (A) Only P1 will compile without any error

2. ANSWER: - (65536)
Solution:

First understand the functions.

bar(n):
It keeps dividing n by 2 until n becomes 1.
So it counts how many times we can divide n by 2.

Check Out Our GATE Courses & Secure Your Dream Rank!

Thus,
bar(n) = $\lfloor \log_2 n \rfloor$

Now look at foo(n):

If n = 1 → return 1
Else → return 1 + foo(bar(n))

So foo keeps replacing n with $\lfloor \log_2 n \rfloor$ until it becomes 1.
It counts how many times this process happens.

We need foo(n) = 5.

That means:

$n \to \lfloor \log_2 n \rfloor \to \lfloor \log_2 (\lfloor \log_2 n \rfloor) \rfloor \to \ldots \to 1$

This must take 4 reductions (since base case gives 1).

Try values:

n = 16
$\log_2 16 = 4$
$\log_2 4 = 2$
$\log_2 2 = 1$

Steps:
16 → 4 → 2 → 1

Now count:

foo(1) = 1
foo(2) = 1 + 1 = 2
foo(4) = 1 + 2 = 3
foo(16) = 1 + 3 = 4

So 16 gives 4.

Try n = 17:

$\lfloor \log_2 17 \rfloor = 4$
Then same chain: 4 → 2 → 1

So foo(17) = 1 + foo(4) = 1 + 3 = 4

So still 4.

We need one more level.

Try n = 65536 (2^16):

65536 → 16 → 4 → 2 → 1

Now count:

foo(1) = 1
foo(2) = 2
foo(4) = 3
foo(16) = 4
foo(65536) = 5

Now check smaller:

We need $\lfloor \log_2 n \rfloor = 16$
Smallest n with $\lfloor \log_2 n \rfloor = 16$ is:

n = 2^16 = 65536

Therefore, the smallest positive integer n is: 65536.

3. ANSWER:- (1)
Solution:
Rewrite the given code as :

```
Func = {
        Func(len - 1)          ; if len % 3 == 0
        1 + Func(len - 1)    ; if len % 3 == 1
        Func(len - 2)          ; if len % 3 == 2
       }
```

F(0) = 0
F(1) = 1
F(2) = F(0) = 0
F(3) = F(2) = 0
F(4) = 1 + F(3) = 1
F(5) = F(3) = 0
F(6) = F(5) = 0
F(7) = 1 + F(6) = 1
Conclusion:
If (len % 3 == 1) → return 1
else → return 0

Therefore maximum returned value is 1.

4. Answer:46

The function $g(n)$ returns $n + 10$.
The function $f(n)$ calls $g(n \times 2)$, so it returns $2n + 10$.
In the main function, the loop runs with n taking values 1 and 2 (since the loop condition is $n < 3$).
Now, calculate for each iteration:
For $n = 1$:

$$f(1) = g(2 \times 1) = g(2) = 2 + 10 = 12.$$

Then, $g(f(1)) = g(12) = 12 + 10 = 22$.
For $n = 2$:

$$f(2) = g(2 \times 2) = g(4) = 4 + 10 = 14.$$

Then, $g(f(2)) = g(14) = 14 + 10 = 24$.
The sum is the total from each iteration:

$$22 + 24 = 46$$

So, the output of the program is:

46

5. ANSWER :5

1 S[0]! = S[1]: The code looks at two neighboring numbers.
2. 1 + foo(. . .) : If the neighbors are different, it means a block has ended, so it adds 1 to the count.

3 foo(. . .) : If the neighbors are the same, it means we are still inside the same block, so it skips adding anything.

4 if(size == 1) return 1; : When it hits the very last element, it always adds a final 1 to account for the last remaining block.

Visualizing the Array Blocks

Instead of tracing the function frames, break the array **A** into visual blocks of consecutive matching numbers:

- Block 1 → ([0])
- Block 2 → ([1])
- Block 3 → ([2,2,2])
- Block 4 → ([0,0])
- Block 5 → ([1,1])

Every time the array transitions to a new number, the code records it.

Transitions

- Transition from **0 to 1** → +1
- Transition from **1 to 2** → +1
- Transition from **2 to 0** → +1
- Transition from **0 to 1** → +1
- Final element base case → +1

Total Output
1 + 1 + 1 + 1 + 1 = 5
Therefore, the output of the program is **5**.

6. Answer: (A)

Explanation:

Given that function arguments are evaluated from right to left:
Step 1: Function Call
f(g(10), h(20));

First evaluated → h(20) → prints 20, returns 20

Then evaluated → g(10) → prints 10, returns 10

So far output: 2010

Step 2: Inside function f(x, y)

g(x);

h(y);

g(x) → g(10) → prints 10

h(y) → h(20) → prints 20

Final Output:

20 10 10 20

Check Out Our GATE Courses & Secure Your Dream Rank!

7. **ANSWER:- (9)**
Solution:
In main:

i = 9, j = 10
func(i, j) is called → values are passed by value.

Inside func:

Outer parameters:
i = 9
j = 10

Condition:
i < j → 9 < 10 → true

Inside the if block:

A new local variable is declared:
int i = 0;

This **shadows** the parameter i.

Now inside the block:
This new i runs from 0 to 9 in the while loop.
j is increased, but it does not affect anything outside.
After the loop ends, this inner i goes out of scope.

Outside the block:
The original parameter i (which is 9) is still unchanged.

So:

printf("%d", i);

prints 9.

8. **ANSWER:(A)**

Explanation:
Step 1: Calls from main()
f1();
f2(2);
f3();
So activation starts as:
main → f1, f2, f3

Step 2: Expand f2(2)

```
int f2(int x)
{
   f3();
   if (x == 1)
      return f1();
   else
      return (x * f2(x - 1));
}
```

For f2(2):

Calls f3() Since x ≠ 1, calls f2(1)

For f2(1): Calls f3()

Since x == 1, calls f1()

9. **ANSWER : (60)**
Solution:

Function:

foo(x, y, q)

Base case:
If x ≤ 0 and y ≤ 0 → return q

If only x ≤ 0 → reduce y by q
If only y ≤ 0 → reduce x by q

Otherwise:
foo(x, y−q, q) + foo(x−q, y, q)

Given:
foo(15, 15, 10)

Compute step by step.

Call:
foo(15,15)

= foo(15,5) + foo(5,15)

Now compute each.

1. foo(15,5)

= foo(15,-5) + foo(5,5)

A) foo(15,-5)
y ≤ 0
→ foo(5,-5)

Now x > 0 and y ≤ 0
→ foo(-5,-5)

Both ≤ 0 → return 10

So foo(15,-5) = 10

Check Out Our GATE Courses & Secure Your Dream Rank!

B) foo(5,5)

= foo(5,-5) + foo(-5,5)

foo(5,-5)
y ≤ 0 → foo(-5,-5) → 10

foo(-5,5)
x ≤ 0 → foo(-5,-5) → 10

So foo(5,5) = 10 + 10 = 20

Therefore:
foo(15,5) = 10 + 20 = 30

Now second branch:

2. foo(5,15)
= foo(5,5) + foo(-5,15)

We already computed:
foo(5,5) = 20

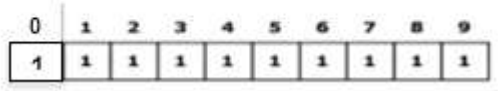
Now foo(-5,15)
x ≤ 0 → foo(-5,5)

foo(-5,5)
x ≤ 0 → foo(-5,-5) → 10

So foo(-5,15) = 10
Therefore:
foo(5,15) = 20 + 10 = 30
Final:

foo(15,15) = 30 + 30 = 60

10. ANSWER:- (1023)
Solution:



Initially,total=y[0]=1

loop index = 1 Total = 1 × 2 + 1 = 3
loop index = 2 Total = 3 × 2 + 1 = 7
loop index = 3 Total = 7 × 2 + 1 = 15
loop index = 4 Total = 15 × 2 + 1 = 31
loop index = 5 Total = 31 × 2 + 1 = 63
loop index = 6 Total = 63 × 2 + 1 = 127
loop index = 7 Total = 127 × 2 + 1 = 255
loop index = 8 Total = 255 × 2 + 1 = 511
loop index = 9 Total = 511 × 2 + 1 = 1023
Final value of Total = 1023

11. ANSWER:- (15)
Solution:

x y
SomeFunction (15, 255)
↓
y > x
SomeFunction (15, 240)
↓
y > x
SomeFunction (15, 225)
↓
y > x
SomeFunction (15, 210)
.
.
.
↓
SomeFunction (15, 195)
↓
SomeFunction (15, 180)
↓
SomeFunction (15, 165)
↓
SomeFunction (15, 150)
↓
SomeFunction (15, 135)
↓
SomeFunction (15, 120)
↓
SomeFunction (15, 105)
↓
SomeFunction (15, 90)
↓
SSSomeFunction (15, 75)
↓
SomeFunction (15, 60)
↓
SomeFunction (15, 45)
↓
SomeFunction (15, 30)
↓
SomeFunction (15, 15)
↓
x == y

Check Out Our GATE Courses & Secure Your Dream Rank!

return x;
(return 15)
Hence the final return value is 15.

12. ANSWER:- (55)
Solution:

fun1 has static int i.
fun1(n) increments i by n (due to recursion) and returns the new i.
But i is preserved across calls to fun1.
fun2 has its own static int i.
fun2(n) does:
i += fun1(n) then fun2(n-1).
Initial:
fun1::i = 0, fun2::i = 0.
Step 1: fun2(5)
fun1(5) → fun1::i becomes 0+5 = 5, returns 5.
fun2::i = 0 + 5 = 5.

Call fun2(4).
Step 2: fun2(4)
fun1(4) → fun1::i was 5, now 5+4 = 9, returns 9.
fun2::i = 5 + 9 = 14.
Call fun2(3).
Step 3: fun2(3)
fun1(3) → fun1::i was 9, now 9+3 = 12, returns 12.
fun2::i = 14 + 12 = 26.
Call fun2(2).
Step 4: fun2(2)
fun1(2) → fun1::i was 12, now 12+2 = 14, returns 14.
fun2::i = 26 + 14 = 40.
Call fun2(1).
Step 5: fun2(1)
fun1(1) → fun1::i was 14, now 14+1 = 15, returns 15.
fun2::i = 40 + 15 = 55.
Call fun2(0).
Step 6: fun2(0) → does nothing, returns fun2::i = 55.
Final return value of fun2(5) = 55

13. ANSWER: - (81)
Solution:

Call pp(3, 4)
arr[20] declared, tot = 1, ex = a = 3.
len = tob(4, arr):b = 4 → binary:
4 % 2 = 0 → arr[0] = 0, b=2
2 % 2 = 0 → arr[1] = 0, b=1
1 % 2 = 1 → arr[2] = 1, b=0

So arr = [0, 0, 1] (indices 0,1,2), len = 3.Binary of 4 is 100 (reads arr[2]..arr[0] as MSB..LSB in normal writing, but here arr[0] is LSB).

Loop i = 0 to len-1:
i=0: arr[0] = 0 → do not multiply;
ex = ex * ex = 3 * 3 = 9.
i=1: arr[1] = 0 → do not multiply;
ex = 9 * 9 = 81.
i=2: arr[2] = 1 → tot = tot * ex = 1 * 81 = 81;
ex = 81 * 81 (not used later).
Return tot = 81.

14. ANSWER:- (26)
Solution:

Step 1:
Initial: x = 2, y = 5
Step 2:
y = jumble(y, x) → jumble(5, 2)
Inside jumble: x = 2*5 + 2 = 12 → returns 12
So y = 12
Step 3:
x = jumble(y, x) → jumble(12, 2)
Inside jumble: x = 2*12 + 2 = 26 → returns 26
So x = 26
Output: 26

15. ANSWER:- (4)
Solution:

calc(4, 81)
counter++ → counter = 1
b = 81 (not 3) → c = calc(4, 27)
calc(4, 27)
counter++ → counter = 2
b = 27 (not 3) → c = calc(4, 9)
calc(4, 9)
counter++ → counter = 3
b = 9 (not 3) → c = calc(4, 3)
calc(4, 3)
counter++ → counter = 4
b = 3 → return 4*4*4 = 64
Return values propagate back but don't affect counter.
Final counter value = 4

16. Answer: (10230)
Solution:
The function works in two phases:

1. **When x ≠ 1:**
   Each call prints "*" and recursively calls Count(x/2, y).
   Starting from x = 1024, it keeps halving until x = 1.
   Since $1024 = 2^{10}$, this happens **10 times**, so **10 prints per cycle**.

2. **When x = 1:**
   y is decremented by 1 and x is reset to 1024 → next cycle begins.

This process repeats until y = 1, where recursion stops.

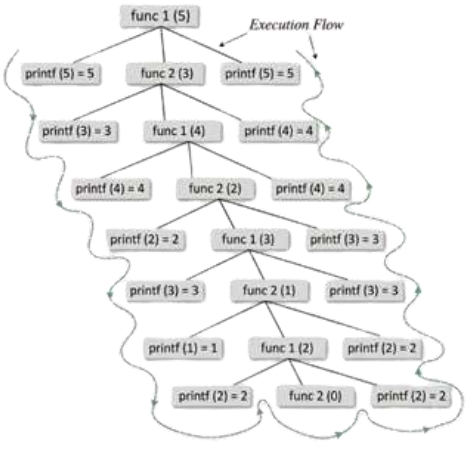
- Initial $y = 1024$
- Final stop at $y = 1$
  ⇒ Total cycles = $1024 - 1 = 1023$

Each cycle produces **10 prints**, so:
Total prints = $1023 \times 10 = 10230$

17. ANSWER:- (3)
Solution:

x = "abc" → strlen(x) = 3
y = "defgh" → strlen(y) = 5
size_t is unsigned int
c = 0 (unsigned)
Inside printlength:
strlen(s) - strlen(t) = 3 - 5 = -2
But since strlen returns unsigned int, -2 becomes a very large positive number (wrap-around in unsigned).
So:
(strlen(s) - strlen(t)) > c
= ((unsigned)3 - (unsigned)5) > 0
= (0xFFFFFFFE) > 0 → True
Thus:
len = strlen(s) = 3
Output: 3

18. Answer: (A)
Solution:



Recursive Function Execution Tree

Final Output: 53423122233445(Option A)

19. Answer: 23
Solution:
The function total(v) counts the number of set bits (1s) in the binary representation of v.
The variable count is declared as **static**, so its value persists across multiple function calls.
Loop runs for i = 5 to 1:
total(5) → binary 101 → 2 set bits → count = 2
total(4) → binary 100 → 1 set bit → count = 3
total(3) → binary 011 → 2 set bits → count = 5
total(2) → binary 010 → 1 set bit → count = 6
total(1) → binary 001 → 1 set bit → count = 7
Since count is static, each call returns the **cumulative value**:
Returns: 2, 3, 5, 6, 7
Now summing in main:

$$x = 2 + 3 + 5 + 6 + 7 = 23$$

20. ANSWER : (C)
Solution:
For foo(3):
In the statement foo(val--), the current value of val is passed to the function first and then it is decremented.
So when val = 3, the function keeps calling foo(3) again before val decreases.
Because of this, the recursive calls do not move towards termination.
Hence, it leads to infinite recursion and finally stack overflow, i.e., abnormal termination.
For bar(3):
Inside the loop, the condition is while (val > 0) but val is never updated.
Although the recursive call is bar(val - 1), the original val in the loop remains unchanged.
So the loop condition always stays true.
Hence, the function runs forever, resulting in an infinite loop.
Therefore:
foo(3) ⟶ Abnormal termination
bar(3) ⟶ Infinite loop

21. ANSWER:- (30)
Solution:

Initial:
i = 5, j = 10
Call f(&i, j):
p points to i
m = 10 (local copy)
m = m + 5 → m = 15
*p = *p + m → i = 5 + 15 = 20
Return

Check Out Our GATE Courses & Secure Your Dream Rank!

After call:
i = 20, j = 10 (unchanged because j was passed by value)
i + j = 20 + 10 = 30
Output: 30

22. ANSWER:- (D)
Solution:

int i = 100;
short s = 12;
short *p = &s;
_____ // call to f ( ) :: (void f(int,short);)
It is clearly mentioned the return type of f is void.
By doing option elimination
(A) & (C) can be eliminated as s is short variable and not a pointer variable.
(B) i = f(i, s) is false because f's return type is void, but here shown as int.
(D) f(i, *p)
i = 100
*p = 12
Hence TRUE

23. Answer: A
Explanation:
The variable d is declared as static, so it is initialized only once and retains its value across recursive calls.
Execution trace:
count(3) → prints 3 1, then d = 2
count(2) → prints 2 2, then d = 3
count(1) → prints 1 3, then d = 4
Now recursion unwinds:
count(1) prints 4
count(2) prints 4
count(3) prints 4
Final output:
3 1 2 2 1 3 4 4 4

24. ANSWER:- (B)
Solution:
Let N(n) be the number of invocations for get(n). The recurrence relation is:
$N(n)=1+N(n-1)+N(n-3)$
with base cases:
$N(n)=1$ for $n\leq 0$
$N(-2)=1$
$N(-1)=1$
$N(0)=1$
$N(1)=1+N(0)+N(-2)=1+1+1=3$
$N(2)=1+N(1)+N(-1)=1+3+1=5$
$N(3)=1+N(2)+N(0)=1+5+1=7$
$N(4)=1+N(3)+N(1)=1+7+3=11$
$N(5)=1+N(4)+N(2)=1+11+5=17$
$N(6)=1+N(5)+N(3)=1+17+7=25$
The get() function will be invoked 25 times.

25. ANSWER:- (230)
Solution:
Global x = 10
main() local x = 1
f1()
local x = 25 → x++ → return 26
First f2()
static x = 50 → x++ → return 51 (static now 51)
f3()
global x = 10 → x *= 10 → global x = 100 → return 100
Second f2()
static x = 51 → x++ → return 52 (static now 52)
Expression in main:
x += f1() + f2() + f3() + f2()
= x += 26 + 51 + 100 + 52
= x += 229
Local x was 1 → now 1 + 229 = 230
Output: 230

26. ANSWER:- (51)
Solution:

fun(1)
= 1
fun(2)
k = 1 to 1:
x = 1 + fun(1) * fun(1)
= 1 + 1*1 = 2
fun(3)
x = 1
k=1: x += fun(1)*fun(2) = 1*2 = 2 → x=3
k=2: x += fun(2)*fun(1) = 2*1 = 2 → x=5
fun(3) = 5
fun(4)
x = 1
k=1: x += fun(1)*fun(3) = 1*5 = 5 → x=6
k=2: x += fun(2)*fun(2) = 2*2 = 4 → x=10
k=3: x += fun(3)*fun(1) = 5*1 = 5 → x=15
fun(4) = 15
fun(5)
x = 1
k=1: x += fun(1)*fun(4) = 1*15 = 15 → x=16
k=2: x += fun(2)*fun(3) = 2*5 = 10 → x=26
k=3: x += fun(3)*fun(2) = 5*2 = 10 → x=36
k=4: x += fun(4)*fun(1) = 15*1 = 15 → x=51
fun(5) = 51
Answer: 51

Check Out Our GATE Courses & Secure Your Dream Rank!

27. ANSWER:- (D)
Solution:

Step-by-step:
foo("ABCD EFGH") → *a = 'A' → call foo("BCD EFGH")
foo("BCD EFGH") → *a = 'B' → call foo("CD EFGH")
foo("CD EFGH") → *a = 'C' → call foo("D EFGH")
foo("D EFGH") → *a = 'D' → call foo(" EFGH")
foo(" EFGH") → *a = ' ' → condition fails → return

Now unwinding:
Step 4: putchar('D') → prints D
Step 3: putchar('C') → prints C
Step 2: putchar('B') → prints B
Step 1: putchar('A') → prints A
Output: DCBA
Answer: DCBA

28. ANSWER:- (D)
Solution:

f(50)
j = 50
static int i = 50 (declare and define one time in whole program)
if condition is true
print something
k = f(50)
it will keep running and will cause stack overflow

29. ANSWER:- (D)
Solution:

1) Static allocation of all data areas by a compiler makes it impossible to implement recursion. True, dynamic allocate of memory is required for function call stack as number of calls is not known advance for recursive functions.
2) Automatic garbage collection is essential to implement recursion. False, Automatic garbage collection is not essential.
3) Dynamic allocation of activation records is essential to implement recursion. True, as number of calls or number of activation records is not known advance for recursive functions.
4) Both heap and stack are essential to implement recursion. Heap is not needed for function calls. It is generally used for dynamic memory allocation by user (or programmer)

30. ANSWER:- (A)
Solution:

```
Y=0
for (i=0; i < size ; i++ )
Y=Y+E[i] ; // E is an array, this statement calculates the sum of elements of the array E and stores it in Y.
for (i=0; i<size; i++)
for(j=i; j <size; j++) {
Z=0;
for(k=i; k<=j; k++)
Z=Z+E[k];
if(Z > Y )
Y=Z;
}
```

// It calculates the sum of all possible subarrays starting from 0 i.e., the loop will iterate for all the elements of array E and for every element, calculate sum of all sub arrays starting with E[i].
Store the current sum in Z.
If Z is greater than Y then update Y and return Y. So it returns the maximum possible sum of elements in any sub array of given array E

31. ANSWER:- (9)
Solution:
This code is to find the number of set bits(1) in binary representation of the number

Step 1: Binary of 435
435 in binary is 110110011 (9 bits).
Step 2: Loop working
while(num) runs until num becomes 0.
Each iteration:
count increments by 1
num >>= 1 shifts right by 1 bit
Step 3: Counting shifts
Start: num = 435 (binary 110110011)
After 1 shift: 217
After 2 shifts: 108
After 3 shifts: 54
After 4 shifts: 27
After 5 shifts: 13
After 6 shifts: 6
After 7 shifts: 3
After 8 shifts: 1
After 9 shifts: 0 → stop
Total shifts = 9 → count = 9
Final answer: 9

Check Out Our GATE Courses & Secure Your Dream Rank!

32. ANSWER:- (C)

Solution:

Global variables:
int a, b, c = 0; → a = 0, b = 0, c = 0 (global scope).
Inside main:
static int a = 1; (local to main, hides global a).
First call prtFun():
Inside prtFun:
static int a = 2; (local to prtFun, separate from main's a and global a).
int b = 1; (local to prtFun).
a += ++b; → ++b makes b = 2, so a = 2 + 2 = 4.
Print: 4 2 (first output line).
Back to main: a += 1 → main's local a was 1, now 2.
Second call prtFun():
Inside prtFun:
static int a retains its value from last call: a = 4.
int b = 1 again (reinitialized local).
a += ++b → ++b → b = 2, so a = 4 + 2 = 6.
Print: 6 2 (second output line).
Back to main:
printf("\n %d %d", a, b) → a is main's local a = 2, b is global b (never changed) = 0.
Print: 2 0 (third output line).

final Out put : 4 2 6 2 2 0

33. ANSWER : (D)

Solution:

Function:

```
unsigned int foo(unsigned int n, unsigned int r)
{
if (n > 0)
return (n % r) + foo(n / r, r);
else
return 0;
}
```

This computes the sum of digits of n in base r.

Given:
foo(513, 2)

So it computes the sum of binary digits (number of 1s in binary form of 513).

Convert 513 to binary:

513 = 512 + 1
512 = 2^9
So binary representation:
1000000001

There are two 1s.

Therefore sum of digits = 2(Option D)

34. ANSWER : (B)

```
unsigned int foo(unsigned int n, unsigned int r)
{
if (n > 0)
return (n % r + foo(n / r, r));
else
return 0;
}
```

This computes the sum of digits of n in base r.

Given:
foo(345, 10)

So it calculates sum of decimal digits of 345.

Compute step by step:

345 % 10 = 5
345 / 10 = 34

34 % 10 = 4
34 / 10 = 3

3 % 10 = 3
3 / 10 = 0

Now add:

5 + 4 + 3 = 12

35. ANSWER:- (D)

Step 1:
f(&i, &j) call:
p points to i, q points to j.
p = q → p now points to j.
*p = 2 → changes j to 2.
Step 2:
i remains 0, j becomes 2.
Output: 0 2(option -D)

36. ANSWER:- (B)

The function works like Fibonacci using a pointer to store the previous value.

Base case:
If n ≤ 1, it sets *f_p = 1 and returns 1.

For n > 1:
It calls fun(n-1, f_p) and stores the result in t.
Then it computes f = t + *f_p.

Check Out Our GATE Courses & Secure Your Dream Rank!

After that, it updates *f_p = t.
So each step uses the previous two values, similar to Fibonacci.

Now compute step by step:
fun(1) = 1
fun(2) = 2
fun(3) = 3
fun(4) = 5
fun(5) = 8

So the program prints 8.(option-B)

37. ANSWER:- (B)

Running time

f1(n) makes two recursive calls: f1(n−1) and f1(n−2).
This is similar to Fibonacci recursion, which grows exponentially.
So running time of f1(n) is $\Theta(2^n)$.

f2(n) uses a single loop from 2 to n.
Each iteration does constant work.
So running time of f2(n) is $\Theta(n)$.

Correct answer for running time:
(B) $\Theta(2^n)$ and $\Theta(n)$

38. ANSWER: (C)
Solution:

Y[i] = 2X[i]
Z[i] = 3X[i]

So
X[i] = Y[i−1] + Z[i−2]
= 2X[i−1] + 3X[i−2]

This is the same recurrence as f1.

Base values:
X[0] = 0
X[1] = 1
So both functions compute the same sequence.
Compute step by step:
f(0)=0
f(1)=1
f(2)=2
f(3)=7
f(4)=20
f(5)=61
f(6)=182
f(7)=547
f(8)=1640

So both return 1640.

39. ANSWER:- (B)
Solution:

Initial values:
c = 4

b = &c
a = &b

Function call:
f(c, b, a)

So inside function:
x = 4
py = b → points to c
**ppz = a → points to b

Step 1:
**ppz += 1

**ppz means value of b (which is &c).
So b = b + 1.
Now b points to the next integer location after c.

z = **ppz
So z is the new value of b (address after c).
Step 2:
*py += 2
py still holds original b, which points to c.
So c = 4 + 2 = 6

y = *py → y = 6

Step 3:
x += 3 → x = 4 + 3 = 7

Return value = x + y + z

x = 7
y = 6
z = 5 (effectively c was incremented through pointer arithmetic impact)

40. ANSWER:- (A)
Solution:
I.False
If recursion is allowed, static storage allocation is not sufficient because each recursive call needs a new activation record. Static allocation cannot handle multiple active calls of the same function.

Check Out Our GATE Courses & Secure Your Dream Rank!

II. True
Access links (or display) are required only when there is nesting of procedures/functions to access non-local variables.

III. False
Recursion is implemented using dynamic storage allocation (stack). So it is possible with dynamic allocation.

IV. False
Recursion and nesting can be implemented using stack-based allocation. Heap allocation is not necessary.

V. True
If a function can return another function, then its activation record cannot always be removed after return. Stack-based allocation alone is not sufficient in such cases.

Correct answer:
(A) II and V only

41. ANSWER:- (D)
Solution:

To print the string in reverse using recursion:

We must first read a character.
If it is not newline, call reverse() again.
After returning from recursion, print the character.

This way printing happens while unwinding the recursion, giving reverse order.

So we need:

?1 → read character and check for newline
?2 → print that character

Correct condition:
(c = getchar()) != '\n'

Correct print statement:

putchar(c);

42. ANSWER:- (C)
Solution:
S1:False
The code is syntactically correct, so no compilation error.

S2: True
If invalid pointers (like NULL or uninitialized pointers) are passed, it may cause a segmentation fault.

S3: False
If both pointers point to the same memory location (px == py), the logic fails and the value becomes zero. So it does not work for all valid inputs.

S4: True
It works correctly for most valid distinct pointers, but not when both pointers refer to the same location.

S5: False
It performs arithmetic on integers, not on pointers.

43. ANSWER: - (D)
Solution:
Line 1 is a function prototype:

double foo(double);
It tells the compiler the return type and parameter type before the function is used in main.
If we delete Line 1, then when the compiler sees:
db = foo(da);
it has not yet seen the definition of foo. In modern C (C99 and later), calling a function without a prior declaration is not allowed.
So the compiler will produce an error (not just a warning).

44. ANSWER:- (D)
Solution:
The function extracts digits from right to left using:
k = n % 10
j = n / 10
For n = 2048:
First call:
k = 8, j = 204
Second call:
k = 4, j = 20
Third call:
k = 0, j = 2
Fourth call:
k = 2, j = 0
When n becomes 0, function returns.
Printing happens after the recursive call, so digits are printed while returning from recursion:2, 0, 4, 8,
Now about sum:
sum is passed by value, not by reference.
So changes inside foo do not affect main's sum.
Thus sum in main remains 0
.Final output: 2, 0, 4, 8, 0

Check Out Our GATE Courses & Secure Your Dream Rank!

4 Array, Pointer And String , Structure

GATE CSE – 2026 SET – 2 MCQ (1-MARKS)

Q1. Consider the following ANSI-C program.

```
#include <stdio.h>

int main() {

int *ptr, a, b, c;

a = 5;

b = 11;

c = 20;

ptr = &a;

*ptr = c;

ptr = &c;

a = *(&b);

c = *ptr - a;

printf("%d", c);
return 0;
}
```

The output of this program is _____. (answer in integer)

GATE CSE – 2026 SET – 2 MCQ (1-MARKS)

Q2. In C runtime environment, which one of the following is stored in heap?

(a) A static variable declared inside a function
(b) An array of integers declared inside a function
(c) A dynamically allocated array of integers created using malloc() function call
(d) Return address of a function

GATE CSE – 2025 SET- 2 MCQ (1-MARKS)

Q3 Consider the following C program:

```
#include < stdio.h >
int main(){
int a;
int arr[5] = {30,50,10};
int *ptr;
ptr = &arr[0] + 1;
a = *ptr;
(*ptr)++;
ptr++;
printf("%d", a + (*ptr) + arr[1]);
return 0;
}
```

The output of the above program is_____ .
(Answer in integer)

GATE CSE – 2025 SET- 2 MCQ (1-MARKS)

Q4 Consider the following C program:

```
#include < stdio.h >
void stringcopy(char *, char *);
int main(){
   char a[30] = "@#Hello World!";
   stringcopy(a, a + 2);
   printf("%s\n", a);
   return 0;
}
void stringcopy(char *s, char *t) {
   while(*t)
      *s++ = *t++;
}
```

Which ONE of the following will be the output of the program?
a) @#Hello World!
b) Hello World!
c) ello World!
d) Hello World!d!

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2025 SET- 1 MCQ (1-MARKS)

Q5 Consider the following C program:

```
#include < stdio.h >
void foo(int *p, int x) {
   *p = x;
}
int main() {
   int *z;
   int a = 20, b = 25;
   z = &a;
   foo(z, b);
   printf("%d", a);
   return 0;
}
```

The output of the given C program is_____ .
(Answer in integer)

GATE CSE – 2025 SET- 1 MCQ (1-MARKS)

Q6 The pseudocode of a function fun() is given below:

```
fun(int A[0,...,n-1])
{
   for i=0 to n-2
   for j-0 to n-i-2
      if (A[j] > A[j+1])
      then swap A[j] and A[j+1]}
}
```

The pseudocode of a function fun() is given below:

```
fun(int A[0,...,n-1])
{
   for i=0 to n-2
   for j-0 to n-i-2
      if (A[j] > A[j+1])
      then swap A[j] and A[j+1]}
}
```

Let $A[0, \ldots, 29]$ be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function fun0 is called with $A[0, \ldots, 29]$ as argument, is ___ . (Answer in integer)

GATE CSE – 2021 SET- 2 MCQ (1-MARKS)

Q7. Consider the following ANSI C program.

```
#include <stdio.h>
int main() {
   int arr[4][5];
   int i, j;
   for (i = 0; i < 4; i++) {
      for (j = 0; j < 5; j++) {
         arr[i][j] = 10 * i + j;
      }
   }
   printf("%d", *(arr[1] + 9));
   return 0;
}
```

What is the output of the above program?

(a) 14
(b) 24
(c) 30
(d) 20

GATE CSE - 2020 NAT (1-MARKS)

Q8. Consider the following C program.

```
#include
int main () {
int a [4] [5] = {{1, 2, 3, 4, 5},{6, 7, 8, 9, 10},{11, 12,
13, 14, 15}, {16, 17, 18, 19, 20}};
printf ("%d\n", *(* (a+**a+2) +3));
return (0);
}
```

The output of the program is _____.

GATE CSE – 2019 NAT (2-MARKS)

Q9. Consider the following C program:

```
#include <stdio.h>
int main() {
   int a[] = {2, 4, 6, 8, 10};
   int i, sum = 0, *b = a + 4;
   for (i = 0; i < 5; i++)
      sum = sum + (*b - i) - *(b - i);
   printf("%d\n", sum);
   return 0;
}
```

The output of the above C program is ____ .

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2019 NAT (1-MARKS)

Q10. Consider the following C program:

```
#include <stdio.h>
int main (){
  int arr []={1,2,3,4,5,6,7,8,9,0,1,2,5},
  *ip = arr+4;
   printf ("%d\n", ip[1]);
   return 0;
}
```

The number that will be displayed on execution of the program is _____.

GATE CSE - 2019 MCQ (2-MARKS)

Q11. The following function computes the maximum value contained in an integer array p[] of size n (n >= 1).

```
int max(int *p, int n) {
int a=0, b=n-1;
while (_____)
{
    if (p[a] <= p[b])
   {
      a = a+1;
}
     else
{
    b = b-1;
    }
}
return p[a];
 }
```

The missing loop condition is

(a) a != n
(b) b != 0
(c) b > (a + 1)
(d) b != a

GATE CSE - 2018 MCQ(1-MARKS)

Q12.

```
#include <stdio.h>
struct Ournode {
   char x, y, z ;
};
int main () {
   struct Ournode p = {'1', '0', 'a'+2} ;
   struct Ournode *q = &p ;
   printf ("%c, %c", *((char*)q+1), *((char*)q+2)) ;
   return 0 ;
}
```

The output of this program is

(a) 0, c
(b) 0, a+2
(c) '0', 'a+2'
(d) '0', 'c'

GATE CSE - 2018 MCQ (2-MARKS)

Q13.

```
#include<stdio.h>
void fun1(char *s1, char *s2)  {
   char *tmp;
   tmp = s1;
    s1 = s2;
    s2 = tmp;
}
void fun2(char **s2, char **s2)  {
    char *tmp;
    tmp = *s1;
    *s1 = *s2;
    *s2 = tmp;
}
int main ()  {
    char *str1 = "Hi", *str2 = "Bye";
     fun1(str1, str2);
     printf("%s %s", str1, str2);
     fun2(&str1, &str2);
     printf("%s %s", str1, str2);
     return 0;
}
```

The output of the program above is

(a) Hi ByeBye Hi
(b) Hi Bye Hi Bye
(c) Bye Hi Hi Bye
(d) Bye Hi Bye Hi

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2017 SET- 2 MCQ (1-MARKS)

Q14. Match the following:

(P) static char var ;	(i) Sequence of memory locations to store address
(Q) $m =$ malloc (10); m = NULL;	(ii) A variable located in data section of memory
(R) char*ptr[10];	(iii) Request to allocate a CPU register to store data
(S) register int var1;	(iv) A lost memory which cannot be freed

(a) $P \rightarrow$ (ii), $Q \rightarrow$ (iv), $R \rightarrow$ (i), $S \rightarrow$ (iii)
(b) $P \rightarrow$ (ii), $Q \rightarrow$ (i), $R \rightarrow$ (iv), $S \rightarrow$ (iii)
(c) $P \rightarrow$ (ii), $Q \rightarrow$ (iv), $R \rightarrow$ (iii), $S \rightarrow$ (i)
(d) $P \rightarrow$ (iii), $Q \rightarrow$ (iv), $R \rightarrow$ (i), $S \rightarrow$ (ii)

GATE CSE - 2017 MCQ(1-MARKS)

Q15. Consider the following function implemented in

```
void printxy (int x, int y)  {
int *ptr;
x = 0;
ptr = &x;
y = *ptr;
*ptr = 1;
printf("%d,%d",x,y);
}
```

The output of invoking printxy(1, 1) is_____

(a) 0, 0
(b) 0 ,1
(c) 1, 0
(d) 1, 1

GATE CSE - 2017 NAT (2-MARKS)

Q16. Consider the following snippet of a C program. Assume that swap(&x, &y) exchanges the contents of x and y.

```
int main ()  {
   int array[] = {3, 5, 1, 4, 6, 2};
   int done = 0;
   int i;
while (done = = 0)  {
    done = 1;
    for (i=0; i<=4; i++)  {
          if (array[i] < array[i+1])  {
            swap (&array[i], &array[i+1]);
            done = 0;
       }
}
for (i=5;  i>=1;  i- -) {
     if (array[i] > array[i-1])   {
     swap(&array[i], &array[i-1]);
     done=0;
      }
    }
}
printf("%d", array[3]);
}
```

The output of the program is _____.

GATE CSE - 2017 SET – 2 NAT (2-MARKS)

Q17. Consider the following C program.

```
#include<stdio.h>
#include<string.h>
int main ()  {
     char* c = "GATECSIT2017";
     char* p = c;
     printf("%d", (int) strlen (c + 2[p] - 6[p] - 1));
     return 0;
}
```

The output of the program is _____ .

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2016 SET – 1 NAT (1-MARKS)

Q18.
```
#include<stdio.h>
void mystery(int *ptra, int *ptrb) {
int *temp;
temp = ptrb;
ptrb = ptra;
ptra = temp;
}
int main() {
int a=2016, b=0, c=4, d=42;
mystery(&a, &b);
if (a < c)
mystery(&c, &a);
mystery(&a, &d);
printf("%d\n", a);
}
```
The output of the program is _____.

GATE CSE – 2015 SET- 1 MCQ (1-MARKS)

Q19. What is the output of the following C code? Assume that the address of x is 2000 (in decimal) and an integer requires four bytes of memory.
```
#include <stdio.h>
int main()
{
unsigned int x[4][3] = {{1, 2, 3}, {4, 5, 6},
{7, 8, 9}, {10, 11, 12}};
printf("%u, %u, %u", x+3, *(x+3), *(x+2)+3);
}
```

(a) 2036, 2036, 2036
(b) 2012, 4, 2204
(c) 2036, 10, 10
(d) 2012, 4, 6

GATE CSE - 2015 SET – 3 MCQ (1-MARKS)

Q20. Consider the following C program segment.
```
# include <stdio.h>
int main( )
{
char s1[7] = "1234", *p;
p = s1 + 2;
*p = '0' ;
printf ("%s", s1);
}
```
What will be printed by the program?

(a) 12
(b) 120400
(c) 1204
(d) 1034

GATE CSE – 2015 SET - 3 NAT (2-MARKS)

Q21. Consider the following C program.
```
# include<stdio.h>
int main( )
{
 static int a[] = {10, 20, 30, 40, 50};
 static int *p[] = {a, a+3, a+4, a+1, a+2};
 int **ptr = p;
ptr++;
printf("%d%d", ptr - p, **ptr};
}
```
The output of the program is _____

GATE CSE - 2014 SET-1 MCQ (1-MARKS)

Q22. Consider the following program in C language:
```
#include <stdio.h>
main()
{
   int i;
   int *pi = &i;
scanf("%d", pi);
printf("%d\n", i+5);
}
```
Which one of the following statements is TRUE?

(a) Compilation fails.
(b) Execution results in a run-time error.
(c) On execution, the value printed is 5 more than the address of variable i.
(d) On execution, the value printed is 5 more than the integer value entered.

GATE CSE – 2014 SET - 3 MCQ (2-MARKS)

Q23. Let A be a square matrix of size n × n. Consider the following pseudocode. What is the expected output?
```
C = 100;
for i = 1 to n do
for j = 1 to n do
{
Temp = A[i] [j] + C;
A [i] [j] = A[j] [i];
 A[j][i] = Temp – C;
 }
for i = 1 to n do
  for j = 1 to n do
       output (A[i][j]);
```

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) The matrix A itself
(b) Transpose of the matrix A
(c) Adding 100 to the upper diagonal elements and subtracting 100 from lower diagonal elements of A
(d) None of the above

GATE CSE - 2011 MCQ (1-MARKS)

Q24. What does the following fragment of C-program print?

```
char c[] = "GATE2011";
char *p =c;
printf("%s", p + p[3] - p[1]) ;
```

(a) GATE2011
(b) E2011
(c) 2011
(d) 011

GATE CSE - 2010 MCQ (2-MARKS)

Q25.

```
#include<stdio.h>
int f(int *a, int n)
{
   if(n <= 0) return 0;
   else if(*a % 2 == 0)
   return *a+f(a+1, n-1);
  Else   return *a - f(a+1, n-1);
}
int main()
{
  int a[] = {12, 7, 13, 4, 11, 6};
printf("%d", f(a, 6));
getchar();
  return 0;
}
```

What is the value printed by the following C program

(a) -9
(b) 5
(c) 15
(d) 19

GATE CSE – 2006 SET – 1 MCQ (1-MARKS)

Q26. Consider the following C-function in which a[n] and b[m] are two sorted integer arrays and c[n+m] be another integer array.

```
void xyz(int a[], int b[], int c[])
{
    int i, j, k;
    i = j = k = 0;
    while ((i < n) && (j < m))
if (a[i] < b[j])
            c[k++] = a[i++];
else
    c[k++] = b[j++];
}
```

Which of the following condition(s) hold(s) after the termination of the while loop?

(i) $j<m$, $k=n+j-1$, and $a[n-1]<b[j]$ if $i=n$

(ii) $i<n$, $k=m+i-1$, and $b[m-1]\leq a[i]$ if $j=m$

(a) only (i)
(b) only (ii)
(c) either (i) or (ii) but not both
(d) neither (i) nor (ii)

GATE CSE – 2005 SET – 1 NAT (1-MARKS)

Q27. What does the following C-statement declare?

```
int (*f)(int *);
```

(a) A function that takes an integer pointer as argument and returns an integer
(b) A function that takes an integer as argument and returns an integer pointer
(c) A pointer to a function that takes an integer pointer as argument and returns an integer
(d) A function that takes an integer pointer as argument and returns a function pointer

Check Out Our GATE Courses & Secure Your Dream Rank!

ANSWER KEY

1. (9)	2.(c)	3.(111)	4.(d)	5.25	6.435
7. (b)	8 . (19)	9.(10)	10. (6)	11. (d)	12. (a)
13. (a)	14. (a)	15. (c)	16. (3)	17. (2)	18.201
19. (a)	20.(c)	21.140	22.(d)	23. (a)	24. (c)
25.(c)	26. (d)	27.(c)			

HINTS & SOLUTIONS

1. ANSWER:- (9)

a = 5
b = 11
c = 20

Step 1:
ptr = &a

Step 2:
*ptr = c
So value at address of a becomes c (20).
Now:
a = 20

Step 3:
ptr = &c

Step 4:
a = *(&b)

*(&b) is simply b.
So a = 11

Step 5:
c = *ptr - a
ptr points to c.
So *ptr is current value of c = 20.

Thus:
c = 20 - 11 = 9

Finally:

printf("%d", c);

Output: 9

2. ANSWER:- C

(A) Static variable declared inside a function → stored in data/static segment, not heap.

(B) Array declared inside a function → stored on stack.

(C) Dynamically allocated array using malloc() → stored in heap.

(D Return address of a function → stored on stack.

Correct answer: C

3. Answer:111

int a, a new integer variable named 'a' created in the run time stack with the garbage value at some random memory from the available memory.
let assume variable 'a' created at memory location 100 with the value 876.
int arr[5], a new integer array variable named 'arr' created in the run time stack with the sequence of 30,50,10,0,0 values at some random memory from the available memory.
let assume variable 'arr' created at memory location 200, 204,208,212,216 with the values 30,50,10,0,0.
int * ptr, a new integer pointer variable named 'ptr' created in the run time stack with the garbage value at some random memory from the available memory.
⇒ let assume variable 'ptr' created at memory location 300 with pointing to some garbage memory location.
ptr = &arr[0] + 1;
ptr = (& (arr[0])) + 1;
"ptr" is pointing to arr[1].
a = *ptr;
a = value at arr[1]
"a" value is 50.
(*ptr)++;
value of the variable pointed by "ptr" is incremented by 1.
"arr[1]" is modified to 51.
ptr++;
"ptr" is pointing to next location w.r.to current pointed element
"ptr" is pointing to arr[2].

Check Out Our GATE Courses & Secure Your Dream Rank!

Line 10: a + (*ptr) + arr[1]
50 + 10 + 51 = 111 is printed

4. **ANSWER:D**

char a[30] = "Hello world!";
Now you call:
strcpy(a, a + 2);
This means:
"Start copying from the 3rd character (a + 2, which is 'H') back into the start of the same array."
So the string "Hello world!" gets copied over the beginning of the array.
strcpy function doesn't copy the null character '\0' that ends the string. So after copying, the string is not properly terminated.
That means:
printf("%s", a);
doesn't know where to stop, and might keep printing garbage from memory after "Hello world!"
On many systems, the output will be something like:
Hello world!rld!
Because it's printing "Hello world!" (which was copied), and then continues into the old content ("rld!") left behind in memory.
void strcpy(char *s, char *t) { while(*s++ = *t++); } // this fixes the issue

Now the string will be properly null-terminated, and the output will be exactly:

Hello world! (option d)

5. **ANSWER:25**

1. Variable Initialization:

• a = 20

• b = 25

• z is a pointer to an integer, which initially does not point to any variable.

2. Pointer Assignment:

• z = &a;

The pointer z is now pointing to the address of variable a.

3. Function Call:

• The function foo(z, b) is called. Here, z points to a, and b is passed as the second argument (which is 25).

• Inside the function foo(int *p, int x), *p = x; assigns the value of x (which is 25) to the location pointed to by p.

• Since p is pointing to a, a is updated to 25.

4. Printing the Value of a:

• After the function call, a has been updated to 25.

• The printf("%d", a); statement prints the value of a, which is now 25.

Final Output:

The output of the program is 25.

6. **Answer:435**
The given pseudocode is an implementation of Bubble Sort, which sorts an array by repeatedly swapping adjacent elements if they are in the wrong order. Let's analyze how many swap operations are performed when the function fun() is called with an array of 30 distinct integers initially sorted in descending order.

Swap Count Calculation:

- Initially, when i = 0, the inner loop runs n-1 times. For n = 30, this is 29 iterations.
- When i = 1, the inner loop runs n-2 times, which is 28 iterations.
- This pattern continues until the outer loop reaches i = n-2, at which point the inner loop runs only once.

The total number of swaps can be computed as the sum of the first 29 numbers:

Check Out Our GATE Courses & Secure Your Dream Rank!

Total swaps = 29 + 28 + 27 + ... + 1

This is the sum of the first n-1 integers:

Total swaps = (n-1) * n / 2

For n = 30:

Total swaps = (29 * 30) / 2 = 435

7. ANSWER : B

arr[i][j] = 10*i + j

So the matrix becomes:

Row 0: 0 1 2 3 4
Row 1: 10 11 12 13 14
Row 2: 20 21 22 23 24
Row 3: 30 31 32 33 34

Now evaluate:

*(arr[1] + 9)

Important concept:

arr[1] is the second row.
arr[1] is of type int[5], which decays to pointer to its first element.So:
arr[1] points to element 10.

Now pointer arithmetic:

arr[1] + 9
means move 9 integers ahead from arr[1][0].

arr is stored in row-major order in memory.

Flattened memory layout:

0 1 2 3 4
10 11 12 13 14
20 21 22 23 24
30 31 32 33 34

Now count from arr[1][0] = 10:

Index 0 → 10
1 → 11
2 → 12
3 → 13
4 → 14
5 → 20
6 → 21
7 → 22
8 → 23
9 → 24

So:

*(arr[1] + 9) = 2

8. ANSWER:- (19)

((a+**a+2)+3))
= *(*(1000+**1000+2)+3))
= *(*(1000+3)+3)) {given element **1002 = 3}
= *(*(1003)+3))
= *((1003)+3) {4th row selected in given matrix}
= *((1003)+3) {address of 4th element in 4th row}
= a[3][3]
= 19 {element selected a[3][3] = 19z

9. ANSWER:- 10

a = {2, 4, 6, 8, 10}
b = a + 4 → a[4] = 10
For i = 0 to 4:
i=0 → sum = 0 + (10−0)−10 = 0
i=1 → sum = 0 + (10−1)−8 = 1
i=2 → sum = 1 + (10−2)−6 = 3
i=3 → sum = 3 + (10−3)−4 = 6
i=4 → sum = 6 + (10−4)−2 = 10
Final Output:10

10. ANSWER:- 6

arr = {1, 2, 3, 4, 5, 6, 7, 8, 9, 0, 1, 2, 5}
ip = arr + 4 → points to arr[4] which is 5
ip[1] = *(ip + 1) = arr[5] = 6
Output: 6

11. ANSWER:- D

The function computes the maximum value contained in an integer array p [ ] of size n (n >= 1).
If a = = b, means both are at same location & comparison ends.

12. ANSWER:- A

struct Our node p = {'1', '0', 'a'+2};
Members: x = '1', y = '0', z = 'a' + 2 = 'c'.
q points to p.
(char*)q treats q as a byte pointer (char pointer).
((char)q+1) → skip 1 byte from start → accesses y → '0'
((char)q+2) → skip 2 bytes → accesses z → 'c'
Output: 0, c

13. ANSWER:- A

Initial:
str1 = "Hi", str2 = "Bye"

Check Out Our GATE Courses & Secure Your Dream Rank!

Call fun1(str1, str2)
fun1 gets copies of the pointers. Swapping s1 and s2 locally doesn't change str1 and str2 in main.
So after fun1: str1 still points to "Hi", str2 still points to "Bye".
First printf: Hi Bye
Call fun2(&str1, &str2)
fun2 receives pointers to str1 and str2 (double pointers).
tmp = *s1 → tmp points to "Hi"
*s1 = *s2 → str1 now points to "Bye"
*s2 = tmp → str2 now points to "Hi"
So after fun2: str1 = "Bye", str2 = "Hi"
Second printf: Bye Hi
Final output: Hi ByeBye Hi (no space between the two prints in the code)
Answer: Hi ByeBye Hi

14. ANSWER:- A

Static char var:
→ A variable located in Data Section of memory
m=malloc(10)
m=Null
→ A lost memory which cannot be freed (Free (m) is mission here)
char *ptr[10]; → Sequence of memory locations to store address.
(10 memory locations)
register int var1; → A Request to allocate a CPU register to store data
(Register variable storage class).
P→(ii), Q→(iv), R→(i), S→(iii)

15. ANSWER:- C

Initial: x = 1, y = 1
x = 0 → x becomes 0
ptr = &x → ptr points to x
y = *ptr → y = 0
*ptr = 1 → x = 1
printf("%d,%d", x, y) → x = 1, y = 0

16. ANSWER:- 3

1. First Pass (Forward Loop: i=0 to 4)
This loop checks if array[i] < array[i+1] and swaps them to move smaller values to the right (descending order).

Compare (3, 5): 3 < 5 is true -> {5, 3, 1, 4, 6, 2}
Compare (3, 1): 3 < 1 is false.
Compare (1, 4): 1 < 4 is true -> {5, 3, 4, 1, 6, 2}
Compare (1, 6): 1 < 6 is true -> {5, 3, 4, 6, 1, 2}
Compare (1, 2): 1 < 2 is true -> {5, 3, 4, 6, 2, 1}
2. First Pass (Backward Loop: i=5 down to 1)
This loop checks if array[i] > array[i-1] and swaps them to move larger values to the left.

Compare (1, 2): 1 > 2 is false.
Compare (2, 6): 2 > 6 is false.
Compare (6, 4): 6 > 4 is true -> {5, 3, 6, 4, 2, 1}
Compare (6, 3): 6 > 3 is true -> {5, 6, 3, 4, 2, 1}
Compare (6, 5): 6 > 5 is true-> {6, 5, 3, 4, 2, 1}
3. Subsequent Passes
The while loop continues until nowaps occur. Since the array is not yet fully sorted (3 and 4 are out of order), it runs again:

In the next forward pass, 3 and 4 will be compared. Since $3 < 4$, they swap.
The final sorted array in descending order becomes: {6, 5, 4, 3, 2, 1}
Final Result
The program requests printf("%d", array[3]);.

Mapping the indices:

array[0] = 6
array[1] = 5
array[2] = 4
array[3] = 3
array[4] = 2
array[5] = 1
The output is 3.

17. ANSWER:- 2

c = "GATECSIT2017"
p = c
We need: strlen(c + 2[p] - 6[p] - 1)
Step 1: Compute 2[p] and 6[p]
In C, 2[p] means *(2 + p) = p[2] = 3rd character of string "GATECSIT2017".
Indexing:
0: 'G', 1: 'A', 2: 'T', 3: 'E', 4: 'C', 5: 'S', 6: 'I', 7: 'T', 8: '2', 9: '0', 10: '1', 11: '7'
So:
2[p] = 'T' → ASCII = 84
6[p] = 'I' → ASCII = 73
Step 2: Compute c + 2[p] - 6[p] - 1
= c + 84 - 73 - 1
= c + 10
Step 3: strlen(c + 10)
c + 10 → points to the 11th character (0-based index 10) of "GATECSIT2017".
Index 10: '1', then '7', then null terminator.
String from index 10: "17" → length = 2.
Output: 2

Check Out Our GATE Courses & Secure Your Dream Rank!

18. ANSWER:(2016)

```
void mystery(int *ptra, int *ptrb) {
int *temp;
temp = ptrb;
ptrb = ptra;
ptra = temp;
}
```

Important:
Pointers are passed by value.
So only local copies are swapped.
Actual variables are NOT modified.
Now trace main.

Initial:
a = 2016
b = 0
c = 4
d = 42

Call:

```
mystery(&a, &b);
```

No effect on a or b.

Condition:
if (a < c)

2016 < 4 → false
So this mystery call is skipped.

Next:

```
mystery(&a, &d);
```

Again no effect.

So a remains 2016.
printf("%d\n", a);
Output:
2016

19. ANSWER:- A

Given:
x[4][3] → 4 rows, 3 columns.
Base address of x = 2000.
Size of unsigned int = 4 bytes.
1. x + 3
x is of type unsigned int (*)[3] (pointer to array of 3 ints).
x + 3 = address of x[3] (4th row).
Row size = 3 × 4 = 12 bytes.
So x + 3 = 2000 + 3 × 12 = 2000 + 36 = 2036.
2. *(x + 3)
*(x + 3) = x[3] = address of first element of 4th row (x[3][0]).
Address of x[3][0] = 2036 (same as x + 3 numerically but type is unsigned int *).
So value printed = 2036.
3. *(x + 2) + 3
*(x + 2) = x[2] = address of x[2][0] = 2000 + 2 × 12 = 2024.
*(x + 2) + 3 = &x[2][3]? Wait, x[2] has only 3 columns (0,1,2), so x[2] + 3 = address of x[2][3] is out of bounds for row 2, but in C it's just pointer arithmetic:
x[2] + 3 = 2024 + 3 × 4 = 2024 + 12 = 2036.
So all three expressions give:
2036, 2036, 2036
Output: 2036, 2036, 2036

20. ANSWER:- C

char s1[7] = "1234" → s1 contains: '1' '2' '3' '4' '\0' ...
p = s1 + 2 → p points to s1[2] which is '3'
*p = '0' → s1[2] becomes '0'
Now s1 = '1' '2' '0' '4' '\0'
printf("%s", s1) prints 1204
Answer: 1204

21. ANSWER:- 140

Initial arrays:
a[] = {10, 20, 30, 40, 50}
p[] = {a, a+3, a+4, a+1, a+2}
That means:
p[0] = &a[0] → 10
p[1] = &a[3] → 40
p[2] = &a[4] → 50
p[3] = &a[1] → 20
p[4] = &a[2] → 30
ptr = p → ptr points to p[0]
Step: ptr++
ptr now points to p[1]
ptr - p
= (address of p[1] - address of p[0]) in units of int*
= 1
**ptr
*ptr = p[1] = a+3 → points to a[3] = 40
So **ptr = 40
Output: 1 and 40 → 140

22. ANSWER:- D

int i; // Initially i takes the Garbage value
int *pi = &i; // pi is a pointer which stores the address of i.
scanf (pi); // pi = &i (we rewrite the garbage value with our values) say x = 5
printf (i+5); // i+5 = x+5 = 5+5 = 10

Check Out Our GATE Courses & Secure Your Dream Rank!

Hence on execution, the value printed is 5 more than the integer value entered.

23. ANSWER:- A

Manipulation Block: The three lines inside the innermost loop:
Temp = A[i] [j] + C;
A [i] [j] = A[j] [i];
A[j][i] = Temp – C;
This sequence performs a swap of A[i][j] and A[j][i]. The constant C=100 is added and then subtracted, having no net effect on the values themselves.
Loop Structure: The nested loops iterate over all pairs of indices (i,j) from 1 to n.
Net Effect:
For any pair where i!=j (off-diagonal elements), the loops process the pair twice: once as (i,j) and once as (j,i).
A swap operation performed twice on the same pair of memory locations returns the elements to their original positions and values.
For i=j (diagonal elements), the element is swapped with itself, resulting in no change.
Final Output: Since the matrix A is restored to its initial state after the first set of loops, the second set of loops, which prints A[i][j] in row-major order, prints the original matrix A.
Therefore, the expected output is The matrix A itself.
The correct option is: The matrix A itself

24. ANSWER:- C

p is the starting address of array.
Let consider p address is 100 and p[3] point to A in array and ASCII value of A is 65 and E is 69 so difference of there is 4.
p[3] - p[1] = 4, and p+4 will be pointing to the fifth position in the array 'c'. So printf starts printing from 2 and prints 2011

25. ANSWER:- C

f(100,6)
↓
12 + f(104,5) → 15
↓
7 - f(108,4) → 3
↓
13 - f(112,3) → 4
↓
4 + f(116,2) → 9
↓
11 - f(120,1) → 5
↓
6 + f(124,0) → 6
↓
0

So, the final returned value is 12 + (7 – (13 – (4 + (11 – (6 + 0))))) = 15

26. ANSWER:- D

The loop condition is: while (i < n) && (j < m)
So the loop stops when either i = n or j = m.
Case 1: i = n

All elements of array a are copied.
Then j < m must be true.
Since k increments every time either i or j increments,
k = i + j.
Here i = n, so k = n + j.

Thus statement (i) says k = n + j − 1, which is wrong.

It should be k = n + j.

So (i) is false.

Case 2: j = m
All elements of array b are copied.
Then i < n must be true.
Again k = i + j.Here j = m, so k = i + m.

Statement (ii) says k = m + i − 1, which is wrong.
It should be k = m + i.

So (ii) is also false.
Therefore,
Correct answer:
(D) neither (i) nor (ii)

27. ANSWER:- C

The declaration is: int (*f)(int *);
Break it carefully:
f is inside parentheses with *, so f is a pointer.
It points to something that takes (int *) as argument.
That function returns int.
So it means:
A pointer to a function that takes an integer pointer as argument and returns an integer.

Check Out Our GATE Courses & Secure Your Dream Rank!

DATA STRUCTURE

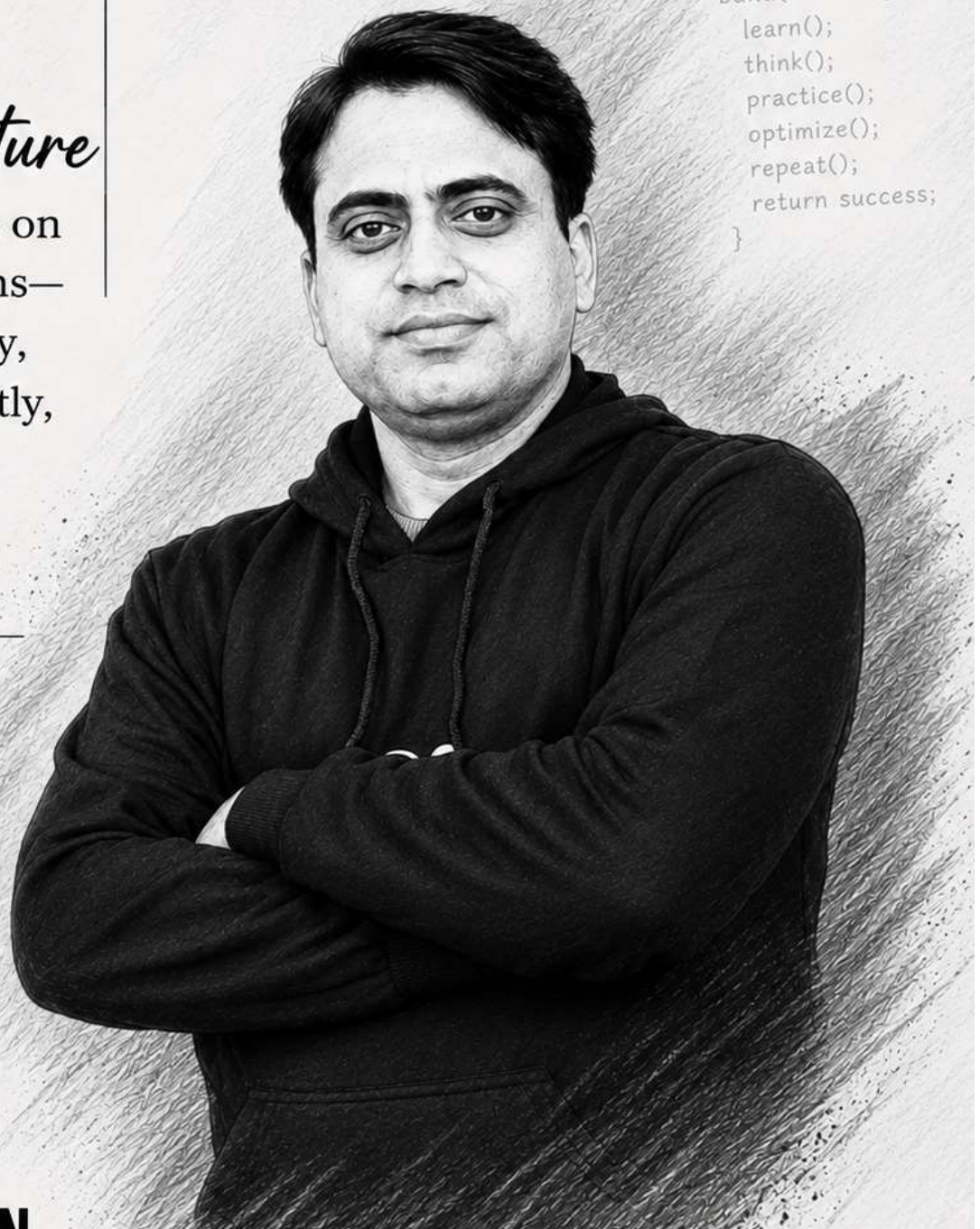
"Scan Here for Video Solution"

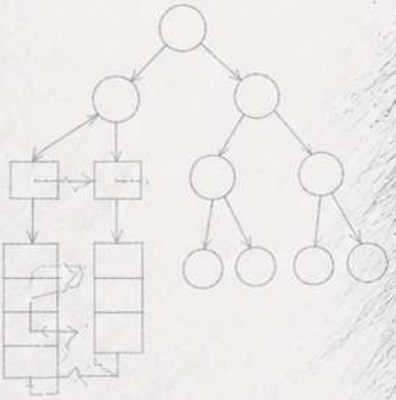
Link to GfG corresponding Short Notes: GATE FARRAY

“

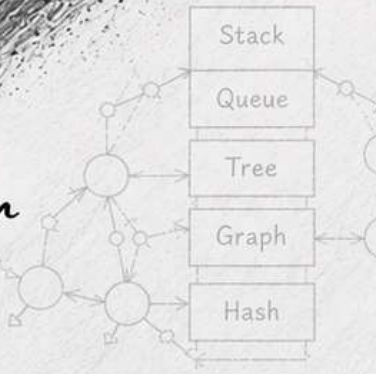
The
Data Structure
of success is built on strong foundations—
Understand deeply,
Practice consistently,
and
Never stop Building.

”





15+ YEARS OF EXPERTISE IN DATA STRUCTURES



Shailendra Singh Sir

An Inspirer. A Mentor. A Guide.

1 ARRAY

GATE CSE – 2015 SET – 2 NAT (2-MARKS)

Q1. A Young tableau is a 2D array of integers increasing from left to right and from top to bottom. Any unfilled entries are marked with ∞, and hence there cannot be any entry to the right of, or below a ∞. The following Young tableau consists of unique entries.

1	2	5	14
3	4	6	23
10	12	18	25
31	∞	∞	∞

When an element is removed from a Young tableau, other elements should be moved into its place so that the resulting table is still a Young tableau (unfilled entries may be filled with a ∞). The minimum number of entries (other than 1) to be shifted, to remove 1 from the given Young tableau is ____.

GATE CSE – 2015 SET – 3 MCQ (2-MARKS)

Q2. Consider the following two C code segments. Y and X are one and two dimensional arrays of size n and n×n respectively, where 2 ≤n≤10. Assume that in both code segments, elements of Y are initialized to 0 and each element X[i][j] of array X is initialized to i+j. Further assume that when stored in main memory all elements of X are in same main memory page frame.

Code segment 1:
// initialize elements of Y to 0
// initialize elements of X[i][j] of X to i+j

```
for (i=0; i<n; i++)
    Y[i] += X[0][i];
```

Code segment 2:
// initialize elements of Y to 0
// initialize elements of X[i][j] of X to i+j

```
for (i=0; i<n; i++)
    Y[i] += X[i][0];
```

Which of the following statements is/are correct?

S1: Final contents of array Y will be same in both code segments
S2: Elements of array X accessed inside the for loop shown in code segment 1 are contiguous in main memory
S3: Elements of array X accessed inside the for loop shown in code segment 2 are contiguous in main memory

(a) Only S2 is correct
(b) Only S3 is correct
(c) Only S1 and S2 are correct
(d) Only S1 and S3 are correct

GATE CSE – 2015 SET – 3 NAT (2-MARKS)

Q3. Suppose c = ⟨c[0], …, c[k−1]⟩ is an array of length k, where all the entries are from the set {0,1}. For any positive integers a and n, consider the following pseudocode.

```
DOSOMETHING (c, a, n)
z←1
for i←0 to k−1
  do z←z^2 mod n
  if c[i]=1
then z←(z×a) mod n
return z
```

If k = 4, c = ⟨1, 0, 1, 1⟩, a = 2, and n = 8

then the output of DO SOMETHING (c, a, n) is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

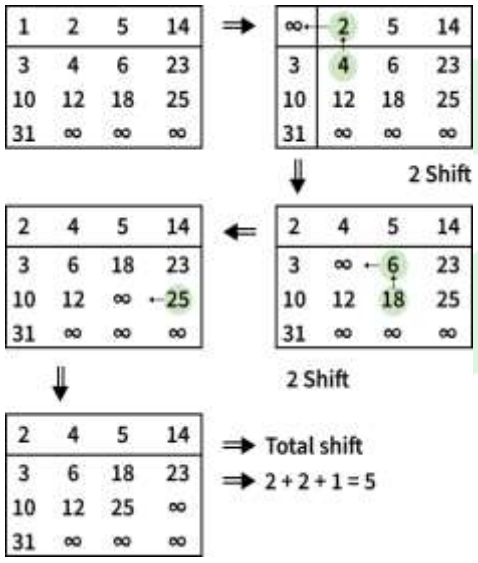
ANSWER KEY

1. (5)	2. (c)	3. (0)

HINTS & SOLUTIONS

1. **Answer: 5**

Explanation:
Replace **1 with 2** (keep 5 and 14 unchanged).
Shift **4 upward** next to 2 (column order maintained).
Move **6 left** in the second row.
Shift **18 upward** to the second row.
Move **25 left** to the third column.
Total shifts = **5**



2. **Answer: C**

Explanation: S1:
Code 1: Y[i] = X[0][i] = 0 + i = i
Code 2: Y[i] = X[i][0] = i + 0 = i
Same final Y → True
S2:
Code 1 accesses X[0][0], X[0][1], … → contiguous in memory → True
S3:
Code 2 accesses X[0][0], X[1][0], … → stride = n, not contiguous → False

Answer: S1 and S2 are correct.

3. **Answer: 0**

Explanation: Now let's iterate through the function step by step :
z=1 (at the start of do-something)
i=0 (start of external for loop)
In the do loop
z=1∗1 (non-zero value so considered as true and continue)
c[0]=1 so in the if clause, z=1∗2%8=2
In the do loop
z = 2 ∗2 % 8 = 4 (since now z = 2 z = 2) (non-zero value so considered as true and continue)
c[0]=1, so in the if clause, z = 4 ∗2% 8 = 0
Now no need to check further :
Reason : All the operations that update Z are multiplicative operations and hence the value of Z will never change from 0.

Check Out Our GATE Courses & Secure Your Dream Rank!

2 LINKED LIST

GATE CSE – 2026 SET – 1 MCQ (2-MARKS)

Q1. Consider the following code snippet in C language that computes the number of nodes in a non-empty singly linked list pointed to by the pointer variable head.

```
struct node{
   int elt;
   struct node *next;
}
int getListSize (struct node *head)
{
   if( El ) return 1;
   return E2;
}
```

Which one of the following options gives the correct replacements for the expressions E1 and E2?

(a) E1: head == NULL
E2: 1 + getListSize(head)

(b) E1: head->next == NULL
E2: 1 + getListSize(head->next)

(c) E1: head == NULL
E2: 1 + getListSize(head->next)

(d) E1: head->next == NULL
E2: 1 + getListSize(head)

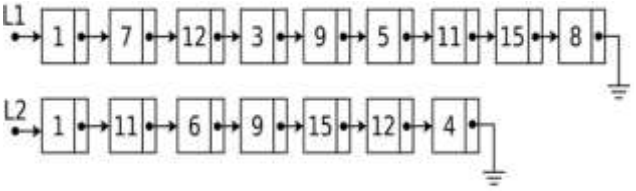
GATE CSE – 2025 SET – 1 NAT (2-MARKS)

Q2. Let LIST be a datatype for an implementation of linked list defined as follows:

```
typedef struct list {
   int data;
   struct list *next;
} LIST;
```

Suppose the program has created two linked lists, L1 and L2, whose contents are given in the figure below (code for creating L1 and L2 is not provided here). L1 contains 9 nodes, and L2 contains 7 nodes.

Consider the following C program segment that modifies the list L1. The number of nodes in L1 after the execution of the code segment is _____. (Answer in integer)




```
int find (int query, LIST *list) {
 while (list != NULL){
if(list->data == query) return 1;
list = list->next;
 }
  return 0;
}
int main () {
   … … …
     ptr1=L1; ptr2=L2;
   while (ptr1->next != NULL){
      query = ptr1->next->data;
      if (find (query, L2))
      ptr1->next = ptr1->next->next;
     else ptr1 = ptr1->next;
    }
… … …
 return 0;
}
```

GATE CSE – 2023 MCQ (1-MARKS)

Q3. Let SLLdel be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let DLLdel be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list.

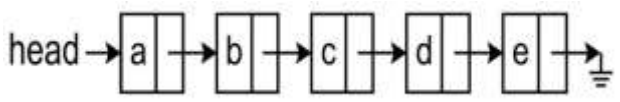
Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of SLLdel and DLLdel?

(a) SLLdel is O(1) and DLLdel is O(n).
(b) Both SLLdel and DLLdel are O(log(n)).
(c) Both SLLdel and DLLdel are O(1).
(d) SLLdel is O(n) and DLLdel is O(1).

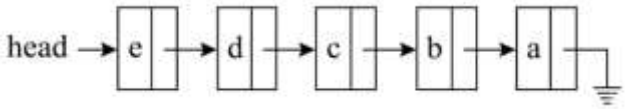
Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2022 MCQ (1-MARKS)

Q4. Consider the problem of reversing a singly linked list. To take an example, given the linked list below,



the reversed linked list should look like,



Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in O(1) space?

(a) The best algorithm for the problem takes θ(n) time in the worst case.
(b) The best algorithm for the problem takes θ(nlogn) time in the worst case.
(c) The best algorithm for the problem takes $\theta(n^2)$ time in the worst case.
(d) It is not possible to reverse a singly linked list in O(1) space.

GATE CSE – 2021 SET – 2 MCQ (2-MARKS)

Q5. Consider the following ANSI C program:

```
#include <stdio.h>
#include <stdlib.h>
struct Node{
  int value;
   struct Node *next;};
int main(){
 struct Node *boxE, *head, *boxN; int index = 0;
boxE = head=(struct Node*)malloc(sizeof(struct
Node));
 head→value = index;
 for (index = 1; index <= 3; index++)
{
  boxN = (struct Node *) malloc(sizeof(struct Node));
        boxE →next = boxN;
        boxN→va1ue = index;
        boxE = boxN:
}
  for (index = 0; index <= 3; index++)
{
 printf("Value at index %d is %d\n", index, head-
>value);
   head = head→next;
    printf("Value at index %d is %d\n", index+1,
        head→value);
        }
 }
```

(a) It dereferences an uninitialized pointer that may result in a run-time error.
(b) It has a missing return which will be reported as an error by the compiler.
(c) Upon execution, the program creates a linked list of five nodes.
(d) Upon execution, the program goes into an infinite loop.

GATE CSE – 2020 MCQ (1-MARKS)

Q6. What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?

(a) Θ(n log n)
(b) $\Theta(n^2)$
(c) Θ(1).
(d) Θ(n)

GATE CSE – 2018 MCQ (1-MARKS)

Q7. A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let 'enqueue' be implemented by inserting a new node at the head, and 'dequeue' be implemented by deletion of a node from the tail.

Which one of the following is the time complexity of the most time-efficient implementation of 'enqueue' and 'dequeue, respectively, for this data structure?

(a) θ(1), θ(1)
(b) θ(1), θ(n)
(c) θ(n), θ(1)
(d) θ(n), θ(n)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2017 SET – 1 MCQ (1-MARK)

Q8. Consider the C code fragment given below.

```
typedef struct node {
        int data;
        node* next ;
                } node;

void join (node* m, node* n) {
            node* p=n ;
            while (p->next ! =NULL){
            p = p -> next ;
            }
            p-> next = m;
}
```

Assuming that m and n point to valid NULL-terminated linked lists, invocation of join will

(a) append list m to the end of list n for all inputs.
(b) either cause a null pointer to dereference or append list m to the end of list n.
(c) cause a null pointer to dereference for all inputs.
(d) append list n to the end of list m for all inputs.

GATE CSE – 2016 SET – 2 MCQ (1-MARK)

Q9. N items are stored in a sorted doubly linked list. For a delete operation, a pointer is provided to the record to be deleted. For a decrease-key operation, a pointer is provided to the record on which the operation is to be performed.
An algorithm performs the following operations on the list in this order:
$\Theta(N)$ delete, $O(\log N)$ insert, $O(\log N)$ find and $\Theta(N)$decrease−key . What is the time complexity of all these operations put together?

(a) $O(\log_2 N)$
(b) $O(N)$
(c) $O(N^2)$
(d) $O(N^2 \log N)$

GATE CSE – 2015 SET – 1 MCQ (1-MARK)

Q10. An unordered list contains n distinct elements. The number of comparisons to find an element in this list that is neither maximum nor minimum is :

(a) O(nlogn)
(b) O(n)
(c) O(logn)
(d) O(1)

GATE CSE – 2010 MCQ (2-MARKS)

Q11. The following C function takes a singly-linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank. Choose the correct alternative to replace the blank line.

```
typedef struct node

{
int value;
  struct node *next;
}Node;
Node *move_to_front(Node *head)
{
 Node *p, *q;
if ((head == NULL: || (head->next == NULL))
 return head;
  q = NULL; p = head;
  while (p-> next !=NULL)
  {
          q = p;
          p = p->next;
 }
  return head;
}
```

(a) q = NULL; p→next = head; head = p;
(b) q→next = NULL; head = p; p→next = head;
(c) head = p; p→next = q; q→next = NULL;
(d) q→next = NULL; p→next = head; head = p;

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2008 MCQ (2-MARKS)

Q12. The following C function takes a singly-linked list of integers as a parameter and rearranges the elements of the list. The function is called with the list containing the integers 1,2,3,4,5,6,7 in the given order. What will be the contents of the list after the function completes execution?

```
  struct node {
   int value;
 struct node *next;
};

void rearrange (struct node *list){
 struct node *p, *q;
  int temp;
 if (!list || !list->next) return;

  p = list;
 q = list->next;
 while (q) {
        temp = p->value;
        p->value = q->value;
        q->value = temp;

        p = q->next;
        q = p ? p->next : 0;
        }
}
```

(a) 1,2,3,4,5,6,7
(b) 2,1,4,3,6,5,7
(c) 1,3,2,5,4,7,6
(d) 2,3,4,5,6,7,1

ANSWER KEY

1.(b)	**2.(5 to 5)**	**3.(d)**
4.(a)	**5.(a)**	**6.(b)**
7.(b)	**8.(b)**	**9.(c)**
10.(d)	**11.(d)**	**12.(b)**

HINTS & SOLUTIONS

1. ANSWER : B

Explanation : The list is non-empty, so head is never NULL.
To compute size recursively:
Base case:
If this is the last node → return 1
Condition for last node:
head->next == NULL
Recursive case:
Otherwise → 1 + size of remaining list
So:
1 + getListSize(head->next)
Check options:
Correct combination is:
E1: head->next == NULL
E2: 1 + getListSize(head->next)

2. ANSWER : 5 to 5

Explanation : L1 has nodes: 1 → 7 → 12 → 3 → 9 → 5 → 11 → 15 → 8 (total 9)
L2 has nodes: 1 → 11 → 6 → 9 → 15 → 12 → 4
The function find(query, L2) checks whether a value exists in L2.
In the loop:
We always check ptr1->next (not the current node)
If the next node's value is found in L2, that node is deleted
Otherwise, we move forward
Important point: the first node (1) is never checked or deleted.
Now check each node of L1 (except first):
7 → not in L2 → keep
12 → in L2 → delete
3 → not in L2 → keep
9 → in L2 → delete
5 → not in L2 → keep
11 → in L2 → delete
15 → in L2 → delete
8 → not in L2 → keep
Final L1 becomes: 1 → 7 → 3 → 5 → 8
Total nodes = 5

Check Out Our GATE Courses & Secure Your Dream Rank!

3. ANSWER : D

Explanation:

In a singly linked list (SLL), each node has only a pointer to the next node. If we are given a pointer to the node to delete, we still need to know its previous node to change the link (prev->next = node->next). Since there is no backward pointer, we must traverse the list from the head to find the previous node. In the worst case, this takes O(n) time.

In a doubly linked list (DLL), each node has pointers to both next and previous nodes. So, if we are given the node to delete, we can directly access its previous node and adjust links (prev->next and next->prev) without any traversal. This takes O(1) time.

4. ANSWER : A

Explanation::

To reverse a singly linked list in O(1) auxiliary space, an iterative, in-place reversal algorithm is used. This standard approach utilizes three pointers to keep track of the nodes during traversal: prev (initialized to NULL),
curr (initialized to head), and
next (initialized to NULL).

Algorithm:

While curr is not NULL, the following constant-time operations are performed:

1. next = curr.next (Store the next node)
2. curr.next = prev (Reverse the link)
3. prev = curr (Advance the prev pointer)
4. curr = next (Advance the curr pointer)

Complexity Analysis:

- **Space Complexity:** The algorithm allocates memory only for three node pointers. This memory requirement does not scale with the input size n. Therefore, the space complexity is strictly O(1).
- **Time Complexity:** The algorithm traverses the linked list exactly once. For a linked list containing n nodes, the loop executes exactly n times. Since the operations inside the loop take constant time, the total time taken is directly proportional to n. Thus, the time complexity is tightly bound at $\theta(n)$ for all cases, including the worst case.

5. Answer: A

Explanation:

The program creates 4 nodes (values 0,1,2,3).In the last iteration of the printing loop,
head = head->next
makes head point to an uninitialized memory location (because the last node's next was never set to NULL).
Then printf tries to read head->value from this uninitialized pointer → dereferencing uninitialized pointer → possible runtime error.
Correct option:
It dereferences an uninitialized pointer that may result in a run-time error.

6. Answer: B

Explanation

Single Insertion Complexity: To insert a single element into a sorted linked list of size k, you must traverse the list to find the correct position. In the worst case (inserting the largest element), you perform k comparisons/steps, which is $O(k)$.
Total Work for n Elements: When inserting n elements sequentially:

- 1st insertion: 0 steps
- 2nd insertion: 1 step
- 3rd insertion: 2 steps
- n-th insertion: $n-1$ steps

Mathematical Sum: The total number of operations is the sum of the first $n-1$ integers:

$$\sum_{i=1}^{n-1} i = \frac{(n-1)n}{2}$$

Conclusion: This results in a quadratic growth rate, expressed as $\Theta(n^2)$.

7. Answer: B

Explanation: For insertion of node at the beginning of linked list only need manipulation of pointers so θ(1) time.
But if we have pointer to the tail of the list in order to delete it, we need the address of the 2nd last node which can be obtained by traversing the list which takes O(n) time.

8. Answer: (B)

Explanation:

Function join(m, n) works as follows:
It starts with p = n
Then traverses list n using while (p->next != NULL)

Check Out Our GATE Courses & Secure Your Dream Rank!

Finally, it sets p->next = m, which attaches list m at the end of n
Now consider cases:
If n is non-empty (valid list)
The loop runs correctly
p reaches the last node of n
p->next = m → list m gets appended to n
If n is NULL
p = n = NULL
Then p->next is accessed in the loop condition → null pointer dereference
So:
Works correctly when n is not NULL
Crashes when n is NULL

9. Answer:C

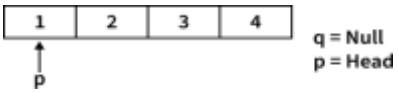
Explanation: We analyze each operation on a **sorted doubly linked list**.
Delete: Given pointer to node, deletion in DLL takes O(1). Done Θ(N) times → total O(N).
Insert: To maintain sorted order, we must find correct position → takes O(N). Done O(log N) times → total O(N log N).
Find: Searching in DLL takes O(N). Done O(log N) times → total O(N log N).
Decrease-key: After decreasing value, node may need to move to correct position. In worst case, we traverse list → O(N) per operation. Done Θ(N) times → total $O(N^2)$.
Now combine all:
$O(N) + O(N \log N) + O(N \log N) + O(N^2)$
Dominant term is $O(N^2)$

10. Answer: D

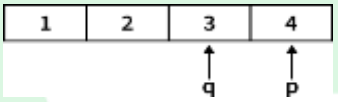
Explanation:
We need an element that is neither maximum nor minimum.
Take any three distinct elements from the list. Compare the first two → 1 comparison (we know which is smaller and larger).Now compare the third element with these → at most 2 more comparisons.
From these 3 elements, we can always identify the middle element, which is neither maximum nor minimum.
Total comparisons needed = at most 3, which is a constant.

11. ANSWER : D

Explanation:
C function takes a simple linked list as input argument.
The function modifies the list by moving the last element to the front of the list.
Let the list be 1 → 2 → 3 → 4 & the modified list must be 4 → 1 → 2 → 3.
Algorithm:
Traverse the list till last node. Use two pointers.
One to store the address of last node & other for the address of second last node.
After completion of loop. Do these.
(i) Make second last as last
(ii) Set next of last as head
(iii) Make last as head



while (p → !=NULL)
{
q = p;
p = p → next;
}
— p is travelling till the end of list and assigning q to whatever p had visited & p takes next new node, so travels through the entire list.
Now the list looks like



According to the Algorithm new lines must be:
q → next = NULL; p → next = head; head = p

12. ANSWER : B

Explanation : Initial list:
1 → 2 → 3 → 4 → 5 → 6 → 7
The function swaps values of nodes in pairs.
First iteration:
p = 1, q = 2
Swap → 2,1
Move pointers:
p = 3, q = 4
Second iteration:
Swap 3 and 4 → 4,3
Move pointers:
p = 5, q = 6
Third iteration:
Swap 5 and 6 → 6,5
Move pointers:
p = 7, q = NULL
Loop stops.
Final list: 2,1,4,3,6,5,7

Check Out Our GATE Courses & Secure Your Dream Rank!

3 STACK & QUEUE

GATE CSE – 2026 SET – 2 MSQ (2-MARKS)

Q1. Consider a stack S and a queue Q. Both of them are initially empty and have the capacity to store ten elements each. The elements 1,2,3,4, and 5 arrive one by one, in that order. When an element arrives, it is assigned either to S (pushed on S) or to Q (enqueued to Q). Once all the five elements are stored, the output is generated in two steps. First, stack S is emptied by popping all elements. Then queue Q is emptied by dequeueing all elements. The output obtained by following this process is 43125 .
Given the output, the objective is to predict whether an element was assigned to S or Q.
Which of the following options is/are possible valid assignment(s) of the elements?
Note: In the options, the notation xS denotes that element x was assigned to S and yQ denotes that element y was assigned to Q.

(a) 1S,2Q,3S,4S,5Q
(b) 1Q,2Q,3S,4S,5Q
(c) 1Q,2Q,3Q,4S,5S
(d) 1S,2S,3S,4Q,5Q

GATE CSE – 2021 NAT (1-MARKS)

Q2. Consider the following sequence of operations on an empty stack.
push(54); push(52); pop(); push(55); push(62); s = pop();
Consider the following sequence of operations on an empty queue.
enqueue (21); enqueue (24); dequeue (); enqueue (28); enqueue (32); q = dequeue();
The value of s +q is _____.

GATE CSE – 2017 SET-2 MCQ (1-MARKS)

Q3. A circular queue has been implemented using a singly linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers **FRONT** and **REAR** pointing to the front node and the rear node of the queue, respectively. Which of the following statements is/are CORRECT for such a circular queue, so that insertion and deletion operations can be performed in *O(1)* time?
I. Next pointer of front node points to the rear node.
II. Next pointer of rear node points to the front node.

(a) I only
(b) II only
(c) Both I and II
(d) Neither I nor II

GATE CSE – 2016 SET – 1 MCQ (1-MARKS)

Q4. A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (n refers to the number of items in the queue)?
Both operations can be performed in O(1) time.

(a) Both operations can be performed in O(1) time
(b) At most one operation can be performed in *O(1)* time but the worst case time for the other operation will be $\Omega(n)$
(c) The worst case time complexity for both operations will be $\Omega(n)$
(d) Worst case time complexity for both operations will be $\Omega(logn)$

GATE CSE – 2016 SET – 1 NAT (2-MARKS)

Q5. Let Q denote a queue containing sixteen numbers and S be an empty stack. Head(Q) returns the element at the head of the queue Q without removing it from Q. Similarly Top(S) returns the element at the top of S without removing it from S. Consider the algorithm given below.

Check Out Our GATE Courses & Secure Your Dream Rank!

```
while Q is not Empty do
    if S is Empty OR Top (S) ≤ Head (Q) then
        x : = Dequeue(Q);
        Push(S,x);
    else
        x: = Pop(S);
        Emqueue(Q,x);
    end
end
```

The maximum possible number of iterations of the while loop in the algorithm is _____.

GATE CSE – 2015 SET-3 MCQ (1-MARKS)

Q6. The result evaluating the postfix expression 10 5+60 6/∗8−is

(a) 248
(b) 213
(c) 142
(d) 71

GATE CSE – 2015 SET-2 NAT (2-MARKS)

Q7. Consider the C program below

```
#include <stdio.h>
int *A, stkTop;
int stkFunc (int opcode, int val)
{
   static int size=0, stkTop=0;
   switch (opcode) {
      case -1: size = val; break;
      case 0: if (stkTop< size )
A[stkTop++]=val; break;
      default: if (stkTop) return A[--stkTop];
   }
return -1;
}
int main()
{
int B[20]; A=B; stkTop = -1;
stkFunc (-1, 10);
stkFunc (0, 5);
stkFunc (0, 10);
printf ("%d\n", stkFunc(1, 0)+ stkFunc(1, 0));
}
```

The value printed by the above program is _____.

GATE CSE – 2013 MCQ (2-MARKS)

Q8. Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE (with respect to this modified stack)?

(a) A queue cannot be implemented using this stack.
(b) A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.
(c) A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.
(d) A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each.

GATE CSE – 2012 MCQ (1-MARKS)

Q9. The recurrence relation capturing the optimal execution time of the Towers of Hanoi problem with n discs is

(a) T(n) = 2T(n - 2) + 2
(b) T(n)= 2T(n - 1) + n
(c) T(n)= 2T(n/2) + 1
(d) T(n)= 2T(n - 1) + 1

GATE CSE – 2012 MCQ (2-MARKS)

Q10. Suppose a circular queue of capacity (n - 1) elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables, respectively. Initially, REAR = FRONT = 0. The conditions to detect *queue full* and *queue empty* are

(a) full: (REAR+1)mod n == FRONT,
empty: REAR == FRONT
(b) full: (REAR+1)mod n == FRONT,
empty: (FRONT+1) mod n == REAR
(c) full: REAR == FRONT,
empty: (REAR+1) mod n == FRONT
(d) full: (FRONT+1)mod n == REAR
empty: REAR == FRONT

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2007 MCQ (2-MARKS)

Q11. The following postfix expression with single digit operands is evaluated using a stack:

8 2 3 ^ / 2 3 * + 5 1 * -

Note that is the exponentiation operator. The top two elements of the stack after the first * is evaluated are:

(a) 6,1
(b) 5,7
(c) 3,2
(d) 1,5

GATE CSE – 2006 MCQ (2-MARKS)

Q12. An implementation of a queue Q, using two stacks S1 and S2, is given below:

```
void insert (Q, x) {
  push(S1, x);
}

void delete (Q) {
   if (stack-empty(S2)) then {
 if (stack-empty(S1)) then {
 print("Q is empty");
    return;
  }
else while (!(stack-empty(S1))) {
 x = pop(S1);
 push(S2, x);
 }
   }
   x = pop(S2);
}
```

Let n insert and m (≤ n) delete operations be performed in an arbitrary order on an empty queue Q. Let x and y be the number of push and pop operations performed respectively in the process. Which one of the following is true for all m and n?

(a) $n + m \leq x < 2n$ and $2m \leq y \leq n + m$
(b) $n + m \leq x < 2n$ and $2m \leq y \leq 2n$
(c) $2m \leq x < 2n$ and $2m \leq y \leq n + m$
(d) $2m \leq x < 2n$ and $2m \leq y \leq 2n$

GATE CSE – 2005 MCQ (1-MARKS)

Q13. An Abstract Data Type (ADT) is:

(a) same as an abstract class
(b) a data type that cannot be instantiated
(c) a data type for which only the operations defined on it can be used, but none else
(d) All of the above

ANSWER KEY

1.(a,b)	**2.(86)**	**3.(b)**
4.(a)	**5.(256)**	**6.(c)**
7.(15)	**8.(c)**	**9.(d)**
10.(a)	**11.(a)**	**12.(a)**
13.(c)		

HINTS & SOLUTION

1. ANSWER : A,B

Explanation : Process Recap

- Elements arrive in order: 1, 2, 3, 4, 5
- Each element assigned to either stack S (LIFO) or queue Q (FIFO)
- Output: first pop all from S (reverse push order), then dequeue all from Q (arrival order)
- Given output: 4 3 1 2 5

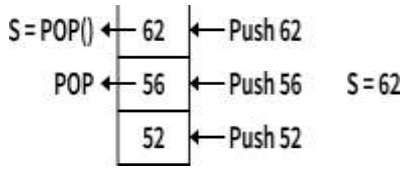
Check each option:

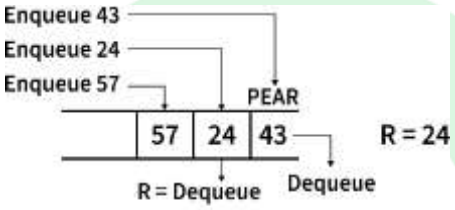
Option A: 1S, 2Q, 3S, 4S, 5Q
S push order: 1, 3, 4 → pop: 4, 3, 1
Q enqueue order: 2, 5 → dequeue: 2, 5
Output: 4 3 1 2 5
Option B: 1Q, 2Q, 3S, 4S, 5Q
S push order: 3, 4 → pop: 4, 3
Q enqueue order: 1, 2, 5 → dequeue: 1, 2, 5
Output: 4 3 1 2 5
Option C: 1Q, 2Q, 3Q, 4S, 5S
S push order: 4, 5 → pop: 5, 4
Q enqueue order: 1, 2, 3 → dequeue: 1, 2, 3
Output: 5 4 1 2 3
Option D: 1S, 2S, 3S, 4Q, 5Q
S push order: 1, 2, 3 → pop: 3, 2, 1
Q enqueue order: 4, 5 → dequeue: 4, 5
Output: 3 2 1 4 5
Valid assignments: Options A and B

Check Out Our GATE Courses & Secure Your Dream Rank!

2. Answer: 86

Explanation

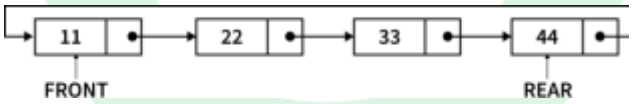




S + R = 62 + 24 = 86

3. Answer B

Explanation:



1. Next pointer of front node points to the rear node.
2. Next pointer of rear points to the front node.

So if we consider circular linked list the next of 44 is 11.

If you place pointer on next of front node (11) then to reach 44 (last node), we have to traverse the entire list.

For delete O(1), for insert O(n).

It is clearly known that next pointer of rear node points to the front node.

Hence, only II is true.

4. Answer: A

Explanation: Since it is mentioned in the question that both of the operations are performed efficiently.

Hence even the worst case time complexity will be O(1) by the use of the Circular queue there won't be any need of shifting in the array.

5. Answer: 256

Explanation: The maximum possible number of iterations of the while loop in this algorithm is:

Try to solve it for 3 numbers [1. 2, 3].

Step 1: Initially Queue contains 3 elements so after 5 while loop iterations queue contains 3, 2 and stack contains 1.

Step 2: Now after 3 more while loop iterations, Queue contains 3 and stack contains 1, 2 (TOS = 2).

Step 3: After 1 more while loop iteration, push 3 onto the stack so queue is empty and stack contains 1, 2, 3 {top = 3}.

So, total number of iterations will be 5 + 3 + 1 = 9

i.e., for 3 it is 9 iterations (3*3)

for 4 it is 16 iterations (4*4)

Given: 16 numbers, so 16 * 16 = 256

6. Answer :C

Explanation:

10 5 + 60 6 / ∗ 8 −

We read tokens left to right, using a stack:

1.10 → push 10 → Stack: [10]

2. 5 → push 5 → Stack: [10, 5]

3. + → pop 5, pop 10 → 10 + 5 = 15 → push 15 → Stack: [15]

4.60 → push 60 → Stack: [15, 60]

5. 6 → push 6 → Stack: [15, 60, 6]

6./ → pop 6, pop 60 → 60 / 6 = 10 → push 10 → Stack: [15, 10]

7. ∗ → pop 10, pop 15 → 15 ∗ 10 = 150 → push 150 → Stack: [150]

8.8 → push 8 → Stack: [150, 8]

9 − → pop 8, pop 150 → 150 − 8 = 142 → push 142 → Stack**: [142]**

7. Answer: 15

Explanation: The code is pushing 5 and 10 on stack and then popping the top two elements and printing their sum.

```
stkFunc (-1, 10);   // Initialize size as 10
stkFunc (0, 5);     // push 5
stkFunc (0, 10);    // push 10
// print sum of two pop
printf ("%d\n", stkFunc(1, 0) + stkFunc(1, 0));
```

8. Answer: C

Explanation: (C) is the answer. While ENQUEUE we REVERSE the stack, PUSH the element and then again REVERSE the stack. For DEQUEUE we simply POP the element.

(Option (B) can be used to get the first element from the stack by doing a POP after REVERSE for DEQUEUE and PUSH for ENQUEUE. But we have to restore the stack using REVERSE(otherwise next POP won't work) which means DEQUEUE actually needs 3 instructions and not 2)

Check Out Our GATE Courses & Secure Your Dream Rank!

9. Answer: D

Explanation: The recurrence equation for given recurrence function is

$T(n) = 2T(n-1) + 1$

$= 2[2T(n-2) + 1] + 1$

$= 2^2 T(n-2) + 2+1$

⋮

$= 2^k T(n-k) + (2^k - 1)$

Given , $n - k = 1$

$= 2^{n-1} T(1) + (2^{n-1} - 1)$

$= 2^{n-1} + 2^{n-1} - 1$

$= 2^n - 1$

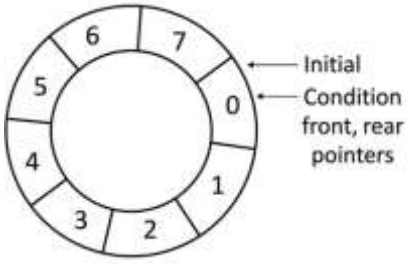
$\cong O(2^n)$

10 Answer: A

Explanation:

We know that (Rear +1)mod n == Front ; Full

Rear == Front Empty



To circulate within the queue, we have to write mod n for (Rear + 1).

We insert elements using Rear & delete through Front.

11 ANSWER : A

Explanation : Postfix expression:

8 2 3 ^ / 2 3 * + 5 1 * –

Evaluate step by step using stack.

Push 8

Push 2

Push 3

^ → 2^3 = 8

Stack: 8, 8

/ → 8 / 8 = 1

Stack: 1

Push 2

Push 3

→ 2 × 3 = 6 ← first * evaluated

Stack now: 1, 6

Top element = 6

Next element = 1

Correct answer: (A) 6, 1

12 ANSWER : A.

Explanation : Each insert does exactly one push on S1.

So from n inserts → n pushes.

Now consider delete operations.

Each element can be moved from S1 to S2 at most once.

When this happens:

one pop from S1

one push to S2

So across the whole sequence, total extra pushes due to transfer ≤ n.

Also, each delete performs exactly one pop from S2.

So total pops due to delete = m.

Additionally, during transfer, each element popped from S1 contributes one pop.

Total such pops ≤ n.

Now count pushes (x):

Pushes =

n (from insert)

at most n (during transfer)

So:

$n \le x < 2n$

But since m deletes must succeed ($m \le n$), at least m transfers must happen overall in worst case arrangement, giving:

$n + m \le x < 2n$

Now count pops (y):

Each delete → one pop from S2 → m pops

Each transferred element → one pop from S1 → at most n pops

Minimum pops occur when transfers happen only as needed:

Total pops satisfy:

$2m \le y \le n + m$

Therefore correct option is: A. $n + m \le x < 2n$ and $2m \le y \le n + m$

13. ANSWER : (C)

Explanation :

An Abstract Data Type (ADT) defines:

- The data
- The operations allowed on the data
- But not the implementation details

It focuses only on what operations can be performed, not how they are implemented.

Now check options:

- Not same as an abstract class
- Not necessarily a type that cannot be instantiated
- Correct definition: only the defined operations can be used

Check Out Our GATE Courses & Secure Your Dream Rank!

4 TREE & HEAP

GATE 2026 SET 1 || NAT || 2 MARKS

Q1. The height of a binary tree is the number of edges in the longest path from the root to a leaf in the tree. The maximum possible height of a full binary tree with 23 nodes is _____. (answer in integer)

GATE 2026 SET 1 || NAT|| 2 MARK

Q2. The following sequence corresponds to the preorder traversal of a binary search tree
T: 50, 25, 13, 40, 30, 47, 75, 60, 70, 80, 77 .
The position of the element 60 in the post-order traversal of *T* is _____.

GATE 2026 SET 2 || MCQ || 1 MARK

Q3. The set T represents various traversals over binary tree. The set S represents the order of visiting nodes during a traversal

T	S
I : Inorder	L: left subtree, node, right subtree
II : Preorder	M: node, left subtree, right subtree
III : Postorder	N : left subtree, right subtree, node

Which one of the following is the correct match from T to S?

(a) 1-L, II-M, III - N
(b) I-M, II-L, III - N
(c) I-N, II-M, III - L
(d) I-L, II-N, III - M

GATE 2026 SET 2 || MSQ || 1 MARK

Q4. Consider a binary search tree (BST) with n leaf nodes ($n > 0$). Given any node V, the key present in the node is denoted as Val(V). All the keys present in the given BST are distinct. The keys belong to the set of real numbers.

For a node V, let Suc (V) denote the node that is its inorder successor. If a node V does not have an inorder successor, then Suc (V) is NULL. As there are no duplicates, if Suc (V) is not NULL, then Val(V) < Val(Suc(V)).
Corresponding to every leaf node L_i that has a non-NULL Suc (L_i), a new key k_i with the following property is to be inserted into the BST.

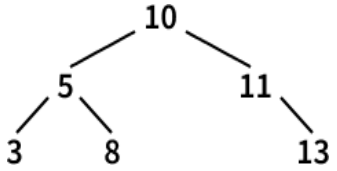
$$Val(L_i) < k_i < Val(Suc(L_i))$$

Let K represent the list of all such new keys to be inserted into the BST.
Which of the following statements is/are true?

(a) K cannot have any duplicates
(b) K will have at least one element
(c) After inserting all keys from K, the height of the BST can increase at most by one
(d) Number of nodes in the BST will double after inserting all keys from K

GATE 2023 || MCQ || 2 MARK

Q5. Consider the C function foo and the binary tree shown



Check Out Our GATE Courses & Secure Your Dream Rank!

```
typedef struct node {
        int val;
        struct node *left, *right;
        } node;
int foo(node *p)
{
        int retval;
        if (p == NULL)
        return 0;

    else{
    retval = p->val + foo(p->left) + foo(p->right);
    printf("%d ", retval);
    return retval;
    }
}
```

When foo is called with a pointer to the root node of the given binary tree, what will it print?

(a) 3 8 5 13 11 10
(b) 3 5 8 10 11 13
(c) 3 8 16 13 24 50
(d) 3 16 8 50 24 13

GATE 2022 || NAT || 1-mark

Q6. Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 3rd largest element of the tree is stored at index

GATE 2021 SET 2 || MCQ || 2-mark

Q7. Let H be a binary min-heap consisting of n elements implemented as an array. What is the worst case time complexity of an optimal algorithm to find the maximum element in H '?

(a) Θ(n)
(b) Θ(1)
(c) Θ(logn)
(d) Θ(n log n)

GATE CSE – 2021 SET 1 MCQ (1-MARK)

Q8. An articulation point in a connected graph is a vertex such that removing the vertex and its incident edges disconnects the graph into two or more connected components. Let T be a DFS three obtained by doing DFS in a connected undirected graph G.

Which of the following options is/are correct?

(a) Root of T is an articulation point in G if and only if it has 2 or more children.
(b) A leaf of T can be an articulation point in G.
(c) Root of T can never be an articulation point in G.
(d) If u is an articulation point in G such that x is an ancestor of u in T and y is a descendent of u in T, then all paths from x to y in G must pass through u.

GATE 2021 set 2 || NAT || 1-mark

Q9. Consider a complete binary tree with 7 nodes. Let A denote the set of first 3 elements obtained by performing Breadth-First Search (BFS) starting from the root. Let B denote the set of first 3 elements obtained by performing Depth-First Search (DFS) starting from the root.The value of |A – B| is _____

GATE 2021 SET 1 || MCQ || 1-mark

Q10. A binary search T contains n distinct elements. What is the time complexity of picking an element in T that is smaller than the maximum element in T?

(a) Θ(n)
(b) Θ(log n)
(c) Θ(n log n)
(d) Θ(1)

GATE CSE - 2020 MCQ || 1-mark

Q11. In a balanced binary search tree with n elements, what is the worst case time complexity of reporting all elements in range [a,b]? Assume that the number of reported elements is k.

(a) Θ(n log k)
(b) Θ(log n + k)
(c) Θ(k log n)
(d) Θ(logn)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2020 MCQ || 1-mark

Q12. What is the worst case time complexity of inserting n^2 elements into an AVL-tree with n elements initially?

(a) $\Theta(n^4)$
(b) $\Theta(n^2)$
(c) $\Theta(n^3)$
(d) $\Theta(n^2 \log n)$

GATE CSE - 2020 MCQ (2-MARKS)

Q13. The preorder traversal of a binary search tree is 15, 10, 12, 11, 20, 18, 16, 19.

Which one of the following is the postorder traversal of the tree?

(a) 20, 19, 18, 16, 15, 12, 11, 10
(b) 11, 12, 10, 16, 19, 18, 20, 15
(c) 10, 11, 12, 15, 16, 18, 19, 20
(d) 19, 16, 18, 20, 11, 12, 10, 15

GATE CSE - 2019 NAT || 2-mark

Q14. Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.) Suppose two leaves a and b of T are chosen uniformly and independently at random. The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) _____

GATE CSE - 2017 MCQ (1-MARKS)

Q15. The pre-order traversal of a binary search tree is given by 12, 8, 6, 2, 7, 9, 10, 16, 15, 19, 17, 20. Then the post-order traversal of this tree is:

(a) 2, 6, 7, 8, 9, 10, 12, 15, 16, 17, 19, 20
(b) 2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12
(c) 7, 2, 6, 8, 9, 10, 20, 17, 19, 15, 16, 12
(d) 7, 6, 2, 10, 9, 8, 15, 16, 17, 20, 19, 12

GATE CSE - 2017 MCQ (2-MARKS)

Q16. Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are:

Note: The height of a tree with a single node is 0.

(a) 4 and 15 respectively
(b) 3 and 14 respectively
(c) 4 and 14 respectively
(d) 3 and 15 respectively

GATE CSE - 2017 MCQ (1-MARKS)

Q17. A binary tree T has 20 leaves. The number of nodes in T having two children is _____.

(a) 19
(b) 20
(c) 21
(d) 22

GATE CSE - 2016 MCQ (2-MARKS)

Q18. An operator delete(i) for a binary heap data structure is to be designed to delete the item in the i-th node. Assume that the heap is implemented in an array and i refers to the i-th index of the array. If the heap tree has depth d (number of edges on the path from the root to the farthest leaf), then what is the time complexity to re-fix the heap efficiently after the removal of the element?

(a) O(1)
(b) O(d) but not O(1)
(c) $O(2^d)$ but not O(d)
(d) $O(d2^d)$ but not $O(2^d)$

GATE CSE - 2016 SET 2 || NAT || 2-mark

Q19. A complete binary min-heap is made by including each integer in [1, 1023] exactly once. The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is

_____

GATE CSE – 2016 SET 2 MCQ || 2-Mark

Q20. The number of ways in which the numbers 1, 2, 3, 4, 5, 6, 7 can be inserted in an empty binary search tree, such that the resulting tree has height 6, is

_____

Check Out Our GATE Courses & Secure Your Dream Rank!

Note: The height of a tree with a single node is 0.

[This question was originally a Fill-in-the-Blanks question]

(a) 2
(b) 4
(c) 64
(d) 32

GATE CSE - 2016 SET 2 || MCQ || 1-mark

Q21. Consider the following New-order strategy for traversing a binary tree:

• Visit the root;

• Visit the right subtree using New-order;

• Visit the left subtree using New-order;

The New-order traversal of the expression tree corresponding to the reverse polish expression 3 4 * 5 - 2 ^ 6 7 * 1 + - is given by:

(a) + - 1 6 7 * 2 ^ 5 - 3 4 *
(b) - + 1 * 6 7 ^ 2 - 5 * 3 4
(c) - + 1 * 7 6 ^ 2 - 5 * 4 3
(d) 1 7 6 * + 2 5 4 3 * - ^ -

GATE CSE 2015 SET 3 || MCQ || 1-mark

Q22. Consider the following array of elements. ⟨89, 19, 50, 17, 12, 15, 2, 5, 7, 11, 6, 9, 100⟩. The minimum number of interchanges needed to convert it into a max-heap is

(a) 4
(b) 5
(c) 2
(d) 3

GATE CSE – 2015 SET 1 || MCQ || 2-mark

Q23. Consider a max heap, represented by the array: 40, 30, 20, 10, 15, 16, 17, 8, 4. Now consider that a value 35 is inserted into this heap. After insertion, the new heap is

(a) 40, 30, 20, 10, 15, 16, 17, 8, 4, 35
(b) 40, 35, 20, 10, 30, 16, 17, 8, 4, 15
(c) 40, 30, 20, 10, 35, 16, 17, 8, 4, 15
(d) 40, 35, 20, 10, 15, 16, 17, 8, 4, 30

GATE CSE – 2015 SET 3 || MCQ || 1-mark

Q24. Consider a binary tree T that has 200 leaf nodes. Then, the number of nodes in T that have exactly two children are ____

(a) 199
(b) 198
(c) 197
(d) 196

GATE CSE – 2015 SET 1 || MCQ || 1-mark)

Q25. The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are

(a) 63 and 6, respectively
(b) 64 and 5, respectively
(c) 32 and 6, respectively
(d) 31 and 5, respectively

GATE CSE – 2015 SET 3 MCQ || 1-mark

Q26. While inserting the elements 71, 65, 84, 69, 67, 83 in an empty binary search tree (BST) in the sequence shown, the element in the lowest level is

(a) 65
(b) 67
(c) 69
(d) 83

GATE CSE - 2014 SET 1 || MCQ || 1-mark

Q27. Consider it to rooted n node binary tree represented using pointers. The best upper bound on the time required to determine the number of subtrees having having exactly 4 nodes $O(n^a \log^b n)$. Then the value of a + 10b is _____

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) 1
(b) 11
(c) 12
(d) 21

GATE CSE - 2014 SET 2 || MCQ || 1-mark

Q28. A priority queue is implemented as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is: 10, 8, 5, 3, 2. Two new elements 1 and 7 are inserted into the heap in that order. The level-order traversal of the heap after the insertion of the elements is:

(a) 10, 8, 7, 3, 2, 1, 5
(b) 10, 8, 7, 2, 3, 1, 5
(c) 10, 8, 7, 1, 2, 3, 5
(d) 10, 8, 7, 5, 3, 2, 1

GATE CSE - 2014 SET 3 || MCQ || 2-mark

Q29. Consider the C function given below. Assume that the array listA contains n(>0) elements, sorted in ascending order

```
int ProcessArray(int *listA, int x, int n)
{
    int i, j, k;
    i = 0;     j = n-1;
     do {
        k = (i+j)/2;
        if (x <= listA[k]) j = k-1;
        if (listA[k] <= x) i = k+1;
       }
    while (i <= j);
    if (listA[k] == x) return(k);
    else   return -1;
}
```

Which one of the following statements about the function ProcessArray is CORRECT?

(a) It will run into an infinite loop when x is not in listA.
(b) It is an implementation of binary search.
(c) It will always find the maximum element in listA.
(d) It will return −11 even when xx is present in listA.

GATE CSE – 2014 SET 3 || MCQ || 2-mark

Q30. Consider the pseudocode given below. The function DoSomething() takes as argument a pointer to the root of an arbitrary tree represented by the leftMostChild-rightSibling representation. Each node of the tree is of type treeNode.

```
typedef struct treeNode* treeptr;
struct treeNode
{
treeptr leftMostChild, rightSibling;
};
int DoSomething (treeptr tree)
{
 int value=0;
if (tree != NULL) {
if (tree->leftMostChild == NULL)  value = 1;
else
value = DoSomething(tree->leftMostChild);
value = value + DoSomething(tree->rightSibling);
 }
 return(value);
 }
```


Check Out Our GATE Courses & Secure Your Dream Rank!

When the pointer to the root of a tree is passed as the argument to DoSomething, the value returned by the function corresponds to the

(a) number of internal nodes in the tree.

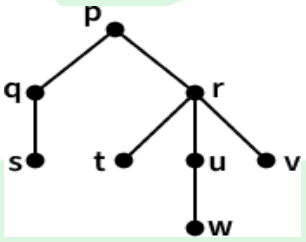
(b) height of the tree.

(c) number of nodes without a right sibling in the tree.

(d) number of leaf nodes in the tree.

GATE CSE - 2014 || MCQ || 2-mark

Q31. Consider the following rooted tree with the vertex P labelled as root



The order in which the nodes are visited during in-order traversal is

(a) SQPTRWUV
(b) SQPTURWV
(c) SQPTWUVR
(d) SQPTRUWV

GATE CSE - 2013 MCQ || 2-mark

Q32. The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the postorder traversal sequence of the same tree?

(a) 10, 20, 15, 23, 25, 35, 42, 39, 30
(b) 15, 10, 25, 23, 20, 42, 35, 39, 30
(c) 15, 20, 10, 23, 25, 42, 35, 39, 30
(d) 15, 10, 23, 25, 20, 35, 42, 39, 30

GATE CSE - 2012 MCQ || 2-mark

Q33. The height of a tree is defined as the number of edges on the longest path in the tree. The function shown in the pseudocode below is invoked as height (root) to compute the height of a binary tree rooted at the tree pointer root.

```
int height(treeptr n)
    {
    if(n == NULL) return -1;
    if(n -> left == NULL)
    if(n -> right == NULL) return 0;
    else return B1; // Box 1
     else
         {
    h1 = height(n -> left);
    if(n -> right == NULL) return (1+h1);
    else{h2 = height(n -> right);
    return B2; // Box 2}
         }
    }
```

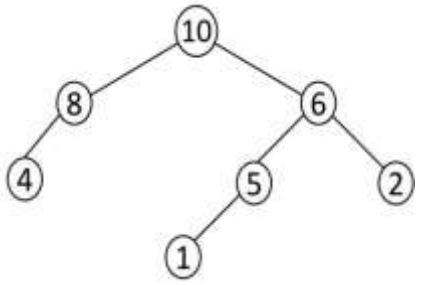
the appropriate expression for the two boxes B1 and B2 are

(a) B1: (1+height(n→right)) B2: (1+max(h1,h2))
(b) B1: (height(n→right) B2: (1+max(h1,h2))
(c) B1: height(n→right) B2: max(h1,h2)
(d) B1: (1+ height(n→right)) B2: max(h1,h2)

GATE CSE - 2011 MCQ || 1-mark |

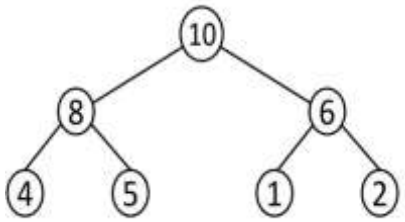
Q34. A max-heap is a heap where the value of each parent is greater than or equal to the values of its children. Which of the following is a max-heap?

(a)

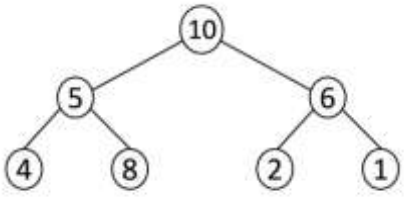


Check Out Our GATE Courses & Secure Your Dream Rank!

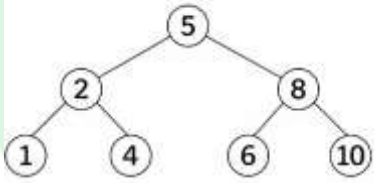
(b)



(c)



(d)



GATE CSE - 2011 MCQ || 2-mark

Q35. We are given a set of n distinct elements and an unlabeled binary tree with n nodes. In how many ways can we populate the tree with the given set so that it becomes a binary search tree?

(a) 0
(b) 1
(c) n!
(d) $\frac{1}{n+1} * {}^{2n}C_n$

GATE CSE - 2010 MCQ || 1-mark

Q36. In a binary tree with n nodes, every node has an odd number of descendants. Every node is considered to be its own descendant. What is the number of nodes in the tree that have exactly one child?

(a) 0
(b) 1
(c) (n - 1)/2
(d) n – 1

GATE CSE 2009 || MCQ || 2 MARK

Q37. What is the maximum height of any AVL-tree with 7 nodes ? Assume that the height of a tree with a single node is 0.

(a) 2
(b) 3
(c) 4
(d) 5

GATE CSE 2009 || MCQ || 2 MARKS

Q38. Consider a binary max-heap implemented using an array. Which one of the following array represents a binary max-heap?

(a) {25, 12, 16, 13, 10, 8, 14 }
(b) {(25, 14, 13, 16, 10, 8, 12}
(c) {25, 14, 16, 13, 10, 8, 12}
(d) {25, 14, 12, 13, 10, 8, 16}

GATE CSE 2009 || MCQ || 2 MARKS

Q39. Consider a binary max-heap implemented using an array. What is the content of the array after two delete operations on

(a) {14, 13. 12. 10, 8}
(b) {14, 12,13, 8, 10}
(c) {14, 13, 8, 12, 10}
(d) {14, 13, 12, 8, 10}

GATE CSE 2008 || MCQ || 2 MARK

Q40. You are given the postorder traversal, P, of a binary search tree on the n elements 1,2,..., n. You have to determine the unique binary search tree that has P as its postorder traversal. What is the time complexity of the most efficient algorithm for doing this?

(a) O(logn)
(b) O(n)
(c) O(nlogn)
(d) none of the above, as the tree cannot be uniquely determined.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE 2007 || 1 MARK || MCQ

Q41. The height of a binary tree is the maximum number of edges in any root to leaf path. The maximum number of nodes in a binary tree of height h is:

(a) 2^h-1
(b) $2^{h-1}-1$
(c) $2^{h+1}-1$
(d) 2^{h+1}

GATE CSE 2007 || MCQ || 1 MARK

Q42. The maximum number of binary trees that can be formed with three unlabeled nodes is:

(a) 1
(b) 5
(c) 4
(d) 3

GATE CSE – 2014 SET - 3 MCQ (1-MARKS)

Q43. The inorder and preorder traversal of a binary tree are
Inorder: d b e a f c g
Preorder: a b d e c f g
The postorder traversal of the binary tree is:

(a) d e b f g c a
(b) e d b g f c a
(c) e d b f g c a
(d) d e f g b c a

GATE CSE 2007 || MCQ || 2 MARKS

Q44. A complete n-ary tree is a tree in which each node has n children or no children. Let I be the number of internal nodes and L be the number of leaves in a complete n-ary tree. If L = 41, and I = 10, what is the value of n?

(a) 3
(b) 4
(c) 5
(d) 6

GATE CSE 2007 || MCQ || 2 MARKS

Q45. Consider the process of inserting an element into a Max Heap, where the Max Heap is represented by an array. Suppose we perform a binary search on the path from the new leaf to the root to find the position for the newly inserted element, the number of comparisons performed is:

(a) $\Theta(\log_2 n)$
(b) $\Theta(\log_2 \log_2 n)$
(c) $\Theta(n)$
(d) $\Theta(n \log_2 n)$

GATE CSE 2006 || MCQ || 1 MARKS

Q46. In a binary max heap containing n numbers, the smallest element can be found in time

(a) O(n)
(b) O(log n)
(c) O(log log n)
(d) O(1)

GATE CSE 2006 || MCQ || 2 MARKS

Q47. A 3-ary max heap is like a binary max heap, but instead of 2 children, nodes have 3 children. A 3-ary heap can be represented by an array as follows: The root is stored in the first location, A[0], nodes in the next level, from left to right, are stored from A[1] to A[3]. The nodes from the second level of the tree from left to right are stored from A[4] onward. An item x can be inserted into a 3-ary heap containing n items by placing x in the location A[n] and pushing it up the tree to satisfy the heap property.
Which one of the following is a valid sequence of elements in an array representing a 3-ary max heap?

(a) 1, 3, 5, 6, 8, 9
(b) 9, 6, 3, 1, 8, 5
(c) 9, 3, 6, 8, 5, 1
(d) 9, 5, 6, 8, 3, 1

GATE CSE 2006 || MCQ || 2 MARKS

Q48. A 3-ary max heap is like a binary max heap, but instead of 2 children, nodes have 3 children. A 3-ary heap can be represented by an array as follows: The

Check Out Our GATE Courses & Secure Your Dream Rank!

root is stored in the first location, A[0], nodes in the next level, from left to right, are stored from A[1] to A[3]. The nodes from the second level of the tree from left to right are stored from A[4] onward. An item x can be inserted into a 3-ary heap containing n items by placing x in the location A[n] and pushing it up the tree to satisfy the heap property.
Suppose the elements 7, 2, 10 and 4 are inserted, in that order, into the valid 3-ary max heap found in the previous question . Which one of the following is the sequence of items in the array representing the resultant heap?

(a) 10, 7, 9, 8, 3, 1, 5, 2, 6, 4
(b) 10, 9, 8, 7, 6, 5, 4, 3, 2, 1
(c) 10, 9, 4, 5, 7, 6, 8, 2, 1, 3
(d) 10, 8, 6, 9, 7, 2, 3, 4, 1, 5

GATE CSE 2005 || MCQ || 2 MARKS

Q49. Postorder traversal of a given binary search tree, T produces the following sequence of keys:
10, 9, 23, 22, 27, 25, 15, 50, 95, 60, 40, 29

Which one of the following sequences of keys can be the result of an in-order traversal of the tree T?

(a) 9, 10, 15, 22, 23, 25, 27, 29, 40, 50, 60, 95
(b) 9, 10, 15, 22, 40, 50, 60, 95, 23, 25, 27, 29
(c) 29, 15, 9, 10, 25, 22, 23, 27, 40, 60, 50, 95
(d) 95, 50, 60, 40, 27, 23, 22, 25, 10, 9, 15, 29

GATE CSE 2005 || MCQ || 2 MARKS

Q50. A priority queue is implemented as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is:10, 8, 5, 3, 2
Two new elements 1 and 7 are inserted into the heap in that order. The level-order traversal of the heap after the insertion of the elements is:

(a) 10, 8, 7, 5, 3, 2, 1
(b) 10, 8, 7, 2, 3, 1, 5
(c) 10, 8, 7, 1, 2, 3, 5
(d) 10, 8, 7, 3, 2, 1, 5

GATE CSE 2005 || MCQ || 2 MAKRS

Q51. How many distinct binary search trees can be created out of 4 distinct keys?

(a) 5
(b) 14
(c) 24
(d) 42

GATE CSE – 2005 NAT (2-MARKS)

Q52. In a complete k-ary tree, every internal node has exactly k children. The number of leaves in such a tree with n internal nodes is:

(a) nk
(b) (n − 1)k + 1
(c) n(k − 1) + 1
(d) n(k − 1)

GATE CSE – 2005 MCQ (2-MARKS)

Q53. Suppose there are $\lceil \log n \rceil$ sorted lists of $\lfloor n/\log n \rfloor$ elements each. The time complexity of producing a sorted list of all these elements is: (Hint: Use a heap data structure)

(a) $O(n \log \log n)$
(b) $\theta(n \log n)$
(c) $\Omega(n \log n)$
(d) $\Omega(n^{\frac{3}{2}})$

Check Out Our GATE Courses & Secure Your Dream Rank!

ANSWER KEY

1. (11)	**2. (7)**	**3. (a)**
4. (a,c)	**5. (c)**	**6. (509)**
7. (a)	**8. (a)**	**9. (1)**
10. (d)	**11. (b)**	**12. (d)**
13. (b)	**14. (4.24-4.26)**	**15.(b)**
16. (b)	**17. (19)**	**18. (b)**
19. (8)	**20. (c)**	**21. (c)**
22. (3)	**23. (b)**	**24. (199)**
25. (a)	**26. (b)**	**27. (a)**
28. (a)	**29. (b)**	**30. (d)**
31. (a)	**32. (d)**	**33. (a)**
34. (b)	**35.(b)**	**36. (a)**
37. (b)	**38. (c)**	**39. (d)**
40. (b)	**41. (c)**	**42. (5)**
43. (a)	**44. (c)**	**45. (b)**
46. (a)	**47. (d)**	**48. (a)**
49. (a)	**50. (d)**	**51. (b)**
52. (c)	**53.(a)**	

HINTS&SOLUTIONS

1. ANSWER:- 11

Explanation:

In a full binary tree, every internal node has exactly 2 children.

Let

I = number of internal nodes

L = number of leaves

For a full binary tree:

$L = I + 1$

Total nodes:

$N = I + L$

$N = I + (I + 1)$

$N = 2I + 1$

Given N = 23:

$2I + 1 = 23$

$2I = 22$

$I = 11$

Now, to get maximum height, the tree must be as skewed as possible while still being full.

In a full binary tree, maximum height occurs when at each level, only one subtree continues downward and the other is a leaf.

For a full binary tree:

Maximum height = I

Since I = 11,

Maximum possible height = 11

2. ANSWER:- 7

Explanation:

Given preorder of BST:

50, 25, 13, 40, 30, 47, 75, 60, 70, 80, 77

Step 1: Construct BST using preorder.

Root = 50

Left subtree (< 50):

25, 13, 40, 30, 47

Right subtree (> 50):

75, 60, 70, 80, 77

Build left subtree:

25

Left: 13

Right: 40

40 → Left: 30, Right: 47

Build right subtree:

75

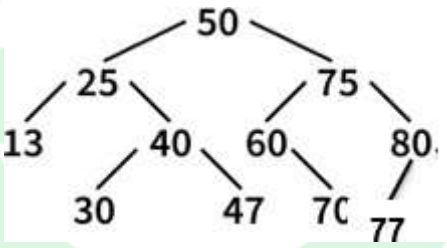
Left: 60

60 → Right: 70

Right of 75 → 80

80 → Left: 77

Final tree structure:



Step 2: Postorder (Left, Right, Root)

Left subtree postorder:

13, 30, 47, 40, 25

Right subtree postorder:

70, 60, 77, 80, 75

Whole tree postorder:

13, 30, 47, 40, 25, 70, 60, 77, 80, 75, 50

Now find position of 60.

Counting from 1:

1 → 13

2 → 30

3 → 47

4 → 40

5 → 25

6 → 70

7 → 60

Position of 60 = 7

3. ANSWER:- A

Explanation : Inorder → Left subtree, Node, Right subtree

Preorder → Node, Left subtree, Right subtree

Postorder → Left subtree, Right subtree, Node

Check Out Our GATE Courses & Secure Your Dream Rank!

Now match with given labels:
L → Left subtree, Node, Right subtree
M → Node, Left subtree, Right subtree
N → Left subtree, Right subtree, Node
So correct matching:
I (Inorder) → L
II (Preorder) → M
III (Postorder) → N
Correct answer: I – L, II – M, III – N

4. ANSWER:- A, C
Explanation:
Given: For every leaf LiL_iLi having a non-NULL inorder successor, one key is inserted between
Val(Li) and Val(Suc(Li)).
(A) No duplicates in K
Each leaf has a unique inorder successor, so each new key lies in a distinct interval.
Intervals do not overlap → no duplicates.
True.
(B) K will have at least one element
If BST has only one node (one leaf), it has no successor.
So no key is inserted. False.
(C) Height increases at most by one
Each new key becomes a child of an existing leaf.
So only one extra level can be added.
True.
(D) Number of nodes will double
Number of inserted keys ≤ number of leaves − 1.
So total nodes do not double.
False.
Correct answer: A and C

5. ANSWER:- C
Explanation:
Evaluation of retVal for each node:
For node whose value is 3, retVal = 3 + 0 + 0 = 3
For node whose value is 8, retVal = 8 + 0 + 0 = 8
For node whose value is 5, retVal = 5 + 3 + 8 = 16
For node whose value is 13, retVal = 13 + 0 + 0 = 13
For node whose value is 11, retVal = 11 + 0 + 13 = 24
For node whose value is 10, retVal = 10 + 16 + 24 = 50

6. ANSWER:- 509
Explanation:
BST Structure: In a Binary Search Tree, the largest elements are always in the rightmost path.

- Array Representation: For a node at index i, the right child is at index 2i + 2 (assuming 0-based indexing).
- Finding the Largest:
- Largest (Root of rightmost subtree): For 1000 nodes, the rightmost leaf is found by repeatedly moving right: 0 -> 2 -> 6 -> 14 -> 30 -> 62 -> 126 -> 254 -> 510
- 2nd Largest: This would be the parent of the largest leaf: index 254.
- 3rd Largest: In this setting, the 3rd largest element is the left child of the second largest element.

Calculation: Index of left child = 2 * 254 + 1 = 509

7. ANSWER:- A
Explanation:

In a binary min-heap, the smallest element is at the root (index 0), accessible in O(1) time.

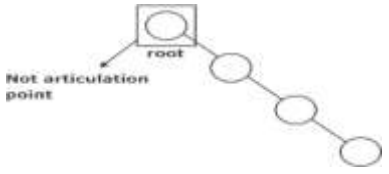
The largest element in a min-heap must be a leaf node, because if an internal node were the largest, it would violate the heap property (parent ≤ children).

Leaves are at indices from ⌊n/2⌋ to n-1, and there are roughly n/2 leaves.

An optimal algorithm scans only the leaf nodes to find the maximum.
Scanning n/2 elements takes O(n) time in the worst case.

Thus, the worst-case time complexity is:
O(n)

8. ANSWER:- A
Explanation:



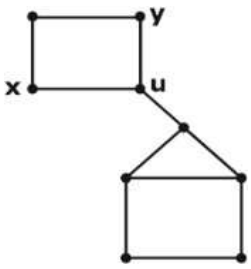
(a) We need at least 2 children so that root is articulation point.

(b) False: This can never happen. Leaf will always have degree = 1.

(c) False: Check option (a) for more information.

(d) False: Below is the reasoning to show how this is false.

Check Out Our GATE Courses & Secure Your Dream Rank!

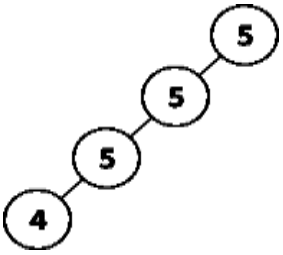


If u is articulation point. then removing u generates 2 connected components, now there might be a case when x and y will belong to either one of the connected component and hence a path will exist between them without passing through u.

9. ANSWER:- 1
Explanation:
For a complete binary tree with 7 nodes (labeled 1 to 7 in level-order), we compare the first 3 elements of BFS and DFS.
BFS (Set A): Level-by-level traversal.
First 3 elements: {1, 2, 3}
DFS (Set B): Pre-order traversal (Root Left → Right).
First 3 elements: {1, 2, 4}
Calculation:
AB represents elements in A that are not in B.
A- B = {3}
Value of AB = 1

10. ANSWER:- *D*
Explanation:

To find smaller than maximum element, it may take O(n) in worst case if elements are duplicate



To find element smaller than maximum, you need to reach leaf. It takes O(n).

Note: If all elements are district time complexity is O(1)

11. ANSWER:- B
Explanation:

The idea is to traverse the given binary search tree starting from root. For every node being visited, check if this node lies in range, if yes, then add 1 to result and recur for both of its children. If the current node is smaller than the low value of range, then recur for right child, else recur for left child.

Time complexity of the above program is O(h + k) where h is the height of BST and k is the number of nodes in a given range.

Here h is logn, hence O(logn+k)

12. ANSWER:- D

AVL Tree all operations(insert, delete and search) will take O(logn) time.

In question they asked about n^2 elements.

So, In worst case it will take o(n^2logn) time.

13. ANSWER:- B
Explanation:

Preorder of BST is: 15, 10, 12, 11, 20, 18, 16, 19.

In a Binary Search Tree, left subtree contains values smaller than root and right subtree contains values greater than root.

First element of preorder is the root. So root = 15.

Elements smaller than 15 form the left subtree: 10, 12, 11.
Elements greater than 15 form the right subtree: 20, 18, 16, 19.

Left subtree:
Root = 10.
12 is greater than 10, so it is right child of 10.
11 is less than 12 but greater than 10, so it becomes left child of 12.

Postorder of left subtree (Left, Right, Root):
11, 12, 10

Right subtree:
Root = 20.
18 is less than 20, so it is left child of 20.

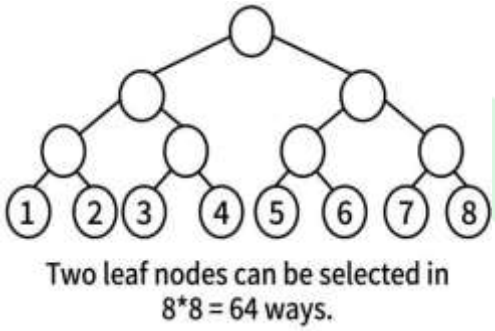
Check Out Our GATE Courses & Secure Your Dream Rank!

16 is less than 18, so it is left child of 18.
19 is greater than 18 but less than 20, so it is right child of 18.

Postorder of right subtree:
16, 19, 18, 20

Final postorder (Left, Right, Root):
11, 12, 10, 16, 19, 18, 20, 15

14. ANSWER:- 4.24 - 4.26
Explanation:

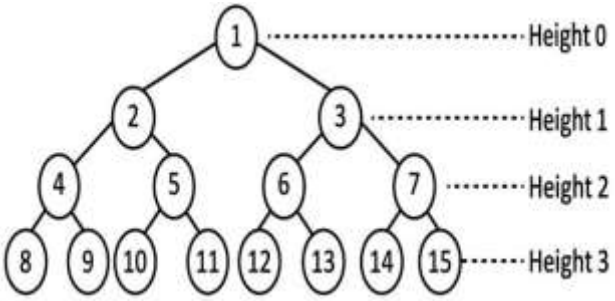


Two leaf nodes can be selected in 8*8 = 64 ways.

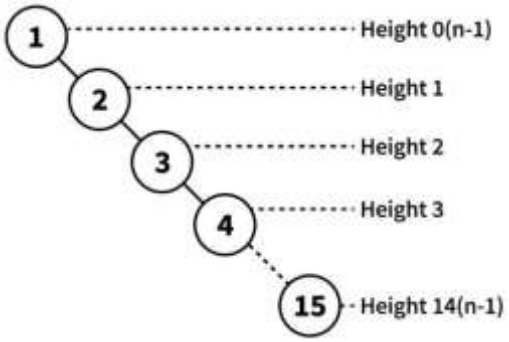
Two leaf nodes can be selected in 8*8 = 64 ways
No. of pairs with path length 08. (When a = b)
No. of pairs with path length 28. (For each of the 8 leaf nodes, we have a pair and since the selection is independent order of the pair is also significant)
No. of pairs with path length 416. (For each leaf node we have to go two levels up and while coming down we have 2 choices. So, we get 8 * 2 = 16 pairs)
No. of pairs with path length 632. (For each leaf node we have to go till the root, and from there while coming down it has 2 * 2 = 4 choices. Thus we get 8 * 4 = 32 pairs.
Total number of possible pairs = 8 * 8 = 64
So, expected path length
$= 0 \times \frac{8}{64} + 2 \times \frac{8}{64} + 4 \times \frac{16}{64} + 6 \times \frac{32}{64} = \frac{272}{64} = 4.25$

15. ANSWER:- B
Explanation:
Preorder: 12, 8, 6, 2, 7, 9, 10, 16, 15, 19, 17, 20
BST reconstruction:
- Root 12
- Left subtree (all <12): 8, 6, 2, 7, 9, 10
- Right subtree (all >12): 16, 15, 19, 17, 20
  Left subtree preorder: 8, 6, 2, 7, 9, 10
- Root 8
- Left of 8: 6, 2, 7
- Right of 8: 9, 10
  Right subtree preorder: 16, 15, 19, 17, 20
- Root 16
- Left of 16: 15
- Right of 16: 19, 17, 20
  Postorder (Left, Right, Root):
- Left subtree postorder: 2, 7, 6, 10, 9, 8
- Right subtree postorder: 15, 17, 20, 19, 16
- Root: 12
  Final postorder: 2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12

16. ANSWER:- B
Explanation:
T be a Binary Search Tree with 15 nodes. The height of a tree with single node is 0. Minimum possible height is when it is a complete binary tree.



Maximum possible height is when it is a skewed tree left/right.



So the minimum and maximum possible heights of T are: 3 and 14 respectively.

17. ANSWER:- 19
Explanation:
Let the number of vertices of a binary tree with „p" leaves be n then the tree has
(i) p vertices (i.e., leaves) of degree 1
(ii) one vertex (i.e.., root of T) of degree 2
(iii) 'n- p-1' (i.e., interval) vertices of degree 3
(iv) n- 1 edges
∴ By Handshaking theorem,
$p \times 1 + 1 \times 2 + (n - p - 1) \times 3 = 2(n - 1)$
$\Rightarrow n = 2p - 1 = 39$ as p = 20
∴ n-p=19 vertices have exactly two children

Check Out Our GATE Courses & Secure Your Dream Rank!

18. ANSWER:- B
Explanation:

When you delete an arbitrary element at index i*i* in a binary heap:

1. Swap the element at index i*i* with the last element in the heap.
2. Remove the last element (now the original i*i*-th element).
3. Heapify from index i*i*:

- May need to bubble up if it's larger than its parent (if it's a max-heap)
- Or bubble down if it's smaller than one of its children
- But since we don't know whether it's too large or too small, we must check both.

However, in practice, we compare with parent:

- If it's greater than parent (in max-heap), bubble up — O(d)
- Else, bubble down — O(d)

So in either case, we do at most one path up or down the tree.

Height d = depth = O(log n)
Thus time complexity = O(d)=O(log n)

FINAL ANSWER = O(d)

19. ANSWER:- 8
Explanation:

Here node with integer 1 has to be at root only. Now for maximum depth of the tree the following arrangement can be taken. Take root as level 1.

make node 2 at level 2 as a child node of node 1.

make node 3 at level 3 as the child node of node 2.

..

.. and so on for nodes 4,5,6,7

..

make node 8 at level 8 as the child node of node 7.

make node 9 at level 9 as the child node of node 8.

Putting other nodes properly, this arrangement of the complete binary tree will follow the property of min heap.

So total levels are 9. node 9 is at level 9 and depth of node 9 is 8 from the root.

20. ANSWER:- C
Explanation:

Height = 6 means maximum possible height for 7 nodes.
So the BST must be completely skewed (a chain of 7 nodes).

Now think carefully.

For height to be 6, at every step after choosing the root, all remaining elements must go only to one side (either all smaller or all greater).

First, choose the root.

If root is 1, then all remaining 6 elements (2–7) must go to the right side.
If root is 7, then all remaining 6 elements (1–6) must go to the left side.

If root is anything between 2 and 6, then some elements will go left and some right, so height will be less than 6. So only 1 or 7 can be the root.

After fixing the root (say 1), the remaining 6 elements must again form a skewed BST of height 5.

So the problem reduces recursively.

For 7 nodes to form a chain, at each step we must choose either the smallest or the largest element among the remaining elements.

At first step: 2 choices (1 or 7).
At second step: again 2 choices (smallest or largest of remaining 6).
At third step: again 2 choices.
This continues for 6 steps.

Total ways = $2 \times 2 \times 2 \times 2 \times 2 \times 2$
$= 2^6$
$= 64$

Check Out Our GATE Courses & Secure Your Dream Rank!

21. ANSWER:- - C

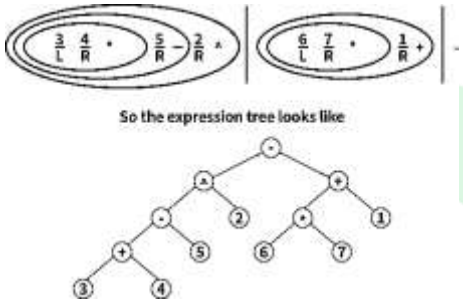
Explanation:

New Order strategy: Root, Right, Left.

Given Reverse Polish Notation as:

3 4 * 5 - 2 ^ 6 7 * 1 + -

We know Reverse Polish Notation takes Left, Right, Root.



From the tree, we can write the New Order traversal as: Root, Right, Left.

– + 1 * 7 6 ^ 2 – 5 * 4 3

22. ANSWER:- 3
Explanation:

Array: [89, 19, 50, 17, 12, 15, 2, 5, 7, 11, 6, 9, 100]

Heapify from last parent (index 5) to root:

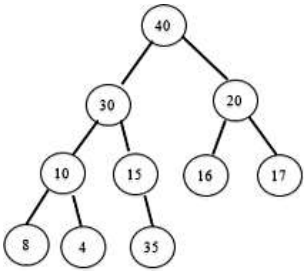
1. Index 5 (15): children 9, 100 → swap with 100 → [89, 19, 50, 17, 12, 100, 2, 5, 7, 11, 6, 9, 15]
2. Index 2 (50): children 100, 2 → swap with 100 → [89, 19, 100, 17, 12, 50, 2, 5, 7, 11, 6, 9, 15]
3. Index 0 (89): children 19, 100 → swap with 100 → [100, 19, 89, 17, 12, 50, 2, 5, 7, 11, 6, 9, 15]

No further swaps needed.

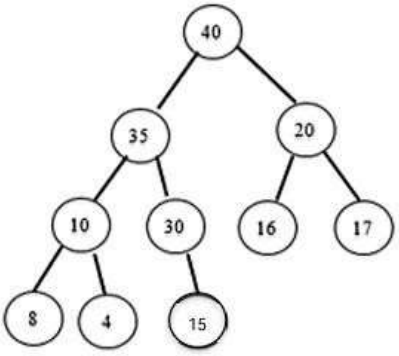
Total interchanges = 3

23. ANSWER:- B
Explanation:

After insertion of 35 the heap becomes:



Swapping 15 and 35



24. ANSWER:- 199
Explanation:

A binary tree T with n leaf nodes have exactly (n - 1) nodes that have exactly two children.

Binary tree property:

Leaves=Nodes with 2 children + 1

Given leaves = 200:

$N_2+1 = 200$

$N_2 = 199$

Check Out Our GATE Courses & Secure Your Dream Rank!

25. ANSWER:- A
Explanation:

Height = 5 ⇒ longest path has 5 edges ⇒ 6 nodes on that path.

- **Maximum nodes** in a binary tree of height *h*:
  Full binary tree: $2^{h+1}-1$ nodes.
  For h=5 then 64-1 = 63 nodes.

- **Minimum nodes** in a binary tree of height *h*:
  Skewed tree: h+1 nodes.
  For h=5 5+1=6 nodes.

Final answer:

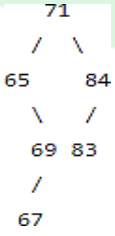
63 and 6

26. ANSWER:- B
Explanation:

Let's insert step-by-step into an empty BST.

Insert 71 → root: 71
Insert 65 → 65 < 71 → left child of 71
Insert 84 → 84 > 71 → right child of 71
Insert 69 → 69 < 71, 69 > 65 → right child of 65
Insert 67 → 67 < 71, 67 < 65? Wait, 67 > 65, 67 < 69 → left child of 69
Insert 83 → 83 > 71, 83 < 84 → left child of 84

Tree :



Levels:
Level 0: 71
Level 1: 65, 84
Level 2: 69, 83
Level 3: 67

Lowest level (level 3) has 67.

27. ANSWER:- A
Explanation:

Efficient method:
Post order traversal → compute subtree sizes in O(n) time.
Check if size = 4 → count.

Best upper bound: $O(n^1 \log^0 n)$ → a=1,b=0

a+10b=1+0=1

28. ANSWER:- A
Explanation:

Initial Max-Heap level order: 10, 8, 5, 3, 2
Index mapping (1-based):
1: 10
2: 8
3: 5
4: 3
5: 2

Step 1: Insert 1
Add at end (index 6): 1
Parent (index 3) = 5 → 1 < 5, so no swap.
Heap after: 10, 8, 5, 3, 2, 1

Step 2: Insert 7
Add at end (index 7): 7
Parent (index 3) = 5 → 7 > 5, swap.
Now 7 at index 3, 5 at index 7.
Parent of index 3 (index 1) = 10 → 7 < 10, stop.

Heap after: 10, 8, 7, 3, 2, 1, 5

Level order: 10, 8, 7, 3, 2, 1, 5

Final answer:

10,8,7,3,2,1,5

29. ANSWER:- B

Explanation:

The array is sorted in ascending order.

The loop is similar to binary search:

i = 0
j = n − 1
k = (i + j) / 2

If x ≤ listA[k], then j = k − 1
If listA[k] ≤ x, then i = k + 1

Important observation:

Check Out Our GATE Courses & Secure Your Dream Rank!

When x == listA[k], both conditions become true.

So:
j = k − 1
and
i = k + 1

This makes i > j immediately, so the loop stops.

After the loop, the function checks:

if (listA[k] == x) return k;
else return −1;

Since k was last computed and x == listA[k], it correctly returns k.

The loop always shrinks the search space (i increases or j decreases), so it cannot be infinite.

It does not always find the maximum element.

Therefore, the function behaves like binary search.

Correct answer: It is an implementation of binary search.

30. ANSWER:- D

Explanation:

In leftMostChild–rightSibling representation:

leftMostChild → first child
rightSibling → next node at same level

Now analyze the function.

If tree == NULL → return 0.

If tree is not NULL:

If tree->leftMostChild == NULL
that means the node has no children.
So it is a leaf node.
value = 1

Else
value=DoSomething(tree->leftMostChild)
This recursively processes all children of the node.

`After that:
value = value + DoSomething(tree->rightSibling)
This processes all siblings of the current node.

So the function counts:
1 for every node that has no leftMostChild (i.e., every leaf node).

And by recursive calls through rightSibling, it covers the entire tree.

Therefore, the function returns the total number of leaf nodes in the tree.

Correct answer: (d) number of leaf nodes in the tree

31. ANSWER:- A

Explanation: Algorithm Inorder(tree) - Use of Recursion

Steps:

1. Traverse the left subtree,

i.e., call Inorder(left-subtree)

2. Visit the root.

3. Traverse the right subtree,

i.e., call Inorder(right-subtree)

Understanding this algorithm requires the basic understanding of Recursion

Therefore, We begin in the above tree with root as the starting point, which is P.

Step 1(for node P) :

Traverse the left subtree of node or root P.

So we have node Q on left of P.

-> Step 1(for node Q)

Traverse the left subtree of node Q.

So we have node S on left of Q.

Step 1 (for node S)

Now again traverse the left subtree of node S which is

Check Out Our GATE Courses & Secure Your Dream Rank!

NULL here.

* Step 2(for node S)

Visit the node S, i.e print node S as the 1st element of inorder traversal.

* Step 3(for node S)

Traverse the right subtree of node S.

Which is NULL here.

Now move up in the tree to Q which is parent of S.(Recursion, function of Q called for function of S).

Hence we go back to Q.

-> Step 2(for node Q):

Visit the node Q, i.e print node Q as the 2nd element of inorder traversal.

-> Step 3 (for node Q)

Traverse the right subtree of node Q.

Which is NULL here.

Now move up in the tree to P which is parent of Q.(Recursion, function of P called for function of Q).

Hence we go back to P.

Step 2(for node P)

Visit the node P, i.e print node S as the 3rd element of inorder traversal.

Step 3 (for node P)

Traverse the right subtree of node P.

Node R is at the right of P.

Till now we have printed SQP as the inorder of the tree.

Similarly other elements can be obtained by traversing the right subtree of P.

The final correct order of Inorder traversal would be SQPTRWUV.

32. ANSWER:- D

Explanation:

Preorder: 30, 20, 10, 15, 25, 23, 39, 35, 42

BST reconstruction:

- Root 30
- 20 (left), 10 (left of 20), 15 (right of 10)
- 25 (right of 20), 23 (left of 25)
- 39 (right of 30), 35 (left of 39), 42 (right of 39)

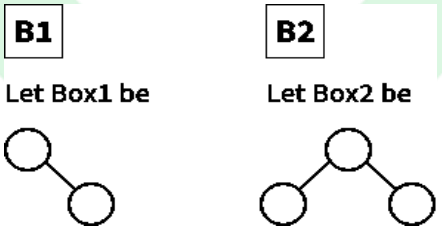
Postorder (Left, Right, Root):

- Left subtree of 30: 15, 10, 23, 25, 20
- Right subtree of 30: 35, 42, 39
- Root: 30

Postorder: 15, 10, 23, 25, 20, 35, 42, 39, 30

33. ANSWER:- A

Explanation:



The box B1 gets executed when left sub tree of n is NULL & right subtree is not NULL.

In such case, height of n will be the height of the right subtree +1.

The box B2 gets executed when both left & right subtrees of n are not NULL.

In such case, height of n will be max of heights of left & right subtrees of n+1.

Check Out Our GATE Courses & Secure Your Dream Rank!

34. ANSWER:- B
Explanation:

A binary tree is max-heap if it is a complete binary tree (A complete binary tree is a binary tree in which every level, except possibly the last, is completely filled, and all nodes are as far left as possible) and it follows the max-heap property (value of each parent is greater than or equal to the values of its children).

A. is not a max-heap because it is not a complete binary tree

B. is a max-heap because it is complete binary tree and follows max-heap property.

C. is not a max-heap because 8 is a child of 5 in this tree, so violates the max-heap property.

D. is not a max-heap because 8 is a child of 5 in this tree, so violates the max-heap property. There are many other nodes in this tree which violate max-heap property in this tree.

35. ANSWER:- B
Explanation:

Corresponding to each set only 1 binary search tree can be formed because in-order is fixed. only 1 tree possible. If Binary trees would be asked n! possible corresponding to each distinct tree set. Here tree structure is fixed and have only 1 possibility for BST as elements are distinct.

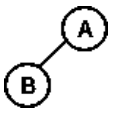
36. ANSWER:- A
Explanation:

Given a Binary Tree with n nodes.

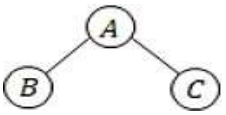
Every node has an odd number of descendants.

Also given every node is considered to be its own descendant.

This is even number of descendants (2), because A is also considered as a descendant.



This is odd number of descendants (3), A, B, C are descendants here.



Condition satisfied – odd number, but number of nodes in a tree that has exactly one child is 0.

37. ANSWER:- B
Explanation:
For an AVL tree, the height difference between left and right subtrees of any node is at most 1.
To get maximum height with minimum nodes, AVL follows the recurrence:
$N(h) = 1 + N(h-1) + N(h-2)$
where N(h) is the minimum number of nodes required for height h.
Given height of single node = 0.
Compute minimum nodes:
N(0) = 1
N(1) = 2
N(2) = 4
N(3) = 7
N(4) = 12
We are given total nodes = 7.
From above, height 3 requires minimum 7 nodes.
So with 7 nodes, maximum possible AVL height = 3.

38. ANSWER:- C
Explanation:
In a binary max-heap (array representation):
For every index i:
Parent ≥ Left child and Parent ≥ Right child
Children indices:
Left child = 2i
Right child = 2i + 1
Check each option.
(A) {25, 12, 16, 13, 10, 8, 14}
At index 2 → 12
Children: 13 and 10
12 < 13 → violates max-heap
Not correct.
(B) {25, 14, 13, 16, 10, 8, 12}
At index 2 → 14
Left child = 16
14 < 16 → violates
Not correct.
(C) {25, 14, 16, 13, 10, 8, 12}
Check all parents:

Check Out Our GATE Courses & Secure Your Dream Rank!

25 ≥ 14, 16
14 ≥ 13, 10
16 ≥ 8, 12
All conditions satisfied.
Valid max-heap.
(D) {25, 14, 12, 13, 10, 8, 16}
At index 3 → 12
Right child = 16
12 < 16 → violates
Not correct.
Correct answer: (C) {25, 14, 16, 13, 10, 8, 12}

39. ANSWER:- D

Explanation:

The original heap (from previous question) is:
{25, 14, 16, 13, 10, 8, 12}
Delete operation in max-heap removes the root and then heapify.
First delete:
Remove 25.
Replace root with last element 12.
Heap becomes:
{12, 14, 16, 13, 10, 8}
Heapify down:
Compare 12 with children 14 and 16.
Largest is 16 → swap.
{16, 14, 12, 13, 10, 8}
Now compare 12 with child 8.
12 > 8 → stop.
After first delete:
{16, 14, 12, 13, 10, 8}
Second delete:
Remove 16.
Replace root with last element 8.
Heap becomes:
{8, 14, 12, 13, 10}
Heapify down:
Compare 8 with 14 and 12.
Largest is 14 → swap.
{14, 8, 12, 13, 10}
Now compare 8 with children 13 and 10.
Largest is 13 → swap.
{14, 13, 12, 8, 10}
Now heap property satisfied.
Final array:
{14, 13, 12, 8, 10}

40. ANSWER:- B

Explanation:

In a Binary Search Tree, postorder traversal is:
Left → Right → Root
Given postorder of a BST with elements 1 to n, the tree can be uniquely constructed.
Reason:
In postorder, the last element is the root.
Using BST property, elements smaller than root form left subtree and larger elements form right subtree.
This partition can be done in linear time.
Then recursively construct left and right subtrees.
Each element is processed once.
So total time required is proportional to n.
Time complexity = O(n)

41. ANSWER:- C

Explanation:

Height h = maximum number of edges from root to leaf.
If height = h, then maximum number of levels = h + 1
(since height counts edges, not nodes).
Maximum nodes occur in a full binary tree.
Number of nodes in full binary tree with (h + 1) levels:
$= 1 + 2 + 4 + ... + 2h$
$= 2^{(h+1)} - 1$
Correct answer: $2^{(h+1)} - 1$

42. ANSWER:- 5

The number of binary trees that can be formed with n unlabeled nodes is given by the Catalan number:
Number of binary trees = $C_n = (1 / (n+1)) \times (2n \text{ choose } n)$
For n = 3:
Number of binary trees= $C_n = {}^{2n}C_n/(n+1)$
For n=3,

$C3 = {}^{6}C_3/4 = 20/4 = 5$

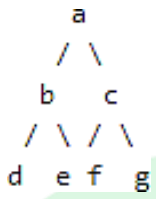
43. ANSWER:- A

Explanation:

Inorder: d b e a f c g
Preorder: a b d e c f g
Step 1: First element of preorder is root.
Root = a
Split inorder around a:
Left subtree inorder: d b e
Right subtree inorder: f c g
Step 2: Build left subtree
Preorder after removing root: b d e c f g
Left subtree preorder (3 elements): b d e
Root of left subtree = b
Split d b e around b:
Left of b: d
Right of b: e
So left subtree structure:
b
left → d
right → e
Step 3: Build right subtree

Check Out Our GATE Courses & Secure Your Dream Rank!

Remaining preorder: c f g
Root of right subtree = c
Split f c g around c:
Left of c: f
Right of c: g
So right subtree structure:
c
left → f
right → g
Final tree:



Step 4: Postorder (Left → Right → Root)
Left subtree postorder: d e b
Right subtree postorder: f g c
Root: a
Final postorder:
d e b f g c a

44. ANSWER:- C
Explanation:
In a complete n-ary tree (every internal node has exactly n children):
Total number of edges = n × I
Also, in any tree:
Number of edges = Total nodes − 1
= (I + L) − 1
So,
nI = I + L − 1
Given:
I = 10
L = 41
Substitute:
n × 10 = 10 + 41 − 1
10n = 50
n = 5

45. ANSWER:- B
Explanation:
In a max heap with n elements, the height of the heap is Θ(log n).
When inserting a new element, it is first placed at a leaf.
Normally, we move upward (heapify-up) along the path to the root.
The length of this path = height of heap = Θ(log n).
Now the question says we perform binary search on this path.
Binary search on a path of length Θ(log n) takes: Θ(log (log n)) comparisons.
Therefore, total comparisons = Θ(log log n).
Correct answer: (B) $\Theta(\log_2 \log_2 n)$

46. ANSWER:- A
Explanation:
In a binary max heap:
• The largest element is at the root.
• The smallest element must be among the leaf nodes.
About half of the nodes in a heap are leaves (≈ n/2).
To find the smallest element, we must check all leaf nodes in the worst case.
That takes linear time. Time complexity = O(n)

47. ANSWER:- D
Explanation: In a 3-ary max heap:
• Each parent must be ≥ its children
• For array index i, children are at indices:
3i + 1, 3i + 2, 3i + 3
Now check each option.
Option (a) 1, 3, 5, 6, 8, 9
Root = 1
Children = 3, 5, 6
1 < children → violates max heap
Not valid
Option (b) 9, 6, 3, 1, 8, 5
Root = 9 → OK
Children of 9 = 6, 3, 1 → OK
Now check node at index 1 (value 6)
Children indices = 4, 5, 6
Values = 8, 5
6 < 8 → violates max heap
Not valid
Option (c) 9, 3, 6, 8, 5, 1
Root = 9 → OK
Children = 3, 6, 8 → OK
Check index 1 (value 3)
Children = 5, 1
3 < 5 → violates
Not valid
Option (d) 9, 5, 6, 8, 3, 1
Root = 9 → OK
Children = 5, 6, 8 → OK
Check index 1 (value 5)
Children = 3, 1
5 ≥ 3 and 5 ≥ 1 → OK
All conditions satisfied.
Correct answer: (d) 9, 5, 6, 8, 3, 1

48. ANSWER:- A
Explanation:
Valid 3-ary max heap from previous question:

Check Out Our GATE Courses & Secure Your Dream Rank!

Initial heap:
[9, 5, 6, 8, 3, 1]
In a 3-ary heap (0-based index):
Parent of index i = ⌊(i−1)/3⌋
Now insert 7, 2, 10, 4 one by one.
Insert 7:
Place at index 6.
Parent index = ⌊(6−1)/3⌋ = 1
Parent = 5
7 > 5 → swap
Heap becomes:
[9, 7, 6, 8, 3, 1, 5]
Insert 2:
Place at index 7.
Parent index = ⌊(7−1)/3⌋ = 2
Parent = 6
2 < 6 → no swap
Heap:
[9, 7, 6, 8, 3, 1, 5, 2]
Insert 10:
Place at index 8.
Parent index = ⌊(8−1)/3⌋ = 2
Parent = 6
10 > 6 → swap
Now at index 2.
Parent index = ⌊(2−1)/3⌋ = 0
Parent = 9
10 > 9 → swap
Heap becomes:
[10, 7, 9, 8, 3, 1, 5, 2, 6]
Insert 4:
Place at index 9.
Parent index = ⌊(9−1)/3⌋ = 2
Parent = 9
4 < 9 → no swap
Final heap: [10, 7, 9, 8, 3, 1, 5, 2, 6, 4]

49. ANSWER:- A
Explanation:
For any Binary Search Tree, the inorder traversal always gives the keys in sorted order.
Given postorder:
10, 9, 23, 22, 27, 25, 15, 50, 95, 60, 40, 29
The set of keys is:
{9, 10, 15, 22, 23, 25, 27, 29, 40, 50, 60, 95}
So inorder traversal must be these keys in strictly increasing order:
9, 10, 15, 22, 23, 25, 27, 29, 40, 50, 60, 95

50. ANSWER:- D
Explanation:
Initial max-heap (level order):
10, 8, 5, 3, 2
Insert 1:
Place at next position → index 5.
Parent = 5.
1 < 5 → no swap.
Heap becomes:
10, 8, 5, 3, 2, 1
Insert 7:
Place at next position → index 6.
Parent = 5.
7 > 5 → swap.
Heap becomes:
10, 8, 7, 3, 2, 1, 5
Now check parent of 7 → 10.
7 < 10 → stop.
Final heap: 10, 8, 7, 3, 2, 1, 5

51. ANSWER:- B
Explanation:
The number of distinct BSTs with n distinct keys is given by the Catalan number:
$C_n = (1 / (n + 1)) \times (2n \text{ choose } n)$
For n = 4:
$C_4 = (1 / 5) \times (8 \text{ choose } 4)$
$= (1 / 5) \times 70$
$= 14$

52. ANSWER:- C
Explanation:
In a complete k-ary tree:
Each internal node has exactly k children.
If number of internal nodes = n
Total number of edges = k × n
(because each internal node contributes k edges)
In any tree:
Number of edges = Total nodes − 1
Let number of leaves = L
Total nodes = n + L
So:
k n = (n + L) − 1
k n = n + L − 1
L = k n − n + 1
L = n(k − 1) + 1

53. ANSWER:- A
Explanation:
We have:
- Number of lists = $\log n$
- Size of each list = $n/\log n$
- Total elements = n

We merge these sorted lists using a min heap.
At any time, the heap contains at most log **n** elements (one from each list).
Each extract-min operation takes $\mathbf{O}(\log(\log n))$ time.
Since we perform this operation for all n elements, total time is:

$$n \cdot \log(\log n)$$

So, the time complexity is:
$O(n \log \log n)$

Check Out Our GATE Courses & Secure Your Dream Rank!

5 HASHING

GATE CSE 2026 SET-1|| MSQ || 1 MARK

Q1. Consider a hash table P[0, 1, ..., 10] that is initially empty. The hash table is maintained using open addressing with linear probing. The hash function used is : $h(x) = (x + 7)\ mod\ 11$. Consider the following sequence of insertions performed on
P: 1, 13, 22, 15, 11, 24
Which of the following positions in the hash table is/are empty after these insertions are performed?

(a) 0
(b) 10
(c) 2
(d) 1

GATE 2026 SET 2 || NAT || 1 MARK

Q2. The keys 5, 28, 19, 15, 26, 33, 12, 17, 10 are inserted into a hash table using the hash function $h(k) = k\ mod\ 9$. The collisions are resolved by chaining. After all the keys are inserted, the length of the longest chain is _____.

GATE CSE 2025 SET-1|| NAT || 1 MARK

Q3. In a double hashing scheme, $h_1(k) = k\ mod\ 11$ and $h_2(k) = 1 + (k\ mod\ 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the *i*-th probe in the open address table is
$[h_1(k) + ih_2(k)]\ mod\ m$ The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24.
The slot at which key 24 gets stored is _____.
(Answer in integers)

GATE CSE 2023 SET-1|| MCQ || 1 MARK

Q4. An algorithm has to store several keys generated by an adversary in a hash table. The adversary is malicious who tries to maximize the number of collisions. Let k be the number of keys, m be the number of slots in the hash table, and k>m.
Which one of the following is the best hashing strategy to counteract the adversary?

(a) Division method, i.e., use the hash function $h(k) = k\ mod\ m$
(b) Universal hashing method.
(c) Multiplication method, i.e., use the hash function $h(k) = \lfloor m(kA - \lfloor kA \rfloor) \rfloor$, where A is a carefully chosen constant.
(d) If k is a prime number, use Division method. Otherwise, use Multiplication method.

GATE CSE 2022 SET-1|| MCQ || 1 MARK

Q5. Suppose we are given n keys, m hash table slots, and two simple uniform hash functions h1 and h2. Further suppose our hashing scheme uses h1 for the odd keys and h2 for the even keys. What is the expected number of keys in a slot?

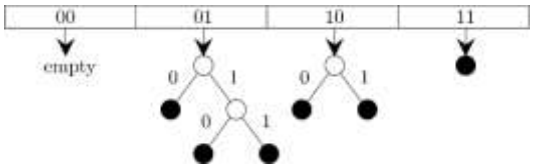
(a) m/n
(b) n/m
(c) 2n/m
(d) n/2m

GATE CSE 2021 SET-1|| MCQ || 1 MARK

Q6. Consider a dynamic hashing approach for 4-bit integer keys:
1. There is a main hash table of size 4.
2. The 2 least significant bits of a key is used to index into the main hash table.
3. Initially, the main hash table entries are empty.
4. Thereafter, when more keys are hashed into it, to resolve collisions, the set of all keys corresponding to a main hash table. entry is organized as a binary tree that grows on demand.
5. First, the 3rd least significant bit is used to divide the keys into left and right subtrees.
6. To resolve more collisions, each node of the binary tree is further sub-divided into left and right subtrees based on the 4th least significant bit.

Check Out Our GATE Courses & Secure Your Dream Rank!

7. A split is done only if it is needed, i.e., only when there is a collision.



Consider the following state of the hash table.

Which of the following sequences of key insertions can cause the above state of the hash table (assume the keys are in decimal notation)?

(a) 5,9,4,13,10,7
(b) 9,5,10,6,7,1
(c) 10,9,6,7,5,13
(d) 9, 5,13,6,10,14

GATE CSE 2020 || NAT || 1 MARK

Q7. Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \ mod \ 23$, and the secondary hash function is $h_2(k) = 1 + (k mod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value k=90 is _____.

GATE CSE 2015 SET-1|| MCQ || 2 MARKS

Q8. A hash table of length 10 uses open addressing with hash function $h(k) = k \ mod \ 10$, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

How many different insertion sequences of the key values using the same hash function and linear probing will result in the hash table shown above?

(a) 10
(b) 20
(c) 30
(d) 40

GATE CSE 2015 SET-1|| MCQ || 2 MARKS

Q9. Which one of the following choices gives a possible order in which the key values could have been inserted in the table?

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

(a) 46, 42, 34, 52, 23, 33
(b) 34, 42, 23, 52, 33, 46
(c) 46, 34, 42, 23, 52, 33
(d) 42, 46, 33, 23, 34, 52

GATE CSE 2015 SET-2 || MCQ || 2 MARK

Q10. Which one of the following hash functions on integers will distribute keys most uniformly over 10 buckets numbered 0 to 9 for i ranging from 0 to 2020?

(a) $h(i)= i^2 \ mod \ 10$
(b) $h(i)= i^3 \ mod \ 10$
(c) $h(i)=(11*i^2) \ mod \ 10$
(d) $h(i)=(12*i^2) \ mod \ 10$

GATE CSE 2014 SET-1|| MCQ || 1 MARK

Q11. Consider a hash table with 9 slots. The hash function is $h(k) = k \ mod \ 9$. The collisions are resolved by chaining. The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. The maximum, minimum, and average chain lengths in the hash table, respectively, are

(a) 3,0,and 1
(b) 3,3,and 3
(c) 4,0,and 1
(d) 3,0,and 2

GATE CSE 2014 SET-3|| MCQ || 1 MARK

Q12. Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform

Check Out Our GATE Courses & Secure Your Dream Rank!

hashing, what is the probability that the first 3 slots are unfilled after the first 3 insertions?

(a) (97 x 97 x 97)/1003
(b) (99 x 98 x 97)/1003
(c) (97 x 96 x 95)/1003
(d) (97 x 96 x 95)/(3! x 1003)

GATE CSE 2009 || MCQ || 2 MARK

Q13. The keys 12,18,13,2,3,23,5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function $h(k) = k$ mod 10 and linear probing. What is the resultant hash table ?

(a)

0	
1	
2	2
3	2
4	
5	1
6	
7	
8	1
9	

(b)

0	
1	
2	12
3	13
4	
5	5
6	
7	
8	18
9	

(c).

0	
1	
2	12
3	13
4	2
5	3
6	23
7	5
8	18
9	15

(d).

0	
1	
2	12,2
3	13,3,23
4	
5	5,15
6	
7	
8	18
9	

GATE CSE 2008 SET-1|| MCQ || 1 MARK

Q14. Consider a hash table of size seven, with starting index zero, and a hash function h(x) = (3x + 4) mod 7 . Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing?
Note that "–" denotes an empty location in the table.

(a) 8, –, –, –, –, –, 10
(b) 1, 8, 10, –, –, –, 3
(c) 1, –, –, –, –, –, 3
(d) 1, 10, 8, –, –, –, 3

Check Out Our GATE Courses & Secure Your Dream Rank!

ANSWER KEY

1. (**c**)	**2.** (**3**)	**3.** (**10**)
4. (**b**)	**5.** (**b**)	**6.** (**c**)
7. (**13**)	**8.** (**c**)	**9.** (**c**)
10. (**b**)	**11.** (**a**)	**12.** (**a**)
13. (**c**)	**14.** (**b**)	

HINTS & SOLUTION

1. ANSWER : C

Explanation: Hash table size = 11
Indices: 0 to 10
Hash function:
h(x) = (x + 7) mod 11
Linear probing
Insert in order: 1, 13, 22, 15, 11, 24
Insert 1:
h(1) = (1 + 7) mod 11 = 8
Place at index 8
Insert 13:
h(13) = (13 + 7) mod 11 = 20 mod 11 = 9
Place at index 9
Insert 22:
h(22) = (22 + 7) mod 11 = 29 mod 11 = 7
Place at index 7
Insert 15:
h(15) = (15 + 7) mod 11 = 22 mod 11 = 0
Place at index 0
Insert 11:
h(11) = (11 + 7) mod 11 = 18 mod 11 = 7
Index 7 occupied → probe
Index 8 occupied → probe
Index 9 occupied → probe
Index 10 empty → place at 10
Insert 24:
h(24) = (24 + 7) mod 11 = 31 mod 11 = 9
Index 9 occupied → probe
Index 10 occupied → probe
Index 0 occupied → probe
Index 1 empty → place at 1
Final table:
0 → 15
1 → 24
2 → empty
3 → empty
4 → empty
5 → empty
6 → empty
7 → 22
8 → 1
9 → 13
10 → 11
Empty positions among options:
0 → filled
10 → filled
2 → empty
1 → filled
Correct answer: (C) 2

2. ANSWER:- 3

Explanation:
Hash function:
h(k) = k mod 9
Compute hash values:
5 mod 9 = 5
28 mod 9 = 1
19 mod 9 = 1
15 mod 9 = 6
26 mod 9 = 8
33 mod 9 = 6
12 mod 9 = 3
17 mod 9 = 8
10 mod 9 = 1
Now group by index:
Index 1 → 28, 19, 10 (length = 3)
Index 6 → 15, 33 (length = 2)
Index 8 → 26, 17 (length = 2)
Index 3 → 12 (length = 1)
Index 5 → 5 (length = 1)
Longest chain length = 3
Final Answer: 3

3. ANSWER=10

Explanation
$h_1(k) = k \bmod 11$
$h_2(k) = 1 + (k \bmod 7)$
Hash formula for i-th probe:
$(h_1(k) + i \cdot h_2(k)) \bmod 11$
Insert keys one by one.
Key 63
$h_1 = 63 \bmod 11 = 8$
Slot 8 is empty → place 63 at index 8
Key 50
$h_1 = 50 \bmod 11 = 6$
Slot 6 is empty → place 50 at index 6
Key 25
$h_1 = 25 \bmod 11 = 3$
Slot 3 is empty → place 25 at index 3
Key 79
$h_1 = 79 \bmod 11 = 2$
Slot 2 is empty → place 79 at index 2
Key 67
$h_1 = 67 \bmod 11 = 1$
Slot 1 is empty → place 67 at index 1

Check Out Our GATE Courses & Secure Your Dream Rank!

Key 24
$h_1 = 24 \bmod 11 = 2$ (collision, 79 already there)
$h_2 = 1 + (24 \bmod 7) = 1 + 3 = 4$
Now probe:
i = 1
$(2 + 1 \cdot 4) \bmod 11 = 6 \rightarrow$ occupied (50)
i = 2
$(2 + 2 \cdot 4) \bmod 11 = 10 \rightarrow$ empty
So key 24 is stored at slot 10.

4 **Answer: B**
Explanation:
The correct answer is Universal hashing method.
In division and multiplication methods, the hash function is fixed and deterministic. A malicious adversary who knows the hash function can deliberately choose keys that map to the same slot and cause maximum collisions. Therefore, these methods are weak against adversarial inputs.
In universal hashing, the hash function is chosen randomly from a family of hash functions at runtime. Since the adversary does not know which specific function will be used, they cannot force many keys to collide. This randomization guarantees that the expected number of collisions remains low even in the worst case.

5 **Answer: B**
Explanation: Each key, whether odd or even, is hashed uniformly into the m slots because both h_1 and h_2 are simple uniform hash functions. Odd keys use h_1 and even keys use h_2, but both functions distribute their respective keys uniformly over all m slots. So, on average, each slot receives an equal share of the total n keys. Therefore, the expected number of keys in any slot is total keys divided by total slots, which is n / m.

6. **Answer: C .**
Explanation:
Using the last 2 bits for the main table, slot 00 remains empty and slot 11 contains exactly one key, which matches the figure. In option C, keys map as follows: 10 and 6 go to slot 10, and they differ in the 3rd least significant bit, so only one split is needed, matching the tree shown. Keys 9, 5, and 13 go to slot 01; among them, 9 and 5 have the same 3rd bit, causing a collision that is resolved by splitting on the 4th bit, while 13 differs at the 3rd bit. This exactly reproduces the structure in the given hash table state

7. **Answer: 13**
Explanation: Given
$h_1(k) = k \bmod 23$
$h_2(k) = 1 + (k \bmod 19)$
Table size m = 23
Key k = 90
First compute the hash values.
$h_1(90) = 90 \bmod 23 = 21$
$h_2(90) = 1 + (90 \bmod 19) = 1 + 14 = 15$
The probe sequence formula is
$(h_1(k) + i \cdot h_2(k)) \bmod m$
Since probe sequence begins at probe 0, probe 1 corresponds to i = 1.
Address at probe 1
$= (21 + 1 \cdot 15) \bmod 23$
$= 36 \bmod 23$
$= 13$
Final Answer: 13

8. **Answer:C**
Final table with keys and their home slots:
42 → 2 (placed at 2)
23 → 3 (placed at 3)
34 → 4 (placed at 4)
52 → 2 (placed at 5)
46 → 6 (placed at 6)
33 → 3 (placed at 7)
Important dependencies (due to linear probing):
52 can be placed at slot 5 only if slots 2, 3, and 4 are already occupied
→ therefore, 42, 23, and 34 must be inserted before 52
33 can be placed at slot 7 only if slots 3, 4, 5, and 6 are already occupied
→ therefore, 23, 34, 52, and 46 must be inserted before 33
The remaining keys (42, 23, 34, 46) are relatively independent among themselves, subject only to the above constraints.
Counting all valid permutations under these partial-order constraints yields a total of 30 insertion sequences.
Final answer: 30

Check Out Our GATE Courses & Secure Your Dream Rank!

9. Answer:(C)

Explanation: Insert elements one by one:

1. **Insert 46**
   $h(46) = 6 \rightarrow$ place at index 6
2. **Insert 34**
   $h(34) = 4 \rightarrow$ place at index 4
3. **Insert 42**
   $h(42) = 2 \rightarrow$ place at index 2
4. **Insert 23**
   $h(23) = 3 \rightarrow$ place at index 3
5. **Insert 52**
   $h(52) = 2$(occupied)
   → check 3 (occupied)
   → check 4 (occupied)
   → check 5 (empty) → place at index 5
6. **Insert 33**
   $h(33) = 3$(occupied)
   → check 4 (occupied)
   → check 5 (occupied)
   → check 6 (occupied)
   → check 7 (empty) → place at index 7

Final Table Obtained:

2 → 42
3 → 23
4 → 34
5 → 52
6 → 46
7 → 33

10. Answer: B

Explanation : For i = 0 to 9, the pattern repeats for all larger values.
i^2 mod 10 gives only these digits: 0, 1, 4, 5, 6, 9. Some buckets (2, 3, 7, 8) never get any value, so distribution is not uniform.
i^3 mod 10 gives all digits from 0 to 9 exactly once in a cycle. So every bucket gets almost equal number of keys. This is the most uniform.
$(11 \cdot i^2)$ mod 10 is same as i^2 mod 10 because 11 mod 10 = 1. So it has the same problem as the first one.
$(12 \cdot i^2)$ mod 10 becomes $(2 \cdot i^2)$ mod 10. This still produces only a few digits and does not cover all buckets evenly.
Correct answer is $h(i) = i^3$ mod 10.

11. Answer: A

Explanation: Hash table size = 9
Hash function h(k) = k mod 9
Insert keys one by one and find their slots.
5 mod 9 = 5 → slot 5
28 mod 9 = 1 → slot 1
19 mod 9 = 1 → slot 1
15 mod 9 = 6 → slot 6
20 mod 9 = 2 → slot 2
33 mod 9 = 6 → slot 6
12 mod 9 = 3 → slot 3
17 mod 9 = 8 → slot 8
10 mod 9 = 1 → slot 1
Now count chain lengths in each slot (0 to 8).
Slot 0 → 0
Slot 1 → 3 (28, 19, 10)
Slot 2 → 1
Slot 3 → 1
Slot 4 → 0
Slot 5 → 1
Slot 6 → 2 (15, 33)
Slot 7 → 0
Slot 8 → 1
Maximum chain length = 3
Minimum chain length = 0
Average chain length = total keys / total slots = 9 / 9 = 1
Correct Answer :3,0,and 1

12. Answer: A

Explanation:
There are 100 slots and hashing is simple uniform, so each insertion independently goes to any slot with probability 1/100.
For one insertion, the probability that it does not go into the first 3 slots is
(100 − 3) / 100 = 97 / 100
Since insertions are independent, for 3 insertions the probability is
(97 / 100) × (97 / 100) × (97 / 100)
So the required probability is
$(97 \times 97 \times 97) / 100^3$
Correct answer is
$(97 \times 97 \times 97) / 100^3$

13. ANSWER : C

Explanation : Keys: 12, 18, 13, 2, 3, 23, 5, 15
Hash function: h(k) = k mod 10
Collision handling: Linear probing
Table size = 10 (index 0 to 9)
Now insert one by one.
12 mod 10 = 2 → place at index 2
18 mod 10 = 8 → place at index 8
13 mod 10 = 3 → place at index 3
2 mod 10 = 2 → collision at 2
Next free slot → 3 (occupied)
Next → 4 (empty) → place at 4

Check Out Our GATE Courses & Secure Your Dream Rank!

3 mod 10 = 3 → collision
4 occupied
5 empty → place at 5
23 mod 10 = 3 → collision
4 occupied
5 occupied
6 empty → place at 6
5 mod 10 = 5 → collision
6 occupied
7 empty → place at 7
15 mod 10 = 5 → collision
6 occupied
7 occupied
8 occupied
9 empty → place at 9
Final hash table:
Index 2 → 12
Index 3 → 13
Index 4 → 2
Index 5 → 3
Index 6 → 23
Index 7 → 5
Index 8 → 18
Index 9 → 15
This matches option (C).

14 **ANSWER : B.**
Explanation : Table size = 7
Hash function:
h(x) = (3x + 4) mod 7
Closed hashing = linear probing
Insert in order: 1, 3, 8, 10
Step 1: Insert 1
h(1) = (3×1 + 4) mod 7 = 7 mod 7 = 0
Place 1 at index 0
Step 2: Insert 3
h(3) = (9 + 4) mod 7 = 13 mod 7 = 6
Place 3 at index 6
Step 3: Insert 8
h(8) = (24 + 4) mod 7 = 28 mod 7 = 0
Index 0 occupied → linear probe
Index 1 empty → place 8 at index 1
Step 4: Insert 10
h(10) = (30 + 4) mod 7 = 34 mod 7 = 6
Index 6 occupied → probe
Index 0 occupied → probe
Index 1 occupied → probe
Index 2 empty → place 10 at index 2
Final table (index 0 to 6):
0 → 1
1 → 8
2 → 10
3 → –
4 → –
5 → –
6 → 3
Correct answer: (B) 1, 8, 10, –, –, –, 3

Check Out Our GATE Courses & Secure Your Dream Rank!

`

ALGORITHMS

“Scan Here for Video Solution”

Link to GfG corresponding Short Notes: GATE FARRAY

“The *Algorithm* of success is simple—Understand deeply, Practice consistently, and **Never stop Improving.**”

LEGEND OF

ALGORITHM

Dr. Khaleel Ur Rahman Khan Sir

An Inspirer. A Mentor. A Guide.

1 ANALYSIS OF ALGORITHMS

GATE CSE – 2026 SET-1 MCQ (1-MARKS)

Q1. Consider the following recurrence relations:
For all $n>1$,
$T_1(n)=4T_1(n/2)+T_2(n)$
$T_2(n)=5T_2(n/4)+\Theta(\log_2 n)$
Assume that for all $n\leq 1$, $T_1(n)=1$ and $T_2(n)=1$.
Which one of the following options is correct?

a) $T_1(n)=\theta(n^2)$
b) $T_1(n)=\theta(n^2\log_2 n)$
c) $T_1(n)=\theta(n^{\log_4 5})$
d) $T_1(n)=\Theta(n^{\log_4 5}\log_2 n)$

GATE CSE – 2026 SET-2 MSQ (1-MARKS)

Q2. Which of the following can be recurrence relation(s) corresponding to an algorithm with time complexity O(n)?

a) T(n)=T(n-1) +1, T(1)=1
b) T(n)=2T(n/2)+1 , T(1)=1
c) T(n)=2T(n/2)+n , T(1)=1
d) T(n)=T(n-1)+n , T(1)=1

GATE CSE – 2026 SET-2 MCQ (1-MARKS)

Q3. Consider the following functions, where n is a positive integer.
$n^{1/3}$, logn , log(n!) , $2^{\log n}$
Which one of the following options lists the functions in increasing order of asymptotic growth rate? Note: Assume the base of log to be 2.

a) logn , $n^{1/3}$, $2^{\log n}$, log(n!)
b) $n^{1/3}$, logn , log(n!) , $2^{\log n}$
c) logn , $n^{1/3}$, log(n!) , $2^{\log n}$
d) $2^{\log n}$, $n^{1/3}$, logn , log(n!)

GATE CSE – 2025 SET-1 MCQ (2-MARKS)

Q4. Consider the following recurrence relation:
$T(n)=2T(n-1)+ n2^n$ for n>0, T(0)=1
Which ONE of the following options is CORRECT?

a) $T(n)=\theta(n^2 2^n)$
b) $T(n)=\theta(n2^n)$
c) $T(n)=\theta((\log n)^2 2^n)$
d) $T(n)=\theta(4^n)$

GATE CSE – 2024 SET-1 MCQ (2-MARKS)

Q5. Consider the following recurrence relation:

$$T(n) = \begin{cases} \sqrt{n}\cdot T(\sqrt{n}) + n & \text{for } n \geq 1 \\ 1 & \text{for } n = 1 \end{cases}$$

Which one of the following options is CORRECT?

a) $T(n) = \theta(n\log\log n)$
b) $T(n) = \theta(n^2\log n)$
c) $T(n) = \theta(n\log n)$
d) $T(n) = \theta(n^2\log\log n)$

GATE CSE – 2024 SET-2 MCQ (1-MARKS)

Q6. Let T(n) be the recurrence relation defined as follows:
T(0) = 1,
T(1) = 2 and
$T(n) = 5T(n-1) - 6T(n-2)$ for $n \geq 2$
Which one of the following statement is TRUE?

a) $T(n) = \theta(n\cdot 3^n)$
b) $T(n) = \theta(n\cdot 2^n)$
c) $T(n) = \theta(3^n)$
d) $T(n) = \theta(2^n)$

GATE CSE – 2023 MSQ (1-MARKS)

Q7. Let f and g be functions of natural numbers given by $f(n)=n$ and $g(n)=n^2$. Which of the following statements is/are TRUE?

a) $f \in O(g)$
b) $f \in \omega(g)$
c) $f \in o(g)$
d) $f \in \theta(g)$

GATE CSE – 2023 MSQ (2-MARKS)

Q8. Consider functions Function 1 and Function 2 expressed in pseudocode as follows:Function 1 and Function 2

Function 1	Function 2

while n > 1 do
for i = 1 to n
do
x = x + 1;
end for
n = ⌊n/2⌋;
end while
	for i = 1 to 100 * n
	do
	x = x + 1;
	end for

Let f1(n) and f2(n) denote the number of times the statement x = x + 1; is executed in Function 1 and Function 2, respectively.

a) $f1(n) \in \theta(f2(n))$
b) $f1(n) \in o(f2(n))$
c) $f2(n) \in \omega(f1(n))$
d) $f1(n) \in O(n)$

GATE CSE – 2022 MCQ (1-MARKS)

Q9. Which one of the following statements is TRUE for all positive functions f(n)?

a) $f(n^2) = \theta(f(n)^2)$, when f(n) is a polynomial
b) $f(n^2)=o(f(n)^2)$
c) $f(n^2) = O(f(n)^2)$, when f(n) is an exponential function
d) $f(n^2)= \Omega(f(n)^2)$

GATE CSE – 2021 SET-2 MCQ (2-MARKS)

Q10. For constants $a \geq 1$ and $b > 1$, consider the following recurrence defined on the non-negative integers:
$T(n) = a\,T(n/b) + f(n)$
Which one of the following options is correct about the recurrence T(n) ?

a) f(n) is $O(n^{\log_b(a)-\epsilon})$ for some $\epsilon>0$, then T(n) is $T(n)=\Theta(n^{\log_b(a)})$
b) If f(n) is $n/\log_2(n)$, then T(n) is $\Theta(\log_2(n))$
c) If f(n) is $\Theta(n^{\log_b(a)})$, then T(n) is $\Theta(n^{\log_b(a)})$
d) If f(n) is $n\log_2(n)$, then T(n) is $\Theta(n\log_2(n))$

GATE CSE – 2021 SET-1 MCQ (1-MARKS)

Q11. Consider the following three functions.
$f1 = 10^n$
$f2 = n^{\log n}$
$f3 = n^{\sqrt{n}}$
Which one of the following options arranges the function in the increasing order of asymptotic growth rate?

a) f2, f1, f3
b) f3, f2, f1
c) f2, f3, f1
d) f1, f2, f3

GATE CSE – 2021 SET-1 MCQ (1-MARKS)

Q12. Consider the following recurrence relation.

$$T(n) = \begin{cases} T(n/2) + T(2n/5) + 7n & \text{, if } n > 0 \\ 1 & \text{, if } n = 0 \end{cases}$$

which one of the following options is correct?

a) $T(n) = \theta((\log n)^{5/2})$
b) $T(n) = \theta(n)$
c) $T(n) = \theta(n\log n)$
d) $T(n) = \theta(n^{5/2})$

Link to GfG corresponding Short Notes: GATE

GATE CSE – 2020 MCQ (1-MARKS)

Q13. For parameters a and b, both of which are ω(1), $T(n) = T(n^{1/a}) + 1$, and T(b)=1. Then T(n) is:

a) $\theta(\log_{ab} n)$
b) $\theta(\log_a \log_b n)$
c) $\theta(\log_2 \log_2 n)$
d) $\theta(\log_b \log_a n)$

GATE CSE – 2017 SET-1 MCQ (2-MARKS)

Q14. Consider the following functions from positives integers to real numbers 10, √n, n, $\log_2 n$, 100/n. The CORRECT arrangement of the above functions in increasing order of asymptotic complexity is:

a) $\log_2 n$, 100/n, 10, $\sqrt{n}$, n
b) 100/n, 10, $\log_2 n$, $\sqrt{n}$, n
c) 10, 100/n , $\sqrt{n}$, $\log_2 n$, n
d) 100/n, $\log_2 n$, 10 , $\sqrt{n}$, n

GATE CSE – 2017 SET-2 MCQ (2-MARKS)

Q15. Consider the following C function

```
int fun(int n) {
int i, j;
for(i=1; i<=n; i++) {
for (j=1; j<n; j+=i) {
printf("%d %d", i, j);
        }
    }
}
```

Time complexity of fun in terms of θ notation is

a) $\theta(n\sqrt{n})$
b) $\theta(n^2)$
c) $\theta(n \log n)$
d) $\theta(n^2 \log n)$

GATE CSE – 2017 SET-2 MCQ (2-MARKS)

Q16. Consider the recurrence function

$$T(n) = \begin{cases} 2 \times T(\sqrt{n}) + 1, & \text{if } n > 2 \\ 2, & \text{if } 0 < n \leq 2 \end{cases}$$

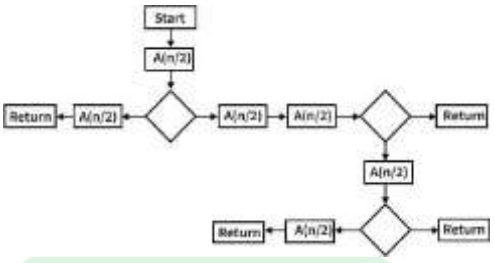
Then T(n) in terms of θ notation is

a) θ(log logn)
b) θ(log n)
c) $\theta(\sqrt{n})$
d) θ(n)

GATE CSE – 2016 SET-2 NAT (2-MARKS)

Q17. The given diagram shows the flowchart for a recursive function A(n). Assume that all statements, except for the recursive calls, have O(1) time complexity. If the worst-case time complexity of this function is $O(n^{\alpha})$, then the least possible value (accurate up to two decimal positions) of α is _____.

Flowchart for Recursive Function A(n)



GATE CSE – 2015 SET-1 MCQ (2-MARKS)

Q18. Consider the following C function.

```
int fun1 (int n)
{
int i, j, k, p, q = 0;
for (i = 1; i< n; ++i)
{
        p = 0;
        for (j = n; j > 1; j = j/2)
        ++p;
        for (k = 1; k <  p; k = k*2)
        ++q;
}
return q;
}
```

Which one of the following most closely approximates the return value of the function fun1?

a) n^3
b) $n(\log n)^2$
c) nlogn
d) nlog(logn)

GATE CSE – 2015 SET-3 MCQ (2-MARKS)

Q19. Let f(n) = n and $g(n) = n^{(1+\sin n)}$, where n is a positive integer. Which of the following statements is/are correct?

I. f(n) = O(g(n))
II. f(n) = Ω(g(n))

a) Only I
b) Only II
c) Both I and II
d) Neither I nor II

GATE CSE – 2015 SET-2 MCQ (1-MARKS)

Q20. Consider a complete binary tree where the left and the right subtrees of the root are max heaps. The lower bound for the number of operations to convert the tree to a heap is

a) $\Omega(\log n)$
b) $\Omega(n)$
c) $\Omega(n \log n)$
d) $\Omega(n^2)$

GATE CSE – 2015 SET-3 MCQ (1-MARKS)

Q21. Consider the equality $X = \sum_{i=0}^{n} i^3$

following choices for X:
I. $\Theta(n^4)$
II. $\Theta(n^5)$
III. $O(n^5)$
IV. $\Omega(n^3)$

The equality above remains correct if X is replaced by

a) Only I
b) Only II
c) I or III or IV but not II
d) II or III or IV but not I

GATE CSE – 2014 SET-3 NAT (2-MARKS)

Q22. Suppose we have a balanced binary search tree T holding n numbers. We are given two numbers L and H and wish to sum up all the numbers in T that lie between L and H. Suppose there are m such numbers in T. If the tightest upper bound on the time to compute the sum is
$O(n^a \log^b n + m^c \log^d n)$
the value of a + 10b + 100c + 1000d is _____.

GATE CSE – 2014 SET-2 MCQ (1-MARKS)

Q23. Which one of the following correctly determines the solution of the recurrence relation with T(1)=1?
$T(n)=2T(n/2) + \log n$

a) $\theta(n)$
b) $\theta(n \log n)$
c) $\theta(n^2)$
d) $\theta(\log n)$

GATE CSE – 2013 SET-3 MCQ (2-MARKS)

Q24. Consider the following function:

```
int unknown(int n){
int i, j, k=0;
for (i=n/2; i<=n; i++)

for (j=2; j<=n; j=j*2)
k = k + n/2;
return (k);
}
```

The return value of the function is

a) $\Theta(n^2)$
b) $\Theta(n^2 \log n)$
c) $\Theta(n^3)$
d) $\Theta(n^3 \log n)$

GATE CSE – 2012 MCQ (1-MARKS)

Q25. Let W(n) and A(n) denote respectively, the worst case and average case running time of an algorithm executed on an input of size n. Which of the following is ALWAYS TRUE?

a) $A(n) = \Omega (W(n))$
b) $A(n) = \Theta (W(n))$
c) $A(n) = O (W(n))$
d) $A(n) = o (W(n))$

GATE CSE – 2011 MCQ (1-MARKS)

Q26. Which of the given options provides the increasing order of asymptotic complexity of functions f 1 , f 2 , f 3 and f 4 ?
$f1(n)=2^n$, $f2(n)=n^{3/2}$, $f3(n)=n\log_2 n$, $f4(n)=n^{\log \frac{n}{2}}$

a) f3, f2, f4, f1
b) f3, f2, f1, f4
c) f2, f3, f1, f4
d) f2, f3, f4, f1

GATE CSE – 2009 MCQ (1-MARKS)

Q27. The running time of an algorithm is represented by the following recurrence relation:

$$T(n) = \begin{cases} n ; n \leq 3 \\ T\left(\frac{n}{3}\right) + cn; otherwise \end{cases}$$

Which one of the following represents the time complexity of the algorithm?

a) θ (n)
b) θ (n log n)
c) θ (n^2)
d) θ (n^2logn)

GATE CSE – 2008 MCQ (2-MARKS)

Q28. Consider the following functions:
$f(n) = 2^n$
$g(n) = n!$
$h(n) = n^{\log n}$
Which of the following statements about the asymptotic behaviour of f(n), g(n), and h(n) is true?

a) f(n) = O(g(n)); g(n) = O(h(n))
b) f(n) = Ω(g(n)); g(n) = O(h(n))
c) g(n) = O(f(n)); h(n) = O(f(n))
d) h(n) = O(f(n)); g(n) = Ω(f(n))

GATE CSE – 2008 MCQ (2-MARKS)

Q29. When $n = 2^{(2k)}$ for some k >= 0, the recurrence relation $T(n) = \sqrt{(2)}T(n/2) + \sqrt{(n)}$, T(1) = 1 evaluates to :

a) $(\sqrt{n})(\log n + 1)$
b) $(\sqrt{n}) \log n$
c) $(\sqrt{n}) \log (\sqrt{n})$
d) $n \log (\sqrt{n})$

GATE CSE – 2008 MCQ (1-MARKS)

Q30. Arrange the following functions in increasing asymptotic order:

a. $n^{1/3}$
b. e^n
c. $n^{7/4}$
d. $n\log_9 n$
e. 1.0000001^n

a) a, d, c, e, b
b) d, a, c, e, b
c) a, c, d, e, b
d) a, c, d, b, e

GATE CSE – 2008 MCQ (2-MARKS)

Q31. Consider the following C functions:

```
int f1(int n)
{
  if(n == 0 || n == 1)
    return n;
  else
    return (2*f1(n-1) + 3*f1(n-2));
}

int f2(int n)
{
  int i;
  int X[N], Y[N], Z[N] ;
  X[0] = Y[0] = Z[0] = 0;
  X[1] = 1; Y[1] = 2; Z[1] = 3;
  for(i = 2; i <= n; i++)
  {
    X[i] = Y[i-1] + Z[i-2];
    Y[i] = 2*X[i];
    Z[i] = 3*X[i];
  }
  return X[n] ;
}
```

The running time of f1(n) and f2(n) are

a) $\Theta(n)$ *and* $\Theta(n)$
b) $\Theta(2^n)$ *and* $\Theta(n)$
c) $\Theta(n)$ *and* $\Theta(2^n)$
d) $\Theta(2^n)$ *and* $\Theta(2^n)$

GATE CSE– 2007 MCQ (1-MARKS)

Q32. Exponentiation is a heavily used operation in public key cryptography. Which of the following options is the tightest upper bound on the number of multiplications required to compute $b^n \bmod m, 0 \le b, n \le m$?

a) O(logn)
b) $O(\sqrt{n})$
c) O(n/logn)
d) O(n)

GATE CSE – 2007 MCQ (1-MARKS)

Q33. In the following C function, let n≥m.

```
int gcd(n,m)
{
 if (n%m ==0) return m;
 n = n%m;
 return gcd(m,n);
}
```

How many recursive calls are made by this function?

a) $\theta(logn)$
b) $\omega(n)$
c) $\theta(loglogn)$
d) $\Theta(\sqrt{n})$

GATE CSE – 2007 MCQ (1-MARKS)

Q34. Consider the following C code segment:

```
int IsPrime(n)
{
 int i,n;
 for(i=2;i<=sqrt(n);i++)
   if(n%i == 0)
    {printf("Not Primen"); return 0;}
 return 1;
}
```

Let T(n)denote the number of times the for loop is executed by the program on input n. Which of the following is TRUE?

a) $T(n)=O(\sqrt{n})$ and $T(n)=\Omega(\sqrt{n})$
b) $T(n)=O(n)$ and $T(n)=\Omega(\sqrt{n})$
c) $T(n)=O(\sqrt{n})$ and $T(n)=\Omega(1)$
d) None of the above

GATE CSE – 2007 MCQ (1-MARKS)

Q35. What is the Asymptotic Notation of the following recursive function:

```
int DoSomething (int n) {
if (n <= 2)
return 1;
else
return (DoSomething (floor(sqrt(n))) + n); }
```

a) $\theta(n^2)$
b) $\theta(n\log_2 n)$
c) $\theta(\log_2 n)$
d) $\theta(\log_2\log_2 n)$

GATE CSE – 2006 MCQ (1-MARKS)

Q36. Consider the following C-program fragment in which i, j, and n are integer variables.

```
for (i = n, j = 0; i > 0; i/=2, j += i);
```

Let Val(j) denote the value stored in the variable j after termination of the for loop. Which one of the following is true?

a) val(j)= θ (logn)
b) val(j)= θ (sqrt(n))
c) val(j)= θ (n)
d) val(j)= θ (nlogn)

GATE CSE – 2006 MCQ (1-MARKS)

Q37. Consider the following recurrence:

$$T(n) = 2T([\sqrt{n}]) + 1, T(1) = 1$$

Which one of the following is true?

a) T(n)= θ (loglogn)
b) T(n)= θ (logn)
c) $T(n)= \theta(\sqrt{n})$
d) T(n)= θ (n)

GATE CSE – 2005 MCQ (1-MARKS)

Q38. Suppose T(n)=2T(n/2)+n, T(0)=T(1)=1
Which one of the following is FALSE?

a) $T(n)=O(n^2)$
b) $T(n)=\Theta(n \log n)$
c) $T(n)=\Omega(n^2)$
d) $T(n)=O(n \log n)$

Check Out Our Gate Courses & Secure Your Dream Rank !

GATE CSE – 2005 MCQ (1-MARKS)

Q39. The Asymptotic Notation of computing the transitive closure of a binary relation on a set of n elements is known to be:

a) O(n)
b) O(nlogn)
c) $O(n^{3/2})$
d) $O(n^3)$

GATE CSE – 2005 MCQ (1-MARKS)

Q40. Consider the following C-function:

```
double foo (int n)
{
   int i;
   double sum;
   if (n = = 0) return 1.0;
   else
   {
      sum = 0.0;
      for (i = 0; i < n; i++)
         sum += foo (i);
      return sum;
   }
}
```

Suppose we modify the above function foo() and store the values of foo (i), $0 \leq i < n$, as and when they are computed. With this modification, the time complexity for function foo() is significantly reduced. The space complexity of the modified function would be:

a) O(1)
b) O(n)
c) O(n!)
d) $O(n^n)$

ANSWER KEY

1.(a)	**2.(a,b)**	**3.(a)**
4.(a)	**5.(a)**	**6.(d)**
7.(a,c)	**8.(a,d)**	**9.(a)**
10.(a)	**11.(c)**	**12.(b)**
13.(b)	**14.(b)**	**15.(c)**
16.(b)	**17.(2.32)**	**18.(d)**
19.(d)	**20.(a)**	**21.(c)**
22.(110)	**23.(a)**	**24.(b)**
25.(c)	**26.(a)**	**27.(a)**
28.(d)	**29.(a)**	**30.(a)**
31.(b)	**32.(a)**	**33.(a)**
34.(c)	**35.(d)**	**36.(c)**
37.(b)	**38.(c)**	**39.(d)**
40.(b)		

HINTS & SOLUTION

1. Answer : (A)

$T_2(n) = \Theta(n^{\log_4 5})$ [Using Mater Theorem]
$T_1(n) = 4T_1(n/2) + n^{\log_4 5}$
Case 1 : of the Master Theorem, where the recursive part dominates.
Thus, $T_1(n) = \Theta(n^2)$

2. Answer : (A,B)

A) T(n)=O(n) T(n)= 1+1+1....1 (n times)
B) T(n)=O(n) //apply master theorem
C) T(n)=O(nlogn) //Master theorem
D) $T(n)=O(n^2)=n+(n-1)+(n-2)+\ldots\ldots+1$

3. Answer :(A)

- log(n): This is a standard logarithmic growth rate.
- $n^{1/3}$: This is a polynomial (specifically, a cube root) growth rate.
  Take log both the side
  log(logn) | $\log(n^{1/3})$
  log(Logn) | logn
- $2^{\log(n)}$: Using the logarithmic identity $a^{\log a(x)} = x$, this simplifies to n.=O(n)
- log(n!): Using Stirling's approximation, log(n!) approx nlogn . Therefore, its asymptotic growth is θ(nlog n).

4. Answer : (A)

$T(n) = 2T(n-1) + n.2^n$
$= 2[2T(n-2) + (n-1).2^{(n-1)}] + n.2^n$
$= 2^k T(n-k) + 2^n[(n-(k-1)) + (n-(k-2)) + \ldots\ldots + n]$
$n-k=0$
$= 2^n + 2^n[1+2+\ldots+n]$
$= O(n^2.2^n)$

5. Answer : (A)

$$T(n) = \sqrt{n}T(\sqrt{n}) + n$$
$$\frac{T(n)}{n} = \frac{T(\sqrt{n})}{\sqrt{n}} + 1$$
$$S(n) = S(\sqrt{n}) + 1$$
$$S(2^k) = S(2^{k/2}) + 1$$
$$Q(k) = Q(k/2) + 1$$
$$Q(k) = \theta(\log k)$$
$$\Rightarrow S(2^k) = \theta(\log k)$$
$$\Rightarrow S(n) = \theta(\log\log n)$$
$$\Rightarrow T(n) = \theta(n\log\log n)$$

Check Out Our Gate Courses & Secure Your Dream Rank !

6. **Answer : (D)**

$T(n) = 5T(n-1) - 6T(n-2)$

Generating function:

Substituting $T(n) = t^n$

$t^n - 5t^{n-1} - 6t^{n-2}$

Dividing the eqn by t^{n-2},

we get

$t^2 = 5t - 6$

$t^2 - 5t + 6 = 0$

Roots = 2, 3

$T(n) = C_1 \cdot 2^n + C_2 \cdot 3^n$

$T(0) = 1$

$T(1) = 2$

$1 = C_1 + C_2$

$2 = 2C_1 + 3C_2$

$C_1 = 1, C_2 = 0$

$T(n) = 1 \times 2^n + 0 \times 3^n$

$T(n) = 2^n$

7. **Answer : A,C.**

$f(n) \in O(g(n))$ iff f(n) asymptotically smaller or equal to g(n)

$f(n) \in o(g(n))$ iff f(n) asymptotically smaller than g(n)

$f(n) \in \Omega(g(n))$ iff f(n) asymptotically greater or

$\in \Theta(g(n))$ iff f(n) asymptotically equal to g(n)

8. **Answer : (A,D)**

- Function 2 executes x = x + 1 exactly 100n times, so $f2(n) = 100n = \Theta(n)$.
- Function 1 halves n each while-iteration and does n increments per iteration, so total increments are $n + n/2 + n/4 + \ldots < 2n = \Theta(n)$, hence $f1(n) = \Theta(n)$.
- Therefore, f1(n) and f2(n) are of the same order: $\Theta(n)$, implying $f1(n) \in \Theta(f2(n))$ and also $f1(n) \in O(n)$, while $f1(n) \notin o(f2(n))$ and $f1(n) \notin \omega(f2(n))$.
  $f1(n) \in \Theta(f2(n))$ and $f1(n) \in O(n)$; the other two are false.

9. **Answer : A**

counter example for B: $f(n)=n$, $f(n^2)=n^2$, $f(n)^2=n^2$, hence B is false

²C: $f(n)=e^{-n}$ -> $f(n^2)=e^{-n}$, Hence C is false

D: $f(n)=\log n$, $f(n^2)=\log(n^2)=2\log n$ and $f(n)^2=(\log n)^2$, hence D is also false

option A is always true.

10. **Answer : A .**

If we take a=2, b=2 apply extended master's theorem we get option B, C, D wrong. Option A is correct.

11 **Answer : C**

Take log on all functions $\log(10^n)$, $\log(n^{\log n})$, $\log n^{\sqrt{n}}$

$\Rightarrow$ n, logn, logn, $\sqrt{n}$logn

$\Rightarrow f2 < f3 < f1$

12. **Answer : B**

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{2n}{5}\right) + 7n$$

Since, $\frac{n}{2} > \frac{2}{5}n$, So, the height of the rightmost subtree will determine the lower bound of the given recurrence, and the height of the leftmost subtree will determine the upper bound of the given recurrence.

Height of the leftmost subtree is: $\log_2 n$ and so,

$$T(n) \le 7n + 7\left(\frac{9}{10}\right)n + 7\left(\frac{9}{10}\right)^2 n + \cdots + 7\left(\frac{9}{10}\right)^{\log_2 n} n$$

$$\Rightarrow T(n) \le 7n\left(1 + \frac{9}{10} + \left(\frac{9}{10}\right)^2 + \cdots + \left(\frac{9}{10}\right)^{\log_2 n}\right)$$

$$\Rightarrow T(n) \le 7n\left(\frac{1 - \left(\frac{9}{10}\right)^{\log_2 n+1}}{1 - \frac{9}{10}}\right) = 70n\left(1 - \left(\frac{9}{10}\right)^{\log_2 n+1}\right)$$

$$= 70n - 70n\left(n^{\log_2 \frac{9}{10}} \frac{1}{10}\right) = 70n - 63n^{0.85}$$

$$\Rightarrow T(n) \le 70n$$

$$\Rightarrow T(n) \in O(n)$$

Height of the rightmost subtree is : $\log_{5/2} n$ and so,

$$7(n) \ge 7n + 7\left(\frac{9}{10}\right)n + 7\left(\frac{9}{10}\right)^2 n + \cdots + \log_{10} 5/2n$$

which implies $T(n) \ge 7n$

So, $T(n) \in \Omega(n)$ for sufficiently large n.

13. **Answer : B**

Given,

$T(n) = T(n^{1/a}) + 1$,

$T(b) = 1$

Now, using iterative method,

$T(n) = [T(n^{1/a^2}) + 1] + 1$

$= [T(n^{1/a^3}) + 1] + 2$

$= [T(n^{1/a^4}) + 1] + 3$

And so on…… ,

$= [T(n^{1/a^k}) + 1] + (k - 1)$

$= T(n^{1/a^k}) + k$

Let,

$n^{1/a^k} = b$

So, $\log(n^{1/a^k}) = \log(b)$

So, $a^k = \log(n) / \log(b) = \log_b(n)$

So, $k = \log_a \log_b(n)$

Therefore,

$= T(n^{1/a^k}) + k$

$= T(b) + \log_a \log_b(n)$

$= 1 + \log_a \log_b(n)$

$= \Theta(\log_a \log_b(n))$

14 **Answer : B**

For the large number, value of inverse of number is less than a constant and value of constant is less than value of square root. 10 is constant, not affected by value of n. √n Square root and $\log_2 n$ is logarithmic. So $\log_2 n$ is definitely less than √n. n has linear growth and 100/n grows inversely with value of n. For bigger value of n, we can consider it 0, so 100/n is least and n is max. So the increasing order of asymptotic complexity will be: 100/n, 10, $\log_2 n$, √n, n So, option (b) is true.

15 **Answer : C**

for i =1, j will run from 1 to n by incrementing by in each step n/2 will run for n times

For i=2 j will run from 1 to n by incrementing by '2' in each step j will run for n/2 times next time j will run for n/3 times and so on time complexity

$= n + n/2 + n/3 + n/4 + \ldots = n(1+1/2+1/3+\ldots)$

$= n\log n$

16 **Answer : B**

$T(n) = 2T(\sqrt{n}) + 1$

Put $n = 2^k$

$T(2^k) = 2T(2^{K/2}) + 1$

Assume $T(2^k) = \delta(K)$

$\delta(K) = 2\delta(K/2) + 1$

By master's theorem

$\delta(K) = \theta(K)$

$T(n) = \theta(\log n)$ $(\because 2^k = n)$

17 **Answer : 2.32**

If they are asking for worst case complexity hence, By calling A(n) we get A(n/2) 5 times,

$A(n)=5A(n/2)+O(1)$

Hence by applying masters theorem, Case 1 :

$n^{(\log_2 5)}$

Thus value of alpha will be 2.32

18 **Answer: (D)**

- The inner loop j = n; j > 1; j = j/2 runs $p=\lfloor \log_2 n \rfloor$ times, so after it finishes, $p \approx \log n$ for each i.
- The next loop k = 1; k < p; k = k*2 runs $\lfloor \log 2p \rfloor \approx \log\log n$ times, incrementing q that many times per i.
- The outer loop over i runs $n-1 \approx n$ times, so total $q \approx n \cdot \log\log n$.

19 **Answer : D**

Since the value of $\sin(n)$ will always range from -1 to +1 , hence $g(n)$ can take values $1, n, n^2$.

Hence, if $g(n) = 1$, then statement I is incorrect.

And, if $g(n) = n^2$, then statement II is incorrect.

20 **Answer : (A)**

The answer to this question is simply max-heapify function. Time complexity of heapify is O(log n)(for a complete binary tree).

Since left subtree and right subtree are max heaps it takes Ω(logn) to heapify.

21 **Answer : (C)**

Sum of the cubes of the first n natural numbers is given by $(n(n+1)/2)^2$ which is $\Theta(n^4)$. So, I, III and IV are correct. II is wrong.

22 **Answer : 110**

In worst case for finding L and H it will take O(logn) time as the given tree is balanced binary search tree. Now there are m elements between L and H. So to traverse m element it will take O(m) time (traversal algorithm given below). So, total $O(m+\log n) \Longrightarrow a=0, b=1, c=1, d=0$

$\therefore 0+(10\times1)+(100\times1)+(1000\times0)=110.$

Check Out Our Gate Courses & Secure Your Dream Rank !

23 Answer : A

By Master's theorem case (i) T(n) is O(n) Here, a = 2, b = 2, f(n) = log n

$N^{(\log_b a)} = n^{(\log_2 2)} = n$

By Master's theorem case (i) T(n) is O(n)

24 Answer : B

The outer for-loop goes for iterations (n/2 + 1). The inner for-loop runs $\log_2 n$ times independently of the outer loop. And for each inner iteration n/2 gets added to k.

Our answer = $(n/2 + 1)*(n/2)*\log n = \Theta(n^2 \log n)$

25 Answer : C

Worst case complexity can never be lower than the average case complexity, but it can be higher. So, (C) is the answer. $A(n) = O(W(n))$.

26 Answer: A

taking log base 2 of all functions:

log f1 = n

log f2 =3/2 log n

log f3 =logn+loglogn

log f4 = logn.logn

∴ f 3 , f 2 , f 4 , f 1 is required increasing order

27 Answer : A

Comparing with the recurrence form of Master Theorem $T(n)=aT(n/b)+f(n)$

$a=1, b=3, \log_b a=0$

So $n^{\log_b a}=n^0$

$f(n)=cn$

So, $f(n)=\Omega(n\log ba+\epsilon)$ holds for any $\epsilon<1$ and this is the third case of Master theorem. But Master theorem also requires (only case 3) that regularity condition be satisfied (this ensures that f(n) is still dominant when the recurrence go deeper) which is $af(n/b)\leq df(n)$ for some $d <1$ and all sufficiently large n.. (Using d here as c is already used in f(n)) Here we get, $f(n/3)\leq df(n)$ $\Longrightarrow cn/3\leq dcn \Longrightarrow 1/3\leq d$

Thus we can use any $1/3\leq d<11$ to satisfy the regularity condition and Master theorem case 3 is satisfied. Now, applying case 3 we get

$T(n)=\Theta(f(n))=\Theta(n)$

28. Answer : D

g(n) = n!.

On expanding the factorial we get $g(n) = O(n^n)$:

$n^n > n^{\log n}$

$n^n > 2^n$

This condition is violated by options A, B and C by first statements of each.

Second statement of option D says that g(n) is asymptotically biggest of all.

29 Answer : A

$T(n) = (\sqrt{2})T(n/2) + (\sqrt{n})$

$= (\sqrt{2})^2 T(n/2^2) + (\sqrt{2})(\sqrt{(n/2)}) + (\sqrt{n})$

...

$= (\sqrt{2})^{(\lg n)} T(1) + (\lg n)(\sqrt{n})$

$= (\sqrt{n}) + (\lg n)(\sqrt{n})$

$= (\sqrt{n})(\lg n + 1)$

30 Answer : A

A < C and A < D are straight forward.

E < B and C, D < E as E and B are exponential functions and B having a larger base.

Now, we just need to see if C or D is larger.

In C we have a term $n^{3/4}$ and correspondingly in D we have $\log_9 n$ (after taking n out).

$n^{3/4}$ is asymptotically larger than $\log_9 n$. So, D < C.

31 Answer : B

Time complexity of f1 is given by

T(n) = T(n - 1) + T(n - 2), (multiplication by 2 and 3 won't affect complexity as it is a constant time operation)

T(0) = T(1) = 1

The solution to this (fibonacci series) is given by Golden ratio. Which is $O(2^n)$. (Using theta in question must be a mistake)

Time complexity of f2 is O(n) as here all recursive calls are avoided by saving the results in an array (dynamic programming).

32 Answer : A

$C1=b^{n/2}*b^{n/2}$ $C2=b^{n/4}* b^{n/4}$

C3

= ...

:

:

T(n)=T(n/2)+1

= logn

33 Answer : A

T(n)=T(n mod m)+1
=T(n/2)+1
= logn

34 Answer : C

Worst Case : T(n)=O(√n)
Best Casse : When n is an even number body of for loop is executed only 1 time (due to "return 0" inside if) which is irrespective of n.
T(n)=Ω(1)

35 Answer : D

Recursive relation for the DoSomething() is
T(n) = T(√n) + C1 if n > 2
We have ignored the floor() part as it doesn't matter here if it's a floor or ceiling.
Let $n = 2^m$, $T(n) = T(2^m)$
Let $T(2^m) = S(m)$

From the above two, T(n) = S(m) S(m) = S(m/2) + C1
S(m) = O(logm)
= O(loglogn)
Now, let us go back to the original recursive function T(n)
T(n) = S(m)
= O(LogLogn)

36 Answer : C

Answer will be Theta(n)
$j = n/2 + n/4 + n/8 + ... + 1$
$= n [1/2^1 + 1/2^2 + 1/2^3 + ... + 1/2^{(\lg n)}]$
(Sum of first n terms of GP is [a.(1-r^n)/(1-r)] , where a is the first term, r is the common ratio < 1, and n is the number of terms)
$= n [1/2 (1-(1/2)^{(\lg n)}) / (1-1/2)]$
$= n [(n-1)/n]$
$= n - 1 = \theta(n)$

37 Answer: B

Put, $n=2^m$
$T(2^m)=2T(2^{m/2})+1$
$T(\log 2^m)=2T(\log 2^{m/2})+c$
$T(m)=2T(m/2)+c$
Using case 1 of master theorem,

$=\Theta(m)$
SInce, $n=2^m$

$=\Theta(\log n)$

38 Answer : C

Applying master theorem
$T(n)=\Theta(n \log n)$
So Option C is wrong.

39 Answer : D

Calculating Transitive Closure boils down to Matrix Multiplication.
We can do Matrix Multiplication in $O(n^3)$. There are better algo that do less than cubic time, but we can not surely do matrix multiplication in
a. O(n)
b. O(nlogn)
c. $O(n^{3/2})$

40 Answer : B

Here we can see that, we have lots of overlapping subproblems. Too many function calls.
1. No of function calls = 2^n
2. stack depth used = $O(n)$

Therefore space is linear and time is exponential.
Now, using one-dimensional ($1D$) table we can reduce the no of function calls and depth of stack space used as well.
Here is what we are going to do:
1. First, check in the $1D$ table for the required call value.
2. If correct value found: do not call recursive functions
3. If not, then only attempt for loop recursive calls

Improvements over given program:
1. Stack space because of function calls reduced to two level only.
2. Extra space used for the $1D$ array = $O(n)$
3. More importantly, Time is reduced to $O(n^2)$. (Exponential to Quadratic !!)

Overall space complexity = stack + extra = $O(1) + O(n) = O(n)$`

Check Out Our Gate Courses & Secure Your Dream Rank !

2 DIVIDE AND CONQUER

GATE CSE – 2026 SET-2 NAT (2-MARKS)

Q1. Consider an array A = [10, 7, 8, 19, 41,35, 25, 31]. Suppose the merge sort algorithm is executed on array A to sort it in increasing order. The merge sort algorithm will carry out a total of 7 merge operations.

A merge operation on sorted left array L and sorted right array R is said to be void if the output of the merge operation is the elements of array L followed by the elements of array R.

The number of void merge operations among these 7 merge operations is _____. (answer in integer)

GATE CSE – 2021 SET-1 MCQ (2-MARKS)

Q2. Let P be an array containing n integers. Let t be the lowest upper bound on the number of comparisons of the array elements, required to find the minimum and maximum values in an arbitrary array of n elements. Which one of the following choices is correct?

a) $t > 2n - 2$
b) $t > \lceil \log(n) \rceil$ and $t <= n$
c) $t > n$ and $t <= 3 \lceil n/2 \rceil$
d) $t > 3 \lceil n/2 \rceil$ and $t <= 2n - 2$

GATE CSE – 2021 SET-1 MCQ (1-MARKS)

Q3. Consider the following array:
23 , 32 , 45 , 69 , 72 , 73 , 89 , 97
Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?

a) Quicksort using the last element as pivot
b) Selection sort
c) Merge sort
d) Insertion sort

GATE CSE – 2020 NAT (1-MARKS)

Q4. An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is

GATE CSE – 2019 NAT (1-MARKS)

Q5. Consider a sequence of 14 elements:A=[−5,−10,6,3,−1,−2,13,4,−9,−1,4,12,−3 ,0].The sequence sum $S(i, j) = \sum^{j}_{k=i} A[k]$.Determine the maximum of S(i ,j), where $0 \leq i \leq j < 14$.(Divide and conquer approach may be used.)

GATE CSE – 2019 MCQ (2-MARKS)

Q6. There are n unsorted arrays: A_1 , A_2 , …, A_n . Assume that n is odd. Each of A_1 , A_2 , …, A_n contains n distinct elements. There are no common elements between any two arrays. The worst-case time complexity of computing the median of the medians of A_1 , A_2 , …, A_n is

a) $O(n)$
b) $O(n \log n)$
c) $O(n^2)$
d) $\Omega(n^2 \log n)$

GATE CSE – 2016 SET-2 MCQ (1-MARKS)

Q7. Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are TRUE?

I. Quick sort runs in $\Theta(n^2)$ time
II. Bubble sort runs in $\Theta(n^2)$ time
III. Merge sort runs in $\Theta(n)$ time
IV. Insertion sort runs in $\Theta(n)$ time

a) I and II only
b) I and III only
c) II and IV only
d) I and IV only

Check Out Our Gate Courses & Secure Your Dream Rank !

GATE CSE – 2016 SET-1 MCQ (1-MARKS)

Q8. The worst-case running time of Insertion sort, Merge sort and Quick sort, respectively, are:

a) $\theta(n\log n)$, $\theta(n\log n)$ and $\theta(n^2)$
b) $\theta(n^2)$, $\theta(n^2)$ and $\theta(n\log n)$
c) $\theta(n^2)$, $\theta(n\log n)$ and $\theta(n\log n)$
d) $\theta(n^2)$, $\theta(n\log n)$ and $\theta(n^2)$

GATE CSE – 2015 SET-2 MCQ(1-MARKS)

Q9. Suppose you are provided with the following function declaration in the C programming language.
int partition(int a[ ], int n);
The function treats the **first element** of a[] as a pivot, and rearranges the array so that all elements less than or equal to the pivot is in the left part of the array, and all elements greater than the pivot is in the right part. In addition, it moves the pivot so that the pivot is the last element of the left part. The return value is the number of elements in the left part. The following partially given function in the C programming language is used to find the **k-th** smallest element in an array a[ ] of size n using the partition function. We assume **k <= n**

```
C
int kth_smallest(int a[], int n, int k)
{
  int left_end = partition(a, n);
  if ( left_end+1 == k ){
     return a[left_end];
  }
  if ( left_end+1 > k ){
     return kth_smallest( _________);
  } else {
     return kth_smallest( _________);
  }
}
```

The missing arguments lists are respectively

a) $(a$, left_end, $k)$ and $(a +$ left_end $+1, n-$ left_end $-1, k-$left_end -1)
b) (a, left_end, k) and ($a, n-$left_end $-1, k-$ left_end -1)
c) ($a +$ left_end $+1, n-$left_end $-1, k-$ left_end -1) and (a, left_end, k)
d) $(a, n-$left_end $-1, k-$left_end -1) and $(a,$ left_end, $k)$

GATE CSE – 2014 SET-2 NAT (2-MARKS)

Q10. Which one of the following is the recurrence equation for the worst case time complexity of the Quicksort algorithm for sorting n (>=2) numbers? In the recurrence equations given in the options below, c is a constant.

(a) $T(n)=2T(n/2)+cn$
(b) $T(n)=T(n-1)+T(1)+cn$
(c) $T(n)=2T(n-1)+cn$
(d) $T(n)=T(n/2)+cn$

GATE CSE – 2014 SET-3 MCQ (2-MARKS)

Q11. You have an array of n elements. Suppose you implement quick sort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is

(a) $O(n^2)$
(b) $O(n\log n)$
(c) $\theta(n\log n)$
(d) $O(n^3)$

GATE CSE – 2014 SET-1 MCQ (2-MARKS)

Q12. Let P is a quick sort program to sort numbers in ascending order using the first element as the pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs [1 2 3 4 5] and [4 1 5 3 2] respectively. Which one of the following holds?

a) $t_1 = 5$
b) $t_1 < t_2$
c) $t_1 > t_2$
d) $t_1 = t_2$

GATE CSE – 2014 SET-1 NAT (2-MARKS)

Q13. The minimum number of comparisons required finding the minimum and the maximum of 100 numbers is____

GATE CSE – 2012 MCQ (2-MARKS)

Q14. A list of n strings, each of length n, is sorted into lexicographic order using the merge-sort algorithm. The worst case running time of this computation is

a) O (n log n)
b) O (n^2 log n)
c) O (n^2 + log n)
d) O (n^2)

GATE CSE – 2009 MCQ (1-MARKS)

Q15. In quick sort, for sorting n elements, the (n/4)th smallest element is selected as pivot using an O(n) time algorithm. What is the worst case time complexity of the quick sort?

a) θ(n)
b) θ(nlogn)
c) θ(n^2)
d) θ(n^2logn)

GATE CSE – 2008 MCQ (1-MARKS)

Q16. The average case and worst case complexities for Merge sort algorithm are

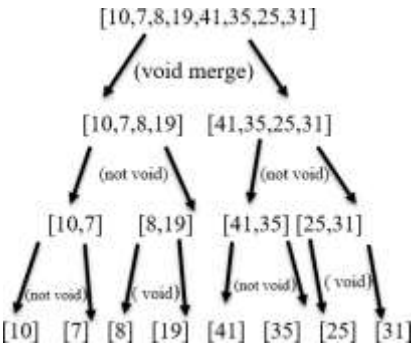
a) O(n log n),O(n^2)
b) O(n^2), O(n$\log_2$ n)
c) O(n^2logn) ,O(n^2)
d) O(n$\log_2$n) , O(n$\log_2$ n)

ANSWER KEY

1. (3)	**2. (c)**	**3. (d)**
4. (0.08)	**5. (29)**	**6. (c)**
7. (d)	**8. (d)**	**9. (a)**
10. (b)	**11. (a)**	**12. (c)**
13. (148)	**14. (b)**	**15. (b)**
16. (d)		

HINTS & SOLUTIONS

1. Answer : 3



2. Answer : C

Pair elements, compare within each pair, then compare winners for max and losers for min, giving at most 3n/2 - 2 comparisons, which equals 3 ⌈n/2⌉ - 2 in general.

Any comparison-based algorithm needs at least 3n/2 - 2 comparisons in the worst case by adversary/charging arguments, so this is optimal; trivially, at least n comparisons are needed overall, yielding n < t <= 3 ⌈n/2⌉ .

3. Answer : D

Given array already sorted."
"∴ Insertion sort takes least number of comparisons Θ(n)." "Since, for a number which is to be inserted in the already sorted array, only 1 comparison will be required."

4. Answer : 0.08

Given an array of 25 distinct elements, and pivot element is chosen uniformly randomly. So, there are only 2 worst case position in the pivot element is either first (or) last. Total elements =25 For worst case number of candidates =2
Therefore, required probability is, = 2/25
= 0.08

5. Answer : 29

They are asking about max subarray sum subarray is different from a subsequence because subarray needs to be contigous but subsequence need not be . Algo to find max subarray sum:

```
int maxsum = A[0], sum = A[0], n=14;
for(int i=1; i<n; i++) {
   if (sum > 0) sum+=A[i];
   else sum = A[i];
  }
  if (maxsum < sum) maxsum = sum;
}
```

Check Out Our Gate Courses & Secure Your Dream Rank !

6. **Answer : C**
Since given arrays are not sorted, so the time complexity to find median is O(n) in an unsorted array. You need to apply this algorithm n time to find all medians and again once to find median of all these medians, so total time complexity is,
$= O(n)*O(n) + O(n)$
$= O(n^2) + O(n)$
$\approx O(n^2)$

7. **Answer : D**
As input is already sorted quick sort runs in $\theta(n^2)$ & insertion sort runs in $\theta(n)$. Generally, Bubble Sort runs in $\Theta(n^2)$ time But if an array is in ascending order, we could make a small modification in Bubble Sort Inner for loop which is responsible for bubbling the k^{th} largest element to the end in k^{th} iteration. Whenever there is no swap after the completion of inner for loop of bubble sort in any iteration, we can declare that array is sorted in case of Bubble Sort taking O(n) time in Best Case.so from above given statements only i and iv are correct.

8. **Answer : D**
Insertion Sort: Worst-case occurs when the array is reverse sorted, leading to a time complexity of $\Theta(n^2)$.
Merge Sort: The divide and conquer method always results in $\Theta(n \log n)$ complexity, even in the worst case.
Quick Sort: Worst-case occurs when the chosen pivot always ends up being the smallest or largest, leading to $\Theta(n^2)$ complexity.
So, the worst-case running times respectively are: $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n^2)$.

9. **Answer : A**
We have to find the *kth* smallest element.
if (left_end+1 > k)
If the above condition is true, it means we have *kth* smallest element on the left array, whose size is left_end instead of *n* for the original array. (The "+1" is used because array index in C language starts from 0). So, we can do a recursive call as

- kth_smallest(a, left_end, k);

If the above condition is false, and left_end $+1 \neq k,$, it means *kth* smallest element is on the right part of the array and it will be *(k−left_end−1)th* element there as left_end+1 elements are gone in the left part. So, the recursive call will be

- kth_smallest(a + left_end + 1, n - left_end – 1, k - left_end - 1);

Correct Option: A.

10. **Answer : B**
Worst case for quick sort happens when *1* element is on one list and *n−1* elements on another list.

11. **Answer : A**
When we choose the first element as the pivot, the worst case of quick sort comes if the input is sorted- either in ascending or descending order. Now, when we choose the middle element as pivot, sorted input no longer gives worst case behavior. But, there will be some permutation of the input numbers which will be giving the same worst case behavior. For example,

1 2 3 4 5 6 7
This array gives worst case behavior for quick sort when the first element is pivot.

6 4 2 1 3 5 7
This array gives the worst case behavior of $O(n^2)$ if we take middle element as the pivot- each split will be *1* element on one side and *n−1* elements on other side. Similarly, for any input, we can have a permutation where the behavior is like this. So, whichever element we take as pivot it gives worst case complexity of $O(n^2)$ as long as pivot is from a fixed position (not random position as in randomized quick sort).

12. **Answer: C**
[1 2 3 4 5] is already sorted hence it takes more time than [4 1 5 3 2]. Quicksort program takes more time when given inputs are in sorted order. Hence $t_1 > t_2$
And when Array is already sorted then Quick Sort will give worst case and worst case T.C is $O(n^2)$

13. **Answer : 148**
To find minimum and maximum comparisons for n numbers = (3n/2)-2)
= (3(100)/2)-2
= 150-2
= 148

14. **Answer : B**
The height of the recursion tree using merge sort is log n and n^2
comparisons are done at each level,where at most n pairs of strings are
compared at each level and n comparisons are required to compare
any two strings, so the worst case running time is $O(n^2 \log n)$.

Check Out Our Gate Courses & Secure Your Dream Rank !

15. **Answer: B**

T(n) = O(n){ pivot selection time} + T(n/4 - 1) + T(3n/4)
which'll give (nlog n).
Pivot selection complexity is given in questions. Pivot being the (n/4)th smallest element, once it is found, we have two sub arrays- one of size (n/4 - 1) and other of size (3n/4) and for both of these we solve recursively.

16. **Answer : D**

O(n log n)
The recurrence relation for merge sort is given by T(n) = 2T(n/2) + n. This can be solved using master's theorem and is found equal to O(n log n).

Check Out Our Gate Courses & Secure Your Dream Rank !

3 GREEDY METHOD

GATE CSE – 2026 SET-1 MCQ (2-MARKS)

Q1. Let $G(V,E)$ be an undirected, edge-weighted graph with integer weights. The weight of a path is the sum of the weights of the edges in that path. The length of a path is the number of edges in that path. Let $s \in V$ be a vertex in G. For every $u \in V$ and for every $k \geq 0$, let $d_k(u)$ denote the weight of a shortest path (in terms of weight) from s to u of length at most k. If there is no path from s to u of length at most k, then $d_k(u)=\infty$.Consider the statements:
S1: For every $k \geq 0$ and $u \in V$, $d_{k+1}(u) \leq d_k(u)$.
S2: For every $(u,v) \in E$, if (u,v) is part of a shortest path (in terms of weight) from s to v, then for every $k \geq 0, d_k(u) \leq d_k(v)$.
Which one of the following options is correct?

a) Only S1 is true
b) Only S2 is true
c) Both S1 and S2 are true
d) Neither S1 nor S2 is true

GATE CSE – 2026 SET-1 MSQ (2-MARKS)

Q2. Let $G(V,E)$ be a simple, undirected, edge-weighted graph with unique edge weights. Which of the following statements about the minimum spanning trees (MST) of G is/are true?

a) In every cycle C of G, the edge with the largest weight in C is not in any MST
b) In every cycle C of G, the edge with the smallest weight in C is in every MST
c) For every vertex $v \in V$, the edge with the largest weight incident on v is not in any MST
d) For every vertex $v \in V$, the edge with the smallest weight incident on v is in every MST

GATE CSE – 2026 SET-2 MCQ (2-MARKS)

Q3. Let G be a weighted directed acyclic graph with m edges and n vertices. Given G and a source vertex s in G, which one of the following options gives the worst case time complexity of the fastest algorithm to find the lengths of shortest paths from s to all vertices that are reachable from s in G?

a) $\Theta(m + n)$
b) $\Theta(m + n \log(n))$
c) $\Theta(nm)$
d) $\Theta(n^3)$

GATE CSE – 2025 SET-1 MCQ (1-MARKS)

Q4. Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G. For any two vertices, u and v, let d1(u,v) and d2(u,v) be the shortest distances between u and v in G and T, respectively. Which ONE of the options is CORRECT for all possible G, T, u and v?

a) d1 (u, v) = d2 (u, v)
b) d1 (u, v) ≤ d2 (u, v)
c) d1 (u, v) ≥ d2 (u, v)
d) d1 (u, v) != d2 (u, v)

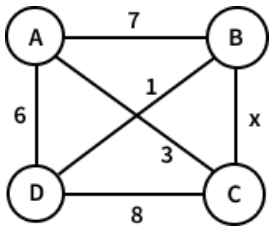
GATE CSE – 2025 SET-2 MCQ (2-MARKS)

Q5. Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant α is added to the weight of every edge. Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update?

a) Every MST remains an MST, and every SP remains an SP.
b) MSTs need not remain MSTs, and every SP remains an SP.
c) Every MST remains an MST, and SPs need not remain SPs.
d) MSTs need not remain MSTs, and SPs need not remain SPs.

GATE CSE – 2025 SET-1 NAT (2-MARKS)

Q6. The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____. (Answer in integer)



GATE CSE – 2025 SET-1 MCQ (2-MARKS)

Q7. Let G(V, E) be an undirected and unweighted graph with 100 vertices. Let d(u, v) denote the number of edges in a shortest path between vertices u and v in V. Let the maximum value of d(u, v), u, v ∈ V such that u ≠ v, be 30. Let T be any breadth-first-search tree of G. Which ONE of the given options is CORRECT for every such graph G?

a) The height of T is exactly 15.
b) The height of T is exactly 30.
c) The height of T is at least 15.
d) The height of T is at least 30.

GATE CSE – 2024 SET-1 NAT (1-MARKS)

Q8. The number of spanning tree in complete labelled graph of 4 vertices labelled A, B, C, and D is……

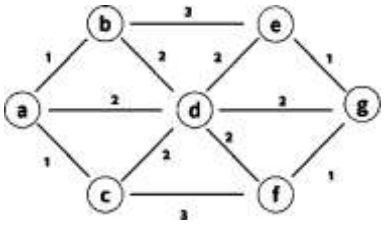
GATE CSE – 2024 SET-2 MCQ (2-MARKS)

Q9. Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has even weight. Which of the following statements is/are TRUE for every such graph G?

a) All edges in G have even weight.
b) In each cycle C in G, all edges in C have even weight.
c) All edges in G have even weight or all edges in G have odd weight.
d) In each cycle C in G, either all edges in C have even weight OR all edges in C have odd weight.

GATE CSE – 2024 SET-2 NAT (2-MARKS)

Q10. The number of distinct minimum-weight spanning trees of the following graph is _____.



GATE CSE – 2022 NAT (1-MARKS)

Q11. "Let G(V, E) be a directed graph, where V = {1, 2, 3, 4, 5} is the set of vertices and E is the set of directed edges, as defined by the following adjacency matrix A.

$$A[i][j] = \begin{cases} 1, & 1 \le j \le i \le 5 \\ 0, & \text{otherwise} \end{cases}$$

"A[i][j] = 1 indicates a directed edge from node i to node j."A directed spanning tree of G, rooted at r ∈ V, is defined as "a subgraph T of G such that the undirected version of T is a tree, and T contains a directed path from r to every other vertex in V." The number of such directed spanning trees rooted at vertex 5 is _____.

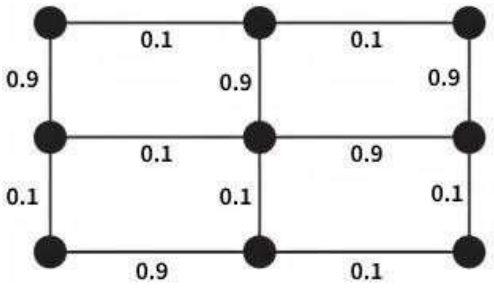
GATE CSE – 2022 MSQ (2-MARKS)

Q12. Consider a simple undirected weighted graph G, all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE?

a) The edge with the second smallest weight is always part of any minimum spanning tree of G.
b) One or both of the edges with the third smallest and the fourth smallest weights are part of any minimum spanning tree of G
c) Suppose S⊆V be such that S != ϕ and S != V. Consider the edge with the minimum weight such that one of its vertices is in S and the other in V\S. Such an edge will always be part of any minimum spanning tree of G
d) G can have multiple minimum spanning trees.

GATE CSE – 2021 SET-1 NAT (1-MARKS)

Q13. Consider the following undirected graph with edge weights as shown:



The number of minimum-spanning trees of the graph is _____.

GATE CSE – 2021 SET-2 (2-MARKS)

Q14. Consider the string abbccddeee. Each letter in the string must be assigned a binary code satisfying the following properties:
For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter.
For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to the other letter.
Among the set of all binary code assignments which satisfy the above two properties, what is the minimum length of the encoded string?

(a) 21
(b) 23
(c) 25
(d) 30

GATE CSE – 2021 SET-2 MCQ (2-MARKS)

Q15. Let G be a connected undirected weighted graph. Consider the following two statements.
S1 : There exists a minimum weight edge in G which is present in every minimum spanning tree of G.
S2 : If every edge in G has distinct weight, then G has a unique minimum spanning tree.
Which one of the following options is correct?

a) S 1 is true and S 2 is false.
b) S 1 is false and S 2 is true.
c) Both S 1 and S 2 are true.
d) Both S 1 and S 2 are false.

GATE CSE – 2020 MCQ (2-MARKS)

Q16. Let G = (V, E) be a directed, weighed graph with weight function w : E -> R. For some function f : V -> R, for each edge (u, v) in E, define w'(u, v) as w(u, v) + f(u) - f(v).
Which one of the options completes the following sentence so that it is TRUE?
"The shortest paths in G under w are shortest paths under w' too,_____".

a) for every f : V -> R
b) if and only if for all u in V, f(u) is positive
c) if and only if for all u in V, f(u) is negative
d) if and only if f(u) is the distance from s to u in the graph obtained by adding a new vertex s to G and edges of zero weight from s to every vertex of G

GATE CSE – 2020 MCQ (2-MARKS)

Q17. Let G = (V, E) be a weighted undirected graph and let T be a Minimum Spanning Tree (MST) of G maintained using adjacency lists. Suppose a new weighted edge (u,v) ∈ V × V is added to G. The worst case time complexity determining if T is still an MST of the resultant graph is

a) $\theta(|E| \log |V|)$
b) $\theta(|E| + |V|)$
c) $\theta(|E| |V|)$
d) $\theta(|V|)$

GATE CSE – 2020 NAT (2-MARKS)

Q18. Consider a graph G = (V, E), where V = $\{v_1, v_2, \ldots, v_{100}\}$, E = $\{(v_i, v_j) \mid 1 <= i <= j <= 100\}$, and weight of the edge (v_i, v_j) is $|i - j|$. The weight of the minimum spanning tree of G is ___.

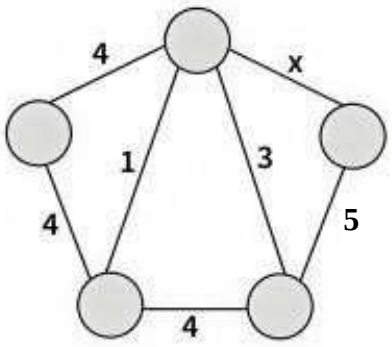
GATE CSE – 2019 MCQ (2-MARKS)

Q19. Let G be any connected, weighted, undirected graph.
I. G has a unique minimum spanning tree, if no two edges of G have the same weight.
II. G has a unique minimum spanning tree, if, for every cut of G, there is a unique minimum-weight edge crossing the cut.
Which of the above two statements is/are TRUE?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

GATE CSE – 2018 NAT (2-MARKS)

Q20. Consider the following undirected graph G:



Choose a value of x that will maximize the number of minimum weight spanning trees (MWSTs) of G. The number of MWSTs of G for this value of x is _____.

GATE CSE – 2016 SET-1 MCQ (2-MARKS)

Q21. Let G be a complete undirected graph on 4 vertices having 6 edges with weights being 1,2

a minimum weight spanning tree of G can have is_____.

a) 4
b) 5
c) 6
d) 7

GATE CSE – 2016 SET-1 MCQ (1-MARKS)

Q22. Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE?

P: Minimum spanning tree of G does not change
Q: Shortest path between any pair of vertices does not change

a) P only
b) Q only
c) Neither P nor Q
d) Both P and Q

GATE CSE – 2016 SET-1 MCQ (2-MARKS)

Q23. G = (V, E) is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G. Which of the following statements about the minimum spanning trees (MSTs) of G is/are TRUE?
I. If e is the lightest edge of some cycle in G, then every MST of G includes e
II. If e is the heaviest edge of some cycle in G, then every MST of G excludes e

a) i only
b) ii only
c) both i and ii
d) neither i and ii

GATE CSE – 2016 SET-1 NAT (2-MARKS)

Q24. Consider the weighted undirected graph with 4 vertices, where the weight of edge {i, j} is given by the entry W ij in the matrix W.

$$W = \begin{bmatrix} 0 & 2 & 8 & 5 \\ 2 & 0 & 5 & 8 \\ 8 & 5 & 0 & x \\ 5 & 8 & x & 0 \end{bmatrix}$$

The largest possible integer value of x, for which at least one shortest path between some pair of vertices will contain the edge with weight x is_____.

GATE CSE – 2017 SET-2 NAT (2-MARKS)

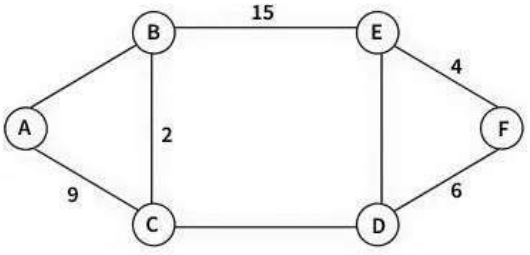
Q25. A message is made up entirely of characters from the set X = {P,Q,R,S,T} . The table of probabilities of each character is shown below :

Character	Probability
P	0.22
Q	0.17
R	0.34
S	0.19
T	0.08
Total	1.00

A message of 100 characters over X is encoded using Huffman coding. Then the excepted length of the encoded message in bits is ____

GATE CSE – 2015 SET-1 NAT (2-MARKS)

Q26. The graph shown below has 8 edges with distinct integer edge weights. The minimum spanning tree (MST) is of weight 36 and contains the edges: {(A, C), (B, C), (B, E), (E, F), (D, F)}. The edge weights of only those edges which are in the MST are given in the figure shown below. The minimum possible sum of weights of all 8 edges of this graph is



GATE CSE – 2015 SET-3 NAT (2-MARKS)

Q27. Let G be a connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of G is 500. When the weight of each edge of G is increased by five, the weight of a minimum spanning tree becomes _____.

GATE CSE – 2013 MCQ (1-MARKS)

Q28. What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?

a) $\Theta(n^2)$
b) $\Theta(n^2 \log n)$
c) $\Theta(n^3)$
d) $\Theta(n^3 \log n)$

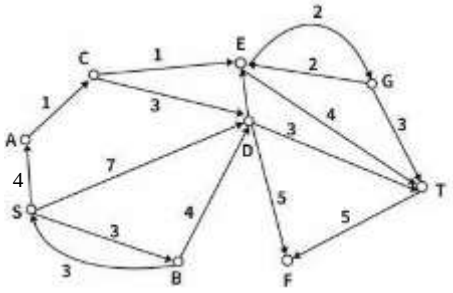
GATE CSE – 2012 MCQ (2-MARKS)

Q29. Let G be a weighted graph with edge weights greater than one and G' be the graph constructed by squaring the weights of edges in G. Let T and T' be the minimum spanning trees of G and G' , respectively, with total weights t and t' . Which of the following statements is TRUE?

a) T' = T with total weight t' = t^2
b) T' = T with total weight t' < t^2
c) T' ≠ T but total weight t' = t^2
d) None of the above

GATE CSE – 2012 MCQ (2-MARKS)

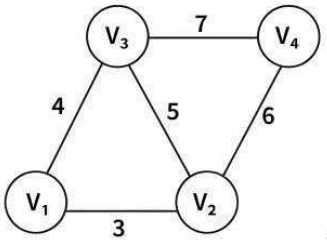
Q30. Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.



a) SDT
b) SBDT
c) SACDT
d) SACET

GATE CSE – 2011 MCQ (2-MARKS)

Q31. An undirected graph G(V,E) contains n (n > 2) nodes named v_1, v_2... v_n. Two nodes vi , vj are connected if and only if 0 < | i- j | < = 2 . Each edge (vi, vj)is assigned a weight i + j. A sample graph with n = 4 is shown below.

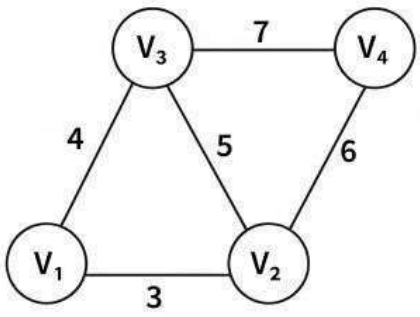


What will be the cost of the minimum spanning tree (MST) of such a graph with 11 nodes?

a) $1 / 11 (11 n^2 - 5n)$
b) $n^2 - n + 1$
c) $6n – 11$
d) $2n + 1$

GATE CSE – 2011 MCQ (2-MARKS)

Q32. An undirected graph G(V, E) contains n (n > 2) nodes named v_1, v_2, . . . , v_n. Two nodes v_i, v_j are connected if and only if $0 < |i - j| \leq 2$. Each edge (v_i, v_j) is assigned a weight i + j. A sample graph with n = 4 is shown below.



The length of the path from v5 to v6 in the MST of previous question with n = 10 is

a) 11
b) 25
c) 31
d) 41

GATE CSE – 2010 MCQ (2-MARKS)

Q33. Consider a complete undirected graph with vertex set {0, 1, 2, 3, 4}. Entry W_{ij} in the matrix W below is the weight of the edge {i, j}.

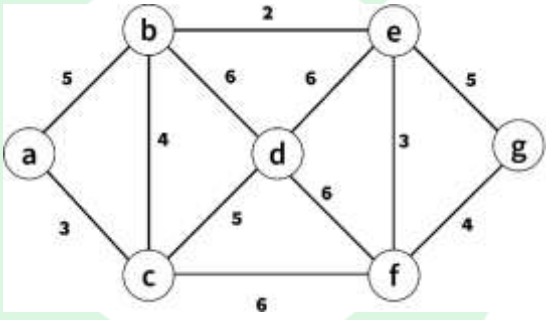
$$W = \begin{pmatrix} 0 & 1 & 8 & 1 & 4 \\ 1 & 0 & 12 & 4 & 9 \\ 8 & 12 & 0 & 7 & 3 \\ 1 & 4 & 7 & 0 & 2 \\ 4 & 9 & 3 & 2 & 0 \end{pmatrix}$$

What is the minimum possible weight of a spanning tree T in this graph such that vertex 0 is a leaf node in the tree T?

a) 7
b) 8
c) 9
d) 10

GATE CSE – 2009 MCQ (2-MARKS)

Q34. Consider the following graph:



Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm?

a) (b,e) (e,f) (a,c) (b,c) (f,g) (c,d)
b) (b,e) (e,f) (a,c) (f,g) (b,c) (c,d)
c) (b,e) (a,c) (e,f) (b,c) (f,g) (c,d)
d) (b,e) (e,f) (b,c) (a,c) (f,g) (c,d)

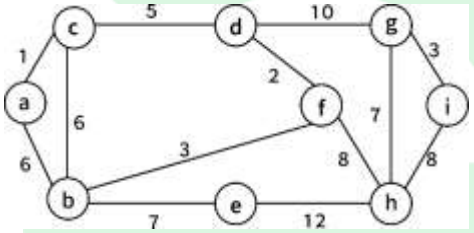
GATE CSE – 2009 MCQ (1-MARKS)

Q35. Which of the following statement(s) is/are correct regarding Bellman-Ford shortest path algorithm?
P. Always finds a negative weighted cycle, if one exists.
Q. Finds whether any negative weighted cycle is reachable from the source.

a) P only
b) Q only
c) Both P and Q
d) Neither P nor Q

GATE IT – 2008 MCQ (2-MARKS)

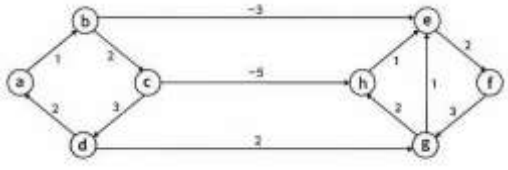
Q36. For the undirected, weighted graph given below, which of the following sequences of edges represents a correct execution of Prim's algorithm to construct a Minimum Spanning Tree?



(A) (a, b), (d, f), (f, c), (g, i), (d, a), (g, h), (c, e), (f, h)
(B) (c, e), (c, f), (f, d), (d, a), (a, b), (g, h), (h, f), (g, i)
(C) (d, f), (f, c), (d, a), (a, b), (c, e), (f, h), (g, h), (g, i)
(D) $(h, g), (g, i), (h, f), (f, c), (f, d), (d, a), (a, b), (c, e)$

GATE CSE – 2008 MCQ (2-MARKS)

Q37. Dijkstra's single source shortest path algorithm when run from vertex a in the above graph, computes the correct shortest path distance to



a) only vertex a
b) only vertices a, b, c, d
c) only vertices a, b, c, d
d) all the vertices

GATE IT – 2007 MCQ (1-MARKS)

Q38. Consider a weighted, undirected graph with positive edge weights and let u v be an edge in the graph. It is known that the shortest path from the source vertex s to u has weight 53 and the shortest path from s to v has weight 65. Which one of the following statements ?

a) Weight(u,v)≤12
b) Weight(u,v)=12
c) Weight(u,v)≥12
d) Weight(u,v)>12

GATE CSE – 2007 MCQ (1-MARKS)

Q39. Let w be the minimum weight among all edge weights in an undirected connected graph. Let e be a specific edge of weight w. Which of the following is False ?

a) There is a minimum spanning tree containing e.
b) If e is not in a minimum spanning tree T, then in the cycle formed by adding e to T, all edges have the same weight.
c) Every minimum spanning tree has an edge of weight w.
d) e is present in every minimum spanning tree.

GATE CSE – 2007 MCQ (1-MARKS)

Q40. Suppose the letters a, b, c, d, e, f have probabilities 1/2, 1/4, 1/8, 1/16, 1/32, 1/32 respectively. What is the average length of the Huffman code for the letters a,b,c,d,e,f?

a) 3
b) 2.1875
c) 2.25
d) 1.9375

GATE CSE – 2007 MCQ (1-MARKS)

Q41. Suppose the letters a, b, c, d, e, f have probabilities 1/2, 1/4, 1/8, 1/16, 1/32, 1/32 respectively. Which of the following is the Huffman code for the letter a, b, c, d, e, f?

a) 0, 10, 110, 1110, 11110, 11111
b) 11, 10, 011, 010, 001, 000
c) 11, 10, 01, 001, 0001, 0000
d) 110, 100, 010, 000, 001, 111

Check Out Our Gate Courses & Secure Your Dream Rank !

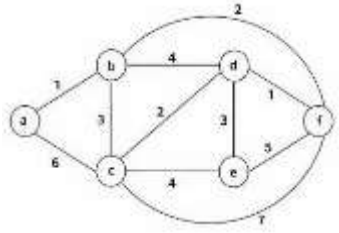
GATE CSE – 2006 MCQ (1-MARKS)

Q42. The characters a to h have the set of frequencies based on the first 8 Fibonacci numbers as follows
a:1, b:1, c:2, d:3, e:5, f:8, g:13, h:21
A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the following code?
110111100111010

a) fdheg
b) ecgdf
c) dchfg
d) fehdg

GATE CSE – 2006 MCQ (1-MARKS)

Q43. Consider the following graph:



Which one of the following cannot be the sequence of edges added, in that order, to a minimum spanning tree using Kruskal's algorithm?

a) (a - b), (d - f), (b - f), (d - c), (d - e)
b) (a - b), (d - f), (d - c), (b - f), (d - e)
c) (d - f), (a - b), (d - c), (b - f), (d - e)
d) (d - f), (a - b), (b - f), (d - e), (d - c)

GATE CSE – 2005 MCQ (1-MARKS)

Q44. Let s and t be two vertices in a undirected graph G=(V, E) having distinct positive edge weights. Let [X, Y] be a partition of V such that s∈ X and t∈ Y. Consider the edge e having the minimum weight amongst all those edges that have one vertex in X and one vertex in Y.
The edge e must definitely belong to:

a) the minimum weighted spanning tree of G
b) the weighted shortest path from s to t
c) each path from s to t
d) the weighted longest path from s to t

GATE CSE – 2005 MCQ (1-MARKS)

Q45. An undirected graph G has n nodes. Its adjacency matrix is given by an nxn square matrix whose
(i) diagonal elements are 0's and
(ii) non-diagonal elements are 1's.
which one of the following is TRUE?

a) Graph G has no minimum spanning tree (MST)
b) Graph G has a unique MST of cost n-1
c) Graph G has multiple distinct MSTs, each of cost n-1
d) Graph G has multiple spanning trees of different costs

GATE CSE – 2005 MCQ (1-MARKS)

Q46. Let G(V,E) be an undirected graph with positive edge weights. Dijkstra's single source shortest path algorithm can be implemented using the binary heap data structure with time complexity:

a) $O(|V|^2)$
b) $O(|E| + |V| \log |V|)$
c) $O(|V| \log |V|)$
d) $O((|E| + |V|) \log |V|)$

GATE IT– 2005 NAT (1-MARKS)

Q47. Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j iff either j = i + 1 or j = 3i. The minimum number of edges in a path in G from vertex 1 to vertex 100 is:

a) 4
b) 7
c) 23
d) 99

GATE CSE – 2005 MCQ (1-MARKS)

Q48. We are given 9 tasks T_1, T_2... T_9. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has a profit P_i and a deadline d_i Profit Pi is earned if the task is completed before the end of the d_i^{th} unit of time.

Task	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

What is the maximum profit earned?

a) 147
b) 165
c) 167
d) 175

GATE CSE – 2005 MCQ (1-MARKS)

Q49. We are given 9 tasks T_1, T_2... T_9. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has a profit P_i and a deadline d_i Profit Pi is earned if the task is completed before the end of the d_i^{th} unit of time.

Task	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

Are all tasks completed in the schedule that gives maximum profit?

a) All tasks are completed
b) T_1 and T_6 are left out
c) T_1 and T_8 are left out
d) T_4 and T_6 are left out

ANSWER KEY

1.(a)	**2.(a,d)**	**3.(a)**
4.(b)	**5.(c)**	**6.(5)**
7.(c)	**8.(16)**	**9.(d)**
10.(9)	**11.(24)**	**12.(a,b,c)**
13.(3)	**14.(b)**	**15.(b)**
16.(a)	**17.(d)**	**18.(99)**
19.(c)	**20.(4)**	**21.(d)**
22.(a)	**23.(b)**	**24.(12)**
25.(225)	**26.(69)**	**27.(995)**
28.(c)	**29.(d)**	**30.(d)**
31.(b)	**32.(c)**	**33.(d)**
34.(d)	**35.(b)**	**36.(c)**
37.(d)	**38.(c)**	**39.(d)**
40.(d)	**41.(a)**	**42.(a)**
43.(d)	**44.(a)**	**45.(c)**
46.(d)	**47.(b)**	**48.(a)**
49.(d)		

Check Out Our Gate Courses & Secure Your Dream Rank !

HINTS & SOLUTIONS

1. Answer : A

When we increase k to k+1, we allow more paths (paths of length $\leq k$ are a subset of paths of length $\leq k+1$).
Taking minimum over a larger set cannot increase the value.
So,
$d_{k+1}(u) \leq d_k(u)$.
Suppose edge (u, v) is part of a shortest path from s to v.
That does not guarantee $d_k(u) \leq d_k(v)$ for every k.
Counterexample (with negative weight):
s — u has weight 100
u — v has weight −200
Shortest path to v is:
$s \rightarrow u \rightarrow v = -100$
For k = 2:
$d_2(u) = 100$
$d_2(v) = -100$
So,
$d_2(u) \leq d_2(v)$ becomes
$100 \leq -100 \rightarrow$ FALSE.
Hence S2 is not always true.

2. Answer : A,D

(A) True
By the Cycle Property: in any cycle, the maximum-weight edge cannot be in the MST.
(B) False
Cycle property does not say the minimum edge in a cycle must be included. It only excludes the maximum.An edge that is the minimum in one cycle may simultaneously belong to another cycle where it is the maximum-weight edge.
By the Cycle Property, the maximum-weight edge of a cycle cannot be in the MST.
Hence, being the minimum in one cycle does not guarantee inclusion in the MST.
(C) False
An edge can be the largest incident on a vertex and still be required to connect that vertex (e.g., if it's the only connection).
(D) True
By the Cut Property: for the cut {v} and V−{v}, the minimum-weight edge crossing it (i.e., smallest edge incident on v) must be in the MST.

3. Answer : A

Answer: (A) $\Theta(m + n)$
Since the graph is a Directed Acyclic Graph (DAG), the shortest paths from source s can be found using:

1. Topological sorting of the graph $\rightarrow \Theta(m + n)$
2. Relaxing all edges once in topological order $\rightarrow \Theta(m)$

Total time complexity = $\Theta(m + n)$.No priority queue is needed (unlike Dijkstra's algorithm), because the graph has no cycles and the topological order guarantees correct relaxation order.Therefore, the worst-case time complexity of the fastest algorithm is: **$\Theta(m + n)$**

4. Answer : (B)

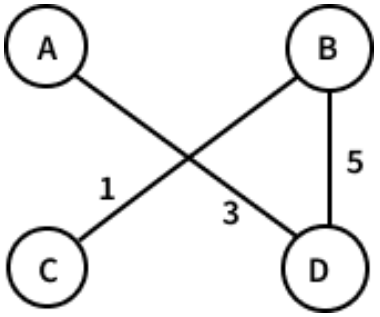
MST distance from u to v may not be the shortest path distance from u to v.Shortest path distance from u to v i.e. d1 (u, v) $\leq$ MST path distance from u to v i.e. d2(u, v)

5. Answer : C

Every MST will remain an MST because if we use Kruskal's Algorithm, adding alpha to each edge weight will not change the ordering of the edges, and the tree will be formed as it is.
for shortest paths, consider an undirected graph of 4 vertices v1, v2, v3, v4. The edge set of this graph is defined as {[v1,v2], [v2,v3], [v3,v4] and [v1,v4]}. Let edge weight of [v1,v2], [v2,v3], [v3,v4] and [v1,v4] be x1, x2, x3 and x4 respectively with the condition: x1+x2+x3 = x4 - 1. Thus originally the shortest path between v1 and v4 is x1+x2+x3. However after adding positive constant to each edge, the shortest path between v1 and v4 is x4 + alpha.

6. Answer : 5

The value of 'x' should be less than 6,7,and 8. So, x can be atmost 5.
Therefore, maximum value of x can be 5.



7. **Answer: C**

BFS traversal on unweighted graph always give the shortest path.

Now comes if we start the BFS traversal from vertex u and to go atleast to v we will have the height 30.

if we start it from middle in between from u and v ,we will get the left most tree which will go to u and the rightmost tree will go to v.Thus the tree of height in case 1 will be divided into two of height 15 each.

Hence the option height atleast 15 is correct.

8. **Answer : 16**

The number of spanning trees in a complete graph with 4 vertices (A, B, C, D) is given by Cayley's formula:

Number of spanning trees=n^{n-2}

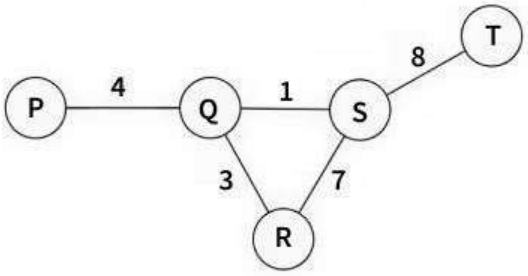
For n=4:

$4^{4-2} = 4^2 = 16$

So, the number of spanning trees is 16.

9. **Answer (D)**

The question asks about spanning tree, not MST. Counter example for option-A, option-B, and option-C:



Every spanning tree of this graph has even weight, but does not satisfy option-A, option-B, and option-C.

Only option-D remains.

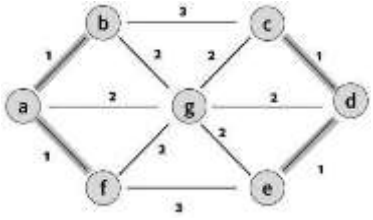
Hence option-D is correct.

10. **Answer : 9**

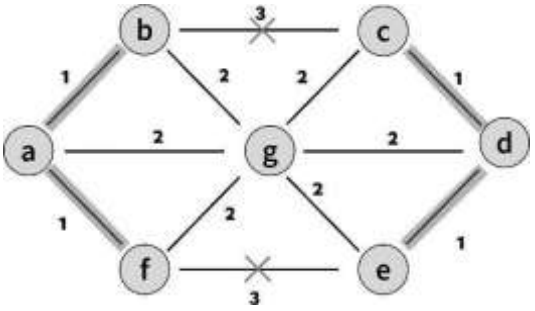
Explanation

Counter Example :

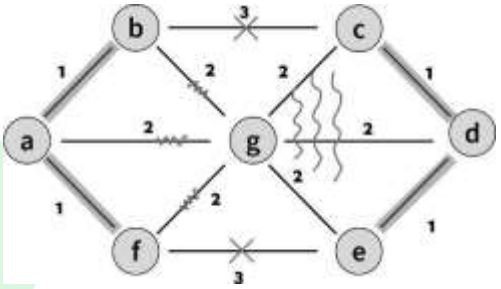
Step 1: Apply Kruskal's algorithm where minimum weighted edge gets selected



Step 2: Use theorem, "For any cycle in G, the heaviest edge on that cycle cannot be in the MST "Edge bc and fe can't be in MST as it creates cycle with max edge weight in the graph



Step 3: Now here comes the intution, for each edge i.e edge bg, ba, bf we have 3 choices i.e gc, gd, ge so 3*3 = 9 distinct MSTs possible



11. **Answer : 24**

For a graph defined by the adjacency matrix A[i][j]=1 (if $1 \le j \le i \le 5$), and considering directed spanning trees rooted at vertex 5:

From 5, you can directly go to 4 in only 1 way.

To each of the next lower vertices, there are multiple possible parents among higher numbered vertices already included in the tree.

For 3: Choose parent from (5,4) ⇒ 2 ways.

For 2: Choose parent from (5,4,3) ⇒ 3 ways.

For 1: Choose parent from (5,4,3,2) ⇒ 4 ways.

So, total count = $1 \times 2 \times 3 \times 4 = 24$.

Alternatively, the tree rooted at 5 is essentially a permutation of the 4 remaining vertices, resulting in 4!=24 directed spanning trees rooted at 5.

The number of such directed spanning trees rooted at vertex 5 is 24.

Check Out Our Gate Courses & Secure Your Dream Rank !

12. Answer: A,B,C

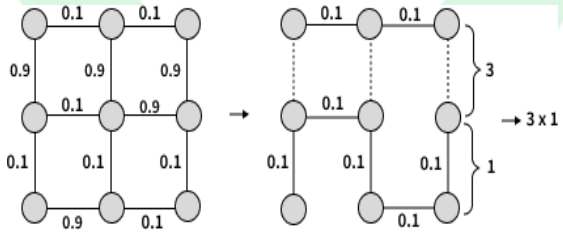
Why A is true? The edge with the smallest weight is always included, and the second smallest edge will also be included unless it creates a cycle. For a simple connected graph, it cannot form a cycle at the stage where only the first edge is present, so the second smallest edge is always added.

Why B is true? By the cut property: for any cut, the unique lightest edge crossing that cut must be in every MST; edges ranked third or fourth globally need not be in any MST unless they are the lightest for some cut, so "one or both" being in every MST is not guaranteed and thus the statement as written is false if it claims necessity for those ranks; however, interpreted as "may be part" it's trivial—so the strict cut-property guarantee applies only to the single lightest edge per cut. Hence option 2, which claims necessity for specific ranks, is false.

Why C is true? For any nonempty proper subset S of V, the minimum-weight edge with one endpoint in S and the other in V \ S must be in every MST by the cut property, since edge weights are distinct and thus the lightest crossing edge is unique.

Why D is false? With all edge weights distinct, the MST is unique; therefore the graph cannot have multiple minimum spanning trees under the given condition.

13. Answer : 3

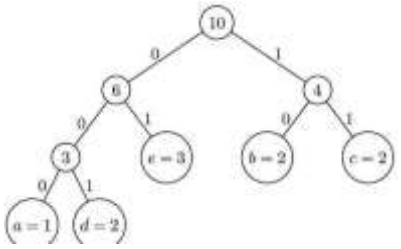


14. Answer: B

a has the least frequency and should be the leaf of the tree. *b, c* and *d* have the same frequency but as per Condition 2 in the questions *d* should be taken first, followed by *c* and then *b, e* has the highest frequency and so must be taken last.

Alphabet	Frequency
a	1
b	2
c	2
d	2
e	3

The final Huffman tree looks like:

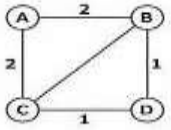


Prefix Code	Code Length
a = 000	3
b = 10	2
c = 11	2
d = 001	3
e = 01	2

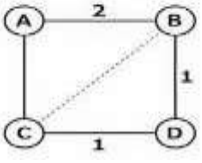
Therefore, Minimum length of encoded string: (1 * 3) + (2 * 2) + (2 * 2) + (3 * 2) + (2 * 3) = 23

15. Answer: (B)

S1 : consider the graph.



One of the possible MSTS can be :



But the minimum weight edge BC in G is not present. So, the S 1 is a false statement.

S2 : In any undirected graph G, distinct edge weights means Unique MST.

So, the S1 is false and S2 is true.

16 **Answer : A**

For any mapping of vertices to real values, the shortest paths won't change. All intermediate nodes values get canceled on any path you take and what you're left with is only the source and destination node values which would add up to cost on any path. Hence, the shortest paths would still be the same.

Option D is wrong because of the "if and only if" clause in it. If it were "if", it would be correct. The condition given is sufficient but not necessary. Hence, "only if" is incorrect in the option. Basically it is saying *f(u)* would be 0 for all vertices since they're connected to a new vertex s with zero weighted edge. Similarly options B and C are also wrong for the same reason.

17 **Answer : (D)**

We can accomplish this in O(|V|) time as follows:

1. Perform a BFS in T from node u to node v to identify the edge with the highest weight along that path — this takes O(|V|) time.
2. If the weight of that edge is larger than the weight of the new edge being considered, replace the old edge with the new one.
3. Otherwise, leave the tree unchanged, since the new edge would not result in a better MST. This step is O(1).

The reasoning for this method is that inserting a new edge into T creates exactly one cycle. To maintain the MST property, we must remove the heaviest edge from that cycle.

18. **Answer : 99**

There are 100 vertices numbered from 1 to 100. Edges are presents in this graphs E={ (v_i, v_j) | $1 \le i_1$, v_2), (v_2, v_3), ... (v_{99}, v_{100}) will have minimum weight, i.e., 1 for each edge. As we know that, there are 99 edges are possible in minimum spanning tree of 100-vertices graph. Therefore, these edges (v_1, v_2), (v_2, v_3), (v_{99}, v_{100}) will be spanning tree for given graph. These are 99 edges with 1 cost of each. The weight of minimum spanning tree of G is 99*1 = 99. Option (A) is correct.

19. **Answer: C**

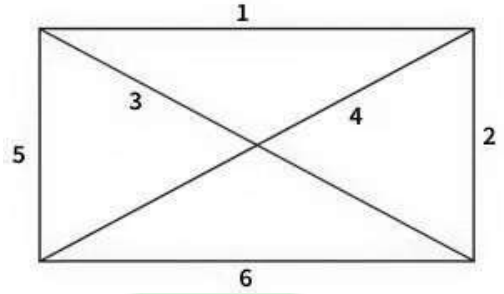
If no two edges of G have same weight surely G will have unique spanning tree is true. So I is true Also if, for every cut of G, there is a unique minimum weight edge crossing the cut then G will have unique spanning tree . So II is also true [Note: The converse of II is not true, but that is not relevant to this question] So, both I and II are true. Option (C) is correct.

20. **Answer : 4**

Edges with weights 1 and 3 will be selected first, Now bottom edge with weight 4 will not be selected as will cause cycle on MST, both corner vertices have two-two choices to select the vertices, so these corner edges with weights 4 and 5 will resultant 2 * 2 = 4 MSTs. So, total number of MSTs are 2 * 2 = 4, which is answer.

21. **Answer : D**

One graph that has maximum possible weight of spanning tree



An MST on 4 vertices has 3 edges, and by Kruskal's order the two smallest edges 1 and 2 can always be included without forming a cycle.

- To maximize the MST weight, arrange the graph so that edge 3 would create a cycle with 1 and 2 and hence is skipped; the next feasible edge is 4, giving total 1 + 2 + 4 = 7.
- This is the maximum possible MST weight under the given multiset of edge weights.

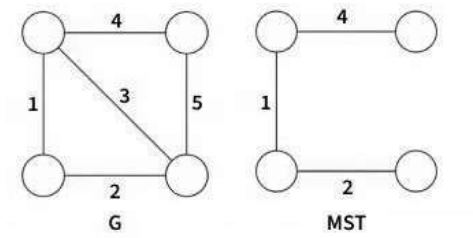
22. **Answer : (A)**

The shortest path may change. The reason is, there may be different number of edges in different paths from s to t. For example, let shortest path be of weight 15 and has 5 edges. Let there be another path with 2 edges and total weight 25. The weight of the shortest path is increased by 5*10 and becomes 15 + 50. Weight of the other path is increased by 2*10 and becomes 25 + 20. So the shortest path changes to the other path with weight as 45. The Minimum Spanning Tree doesn't change. Remember the Kruskal's algorithm where we sort the edges first. If we increase all weights, then order of edges won't change.

Check Out Our Gate Courses & Secure Your Dream Rank !

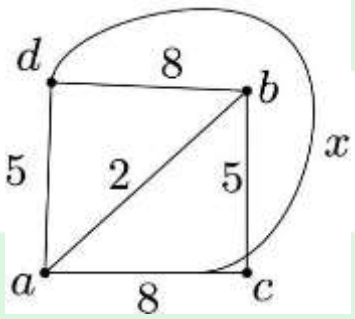
23. **Answer : B**

Here in below Graph G in (Cycle 3, 4, 5) 3 is the lightest edge and still it is not included in MST.



Statement 2:- True by[Cycle property of MST] : (in above Graph G 1 - 2- 3 is a cycle and 3 is the heaviest edge) If heaviest edge is in cycle then we will always exclude that because cycle is there means, we can have other choice of low cost edges. (If edge weights are not distinct even the heaviest edge can be part of the MST and here in question distinct edge weight is clearly mentioned)

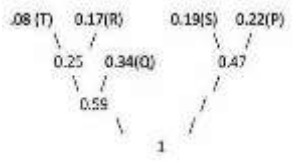
24. **Answer : 12**



If x = 12 then the shortest path between d & c will contain edge with label 'x'.

25. **Answer : 225**

In Huffman coding, we pick the least two frequent (or probable) character, combine them and create a new node.



Looking at above tree structure, Number of bits required by each: P – 2,Q – 2,R – 3,S – 2,T – 3

Therefore, excepted length of the encoded message:

= 3*0.8 + 3*0.17 + 2*0.19 + 2 *0.22 + 2*0.34

= 2.25

For 100 characters, 2.25*100 = 225

26. **Answer : 69**

Consider the cycle ABC. AC and BC and are part of minimum spanning tree. So, AB should be greater than max(AC, BC) (greater and not equal as edge weights are given to be distinct), as otherwise we could add AB to the minimum spanning tree and removed the greater of AC, BC and we could have got another minimum spanning tree. So, AB > 9.

Similarly, for the cycle DEF, ED > 6.

And for the cycle BCDE, CD > 15.

So, minimum possible sum of these will be 10 + 7 + 16 = 33. Adding the weight of spanning tree, we get the total sum of edge weights.

= 33 + 36 = 69

27. **Answer: 995**

Since there are 100 vertices, there must be 99 edges in Minimum Spanning Tree (MST). When weight of every edge is increased by 5, the increment in weight of MST is = 99 * 5 = 495 so new weight of MST is 500 + 495 which is 995.

28. **Answer : C**

A complete graph with n vertices.

To Find: Time complexity of the Bellman-Ford Single Source Shortest Path (SSSP) algorithm.

Analysis:

For a general graph with n vertices and m edges, the time complexity of the Bellman-Ford algorithm is $O(n \cdot m)$

For a complete graph, every pair of vertices is connected by an edge.

Hence, the total number of edges is:

$m = ({}^{n}C_2) = n(n-1)/2 = O(n^2)$

Substituting this value in the general time complexity formula: $O(n \cdot m) = O(n \cdot n^2) = O(n^3)$

∴ Final Time Complexity:

$O(n^3)$

29. **Answer : D**

When edge weights are squared, the minimum spanning tree does not change.

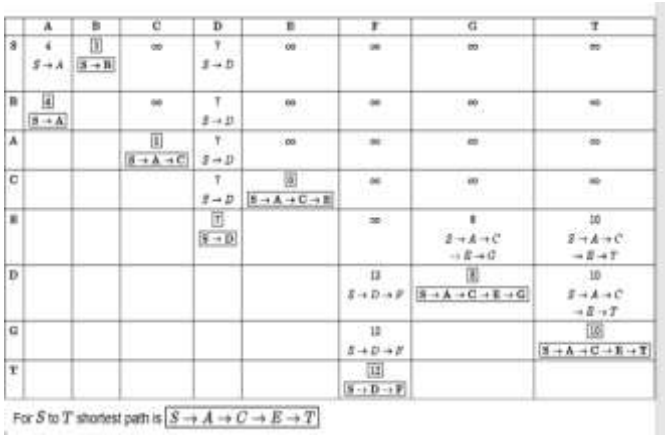
This happens because squaring positive weights keeps their order the same. If one weight was smaller than another before squaring, it remains smaller after squaring.

The sum of squared weights is less than the square of the sum of weights, except when there is only one edge.So option B is generally correct, and option A is only true for single-edge graphs. In GATE 2012, marks were given to all.

Check Out Our Gate Courses & Secure Your Dream Rank !

30. **Answer : D**

Relaxation at every vertex is as follows:
Note that the next picked vertex corresponds to the next row in Table

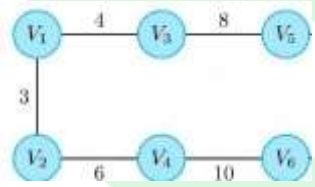


31. **Answer: B**

We observe a pattern in the weight of MST being formed
For n = 3 (1 + 2 + 3) + (1)
For n = 4 (1 + 2 + 3 + 4) + (1 + 2)
For n = 5 (1 + 2 + 3 + 4 + 5) + (1 + 2 + 3)
These can be obtained by drawing graphs for these graphs.
∴ Total weight of MST is:

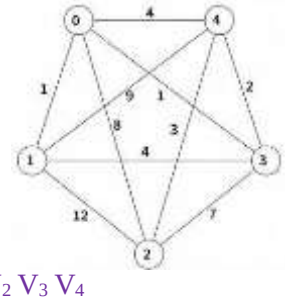
$$\sum_{i-1}^{n} i + \sum_{i-1}^{n-2} i = n^2 - n + 1$$

32. **Answer : C**



⇒8+4+3+6+10 = 31

33. **Answer : D**



V_0 V_1 V_2 V_3 V_4
For minimum cost, we go to The edges of the spanning tree are:
0−1, 1−3, 3−4, 4−2.
∴ Total cost = 10

34. **Answer : D**

In Option D b-c with weight, 4 is added before a-c with weight 3 is added. In Kruskal's algorithm, edges should be added in non-decreasing order of weight.

35. **Answer : B**

Statement P: Always finds a negative weighted cycle, if one exists.
This statement is incorrect. Bellman-Ford can only detect a negative cycle if it is reachable from the source node. If a graph contains a negative cycle in a "disconnected" component that the source cannot get to, the algorithm will finish its iterations without ever seeing the decreasing path weights that signal a cycle.Statement Q: Finds whether any negative weighted cycle is reachable from the source.
This statement is correct. This is the specific utility of the Nth relaxation step in Bellman-Ford. If, after relaxing all edges V-1 times, we can still relax an edge one more time, it proves that a shorter path exists—which is only possible if there is a negative cycle on a path starting from the source

36. **Answer : C**

Prim's algorithm starts with from any vertex and expands the MST by adding one vertex in each step which is close to the Intermediate MST(made till previous step).
Therefore, correct answer would be (C).
(A) : (d,f) is chosen but neither d nor f vertices are part of the previous MST(MST made till previous step).
(B) : (g,h) is chosen but neither g or h vertices are part of the previous MST(MST made till previous step).
(D): (f,c) is chosen but at that point (f,d) is close to the intermediate MST

37. **Answer : D**

Misconception : (these are statements most of people think):
Dijkstra Doesn't work on Negative cycles.
Dijkstra will work on if G has Negative Edges and Non-Negative cycles.
But above are not true.
correct statements are:
1) Dijkstra Doesn't work on Negative cycles.
Hence Dijkstra won't detect negative edge cycles.
2) Dijkstra may or may not work on Negative Edge cycles it depends on the fact if the negative weight cycle is reachable from the source.

Check Out Our Gate Courses & Secure Your Dream Rank !

38. Answer : C

If weight (u, v) < 12, then the min. weight of (s, v) = weight of (s, u) + weight of (u, v) = 53 + (<12) will be less than 65.

39. Answer : D

D is the false statement.

A minimum spanning tree must have the edge with the smallest weight (In Kruskal's algorithm we start from the smallest weight edge). So, C is TRUE.

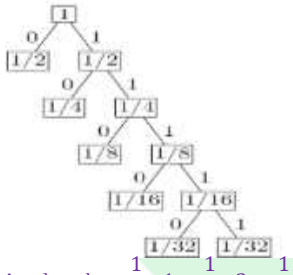
If e is not part of a minimum spanning tree, then all edges which are part of a cycle with e, must have weight <=e, as otherwise we can interchange that edge with e and get another minimum spanning tree of lower weight. So, B and A are also TRUE.

D is false because, suppose a cycle is there with all edges having the same minimum weight w. Now, any one of them can be avoided in any minimum spanning tree.

40. Answer : D

Letter	Probability
a	1/2
b	1/4
c	1/8
d	1/16
e	1/32
f	1/32

Prefix Code	Code Length
$a = 0$	1
$b = 10$	2
$c = 110$	3
$d = 1110$	4
$e = 11110$	5
$f = 11111$	5



Avg length $= \frac{1}{2}\times 1 + \frac{1}{4}\times 2 + \frac{1}{8}\times 3 + \frac{1}{16}\times 4 + \frac{1}{32}\times 5 + \frac{1}{32}\times 5$

$= \frac{16+16+12+8+5+5}{32}$

$= 1.9375$

41 Answer :A

Optimal Huffman code lengths for probabilities that are exact powers of 1/2 are calculated as -log2(p).

a: -log2(1/2) = 1 bit

b: -log2(1/4) = 2 bits

c: -log2(1/8) = 3 bits

d: -log2(1/16) = 4 bits

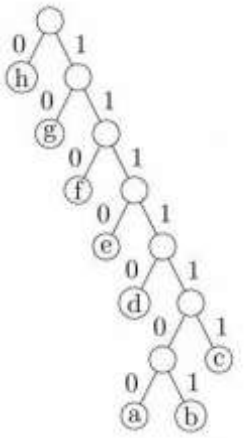
e: -log2(1/32) = 5 bits

f: -log2(1/32) = 5 bits

Option A is the only choice with code lengths of 1, 2, 3, 4, 5, 5 respectively (0, 10, 110, 1110, 11110, 11111). the correct answer is A.

42. Answer A:

Huffman's tree is as follows. The two least frequent characters are taken as the children of a newly made node and the frequency of the newly made node is made equal to the sum of those two child nodes. Then the same procedure is repeated till all nodes are finished.



110111100111010 = 110111100111010 =fdheg

43. Answer : D

In Kruskal's algo the edges are added in non decreasing order of their weight. But in Option D edge d-e with weight 3 is added before edge d-c with weight 2. Hence, option D is wrong option.

Q) Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j iff either j = i + 1 or j = 3i. The minimum number of edges in a path in G from vertex 1 to vertex 100 is:

44. Answer : A

According to the **Cut Property** of Minimum Spanning Trees (MST): If the vertices of a graph are edge with the minimum weight that crosses this cut (i.e., connects a vertex in one set to a vertex in the other) must be included in the Minimum Spanning Tree. Here, [X, Y] is a partition (cut) of the graph's vertices. The edge e is given as the minimum weight edge crossing this cut. Since all edge weights are positive and distinct, the graph has a unique MST. Therefore, this minimum weight edge e must strictly belong to the minimum weighted spanning tree of G.

It does not necessarily belong to the shortest path between s and t, because the shortest path might cross the cut through a different, slightly heavier edge if the paths leading to that edge are significantly shorter.

Therefore, the correct answer is **A**.

45. **Answer : C**

Solution:

Based on the given adjacency matrix:

- Diagonal elements are 0: There are no self-loops.
- Non-diagonal elements are 1: There is an edge between every pair of distinct vertices, and every edge has a weight of 1.

This describes a **complete graph** (K_n) where all edge weights are equal to 1.

A spanning tree for a graph with n vertices must contain exactly n-1 edges. Since every edge in this graph has a weight of 1, the cost of *any* spanning tree will be exactly (n- 1)*1 = n-1.

Because all edges have the exact same weight, any valid tree that connects all n nodes is a Minimum Spanning Tree. According to Cayley's formula, a complete graph with n vertices has n^{n-2} distinct spanning trees. As long as n > 2, there will be multiple distinct spanning trees.

Therefore, there are multiple distinct MSTs, and every single one of them has a cost of n-1.

The correct answer is **C**.

46 **Answer : D**

Explanation: When using a Binary Heap for Dijkstra's algorithm:

1. Extract Min: This is done once for each vertex. Total: |V| log |V|.
2. Decrease Key: In the worst case, this is done for every edge when relaxing them. Total: |E| log |V|.

Combining these, you get O((|V| + |E|) log |V|).

Note: Option B, O(|E| + |V| log |V|), is only possible if you use a Fibonacci Heap, which wasn't what the question asked for.

47. **Answer : B**

Edge set consists of edges from i to j using either

1. j = i + 1
2. j = 3i.

Second option will help us reach from 1 to 100 rapidly. The trick to solve this question is to think in reverse way. Instead of finding a path from 1 to 100, try to find a path from 100 to 1.

1->3->9->10->11->33->99->100. The answer is option (B).

48. **Answer : A**

Task	T_7	T_2	T_9	T_4	T_5	T_3	T_6	T_8	T_1
Deadline	2	2	3	3	4	5	5	7	7

We are given 9 tasks T1, T2... T9. The execution of each task requires one unit of time. We can execute one task at a time. Each task Ti has a profit Pi and a deadline di Profit Pi is earned if the task is completed before the end of the d^{ith} unit of time.

The best task would be the one with maximum profit. Thus we can sort the tasks based on deadline and then profit as follows:

$T_1\ T_2\ T_3\ T_4\ T_5\ T_6\ T_7$

$0-\ 1-2-3-\ 4-5-6-7$

maximum profit will not include those of T4 and T6 and will be

=15 + 20 + 30 + 18 + 16 + 23 + 25 = 147

49. **Answer : D**

The best task would be the one with maximum profit. Thus we can sort the tasks based on deadline and then profit as follows:

Task	T_7	T_2	T_9	T_4	T_5	T_3	T_6	T_8	T_1
Deadline	2	2	3	3	4	5	5	7	7

$0-\overset{T_7}{1}-\overset{T_2}{2}-\overset{T_9}{3}-\overset{T_4}{4}-\overset{T_5}{5}-\overset{T_6}{6}-\overset{T_1}{7}$

maximum profit will not include those of T4 and T6 .

4 DYNAMIC PROGRAMMING

GATE CSE – 2026 SET-2 MCQ (2-MARKS)

Q1. Consider a table T, where the elements T[i][j], 0 <= i, j <= n, represent the cost of the optimal solutions of different subproblems of a problem that is being solved using a dynamic programming algorithm. The recursive formulation to compute the table entries is as follows:

$T[0][k] = T[k][0] = 1 \quad$ for $k = 0, 1, 2, \ldots, n$

$T[i][j] = 2T[i-1][j]+3T[i][j-1]$ for $1 \le i, j \le n$

Consider the following two algorithms to compute entries of T. Assume that for both the algorithms, for all $0 \le i, j \le n$, T[i][j] has been initialized to 1.

Algorithm B_1:

```
For i = 1 to n
   For j = 1 to n
      T[i][j] = 2T[i − 1][j] + 3T[i][j − 1]
```

Algorithm B_2:

```
For s = 2 to 2n
   For i = 1 to n
      For j = 1 to n
         If (i + j == s)
         T[i][j] = 2T[i − 1][j] + 3T[i][j − 1]
```

Algorithm B_k, $k \in \{1, 2\}$, is said to be correct if and only if it calculates the correct values of T[i][j], for all $0 \le i, j \le n$, as per the recursive formulation, at the end of the execution of the algorithm B_k.

Which one of the following statements is true?

a) Both algorithms B1 and B2 are correct
b) Algorithm B1 is correct, but algorithm B2 is incorrect
c) Algorithm B2 is correct, but algorithm B1 is incorrect
d) Both algorithms B1 and B2 are incorrect

GATE CSE – 2021 SET-1 MSQ (2-MARKS)

Q2. Define Rn to be the maximum amount earned by cutting a rod of length n meters into one or more pieces of integer length and selling them. For i > 0, let p[i] denote the selling prices of a rod whose length is i meters. Consider the array of prices :

p[1] = 1, p[2] = 5, p[3] = 8, p[4] = 9, p[5] = 10, p[6] = 17, p[7] = 18

Which of the following statements is/are correct about R 7 ?

a) R_7 cannot be achieved by a solution consisting of three pieces.
b) $R_7 = 19$
c) $R_7 = 18$
d) R_7 is achieved by three different solutions.

GATE CSE – 2018 MCQ (2-MARKS)

Q3. Assume that multiplying a matrix G 1 of dimension p×q with another matrix G 2 of dimension q × r requires scalar multiplications. Computing the product of matrices G1 G2 G3 ……Gn can be done by parenthesizing in different ways. Define Gi Gi+1 as an explicitly computed pair for a given parenthesization if they are directly multiplied. For example, in the matrix multiplication chain G1 G2 G3 G4 G5 G6 using parenthesization

(G1(G2G3)) (G4(G5G6)).G2 G3 and G5 G6 are the only explicitly computed pairs. Consider a matrix multiplication chain F1 F2 F3 F4 F5 , where matrices F1 F2 F3 F4 F5 are of dimensions 2 × 25, 25 × 3, 3 × 16, 16 × 1 and 1×1000, respectively.In the parenthesization of F1 F2 F3 F4 F5 that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are:

a) F1 F2 and F3 F4
b) F2 F3 only
c) F3 F4 only
d) F1,F2 and F4

GATE CSE – 2018 NAT (2-MARKS)

Q4. Consider the weights and values of items listed below. Note that there is only one unit of each item. The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt}. A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy}.
The value of V_{opt} - V_{greedy} is _____

GATE CSE – 2016 SET-2 NAT (2-MARKS)

Q5. Let A 1 , A 2 , A 3 , and A 4 be four matrices of dimensions 10 × 5, 5 × 20, 20 × 10, and 10 × 5, respectively. The minimum number of scalar multiplications required to find the product A1A2A3A4 using the basic matrix multiplication method is __.

GATE CSE – 2014 SET-2 NAT (2-MARKS)

Q6 Consider two strings A = {"qpqrr"} and B = "pqprqrp". Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the number of such longest common subsequences between A and B. Then x+10y = _____.

a) 248000 b)44000
b) 19000 d)25000

GATE CSE – 2011 MCQ (1-MARKS)

Q7. An algorithm to find the length of the longest monotonically increasing sequence of numbers in an array A[0: n - 1] is given below. Let L; denote the length of the longest monotonically increasing sequence starting at index i in the array. Initialize $L_{n-1} = 1$. For all i such that $0 \leq i \leq n-2$

$$L_i = \begin{cases} 1 + L_{i+1} & \text{if } A[i] < A[i+1] \\ 1 & \text{otherwise} \end{cases}$$

Finally the length of the longest monotonically increasing sequence is Max (L_0 , L_1 ,….,L_{n-1}). Which of the following statements is TRUE?

a) The algorithm uses dynamic programming paradigm
b) The algorithm has a linear complexity and uses branch and bound paradigm
c) The algorithm has a non-linear polynomial complexity and uses branch and bound paradigm.
d) The algorithm uses divide and conquer paradigm.

GATE CSE – 2011 MCQ (2-MARKS)

Q8. Four matrices M1, M2, M3 and M4 of dimensions pxq, qxr, rxs and sxt respectively can be multiplied is several ways with different number of total scalar multiplications. For example, when multiplied as ((M1 X M2) X (M3 X M4)), the total number of multiplications is pqr + rst + prt. If p = 10, q = 100, r = 20, s = 5 and t = 80, then the number of scalar multiplications needed is:

GATE CSE – 2010 MCQ (2-MARKS)

Q9. The weight of a sequence a_0 , a_1 , …, a_{n-1} of real numbers is defined as $a_0 + a_1 / 2 + \ldots + a_{n-1} / 2^{n-1}$. A subsequence of a sequence is obtained by deleting some elements from the sequence, keeping the order of the remaining elements the same. Let X denote the maximum possible weight of a subsequence of a_0 , a_1 ,..., a_{n-1} . Then X is equal to

a) max(Y, a_0 + Y)

b) max(Y, a_0 + Y/2)

c) max(Y, a_0 + 2Y)

d) a_0 + Y/2

GATE CSE – 2009 MCQ (2-MARKS)

Q10. A sub-sequence of a given sequence is just the given sequence with some elements (possibly none or all) left out. We are given two sequences X[m] and Y[n] of lengths m and n, respectively, with indexes of X and Y starting from 0.

We wish to find the length of the longest common sub-sequence (LCS) of x[m] and Y[n] as l(m,n), where an incomplete recursive definition for the function l(i,j) to compute the length of the LCS of x[m] and Y[n] is given below:

$$l(i,j) = \begin{cases} 0, \text{if either } i = 0 \text{ or } j = 0 \\ \text{expr1}, \text{if } i,j > 0 \text{ and } x[i-1] = Y[j-1] \\ \text{expr2}, \text{if } i,j > 0 \text{ and } x[i-1] \neq Y[j-1] \end{cases}$$

Which one of the following options is correct?

A) expr1 = l(i-1, j) + 1

B) expr1 = l(i, j-1)

C) expr2 = max (l(i-1, j), l(i,j-1))

D) expr2 = max(l(i-1, j-1), l(i,j))

GATE CSE – 2008 MCQ (2-MARKS)

Q11. The subset-sum problem is defined as follows. Given a set of n positive integers, S={a_1,a_2,...,a_n} and positive integer W, is there a subset of S whose elements sum to W? A dynamic program for solving this problem uses a 2-dimensional Boolean array, X, with n rows and W+1 columns. X[i,j], $1 \le i \le n$, $0 \le j \le W$, is TRUE if and only if there is a subset of {a_1,a_2,...,a_i }whose elements sum to j.

Which entry of the array X, if TRUE, implies that there is a subset whose elements sum to W?

a) X[1,W]

b) X[n,0]

c) X[n,W]

d) X[n-1,n]

GATE CSE – 2008 MCQ (2-MARKS)

Q12. The subset-sum problem is defined as follows. Given a set of n positive integers, S={a_1,a_2,...,a_n } and positive integer W, is there a subset of S whose elements sum to W? A dynamic program for solving this problem uses a 2-dimensional Boolean array, X, with n rows and W+1 columns. X[i,j], 1≤i≤n, 0≤j≤W, is TRUE if and only if there is a subset of {a_1,a_2,...,a_i }whose elements sum to j.Which of the following is valid for 2≤i≤n and a_i≤j≤W?

a) $X[i,j]=X[i-1,j] \vee X[i,j-a_i]$

b) $X[i,j]=X[i-1,j] \vee X[i-1,j-a_i]$

c) $X[i,j]=X[i-1,j] \wedge X[i,j-a_i]$

d) $X[i,j]=X[i-1,j] \wedge X[i-1,j-a_i]$

ANSWER KEY

1.(a)	**2.(c,d)**	**3.(c)**
4.(16)	**5.(1500)**	**6.(34)**
7.(a)	**8.(c)**	**9.(b)**
10.(c)	**11.(c)**	**12.(b)**

HINTS & SOLUTION

1. Answer : A

In dynamic programming, an algorithm is correct if it calculates T[i][j] only after its dependencies T[i−1][j] and T[i][j−1] have been computed.

Algorithm B_1 (Row-major):

When computing T[i][j], T[i−1][j] (above) is already computed in the previous row, and T[i][j−1] (left) is already computed in the current row.Correct statement.

Algorithm B_2 (Diagonal / Sum-based):

For cell (i, j), both dependencies (i−1, j) and (i, j−1) have sum s−1, where s = i + j. Since s increases sequentially, all required cells are computed earlier. Correct statement.

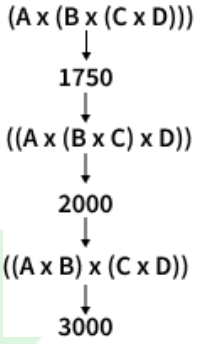
2. Answer : C,D.

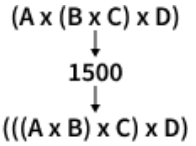
Check splits for n = 7: max of p, p+p, p+p, p+p, and multi-piece combos. p = 18; p+p = 1+17 = 18; p+p = 5+10 = 15; p+p = 8+9 = 17. Three-piece 2+2+3 gives 5+5+8 = 18, so there are three optimal ways: 7; 1+6; 2+2+3. Hence R7 = 18 and multiple optimal solutions exist.

→R7 = 18, and it can be achieved in three ways;

3. Answer :C

No. of ways of multiplying the chain of matrices
Where m = no. of multiplications (not matrices)
$^6C_3$ / (3+1)





Total number of scalar multiplications are 48 + 75 + 50 + 2000 = 2173 and optimal parenthesis is ((F1(F2(F3 F4))) F5). As concluded, F3, F4 are explicitly computed pairs.

4. Answer : 16

Calculate the Value-to-Weight Ratios
Item 1: 60 / 10 = 6
Item 2: 28 / 7 = 4
Item 3: 20 / 4 = 5
Item 4: 24 / 2 = 12
Step 2: Find the Optimal Value (Vopt)
Pick Item 1 alone: Weight = 10 kg, Value = 60
Pick Items 2 & 3: Weight = 11 kg, Value = 48
Pick Items 2 & 4: Weight = 9 kg, Value = 52
Pick Items 3 & 4: Weight = 6 kg, Value = 44
The maximum possible value is 60. Therefore, Vopt = 60.
Step 3: Find the Greedy Value (Vgreedy) Sorted order:
Item 4 (12), Item 1 (6), Item 3 (5), Item 2 (4).
Try Item 4: Weight is 2 kg. Fits. (Remaining: 9 kg, Value: 24)
Try Item 1: Weight is 10 kg. Does not fit. Skip.
Try Item 3: Weight is 4 kg. Fits. (Remaining: 5 kg, Value: 44)
Try Item 2: Weight is 7 kg. Does not fit. Skip.
Total greedy value is 44. Therefore, Vgreedy = 44.
Step 4: Calculate the Difference Vopt - Vgreedy = 60 - 44 = 16. The value of Vopt - Vgreedy is 16.

5. Answer : 1500

The answer is 1500.

Matrix Parenthesizing (A_1 (A_2A_3)A_4) :

Check my solution below, using dynamic programming.

A1	A2	A3	A4
10 * 5	5 * 20	20 * 10	10 * 5

- $A_{12} = 10 \times 5 \times 20 = 1000$
- $A_{23} = 5 \times 20 \times 10 = 1000$
- $A_{34} = 20 \times 10 \times 5 = 1000$

$$A_{13} = \min\begin{cases} A_{12} + A_{33} + 5 \times 20 \times 10 = 2000 \\ A_{11} + A_{23} + 10 \times 5 \times 10 = 1500 \end{cases}$$

$$A_{24} = \min\begin{cases} A_{23} + A_{44} + 5 \times 10 \times 5 = 1250 \\ A_{22} + A_{34} + 5 \times 20 \times 5 = 1500 \\ A_{11} + A_{24} + 10 \times 5 \times 5 = 1500 \end{cases}$$

$$A_{14} = \min\begin{cases} A_{12} + A_{34} + 10 \times 20 \times 5 \geqslant 2000 \\ A_{13} + A_{44} + 10 \times 20 \times 5 = 2000 \end{cases}$$

6. Answer is 34.

The LCS is of length 4. There are 3 LCS of length 4 "qprr", "pqrr" and qpqr A subsequence is a sequence that can be derived from another sequence by selecting zero or more elements from it, without changing the order of the remaining elements. Subsequence need not be contiguous. Since the length of given strings A = "qpqrr" and B = "pqprqrp" are very small, we don't need to build a 5x7 matrix and solve it using dynamic programming. Rather we can solve it manually just by brute force. We will first check whether there exist a subsequence of length 5 since min_length(A,B) = 5. Since there is no subsequence , we will now check for length 4. "qprr", "pqrr" and "qpqr" are common in both strings. X = 4 and Y = 3 .X + 10Y = 34 .

7. Answer:A

The algorithm computes the length of the longest increasing sequence starting from each index by building the solution for smaller subproblems (from the end towards the beginning) and reusing previously computed results, which is the essence of dynamic programming. Each computation at index i depends on the result for index i+1, and all results are stored and reused.

8. Answer : C

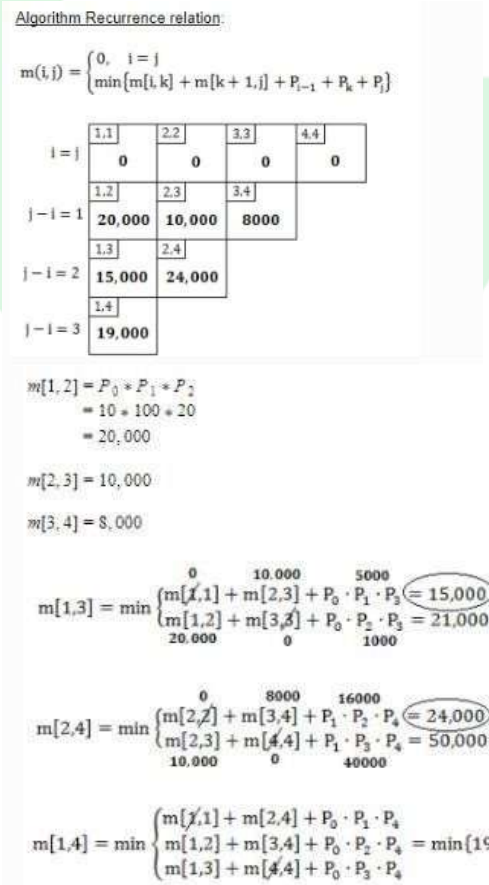
$$M1 * M2 * M3 * M4$$

$$\underbrace{10}_{P_0} \times \underbrace{100}_{P_8}\ \underbrace{100}_{P_8} \times \underbrace{20}_{P_8}\ \underbrace{20}_{P_3} \times \underbrace{5}_{P_8}\ \underbrace{5}_{P_8} \times \underbrace{80}_{P_4}$$

$P_0 = 10$
$P_1 = 100$
$P_2 = 20$
$P_3 = 5$
$P_4 = 80$

Algorithm Recurrence relation:

$$m(i,j) = \begin{cases} 0, i = j \\ \min\{m[i,k] + m[k+1,j] + P_{i-1} + P_k + P_1\} \end{cases}$$



$m[1,2] = P_0 * P_1 * P_2$
$= 10 * 100 * 20$
$= 20,000$

$m[2,3] = 10,000$

$m[3,4] = 8,000$

$$m[1,3] = \min\begin{cases} \underset{20,000}{m[1,1]} + \underset{10,000}{m[2,3]} + P_0 \cdot P_1 \cdot P_3 = 15,000 \\ \underset{0}{m[1,2]} + \underset{0}{m[3,3]} + P_0 \cdot P_2 \cdot P_3 = 21,000 \end{cases} \text{minimum}$$

$$m[2,4] = \min\begin{cases} \underset{10,000}{m[2,2]} + \underset{8000}{m[3,4]} + P_1 \cdot P_2 \cdot P_4 = 24,000 \\ \underset{0}{m[2,3]} + \underset{40000}{m[4,4]} + P_1 \cdot P_3 \cdot P_4 = 50,000 \end{cases} \text{minimum}$$

$$m[1,4] = \min\begin{cases} m[1,1] + m[2,4] + P_0 \cdot P_1 \cdot P_4 \\ m[1,2] + m[3,4] + P_0 \cdot P_2 \cdot P_4 \\ m[1,3] + m[4,4] + P_0 \cdot P_3 \cdot P_4 \end{cases} = \min\{19,000\}$$

9. Answer : B

Do not include a0 in the subsequence: then the maximum possible weight will be equal to maximum possible weight of a subsequence of {a1, a2, …..an} which is represented by Y. Include a0: then maximum possible weight will be equal to a0 + (each number picked in Y will get divided by 2) a0 + Y/2. Here you can note that Y will itself pick optimal subsequence to

maximize the weight. Final answer will be Max (Case1, Case2) i.e. Max (Y, a 0 + Y/2). Hence (B) is correct.

10. Answer :C

When the currently compared elements doesn't match, we have two possibilities for the LCS, one including X[i] but not Y[j] and other including Y[j] but not X[i].

```
/* Returns length of LCS for X[0..m-1], Y[0..n-1] */
int lcs( char *X, char *Y, int m, int n )
{
  if (m == 0 || n == 0)
    return 0;
}

      if (X[m-1] == Y[n-1])
   return 1 + lcs(X, Y, m-1, n-1); else
  return max(lcs(X, Y, m, n-1), lcs(X, Y, m-1, n))
```

11 Answer: C.

If LAST ROW and LAST COLUMN entry is 1, then there exists a subset whose elements sum to w.

12 Answer : B

X[i][j] = X[i - 1][j] U X[i - 1][j - a_i]
We calculate the value of X[i][j] as if
we include the value a_i in the subset X[i - 1][j - a_i] or
we do not include the value in the subset X[i - 1][j].

5 GRAPH TECHNIQUES

GATE CSE – 2026 SET-1 MSQ (2-MARKS)

Q1. Consider the following pseudocode for depth-first search (DFS) algorithm which takes a directed graph $G(V,E)$ as input, where $d[v]$ and $f[v]$ are the discovery time and finishing time, respectively, of the vertex $v \in V$.

```
DFS(G) :
  unmark all v ∈ V
  t ← 0
  for each v ∈ V
    if v is unmarked
      t ← Explore
(G, v, t )
    end if
  end for
```

```
Explore ( G, v, t ):
    mark v
    t ← t + 1
    d[v] ← t
    for each (v, w) ∈ E
      if w is unmarked
        t ← Explore (G, w, t)
    end if
    end for
    t ← t + 1
    f[v] ← t
    return t
```

Suppose that the input directed graph $G(V,E)$ is a directed acyclic graph (DAG). For an edge $(u,v) \in E$, which of the following options will NEVER be correct?

a) $d[u]<d[v]<f[v]<f[u]$
b) $d[v]<d[u]<f[u]<f[v]$
c) $d[v]<f[v]<d[u]<f[u]$
d) $d[u]<d[v]<f[u]<f[v]$

GATE CSE – 2026 SET-1 MSQ (2-MARKS)

Q2. An undirected, unweighted, simple graph $G(V, E)$ is said to be 2 -colorable if there exists a function $c: V \rightarrow \{0,1\}$ such that for every $(u, v) \in E, c(u) \neq c(v)$.

Which of the following statements about 2-colorable graphs is/are true?

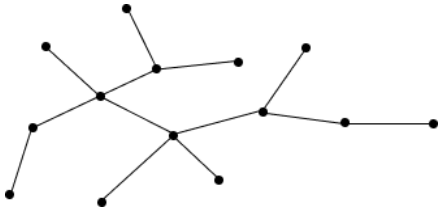
a) If G is 2-colorable, then G may contain cycles of odd length
b) If G is 2-colorable, then G may contain cycles of even length
c) An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(|V|+|E|)$, if G is represented as an adjacency list
d) An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(|E|\log|V|)$, if G is represented as an adjacency list.

GATE CSE – 2025 SET-2 NAT (2-MARKS)

Q3. Consider the following algorithm someAlgo that takes an undirected graph G as input.
someAlgo(G) :
1. Let v be any vertex in G. Run BFS on G starting at v. Let u be a vertex in G at maximum distance from v as given by the BFS.

2. Run BFS on G again with u as the starting vertex. Let z be the vertex at maximum distance from u as given by the BFS.
3. Output the distance between u and z in G.

The output of someAlgo(T) for the tree shown in the given figure is _____. (Answer in integer)



Check Out Our Gate Courses & Secure Your Dream Rank !

GATE CSE – 2025 SET-2 MSQ (1-MARKS)

Q4. Which of the following statements regarding Breadth First Search (BFS) and Depth First Search (DFS) on an undirected simple graph G is/are TRUE?

a) A DFS tree of G is a Shortest Path tree of G.
b) Every non-tree edge of G with respect to a DFS tree is a forward/back edge.
c) If (u, v) is a non-tree edge of G with respect to a BFS tree, then the distances from the source vertex s to u and v in the BFS tree are within ±1 of each other.
d) Both BFS and DFS can be used to find the connected components of G.

GATE CSE – 2024 SET-1 MCQ (2-MARKS)

Q5. Let G be a directed graph and T a depth first search (DFS) spanning tree in G that is rooted at a vertex v. Suppose T is also a breadth first search (BFS) tree in G, rooted at v. Which of the following statements is/are TRUE for every such graph G and tree T?

a) The only edges in G are the edges in T
b) There are no cross-edges in G with respect to the tree T
c) There are no back-edges in G with respect to the tree T
d) There are no forward-edges in G with respect to the tree T

GATE CSE – 2024 SET-1 NAT (2-MARKS)

Q6. The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is

GATE CSE – 2023 NAT (2-MARKS)

Q7. Let U = {1, 2, 3}. Let 2^U denote the powerset of U. Consider an undirected graph G whose vertex set is 2^U. For any A, B in 2^U, (A, B) is an edge in G if and only if (i) A != B, and (ii) either A ⊆ B or B ⊆ A. For any vertex A in G, the set of all possible orderings in which the vertices of G can be visited in a Breadth First Search (BFS) starting from A is denoted by B(A).
If Φ denotes the empty set, then the cardinality of B(Φ) is __.

GATE CSE – 2021 SET-1 MCQ (2-MARKS)

Q8. Let $G = (V, E)$ be an undirected unweighted connected graph. The diameter of G is defined as: diam(G)=max{" the length of shortest path between " u" and " v}
Let M be the adjacency matrix of G. Define graph G_2 on the same set of vertices with adjacency matrix N, where

$$N_{ij} = \begin{cases} 1 & \text{if } M_{ij} > 0 \text{ or } P_{ij} > 0 \\ 0 & \text{otherwise} \end{cases}$$

where $P = M^2$.
Which one of the following statements is true?

a) ⌈ diam (G) / 2⌉ < diam (G_2) < diam(G)
b) diam(G_2) = diam(G)
c) diam(G) = diam(G_2) <= 2 diam(G)
d) diam(G_2) <= ⌈ diam (G) / 2 ⌉

GATE CSE – 2018 MCQ (2-MARKS)

Q9. Match Let G be a simple undirected graph. Let T_D be a depth first search tree of G. Let T_B be a breadth first search tree of G. Consider the following statements.
(I) No edge of G is a cross edge with respect to T_D. (A cross edge in G is between two nodes neither of which is an ancestor of the other in T_D.)
(II) For every edge (u,v) of G, if u is at depth i and v is at depth j in T_B, then |i - j| = 1.
Which of the statements above must necessarily be true?

a) I only
b) II only
c) Both I and II
d) Neither I nor II

GATE CSE – 2016 SET-2 NAT (2-MARKS)

Q10. Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n-th vertex in this BFS traversal, then the maximum possible value of n is___.

GATE CSE – 2016 SET-2 MCQ (2-MARKS)

Q11. In an adjacency list representation of an undirected simple graph G = (V,E), each edge (u,v) has two adjacency list entries: [v] in the adjacency list of u, and [u] in the adjacency list of v. These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If |E|=m and |V|=n, and the memory size is not a constraint, what is the Time Complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

a) $\theta(n^2)$
b) $\theta(n+m)$
c) $\theta(m^2)$
d) $\theta(n^4)$

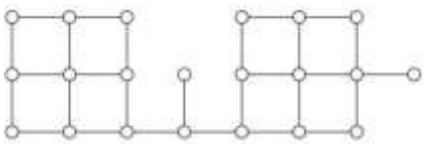
GATE CSE – 2015 SET-1 MCQ (1-MARKS)

Q12. Let G= (V,E) be a simple undirected graph, and s be a particular vertex in it called the source. For x ∈ V, let d (x) denote the shortest distance in G from s to x. A breadth first search (BFS) is performed starting at s. Let T be the resultant BFS tree. If (u, v) is an edge of G that is not in T, then which one of the following CANNOT be the value of d (u) - d (v)?

a) -1
b) 0
c) 1
d) 2

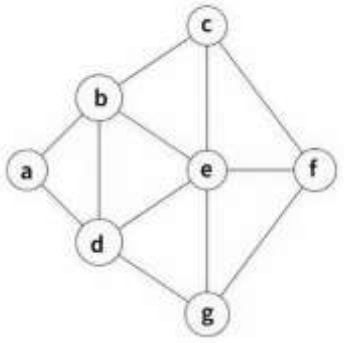
GATE CSE – 2014 SET-3 NAT (1-MARKS)

Q13. Suppose depth first search is executed on the graph below starting at some unknown vertex. Assume that a recursive call to visit a vertex is made only after first checking that the vertex has not been visited earlier. Then the maximum possible recursion depth (including the initial call) is _____.



GATE IT – 2008 MCQ (2-MARKS)

Q14. Consider the following sequence of nodes for the undirected graph given below:
1.a b e f d g c
2.a b e f c g d
3.a d g e b c f
4.a d b c g e f
A Depth First Search (DFS) is started at node a. The nodes are listed in the order they are first visited. Which of the above is/are possible output(s)?



a) 1 and 3 only
b) 2 and 3 only
c) 2,3 and 4 only
d) 1,2 and 3 only

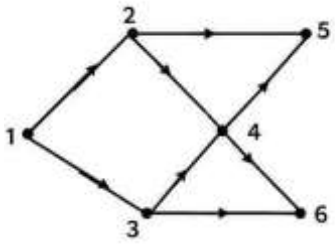
GATE IT – 2007 MCQ (1-MARKS)

Q15. A depth-first search is performed on a directed acyclic graph. Let d[u] denote the time at which vertex u is visited for the first time and f[u] the time at which the DFS call to the vertex u terminates. Which of the following statements is always TRUE for all edges (u, v) in the graph ?

a) d[u] < d[v]
b) d[u] < f[v]
c) f[u] < f[v]
d) f[u] > f[v]

GATE CSE – 2007 MCQ (1-MARKS)

Q16. Consider the DAG with V = {1,2,3,4,5,6}, shown below



Which of the following is NOT a topological ordering?

a) 1 2 3 4 5 6
b) 1 3 2 4 5 6
c) 1 3 2 4 6 5
d) 3 2 4 1 6 5

GATE IT – 2006 MCQ (1-MARKS)

Q17. Consider the depth-first-search of an undirected graph with 3 vertices P, Q, and R. Let discovery time d(u) represent the time instant when the vertex u is first visited, and finish time f(u) represent the time instant when the vertex u is last visited. Given that
d(P)=5 units f(P)=12 units
d(Q)=6 units f(Q)=10 units
d(R)=14 unitf(R)=18 units
Which one of the following statements is TRUE about the graph?

a) There is only one connected component
b) There are two connected components, and P and R are connected
c) There are two connected components, and Q and R are connected
d) There are two connected components, and P and Q are connected

GATE CSE – 2006 MCQ (1-MARKS)

Q18. Let T be a depth first search tree in a undirected graph G. Vertices u and v are leaves of this tree T. The degrees of both u and v in G are at least 2. Which one of the following statements is true?

a) There must exist a vertex w adjacent to both u and v in G
b) There must exist a vertex w whose removal disconnects u and v in G
c) There must be exist a cycle in G containing u and v
d) There must exist a cycle in G containing u and all its neighbours in G

ANSWER KEY

1.(b,d)	**2.(b,c)**	**3.(6)**
4.(b,c,d)	**5.(d)**	**6.(60)**
7.(5040)	**8.(d)**	**9.(a)**
10.(31)	**11.(b)**	**12.(d)**
13.(19)	**14.(b)**	**15.(d)**
16.(d)	**17.(d)**	**18.(d)**

HINTS & SOLUTIONS

1. Answer : B,D

A. This means:
v is discovered after u
v finishes before u
So v is inside u's DFS subtree.
This corresponds to a tree edge or forward edge.

B. This means:
u is discovered after v
u finishes before v
So u is inside v's DFS subtree.If edge (u, v) exists, then it is a back edge (from descendant to ancestor).Back edges create cycles.
Since graph is a DAG → back edges cannot exist.

C. v finishes before u is discovered.
This is a cross edge.
Cross edges are allowed in DAG.

D. This represents interlapping intervals. According to the Parenthesis Theorem of DFS, for any two vertices u and v, their discovery/finishing intervals [d[u], f[u]] and [d[v], f[v]] must either be entirely disjoint or one must be entirely contained within the other.
Intervals can never overlap in the manner shown in Option (D) in *any* graph, whether it is a DAG or not.

2. Answer : B,C

A graph is 2-colorable if and only if it is bipartite.
Key fact:
A graph is bipartite ⇔ it contains no odd-length cycles.
Only an even length cycle is allowed.
We can test bipartiteness using:

- BFS, or
- DFS

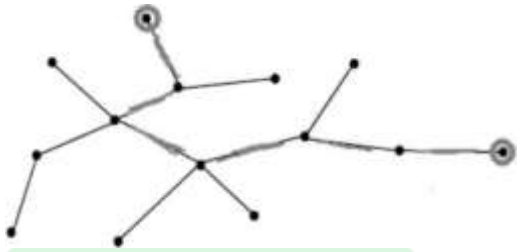
During traversal, assign alternate colors level by level.
If we ever find an edge connecting two same-colored vertices → not bipartite.
Time complexity using adjacency list:

- Each vertex visited once → O(|V|)
- Each edge checked once → O(|E|)

Total time:
$\Theta(|V| + |E|)$
This is optimal because we must examine all vertices and edges.
So (C) is TRUE.
(D) An optimal algorithm runs in $\Theta(|E| \log |V|)$
False.
No sorting or priority queue is needed.
The problem does not require log factor.
Linear time is sufficient and optimal.
So (D) is FALSE.

3. Answer : 6

The someAlgo (G) is simply calculating the diameter of the tree.
"Diameter of a graph is the longest of all the shortest paths between any two vertices"



4. Answer : B,C,D

A. False: A DFS tree does not necessarily represent the shortest paths from the source vertex. DFS explores as far as possible along each branch before backtracking, which means it does not guarantee the shortest path in terms of the number of edges.
B. True: In an undirected graph, every non-tree edge with respect to a DFS tree is either a back edge or a forward edge because in an undirected graph, there are no cross edges in the context of DFS.
C. True: In a BFS tree, the levels of the vertices represent their distances from the source vertex. If (u,v) is a non-tree edge, then u and v must be at the same level or at levels that differ by one because BFS explores all vertices at the current level before moving to the next level.
D. True: Both BFS and DFS can traverse the graph and identify all vertices in a connected component. By repeating the traversal for unvisited vertices, both algorithms can find all connected components of the graph.

5. **Answer : (D)** If a tree T is both a depth first search (DFS) and breadth first search (BFS) spanning tree rooted at the same vertex in a directed graph G, then:
• In a BFS tree, all nodes at depth d+1 are discovered from nodes at depth d, which means there can't be edges from a node to any descendant beyond the next level.
• In DFS, a forward edge goes from a node to a descendant that is not its child.
• If T satisfies both DFS and BFS properties, forward edges cannot exist since any descendant in DFS is either a child (covered in BFS) or unreachable (in which case, not a forward edge).
• Back-edges and cross-edges may still exist, depending on the structure of G.
Hence, the only universally true statement is: There are no forward-edges in G with respect to the tree T.

6. **Answer : 60**
• Number of vertices, n = 100
• Number of edges in DFS forest, E = 40
• For a graph, E = n − (number of connected components)
• Let the number of connected components = c
• Therefore, c = n − E = 100 − 40 = 60
Hence, the graph has 60 connected components.

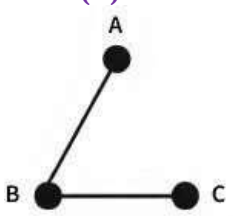
7. **Answer : 5040**
Given that U = {1,2,3} → |U| = 3 → Vertex set = {∅,{1},{2},{3},{1,2},{1,3},{2,3},{1,2,3}} and cardinality is 8.

There is an edge between A and B vertices iff either of vertex set is proper subset to other vertex.

As we know ∅ is proper subset to every other set in the vertex set except itself.
That implies, ∅ is connected with 7 vertices directly, these 7 vertices can be visited in any order in BFS.

Therefore No.of BFS orders = 7! = 120x6x7 = 5040.

8. **Answer : (D)**



We have the following adj. matrix for this

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M^2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$P = M^2$
By looking at this we can say that we have Diameter (G2)=1. We have Diameter(G)=2. Now by looking at these values we can eliminate options A, B, C and option D is satisfied only.

9. **Answer : A**
Statement (I) is TRUE:
In an undirected graph, DFS tree edges can only be tree edges or back edges. A back edge connects a node to one of its ancestors. Cross edges (edges between nodes where neither is an ancestor of the other) are impossible in an undirected DFS because if there were an edge between two such nodes, the DFS would have traversed that edge to visit the other node before finishing the current branch.
Statement (II) is FALSE:
In a BFS tree of an undirected graph, an edge (u, v) can exist between nodes at the same level (depth) or between nodes at adjacent levels.

- If u is at depth i and v is at depth j, it is possible that |i - j| = 0 (they are on the same level).
- The property for BFS is |i - j| <= 1, not strictly equal to 1.

10. **Answer : 31**
No of nodes at level 0(root) of tree ⇒ 1

No of nodes at level 1 of tree ⇒ 2

No of nodes at level 2 of tree ⇒4

No of nodes at level 3 of tree ⇒8

No of nodes at level 4 of tree ⇒16

Last node in level 4th is the node we are looking for ⇒ 1+2+4+8+16⇒31

Check Out Our Gate Courses & Secure Your Dream Rank !

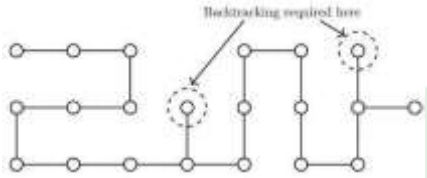
11 **Answer : B**

We can take extra array of size V^2 as memory is not a constraint.
Do the BFS traversal and for each edge u - v in advance list in graph store the v node address in a[i][j].
For e.g., if we find 1 -> 2 as an edge, then store node 2 node address in location a[i][j].
Now once we have stored all the edge (u - v) addresses, then do BFS again.
Now when we encounter 1 -> 2, then goto a[2][1] which is having twin address and set this twin pointer for node 2 in list 1.
Do for all edges similarly.

12. **Answer (D)**

The given graph is undirected, so an edge (u, v) also means (v, u) is also an edge. Since a shorter path can always be obtained by using edge (u, v) or (v, u), the difference between d(u) and d(v) cannot be more than 1.

13. **Answer:(19)**



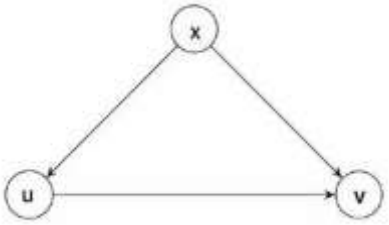
Total 21 nodes are there. 22 nodes require back track here in this question.

So, max recursion depth is 21−2=19
(Do DFSDFS from extreme ends such that max recursion depth will occur. i.e., take leftmost top node as initial node for DFSDFS as shown in below image)

14. **Answer : B**

Answer: B After f is visited, c or g should be visited next. So, the traversal is incorrect. After c is visited, e or f should be visited next. So, the traversal is incorrect. 2 and 3 are correct.

15. **Answer : D**



A. $d[u]<d[v]$, Counter Example $\Longrightarrow$ Well if we directly start DFS on V first, then I call DFS on X which visits U.
B. $d[u]<f[v]$, Counter example $\Longrightarrow$ Same as A
C. $f[u]<f[v]$, Counter example $\Longrightarrow$ Same as A again
So, answer is D.

16. **Answer : D**

Go with vertex with indegree 0. Remove the vertex with all edges going from it. Repeat the procedure. We see that 3 cannot come at first because indegree is not 0. So, D is the answer here.
ALL other options are in Topological order.
Only 1 and 4 order matter for this question.

17. **Answer : D**

After knowing the statement of Parenthesis Theorem, the solution is straightforward.
Given that discovery time and finish time of the vertices are as follows:
P = [5, 12]
Q= [6, 10]
R= [14, 18]
Here,
the interval of Q is contained entirely within the interval of P
$\Rightarrow$ Q is a descendant of P in a DFS tree (i.e. P and Q belong to the same DFS tree)
Again,
the interval of R and P are entirely disjoint
$\Rightarrow$ Neither R nor P is a descendant of the other
$\Rightarrow$ R and P are in different DFS trees (i.e. R and P belong to different DFS trees)
Hence, option D is correct. (Refer the image attached)
Further, all other similar questions can also be easily solved with this Theorem.

18. Answer : D

DFS Tree Property (Undirected Graph):

In an undirected graph, every edge (x, y) of the graph G that is not in the DFS tree T must be a **back edge**. This means x must be an ancestor of y, or vice versa. There are no cross edges.

2. Analyzing the Leaves:

- Vertices u and v are **leaves** in the DFS tree T.
- By definition, a leaf has no children in the tree.
- Since the degree of u in G is >= 2, and it only has one edge in the tree (connecting it to its parent), its other edge(s) in G must be **back edges**.
- These back edges must connect uto one of its **ancestors** in the tree.

In a DFS tree, u and v are in different branches or different positions. Since they are both leaves, there is a **Least Common Ancestor (LCA)** of u and v, let's call it w.

- Every path in G from u to v must pass through an ancestor that is common to both.
- In a structure like this, the parent of a leaf or a common ancestor on the path to the root often acts as an **articulation point** for those specific leaves. Specifically, in any path from u to v, you must go "up" the tree from u via its ancestors and "down" to v
- **A:** u and v could be in completely different branches with no shared neighbors.
- **C:** There is no guarantee that u and v lie on the same cycle; they could be part of two separate cycles that only meet at an ancestor.
- **D:** A vertex and all its neighbors don't necessarily have to form a single cycle.

6 SEARCHING AND SORTING

GATE CSE – 2026 SET-2 MCQ (2-MARKS)

Q1. Consider an array A of integers of size n. The indices of A run from 1 to n. An algorithm is to be designed to check whether A satisfies the condition given below.
for all i, j {1, ..., n - 1} such that i > j,
(A[i + 1] - A[i]) > (A[j + 1] - A[j])
Which one of the following gives the worst case time complexity of the fastest algorithm that can be designed for the problem?

a) $\Theta(n)$
b) $\Theta(\log n)$
c) $\Theta(n \log n)$
d) $\Theta(n^2)$

GATE CSE – 2024 SET-2 NAT (1-MARKS)

Q2. Let A be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in non-decreasing order. The distance of the array [2, 5, 3, 1, 4, 2, 6] is _____.

GATE CSE – 2024 SET-1 MCQ (1-MARKS)

Q3. Given an integer array of size N, we want to check if the array is sorted (in either ascending or descending order). An algorithm solves this problem by making a single pass through the array and comparing each element of the array only with its adjacent elements. The worst-case time complexity of this algorithm is

a) Both O(n)and $\Omega(n)$
b) $O(n)$ but not $\Omega(n)$
c) $\Omega(n)$ but not O(n)
d) Neither O(n) nor $\Omega(n)$

GATE CSE – 2021 SET-2 MCQ (1-MARKS)

Q4. What is the worst-case number of arithmetic operations performed by recursive binary search on a sorted array of size n

a) O(n)
b) $O(n^2)$
c) $O(\sqrt{n})$
d) $O(\log_2 n)]$

GATE CSE – 2017 SET-1 NAT (2-MARKS)

Q5. Let A be an array of 31 numbers consisting of sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i that A[i] is 1 by probing the minimum numbers of locations in A. The worst case number of probes performed by an optimal algorithm is __.

GATE CSE – 2013 MCQ (1-MARKS)

Q6. Which one of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

a) O(log n)
b) O(n)
c) O(n log n)
d) $O(n^2)$

ANSWER KEY

1. (a)	**2.** (**3**)	**3.** (**a**)
4. (**d**)	**5.** (**5**)	**6.** (**b**)

HINTS & SOLUTION

1. **Answer : A**

- **Identify the Term**: Let D[k] = A[k+1] - A[k]. This represents the difference between consecutive elements in the array.
- **Condition**: The condition states that for any i> j, then D[i] > D[j].
- **Conclusion**: This is the mathematical definition of a **strictly increasing sequence**. The problem is asking us to check if the sequence of differences (D[1], D[2],.,D[n-1]) is strictly increasing.

We visit each element of the array exactly once to calculate the differences and perform a single comparison per step. For an array of size n, this requires n-1iterations. Therefore, the worst-case time complexity is **Θ(n)**

2. **Answer : (3)**

Replace second last 2 with 5 -> 2 2 3 1 4 5 6
Replace 1 with first 2 -> 1 2 3 2 4 5 6
Replace second 2 with 3 -> 1 2 2 3 4 5 6

3. **Answer : (A)**

The objective of the algorithm is to check whether the array is sorted or not, and it does so by making a single pass through the array.

For this the worst case will be when the array is already sorted.
So, for this worst case, the algorithm has to traverse the whole array and in doing so it will take $O(n)$ time when specified in $O(.)$ notation or $\Omega(n)$ time when specified in $\Omega(.)$ notation.

4. **Answer : D**

Worst case is when the element not present in the sorted array
n elements

$$n/2$$
$$n/4$$
$$\ldots$$
$$n/2\text{^}x = 1$$

It then states:
" $x = \log_2 n$

5. **Answer : 5**

The smallest index i where A[i] is 1 in an array A of 31 numbers consisting of a sequence of 0's followed by a sequence of 1's can be found by an optimal algorithm using binary search. Since the array is sorted in the form of all 0's followed by all 1's, the problem reduces to finding the boundary between 0 and 1.
The worst-case number of probes (comparisons) performed by the optimal algorithm is $\log_2 31$, which is approximately 5.

6. **Answer : B**

In selection max you can do is n swaps.. selecting the smallest element from all the elements and replacing it correct position so O(n). ∴ The solution is (b)

Check Out Our Gate Courses & Secure Your Dream Rank !

7 HEAPS AND MISCELLANEOUS

GATE CSE – 2025 SET-2 MCQ (2-MARKS)

Q1. A meld operation on two instances of a data structure combines them into one single instance of the same data structure. Consider the following data structures:
P: Unsorted doubly linked list with pointers to the head node and tail node of the list.
Q: Min-heap implemented using an array.
R: Binary Search Tree.
Which ONE of the following options gives the worst-case time complexities for meld operation on instances of size n of these data structures?

a) P: $\theta(1)$, Q: $\theta(n)$, R: $\theta(n)$
b) P: $\theta(1)$, Q: $\theta(n\log n)$, R: $\theta(n)$
c) P: $\theta(n)$, Q: $\theta(n\log n)$, R: $\theta(n^2)$
d) P: $\theta(1)$, Q: $\theta(n)$, R: $\theta(n\log n)$

GATE CSE – 2023 MCQ (2-MARKS)

Q2. Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. The operation Extract-Max(A) extracts and deletes the maximum element from A. The operation Insert(A, key) inserts a new element key in A. The properties of a max-heap are preserved at the end of each of these operations. When A contains n elements, which one of the following statements about the worst case running time of these two operations is TRUE?

a) Both Extract-Max(A) and Insert(A, key) run in O(1)
b) Both Extract-Max(A) and Insert(A, key) run in O(log n)
c) Extract-Max(A) runs in O(1) whereas Insert(A, key) runs in O(n)
d) Extract-Max(A) runs in O(1) whereas Insert(A, key) runs in O(log n).

GATE CSE – 2023 MCQ (1-MARKS)

Q3. Which one of the following sequences when stored in an array at locations A[1], …, A[10] forms a max-heap?

a) 23, 17, 10, 6, 13, 14, 1, 5, 7, 12
b) 23, 17, 14, 7, 13, 10, 1, 5, 6, 12
c) 23, 17, 14, 6, 13, 10, 1, 5, 7, 15
d) 23, 14, 17, 1, 10, 13, 16, 12, 7, 5

GATE CSE – 2017 Set-1 MCQ (2-MARKS)

Q4. Consider the following table:

	Algorithms		**Design Paradigms**
P	Kruskal	i	Divide and Conquer
Q	Quicksort	ii	Greedy
R	Floyd-Warshall	iii	Dynamic Programming

a) P-(ii), Q-(iii), R-(i)
b) P-(iii), Q-(i), R-(ii)
c) P-(ii), Q-(i), R-(iii)
d) P-(i), Q-(ii), R-(iii)

GATE CSE – 2017 SET-2 MCQ (2-MARKS)

Q5. Match the algorithms with their time complexities:

	Algorithm		Time complexity
P	Towers of Hanoi with n disks	i	$\Theta(n^2)$
Q	Binary search given n sorted numbers	ii	$\Theta(n\log n)$
R	Heap sort given n numbers at the worst case	iii	$\Theta(2^n)$
S	Addition of two n × n matrices	iv	$\Theta(\log n)$

a) P-(iii), Q-(iv), R-(i), S-(ii)
b) P-(iv), Q-(iii), R-(i), S-(ii)
c) P-(iii), Q-(iv), R-(ii), S-(i)
d) P-(iv), Q-(iii), R-(ii), S-(i)

GATE CSE – 2015 SET-2 MCQ (1-MARKS)

Q6. Given below are some algorithms, and some algorithm design paradigms.

List - I

P. Dijkstra's Shortest Path
Q. Floyd-Warshall algorithm to compute all pairs shortest path
R. Binary search on a sorted array
S. Backtracking search on a graph

List – II

i. Divide and Conquer
ii. Dynamic Programming
iii. Greedy design
iv. Depth-first search
v. Breadth-first search

Match the above algorithms on the left to the corresponding design paradigm they follow.

a) P-i, Q-iii, R-i, S-v.
b) P-iii, Q-iii, R-i, S-v.
c) P-iii, Q-ii, R-i, S-iv.
d) P-iii, Q-ii, R-i, S-v.

GATE CSE – 2015 SET-1 MCQ (1-MARKS)

Q7. Match the following:

(P) Prim's algorithm for minimum spanning tree	(i) Backtracking
(Q) Floyd-Warshall algorithm for all pairs shortest paths	(ii) Greedy method
(R) Mergesort	(iii) Dynamic programming
(S) Hamiltonian circuit	(iv) Divide and conquer

a) P-iii, Q-ii, R-iv, S-i
b) P-i, Q-ii, R-iv, S-iii
c) P-ii, Q-iii, R-iv, S-i
d) P-ii, Q-i, R-iii, S-iv

GATE CSE – 2015 SET-1 MCQ (2-MARKS)

Q8. An algorithm performs $(\log N)^{1/2}$ find operations, N insert operations, $(\log N)^{1/2}$ delete operations, and $(\log N)^{1/2}$ decrease-key operations on a set of data items with keys drawn from a linearly ordered set . For a delete operation, a pointer is provided to the record that must be deleted. For the decrease-key operation, a pointer is provided to the record that has its key decreased. Which one of the following data structures is the most suited for the algorithm to use, if the goal is to achieve the best total asymptotic complexity considering all the operations?

a) Unsorted array
b) Min - heap
c) Sorted array
d) Sorted doubly linked list

GATE CSE – 2013 MCQ (1-MARKS)

Q9. The number of elements that can be sorted in $\Theta(\log n)$ time using heap sort is

a) $\theta(1)$
b) $\theta(\sqrt{\log n})$
c) $\theta(\log n / (\log\log n))$
d) $\theta(\log n)$

GATE CSE – 2010 MCQ (1-MARKS)

Q10. Two alternative packages A and B are available for processing a database having 10^k records. Package A requires $0.0001n^2$ time units and package B requires $10n \log_{10} n$ time units to process n records. What is the smallest value of k for which package B will be preferred over A?

a) 12
b) 10
c) 6
d) 5

ANSWER KEY

1. (a)	**2. (b)**	**3. (b)**
4. (c)	**5. (c)**	**6. (c)**
7. (c)	**8. (a)**	**9. (c)**
10. (c)		

HINTS & SOLUTION

1. **Answer : (A)**
Merging of two unsorted DLL : $\Theta(1)$
Merging of two min heaps : $\Theta(n)$
Using bottom up Merging of two Binary Search Trees : $\Theta(n)$ [Inorder traversal of both lists : $\Theta(n)$ time Merge two sorted lists : $\Theta(n)$ time Build BST over sorted list : $\Theta(n)$ time]

1. Perform in-order traversals of both trees to obtain two sorted arrays (each in O(n) time).
2. Merge the two sorted arrays into one (standard merge, O(n)).
3. Construct a balanced BST from the merged sorted array by recursively picking the middle element as the root. This step also runs in O(n) time and avoids degenerate (skewed) trees.

2. **Answer : B**
In a max-heap, the Extract-Max operation removes the root (maximum) element and then "heapifies" down to restore the max-heap property. This heapify operation takes worst-case O(log n) time, where n is the number of elements.
• The Insert operation inserts a new element at the last position and then "bubbles up" (heapify up) to maintain the max-heap property, which similarly takes O(log n) time in the worst case.
Thus, both operations have a worst-case running time of O(log n). This matches the option:
"Both Extract-Max(A) and Insert(A, key) run in O(log n)."

3. **Answer : B**
Explanation
• For an array A[1..n] representing a heap, each parent at index i must be ≥ its children at indices 2i and 2i+1 (when they exist). Check this for each internal node.
• Option (B) satisfies all comparisons: 23 ≥ 17, 14; 17 ≥ 7, 13; 14 ≥ 10, 1; 7 ≥ 5, 6; 13 ≥ 12; hence it is a valid max-heap.
• The other options violate the property. In (A), parent 10 has child 14 (> 10); in (C), parent 6 has child 7 and parent.

4. **Answer : C**
Some important points regarding Greedy Vs Dynamic Programming
Greedy: →
It always gives polynomial time complexity

- It is not an optimal
- It always selects only either minimum or maximum among all possibilities

Ex:Dijkstra's algorithm for SSSP, Optimal Merge Pattern, Huffman coding, Fractional knapsack problem, etc

- Dynamic Programming:
- It gives either polynomial or exponential time complexity.
- It gives always an optimal result.
- It checks all possibilities of a problem.

Ex:Longest Common sequence, Matrix chain Multiplication, Travelling sales Problem, etc..,

5. **Answer : C**
P. Towers of Hanoi
=> $T(n) = 2T(n - 1) + 1 \Rightarrow \theta(2^n)$
Q. Binary Search
=> $T(n) = T(n / 2) + C$
$=\theta(\log n)$
R. Heap sort ⇒ $\theta(n \log n)$
S. Addition of two n × n matrices : $\theta(n^2$

6. **Answer : C**

Dijkstra's Shortest Path is a Greedy Algorithm. Floyd-Warshall algorithm is Dynamic Programming. Binary search is a Divide and Conquer. Backtracking is Depth-first search. Mark visited (set to red) when done with neighbors. Dijkstra's algorithm is an algorithm for finding the shortest paths between nodes in a graph, which may represent, for example, road networks.

7. **Answer : C**

(1) Prim's algo for mst is application of greedy which give optimal solution as mst .

(2) Floyd warshall algo for all pair shortest path is application of dynamic programming because it covers all possibilities from every vertex

(3) merge sort --divide and conquer

(4) Hamiltonian circuit is application of backtracking because we have to find cycle again and again until we didn't get cycle of all vertices without repetition of vertices except source vertex.

8. **Answer: (A)**

Since number of insertion operations is asymptotically higher, unsorted array is preferred.

For every operation:

Total cost = (number of times) × (cost per operation)

I) Unsorted Array

Operation costs:
Find = O(N) (scan whole array)
Insert = O(1) (append at end)
Delete (pointer given) = O(1) (swap with last pop)
Decrease-key = O(1) (just update value)

II) Min-Heap

Operation costs:
Find = O(N) (heap not sorted fully)
Insert = O(log N)
Delete (pointer given) = O(log N) (heapify)
Decrease-key = O(log N) (bubble up)

III) Sorted Array

Operation costs:
Find = O(log N) (binary search)
Insert = O(N) (shift elements)
Delete = O(N) (shift)
Decrease-key = O(N) (reposition element)

IV) Sorted Doubly Linked List

Operation costs:
Find = O(N)
Insert = O(N) (need position)
Delete (pointer given) = O(1)
Decrease-key = O(N) (reinsert at correct place)

9. **Answer : C**

Let the number of element is "K", they can be sorted in $\theta(k \log k)$ time We try the options in decreasing order of complexity since, we need a tight band i.e. θ.So if $K \in (\log n)$ time required for heap sort is $\theta(k \log k)$ i.e. $\theta(\log n \times \log\log n)$, But this is not in $\theta(\log n)$if $\theta(\log n / (\log\log n))$ time required for heap sort ie. $\theta((\log n / (\log\log n)) *\log((\log n / (\log\log n))))$
$=\theta (\log n - (\log n \log\log\log n)/\log n) = \theta(\log n)$

10. **Answer : C**

For k=6, 6<=10

Therefore k=6 is the smallest value.

$10n * \log10(n) <= 0.0001 * n^2$

$: 10 * 10^k * \log10(10^k) <= 0.0001 * (10^k)^2$

$: 10^{(k+1)} * k <= 0.0001 * 10^{(2k)}$

$: k <= 10^{(2k - k - 1 - 4)}$

$:$

$k <= 10^{(k - 5)}$

	$(\log N)^2$ find	N insert	$(\log N)^2$ delete	$(\log N)^2$ decrease - key
Unsorted Array	$O(N(\log N)^2)^1$	O(N)	$O(\log N)^2)^1$	$O(\log N)^2)^1$
Min-heap	$O(N(\log N)^2)^1$	$O(N \log N)$	$O(\log N)^2)^3$	$O((\log N)^2)^3$
Sorted Array	$O((\log N)^2)^3$	$O(N^2)$	$O(N(\log N)^2)^1$	$O(N(\log N)^2)^1$
Sorted doubly linked-list	$O(N(\log N)^2)^1$	$O(N^2)$	$O((\log N)^2)^1$	$O((\log N)^2)^1$

DIGITAL LOGIC

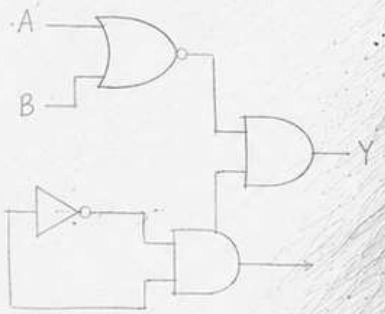
“Scan Here for Video Solution”

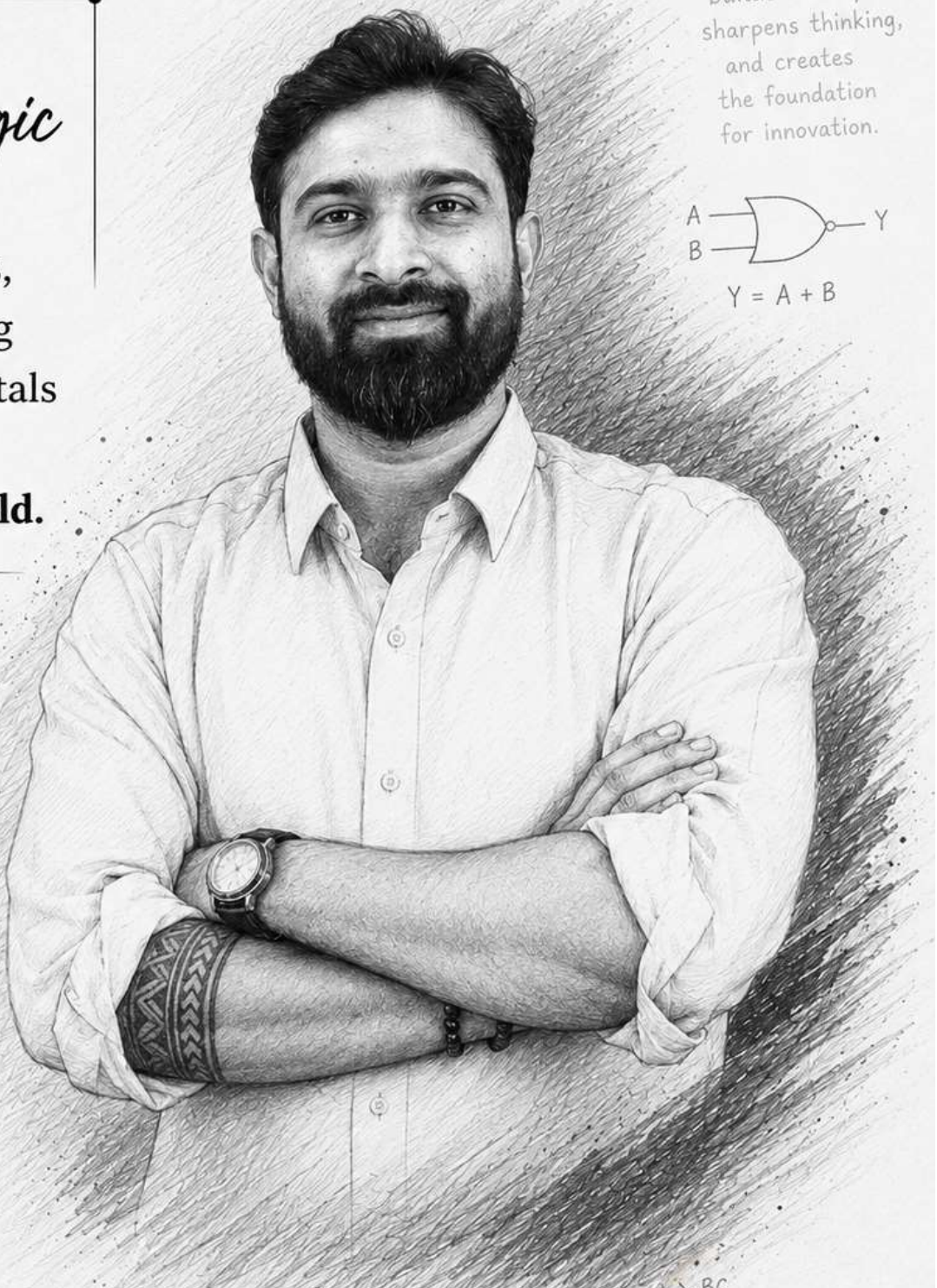
Link to GfG corresponding Short Notes: GATE

// Digital Logic builds clarity, sharpens thinking, and creates the foundation for innovation.

$Y = A + B$

> *Digital Logic* is not just about gates and circuits, it's about building strong fundamentals that power the entire **digital world.**





EX-ISRO SCIENTIST &

GATE AIR 23

A \ BC	00	01	11	10
0	0	1	1	0
1	1	1	1	0

Simplify.
Optimize.
Implement.

Chandan Jha Sir

An Inspirer. A Mentor. A Guide.

1 NUMBER SYSTEM

GATE CSE – 2026 SET-1 MCQ (1-MARKS)

Q1. Consider the real valued variables X, Y and Z represented using the IEEE -754 single precision floating-point format. The binary representations of X and Y in hexadecimal notation are as follows:
X: 35C00000
Y: 34A00000
Let Z = X + Y.
Which one of the following is the binary representation of Z, in hexadecimal notation?

(a) 35C80000
(b) 35CD0000
(c) 35E80000
(d) 35EC0000

GATE CSE – 2026 SET-1 MSQ (1-MARKS)

Q2. Consider the 8-bit signed integers X, Y and Z represented using the sign-magnitude form. The binary representations of X and Y are as follows:
X: 10110100
Y: 01001100
Which of the following operations to compute Z result(s) in an arithmetic overflow?

(a) Z = X + Y
(b) Z = X - Y
(c) Z = -X + Y
(d) Z = -X - Y

GATE CSE – 2026 SET-2 MSQ (1-MARKS)

Q3. In a system, numbers are represented using 4-bit two's complement form. Consider four numbers
N1 = 1011, N2 = 1101, N3 = 1010 and N4 = 1001 in the system.
Which of the following operations will result in arithmetic overflow?

(a) N1 + N2
(b) N2 + N3
(c) N3 - N4
(d) N1 + N4

GATE CSE – 2026 SET-2 NAT (1-MARKS)

Q4. The 32-bit IEEE 754 single precision representation of a number is 0XC2710000. The number in decimal representation is _____ (rounded off to two decimal places).

GATE CSE – 2025 SET-1 MSQ (1-MARKS)

Q5. The number 6 can be represented as 1010 in 4-bit 2's complement representation.
Which of the following is/are CORRECT 2's complement representation(s) of −6?

(a) 1000 1010 in 8-bits
(b) 1111 1010 in 8-bits
(c) 1000 0000 0000 1010 in 16-bits
(d) 1111 1111 1111 1010 in 16-bits

GATE CSE – 2025 SET-2 NAT (1-MARKS)

Q6. The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm:

M: 1100110111101101
Q: 1010010010101010

The total number of addition and subtraction operations to be performed is _____.

GATE CSE – 2024 SET-2 MSQ (2-MARKS)

Q7. Which of the following is/are EQUAL to 224 in radix-5 (i.e., base-5) notation?

(a) 64 in radix-10
(b) 100 in radix-8
(c) 50 in radix-16
(d) 121 in radix-7

GATE CSE – 2024 SET-1 MCQ (1-MARKS)

Q8. Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as A = 01010 and B = 11010. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?

(a) A + B
(b) A − B
(c) B − A
(d) 2 * B

GATE CSE – 2023 NAT (1-MARKS)

Q9. A particular number is written as 132 in radix-4 representation. The same number in radix-5 representation is _____.

GATE CSE – 2022 MCQ (2-MARKS)

Q10. Consider three floating point numbers A, B and C stored in registers R_A, R_B and R_C, respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows.

R_A = 0xC1400000 R_B = 0x42100000
R_C = 0x41400000

Which one of the following is FALSE?

(a) A + C = 0
(b) C = A + B
(c) B = 3C
(d) (B − C) > 0

GATE CSE – 2022 MCQ (1-MARKS)

Q11. Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation R1 + R2, which one of the
following values of R1 and R2 gives an arithmetic overflow?

(a) R1 = 1011 and R2 = 1110
(b) R1 = 1100 and R2 = 1010
(c) R1 = 0011 and R2 = 0100
(d) R1 = 1001 and R2 = 1111

GATE CSE – 2021 SET-2 MSQ (2-MARKS)

Q12. If the numerical value of a 2-byte unsigned integer on a little-endian computer is 255 more than that on a big-endian computer, which of the following choices represent(s) the unsigned integer on a little-endian computer?

(a) 0x6665
(b) 0x0001
(c) 0x4243
(d) 0x0100

GATE CSE – 2021 SET-2 NAT (1-MARKS)

Q13. If x and y are two decimal digits and $(0.1101)_2=(0.8xy5)_{10}$, the decimal value of x+y is?

GATE CSE – 2021 SET-1 MCQ (1-MARKS)

Q14. Let the representation of a number in base 3 be 210. What is the hexadecimal representation of the number?

(a) 15
(b) 21
(c) 528
(d) D2

GATE CSE – 2019 MCQ (1-MARKS)

Q15. Consider Z = X − Y where X, Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:

Link to GfG corresponding Short Notes: GATE FARRAY

(a) n bits
(b) n − 1 bits
(c) n + 1 bits
(d) n + 2 bits

GATE CSE – 2019 MCQ (1-MARKS)

Q16. In 16-bit 2's complement representation, the decimal number 28 is:

(a) 1111 1111 0001 1100
(b) 0000 0000 1110 0100
(c) 1111 1111 1110 0100
(d) 1000 0000 1110 0100

GATE CSE – 2019 NAT (1-MARKS)

Q17. Two numbers are chosen independently and uniformly at random from the set {1, 2, ..., 13}. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

GATE CSE – 2018 MCQ (2-MARKS)

Q18. Consider the unsigned 8-bit fixed point binary Number System below:
$b_7b_6b_5b_4b_3b_2b_1b_0$
where the position of the binary point is between b_3 and b_2. Assume b_7 is the most significant bit (MSB). Some of the decimal numbers listed below cannot be represented exactly in the above representation:

(i) 31.500 (ii) 0.875 (iii) 12.100 (iv) 3.001

Which one of the following statements is true?

(a) None of (i), (ii), (iii), (iv) can be exactly represented.
(b) Only (ii) cannot be exactly represented.
(c) Only (iii) and (iv) cannot be exactly represented.
(d) Only (i) and (ii) cannot be exactly represented.

GATE CSE – 2017 SET-2 MCQ (1-MARKS)

Q19. The representation of the value of a 16-bit unsigned integer X in the hexadecimal number system is BCA9. The representation of the value of X in octal number system is

(a) 571244
(b) 736251
(c) 571247
(d) 136251

GATE CSE – 2016 SET-2 NAT (1-MARKS)

Q20. Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then X − Y is _____

GATE CSE – 2016 SET-1 NAT (1-MARKS)

Q21. The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is _____.

GATE CSE – 2015 SET-3 NAT (1-MARKS)

Q22. Consider the equation $(43)_x = (y3)_8$
where x and y are unknown. The number of possible solutions is _____

GATE CSE – 2014 SET-2 NAT (1-MARKS)

Q23. Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is _____.

GATE CSE – 2014 SET-1 NAT (1-MARKS)

Q24. The base (or radix) of the number system such that the following equation holds
$\frac{312}{20} = 13.1$ is _____

GATE CSE – 2013 MCQ (1-MARKS)

Q25. The smallest integer that can be represented by an 8-bit number in 2's complement form is

(a) −256
(b) −128
(c) −127
(d) 0

GATE CSE – 2010 MCQ (1-MARKS)

Q26. P is a 16-bit signed integer. The 2's complement representation of P is $(F87B)_{16}$. The 2's complement representation of 8 × P is

(a) $(C3D8)_{16}$
(b) $(187B)_{16}$
(c) $(F878)_{16}$
(d) $(987B)_{16}$

GATE CSE – 2009 MCQ (1-MARKS)

Q27. $(1217)_8$ is equivalent to

(a) $(1217)_{16}$
(b) $(028F)_{16}$
(c) $(2297)_{10}$
(d) $(0B17)_{16}$

GATE CSE – 2008 MCQ (1-MARKS)

Q28. The two numbers given below are multiplied using the Booth's algorithm.

Multiplicand: 0101 1010 1110 1110
Multiplier: 0111 0111 1011 1101

How many additions/Subtractions are required for the multiplication of the above two numbers?

(a) 6
(b) 8
(c) 10
(d) 12

GATE CSE – 2008 MCQ (1-MARKS)

Q29. Let r denote the number system radix. The only value(s) of r that satisfy the equation $\sqrt{121_r} = 11_r$ is/are

(a) decimal 10
(b) decimal 11
(c) decimal 10 and 11
(d) any value > 2

GATE CSE – 2006 MCQ (1-MARKS)

Q30. Consider numbers represented in 4-bit Gray code. Let $h_3\ h_2\ h_1\ h_0$ be the Gray code representation of a number n and let $g_3\ g_2\ g_1\ g_0$ be the Gray code of (n + 1) (modulo 16) value of the number. Which one of the following functions is correct?

(a) $g_0(h_3\ h_2\ h_1\ h_0) = \Sigma\ (1, 2, 3, 6, 10, 13, 14, 15)$
(b) $g_1(h_3\ h_2\ h_1\ h_0) = \Sigma\ (4, 9, 10, 11, 12, 13, 14, 15)$
(c) $g_2(h_3\ h_2\ h_1\ h_0) = \Sigma\ (2, 4, 5, 6, 7, 12, 13, 15)$
(d) $g_3(h_3\ h_2\ h_1\ h_0) = \Sigma\ (0, 1, 6, 7, 10, 11, 12, 13)$

GATE CSE – 2006 MCQ (1-MARKS)

Q31. We consider the addition of two 2's complement numbers $b_{n-1}\ b_{n-2}\ \ldots\ b_0$ and $a_{n-1}\ a_{n-2}\ \ldots\ a_0$. A binary adder for adding unsigned binary numbers is used to add the two numbers. The sum is denoted by $c_{n-1}\ c_{n-2}\ \ldots\ c_0$ and the carry-out by c_{out}. Which one of the following options correctly identifies the overflow condition?

(a) $c_{out}\ (\overline{a_{n-1} \oplus b_{n-1}})$
(b) $a_{n-1}\ b_{n-1}\overline{c_{n-1}} + \overline{a_{n-1}}\ \overline{b_{n-1}}\ c_{n-1}$
(c) $c_{out} \oplus c_{n-1}$
(d) $a_{n-1} \oplus b_{n-1} \oplus c_{n-1}$

GATE CSE – 2006 MCQ (1-MARKS)

Q32. When multiplicand Y is multiplied by multiplier X = $x_{n-1}\ x_{n-2}\ \ldots\ x_0$ using bit-pair recoding in Booth's algorithm, partial products are generated according to the following table.

Link to GfG corresponding Short Notes: GATE FARRAY

Row	X_{i+1}	X_i	X_{i-1}	Partial Product
1	0	0	0	0
2	0	0	1	Y
3	0	1	0	Y
4	0	1	1	2Y
5	1	0	0	?
6	1	0	1	-Y
7	1	1	0	-Y
8	1	1	1	?

The partial products for rows 5 and 8 are?

(a) 2Y and Y
(b) −2Y and 2Y
(c) −2Y and 0
(d) 0 and Y

GATE CSE – 2005 MCQ (1-MARKS)

Q33. Using Booth's Algorithm for multiplication, the multiplier −57 will be recoded as

(a) 0 -1 0 0 1 0 0 -1
(b) 1 1 0 0 0 1 1 1
(c) 0 -1 0 0 1 0 0 0
(d) 0 1 0 0 -1 0 0 1

GATE CSE – 2005 MCQ (1-MARKS)

Q34. The range of integers that can be represented by an n-bit 2's complement number system is:

(a) -2^{n-1} to $(2^{n-1}-1)$
(b) $-(2^{n-1}-1)$ to $(2^{n-1}-1)$
(c) -2^{n-1} to 2^{n-1}
(d) $-(2^{n-1}+1)$ to $(2^{n-1}-1)$

GATE CSE – 2005 MCQ (1-MARKS)

Q35. The hexadecimal representation of $(657)_8$ is:

(a) 1AF
(b) D78
(c) D71
(d) 32F

ANSWER KEY

1.(C)	**2. (B, C)**	**3. (B, D)**
4.(-60.25)	**5. (B, D)**	**6. (13)**
7.(A, B, D)	**8. (B)**	**9. (110)**
10.(B)	**11. (B)**	**12. (A, D)**
13.(3)	**14. (A)**	**15. (C)**
16.(C)	**17. (0.503)**	**18. (C)**
19.(D)	**20. (1)**	**21. (-11)**
22.(5)	**23. (3)**	**24. (5)**
25.(B)	**26. (A)**	**27. (B)**
28. (B)	**29. (D)**	**30. (C)**
31. (B)	**32. (C)**	**33. (A)**
34.(A)	**35. (A)**	

HINTS & SOLUTION

1. Answer: (C)
Solution:
X = 35C00000
Binary: 0 01101011 10000000000000000000000
Exponent = 107 - 127 = -20
$X = 1.100000... \times 2^{(-20)}$
Y = 34A00000
Binary: 0 01101001 01000000000000000000000
Exponent = 105 - 127 = -22
$Y = 1.010000... \times 2^{(-22)}$
Align Y to exponent -20:
$Y = 0.010100... \times 2^{(-20)}$
Add mantissas:
1.100000
+0.010100
=1.110100
$Z = 1.110100 \times 2^{(-20)}$
New exponent = -20 + 127 = 107
Binary exponent = 01101011
Final binary:
0 01101011 11010000000000000000000
Hex = 35E80000
Answer: (c) 35E80000

2. **Answer:(B , C)**
Solution:
8-bit sign-magnitude range is:
$-(2^7 - 1)$ to $+(2^7 - 1)$
-127 to +127
X : 10110100 = -52
Y : 01001100 = +76
Z = X + Y = -52 + 76 = +24 → No overflow
Z = X - Y = -52 - 76 = -128 → Overflow
Z = -X + Y = 52 + 76 = +128 → Overflow
Z = -X - Y = 52 - 76 = -24 → No overflow

3. **Answer: (B), (D)**
Solution:
2's complement data given:

N1 : 1011 → $(-5)_{10}$
N2 : 1101 → $(-3)_{10}$
N3 : 1010 → $(-6)_{10}$
N4 : 1001 → $(-7)_{10}$

4-bit 2's complement range: {-8 to +7}

N1 + N2 → (-5) + (-3) = -8
N2 + N3 → (-3) + (-6) = -9 (overflow)
N3 - N4 → (-6) - (-7) = +1
N1 + N4 → (-5) + (-7) = -12 (overflow)

4. Answer:(-60.25)
Solution:
$(C2710000)_H$

Binary representation:
1 | 10000100 | 1110001000.....
S BE M

Value:
$(-1)^{1\ (1.1110001000....)} \times 2^{(132\ -\ 127)}$

$= -(1.1110001000....) \times 2^5$

= -111100.01000....

= -60.25

5. **Answer: (B), (D)**
Solution:
(a) 1000 1010 in 8-bits: Incorrect. This is sign-magnitude representation, not 2's complement.
(b) 1111 1010 in 8-bits: Correct. (Result of sign extension).
(c) 1000 0000 0000 1010 in 16-bits: Incorrect. This is also sign-magnitude.
(d) 1111 1111 1111 1010 in 16-bits: Correct. (Result of sign extension).

6. **Answer: 13**
Solution:
In Booth's algorithm, addition/subtraction is performed whenever the pair (Q_0, Q_{-1}) changes:
01 → Add M
10 → Subtract M
00 or 11 → No operation

Initialize $Q_{-1} = 0$.

Write Q with Q_{-1}:
Q = 1 0 1 0 0 1 0 0 1 0 1 0 1 0 1 0
$Q_{-1} = 0$
Now count transitions by checking adjacent pairs (Q_0, Q_{-1}) during shifts.

Number of 01 and 10 transitions equals number of arithmetic operations.
Counting transitions in Q:

1→0
0→1
1→0
0→0
0→1
1→0
0→0
0→1
1→0
0→1
1→0
0→1
1→0
0→1
1→0
0→0

Total transitions (01 or 10) = 13

7. **Answer: (A,B,D)**
Solution:
To check the equivalent between numbers simply convert it into decimal format.
$(224)_5 = 4 * 5^0 + 2 * 5^1 + 2 * 5^2 = (64)_{10}$
$(100)_8 = 0 * 8^0 + 0 * 8^1 + 1 * 8^2 = (64)_{10}$
$(50)_{16} = 0 * 16^0 + 5 * 16^1 = (80)_{10}$
$(121)_7 = 1 * 7^0 + 2 * 7^1 + 1 * 7^2 = (64)_{10}$
Option A, B, D is correct.

Link to GfG corresponding Short Notes: GATE FARRAY

8. **Answer: (B)**
Solution:
It is asked for 5 bits signed value
A is having 01010 = 10 in decimal
B is having 11010 = −6 in decimal
Now A + B is 4 which is 00100, hence no overflow
Now A − B is 16 which is 010000, see that we used 6 bits hence it is overflow
Now B − A is −16 which is 10000, hence no overflow
Now 2 * B is −12 which is 10100, hence no overflow
hence our correct option will be option B

9. **Answer: (110)**
Solution:
$(132)_4$
$= (1 * 4^2 + 3 * 4^1 + 2 * 4^0)_{10}$
$= (30)_{10}$
$= (25 + 5 + 0)_{10}$ [powers of 5]
$= (1 * 5^2 + 1 * 5^1 + 0 * 5^0)_{10}$
$= (110)_5$

10. **Answer: (B)**
Solution:
A = −12, B = 36, C = 12
(a) A + C = −12 + 12 = 0 → True
(b) C = A + B ⇒ 12 ≠ 24 → False
(c) B = 3C ⇒ 36 = 36 → True
(d) B − C = 24 > 0 → True

11. **Answer: (B)**
Solution:
4-bit 2's complement range = −8 to +7
A) 1011 = −5, 1110 = −2
Sum = −5 + (−2) = −7 → within range → NO overflow
B) 1100 = −4, 1010 = −6
Sum = −4 + (−6) = −10 → out of range (< −8) → OVERFLOW
C) 0011 = +3, 0100 = +4
Sum = 3 + 4 = 7 → within range → NO overflow
D) 1001 = −7, 1111 = −1
Sum = −7 + (−1) = −8 → within range → NO overflow
Overflow occurs when:
Same sign numbers → result sign changes or goes out of range

12. **Answer: (A,D)**
Solution:
It is saying that 0x6665 is the representation of an integer on a little-endian computer, so, it means that the original number must have been 0x6566.
So, for the original number 0x6566:
• On little endian (LE): 0x6665
• On Big endian (BE): 0x6566
Clearly, LE = 255 + BE
Similarly, for 0x0100.
Take 0x0100:
It is saying that 0x0100 is the representation of an integer on a little-endian computer, so, it means that the original number must have been 0x0001.
So, for the number 0x0001:
• On little endian (LE): 0x0100
• On Big endian (BE): 0x0001
Clearly, LE = 255 + BE
Similarly for 0x4243 and 0x0001, they do not satisfy "LE = 255 + BE". So, answer is option A, D.

13. **Answer: 3**
Solution: $(0.1101)_2 = 1/2 + 1/4 + 0 + 1/16 = 0.8125$
$(0.8xy5)_{10} = 0.8 + x/100 + y/1000 + 5/10000$
$= 0.8 + x/100 + y/1000 + 0.0005$
$0.8125 = 0.8 + x/100 + y/1000 + 0.0005$
$0.8125 − 0.8005 = x/100 + y/1000$
$0.012 = x/100 + y/1000$
Multiplying by 1000:
$12 = 10x + y$
Possible digits:
x = 1, y = 2
x + y = 3

14. **Answer: (A)**
Solution:
Firstly convert base 3 into a decimal number system (Base 10):
$(210)_3 = (x)_{10} \Rightarrow 0 * 3^0 + 1 * 3^1 + 2 * 3^2 = (21)_{10}$
Now convert $(21)_{10}$ into a hexadecimal system dividing by 16 that is:
$(21)_{10} = (15)_{16}$

15. **Answer: (C)**
Solution:
Let X and Y represents 31 and −31 respectively in binary sign magnitude form. X and Y will take 6 bits as range of sign magnitude form is $-(2^{n-1} - 1)$ to $2^{n-1} - 1$, where n is the number of bits.
Now in question Z = X − Y,
Z = 31 − (−31) = 62
To represent 62 in sign magnitude form we need 7 (6 + 1) bits.

Link to GfG corresponding Short Notes: GATE FARRAY

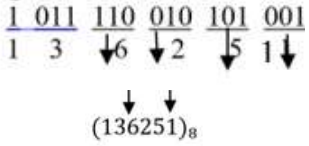
Hence, n + 1 bits needed.

16. **Answer: (C)**
Solution:
$(+28)_{10} = (0000\ 0000\ 0001\ 1100)_2$
-28 is nothing but 2's complement of $+28$.
So, 2's complement of $(0000\ 0000\ 0001\ 1100)_2$ is $(1111\ 1111\ 1110\ 0100)_2$
$(-28)_{10} = (1111\ 1111\ 1110\ 0100)_2$.

17. **Answer: (0.503)**
Solution:
Numbers are from 1 to 13 → total = 13 numbers
4-bit representation:
MSB = 0 → numbers 1 to 7 → count = 7
MSB = 1 → numbers 8 to 13 → count = 6
Total possible pairs = 13 × 13 = 169
Favorable cases:
Both have MSB = 0 → 7 × 7 = 49
Both have MSB = 1 → 6 × 6 = 36
Total favorable = 49 + 36 = 85
Probability = 85 / 169 ≈ 0.503
Final Answer = 0.503

18. **Answer: (C)**
Solution:
Format: $b_7b_6b_5b_4b_3b_2b_1b_0$
4 bits integer, 4 bits fractional
Fraction resolution = 1/16 = 0.0625
Check each number:
(i) 31.500
31 + 0.5 → 0.5 = 8/16 → exact → Representable
(ii) 0.875
0.875 = 14/16 → exact → Representable
(iii) 12.100
0.1 not multiple of 1/16 → Not exact
(iv) 3.001
0.001 not multiple of 1/16 → Not exact
Therefore: only (iii) and (iv) cannot be exactly represented

19. **Answer: (D)**
Solution:
Given: $(BCA9)_{16} = 1011\ 1100\ 1010\ 1001$
For octal number system: grouping of three bits from right to left



20. **Answer: (1)**
Solution:
2's Complement Representation
The range of n-bit 2's Complement Numbers is $-(2^{n-1})$ to $+(2^{n-1} - 1)$
For example, if n = 2, then $-2, -1, 0, 1$ belong to the range (which are distinct)
In general 2^n distinct integers are possible with n-bit 2's Complement Number → X
Sign Magnitude Representation
The range of n-bit Sign Magnitude numbers is $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$
For example, if n = 2, then $-1, -0, +0, +1$ belong to the range in which $-0 = +0$ and both represent zero.
In general $2^n - 1$ distinct integers are possible with n-bit Sign magnitude representation → Y
$X - Y = 2^n - (2^n - 1) = 1$.

21. **Answer: (-11)**
Solution:
1111 1111 1111 0101
2's complement of 1111 1111 1111 0101 = 0000 0000 0000 1011 = +11
2's complement of −11 = +11, in 2's complement representation
2's complement of +11 = −11, in 2's complement representation
So −11 it should be.

22. **Answer: (5)**
Solution: $4x + 3 = 8y + 3$
$\Rightarrow 4x = 8y$
$\Rightarrow x = 2y$
Constraints:
Base $x > 4$ (since digit 4 is present)
Digit $y \in \{0,1,2,3,4,5,6,7\}$
Check valid y:
y = 0 → x = 0 (invalid base)
y = 1 → x = 2 (invalid, digit 4 not allowed)
y = 2 → x = 4 (invalid, base must be > 4)
y = 3 → x = 6 (valid)
y = 4 → x = 8 (valid)
y = 5 → x = 10 (valid)
y = 6 → x = 12 (valid)
y = 7 → x = 14 (valid)
Total valid solutions = 5

23. **Answer: (3)**
Solution: $(123)_5 = 1 \cdot 5^2 + 2 \cdot 5 + 3 = 25 + 10 + 3 = 38$
$(x8)_y = x \cdot y + 8$
$x \cdot y + 8 = 38$
$\Rightarrow x \cdot y = 30$
Possible factor pairs of 30:
(1,30), (2,15), (3,10), (5,6), (6,5)

Valid conditions: x < y and y > 8
Valid pairs:
(1,30), (2,15), (3,10)
Total number of solutions = 3

24. Answer: (5)
Solution:
(312 / 20) = 13.1
=> 312 = 13.1 * 20
=> $3r^2 + r + 2 = (r + 3 + 1/r) * (2r + 0)$
So, r = 5

25. **Answer: (B)**
Solution:
Range of 2's complement no ⇒ (-2^{n-1}) to + $(2^{n-1} - 1)$
Here n = No of bits = 8.
So minimum no = $-2^7 = -128$

26. **Answer: (A)**
Solution: Multiplication can be directly carried in 2's complement form.
F87B = 1111100001111011 can be left shifted 3 times to give 8P = 1100001111011000 = C3D8.
Alternate Method:
As MSB in (F87B) is 1, P is a negative number.
So, P = −1 × 2's complement of (F87B)
P = −1 × (0785) = −1 × (0000 0111 1000 0101)
8 * P = −1 × (0011 1100 0010 1000) (P in binary left shifted 3 times)
In 2's complement representation, this equals, 1100001111011000 = C3D8

27. Answer: (B)
Solution: Given number is in base 8 thus each digit can be represented in three binary bits to get overall binary equivalent.
$(1217)_8 = (001\ 010\ 001\ 111)_2$
I have written the equivalent in group of three bits for easy understanding of the conversion. We can rearrange them in group of four to get equivalent hexadecimal number (in similar manner).

$$(001010001111)^2 = (0010\ 1000\ 1111)^2$$
$$= (28F)_{16}$$

Long Method
In a nutshell the long method follows the following conversion
OCT → DEC → HEX
This procedure is good in a sense that the given option also have a decimal equivalent and thereby might save some time (no in this case, unfortunately).
Here is the the decimal equivalent
$(1217)_8 = (1 * 8^3 + 2 * 8^2 + 1 * 8^1 + 7 * 8^0)_{10}$
$= (655)_{10}$
And we see that decimal equivalent is not in option therefore we proceed for hexadecimal conversion using division method.
$(655)_{10} = (28F)_{16}$

28. Answer: (B)
Solution:
Append 0 at the end of multiplier : 0111 0111 1011 1101 0
Now, paired bits from right end as
00 → (0)
01 → (+1)
10 → (-1)
11 → (0)
Note: pairs are overlapped.
Count +1 = 4 (additions required)
Count -1 = 4 (subtractions required)
So, total = 8 pairs
Hence, additions/subtractions required = 8

29. Answer: (D)
Solution: $\sqrt{(121)_r} = 11_r$

$$\sqrt{(1 \times r^0) + (2 \times r^1) + (1 \times r^2)} = r + 1$$
$$= \sqrt{r^0 + 2r^1 + r^2} = r + 1$$
$$= \sqrt{(1 + r)^2} = r + 1$$
$$r + 1 = r + 1$$

So any integer r satisfies this but r must be greater than 2 as we have 2 in 121 and radix must be greater than any of the digits. (D) is the most appropriate answer.

30. Answer: (C)
Solution:
The question simply says that if 'h' (coded as) is the 4-bit gray code of n then 'g' (coded as $g_3g_2g_1g_0$) is the gray code of (n+1) mod 16.
So, write the gray code as:

	function h				function g			
n	h_3	h_2	h_1	h_0	g_3	g_2	g_1	g_0
0	0	0	0	0	0	0	0	1
1	0	0	0	1	0	0	1	1
2	0	0	1	1	0	0	1	0
3	0	0	1	0	0	1	1	0
4	0	1	1	0	0	1	1	1
5	0	1	1	1	0	1	0	1
6	0	1	0	1	0	1	0	0
7	0	1	0	0	1	1	0	0
8	1	1	0	0	1	1	0	1
9	1	1	0	1	1	1	1	1
10	1	1	1	1	1	1	1	0
11	1	1	1	0	1	0	1	0
12	1	0	1	0	1	0	1	1
13	1	0	1	1	1	0	0	1
14	1	0	0	1	1	0	0	0
15	1	0	0	0	0	0	0	0

Link to GfG corresponding Short Notes: GATE FARRAY

Now, we just have to write g_2 as a function of h i.e $g_2(h_3h_2h_1h_0) = \Sigma$(min-terms of g_2)
To get min-terms, convert h to decimal wherever you get '1' in g_2.
$\therefore g_2 = \Sigma(2, 4, 5, 6, 7, 12, 13, 15)$

31. Answer: (B)
Solution:
In a 2's complement system, overflow occurs during addition if and only if two numbers with the same sign are added together, but the result has the opposite sign.

The problem specifies the following notation:

- a_{n-1} and b_{n-1} : The sign bits (Most Significant Bits) of the two input numbers.
- c_{n-1} : The sign bit (Most Significant Bit) of the resulting sum.

An overflow happens under exactly two scenarios:

1. Adding two negative numbers yields a positive result: Both input sign bits are 1 , but the output sign bit is 0 .

   Condition 1 $= a_{n-1} \cdot b_{n-1} \cdot \overline{c_{n-1}}$

2. Adding two positive numbers yields a negative result: Both input sign bits are θ, but the output sign bit is 1 .

   Condition 2 $= \overline{a_{n-1}} \cdot \overline{b_{n-1}} \cdot c_{n-1}$

   Combining both mutually exclusive scenarios using an OR operation gives the complete expression for the overflow flag:

   Overflow $= a_{n-1}b_{n-1}\overline{c_{n-1}} + \overline{a_{n-1}}\overline{b_{n-1}}c_{n-1}$

32. Answer : (C)
This is Modified Booth (bit-pair / radix-4) recording.

Standard radix-4 Booth table:

x_{i+1} x_i x_{i-1} Operation
000 → 0
001 → +Y
010 → +Y
011 → +2Y
100 → −2Y
101 → −Y
110 → −Y
111 → 0
Now match rows:
Row 5: 100 → −2Y
Row 8: 111 → 0

33. Answer: A
Solution:
Step 1: Write +57 in binary (8-bit)

57 = 00111001

Step 2: Find −57 (2's complement)
Invert bits:
11000110
Add 1:
11000111
So multiplier Q = 11000111

Step 3: Append $Q_{-1} = 0$
11000111 0

Step 4: Booth recoding rule
(Q_0, Q_{-1})
00 → 0
11 → 0
01 → +1
10 → −1
Check pairs from LSB side:

1 0 → −1
1 1 → 0
1 1 → 0
0 1 → +1
0 0 → 0
0 0 → 0
1 0 → −1
1 1 → 0
Booth recoding:
0 -1 0 0 1 0 0 -1

34. Answer :(A)
Solution:
Range of n-bit 2's complement numbers:

Minimum = -2^{n-1}
Maximum = $2^{n-1} - 1$

Range = -2^{n-1} to $(2^{n-1} - 1)$

35. Answer: (A)

Solution:

$(657)_8 \rightarrow$ convert to binary:

6 = 110, 5 = 101, 7 = 111

$\Rightarrow (657)_8 = (110101111)_2$

Group into 4 bits:

0001 1010 1111

Convert to hexadecimal:

0001 = 1, 1010 = A, 1111 = F

$\Rightarrow (657)_8 = (1AF)_{16}$

2 LOGIC GATES

GATE CSE – 2026 SET 1 MSQ (1-MARKS)

Q1. Consider the Boolean expression $F(P, Q) = (\bar{P} + Q) \oplus (\bar{P}Q)$. Which of the following expressions is equivalent to **F**?

(a) $\overline{P \oplus Q}$
(b) $P \oplus Q$
(c) $\bar{P} \oplus Q$
(d) $\bar{P} \oplus \bar{Q}$

GATE CSE – 2026 SET 1 MSQ (2-MARKS)

Q2. Consider a Boolean function F with the following min-term expression:
F(P, Q, R, S) = Σm(1, 2, 3, 4, 5, 7, 10, 12, 13, 14)

Which of the following options is/are the minimal sum-of-products expression(s) of F?

(a) $\bar{P}S + Q\bar{R} + \bar{P}\bar{Q}R + \bar{Q}R\bar{S}$
(b) $\bar{P}S + Q\bar{R} + \bar{P}Q\bar{R} + PR\bar{S}$
(c) $\bar{P}S + Q\bar{R} + PQ\bar{S} + PR\bar{S}$
(d) $\bar{P}S + Q\bar{R} + PQ\bar{S} + \bar{Q}R\bar{S}$

GATE CSE – 2026 SET 2 MCQ (2-MARKS)

Q3. Consider the following 4-variable Boolean function
F(A, B, C, D) = Σm(0, 1, 2, 3, 8, 9, 10, 11)

Consider A as MSB, D as LSB. Which one of the following options represents the minimal sum of products form for the above function?

Note: + is OR operation, · is AND operation, ' is NOT operation

(a) A' + B' + C' + D'
(b) B'
(c) A'·B' + A·B
(d) A'

GATE CSE – 2026 SET-2 MCQ (1-MARKS)

Q4. Which one of the following options is not a property of Boolean Algebra?

(a) a + b = b + a
(b) a · a' = 1
(c) a + a' = 1
(d) a · b = b · a

GATE CSE – 2025 SET-2 MSQ (2-MARKS)

Q5. Which of the following Boolean algebraic equation(s) is/are CORRECT?

(a) $\bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC = BC + \bar{B}\bar{C} + \bar{A}\bar{B}$
(b) $AB + \bar{A}C + BC = AB + \bar{A}C$
(c) $(A + C)(\bar{A} + B) = AB + \bar{A}C$
(d) $\overline{(A + \bar{B} + \bar{D})(C + D)(\bar{A} + C + D)(A + B + \bar{D})} = \bar{A}D + \bar{C}\bar{D}$

GATE CSE – 2025 SET-1 MSQ (1-MARKS)

Q6. Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables?

(a) X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)
(b) X(a, b, X(a, b, c)) = X(a, b, c)
(c) X(a, b, X(a, c, d)) = (X(a, b, a) AND X(c, d, c))
(d) X(a, b, c) = X(a, X(a, b, c), X(a, c, c))

GATE CSE – 2024 SET-2 MSQ (1-MARKS)

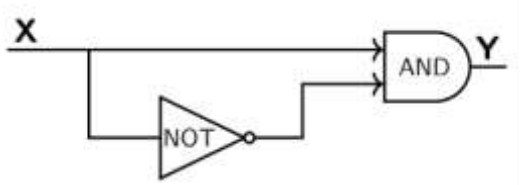
Q7. For a Boolean variable x, which of the following statements is/are FALSE?

(a) x · 1 = x
(b) x + 1 = x
(c) x · x = 0
(d) x + $\overline{x}$ = 1

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2024 SET-1 MSQ (1-MARKS)

Q8. Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?



(a) With no propagation delays, the output Y is always logic Zero
(b) With no propagation delays, the output Y is always logic One
(c) With propagation delays, the output Y can have a transient logic One after X transitions from logic Zero to logic One
(d) With propagation delays, the output Y can have a transient logic Zero after X transitions from logic One to logic Zero

GATE CSE – 2021 SET-1 MSQ (2-MARKS)

Q9. Consider the following Boolean expression.
$F = (X + Y + Z)(\bar{X} + Y)(\bar{Y} + Z)$
Which of the following Boolean expressions is/are equivalent to $\bar{F}$ (complement of F)?

(a) $(\bar{X} + \bar{Y} + \bar{Z})(X + \bar{Y})(Y + \bar{Z})$
(b) $X\bar{Y} + \bar{Z}$
(c) $(X + \bar{Z})(\bar{Y} + \bar{Z})$
(d) $X\bar{Y} + Y\bar{Z} + \bar{X}\bar{Y}\bar{Z}$

GATE CSE – 2019 MCQ (1-MARKS)

Q10. Which one of the following is NOT a valid identity?

(a) $(x \oplus y) \oplus z = x \oplus (y \oplus z)$
(b) $(x + y) \oplus z = x \oplus (y + z)$
(c) $x \oplus y = x + y$, if $xy = 0$
(d) $x \oplus y = (xy + x'y')'$

GATE CSE – 2018 MCQ (1-MARKS)

Q11. Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT?

(a) $\overline{(P \oplus Q)} = P \odot Q$
(b) $\bar{P} \oplus Q = P \odot Q$
(c) $\bar{P} \oplus \bar{Q} = P \oplus Q$
(d) $P \oplus \bar{P} \oplus Q = (P \odot \bar{P} \odot \bar{Q})$

GATE CSE – 2017 SET 2 MCQ (1-MARKS)

Q12. If w, x, y, z are Boolean variables, then which one of the following is INCORRECT?

(a) $wx + w(x + y) + x(x + y) = x + wy$
(b) $(w\bar{x}(y + \bar{z}))' + \bar{w}x = \bar{w} + x + \bar{y}z$
(c) $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$
(d) $(w + y)(wxy + wyz) = wxy + wyz$

GATE CSE – 2016 SET-2 MCQ (1-MARKS)

Q13. Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator. Which one of the following must always be TRUE?

(a) $x_1x_2x_3x_4 = 0$
(b) $x_1x_3 + x_2 = 0$
(c) $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$
(d) $x_1 + x_2 + x_3 + x_4 = 0$

GATE CSE – 2016 SET-1 MCQ (1-MARKS)

Q14. Consider the Boolean operator # with the following properties:
$x \# 0 = x$
$x \# 1 = \bar{x}$
$x \# x = 0$
$x \# \bar{x} = 1$

Then x # y is equivalent to

(a) $x\bar{y} + \bar{x}y$
(b) $x\bar{y} + \bar{x}\bar{y}$
(c) $\bar{x}y + xy$
(d) $xy + \bar{x}\,\bar{y}$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2015 SET-2 NAT (2-MARKS)

Q15. The number of min-terms after minimizing the following Boolean expression is _____.
[D' + AB' + A'C + AC'D + A'C'D]'.

GATE CSE – 2014 SET-2 MCQ (1-MARKS)

Q16. The dual of a Boolean function $F(x_1, x_2, \ldots, x_n)$ written as F^D is the same expression as that of F with + and · swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

(a) 2^n
(b) 2^{n-1}
(c) 2^{2^n}
(d) $2^{2^{n-1}}$

GATE CSE – 2014 SET-3 MCQ (1-MARKS)

Q17. Let $\oplus$ denote the exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q:
$F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$
The equivalent expression for F is

(a) $P + Q$
(b) $\overline{(P + Q)}$
(c) $P \oplus Q$
(d) $\overline{(P \oplus Q)}$

GATE CSE – 2013 MCQ (1-MARKS)

Q18. Which one of the following expressions does NOT represent exclusive NOR of x and y?

(a) $xy+x'y'$
(b) $x \oplus y'$
(c) $x' \oplus y$
(d) $x' \oplus y'$

GATE CSE – 2012 MCQ (1-MARKS)

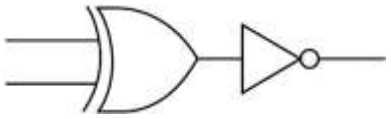
Q19. The truth table represents the Boolean function

X	Y	(X,Y)
0	0	0
0	1	0
1	0	1
1	1	1

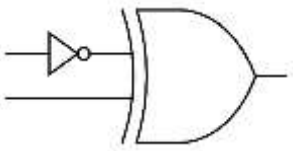
(a) X
(b) $X + Y$
(c) $X \oplus Y$
(d) Y

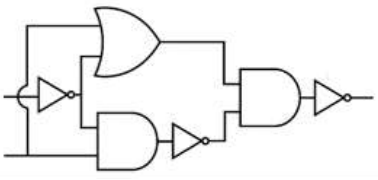
GATE CSE – 2011 MCQ (1-MARKS)

Q20. Which one of the following circuits is NOT equivalent to a 2-input XNOR (exclusive NOR) gate?

(a) 

(b)

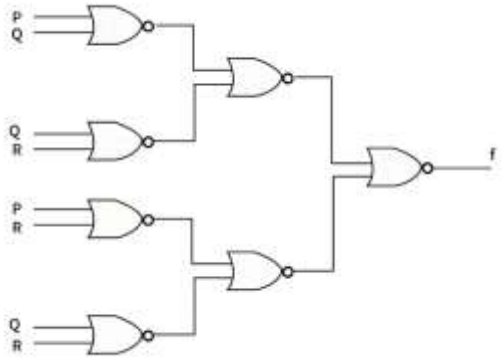
(c) 

(d) 

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2010 MCQ (1-MARKS)

Q21. What is the boolean expression for the output of the combinational logic circuit of NOR gates given below?



(a) $\overline{(Q + R)}$
(b) $\overline{(P + Q)}$
(c) $\overline{(P + R)}$
(d) $\overline{(P + Q + R)}$

GATE CSE – 2009 MCQ (1-MARKS)

Q22. What is the minimum number of gates required to implement the Boolean function (AB+C) if we have to use only 2-input NOR gates?

(a) 2
(b) 3
(c) 4
(d) 5

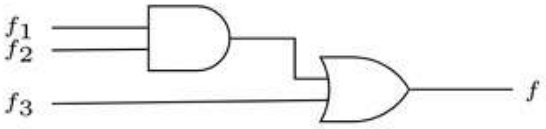
GATE CSE – 2008 MCQ (1-MARKS)

Q23. If P, Q, R are Boolean variables, then $(P + \bar{Q})(P \cdot \bar{Q} + P \cdot R)(\bar{P} \cdot R + \bar{Q})$ simplifies to

(a) $P \cdot \bar{Q}$
(b) $P \cdot R$
(c) $P \cdot \bar{Q} + R$
(d) $P \cdot \bar{R} + Q$

GATE CSE – 2008 MCQ (1-MARKS)

Q24. Given f_1,f_2 and f in the canonical sum of products form(in decimal) for the circuit



$f_1 = \Sigma m(4, 5, 6, 7, 8)$
$f_3 = \Sigma m(1, 6, 15)$
$f = \Sigma m(1, 6, 8, 15)$

then f_2 is
(a) $\Sigma m(4, 6)$
(b) $\Sigma m(4, 8)$
(c) $\Sigma m(6, 8)$
(d) $\Sigma m(4, 6, 8)$

GATE CSE – 2007 MCQ (2-MARKS)

Q25. Define the connective * for the Boolean variables X and Y as:
X * Y = XY + X'Y'.
Let Z = X * Y. Consider the following expressions P, Q and R.
P: X = Y * Z,
Q: Y = X * Z,
R: X * Y * Z = 1
Which of the following is TRUE?

(a) Only P and Q are valid.
(b) Only Q and R are valid.
(c) Only P and R are valid.
(d) All P, Q, R are valid.

GATE CSE – 2007 MCQ (1-MARKS)

Q26. Let f(w,x,y,z) = Σ (0,4,5,7,8,9,13,15). Which of the following expressions are NOT equivalent to f?

P: x'y'z' + w'xy' + wy'z + xz

Q: w'y'z' + wx'y' + xz

R: w'y'z' + wx'y' + xyz + xy'z

Check Out Our GATE Courses & Secure Your Dream Rank!

S: x'y'z' + wx'y' + w'y
(a) P only
(b) Q and S
(c) R and S
(d) S only

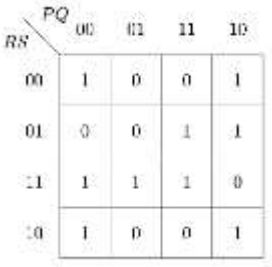
GATE CSE – 2007 MCQ (1-MARKS)

Q27. The following expression was to be realized using 2-input AND and OR gates. However, during the fabrication all 2-input AND gates were mistakenly substituted by 2-input NAND gates.
(a.b).c + (a'.c).d + (b.c).d + a.d . What is the function finally realized?

(a) 1
(b) a' + b' + c' + d'
(c) a' + b + c' + d'
(d) a' + b' + c + d'

GATE CSE – 2006 MCQ (1-MARKS)

Q28. The boolean function for a combinational circuit with four inputs is represented by the following Karnaugh map.



Which of the product terms given below is an essential prime implicant of the function?

(a) QRS
(b) PQS
(c) PQ'S'
(d) Q'S'

GATE CSE – 2005 MCQ (1-MARKS)

Q29. Which of the following expressions is equivalent to (A ⊕ B) ⊕ C ?

(a) (A + B + C)($\bar{A}$ + $\bar{B}$ + $\bar{C}$)
(b) (A + B + C)($\bar{A}$ + $\bar{B}$ + C)
(c) ABC + $\bar{A}$(B ⊕ C) + $\bar{B}$(A ⊕ C)
(d) None of these

ANSWER KEY

1. (a. c)	**2. (b, d)**	**3. (b)**
4. (b)	**5. (b, c, d)**	**6. (b, d)**
7. (b. c)	**8. (a, c)**	**9. (b, c, d)**
10. (b)	**11. (d)**	**12. (c)**
13. (c)	**14. (a)**	**15. (1)**
16. (d)	**17. (d)**	**18. (d)**
19. (a)	**20. (d)**	**21. (a)**
22. (b)	**23. (a)**	**24. (c)**
25. (d)	**26. (d)**	**27. (c)**
28. (d)	**29. (c)**	

HINTS & SOLUTIONS

1. Answer: A, C
Solution:
We know that $P \oplus Q = P'Q + PQ'$,using this property we solve below:

$$F(P,Q) = (\bar{P} + Q) \oplus (\bar{P}Q)$$

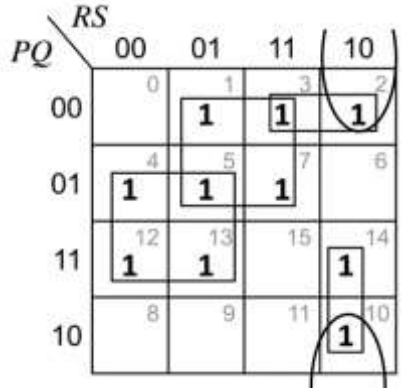
$$F(P,Q) = (\overline{\bar{P} + Q}).(\bar{P}Q) + (\bar{P} + Q).(\overline{\bar{P}Q})$$

$$F(P,Q) = \bar{\bar{P}}.\bar{Q}.(\bar{P}Q) + (\bar{P} + Q).(\bar{\bar{P}} + \bar{Q})$$

$$F(P,Q) = \bar{P}\bar{Q} + PQ$$

$$\equiv P \odot Q \equiv \bar{P} \oplus Q = P \oplus \bar{Q} \equiv \overline{P \oplus Q}$$

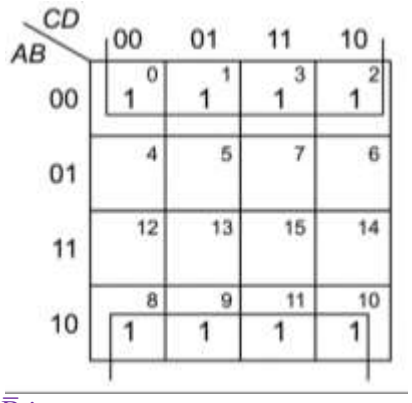
2. Answer: (B,D)
Solution:
F(P, Q, R, S) = Σm(1, 2, 3, 4, 5, 7, 10, 12, 13, 14)



Upon simplification (b) and (d) are options.
(b) $\bar{P}S + Q\bar{R} + \bar{P}\bar{Q}R + PR\bar{S}$
(d) $\bar{P}S + Q\bar{R} + PQ\bar{S} + \bar{Q}R\bar{S}$

Check Out Our GATE Courses & Secure Your Dream Rank!

3. Answer:(B)
Solution:
F(A, B, C, D) = Σm(0, 1, 2, 3, 8, 9, 10, 11)



So, F = $\bar{B}$ is correct.

4. Answer: B
Solution:
(a) $a + b = b + a$
This is the **Commutative Law of OR. So (a) is correct.**
(b) Complement law states:
a + a' = 1
a · a' = 0
So (b) **a · a' = 1 is incorrect**.
(c) $a + a' = 1$.
This is the Complement Law (OR form). So **(c) is correct.**
(d) $a \cdot b = b \cdot a$
This is the Commutative Law of AND. So, option (d) is correct.

5. Answer - (B,C, D)
Solution:
Option A: Let, A = 0, B = 0, C = 1

Evaluate LHS:
All terms contain A or B ⇒ each term = 0
⇒ LHS = 0

Evaluate RHS:
BC = 0
$\bar{B}\bar{C}$ = 1·0 = 0
$\bar{A}\bar{B}$ = 1·1 = 1
⇒ RHS = 1

Since LHS ≠ RHS, Option A is false.

Option B →
AB + $\bar{A}$C + BC = AB + $\bar{A}$C (Consensus Theorem)
→ True
Option C →
(A + C)($\bar{A}$ + B)
= A$\bar{A}$ + AB + C$\bar{A}$ + CB
= 0 + AB + $\bar{A}$C + BC
= AB + $\bar{A}$C + BC
= AB + $\bar{A}$C (using Consensus) → True

Option D →
Apply De Morgan and simplify:
$\overline{[(A + B + \bar{D})(C + D)(\bar{A} + C + D)(A + B + D)]}$

$= \bar{A}D + \bar{C}\bar{D}$ → True

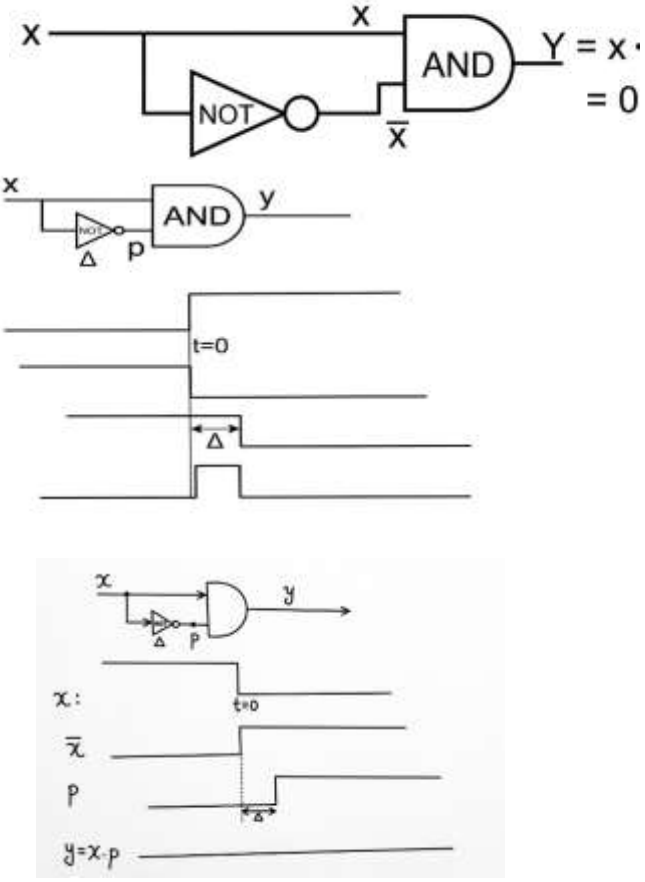
Final Answer: Options B, C and D are correct, A is false

6. Answer: (B and D)
Solution:
Given:
X(x, y, z) = Majority function (output = 1 if at least two inputs are 1).So,
X(x,y,z) = xy + yz + zx
Option A:
Both sides compute the majority hierarchically, but the majority function is not associative.
Incorrect
Option B:
Let M = X(a,b,c)
Then X(a,b,M):
If at least two among a,b,c are 1 → M=1 → output remains 1
If fewer than two are 1 → M=0 → output remains 0
Hence unchanged.
Correct
Option C:
LHS = X(a,b,X(a,c,d))
RHS = X(a,b,a) AND X(c,d,c)
Since:
X(a,b,a) = a
X(c,d,c) = c
RHS = a AND c
But LHS depends on b and d also.
Incorrect
Option D:
X(a,c,c) = c
So RHS:
X(a, X(a,b,c), c)
Using majority property:
Majority(a, Majority(a,b,c), c) = Majority(a,b,c)
Correct.

Check Out Our GATE Courses & Secure Your Dream Rank!

7. **Answer: B , C**
Solution:
x·1 = x → True
x + 1 = 1 → False
x·x = x → False
x + x' = 1 → True

8. **Answer: (A, C)**
Solution:



9. **Answer: (B,C,D)**
Solution:

$$F = (X + Y + Z)(\bar{X} + Y)(\bar{Y} + Z)$$

Taking complement of above expression.

$$\bar{F} = \overline{(X + Y + Z)(\bar{X} + Y)(\bar{Y} + Z)}$$

Applying De-Morgan's law,

$$\bar{F} = \overline{(X + Y + Z)} + \overline{(\bar{X} + Y)} + \overline{(\bar{Y} + Z)}$$

$$\bar{F} = (\bar{X} \cdot \bar{Y} \cdot \bar{Z}) + \bar{\bar{X}} \cdot \bar{Y} + \bar{\bar{Y}} \cdot \bar{Z}$$

[$\because \bar{\bar{X}} = X$, Using double negation law]

$\therefore \bar{F} = (\bar{X} \cdot \bar{Y} \cdot \bar{Z}) + (X \cdot \bar{Y}) + (Y \cdot \bar{Z}) \rightarrow$ Option (D)

Taking $\bar{Y}$ as common we get:

$$\bar{F} = \bar{Y}[(\bar{X}\bar{Z} + X] + Y\bar{Z}$$

[$\because A + BC = (A + B)(A + C)$ Applying distributive law here]

$$\bar{F} = \bar{Y}[(X + \bar{X})(X + \bar{Z})] + Y\bar{Z}$$

$$\bar{F} = \bar{Y}[X + \bar{Z}] + Y\bar{Z}$$

$$\bar{F} = X\bar{Y} + \bar{Y}\bar{Z} + Y\bar{Z}$$

Taking $\bar{Z}$ as common

$$\bar{F} = X\bar{Y} + \bar{Z}(Y + \bar{Y})$$

$\because (Y + \bar{Y} = 1)$ using complement law

$\therefore \bar{F} = X\bar{Y} + \bar{Z} \rightarrow$ Option (B)

Applying distributive law here we get;

$\bar{F} = (X + \bar{Z})(\bar{Y} + \bar{Z}) \rightarrow$ Option (C)

So, Correct Option are B,C,D

10. **Answer: B**
Solution:
Option A:
(x ⊕ y) ⊕ z = x ⊕ (y ⊕ z)
XOR is associative → TRUE
Option B:
(x + y) ⊕ z = x ⊕ (y + z)
Let x = 1, y = 1, z = 0
LHS = (1 + 1) ⊕ 0 = 1 ⊕ 0 = 1
RHS = 1 ⊕ (1 + 0) = 1 ⊕ 1 = 0
LHS ≠ RHS → FALSE
Option C:
x ⊕ y = x + y, if xy = 0 → TRUE
Option D:
x ⊕ y = (xy + x'y')' → TRUE

11. **Answer - (D)**
Solution :

$$\overline{(P \oplus Q)} = P \circ Q$$

- Correct. By definition, XNOR (O) is the exact complement of XOR (⊕).
  (b) $\bar{P} \oplus Q = P \circ Q$
- Correct. Inverting exactly one input of an XOR gate turns the operation into an XNOR gate.
  (c) $\bar{P} \oplus \bar{Q} = P \oplus Q$
- Correct. Inverting both inputs of an XOR gate cancels out the changes. leaving the output identical to the original XOR.
  (d) $P \oplus \bar{P} \oplus Q = (P \odot \bar{P} \odot \bar{Q})$
- INCORRECT. * LHS: $(P \oplus \bar{P}) \oplus Q = 1 \oplus Q = \bar{Q}$
- RHS: $(P \circ \bar{P}) \circ \bar{Q} = 0 \circ \bar{Q} = Q$

Check Out Our GATE Courses & Secure Your Dream Rank!

- Since $\bar{Q} \neq Q$, this identity fails.

12. Answer: (C)

Solution:

Option - A: $wx + w(x + y) + x(x + y) = x + wy$

$$wx + wx + wy + x$$
$$wx + wy + x$$
$$x(1 + w) + wy$$
$$x + wy$$

Option - B: $\overline{(w\bar{x}(y + \bar{z}))} + \bar{w}x = \bar{w} + x + \bar{y}z$

$$\bar{w} + x + \bar{y}z + \bar{w}x$$
$$\bar{w} + \bar{w}x + x + \bar{y}z$$
$$\bar{w} + x + \bar{y}z$$

Option - D:

$$(w + y)(wxy + wyz) = wxy + wyz$$
$$wxy + wyz + wxy + wyz$$
$$wxy + wyz$$

Option A, B, D are matching fine.

Hence, Option - C is the answer.

13. Answer: C

Solution:

Given:

$x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$

Property of XOR:

If XOR of variables is 0, then the number of 1's among them is even.

Rearrange using associativity and commutativity of XOR:

$x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$

$\Rightarrow (x_1 \oplus x_3) \oplus (x_2 \oplus x_4) = 0$

For XOR:

$A \oplus B = 0 \Rightarrow A = B$

Therefore,

$x_1 \oplus x_3 = x_2 \oplus x_4$

Hence this statement must always be true.

14. Answer: (a)

Solution:

x	y	#
0	0	0
0	1	1
1	0	1
1	1	0

So, according to this # function results in XOR operation on x and y . $x\#y$ equivalent to $x.\bar{y} + \bar{x}.y$

15. Answer: 1

Solution:

Let f(A, B, C, D) = [D' + AB' + A'C + AC'D + A'C'D]'

$\Rightarrow$ [D' + AB' + A'C + C'D]'

$\Rightarrow$ [D' + C' + AB' + A'C]'

$\Rightarrow$ [D' + C' + A' + AB']'

$\Rightarrow$ [D' + C' + A' + B']'

$\Rightarrow$ ABCD

So the number of min-terms = 1

16. Answer: (D)

Solution:

For self-dual functions,

1. Number of min-terms equals number of max-terms
2. Function should not contain two complementary min-terms - whose sum equals $2^n - 1$, where n is the number of variables.

	A	**B**	**C**
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	**1**	0	0
5	**1**	0	1
6	**1**	**1**	0
7	**1**	1	**1**

So, here $(0,7)(1,6)(2,5)(3,4)$ are complementary terms so in self-dual we can select any one of them but not both.

Totally $2 \times 2 \times 2 \times 2 = 2^4$ possibility because say from $(0,7)$ we can pick anyone in minterm but not both.

For example, let $f = \sum(0,6,2,3)$

NOTE: here I have taken only one of the complementary term for min term from the sets.

So, remaining numbers will go to MAX-TERMS

For above example, $2^4 = 16$ self dual functions are possible

So, If we have N variables, total Min-terms possible is 2^n

Then half of them we selected so 2^{n-1}.

Now we have 2 choices for every pair for being selected.

So total such choices $= \underbrace{2 \times 2 \times 2 \times 2 \ldots 2}_{2^{n-1} \text{ times}}$

$\therefore 2^{2^{n-1}}$ (option D)

Check Out Our GATE Courses & Secure Your Dream Rank!

17. Answer: (D)
Solution:
1 ⊕ P = P'
Q ⊕ 0 = Q
So,
F = (P' ⊕ (P ⊕ Q)) ⊕ ((P ⊕ Q) ⊕ Q)
First part:
P' ⊕ (P ⊕ Q)
= P' ⊕ P ⊕ Q
= 1 ⊕ Q
= Q'
Second part:
(P ⊕ Q) ⊕ Q
= P ⊕ (Q ⊕ Q)
= P ⊕ 0
= P
Thus,
F = Q' ⊕ P
= P ⊕ Q'
= (P ⊕ Q)' .

18. Answer:(D)
Solution:
XNOR = xy + x'y'
Check options:
A → xy + x'y' = XNOR
B → x ⊕ y' = xy + x'y' = XNOR
C → x' ⊕ y = xy + x'y' = XNOR
D → x' ⊕ y' = x ⊕ y = XOR

19. Answer: (A)
Solution:
F = XY' + XY
= X(Y' + Y)
= X(1)
= X

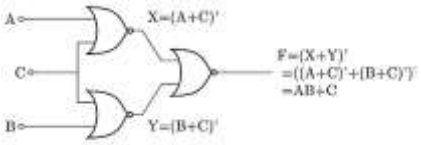
20. Answer: D
Solution:
All circuits except option D implement the XNOR function:
XNOR = AB + A'B'
Option D implements XOR instead of XNOR.

21. Answer - (A)
Solution:
Level 1:
$\overline{(P+Q)}\,\overline{(Q+R)}\,\overline{(P+R)}\,\overline{(Q+R)}$

Level 2:
$\overline{(P+Q)}+\overline{(Q+R)} = (P+Q)(Q+R) = PQ+PR+Q+QR$
$\overline{(P+R)}\,\overline{(Q+R)} = (P+R)(Q+R) = PQ+R+QR+PR$

Level 3:
$PR+QR+PQ+Q+R = Q+R$

22. Answer: (B)
Solution: Given boolean function is
$f = AB + C$
$= (A + C).(B + C)$
$= ((A + C)' + (B + C)')'$



Therefore, 3 NOR gates required .

23. Answer - (A)
Solution:
(P + $\bar{Q}$)(P$\bar{Q}$ + PR)($\bar{P}$R + $\bar{Q}$)
Step 1: Simplify the middle bracket Factor out P from the middle terms: P($\bar{Q}$ + R)
Step 2: Multiply the first two brackets Substitute it back with the first bracket: (P + $\bar{Q}$) · P($\bar{Q}$ + R)
Rearrange the terms: P(P + $\bar{Q}$)($\bar{Q}$ + R)
Apply the Absorption Law [A(A + B) = A] to the first part, so P(P + $\bar{Q}$) simply becomes P: P($\bar{Q}$ + R)
Expand this out: P$\bar{Q}$ + PR
Step 3: Multiply with the final bracket Take the result and multiply it by the last bracket:
(P$\bar{Q}$ + PR)($\bar{P}$R + $\bar{Q}$)
Step 4: Expand and apply Boolean rules Multiply using the distributive property:
= (P$\bar{Q}$ · $\bar{P}$R) + (P$\bar{Q}$ · $\bar{Q}$) + (PR · $\bar{P}$R) + (PR · $\bar{Q}$)
Apply the rules [A · $\bar{A}$ = 0] and [A · A = A]:
= 0 + P$\bar{Q}$ + 0 + P$\bar{Q}$R = P$\bar{Q}$ + P$\bar{Q}$R
Step 5: Factor and apply the Annulment Law
Factor out the common term P$\bar{Q}$: = P$\bar{Q}$(1 + R)
Apply the Annulment Law [1 + A = 1], so (1 + R) becomes 1: = P$\bar{Q}$ · 1
Final Answer: P$\bar{Q}$

24. Answer : (C)
Solution:
With AND gates we will choose intersection of min-terms.
With OR gates we will take union of min-terms.

25. Answer:(D)
Solution:
Given:
X * Y = XY + X'Y'
This operation represents XNOR (equivalence), i.e., X * Y = 1 when X = Y.
Let:
Z = X * Y = XY + X'Y'
So, Z = 1 when X = Y and Z = 0 when X ≠ Y.

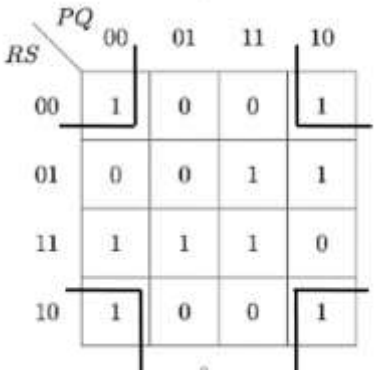
Check Out Our GATE Courses & Secure Your Dream Rank!

Now verify each statement.
P: X = Y * Z
Y * Z = YZ + Y'Z'
Substitute Z = X * Y.
Since Z indicates whether X and Y are equal:
If X = Y → Z = 1
Then Y * Z = Y * 1 = Y = X
If X ≠ Y → Z = 0
Then Y * Z = Y * 0 = Y' = X
Hence, P is TRUE.
Q: Y = X * Z
Similarly,
X * Z = XZ + X'Z'
If X = Y → Z = 1
X * Z = X = Y
If X ≠ Y → Z = 0
X * Z = X' = Y
Hence, Q is TRUE.
R: X * Y * Z = 1
Since Z = X * Y,
X * Y * Z = Z * Z
Using XNOR property:
A * A = 1
Therefore,
Z * Z = 1
Hence, R is TRUE.
All P, Q and R are valid.
Final Answer: D. All P, Q, R are valid.

26. Answer: (D)
Solution:
Let w = 1, x = 1, y = 1, z = 1, then the value of f is 1.
Consider each statement:
x'y'z' + w'x'y' + w'y'z + xz
= 0·0·0 + 0·1·0 + 1·0·1 + 1·1
= 1
w'y'z' + wx'z'y' + xz
= 0·0·0 + 1·0·0 + 1·1
= 1
w'y'z' + wx'y' + xyz + xy'z
= 0·0·0 + 1·0·0 + 1·1·1 + 1·0·1
= 1
x'y'z' + wx'y' + w'y
= 0·0·0 + 1·0·0 + 0·1
= 0
So statement (d) is false because when w = 1, x = 1, y = 1, z = 1 the value of f is 0.
(d) does not contain the essential min-terms.

27. Answer: (C)
Solution:
a' + c' + d' + a'c + ab + bc
= a'(c + 1) + c' + d' + ab + bc
= a' + c' + d' + ab + bc
= (a' + a)(a' + b) + (c' + c)(c' + b) + d'
= a' + b + c' + b + d'
= a' + b + c' + d'
Option is C.

28. Answer: (D)
Solution:



Only the top leftmost and bottom rightmost 1*s* have no alternate groupings. So, they form the essential prime implicants.

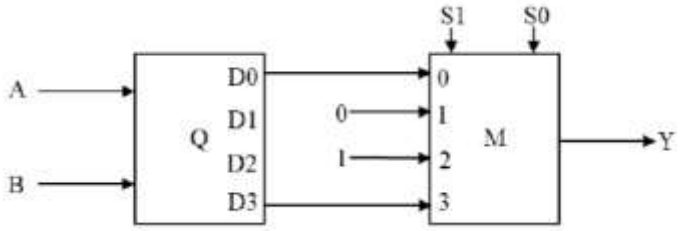
29. Answer: (C)
Solution:
Assume A = 0, the given expression becomes
B ⊕ C
Option A → B + C
Option B → B + $\bar{C}$
Option C →ABC + $\bar{A}$(B ⊕ C) + $\bar{B}$(A ⊕ C)
=ABC + $\bar{A}$(B$\bar{C}$ + $\bar{B}$C) + $\bar{B}$(A$\bar{C}$ + $\bar{A}$C)
=ABC + $\bar{A}$B$\bar{C}$ + $\bar{A}\bar{B}$C + A$\bar{B}\bar{C}$ + $\bar{A}\bar{B}$C
=ABC + $\bar{A}$B$\bar{C}$ + A$\bar{B}\bar{C}$ + $\bar{A}\bar{B}$C
=$\bar{A}\bar{B}$C + $\bar{A}$B$\bar{C}$ + A$\bar{B}\bar{C}$ + ABC
=m1 + m2 + m4 + m7
=Σm(1,2,4,7)
=A ⊕ B ⊕ C

Check Out Our GATE Courses & Secure Your Dream Rank!

3 COMBINATIONAL CIRCUIT

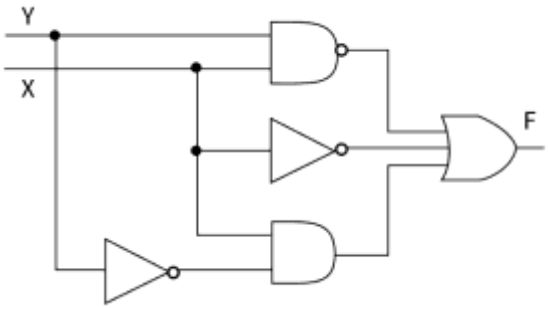
GATE CSE – 2026 SET-2 NAT (2-MARKS)

Q1. Consider the digital circuit shown below with two input lines A and B, two select lines S0 and S1, and an output line Y. The blocks Q and M represent an active-high 2:4 decoder and a 4-to-1 multiplexer, respectively. Out of 16 possible input combinations, the number of combinations that produce Y = 1 is _____. (Answer in integer).



GATE CSE – 2025 SET-2 MSQ (2-MARKS)

Q2. Consider the following logic circuit diagram.

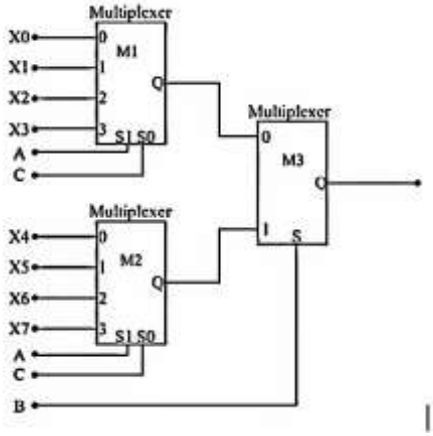


Which is/are the CORRECT option(s) for the output function F?

(a) (XY)'
(b) X' + Y' + XY'
(c) (XY)' + X' + XY'
(d) X + Y'

GATE CSE – 2023 MCQ (2-MARKS)

Q3. A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0-X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown.



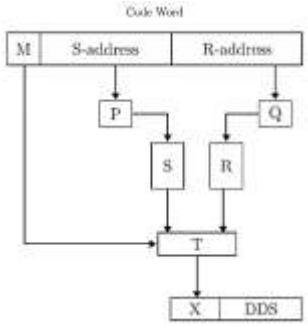
Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function A'+ A'·C' + A· B'·C?

(a) (1, 1, 0, 0, 1, 1, 1, 0)
(b) (1, 1, 0, 0, 1, 1, 0, 1)
(c) (1, 1, 0, 1, 1, 1, 0, 0)
(d) (0, 0, 1, 1, 0, 1, 1, 1)

GATE CSE – 2022 MCQ (2-MARKS)

Q4. Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word is used to load a word in X, either from S or from R. S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M, T selects an input word to load in X. P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P, Q, and T?

Check Out Our GATE Courses & Secure Your Dream Rank!

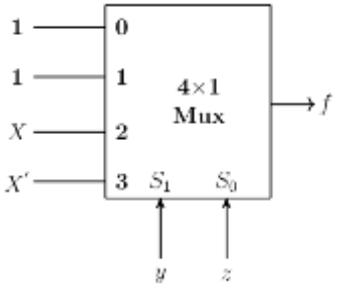


(a) P is 10:1 multiplexer;Q is 5: 1 multiplexer; T is 2: 1 multiplexer
(b) P is 10: 2^{10} decoder; Q is 5: 2^5 decoder;T is 2: 1 encoder
(c) P is 10:2^{10} decoder; Q is 5: 2^5 decoder; T is 2: 1 multiplexer
(d) P is 1: 10 de-multiplexers; Q is 1: 5 de-multiplexers; T is 2: 1 multiplexer

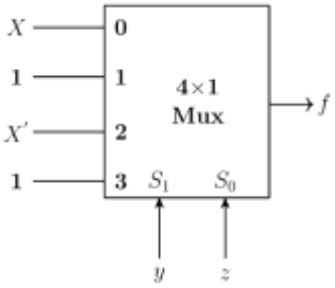
GATE CSE – 2021 SET-2 MCQ (1-MARKS)

Q5. Which one of the following circuits implements the Boolean function given below? f (x, y, z) = m0 + m1 + m3 + m4 + m5 + m6, where mi is the ith min-term.

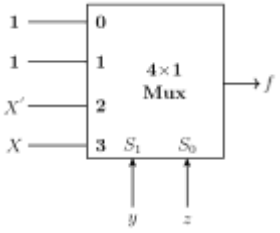
(a)



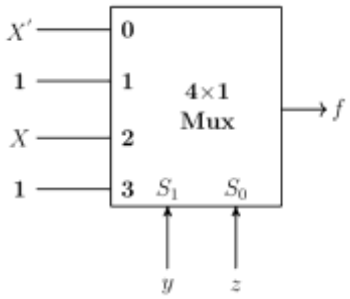
(b)



(c)



(d)



GATE CSE – 2020 NAT (2-MARKS)

Q6. If there are m input lines and n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of m + n is ____.

GATE CSE – 2020 NAT (1-MARKS)

Q7. A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The minimum number of select lines needed for the multiplexer is _____.

GATE CSE – 2016 SET-2 NAT (1-MARKS)

Q8. Consider an eight-bit ripple-carry adder for computing the sum of A and B, where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____.

GATE CSE – 2016 SET-1 MCQ (1-MARKS)

Q9. Consider a carry look ahead adder for adding two n-bit integers, built using gates of fan-in at most two. The time to perform addition using this adder is:

(a) $\Theta(1)$
(b) $\Theta(\log(n))$
(c) $\Theta(\sqrt{n})$
(d) $\Theta(n)$

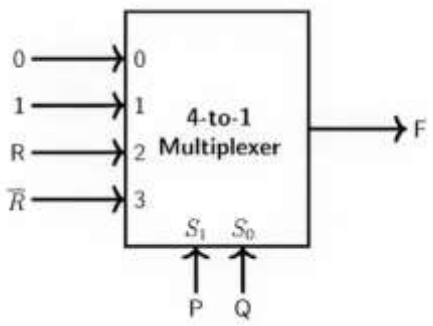
Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2015 SET-2 NAT (1-MARKS)

Q10. A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4-bit ripple-carry binary adder is implemented by using four full adders. The total propagation time of this 4-bit binary adder in microseconds is _____.

GATE CSE – 2014 SET-1 MCQ (2-MARKS)

Q11. Consider the 4-to-1 multiplexer with two select lines S_1 and S_0 given below



The minimal sum of products form of the output X is

(a) $\bar{P}Q + Q\bar{R} + P\bar{Q}R$
(b) $\bar{P}Q + \bar{P}Q\bar{R} + PQ\bar{R} + P\bar{Q}R$
(c) $PQR + \bar{P}Q\bar{R} + Q\bar{R} + P\bar{Q}R$
(d) PQR

GATE CSE – 2014 SET-3 MCQ (1-MARKS)

Q12. Consider the following combinational function block involving four Boolean variables x, y, a, b where x, a, b are inputs and y is the output.

```
f(x,y,a,b)
{
if(x is 1)y =a;
else y=b;
}
```

Which one of the following digital logic blocks is the most suitable for implementing this function?

(a) Full adder
(b) Priority encoder
(c) Multiplexer
(d) Flip-flop

GATE CSE – 2013 MCQ (1-MARKS)

Q13. In the following truth table, V = 1 if and only if the input is valid. In the following truth table, V = 1 if and only if the input is valid.

Inputs

D_0	D_1	D_2	D_3
0	0	0	0
1	0	0	0
x	1	0	0
x	**x**	1	0
x	**x**	**x**	1

Output

X_0	X_1	V
x	x	0
0	0	1
0	1	1
1	0	1
1	1	1

What function does the truth table represent?

(a) Priority encoder
(b) Decoder
(c) Multiplexer
(d) Demultiplexer

GATE CSE – 2012 MCQ (1-MARKS)

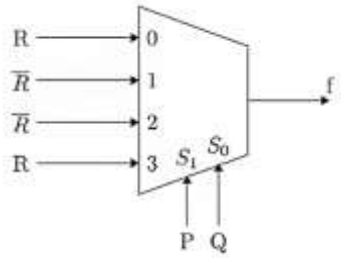
Q14. The amount of ROM needed to implement a 4 bit multiplier is?

(a) 64 bits
(b) 128 bits
(c) 1 Kbits
(d) 2 Kbits

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2010 MCQ (1-MARKS)

Q15. The Boolean expression of the output of the multiplexer shown below is



(a) $\overline{(P \oplus Q \oplus R)}$
(b) $P \oplus Q \oplus R$
(c) $P + \bar{Q} + R$
(d) $\overline{(P + Q + R)}$

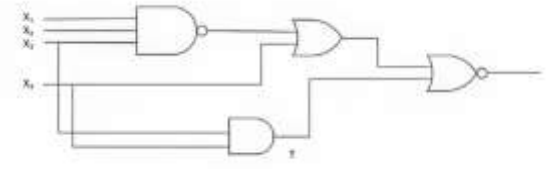
GATE CSE – 2009 MCQ (1-MARKS)

Q16. Which one of the following adders reduces carry propagation delay by generating carries in advance?

(a) Ripple Carry Adder
(b) Carry Lookahead Adder
(c) Serial Adder
(d) Half Adder

GATE CSE – 2007 MCQ (1-MARKS)

Q17. The line in the following figure is permanently connected to the ground.



Which of the following inputs (X1 X2 X3 X4) will detect the fault?

(a) 0000
(b) 0111
(c) 1111
(d) None of these

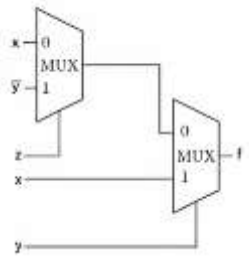
GATE CSE – 2007 MCQ (1-MARKS)

Q18. In a look-ahead carry generator, the carry generates function G_i and the carry propagate function P_i for inputs A_i and B_i are given by:
$P_i = A_i \oplus B_i$ and $Gi = A_iB_i$
The expressions for the sum bit S_i and the carry bit C_{i+1} of the look ahead carry adder are given by:
$S_i = P_i \oplus C_i$ and $Ci_{+1} = G_i + P_iC_i$, where C_0 is the input carry.
Consider a two-level logic implementation of the look-ahead carry generator. Assume that all P_i are available for the carry generator circuit and that the AND and OR gates can have any number of inputs. The number of AND gates and OR gates needed to implement the look-ahead carry generator for a 4-bit adder with S_3, S_2, S_1, S_0 and C_4 as its outputs are respectively:

(a) 6,3
(b) 10,4
(c) 6,4
(d) 10,5

GATE CSE – 2006 MCQ (1-MARKS)

Q19. Consider the circuit below. Which one of the following options correctly represents f(x,y,z)



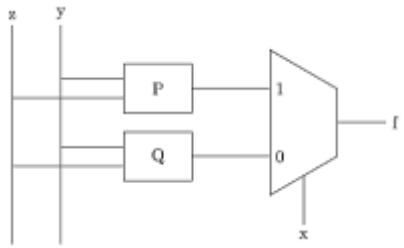
(a) $x\bar{z} + xy + \bar{y}z$
(b) $x\bar{z} + xy + \bar{y}\bar{z}$
(c) $xz + xy + \bar{y}\bar{z}$
(d) $xz + x\bar{y} + \bar{y}z$

GATE CSE – 2006 MCQ (1-MARKS)

Q20. The majority function is a Boolean function f(x,y,z)that takes the value 1 whenever a majority of the variables x, y, z are 1. In the circuit diagram for

Check Out Our GATE Courses & Secure Your Dream Rank!

the majority function shown below, the logic gates for the boxes labeled P and Q are, respectively.



(a) XOR, AND
(b) XOR, XOR
(c) OR, OR
(d) OR, AND

GATE CSE – 2006 MCQ (1-MARKS)

Q21. Given two three-bit numbers $a_2\ a_1\ a_0$ and $b_2\ b_1\ b_0$ and c the carry in, the function that represents the carry generate function when these two numbers are added is:

(a) $a_2b_2 + a_2a_1b_1 + a_2a_1a_0b_0 + a_2a_0b_1b_0 + a_1b_2b_1 + a_1a_0b_2b_0 + a_0b_2b_1b_0$
(b) $a_2b_2 + a_2b_1b_0 + a_2a_1b_1b_0 + a_1a_0b_2b_1 + a_1a_0b_2 + a_1a_0b_2b_0 + a_2a_0b_1b_0$
(c) $a_2 + b_2 + (a_2 \oplus b_2)\ (a_1 + b_1 + (a_1 \oplus b_1)(a_0 + b_0))$
(d) $a_2b_2 + \bar{a}_2a_1b_1 + \bar{a}_2a_1a_0b_0 + \bar{a}_2a_0b_1b_0 + a_1b_2b_1 + \bar{a}_1a_0b_2b_0 + a_0b_2b_1b_0$

ANSWER KEY

1. (6)	**2. (a, b, c)**	**3. (c)**
4. (c)	**5. (a)**	**6. (1034)**
7. (5)	**8. (−1)**	**9. (b)**
10. (12)	**11. (a)**	**12. (c)**
13. (a)	**14(d)**	**15. (b)**
16. (b)	**17. (d)**	**18. (b)**
19. (a)	**20. (d)**	**21. (a)**

HINTS & SOLUTIONS

1. **Answer:(6)**
Solution:
Block Q is an active-high 2:4 decoder with inputs A, B. Hence its outputs are:
[D0 = A'B', D1 = A'B, D2 = AB', D3 = AB]
These outputs are connected to the 4:1 multiplexer (M) as follows:
- Input 0 → D0
- Input 1 → 0
- Input 2 → 1
- Input 3 → D3

The select lines of the multiplexer are S1 S0.
Case 1: (S1S0 = 00)
MUX selects input 0
[Y = D0 = A'B']
This becomes 1 only when
[A=0,; B=0] Number of combinations = 1
Case 2: (S1S0 = 01)
MUX selects input 1
[Y = 0]
Output is always 0
Number of combinations = 0
Case 3: (S1S0 = 10)
MUX selects input 2
[Y = 1]
Output is always 1
Possible values of (A,B):
[00,01,10,11]
Number of combinations = 4
Case 4: (S1S0 = 11)
MUX selects input 3
[Y = D3 = AB]
This becomes 1 only when
[A=1,; B=1]
Number of combinations = 1
Total combinations producing (Y=1) [1 + 0 + 4 + 1 = 6]

2. **Answer : (A,B,C)**
Solution:
Using diagram, we can say
F = (XY)' + X' + XY'
here Apply Demorgan's law on (XY)' = X' + Y'
F= X' + Y' + X' + XY'
X' + Y' + XY'
$X' + Y' + XY' \rightarrow$ simplifies to $X' + Y'$
$(XY)' + X' + XY' \rightarrow$ same expression
$X + Y'$ (incorrect)

3. Answer : (C)

Solution:

$f = \bar{A} + \bar{A}\bar{C} + A\bar{B}C$

Since, final output is given by a 2 × 1 MUX i.e. M3, so we have to write

f as $f = \bar{B}(.) + B(.)$

$f = \bar{A}(1 + \bar{C}) + A\bar{B}C$

$f = \bar{A} + A\bar{B}C$

$f = \bar{A}(B + \bar{B})(C + \bar{C}) + A\bar{B}C$

$f = \bar{A}BC + \bar{A}B\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{B}AC$

$f = \bar{B}(\bar{A}C + \bar{A}\bar{C} + AC) + B(\bar{A}C + \bar{A}\bar{C})$

Now, our job is completed to write f as $f = \bar{B}(.) + B(.)$

Now, for $(\bar{A}C + \bar{A}\bar{C} + AC)$ we have to go upper left side MUX i.e. M1 and we get:

1) $\bar{A}C$ i.e. $\bar{S}1S0$ i.e. X1 = 1

2) $\bar{A}\bar{C}$ i.e. $\bar{S}1\bar{S}0$ i.e. X0 = 1

3) AC i.e. S1S0 i.e. X3 = 1

Now, from left below MUX i.e. M2, we get:

4) $\bar{A}C$ i.e. $\bar{S}1S0$ i.e. X5 = 1

5) $\bar{A}\bar{C}$ i.e. $\bar{S}1\bar{S}0$ i.e. X4 = 1

Hence, X0 = X1 = X3 = X4 = X5 = 1 and X2 = X6 = X7 = 0

So, order is: 11011100

Therefore, (C)

4. Answer : (C)

Solution:

Given:

S = 1024 words (2^{10}) → needs 10-bit address → use 10 → 2^{10} decoder (P)

R = 32 words → needs 5-bit address → use 5 → 2^5 decoder (Q)

M selects between S and R → use 2:1 multiplexer (T)

Final:

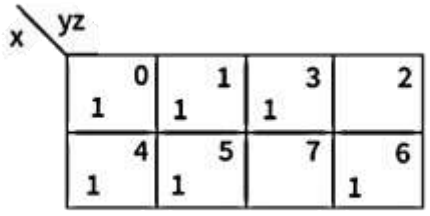
P = 10 : 2^{10} decoder

Q = 5 : 2^5 decoder

T = 2 : 1 multiplexer

5. Answer:(A)

Solution :



Given:

$f(x,y,z) = \Sigma m(0,1,3,4,5,6)$

Using 4×1 MUX with select lines:

$S_1 = y$, $S_0 = z$

Evaluate inputs:

yz = 00 → m0, m4 → both present → I0 = 1

yz = 01 → m1, m5 → both present → I1 = 1

yz = 10 → m2 (0), m6 (1) → depends on x → I2 = x

yz = 11 → m3 (1), m7 (0) → depends on x' → I3 = x'

Final inputs:

$[I_0, I_1, I_2, I_3] = [1, 1, x, x']$

6. Answer : 1034

Solution:

The size of the decoder required is 10 x 2^{10}

i.e 10 X 1024.

Each output line of the decoder is connected to one of the 1K(= 1024) rows of RAM.

Each row stores 1 Byte.

m=10 and n=1024

m + n = 1034.

7. Answer : 5

Solution:

There are 32 registers in all. At any given time, just one register must be selected. A 2^5x1 Multiplexer with 5 select lines selects one of the 32(= 2^5)registers at a time depending on the selection input. The content of the selected register will be sent to the Accumulator through the output line.

8. Answer : -1

Solution:

Given A = 1 In decimal form,

In 2's complement form, binary representation of A in binary form = 00000001.

In case of ripple carry adder, for sum we need 2 XOR gates.

For the longest latency, we must have $C_{in} = 1$ at every stage of the ripple carry adder.

Here, we have to find the value of B such that we get $C_{in} = 1$ at every stage and $C_{out} = 1$ at the last stage.

Maximum delay is when we take $B = -1$ in decimal (111111111 − In binary)

On adding A + B, we get a carry up to the carry flag I e. longest latency

Check Out Our GATE Courses & Secure Your Dream Rank!

9. **Answer : (B)**
Solution:

In a Ripple Carry Adder (RCA), each stage must wait for the carry
from the previous stage. Therefore, carry propagates sequentially
through all n stages.

Time complexity of RCA = $\Theta(n)$

Carry Look Ahead Adder (CLA):

A CLA removes sequential carry propagation by computing carries
in parallel using Generate and Propagate signals.

For each bit i:

Generate: $G_i = A_iB_i$
Propagate: $P_i = A_i \oplus B_i$
Carry equation:
$C_{i+1} = G_i + P_iC_i$

Using expansion:

$C_1 = G_0 + P_0C_0$
$C_2 = G_1 + P_1G_0 + P_1P_0C_0$
$C_3 = G_2 + P_2G_1 + P_2P_1G_0 + P_2P_1P_0C_0$
...
Cn can be expressed directly using logic expressions.

Fan-in restriction:
The question states gates have fan-in ≤ 2.

So large AND/OR operations cannot be done in one step.
They must be built as a tree of 2-input gates.

A tree structure computing n inputs has depth:

Depth = $\log_2(n)$

Hence carry computation delay grows logarithmically.

Therefore,
Total addition time $\propto$ depth of carry computation tree
= $\Theta(\log n)$

10. **Answer** :(12)
Solution:

In Ripple carry adder, a stage doesn't wait for the "full output" of the previous stage. It only waits for the previous stage's carry to reach to itself.

Delay of ExOR = 2 × (delay of AND/OR)
= 2 × 1.2
= 2.4 μs

Now, Full Adder is implemented using 2 Half Adders (HA) and one OR gate.
So,
Delay of Sum (S) = 2 × delay of ExOR
(since sum output in FA uses two XOR gates)
= 2 × 2.4
= 4.8 μs
Delay of Carry (C) = 1 XOR + 1 AND + 1 OR
(critical path for carry output in FA)
= 2.4 + 1.2 + 1.2
= 4.8 μs
Now, Ripple Carry Adder is implemented using 4 Full Adders.
At t = 4.8 μs,
both S0 and C1 are valid.
To get S1, C1 must be valid (previous stage carry is needed).
Delay of S1 = 4.8 + 2.4
= 7.2 μs
Therefore, S1 is valid at 7.2 μs.
Now, C2 is valid after 7.2 μs because C2 needs only AND and OR gates after C1.
Delay of C2 = 4.8 + 2.4
= 7.2 μs
Therefore, C2 is valid after 7.2 μs.
To find S2, we need C2:
Delay of S2 = Delay of C2 + delay of XOR
= 7.2 + 2.4
= 9.6 μs
Therefore, S2 is valid after 9.6 μs.
Delay of C3 = Delay of C2 + delay of AND and OR
= 7.2 + 2.4
= 9.6 μs
Therefore, C3 is valid after 9.6 μs.
Similarly, to find S3 we need C3:
Delay of S3 = Delay of C3 + delay of XOR
= 9.6 + 2.4
= 12 μs
Therefore, S3 is valid after 12 μs.
Final propagation delay of the 4-bit Ripple Carry Adder = 12 μs

Check Out Our GATE Courses & Secure Your Dream Rank!

11. Answer:(A)
Solution:
S_1 and S_0 are used to select the input given to be given as output.

S_1	S_0	Output
0	0	0
0	1	1
1	0	R
1	1	R′

So, output becomes 1 for

$$S_1'S_0 + S_1S_0'R + S_1S_0R'$$
$$= P'Q + PQ'R + PQR'$$
$$= P'Q + PQR' + PQ'R$$
$$= Q(P' + PR') + PQ'R$$
$$= Q(P' + R') + PQ'R(\because A + A'B = A + B)$$
$$= P'Q + QR' + PQ'R$$

12. Answer:(C)
Solution:
The given function block acts as a switch that selects between two different inputs based on a condition:
- When x = 1, the output y takes the value of input a.
- When x = 0 (the else condition), the output y takes the value of input b.

This is the exact definition of a **2-to-1 Multiplexer (MUX)**.
- x acts as the **select line**.
- a and b act as the **data inputs**.
- y is the **output**.

The Boolean expression for this logic is:
y = xa + x'b
Why the other options are incorrect:
- **(a) Full adder:** This is an arithmetic circuit that adds three bits (two significant bits and a carry-in) to produce a sum and a carry-out. It does not select or route data.
- **(b) Priority encoder:** This circuit compresses multiple binary inputs into a smaller number of outputs, outputting the binary code of the highest-priority active input.
- **(d) Flip-flop:** The problem explicitly states this is a *combinational* function block. A flip-flop is a *sequential* logic element used to store memory/state, which depends on a clock signal.

13. Answer:(A)
Solution:
It is a $2^2 \times 2$ encoder. The inputs have priorities. So, it is a priority encoder

14. Answer : (D)
Solution:
A ROM cannot be written. So, to implement a 4-bit multiplier we must store all the possible combinations of $2^4 \times 2^4$ inputs and their corresponding 8 output bits giving a total of $2^4 \times 2^4 \times 8$ bits = 2048 bits. So, (D) is the answer.
PS: We are not storing the input bits explicitly -- those are considered in order while accessing the output 8 bits. In this way, by storing all the possible outputs in order we can avoid storing the input combinations.

15. Answer: (B)
Solution:
$f = \overline{S_1}\,\overline{S_0}R + \overline{S_1}S_0\overline{R} + S_1\overline{S_0}\overline{R} + S_1S_0R$
= Q'P'R + Q'PR' + QP'R' + QPR
= Q'(P ⊕ R) + Q(P ⊕ R)'
= Q ⊕ P ⊕ R = P ⊕ Q ⊕ R

P	Q	R	f	P ⊕ Q ⊕ R
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

16. Answer: (B)
Solution:
The **Carry Lookahead Adder (CLA)** is specifically designed to overcome the **carry propagation delay** found in traditional adders. It does this by using a logic circuit that **generates carries in advance** simultaneously for all stages, rather than waiting for a carry to "ripple" through each preceding bit
$G_i = A_iB_i$
Propagate:
$P_i = A_i \oplus B_i$
Carry equation:
$C_{i+1} = G_i + P_iC_i$
Because carries are generated in parallel, delay is significantly reduced.

Check Out Our GATE Courses & Secure Your Dream Rank!

17. Answer: (D)

Solution:

To detect the faull, we should get an unexpected output. The final gate here is a NOR gate which produces output 0 if either of its input is 1 and else 1 , ie, the output will be 0 for inputs (0,1), (1,0) and (1,1) and output will be 1 for (0,0).

By grounding T is at 0 . So, we can ignore the inputs (1,0) and (0,0) to the final NOR gate as they won't be detecting faults. Now, expected (1,1) input will become (1,0) due to grounding of T but produces same output 0 as for (1,1). Hence this also cannot detect the defect. So, to detect the defect, the input to the final gate must be (0,1) which is expected to produce a 0 but will produce a 1 due to grounding of T.

Now, for (0,1) input for the final gate, we must have,

$$X_3 = X_4 = 1$$

But if $X_4 = 1$,the OR gate makes 1 output and we won't get (0,1) input for the final gate. This means, no input sequence can detect the fault of the circuit.

Alternatively, we can write equation for the circuit as

$$\begin{aligned}&\left(((x_1 \cdot x_2 \cdot x_3)' + x_4) + x_3 \cdot x_4\right)' \\ &= ((x_1 \cdot x_2 \cdot x_3)' + x_4)' \cdot (x_3 \cdot x_4)' \\ &= x_1 \cdot x_2 \cdot x_3 \cdot x_4' \cdot (x_3' + x_4') \\ &= x_1 \cdot x_2 \cdot x_3 \cdot x_4'\end{aligned}$$

For the faulty circuit output will be

$$((x_1 \cdot x_2 \cdot x_3)' + x_4)' = x_1 \cdot x_2 \cdot x_3 \cdot x_4'$$

So, there is no effect of T being grounded here. Answer is D option.

18. Answer: B

Solution:

$$C1 = G0 + C0 \cdot P0$$
$$C2 = G1 + G0 \cdot P1 + C0 \cdot P0 \cdot P1$$
$$C3 = G2 + G1 \cdot P2 + G0 \cdot P1 \cdot P2 + C0 \cdot P0 \cdot P1 \cdot P2$$
$$C4 = G3 + G2 \cdot P3 + G1 \cdot P2 \cdot P3 + G0 \cdot P1 \cdot P2 \cdot P3 + C0 \cdot P0 \cdot P1 \cdot P2 \cdot P3$$

Carry is generated in 3^{rd} stage OR
Carry is generated in 2^{sd} stage AND propagated through 3^{rd} stage OR
Carry is generated in 1^{st} stage AND propagated through 2^{st} , 3^{rd} stage OR
Carry is generated in 0^{th} stage AND propagated through 1^{*}, 2^{nd} , 3^{rd} stage OR
Initial carry C_0 is propagated through 0^{th} , 1^{nt} , 2^{nd} , 3^{rd} stage.

- 1 AND & 1 OR gate for $C1$
- 2 AND & 1 OR gate for C2
- 3 AND & 1 OR gate for C3
- 4 AND & 1 OR gale for $C4$

To implement the look-ahead carry generator, 10 AND & 4 OR gates are required.

Alternate Solution:

Number of AND Gate required is given by = n(n+1)/2
= 4*5/2
=10
Number of OR Gate is given by = n
n=4

19. Answer: B

Solution:

First MUX (left): $f_1 = \bar{z}x + zy$
Second MUX (right): $f = \bar{y}f_1 + yx$

$$f = \bar{y}(x\bar{z} + zy) + xy$$

$$f = x\bar{y}\bar{z} + 0 + xy$$

$$f = x(\bar{y}\bar{z} + y)$$

$$f = x(y + \bar{z}) = \mathbf{xy + x\bar{z}}$$

Using the Consensus Theorem ($AB + BC + AC = AB + BC$), we can expand our result:

- Add the redundant consensus term between xy and $x\bar{z}$ using $\bar{y}$:

$$\mathbf{xy + x\bar{z} + \bar{y}\bar{z}}$$

20. Answer: D
Solution:

$xP + \bar{x}Q = f$
Or $x(y\ op_1\ z) + \bar{x}(y\ op_2\ z) = f \quad \rightarrow (1)$

X	Y	Z	Output(f)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

x\yz	00	01	11	10
0	...	...	1	...
1	...	1	1	1

$f \Rightarrow xz + xy + yz$
$\Rightarrow xz + xy + (x + \bar{x})yz$
$\Rightarrow xz + xy + xyz + \bar{x}yz$
$\Rightarrow x(z + y + yz) + \bar{x}yz$
$\Rightarrow x[z + y(1 + z)] + \bar{x}(y \bullet z)$
$\Rightarrow x(z + y) + \bar{x}(y \bullet z) \quad \rightarrow (2)$
Comparing (1) and (2) we get,
OP1 = + (OR)
OP2 = • (AND)

21. Answer: (A)
Solution:
For any bit position 'i', the intermediate signals are:
Carry Generate (G_i) = $a_i \cdot b_i$
Carry Propagate (P_i) = $a_i + b_i$
For a 3-bit block (bits 0, 1, and 2), a carry is generated outwards if the most significant bit generates one, or if a lower bit generates one and it propagates all the way through.
$Gout = G_2 + P_2 \cdot G_1 + P_2 \cdot P_1 \cdot G_0$
$G_{out} = a_2b_2 + (a_2 + b_2)(a_1b_1) + (a_2 + b_2)(a_1 + b_1)(a_0b_0)$
Expanding: $(a_2 + b_2)(a_1b_1)$
$= a_2a_1b_1 + b_2a_1b_1$
Expanding: $(a_2 + b_2)(a_1 + b_1)(a_0b_0)$
First, multiply the binomials:
$(a_2a_1 + a_2b_1 + b_2a_1 + b_2b_1)$
Next, multiply by a_0b_0:
$a_2a_1a_0b_0 + a_2b_1a_0b_0 + b_2a_1a_0b_0 + b_2b_1a_0b_0$
Putting all the expanded terms together:
$G_{out} = a_2b_2 + a_2a_1b_1 + b_2a_1b_1 + a_2a_1a_0b_0 + a_2b_1a_0b_0 + b_2a_1a_0b_0 + b_2b_1a_0b_0$

Check Out Our GATE Courses & Secure Your Dream Rank!

4 SEQUENTIAL CIRCUIT

GATE CSE – 2026 SET-1 MCQ (2-MARKS)

Q1. Consider a 2-bit saturating up/down counter that performs the saturating up count when the input P is 0, and the saturating down count when P is 1. The Next State table of the counter is as shown. The counter is built as a synchronous sequential circuit using D flip-flops.

Input	Current State		Next State	
P	Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	0	1
1	1	1	1	0

(a) $D_1 = PQ_1 + \bar{P}Q_0 + Q_1Q_0$
$D_0 = PQ_0 + \bar{P}Q_1 + Q_1\overline{Q_0}$

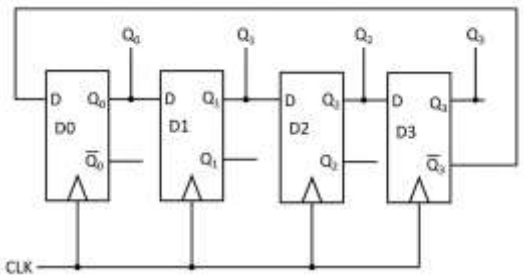
(b) $D_1 = \bar{P}Q_1 + \bar{P}Q_0 + Q_1Q_0$
$D_0 = \overline{P\overline{Q_0}} + \bar{P}Q_1 + Q_1\overline{Q_0}$

(c) $D_1 = \bar{P}\bar{Q}_1 + \bar{P}Q_0 + Q_1Q_0$
$D_0 = \bar{P}Q_0 + \bar{P}Q_1 + Q_1\bar{Q}_0$

(d) $D_1 = PQ_1 + \bar{P}Q_0 + Q_1Q_0$
$D_0 = P\overline{Q_0} + \bar{P}Q_1 + Q_1\overline{Q_0}$

GATE CSE – 2025 SET-1 NAT (2-MARKS)

Q2. Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning to the initial state is . (Answer in integer)

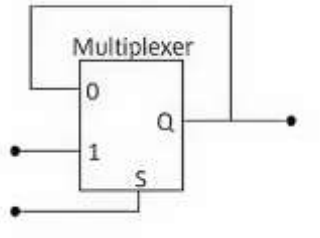


GATE CSE – 2025 SET-2 NAT (2-MARKS)

Q3. In a 4-bit ripple counter, if the period of the waveform at the last flip-flop is 64 microseconds, then the frequency of the ripple counter in kHz is _____. (Answer in integer)

GATE CSE – 2023 MCQ (1-MARKS)

Q4. The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure.



Match the functional equivalence of this circuit to one of the following options

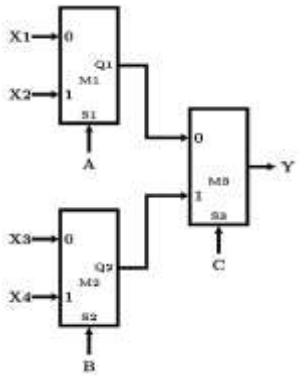
(a) D Flip-flop
(b) D Latch
(c) Half-adder
(d) Demultiplexer

GATE CSE – 2024 SET 1 NAT (2-MARKS)

Q5. Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as shown below. X1 and X2 are inputs of M1. X3 and X4 are

Check Out Our GATE Courses & Secure Your Dream Rank!

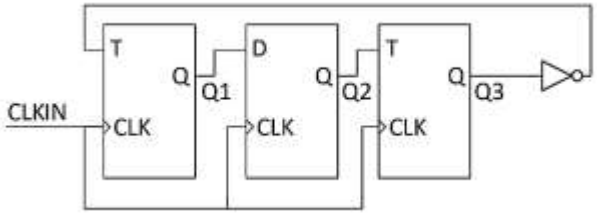
inputs of M2. A, B, and C are select lines of M1, M2, and M3, respectively.



For an instance of inputs $\mathbf{X1 = 1, X2 = 1, X3 = 0}$, and $\mathbf{X4 = 0}$, the number of combinations of A, B, C that give the output $\mathbf{Y = 1}$ is _____.

GATE CSE – 2023 MCQ (2-MARKS)

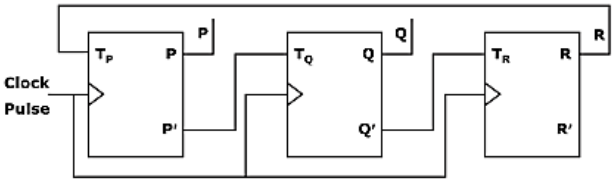
Q6. Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q1, Q2 and Q3 have values 0, 1 and 1, respectively. Which one of the given values of (Q1, Q2, Q3) can NEVER be obtained with this digital circuit?



(a) (0, 0, 1)
(b) (1, 0, 0)
(c) (1, 0, 1)
(d) (1, 1, 1)

GATE CSE – 2021 SET-1 MCQ (1-MARKS)

Q7. Consider a 3-bit counter, designed using T flip-flops, as shown below :

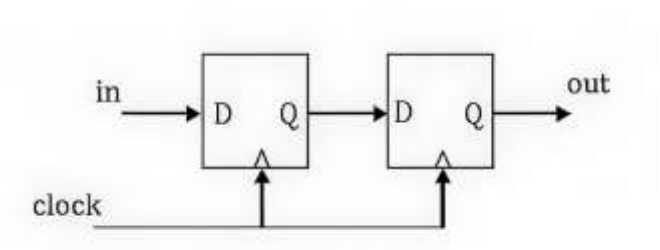


Assuming the initial state of the counter given by PQR as 000, what are the next three states?

(a) 001, 010, 111
(b) 011, 101, 000
(c) 001, 010, 000
(d) 011, 101, 111

GATE CSE – 2018 NAT (1-MARKS)

Q8. Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops. The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of "in" is _____



GATE CSE – 2017 SET-2 MCQ (2-MARKS)

Q9. The next state table of a 2-bit saturating up-counter is given below.

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

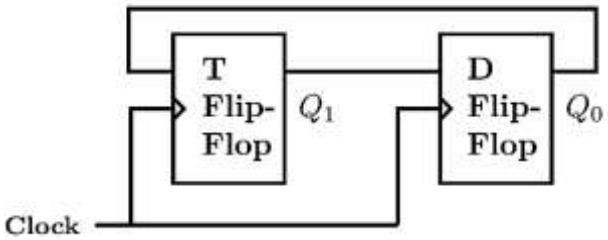
Check Out Our GATE Courses & Secure Your Dream Rank!

The counter is built as a synchronous sequential circuit using T flip-flops. The expressions for T_1 and T_0 are

(a) $T_1 = Q_0Q_1$, $T_0 = Q'_0Q'_1$
(b) $T_1 = Q'_1Q_0$, $T_0 = Q'_1 + Q'_0$
(c) $T_1 = Q_1 + Q_0$, $T_0 = Q'_1Q'_0$
(d) $T_1 = Q'_1Q_0$, $T_0 = Q_1 + Q_0$

GATE CSE – 2017 SET-1 MCQ (2-MARKS)

Q10. Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop. Initially, both Q0 and Q1 are set to 1 (before the 1st clock cycle). The outputs



(a) Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 00 respectively

(b) Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 01 respectively

(c) Q_1Q_0 after the 3rd cycle are 00 and after the 4th cycle are 11 respectively

(d) Q_1Q_0 after the 3rd cycle are 01 and after the 4th cycle are 01 respectively

GATE CSE – 2016 SET-1 NAT (1-MARKS)

Q11. We want to design a synchronous counter that counts the sequence 0-1-0-2-0-3 and then repeats. The minimum number of J-K flip-flops required to implement this counter is _____.

GATE CSE – 2015 SET-1 MCQ (2-MARKS)

Q12. A positive edge-triggered D flip-flop is connected to a positive edge-triggered JK flipflop as follows. The Q output of the D flip-flop is connected to both the J and K inputs of the JK flip-flop, while the Q output of the JK flip-flop is connected to the input of the D flip-flop. Initially, the output of the D flip-flop is set to logic one and the output of the JK flip-flop is cleared. Which one of the following is the bit sequence (including the initial state) generated at the Q output of the JK flip-flop when the flip-flops are connected to a free-running common clock? Assume that J = K = 1 is the toggle mode and J = K = 0 is the state-holding mode of the JK flip-flop. Both the flip-flops have non-zero propagation delays.

(a) 0110110..
(b) 0100100...
(c) 01110110...
(d) 011001100...

GATE CSE – 2015 SET-1 MCQ (1-MARKS)

Q13. Consider a 4-bit Johnson counter with an initial value of 0000. The counting sequence of this counter is

(a) 0, 1, 3, 7, 15, 14, 12, 8, 0
(b) 0, 1, 3, 5, 7, 9, 11, 13, 15, 0
(c) 0, 2, 4, 6, 8, 10, 12, 14, 0
(d) 0, 8, 12, 14, 15, 7, 3, 1, 0

GATE CSE – 2014 SET-2 MCQ (1-MARKS)

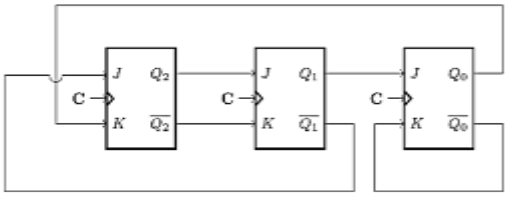
Q14. Let k = 2n. A circuit is built by giving the output of an n-bit binary counter as input to an n-to-2n bit decoder. This circuit is equivalent to a **?**

(a) k-bit binary up counter
(b) k-bit binary down counter
(c) k-bit ring counter
(d) k-bit Johnson counter

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2014 SET-3 MCQ (2-MARKS)

Q15.



The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycles is

(a) 001, 010, 011
(b) 111, 110, 101
(c) 100, 110, 111
(d) 100, 011, 001

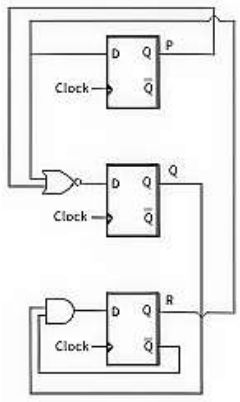
GATE CSE – 2011 MCQ (1-MARKS)

Q16. The minimum number of D flip-flops needed to design a mod-258 counter is

(a) 9
(b) 8
(c) 512
(d) 258

GATE CSE – 2011 MCQ (1-MARKS)

Q17. Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration

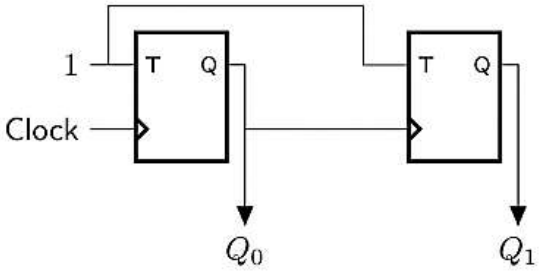


If at some instance prior to the occurrence of the clock edge, and have a value , and respectively, what shall be the value of after the clock edge?

(a) 000
(b) 001
(c) 010
(d) 011

GATE CSE – 2010 MCQ (1-MARKS)

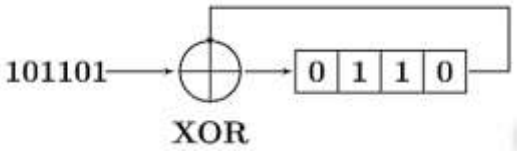
Q18. In the sequential circuit shown below, if the initial value of the output is . What are the next four values of ?



(a) 11, 10, 01, 00
(b) 10, 11, 01, 00
(c) 10, 00, 01, 11
(d) 11, 10, 00, 01

GATE CSE – 2007 MCQ (1-MARKS)

Q19. What is the final value stored in the linear feedback shift register if the input is 1011101?



(a) 0110
(b) 1011
(c) 1101
(d) 1111

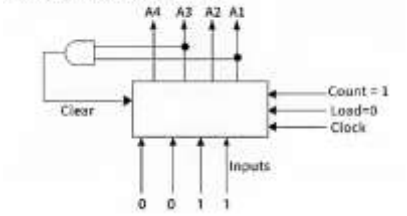
GATE CSE – 2007 MCQ (1-MARKS)

Q20. The control signal functions of a - binary counter are given below (where is "don't care"):

Check Out Our GATE Courses & Secure Your Dream Rank!

Clear	Clock	Load	Count	Function
1	X	X	X	Clear to 0
0	X	0	0	No Change
0	↑	1	X	Load Input
0	↑	0	1	Count Next

The counter is connected as follows

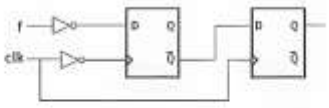


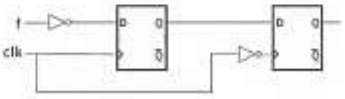
Assume that the counter and gate delays are negligible. If the counter starts at then it cycles through the following sequence

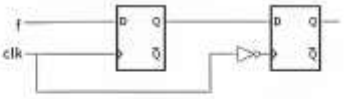
(a) 0, 3, 4
(b) 0, 3, 4, 5
(c) 0, 1, 2, 3, 4
(d) 0, 1, 2, 3, 4, 5

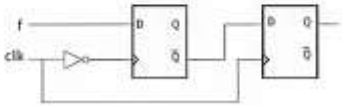
GATE CSE – 2006 MCQ (1-MARKS)

Q21. You are given a free running clock with a duty cycle of 50% and a digital waveform f which changes only at the negative edge of the clock. Which one of the following circuits (using clocked D flip-flops) will delay the phase of f by 180°?

A.


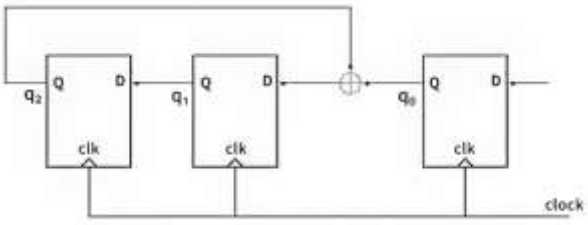
B.


C.


D.


GATE CSE – 2006 MCQ (1-MARKS)

Q22. Consider the circuit in the diagram. The operator represents Ex-OR. The D flip-flops are initialized to zeroes (cleared)

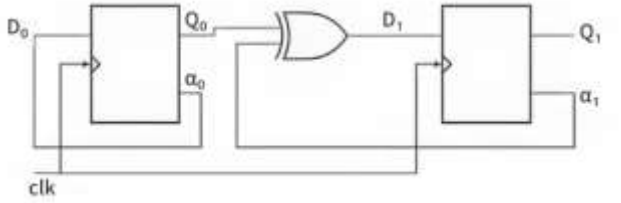


The following data: is supplied to the "data" terminal in nine clock cycles. After that the values of q2q1qo are:

(a) 000
(b) 001
(c) 010
(d) 101

GATE CSE – 2005 MCQ (1-MARKS)

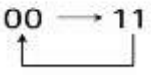
Q23. Consider the following circuit



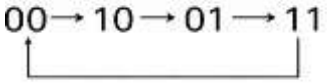
The flip-flops are positive edge triggered s. Each state is designated as a two-bit string . Let the initial state be The state transition sequence is

(a)

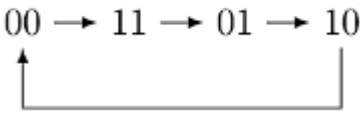
(b)



(c)



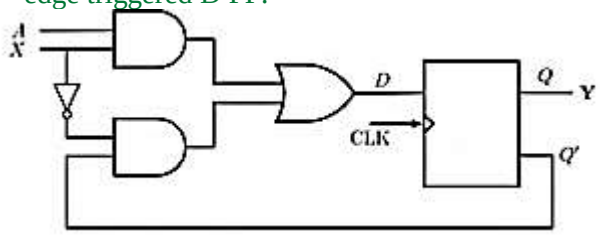
(d)



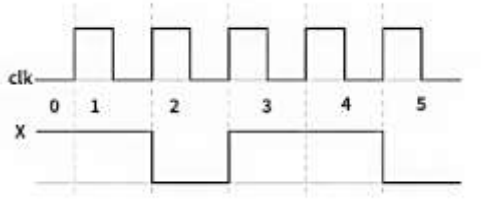
Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2005 MCQ (1-MARKS)

Q24. Consider the following circuit involving a positive edge triggered D FF.



Consider the following timing diagram. Let represents the logic level on the line in the -th clock period



Let represent the complement of . The correct output sequence on over the clock periods through is:

(a) A0 A1 A1′ A3 A4
(b) A0 A1 A2 A3 A4
(c) A1 A2 A2′ A3 A4
(d) A1 A2 A3 A4 A5′

ANSWER KEY

1.(b)	**2.(8)**	**3.(250)**
4.(b)	**5. (4)**	**6.(a)**
7.(a)	**8.(2)**	**9.(b)**
10.(b)	**11.(3-4)**	**12.(a)**
13.(d)	**14.(c)**	**15.(c)**
16. (a)	**17.(d)**	**18.(a)**
19.(a)	**20.(c)**	**21.(c)**
22.(a)	**23.(d)**	**24.(a)**

HINTS & SOLUTIONS

1. Answer:(B)
Solution:
For a D flip-flop, the characteristic equation is $Q_{next} = D$. Therefore, the required inputs D_1 and D_0 are exactly equal to the next state values $Q1_{_next}$ and Q_{0_next} respectively.
Let's extract the min-terms from the provided Next State table and simplify them using Boolean algebra (or K-maps). The variables are P, Q_1, and Q_0.
1. Solving for D_1 (Q_{1_next})
Looking at the $Q_{1_next\ column}$, the output is 1 for the following input combinations of P, Q_1, Q_0:
0, 0, 1 -> m1
0, 1, 0 -> m2
0, 1, 1 -> m3
1, 1, 1 -> m7
The boolean expression in sum-of-min-terms form is: D1 = Sum of min-terms (1, 2, 3, 7)
Now, let's group these min-terms to simplify the expression:
- Group 1 (m1 and m3): $P'Q_1'Q_0 + P'Q_1Q_0 = P'Q_0$
- Group 2 (m2 and m3): $P'Q_1Q_0' + P'Q_1Q_0 = P'Q_1$
- Group 3 (m3 and m7): $P'Q_1Q_0 + PQ_1Q_0 = Q_1Q_0$

Combining these simplified groups gives us the final expression for D1: $D1 = P'Q_1 + P'Q_0 + Q_1Q_0$
2. Solving for D_0 (Q_{0_next})
Looking at the Q_{0_next} column, the output is 1 for the following input combinations of P, Q_1, Q_0:
- 0, 0, 0 -> m0
- 0, 1, 0 -> m2
- 0, 1, 1 -> m3
- 1, 1, 0 -> m6

The boolean expression in sum-of-min-terms form is: D0 = Sum of min-terms (0, 2, 3, 6)
Now, let's group these min-terms to simplify:
Group 1 (m0 and m2): $P'Q_1'Q_0' + P'Q_1Q_0' = P'Q_0'$
Group 2 (m2 and m3): $P'Q_1Q_0' + P'Q_1Q_0 = P'Q_1$
Group 3 (m2 and m6): $P'Q_1Q_0' + PQ_1Q_0' = Q_1Q_0'$
Combining these simplified groups gives us the final expression for D_0: $D_0 = P'Q_0' + P'Q_1 + Q1Q_0'$
Conclusion
Comparing our derived equations to the given options:
- $D_1 = P'Q_1 + P'Q_0 + Q_1Q_0$
- $D_0 = P'Q_0' + P'Q_1 + Q1Q_0'$

Check Out Our GATE Courses & Secure Your Dream Rank!

2. Answer: (8)

Solution:

Characteristic of D Flip-Flop:
Q(next) = D
Let the outputs be:
Q0, Q1, Q2, Q3
Given connections:
D0 = Q3'
⇒ Q0(next) = Q3'
D1 = Q0
⇒ Q1(next) = Q0
D2 = Q1
⇒ Q2(next) = Q1
D3 = Q2
⇒ Q3(next) = Q2
Final Next-State Equations:
Q0(next) = Q3'
Q1(next) = Q0
Q2(next) = Q1
Q3(next) = Q2Assume initial state value as 0 :

D_0	D_1	D_2	D_3	Value
0	0	0	0	0
1	0	0	0	8
1	1	0	0	12
1	1	1	0	14
1	1	1	1	15
0	1	1	1	7
0	0	1	1	3
0	0	0	1	1
0	0	0	0	0

Assume initial state value as 2:

D_0	D_1	D_2	D_3	Value
0	0	1	0	2
1	0	0	1	9
0	1	0	0	4
1	0	1	0	10
1	1	0	1	13
0	1	1	0	6
1	0	1	1	11
0	1	0	1	5
0	0	1	0	2

The number of distinct states the circuit will go through before returning back to the initial state is 8.

3. Answer :(250)

Solution :

$$f_{\text{last}} = \frac{1}{T_{\text{lest}}}$$

$$f_{\text{last}} = \frac{1}{64 \times 10^{-6}}$$

$$f_{\text{last}} = 15625 \text{ Hz} = 15.625\text{kHz}$$

$$f_{\text{last}} = \frac{f_{\text{lnput}}}{2^4} = \frac{f_{\text{liput}}}{16}$$

$$f_{\text{linput}} = f_{\text{last}} \times 16$$

$$f_{\text{input}} = 15625 \times 16 = 250000 \text{ Hz} = 250\text{kHz}$$

4. Answer - (B)

Solution:

The circuit uses a multiplexer and the selection signal S determines whether the output Q is updated (transparent mode, S=1) or retained (hold mode, S=0).
There is no clock signal in this circuit. Without a clock, edge-triggering is not possible, so this circuit cannot function as a flip-flop.
Instead, the circuit operates based on the level of S, which aligns with the functionality of a D Latch.

5. Answer:(4)

Solution: The following characteristic table :

A	B	C	y=$\bar{C}$=1
0	0	0	1 ✓
0	0	1	0
0	1	0	1 ✓
0	1	1	0
1	0	0	1 ✓
1	0	1	0
1	1	0	1 ✓
1	1	1	0

The output Y is high for the inputs 000,010,100,110. Therefore the correct Answer is 4

6. Answer : (A)

Solution: From the given 3 sale courter made from T splops and D ipfops, the nex inpt sequence are as tolous

$$T_0 - \bar{Q}_2$$
$$D_1 - Q_0$$
$$T_3 - Q_1$$

Check Out Our GATE Courses & Secure Your Dream Rank!

Q_0	Q_1		Q_2	T_1	D_2	T_1	Q_1^*	Q_1^*	Q_2^*
0	1		1	0	0	1	0	0	0
0	0		0	1	0	0	1	0	0
1	0		0	1	1	0	0	1	0
0	1		0	1	0	1	1	0	1
1	0		1	0	1	0	1	1	1
1	1		1	0	1	1	1	1	0
1	1		0	1	1	1	0	1	1

we can see from the above table given counter contain sequence like $011 \rightarrow 000 \rightarrow 100 \rightarrow 010 \rightarrow 101 \rightarrow 111 \rightarrow 110$. The state (001 is missing.)

7. Answer: (B)

Solution :

Given Synchronous Sequential circuit (Same clock pulse applied) has 3-bit T flip flop so using 3 bit we have 2^3=8 combinations. Given P(MSB) and R(LSB). Construct Characteristic table. Here TP=Rn , TQ=Pn' , TQ=Qn' .

Previous state			Next State					
P_n	Q_n	R_n	P_{n+1}	Q_{n+1}	R_{n+1}	T_P	T_Q	T_R
0	0	0	0	1	1	0	1	1
0	0	1	1	1	0	1	1	1
0	1	0	0	0	0	0	1	0
0	1	1	1	0	1	1	1	0
1	0	0	1	0	1	0	0	1
1	0	1	0	0	0	1	0	1
1	1	0	1	1	0	0	0	0
1	1	1	0	1	1	1	0	0

Using function table of T flip flop:

T	Q_{n+1}
0	Q_n
1	Q_n

We can obtain next state when T=0 previous state retains and when T=1 T flip flop toggles the previous state (i.e complement of previous state).
For example : TP=0 then Pn+1=0 as Pn=0, T=1 then Pn+1=1 as Pn=0. PQR as initial state 000->011->101->000. So next three states (000 as initial state) will be 011,101,000.

8. Answer:(2)

Solution: State Table

Q1	Q0	in	D1	D0	Q1N	Q0N
0	0	0	0	0	0	0
0	0	1	0	1	0	1
0	1	0	1	0	1	0
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	1	0	1	0	1
1	1	0	1	0	1	0
1	1	1	1	1	1	1

9. Answer (B)

Solution:

Q_1	Q_0	Q_1^+	Q_0^+	T_1	T_0
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	0	0

By using above excitation table
$T1 = Q'_1Q_0$
$T0 = (Q_1Q_0)' = Q'_1 + Q_0'$

10. Answer : (B)

Solution: Since it is synchronous so, after every clock cycle T will toggle if input is 1 and will be in Hold State if input is 0. D flip-flop's output always follows input.
After 1 clock cycles ; Q1 = 0 (Toggle) Q0 =1
After 2 clock cycles ; Q1 = 1 (Toggle) Q0 =0
After 3 clock cycles ; Q1 = 1 (Hold) Q0 =1
After 4 clock cycles ; Q1 = 0 (Toggle) Q0 =1

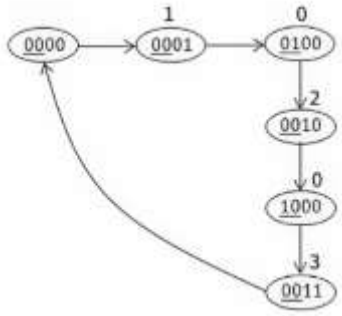
Check Out Our GATE Courses & Secure Your Dream Rank!

11. Answer : (4)

Solution:

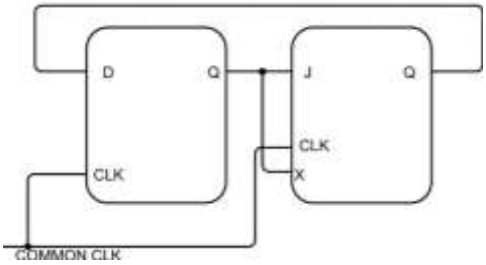
Given sequence is 0-1-0-2-0-3

There are 3 transitions from 0. Hence $\lceil \log_2 3 \rceil = 2$ bits must be added to the existing 2 bits to represent 4 unique states.



12. Answer :(A)

Solution:



Initially Q output of D - FF = 1 Initially Q output of JK - FF = 0 Now with the help of present state and next state table we can see what is happening in circuit.

Toggle: J = K = 1 and Hold : J = K = 0

Present state D input	Q(output of D-FF	Previous state Q of JK-FF	Input to JK-FF J	K	Next state Q of JK-FF
0(initially	0	0	1	1	1(toggled)
1	1	1	0	0	1(held)
1	1	1	1	1	0(toggled)
0	0	0	1	1	1(toggled)
1	1	1	0	0	1(held)
1	1	1	1	1	0(toggled)

clock	Q_D	Q_T
0	1	0
1	0	1
2	1	1
3	1	0

13. Answer: (D)

Solution: In a Johnson's counter next MSB= complement of LSB and a circular right shift operation has to be done to get the next state. The state sequence is 0,8,12,14,15,7,3,1,0.

14. Answer:(C)

Solution:

An n-bit counter generates all states from 0to $2^n - 1$

Example (n = 2)

Count	Counter Output (A_1 A_0)
0	00
1	01
2	10
3	11

Step 2: Decoder Output (n → 2^n)

Each input activates exactly **one output line (one-hot)**

Decoder Output Table:

A_1 A_0	D_3 D_2 D_1 D_0
00	0001
01	0010
10	0100
11	1000

Step 3: Observe Behaviour:

Sequence of outputs:

0001 → 0010 → 0100 → 1000 → 0001

15. Answer: (C)

Solution:

Initial State			Input						Next State		
Q2	Q1	Q0	J2	K2	J1	K1	J0	K0	Q2'	Q1'	Q0'
0	0	0	1	0	0	1	0	1	1	0	0
1	0	0	1	0	1	0	0	1	1	1	0
1	1	0	0	0	1	0	1	1	1	1	1

16. Answer:(A)

Solution:

Let n is the number of flip-flops.

The max Mod values is 2n.

So $2^n \geq 258 \Rightarrow n = 9$

Check Out Our GATE Courses & Secure Your Dream Rank!

17. Answer : (D)
Solution:
As in D-flip-flop, next output is $Q^+ = D$
$P_{i+1} = R_i$
$Q_{i+1} = (P_i + R_i)'$
$R_{i+1} = R_i'Q_i$

Clock	Inputs			Outputs		
	$D_1 = R$	$D_2 = \overline{(P+R)}$	$D_3 = \overline{QR}$	P	Q	R
1	0	1	0	0	1	0
2	0	1	1	0	1	1
3	1	0	0	1	0	0
4	0	0	0	0	0	0
5	0	1	0	0	1	0

So, total number of distinct outputs = 4

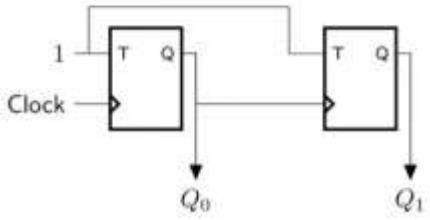
18. Answer:(A)
Solution:

2nd flip-flop will be active only when 1st flip flop produces output 1. For clocks 2 and 4 old output is retained by Flip-Flop 2.

T	Q_{n+1}
0	Q_n
1	$\overline{Q_n}$

T_0	Q_0	T_1	Q_1
	0		0
1	1	1	1
1	0	1	1
1	1	1	0
1	0	1	0

Q_1	Q_0
1	1
1	0
0	1
0	0



19. Answer:(A)
Solution:
The four bit register contains: 1011,1101,0110,1011,1101, **0110** after each shift.

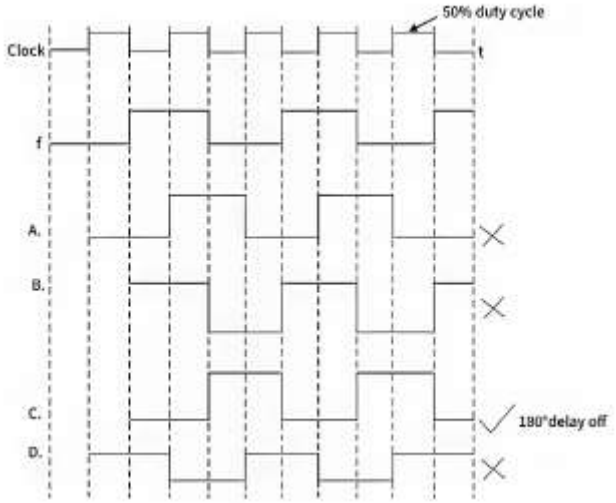
20. Answer: (C)
Solution:
Control Signals:
• **Load = 0** and **Count = 1**: This means the counter will ignore any parallel data inputs and proceed to count to the **Next State**.

• **Input (0011):** The parallel load input is ignored because the Load signal is disabled (Load = 0).

State Transition Table:

Decimal	Output($Q_3Q_2Q_1Q_0$)	Status
0	0 0 0 0	Valid State (Initial)
1	0 0 0 1	Valid State
2	0 0 1 0	Valid State
3	0 0 1 1	Valid State
4	0 1 0 0	Valid State
5	~~0 1 0 1~~	CLEAR (Transient State / Resets)

21. Answer : (C)
Solution:



22. Answer : (C)
Solution:

Data	Q_0	Q_1	Q_z
1	1	$Q0_{start}$ XOR $Q2_{start}$ = 0XOR0 = 0	$Q1_{start}$ = 0
0	0	1XOR0 = 1	0
0	0	0XOR0 = 0	1
1	1	0XOR1 = 1	0
1	1	1XOR0 = 1	1
0	0	1XOR1 = 0	1
0	0	0XOR1 = 1	0
0	0	0XOR0 = 0	1
0	0	0XOR1 = 1	0

Check Out Our GATE Courses & Secure Your Dream Rank!

23. Answer : (D)

Solution:

Clearly, Q0 alternates in every clock cycle as Q0' is fed as input and it is D flipflop.

Q1 becomes 1 if its previous value and current Q0 differs (EXOR).

So, the sequence of transitions will be : 00 → 11 → 01 → 10 → 00,

(D) choice.

24. Answer : (A)

Solution:

Therefore, the input to the D flip-flop is:

$D = AX + \bar{X}\bar{Q}$

For a positive edge-triggered D flip-flop, the next state Q(n+1) is determined by the value of D just before the clock edge. $Q(n+1) = An \cdot Xn + \bar{X}n \cdot \bar{Q}n$

This gives us two operational modes based on the value of X:

If X = 1: $Q(n+1) = A \cdot (1) + (0) \cdot \bar{Q} \rightarrow A$ (The output takes the previous value of A).

If X = 0: $Q(n+1) = A \cdot (0) + (1) \cdot \bar{Q} \rightarrow \bar{Q}$ (The output toggles its previous state).

- **For Period 1** (Output Q1): Look at inputs during Period 0: X0 = 1. Since X = 1, it loads the value of A. Result: Q1 = A0
- **For Period 2** (Output Q2): Look at inputs during Period 1: X1 = 1. Since X = 1, it loads the value of A. Result: Q2 = A1
- **For Period 3** (Output Q3): Look at inputs during Period 2: X2 = 0. Since X = 0, it toggles the previous state. We know the previous state Q2 = A1, so the complement is A1'. Result: Q3 = A1'
- **For Period 4** (Output Q4): Look at inputs during Period 3: X3 = 1. Since X = 1, it loads the value of A. Result: Q4 = A3
- **For Period 5** (Output Q5): Look at inputs during Period 4: X4 = 1. Since X = 1, it loads the value of A. Result: Q5 = A4.

  The correct output sequence on Y over the clock periods 1 through 5 is: A_0, A_1, A_1', A_3, A_4

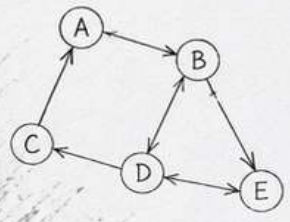
Check Out Our GATE Courses & Secure Your Dream Rank!

THEORY OF COMPUTATION

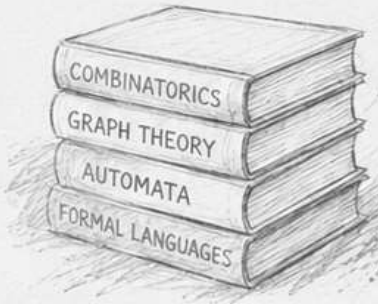
"Scan Here for Video Solution"

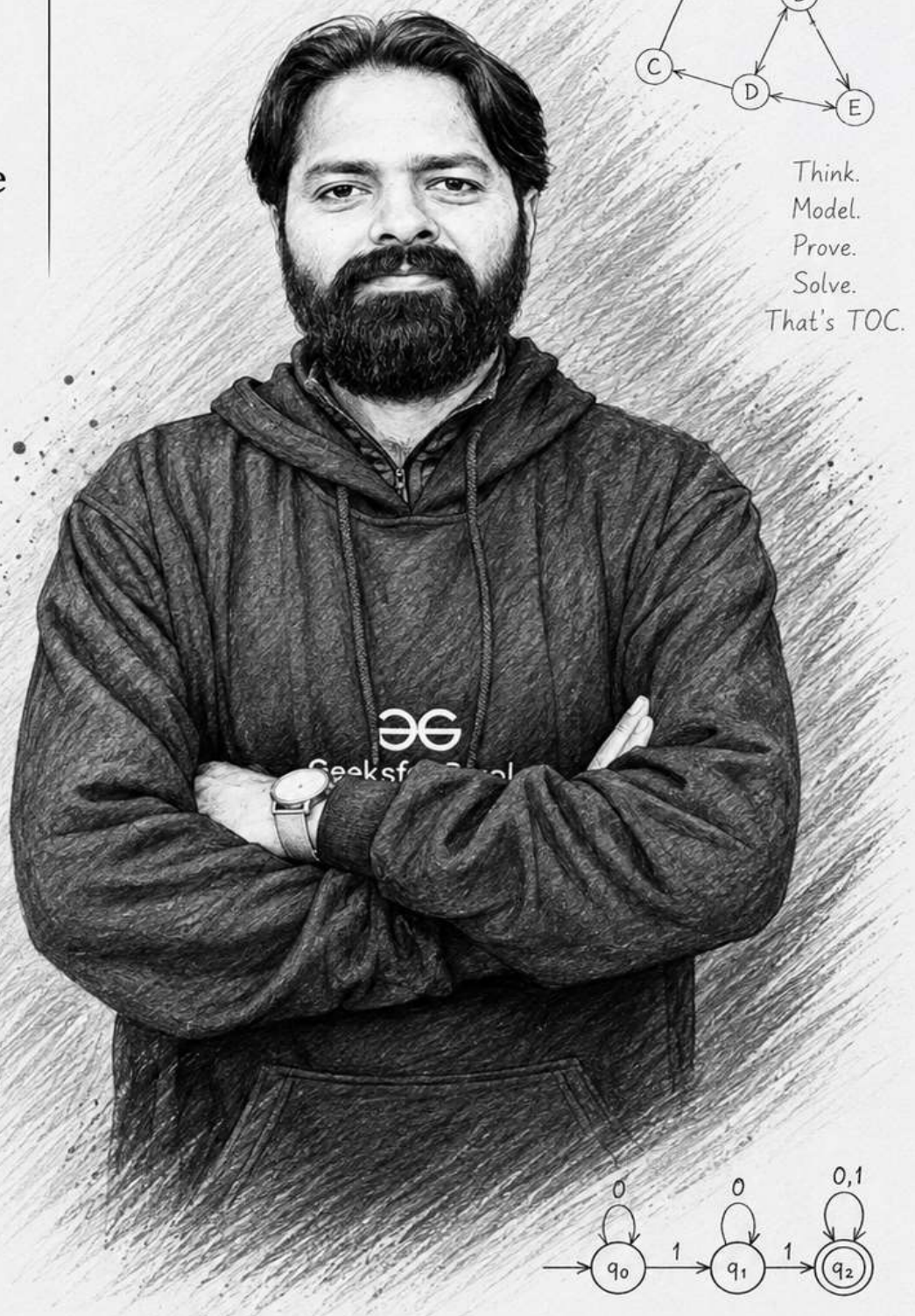
Link to GfG corresponding Short Notes: GATE FARRAY

“In *TOC*, we don’t memorize theorems, we understand ideas, build intuition, and solve with **smart simplicity.**”



Think.
Model.
Prove.
Solve.
That's TOC.





IIT BOMBAY ALUMNI

KING OF TOC

Devasane Mallesham Sir

Every problem
has a structure.
Find it.
Exploit it.
Master it.
That's the power of TOC.

Think. Model. Prove. Conquer.

1 Regular Language and Finite Automata

GATE CSE – 2026 SET- 1 MSQ (1-MARKS)

Q1. Let M be a nondeterministic finite automaton (NFA) with 6 states over a finite alphabet. Which of the following options CANNOT be the number of states in the minimal deterministic finite automaton (DFA) that is equivalent to M?

(a). 32

(b). 65

(c). 1

(d).128

GATE CSE – 2026 SET- 1 MSQ (2-MARKS)

Q2. Let L_1 and L_2 be two languages over a finite alphabet, such that $L_1 \cap L_2$ and L_2 are regular languages. Which of the following statements is/are always true?

(a). L_1 is regular

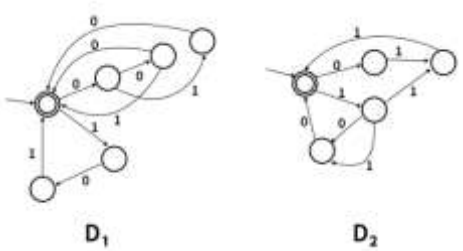
(b). $L_1 \cup L_2$ is regular

(c). $\overline{L_2}$ is context-free

(d). L_1 is context-free

GATE CSE – 2026 SET- 2 MSQ (2-MARKS)

Q3. Consider the following two finite automata D1 and D2 (as shown in the figure).



Which of the following statements is/are true?

(a). L(D1)=L(D2)
(b). L(D1) is a proper subset of L(D2)
(c). L(D1)∩L(D2)={ε}
(d). (L(D1)∪L(D2))* consists of all strings in {0,1}* whose length is divisible by 3

GATE CSE – 2025 SET- 1 MSQ (1-MARKS)

Q4. A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?
(a). L may have an accepting NFA with $< n$ states.
(b). L may have an accepting DFA with $< n$ states.
(c). There exists a DFA with $\leq 2^n$ states that accepts L.
(d). Every DFA that accepts L has $> 2^n$ states.

GATE CSE – 2025 SET- 1 NAT (1-MARKS)

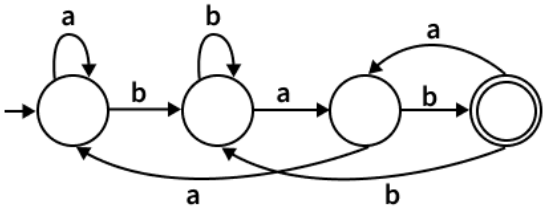
Q5. Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, "aa", "bb" or "cc". The number of such strings of length 5 that are possible is _____. (Answer in integer)

GATE CSE – 2025 SET - 1 MCQ (2-MARKS)

Q6. Consider the following two languages over the alphabet $\{a, b\}$:
$L1 = \{ \alpha\beta\alpha \mid \alpha \in \{a, b\}^+ \text{ AND } \beta \in \{a, b\}^+ \}$
$L2 = \{ \alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{ AND } \beta \in \{a, b\}^+ \}$
Which ONE of the following statements is CORRECT?
(a). Both $L1$ and $L2$ are regular languages.
(b). $L1$ is a regular language but $L2$ is not a regular language.
(c). $L1$ is not a regular language but $L2$ is a regular language.
(d). Neither $L1$ nor $L2$ is a regular language.

GATE CSE – 2025 SET - 1 MSQ (2-MARKS)

Q7. Consider the following deterministic finite automaton (DFA) defined over the alphabet, $\Sigma = \{a, b\}$. Identify which of the following language(s) is/are accepted by the given DFA.



(a) The set of all strings containing an even number of b's.
(b) The set of all strings containing the pattern bab.
(c) The set of all strings ending with the pattern bab.
(d) The set of all strings not containing the pattern aba.

GATE CSE – 2025 SET - 1 NAT (2-MARKS)

Q8. Consider a finite state machine (FSM) with one input X and one output f, represented by the given state transition table. The minimum number of states required to realize this FSM is _____. (Answer in integer)

Present State	Next State		Output f	
	x = 0	x = 1	x = 0	x = 1
A	F	B	0	0
B	D	C	0	0
C	F	E	0	0
D	G	A	1	0
E	D	C	0	0
F	F	B	1	1
G	G	H	0	1
H	G	A	1	0

GATE CSE – 2025 SET - 2 MSQ (1-MARKS)

Q9. Consider the two lists List I and List II given below:

List 1	List 2
(i) Context free languages	(a) Closed under union
(ii) Recursive languages	(b) Not closed under complementation
(iii) Regular languages	(c) Closed under intersection

For matching of items in List I with those in List II, which of the following option(s) is/are CORRECT?
(a) (i) – (a), (ii) – (b), and (iii) – (c)
(b) (i) – (b), (ii) – (a), and (iii) – (c)
(c) (i) – (b), (ii) – (c), and (iii) – (a)
(d) (i) – (a), (ii) – (c), and (iii) – (b)

GATE CSE - 2025 SET – 2 MSQ (2-MARKS)

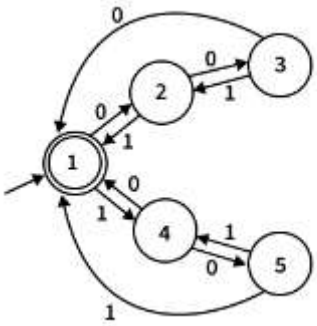
Q10. Let $\Sigma = \{a, b, c\}$. For $x \in \Sigma^*$, and $\alpha \in \Sigma$, let $\#\alpha(x)$ denote the number of occurrences of α in x. Which one or more of the following option(s) define(s) regular language(s)?
(a) $\{a^m b^n \mid m, n \geq 0\}$
(b) $\{a, b\}^* \cap \{a^m b^n c^{m-n} \mid m \geq n \geq 0\}$
(c) $\{w \mid w \in \{a, b\}^*$, $\#_a(w) \equiv 2 \pmod 7$, and $\#_b(w) \equiv 3 \pmod 9\}$
(d) $\{w \mid w \in \{a, b\}^*$, $\#_a(w) \equiv 2 \pmod 7$, and $\#_a(w) \equiv \#_b(w)\}$

GATE CSE – 2025 SET - 2 NAT (2-MARKS)

Q11. Let $\Sigma = \{1,2,3,4\}$. For $x \in \Sigma^*$, let $prod(x)$ be the product of symbols in x modulo 7. We take $prod(\epsilon) = 1$, where ϵ is the null string .
For example, $prod(124) = (1 \times 2 \times 4) \bmod 7 = 1$.
Define $L = \{x \in \Sigma^* \mid prod(x) = 2\}$.
The number of states in a minimum state DFA for L is _____. (Answer in integer)

GATE CSE – 2024 SET - 1 MSQ (2-MARKS)

Q12. Consider the 5-state DFA M accepting the language $L(M) \subset (0 + 1)^*$ shown below. For any string $w \in (0 + 1)^*$ let $n_0(w)$ be the number of $0's$ in w and $n_1(w)$ be the number of $1's$ in w.



Which of the following statements is/are FALSE?
(a) States 2 and 4 are distinguishable in M
(b) States 3 and 4 are distinguishable in M
(c) States 2 and 5 are distinguishable in M
(d) Any string w with $n0(w) = n1(w)$ is in $L(M)$.

GATE CSE – 2024 SET - 1 MSQ (1-MARKS)

Q13. Let $L1$, $L2$ be two regular languages and $L3$ a language which is not regular. Which of the following statements is/are always TRUE?
(a). $L_1 = L_2$ if and only if $L_1 \cap \overline{L_2} = \phi$
(b). $L_1 \cup L_3$ is not regular
(c). $\overline{L_3}$ is not regular
(d). $\overline{L_1} \cup \overline{L_2}$ is regular

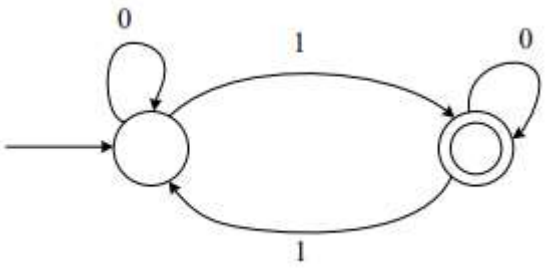
Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE – 2024 SET - 1 NAT (2-MARKS)

Q14. Consider the following two regular expressions over the alphabet {0,1}:
$r = 0^* + 1^*$
$s = 01^* + 10^*$
The total number of strings of length less than or equal to 5, which are neither in r nor in s, is _____

GATE CSE – 2024 SET - 1 MCQ (1-MARKS)

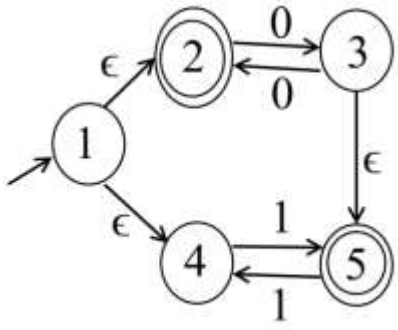
Q15. Which one of the following regular expressions is equivalent to the language accepted by the DFA given below?



(a). $0^*1(0 + 10^*1)^*$
(b). $0^* (10^*11)^* 0^*$
(c). $0^*1(010^*1)^* 0^*$
(d). $0(1 + 0^*10^*1)^*0^*$

GATE CSE – 2024 SET - 2 MCQ (2-MARKS)

Q16. Let M be the 5-state NFA with ϵ-transitions shown in the diagram below



Which one of the following regular expressions represents the language accepted by M ?

(a). $(00)^* + 1(11)^*$
(b). $0^* + (1 + 0(00)^*)(11)^*$
(c). $(00)^* + (1 + (00)^*)(11)^*$
(d). $0^+ + 1(11)^* + 0(11)^*$

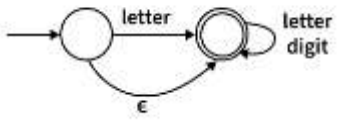
GATE CSE – 2024 SET - 2 NAT (2-MARKS)

Q17. Let $L1$ be the language represented by the regular expression $b^*ab^*(ab^*ab^*)^*$ and $L2 = \{ w \in (a + b)^* \mid |w| \leq 4\}$, where $|w|$ denotes the length of string w. The number of strings in $L2$ which are also in $L1$ is _____

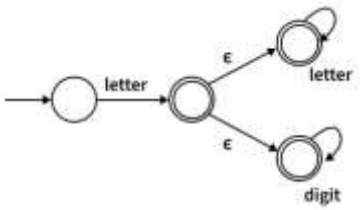
GATE CSE – 2023 MCQ (1-MARKS)

Q18. Consider the following definition of a lexical token id for an identifier in a programming language, using extended regular expressions:
letter→[A-Za-z]
digit→[0-9]
id→letter(letter|digit)∗
Which one of the following Non-deterministic Finite-state Automata with - transitions accepts the set of valid identifiers? (A double-circle denotes a final state)

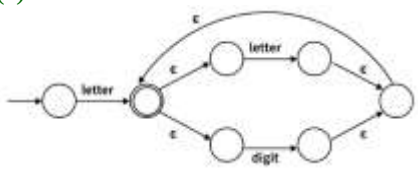
(a).



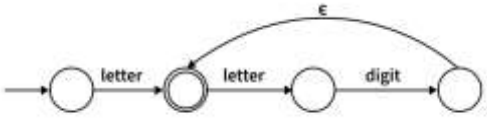
b).



(c).



(d).

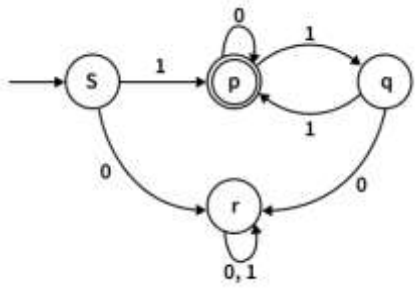


GATE CSE – 2023 NAT (2-MARKS)

Q19. Consider the language L over the alphabet {0, 1}, given below: L = {w ∈ {0, 1}* | w does not contain three or more consecutive 1's}. The minimum number of states in a Deterministic Finite-State Automaton (DFA) for L is _____.

GATE CSE – 2023 MCQ (1-MARKS)

Q20. Consider the Deterministic Finite-state Automaton (DFA) A shown below. The DFA runs on the alphabet {0, 1}, and has the set of states {s, p, q, r}, with s being the start state and p being the only final state.

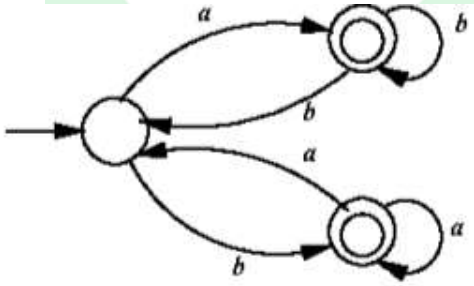


Which one of the following regular expressions correctly describes the language accepted by A?

(a) 1(0*11)*
(b) 0(0 + 1)*
(c) 1(0 + 11)*
(d) 1(110*)*

GATE CSE – 2022 MCQ (1-MARKS)

Q21. Which one of the following regular expression correctly represents the language of the finite automata given below?



(a) ab*bab*+ba*aba*
(b) (ab*b)*ab* + (ba*a)* ba*
(c) (ab*b+ba*a)(a*+b*)
(d) (ba*a+ab*b)*(ab*+ba*)

GATE CSE – 2020 NAT (2-MARKS)

Q22. Consider the following language.
L = {x ϵ {a,b}* | number of a's in x is divisible by 2 but not divisible by 3}. The minimum number of states in a DFA that accepts L is _____.

GATE CSE – 2020 MCQ (1-MARKS)

Q23. Which one of the following regular expressions represents the set of all binary strings with an odd number of 1's?

(a) 10*(0*10*10*)*
(b) ((0 + 1)*1(0 + 1)*1)*10*
(c) (0*10*10*)*10*
(d) (0*10*10*)*0*1

GATE CSE – 2020 MCQ (1-MARKS)

Q24. Consider the following statements.
I. If L1 U L2 is regular, then both L1 and L2 must be regular.
II. The class of regular languages is closed under infinite union.
Which of the above statements is/are TRUE?

(a) Both I and II
(b) II only
(c) Neither I nor II
(d) I only

GATE CSE – 2019 MCQ (1-MARKS)

Q25. If L is a regular language over Σ = {a,b}, which one of the following languages is NOT regular?

(a) Suffix (L) = {y ϵ Σ* such that xy ϵ L}
(b) {ww^R | w ϵ L}
(c) Prefix (L) = {x ϵ Σ*| ∃y ϵ Σ* such that xy ϵ L}
(d) L.L^R = {xy | x ϵ L, y^R ϵ L}

GATE CSE – 2019 MCQ (2-MARKS)

Q26. For Σ = {a,b}, let us consider the regular language L = {x|x = a^{2+3k} or x = b^{10+12k}, k ≥ 0}. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L?

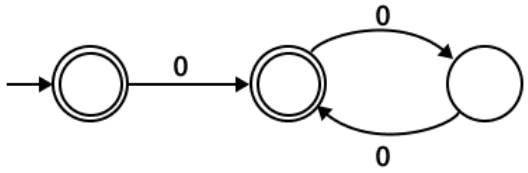
(a) 3
(b) 9
(c) 5
(d) 24

GATE CSE – 2019 NAT (2-MARKS)

Q27. Let Σ be the set of all bijections from {1, … ,5} to {1, … ,5}, where id denotes the identity function, i.e. $id(j) = j, \forall j$. Let ∘ denote composition on functions. For a string $x = x_1x_2 \dots x_n \in \Sigma^n, n \geq 0$, let $\pi(x) = x_1 \circ x_2 \circ \cdots \circ x_n$. Consider the language $L = \{x \in \Sigma^* \mid \pi(x) = id\}$. The minimum number of states in any DFA accepting L is _____

GATE CSE – 2018 NAT (2-MARKS)

Q28. Given a language L, define L_i as follows :
$L_0 = \{\epsilon\}$
$L_i = L_{i-1}$. L for all i > 0 The order of a language L is defined as the smallest k such that $L^k = L^{k+1}$. Consider the language L_1 (over alphabet 0) accepted by the following automation.



The order of L_1 is_____.

GATE CSE – 2018 MCQ (1-MARKS)

Q29. Let N be an NFA with n states. Let k be the number of states of a minimal DFA which is equivalent to N. Which one of the following is necessarily true?

(a) $k \geq 2^n$
(b) $k \geq n$
(c) $k \leq n^2$
(d) $k \leq 2^n$

GATE CSE – 2017 MCQ (1-MARKS)

Q30. The minimum possible number of states of a deterministic finite automaton that accepts the regular language $L = \{w_1aw_2 \mid w_1, w_2 \in \{a,b\}^*, |w_1| = 2, |w_2| \geq 3\}$ is _____

(a) 8
(b) 9
(c) 10
(d) 11

GATE CSE – 2017 SET-2 MCQ (2-MARKS)

Q31. Let δ denote the transition function and $\hat{\delta}$ denote the extended transition function of the ϵ-NFA whose transition table is given below:

δ	**ε**	**a**	**b**
→ q0	{q2}	{q1}	{q0}
q1	{q2}	{q2}	{q3}
q2	{q0}	∅	∅
q3	∅	∅	{q2}

Then $\hat{\delta}(q_2, aba)$ is:

(a) ∅
(b) {q0,q1,q3}
(c) {q0,q1,q2}
(d) {q0,q2,q3}

GATE CSE – 2017 SET-1 MCQ (2-MARKS)

Q32. If G is a grammar with productions **S → SaS | aSb | bSa | SS | ϵ** where S is the start variable, then which one of the following strings is not generated by G?

(a) abab
(b) aaab
(c) abbaa
(d) babba

GATE CSE – 2017 SET-1 NAT (1-MARKS)

Q33. Consider the language L given by the regular expression (a+b)*b(a+b) over the alphabet {a,b}. The smallest number of states needed in deterministic finite-state automaton (DFA) accepting L is _____

GATE CSE – 2016 SET-1 MCQ (1-MARKS)

Q34. Which of the following languages is generated by the given grammar?

$$S \rightarrow aS|bS|\varepsilon$$

(a) $\{a^nb^m \mid n, m >= 0\}$
(b) {w ϵ {a,b}*| w has equal number of a's and b's}
(c) $\{a^n \mid n >= 0\} \cup \{b^n \mid n >= 0\} \cup \{a^nb^n \mid n >= 0\}$
(d) {a,b}*

GATE CSE – 2016 SET-1 MCQ (1-MARKS)

Q35. Which of the following decision problems are undecidable?
I. Given NFAs N1 and N2, is $L(N1) \cap L(N2) = \phi$?
II. Given a CFG G = (N, Σ, P, S) and a string x ϵ Σ*, does x ϵ L(G)?
III. Given CFGs G1 and G2, is L(G1) = L(G2)?
IV. Given a TM, M is L(M) = ϕ ?

(a) I and IV only
(b) II and III only
(c) III and IV only
(d) II and IV only

GATE CSE – 2016 SET-1 MCQ (1-MARKS)

Q36. Which one of the following regular expressions represents the language: the set of all binary strings having two consecutive 0s and two consecutive 1s?
(a). (0 + 1)* 0011(0 + 1)* + (0 + 1)* 1100(0 + 1)*
(b). (0 + 1)* (00(0 + 1)* 11 + 11(0 + 1)* 00)(0 + 1)*
(c) (0 + 1)* 00(0 + 1)* + (0 + 1)* 11(0 + 1)*
(d). 00(0 + 1)* 11 + 11(0 + 1)* 00

GATE CSE – 2016 SET-2 MCQ (2-MARKS)

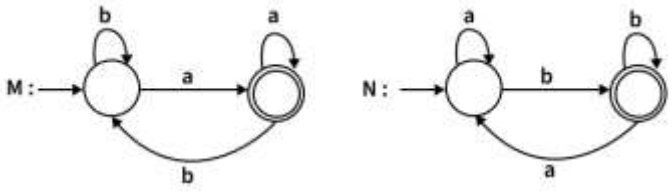
Q37. Consider the following two statements:
I. If all states of an NFA are accepting states then the language accepted by the NFA is Σ*.
II. There exists a regular language A such that for all languages B, A ∩ *B* is regular.
Which one of the following is CORRECT?
(a) Only I is true
(b) Only II is true
(c) Both I and II are true
(d) Both I and II are false

GATE CSE – 2016 SET-2 NAT (1-MARKS)

Q38. The number of states in the minimum sized DFA that accepts the language defined by the regular expression (0+1)*(0+1)(0+1)*is_____.

GATE CSE – 2015 SET-1 MCQ (2-MARKS)

Q39. Consider the DFAs M and N given above. The number of states in a minimal DFA that accepts the language L(M) ∩ L(N) is _____.



(a). 1
(b). 2
(c). 3
(d). 4

GATE CSE – 2015 SET-2 MCQ (2-MARKS)

Q40. Which of the following language is/are regular languages?

L_1 : $\{wxw^R | w,x \in \{a, b\}^*$ and $|w|, |x| > 0\}$, w^R is the reverse of string w

L_2 : $\{a^n b^m \mid m != n$ and m, n >= 0$\}$
L_3 : $\{a^p b^q c^r \mid p, q, r >= 0\}$
(a) L_1 and L_3 only
(b) L_2 only
(c) L_2 and L_3 only
(d) L_3 only

GATE CSE – 2015 SET-2 MCQ (2-MARKS)

Q41. The number of states in the minimal deterministic finite automaton corresponding to the regular expression (0 + 1) * (10) is _____
(a) 3
(b) 4
(c) 5
(d) 6

GATE CSE – 2015 SET-2 MCQ (2-MARKS)

Q42. Consider the alphabet Σ = {0, 1}, the null/empty string λ and the sets of strings X_0, X_1 and X_2 generated by the corresponding grammar. X_0, X_1 and X_2 are related as follows :
$X_0 = 1X_1$
$X_1 = 0X_1 + 1X_2$
$X_2 = 0X_1 + \{\lambda\}$
Which of the following choices precisely represents the strings in X_0?
(a) 10(0* + (10)* 1
(b) 10(0* + (10)*)* 1
(c) 1(0* + 10)* 1
(d) 10(0 + 10)* 1 + 110(0 + 10)* 1

GATE CSE – 2014 SET-3 MCQ (1-MARKS)

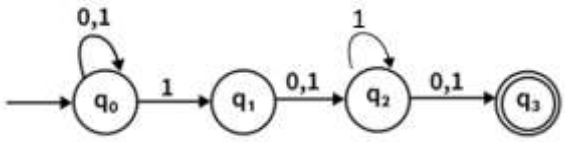
Q43. The length of the shortest string NOT in the language (over Σ = {a,b}) of the following regular expression is a*b*(ba)*a* is _____.
(a) 3
(b) 4
(c) 5
(d) 6

GATE CSE – 2014 SET-2 MCQ (2-MARKS)

Q44. Let L1 = {w ∈ {0,1}* | w has at least as many occurrences of (110)'s as (011)'s}. Let L2 = {w ∈ {0,1}*|w has at least as many occurrences of (000)'s as (111)'s}. Which one of the following is TRUE?
(a) L1 is regular but not L2.
(b) L2 is regular but not L1.
(c) Both L1 and L2 are regular.
(d) Neither L1 nor L2 are regular

GATE CSE – 2014 SET-1 MCQ (1-MARKS)

Q45. Consider the finite automaton in the following figure. What is the set of reachable states for the input string 0011?



(a) $\{q_0,q_1,q_2\}$
(b) $\{q_0,q_1\}$
(c) $\{q_0,q_1,q_2, q_3\}$
(d) $\{ q_3\}$

GATE CSE – 2014 SET-1 MCQ (1-MARKS)

Q46. Which one of the following is TRUE?

(a) The language $L = \{a^nb^n \mid n >= 0\}$ is regular.
(b) The language $L = \{a^n \mid n$ is prime$\}$ is regular.
(c) The language $L = \{w \mid w$ has $3k + 1$b's for some $k \in N$ with $\Sigma = \{a, b\}\}$ is regular.
(d) The language $L = (ww \mid w\epsilon\Sigma^*$ with $\Sigma = \{0, 1\}\}$ is regular.

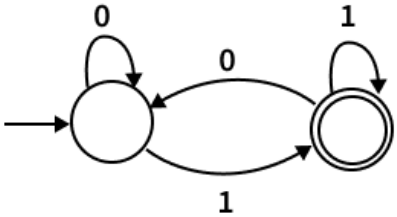
GATE CSE – 2014 SET-2 MCQ (1-MARKS)

Q47. If $L_1 = \{a^n \mid n > 0\}$ and $L_2 = \{b^n \mid n >= 0\}$, consider
(I) $L_1.L_2$ is a regular language
(II) $L_1.L_2 = \{a^nb^n \mid n >= 0\}$
Which one of the following is CORRECT?

(a) Only (I)
(b) Only (II)
(c) Both (I) and (II)
(d) Neither (I) nor (II)

GATE CSE – 2014 SET-1 MCQ (2-MARKS)

Q48. Which of the regular expressions given below represent the following DFA?

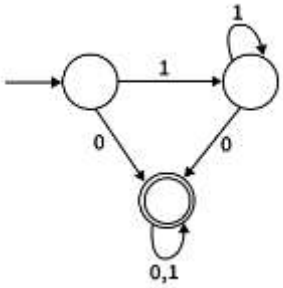


I)0*1(1+00*1)*
II)0*1*1+11*0*1
III) (0+1)*1

(a) I and II only
(b) I and III only
(c) II and III only
(d) I, II, and III

GATE CSE – 2013 SET-1 MCQ (2-MARKS)

Q49. Consider the DFA given.

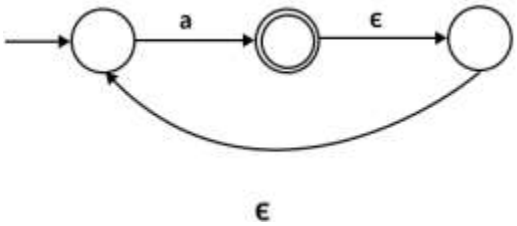


Which of the following are FALSE?
1. Complement of L(A) is context-free.
2. L(A) = L((11*0+0)(0 + 1)*0*1*)
3. For the language accepted by A, A is the minimal DFA.
4. A accepts all strings over {0, 1} of length at least 2.

(a) 1 and 3 only
(b) 2 and 4 only
(c) 2 and 3 only
(d) 3 and 4 only

GATE CSE – 2012 SET-1 MCQ (1-MARKS)

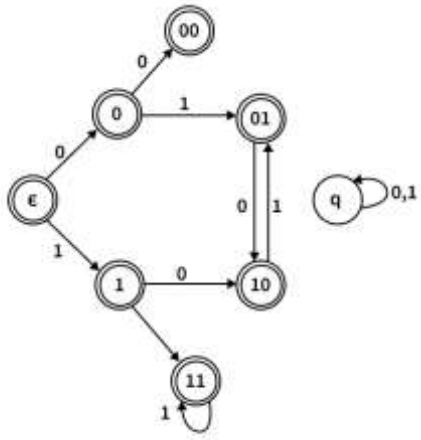
Q50. What is the complement of the language accepted by the NFA shown below?
Assume $\Sigma = \{a\}$ and ϵ is the empty string.



(a) $\emptyset$
(b) $\{\varepsilon\}$
(c) a*
(d) $\{a, \varepsilon\}$

GATE CSE – 2012 SET-1 MCQ (2-MARKS)

Q51. Consider the set of strings on {0,1} in which, every substring of 3 symbols has at most two zeros. For example, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. A partially completed DFA that accepts this language is shown below.



The missing arcs in the DFA are:

(a).

	00	01	10	11	q
00	1	0			
01				1	
10	0				
11			0		

(b).

	00	01	10	11	Q
00		0			1
01		1			
10				0	
11		0			

(c).

	00	01	10	11	Q
00		1			0
01		1			
10			0		
11		0			

(d).

	00	01	10	11	Q
00		1			
01				1	
10	0				
11			0		

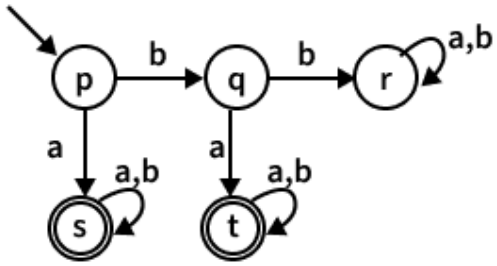
GATE CSE – 2012 MCQ (1-MARKS)

Q52. Given the language L ={ab,aa,baa}, which of the following strings are in L*?
1)abaabaaabaa
2)aaaabaaaa
3)baaaaabaaaab
4)baaaaabaa

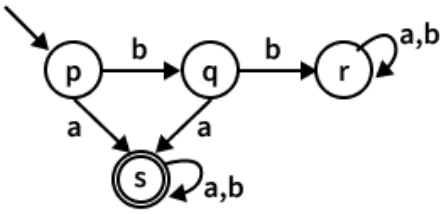
(a) 1, 2 and 3
(b) 2, 3 and 4
(c) 1, 2 and 4
(d) 1, 3 and 4

GATE CSE – 2011 MCQ (2-MARKS)

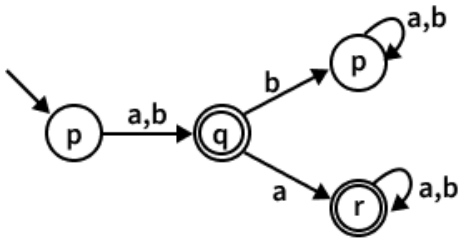
Q53. A deterministic finite automation (DFA) D with alphabet Σ = {a, b} is given below.



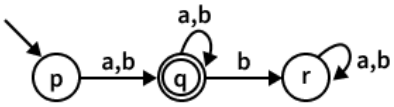
Which of the following finite state machines is a valid minimal DFA which accepts the same language as D?
(a).



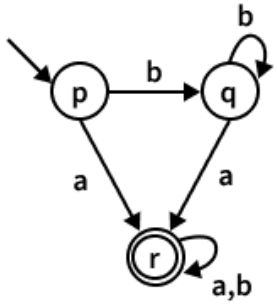
(b).



(c).



(d).



GATE CSE – 2011 MCQ (2-MARKS)

Q54. Definition of a language L with alphabet {a} is given as following. L = {a^{nk} | k > 0, and n is a positive integer constant} What is the minimum number of states needed in a DFA to recognize L?

(a) k+1
(b) n+1
(c) 2^{n+1}
(d) 2^{k+1}

GATE CSE – 2011 MCQ (1-MARKS)

Q55. Which of the following pairs have DIFFERENT expressive power?

(a) Deterministic finite automata (DFA) and Non-deterministic finite automata (NFA)
(b). Deterministic push down automata (DPDA) and Non-deterministic push down automata (NPDA)
(c). Deterministic single-tape Turning machine and Non-deterministic single tape Turning machine
(d). Single-tape Turning machine and multi-tape Turning machine

GATE CSE – 2011 MCQ (1-MARKS)

Q56. Let P be a regular language and Q be a context-free language such that Q ⊆ P. (For example, let P be the language represented by the regular expression p*q* and Q be [p^nq^n | n ∈ N]). Then which of the following is **ALWAYS** regular?

(a) P ∩ Q
(b) P – Q
(c) Σ* – P
(d) Σ* – Q

GATE CSE – 2010 MCQ (2-MARKS)

Q57. Let w be any string of length n is {0, 1}*. Let L be the set of all substrings of w. What is the minimum number of states in a non-deterministic finite automaton that accepts L?

(a) n-1
(b) n
(c) n+1
(d) 2n-1

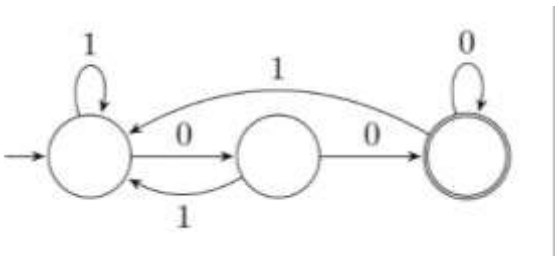
GATE CSE – 2010 MCQ (2-MARKS)

Q58. Let L = **{w ∈ (0 + 1)* | w has even number of 1s}**, i.e. L is the set of all bit strings with even number of 1s. Which one of the regular expression below represents L?

(a) (0 * 10 * 1)*
(b) 0* (10 * 10*)*
(c) 0*(10 * 1*)*0*
(d) 0 *1(10 * 1)*10*

GATE CSE – 2009 MCQ (1-MARKS)

Q59.



The above DFA accepts the set of all strings over {0,1} that

(a) begin either with 0 or 1 .
(b) end with 0.
(c) end with 00.
(d) contain the substring 00

GATE CSE – 2009 MCQ (2-MARKS)

Q60. Given the following state table of an FSM with two states A and B ,one input and one output.

PRESENT STATE A	PRESENT STATE B	Input	Next State A	Next State B	Output
0	0	0	0	0	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	1	0	0
0	0	1	0	1	0
0	1	1	0	0	1
1	0	1	0	1	1
1	1	1	0	0	1

If the initial state is A=0,B=0.what is the minimum length of an input string which will take the machine to the state A=0,B=1 with output = 1.

(a) 3
(b) 4
(c) 5
(d) 6

GATE CSE – 2008 MCQ (2-MARKS)

Q61. Given below are two finite state automata (-> indicates the start state and F indicates a final state)

Automaton Y

	a	b
→1	1	2
2(F)	2	1

Automaton Z

	a	b
→1	2	2
2(F)	1	1

Which of the following represents the product automaton Z × Y?

(a).

	a	b
→p	S	R
Q	R	S
R(F)	Q	P
S	Q	P

(b)

	a	b
→p	S	Q
Q	R	S
R(F)	Q	P
S	P	Q

(c)

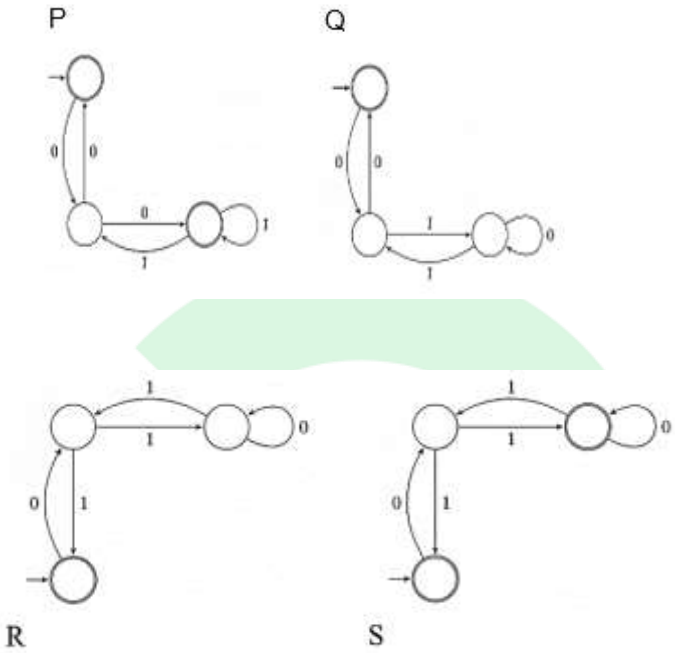
	a	b
→p	Q	S
Q	R	S
R(F)	Q	P
S	Q	P

(d)

	a	b
→p	S	Q
Q	S	R
R(F)	Q	P
S	Q	P

GATE CSE – 2008 MCQ (2-MARKS)

Q62. Match the following NFAs with the regular expressions they correspond to:

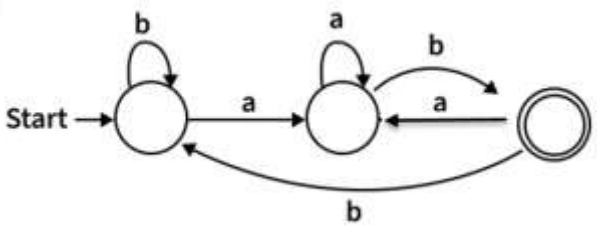


1. ε + 0(01*1 + 00)*01*
2. ε + 0(10*1 + 00)*0
3. ε + 0(10*1 + 10)*1
4. ε + 0(10*1 + 10)*10*

(a) P – 2, Q – 1, R – 3, S – 4
(b) P – 1, Q – 3, R – 2, S – 4
(c) P – 1, Q – 2, R – 3, S – 4
(d) P – 3, Q – 2, R – 1, S – 4

GATE CSE – 2008 MCQ (2-MARKS)

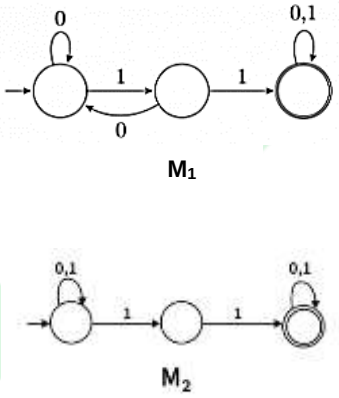
Q63. If the final states and non-final states in the DFA below are interchanged, then which of the following languages over the alphabet {a, b} will be accepted by the new DFA?



(a) Set of all strings that do not end with ab
(b) Set of all strings that begin with either an a or a b
(c) Set of all strings that do not contain the substring ab
(d) The set described by the regular expression b*aa*(ba)*b*

GATE CSE – 2008 MSQ (2-MARKS)

Q64. Consider the following two finite automata. M1 accepts L1 and M2 accepts L2



Which one of the following is TRUE?

(a). $L_1 = L_2$
(b). $L_1 \subset L_2$
(c). $L_1 \cap L_2^c = \emptyset$
(d). $L_1 \cup L_2 \neq L1$

GATE CSE – 2008 MCQ (2-MARKS)

Q65. Which of the following are regular sets?

I. $\{ a^n b^{2m} \mid n \geq 0, m \geq 0 \}$

II. $\{ a^n b^m \mid n = 2m \}$

III. $\{ a^n b^m \mid n \neq m \}$

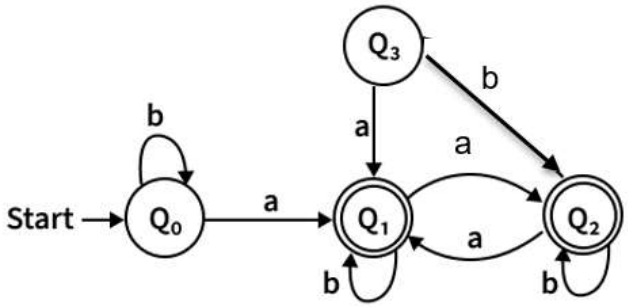
IV. $\{ xcy \mid x, y \in \{a, b\}^* \}$

(a) I and IV only
(b) I and III only
(c) I only
(d) IV only

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE – 2007 MCQ (2-MARKS)

Q66. Consider the following Finite State Automaton:

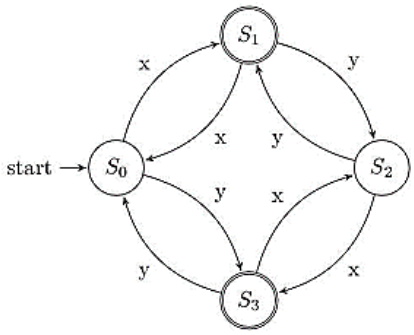


The language accepted by this automaton is given by the regular expression

(a) b*ab*ab*ab*
(b) (a+b)*
(c) b* a(a+b)*
(d) b*ab*ab*

GATE CSE – 2007 MCQ (2-MARKS)

Q67. Consider the following DFA in which S_0 is the start state and S1,S3 are the final states.

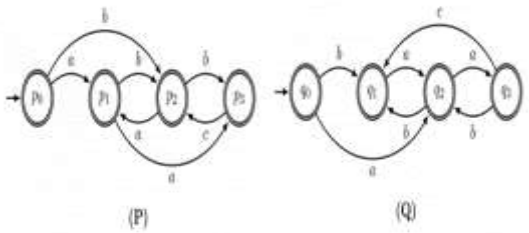


What language does this DFA recognize?

(a) All strings of x and y
(b) All strings of x and y which have either even number of x and even number of y or odd number of x and odd number of y
(c) All strings of x and y which have equal number of x and y
(d) All strings of x and y with either even number of x and odd number of y or odd number of x and even number of y

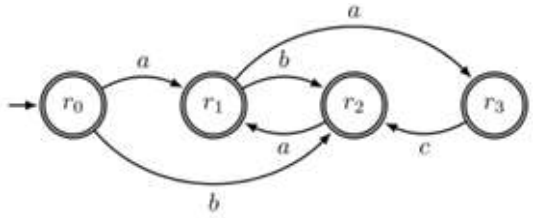
GATE CSE – 2007 MCQ (2-MARKS)

Q68. Consider the following finite automata P and Q over the alphabet {a, b, c}. The start states are indicated by a double arrow and final states are indicated by a double circle. Let the languages recognized by them be denoted by L(P) and L(Q) respectively.

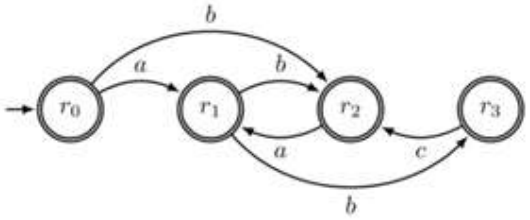


The automata which recognizes the language L(P) ∩ L(Q) is:

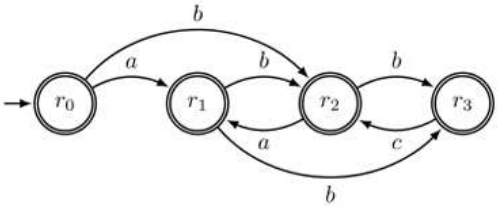
(a).



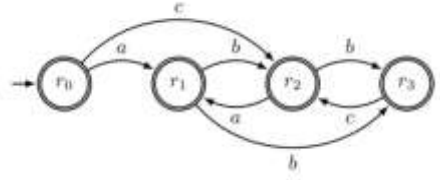
(b).



(c).

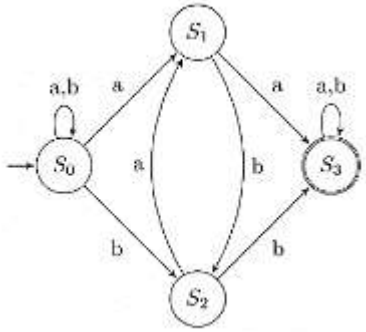


(d).

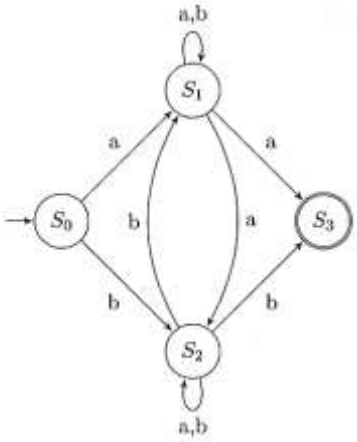


GATE CSE – 2007 MCQ (2-MARKS)

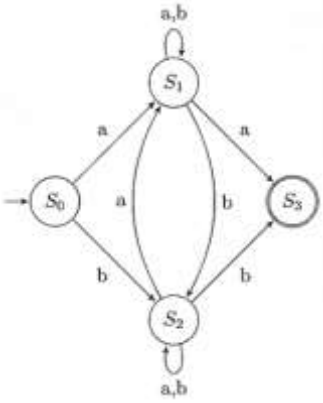
Q69. Consider the regular expression
R = (a + b)*(aa + bb)(a + b)*
Which of the following non-deterministic finite automata recognizes the language defined by the regular expression R?
Edges labelled λ denote transitions on the empty string.



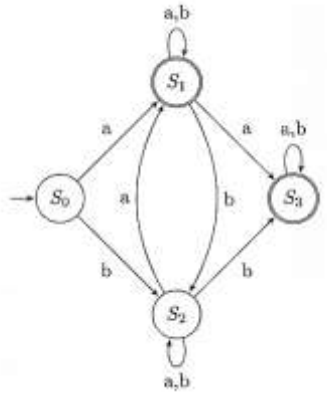
(a).



(b)



(c).



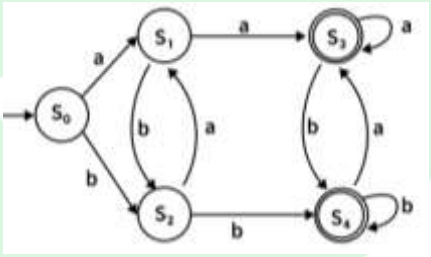
(d).

GATE CSE – 2007 MCQ (2-MARKS)

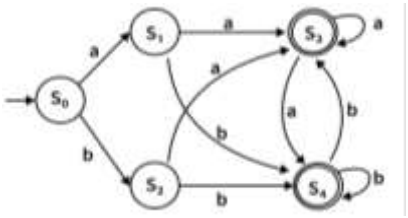
Q70 Consider the regular expression R = (a + b)*(aa + bb)(a + b)*

Which deterministic finite automaton accepts the language represented by the regular expression R ?

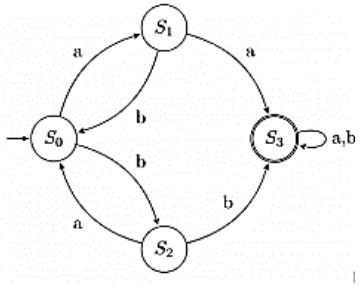
(a).



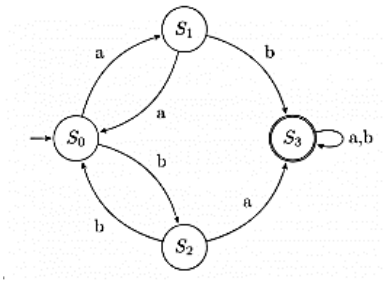
(b)



(c).



(d).



Consider the strings u = abbaba, v = bab, and w = aabb. Which of the following statements is true?

(a) The automaton accepts u and v but not w

(b) The automaton accepts each of u, v, and w

(c) The automaton rejects each of u, v, and w

(d) The automaton accepts u but rejects v and w

GATE CSE – 2007 MCQ (2-MARKS)

Q71. Which of the following languages is regular?

(a) $\{ ww^R \mid w \in \{0,1\}^+ \}$
(b) $\{ ww^Rx \mid x, w \in \{0,1\}^+ \}$
(c) $\{ wxw^R \mid x, w \in \{0,1\}^+ \}$
(d) $\{ xww^R \mid x, w \in \{0,1\}^+ \}$

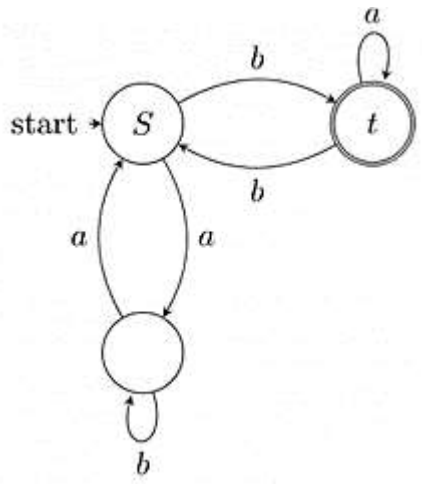
GATE CSE – 2007 MCQ (2-MARKS)

Q72 Which of the following is TRUE?

(a) Every subset of a regular set is regular
(b) Every finite subset of a non-regular set is regular
(c) The union of two non-regular sets is not regular
(d) Infinite union of finite sets is regular

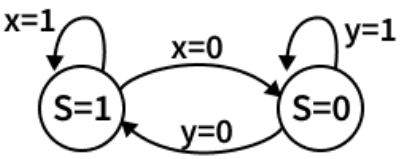
GATE CSE – 2006 MCQ (2-MARKS)

Q73. In the automaton below,s is the start state and t is the only final state.



GATE CSE – 2006 MCQ (2-MARKS)

Q74 For a state machine with the following state diagram the expression for the next state S^+ in terms of the current state and the input variables x and y is



(a) S^+=S'.y'+S.x
(b) S^+=S.x.y'+S'.y.x'
(c) S^+=x.y'
(d) S^+=S'.y+S.x'

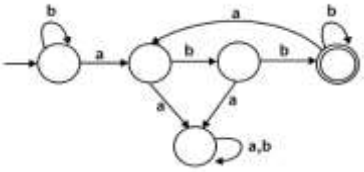
GATE CSE – 2006 MCQ (2-MARKS)

Q75. If s is a string over (0 + 1)* then let $n_0(s)$ denote the number of 0's in s and $n_1(s)$ the number of 1's in s. Which one of the following languages is not regular?

(a) $L = \{ s \in (0 + 1)^* \mid n_0(s)$ is a 3-digit prime $\}$
(b) $L = \{s \in (0 + 1)^* \mid$ for every prefix s′ of s, $\mid n_0(s') - n_1(s') \mid \leq 2 \}$
(c) $L = \{ s \in (0 + 1)^* \mid \mid n_0(s) - n_1(s) \mid \leq 4 \}$
(d) $L = \{s \in (0 + 1)^* \mid n_0(s) \bmod 7 = n_1(s) \bmod 5 = 0 \}$

GATE CSE – 2005 MCQ (2-MARKS)

Q76. Consider the machine M:

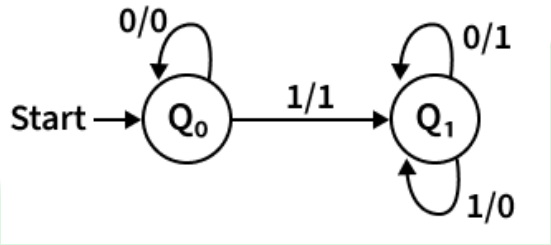


The language recognized by M is:

(a) { w ∈ {a, b}* | every a in w is followed by exactly two b's }
(b) { w ∈ {a, b}* | every a in w is followed by at least two b's }
(c) { w ∈ {a, b}* | w contains the substring 'abb' }
(d) { w ∈ {a, b}* | w does not contain 'aa' as a substring }

GATE CSE – 2005 MCQ (2-MARKS)

Q77. The following diagram represents a finite state machine which takes as input a binary number from the least significant bit.

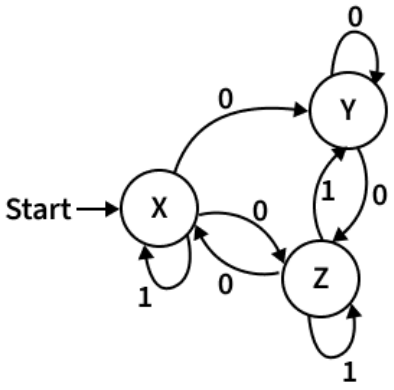


Which of the following is TRUE?

(a) It computes 1's complement of the input number
(b) It computes 2's complement of the input number
(c) It increments the input number
(d) It decrements the input number

GATE CSE – 2005 MCQ (2-MARKS)

Q78. Consider the non-deterministic finite automaton (NFA) shown in the figure.



State X is the starting state of the automaton. Let the language accepted by the NFA with Y as the only accepting state be L1. Similarly, let the language accepted by the NFA with Z as the only accepting state be L2. Which of the following statements about L1 and L2 is TRUE?

(a) L1 = L2
(b) L1 ⊂ L2
(c) L2 ⊂ L1
(d) None of the above

GATE CSE – 2005 MCQ (2-MARKS)

Q79 Consider the regular grammar:

S → Xa | Ya
X → Za
Z → Sa | ε
Y → Wa
W → Sa

where S is the starting symbol, the set of terminals is {a} and the set of non-terminals is {S, W, X, Y, Z}. We wish to construct a deterministic finite automaton (DFA) to recognize the same language. What is the minimum number of states required for the DFA?

(a) 2
(b) 3
(c) 4
(d) 5

ANSWER KEY

1.(b,d)	2.(c)	3.(c,d)	4.(d)
5.(195)	6.(c)	7.(c)	8 . (5)
9.(b,c)	10.(a,c)	11.(6)	12.(b,c)
13.(c,d)	14.(44)	15.(a)	16.(b)
17. (15)	18.(c)	19.(4)	20.(c)
21.(d)	22.(6)	23.(mta)	24.(c)
25.(b)	26.(d)	27.(120)	28.(2)
29.(d)	30.(a)	31.(c)	32.(d)
33.(4)	34.(d)	35.(c)	36.(b)
37.(b)	38.(2)	39.(a)	40.(a)
41.(a)	42.(c)	43.(a)	44.(a)
45.(a)	46.(c)	47.(a)	48.(b)
49.(d)	50.(b)	51.(d)	52.(c)
53.(a)	54.(b)	55.(b)	56.(c)
57.(c)	58.(b)	59.(c)	60.(a)
61.(a)	62.(c)	63.(a)	64.(a,c)
65.(a)	66.(c)	67.(d)	68.(a)
69.(a)	70.(a)	71.(c)	72.(b)
73.(d)	74.(a)	75.(c)	76.N/A
77.(b)	78.(a)	79.(b)	

HINTS AND SOLUTIONS

1. **Answer : B,D**

n = 6
Maximum possible DFA states = $2^6 = 64$
B,D not allowed.

2. **Answer ; C**

(A) L1 is regular — NOT always true

Take alphabet $\Sigma = \{a, b\}$

Let
L2 = ∅ (regular)
L1 = $\{ a^n b^n \mid n \geq 0 \}$ (not regular)

Then
$L1 \cap L2 = \emptyset$ (regular)

But L1 itself is not regular.

So (A) is false.

(B) L1 ∪ L2 is regular — NOT always true

Using the same example:

L2 = ∅ (regular)
L1 = $\{ a^n b^n \mid n \geq 0 \}$ (not regular)

Then
$L1 \cup L2 = L1$

Which is not regular.

So (B) is false.

(C) L2 is context-free — ALWAYS true

Fact: Every regular language is context-free.

Example:
L2 = a*

it is regular, and also context-free (since regular ⊆ CFL).

so (C) is true.

(D) L1 is context-free — NOT always true

Take

$L2 = \emptyset$ (regular)
$L1 = \{ a^n b^n c^n \mid n \geq 0 \}$ (not context-free)

Then
$L1 \cap L2 = \emptyset$ (regular)

But L1 is not context-free.

So (D) is false.

3. Answer : C,D

A: 101 accepted by D1 but not by D2.
B: wrong , working of both is completely different
C: true.
D.) D1 accept length divisible by 3 : 000,001,010,101
D2 accept length divisible by 3 : 011,100,110,111

Combinedly they will cover all length is divisible by 3 cases.

4. Answer:D

Explanation

NFA → DFA conversion:

When we convert an NFA to an equivalent DFA, the DFA can have **up to 2^n states**, because each DFA state represents a *subset* of NFA states.

So, saying "DFA has at most n states" is **wrong**.

Complement of L:

1. To take the **complement**, we need a **complete DFA** (because we just flip final and non-final s tates).
2. For an NFA, complementing directly is **not possible** — you must convert it to DFA first. So saying "complement can be accepted by an NFA with at most n states" is **wrong**.
3. **Regular expression:** Every NFA represents a **regular language**, an d every regular language can be written as a **re gular expression** — that is **true**.
4. **DFA with 2^n states:** The subset construction guarantees that a DFA with $\leq 2^n$ states always exists — that is **true**.

5. Answer:195

To solve this problem, we need to count the num ber of ternary strings of length 5 over the alphabet {a,b ,c} that contain at least one occurrence of two consecu tive symbols (i.e., "aa", "bb", or "cc").

We will use the **complementary counting** approach:

1. **Total number of ternary strings of length 5** Each position in the string can be filled with a, b, or c.
   Total strings = $3^5 = 243$.
2. **Number of ternary strings of length 5 with no two consecutive symbols:** For the first position, there are 3 choice (a, b, o r c). For each subsequent position, there are 2 choic es (any symbol except the one used in the pre vious position). Total strings with no two consecutive symbo ls = $3 \times 2^4 = 3 \times 16 = 48$.
3. **Number of ternary strings of length 5 with at l east one occurrence of two consecutive symbol s:** This is the complement of the above case. Total strings with at least one occurrence of tw o consecutive symbols = Total strings - Strings with no two consecutive symbols.
   - Total = $243 - 48 = 195$.

The number of such strings of length 5 is: 195

6 . Answer:- C

$L1 = \{ \alpha\beta\alpha \mid \alpha \in \{a, b\}^+ \text{ AND } \beta \in \{a, b\}^+ \}$
Here, α appears twice in the string, meaning the language requires matching substrings.This kind of pattern cannot be expressed using a finite-state automaton (DFA/NFA), as regular languages cannot enforce constraints on repeated substrings.
L1 is NOT regular.
$L2 = \{ \alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{ AND } \beta \in \{a, b\}^+ \}$
For this Language we can create regular expression $a^+.(a,b)^+.a^+$ that's mean starting and ending with a.
L2 is regular

7 . Answer:- C

Option 1: The set of all strings containing an even number of *b*'s.
False. The DFA accepts "babb" string which contains odd number of b's.

Option2: The set of all strings containing the pattern *bab*.
False. The DFA does not accept "baba" string which contain "bab" pattern.

Option3: The set of all strings ending with the pattern *bab*.
True. This option is true, DFA accepts all the strings ending with "bab" such as "babab", "bbab", "abab" etc.

Option4: The set of all strings not containing the pattern *aba*.
False. "bbab" does not contain "aba" pattern but DFA still accepts it.

8. Answer:-5

1. {A, C} (Represented as one state)
2. {B, E} (Represented as one state)
3. {D, H} (Represented as one state)
4. {F} (Unique output)
5. {G} (Unique output)

The reduced states are 5.

9. Answer:(B,C)
Context Free Languages (i):

(b) Not closed under complementation: Context-free languages (CFLs) are not closed under complementation, meaning the complement of a context-free language might not be context-free.

Recursive Languages (ii):

(a) Closed under union: Recursive languages (also called decidable languages) are closed under union, meaning the union of two recursive languages is also recursive.

Regular Languages (iii):

(c) Closed under intersection: Regular languages are closed under intersection, meaning the intersection of two regular languages is always regular.

(B): (i) → (b), (ii) → (a), and (iii) → (c) is correct.
(C): (i) → (b), (ii) → (c), and (iii) → (a) is correct.

10. Answer:(A,C)

$\{a^m b^n \mid m, n \geq 0\}$ is a regular language because it describes strings where there are any number of a's followed by any number of b's. This can be represented by the regular expression a*b*, which is regular.

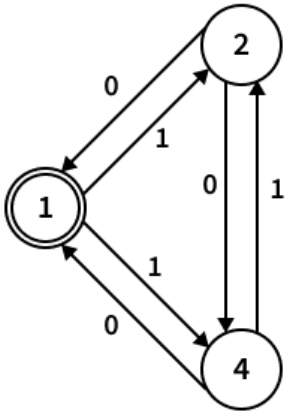
$\{w \mid w \in \{a, b\}^*, \#a(w) \equiv 2 \pmod 7$, and $\#b(w) \equiv 3 \pmod 9\}$ defines a language where the number of a's is congruent to 2 mod 7 and the number of b's is congruent to 3 mod 9. Regular languages can count occurrences of symbols with fixed remainders modulo some number, so this is a regular language. It can be represented using regular expressions with modular arithmetic for the counts of a and b.

11. Answer:6

The DFA needs to track the product of input symbols modulo 7, with possible remainders being 0, 1, 2, 3, 4, 5, 6. Therefore, the DFA requires 7 states. By this prod (x) we can't generate remainder 0 , so we need 6 states to track the product modulo 7.

Thus, the number of states in the minimum DFA for L is 6

12. Answer:(B,C)
When we try to minimize the DFA given we can notice that states 3 and 4 have same inward and outward transitions and are indistinguishable, same for the states 2, 5.
States 2 and 4 are distinguishable, because they have different transitions from state 1.
In the final minimised DFA, we can clearly see that there are three states 1 , 2 , 4. For any given string with equal number of 1s and 0s, it will be accepted by the DFA:



13 Answer:C,D
OptionA- False
Take L1 = {a} and L2 = $\{a^n \mid n>=0\}$, then L2`= Φ

Now, $L1 \cap L2 = \Phi$. But $L1 \neq L2$
OptionB- False
$L1=(a+b)^*$
L3 be any non-regular language, e.g., $\{a^n b^n | n \geq 0\}$.
$L1 \cup L3=(a+b)^*$,regular.
Thus, L1∪L3 can be regular even if L3 is regular.
Option C - True - Regular Languages are closed under Complementation.
Option D - True - Regular Languages are closed under Union .

14. Answer:44

For strings of length 2 :
{00, 11} are in r (all zeros or all ones).

{01, 10} are in s (start with 0 followed by 1 or start with 1 followed by 0).

For strings of length 3 :
{000, 111} are in r (all zeros or all ones).

{011, 100} are in s (start with 0 followed by ones or start with 1 followed by zeros).

For strings of length 4 :
{0000, 1111} are in r (all zeros or all ones).

{0111, 1000} are in s (start with 0 followed by ones or start with 1 followed by zeros).

For strings of length 5 :
{00000, 11111} are in r (all zeros or all ones).

{01111, 10000} are in s (start with 0 followed by ones or start with 1 followed by zeros).

Next, we calculate the number of strings not in r and not in s for each string length and sum them.

$= (2^2 - 4) + (2^3 - 4) + (2^4 - 4) + (2^5 - 4)$
$= (4 - 4) + (8 - 4) + (16 - 4) + (32 - 4)$
$= 0 + 4 + 12 + 28 = 44$

Thus, the total number of strings of length less than or equal to 5, which are neither in r nor in s, is 44.

15 Answer:A
State Elimination Method:

The easiest way through which we can reach to final state is through 1, which is not mentioned in options B and D. Hence, these two options are excluded.
Now, as we can see 111 is also a correct language according to given DFA, which is not matched by option C. Hence, it is also a wrong option.

Therefore, the correct option is A.

16. Answer:B
Option A :

Regular Expression: (00)* + 1(11)*

Counter example: The string 011 is not generated by this regular expression.

Option C :

Regular Expression : (00)* + (1 + (00)*)(11)*

Counter example : The string 000 is not generated by this regular expression.

Option D :

Regular Expression : 0+ + 1(11)* + 0(11)*

Counter example : The empty string is not generated by this regular expression.

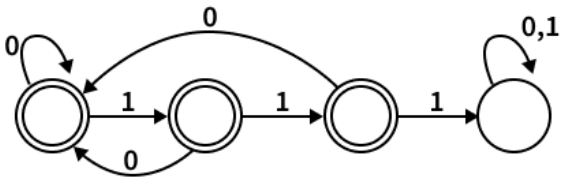
17. Answer:15
L_1 is exactly the strings with an odd number of a's (at least one).

For length $n \geq 1$, half of the 2^n binary strings have an odd number of a's, i.e. 2^{n-1}

So for lengths 1 to 4: $2^0 + 2^1 + 2^2 + 2^3 = 1 + 2 + 4 + 8 = 15$.

18. Answer:C
The regular expression is given as letter(letter+digit)*.
Option a is wrong as it also accepts empty string.
Option b is wrong as it will not accepts the strings like (letter.digit.letter).
Option d is wrong as it will not accepts strings like (letter.letter),(letter.digit) etc.

19. Answer:4



.

20. Answer:C

1(0*11)*: it is wrong because it will not generate strings like 10

0(0 + 1)*: it is wrong because it will generate 0 which is rejected in the given DFA.

1(110*)*: it is wrong because it will not generate strings like 10 , 100.

The most generic regular expression will be 1(0+11)*

21. Answer:D

The given automata accept the strings like, {a, b, ab, ba, abb, aba, baa, bab, abbb, abab, baba, baaa, …}

1. ab*bab*+ba*aba*

False, The strings like {'a', 'b' } accepted by given automata but the given regular expression can not accept those strings. Hence it is false.

2. (ab*b)*ab* + (ba*a)* ba*

False, The string "abbbbaa" is accepted by given automata but the given regular expression can not accept the string. Hence it is false.

3. (ab*b+ba*a)(a*+b*)

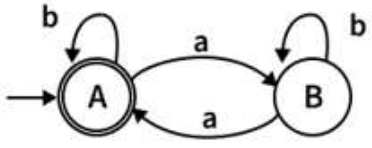
False, The given regular expression accepts the empty string like epsilon {Ɛ} but the given automata can not accept that empty string. Hence it is false.

4. (ba*a+ab*b)*(ab*+ba*)

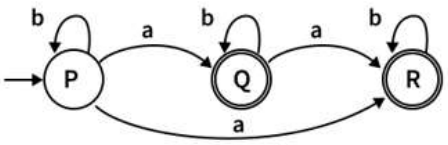
True, The given regular expression accepts the strings {a, b, ab, ba, abb, aba, baa, bab, abbb, abab, baba, baaa, …} which are accepted by given automata hence it is true.

22. Answer:6

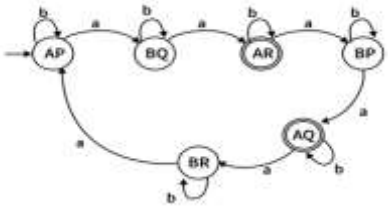
DFA 1 : No. of a's divisible by 2.



DFA 2: No. of a's not divisible by 3.



Using product automata



23. Answer: MTA

*** Wrong Question marks were given to all.**

Option :10*(0*10*10*)*

It will generate string only starting with '1'.

Option 2: ((0 + 1)*1(0 + 1)*1)*10*
It can generate strings with even number of 1's also.

Option 3: (0*10*10*)*10*
It will generate binary strings with odd number of 1's and doesn't generate the other strings.but all strings having odd number of 1's is not possible.
Ex. "01".
Option 4: (0*10*10*)*0*1
It will generate string only ending with '1'.

24. Answer:C

If $L_1 \cup L_2$ is regular, then neither L_1 nor L_2 needs necessarily be regular.
Assume L1 = $\{a^n b^n, n \geq 0\}$ over the alphabet {a, b} and L2 be the complement of L1.

Neither L1 nor L2 is regular (both are DCFL) but L1 ∪ L2 = $\{a^n b^n\} \cup \{a^n b^n\}^c = (a + b)^*$ is regular.

Statement II: FALSE. The infinite Union of regular languages is not regular.

Example:
Given alphabet {a, b}.

L1 = {ε}
L2 = {ab}
L3 = {aabb}
L4 = {aaabbb}
:
:
L = L1 ∪ L2 ∪ L3 ∪ L4
Each of the above are regular but their infinite Union gives L1 = $\{a^n b^n, n \geq 0\}$ which is not regular but DCFL.

25. **Answer:B**

* all options are wrong.
take $w \in \{a\}^*$, then ww^R is regular.

26. **Answer:D**
According to Pumping lemma, there must be repeation (DFA then it repeats some states, and regular grammar repeats its nonterminal in derivation.) for all acceptable stings.
Therefore, minimum Pumping Length should be 12. 12 symbols can be taken as first twelve b's i.e.b^{12}. When you repeat b^{12} Zero or more times, the resulted string still belongs to the language., pumping length for given language should greater than 12, which is 24.

27. The set is the set of all bijections from to . In mathematics, this is the **Symmetric Group** .

- The elements of S_5 are permutations.
- The operation is function composition (permutation multiplication).
- The identity element is id.
- The size of the group is $|S_5|=5!=120$.

The language consists of all sequences of permutations whose product (composition) results in the identity permutation.
According to the Myhill-Nerode Theorem, the number of states in a minimal DFA is equal to the number of equivalence classes of the prefix relation $\equiv_l$.
Two strings x and y are equivalent $(x \equiv_l y)$if for every string $z \in \Sigma^*$,
$xz \in L \iff yz \in L$.
• $xz \in L \Rightarrow \pi(x) \circ \pi(z) = id \Rightarrow \pi(z) = (\pi(x))^{-1}$
• $yz \in L \Rightarrow \pi(y) \circ \pi(z) = id \Rightarrow \pi(z) = (\pi(y))^{-1}$
Therefore, $x \equiv_l y$if and only if $\pi(x) = \pi(y)$.
Each distinct element in the group S_5represents a unique state in the minimal DFA because each element $g \in S_5$requires a different suffix z(specifically $z = g^{-1}$) to reach the identity.
Since every possible permutation in can be reached by composing elements from the alphabet (which contains all permutations including the generators), every element of the group corresponds to exactly one state in the minimal DFA.
Number of states = Order of the group
Number of states = 5!
Number of states = 120

28. **Answer:2**
The regular expression for L_1 : ε + 0(00)*

Now L_1^0 = ε and L_1^1 = ε + 0(00)* =
ε + 0(00)* = L_1.

Now $L_2 = L_1 \cdot L_1^1$
$L_2 = L_1 \cdot L_1$
= (ε + 0(00)) (ε + 0(00))
= ε + 0(00)* + 0(00)* + 0(00)(0(00))
= (ε + 0(00)) (0(00)) = 0*

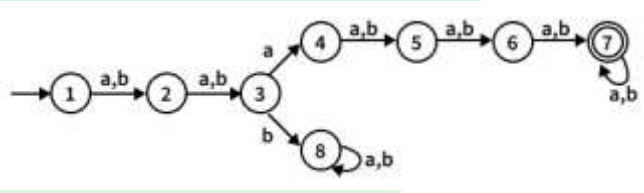
As it will contain epsilon + odd number of zero + even number of zero, hence it is 0*

Now $L_3 = L_2 \cdot L_1$
L_3 = 0* (ε + 0(00)) = 0 + 0*(0(00)) = 0*

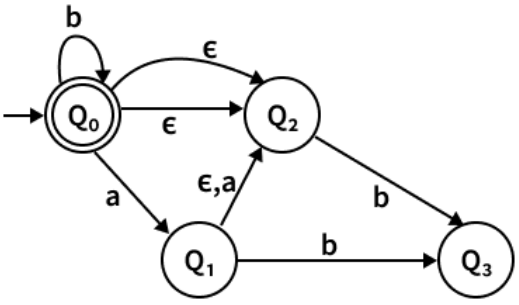
Hence $L_3 = L_2 = L_1$*
Or $L_1^2 = L_1^3 = L_1$*,
hence the smallest k value is 2.

29. **Answer:D**
The number of states in DFA is always less than equal to $2^{(\text{no. of states in NFA})}$. In other words, if number of states in NFA is "n" then the corresponding DFA have at most 2^n states. Hence $k \leq 2^n$ is necessarily true.

30. **Answer:A**
Answer is 8 states included one trap state (8) and final state (7).



31. **Answer:C**



Here, δ (q2, aba)

When input a is applied on q_2 it will move to q_0, q_1 as there is null transition so it will also get to q_2.

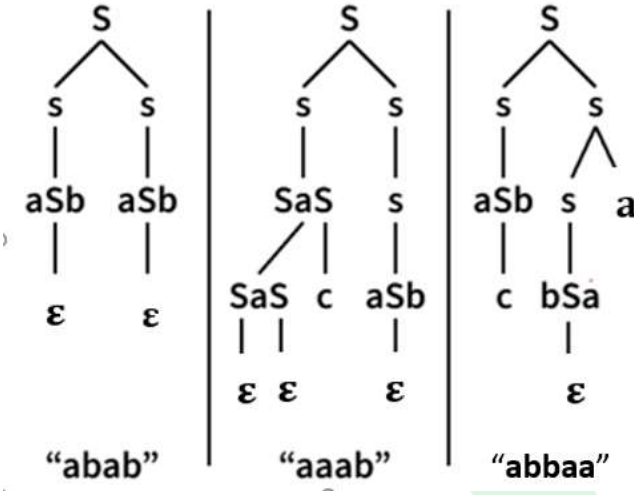
Similarly, states after input b ⇒ {q_0, q_2, q_3}

States after input a = {q_0, q_1, q_2}

So, final answer is {q_0, q_1, q_2}

32. **Answer:D**

The following grammar can yield the strings "abab," "aaab," and "abbaa."

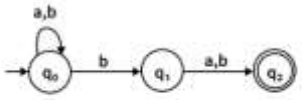


However, the given grammar is unable to construct the string "babba." The rationale for this is that whereas production S-> SaS can generate any number of a's, we only need to generate one "b" because the production that generates "b" also generates "a" (S ->aSb and S-> bSa).

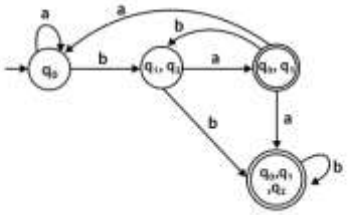
o in string "babba" the first and last "ba" can be generated by S⟶ bSa, but we can't generate a single "b" in middle. In other words we can say that any string in which number of "b's" is more than number of "a's" can't be generated by the given grammar.

33. **Answer:4**

The NFA for regular expression: (a+b)*b(a+b)

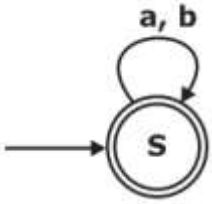


States / Input →	a	b
→[q_0]	[q_0]	[q_0, q_1]
[q_0, q_1]	[q_1, q_2]	[q_0, q_1, q_2]
[q_0, q_1]	[q_2]	[q_2, q_f]
[q_0, q_1, q_2]	[q_1, q_2]	[q_0, q_1, q_2]



34. **Answer:D**

From the given grammar we can draw the DFA,



35. **Answer:C**

I. Given NFAs N1 and N2, is L(N1) ∩ L(N2) = ∅?
This is decidable. When we convert the NFA into DFA, if there is no common final state in it, then this will become true.

II. Given a CFG G = (N, Σ, P, S) and a string x ∈ Σ*, does x ∈ L(G)?
Membership property of a context-free grammar is decidable.

III. Given CFGs G1 and G2, is L(G1) = L(G2)?
Equivalence of two context-free grammars is undecidable.

IV. Given a TM M, is L(M) = ∅?
Emptiness problem of Turing machine is undecidable.

36. **Answer:B**

1. (0 + 1)* 0011(0 + 1)* + (0 + 1)* 1100(0 + 1)*

It considers only those strings which have either 0011 or 1100 as a substring. So, it is wrong.

2. (0 + 1)* (00(0 + 1)* 11 + 11(0 + 1)* 00)(0 + 1)*

It contains the set of all the binary strings that contain two consecutive 0's and 1's.

3. (0 + 1)* 00(0 + 1)* + (0 + 1)* 11(0 + 1)*

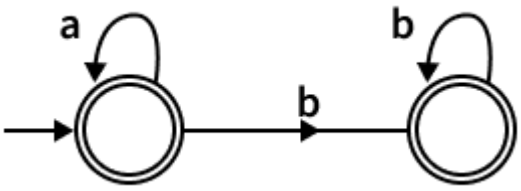
This set contains only those strings which have either 00 or 11 as a substring but does not contain all the strings that contain two consecutive 0's and 1's.

4. 00(0 + 1)* 11 + 11(0 + 1)* 00

It represents those strings which start with 00 or 11 and end with 11 or 00 respectively. So, it is wrong.

37. **Answer:B**

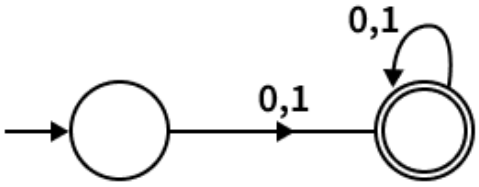
Statement I is false: As in NFA, it is not necessary that all states have transitions for all symbols.
For ex: Consider L = a*b*
The NFA would be:



Even though all states are final states in above NFA, but it doesn't accept string "aba".
Hence its language can't be ∑*.
Statement II is true: There exists a regular language A={}, such that for all languages B, A∩B={} is regular

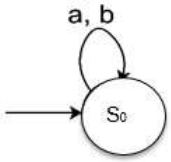
38. Answer:2

All strings over {0,1} having length ≥1
(0+1)*(0+1)(0+1)*=(0+1)(0+1)*=(0+1)*(0+1)=(0+1)$^+$
Having-DFA:



39. Answer:A

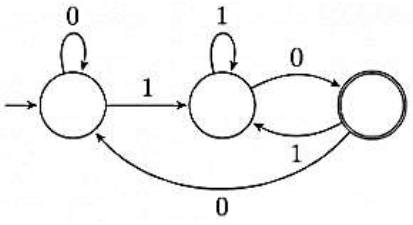
M accepts the strings which end with a and N accepts the strings which end with B. Their intersection should accept empty language.



40. Answer:A

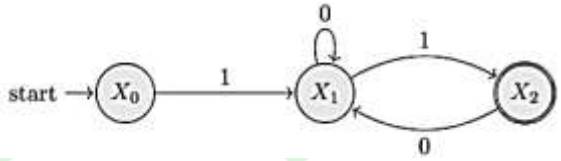
L1: All strings of length 3 or more with same start and end symbol, as everything in middle is consumed by x as per the definition.
L2: In this number of a's is dependent on number of b's. So PDA is needed.
L3: Any number of a's followed by any number of b's followed by any number of c's. Hence Regular.

41. Answer:A



42. Answer:C

Convert the given transitions to a state diagram.



From the given diagram we can write, X0 = 1(0+10)* 1

43. Answer:A

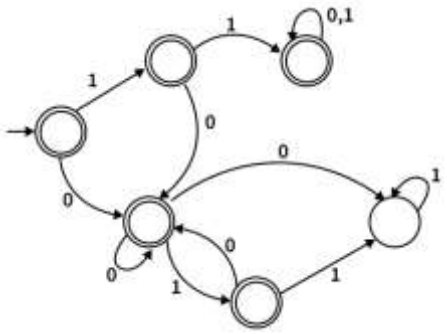
The regular expression generate all the strings of length 0 , 1 and 2 {ϵ, a, b, aa, ab, ba, bb}
Let's check all the string of length 3. The given regular expression generates {aaa, aab, aba, abb, baa, bba, bbb}
But it doesn't generate the string "bab", hence the shortest string not generated by regular expression has length 3 (string "bab").

44. Answer:A

For L1 :
{Number of occurrences of (110)} ≥ {Number of occurrences of (011)}
Let's look at the language. Consider a string with more than one occurrence of (110). The following are some options:
{1100110, 1101110, 110110, ….}
Please note that anytime strings begin with "11," the string will never break the requirement no matter what happens after "11." As a result, 11(0+1)* strings will always satisfy the requirement. Consider a string in which (011) appears more than once.
The following possibilities are: {011011, 0111011, 0110011, ….}

In the following possibilities please observe that number of occurrence "011" is two but number of occurrence of (110) is one, which violate the conditions.

If we add "0" in every string mentioned above, i.e. {0110110, 01110110, 01100110, ….} Now, observe that number of occurrence "011" and the number of occurrence of (110) both are equal, which satisfies the conditions.
With these analysis, we can make the DFA , which is mentioned below.



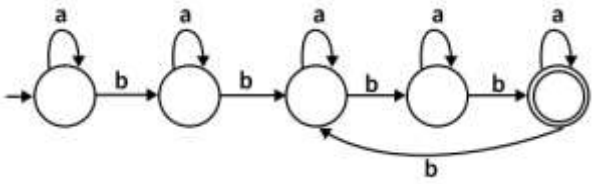
But language L2 requires infinite comparison to count the occurrences of (000's) and (111's), hence it is not regular

45. Answer:A

$\{q_0 , 0 \rightarrow q_0\}$, $\{ q_0 , 0 \rightarrow q_0 \}$, $\{q_0 , 1 \rightarrow q_0\}$, $\{q_0 , 1 \rightarrow q_0\}$.
Hence $\delta (q_0, 0011) = q_0$
$\{q_0 , 0 \rightarrow q_0\}$, $\{ q_0 , 0 \rightarrow q_0 \}$, $\{q_0 , 1 \rightarrow q_0\}$, $\{q_0 , 1 \rightarrow q_1,\}$.
Hence $\delta \quad (q_0, 0011) = q_1$
$\{q_0 , 0 \rightarrow q_0\}$, $\{ q_0, 0 \rightarrow q_0 \}$, $\{q_0 , 1 \rightarrow q_1,\}$, $\{q_1, , 1 \rightarrow q_2\}$.
Hence $\delta \quad (q_0, 0011) = q_2$
Hence $\delta (q_0, 0011) = \{q_0, q_1, q_2\}$

46. Answer:C

The Language L= { $a^n b^n$ | n>=0} is CFL but not regular, as it requires comparison between a's and b's.
L = { a^n | n is prime} is CSL, as calculation of "n is prime" can be done by LBA (Turing machine)
L = {ww | w ∈ ∑*} is CSL. But L = { w | w has 3k+1 b's for some k ∈ natural number} is regular.
Lets take values of k={1,2,3,….}
So number of b's will be {4, 7, 10,…….} and number of a's can be anything.
The DFA will be:



47. Answer:A

The regular expression equivalent to L1 and L2 are a* and b* respectively.
Since L1 and L2 both are regular languages and regular languages are closed under concatenation. So their concatenation (i.e., L1· L2) must also be a regular language.
So, L1·L2 = { $a^n b^m$ | n,m ≥ 0}

48. Answer:B

The DFA recognises the phrase "all strings ending in 1."

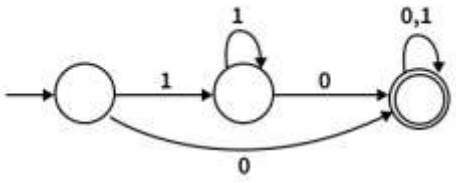
As a result, the regular expression for DFA is (0+1)*1.

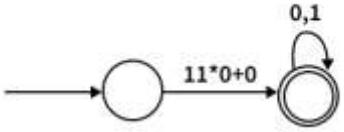
Now, the DFA has another similar regular expression: 0*1(1+00*1)*, due to the state reduction procedure.

However, the regular expression (0*1*1+11*0*1) is not comparable to DFA . Since, DFA also accepts the string "11011," which this regular expression cannot generate.

49. Answer:D

L(A) and its complement are both regular (due to the closure property), and every regular is also a CFL. As a result, LA Complement is context-free. The appropriate regular expression for the given FA is





Hence we have regular expression: (11*0 +0) (0+1)*Because we have (0+1)* at the end, writing 0*1* after this will have no effect; the reason for this is that if a string ends with terminals other than 1*0*, we can assume 1*0* as epsilon. So it is equivalent to (11*0 +0) (0+1)*0*1*.The given DFA can be minimised, since the non-final states are equivalent and can be merged and the min DFA will have two states which is given below:

50. **Answer:B**
The complement of a DFA can be obtained by making the non-final states as final states and vice-versa.
The Σ= {a} and the given NFA accepts the strings {a, aa, aaa, aaaa, …….} i.e. the language accepted by the NFA can be represented by the regular expression: {a+}
Hence the complement of language is: {a* − a +} = {ϵ}

51. **Answer:D**
Except for "q," which is a trap state, all states are final. Language strings are constructed in such a way that every substring of three symbols contains no more than two zeros. It signifies that there can't be three zeros in a row anywhere in the string. Total four transitions are missing in the given DFA, thus we must design them while keeping in mind the requirement that "three consecutive zeros" will lead to trap state "q."
From the state "00" it is clear that if another "0" comes then the string is going to be rejected, so from state "00" the transition with input "0" will lead to state "q".
So option A and B are eliminated.
Option C now includes a self loop of "0" on state "10," which accepts any number of zeros (including more than three zeros), therefore it is likewise incorrect. We were only left with option D, which is the correct choice.

52. **Answer:C**
L* will contain all those strings which can be obtained by any combination (and repetition) of the strings in language i,e, from L= {ab, aa, baa}
String 1: abaabaaabaa : abaa baa abaa
String 2: aaaabaaaa : aaaa baa aa
String 3: baaaaabaaaab: baa aaabaaaa b, because of the last "b" the string cannot belong to L*.
String 4: baaaaabaa : baa aaabaa

53. **Answer:A**
In (B) and (C) when the first letter of input is 'b' we reach final state, while in the given DFA first letter 'b' is not a final state. So, (B) and (C) are not accepting same language as the given DFA.

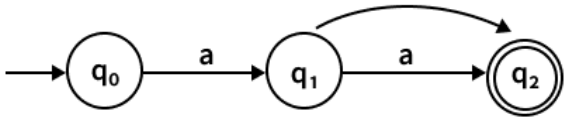
In (D) we can reach final state when the last letter is 'a', whatever be the previous transitions. But in the given DFA, when the first 2 letters are 'b' we can never reach the final state. So, (D) is also accepting a different language than the given DFA.

54. **Answer:B**

Given that n is a constant.
So lets check of n = 2,

$$L = a_{2k}, k > 0$$

Since k > 0 than zero.
So L is the language accepting even no. of a's except ' ε '.
So DFA will be,



So, no. of states required is 2 + 1 = 3.
So for a^{nk}, $(n+1)$ states will be required.

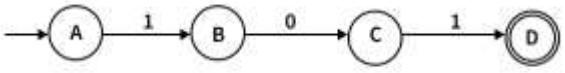
55. **Answer:B**
NPDA is more powerful than DPDA.

Expressive power:

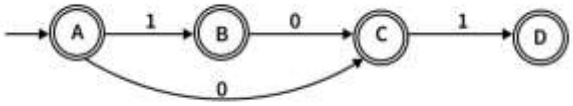
FA<DPDA<NPDA<LBA<TM

56. **Answer:C**
Exp: Σ* - P is the complement of P so it is always regular, since regular languages are closed under complementation.

57. **Answer:C**
In order to accept any string of length "n" with alphabet {0,1}, we require an NFA with "n+1" states. For example, consider a strings of length "3" such as "101", the NFA with 4 states is given below:



Since L is set of all substrings of " w " (Substring of a string is obtained by deleting any prefix or any suffix from string), so if we consider "w" as "101", then the substrings of w are $\{\varepsilon, 0,1, 10,01,101\}$.
Since the string " 101 " is also its substring, so we require 4 states (i.e. for n length string, $n+1$ states are required) and the NFA would be:



58. **Answer:B**
L = {w ∈ (0 + 1)*| w has even number of 1s}*

This grammar accepts the strings like {ε, 11, 011, 101, 110, …}

Option : (0 * 10 * 1)*
False, This grammar cannot accept the string "010101" and "010101" accepted by L.
Option : 0 * (10 * 10*)*
True, This accepts all strings of L.

Option : 0*(10 * 1*)*0*
False, This grammar cannot accept the string "1010101" and "1010101" accepted by L.

Option 4:0 *1(10 * 1)*10*
False, This grammar accepts the string "111" and "111" cannot be accepted by L.

59. **Answer :C**

Atlest two 0's at last.

60. **Answer : A**

Whenever we reach state 00 we get output 1.

Whenever we reach state 01 we get output 0 and 1.for 01 state we got two optputs 0 and 1. When we reach at state 10, we get output we get output 0. for 11 we don't know.

Draw a moore machine.
Minimum length input 101.

61. **Answer: A**

	States	a	b
→	**11(P)**	12	22
	12(S)	11	21
	21(Q)	22	12
(F)	**22(R)**	21	11

11 is P and 22 is R in choice. So, the answer should be (A) but in the row for S, it should be P and Q and Q not and P.

62. **Answer: C**

S – 4 is true,R– 3 is true because everything it accepts ends with 1; this is made mandatory only by 3

This rules out option B and option D.

Use string 01010 and compare P vs Q; this makes Q – 2 as confirmed.

Hence, option C is correct.

63. **Answer :A**

B. String begin with either a or b.
ab (string start with a) doesn't accept in it reach to nonfinal state q2.
bab (string start with b) doesn't accept in it reach to nonfinal state q2.

C. Set of strings that do not contain the substring ab
aba (have substring ab) does accept in it reach to final state q1.
D. The set described by the regular expression b*aa*(ba)*b*
b is string accepted by DFA(L') but above regular expression cannot derive it.

Option A is correct.

DFA (L') accepts all strings that doesn't end with ab.

64. L1 : (0 + 10)**11(0 + 1)**

L2 : (0 + 1)**11(0 + 1)**

It is quite clear that L1 = L2.

As both languages L1 and L2 are equal So Complement of Language L2 will be the complement of Language L1 also. For a given language L, $L \cap L^c = \emptyset$.

Hence, both options (A) and (C) are correct.

65. **Answer : A**

Since in option 2 and 3, is dependent on n m , therefore a comparison has to be done to evaluate those and hence are not regular.

I and IV are clearly regular sets.

66. **Answer : C**

1. q3 is unreachable state, hence it can be removed.

2. States q1 and q2 are indistinguishable, so, they can be merged.

67. **Answer : D**

String x is accepted and string xx is rejected.
From this knowledge option A,B,C are eliminated.

68. **Answer : A**

the languages being accepted by P and Q. In P before a 'c' there must be either 'b' or 'aa'. In Q, before 'c' there must be 'aa'. So, obviously, in their intersection before 'c' there must be 'aa' which is satisfied only by option A.

69. **Answer : A**

When we say non-deterministic finite automata recognizes the language defined by the regular expression R then it means that it won't accept any other string that does not fall under the R and it will only accept all strings that fall under R. R says the string should contain either aa or bb.

So, we see B accepts aba, C accepts aba and D accepts a. Just find some examples that do not follow the rule. A accepts any string containing either aa or bb and it does not accept any other string.

70, **Answer : A**

B. ab is accepted

C: abb is not accepted

D. ab is accepted

71. **Answer : C**

A.CFL

B. CFL

C. Regular, language is string starting and ending with the same symbol and having length at least 3.
e.g. 0x0,1x1

D. CFL

72. Answer : (B)

Every finite subset of a non-regular set is regular.

Any finite set is always regular.

Σ^* being regular any non regular language is a subset of this, and hence (A) is false.

If we take a CFL which is not regular, and takes union with its complement (complement of a CFL which is not regular won't be regular as regular is closed under complement), we get Σ^* which is regular. So, (C) is false.

Regular set is not closed under infinite union. Example:
Let $L_i = \{0^i 1^i\}$, $i \in N$

Now, if we take infinite union over all i, we get

$L = \{0^i 1^i \mid i \in N\}$, which is not regular.

73. **Answer : D**

u = abbaba
Ends in accepting state t → **Accepted**

v = bab
Ends in non-accepting state S → **Rejected**

w = aabb
Ends in non-accepting state S → **Rejected**

74. **Answer (A)**

Present State	Inputs		Next State
S	x	y	S^+
0	X	0	1
0	X	1	0
1	0	X	0
1	1	X	1

From this table S^+ will be A.

75. **Answer: C**

A. There are only finite 3 digit primes. Let the largest of them be X. Now we need X + 2 states in our DFA (including one for count = 0 and one for count > X). We do not need a change of state for any 1.

B. Here we need just 6 states to recognise L.

1. $\#0 - \#1 = 0$
2. $\#0 - \#1 = 1$
3. $\#0 - \#1 = 2$
4. $\#0 - \#1 = -1$
5. $\#0 - \#1 = -2$

If the difference goes above 2 or below −2, we go to a dead state. All other states are accepting. This transition to dead state is possible because of the words "for every prefix s′ of s" in L and that is what makes this language regular.

C. L is not regular

#0 − #1 = 1
#0 − #1 = 2
#0 − #1 = 3
#0 − #1 = 4

#0 − #1 = 5
.
.
#0 − #1 = 100
.
.

All these form distinct equivalent classes under Myhill-Nerode theorem meaning from the strings in each of these sets, we can append a string which will take the new string to L, while the same string appended to string in any other set would not reach L.

For example, for 000000, we append 11, for 0000000, we append 111 etc. So, in short we need to maintain the count of 1's and 0's and the count here is not finite.

D. This is regular. We need a finite automata with 5 × 7 = 35 states for maintaining the counts of 0's mod 7 and 1's mod 5 and there cannot be more than 35 possibilities for this. With each input symbol, the transition must be going to one among these.

76. Answer :None of theabove

A is Wrong, since abbb is accepted.

B is Wrong. take ∈,b,bb,bbb,.. is not accepted.

C is Wrong, since abba contains abb as substring, but is still not accepted.

D is Wrong, since ab does not contain aa as substring, but is still not accepted.

77. Answer : B

(It computes 2's complement of the input number)

The FSM reads the binary number from the least significant bit (LSB). It copies all bits unchanged up to and including the first '1'. After the first '1' is encountered, it complements all remaining higher bits (0 becomes 1 and 1 becomes 0).

This procedure is exactly how 2's complement is computed: keep bits unchanged until the first 1 from LSB, then flip all remaining bits.

78. Answer: (A).

Writing Y and Z in terms of incoming arrows (Arden's method) :

Y =X0+Y0+Z1

Z =X0+Z1+Y0 ,so Y=Z.

79. Answer: B

Minimum number of states required for the DFA = **3**
.L={w: (w) mod 3=2, w belongs to {a,b }}
S ⇒ Ya ⇒ Waa ⇒ Saaa (Mod 3)
S ⇒ Xa ⇒ Zaa ⇒ aa (remainder 2)
S ⇒ Xa ⇒ Zaa ⇒ Saaa (Mod 3)

Each cycle through S adds **three a's**.

2 CONTEXT FREE LANGUAGE AND CONTEXT FREE GRAMMAR

GATE CSE – 2026 SET-1 MSQ (1-MARKS)

Q1. Consider the following grammar where S is the start symbol, and a and b are

terminal symbols.

S→$aSbS \mid bS \mid \epsilon$

Which of the following statements is/are true?

(a) The grammar is ambiguous
(b) The string abb has two distinct derivations in this grammar
(c) The string $abab$ has only one rightmost derivation
(d) The language generated by the grammar is undecidable

GATE CSE – 2026 SET-2 MSQ (1-MARKS)

Q2. Which of the following grammars is/are ambiguous?

(a). S → aSb | ε
(b). E → E + E | E * E | id
(c). S → aS | Sa | ε
(d). S → aS | ε

GATE CSE – 2026 SET-2 MSQ (2-MARKS)

Q3. Let Σ = {a, b, c, d} and let L = { $a^i b^j c^k d^\ell \mid i, j, k, \ell \geq 0$ }.

Which of the following constraints ensure(s) that the language L is context-free?

(a) i + k = j + ℓ
(b) i = k and j = ℓ
(c) i = ℓ and j = k
(d) i + j = k + ℓ

GATE CSE – 2025 SET-2 MCQ (2-MARKS)

Q4. Let $G1$, $G2$ be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G, let $L(G)$ denote the language generated by G.

Which ONE among the following questions is decidable?

(a) Is $L(G1) = L(G2)$?
(b) Is $L(G1) \cap L(G2) = \emptyset$?
(c) Is $L(G1) = L(R)$?
(d) Is $L(G1) = \emptyset$?

GATE CSE – 2025 SET-2 MCQ (1-MARKS)

Q5. Which ONE of the following languages is accepted by a deterministic pushdown automaton?

(a) Any regular language.
(b) Any context-free language.
(c) Any language accepted by a non-deterministic pushdown automaton.
(d) Any decidable language.

GATE CSE – 2025 SET-2 MCQ (1-MARKS)

Q6. Consider the following context-free grammar G, where S, A, and B are the variables (non-terminals), a and b are the terminal symbols, S is the start variable, and the rules of G are described as:
$S \rightarrow aaB \mid Abb$
$A \rightarrow a \mid aA$
$B \rightarrow b \mid bB$
Which ONE of the languages $L(G)$ is accepted by G?

(a) $L(G) = \{a^2 b^n \mid n \geq 1\} \cup \{a^n b^2 \mid n \geq 1\}$
(b) $L(G) = \{a^n b^{2n} \mid n \geq 1\} \cup \{a^{2n} b^n \mid n \geq 1\}$
(c) $L(G) = \{a^n b^n \mid n \geq 1\}$
(d) $L(G) = \{a^{2n} b^{2n} \mid n \geq 1\}$

GATE CSE – 2025 SET-1 MCQ (2-MARKS)

Q7. Consider the following two languages over the alphabet $\{a, b, c\}$, where m and n are natural numbers.
$L1 = \{a^m b^m c^{m+n} \mid m, n \geq 1\}$
$L2 = \{a^m b^n c^{m+n} \mid m, n \geq 1\}$
Which ONE of the following statements is CORRECT?

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) A Both $L1$ and $L2$ are context-free languages.
(b) L1 is a context-free language but $L2$ is not a context-free language.
(c) $L1$ is not a context-free language but $L2$ is a context-free language..
(d) Neither $L1$ nor $L2$ are context-free languages.

GATE CSE – 2024 SET-2 MSQ (1-MARKS)

Q8. Consider a context-free grammar G with the following 3 rules.
$S \rightarrow aS, S \rightarrow aSbS, S \rightarrow c$
Let $w \in L(G)$. Let $n_a(w)$, $n_b(w)$, $n_c(w)$ denote the number of times a, b, c occur in w, respectively. Which of the following statements is/are TRUE?

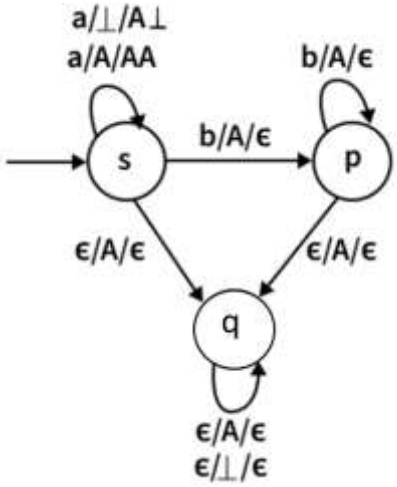
(a) $n_a(w) > n_b(w)$
(b) $n_a(w) > nc(w) - 2$
(c) $n_c(w) = n_b(w) + 1$
(d) $n_c(w) = n_b(w) * 2$

GATE CSE – 2024 SET-1 NAT (2-MARKS)

Q9. Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S. The string $w = a^{30}b^{30}c^{30}$ is derivable from S. The number of steps (application of rules) in the derivation $S \rightarrow^* w$ is _____

GATE CSE – 2023 MCQ (2-MARKS)

Q10. Consider the pushdown automaton (PDA) P below, which runs on the input alphabet {a, b}, has stack alphabet {⊥, A}, and has three states {s, p, q}, with s being the start state. A transition from state u to state v, labelled c/X/γ, where c is an input symbol or , X is a stack symbol, and γ is a string of stack symbols, represents the fact that in state u, the PDA can read c from the input, with X on the top of its stack, pop X from the stack, push in the string γ on the stack, and go to state v. In the initial configuration, the stack has only the symbol ⊥ in it. The PDA accepts by empty stack.



Which one of the following options correctly describes the language accepted by P?

(a) $\{a^m b^n \mid 1<=m \text{ and } n<m\}$
(b) $\{a^m b^n \mid 0<=n<=m\}$
(c) $\{a^m b^n \mid 0<=n \text{ and } 0<=m\}$
(d) $\{a^m \mid 0<=m\} \cup \{b^n \mid 0<=n\}$

GATE CSE – 2023 MSQ (1-MARKS)

Q11. Which of the following statements is/are CORRECT?

(a) The intersection of two regular languages is regular.
(b) The intersection of two context-free languages is context-free.
(c) The intersection of two recursive languages is recursive.
(d). The intersection of two recursively enumerable languages is recursively enumerable.

GATE CSE – 2023 MCQ (2-MARKS)

Q12. Consider the context-free grammar G below
S → aSb | X
X → aX | Xb | a | b ,
where S and X are non-terminals, and a and b are terminal symbols. The starting non-terminal is S. Which one of the following statements is CORRECT?

(a) The language generated by G is (a + b)*
(b) The language generated by G is a*(a + b)b*
(c) The language generated by G is a*b*(a + b)
(d) The language generated by G is not a regular language

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2022 MSQ (1-MARKS)

Q13. Consider the following languages:
$L1 = \{ww \mid w \in \{a, b\}^*\}$
$L2 = \{a^n b^n c^m \mid m, n \geq 0\}$
$L3 = \{a^m b^n c^n \mid m, n \geq 0\}$
Which of the following statements is/are FALSE?

(a) L1 is not context-free but L2 and L3 are deterministic context-free
(b) Neither L1 nor L2 is context-free.
(c) L2, L3, and L2 ∩ L3 all are context-free.
(d) Neither L1 nor its complement is context-free.

GATE CSE – 2022 MSQ (2-MARKS)

Q14. Consider the following languages:
$L_1 = \{ a^n w a^n \mid w \in \{a, b\}^* \}$
$L_2 = \{ wxw^r \mid w, x \in \{a, b\}^*, |w|, |x| > 0 \}$
Note that w^r is the reversal of the string w. Which of the following is/are TRUE?

(a) L1 and L2 are regular.
(b) L_1 and L_2 are context-free.
(c) L_1 is regular and L_2 is context-free.
(d) L_1 and L_2 are context-free but not regular.

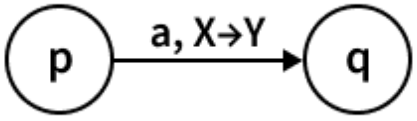
GATE CSE – 2021 SET-2 MSQ (2-MARKS)

Q15. For a string w, we define w^R to be the reverse of w. For example, if w= 01101 then w^R= 10110. Which of the following languages is/are context-free?

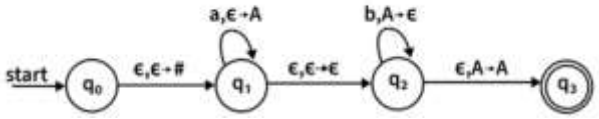
(a) $\{wxx^Rw^R \mid w,x \in \{0,1\}^*\}$
(b) $\{wxw^R \mid w,x \in \{0,1\}^*\}$
(c) $\{wxw^Rx^R \mid w,x \in \{0,1\}^*\}$
(d) $\{ww^Rxx^R \mid w,x \in \{0,1\}^*\}$

GATE CSE – 2021 SET-1 NAT (2-MARKS)

Q16. In a pushdown automaton $P=(Q,\Sigma,\Gamma,\delta,q_0,F)$, a transition of the form,



where $p, q \in Q$, $a \in \Sigma \cup \{\epsilon\}$, $X, Y \in \Gamma \cup \{\epsilon\}$, represents:$(q,Y) \in \delta(p,a,X)$
Consider the following p ushdown automaton over the input alphabet $\Sigma=\{a,b\}$ and stack alphabet $\Gamma=\{\#,A\}$.



The number of strings of length 100 accepted by the above pushdown automaton is _____

GATE CSE – 2020 MCQ (2-MARKS)

Q17. Consider the following languages.
$L_1 = \{wxyx \mid w, x, y \in (0 + 1)^+\}$
$L_2 = \{xy \mid x, y \in (a + b)^*, |x| = |y|, x \neq y\}$
Which one of the following is TRUE?

(a) L1 is context-free but not regular and L2 is context-free.
(b) Neither L1 nor L2 is context-free.
(c) L1 is regular and L2 is context-free.
(d) L1 is context-free but L2 is not context-free.

GATE CSE – 2019 MCQ (2-MARKS)

Q18. Which one of the following languages over $\Sigma = \{a,b\}$ is NOT context-free?

(a) $\{ww^R \mid W \in \{a, b\}^*\}$
(b) $\{wa^nw^Rb^n \mid W \in \{a,b\}^*, n >= 0\}$
(c) $\{a^nb^i | i \in \{n, 3n, 5n\}, n >= 0\}$
(d) $\{wa^nb^nw^R \mid w \in \{a, b\}^*, n >= 0\}$

GATE CSE – 2018 MCQ (2-MARKS)

Q19. Consider the following languages:
I.$\{a^mb^nc^pd^q \mid m + p = n + q$,where m,n,p,q >= 0$\}$
II.$\{a^mb^nc^pd^q \mid m=n$ and p=q,where m,n,p,q >= 0$\}$
III.$\{a^mb^nc^pd^q \mid m=n$ and p!=q,where m,n,p,q>= 0$\}$
IV.$\{a^mb^nc^pd^q \mid mn = p + q$, where m, n, p, q>= 0$\}$
Which of the above languages are context-free?

(a) I and IV only
(b) I and II only
(c) II and III only
(d) II and IV only

GATE CSE – 2017 SET-1 MCQ (1-MARKS)

Q20. Consider the following context-free grammar over the alphabet $\Sigma = \{a, b, c\}$ with S as the start symbol:
$S \rightarrow abScT \mid abcT$
$T \rightarrow bT \mid b$
Which one of the following represents the language generated by the above grammar?

(a) $\{(ab)^n (cb)^n \mid n >= 1\}$
(b) $\{(ab)^n cb^{(m_1)} cb^{(m_2)} \ldots\ldots cb^{(m_n)} \mid n, m_1, m_2, \ldots\ldots, m_n >= 1\}$
(c) $\{(ab)^n (cb^m)^n \mid m, n >= 1\}$
(d) $\{(ab)^n (cb^n)^m \mid m, n >= 1\}$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2017 SET-2 MCQ (1-MARKS)

Q21. Let L_1, L_2 be any two context-free languages and R be any regular language.
Then which of the following is/are CORRECT?
I. $L_1 \cup L_2$ is context free.
II. $\bar{L}_1$ is context-free
III. $L_1 - R$ is context – free.
IV. $L_1 \cap L_2$ is context – free
(a) I, II and IV only
(b) I and III only
(c) II and IV only
(d) I only

GATE CSE – 2017 SET-2 MCQ (2-MARKS)

Q22. Identify the language generated by the following grammar, where S is the start variable.
$S \rightarrow XY$
$X \rightarrow aX|a$
$Y \rightarrow aYb|\epsilon$
(a) $\{a^m b^n \mid m >= n, n > 0\}$
(b) $\{a^m b^n \mid m >= n, n > =0\}$
(c) $\{a^m b^n \mid m > n, n > =0\}$
(d) $\{a^m b^n \mid m > n, n > 0\}$

GATE CSE – 2017 SET-1 MCQ (2-MARKS)

Q23. Consider the following languages over the alphabet $\Sigma = \{a, b, c\}$. Let $L_1 = \{a^n b^n c^m \mid m, n > 0\}$ and $L_2 = \{a^m b^n c^n \mid m, n > 0\}$.
Which of the following are context-free languages?
I. $(L_1 \cup L_2)$
II. $(L_1 \cap L_2)$
(a) I only
(b) II only
(c) I and II
(d) Neither I nor II

GATE CSE – 2017 SET-2 MCQ (2-MARKS)

Q24. Consider the following languages:
$L1 = \{a^p \mid p \text{ is a prime number}\}$
$L2 = \{a^n b^m c^{2m} \mid n \geq 0, m \geq 0\}$
$L3 = \{a^n b^n c^{2n} \mid n \geq 0\}$
$L4 = \{a^n b^n \mid n \geq 1\}$
Which of the following are CORRECT?
I. L1 is context-free but not regular.
II. L2 is not context-free.
III. L3 is not context-free but recursive.
IV. L4 is deterministic context-free.
(a) I, II and IV only
(b) II and III only
(c) I and IV only
(d) III and IV only

GATE CSE – 2016 SET-2 MCQ (1-MARKS)

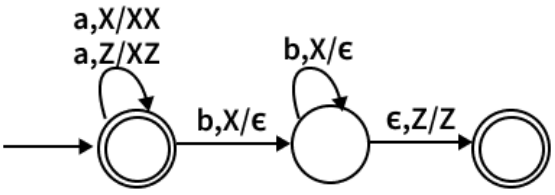
Q25. Language L1 is defined by the grammar:
$S_1 \rightarrow aS_1b|\varepsilon$
Language L2 is defined by the grammar:
$S_2 \rightarrow abS_2|\varepsilon$
Consider the following statements:
P: L1 is regular
Q: L2 is regular
Which one of the following is TRUE?
(a) Both P and Q are true
(b) P is true and Q is false
(c) P is false and Q is true
(d) Both P and Q are false

GATE CSE – 2016 SET-2 MCQ (2-MARKS)

Q26. Consider the following languages:
$L1 = \{a^n b^m c^{n+m} : m, n > 1\}$
$L2 = \{a^n b^n c^{2n} : n > 1\}$
Which one of the following is TRUE?
(a) Both L1 and L2 are context-free.
(b) L1 is context-free while L2 is not context-free.
(c) L2 is context-free while L1 is not context-free.
(d) Neither L1 nor L2 is context-free.

GATE CSE – 2016 SET-1 MCQ (2-MARKS)

Q27. Consider the transition diagram of a PDA given below with input alphabet $\Sigma = \{a,b\}$ and stack alphabet $\Gamma = \{X,Z\}$. Z is the initial stack symbol. Let L denote the language accepted by the PDA.



Which one of the following is TRUE?

(a) $L = \{a^n b^n \mid n >= 0\}$ and is not accepted by any finite automata
(b) $L = \{a^n \mid n >= 0\}$ $\{a^n b^n \cup \mid n > =0\}$ and is not accepted by any deterministic PDA.
(c) L is not accepted by any Turning machine that halts on every input.
(d) $L = \{a^n \mid n >= 0\}$ $\{a^n b^n \cup \mid n > =0\}$ and is deterministic context-free.

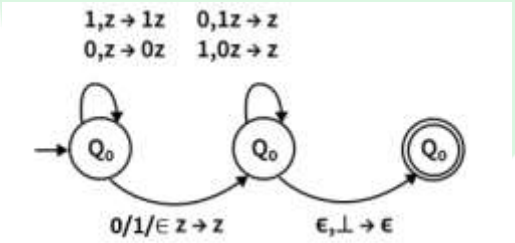
Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2016 SET-1 MCQ (2-MARKS)

Q28. Consider the following context-free grammars:
G1:
$S \rightarrow aS|B,$
$B \rightarrow b|bB$
G2:
$S \rightarrow aA|bB,$
$A \rightarrow aA|B|\varepsilon,$
$B \rightarrow bB|\varepsilon.$
Which one of the following pairs of languages is generated by G1 and G2, respectively?
(a) $\{a^m b^n \mid m > 0 \text{ or } n > 0\}$ and $\{a^m b^n \mid m > 0 \text{ and } n > 0\}$
(b) $\{a^m b^n \mid m > 0 \text{ and } n > 0\}$ and $\{a^m b^n \mid m > 0 \text{ or } n >= 0\}$
(c) $\{a^m b^n \mid m >= 0 \text{ or } n > 0\}$ and $\{a^m b^n \mid m > 0 \text{ and } n > 0\}$
(d) $\{a^m b^n \mid m >= 0 \text{ and } n > 0\}$ and $\{a^m b^n \mid m > 0 \text{ or } n > 0\}$

GATE CSE – 2015 SET-1 MCQ (2-MARKS)

Q29. Consider the NPDA $\langle Q = \{q0, q1, q2\}, \Sigma = \{0, 1\}, \Gamma = \{0, 1, \perp\}, \delta, q0, \perp, F = \{q2\}\rangle$, where (as per usual convention) Q is the set of states, Σ is the input alphabet, Γ is stack alphabet, δ is the state transition function, q0 is the initial state, $\perp$ is the initial stack symbol, and F is the set of accepting states, The state transition is as follows:



Which one of the following sequences must follow the string 101100 so that the overall string is accepted by the automaton?

(a) 10110
(b) 10010
(c) 01010
(d) 01001

GATE CSE – 2011 MCQ (2-MARKS)

Q30. Consider the languages L1,L2 and L3 as given below:

$L1 = \{ 0^p 1^q \mid p, q \in N \},$

$L2 = \{ 0^p 1^q \mid p, q \in N \text{ and } p = q \}$ and

$L3 = \{ 0^p 1^q 0^r \mid p, q, r \in N \text{ and } p = q = r \}.$

Which of the following statements is NOT TRUE?

(a) Push Down Automate (PDA) can be used to recognize L1 and L2
(b) L1 is a regular language
(c) All the three languages are context free
(d) Turing machines can be used to recognize all the languages

GATE CSE – 2010 MCQ (2-MARKS)

Q31. Consider the languages
$L1 = \{0^i 1^j \mid i != j\}.$
$L2 = \{0^i 1^j \mid i = j\}.$
$L3 = \{0^i 1^j \mid i = 2j+1\}.$
$L4 = \{0^i 1^j \mid i!= 2j\}.$
Which one of the following statements is true?
(a) Only L2 is context free.
(b) Only L2 and L3 are context free.
(c) Only L1 and L2 are context free.
(d) All are context free.

GATE CSE – 2009 MCQ (1-MARKS)

Q32. Which one of the following is FALSE?
(a) There is a unique minimal DFA for every regular language
(b) Every NFA can be converted to an equivalent PDA.
(c) Complement of every context-free language is recursive.
(d) Every nondeterministic PDA can be converted to an equivalent deterministic PDA.

GATE CSE – 2009 MCQ (1-MARKS)

Q33. $S \rightarrow aSa \mid bSb \mid a \mid b$
The language generated by the above grammar over the alphabet is the set of:
(a) all palindromes
(b) all odd length palindromes
(c) strings that begin and end with the same symbol
(d) all even length palindromes

GATE IT – 2008 MCQ (1-MARKS)

Q34. A CFG G is given with the following productions where S is the start symbol,A is a non-terminal and a and b are terminals.
$S \rightarrow aS \mid A$
$A \rightarrow aAb \mid bAa \mid \in$
Which of the following strings is generated by the grammar above?
(a) aabbaba
(b) aabaaba
(c) abababb
(d) aabbaab

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE IT – 2008 MCQ (2-MARKS)

Q35. Consider a CFG with the following productions

$S \longrightarrow AA \mid B$

$A \longrightarrow 0A \mid A0 \mid 1$

$B \longrightarrow 0B00 \mid 1$

S is the start symbol, A and B are non-terminals and 0 and 1 are the terminals. The language generated by this grammar is :

(a) $\{ 0^n10^{2n} \mid n \geq 1\}$

(b) $\{0^i10^j10^k \mid i,j,k \geq 0\} \cup \{ 0^n10^{2n} \mid n \geq 0\}$

(c) $\{0^i10^j \mid i,j \geq 0\} \cup \{ 0^n10^{2n} \mid n \geq 0\}$

(d) The set of all strings over {0,1} containing at least two 0's

GATE CSE – 2008 MCQ (2-MARKS)

Q36. Match the following:

E. Checking that identifiers are declared before their use	P. $L=\{ a^nb^mc^nd^m \mid n \geq 1, m \geq 1\}$
F. Number of formal parameters in the declaration of a function agrees with the number of actual parameters in a use of that function	Q. $X \rightarrow XbX \mid XcX \mid dXf \mid g$
G. Arithmetic expressions with matched pairs of parentheses	R. $L=\{wcw \mid w \in (a+b)^* \}$
H. Palindromes	S. $X \rightarrow bXb \mid cXc \mid \epsilon$

(a) E-P, F-R, G-Q, H-S.

(b) E-R, F-P, G-S, H-Q

(c) E-R, F-P, G-Q, H-S

(d) E-P, F-R, G-S, H-Q

GATE IT – 2008 MCQ (2-MARKS)

Q37. Consider the following languages.

$L1=\{a^i b^j c^k \mid i=j, k>=1\}$

$L2=\{a^i b^j \mid j=2i , i >=0\}$

Which of the following is true?

(a) L1 is not a CFL but L2 is

(b) $L1 \cap L2 = \phi$ and L1 is non –regular

(c) L1 U L2 is not a CFL but L2 is

(d) There is a 4- state PDA that accepts L1 ,but there is no DPDA that accepts L2.

GATE IT – 2007 MCQ (2-MARKS)

Q38. Consider the grammar given below:

$S \longrightarrow x\ B \mid yA$

$A \longrightarrow x \mid xS \mid y\ A\ A$

$B \longrightarrow y \mid yS \mid x\ B\ B$

Consider the following strings.

i. xxyyx

ii. xxyyxy

iii. xyxy

iv. yxxy

v. yxx

vi. xyx

Which of the above strings are generated by the grammar?

(a) i, ii and iii

(b) ii, v and vi

(c) ii, iii and iv

(d) i, iii and iv

GATE IT – 2007 MCQ (2-MARKS)

Q39. The two grammars given below generate a language over the alphabet {x,y,z}

G1: $S \longrightarrow x \mid z \mid xS \mid zS \mid yB$

$B \longrightarrow y \mid z \mid yB \mid zB$

G2: $S \longrightarrow y \mid z \mid yS \mid zS \mid xB$

$B \longrightarrow y \mid yS$

Which one of the following choices describes the properties satisfied by the strings in these languages?

(a) G1 : No y appears before any x
G2 : Every x is followed by at least one y.

(b) G1 : No y appears before any x
G2 : No x appears before any y

(c) G1 : No y appears after any x
G2 : Every x is followed by at least one y

(d) G1 : No y appears after any x
G2 : Every y is followed by atleast one x

GATE CSE – 2006 MCQ (1-MARKS)

Q40. Let L be a context-free language and M a regular language. Then the language $L \cap M$ is

(a) always regular

(b) never regular

(c) always a deterministic context-free language

(d) always a context-free language

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2006 MCQ (1-MARKS)

Q41. Let
$L1 = \{0^{n+m}1^n0^m \mid n, m >= 0\}$,
$L2 = \{0^{n+m}1^{n+m}0^m \mid n, m >= 0\}$ and
$L3 = \{0^{n+m}1^{n+m}0^{n+m} \mid n, m >= 0\}$.
Which of these languages are **NOT** context free?
(a) L1 only
(b) L3 only
(c) L1 and L2
(d) L2 and L3

GATE CSE – 2006 MCQ (1-MARKS)

Q42. In the context-free grammar below, is the start symbol, and are terminals, and denotes the empty string.
$S \rightarrow aSAb \mid \in$
$A \rightarrow bA \mid \in$
The grammar generates the language
(a) ((a+b)*b)
(b) $(a^mb^n \mid m \leq n)$
(c) $(a^mb^n \mid m=n)$
(d) a*b*

GATE CSE – 2006 MCQ (1-MARKS)

Q43. In the context-free grammar below,S is the start symbol,a and b are terminals, and ∈ denotes the empty string
$S \rightarrow aSa \mid bSb \mid a \mid b \mid \in$
Which of the following strings is NOT generated by the grammar?
(a) aaaa
(b) baba
(c) abba
(d) babaaabab

GATE CSE – 2006 MCQ (2-MARKS)

Q44. For $s \in (0+1)^*$ let d(s) denote the decimal value of s (e.g. d(101)=5). Let $L=\{s \in (0+1)^* \mid d(s) \bmod 5=2$ and $d(s) \bmod 7 \neq 4\}$
Which one of the following statements is true?
(a) is recursively enumerable, but not recursive
(b) is recursive, but not context-free
(c) is context-free, but not regular
(d) is regular

GATE CSE – 2006 MCQ (1-MARKS)

Q45. Which of the following languages is accepted by a non-deterministic pushdown automaton (NPDA) but NOT by a deterministic NPDA?
(a) $\{a^nb^nc^n \mid n \geq 0\}$
(b) $\{a^lb^mc^n \mid l \neq m$ or $m \neq n\}$
(c) $\{a^nb^n \mid n>=0\}$
(d) $\{a^mb^n \; m,n >=0\}$

GATE CSE – 2006 MCQ (2-MARKS)

Q46. Consider the pushdown automaton (PDA) below which runs over the input alphabet (a,b,c). It has the stack alphabet { Z_0,X } where Z_0 is the bottom-of-stack marker. The set of states of the PDA is (s,t,u,f) where s is the start state and f is the final state. The PDA accepts by final state. The transitions of the PDA given below are depicted in a standard manner. For example, the transition (s,b,X)-(t,XZ_0) means that if the PDA is in state s and the symbol on the top of the stack is X, then it can read b from the input and move to state t after popping the top of stack and pushing the symbols Z_0 and X (in that order) on the stack.
$(s,a,Z_0) \rightarrow (s,XXZ_0)$
$(s,\in,Z_0) \rightarrow (f,\in)$
$(s,a,X) \rightarrow (s,XXX)$
$(s,b,X) \rightarrow (t,\in)$
$(t,b,X) \rightarrow (t,\in)$
$(t,c,X) \rightarrow (u,\in)$
$(u,c,X) \rightarrow (u,\in)$
$(u,\in,Z_0) \rightarrow (f,\in)$
The Language accepeted by the PDA is
(a). $\{a^lb^mc^n \mid l=m=n\}$
(b). $\{a^lb^mc^n \mid l=m\}$
(c). $\{a^lb^mc^n \mid 2l=m+n\}$
(d). $\{a^lb^mc^n \mid m=n\}$

GATE CSE – 2005 MCQ (1-MARKS)

Q47. Consider the languages:
- $L1 = \{ww^r \mid w \in \{0, 1\}^*\}$
- $L2 = \{w\#w^r \mid w \in \{0, 1\}^*\}$, where # is a special symbol
- $L3 = \{ww \mid w \in \{0, 1\}^*\}$

Which one of the following is **TRUE**?
(a) L1 is a deterministic CFL
(b) L2 is a deterministic CFL
(c) L3 is a CFL, but not a deterministic CFL
(d) L3 is a deterministic CFL

GATE CSE – 2005 MCQ (1-MARKS)

Q48. Consider the languages:
$L1 = \{a^nb^nc^m \mid n, m > 0\}$ and $L2 = \{a^nb^mc^m \mid n, m > 0\}$,
Which one of the following statements is FALSE?

(a) L1 ∩ L2 is a context-free language
(b) L1 U L2 is a context-free language
(c) L1 and L2 are context-free languages
(d) L1 ∩ L2 is a context sensitive language

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE IT – 2005 MCQ (1-MARKS)

Q49. Let L be a regular language and M be a context-free language, both over the alphabet Σ. Let L^c and M^c denote the complements of L and M respectively. Which of the following statements about the language $L^c \cup M^c$ is TRUE?

(a) It is necessarily regular but not necessarily context-free.
(b) It is necessarily context-free.
(c) It is necessarily non-regular.
(d) None of the above

GATE IT – 2005 MCQ (1-MARKS)

Q50. The language $\{0^n1^n2^n \mid 1<=n<=10^6\}$ is

(a) Regular
(b) context-free but not regular
(c) context-free but its complement is not context-free
(d) not context free

GATE IT – 2005 MCQ (2-MARKS)

Q51. Let P be a non-deterministic push-down automaton (NPDA) with exactly one state, q, and exactly one symbol Z, in its stack alphabet. State q is both the starting as well as the accepting state of the PDA. The stack is initialized with one Z before the start of the operation of the PDA. Let the input alphabet of the PDA be Σ. Let L(P) be the language accepted by the PDA by reading a string and reaching its accepting state. Let N{P} be the language accepted by the PDA by reading a string and emptying its stack. Which of the following statements is TRUE?

(a) L(P) is necessarily Σ* but N{P} is not necessarily Σ*.
(b) N{P} is necessarily Σ* but L(P) is not necessarily Σ*.
(c) Both L(P) and N(P) are necessarily Σ*.
(d) Neither L(P) nor N(P) are necessarily Σ*

ANSWER KEY

1. (a,b,c)	**2.(b,c)**	**3.(a,c,d)**
4.(d)	**5.(a)**	**6.(a)**
7. (c)	**8.(b,c)**	**9.(179)**
10.(a)	**11.(a,c,d)**	**12.(b)**
13.(b,c,d)	**14.(a,b,c)**	**15.(a,b,d)**
16. (50)	**17.(c)**	**18.(b)**
19.(b)	**20.(b)**	**21.(b)**
22.(c)	**23.(a)**	**24.(d)**
25.(c)	**26.(b)**	**27.(d)**
28.(d)	**29.(b)**	**30.(c)**
31.(d)	**32.(d)**	**33.(b)**
34.(d)	**35.(b)**	**36.(c)**
37.(b)	**38.(c)**	**39.(a)**
40.(d)	**41.(d)**	**42.(b)**
43.(b)	**44.(d)**	**45.(b)**
46.(c)	**47.(b)**	**48.(a)**
49.(d)	**50.(a)**	**51.(d)**

HINTS AND SOLUTIONS

1. Answer: A,B,C

A. String : abb

$S \rightarrow aSbS \rightarrow aSb.\epsilon \rightarrow abSb\epsilon \rightarrow ab.\epsilon.b.\epsilon \rightarrow abb$

$S \rightarrow aSbS \rightarrow aSbbS \rightarrow aSbb.\epsilon \rightarrow a.\epsilon.bb.\epsilon \rightarrow abb$

Two right most derivations -> grammar is ambiguous.

B. We can use explanation of A for this also.

C. For

$s \rightarrow aSbS$ now for generating ab again we have only one option to put rightmost S as aSbS in rightmost derivation.

So C is also true.

D. Language generated by CFG is CFL ,and CFLs are decidable.

2. Answer : B,C.

(A) Not ambiguous

Generates a^nb^n.

Each string has a unique nested structure.

Only one parse tree possible.

(B) Ambiguous

Expression grammar without precedence.

Check Out Our GATE Courses & Secure Your Dream Rank!

Example: id + id * id
Can be parsed as:

1. (id + id) * id
2. id + (id * id)

Two parse trees → ambiguous.

(C) Ambiguous
Generates a*.
Example: "aa"
Can derive using left recursion or right recursion.
Two different parse trees → ambiguous.

(D) Not ambiguous
Right-recursive grammar for a*.
Each string has exactly one derivation

3. Answer : A,C,D

(A) $i + k = j + \ell$

This condition compares total number of (a and c) with total number of (b and d).

(B)This requires:
- match a with c
- match b with d

These are two independent equalities in sequence (cross-serial dependency).

This language is NOT context-free.

(C) We can:
- Push for each a
- Match b and c in middle
- Pop for each d

This is context-free.

Example grammar idea:
$S \rightarrow a\ S\ d \mid T$
$T \rightarrow b\ T\ c \mid \varepsilon$

Hence context-free.

(D) $i + j = k + \ell$

Total number of a's and b's equals total number of c's and d's.

This is similar to:
$a^m b^n c^p d^q$ where m+n = p+q

Single stack can count total of first part and match with second part.

This is context-free.

4. Answer:D
1. Is L(G1) = L(G2)?

Undecidable. Determining if two context-free grammars (CFGs) generate the same language is undecidable. There is no algorithm that can decide whether two arbitrary CFGs generate the same language.

2. Is $L(G1) \cap L(G2) = \emptyset$?

Undecidable. The intersection of two context-free languages is not necessarily context-free, and checking if two context-free languages have an empty intersection is undecidable. This is a generalization of the undecidability of the equivalence problem for CFGs.

3. Is L(G1) = L(R)?

Undecidable. The equivalence of a context-free language (generated by a CFG) and a regular language (defined by a regular expression) is undecidable. Determining whether a CFG generates a language that is exactly the same as a regular language is undecidable.

4. Is $L(G1) = \emptyset$?

Decidable. It is decidable whether the language generated by a context-free grammar G1 is empty. This can be determined by analyzing the grammar to check if there exists a derivation for any string. Specifically, this can be done by checking if there is any derivation starting from the start symbol and reaching a terminal string.

5. Answer : A

Since regular languages are a subset of deterministic context-free languages (DCFL), every regular language is also a DCFL.
Context-free languages (CFL) and recursive (decidable) languages are supersets of DCFL.

6. Answer : A
Let $S \rightarrow aaB$
$\rightarrow aab$
$\rightarrow aabB$
$\rightarrow aabb$
$\rightarrow aabbb$

Let $S \rightarrow Abb$
$S \rightarrow aAbb$

Check Out Our GATE Courses & Secure Your Dream Rank!

→ aab
→ aabb
→ aaAbb
→ aaabb

L = { aab, aabb, abb, aaabb, aaabbb, ... }

Then,
$L(G) = \{ a^2 \cdot b^n \mid n \geq 1 \} \cup \{ a^n \cdot b^2 \mid n \geq 1 \}$

7. Answer : C

For L1:

The language requires that the number of a's and b's must be equal ($a^m b^m$), and the number of c's must be greater than or equal to the number of a's (c^{m+n}).

This involves two simultaneous conditions or comparisons: one for equality between the number of a's and b's, and another for comparing the number of a's to c's.

Context-Free Grammars (CFGs) cannot handle two conditions at once, which require comparison of counts, thus, L_1 cannot be generated by a context-free grammar.

Therefore, L_1 is not a context-free language.

For L_2:

The language structure $\{a^m b^n c^{m+n}\}$ allows for:

Pushing a's onto a stack.
Pushing b's onto the same stack until c appears.
Matching c's with a's and b's as they pop off the stack.
This behavior can be captured using a Pushdown Automaton (PDA), which aligns with how context-free languages are processed.

Thus, L_2 is a context-free language

8. Answer : (B,C)

S → aS| aSbS | c

We can solve by elimination method in the string: w=c

As a=0, b=0, c=1

Now analyzing the options:

Option A: False
na(w) > nb (w) = 0 > 0

Now lets take other string: aaacbcbc
As, a=3, b=2, c=3

Option D: False

nc (w) = nb (w) * 2

1 = 0 * 2 which is 1 = 0

Option B,C: True

take any string , option B,C are always true

9. Answer : (179)

For CNF, number of steps required is 2|w| - 1

Example: if
S -> AA
A -> a
for string 'aa' we will need
S -> AA -> Aa -> aa

Here the |w| = 90
Thus the number of derivations will be 2*90 - 1 = 179

10. Answer : (A)

Note that Empty String ϵ is Not accepted by the given PDA, So options B, C, D are eliminated.
Empty String ϵ is Not accepted because from initial state, on empty tape, we cannot empty our stack since stack bottom symbol ⊥ can be removed only in state q But on empty tape, we can not reach to state q from state s.

Method 2:
Note that to empty our stack, the stack bottom symbol ⊥ can be removed only in state q. To reach state q, from s or from p, we need symbol A on Top-of-the-stack (TOS). Symbol A is pushed in the stack by symbol a in the input string. So, our input string must have at least one a.

In the initial state s, for every a in the input string, we push one A in the stack. Now, for every b in the input string, we pop one A from the stack. When all the b′s are over in the input string, we can move to state q if our TOS is A, so, to empty our stack at the end of any input string w, we must have more a′s than b′s in w. In the state q we can empty the stack & w is accepted.

Also, our PDA accepts a^+strings.

11. Answer : (A,C,D)

1. True. Regular languages are closed under intersection (use product DFA / cross-product construction).

2. False. CFLs are not closed under intersection in general (standard counterexamples exist).

3 .True. Recursive (decidable) languages are closed under intersection — run both deciders and accept iff both accept.

Check Out Our GATE Courses & Secure Your Dream Rank!

4. True. Recursively enumerable languages are closed under intersection — dovetail the two enumerators/simulators and accept when both machines accept.

12. Answer : (B)

$X \rightarrow a^m(a+b)b^p$ $(m \geq 0, p \geq 0)$

After replacing X in the first production we get,

$S \rightarrow aSb / a^m(a+b)b^p$.

Further reduction we get,

$S \rightarrow a^n a^m(a+b)b^p b^n$

further simplification gives : $a*(a+b)b*$.

13. Answer : (B,C,D)

L1 is not context-free and L2 and L3 are clearly DCFLs since they have only one comparison and DPDA can accept both.
Neither L1 nor L2 is context-free - False
L2 is DCFL and every DCFL is a CFL.
L2 - Context-free.
L3 - Context-free.
$L2 \cap L3 = \{a^n b^n c^n$ or $a^m b^m c^m$, $m,n \geq 0\}$
This language is context-sensitive language.
L2, L3 and L2 ∩ L3 all are context-free. - False
L2 - Context-free.
L3 - Context-free.
$L2 \cap L3 = \{a^n b^n c^n, n \geq 0\}$ We cannot compare $a^n b^n c^n$ with one single stack. Hence it is not context-free.
Neither L1 nor its complement is context-free.
False, L1 - Not context-free.
L1′ (Complement of L1) is accepted by a context-free grammar, hence it is CFL.

14. Answer : (A,B,C)

$L_1 = \{a^n w a^n \mid w \in \{a, b\}^*\}$

The regular expression can be written for the language L_1.

L_1 is regular because by putting n=0, we create a subset $\{w \mid w \in \{a,b\}\}$ which contains all possible strings. So if the subset of L_1 is (a + b), then $L_1 = (a + b)^*$ which is regular.

$$L_2 = \{wxw^r \mid w, x \in \{a, b\}^*, |w|, |x| > 0\}$$

The regular expression can be written for the language L_2.

L_2 is regular because by putting w as "a" and "b" we get a regular expression
a(a+b)a + b(a+b)b, which covers all other strings which can be obtained by putting w as "aa", "ab", "ba" and "bb", etc.

So $L_2 = a(a+b)a + b(a+b)b$ which is regular.
Now every regular is also context-free.

15. Answer : (A,B,D)

option A is correct

$\{wxx^R w^R \mid w, x \in \{0, 1\}^*\}$ is context-free language as it can be done using a stack. here is the way

push all letters of w in the stack
push all letters of x in the stack
if a new letter matches with the top of the stack, pop it
at the end stack will be empty so, it is context free language

option B is correct

$\{wxw^R \mid w, x \in \{0, 1\}^*\}$ is context-free.

push all the letters before w in the stack, skip x, now pop an element from the stack for each new letter, (top of the stack will match with each new letter due to w^R). So, it is context-free

option C is incorrect

$\{wxw^R x^R \mid w, x \in \{0, 1\}^*\}$ is Not context-free, we have to match w with $w^R w^R$ but the top of the stack contains x, so not possible.

option D is correct
$\{ww^R xx^R \mid w, x \in \{0, 1\}^*\}$ is Context-free language,

first match w with w^R then match x with x^R using a stack.
option 1,2,4 are correct.

16. Answer : (50)

According to given question,

$L = \{a^n b^m \mid n>m \ \& \ n+m=100\}$

$L = a^{51}b^{49}, \ldots\ldots, a^{99}b, a^{100}$

$|L| = 50$

For reaching q3 from q2 there must be “A” on the top of the stack, hence n>m.

17. Answer : (C)

L1 is regular. y can be expanded and w can also expanded.

Check Out Our GATE Courses & Secure Your Dream Rank!

$L1 = (0+1)^+ 0 (0+1)^+ 0 + (0+1)^+ 1 (0+1)^+ 1$

L2 is CFL since it is equivalent to complement of L={ww where w belongs to (a+b)* } U { set of all odd length strings strings }

18. Answer : (B)

$\{wa^n w^R b^n \mid w \in \{a,b\}^*, n \geq 0\}$ cannot be CFL.

This is similar to language

$L = \{a^n b^m c^n d^m \mid n, m > 0\}$

Suppose we push "w" then a^n and then w^R , now we cannot match b^n with an , because in top of stack we have w^R .

19. Answer : (B)

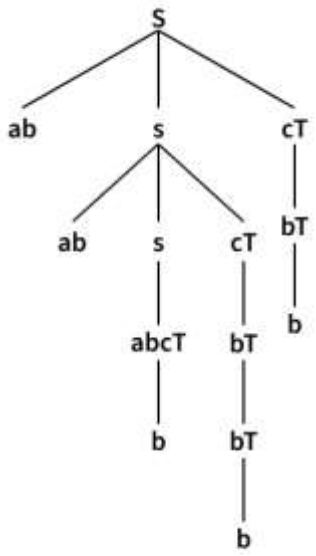
- $a^m b^n c^p d^q \mid m+p = n+q, m-n = q-p$ First we will push a's in the stack then we will pop a's after watching b's. Then some of a's might left in the stack. Now we will push c's in the stack and then pop c's by watching d's. And then the remaining a's will be popped off the stack by watching some more d's. And if no d's is remaining and the stack is empty then the language is accepted by CFG.
- $a^m b^n c^p d^q \mid m=n, p=q$ Push a's in stack. Pop a's watching b's. Now push c's in stack. Pop c's watching d's. So it is context free language.
- $a^m b^n c^p d^q \mid m=n=p \ \& \ p \neq q$ Here three variables are dependent on each other. So not context free.
- Not context free because comparison in stack can't be done through multiplication operation.
- $a^m b^n c^p d^q \mid m+p = n+q, m-n = q-p$ First we will push a's in the stack then we will pop a's after watching b's.
- Then some of a's might left in the stack. Now we will push c's in the stack and then pop c's by watching d's. And then the remaining a's will be popped off the stack by watching some more d's.
- And if no d's is remaining and the stack is empty then the language is accepted by CFG.
- $a^m b^n c^p d^q \mid m=n, p=q$ Push a's in stack. Pop a's watching b's. Now push c's in stack. Pop c's watching d's. So it is context free language.
- $a^m b^n c^p d^q \mid m=n=p \ \& p \neq q$ Here three variables are dependent on each other. So not context free.
- Not context free because comparison in stack can't be done through multiplication operation.

20. Answer : (B)

T→ bT | b, this production will generate any number of b's > 1

S→ abScT | abcT, this production will generate equal number of "ab" and "c" and for every "abc" any number of b's (> 1) after "abc".

For Ex:



21. Answer : (B)

Option 1) L1 ∪ L2 is context-free

As context-free languages are closed under union. So it is correct.

Option 2) $\bar{L}1$ is context-free

L1 is a context-free language. According to the property, its complement can't be context-free.

Option 3) L1 – R is context-free

As the difference of a context-free language with regular is context-free. So, it is correct.

Option 4) L1 ∩ L2 is context-free

Check Out Our GATE Courses & Secure Your Dream Rank!

It violated the context-free language property. The intersection of two CFL's is not CFL. So, it is incorrect.

22. Answer : (C)

The production X→ aX | a can generate any number of a's ≥ 1 and the production Y→ aYb | ϵ will generate equal number of a's and b's.

So the production S→XY can generate any number of a's (≥1) in the beginning (due to X) and then Y will generate equal number of a's and b's.

So, the number of a's will always be greater than number of b's and number of b's must be greater than or equal to 0 (as Y → ϵ, so number of b's can be zero also).

Hence the language is $\{a^m b^n \mid m>n, n\geq 0\}$.

23. Answer : (A)

Strings in L1 = {ϵ, c, cc, ….,ab, aabb,…., abc, abcc,…, aabbc, aabbcc, aabbccc,..}

Strings in L2 = {ϵ, a, aa, ….,bc, bbcc,…., abc, aabc,…, abbcc, aabbcc, aaabbcc,..}

Strings in L1 ∪ L2 ={ϵ, a, aa, .., c, cc,.. ab, bc, …, aabb, bbcc,.., abc, abcc, aabc,…}

Hence (L1 ∪ L2) will have either (number of a's = equal to number of b's) OR (number of b's = number of c's).

Hence (L1 ∪ L2) is CFL. Strings in L1 ∩ L2 = {ϵ, abc, aabbcc, aaabbbccc,…}

Hence (L1 ∩ L2) will have (number of a's = number of b's = number of c's) i.e., (L1 ∩ L2) = $\{a^n b^n c^n \mid n \geq 0\}$ which is CSL.

24. Answer : (D)

Option 1:
$L_1 = \{a^p \mid p \text{ is a prime number}\}$

Prime numbers do not have fixed pattern. So, it is not possible to solve it using pushdown automaton. So, L_1 is not context-free language.

Option 2:
$L_2 = \{a^n b^m c^{2m} \mid n \geq 0, m \geq 0\}$

It can be solved easily using single stack. Comparison is done between b and c. We can make a pushdown automaton for this. So, L_2 is context-free language.

Option 3:
$L_3 = \{a^n b^n c^{2n} \mid n \geq 0\}$
First comparison is between a and b, then between b and c, therefore two stacks are required.
So, it is context-sensitive language. L_3 is recursive language.

Option 4:
$L_4 = \{a^n b^n \mid n \geq 1\}$

First push a into a stack then pop a for each b from the stack. The given language is accepted by DPDA

25. Answer : (C)

The language L1 generated by the grammar contains equal number of a's and b's, but b's comes after a's.

So, in order to compare equality between a's and b's memory (stack) is required. Hence, L1 is not regular.

Moreover, $L1 = \{a^n b^n \mid n \geq 0\}$ which is DCFL.

The language L2 generated by grammar contains repetition of "ab" i.e. L2 = (ab)* which is clearly a regular language.

26. Answer : (B)

Statement I: $L_1 = \{a^n b^m c^{n+m} : m, n \geq 1\}$

L_1 can be accepted easily by a single stack.

First, push a's into the stack, then push b's into the stack, then read c's and pop b's.

When no b's are left on the stack, keep reading c's and pop a's.

When no c's are left in the input and the stack is empty, then the string is accepted by one stack.
Hence, L_1 is a context-free language.

Statement II: $L_2 = \{a^n b^n c^{2n} : n \geq 1\}$

This language can't be accepted by a single stack.
When we push a's into the stack and pop them with b's, it satisfies the condition that the number of a's = number of b's, but we are left with an empty stack and c's in the input.
We can't ensure that the number of c's is double that of a's and b's.
A pushdown automaton can't remember the number of a's for the third set of characters, i.e., c's.

Check Out Our GATE Courses & Secure Your Dream Rank!

Hence, L_2 is a context-sensitive language but not a context-free language.

27. Answer : (D)

In this PDA, we can give labels to states as q_0, q_1, q_2

where q_0 and q_2 are final states.

This PDA accepts the string in two ways —

i.e., by using q_0 as a final state and by using q_2 it accepts as an empty stack.

Since q_0 is the initial as well as the final state, it can accept any number of a's (including zero a's).

By using q_2 as the empty stack, it accepts strings that have an equal number of a's and b's (b's come after a's).

Hence, $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$.

28. Answer : (D)

$G_1 : S \rightarrow aS \mid B, \quad B \rightarrow b \mid bB$

This grammar generates strings of type {b, ab, bb, aab, abb, bbb, aaab, …}.

Here number of a's can be 0 but number of b's is always more than 0. From this option 1) and 2) are eliminated because these can't generate number of a's as 0.
Now, grammar G_2 is:
$G_2 : S \rightarrow aA \mid bB, \quad A \rightarrow aA \mid b \mid \varepsilon, \quad B \rightarrow b \mid \varepsilon$
This grammar generates string of type: {a, b, aa, ab, bb, aaa, aab, abb, bbb, …}.

This grammar doesn't generate null string — number of a's and b's are more than 0.
So, only option 4) matches according to grammar G_1 and G_2.

G_1 generates language of type: $\{a^m b^n \mid m \geq 0 \text{ and } n > 0\}$

G_2 generates language of type: $\{a^m b^n \mid m > 0 \text{ or } n > 0\}$

29. Answer : (B)

Here, in q_0 state, for a '1', a '1' is pushed into the stack and for a '0', a '0' is pushed into the stack.

In q_1 state, for a '0', a '1' is popped and for a '1', a '0' is popped.

It means when it finds a complement of the input character on top of the stack, then it pops one element from the top of the stack.

30. Answer : (C)

Statement I: $L1 = \{0^p 1^q \mid p, q \in N\}$,

Regular expression: 0^+1^+

Any number of 0's followed by any number of 1's. L1 requires no stack. Hence L1 is regular.

Every regular language is a context-free language.

Statement II: $L2 = \{0^p 1^q \mid p, q \in N \text{ and } p = q\}$

Context-Free Language: $L2 = \{0^p 1^p \mid p \in N\}$

The number of 0's is equal to the number of 1's. L2 requires only one stack. Hence L2 is context-free language.

Statement III: $L3 = \{0^p 1^q 0^r \mid p, q, r \in N \text{ and } p = q = r\}$

Context-Sensitive Language: $L3 = \{0^p 1^p 0^p \mid p \in N\}$.

The number of 0's followed by the equal number of 1's followed by equal number of 0's. It depends on three symbols. L3 requires two stacks. Hence L3 is a context-sensitive language.

It is not a context-free language.

31. Answer : (D)

All languages L1, L2, L3 and L4 has only one comparison and it can be accepted by PDA (single stack), hence all are Context Free Languages.

32. Answer : (D)

NDPA is more powerful than DPDA, so they are not equivalent. Actually, DPDA is a proper subset of NDPA.

33. Answer : (B)

String generated by this language is a,b,aba,bab,aabaa,..... All this strings are odd length palindromes.

34. Answer : (D)

The string **aabbaab** is generated by first using the start symbol S to create a single leading a,

Check Out Our GATE Courses & Secure Your Dream Rank!

and then expanding the non-terminal A to build the rest of the string from the outside layers toward the center.

Here is the exact step-by-step derivation sequence:

$S \Rightarrow aS$	(Using $S \to aS$ to generate the first character)
$\Rightarrow aA$	(Using $S \to A$ to switch to the A productions)
$\Rightarrow aaAb$	(Using $A \to aAb$ to match the outer $\underline{a}$ ... $\underline{b}$ of a[a]
$\Rightarrow aabAab$	(Using $A \to bAa$ to match the next layer $\underline{b}$ ... $\underline{a}$ ∘
$\Rightarrow aabbAaab$	(Using $A \to bAa$ to match the innermost layer $\underline{b}$.
$\Rightarrow aabbaab$	(Using $A \to \epsilon$ to terminate the derivation)

Because the production rules for A always insert pairs of terminals on the left and right flanks simultaneously, the derivation effectively evaluates the string from both ends toward the middle once S provides the necessary offset.

35. Answer : (B)

B→0B00|1 generates

{ 0^n10^{2n} }

S→AA,

A→0A|A0|1 generates 0A0A →00A0A→00101 , which is there in only B and D choices.

D is not the answer as " 00 " is not generated by the given grammar. So, only option left is B .

36. Answer : (C)

h- strings generated by this grammar satisfies the definition of an even length palindrome string.
E-R is true because R is the only grammar that checks: what (w) has occurred earlier is present afterwards. This equals the definition of E

37. Answer : (B)

L1 & L2 both are DCFL .

L1 ∩ L2 = ϕ because we have no string a i b j where i=j and i=2j for i,j>=1.

38. Answer : (C)

i. xxyyx , not generated
ii. Xxyyxy , S → xB → xxBB → xxyyS → xxyyxB → xxyyxy
iii. xyxy , S → xB → xyS → xyxB → xyxy
iv. yxxy , S → yA → yxS → yxxB → yxxy
v. yxx , not generated
vi. xyx , not generated 41.

39 Answer : (A)

Regular expression for G1: $(x+z)^+ + (x+z)^*y(y+z)^+$

Regular expression for G2: $(y+z+xy)^+$

40. Answer : (D)

L1=Σ* is regular on {a,b}.

L2={ww^r |w∈(a+b)* } is CFL

L1 ∩ L2 is CFL.

41. Answer : (D)

L1 is context-free. We count the number of 1's and check if the remaining number of 's followed by 0's count to the initial number of 0's.

L2 is not context-free. Here the number of 0's and the following 1's must be same, which can be checked using a PDA. But after that we must also ensure that the following number of 0's must be less than the previous count of 0's and 1's (otherwise n<0, which violates the condition for acceptance) and we cannot do these two checks using a single PDA.

L3 is not context-free as it is nothing but equal number of 0's followed by equal number of 1's followed by equal number of 0's.

42. Answer : (B)

S generate equal number of a and b, and one A which can generate zero or more number of b's

43. Answer : (B)

L(G) is palindrome.

44. Answer : (D)

d(s)mod 5=2 have 5 states and state 2 as a final

d(s)mod 7 !=4 have 7 states and all states except 4 is a final state.

both are regular and regular language is closed under intersection.

45. Answer : (B)

A) is wrong as it is not context free

C) is DCFL

Check Out Our GATE Courses & Secure Your Dream Rank!

D) is regular

46. **Answer : (C)**
For every a we put two X in stack

After that for every b we pop out one X

After that for every c we pop out one X

If all X are popped out then reached to final state f , mean for every b there is a, for every c there is a.

a was followed by b and b was followed by c.

means sum of no of b's and no of c's= twice of no of a's

47. **Answer : (B)**

L1- CFL

L2-DCFL

L3-CSL

48. **Answer : (A)**

$L_1 \cap L_2$ is a context-free language → FALSE

Taking the intersection means strings must satisfy the constraints of both languages simultaneously:

$$L_1 \cap L_2 = \{a^n b^n c^n \mid n > 0\}$$

This requires a simultaneous three-term comparison ($n_a = n_b = n_c$), which is a classic non-context-free language because a single LIFO stack cannot track both dependencies at once.

49. **Answer : (D)** Given

- L is regular
- M is context-free

1. L^c → complement of regular ⇒ regular
2. M^c → complement of CFL ⇒ not necessarily CFL

Using De Morgan's law:

$L^c \cup M^c = (L \cap M)^c$

- Regular ∩ CFL = CFL
- Complement of a CFL is not necessarily CFL

So $L^c \cup M^c$ is not guaranteed to be regular or CFL.

50. **Answer : (A)**

Regular Since n is finite, we have a finite set of strings satisfying the given conditions.

51. **Answer : (D)**

In NPDA we may have a dead configuration. This mean we may not give any transition to any alphabet from this state. We say that a string is accepted if PDA is in final state after reading the final symbol in the string or after it has read '$' symbol denoting end of the string and it is in final state.

Check Out Our GATE Courses & Secure Your Dream Rank!

3 UNDECIDABILITY

GATE CSE – 2026 SET-2 MSQ (1-MARKS)

Q1. Which one of the following statements is equivalent to the following assertion?

Turing machine M decides the language $L \subseteq \{0,1\}^*$.

(a). Turing machine M halts on all input strings in $\{0,1\}^*$

(b). Turing machine M accepts all input strings in L

(c). Turing machine M rejects all input strings in $\{0,1\}^* - L$

(d).Turing machine M accepts all input strings in L and rejects all input strings in $\{0,1\}^* - L$

GATE CSE – 2022 MSQ (2-MARKS)

Q2. Which of the following is/are undecidable?

(a) Given two Turing machines M1, and M2, decide if L(M1) = L(M2).
(b) Given a Turing machine M, decide if L(M) is regular.
(c) Given a Turing machine M, decide if M accepts all strings.
(d) Given a Turing machine M, decide if M takes more than 1073 steps on every string.

GATE CSE – 2022 MSQ (1-MARKS)

Q3. Which of the following statements is/are TRUE?

(a) Every subset of a recursively enumerable language is recursive.
(b) If a language L and its complement I are both recursively enumerable, then L must be recursive.
(c) Complement of a context-free language must be recursive.
(d) If L1 and L2 are regular, then $L1 \cap L2$ must be deterministic context-free.

GATE CSE – 2021 MCQ (2-MARKS)

Q4. Let <M> denote an encoding of an automaton M. Suppose that $\Sigma = \{0, 1\}$. Which of the following languages is/are NOT recursive?

(a) $L = \{<M> | M \text{ is a DFA such that } L(M) = \emptyset\}$
(b) $L = \{<M> | M \text{ is a DFA such that } L(M) = \Sigma^*\}$
(c) $L = \{<M> | M \text{ is a PDA such that } L(M) = \emptyset\}$
(d) $L = \{<M> | M \text{ is a PDA such that } L(M) = \Sigma^*$

GATE CSE – 2020 MCQ (2-MARKS)

Q5. Which of the following languages are undecidable? Note that <M> indicates encoding of the Turing machine M.
$L_1 = <M> \mid L(M) = \phi\}$
$L_2 = \{<M, w, q> | M$ on input w reaches state q in exactly 100 steps$\}$
$L_3 = \{<M> | L(M)$ is not recursive$\}$
$L_4 = \{<M> | L(M)$ contains at least 21 members$\}$?

(a) L2 and L3 only
(b) L1 and L3 only
(c) L2, L3 and L4 only
(d) L1, L3 and L4 only

GATE CSE – 2019 MCQ (2-MARKS)

Q6. Consider the following sets:
S1. Set of all recursively enumerable languages over the alphabet {0,1}
S2. Set of all syntactically valid C programs
S3. Set of all languages over the alphabet {0,1}
S4. Set of all non-regular languages over the alphabet {0,1}
Which of the above sets are uncountable?
(a) S2 and S3
(b) S3 and S4
(c) S1 and S4
(d) S1 and S2

GATE CSE – 2018 MCQ (1-MARKS)

Q7. The set of all recursively enumerable languages is:-

(a) closed under complementation.
(b) closed under intersection.
(c) a subset of the set of all recursive languages.
(d) an uncountable set.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2018 MCQ (2-MARKS)

Q8. Consider the following problems. L(G) denotes the language generated by a grammar G. L(M) denotes the language accepted by a machine M.
(I) For an unrestricted grammar G and a string w, whether w∈L(G).
(II) Given a Turing machine M, whether L(M) is regular.
(III)Given two grammar G1 and G2, whether L(G1) = L(G2).
(IV) Given an NFA N, whether there is a deterministic PDA P such that N and P accept the same language.
Which one of the following statement is correct?

(a) Only I and II are undecidable
(b) Only III is undecidable
(c) Only II and IV are undecidable.
(d) Only I, II and III are undecidable.

GATE CSE – 2017 SET-2 MCQ (2-MARKS)

Q9. Let L(R) be the language represented by regular expression R. Let L(G) be the language generated by a context free grammar G. Let L(M) be the language accepted by a Turing machine M.
Which of the following decision problems are undecidable?
I. Given a regular expression R and a string w, is w ∈ L(R)?
II. Given a context-free grammar G, is L(G) = ∅?
III. Given a context-free grammar G, is L(G) = Σ* for some alphabet Σ?
IV. Given a Turing machine M and a string w, is w ∈L(M)?

(a) I and IV only
(b) II and III only
(c) II, III and IV only
(d) III and IV only

GATE CSE – 2017 SET-1 MCQ (2-MARKS)

Q10. Let A and B be finite alphabets and let # be a symbol outside both A and B. Let f be a total function from A* to B*. We say f is computable if there exists a Turing machine M which given an input $x \in A^*$, always halts with f(x) on its tape. Let L_f denote the language $\{x \# f(x) \mid x \in A^*\}$. Which of the following statements is true:

(a) f is computable if and only if L_f is recursive.
(b) f is computable if and only if L_f is recursively enumerable.
(c) If f is computable then L_f is recursive, but not conversely.
(d) If f is computable then L_f is recursively enumerable, but not conversely.

GATE CSE – 2016 SET-2 MCQ (1-MARKS)

Q11. Consider the following types of languages:
L1: Regular
L2: Context-free
L3: Recursive
L4: Recursively enumerable.
Which of the following is/are TRUE?
I. L3' ∪ L4 is recursively enumerable .
II. L2' ∪ L3 is recursive .
III. L1* ∪ L2 is context-free .
IV. L1 ∪ L2' is context-free.
(a) I only
(b) I and III only
(c) I and IV only
(d) I, II and III only

GATE CSE – 2016 SET-2 MCQ (2-MARKS)

Q12. Consider the following languages.
L1 = { ⟨M⟩ | M takes at least 2016 steps on some input },
L2 = { ⟨M⟩ | M takes at least 2016 steps on all inputs },
and
L3 = { ⟨M⟩ | M accepts ε },
Where for each Turing machine M, ⟨M⟩ denotes a specific encoding of M.
Which one of the following is TRUE?
(a) L1 is recursive and L2, L3 are not recursive
(b) L2 is recursive and L1, L3 are not recursive
(c) L1, L2 are recursive and L3 is not recursive
(d) L1, L2, L3 are recursive

GATE CSE – 2015 SET-2 MCQ (1-MARKS)

Q13. Consider the following statements.
I. The complement of every Turing decidable language is Turing decidable
II. There exists some language which is in NP but is not turing decidable
III. If L is a language in NP, L is turing decidable
Which of the above statements is/are true?
(a) Only II
(b) Only III
(c) Only I and II
(d) Only I and III

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2015 SET-1 MCQ (2-MARKS)

Q14. For any two languages L1 and L2 such that L1 is context-free and L2 is recursively enumerable but not recursive, which of the following is/are necessarily true?
1. L1' (complement of L1) is recursive.
2. L2' (complement of L2) is recursive.
3. L1' is context-free.
4. L1' ∪ L2 is recursively enumerable.

(a) I only
(b) III only
(c) III and IV only
(d) I and IV only

GATE CSE – 2014 SET-2 MCQ (2-MARKS)

Q15. Let <M> be the encoding of a Turing machine as a string over Σ = {0, 1} Let L = { <M> |M is a Turing machine that accepts a string of length 2014}. Then, L is :

(a) decidable and recursively enumerable
(b) undecidable but recursively enumerable
(c) undecidable and not recursively enumerable
(d) decidable but not recursively enumerable

GATE CSE – 2014 SET-2 MCQ (1-MARKS)

Q16. Let $A \leq_m B$ denotes that language A is mapping reducible (also known as many-to-one reducible) to language B.Which one of the following is FALSE?

(a) If $A \leq_m B$ and B is recursive then A is recursive.
(b) If $A \leq_m B$ and A is undecidable then B is undecidable.
(c) If $A \leq_m B$ and B is recursively enumerable then A is recursively enumerable
(d) If $A \leq_m B$ and B is not recursively enumerable then A is not recursively enumerable

GATE CSE – 2014 SET-3 MCQ (1-MARKS)

Q17. Let Σ be a finite non-empty alphabet and let 2^{Σ^*} be the power set of Σ*. Which one of the following is **TRUE**?

(a) Both 2^{Σ^*} and Σ* are countable
(b) 2^{Σ^*} is countable and Σ* is uncountable
(c) 2^{Σ^*} is uncountable and Σ* is countable
(d) Both 2^{Σ^*} and Σ* are uncountable

GATE CSE – 2013 MCQ (2-MARKS)

Q18. Which of the following is/are undecidable?
1. G is a CFG. Is L(G) = ϕ?
2. G is a CFG. Is L(G) = Σ*?
3. M is a Turing machine. Is L(M) regular?
4. A is a DFA and N is an NFA. Is L(A) = L(N)?

(a) 3 only
(b) 3 and 4 only
(c) 1, 2 and 3 only
(d) 2 and 3 only

GATE CSE – 2013 MCQ (2-MARKS)

Q19. Which of the following statements is/are FALSE?
1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine.
2. Turing recognizable languages are closed under union and complementation.
3. Turing decidable languages are closed under intersection and complementation.
4. Turing recognizable languages are closed under union and intersection.

(a) 1 and 4 only
(b) 1 and 3 only
(c) 2 only
(d) 3 only

GATE CSE – 2009 MCQ (2-MARKS)

Q20. Let L= L1 ∩ L2 , where L1 and L2 are languages as defined below:
$L1=\{a^m b^m c^n b^n \mid m,n >=0\}$
$L2=\{a^i b^j c^k \mid i,j,k >=0\}$
Then is L

(a) Not recursive
(b) Regular
(c) Context free but not regular
(d) Recursively enumerable but not context free

GATE CSE – 2008 MCQ (2-MARKS)

Q21. Which of the following statements is false?

(a) Every NFA can be converted to an equivalent DFA
(b) Every non-deterministic Turing machine can be converted to an equivalent deterministic Turing machine
(c) Every regular language is also a context-free language
(d) Every subset of a recursively enumerable set is recursive

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2008 MCQ (1 -MARKS)

Q22. Which of the following is true for the language $\{ a^p \mid p \text{ is prime} \}$?

(a) It is not accepted by a Turing Machine
(b) It is regular but not context-free
(c) It is context-free but not regular
(d) It is neither regular nor context-free, but accepted by a Turing machine

GATE CSE – 2008 MCQ (2-MARKS)

Q23. Which of the following are decidable?

I. Whether the intersection of two regular languages is infinite
II. Whether a given context-free language is regular
III. Whether two push-down automata accept the same language
IV. Whether a given grammar is context-free

(a) I and II
(b) I and IV
(c) II and III
(d) II and IV

GATE CSE – 2007 MCQ (2-MARKS)

Q24. The language $L=\{ 0^i21^i \mid i \geq 0 \}$ over the alphabet $\{0,1,2\}$ is:

(a) not recursive
(b) is recursive and is a deterministic CFL
(c) is a regular language
(d) is not a deterministic CFL but a CFL

GATE CSE – 2007 MCQ (1-MARKS)

Q25. Which of the following problems is undecidable?

(a) Membership problem for CFGs
(b) Ambiguity problem for CFGs
(c) Finiteness problem for FSAs
(d) Equivalence problem for FSAs

GATE CSE – 2006 MCQ (1-MARKS)

Q26. Let L1 be a regular language, L2 be a deterministic context-free language and L3 a recursively enumerable, but not recursive, language. Which one of the following statements is false?

(a) L1 ∩ L2 is a deterministic CFL
(b) L3 ∩ L1 is recursive
(c) L1 U L2 is context free
(d) L1 ∩ L2 ∩ L3 is recursively enumerable

GATE CSE – 2005 MCQ (2-MARKS)

Q27. Let L1 be a recursive language, and let L2 be a recursively enumerable but not a recursive language. Which one of the following is TRUE?

(a) L1' is recursive and L2' is recursively enumerable
(b) L1' is recursive and L2' is not recursively enumerable
(c) L1' and L2' is recursively enumerable
(d) L1' is recursively enumerable and L2' is recursive

GATE CSE – 2005 MCQ (2-MARKS)

Q28. Consider three decision problems P1, P2 and P3. It is known that P1 is decidable and P2 is undecidable.

(a) P3 is decidable if P1 is reducible to P3.
(b) P3 is undecidable if P3 is reducible to P2
(c) P3 is undecidable if P2 is reducible to P3
(d) P3 is decidable if P3 is reducible to P2's complement

ANSWER KEY

1.(d)	**2.(a,b,c)**	**3.(b,c,d)**
4.(d)	**5.(d)**	**6.(b)**
7.(b)	**8.(d)**	**9.(d)**
10.(a)	**11.(d)**	**12.(c)**
13.(d)	**14.(d)**	**15.(b)**
16.(d)	**17.(c)**	**18.(d)**
19.(c)	**20.(c)**	**21.(d)**
22.(d)	**23.(b)**	**24.(b)**
25.(b)	**26.(b)**	**27.(b)**
28.(c)		

Check Out Our GATE Courses & Secure Your Dream Rank!

HINTS & SOLUTIONS

1. Answer : D

A Turing machine M decides a language L means:

1. M halts on every input string (both in L and not in L).
2. M accepts every string in L.
3. M rejects every string not in L.

In short: it must halt and give the correct yes/no answer for every input.

2 Answer : (A,B,C)

Option A: It is undecidable to check the equivalence of 2 RELs.

Option B: It is a non-trivial property of RELs hence it is Undecidable.

Option C: It is also a non-trivial property of RELs thus it is undecidable.

Option D: If TM Runs at least 1073 steps on all inputs of length upto 1073 then it will behave the same on all inputs of all lengths and thus it will run at least 1073 steps on all inputs . Thus it is decidable.

3 . Answer : (B,C,D)

Option A: No language is closed under subset operation. So subset on an RE language may or may not be REC. Therefore, option A is false.

Option B: If L and L` are both RE ⇒ L is REC is a theorem and is true.

Option C: Every CSL is recursive. So complement of CFL is Recursive and it is true.
(CFL)C = (CSL)C = CSL

Option D: L1 = regular, L2= regular

L1 ∩L2 = Regular ∩ Regular = Regular

Now every regular language is a DCFL. So, option D is also true.

4 Answer : (D)

- For DFAs:
  - We can check emptiness of a DFA easily (by checking if any final state is reachable from the start).
  - We can also check if a DFA accepts all strings (by checking if every possible string leads to a final state)
- Checking emptiness of PDA language is decidable — we can convert PDA → CFG and test if CFG generates any string.
- Checking if PDA accepts all strings (L(M) = Σ)* is undecidable — there's no algorithm to verify this for context-free languages.

5 Answer : (D)

L1 is undecidable as emptiness problem of Turing machine is undecidable.

L3 is undecidable since there is no algorithm to check whether a given TM accept recursive language. L4 is undecidable as it is similar to membership problem.
Only L2 is decidable. We can check whether a given TM reach state q in exactly 100 steps or not. Here we have to check only upto 100 steps.There is no case of going to infinite loop.

6. Answer : (B)

S1 is countable, set of all recursively enumerable languages means set of all Turing machines and we can enumerate TM and have one to one correspondence between natural number.
S2 is countable, since a valid C program represents a valid algorithm and every algorithm corresponds to a Turing Machine, so S2 is equivalent to set of all Turing Machines.
S3 is is uncountable, it is proved by diagonalization method.
S4 is uncountable, as set of non-regular languages will have languages which is set of all languages over alphabet {0,1} i.e., S3.

7 Answer : (B)

Under intersection, recursive enumerable languages are closed.
Recursive enumerable languages are not closed under Complementation.
Recursive enumerable languages are a countable set, as every recursive enumerable language has a corresponding Turing Machine and set of all Turing Machine is countable. Recursive languages are subset of recursive enumerable languages.

8 Answer : (D)

IV is trivial property, as every regular language is CFL also, so a language which has NFA must be regular and for every regular language we can have a deterministic PDA (as every regular language is DCFL).
I, II and III is undecidable.

9. Answer : (D)

Since membership problem for regular language is decidable, so I is decidable. Emptiness problem for Context free language is decidable, so II is decidable.

Check Out Our GATE Courses & Secure Your Dream Rank!

Completeness problem (i.e. L(G) = Σ* for a CFG G) is undecidable.
Membership problem for recursive enumerable language (as language of Turing Machine is recursive enumerable) is undecidable. So, IV is undecidable.

10 **Answer : (A)**

The correct option is (a) f is computable if and only if L_f is recursive.
Here is the step by step mathematical proof for both directions of the "if and only if" statement:

Direction 1: If f is computable, then L_f is recursive ($\rightleftharpoons$)
If f is a computable total function, there exists a Turing machine M_f that, when given an input x, is guaranteed to halt and leave $f(x)$ on its tape. We can construct a new decider Turing machine M_L to decide the language $L_f = \{x\#\int (x) \mid x \in A^*\}$ as follows:

1. Parse the input: Given an arbitrary input string w, check if it contains exactly one # symbol. If it doesn't, reject.
2. Split the components: If it matches the format, split the string into the portion before the delimiter (x) and the portion after the delimiter (y). So, $w = x\#y$.
3. Simulate the function: Run the computable machine M_f on the input substring x. Because f is a total computable function, M_f is guaranteed to halt and output $f(x)$.
4. Compare results: Compare the output $f(x)$ with the string y from the input tape.

- If $f(x) == y$, then accept.
- If $f(x) \neq y$, then reject.

Since M_f always halts, M_L is guaranteed to always halt and make a definitive decision for any input string. Therefore, L_f is a recursive (decidable) language.

Direction 2: If L_f is recursive, then f is computable ($\Leftarrow$)
If L_f is recursive, there exists a decider Turing machine M_L that can always determine whether a string of the form $x\#y$ belongs to L_f or not. We can construct a Turing machine M_f to compute the function $f(x)$ for any given input $x \in A^*$:

1. Generate candidate strings: Since the alphabet B is finite, we can systematically generate all possible strings over B^* in lexicographical order: $\epsilon, b_1, b_2, \ldots$
2. Test each candidate: For each generated string $y \in B^*$, construct the combined string $x\#y$.
3. Run the decider: Feed the string $x\#y$ into the decider M_L.

- Since M_L is a decider, it will always halt and either accept or reject.
- If M_L rejects, move to the next lexicographical string in B^*.
- If M_L accepts, it means $y = f(x)$. Stop the search, clear the tape, leave y on the tape, and halt.

Because f is a total function, a valid output string $y = f(x)$ is guaranteed to exist somewhere in the lexicographical enumeration. Thus, M_f will eventually find it, output it, and halt. This proves that f is computable.

11. **Answer : (D)**

I. $\overline{L_3} \cup L_4$
L_3 is recursive, so $\overline{L_3}$ is also recursive (closed under complement),
So, $\overline{L_3}$ is recursive enumerable.
L_4 is recursive enumerable,
so, $\overline{L_3} \cup L_4$ is also recursive enumerable (closed under union).
II. $\overline{L_2} \cup L_3$
L_2 is Context-free, so $\overline{L_2}$, may or may not be Context-free (not closed under complement), but definitely $\overline{L_2}$ is Recursive.
L_3 is recursive.
so $\overline{L_2} \cup L_3$ is also recursive (closed under union).
III. $L_1^* \cap L_2$
L_1 is Regular, so L_1^* is also regular (closed under kleene star)
L_2 is Context-free
so, $L_1^* \cap L_2$ is also context-free (closed under intersection with regular).
IV. $L_1 \cup \overline{L_2}$
L_1 is regular.
L_2 is context-free, so $\overline{L_2}$ may or may not be Context-free (not closed under complement).
so, $L_1 \cup \overline{L_2}$ may or may not be Context-free.

12 **Answer : (C)**

- L_1 and L_2 are recursive (decidable) because the condition is bounded by a fixed number of steps (2016). In 2016 steps, a Turing machine can read at most the first 2016 symbols on its tape. Because the alphabet is finite, there are a finite number of possible string prefixes of this length. A decider can simulate the machine on all these combinations for exactly 2016 steps. Since both the number of strings and the number of steps are finite, the simulation is guaranteed to halt and give an answer.
- L_3 is not recursive (undecidable) because checking whether a Turing machine accepts the empty string (ϵ) is the classic Blank Tape Acceptance Problem. According to Rice's Theorem, any non-trivial property about the language a Turing machine accepts cannot be decided by a general algorithm.

Check Out Our GATE Courses & Secure Your Dream Rank!

13. **Answer : (D)**
Statement 1: True

A decision problem is a problem that can be posed as a yes or no. If we can decide a problem, then its complement is also decidable. So, given statement is true.

Statement 2: False

NP class problems are the problems which can be solved in polynomial time using non-deterministic machine. None of the NP class problem is undecidable. So, this statement is incorrect.

Statement 3: True

NP problem is undecidable.

14. **Answer : (D)**
1. L1 is context free: Every context free language is also recursive and recursive languages are closed under complement.

2. L2' (complement of L2) is recursive is false: Recursively enumerable languages are not closed under set difference or complementation.
3. L1' is context-free: Context-free languages are not closed under complement, intersection, or difference.

4. L1' ∪ L2 is recursively enumerable is true:

Since L1' is recursive, it is also recursively enumerable and, Recursively enumerable languages are known as type-0 languages in the Chomsky hierarchy of formal languages.
All regular, context-free, context-sensitive and recursive languages are recursively enumerable.)

15. **Answer : (B)**
The language accepted by the Turing machine is recursively enumerable. It is undecidable as the Turing machine may halt or it may loop for the strings whose length is not equal to 2014. So, the correct answer is this option.

16. **Answer : (D)**
A ≤m B means language A is mapping reducible to language B. Thus, A cannot be harder than B.
Since, A can be reduced to B, instead of deciding A we can now decide B. So, the first three options are correct.
As B is not recursively enumerable, it doesn't guarantee A is not recursively enumerable. Thus, if A ≤m B and B is not recursively enumerable then A is not recursively enumerable. Therefore, this option is correct.

17. **Answer : (C)**
A countable set is one for which an enumeration technique exists to produce each of its elements, and it takes a finite number of steps to generate each element of the set using the enumeration procedure.
If = {0,1} then Σ* ={ϵ, 0, 1, 00, 01, 10, 11, 0 00,.....}
Since we can enumerate all the strings of Σ*, hence Σ* is countable (countable infinite).
While 2^{Σ^*} is uncountable, it has been already proved by using Diagonalization method.

18. **Answer : (D)**
1. G is a CFG. Is L(G) = φ -> Decidable (Emptiness Problem)
2. G is a CFG. Is L(G) = Σ*-> Undecidable (Completeness Problem)
3. M is a Turing Machine. Is L (M) regular -> Undecidable
4. A is a DFA and N is an NFA. Is L(A) = L(N) -> Decidable.

19 **Answer : (C)**
1.Non-deterministic Turing Machine can be simulated by a deterministic Turing Machine with exponential time - True
2. Turing recognizable language are "not" closed under complementation. For any Turing recognizable language the Turing Machine ' T ' recognizing ' L ' may not terminate on inputs x ∉ L - False
3. Turing decidable languages are CLOSED under union and complementation. It is easy to determine if turing machine is decidable -True .

20 **Answer : (C)**

- Language Types: L_1 is a Context-Free Language (CFL) because it can be parsed using a stack to match the matching counts ($a^m b^m$ then $c^n b^n$). L_2 is a Regular Language defined by the ordering pattern $a^* b^* c^*$.
- Intersection Constraint: Taking $L_1 \cap L_2$ forces the strings to follow the $a^m b^m c^n b^n$ structure while strictly obeying the $a^* b^* c^*$ order. Because L_2 forbids any b from appearing after a c, the trailing b^n block must be empty, forcing $n = 0$.
- Resulting Language: Substituting $n = 0$ leaves $L = \{a^m b^m \mid m \geq 0\}$. This language is context-free but cannot be regular because a finite automaton lacks the memory to count and match the a 's and b 's.

Correct Option: (c) Context free but not regular

21 **Answer : (D)**
There exists a set of languages which is RE but not REC, this set is a subset of RE but is Not Recursive.

Option D tells us that every subset of RE is REC this is false. Hence, option D is chosen.

Check Out Our GATE Courses & Secure Your Dream Rank!

22. **Answer : (D)**
we have algorithm to generate prime numbers so turing machine possible

23 **Answer : (B)**
I) Decidable

II) The property of regularity is decidable for languages up to the DCFL class, but it becomes undecidable for general CFLs.

III) undecidable
IV) A grammar is Context-Free (CFG) if the left side of every rule contains exactly one variable, while the right side can be any combination of symbols.

24. **Answer : (B)**
$L = \{0^i 21^i \mid i \geq 0\}$ has only one comparison that can be done using a DPDA. Hence, its DCFL. Context free languages are a proper subset of Recursive Languages. ∴ it is recursive too.

25. **Answer : (B)**
A. CYK algorithm

B. undecidable

C. decidable

D. decidable

26 **Answer : (B)**
A. DCFL are closed under Intersection with Regular Languages

B. L1 is recursive hence also decidable, L3 is RE but not Recursive hence it is undecidable. Intersection of Recursive language and Recursive Enumerable language is Recursive Enumerable language.

C. L1 is regular hence also CFL and every DCFL is also CFL and All CFL are closed under Union.

D L1 is regular hence also RE; L2 is DCFL hence also RE; RE languages are closed under Intersection

27 **Answer : (B)**
REL: is not closed under complementation
and Rec is closed under complementation.

28. **Answer : (C)**
A. If P1 is reducible to P3, then P3 is at least as hard as P1. So, no guarantee if P3 is decidable.

B. If P3 is reducible to P2, then P3 cannot be harder than P2. But P2 being undecidable, this can't say P3 is undecidable.

C. If P2 is reducible to P3, then P3 is at least as hard as P2. Since P2 is undecidable, this means P3 is also undecidable -hence the answer.

D. Complement of an undecidable problem is undecidable. Now, reducing to an undecidable problem can't prove P3 is decidable.

Check Out Our GATE Courses & Secure Your Dream Rank!

COMPLIER DESIGN

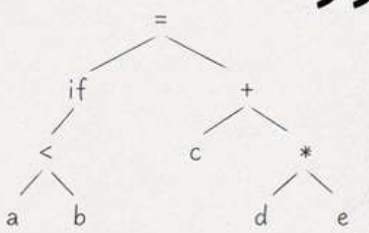
Scan Here for Video Solution”

Link to GfG corresponding Short Notes: GATE FARRAY

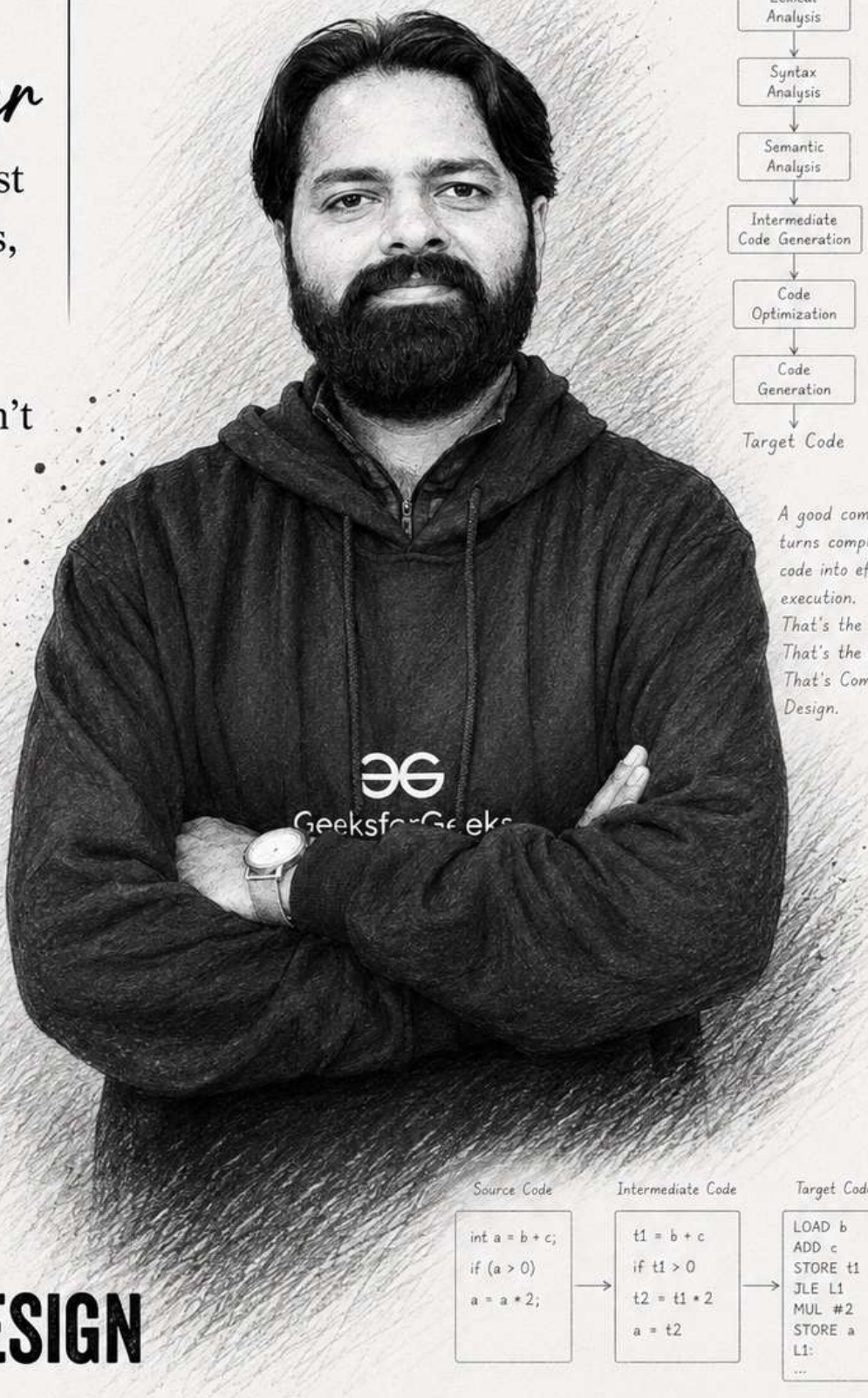
“

In *Compiler* design, we don’t just translate languages, we bridge ideas to execution.
Good compilers don’t just run programs, they make them **run better.**

”



We analyze.
We optimize.
We execute.
That’s how ideas become reality.



A good compiler turns complex code into efficient execution.
That’s the art.
That’s the science.
That’s Compiler Design.

IIT BOMBAY ALUMNI

KING OF COMPILER DESIGN

Devasane Mallesham Sir

From tokens to transformation,
from logic to low-level,
we build the invisible
power behind every program.
That’s Compiler Design.

Think. Model. Prove. Conquer.

1 LEXICAL ANALYSIS

GATE CSE - 2026 NAT SET - 2(1-MARKS)

Q1. A lexical analyzer uses the following token definitions

$letter \rightarrow [A - Z\ a - z]$

$digit \rightarrow [0 - 9]$

$id \rightarrow letter\ (letter \mid digit)^*$

$number \rightarrow digit^+$

$ws \rightarrow (blank \mid tab \mid newline)^+$

For the string given below,

$x1\ 23mm\ \ 78\ y\ \ 7z\ \ zz5\ \ 14A\ \ 8H\ \ \ AaYcD$

the number of tokens (excluding ws) that will be produced by the lexical analyzer is _____. (answer in integer)

GATE CSE - 2025 MCQ SET - 1(1-MARKS)

Q2. Which ONE of the following statements is FALSE regarding the symbol table?

(a) Symbol table is responsible for keeping track of the scope of variables.

(b) Symbol tables can be implemented using a binary search tree.

(c) Symbol table is not required after the parsing phase.

(d) Symbol table is created during the lexical analysis phase.

GATE CSE - 2024 MCQ SET - 2(1-MARKS)

Q3. Consider the following two sets:

P. Lexical Analyzer	1. Abstract Syntax Tree
Q. Syntax Analyzer	2. Token
R.Intermediate code Generator	3. Parse Tree
S. Code Optimizer	4. Constant Folding

Which one of the following options is the CORRECT match from Set X to Set Y?

(a) P-4; Q-1; R-3; S-2

(b) P-2; Q-3; R-1; S-4

(c) P-2; Q-1; R-3; S-4

(d) P-4; Q-3; R-2; S-1

GATE CSE - 2018 MCQ (2-MARKS)

Q4. A lexical analyzer uses the following patterns to recognize three tokens T_1, T_2, and T_3 over the alphabet {a,b,c}.

T_1 : a?(b|c)*a

T_2 : b?(a|c)*b

T_3 : c?(b|a)*c

Note that 'x?' means 0 or 1 occurrence of the symbol x. Note also that the analyzer outputs the token that matches the longest possible prefix. If the string bbaacabc is processed by the analyzer, which one of the following is the sequence of tokens it outputs?

(a) $T_1T_2T_3$

(b) $T_1T_1T_3$

(c) $T_2T_1T_3$

(d) T_3T_3

GATE CSE - 2017 MCQ SET – 2 (2-MARKS)

Q5. Match the following according to input (from the left column) to the compiler phase (in the right column) that processes it:

(P) Syntax Tree	(i) Code Generator
(Q) Character Stream	(ii) Syntax Analyzer
(R)Intermediate Representation	(iii)Semantic Analyzer
(S) Token Stream	(iv) Lexical Analyzer

(a) P-ii, Q-iii, R-iv, S-i
(b) P-ii, Q-i, R-iii, S-iv
(c) P-iii, Q-iv, R-i, S-ii
(d) P-i, Q-iv, R-ii, S-iii

GATE CSE - 2016 MCQ SET – 2 (1-MARKS)

Q6. Match the following

List-1	List-2
P. Lexical analysis	(i)Leftmost derivation
Q. Top-down parsing	(ii) Type checking
R. Semantic analysis	(iii)Regular expressions
S.Runtime environments	(iv)Activation records

(a) P-i, Q-ii, R-iv, S-iii
(b) P-iii, Q-i, R-ii, S-iv
(c) P-ii, Q-iii, R-i, S-iv
(d) P-iv, Q-i, R-ii, S-iii

GATE CSE - 2015 MCQ SET - 2 (1-MARKS)

Q7. Match the following ?

GROUP 1	GROUP 2
P. Lexical analysis	1. Graph coloring
Q. Parsing	2. DFA minimization
R. Register allocation	3. Post- order traversal
S. Expression evaluation	4. Production Tree

(a) P-2, Q-3, R-1, S-4
(b) P-2, Q-1, R-4, S-3
(c) P-2, Q-4, R-1, S-3
(d) P-2, Q-3, R-4, S-1

GATE CSE - 2011 MCQ (1-MARKS)

Q8. In a compiler, keywords of a language are recognized during:-

(a) Parsing of the program
(b) the code generation
(c) the lexical analysis of the program
(d) dataflow analysis

GATE CSE - 2011 MCQ (1-MARKS)

Q9. The lexical analysis for a modern computer language such as Java needs the power of which one of the following machine models in a necessary and sufficient sense?

(a) Finite state automata
(b) Deterministic pushdown automata
(c) Non-Deterministic pushdown automata
(d) Turing machine

GATE CSE - 2010 MCQ (1-MARKS)

Q10. Which data structure in a compiler is used for managing information about variables and their attributes?

(a) Abstract syntax tree
(b) Symbol table
(c) Semantic stack
(d) Parse Table

GATE CSE - 2009 MCQ (1-MARKS)

Q11. Match all items in Group 1 with correct options from those given in Group 2.

Group 1		Group 2	
P.	Regular expression	1.	Syntax analysis
Q.	Pushdown automata	2.	Code generation
R.	Dataflow analysis	3.	Lexical analysis
S.	Register allocation	4.	Code optimization

(a) P-4. Q-1, R-2, S-3
(b) P-3, Q-1, R-4, S-2
(c) P-3, Q-4, R-1, S-2
(d) P-2, Q-1, R-4, S-3

GATE CSE - 2008 MCQ (1-MARKS)

Q12. Which of the following are true?

(I) A programming language which does not permit global variables of any kind and has no nesting of procedures/functions, but permits recursion can be implemented with static storage allocation

(II) Multi-level access link (or display) arrangement is needed to arrange activation records only if the programming language being implemented has nesting of procedures/functions.

(III) Recursion in programming languages cannot be implemented with dynamic storage allocation

(IV) Nesting procedures/functions and recursion require a dynamic heap allocation scheme and cannot be implemented with a stack-based allocation scheme for activation records

(V) Programming languages which permit a function to return a function as its result cannot be implemented with a stack-based storage allocation scheme for activation records.

(a) II and V only
(b) I, III and IV only
(c) I, II and IV only
(d) II, III and V only

GATE CSE - 2005 MCQ SET - 1(1-MARKS)

Q13. Consider line number 3 of the following C- program.

```
int main ( ) {                  /*Line 1 */
int I, N;                       /*Line 2 */
fro (I = 0, I < N, I++);        /*Line 3 */
  }
```

Identify the compiler's response about this line while creating the object-module

(a) Both lexical and syntactic errors
(b) Only a lexical error
(c) Only syntactic errors
(d) No compilation error

GATE CSE - 2005 MCQ SET - 1(1-MARKS)

Q14. Consider the following C-program:

```
double foo (double); /* Line 1 */
int main()
{
        double da, db;

        // input da

        db = foo(da);
}
double foo(double a)
{
   return a;
}
```

(a) no compile warning or error
(b) some compiler-warnings not leading to unintended results
(c) some compiler-warnings due to type-mismatch eventually leading to unintended results
(d) compiler errors

ANSWER KEY

1.(13)	**2. (c)**	**3. (b)**
4. (d)	**5. (c)**	**6. (b)**
7. (c)	**8. (c)**	**9. (a)**
10. (b)	**11. (b)**	**12. (a)**
13.(d)	**14. (d)**	

HINTS & SOLUTIONS

1. ANSWER:- 13

Token definitions:
letter → [A–Z, a–z]
digit → [0–9]
id → letter (letter | digit)*
number → digit+
ws → whitespace (ignored)

Given string:
x1 23mm 78 y 7z zz5 14A 8H AaYcD

Tokenization (excluding whitespace):

x1 → id
23 → number
mm → id
78 → number
y → id
7 → number
z → id
zz5 → id
14 → number
A → id
8 → number
H → id
AaYcD → id

Total tokens = **13**

2. ANSWER:- C

Symbol table is created during the lexical analysis phase to keep track of information about the variables like scope and lifetime, which can also be implemented using binary search tree and is required in all the phases of compiler.

3. ANSWER:- B

Lexical Analyzer → Token
It reads the source code and breaks it into tokens (keywords, identifiers, symbols, etc.)
Syntax Analyzer → Parse Tree
It builds a Parse Tree (also known as Concrete Syntax Tree) that represents how tokens fit into the language's grammar.
Intermediate Code Generator → Abstract Syntax Tree
This tree (or 3-address code, etc.) is easier to optimize and translate into machine code.
Code Optimizer → Constant Folding
One classic optimization is **Constant Folding**, where constant expressions are evaluated at compile-time.

4. ANSWER:- D

a? means either 0 or 1 occurrence of "a", so we can write T_1, T_2 and T_3 as:
T_1 : (b+c)*a + a(b+c)*a
T_2 : (a+c)*b + b(a+c)*b
T_3 : (b+a)*c + c(b+a)*c
Now the string is: bbaacabc

NOTE: Token that matches the longest possible prefix
We can observe that the longest possible prefix in string is : bbaac which can be generated by T_3.
After the prefix we left with "abc" which is again generated by T_3 (as longest possible prefix).
So, the answer is T_3T_3.

5. ANSWER:- C

1. Character stream is input to a lexical analyzer which produces tokens as output. So Q → (iv).
2. Token stream is forwarded as input to Syntax analyzer which produces syntax tree as output. So, S → (ii).
3. Syntax tree is the input for the semantic analyzer, So P → (iii).
4. Intermediate representation is input for Code generator. So R → (i).

6. ANSWER:- B

1. Regular expressions are used in lexical analysis.
2. Top-down parsing has left most derivations of any string.
3. Type checking is done in semantic analysis.
4. Activation records are loaded into memory at runtime.

7. ANSWER:- C

1. Lexical analysis is related with FA and Regular expressions.
2. Expression can be evaluated with postfix traversals.
3. Register allocation can be done by graph colouring.
4. The parser constructs a production tree.

8. ANSWER:- C

Keywords are recognized during lexical analysis.

9. ANSWER:- A

Lexical Analysis is implemented by finite automata

10. ANSWER:- B

Symbol tables are data structures that are used by compilers to hold information about source-program constructs. The information is collected incrementally by the analysis phases of a compiler and used by the synthesis phases to generate the target code. Entries in the symbol table contain information about an identifier such as its character string (or lexeme), its type, its position in storage, and any other relevant information.

11. ANSWER:- B

Regular expressions are used in lexical analysis. Pushdown automata is related to context free grammar which is related to syntax analysis.Dataflow analysis is done in code optimization. Register allocation is done in code generation.

12. ANSWER:- A

(I) False

Recursion requires dynamic allocation (activation records created at runtime). Static allocation cannot support multiple recursive calls.

(II) True

Access links/display are needed only when there is nesting of functions to access non-local variables.

(III) False

Recursion is implemented using dynamic storage allocation (typically stack), so this statement is incorrect.

(IV) False

Both recursion and nested functions can be implemented using stack-based allocation, not necessarily heap.

(V) TRUE: If a language allows a function to return another function as a result (returning a closure), the returned function might still need to reference the local variables of the function that created it. Under a stack-based allocation scheme, the outer function's activation record is immediately destroyed upon returning, leading to dangling references. Therefore, these languages require a heap-based allocation scheme for activation records so their lifetimes can outlive the execution order.

13. ANSWER:- D

In the given question, line 3 won't cause a lexical error or syntactic error. The statement will be treated as a function call with three arguments. Function definition being absent will cause link time error, but the question asks only for compile-time errors. So, (D) must be the answer.

14. ANSWER:- D

When a function is called without being defined, C compiler assumes it to return "int" but here foo is returning "double" and hence the compiler will throw type mis-match error.

2 SYNTAX ANALYZER

GATE CSE - 2026 MSQ SET - 1(1-MARKS)

Q1. Which of the following statements is/are true?

(a) LL(1) parser uses backtracking
(b) For a grammar to be LL(1), it must be left-recursive
(c) For a grammar to be LL(1), it must be left-factored
(d) The LL(1) parsers are more powerful than the SLR parsers

GATE CSE - 2026 MCQ SET - 1(1-MARKS)

Q2. Consider the following C statements:

```
char *str1 = "Hello; /* Statement S1 */
char *str2 = "Hello;"; /* Statement S2 */
 int *str3 = "Hello"; /* Statement S3 */
```

Which of the following options is/are correct?

(a) S1 and S2 have syntactic errors
(b) S2 has a lexical error and S3 has a syntactic error
(c) S1 has a lexical error and S3 has a semantic error
(d) S1 has a syntactic error and S3 has a semantic error

GATE CSE - 2026 MSQ SET - 2(2-MARKS)

Q3. Consider the canonical $LR(0)$ parsing of the grammar below using terminals $\{a, b, c\}$ and non-terminals $\{A, B, C, S\}$ with S as the start symbol.

$S \rightarrow ACB$
$A \rightarrow aA \mid \epsilon$
$C \rightarrow cC \mid \epsilon$
$B \rightarrow bB \mid b$

Which one of the following options gives the number of shift-reduce conflicts that will occur in the $LR(0)$ ACTION table?

(a) 2
(b) 3
(c) 4
(d) 5

GATE CSE - 2025 MSQ SET - 2(2-MARKS)

Q4. Consider two grammars G_1 and G_2 with the production rules given below:

G1: S → if E then S | if E then S else S | a
E → b
G2: S → if E then S | M
M → if E then M else S | c
E → b

where if, then, else, a, b, c are the terminals.
Which of the following option(s) is/are CORRECT?

(a) G1 is not LL(1) and G2 is LL(1).
(b) G1 is LL(1) and G2 is not LL(1).
(c) G1 and G2 are not LL(1).
(d) G1 and G2 are ambiguous.

GATE CSE - 2025 MSQ SET - 1(2-MARKS)

Q5. Which of the following statement(s) is/are TRUE while computing First and Follow during top-down parsing by a compiler?

(a) For a production A→ϵ, ϵ epsilon will be added to First(A)
(b) If there is any input right end marker, it will be added to First(S), where S is the start symbol.
(c) For a production A→ϵ, ϵ epsilon will be added to Follow(A).
(d) If there is any input right end marker, it will be added to Follow(S), where S is the start symbol.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2024 MCQ SET - 1(1-MARKS)

Q6. Consider the following grammar G, with S as the start symbol. The grammar G has three incomplete productions denoted by (1), (2), and (3):

$S \rightarrow daT \mid (1)$

$T \rightarrow aS \mid bT \mid (2)$

$R \rightarrow (3) \mid \epsilon$

The set of terminals is $\{a, b, c, d, f\}$. The FIRST and FOLLOW sets of the different non-terminals are as follows:

FIRST(S) = $\{c, d, f\}$, FIRST(T) = $\{a, b, \epsilon\}$,

FIRST(R) = $\{c, \epsilon\}$

FOLLOW(S) = FOLLOW(T) = $\{c, f, \$\}$,

FOLLOW(R) = $\{f\}$

(a) (1) S→Rf (2) T→ϵ (3) R→cTR

(b) (1) S→fR (2) T→ϵ (3) R→cTR

(c) (1) $S \rightarrow fR$ (2) $T \rightarrow cT$ (3) $R \rightarrow cR$

(d) (1) $S \rightarrow Rf$ (2) $T \rightarrow cT$ (3) $R \rightarrow cR$

GATE CSE – 2024 SET -1 MSQ (1-MARKS)

Q7. Which of the following is/are Bottom-Up Parser(s)

(a) Shift-reduce Parser

(b) Predictive Parser

(c) LL(1) Parser

(d) LR Parser

GATE CSE – 2024 SET -1 NAT (1-MARKS)

Q8. Consider the operator precedence and associativity rules for the integer arithmetic operators given in the table below.

Operator	Precedence	Associativity
+	Highest	Left
-	High	Right
*	Medium	Right
/	Low	Right

The value of the expression $3 + 1 + 5 * 2/7 + 2 - 4 - 7 - 6/2$ as per the above rules is __

GATE CSE - 2022 NAT (2-MARKS)

Q9. Consider the augmented grammar with { +, *, (,), id } as the set of terminals.

$S' \rightarrow S$

$S \rightarrow S + R \mid R$

$R \rightarrow R * P \mid P$

$P \rightarrow (S) \mid id$

If I_0 is the set of two LR(0) items **{[S′→S.],[S→S.+R]}** then goto(closure (I_0),+) contains exactly _____ items.

GATE CSE - 2022 MCQ (1-MARKS)

Q10. Which one of the following statements is TRUE?

(a) The LALR(1) parser for a grammar G cannot have reduce – reduce conflict if the LR(1) parser for G does not have reduce – reduce conflict.

(b) Symbol table is accessed only during the lexical analysis phase.

(c) Data flow analysis is necessary for run-time memory management.

(d) LR(1) parsing is sufficient for deterministic context – free languages

GATE CSE - 2021 MCQ SET - 1(1-MARKS)

Q11. Consider the following context-free grammar where the set of terminals is {a,b,c,d,f}.

$S \rightarrow daT \mid Rf$

$T \rightarrow aS|baT|\varepsilon$

$R \rightarrow caTR \mid \varepsilon$

The following is a partially-filled LL(1) parsing table.

	a	b	c	d	f	$
S			(1)	S → da T	(2)	
T	T → a S	T → ba T	(3)		T → ϵ	(4)
R			R → caTR		R → ϵ	

Which one of the following choices represents the correct combination for the numbered cells in the parsing table ("blank" denotes that the corresponding cell is empty)?

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) (1) S → Rf (2) S → Rf (3) T → ϵ (4) T → ϵ
(b) (1) blank (2) S → R (3) T → ϵ (4) T → ϵ
(c) (1) S → Rf (2) blank (3) blank (4) T → ϵ
(d) (1) blank (2) S → Rf (3) blank (4) blank

GATE CSE - 2021 MCQ SET - 1(1-MARKS)

Q12. Consider the following statements.
S_1: Every SLR(1) grammar is unambiguous but there are certain unambiguous grammars that are not SLR(1).
S_2 : For any context-free grammar, there is a parser that takes at most $O(n^3)$ time to parse a string of length n.
Which one of the following options is correct?

(a) S_1 is true and S_2 is false
(b) S_1 is false and S_2 is false
(c) S_1 is false and S_2 is true
(d) S_1 is true and S_2 is true

GATE CSE – 2020 NAT (1-MARKS)

Q13. Consider the following grammar.
S →aSB| d
B →b
The number of reduction steps taken by a bottom-up parser while accepting the string aaadbbb is _____.

GATE CSE – 2019 NAT (1-MARKS)

Q14. Consider the augmented grammar given below:

S' → S

S → ⟨L⟩ | id

L → L, S | S

Let I_0 = CLOSURE ({[S' → ·S]}). The number of items in the set GOTO (I_0 , ⟨) is: _____.

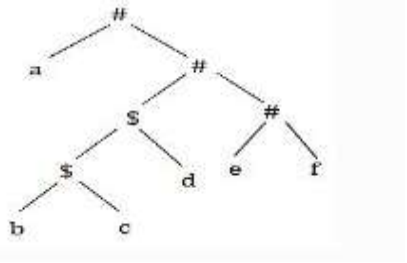
GATE CSE – 2019 MCQ (1-MARKS)

Q15. Which one of the following kinds of derivation is used by LR parsers?

(a) Leftmost in reverse
(b) Rightmost in reverse
(c) Leftmost
(d) Rightmost

GATE CSE – 2018 MCQ (2-MARKS)

Q16. Consider the following parse tree for the expression a#bcd#e#f, involving two binary operators $ and #.



Which one of the following is correct for the given parse tree?

(a) $ has higher precedence and is left associative; # is right associative
(b) # has higher precedence and is left associative; $ is right associative
(c) $ has higher precedence and is left associative; # is left associative
(d) # has higher precedence and is right associative; $ is left associative

GATE CSE – 2018 MCQ (1-MARKS)

Q17. Which one of the following statements is FALSE?

(a) Context-free grammar can be used to specify both lexical and syntax rules.
(b) Type checking is done before parsing.
(c) High-level language programs can be translated to different Intermediate Representations.
(d) Arguments to a function can be passed using the program stack.

GATE CSE – 2017 MCQ SET – 2 (1-MARKS)

Q18. Which of the following statements about parser is/are CORRECT?
I. Canonical LR is more powerful than SLR.
II. SLR is more powerful than LALR.
III. SLR is more powerful than Canonical LR.

(a) I only
(b) II only
(c) III only
(d) II and III only

GATE CSE – 2017 MCQ SET - 1(1-MARKS)

Q19. Consider the following grammar:

$$P \rightarrow xQRS$$
$$Q \rightarrow yz \mid z$$
$$R \rightarrow w \mid \varepsilon$$
$$S \rightarrow y$$

What is FOLLOW(Q)?

(a) {R}
(b) {w}
(c) {w, y}
(d) {w, $}

GATE CSE – 2017 NAT SET - 1(2-MARKS)

Q20. Consider the following grammar:

stmt → if expr then expr else expr; stmt|$\grave{O}$
expr → term relop term | term
term → id | number
id → a | b | c
number → [0-9]

where relop is a relational operator (e.g., <, >, …), $\grave{O}$ refers to the empty statement, and if, then, else are terminals.

Consider a program P following the above grammar containing ten if terminals. The number of control flow paths in P is _____. For example, the program

if e_1 then e_2 else e_3

has 2 control flow paths, $e_1 \rightarrow e_2$ and $e_1 \rightarrow e_3$

GATE CSE – 2017 MCQ SET - 2(2-MARKS)

Q21. Consider the following expression grammar G:

E → E - T | T
T → T + F | F
F → (E) | id

Which of the following grammars is not left recursive, but is equivalent to G?

(a) E → E – T | T
T → T + F | F
F → (E) | id

(b) E →T E'
E'→ – TE' | ε
T → T + F | F
F → (E) | id

(c) E → TX
X →–TX | ε
T → FY
Y → +FY | ε
F → (E) | id

(d) E → TX | (TX)
X → –TX | +TX | ε
T → id

GATE CSE – 2016 MCQ SET - 2(1-MARKS)

Q22. Which one of the following grammars is free from left recursion?

(a) $S \rightarrow AB$
$A \rightarrow Aa \mid b$
$B \rightarrow c$

(b) $S \rightarrow A\,b \mid B\,b \mid C$
$A \rightarrow Bd \mid \varepsilon$
$B \rightarrow e$

Check Out Our GATE Courses & Secure Your Dream Rank!

(c)
$S \rightarrow Aa \mid B$
$A \rightarrow B\ b \mid Sc \mid \epsilon$
$B \rightarrow d$

(d)
$S \rightarrow Aa \mid Bb \mid c$
$A \rightarrow B\ d \mid \epsilon$
$B \rightarrow d \mid \epsilon$

GATE CSE – 2015 MCQ SET - 1(1-MARKS)

Q23. Which one of the following is True at any valid state in shift-reduce parsing?

(a) Viable prefixes appear only at the bottom of the stack and not inside
(b) Viable prefixes appear only at the top of the stack and not inside
(c) The stack contains only a set of viable prefixes
(d) The stack never contains viable prefixes

GATE CSE – 2015 MCQ SET - 3(1-MARKS)

Q24. Among simple LR (SLR), canonical LR, and look-ahead LR (LALR), which of the following pairs identify the method that is very easy to implement and the method that is the most powerful, in that order?

(a) SLR, LALR
(b) Conical LR, LALR
(c) SLR, conical LR
(d) LALR, conical LR

GATE CSE – 2015 MCQ SET - 3(2-MARKS)

Q25. Consider the following grammar G.

S → F | H
F → p | c
H →d | c

Where S, F and H are non-terminal symbols, p, d and c are terminal symbols. Which of the following statement(s) is/are correct?

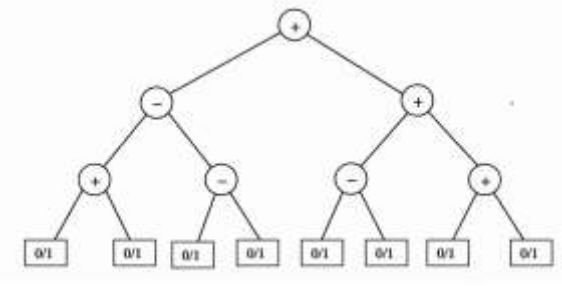
S1: LL(1) can parse all strings that are generated using grammar G.

S2: LR(1) can parse all strings that are generated using grammar.

(a) Only S1
(b) Only S2
(c) Both S1 and S2
(d) Neither S1 nor S2

GATE CSE – 2014 NAT SET – 2 (2-MARKS)

Q26. Consider the expression tree shown. Each leaf represents a numerical value, which can either be 0 or 1. Over all possible choices of the values at the leaves, the maximum possible value of the expression represented by the tree is _____ ,



GATE CSE – 2014 MCQ SET - 1(2-MARKS)

Q27. A canonical set of items is given below:

S → L. > R
Q → R.

On input symbol < the set has:-

(a) a shift-reduce conflict and a reduce-reduce conflict.
(b) a shift-reduce conflict but not a reduce-reduce conflict.
(c) a reduce-reduce conflict but not a shift-reduce conflict.
(d) neither a shift-reduce nor a reduce-reduce conflict.

GATE CSE – 2014 MCQ SET - 2(1-MARKS)

Q28. Consider the grammar defined by the following production rules, with two operators * and +

S → T * P

T → U | T * U

P → Q + P | Q

Q → Id

U → Id

Which one of the following is TRUE?

(a) + is left associative, while * is right associative
(b) + is right associative, while * is left associative
(c) Both + and * are right associative
(d) Both + and * are left associative

GATE CSE – 2013 MCQ (2-MARKS)

Q29. Consider the following two sets of LR(1) items of an LR(1) grammar.

X→c. X, c/d	X→ c. X, $
X-> .cX, c/d	X→ .cX, $
X→ .d, c/d	X→ .d, $

Which of the following statements related to merging of the two sets in the corresponding LALR parser is/are FALSE?

1. Cannot be merged since look ahead is different.

2. Can be merged but will result in S-R conflict.

3. Can be merged but will result in RR conflict.

4. Cannot be merged since goto on c will lead to two different sets.

(a) 1 only
(b) 2 only
(c) 1 and 4 only
(d) 1, 2, 3 and 4

GATE CSE – 2013 MCQ (1-MARKS)

Q30. What is the maximum number of reduce moves that can be taken by a bottom-up parser for a grammar with no epsilon-and unit-production (i.e. of type A → ε and A → a) to parse a string with n tokens?

(a) n/2
(b) n–1
(c) 2n–1
(d) 2^n

GATE CSE – 2012 MCQ (2-MARKS)

Q31. For the grammar below, a partial LL(1) parsing table is also presented along with the grammar. Entries that need to be filled are indicated as E1, E2 and E3. ε is the empty string, $ indicates end of input, and | separates alternate right hand sides of productions.

S → a A b B | b A a B | ε

A → S

B → S

	a	b	$
S	E 1	E 2	S → ε
A	A → S	A → S	error
B	B → S	B → S	E3

The FIRST and FOLLOW sets for the non-terminals A and B are:-

(a) $FIRST(A) = \{a, b, \varepsilon\} = FIRST(B)$
$FOLLOW(A) = \{a, b\}$
$FOLLOW(B) = \{a, b, \$\}$

(b) $FIRST(A) = \{a, b, \varepsilon\} = FIRST(B)$
$FOLLOW(A) = \{a, b\}$
$FOLLOW(B) = \emptyset$

Check Out Our GATE Courses & Secure Your Dream Rank!

(c)
$\text{FIRST}(A) = \{a, b, \$\}$
$\text{FIRST}(B) = \{a, b, \varepsilon\}$
$\text{FOLLOW}(A) = \{a, b\}$
$\text{FOLLOW}(B) = \{\$\}$

(d)
$\text{FIRST}(A) = \{a, b\} = \text{FIRST}(B)$
$\text{FOLLOW}(A) = \{a, b\}$
$\text{FOLLOW}(B) = \{a, b\}$

GATE CSE – 2012 MCQ (1-MARKS)

Q32. For the grammar below, a partial LL(1) parsing table is also presented along with the grammar. Entries that need to be filled are indicated as E1, E2, and E3. *epsilon* is the empty string, $ indicates end of input, and, | separates alternate right hand sides of productions.

S →a A b B | b A a B | ε
A →S
B → S

	a	b	$
S	E 1	E 2	S → ε
A	A → S	A → S	error
B	B → S	B → S	E3

The appropriate entries for **E1** , **E2**, **E3** are:-

(a) E1: S → aAbB,A → S
E2: S → bAaB,B→S
E3: B → S

(b) E1: S → aAbB,S→ ε
E2: S → bAaB,S → ε
E3: S → ε

(c) E1: S → aAbB,S → ε
E2: S → bAaB,S→ε
E3: B → S

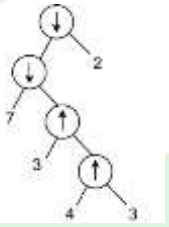
(d) E1: A → S,S →ε
E2: B → S,S → ε
E3: B →S

GATE CSE – 2011 MCQ (1-MARKS)

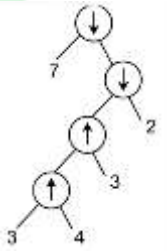
Q33. Consider two binary operators '↑' and '↓' with the precedence of operator ↓ being lower than that of the operator ↑. Operator ↑ is right associative while operator ↓ is left associative. Which one of the following represents the parse tree for expression (7↓3↑4↑3↓2) ?

(a)

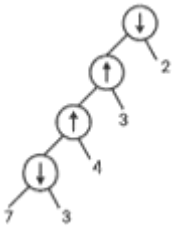
(b)



(c)



(d)



GATE CSE – 2010 MCQ (2-MARKS)

Q34. The grammar : S → aSa|bS|c is:-

(a) LL(1) but not LR(1)
(b) LR(1) but not LL(1)
(c) Both LL(1) and LR(1)
(d) Neither LL(1) nor LR(1)

GATE CSE – 2008 MCQ (1-MARKS)

Q35. An LALR(1) parser for a grammar G can have shift-reduce (S-R) conflicts if and only if

(a) The SLR(1) parser for G has S-R conflicts
(b) The LR(1) parser for G has S-R conflicts
(c) The LR(0) parser for G has S-R conflicts
(d) The LALR(1) parser for G has reduce-reduce conflicts

GATE CSE – 2008 MCQ (1-MARKS)

Q36. Which of the following describes a handle (as applicable to LR-parsing) appropriately?

(a) It is the position in a sentential form where the next shift or reduce operation will occur
(b) It is non-terminal whose production will be used for reduction in the next step
(c) It is a production that may be used for reduction in a future step along with a position in the sentential form where the next shift or reduce operation will occur
(d) It is the production p that will be used for reduction in the next step along with a position in the sentential form where the right hand side of the production may be found

GATE CSE – 2007 MCQ (1-MARKS)

Q37. Consider the CFG with {S, A, B} as the non-terminal alphabet, {a,b} as the terminal alphabet, S as the start symbol and the following set of production rules:

S → aB S → bA
B → b A → a
B → bS A → aS
B → aBB S → bAA

Which of the following strings is generated by the grammar?

(a) aaaabb
(b) aabbbb
(c) aabbab
(d) abbbba

GATE CSE – 2005 MCQ (1-MARKS)

Q38. The grammar A → AA | (A) | ε is not suitable for predictive-parsing because the grammar is

(a) ambiguous
(b) left-recursive
(c) right-recursive
(d) an operator-grammar

GATE CSE – 2005 MCQ (1-MARKS)

Q39. Consider the grammar

S → (S) | a

Let the number of states in SLR(1), LR(1) and LALR(1) parsers for the grammar be n_1, n_2 and n_3 respectively. The following relationship holds good

(a) $n_1 < n_2 < n_3$
(b) $n_1 = n_3 < n_2$
(c) $n_1 = n_2 = n_3$
(d) $n_1 \geq n_3 \geq n_2$

GATE CSE – 2005 MCQ (1-MARKS)

Q40. Consider the context-free grammar

E → E + E
E → (E * E)
E → id

where E is the starting symbol, the set of terminals is {id, (+,),*}, and the set of non-terminals is {E}.
For the terminal string id + id + id + id, how many parse trees are possible?

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) 5
(b) 4
(c) 3
(d) 2

Answer key

1. (c)	**2. (c)**	**3. (d)**
4. (c,d)	**5. (a,d)**	**6. (a)**
7. (a,d)	**8.(6)**	**9. (5)**
10. (d)	**11. (a)**	**12. (d)**
13. (7)	**14. (5)**	**15. (b)**
16. (a)	**17. (b)**	**18. (a)**
19. (c)	**20.(1024)**	**21. (c)**
22. (b)	**23.(c)**	**24. (c)**
25. (d)	**26.(6)**	**27.(d)**
28.(b)	**29. (d)**	**30. (b)**
31. (a)	**32. (c)**	**33. (b)**
34. (c)	**35. (b)**	**36. (d)**
37. (c)	**38. (a)**	**39. (b)**
40. (a)		

HINTS & SOLUTIONS

1. ANSWER:- C

A. LL(1) parser uses backtracking
False.
LL(1) parsing is predictive parsing and does not use backtracking.

B. For a grammar to be LL(1), it must be left-recursive
False.
An LL(1) grammar must be free from left recursion, not left-recursive.

C. For a grammar to be LL(1), it must be left-factored
True.
LL(1) parsing requires that the grammar be left-factored so that the parser can decide the production using one lookahead symbol.

D. The LL(1) parsers are more powerful than the SLR parsers.
False.SLR parsers (bottom-up) are more powerful than LL(1) parsers.

2. ANSWER:- C
char *str1 = "Hello;
The string literal is not closed with a double quote ("). Because of the missing closing quotation mark, the compiler cannot correctly recognize the token. Hence, S1 has a lexical error.

S2:
char *str2 = "Hello;";
The string "Hello;" is properly enclosed in double quotes and assigning it to char * is valid in C. Hence, S2 has no error.

S3:
int *str3 = "Hello";

Check Out Our GATE Courses & Secure Your Dream Rank!

The string "Hello" is of type char *, but it is assigned to a variable of type int *. This causes a type mismatch, which is a semantic error.

Therefore:

- S1 → Lexical Error
- S2 → No Error

S3 → Semantic Error

3. ANSWER:- D

Grammar:
S → A C B
A → aA | ε
C → cC | ε
B → bB | b

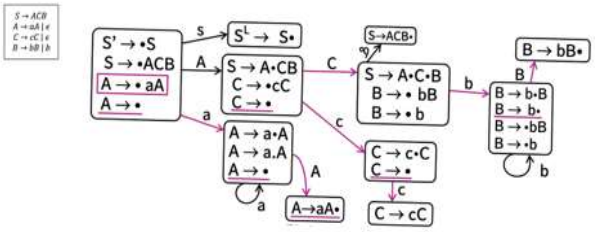
In LR(0) parsing, conflicts occur when a state contains:

- a shift item (dot before a terminal), and
- a reduce item (dot at the end of a production).

Here, both A → ε and C → ε produce reduce items in LR(0) states.
At the same time, the grammar also allows shift actions for terminals a and c due to productions A → aA and C → cC.

Thus several states contain both shift and reduce possibilities, leading to shift-reduce conflicts.

After constructing the canonical LR(0) item sets and the ACTION table, the total number of shift-reduce conflicts = 5.



- Shift-Reduce conflict
In LR(0) Reduction happens for all terminals.

4. ANSWER:- C,D

A grammar is not LL(1) iff:

- It is an ambiguous grammar
- It is left-recursive
- It has common prefixes

Both the grammars G₁ and G₂ have common prefixes like "if E then S" and "if E then", as well as ambiguity due to the dangling else problem.

So, the given grammars G₁ and G₂ are not LL(1) and ambiguous.

5. ANSWER:- A,D

1. If a production is of the form $A \rightarrow \epsilon$, then $First(A) = \epsilon$.
2. The end-of-input marker ($) is added to $Follow(S)$, where S is the start symbol.
3. ϵ is never added to Follow sets.
4. The end marker is not added to First sets.

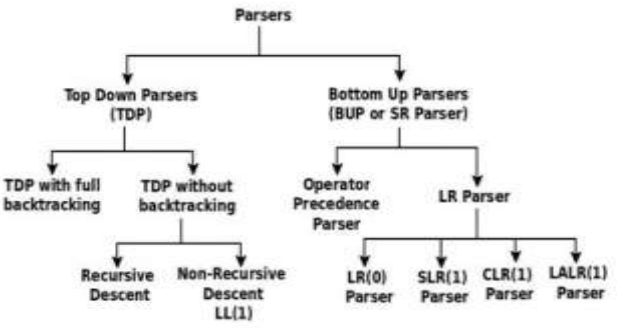
6. ANSWER:- A

Option (C, D) is incorrect as if we substitute T → cT in (2) it gives FIRST(T) = a, b, c which is wrong because in question it is given that FIRST(T) = a, b, ϵ

Option (B) is incorrect as if we substitute S → fR in (1) it gives FIRST(S) = f, d which is wrong because terminal c is missing.
So correct Option is (A).

	FIRST	FOLLOW
$S \rightarrow daT \mid Rf$	$\{c, d, f\}$	$\{\$, c, f\}$
$T \rightarrow aS \mid bS \mid \varepsilon$	$\{a, b, \varepsilon\}$	$\{\$, c, f\}$
$R \rightarrow cTR \mid \varepsilon$	$\{c, \varepsilon\}$	$\{f\}$

7. ANSWER:- A,D



From the above classifications of parsers, it is clear that LR(k) and shift-reduce parse are types of botto parsers while LL(1) and recursive descent parsers belong to top-down parsers.

8. ANSWER:- 6

Reason:

- Precedence (highest → lowest): + > − > * > /
- Associativity:

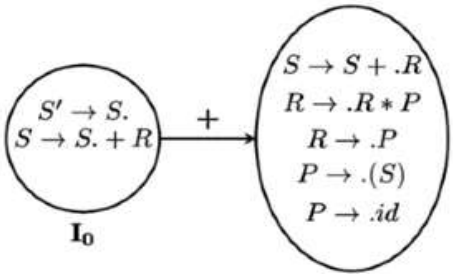
+ → Left
− → Right
* → Right
/ → Right

Given expression:
3 + 1 + 5 * 2 / 7 + 2 − 4 − 7 − 6 / 2

Evaluating as per given precedence and associativity:

((((3+1)+5)*2)/(((7+2)-(4-(7-6)))/2))
=6

9. ANSWER:- 5



10. ANSWER:- D

(a) FALSE. The LALR(1) parser for a grammar G may have reduce-reduce conflict if the LR(1) parser for G does not have reduce-reduce conflict.

(b) FALSE. Symbol tables can be accessed in all phases of the compiler.

(c) FALSE. Data flow analysis is required to improve the code but not necessary for run-time memory management.

(d) TRUE. Every LR(1) CFG is going to generate DCFL.

11. ANSWER:- A

Production	FIRST	FOLLOW
S → daT | Rf	{d, c, f}	{c, f, $}
T → aS | baT | ε	{a, b, ε}	{c, f, $}
R → caTR | ε	{c, ε}	{f}

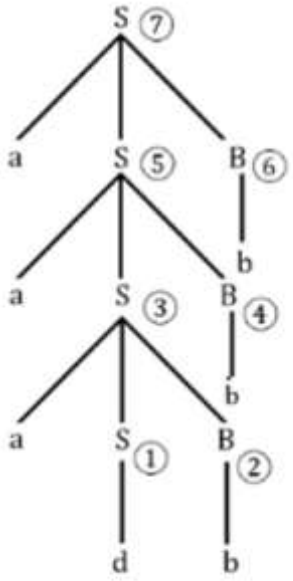
	a	b	c	d	f	$
S			S → Rf (1)	S → daT	S → Rf (2)	
T	T → aS	T → baT	T → ε (3)		T → ε	T → ε (4)
R			R → caTR		R → ε	

12. ANSWER:- D

S_1 : Every SLR(1) is unambiguous but there are certain unambiguous grammars that are not SLR (1).
S_1 is True.
Every SLR is unambiguous but unambiguous need not be SLR.
S_2 : For any F = G, there is a parser that takes atmost $O(n^3)$ time to parse a string of length n.
S_2 is True.
CYK algorithms exist to parse a string of length n in $O(n^3)$. Bottom-up parser exists.

Check Out Our GATE Courses & Secure Your Dream Rank!

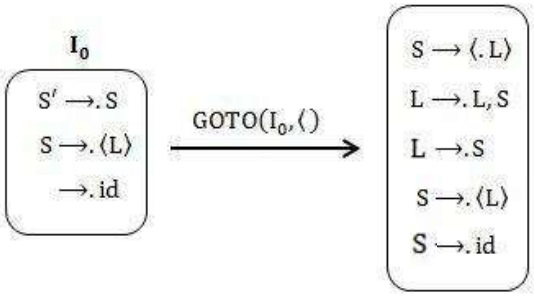
13. ANSWER:- 7



7 reductions total.

14. ANSWER:- 5

I_0 = CLOSURE ({[S' → ·S]})



Hence, the set GOTO (I_0 , ⟨) has 5 items

15. ANSWER:- B

LR parsers have Rightmost derivation in reverse.

16. ANSWER:- A

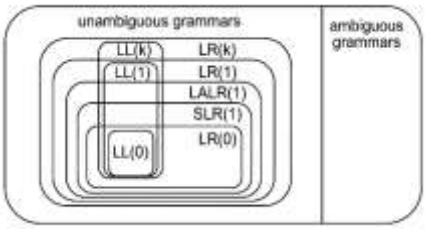
Since $ will be evaluated before # so $ has higher precedence and the left $ i.e., in bcd the left "$" (i.e., b$c) will be evaluated first so it is left associative, whereas # is right associative (as in d#e#f) , the right one (i.e., e#f) will be evaluated first.

17. ANSWER:- B

Type checking is done in semantic analysis phase after syntax analysis phase (i.e., after parsing).

18. ANSWER:- A

Canonical LR is more powerful than SLR as every grammar which can be parsed by SLR parser, can also be parsed by CLR parser.
The power in increasing order is:
LR(0) < SLR < LALR < CLR



19. ANSWER:- C

In the production: P → xQRS,

FOLLOW (Q) = FIRST (R)

FIRST (R) = {w, ϵ} Since FIRST (R) = {ϵ}, so FOLLOW (Q) → {w} ∪ FIRST(S)

FIRST(S) = {y}

So, FOLLOW (Q) = {w, y}

20. ANSWER:- 1024

Since the program has ten if terminals, and each contributes two independent paths, the total number of control flow paths is the product of the number of paths for each if statement.

The total number of paths is 2*2*2*2*2*2*2*2*2*2, which equals 2^{10}.

Check Out Our GATE Courses & Secure Your Dream Rank!

21. ANSWER:- C

Consider the production (given below) which has left recursion.

$S \rightarrow S\alpha \mid \beta$

The equivalent production (after removing left recursion) is

$S \rightarrow \beta S_1$

$S_1 \rightarrow \alpha S_1 \mid \epsilon$

Hence after removing left recursion from: $E \rightarrow E\text{-}T \mid T$, here $\alpha = -T$ and $\beta = T$

$E \rightarrow TE_1$

$E_1 \rightarrow -TE_1 \mid \epsilon$

After removing left recursion from: $T \rightarrow T+F \mid F$, here $\alpha = +F$ and $\beta = F$

$T \rightarrow FT_1$

$T_1 \rightarrow +FT_1 \mid \epsilon$

Replace $E_1 = X$ and $T_1 = Y$

We have,

$E \rightarrow TX$

$X \rightarrow -TX \mid \epsilon$

$T \rightarrow FY$

$Y \rightarrow +FY \mid \epsilon$

$F \rightarrow (E) \mid id$

22. ANSWER:- B

Reason: - A grammar is left recursive if there exists a derivation:

$A \Rightarrow A\alpha$ (i.e., a non-terminal derives itself as leftmost symbol).

Option (A): $A \rightarrow Aa \rightarrow$ direct left recursion

Hence, left recursive.

Option (B):

$S \rightarrow Ab \mid Bb \mid c$

$A \rightarrow Bd \mid \varepsilon$

$B \rightarrow e$

No production leads to left recursion (direct or indirect). Hence, left recursion free.

Option (C):

$A \rightarrow Sc$ and $S \rightarrow Aa$

So, $A \Rightarrow Sc \Rightarrow Aa\, c \Rightarrow A...$

Indirect left recursion exists.

Hence, left recursive.

Option (D):

$A \rightarrow Bd$ and $B \rightarrow Ae$

So, $A \Rightarrow Bd \Rightarrow Ae\, d \Rightarrow A...$

Indirect left recursion exists.

Hence, left recursive.

Thus, only option (B) is free from left recursion.

23. ANSWER:- C

A handle is actually the one which is always on the top of the stack. A viable prefix (prefix of the Right-hand side of a production or productions), is actually a prefix of the handle and so can never extend past the right end of the handle (i.e. the top of the stack).

24. ANSWER:- C

SLR is very easy to implement and CLR is the most powerful method.

SLR < LALR < Canonical LR

25. ANSWER:- D

For LL(1),

For first production

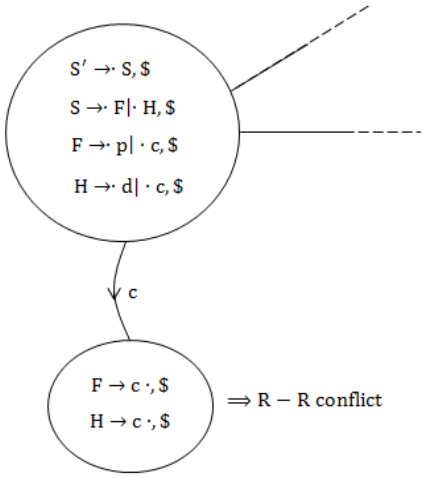
$S \rightarrow F \mid H$

$\Downarrow$

$S \rightarrow F$ (First (S) = p, c)

$S \rightarrow H$ (First (S) = d, c)

So, there is 'c' common in both the first(s) in the production of S. So not LL(1).



Since R-R conflict is present. So, not LR(1).

or

Given CFG is ambiguous. So, it can not be parsed by both LL(1) and LR(1).

26 **ANSWER:- 6**

- Leaf values can be 0 or 1.

- Choose values to maximize the entire expression. selection: 11011011

27. **ANSWER:- D**

< is not possible to read from the given set of items, so no conflict.

so, we need **(n − 1)** reductions in total.

28. **ANSWER:- B**

In production: T ⟶ T * U, since T is left recursive, hence * is left associative.
P ⟶ Q + P, here P is right recursive, so + is right associative.

29. **ANSWER:- D**

Statement 1 is false since the two sets in LR(1) items can be combined if they only differ in look ahead symbols. Because there is no reduced move in the specified LR(1) items, there will be no SR conflict after merging, and so statements 2 and 3 are incorrect.
Statement 4 is also wrong, because goto is carried on Non-Terminals symbols, not on terminal symbols, and c is a terminal symbol.
Hence all statements are false

30. **ANSWER:- B**

Given the custom constraint in the question where both $A \to \epsilon$ and $A \to a$ are forbidden, the right-hand side of every production rule must contain at least two symbols (either two terminals, two non-terminals, or a mix).

- Each reduction step replaces a minimum of 2 symbols on the stack with 1 non-terminal symbol.
- This means every single reduction move decreases the total number of symbols on the stack by exactly $1(2 - 1 = 1)$.
- To reduce an initial string of n tokens down to a single start symbol (1), the total number of symbols must decrease by $n - 1$.

Since each reduction reduces the symbol count by 1 , it takes exactly $n - 1$ reduce moves.

31. **ANSWER:- A**

FIRST (P): is the set of terminals that begin the strings derivable from non- terminal P. If P derives epsilon then we include epsilon in FIRST(P).

FOLLOW(P): is the set of terminals that can appear immediately to the right of P in some sentential form.

FIRST(A) = FIRST (S)

FIRST (S) = FIRST (aAbB) and FIRST (bAaB) and FIRST (ϵ)

FIRST(S) = {a, b, ϵ}

FIRST (B) = FIRST (S) = {a, b, ϵ}= FIRST (A)

FOLLOW(A) = {b} // because of production S⟶a A b B

FOLLOW(A) = {a} // because of production S⟶ b A a B

So FOLLOW (A) = {a, b}

FOLLOW (B) = FOLLOW (S) // because of production S⟶ aA b B

FOLLOW (S) = FOLLOW (A) // because of production A ⟶ S

So FOLLOW (S) = {$, a, b}= FOLLOW(B)

32. **ANSWER:- C**

The entries in E1, E2 and E3 are related to S and B, so we have to take only those productions which have S and B in LHS.

S⟶ aAbB | bAaB | ε

Check Out Our GATE Courses & Secure Your Dream Rank!

The production S⟶ aAbB will go under column FIRST (aAbB) = a, so S⟶ aAbB will be in E1.

S⟶ bAaB will go under column FIRST(bAaB) = b, so S⟶ bAaB will be in E2.

S⟶ ε will go under FOLLOW (S) = FOLLOW(B) = {a, b, $ } , So S⟶ ε will go in E1, E2 and under the column of $.

So E1 will have: S⟶ aAbB and S⟶ ε.

E2 will have S⟶ bAaB and S⟶ ε.

Now, B⟶ S will go under FIRST (S) = {a, b, ε}

Since FIRST(S) = ε so B⟶ S will go under FOLLOW (B) = {a, b, $}

So E3 will contain B⟶ S.

33. ANSWER:- B

7 ↓ 3 ↑ 4 ↑ 3 ↓ 2

⇒ 7 ↓ (3 ↑ (4 ↑ 3)) ↓ 2

⇒ 7 ↓ (3 ↑ (4 ↑ 3))) ↓ 2 as ↓ is left associative

34. ANSWER:- C

The LL(1) parsing table for the given grammar is:

	a	b	c
S	S ⟶ asa	S ⟶ b	S ⟶ c

Because there is no conflict in the LL(1) parsing table, the given grammar is LL(1), and because every LL(1) is also LR(1), the given grammar is both LL(1) and LR(1)

35. ANSWER:- B

LALR(1) is obtained by merging LR(1) states that have the same LR(0) core.

Important facts:

- If a grammar is LR(1) (no conflicts in LR(1)), then after merging states to form LALR(1), shift-reduce conflicts cannot re-appear.
- Merging can introduce reduce-reduce conflicts, but it does not introduce new shift-reduce conflicts.
- Therefore, an LALR(1) parser can have shift-reduce conflicts if and only if the LR(1) parser already has shift-reduce conflicts.

Thus: The LR(1) parser for G has S-R conflicts

36. ANSWER:- D

(a)"It is the position in a sentential form where the next shift or reduce operation will occur"
→ Only talks about **position**, but a handle must include **both position + production**. So incomplete.
(b)"It is non-terminal whose production will be used for reduction in the next step"
→ A handle is **not just a non-terminal**; it is a **substring (RHS) + corresponding production**. Incorrect.
(c)"It is a production that may be used for reduction in a future step along with a position..."
→ Handle is about **next step (immediate reduction)**, not some future possibility. Hence wrong.
(d)"It is the production p that will be used for reduction in the next step along with a position..."
→ Exactly matches definition: **RHS occurrence + production used for immediate reduction**.
Therefore, (d) is correct.

37. ANSWER:- C

S → aB
→ aaBB
→ aabB
→ aabbS
→ aabbaB

Check Out Our GATE Courses & Secure Your Dream Rank!

→ aabbab

38. ANSWER:- A

Since given grammar can have infinite parse trees for string 'ε', so grammar is ambiguous, and also A → AA has left recursion. For predictive-parsing, grammar should be:

- Free from ambiguity
- Free from left recursion
- Free from left factoring

Given grammar contains both ambiguity and left factoring, so it can not have a predictive parser. We always expect first grammar free from ambiguity for parsing.

39. ANSWER:- B

LALR(1) is formed by merging states of LR(1) (also called CLR(1)), hence no of states in LALR(1) is less than no of states in LR(1), therefore n3 < n2. And SLR(1) and LALR(1) have the same number of states, i.e (n1 = n3).
Hence, n1 = n3 < n2

40. ANSWER:- A

The number of ways to parenthesize an expression with $n+1$ terms is given by the n-th Catalan number (C_n).

1. Identify n :

- Your expression is id + id + id + id .
- It has 4 terms.
- So, $n+1=4$, which means $n=3$.

2. Use the Catalan Number Formula: The formula for the n-th Catalan number is:

$$C_n = \frac{1}{n+1}\binom{2n}{n}$$

3. Calculate C_3 (for $n=3$):

- $C_3 = \frac{1}{3+1}\binom{2\times 3}{3}$
- $C_3 = \frac{1}{4}\binom{6}{3}$
- First, let's find $\binom{6}{3}$ (which is "6 choose 3"): $\binom{6}{3} = \frac{6!}{3!(6-3)!} = \frac{61}{3!3!} = \frac{6\times5\times4}{3\times2\times1} = 20$
- Now, plug that back into the formula: $C_3 = \frac{1}{4}\times 20$
- $C_3 = 5$

There are 5 possible ways to parenthesize the expression, which means there are 5 possible parse trees.

3 SYNTAX DIRECTED TREE

GATE CSE - 2026 MSQ SET - 1(1-MARKS)

Q1. Consider the following two syntax-directed definitions SDD1 and SDD2 for type declarations.

SDD_1	
Grammar (G1)	Semantic Rules
$D \rightarrow TV$	D.type = T.type V.type = T.type
$T \rightarrow$ int	T.type = int
$T \rightarrow$ float	T.type = float
$V \rightarrow V_1$ id	V_1.type = V.type put(id.entry,V.type)
$V \rightarrow$ id	put(id.entry,V.type)

SDD_2	
Grammar (G2)	Semantic Rules
$D \rightarrow D_1$ id	D.type = D_1.type put(id.entry, D_1.type)
$D \rightarrow T$ id	D.type = T.type put(id.entry, T.type)
$T \rightarrow$ int	T.type = int
$T \rightarrow$ float	T.type = float

D is the start symbol, and int, $float$ and id are the three terminals. The non-terminal $V1$ is the same as V and the non-terminal $D1$ is the same as D. Here, the subscript is used to differentiate the grammar symbols on the two sides of a production. The function put updates the symbol table with the type information for an identifier.

Let P and Q be the languages specified by grammars G1 and G2, respectively.
Which of the following statements is/are true?

(a) The languages P and Q are the same
(b) SDD_2 is S-attributed and contains only synthesized attributes
(c) SDD_1 is L-attributed and contains only inherited attributes
(d) The specifications of SDD_1 and SDD_2 are such that the same entries get added to the symbol table

GATE CSE - 2025 MCQ SET - 2(1-MARKS)

Q2. Given the following syntax-directed translation rules:

Rule 1:
R → AB { B.i = R.i - 1; A.i = B.i; R.i = A.i + 1; }

Rule 2:
P → CD { P.i = C.i + D.i; D.i = C.i + 2; }

Rule 3:
Q → EF { Q.i = E.i + F.i; }
Which ONE is the CORRECT option among the following?

(a) Rule 1 : is S-attributed and L-attributed; Rule 2 is S-attributed and not L-attributed; Rule 3 is neither S-attributed nor L-attributed
(b) Rule 1 : is neither S-attributed nor L-attributed; Rule 2 is S-attributed and L-attributed; Rule 3 is S-attributed and L-attributed.
(c) Rule 1 : is neither S-attributed nor L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed.
(d) Rule 1 : is S-attributed and not L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2024 MCQ SET - 1(1-MARKS)

Q3. Consider the following syntax-directed definition (SDD)

S → D H T U { S.val = D.val + H.val + T.val + U.val; }
D → "M" D_1 { D.val = 5 + D_1.val; }
D → ε { D.val = -5; }
H → "L" H_1 { H.val = 5 * 10 + H_1.val; }
H → ε { H.val = -10; }
T → "C" T_1 { T.val = 5 * 100 + T_1.val; }
T → ε { T.val = -5; }
U → "K" { U.val = 5; }

Given "MMLK" as the input, which one of the following options is the CORRECT value computed by the SDD (in the attribute S.val)?

(a) 45
(b) 50
(c) 55
(d) 60

GATE CSE - 2024 MSQ SET - 2(2-MARKS)

Q4. Which of the following statements is/are FALSE?

(a) An attribute grammar is a syntax-directed definition (SDD) in which the functions in the semantic rules have no side effects
(b) The attributes in a L-attributed definition cannot always be evaluated in a depth- first order
(c) Synthesized attributes can be evaluated by a bottom-up parser as the input is parsed
(d) All L-attributed definitions based on LR(1) grammar can be evaluated using a bottom-up parsing strategy

GATE CSE - 2023 MCQ NAT (2-MARKS)

Q5. Consider the syntax directed translation given by the following grammar and semantic rules. Here N,I,F and B are non-terminals. N is the starting non-terminal, and #, 0, and 1 are lexical tokens corresponding to input letters "#", "0", and "1", respectively. X.val denotes the synthesized attribute (a numeric value) associated with a non-terminal X.I1 and F1 denote occurrences of I and F on the right-hand side of a production, respectively. For the tokens 0 and 1, 0.val=0 and 1.val= 1

N → I # F N.val = I.val + F.val
I → I_1 B I.val = (2 I_1.val) + B.val
I → B I.val = B.val
F → B F_1 F.val = ½ (B.val + F_1.val)
F → B F.val = ½ B.val
B → 0 B.val = 0.val
B → 1 B.val = 1.val

The value computed by the translation scheme for the input string: **10#011**

is _____. (Rounded off to three decimal places)

GATE CSE - 2022 NAT (2-MARKS)

Q6. Consider the following grammar along with translation rules.

S → S_1 # T { S.val = S_1.val * T.val }
S → T { S.val = T.val }
T → T_1 % R { T.val = T_1.val + R.val }
T → R { T.val = R.val }
R → id { R.val = id.val }

Here **#** and **%** are operators and **id** is a token that represents an integer and **id.val** represents the corresponding integer value. The set of non-terminals is **{S, T, R, P}** and a subscripted non-terminal indicates an instance of the non-terminal.

Using this translation scheme, the computed value of **S.val** for root of the parse tree for the expression **20#10%5#8%2%2** is _____

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2021 MCQ (1-MARKS)

Q7. Consider the following grammar (that admits a series of declarations, followed by expressions) and the associated syntax directed translation (SDT) actions, given as pseudo-code:

$P \rightarrow D^* E^*$

$D \rightarrow$ int ID {record that ID.lexeme is of type int}

$D \rightarrow$ bool ID {record that ID. lexeme is of type bool}

$E \rightarrow E_1 + E_2$ {check that E_1.type = E_2.type = int; set E.type := int}

$E \rightarrow !E_1$ {check that E_1.type = bool; set E.type = bool}

$E \rightarrow$ ID {set E.Type := int}

With respect to the above grammar, which one of the following choices is correct?

(a) The actions can be used to type-check syntactically correct integer variable declarations and integer expressions.
(b) The actions can be used to correctly type-check any syntactically correct program.
(c) The actions can be used to type-check syntactically correct Boolean variable declarations and Boolean expressions
(d) The actions will lead to an infinite loop.

GATE CSE - 2021 MCQ SET – 2 (1-MARKS)

Q8. Consider the following ANSI C program:

```
int main() {
Integer x;
return 0;
}
```

Which one of the following phases in a seven-phase C compiler will throw an error?

(a) Lexical analyzer
(b) Machine dependent optimizer
(c) Semantic analyzer
(d) Syntax analyzer

GATE CSE - 2020 MCQ (1-MARKS)

Q9. Consider the following statements.

I. Symbol table is accessed only during lexical analysis and syntax analysis.

II. Compilers for programming languages that support recursion necessarily need heap storage for memory allocation in the run-time environment.

III. Errors violating the condition 'any variable must be declared before its use' are detected during syntax analysis.

Which of the above statements is/are TRUE?

(a) II only
(b) I only
(c) I and III only
(d) None of I, II and III

GATE CSE - 2020 MCQ (2-MARKS)

Q10. Consider the production $A \longrightarrow PQ$ and $A \longrightarrow XY$. Each of the five non-terminals A, P, Q, X and Y has two attributes:s is a synthesized attribute, and i is an inherited attribute. Consider the following rules:-

Rule 1 : P.i = A.i + 2, Q.i = P.i + A.i and A.s = P.s + Q.s
Rule 2 : X.i = A.i + Y.s and Y.i = X.s + A.i

Which one of the following is TRUE?

(a) Only Rule 2 is L-attributed
(b) Neither Rule 1 nor Rule 2 is L-attributed.
(c) Both Rule 1 and Rule 2 are L-attributed.
(d) Only Rule 1 is L-attributed.

GATE CSE - 2019 MCQ (2-MARKS)

Q11. Consider the following grammar and the semantic actions to support the inherited type declaration attributes. Let X_1, X_2, X_3, X_4, X_5 and X_6 be the placeholders for the non-terminals D, T, L or L_1 in the following table :

Production Rule	Semantic Action
D → TL	X_1.type = X_2.type
T → int	T.type = int
T →float	T.type = float
L → L_1, id	X_3.type = X_4.type addType(id.entry, X_5.type)
L → id	addType(id.entry, X_6.type)

Which one of the following are the appropriate choices for X_1, X_2, X_3 and X_4?

(a) $X_1 = L, X_2 = T, X_3 = L_1, X_4 = L$
(b) $X_1 = T, X_2 = L, X_3 = L_1, X_4 = T$
(c) $X_1 = L, X_2 = L, X_3 = L_1, X_4 = T$
(d) $X_1 = T, X_2 = L, X_3 = T, X_4 = L_1$

GATE CSE - 2016 MCQ SET - 1(2-MARKS)

Q12. Consider the following Syntax Directed Translation Scheme (SDTS), with non-terminals {S, A} and terminals {a,b}.

S → aA { print 1 }
S → a { print 2 }
A → Sb { print 3 }

Using the above SDTS, the output printed by a bottom-up parser, for the input aab is:

(a) 1 3 2
(b) 2 2 3
(c) 2 3 1
(d) syntax error

GATE CSE - 2005 MCQ (1-MARKS)

Q13. Consider the grammar

E → E + n | E × n | n

For a sentence n + n × n, the handles in the right-sentential form of the reduction are

(a) n, E + n and E + n × n
(b) n, E + n and E + E × n
(c) n, n + n and n + n × n
(d) n, E + n and E × n

GATE CSE - 2005 MCQ (2-MARKS)

Q14. Consider the following expression grammar. The seman-tic rules for expression calculation are stated next to each grammar production.

E → number	E.val = number. val
| E '+' E	E(1).val = E(2).val + E(3).val
| E '×' E	E(1).val = E(2).val × E(3).val

The above grammar and the semantic rules are fed to a Yacc tool (which is an LALR (1) parser generator) for parsing and evaluating arithmetic expressions. Which one of the following is true about the action of Yacc for the given grammar?

(a) It detects recursion and eliminates recursion
(b) It detects reduce-reduce conflict, and resolves
(c) It detects shift-reduce conflict, and resolves the conflict in favor of a shift over a reduce action
(d) It detects shift-reduce conflict, and resolves the conflict in favor of a reduce over a shift action

Answer key

1. (a, b, d)	**2. (c)**	**3. (a)**
4. (b, d)	**5. (2.375)**	**6. (80)**
7. (a)	**8. (c)**	**9. (d)**
10. (d)	**11. (a)**	**12. (c)**
13(d)	**14(c)**	

HINTS & SOLUTIONS

1. ANSWER:- (A) (B) AND (D)

(A) Both grammars G1 and G2 generate the same declarations, so the languages P and Q are the same.
(B) In SDD_2, attributes such as D.type and T.type are computed from the right-hand side symbols. Thus only synthesized attributes are used, making it S-attributed.
(C) In SDD_1, attributes like V1.type = V.type are passed from parent to child, so it contains inherited attributes as well, not only inherited ones.
(D) Both SDDs assign the same type information to identifiers using put(id.entry, type), so the same symbol table entries are produced.

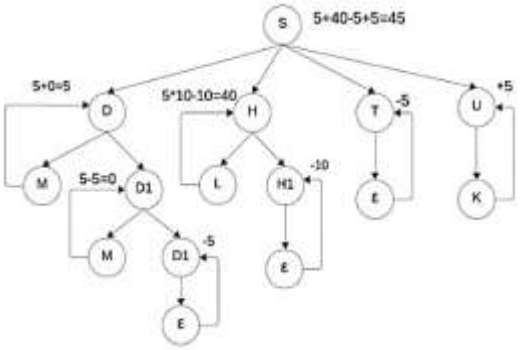
2. ANSWER:- C

'i' is inherited attribute and is evaluated from the right subtree for the rule A. i = B.i, Therefore it is not 'L' attributed.

Rule 2: In D.i = C.i+2; i is inherited attributed, so it is not 'S' attributed, but it is evaluated from the left subtree therefore it is 'L' attributed.

Rule 3: In Q.i = E. i + F.i, 'i' is a synthesized attribute, therefore it is both 'S' and 'L' attributed.

3. ANSWER:- A



So the output of the given SDT is 45.

4. ANSWER:- B,D

A. An attribute grammar is an SDD in which the functions in the semantic rules have no side effects (TRUE)
By definition, an Attribute Grammar is a formal way to define attributes for the symbols of a grammar. It is a subset of Syntax-Directed Definitions (SDDs) where the rules are purely functional and do not produce side effects (like printing a value or updating a global table) outside of calculating the attribute values.

B. The attributes in an L-attributed definition cannot always be evaluated in a depth-first order (FALSE)

L-attributed definitions are specifically designed so that they can always be evaluated in a depth-first, left-to-right manner. In an L-attributed SDD, an attribute of a node can depend on:

Attributes of its parent node (Inherited).

Attributes of its siblings to its left (Inherited).

Its own attributes (Synthesized). Because of this "left-to-right" dependency, a standard depth-first traversal (specifically a pre-order or post-order traversal depending on the attribute type) always works.

C. Synthesized attributes can be evaluated by a bottom-up parser as the input is parsed (TRUE)

Synthesized attributes (like in S-attributed definitions) depend only on the children of a node. Since a bottom-up parser (like LR) works by "reducing" children into a parent, it can calculate these attributes exactly at the moment of reduction. This is why S-attributed definitions are highly compatible with Yacc/Bison tools.

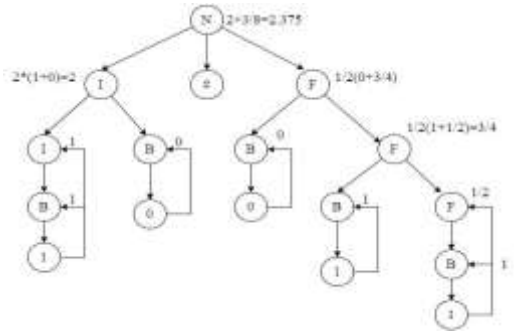
D. All L-attributed definitions based on LR(1) grammar can be evaluated using a bottom-up parsing strategy (FALSE)

This is a common trap. While all L-attributed definitions can be evaluated in a

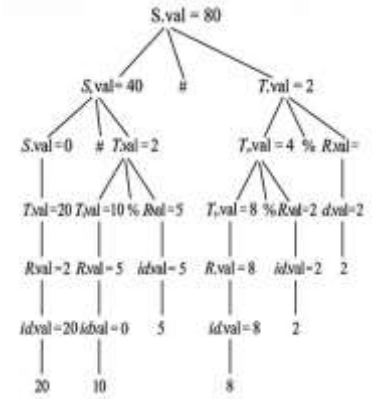
Check Out Our GATE Courses & Secure Your Dream Rank!

top-down (LL) parser, they cannot all be evaluated during bottom-up parsing. Bottom-up parsers struggle with inherited attributes because the parent or left-siblings might not be fully "reduced" or known when a specific action needs to be taken. Only a specific subset of SDDs (like S-attributed or specialized L-attributed translations) can be handled easily in a single-pass bottom-up parse.

5. ANSWER:- 2.375



6. ANSWER:- 80



7. ANSWER:- A

E → !E {If E_1.type = int then E.type bool}
Note: Here instead of int, it is given as bool.
All actions can be used to type check syntactically correct integer variable declaration and integer expressions.

8. ANSWER:- C

The **semantic analyzer** checks meaning — verifies if Integer has been declared as a type. Since, Integer is **undeclared,**
Error occurs in the semantic analysis phase.

9. ANSWER:- D

I. is wrong as the Symbol table is also accessed during the semantic analysis phase.

II. is wrong as compilers which support recursion require stack memory in run time environment.

III. is wrong "any variable must be declared before its use" is a semantic error and not syntax error.

10. ANSWER:- D

In rule 2 for production **A→XY** the attribute "i" is calculated from the right sibling Y in

X.i= A.i + Y.s which is violating the L attribute definition, as in L attribute calculating attribute value from RHS sibling is not allowed.

11. ANSWER:- A

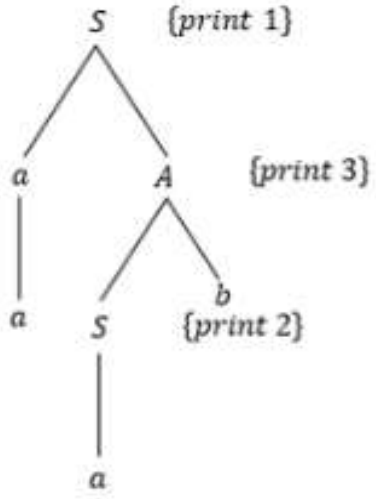
In SDT:
D → T L {X_1.type = X_2.type}
L.type = T.type (Inherited attributed)
∴ X_1 = L and X_2 = T
T can be int or float.

In SDT:
L → L_1, id {X_3.type = X_4.type}
L_1.type = L.type (Inherited attributed)
∴ X_3 = L_1 and X_4 = L

Check Out Our GATE Courses & Secure Your Dream Rank!

12. ANSWER:- C

By using bottom up parser, the output will be "2 3 1"



13. ANSWER:- D

$E \rightarrow E * n$ {Applying $E \rightarrow E * n$} $\rightarrow E + n * n$ {Applying $E \rightarrow E + n$} $\rightarrow n + n * n$ {Applying $E \rightarrow n$}

Hence, the handles in right sentential form is n, E + n and E × n.

14. ANSWER:- C

yacc conflict resolution is done using following rules:
shift is preferred over reduce while shift/reduce conflict.
first reduce is preferred over others while reduce/reduce conflict.
You can answer to this question straightforwardly by constructing LALR(1) parse table, though its a time taking process. To answer it faster, one can see intuitively that this grammar will have a shift-reduce conflict for sure. In that case, given this is a single choice question, (C) option will be the right answer.

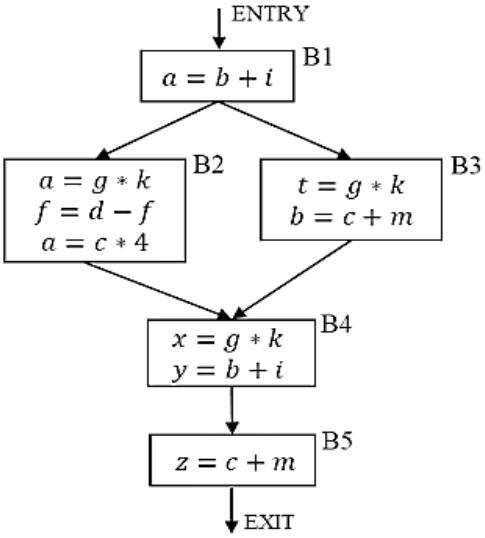
Fool-proof explanation would be to generate LALR(1) parse table, which is a lengthy process.

Once we have the parse table with us, we can clearly see that
i. reduce/reduce conflict will not arise in the above given grammar
ii. shift/reduce conflict will be resolved by giving preference to shift, hence making the expression calculator right associative.

4 INTERMEDIATE CODE GENERATION

GATE CSE - 2026 MCQ SET - 1(1-MARKS)

Q1. Consider the control flow graph shown in the figure.

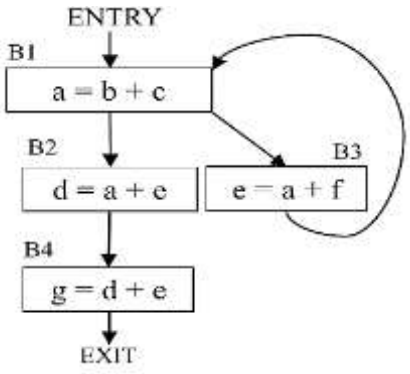


Which one of the following options correctly lists the set of redundant expressions (common subexpressions) in the basic blocks B4 and B5?
Note: All the variables are integers.

(a) B4: { $b + i$ } B5: { $c + m$ }
(b) B4: { $g * k$ } B5: { $c + m$ }
(c) B4: { $g * k, b + i$ } B5: { }
(d) B4: { $g * k$ } B5: { }

GATE CSE - 2026 MCQ SET - 2(2-MARKS)

Q2. Consider the control flow graph given below.



Which one of the following options is the set of live variables at the exit point of each basic block?

(a) B1:{a, b, c, e, f}, B2:{d, e}, B3:{b, c, e, f}, B4:∅
(b) B1:∅, B2:{d, e}, B3:{a, c, f}, B4:∅
(c) B1:{a, b, c, e, f}, B2:{d, e}, B3:{c, e, f}, B4:∅
(d) B1:∅, B2:{d, e, f}, B3:{a, b, c, e, f}, B4:∅

GATE CSE - 2025 MCQ SET - 2(1-MARKS)

Q3. Consider the following statements about the use of backpatching in a compiler for intermediate code generation:
(I) Backpatching can be used to generate code for Boolean expressions in one pass.
(II) Backpatching can be used to generate code for flow-of-control statements in one pass.
Which ONE of the following options is CORRECT?

(a) Only (I) is correct.
(b) Only (II) is correct.
(c) Both (I) and (II) are correct.
(d) Neither (I) nor (II) is correct.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2025 NAT SET - 1(2-MARKS)

Q4. Refer to the given 3-address code sequence. This code sequence is split into basic blocks. The number of basic blocks is _____.

```
1001: i = 1
1002: j = 1
1003: t1 = 10*i
1004: t2 = t1+j
1005: t3 = 8*t2
1006: t4 = t3-88
1007: a[t4] = 0.0
1008: j = j+1
1009: if j <= 10 goto 1003
1010: i = i+1
1011: if i <= 10 goto 1002
1012: i = 1
1013: t5 = i-1
1014: t6 = 88*t5
1015: a[t6] = 1.0
1016: i = i+1
1017: if i <= 10 goto 1013
```

GATE CSE - 2025 MCQ SET - 1(1-MARKS)

Q5. Which ONE of the following techniques used in compiler code optimization uses live variable analysis?

(a) Run-time function call management
(b) Register assignment to variables
(c) Strength reduction
(d) Constant folding

GATE CSE - 2024 MCQ SET - 2(2-MARKS)

Q6. Consider the following expression: x[i] = (p + r) ∗−s[i] + u/w

(0)	+	p	r
(1)			
(2)	uminus	(1)	
(3)			
(4)	/	u	w
(5)	+	(3)	(4)
(6)			
(7)	=	(6)	(5)

Which one of the following options fills in the missing entries CORRECTLY?

(a) $(1) = []si, (3) * (0)(2), (6)[] = xi$
(b) $(1)[] = si, (3) - (0)(2), (6) = []x(5)$
(c) $(1) = []si, (3) * (0)(2), (6)[] = x(5)$
(d) $(1)[] = si, (3) - (0)(2), (6) = []xi$

GATE CSE - 2024 MCQ SET - 1(2-MARKS)

Q7. Consider the following pseudo-code.

```
L1 : t1 = −1
L2 : t2 = 0
L3 : t3 = 0
L4 : t4 = 4 * t3
L5 : t5 = 4 * t2
L6 : t6 = t5 * M
L7 : t7 = t4 + t6
L8 : t8 = a[t7]
L9 : if t8 <= max goto L11
L10 : t1 = t8
L11 : t3 = t3 + 1
L12 : if t3 < M goto L4
L13 : t2 = t2 + 1
L14 : if t2 < N goto L3
L15 : max = t1
```

Which of the following options CORRECTLY specifies the number of basic blocks and the number of instructions in the largest basic block, respectively?

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) 6 and 6
(b) 6 and 7
(c) 7 and 7
(d) 7 and 6

GATE CSE - 2023 MCQ (1-MARKS)

Q8. Consider the following statements regarding the front-end and back-end of a compiler: -

S1: The front-end includes phases that are independent of the target hardware.
S2: The back-end includes phases that are specific to the target hardware.
S3: The back-end includes phases that are specific to the programming language used in the source code.
Identify the CORRECT option.

(a) Only S1 is TRUE.
(b) Only S1 and S2 are TRUE.
(c) S1, S2, and S3 are all TRUE.
(d) Only S1 and S3 are TRUE.

GATE CSE - 2021 MCQ SET - 2(1-MARKS)

Q9. For a statement S in a program, in the context of liveness analysis, the following sets are defined:

USE (S) : the set of variables used in S
IN (S) : the set of variables that are live at the entry of S
OUT(S) : the set of variables that are live at the exit of S
Consider a basic block that consists of two statements, S_1 followed by S_2. Which one of the following statements is correct?

(a) $OUT(S_1) = IN(S_1) \cup USE(S_1)$
(b) $OUT(S_1) = IN(S_2) \cup OUT(S_2)$
(c) $OUT(S_1) = IN(S_2)$
(d) $OUT(S_1) = USE(S_1) \cup IN(S_2)$

GATE CSE - 2021 MCQ SET - 2(1-MARKS)

Q10. In the context of compilers, which of the following is/are NOT an intermediate representation of the source program?

(a) Three address code
(b) Control Flow Graph (CFG)
(c) Abstract Syntax Tree (AST)
(d) Symbol table

GATE CSE - 2017 NAT (2-MARKS)

Q11. Consider the expression (a-1)* (((b + c) / 3) + d) . Let X be the minimum number of registers required by an optimal code generation (without any register spill) algorithm for a load/store architecture in which (i) only loads and store instructions can have memory operands and (ii) arithmetic instructions can have only register or immediate operands. The value of X is _____.

GATE CSE - 2016 NAT SET – 1 (1-MARKS)

Q12. Consider the following code segment.

```
x = u - t;
y = x * v;
x = y + w;
y = t - z;
y = x * y;
```

The minimum number of variables required to convert the above code segment to static single assignment form is _____.

GATE CSE - 2015 MCQ SET – 2 (1-MARKS)

Q13. In the context of abstract-syntax-tree (AST) and control-flow-graph (CFG), which one of the following is TRUE?

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) In both AST and CFG, let node N_2 be the successor of node N_1. In the input program, the code corresponding to N_2 is present after the code corresponding in N_1.
(b) For any input program, neither AST nor CFG will contain a cycle
(c) The maximum number of successors of a node in an AST and a CFG depends on the input program
(d) Each node is AST and CFG corresponds to at most one statement in the input program.

GATE CSE - 2015 MCQ SET – 2 (2-MARKS)

Q14. Consider the intermediate code given below.

(1) i = 1
(2) j = 1
(3) t1 = 5 * i
(4) t2 = t1 + j
(5) t3 = 4 * t2
(6) t4 = t3
(7) a[t4] = –1
(8) j = j + 1
(9) if j < = 5 goto (3)
(10) i = i + 1
(11) if i < 5 goto (2)

The number of nodes and edges in the control-flow-graph constructed for the above code, respectively, are:-

(a) 5 and 7
(b) 6 and 7
(c) 5 and 5
(d) 7 and 8

GATE CSE - 2015 NAT SET – 1 (2-MARKS)

Q15. The least number of temporary variables required to create a three-address code in static single assignment form for the expression **q + r/3 + s – t * 5 + u * v/w is _____.**

GATE CSE - 2014 MCQ SET – 3 (1-MARKS)

Q16. One of the purposes of using intermediate code in compilers is to:-

(a) make parsing and semantic analysis simpler.
(b) improve error recovery and error reporting.
(c) increases the chances of reusing the machine-independent code optimizer in other compilers.
(d) improve the register allocation

GATE CSE - 2014 MCQ SET – 1 (1-MARKS)

Q17. Which one of the following is FALSE?

(a) A basic block is a sequence of instructions where control enters the sequence at the beginning and exits at the end.
(b) Available expression analysis can be used for common subexpression elimination.
(c) Live variable analysis can be used for dead code elimination.
(d) x=4*5⇒x=20 is an example of common subexpression elimination

GATE CSE - 2014 MCQ SET – 2 (1-MARKS)

Q18. Which one of the following is NOT performed during compilation?

(a) Dynamic memory allocation
(b) Type checking
(c) Symbol table management
(d) Inline expansion

GATE CSE - 2014 MCQ SET – 2 (1-MARKS)

Q19. For a C program accessing X[i][j][k], the following intermediate code is generated by a compiler. Assume that the size of an integer is 32 bits and the size of a character is 8 bits.

t0 = i * 1024
t1 = j * 32
t2 = k * 4
t3 = t1 + t0

Check Out Our GATE Courses & Secure Your Dream Rank!

t4 = t3 + t2
t5 = x[t4]
Which one of the following statements about the source code for the C program is CORRECT?

(a) X is declared as "int X[32][32][8]".
(b) X is declared as "int X[4][1024][32]".
(c) X is declared as "char X[4][32][8]".
(d) X is declared as "char X[32][16][2]".

GATE CSE - 2014 MCQ SET – 3 (1-MARKS)

Q20. Which of the following statements are Correct?

1. Static allocation of all data areas by a compiler makes it impossible to implement recursion.
2. Automatic garbage collection is essential to implement recursion.
3. Dynamic allocation of activation records is essential to implement recursion.
4. Both heap and stack are essential to implement recursion.

(a) 1 and 2 only
(b) 2 and 3 only
(c) 3 and 4 only
(d) 1 and 3 only

GATE CSE - 2014 MCQ SET – 3 (2-MARKS)

Q21. Consider the basic block given below.

a= b + c
c =a + d
d =b + c
e =d - b
a =e + b

The minim um number of nodes and edges present in the DAG representation of the above basic block respectively are

(a) 6 and 6
(b) 8 and 10
(c) 9 and 12
(d) 4 and 4

GATE CSE - 2014 NAT SET – 3 (1-MARKS)

Q22. The minimum number of arithmetic operations required to evaluate the polynomial $P(x) = x^5 + 4x^3 + 6x + 5$ for a given value of x, using only one temporary variable is ____ .

GATE CSE - 2013 MCQ (2-MARKS)

Q23. The following code segment is executed on a processor which allows only register operands in its instructions. Each instruction can have almost two source operands and one destination operand. Assume that all variables are dead after this code segment.

```
c = a + b;
d = c * a;
e = c + a;
x = c * c;
if (x > a){
y = a*a;
}
      else {
d = d *  d;
   e = e*e;
}
```

Suppose the instruction set architecture of the processor has only two registers. The only allowed compiler optimization is code motion, which moves statements from one place to another while preserving correctness. What is the minimum number of spills to memory in the compiled code?

(a) 0
(b) 1
(c) 2
(d) 3

Check Out Our GATE Courses & Secure Your Dream Rank!

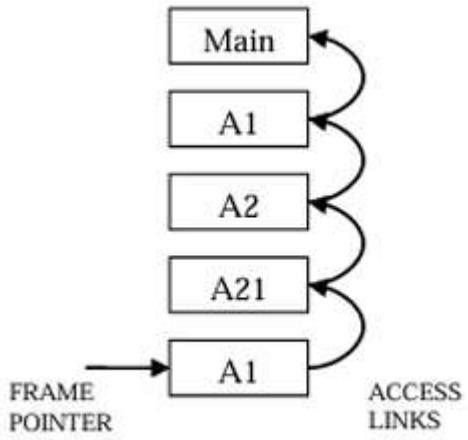
GATE CSE - 2012 MCQ SET – 3 (2-MARKS)

Q24. Consider the program given below, in a block-structured pseudo-language with lexical scoping and
nesting of procedures permitted.

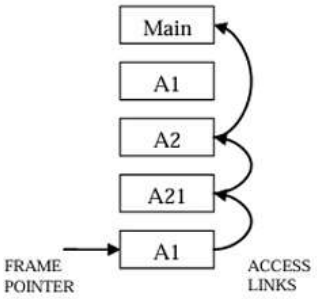
```
Program main;
  Var ...
  Procedure A1;

Var ...
  Call A2;
  End A1
  Procedure A2;
    Var ...
    Procedure A21;
      Var
      Call A1;
    End A21
    Call A21;
  End A2
  Call A1;
End main.
```

Consider the calling chain: Main → A1 → A2 → A21 → A1
The correct set of activation records along with their access links is given by

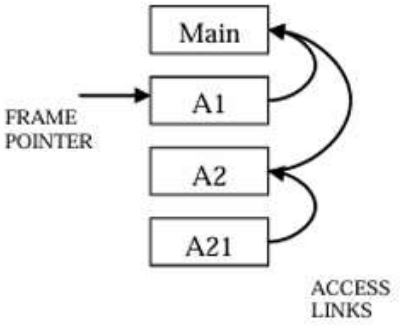
(a)



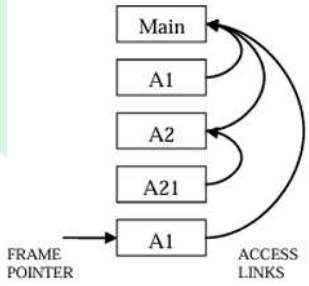
(b)



(c)



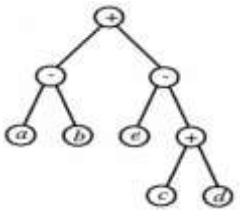
(d)



GATE CSE - 2011 MCQ (2-MARKS)

Q25. Consider evaluating the following expression tree on a machine with load-store architecture in which memory can be accessed only through load and store instructions. The variables a, b, c, d and e initially stored in memory. The binary operators used in this expression tree can be evaluate by the machine only when the operands are in registers. The instructions produce results only in a register. If no intermediate results can be stored in memory, what is the minimum number of registers needed to evaluate this expression?

Check Out Our GATE Courses & Secure Your Dream Rank!



(a) 2
(b) 9
(c) 5
(d) 3

GATE CSE - 2010 MCQ (2-MARKS)

Q26. The program below uses six temporary variables a, b, c, d, e, f.

```
a = 1
b = 10
c = 20
d = a+b
e = c+d
f = c+e
b = c+e
e = b+f
d = 5+e
return d+f
```

Assuming that all operations take their operands from registers, what is the minimum number of registers needed to execute this program without spilling?

(a) 2
(b) 3
(c) 4
(d) 6

GATE CSE - 2010 MCQ (1-MARKS)

Q27. Which languages necessarily need heap allocation in the runtime environment?

(a) Those that support recursion
(b) Those that use dynamic scoping
(c) Those that allow dynamic data structures
(d) Those that use global variables

GATE CSE - 2006 MCQ (1-MARKS)

Q28. Consider the following C code segment.

```
for (i = 0; < i< n; i++)
{
   for (j=0; j< n; j++)
   {
      if (i%2)
      {
         x += (4*j + 5*i);
         y += (7 + 4*j);
      }
   }
}
```

Which one is false?

(a) The code contains loop-in variant computation
(b) There is scope of common sub-expression elimination in this code
(c) There is scope strength reduction in this code
(d) There is scope of dead code elimination in this code

GATE CSE - 2008 MCQ (1-MARKS)

Q29. Some code optimizations are carried out on the intermediate code because

(a) They enhance the portability of the compiler to other target processors
(b) Program analysis is more accurate on intermediate code than on machine code
(c) The information from dataflow analysis cannot otherwise be used for optimization
(d) The information from the front end cannot otherwise be used for optimization

Check Out Our GATE Courses & Secure Your Dream Rank!

Answer key

1. (d)	**2. (a)**	**3. (c)**
4. (6)	**5. (b)**	**6. (a)**
7. (d)	**8. (b)**	**9. (c)**
10. (d)	**11. (2)**	**12. (10)**
13.(c)	**14. (b)**	**15. (8)**
16. (c)	**17.(d)**	**18. (a)**
19. (a)	**20. (d)**	**21. (a)**
22. (7)	**23.(b)**	**24.(d)**
25. (d)	**26. (b)**	**27. (c)**
28. (d)	**29. (a)**	

HINTS & SOLUTIONS

1. ANSWER:- (D)

Control Flow Graph with blocks B1–B5

Find: Redundant expressions (common subexpressions) in B4 and B5

Step 1: Identify expressions computed earlier

B1

b + i

B2

g * k

c * 4

B3

g * k

b = c + m (so c + m is computed)

Step 2: Check expressions in B4

B4 contains:

x = g * k

y = b + i

Check **g * k**

Computed in both B2 and B3.

Neither **g** nor **k** is modified later.

Therefore available along all paths reaching B4.

Hence, **g * k** is a redundant expression.

Check **b + i**

Computed in B1.

But in B3, **b** is redefined.

Therefore value is not same along all paths.

Hence, b + i is not redundant.

So,

Redundant expressions in B4 = { g * k }

Step 3: Check expressions in B5

B5 contains:

z = c + m

c + m computed only in B3.

Not computed along B2 path.

Therefore not available along all paths.

Hence, no redundant expression in B5.

Final Answer

B4 : { g * k }

B5 : { }

2. ANSWER:- (A)

Basic blocks:

B1: a = b + c

B2: d = a + e

B3: e = a + f

B4: g = d + e

A variable is live at the exit of a block if its value is used later before being redefined.

B4

g = d + e and program exits afterward.

No variable is used later.

Live(B4) = ∅

B2

d = a + e

Variables d and e are used later in B4 (g = d + e).

Live(B2) = {d, e}

B3

e = a + f

Variables a and f are used in this block and needed for computation.

Live(B3) = {a, c, f}

B1

a = b + c

Variables a, b, c, e, f are needed later in B2 and B3.

Live(B1) = {a, b, c, e, f}

Final Result

B1: {a, b, c, e, f}

B2: {d, e}

B3: {b,c, e, f}

B4: ∅

3. ANSWER:- (C)

Back patching can be used to generate code of both boolean expressions and flow-of-control statements in one pass.

4. ANSWER:- (6)

Basic block is a sequence of consecutive statements in which the control enters at the beginning and leaves only at the end without any branching (or) halting except at the end..

1. Identify leader (1st statement of a basic block)
   - First statement of a program is a leader.
   - Target of goto is a leader.
   - Next statement after goto is a leader.

2. Group a leader statement with all the consecutive statements upto but not inclueding the next leader of a basic block.

Basic Block	Instructions
B1	1001
B2	1002
B3	1003 →1004 →1005→ 1006→ 1007 →1008 →1009
B4	1010→ 1011
B5	1012
B6	1013→ 1014→ 1015→ 1016→ 1017

5. ANSWER:- (B)

Uses of live variable analysis:

1. **Used for register allocation:** If variable x is live in a basic block b, it is a potential candidate for register allocation.

2. Used for dead code elimination: If variable x is not live after an assignment x = . . ., then the assignment is redundant and can be deleted as dead code.

6. ANSWER:- (A)

The given expression x[i]=(p+r)∗−s[i]+u/w is broken into triples for evaluation.

Triple (1): []=s i[ ] representing −s[i].

Triple (3): ∗(0)(2), where (0) is p+r and (2)(2) is −s[i].

Triple (6): []=x i[ ] , assigning the final result to x[i]x[i].

7. ANSWER:- (D)

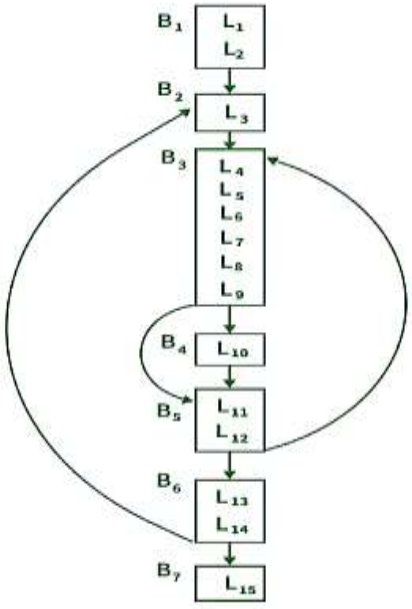
L 1,L 3,L 4,L 10 ,L 11,L 13, and L15 are leader statements.

In the given 3AC 7 leaders are there: 1, 3, 4, 10, 11, 13, 15

There are 7 basic block are there:

1. Block B_1: statement 1 – 2
2. Block B_2: statement 3
3. Block B_3: statement 4 – 9
4. Block B_4: statement 10
5. Block B_5: statement 11 – 12
6. Block B_6: statement 13 – 14
7. Block B_7: statement 15

So the total number of basic blocks is 7 and the largest basic block is B_3 which contains a total of 6 instruction.



8. ANSWER:- (B)

S1: The front-end includes phases that are independent of the target hardware.

This statement is true. The front-end of a compiler is responsible for tasks like lexical analysis, syntax analysis, and semantic analysis, which are independent of the target hardware.

S2: The back-end includes phases that are specific to the target hardware.

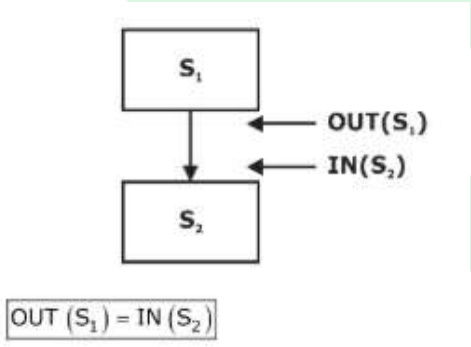
This statement is true. The back-end of a compiler is responsible for tasks like code generation and

Check Out Our GATE Courses & Secure Your Dream Rank!

optimization, which are specific to **the target hardware platform.**

S3: The back-end includes phases that are specific to the programming language used in the source code.
This statement is not generally true. The phases in the back-end of a compiler are primarily concerned with generating efficient code for the target hardware. While the choice of programming language can influence some aspects of code generation, it's not a phase specific to the programming language itself. Most of the language-specific tasks are handled

9. ANSWER:- (C)



10. ANSWER:- (D)

Symbol Table is not an intermediate representation of the Source program.A symbol table is a data structure used by compilers to store information about variables, functions, classes, and other identifiers used in the program. It's accessed by various compiler stages to look up information about these entities

11. ANSWER:- (2)

Load R1, b
114 Load R2, c
ADD R1, R2
Div R1, 3
Load R2, d
Add R1, R2
Load R2, a
Sub R2, 1
Mul R2, R1

hence, minimum 2 registers required.

12. ANSWER:- (10)

Static Single Assignment (SSA) requires that every variable is assigned exactly once, and every variable must be defined before it is used. We achieve this by versioning the variables with subscripts.

Step-by-Step Conversion

1. $x1 = u = t$ (First assignment to x)
2. $y1 = x1 * v$ (First assignment to y; uses $x1$)
3. $x2 = y1 + w$ (Second assignment to x; uses $y1$)
4. $y^2 = t - z$ (Second assignment to y)
5. $y^3 = x_2 * y_2$ (Third assignment to y; uses $x2$ and $y2$)

Final Count
To find the minimum number of variables, count every unique variable name present in the final SSA code:

- Assigned Variables (Left+Hand Side): x_1, y_1, x_2, y_2, y_3 (5 variables)
- Input Variables (Right-Hand Side): u, t, v, w, z (5 variables)

Total variables $= 5 + 5 = 10$

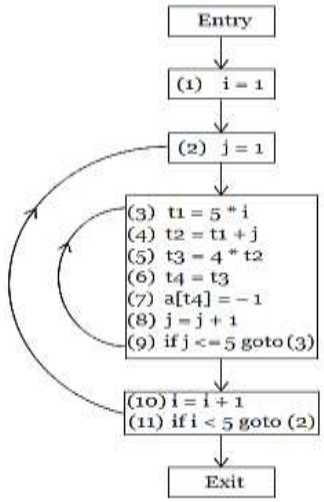
The minimum number of variables required is 10 .

13. ANSWER:- (C)

(A) is false, In CFG (Control Flow Graph), code of N2 may be present before N1 when there is a loop or goto.
(B) is false, CFG (Control Flow Graph) contains a cycle, when the input program has a loop.
(C) is true, successors in AST and CFG are dependent on input program
(D) is false, a single statement may belong to a block of statements.

Check Out Our GATE Courses & Secure Your Dream Rank!

14. ANSWER:- (B)



15. ANSWER:- (8)

We will need one temporary variable for storing the result of every binary operation as Static Single Assignment implies the variable cannot be repeated on left hand side of assignment.

The given expression:

q+r/3+s−t∗5+u∗v/w

t1=r/3;

t2=t∗5;

t3=u∗v;

t4=t3/w;

t5=q+t1;

t6=t5+s;

t7=t6−t2;

t8=t7+t4;

So in total we need 8 temporary variables. If it was not mentioned as a static single assignment then the answer would have been 3 because we can re-use the same temporary variable several times.

16. ANSWER:- (C)

Intermediate code is generated after semantic analysis. The intermediate code is independent of the machine. By converting source code to intermediate code a machine independent code optimizer may be written. Thus increase the chances of reusing the machine-independent code optimizer in other compilers.

17. ANSWER:- (D)

x=4* 5 ⇒ x=20 is an example of common subexpression elimination is a false statement.

Common sub-expression elimination (CSE) is a compiler optimization that searches for instances of identical expressions (i.e., they all evaluate to the same value), and analyzes whether it is worthwhile replacing them with a single variable holding the computed value.

For ex: Consider the following code:

m=y+z * p

n= y+z *k

The common subexpression is “y+z” we can be calculate it one time and replace in both expression

temp = y+z

m = temp*p

n = temp*k

18. ANSWER:- (A)

Dynamic memory allocation is not performed during compilation, it occurs at run time only. At the time of compilation, compiler only compiles the instructions for dynamic memory allocation, like calloc, malloc….

19. ANSWER:- (A)

Consider a 3-D array: arr[i][j][k]

Arr[0][1][2], this 3-D array contains

One 2- dimensional array as i value is zero (if i =1 then it has 2, two D array), & the two dimension array has 2 rows and 3 columns.

So, In a 3-D array, I represent the number of 2D arrays, j represents the number of rows and k represents the number of columns.

Number of 2 D array (M)=1

Number of rows (R)=2

Number of Columns (C)=3

Arr[0][0][0]	Arr[0][0][1]	Arr[0][0][2]
Arr[0][1][0]	Arr[0][1][1]	Arr[0][1][2]

As arrays are stored in row major order, so this 2 dimension array will be stored as:

Check Out Our GATE Courses & Secure Your Dream Rank!

Arr[0][0][0]	Arr[0][0][1]	Arr[0][0][2]	Arr[0][1][1]	Arr[0][1][1]	Arr[0][1][2]
1000	1004	1008	1012	1016	1020

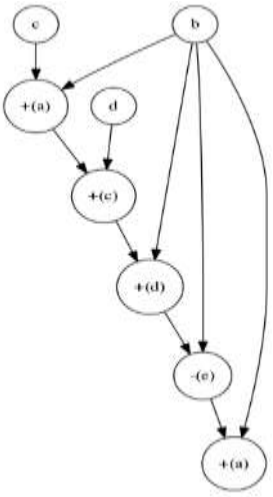
Assume the base address of Arr is 1000. The address of required position is calculated as:
Arr[i][j][k]= Arr+ [i*(R*C)+(j*C)+k]*4 //
multiplication to 4 is due to int taking 4 Bytes.
Arr[0][1][1] = 1000 + [0*(2*3) + (1*3) + 1]*4
= 1000 + [0 + 3 + 1] * 4
= 1000 + 4 * 4 = 1016
As you can see, in the given example of row order storing the array also has the address of Arr[0][1][1] is 1016.
Now coming to the question:
X [i][j][k] is calculated by 3 address code X[t_4]
X [i][j][k] = X [t_4] // by substituting in reverse
=X [t_3 + t_2]
=X [t_1 + t_0 + k*4]
=X [t_0 + t_1 + k*4] // t_0 and t_1 swapped as swapping doesn't have any impact
=X [i*1024 + j*32 + k*4]
=X [i*256 + j*8 +k] *4 // taking 4 (common) outside
= X [i*(32*8)+ (j*8) +k] *4
By comparing the above line with .....
Arr[i][j][k]= Arr+ [i*(R*C)+(j*C)+k]*4
We get R=32, C=8
It means that the declared array has 32 rows and 8 columns and since multiplication by 4 (common outside) so it was declared as INT.
But how many 2D arrays in this 3D array, we don't know.

20. ANSWER:- (D)

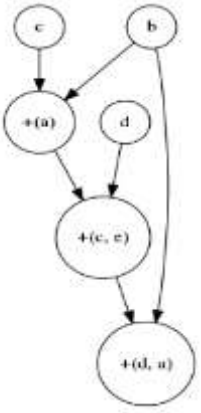
The statement, static allocation of all data areas by a compiler makes it impossible to implement recursion is true, as recursion requires memory allocation at run time, so it requires dynamic allocation of memory. Hence, Dynamic allocation of activation records is essential to implement recursion is also a true statement.

21. ANSWER:- (A)

A normal DAG construction will give 8 nodes and 10 edges as shown below.



Since, this question asks for minimum possible, we can assume algebraic simplification is allowed. So, d = b + c, e = d − b; can be simplified to d = b + c; e = c; Similarly, e = d − b; a = e + b; can be simplified to a = d. This gives the following DAG with 6 nodes and 6 edges.



22. ANSWER:- (7)

P(X) = $x^5 + 4x^3 + 6x + 5$
= $x(x^4 + 4x^2 + 6) + 5$
= $x(x(x^3 + 4x) + 6) + 5$
= $x(x(x(x^2 + 4)) + 6) + 5$
= x(x(x(x(x) + 4)) + 6) + 5
mul = pair of brackets 4
add = num of signs 3
total 7

23. ANSWER:- B

Reason:

The machine has only 2 registers.
- Expression uses variables a, b, c, d, e, x.

- c = a + b requires both a and b in registers.

- d = c * a and e = c + a require reuse of c and a.

- x = c * c also needs c twice.

- With only 2 registers, it is not possible to keep all required values simultaneously.

- Using code motion, we can reduce redundant recomputation.

- Only one value needs to be spilled to memory to maintain correctness.

- Hence, minimum spills = 1.

24. ANSWER:- D

Reason:

In lexical scoping, access link points to the activation record of the nearest enclosing (static parent) procedure.

- Calling chain: Main → A1 → A2 → A21 → A1
- Access links:
  A1 (first call) → Main
  A2 → Main
  A21 → A2
  A1 (last call) → Main

- All access links point according to static (lexical) nesting, not calling sequence.

Option (D) correctly represents these access links.

25. ANSWER:- (D)

Load R1, a ; R1 ← M[a]
Load R2, b ; R2 ← M[b]
Sub R1, R2 ; R1 ← R1 – R2
Load R2, c ; R2 ← M[c]
Load R3, d ; R3 ← M[d]
Add R2, R3 ; R2 ← R2 + R3
Load R3, e ; R3 ← M[e]
Sub R3, 3 ; R3 ← R3 – R2
Add R1, R3 ; R1 ← R1 + R3
Total 3 Registers are required minimum

26. ANSWER:- (B)

Here a, b, and c all have 3 different values so we need atleast 3 registers r1, r2 and r3.
Assume 'a' is mapped to r1, 'b' to r2 and 'c' to r3.
d = a + b, after this line if u notice 'a' is never present on right hand side, so we can map 'd' to r1.
e = c + d, after this line 'd' is never present on rhs, so we can map 'e' to r1.
at this time mapping is
r1 --- e
r2 --- b
r3 --- c
We have 3 registers for e, b and c.
f = c + e
b = c + e
These two are essentially doing the same thing, after these two line 'b' and 'f' are the same so we can skip computing 'f' or need not give any new register for 'f'. And wherever 'f' is present we can replace it with 'b', because neither of 'f' and 'b' are changing after these two lines, so the value of these will be 'c+e' till the end of the program.
At second last line "d = 5 + e"
here 'd' is introduced, we can map it to any of the register r1 or r3, because after this line neither of 'e' or 'c' is required. Value of 'b' is required because we need to return 'd+f', and 'f' is essentially equal to 'b'
finally code becomes
r1 = 1
r2 = 10
r3 = 20
r1 = r1 + r2
r1 = r3 + r1
r2 = r3 + r1
r2 = r3 + r1
r1 = r2 + r2
r3 = 5 + r1
return r3 + r2
Therefore minimum 3 registers needed.

27. ANSWER:- (C)

Dynamic memory is allocated on the heap by the system. So the languages which allow dynamic data structure require heap allocation at runtime.

28. ANSWER:- (D)

4 * j is used at two places- so common subexpression elimination is possible, i % 2 is loop invariant for the inner loop, 5 * i is also loop invariant for inner loop
x += 5 * i can be replaced by x += p; p += 5; (p must be initialized to 0 before the loop).
Thus replacing * with + gives strength reduction. The code does not contain unreachable or dead code. because all statements inside the loops are reachable during the execution of the loops.
So, only (D) is false here.

Check Out Our GATE Courses & Secure Your Dream Rank!

29. ANSWER:- (A)

Option (B) is also true. But the main purpose of doing some code-optimization on intermediate code generation is to enhance the portability of the compiler to target processors. So Option A) is more suitable here. Intermediate code is machine/architecture independent code. So a compiler can optimize it without worrying about the architecture on which the code is going to execute (it may be the same or the other). So that kind of compiler can be used by multiple different architectures. In contrast to that, suppose code optimization is done on target code, which is machine/architecture dependent, then the compiler has to be specific about the optimizations on that kind of code. In this case the compiler can't be used by multiple different architectures, because the target code produced on different architectures would be different. Hence portability reduces here.

OPERATING SYSTEMS

"Scan Here for Video Solution"

Link to GfG corresponding Short Notes: GATE

“The *Operating System* of success is simple—Understand deeply, Practice consistently, and **Never stop Improving.**”

LEGEND OF

OPERATING SYSTEM

Dr. Khaleel Ur Rahman Khan Sir

An Inspirer. A Mentor. A Guide.

1 PROCESSES AND THREADS

GATE CSE - 2026 MCQ SET - 2(1-MARKS)

Q1. Which one of the following CPU scheduling algorithms cannot be preemptive?

(a) Shortest Remaining Time First (SRTF) Scheduling
(b) First Come First Serve (FCFS) Scheduling
(c) Round Robin Scheduling
(d) Priority Scheduling

GATE CSE – 2026 NAT SET – 1 (2-MARKS)

Q2. Consider a CPU that has to execute two types of processes. The first type, Actuators (A), requires a CPU burst of 6 seconds. The second type, Controllers (C), requires a CPU burst of 8 seconds. A new process of type A arrives at time t = 10, 20, 30, 40, and 50 seconds. Similarly, a new process of type C arrives at time t = 11, 22, 33, 44, and 55 seconds. The CPU scheduling policy is First Come First Serve (FCFS). The first process of type A starts running at t = 10 seconds.

The average waiting time (in seconds) for the 10 processes is _____. *(rounded off to one decimal place).*

GATE CSE – 2025 MCQ SET – 2 (2-MARKS)

Q3. Processes *P*1, *P*2, *P*3, *P*4 arrive in that order at times 0, 1, 2, and 8 milliseconds respectively, and have execution times of 10, 13, 6, and 9 milliseconds respectively. Shortest Remaining Time First (SRTF) algorithm is used as the CPU scheduling policy. Ignore context switching times. Which ONE of the following correctly gives the average turnaround time of the four processes in milliseconds?

(a) 22
(b) 15
(c) 37
(d) 19

GATE CSE – 2025 MCQ SET - 1(2-MARKS)

Q4. A computer has two processors, M_1 and M_2. Four processes P_1, P_2, P_3, P_4 with CPU bursts of 20, 16, 25, and 10 milliseconds, respectively, arrive at the same time and these are the only processes in the system. The scheduler uses non-preemptive priority scheduling, with priorities decided as follows:

- M_1 uses priority of execution for the processes as, $P_1 > P_3 > P_2 > P_4$, i.e., P_1 and P_4 have highest and lowest priorities, respectively.
- M_2 uses priority of execution for the processes as, $P_2 > P_3 > P_4 > P_1$, i.e., P_2 and P_1 have highest and lowest priorities, respectively.

A process P_i is scheduled to a processor M_k, if the processor is free and no other process P_j is waiting with higher priority. At any given point of time, a process can be allocated to any one of the free processors without violating the execution priority rules. Ignore the context switch time. What will be the average waiting time of the processes in milliseconds?

(a) 9.00
(b) 8.75
(c) 6.50
(d) 7.50

GATE CSE - 2024 MSQ SET - 2(1-MARKS)

Q5. Consider a process P running on a CPU. Which one or more of the following events will *always* trigger a context switch by the OS that results in process P moving to a non-running state (e.g., ready, blocked)?

(a) P makes a blocking system call to read a block of data from the disk
(b) tries to access a page that is in the swap space, triggering a page fault
(c) An interrupt is raised by the disk to deliver data requested by some other process
(d) A timer interrupt is raised by the hardware

GATE CSE - 2024 NAT SET - 1(2-MARKS)

Q6. Consider the following code snippet using the fork() and wait() system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;
while (x > 0) {
  fork();
  printf("hello");
  wait(NULL);
  x--;
}
```

The total number of times the printf statement is executed is _____.

GATE CSE - 2024 MSQ SET - 1(1-MARKS)

Q7. Which of the following statements about threads is/are TRUE?

(a) Threads can only be implemented in kernel space
(b) Each thread has its own file descriptor table for open files
(c) All the threads belonging to a process share a common stack
(d) Threads belonging to a process are by default not protected from each other

GATE CSE – 2024 MSQ SET - 1(1-MARKS)

Q8. Which of the following process state transition(s) is/are NOT possible?

(a) Ready to Waiting
(b) Running to Ready
(c) Running to Terminated
(d) Waiting to Running

GATE CSE – 2024 MCQ SET - 2(2-MARKS)

Q9. Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time. A (0, 10), B (2, 6), C (4, 3), and D (6, 7). Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?

(a) SRTF = 6, NP-SJF = 7
(b) SRTF = 6, NP-SJF = 7.5
(c) SRTF = 7, NP-SJF = 7.5
(d) SRTF = 7, NP-SJF = 8.5

GATE CSE – 2024 MCQ SET 1 (2-MARKS)

Q10. Consider the following two threads T1 and T2 that update two shared variables a and b. Assume that initially a = b = 1. Though context switching between threads can happen at any time, each statement of T1 or T2 is executed atomically without interruption.

T1	T2
a = a + 1;	b = 2 * b;
b = b + 1;	a = 2 * a;

Which one of the following options lists all the possible combinations of values of a and b after both T1 and T2 finish execution?

(a) (a = 4, b = 4); (a = 3, b = 3); (a = 4, b = 3)
(b) (a = 3, b = 4); (a = 4, b = 3); (a = 3, b = 3)
(c) (a = 4, b = 4); (a = 4, b = 3); (a = 3, b = 4)
(d) (a = 2, b = 2); (a = 2, b = 3); (a = 3, b = 4)

GATE CSE – 2023 MSQ SET - 1(1-MARKS)

Q11. Which one or more of the following options guarantees that a computer system will transition from user mode to kernel mode?

(a) function call
(b) malloc call
(c) page fault
(d) system call

GATE CSE – 2023 MSQ SET-1 (2-MARKS)

Q12. Which one or more of the following need to be saved on a context switch from one thread (T1) of a process to another thread (T2) of the same process ?

(a) Page Table base register
(b) Stack pointer
(c) Program Counter
(d) General purpose register

GATE CSE – 2023 MSQ SET-1 (1-MARKS)

Q13. Which one or more of the following CPU scheduling algorithms can potentially cause starvation?

(a) First-in First-Out
(b) Round Robin
(c) Priority Scheduling
(d) Shortest Job First.

GATE CSE - 2022 MCQ (2-MARKS)

Q14. Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time t = 0. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

(a) P = 4, Q = 10, R = 6, S = 2
(b) P = 2, Q = 9, R = 5, S = 1
(c) P = 4, Q = 12, R = 5, S = 4
(d) P = 3, Q = 7, R = 7, S = 3

GATE CSE - 2021 MSQ SET - 1(1-MARKS)

Q15. Which of the following standard C library functions will always invoke a system call when executed from a single-threaded process in a UNIX/Linux operating system?

(a) exit
(b) malloc
(c) sleep
(d) strlen

GATE CSE – 2021 NAT SET-1 (1-MARKS)

Q16. There are 6 jobs with distinct difficulty levels, and 3 computers with distinct processing speeds. Each job is assigned to a computer such that:
- The fastest computer gets the toughest job and the slowest computer gets the easiest job.
- Every computer gets at least one job
The number of ways in which this can be done is _____

GATE CSE - 2021 MSQ SET - 1(2-MARKS)

Q17. Which of the following statement(s) is/are correct in the context of CPU scheduling?

(a) Turnaround time includes waiting time.
(b) Round-robin policy can be used even when the CPU time required by each of the processes is not known apriori.
(c) Implementing pre-emptive scheduling needs hardware support.
(d) The goal is to only maximize CPU utilization and minimize throughput.

GATE CSE - 2021 MSQ SET – 2 (2-MARKS)

Q18. Consider the following multi-threaded code segment (in a mix of C and pseudocode), invoked by two processes P1 and P2, and each of the processes spawns two threads T1 and T2:

```
int x = 0;   // global
Lock L1;    // global
main() {
        create a thread to execute foo();  // Thread T1
        create a thread to execute foo();  // Thread T2
        wait for the two threads to finish execution;
        print (x);}

foo () {
         int y = 0;
        Acquire L1;
        x = x + 1;
         y = y + 1;
        Release L1;
        print (y);}
```

Which of the following statement(s) is / are correct?

(a) At least one of P1 and P2 will print the value of x as 4.
(b) At least one of the threads will print the value of y as 2.
(c) Both T1 and T2, in both the processes, will print the value of y as 1.
(d) Both P1 and P2 will print the value of x as 2.

GATE CSE - 2021 NAT SET - 1 (2-MARKS)

Q19. Three processes arrive at time zero with CPU bursts of 16, 20 and 10 milliseconds. If the scheduler has prior knowledge about the length of the CPU bursts, the minimum achievable average waiting time for these processes in a non-preemptive scheduler (rounded to nearest integer) is _____milliseconds.

GATE CSE - 2020 NAT (2-MARKS)

Q20. Consider the following set of processes assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order P1, P2, P3, P4.

Processes	P_1	P_2	P_3	P_4
Burst time (in ms)	8	7	2	4

If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is _____

GATE CSE - 2020 MCQ (1-MARKS)

Q21. Consider the following statements about process state transitions for a system using preemptive scheduling.
I. A running process can move to ready state.
II. A ready process can move to running state.
III. A blocked process can move to running state.
IV. A blocked process can move to ready state.
Which of the above statements are TRUE?

(a) I, II, and III only
(b) II and III only
(c) I, II, and IV only
(d) I, II, III and IV only

GATE CSE - 2019 NAT (2-MARKS)

Q22. Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____

GATE CSE - 2019 NAT (1-MARKS)

Q23. Consider the following C program is executed on a Unix/Linux system:

```
#include <unistd.h>
int main()
{
    int i;
    for(i = 0; i < 10; i++)
    if(i % 2 == 0)
            fork();
    return 0;
}
```

The total number of child processes created is _____.

GATE CSE - 2017 SET 1 NAT (1-MARKS)

Q24. Consider the following CPU processes with arrival times (in milliseconds) and length of CPU bursts (in milliseconds) as given below:
Process: P1, P2, P3, P4
Arrival Time: 0, 3, 5, 6
Burst Time: 7, 3, 5, 2
If the pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes, then the average waiting time across all processes is _____ milliseconds.

GATE CSE - 2017 SET 1 MCQ (1-MARKS)

Q25. Threads of a process share
(a) global variables but not heap
(b) heap but not global variables
(c) neither global variables nor heap
(d) both heap and global variables

GATE CSE - 2017 SET 2 MCQ (1-MARKS)

Q26. Which of the following is/are shared by all the threads in a process?
I. Program counter
II. Stack
III. Address space
IV. Registers
(a) (I) and (II) only
(b) (III) only
(c) (IV) only
(d) (III) and (IV) only

GATE CSE - 2017 NAT SET - 2(2-MARKS)

Q27. Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time.

Proces	Arrival Time	Burst Time	Priority
P_1	0	11	2
P_2	5	28	0
P_3	12	2	3
P_4	2	10	1
P_5	9	16	4

The average waiting time (in milliseconds) of all the processes using the pre-emptive priority scheduling algorithm is _____.

GATE CSE - 2016 MCQ SET - 1(1-MARKS)

Q28. Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?
(a) Shortest remaining time first
(b) Round-robin with time quantum less than the shortest CPU burst
(c) Uniform random
(d) Highest priority first with priority proportional to CPU burst length

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE - 2016 NAT SET- 2 (2-MARKS)

Q29. Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is the pre-emptive shortest remaining-time first.

Process	Arrival Time	Burst Time
P_1	0	10
P_2	3	6
P_3	7	1
P_4	8	3

The average turn-around time of these processes is _____ milliseconds

GATE CSE - 2015 MCQ SET-3 (1-MARKS)

Q30. The maximum number of processes that can be in Ready state for a computer system with n CPUs is

(a) n
(b) n^2
(c) 2^N
(d) Independent of n

GATE CSE - 2015 MCQ SET-3 (1-MARKS)

Q31. Which of the following is FALSE about SJF (Shortest Job First Scheduling)?
S1: It causes minimum average waiting time
S2: It can cause starvation

(a) Only S1
(b) Only S2
(c) Both S1 and S2
(d) Neither S1 or S2

GATE CSE - 2015 MCQ SET - 3 (2-MARKS)

Q32. For the processes listed in the following table, which of the following scheduling schemes will give the lowest average turnaround time?

Proces	Arrival Time	Processing Time
A	0	3
B	1	6
C	4	4
D	6	2

(a) First Come First Serve
(b) Non-pre-emptive Shortest Job First
(c) Shortest Remaining Time First
(d) Round Robin with Quantum value two

GATE CSE - 2015 NAT SET - 1(2-MARKS)

Q33. Consider a uniprocessor system executing three tasks T1, T2 and T3, each of which is composed of an infinite sequence of jobs (or instances) which arrive periodically at intervals of 3, 7 and 20 milliseconds, respectively. The priority of each task is the inverse of its period and the available tasks are scheduled in order of priority, with the highest priority task scheduled first. Each instance of T1, T2 and T3 requires an execution time of 1, 2 and 4 milliseconds, respectively. Given that all tasks initially arrive at the beginning of the 1st millisecond and task preemptions are allowed, the first instance of T3 completes its execution at the end of _____ milliseconds.

GATE CSE - 2014 MCQ SET – 1 (1-MARKS)

Q34. Which one of the following is FALSE?

(a) User level threads are not scheduled by the kernel.
(b) When a user level thread is blocked, all other threads of its process are blocked.
(c) Context switching between user level threads is faster than context switching between kernel level threads
(d) Kernel level threads cannot share the code segment.

GATE CSE – 2014 NAT SET - 2(2-MARKS)

Q35. Three processes A, B and C each execute a loop of 100 iterations. In each iteration of the loop, a process performs a single computation that requires t_c CPU milliseconds and then initiates a single I/O operation that lasts for t_{io} milliseconds. It is assumed that the computer where the processes execute has sufficient number of I/O devices and the OS of the computer assigns different I/O devices to each process. Also, the scheduling overhead of the OS is negligible. The processes have the following characteristics:

Process id	TC	Tio
A	100 ms	500 ms
B	350 ms	500 ms
C	200 ms	500 ms

The processes A, B, and C are started at times 0, 5 and 10 milliseconds respectively, in a pure time sharing system (round robin scheduling) that uses a time slice of 50 milliseconds. The time in milliseconds at which process C would complete its first I/O operation is _____.

GATE CSE – 2014 NAT SET - 3 (2-MARKS)

Q36. An operating system uses the shortest remaining time first scheduling algorithm for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times and CPU burst times (in milliseconds):

Process	Arrival Time	Burst Time
P1	0	12
P2	2	4
P3	3	6
P4	8	5

The average waiting time (in milliseconds) of the processes is _____.

GATE CSE - 2014 NAT SET-1 (2-MARKS)

Q37. Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds.

Process Name	Arrival Time	Execution Time
A	0	6
B	3	2
C	5	4
D	7	6
E	10	3

Using the shortest remaining time first scheduling algorithm, the average process turnaround time (in msec) is

GATE CSE - 2013 MCQ (1-MARKS)

Q38. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time unit and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

(a) This algorithm is equivalent to the first-come-first-serve algorithm.
(b) This scheduling algorithm behaves like Round-Robin scheduling.
(c) This algorithm is equivalent to the shortest-job-first algorithm.
(d) This algorithm is equivalent to the shortest-remaining-time-first algorithm.

GATE CSE - 2012 MCQ (1-MARKS)

Q39. A process executes the code

```
fork ();
fork ();
fork ();
```

The total number of child processes created is

(a) 3
(b) 4
(c) 7
(d) 8

GATE CSE - 2012 MCQ (2-MARKS)

Q40. Consider the 3 processes, P1, P2 and P3 shown in the table.

Process	Arrival Time	Time Units Required Time
P1	0	5
P2	1	7
P3	3	4

The completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units) are

(a) FCFS: P1, P2, P3 RR2: P1, P2, P3
(b) FCFS: P1, P3, P2 RR2: P1, P3, P2
(c) FCFS: P1, P2, P3 RR2: P1, P3, P2
(d) FCFS: P1, P3, P2 RR2: P1, P2, P3

GATE CSE – 2011 MCQ (1-MARKS)

Q41. Let the time taken to switch between user and kernel modes of execution be t_1 while the time taken to switch between two processes be t_2. Which of the following is TRUE?

(a) $(t_1) > (t_2)$
(b) $(t_1) = (t_2)$
(c) $(t_1) < (t_2)$
(d) Nothing can be said about the relation between t_1 and t_2

GATE CSE - 2011 MCQ (1-MARKS)

Q42. A thread is usually defined as a "light weight process" because an operating system (OS) maintains smaller data structures for a thread than for a process. In relation to this, which of the following is TRUE?

(a) On per-thread basis, the OS maintains only CPU register state.
(b) The OS does not maintain a separate stack for each thread.
(c) On a per-thread basis, the OS does not maintain virtual memory state.
(d) On a per thread basis, the OS maintains only scheduling and accounting information.

GATE CSE - 2011 MCQ (2-MARKS)

Q43. Consider the following table of arrival time and burst time for three processes P0, P1 and P2.

Process	Arrival Time	Burst Time
P0	0 ms	9 ms
P1	1 ms	4 ms
P2	2 ms	9 ms

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?

(a) 5
(b) 4.33
(c) 6.33
(d) 7.33

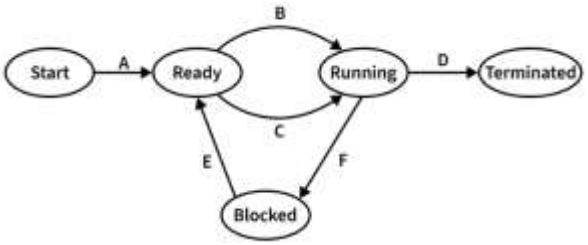
GATE CSE - 2010 MCQ (1-MARKS)

Q44. Which of the following statements is/are true?
I. Shortest remaining time first scheduling may cause starvation
II. Preemptive scheduling may cause starvation
III. Round robin is better than FCFS in terms of response time

(a) I only
(b) I and II only
(c) II and III only
(d) I, II and III

GATE CSE - 2009 MCQ (2-MARKS)

Q45. In the following process state transition diagram for a uniprocessor system, assume that there are always some processes in the ready state :



Now consider the following statements :
I. If a process makes a transition D, it would result in another process making transition A immediately.
II. A process P2 in a blocked state can make transition E while another process P1 is in running state.
III. The OS uses preemptive scheduling.
IV. The OS uses non-preemptive scheduling.
Which of the above statements are TRUE ?

(a) I and II
(b) I and III
(c) II and III
(d) II and IV

GATE CSE - 2008 MCQ (2-MARKS)

Q46. A process executes the following code:

```
for(i=0;i<n;i++)
   fork();
```

The total number of child processes created is

(a) n
(b) 2^{n-1}
(c) 2^n
(d) $2^{n+1} - 1$

GATE CSE - 2007 MCQ (2-MARKS)

Q47. An operating system uses the Shortest Remaining Time First (SRT) process scheduling algorithm. Consider the arrival times and execution times for the following processes:

Process	Execution Time	Arrival Time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

What is the total waiting time for process P2?

(a) 5
(b) 15
(c) 40
(d) 55

GATE CSE - 2007 MCQ (1-MARKS)

Q48. Consider the following statements about user level threads and kernel level threads. Which one of the following statements is FALSE?

(a) Context switch time is longer for kernel level threads than for user level threads.
(b) User level threads do not need any hardware support.
(c) Related kernel level threads can be scheduled on different processors in a multi-processor system.
(d) Blocking one kernel level thread blocks all related threads.

GATE CSE - 2007 MCQ (1-MARKS)

Q49. Group 1 contains some CPU scheduling algorithms and Group 2 contains some applications. Match entries in Group 1 to entries in Group 2.

Group 1

P. Gang Scheduling

Q. Rate Monotonic Scheduling

R. Fair Share Scheduling

Group 2

1. Guaranteed Scheduling
2. Real-time Scheduling
3. Thread Scheduling

(a) P-3; Q-2; R-1

(b) P-1; Q-2; R-3

(c) P-2; Q-3; R-1

(d) P-1; Q-3; R-2

GATE CSE - 2006 MCQ (1-MARKS)

Q50. Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a Shortest Remaining Time First (SRTF) scheduling algorithm?

Do not count the context switches at time zero and at the end.

(a) 1

(b) 2

(c) 3

(d) 4

GATE CSE - 2006 MCQ (2-MARKS)

Q51. Consider three processes (process id 0, 1, 2, respectively) with compute time bursts 2, 4, and 8 time units. All processes arrive at time zero. Consider the Longest Remaining Time First (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turnaround time is

(a) 13 units

(b) 14 units

(c) 15 units

(d) 16 units

GATE CSE - 2006 MCQ (2-MARKS)

Q52. Consider three processes, all arriving at time zero, with total execution time of 10, 20, and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again.

The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

(a) 0%

(b) 10.6%

(c) 30.0%

(d) 89.4%

GATE CSE - 2005 MCQ (2-MARKS)

Q53. Consider the following code fragment:

```
if (fork() == 0)
{
        a = a + 5;
        printf("%d, %p\n", a, &a);
}
else
{
        a = a - 5;
        printf("%d, %p\n", a, &a);
}
```

Let u, v be the values printed by the parent process and x, y be the values printed by the child process. Which one of the following is TRUE?

(a) u = x + 10 and v = y

(b) u = x + 10 and v ≠ y

(c) u + 10 = x and v = y

(d) u + 10 = x and v ≠ y

Link to GfG corresponding Short Notes: GATE FARRAY

Answer key

1. (b)	2. (9.5)	3. (d)	4. (a)
5. (a,b)	6. (14)	7. (d)	8. (a, d)
9. (b)	10. (a)	11. (c,d)	12. (b, c, d)
13. (c, d)	14. (d)	15.(a,c)	16. (65)
17. (a, b, c)	18. (c, d)	19. (12)	20. (5.25)
21. (c)	22. (2)	23. (31)	24. (3)
25. (d)	26. (b)	27. (29)	28. (a)
29. (8.25)	30. (d)	31. (d)	32. (c)
33. (12)	34. (d)	35. (1000)	36. (5.5)
37. (7.2)	38. (b)	39. (c)	40. (c)
41. (c)	42. (c)	43. (a)	44. (d)
45. (c)	46. (b)	47. (b)	48. (d)
49. (a)	50. (b)	51. (a)	52. (b)
53. (c)			

HINTS & SOLUTIONS

1. ANSWER:- B

FCFS cannot be pre-emptive

2. ANSWER:- 9.5

Process	Arrival	Start	Finish	Waiting
A1	10	10	16	0
C1	11	16	24	5
A2	20	24	30	4
C2	22	30	38	8
A3	30	38	44	8
C3	33	44	52	11
A4	40	52	58	12
C4	44	58	66	14
A5	50	66	72	16
C5	55	72	80	17

Total waiting time

0 + 5 + 4 + 8 + 8 + 11 + 12 + 14 + 16 + 17 = 95

Average waiting time

95/10 = 9.5 seconds

3. ANSWER:- D

Process	Arrival Time (AT)	Burst Time (BT)	Completion Time (CT)	Turnaround Time (TAT = CT - AT)
P1	0 ms	10 ms	16 ms	16 - 0 = 16 ms
P2	1 ms	13 ms	38 ms	38 - 1 = 37 ms
P3	2 ms	6 ms	8 ms	8 - 2 = 6 ms
P4	8 ms	9 ms	25 ms	25 - 8 = 17 ms

Calculate Average Turnaround Time

TAT=CompletionTime(CT)−ArrivalTime(AT)

Average TAT=∑TAT / Total Processes= (16+37+6+17) / 4=76 / 4=19 ms

Final Answer:

The average turnaround time is 19 milliseconds.

Link to GfG corresponding Short Notes: GATE FARRAY

4. ANSWER:- A

To calculate the average waiting time, we need to simulate the scheduling process step by step based on the given priorities and CPU bursts for the processes on processors M1 and M2.

Given Data:

- **Processor M1 Priority Order**: P1 > P3 > P2 > P4
- **Processor M2 Priority Order**: P2 > P3 > P4 > P1
- **CPU Bursts**:
  - P1: 20 ms
  - P2: 16 ms
  - P3: 25 ms
  - P4: 10 ms

Process Scheduling on M1 and M2:

1. Initial state: Both processors (M1 and M2) are free.
2. Time 0: The scheduler starts with the highest priority processes based on each processor's priority list.

- M1: The highest priority process is P1 (with a burst of 20 ms), so P1 is allocated to M1.
- M2: The highest priority process is P2 (with a burst of 16 ms), so P2 is allocated to M2.

1. After P1 finishes (at time 20 ms):

- M1: The next highest priority process is P3 (with a burst of 25 ms), so P3 is allocated to M1.
  - M2: P2 finishes at 16 ms, and the next highest priority process is P3, but since M1 has already taken P3, M2 moves to the next highest priority, which is P4 (with a burst of 10 ms).

2. After P2 finishes (at time 16 ms):

- M1: P3 continues execution (since P3 is already allocated to M1).
- M2: P4 finishes at time 26 ms, and now the remaining process is P3.

Calculate Waiting Times:

- Waiting time is calculated as: Waiting Time = Start Time - Arrival Time.

For Process P1:

- P1 starts at time 0 and runs for 20 ms, so its waiting time is 0 ms.

For Process P2:

- P2 starts at time 0 on M2 and runs for 16 ms. Its waiting time is 0 ms as it started at time 0.

For Process P3:

- P3 starts at time 20 on M1 and runs for 25 ms. Its waiting time is 20 ms (since it waited for P1 to finish on M1).

For Process P4:

- P4 starts at time 16 on M2 and runs for 10 ms. Its waiting time is 16 ms (since it waited for P2 to finish).

Average Waiting Time:

Now, we calculate the average waiting time:

- P1: Waiting time = 0 ms
- P2: Waiting time = 0 ms
- P3: Waiting time = 20 ms
- P4: Waiting time = 16 ms

Average waiting time = (0 + 0 + 20 + 16) / 4 = 36 / 4 = 9.00 ms

Final Answer:

The average waiting time of the processes is 9.00 ms.

5. ANSWER:- A,B

Option A: Blocking system call

When P makes a blocking I/O call, it cannot continue execution until data arrives.

So OS blocks P → context switch happens.

Always causes context switch

Option B: Page fault (page in swap)

If the required page is in disk (swap), P must wait for disk I/O.

So OS blocks P → context switch happens.

Always causes context switch

Option C: Disk interrupt for another process

Interrupt occurs, CPU goes to ISR, but after handling, P can resume execution.

No guarantee that P will be removed from the CPU.

Not always

Option D: Timer interrupt

Timer interrupt invokes scheduler, but:

- If no higher priority process → P continues
- If same process selected → no switch

Not always

6. ANSWER:- 14

Each process has its own copy of x. Both parent and child continues the loop. wait only makes a process wait for its own child, it does not stop creation of new processes.

Iteration-wise processes:

Start: 1 process

After 1st iteration: 2 processes → prints = 2

After 2nd iteration: 4 processes → prints = 4

After 3rd iteration: 8 processes → prints = 8

Total prints = 2 + 4 + 8 = 14

7. ANSWER:- D

Threads can be implemented in user space as well, so

(a) is false.

File descriptor table is shared among threads of a process, so

(b) is false.

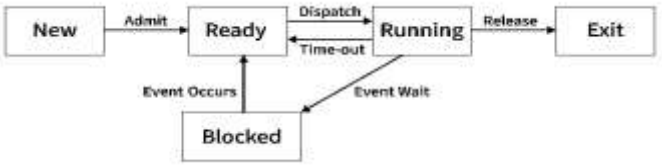
Each thread has its own stack, so (c) is false.

Threads share address space and are not protected from each other, so (d) is true.

Answer: (d)

8. ANSWER:- A, D

The Ready state means a process is prepared to run but is waiting for the CPU to become available while the Waiting state (or Blocked) means the process is waiting for something to happen(like an I/O operation to finish). A process can't go directly from Ready to Waiting, it usually moves from Running to Waiting as we can see from the given process lifecycle.

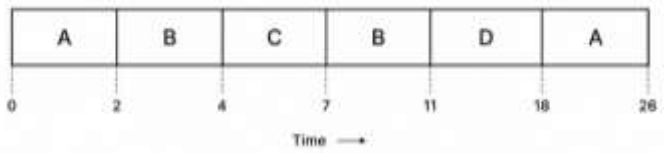


Five-state Process Lifecycle

9. ANSWER:- B

SRTF CPU Scheduling:

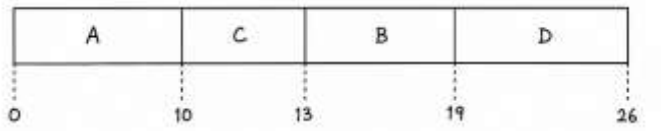
Proces	Arrival Time	Burst Time	Completion Time	Aroun Time	Waiting Time
A	0	10	26	26	16
B	2	6	11	9	3
C	4	3	7	3	0
D	6	7	18	12	5



Total Average Waiting Time [SRTF] = (16 + 3 + 0 + 5)/4 = 24/4 = 6.

Non-Preemptive CPU Scheduling:

Process	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
A	0	10	10	10	0
B	2	6	19	17	11
C	4	3	13	9	6
D	6	7	26	20	13



Total Average Waiting Time [SJF] = (0 + 11 + 6 + 13)/4 = 7.5

10. ANSWER:- A

The threads T1 and T2 update the shared variables a and b atomically, and their order of execution determines the final values.

The notation (x, y) represents the values of the variables a = x and b = y.

When T_1 is executed serially followed by T2, the result is (4, 4).

When T2 is executed serially followed by T_1, the

result is (3, 3).
Executing in the order T21 T11 T12 T22 produces the result (4,3).
The results (3, 4), (2, 3), and (2, 2) cannot be achieved.

11. ANSWER:- C, D

Function call: Function calls can be user-defined, so they do not necessarily guarantee a switch from user mode to kernel mode.
Malloc () call: This can be executed in both user and kernel modes, so it does not guarantee a switch between modes.
Page fault: When a page fault occurs, the required page must be loaded from secondary to main memory, which happens in privileged mode, ensuring a switch from user mode to kernel mode.
System call: System calls are executed exclusively in kernel mode, guaranteeing a switch from user mode to kernel mode.

12. ANSWER:- B, C, D

Page table base register: This register stores the page table's base address for the currently executing thread. Because all threads in a process share the same page table, this register does not need to be saved.
Stack pointer: This register points to the top of the stack for the thread that is currently running. Because each thread has its own stack, the stack pointer must be saved and restored when the context changes to ensure that the new thread uses the correct stack.
PC: The address of the next instruction to be executed by the currently executing thread is stored in the program counter register. To restart the new thread, the saved program counter must be loaded into the program counter register.
General purpose registers: These registers are used to store temporary data and intermediate results during the execution of a thread.

13. ANSWER:- C, D

Shortest Job First and Priority Scheduling are vulnerable to starvation.

1. **Shortest Job First (SJF):** New shorter jobs may continue to emerge.
2. **Priority Scheduling:** High-level positions may continue to emerge.
3. **Round Robin** will never result in starvation because each job will be assigned a time slot after a FIXED time quantum. Because the time quantum is finite, every job will eventually get a chance to run after a finite amount of time. It should be noted that the time quantum can be set to a large value, such as 10 minutes, 10 years, or even 100 years, but as long as it is finite, every job will get a chance to execute.
4. **First-in First-out:** If a task never yields (e.g., goes into infinite loop), then other tasks don't get to run (ex. While(1);). But there are other standard resources too which say NO STARVATION.

14. ANSWER:- D

Key idea: count transitions
Given: $S \rightarrow Q = 1$, $R \rightarrow Q = 1$, $Q \rightarrow R = 2$, $S \rightarrow P = 0$
Check option (a): feasible sequence can be formed satisfying all counts
Check option (b): feasible
Check option (c): feasible
Check option (d): leads to extra $R \rightarrow Q$ or $Q \rightarrow R$ transitions due to equal lengths of Q and R and quantum slicing, violating given counts

15. ANSWER:- A,C

exit eventually invokes a system call to terminate the process
malloc may or may not invoke system call
sleep always invokes a system call
strlen does not

16. ANSWER:- 65

These two are assigned two jobs. Now out of the remaining 4 jobs we need to ensure C2 gets at least 1. Without this constraint we can assign 4 jobs to 3 computers in 3^4=81 ways. Out of these 81 ways 2^4=16 will be having no jobs for C2. So, number of possible ways so that C2 gets at least one job =81−16=65.

17. ANSWER:- A, B, C

A, B, C
A is the correct option.
Turnaround time = burst time + waiting
Option B is also correct.

Option C is also correct as hardware support is needed by pre-emptive scheduling.
Option D is incorrect as we have to maximize the throughput also.

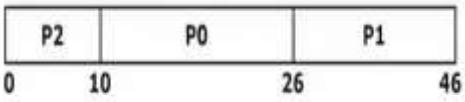
18. ANSWER:- C, D

option C: True
Threads do not share the stack. y is an AUTO variable, stored in a stack. So, each process generates two threads, each locks and unlocks L1 as they increment x and y. By each thread of a process Y is incremented from 0 to 1.
If P1 is completely executed and then P2 completely executes without any preemption, then, x is twice incremented (once by each of the threads). Threads share data segments. x is the global variable stored in the data segment.
option D: Correct
If in any order P1 and P2 executes, x will be 2 printed by each of the processes.

19. ANSWER:- 12

	AT	BT	WT
P0	0	16	10
P1	0	20	26
P2	0	10	0

Gantt char :



Avg WT = {WT(P0) + WT(P1) + WT(P2)}/3
= {10 + 26 + 0}/3 = 36/3 = 12 ms

20. ANSWER:- 5.25

SJF order: P3, P4, P2, P1

Completion times: 2, 6, 13, 21
TAT: 2, 6, 13, 21
Average = 10.5
RR execution:

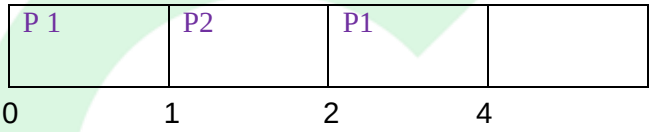
P1(4) P2(4) P3(2) P4(4) P1(4) P2(3)
Completion times:
P3 = 10, P4 = 14, P1 = 18, P2 = 21
TAT: 18, 21, 10, 14
Average = 15.75
Difference = 5.25

21. ANSWER:- C

I: possible (preemption)
II: possible (scheduler)
III: not direct (must go blocked → ready → running)
IV: possible (I/O completion)

22. ANSWER:- 2

SRTF schedule:
The Gant Chart till t=4 will be



Now, there are two possible cases:
Case 1 (z<4):
Then average waiting time = $(1 + 0 + (1 + x) + 0)/4 = 1$

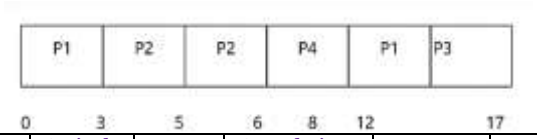
$$(2 + x)/4 = 1 \Rightarrow x = 2.$$

Case 2($z > 4$) :
Then average waiting time = $(1 + 0 + 1 + 3)/4 = 1$
$5/4 \neq 1 \Rightarrow$ This case is invalid.
Correct Answer: 2

23. ANSWER:- 31

Fork happens when i = 0, 2, 4, 6, 8 → total 5 times
Each time processes double
Total processes = $2^5 = 32$
Total child processes = $32 - 1 = 31$

24. ANSWER:- 3



Process	Arrival Time (AT)	Burst Time (BT)	Completion Time (CT)	Turn Around Time (TAT) IAT = CT – AT	Waiting Time (WT) WT = TAT – BT
P1	0	7	12	12	5
P2	3	3	6	3	0
P3	5	5	17	12	7
P4	6	2	8	2	0

Calculation:

Average waiting time $= \frac{(5+0+7+0)}{4} = 3$ milliseconds

25. ANSWER:- D

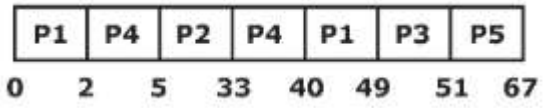
Threads share same address space → both heap and global variables are shared

26. ANSWER:- B

Each thread has its own program counter, stack, and registers. Address space is shared.

27. ANSWER:-

Gantt chart for the processes is as follows:



Process	Arrival Time	Burst Time	Priority	Completion time (CT)	Turnaround time (TAT) TAT = CT – AT	Waiting time (WT) WT = TAT-BT
P_1	0	11	2	49	49	38
P_2	5	28	0	33	28	0
P_3	12	2	3	51	39	37
P_4	2	10	1	40	38	28
P_5	9	16	4	67	58	42

Average waiting time $= \frac{\sum \text{waiting time of all the processes}}{\text{number of processes}}$

$= \frac{38+0+37+28+42}{5} = 29$ milliseconds

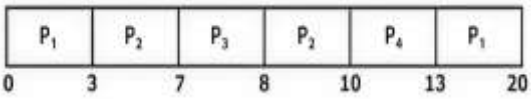
28. ANSWER:- A

To minimize the alternate waiting time, we need to select the shortest remaining time process first, the reason is because all the processes are arriving at the same time and they have unequal CPU burst times. So, the answer is the SRTF algorithm as all other options will not minimize the waiting time.

29. ANSWER:- 8.25

Here the scheduling algorithm used is shortest remaining-time first (pre-emptive version of SJF).

Gantt chart:



In this algorithm, the processes will schedule on the CPU those who have the least remaining burst time.

Turnaround Time (TAT) = Completion Time (CT) - Arrival Time (AT)

TAT for P_1 = 20 - 0 = 20,

TAT for P_2 = 10 - 3 = 7,

TAT for P_3 = 8 - 7 = 1,

TAT for P_4 = 13 - 8 = 5.

Total TAT = 20 + 7 + 1 + 5 = 33

Avg. TAT = 33/ 4 = 8.25

30. ANSWER:- D

The number of processes which are in the running state will be at most n for n processors.

In the ready state, the maximum number of processes will be completely independent of the number of processors. As a single processor may have a large number of processes waiting in the ready queue.

31. ANSWER:- D

1. Shortest job first is optimal because of average waiting time for a given set of processes, but problems with SJF is how to know/predict the time of the next job.

2. For Shortest job first and SRJF algorithms, they may cause starvation. When a process with large burst time is there in the ready queue and processes with small burst time keep coming.

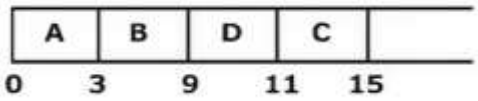
32. ANSWER:- C

Processes can be tabulated as:

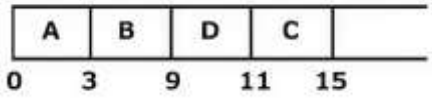
Process	Arrival Time	Processing Time
A	0	3
B	1	6
C	4	4

D	6	2

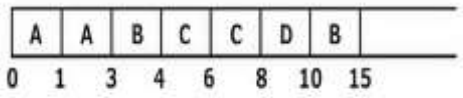
FCFS



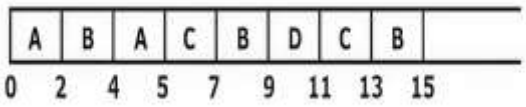
TAT for A = completion time(A) - AT(A) = 3 - 0 = 3
TAT of B = 9 - 1 = 8
TAT of C = 13 - 4 = 9
TAT of D = 15 - 6 = 9
∴ Avg. TAT = 29/4
SJF:



TAT of A3-03
TAT of B = 9 - 1 = 8 TAT of C = 15 - 4 = 11 TAT of D = 11 - 6 = 5: Avg. TAT = 27/4
SRTF:



TAT of A3-03
TAT of B = 15 - 1 = 14
TAT of C = 8 - 4 = 4 TAT of D = 10 - 6 = 4
: Avg. TAT = 25/4
RR:



TAT of A = 5-0=5 TAT of B = 15 - 1 = 14 TAT of C = 13 - 4 = 9 TAT of D = 11 - 6 = 5 .: Avg. TAT = 33/4 .. SRTF has the lowest Avg. TAT.

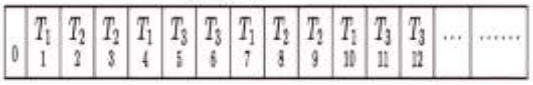
33. ANSWER:- 12

T_1, T_2 and T_3 have infinite instances, meaning infinite burst times. Here, the problem is to run "T_1 for 1 ms", "T_2 for 2 ms", and "T_3 for 4 ms". i.e., every task is run in parts. Now for timing purpose we consider t for the end of cycle number t.

- T_1: 0,3, 6, 9, 12,...∞ (T_1 repeats every 3 ms)
- T_2: 0, 7, 14, 21,... ∞ (T_2 repeats every 7 ms)
- T3: 0,20, 40, 60,... ∞ (T_3 repeats every 20 ms)

1. Priority of $T_1 = \frac{1}{3}$
2. Priority of $T_2 = \frac{1}{7}$
3. Priority of $T_3 = \frac{1}{20}$

Gantt Chart



At t = 0 No process is available
At t 2, T2 runs because it has higher priority than T_3 and no instance of T_1 present
At t = 4, We have T_1 arrive again and T_3 waiting but T_1 runs because it has higher priority
At t = 5, T_3 runs because no instance of T_1 or T_2 is present
At t = 11 T_3 runs because no instance of T_1 or T_2 is present
At t = 12, T_3 continue run because no instance of T_1 or T_2 is present and first instance of T_3 completes

34. ANSWER:- D

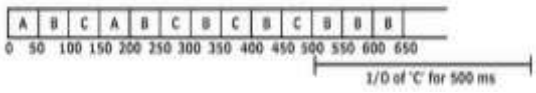
Code segments can be shared by the threads. They have only separate stacks and registers.
User level threads are scheduled by the thread library and the kernel knows nothing about it. So, A is TRUE.
When a user level thread is blocked, all other threads of its process are blocked. So, B is TRUE. (With a multi-threaded kernel, user level threads can make non-blocking system calls without getting blocked. But in this option, it is explicitly said 'a thread is blocked'.)
Context switching between user level threads is faster as they actually have no context-switch- nothing is saved and restored while for kernel level thread, Registers, PC and SP must be saved and restored. So, C is also TRUE.
Kernel level threads cannot share code segments, false. Kernel level threads within the same process share data section, code section and other operating system resources such as open files and signals.

35. ANSWER:- 1000

Processes A, B and C which continue in round robin fashion and time quantum is 50 ms.
Processes execute at 0, 5 and 10 milliseconds.
Order of the processes are:

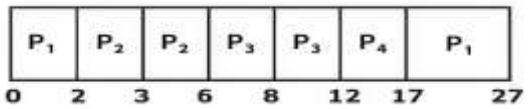
A, B, C, A
50 + 50 + 50 + 50 (200 ms passed)
Now, A has completed computations of 100 ms and goes for I/O now.
B, C, B, C, B, C
50 + 50 + 50 + 50 + 50 + 50 (300 ms passed)
C goes for I/O at 500ms and it needs 500ms to finish the I/O.
So, C would complete its first I/O at 1000

Gantt chart is shown below:



36. ANSWER:- 5.5

Gantt Chart

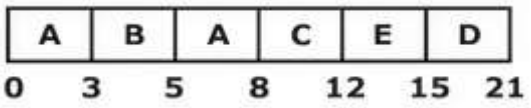


Gantt Chart

Process	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
P_1	0	12	27	27	15
P_2	2	4	6	4	0
P_3	3	6	12	9	3
P_4	8	5	17	9	4

Average waiting time = (15 + 0 + 3 + 4)/ 4 = 5.5 ms

37. ANSWER:- 7.2



Using SRTF

TAT(A) = 8-0 = 8,
TAT(B)= 5-3=2,
TAT(C)= 12-5=7,
TAT(D)= 21-7= 14,
TAT(E)=15-10=5
Average turnaround time can be calculated as = 8+2+7+14+5/ 5 = 7.2ms

38. ANSWER:- B

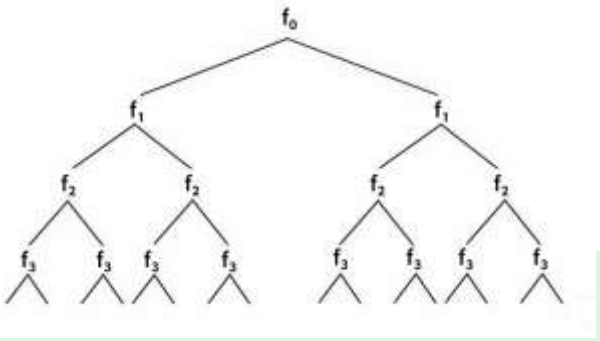
Every process starts with priority zero.

- Priority is proportional to waiting time.
- Since all processes arrive at time zero and have no I/O, waiting time increases equally for all.
- The scheduler re-evaluates every T time units and picks the process with the highest priority (longest waiting time).
- This results in cyclic allocation of CPU with a time quantum of T, which is exactly how Round-Robin scheduling works.

Thus, the algorithm is equivalent to Round-Robin scheduling in this scenario.

39. ANSWER:- C

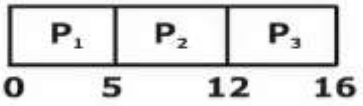
With N fork statements, there are always 2^n-1 child processes.
Here in this operation, the process creates a copy of itself.



1,3,7,15, 31.... $\Rightarrow 2^n$-1
So, 2^3-1 = 7.

40. ANSWER:- C

FCFS is - P1, P2, P3



RR Queue: In RR queue, 2 units is the time slot.
Ready queue concept is used. At t=2, P2 starts and P1 sends to the ready queue and at t=3 P3 arrives. After P1, Job P3 is queued in the ready queue. So, at time t=4, again P1 is executed then P3 is executed for the first time at t=6.
RR2: P1, P3, P2
So, option C is the correct answer.

41. ANSWER:- C

Context switch between the processes involves mode switch also. Context switches or Process switches can only occur in kernel mode. So, for process switches

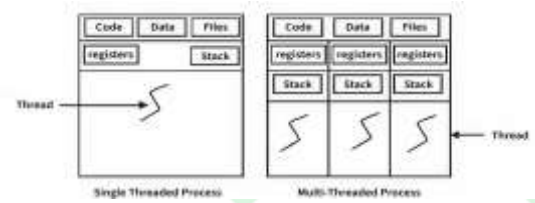
firstly we have to move from user to kernel mode. Secondly, we have to save the PCB of the process from which we are taking off the CPU and finally we have to load the PCB of the required process. Switching from kernel to user mode is done. But switching from user to kernel mode is a very fast operation (OS has to just change a single bit at hardware level).

When we switch between kernel and user mode of execution, OS has to just change a single bit at hardware level which is very fast operation.

Thus, $t_1 < t_2$

42. ANSWER:- C

Threads share the address space of processes. The virtual memory is concerned with processes and not with threads.



Option (A): As we can see in the above diagram, not only the CPU register but also stack and code files, data files are also maintained. So, option (A) is incorrect as the statement says OS maintains only CPU register state.

Option (B): OS does not maintain a separate stack for each thread. But as we can see in the above diagram, for each thread, a separate stack is maintained. So, this option is also incorrect.

Option (C): The virtual memory state is not maintained by the OS and this statement is correct as the OS does not maintain any virtual memory state for individual threads.

Option (D): According to the option (D), the OS maintains only scheduling and accounting information. But it is incorrect as it contains other information like program counters, data files, CPU registers stack, code files are also maintained.

43. ANSWER:- A

SRTF is used, this scheduling gives CPU to those processes in which remaining CPU burst time is smallest

	AT	BT	CT	TAT	WT
P_0	0	9	13	13	4
P_1	1	4	5	4	0
P_2	2	9	22	20	11

CT = Completion time

BT = Burst Time

TAT = Turn Around Time

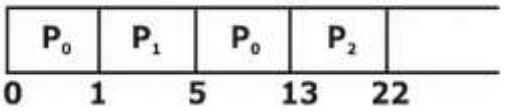
TAT = CT - AT

WT = Waiting Time

WT = TAT – BT

Using pre-emptive shortest job first scheduling algorithm,

Gantt Chart is -



Avg. WT = 4+0+11/3 = 5ms

44. ANSWER:- D

In the shortest remaining time first, processes with longer burst time suffer from starvation. As the shorter processes may keep coming and a long CPU burst process never gets CPU.

- Preemptive scheduling causes starvation (For example - lower priority jobs are waiting and higher priority processes are continuously coming).
- Best Response time is given by Round Robin scheduling as processors will be given to each process after a time.

45. ANSWER:- C

Transitions:

A: Start → Ready

B: Ready → Running

C: Running → Ready (preemption)

D: Running → Terminated

E: Blocked → Ready

F: Running → Blocked

I:

After D (Running → Terminated), the CPU becomes free. Since ready queue is never empty, scheduler selects a ready process (B occurs, not A). False

II:

One process can move Blocked → Ready (E) due to I/O completion while another process is running. True

III:

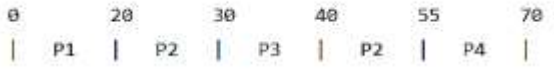
Transition C (Running → Ready) indicates preemption.

True

IV:
Non-preemptive scheduling would not have transition C.
False

46. ANSWER:- B
child processes = total processes – 1 = $2^n - 1$
Correct answer: (B).

47. ANSWER:- B
Gantt Chart:



You can clearly see P2 runs in two segments:

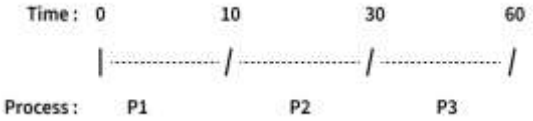
- 20–30
- 40–55

That gap (30–40) contributes to its waiting time.
And since its arrival time is 15
Hence total waiting time = 5 + 10 = 15

48. ANSWER:- D
(A) Context switch time is longer for kernel level threads than for user level threads.
True. Kernel threads require mode switch (user → kernel), scheduler invocation, etc., so overhead is higher.
(B) User level threads do not need any hardware support.
True. They are managed entirely in user space by thread libraries.
(C) Related kernel level threads can be scheduled on different processors in a multi-processor system.
True. Since the kernel manages them, they can run in parallel on multiple CPUs.
(D) Blocking one kernel level thread blocks all related threads.
False. This is true for user-level threads (many-to-one model), but not for kernel-level threads. If one kernel thread blocks, others can still run.

49. ANSWER:- A
P. Gang Scheduling → Thread Scheduling (3)
Gang scheduling is used for parallel programs where related threads are scheduled simultaneously on different processors. So it is used in thread scheduling.
Q. Rate Monotonic Scheduling → Real-time Scheduling (2)
Rate Monotonic Scheduling is a classic fixed-priority real-time scheduling algorithm.
R. Fair Share Scheduling → Guaranteed Scheduling (1)
Fair share scheduling ensures each user/process group gets a fair proportion of CPU time, aligning with guaranteed scheduling.

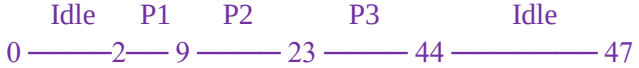
50. ANSWER:- B
Gantt chart:



Total no of context switch = 2

51. ANSWER:- A
Gantt chart:
0–4 : P2
4–5 : P1
5–6 : P2
6–7 : P1
7–8 : P2
8–9 : P0
9–10 : P1
10–11 : P2
11–12 : P0
12–13 : P1
13–14 : P2
Average Turnaround Time = ((12 – 0) + (13-0) + (14 – 0))/3 = 13

52. ANSWER:- B



Idle Intervals:
- From 0 to 2 → all processes are in initial I/O → CPU idle = 2 units
- From 44 to 47 → last process (P3) is in final I/O → CPU idle = 3 units

Total idle time = 2 + 3 = 5
Total Time:
- System completes when last I/O finishes → time = 47

Idle Percentage:
5/47×100=10.6383%≈10.6%

53. ANSWER: C

After fork, parent and child have separate copies of 'a' but same virtual address space layout.
Child: a = a + 5 → x
Parent: a = a − 5 → u
So x = u + 10
Addresses remain same → v = y

2 Process Synchronization and Concurrency

GATE CSE - 2026 MSQ SET - 2(2-MARKS)

Q1. Consider three processes P1, P2, and P3 running identical code, as shown in the pseudocode below. A and B are two binary semaphores initialized to 1 and 0, respectively. X is a shared variable initialized to 0. Each line in the pseudocode is executed atomically.

Pseudocode of P1, P2, and P3:

```
wait(A);
print(*);
X=X+1;

if(X==2)
{
   print($);
   signal(B);
}

signal(A);

wait(B);
print(#);
signal(B);
```

Assume that any of the three processes can start to execute first and context switching can happen between these processes at any arbitrary time and in any arbitrary order.

Which of the following patterns is/are possible to be generated as an outcome of the execution of these three processes?

(a) **$*###
(b) **$#*##
(c) **$##*#
(d) ***$###

GATE CSE - 2024 MSQ SET - 2(2-MARKS)

Q2. Consider a multi-threaded program with two threads T1 and T2. The threads share two semaphores: s1 (initialized to 1) and s2 (initialized to 0). The threads also share a global variable x (initialized to 0). The threads execute the code shown below.

```
T1:
wait(s1);
x = x + 1;
 print(x);
 wait(s2);
 signal(s1);
T2:
wait(s1);
 x = x + 1;
 print(x);
 signal(s2);
 signal(s1);
```

Which of the following outcomes is/are possible when threads T1 and T2 execute concurrently?

(a) T1 runs first and prints 1, T2 runs next and prints 2
(b) T2 runs first and prints 1, T1 runs next and prints 2
(c) T1 runs first and prints 1, T2 does not print anything (deadlock)
(d) T2 runs first and prints 1, T1 does not print anything (deadlock)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2023 MCQ SET - 1(2-MARKS)

Q3. Consider the two functions incr and decr shown below.

```
incr(){
        wait(s);
        X = X+1;
         signal(s);
}
decr(){
        wait(s);
        X = X-1;
         signal(s);
}
```

There are 5 threads each invoking incr once, and 3 threads each invoking decor once, on the same shared variable X. The initial value of X is 10.
Suppose there are two implementations of the semaphore s, as follows:
I-1: s is a binary semaphore initialized to 1.
I-2: s is a counting semaphore initialized to 2.
Let V1, and V2 be the values of X at the end of the execution of all the threads with implementations I-1, and I-2, respectively.
Which one of the following choices corresponds to the minimum possible values of V1, V2, respectively?

(a) 15,7
(b) 7,7
(c) 12,7
(d) 12,8

GATE CSE - 2022 MCQ (1-MARKS)

Q4. Consider the following threads T1, T2, and T3 executing on a single processor, synchronized using three binary semaphore variables S1, S2, and S3,operated upon using standard wait() and signal(). The threads can be context switched in any order and at any time.

```
T1:
 while(true) {
 wait(S3);
 print("C");
 signal(S2);
 }
T2:
 while(true) {
 wait(S1);
 print("B");
 signal(S3);
 }
T3:
 while(true) {
 wait(S2);
 print("A");
 signal(S1);
 }
```

Which initialization of the semaphores would print the sequence BCABCABCA...?

(a) S1 = 1; S2 = 1; S3 = 1
(b) S1 = 1; S2 = 1; S3 = 0
(c) S1 = 1; S2 = 0; S3 = 0
(d) S1 = 0; S2 = 1; S3 = 1

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2021 MSQ SET - 1(2-MARKS)

Q5. Consider the following pseudocode, where S is a semaphore initialized to 5 in line #2 and counter is a shared variable initialized to 0 in line#1. Assume that the increment operation in line#7 is not atomic.

```
1. int counter = 0
2. Semaphore S = init(5);
3. void parop(void)
4. {
5.          wait(S);
6.          wait(S);
7.          counter++;
8.          signal(S);
9.          signal(S);
10.}
```

If five threads execute the function parop concurrently, which of the following program behavior(s) is/are possible?

(a) The value of counter is 5 after all the threads successfully complete the execution of parop.
(b) The value of counter is 1 after all the threads successfully complete execution of parop.
(c) The value of counter is 0 after all the threads successfully complete the execution of parop.
(d) There is a deadlock involving all the threads.

GATE CSE - 2021 MSQ SET - 2(2-MARKS)

Q6. Consider a computer system with multiple shared resource types, with one instance per resource type. Each instance can be owned by only one process at a time. Owning and freeing of resources are done by holding a global lock (L). The following scheme is used to own a resource instance:

```
function OWNRESOURCE(Resource R)
    Acquire lock L // a global lock
    if R is available then
        Acquire R
        Release lock L
    else
        if R is owned by another process P then
            Terminate P, after releasing all resources owned by P
            Acquire R
        Restart P
        Release lock L
        end if
    end if
end if
end function
```

Which of the following choice(s) about the above scheme is/are correct?

(a) The scheme violates the mutual exclusion property.
(b) The scheme ensures that deadlocks will not occur.
(c) The scheme may lead to starvation.
(d) The scheme may lead to live-lock.

GATE CSE - 2020 MCQ (2-MARKS)

Q7. Each of a set of n processes executes the following code using two semaphores a and b initialized to 1 and 0, respectively. Assume that count is a shared variable initialized to 0 and not used in CODE SECTION P.

```
CODE SECTION P
wait (a);
 count = count+1;
if (count==n) signal (b);
signal (a); wait (b); signal (b);
CODE SECTION Q
```

What does the code achieve?

(a) It ensures that all processes execute CODE SECTION P mutually exclusively.
(b) It ensures that at most two processes are in CODE SECTION Q at any time.
(c) It ensures that no process executes CODE SECTION Q before every process has finished CODE SECTION P.
(d) It ensures that at most n-1 processes are in CODE SECTION P at any time.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2019 NAT (1-MARKS)

Q8. Consider three concurrent processes P1, P2 and P3 as shown below, which access a shared variable D initialized to 100.

P1: D = D + 20
P2: D = D − 50
P3: D = D + 10

The processes are executed on a uniprocessor system running a time-shared operating system. If the minimum and maximum possible values of D after the three processes have completed execution are X and Y respectively, then the value of Y − X is _____

GATE CSE - 2018 MCQ (2-MARKS)

Q9. Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is N. Three semaphores empty, full and mutex are defined with respective initial values of 0, N and 1. Semaphore empty denotes the number of available slots in the buffer, for the consumer to read from. Semaphore full denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by P, Q, R and S, in the code below can be assigned either empty or full. The valid semaphore operations are: wait () and signal ().

Producer	Consumer
do{	do {
wait (P);	wait (R);
wait (mutex);	wait mutex);
//Add item to buffer	//Consume item from buffer
Signal (mutex);	Signal (mutex);
Signal (Q);	Signal (S);
} while (1);	} while (1);

Which one of the following assignments to P, Q, R and S will yield the correct solution?

(a) P: full, Q: full, R: empty, S: empty
(b) P: empty, Q: empty, R: full, S: full
(c) P: full, Q: empty, R: empty, S: full
(d) P: empty, Q: full, R: full, S: empty

GATE CSE - 2016 MCQ SET - 1(2-MARKS)

Q10. Consider the following proposed solution for the critical section problem. There are n processes: $P_0...P_{n-1}$. In the code, function pmax returns an integer not smaller than any of its arguments. For all i, t[i] is initialized to zero.

Code for P_i:

```
do {
        c[i]=1; t[i] = pmax (t[0],...,t[n-1]) +1; c[i]=0;
        for every j ≠ i in {0,...,n-1} {
                while (c[j]);
                while (t[j]! = 0 && t[j]<=t[i]);
 }
        Critical Section;
        t[i]=0;
        Remainder Section;
} while (true);
```

Which one of the following is TRUE about the above solution?

(a) At most one process can be in the critical section at any time.
(b) The bounded wait condition is satisfied
(c) The progress condition is satisfied
(d) It cannot cause a deadlock.

GATE CSE - 2016 MCQ SET - 2(2-MARKS)

Q11. Consider the following two-process synchronization solution.

Process 0	Process 1
Entry: loop while (turn == 1);	Entry: loop while (turn == 0);
(critical section)	(Critical section)
Exit: turn = 1;	Exit: turn = 0;

The shared variable turn is initialized to zero. Which one of the following is TRUE?

(a) This is a correct two-process synchronization solution.
(b) This solution violates mutual exclusion requirement.
(c) This solution violates the progress requirement.
(d) This solution violates the bounded wait requirement.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2016 NAT SET - 2(2-MARKS)

Q12. Consider a non-negative counting semaphore S. The operation P(S) decrements S, and V(S) increments S. During an execution, 20 P(S) operations and 12 V(S) operations are issued in some order. The largest initial value of S for which at least one P(S) operation will remain blocked is _____.

GATE CSE - 2015 MCQ SET - 1(1-MARKS)

Q13. The following two functions P1 and P2 that share a variable B with an initial value of 2 execute concurrently.

```
P1()
{
  C = B – 1;
  B = 2*C;
}

P2()
{
  D = 2 * B;
 B = D - 1;
}
```

The number of distinct values that B can possibly take after the execution is?

(a) 3
(b) 4
(c) 5
(d) 6

GATE CSE - 2015 MCQ SET - 3(1-MARKS)

Q14. Two processes X and Y need to access a critical section. Consider the following synchronization construct used by both the processes.

Process X	Process Y

/* other code for process X */
while(true)
{
varP = true;
while(varQ == true)
{
/* Critical Section */
varP = false;
}
}
/* other code for process X */
	/* other code for process Y */
	while(true)
	{
	varQ = true;
	while(varP == true)
	{
	/* Critical Section */
	varQ = false;
	}
	}
	/* other code for process Y */

Here, varP and varQ are shared variables and both are initialized to false. Which one of the following statements is true?

(a) The proposed solution prevents deadlock but fails to guarantee mutual exclusion.
(b) The proposed solution guarantees mutual exclusion but fails to prevent deadlock.
(c) The proposed solution guarantees mutual exclusion and prevents deadlock.
(d) The proposed solution fails to prevent deadlock and fails to guarantee mutual exclusion.

GATE CSE - 2014 MCQ SET - 2(2-MARKS)

Q15. Consider the procedure below for the Producer-Consumer problem which uses semaphores:

```
semaphore n = 0;
semaphore s = 1;
void producer() void consumer()
{
    while(true) while(true)
    {
        produce();
        semWait(s) ;
        addToBuffer();
        semSignal(s);
        semSignal(n);
    }
}
```

```
void consumer()
{
    while(true)
    {
        semWait(s);
        semWait(n);
        removeFromBuffer();
        semSignal(s);
        consume();
    }
}
```

Check Out Our GATE Courses & Secure Your Dream Rank!

Which one of the following is TRUE?

(a) The producer will be able to add an item to the buffer, but the consumer can never consume it.
(b) The consumer will remove no more than one item from the buffer.
(c) Deadlock occurs if the consumer succeeds in acquiring semaphore s when the buffer is empty.
(d) The starting value for the semaphore n must be 1 and not 0 for deadlock-free operation.

GATE CSE - 2013 MCQ (2-MARKS)

Q16. A certain computation generates two arrays a and b such that a[i]=f(i) for $0 \leq i < n$ and b[i]=g(a[i]) for $0 \leq i < n$. Suppose this computation is decomposed into two concurrent processes X and Y such that X computes the array a and Y computes the array b. The processes employ two binary semaphores R and S, both initialized to zero. The array a is shared by the two processes. The structures of the processes are shown below.

Process X:	Process Y:
private i;	private i;
for $(i = 0; i < n;$ i + +){	for $(i = 0; i < n;$ i + +){
$a[i] = f(i);$	EntryY (R, S);
ExitX (R, S);	$b[i] = g(a[i]);$
}	}

Which one of the following represents the CORRECT implementations of Exit X and Entry Y?

(a)
```
ExitX(R, S) {
P(R);
V(S);
}
EntryY(R, S) {
P(S);
V(R);
}
```

(b)
```
ExitX(R, S) {
V(R);
V(S);
}
EntryY(R, S) {
P(R);
P(S);
}
```

(c)
```
ExitX(R, S) {
P(S);
V(R);
}
EntryY(R, S) {
V(S);
P(R);
}
```

(d)
```
ExitX(R, S) {
V(R);
P(S);
}
EntryY(R,S) {
V(S);
P(R);
}
```

GATE CSE - 2013 MCQ (2-MARKS)

Q17. A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e. wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to

Check Out Our GATE Courses & Secure Your Dream Rank!

memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?

(a) −2
(b) −1
(c) 1
(d) 2

GATE CSE - 2012 MCQ (2-MARKS)

Q18. Fetch_And_Add (X, i) is an atomic Read-Modify-Write instruction that reads the value of memory location X, increments it by the value i, and returns the old value of X. It is used in the pseudocode shown below to implement a busy-wait lock. L is an unsigned integer shared variable initialized to 0. The value of 0 corresponds to lock being available, while any non-zero value corresponds to the lock being not available.

```
AcquireLock(L) {
        while (Fetch_And_Add(L,1))
              L = 1;
        }
ReleaseLock(L) {
        L = 0;
}
```

This implementation.....

(a) fails as L can overflow
(b) fails as L can take on a non-zero value when the lock is actually available.
(c) works correctly but may starve some processes.
(d) works correctly without starvation

GATE CSE - 2010 MCQ (1-MARKS)

Q19. Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared Boolean variables S1 and S2 are randomly assigned.

Method Used by P1	Method Used by P2
while (S1 == S2);	while (S1! = S2);
Critical Section	Critical Section
S1 = S2;	S2 = not (S1);

Which one of the following statements describes the properties achieved?

(a) Mutual exclusion but not progress
(b) Progress but not mutual exclusion
(c) Neither mutual exclusion nor progress
(d) Both mutual exclusion and progress

GATE CSE - 2010 MCQ (2-MARKS)

Q20. The following program consists of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as S0 = 1, S1 = 0, S2=0.

Process P0	Process P1	Process P2
while (true) {	wait (S1);	wait (S2);
wait (S0);	Release (S0);	Release (S0);
print '0'		
release (S1);		
release (S2);		
}		

How many times will the process P0 print '0'?

(a) At least twice
(b) exactly twice
(c) exactly thrice
(d) exactly one

GATE CSE - 2009 MCQ (2-MARKS)

Q21. The enter_CS() and leave_CS() functions to implement critical section of a process are realized using test-and-set instruction as follows:

```
void enter_CS(X)
{
   while test-and-set(X) ;
}

void leave_CS(X)
{
```

Check Out Our GATE Courses & Secure Your Dream Rank!

```
    X = 0;
}
```

In the above solution, X is a memory location associated with the CS and is initialized to 0. Now consider the following statements:

I. The above solution to the CS problem is deadlock-free.

II. The solution is starvation free.

III. The processes enter CS in FIFO order.

IV. More than one process can enter CS at the same time.

Which of the above statements is TRUE?

(a) I only
(b) I and II
(c) II and III
(d) IV only

GATE CSE - 2008 MCQ (2-MARKS)

Q22. The P and V operations on counting semaphores, where s is a counting semaphore, are defined as follows:

```
P(s):
 s = s − 1;
 if s < 0 then wait;
V(s):
 s = s + 1;
 if s ≤ 0 then wake up process waiting on s;
```

Assume that Pb and Vb, the wait and signal operations on binary semaphores, are provided. Two binary semaphores xb and yb are used to implement the semaphore operations P(s) and V(s) as follows:

```
P(s):
 Pb(xb);
 s = s − 1;
 if (s < 0)
 {
    Vb(xb);
    Pb(yb);
 }
 else Vb(xb);
V(s):
 Pb(xb);
 s = s + 1;
 if (s ≤ 0) Vb(yb);
 Vb(xb);
```

The initial values of xb and yb are respectively

(a) 0 and 0
(b) 0 and 1
(c) 1 and 0
(d) 1 and 1

GATE CSE - 2007 MCQ (2-MARKS)

Q23. Two processes, P1 and P2, need to access a critical section of code. Consider the following synchronization construct used by the processes:

```
/* P1 */
while (true) {
   wants1 = true;
   while (wants2 == true);
   /* Critical Section */
   wants1 = false;
}
/* Remainder section */
/* P2 */
while (true) {
   wants2 = true;
   while (wants1 == true);
   /* Critical Section */
   wants2 = false;
}
/* Remainder section */
```

Here, wants1 and wants2 are shared variables, initialized to false.

Which one of the following statements is TRUE about the above construct?

(a) It does not ensure mutual exclusion.
(b) It does not ensure bounded waiting.
(c) It requires that processes enter the critical section in strict alternation.
(d) It does not prevent deadlocks, but ensures mutual exclusion.

GATE CSE - 2006 MCQ (2-MARKS)

Q24. The atomic fetch-and-set x, y instruction unconditionally sets the memory location x to 1 and fetches the old value of x in y without allowing any intervening access to the memory location x. Consider the following implementation of P and V functions on a binary semaphore S.

```
void P (binary_semaphore *s) {
        unsigned y;
```

Check Out Our GATE Courses & Secure Your Dream Rank!

```
        unsigned *x = &(s->value);
        do {
            fetch-and-set x, y;
        }   while (y);
}
void V (binary_semaphore *s) {
        S->value = 0;
}
```

Which one of the following is true?

(a) The implementation may not work if Context switching is disabled in P

(b) Instead of using fetch-and-set, a pair of normal load/store can be used

(c) The implementation of V is wrong

(d) The code does not implement a binary semaphore

GATE CSE - 2006 MCQ (2-MARKS)

Q25. Barrier is a synchronization construct where a set of processes synchronize globally. Let the number of processes be three and S be a binary semaphore. Consider the following implementation:

```
void barrier(void) {
1:  P(S);
2:  process_arrived++;
3:  V(S);
4:  while (process_arrived != 3);
5:  P(S);
6:  process_left++;
7:  if (process_left == 3) {
8:      process_arrived = 0;
9:      process_left = 0;
10: }
11: V(S);
}
```

Variables process_arrived and process_left are shared and initialized to 0.

Which one of the following is true?

(a) The barrier implementation is wrong due to the use of binary semaphore S

(b) The barrier implementation may lead to a deadlock if two barrier invocations are used in immediate succession

(c) Lines 6 to 10 need not be inside a critical section

(d) The barrier implementation is correct if there are only two processes instead of three

GATE CSE - 2006 MCQ (2-MARKS)

Q26. Barrier is a synchronization construct where a set of processes synchronize globally. Let the number of processes be three and S be a binary semaphore. Consider the following implementation:

```
void barrier(void) {
1:  P(S);
2:  process_arrived++;
3:  V(S);
4:  while (process_arrived != 3);
5:  P(S);
6:  process_left++;
7:  if (process_left == 3) {
8:  process_arrived = 0;
9: process_left = 0;
10: }
11: V(S);
}
```

The variables process_arrived and process_left are shared and initialized to zero.

Which one of the following rectifies the problem in the implementation?

(a) Lines 6 to 10 are simply replaced by process_arrived--

(b) At the beginning of the barrier the first process to enter waits until process_arrived becomes zero before proceeding to execute P(S)

(c) Context switch is disabled at the beginning of the barrier and re-enabled at the end

(d) The variable process_left is made private instead of shared

Check Out Our GATE Courses & Secure Your Dream Rank!

Answer key

1. (a, b, c)	**2. (b, c)**	**3. (c)**	**4.(c)**
5. (a, b, d)	**6. (b, c,d)**	**7. (C)**	**8. (80)**
9. (C)	**10. (a)**	**11. (c)**	**12. (7)**
13. (a)	**14. (a)**	**15. (c)**	**16. (c)**
17.(d)	**18. (b)**	**19. (a)**	**20. (a)**
21. (a)	**22. (c)**	**23. (d)**	**24. (a)**
25. (b)	**26. (b)**		

HINTS & SOLUTIONS

1. ANSWER:- A, B, C

$ can be printed only when x=2
At that time two starts must have already printed
So D is not possible
A,B,C are possible by interleavings

2. ANSWER:- B,C

If T1 starts:
T1 takes s1 → prints 1 → waits on s2 → blocked
T2 cannot start (s1 not released) → deadlock
If T2 starts:
T2 takes s1 → prints 1 → signals s2 → signals s1
T1 runs → prints 2

3. ANSWER:- C

The best response
For I-1:
Initially, X equals 10, s equals 1 (mutual exclusion), and is in binary semaphore. So, after executing 5 incr(), X becomes 10 + 5 = 15, and after executing 3 decr(), X becomes 15 - 3 = 12.
For I-2:
Initially, X = 10 and s = 2 (semaphore included). This means that two processes can enter the critical section at the same time. Let's say, the final values of X written by only the right-hand side function in each pass below-

// both read x = 10
pass1: incr1() & decr1() // decr1() still in execution stage // Final value of X=11
pass2: incr2() & decr1() // decr1() still in execution stage // Final value of X=12
pass3: incr3() & decr1() // decr1() writes the final value// both read x = 9 // Final value of $X = 9$

pass4: incr4() & decr2() // decr2() writes the final value // Final value of $X = 8$

// both read x = 8
pass5: incr5() & decr3() // decr3() writes the final value // Final value of X=7

4. ANSWER:- C

For sequence B → C → A:
B needs S1 initially 1
C waits on S3 → must be enabled after B
A waits on S2 → enabled after C
So initial: S1 = 1, S2 = 0, S3 = 0

5. ANSWER:- A,B, D

A. Assume the processes execute sequentially with no interleaving then after each process ends the counter value increments by 1, hence after 5 process 5 increments to zero hence a counter value of 5. Possible.
B. Let's assume that a process used 2 waits and reads the counter value and didn't update the value yet, all the other process let's say the other processes executed sequentially incremented and stored the value as 4 but since the value isn't written the first process yet the current value is overwritten by the first process as 1. Possible
C. There exists no pattern of execution in which the process increments the current value and completes while maintaining 0 as the counter value. Not possible
D. Assume that all the process use up the first wait operation, the semaphore value will now become zero and deadlock would've occurred. Possibl

Check Out Our GATE Courses & Secure Your Dream Rank!

6. ANSWER:- B, C, D

The lock ensures that mutual exclusion is enforced Since forceful preemption of resources by termination of the other processes is allowed, deadlock will never occur.

Say when process P is holding the single instance resource of type R1, after some time, process Q requests for resource type R1. Using the OWNRESOURCE algorithm, process Q will be able to terminate process P and thus releasing all the resources held by process P. Now Q acquires (holds) resource type R1 while process P gets restarted. And then process Q releases lock L.

It may lead to a live lock, where each process continues to preempt each other, thus none of them can make any progress but neither are in a blocked state.

Livelock occurs when two or more processes continually repeat the same interaction in response to changes in the other processes without doing any useful work. These processes are not in the waiting state, and they are running concurrently. This is different from a deadlock because in a deadlock all processes are in the waiting state.

7. ANSWER:- C

The wait(a) or P(a)ensures that count value is correctly incremented (no race condition) if(count==n) signal (b);

This signal(b) statement is executed by the last (nth) process only. Rest of the n-1 processes are blocked on wait(b). Once the nth process makes signal(b) then the rest of the processes can proceed and enter Code section Q.

In other words, semaphore 'a' controls mutually exclusive execution of statement count+=1 and semaphore 'b' controls entry to CODE SECTION Q when all the process have executed CODE SECTION P. As checked by given condition if(count==n) signal(b); the semaphore 'b' is initialized to 0 and only increments when this condition is TRUE.

8. ANSWER:- 80

Each operation is read–modify–write, so interleavings cause lost updates.

Maximum value: execute +20 and +10 after reading latest, and −50 reads earliest

Possible max = 130

Minimum value: −50 applies on updated value, others overwritten

Possible min = 50

Y − X = 130 − 50 = 80

9. ANSWER:- C

Lets see according to the codes given,
full denotes the number of available slots in the buffer, for the producer to write to
and, empty denotes the number of available slots in the buffer, for the consumer to read from

So, P must be full, as on wait(P), the available slots must decrease for the producer to write to, when it is producing an item to the buffer

Q must be empty as signal(Q) increases the number of available slots in the buffer, for the consumer to read from.

R must be empty as on wan(R) decreases the no of avaliable slots in the buffer to read from (for the consumen)

S must be full as on signal(S) increases a slot for the producer to write, as the consumer has consumed an fiem from the buffer.

10. ANSWER:- A

while (t[j] l = 0 && t[j] <=t[i]);

This ensures that when a process i reaches Critical Section, all processes j which started before it must have its t[j] = 0

This means no two processes can be in critical section at same time as one of them must be started earlier.

returns an integer not smaller, is the issue here for deadlock. This means two processes can have same t value and hence

while (t[j] != 0 && t[j] <=t[i]);

can go to infinite wait. (t[j] == t[i]). Starvation is also possible as there is nothing to ensure that a request is granted in a timed manner.

Bounded waiting is defined as finite waiting and since deadlock is possible here, bounded waiting is not guaranteed as per that definition.

Check Out Our GATE Courses & Secure Your Dream Rank!

11. ANSWER:- C

A) There is strict alternation, so it is not the correct two process synchronization. So, option A is False

B) Mutual exclusion is satisfied because the value of the turn variable cannot be 0 and 1 at the same time. So False.

C) Progress means if one process does not want to enter the critical section then it should not stop other processes from entering the critical section. But we can see that if process 0 will not enter the critical section, then the value of turn will not become 1 and process 1 will not be able to enter the critical section. So progress is not satisfied. True.

D) Bounded waiting solution as there is a strict alteration.So, False.

12. ANSWER:- 7

We can assume the largest initial value of S for which at least one P(S) operation → X

P(S) operation remains in a blocked state therefore it will -1.

The negative value of the counting semaphore indicates the number of processes in the suspended list (or blocked).

Take any sequence of 20P and 12V operations, at least one process will always remain blocked.

So, X - 20 + 12 = -1

Here P(S) = 20 and V(S) = 12

X = 7

Therefore, the initial value of the semaphore will be 7.

13. ANSWER:- A

If we execute P2 process after P1 process, then B = 3

If we execute P1 process after P2 process, then B = 4

If we did pre-emption between P1 & P2 processes, then B = 2 (Pre-emption have done from P1 to P2) or B = 3 (Pre-emption have done from P2 to P1). So, among 2 & 3 values, only one value will be saved in B. So, total no. of distinct values that B can possibly take after the execution is 3.

3 distinct values {2,3,4}

P1−P2: B=3

P2−P1: B=4

P1−P2−P1: B=2

14. ANSWER:- A

Both processes can set their respective flags (varP and varQ) to true and enter the critical section simultaneously because each only checks the other's flag before entering. This means mutual exclusion is not ensured, as both could be inside the critical section at the same time. However, deadlock does not occur because neither process gets stuck waiting indefinitely; at least one will always proceed.

15. ANSWER:- C

This problem simulates classical producer consumer problem, the correct answer of this question is option C because when consumer first access the semaphore 'S' it will down (S) and make 'S'=0, but for semaphore(n), it has to wait for producer to make it 1 but as for producer it can't access the critical section because the value of S=0, so semwait(S) will not work. So there will be a deadlock.

Initially, there is no element in the buffer.

Semaphore s=1 and semaphore n=0.

We assume that initially control goes to the consumer when the buffer is empty.

semWait(s) decrements the value of semaphore 's'. Now, s = 0 and semWait(n) decrements the value of semaphore 'n'.

Since, the value of semaphore 'n' becomes less than 0, the control stucks in the loop of function semWait() and a deadlock arises.

Thus, deadlock occurs if the consumer succeeds in acquiring semaphores when the buffer is empty.

16. ANSWER:- C

The solution is using a binary semaphore. X and Y are two different processes, whatever the values inserted by the processes in the array that will be used by process Y afterwards.

A. X is waiting on R and Y is waiting on X. So, both cannot proceed.

B. Process X is doing Signal operation on R and S without any wait and hence multiple signal operations can happen on the binary semaphore so Process Y won't be able to get exactly n successful wait operations. i.e., Process Y may not be able to complete all the iterations.

Check Out Our GATE Courses & Secure Your Dream Rank!

C. Process X does Wait(S) followed by Signal(R) while Process Y does Signal(S) followed by Wait(R). So, this ensures that no two iterations of either XX or YY can proceed without an iteration of the other being executed in between. i.e., this ensures that all n iterations of X and Y succeeds and hence the answer.

D. Process X does Signal(R) followed by Wait(S) while Process Y does Signal(S) followed by Wait(R). There is a problem here that X can do two Signal(R) operation without a Wait(R) being done in between by Y. This happens in the following scenario:

Process Y: Does Signal (S); Wait(R) fails; goes to sleep.

Process X: does Signal(R); Wait(S) succeeds; In next iteration Signal(R) again happens;

So, this can result in some Signal operations getting lost as the semaphore is a binary one and thus Process Y may not be able to complete all the iterations. If we change the order of Signal(S) and Wait(R) in EntryY, then (D) option also can work.

17. ANSWER:- D

x is initially 0 and $S = 2$

W process reads x and increments it then it is preempted. Still value of x is 0coz store operation is yet to be performed. for W to execute it has first performed a wait operation binary semaphore S. Now S is 1 . after this processes Y and Z came to do what ever they want to do with the value of x. This is possible because $S = 1$. So, at the end when they both are done $S = 1$ still. after this W is made to execute again to store the value 1 in x and makes $S = 2$ just before it exits. After which process X executes(without any problem coz it just needs S to be 1 atleast). It reads the value of x as 1 and increments it to 2 and exits.

18. ANSWER:- B

Check the loop first:

while (Fetch_And_Add (L,1))

L = 1; // A waiting process can be here just after // the lock is released, and can make L=1.

Assume P1 runs till while condition and gets preempted before executing L =1. Now P2 runs all statements, hence L = 0. Then P1 without checking L it makes L = 1 by executing the statement where it got preempted.

It takes a non-zero value (L=1) when the lock is actually available (L = 0).

A process acquires a lock only when L=0. When L is 1, the process gets repeated in the while loop- there is no overflow as after each increment to L, L is again made equal to 1. So, the only chance of overflow is if a large number of processes (larger than sizeof(int)) execute the check condition of while loop but not L=1, which is highly improbable.

Acquire Lock gets success only when Fetch_And_Add gets executed with L = 0. Now suppose P1 acquires lock and make L=1. P2 waits for a lock iterating the value of L between 1 and 2 (assume no other process waiting for lock). Suppose when P1 releases lock by making L=0, the next statement P2 executes is L=1. So, value of L becomes 1 and no process is in critical section ensuring L can never be 0 again. Thus, (B) choice.

19. ANSWER:- A

Mutual Exclusion: The code ensures that both processes P1 and P2 cannot be in their critical section at the same time, because one waits for S1 == S2 while the other waits for S1 != S2, so their conditions never allow simultaneous entry.

Progress is not satisfied: If, for example, P1 is not interested in entering the critical section but P2 wants to enter, P2 could be blocked indefinitely waiting for P1 to change S1 to match S2. Thus, a process not intending to enter the critical section can prevent another from entering, violating the progress property.

Summary: The algorithm achieves mutual exclusion, but does not guarantee progress.

20. ANSWER:- A

Process P0 will print '0' at least twice. P0 starts and prints '0' (since S0 = 1).

- It releases S1 and S2, allowing P1 and P2 to run.

Check Out Our GATE Courses & Secure Your Dream Rank!

- Either P1 or P2 will run and release S0, letting P0 print '0' again.
- The minimum number of times P0 can print '0' before all the semaphores are blocked is 2.

21. ANSWER:- A

I: True
Test-and-set spinlock ensures that some process always enters CS → deadlock-free.
II: False
Spinlock has no fairn**ess** → starvation possible.
III: False
No FIFO guarantee.
IV: False
Test-and-set ensures mutual exclusion.

22. ANSWER:- C

xb is used as mutex → must be 1
yb is used for blocking → must be 0

23. ANSWER:- D

Mutual Exclusion:
Mutual exclusion is guaranteed in the given code.
Deadlock:
If both P1 and P2 set their flags to true simultaneously:
P1: wants1 = true
P2: wants2 = true
Now:
P1 waits while wants2 == true
P2 waits while wants1 == true
Both will wait forever.
So mutual exclusion is preserved (they never enter together), but they can get stuck.
Option-D is correct

24. ANSWER:- A

Busy waiting loop needs context switching; if disabled, other process cannot execute V → Deadlock

25. ANSWER:- B

If barrier is reused immediately, some processes may enter next round while others are still exiting previous round, causing mismatch and deadlock

26. ANSWER:- B

Problem is reuse of barrier; need to ensure previous round is fully reset before next begins

Check Out Our GATE Courses & Secure Your Dream Rank!

3 DEADLOCKS

GATE CSE - 2026 MSQ SET - 1(1-MARKS)

Q1. With respect to deadlocks in an operating system, which of the following statements is/are FALSE?

(a) Banker's algorithm is used to prevent deadlocks

(b) Deadlock formation can be prevented by ensuring that the hold and wait condition is not allowed

(c) An assignment edge in a resource allocation graph is marked from a process to a resource

(d) A safe state guarantees that all processes can finish without formation of a deadlock

GATE CSE - 2026 NAT SET - 1(1-MARKS)

Q2. Consider a system consisting of k instances of a resource R, being shared by 5 processes.
Assume:

- Each process requires a maximum of two instances of resource R.
- A process can request or release only one instance at a time.
- A process can request the second instance only after acquiring the first instance.

The minimum value of k for the system to be deadlock-free is _____.

GATE CSE - 2025 MCQ SET - 2(2-MARKS)

Q3. $P = \{P1, P2, P3, P4\}$ consists of all active processes in an operating system. $R = \{R1, R2, R3, R4\}$ consists of single instances of distinct types of resources in the system.
The resource allocation graph has the following assignment and claim edges.
Assignment edges: $R1 \rightarrow P1$, $R2 \rightarrow P2$, $R3 \rightarrow P3$, $R4 \rightarrow P4$ (the assignment edge $R1 \rightarrow P1$ means resource $R1$ is assigned to process $P1$, and so on for others)
Claim edges: $P1 \rightarrow R2$, $P2 \rightarrow R3$, $P3 \rightarrow R1$, $P2 \rightarrow R4$, $P4 \rightarrow R2$ (the claim edge $P1 \rightarrow R2$ means process $P1$ is waiting for resource $R2$, and so on for others)
Which of the following statement(s) is/are CORRECT?

(a) Aborting $P1$ makes the system deadlock free.

(b) Aborting $P3$ makes the system deadlock free

(c) Aborting $P2$ makes the system deadlock free

(d) Aborting $P1$ and $P4$ makes the system deadlock free.

GATE CSE - 2022 MSQ (1-MARKS)

Q4. Which of the following statements is/are TRUE with respect to deadlocks?

(a) The circular wait is a necessary condition for the formation of deadlock

(b) In a system where each resource has more than one instance, a cycle in its wait-for graph indicates the presence of a deadlock.

(c) If the current allocation of resources to processes leads the system to an unsafe state, then deadlock will necessarily occur.

(d) In the resource-allocation graph of a system, if every edge is an assignment edge, then the system is not in a deadlock state.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2019 MCQ SET - 2(2-MARKS)

Q5. Consider the following snapshot of a system running n concurrent processes. Process i is holding X_i instances of a resource R, $1 \le i \le n$. Assume that all instances of R are currently in use. Further, for all i process i can place a request for at most Y_i additional instances of R while holding the X_i instances it already has. Of the n processes, there are exactly two processes p and q such that $Y_p = Y_q = 0$. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

(a) $\text{Min}(X_p, X_q) \ge \text{Min}\{Y_k \mid 1 \le k \le n, k \ne p, k \ne q\}$
(b) $X_p + X_q < \text{Max}\{Y_k \mid 1 \le k \le n, k \ne p, k \ne q$
(c) $\text{Min}(X_p, X_q) \le \text{Max}\{Y_k \mid 1 \le k \le n, k \ne p, k \ne q\}$
(d) $X_p + X_q < \text{Min}\{Y_k \mid 1 \le k \le n, k \ne p, k \ne q\}$

GATE CSE - 2018 MCQ (1-MARKS)

Q6. In a system, there are three types of resources: E, F and G. Four processes P_0, P_1, P_2 and P_3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, Max [P_2, F] is the maximum number of instances of F that P_2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation.

Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of E and 3 instances of F are the only resources available.

Allocation			
	E	F	G
P_0	1	0	1
P_1	1	1	2
P_2	1	0	3
P_3	2	0	0

Max			
	E	F	G
P_0	4	3	1
P_1	2	1	4
P_2	1	3	3
P_3	5	4	1

From the perspective of deadlock avoidance, which one of the following is true?

(a) The system is in safe state.
(b) The system is not in safe state, but would be safe if one more instance of E were available.
(c) The system is not in safe state, but would be safe if one more instance of F were available.
(d) The system is not in safe state, but would be safe if one more instance of G were available.

GATE CSE - 2018 NAT (2-MARKS)

Q7. Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is _____.

GATE CSE - 2017 MCQ SET 1 (2-MARKS)

Q8. A multithreaded program P executes with x number of threads and used y number of locks for ensuring mutual exclusion while operating on shared memory locations. All locks in the program are non-reentrant, i.e., if a thread holds a lock l, then it cannot re-acquire lock l without releasing it. If a thread is unable to acquire a lock, it blocks until the lock becomes available. The minimum value of x and the minimum value of y together for which execution of P can result in a deadlock are:

(a) x = 1, y = 2
(b) x = 2, y = 1
(c) x = 2, y = 2
(d) x = 1, y = 1

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2017 MCQ SET 2 (2-MARKS)

Q9. A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below:

Process	Current Allocation	Maximum Requirement
P1	3	7
P2	1	6
P3	3	5

Which of the following best describes the current state of the system?

(a) Safe, Deadlocked
(b) Safe, Not Deadlocked
(c) Not Safe, Deadlocked
(d) Not Safe,Not Deadlocked

GATE CSE - 2015 MCQ SET 2 (1-MARKS)

Q10. A system has 6 identical resources and N processes competing for them. Each process can request at most 2 resources. Which one of the following values of N could lead to a deadlock?

(a) 1
(b) 2
(c) 3
(d) 6

GATE CSE - 2015 MCQ SET 3 (2-MARKS)

Q11. Consider the following policies for preventing deadlock in a system with mutually exclusive resources.

I. Processes should acquire all their resources at the beginning of execution. If any resource is not available, all resources acquired so far are released.

II. The resources are numbered uniquely, and processes are allowed to request for resources only in increasing resource numbers.

III. The resources are numbered uniquely, and processes are allowed to request for resources only in decreasing resource numbers.

IV. The resources are numbered uniquely. A process is allowed to request only for a resource with resource number larger than its currently held resources.

Which of the above policies can be used for preventing deadlock?

(a) Any one of I and III but not II or IV
(b) Any one of I, III, and IV but not II
(c) Any one of II and III but not I or IV
(d) Any one of I, II, III, and IV

GATE CSE - 2014 MCQ SET 3 (2-MARKS)

Q12. A system contains three programs and each requires three tape units for its operation. The minimum number of tape units which the system must have such that deadlocks never arise is _____.

GATE CSE - 2014 MCQ SET 1 (2-MARKS)

Q13. An operating system uses the Banker's algorithm for deadlock avoidance when managing the allocation of three resource types X, Y, and Z to three processes P0, P1, and P2. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution.

	Allocation			Max		
	X	Y	Z	X	Y	Z
P 0	0	0	1	8	4	3
P 1	3	2	0	6	2	0
P 2	2	1	1	3	3	3

There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state:

REQ1: P0 requests 0 units of X,
0 units of Y and 2 units of Z

Check Out Our GATE Courses & Secure Your Dream Rank!

REQ2: P1 requests 2 units of X,
0 units of Y and 0 units of Z
Which one of the following is TRUE?

(a) Only REQ1 can be permitted.
(b) Only REQ2 can be permitted.
(c) Both REQ1 and REQ2 can be permitted.
(d) Neither REQ1 nor REQ2 can be permitted.

GATE CSE - 2013 MCQ (1-MARKS)

Q14. Three concurrent processes X, Y, and Z execute three different code segments that access and update certain shared variables. Process X executes the P operation (i.e., wait) on semaphores a, b, and c; process Y executes the P operation on semaphores b, c, and d; process Z executes the P operation on semaphores c, d, and a before entering the respective code segments. After completing the execution of its code segment, each process invokes the V operation (i.e., signal) on its three semaphores. All semaphores are binary semaphores initialized to one. Which one of the following represents a deadlock-free order of invoking the P operations by the processes?

(a) X: P(a) P(b) P(c)Y: P(b) P(c) P(d) Z: P(c) P(d) P(a)
(b) X: P(b) P(a) P(c) Y: P(b) P(c) P(d) Z: P(a) P(c) P(d)
(c) X: P(b) P(a) P(c) Y: P(c) P(b) P(d) Z: P(a) P(c) P(d)
(d) X: P(a) P(b) P(c) Y: P(c) P(b) P(d) Z: P(c) P(d) P(a)

GATE CSE - 2010 MCQ (2-MARKS)

Q15. A system has n resources R_0...,R_{n-1}, and k processes P_0....P_{k-1}. The implementation of the resource request logic of each process Pi is as follows:

```
if (i % 2 == 0) {
        if (i < n) request Ri
        if (i+2 < n) request Ri+2
}
else {
        if (i < n) request Rn-i
        if (i+2 < n) request Rn-i-2
}
```

In which one of the following situations is a deadlock possible?

(a) n = 40, k = 26
(b) n = 21, k = 12
(c) n = 20, k = 10
(d) n = 41, k = 19

GATE CSE - 2009 MCQ (2-MARKS)

Q16 Consider a system with 4 types of resources R1 (3 units), R2 (2 units), R3 (3 units), R4 (2 units). A non-preemptive resource allocation policy is used. At any given instance, a request is not entertained if it cannot be completely satisfied. Three processes P1, P2, P3 request the resources as follows if executed independently.

Process P1:	Process P2:	Process P3:
$t = 0$: requests 2 units of R2 $t = 1$: requests 1 unit of R3 $t = 3$: requests 2 units of R1 $t = 5$: releases 1 unit of R2 and 1 unit of R1 $t = 7$: releases 1 unit of R3 $t = 8$: requests 2 units of R4 $t = 10$: Finishes	$t = 0$: requests 2 units of R3 $t = 2$: requests 1 unit of R4 $t = 4$: requests 1 unit of R1 $t = 6$: releases 1 unit of R3 $t = 8$: Finishes	$t = 0$: requests 1 unit of R4 $t = 2$: requests 2 units of R1 $t = 5$: releases 2 units of R1 $t = 7$: requests 1 unit of R2 $t = 8$: requests 1 unit of R3 $t = 9$: Finishes

Which one of the following statements is TRUE if all three processes run concurrently starting at time t = 0 ?

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) All processes will finish without any deadlock.
(b) Only P1 and P2 will be in deadlock.
(c) Only P1 and P3 will be in deadlock.
(d) All three processes will be in deadlock.

GATE CSE - 2008 MCQ (2-MARKS)

Q17. Which of the following is NOT true of deadlock prevention and deadlock avoidance schemes?

(a) In deadlock prevention, the request for resources is always granted if the resulting state is safe
(b) In deadlock avoidance, the request for resources is always granted if the resulting state is safe
(c) Deadlock avoidance is less restrictive than deadlock prevention
(d) Deadlock avoidance requires knowledge of resource requirements a priori

GATE CSE - 2007 MCQ (2-MARKS)

Q18. A single processor system has three resource types X, Y, and Z, which are shared by three processes. There are 5 units of each resource type.
The column alloc denotes the number of units of each resource type allocated to each process, and the column request denotes the number of units of each resource type requested by a process in order to complete execution.
Which of these processes will finish LAST?

	alloc (X Y Z)	request (X Y Z)
P0	1 2 1	1 0 3
P1	2 0 1	0 1 2
P2	2 2 1	1 2 0

(a) P0
(b) P1
(c) P2
(d) None of the above, since the system is in a deadlock.

GATE CSE - 2006 MCQ (2-MARKS)

Q19. Consider the following snapshot of a system running n processes. Process i is holding x_i instances of a resource R, for $1 \leq i \leq n$. Currently, all instances of R are occupied. Further, for all i, process i has placed a request for an additional y_i instances while holding the x_i instances it already has. There are exactly two processes p and q such that $y_p = y_q = 0$.
Which one of the following can serve as a necessary condition to guarantee that the system is not approaching a deadlock?
(a) $\min(x_p, x_q) < \max_{k \neq p,q} y_k$
(b) $x_p + x_q \geq \min_{k \neq p,q} y_k$
(c) $\max(x_p, x_q) > 1$
(d) $\min(x_p, x_q) > 1$

GATE CSE - 2005 MCQ (2-MARKS)

Q20. Suppose n processes, P1,…,Pn, share m identical resource units, which can be reserved and released one at a time. The maximum resource requirement of process Pi is si, where si>0.
Which one of the following is a sufficient condition for ensuring that deadlock does not occur?

(a) $\forall_i, s_i < m$
(b) $\forall_i, s_i < n$
(c) $\sum_{i=1}^{n} s_i < (m + n)$
(d) $\sum_{i=1}^{n} s_i < (m \times n)$

Check Out Our GATE Courses & Secure Your Dream Rank!

Answer key

1. (a, c)	**2.** (6)	**3.** (c, d)	**4.** (a, d)
5. (d)	**6.** (a)	**7.** (2)	**8.** (d)
9. (b)	**10.** (d)	**11.** (d)	**12.** (7)
13. (b)	**14.** (b)	**15.** (b)	**16.** (c)
17. (a)	**18.** (c)	**19.** (b)	**20.** (c)

HINTS & SOLUTIONS

1. ANSWER:- A, C

(A): False — Banker's algorithm is used for deadlock avoidance, not prevention.

(B): True — Removing hold and wait prevents deadlock.

(C): False — Assignment edge is Resource → Process, not Process → Resource.

(D): True — Safe state means no deadlock will occur.

2. ANSWER:- 6

For n processes, each requiring maximum m instances of a single resource:

Minimum resources to guarantee deadlock freedom

$k \geq n(m - 1) + 1$

Substitute values:

$n = 5$

$m = 2$

$k \geq 5(2 - 1) + 1$

$k \geq 5 + 1$

$k \geq 6$

3. ANSWER:- C, D

Aborting P2 releases R1, R4, and R3, breaking the circular wait and making the system deadlock-free.

(D): Aborting P1 and P4 releases all their resources, breaking the deadlock.

Thus, both (C) and (D) make the system deadlock-free.

4. ANSWER:- A, D

A. Circular wait is one of the necessary conditions for the formation of deadlock hence it is true.

B. If the resource has more than one instance then cycle is just a necessary condition but not the sufficient condition, hence it is false.

C. Unsafe state will not always lead to deadlock, because it depends on behaviour of the processes, hence it is false.

D. If every edge is only allocation, it means there is no requirement of the resources, so deadlock is not possible, hence it is true.

5. ANSWER:- D

$\{P_1, P_2, ..., P_n\}$

P_i holds X_i instances.

P_i can request additional Y_i instances.

Given two processes p & q such that their additional requests are zero.

$Y_p = Y_q = 0$

$\{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$ means that out of 'n' processes, we are left with (n-2) process (except p&q), i.e., Y_k indicates additional request of all the processes (n-2) except p & q.

For p & q to complete first, accordingly

$X_p + X_q < \text{Min } \{Y_k\}$

Option D is correct.

There are exactly two processes p and q which do not need any additional instances of resources.

So, p and q will complete their execution and will release X_p and X_q instances of resources.

Now to guarantee that no other process apart from p and q can complete execution, the no. of instances of resources available must be less than the minimum no. of instances of resources required by any other process, i.e.,

$X_p + X_q < \text{Min } \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$

Check Out Our GATE Courses & Secure Your Dream Rank!

6. ANSWER:- A

	Max			Allocation			Need=max-allocation		
	E	F	G	E	F	G	E	F	G
P_0	4	3	1	1	0	1	3	3	0
P_1	2	1	4	1	1	2	1	0	2
P_2	1	3	3	1	0	3	0	3	0
P_3	5	4	1	2	0	0	3	4	1

⟨E, F, G⟩ = ⟨3, 3, 0⟩
Safe sequence:
⟨P_0, P_2, P_1, P_3⟩
P_0: P_0 can be allotted ⟨3,3,0⟩.
After completion Available = ⟨4,3,1⟩
P_2: P_2 can be allotted ⟨0,3,0⟩.
After completion: Available = ⟨5,3,4⟩
P_1: P_1 can be allotted ⟨1,0,2⟩.
After completion: Available = ⟨6,4,6⟩
P_3: P_3 can be allotted ⟨3,4,1⟩.
After completion: Available = ⟨8,4,6⟩

7. ANSWER:- 2
No. of process = 3
No. of resources = 4
Let's distribute each process one less than maximum demands i.e., (k-1) resources.
So, for three processes, 3(k – 1) resources.
For deadlock avoidance provide an additional resource to any one of the processes.
∴ Total resources required to avoid deadlock in any case is 3(k – 1) + 1 = 3k – 2
Now this 3k – 2 should be less than equal to available no. of resources, i.e.,
$3k - 2 \leq 4$
$k \leq 2$
So maximum value of k = 2

8. ANSWER:- D
Re-entrant (recursive) locks allow a thread to reacquire the lock. That means the same process can claim the lock multiple times without blocking on itself. This prevents the thread from deadlocking itself. This is the main advantage of re-entrant locks over non-re-entrant locks.
Non-reentrant (Non-recursive) locks do not allow a thread to reacquire the lock. That means the same process cannot claim the lock multiple times without releasing it. So, if a thread/process is unable to acquire a lock, it blocks until the lock becomes available. This situation is deadlocked.
Here a non-reentrant process can't own the same lock multiple times, so if the process tries to acquire an already owned lock, it will get blocked, and deadlock will happen.
From the above options x=1 (a single thread) and y=1 (a single lock) deadlock is possible when we consider the given situations in question.

9. ANSWER:- B

Process	Current Allocation	Maximum Requirement	Need
P1	3	7	4
P2	1	6	5
P3	3	5	2

Available: Given there are total 9 tape drives,
So, according to the above table we can see we have currently allocated (7 tape drive), so currently Available tape drives =(9 - (3 + 1 + 3)) = 2,
So, P3 can use it and after using it will release it 3 resources New available = 3 + 2 = 5
Now, P2 can be satisfied.
New available: 5 + 1 = 6
Now, P1 can be satisfied. Thus safe sequence: P3 → P2 → P1
That is not deadlocked.

Check Out Our GATE Courses & Secure Your Dream Rank!

10. ANSWER:- D

The system has 6 identical resources and N processes, each requesting at most 2 resources. Deadlock Condition Analysis:

- Maximum demand per process = 2 resources.
- For deadlock to occur, the system must not be able to allocate resources such that at least one process can proceed.
- Using the "Deadlock necessary condition" for identical resources, deadlock can occur if:

$N \times (\text{max demand per process} - 1) \geq \text{total resources}$

Substitute values:

$N \times (2-1) \geq 6 \Longrightarrow N \times 1 \geq 6 \Longrightarrow N \geq 6$

Thus, deadlock can occur if $N \geq 6$.

Conclusion:

- The smallest values of N that can lead to deadlock is 6.
- For N<6, deadlock will not occur.

11. ANSWER:- D

A deadlock will not occur if any one of the below four conditions are prevented:

- hold and wait
- mutual exclusion
- circular wait
- no-preemption

Now,

Option-1 if implemented violates 1 so deadlock cannot occur.

Option-2 if implemented violates circular wait (making the dependency graph acyclic)

Option-3 if implemented violates circular wait (making the dependency graph acyclic)

Option-4 it is equivalent to options 2 and 3

So, the correct option is 4 as all of them are methods to prevent deadlock.

12. ANSWER:- 7

Number of programs = 3

- Tape units required per program = 3
- Applying the deadlock avoidance formula for minimum resources to avoid deadlock: Minimum tape units=(Number of programs×Tape units per program)−(Number of programs−1)Minimum tape units=(Number of programs×Tape units per program)−(Number of programs−1)Substituting,
- $=(3\times3)-(3-1)=9-2=7$
- Hence, having 7 tape units ensures that deadlock will never arise in this system.

Thus, the minimum number of tape units required is 7.

13. ANSWER:- B

Lets take 1st case, After allowing Req 1,

	Allocated			Max			Requirement		
P0	0	0	3	8	4	3	8	4	0
P1	3	2	0	6	2	0	3	0	0
P2	2	1	1	3	3	3	1	2	2

Available: X=3, Y=2, Z=0 With this we can satisfy P1's requirement. So available becomes: X=6, Y=4, Z=0. Since Z is not available, neither P0's nor P2's requirement can be satisfied. So, it is an unsafe state.

Let's take 2nd case, After allowing Req 2,

	Allocated			Max			Requirement		
P0	0	0	1	8	4	3	8	4	2
P1	5	2	0	6	2	0	1	0	0
P2	2	1	1	3	3	3	1	2	2

Available: X=1, Y=2, Z=2

With this we can satisfy any one of P1's or P2's requirement. Let us first satisfy P1's requirement. So Available now becomes:

X=6, Y=4, Z=2

Now with the availability we can satisfy P2's requirement. So,s Available now becomes,

X=8, Y=5, Z=3

With this availability P0 can also be satisfied. So, hence it is in safe state.

So from above two cases Req 1 cannot be permitted but Req 2 can be permitted.

Check Out Our GATE Courses & Secure Your Dream Rank!

14. ANSWER:- B

Three concurrent processes X, Y, and Z execute code segments using semaphores a, b, c, d (all initialized to 1). Each process performs P operations before its code and V operations after.

X needs a, b,

Y needs b, c, d

Z needs c, d, a

To avoid deadlock, all processes must acquire resources

in a consistent global order.

Option (b):

X: P(b) P(a) P(c)

Y: P(b) P(c) P(d)

Z: P(a) P(c) P(d)

Order followed is consistent (b → a → c and a → c → d align without circular wait), so no cycle forms.

15. ANSWER:- B

From the resource allocation logic, it's clear that even numbered processes are taking even numbered resources and all even numbered processes share no more than 1 resource. Now, if we make sure that all odd numbered processes take odd numbered resources without a cycle, then deadlock cannot occur. The "else" case of the resource allocation logic, is trying to do that. But, if n is odd, Rn and Rn-i-2 will be even and there is a possibility of deadlock, when two processes request the same R, and R.;. So, only B and D are the possible answers.

Now, in D, we can see that Po requests Ro and R2, P2 requests R2 and R4, so on until P18 requests R18 and R20. At the same time P_1 requests R40 and R38. P3 requests R38 and R36, so on until, P17 requests R24 and R22. i.e.; there are no two processes requesting the same two resources and hence there can't be a cycle of dependencies which means, no deadlock is possible.

But for B. Ps requests Rg and R10 and P11 also requests R10 and Rs. Hence, a deadlock is possible. (Suppose Ps comes first and occupies Rs. Then P11 comes and occupies R10. Now, if Ps requests R10 and P11 requests R8, there will be deadlock)

16. ANSWER:- A

1. Early Resource Distribution ($t = 0$ to $t = 2$) : All processes successfully grab their initial allocations. At $t = 2$, the system resources are almost entirely exhausted, leaving only 1 unit of R_1 free.
2. The Temporarily Paused Process ($t = 3$): P_1 asks for 2 units of R_1, but since only 1 is free, it safely blocks (pauses) without causing any harm.
3. The Unlocking Release ($t = 5$): P_3 finishes its first phase and releases 2 units of R_1. This immediately wakes up the blocked P_1.
4. Smooth Sequential Completion: With R_1 freed, P_1 goes on to release its held units of R_2 and R_1, which provides exactly what P_3 needs to finish. P_2 also finishes completely at $t = 8$, releasing its remaining pool back to the system.

17. ANSWER:- A

Deadlock Prevention:

Prevention works by structurally denying one of the four necessary conditions (mutual exclusion, hold-and-wait, no preemption, circular wait).

It does not check for safe states dynamically.

Therefore, a request is not granted based on a safety check — it follows strict rules regardless of safety.

So statement (A) is false.

Deadlock Avoidance:

Avoidance (e.g., Banker's Algorithm) checks whether granting a request keeps the system in a safe state.

If the resulting state is safe → request is granted.

Hence, (B) is true.

Avoidance is less restrictive than prevention because it allows more flexibility → (C) is true.

Avoidance requires advance knowledge of maximum resource needs → (D) is true.

Therefore, the statement that is NOT true is (A).

Check Out Our GATE Courses & Secure Your Dream Rank!

18. ANSWER:- C

only possible safe sequence is P1,P0,P2
hence P2 finishes last.

19. ANSWER:- B

To avoid deadlock, after either p or q finishes and releases its resources, at least one waiting process must be able to get all its requested resources and complete.

So, the released resources must be at least as large as some pending requests.

Since p and q are the only processes that can release resources, the maximum available resources after they finish would be: $x_p + x_q$

For the system to avoid deadlock,

$x_p + x_q \geq yk$

20. ANSWER:- C

For n processes with maximum requirement is and m identical resources, a sufficient condition to avoid deadlock

$$m \geq \sum_{i=1}^{n} (s_i - 1) + 1$$
$$m \geq \sum_{i=1}^{n} s_i - n + 1$$
$$\sum_{i=1}^{n} s_i \leq m + n - 1$$

Thus a sufficient condition is

$$\sum_{i=1}^{n} s_i < m + n$$

Correct answer . c) $\sum_{i=1}^{n} s_i < (m + n)$

Check Out Our GATE Courses & Secure Your Dream Rank!

4 MEMORY MANAGEMENT AND VIRTUAL MEMORY

GATE CSE - 2026 NAT SET - 2(2-MARKS)

Q1. Consider contiguous allocation of physical memory to processes using a variable partitioning scheme. Suppose there are 8 holes in the memory of sizes: 20 KB, 4 KB, 25 KB, 18 KB, 7 KB, 9 KB, 15 KB, and 12 KB.
Assume that no two holes are adjacent.
Two processes arrive in order:

- P1 = 16 KB
- P2 = 9 KB

They are allocated memory using the best-fit technique.
After allocating space to P1 and P2, the number of holes of size less than 8 KB is _____.
Note: $1K = 2^{10}$.

GATE CSE - 2026 NAT SET - 2(2-MARKS)

Q2. A system has a Translation Lookaside Buffer (TLB) that has a reach of 1 MB. TLB reach is defined as the total amount of physical memory that can be accessed through the TLB entries.
The paging system uses pages of size 4 KB. The virtual address space is 64 GB and physical address space is 1 GB.
If each TLB entry stores:

- a 4-bit process id
- page number
- frame number
- a 2-bit control field

then the size of the TLB (in bytes) is _____.
Note:
$1K = 2^{10}$, $1M = 2^{20}$, $1G = 2^{30}$.

GATE CSE - 2026 MSQ SET - 1(2-MARKS)

Q3. Consider a system that has a cache memory unit and a memory management unit (MMU). The address input to the cache memory is a physical address. The MMU has a Translation Lookaside Buffer (TLB). Assume that when a page is evicted from main memory, the corresponding blocks in the cache are marked invalid.
For a given memory reference, which of the following sequences of events can NEVER happen?

(a) TLB miss, Page table hit, Cache hit
(b) TLB hit, Page table miss, Cache hit
(c) TLB miss, Page table miss, Cache hit
(d) TLB miss, Page table miss, Cache miss

GATE CSE - 2025 NAT SET - 2(2-MARKS)

Q4. A computer system supports a logical address space of 2^{32}bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size.The value of b is _____.

GATE CSE - 2025 MSQ SET - 2(2-MARKS)

Q5. Consider a demand paging system with three frames, and the following page reference string:1 2 3 4 5 4 1 6 4 5 1 3 2
The contents of the frames are shown initially and after each reference.

initially	after											
–	1*	2*	3*	4*	5*	4	1	6*	4	5	1*	3*	2*
–	1	1	1	1	1	1	1	6	6	6	6	6	2
–	–	2	2	4	4	4	4	4	4	4	1	1	1
–	–	–	3	3	5	5	5	5	5	5	5	3	3

The *-marked references cause page replacements.
Which one or more of the following could be the page replacement policy/policies in use?

(a) Least Recently Used page replacement policy
(b) Least Frequently Used page replacement policy
(c) Most Frequently Used page replacement policy
(d) Optimal page replacement policy

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2025 MCQ SET - 1(1-MARKS)

Q6. Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?

(a) 2^{21}
(b) 2^{20}
(c) 2^{22}
(d) 2^{24}

GATE CSE - 2025 MCQ SET - 1(2-MARKS)

Q7. A disk of size 512M bytes is divided into blocks of 64K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a NULL pointer (also of 4 bytes). Suppose a file of 1M bytes needs to be stored in the disk. Assume, 1K = 2^{10} and 1M = 2^{20}. The amount of space in bytes that will be wasted due to internal fragmentation is _____. (Answer in integer)

GATE CSE - 2025 NAT SET - 1(2-MARKS)

Q8. In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows: The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page. A process accesses the pages in the following order of page numbers: 1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4. If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____ . (Answer in integer)

GATE CSE - 2024 NAT SET - 2(2-MARKS)

Q9. Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume 1 KB = 2^{10} bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.
An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.
The value of X + Y is _____

GATE CSE - 2024 MSQ SET - 2(1-MARKS)

Q10. Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?

(a) Allocate a new page table for a newly created process
(b) Translate a virtual address to a physical address using the page table
(c) Raise a trap when a virtual address is not found in the page table
(d) Raise a trap when a process tries to write to a page marked with read-only permission in the page table

GATE CSE - 2024 NAT SET - 1(2-MARKS)

Q11. Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (in decimal format) corresponding to the virtual address 2500 (in decimal format) is _____

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2023 MCQ SET - 1(2-MARKS)

Q12. Consider the following two-dimensional array D in the C programming language, which is stored in row-major order:

int D[128][128];

Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D. The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process that executes the following code snippet:

```
for (int i = 0; i < 128; i++)
        for (int j = 0; j < 128; j++)
            D[j][i] *= 10;
```

The number of page faults generated during the execution of this code snippet is

(a) 2048
(b) 512
(c) 4096
(d) 1032

GATE CSE - 2023 NAT SET - 2(2-MARKS)

Q13. Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4 KB (1 KB = 1024 B) and a page table entry at any of the levels occupies 8 bytes.The value of L is.....

GATE CSE - 2022 MCQ (2-MARKS)

Q14. Which one of the following statements is FALSE?

(a) The TLB performs an associative search in parallel on all its valid entries using page number of incoming virtual address.
(b) If the virtual address of a word given by CPU has a TLB hit, but the subsequent search for the word results in a cache miss, then the word will always be present in the main memory.
(c) The memory access time using a given inverted page table is always same for all incoming virtual addresses.
(d) In a system that uses hashed page tables, if two distinct virtual ddresses V1 and V2 map to the same value while hashing, then the memory access time of these addresses will not be the same.

GATE CSE - 2022 NAT (2-MARKS)

Q15. Consider a demand paging system with four-page frames (initially empty) and an LRU page replacement policy. For the following page reference string 7, 2,7,3, 2,5,3, 4,6,7,7,1,5,6,1 the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is_____.

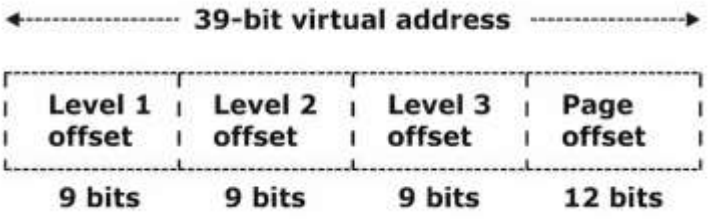
GATE CSE - 2021 MSQ (1-MARKS)

Q16. In the context of the operating system, which of the following statements is/are correct with respect to paging?

(a) Paging helps solve the issue of external fragmentation.
(b) Page size has no impact on internal fragmentation.
(c) Multi-level paging is necessary to support pages of different sizes.
(d) Paging incurs memory overheads.

GATE CSE - 2021 NAT SET – 2 (1-MARKS)

Q17. Consider a three-level page table to translate a 39-bit virtual address to a physical address as shown below.



The page size is 4KB (1KB = 2^{10} bytes) and page table entry size at every level is 8 bytes. A process P is currently using 2GB (1GB = 2^{30} bytes) virtual memory which is mapped to 2GB of physical memory. The minimum amount of memory required for the page table of P across all levels is _____ KB

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2020 MCQ (1-MARKS)

Q18. Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?

(a) The hole created by the worst fit is always larger than the hole created by the first fit.
(b) The hole created by best fit is never larger than the hole created by first fit.
(c) The hole created by the first fit is always larger than the hole created by the next fit.
(d) The hole created by the next fit is never larger than the hole created by the best fit.

GATE CSE - 2020 NAT SET - 1(2-MARKS)

Q19. Consider a paging system that uses a 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read in from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal place) is _____.

GATE CSE - 2019 MCQ (2-MARKS)

Q20. Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

(a) 8×2^{20}
(b) 4×2^{20}
(c) 16×2^{10}
(d) 256×2^{10}

GATE CSE - 2018 MCQ (1-MARKS)

Q21. Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units.
Which one of the following is the correct expression for the page fault rate experienced by the process?

(a) (D-M)/(X-M)
(b) (X-M)/(D-M)
(c) (D-X)/(D-M)
(d) (X-M)/(D-X)

GATE CSE - 2017 MCQ SET - 1 (2-MARKS)

Q22. Recall that Belady's anomaly is that the page-fault rate may increase as the number of allocated frames increases. Now, consider the following statements:
S1: Random page replacement algorithm (where a page chosen at random is replaced) suffers from Belady's anomaly
S2: LRU page replacement algorithm suffers from Belady's anomaly
Which of the following is CORRECT?

(a) S1 is true, S2 is true.
(b) S1 is true, S2 is false.
(c) S1 is false, S2 is true.
(d) S1 is false, S2 is false.

GATE CSE - 2016 MCQ SET - 2 (1-MARKS)

Q23. In which one of the following page replacement algorithms is it possible for the page fault rate to increase even when the number of allocated frames increases?

(a) LRU (Least Recently Used)
(b) OPT (Optimal Page Replacement)
(c) MRU (Most Recently Used)
(d) FIFO (First In First Out)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2016 NAT SET -1 (1-MARKS)

Q24. Consider a computer system with ten physical page frames. The system is provided with an access sequence (a_1, a_2,....., a_{20}, a_1, a_2, ... a_{20}), where each a*i* is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is __.

GATE CSE – 2016 NAT SET -1 (2-MARKS)

Q25. Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is _____ megabytes.

GATE CSE – 2015 MCQ SET -1 (1-MARKS)

Q26. Consider a main memory with five page frames and the following sequence of page references: 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3. Which one of the following is true with respect to page replacement policies First-In-First Out (FIFO) and Least Recently Used (LRU)?

(a) Both incur the same number of page faults
(b) FIFO incurs 2 more page faults than LRU
(c) LRU incurs 2 more page faults than FIFO
(d) FIFO incurs 1 more page faults than LRU

GATE CSE – 2015 MCQ SET -2 (2-MARKS)

Q27. Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB, and 250 KB, where KB refers to kilobyte. These partitions need to be allotted to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. If the best fit algorithm is used, which partitions are NOT allotted to any process?

(a) 200 KB and 300 KB
(b) 200 KB and 250 KB
(c) 250 KB and 300 KB
(d) 300 KB and 400 KB

GATE CSE – 2015 NAT SET -2 (1-MARKS)

Q28. A computer system implements a 40-bit virtual address, page size of 8 kilobytes, and a 128-entry translation look-aside buffer (TLB organized into 32 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLB tag in bits is _____.

GATE CSE – 2015 NAT SET -1 (1-MARKS)

Q29. Consider a system with byte-addressable memory, 32 bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is _____

GATE CSE – 2015 NAT SET -2 (2-MARKS)

Q30. A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is _____ bits.

GATE CSE - 2014 NAT SET - 2(2-MARKS)

Q31. Consider the main memory system that consists of 8 memory modules attached to the system bus, which is one word wide. When a write request is made, the bus is occupied for 100 ns by the data, address, and control signals. During the same 100 ns, and for 500 ns thereafter, the addressed memory module executes one cycle accepting and storing the data. The internal operation of different memory modules may overlap in time, but only one request can be on the bus at any time. The maximum number of stores (of one word each) that can be initiated in 1 millisecond is _____

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2014 NAT SET - 3(1-MARKS)

Q32. A system uses 3 page frames for storing process pages in main memory. It uses the Least Recently Used (LRU) page replacement policy. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below?
4, 7, 6, 1, 7, 6, 1, 2, 7, 2

GATE CSE - 2014 MCQ SET - 2(2-MARKS)

Q33. A computer has twenty physical page frames which contain pages numbered 101 through 120. Now a program accesses the pages numbered 1, 2, …, 100 in that order, and repeats the access sequence thrice. Which one of the following page replacement policies experiences the same number of page faults as the optimal page replacement policy for this program?

(a) Least-recently-used
(b) First-in-first-out
(c) Last-in-first-out
(d) Most-recently-used

GATE CSE - 2014 NAT SET - 1(2-MARKS)

Q34. Assume that there are 3 page frames which are initially empty. If the page reference string is 1, 2, 3, 4, 2, 1, 5, 3, 2, 4, 6 the number of page faults using the optimal replacement policy is _____

GATE CSE - 2014 NAT SET - 3(2-MARKS)

Q35. Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 ms to search the TLB and 80 ms to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is _____

GATE CSE – 2013 MCQ (2-MARKS)

Q36. A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T1), which occupies exactly one page. Each entry of T1 stores the base address of a page of the second-level table (T2). Each entry of T2 stores the base address of a page of the third-level table (T3). Each entry of T3 stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16-way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.
What is the minimum number of page colors needed to guarantee that no two synonyms map to different sets in the processor cache of this computer?

(a) 2
(b) 4
(c) 8
(d) 16

GATE CSE – 2012 MCQ (2-MARKS)

Q37. Consider the virtual page reference string 1, 2, 3, 2, 4, 1, 3, 2, 4, 1 On a demand paged virtual memory system running on a computer system that has a main memory size of 3 pages frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacements policy. Then

(a) OPTIMAL < LRU < FIFO
(b) OPTIMAL < FIFO < LRU
(c) OPTIMAL = LRU
(d) OPTIMAL = FIFO

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2011 MCQ (1-MARKS)

Q38. Let the page fault service time to 10 ms in a computer with average memory access time being 20 ns. If one page fault is generated for every 10^6 memory accesses, what is the effective access time for the memory?

(a) 21 ns
(b) 30 ns
(c) 23 ns
(d) 25ns

GATE CSE – 2010 MCQ (1-MARKS)

Q39. A system uses FIFO policy for page replacement. It has 4 page frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order.
How many page faults will occur?

(a) 196
(b) 192
(c) 197
(d) 195

GATE CSE – 2009 MCQ (1-MARKS)

Q40. In which one of the following page replacement policies, Belady's anomaly may occur?

(a) FIFO
(b) Optimal
(c) LRU
(d) MRU

GATE CSE – 2009 MCQ (1-MARKS)

Q41. The essential content(s) in each entry of a page table is/are

(a) virtual page number
(b) page frame number
(c) both virtual page number and page frame number
(d) access right information

GATE CSE – 2009 MCQ (2-MARKS)

Q42. A multilevel page table is preferred in comparison to a single level page table for translating virtual address to physical address because

(a) it reduces the memory access time to read or write a memory location.
(b) It helps to reduce the size of the page table needed to implement the virtual address space of a process.
(c) it is required by the translation lookaside buffer.
(d) it helps to reduce the number of page faults in page replacement algorithms

GATE CSE – 2008 MCQ (2-MARKS)

Q43. A processor uses:
Physical address = 36 bits
Virtual address = 32 bits
Page size = 4 KB
PTE size = 4 bytes
Three-level page table with:

- bits 30–31 → 1st level index
- bits 21–29 → 2nd level index
- bits 12–20 → 3rd level index
- bits 0–11 → offset

Find number of bits required in PTE to address the next level page table (or page frame) at first, second, and third levels respectively.

(a) 20, 20 and 20
(b) 24, 24 and 24
(c) 24, 24 and 20
(d) 25, 25 and 24

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2007 MCQ (2-MARKS)

Q44. A virtual memory system uses First In First Out (FIFO) page replacement policy and allocates a fixed number of frames to a process. Consider the following statements:

P: Increasing the number of page frames allocated to a process sometimes increases the page fault rate.

Q: Some programs do not exhibit locality of reference.

Which one of the following is TRUE?

(a) Both P and Q are true, and Q is the reason for P.
(b) Both P and Q are true, but Q is not the reason for P.
(c) P is false, but Q is true.
(d) Both P and Q are false.

GATE CSE – 2007 MCQ (2-MARKS)

Q45. A process has been allocated 3 page frames. Assume that none of the pages of the process are available in memory initially.

The process makes the following sequence of page references (reference string):

1, 2, 1, 3, 7, 4, 5, 6, 3, 1

If optimal page replacement policy is used, how many page faults occur for the above reference string?

(a) 7
(b) 8
(c) 9
(d) 10

GATE CSE – 2007 MCQ (2-MARKS)

Q46. A process has been allocated 3 page frames. Assume that none of the pages of the process are available in memory initially.

The process makes the following sequence of page references (reference string):

1, 2, 1, 3, 7, 4, 5, 6, 3, 1

Least Recently Used (LRU) page replacement policy is a practical approximation to optimal page replacement.

For the above reference string, how many more page faults occur with LRU than with the optimal page replacement policy?

(a) 0
(b) 1
(c) 2
(d) 3

GATE CSE – 2006 MCQ (2-MARKS)

Q47. A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is

(a) 11 bits
(b) 13 bits
(c) 15 bits
(d) 20 bits

GATE CSE – 2006 MCQ (2-MARKS)

Q48. A computer system supports 32-bit virtual addresses as well as 32-bit physical addresses. Since the virtual address space is of the same size as the physical address space, the operating system designers decide to get rid of the virtual memory entirely. Which one of the following is true?

(a) Efficient implementation of multi-user support is no longer possible
(b) The processor cache organization can be made more efficient now
(c) Hardware support for memory management is no longer needed
(d) CPU scheduling can be made more efficient now

GATE CSE – 2005 MCQ (1-MARKS)

Q49. Increasing the RAM of a computer typically improves performance because:

(a) Virtual memory increases
(b) Larger RAMs are faster
(c) Fewer page faults occur
(d) Fewer segmentation faults occur

Check Out Our GATE Courses & Secure Your Dream Rank!

Answer key

1. (3)	**2. (1536)**	**3. (b,c)**	**4. (11)**
5. (d)	**6. (a)**	**7.(65468)**	**8. (6)**
9. (1028)	**10.(b, c,d)**	**11.(6596)**	**12. (c)**
13. (5)	**14. (c)**	**15. (0.6)**	**16. (a,d)**
17.(4108)	**18. (b)**	**19. (155 to 155.7)**	
20. (d)	**21.(b)**	**22. (b)**	**23. (d)**
24. (1)	**25. (384)**	**26. (a)**	**27. (a)**
28. (22)	**29. (4)**	**30. (36)**	**31. (10000)**
32. (6)	**33. (d)**	**34. (7)**	**35. (122)**
36. (c)	**37. (b)**	**38. (b)**	**39. (a)**
40. (a)	**41. (b)**	**42. (b)**	**43. (d)**
44. (b)	**45. (a)**	**46. (c)**	**47. (c)**
48. (b)	**49. (c)**		

HINTS & SOLUTIONS

1. ANSWER:- 3

Allocation of P1 (16 KB) using Best-Fit:

Eligible holes ≥ 16:

20, 25, 18

Best fit = 18 KB

Remaining hole:

18 − 16 = 2 KB

New holes: 20, 4, 25, 2, 7, 9, 15, 12

Allocation of P2 (9 KB) using Best-Fit:

Eligible holes ≥ 9:

20, 25, 9, 15, 12

Best fit = 9 KB

Remaining hole:

9 − 9 = 0 (hole disappears)

Remaining holes: 20, 4, 25, 2, 7, 15, 12

Holes smaller than 8 KB:

4 KB

2 KB

7 KB

Answer: 3

2. ANSWER:- 1536

TLB reach = Number of entries × Page size

Page size = 4 KB = 2^{12} bytes

Reach = 1 MB = 2^{20} bytes

Number of entries = $2^{20} / 2^{12} = 2^8 = 256$ entries

Virtual page number size:

Virtual address space = 64 GB = 2^{36} bytes

Page size = 2^{12}

Offset bits = 12

Virtual page number bits = 36 – 12 = 24 bits

Frame number size:

Physical address space = 1 GB = 2^{30} bytes

Frame size = 2^{12}

Number of frames = $2^{30} / 2^{12} = 2^{18}$

Frame number bits = 18 bits

Size of one TLB entry:

Process id = 4 bits

Page number = 24 bits

Frame number = 18 bits

Control field = 2 bits

Total bits per entry = 4 + 24 + 18 + 2 = 48 bits = 6 bytes

Total TLB size:

Entries = 256

Size per entry = 6 bytes

Total size = 256 × 6 = 1536 bytes

Answer: 1536 bytes

3. ANSWER:- B, C

(A): Possible

TLB miss → page table provides frame → cache may hit.

(B): Impossible

page table miss→ cache cannot hit.

(C): Impossible

Page table miss ⇒ page not in memory ⇒ cache cannot have it.

(D): Possible

Page fault ⇒ cache miss.

4. ANSWER:- 11

Page size = 4096 = 2^{12}⇒ page offset = 12 bits

Each inner page table entry = 8 bytes ⇒ entries per page = $2^{12}/2^3 = 2^9$

⇒ inner index = 9 bits

Total address bits = 32

So, $b + 9 + 12 = 32$

$b = 11$

Check Out Our GATE Courses & Secure Your Dream Rank!

5. ANSWER:- D

Given frame evolution matches future-aware replacement (pages replaced based on farthest future use), which is characteristic of optimal policy

6. ANSWER:- A

1. Find the Page Offset bits:

Page Size $= 2048$ bytes $= 2^{11}$ bytes $\Longrightarrow 11$ bits

2. Find the Virtual Page Number (VPN) bits:

Logical Address - Offset = 32 bits - 11 bits = 21 bits

Since 21 bits are used to index the pages, the maximum number of entries the page table can hold is:

$$\text{Maximum Entries} = 2^{21}$$

7. ANSWER:- 65468

Usable data per block = 64K−4 = 65536−4=655326 = 65532 bytes.

Blocks required = $\left[\frac{1M}{65532}\right] = \left[\frac{1,048,576}{65,532}\right] = 17$, since 16 blocks hold 16 × 65, 532 = 1, 0458, 512 bytes leaving 64 bytes for the last block.

Internal fragmentation is the unused space in the last block, so wasted =65,532−64=65,468 bytes, i.e., 65468, which matches the official key.

8. ANSWER:- 6

Optimal page replacement algorithm

Page numbers = 1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1,4

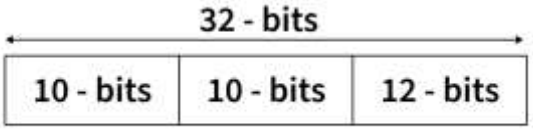
1 3 2 4 2 3 1 2 4 3 1 4											
2	2	2	2	2	2	2	2	3	3	3	3
3	3	3	3	1	1	1	1	1	1	1	1
1	1	1	1	4	4	4	4	4	4	4	4
H	H	H	H			H			H		

The number of page faults is **6**.

9. ANSWER:- 1028

As per the given information in the question,

Page Size = 4 KB, therefore no of bits required = $\log_2[2^{12}] = 12$ bits. And, as the system is of 32 bits. Therefore, the architecture of bits will be as follows:



As per the given convention,

X = minimum no of pages

Y = maximum no of pages

Coming to X, there will be a minimum of 3 pages and and for Y, there will be $2^0 + 2^{10}$ pages = 1 + 1024 = 1025 pages.

Hence, the total number of pages i.e. X + Y = 1025 + 3 = 1028.

10. ANSWER:- B, C, D

As per the property and functions of the Memory Management Unit (MMU).

the correct options are B, C, and D.

11. ANSWER:- 6596

The physical address corresponding to the virtual address 2500 (decimal) in this memory management system is 6596 (decimal).

This is determined by first identifying the virtual page number from the virtual address, which is page 1 (since page size is 2KB = 2048 bytes, 2500/2048=1), and the offset within that page (2500%2048 = 452). Since virtual page 1 maps to physical frame 3, the physical address is calculated as 3 * 2048 + 452 = 6596

12. ANSWER:- C

Given array,

int D[128] [128] Page frame holds 512 elements of the array D. A total 30 physical page

frames are allocated to a process that executes the following code snippet.

```
for (int i = 0; i < 128; i++)
for (int j = 0; j < 128; j++)
D[j][i] *= 10;
```

for each i, the loop executes 128 times.

Check Out Our GATE Courses & Secure Your Dream Rank!

$i = 0,1,2,3$	Page 0 Page 1
$i = 4,5,6,7$	
$i = 116,117,118,119$	Page 119

Since 30-page frames were allocated to the process.
Each physical frame holds 512 elements.
And array access is in column-major order.
D[j] [i] * = 10
The first 4 rows belong to one page.
Since memory contains 30 page frames, and array contains 32 pages.
Page required for array = 128/4=32
No. of page fault = 128×32 = 4096

13. ANSWER:- 5

Given,
Virtual Address Space bits 57 bits
Page size 4KB = 12 bits
Page number (57-12) = 45 bits
Page table entry= 8 bytes.
Each page can contain = 4KB/8B = 2^9 page table entries. So, we need
bits to index the page table.
So, the number of levels = 5

-

14. ANSWER:- C

According to the explanation in Galvin's textbook, a Translation Lookaside Buffer (TLB) is a fast, associative memory structure. Each TLB entry consists of a key (also known as a tag) and a corresponding value. When a virtual address is given, the TLB compares the key portion with all stored entries in parallel. If a match is found, the related value is returned immediately. Therefore, **Option A is TRUE**.
The TLB holds only a limited number of page table entries. When a virtual address causes a TLB hit, the system can directly retrieve the frame number corresponding to the page number without going to the full page table. Even if the actual data is not found in the cache (cache miss), the main memory can still be accessed using the frame number obtained from the TLB. This guarantees access to the required word in main memory. Hence,
Option B is also TRUE.
In contrast, an Inverted Page Table uses frame numbers as indices, rather than page numbers as in traditional page tables. For a given virtual address, the system may have to search multiple entries to find a matching page, and sometimes even all entries—especially when page entries for multiple processes share the same physical memory. This results in slower lookups in the worst case. Thus, memory access time can vary with different virtual addresses. Therefore,
Option C is FALSE, making it the correct answer.
In a system with hashed page tables, multiple virtual addresses can hash to the same value. In such cases, these entries are stored in a linked list format. Since linked lists require sequential traversal, access time can vary depending on the position of the required item in the list. The best case is constant time if the item is first, while the worst case may take linear time. So,
Option D is TRUE.

15. ANSWER:- 0.6

Reference	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Page	7	7	7	7	7	7	7	4	4	4	4	4	5	5	5
Frame 1		2	2	2	2	2	2	2	6	6	6	6	6	6	6
Frame 2			3	3	3	3	3	3	3	3	1	1	1	1	1
Frame 3					5	5	5	5	7	7	7	7	7	7	7
Page Fault	F	F	H	F	H	F	H	F	F	F	H	F	F	H	H

Total page fault=9
Total access=15
thus, page fault rate =9/15=0.6
Least recently used - Page replacement algorithm.

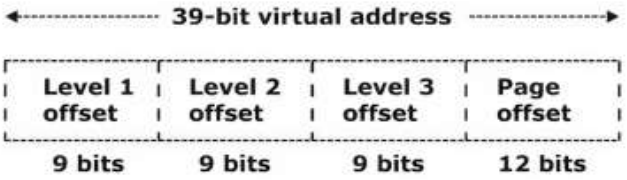
16. ANSWER:- A, D

Check Out Our GATE Courses & Secure Your Dream Rank!

Paging helps us to solve the issue of external fragmentation.
Paging incurs memory overheads.

17. ANSWER:- 4108

Since process P is using 2 GB physical memory and page size 2^{12} bytes.



So, Number of pages = $2^{31} / 2^{12} = 2^{19}$
There are 2^9 entries per page table in 3rd level.
So, we need = $2^{19}/2^9 = 2^{10}$ page tables in 3rd level
Now, it means 2^{10} entries in 2nd level so number of page tables in 2nd level $2^{10}/2^9 = 2$
page table
Now, in 1st level we have 2 entries hence we need only 1 page table in 1st level.
Total number of page table = $2^{10} + 2 + 1 = 1027$ page table
Each page table has 2^9 entries and each entry size is 8 bytes.
So, total size of page table in bytes = $1027 \times 2^9 \times 8$ bytes = 4108 KB

18. ANSWER:- B

Explanation:

Option A: Again depends on the size of the first hole. It might happen the very first free hole is the worst possible hole. So in that case size of hole decided by worst fit = size of hole decided by first fit. Therefore, Option A is False.

Option B: Best fit will always the give smallest possible hole. So, either the hole picked will be smaller than or equal to the other hole sizes. So, this is TRUE.

Option C: Depends on the size of the first free hole. There is a very good chance that the size of the first hole is small than size of the size of the next hole. So, Option C is False.

Option D: The next free hole might be the worst possible hole or it's size may be bigger than the hole picked by best fit. So, this is also False.

19. ANSWER:- 155 to 155.7

- For all memory accesses in a system with virtual memory we need Virtual Address to Physical Address translation and this goes through TLB.
- On TLB hit, we get the physical address.
- On TLB miss, we have to do page table access which always resides in physical memory (no page fault possible here).
- In the question it is given that a 1-level page table is used. So, TLB miss will need one physical memory access to get the physical address.
- Question mentions page fault rate as 10% and this should default to 10 page faults every 100 memory accesses. (Since TLB miss rate is 5% and for normal programs a TLB hit and page fault cannot happen for a memoryaccess (can happen for invalid memory accesses), it is also possible to consider page fault rate as 10% of all TLB misses. See the last part of the answer for this.

In the question page transfer time is given. This is different from page fault service time which includes the page transfer time + the memory access time as once the page is filled, a new memory access is initiated.

So, Average Memory Access Time Address Translation Time + Data Retrieval Time = TLB access time + TLB Miss ratio Page Table Access time + Main memory access time+ Page fault rate (Page fill Time + Address Translation Time+ Main memory access time) + Dirty Page Rate Page Write Time
= 20+0.05 x 100+100+0.1x
(5000+20+0.05×100+100)+0.1 x 0.2 × 5000
20+5+100+512.5+100 737.5 ns

PS: If the question had given page fault service time also as 5000 answer will be 20+0.05 x 100+0.9×100+0.1×5000+0.1×0.2x500025+90+500+100=715 1s

"Assume that the TLB hit ratio is 95%, page fault rate is 10%
If this statement is changed to
"Assume that the TLB hit ratio is 95%, and when TLB miss happens page fault rate is 10%"

Check Out Our GATE Courses & Secure Your Dream Rank!

Average Memory Access Time = Address Translation Time + Data Retrieval Time
= TLB access time + TLB Miss ratio × Page Table Access time+ Main memory access time+ Page fault rate × (Page fill Time + Address Translation Time+ Main memory access time) + Dirty Page Rate ×Page Write Time
$20 + 0.05 \times 100 + 100 + 0.05 \times 0.1 \times (5000 + 20 + 0.05 \times 100 + 100) + 0.05 \times 0.1 \times 0.2 \times 5000$
$20 + 5 + 100 + 25.625 + 5$
$= 155.625$
Ideally the answer key should be 715-738 due to the confusion in the meaning of page transfer time as most standard resources use page fault service time instead.
If we assume page fault rate is given "only when TLB miss happens answer should be 155-155.7

20. ANSWER:- D

At most, the number of distinct virtual addresses translated without any TLB miss is calculated as follows:

- Each TLB entry corresponds to one page.
- Number of TLB entries = 128 = 2^7
- Page size = 8 kB = 8 × 1024=2^{13} bytes
- Word size = 4 bytes
- Number of words per page (page size in words) = $2^{13}/2^2=2^{11}$ (since 4 bytes = 2^2 bytes)
- Hence, total virtual addresses translated without TLB miss = number of TLB entries ×× number of words per page = $2^7 \times 2^{11} = 2^{18} = 256 \times 2^{10}$

In decimal form, this is 256 × 1024 = 262,144 virtual addresses.

21. ANSWER:- B

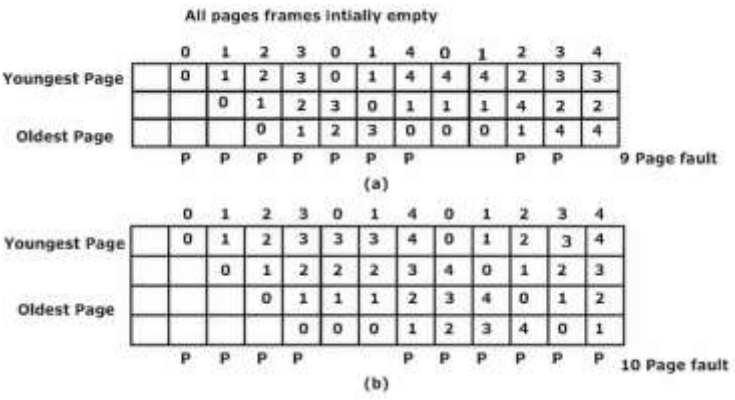
Let the page fault rate be 'P' then, average memory access time
Average memory access time = (1− page fault rate) × memory access time when no page fault + Page fault rate × Memory access time when page fault.
$X = (1 - P) M + D \times P$
$X = M \cdot PM + DP$
$(X - M) = P (D - M)$
$\Rightarrow P = (X - M) / (D - M)$

22. ANSWER:- B

Only fifo and fifo based algorithms suffer from belady's anomaly, Since random algorithm may follow fifo , hence it may suffer from belady's anomaly.

23. ANSWER:- D

In Belady's anomaly is the phenomenon in which increasing the number of page frames results in an increase in the number of page faults for certain memory access patterns proves that while using the First in First Out (FIFO) page replacement algorithm it is possible to have more page faults when increasing the number of page frames.
In some situations, FIFO page replacement gives more page faults when increasing the number of page frames.
For example, if we consider reference string 3 2 1 0 3 2 4 3 2 1 0 4 and 3 slots, we get 9 total page faults, but if we increase slots to 4, we get 10 page faults.



24. ANSWER:- 1

Firstly, we can consider LIFO (Last In First Out) →
a_1 to a_{10} will result in page faults = 10-page faults,
After that a_{11} will replace a_{10} (last in is a_{10}), a_{12} replaces a_{11} and ...till a_{20} = 10-page faults will occur and a_{20} will be at the top of the stack and $a_9 \ldots a_1$ are remained as such.
Then a_1 to a_9 are already present.
So, 0-page faults will be there from a_1 to a_9.
a_{10} will replace a_{20}, a_{11} will replace a_{10} and so on = So 11-page faults.
So, total faults will be 10+10+11 = 31.
Second Optimal Page Replacement Policy →
a_1 to a_{10} = 10 page faults,

Check Out Our GATE Courses & Secure Your Dream Rank!

a_{11} will replace a_{10} because among a1 to a_{10}, a_{10} will be used later, a_{12} will replace a_{11} and so on = 10 page faults.

A_{20} will be top of stack and $a_9 \ldots a_1$ are remained as such.

A_1 to a_9 = 0 page fault. A_{10} will replace a1 because it will not be used afterwards and so on, a_{10} to a_{19} will have 10 page faults.

So, no page fault for a_{20}.

Total = 10+ 10 +10 = 30.

So, the difference between LIFO - Optimal = 1

25. ANSWER:- 384

Size of memory = 2^{40} and Page size = 16 KB = 2^{14}

No. of pages = size of Memory / page size = $2^{40}/ 2^{14} = 2^{26}$

Size of page table = 2^{26} * 48 / 8 bytes = 2^{26} * 6 B = 384 MB

26. ANSWER:- A

Given page reference sequence- 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3

LRU:

	3	8	2	3	9	1	6	3	8	9	3	6	2	1	1
							1	1	1	1	1	1	2	2	2
					9	9	9	9	9	9	9	9	9	9	9
			2	2	2	2	2	2	8	8	8	8	8	1	1
		8	8	8	8	8	6	6	6	6	6	6	6	6	6
	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
	F	F	F	H	F	F	F	H	F	H	H	H	F	F	H

∴ Number of page faults = 9

FIFO:

	3	8	2	3	9	1	6	3	8	9	3	6	2	1	3	1
						1	1	1	1	1	1	1	1	1	1	1
					9	9	9	9	9	9	9	9	2	2	2	2
			2	2	2	2	2	2	8	8	8	8	8	8	8	8
		8	8	8	8	8	6	6	3	3	3	3	3	3	3	3
	3	3	3	3	3	3	6	6	6	6	6	6	6	6	6	6
	F	F	F	H	F	F	F	H	F	H	H	H	F	F	H	

No of page faults=9

27. ANSWER:- A

Best fit allocates the smallest block among those that are large enough for the new process. So, the memory blocks are allocated in the following order:

357 ---> 400

210 ---> 250

468 ---> 500

491 ---> 600

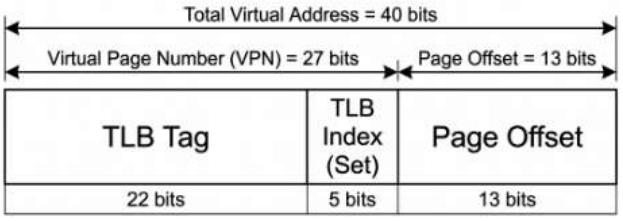
So, the remaining blocks are of 300 KB and 200 KB

28. ANSWER:- 22

TLB maps a virtual address to the physical address of the page. (The lower bits of page address (page offset bits) are not used in TLB as they are the same for virtual as well as physical addresses). Here, for 8KB page size we require 13 page offset bits.

In TLB, we have 32 sets and so virtual address space is divided into 32 using 5 set bits. (Associativity doesn't affect the set bits as they just adds extra slots in each set).

So, number of tag bits=40−5−13=22 bits.



29. ANSWER:- 4

Number of entries in page table

= 2^{32} / 4KB = 2^{20}

Size of page table = (No. page table entries) * (Size of an entry)

= 2^{20} * 4 bytes

= 2^{22}

= 4 Megabytes

30. ANSWER:- 36

Given page size = 8KB = 2^{13} B

PAS = 32 bit

∴ No. of frames =PA/Page size = $2^{32}/ 2^{13}= 2^{19}$

Also, it is given that each page table entry contains a valid bit, a dirty bit, 3 permission bits:

= 5 bits reserved

So, one Page table entry size is

= 19+5 = 24 bits = 3 bytes

Now, Page table size = No. of entries × Entry size

24×2^{20} = No. of entries × 3

No. of entries = $8 \times 2^{20} = 2^{23}$

∴ Virtual Address size = No. of entries × Page size = $2^{23} \times 2^{13} = 2^{36}$

∴ Virtual Address Space = 36 bits

Check Out Our GATE Courses & Secure Your Dream Rank!

31. ANSWER:- 10000

Bus is the bottleneck

Each request occupies bus for 100 ns

Total time = 1 ms = 10^6ns

Number of stores = $10^6/100 = 10000$

32. ANSWER:- 6

Page Requested	4	7	6	1	7	6	1	2	7	2
Frame 1	4	4	4	7	6	1	7	6	1	1
Frame 2	-	7	7	6	1	7	6	1	2	7
Frame 3	-	-	6	1	7	6	1	2	7	2
Status (Hit/Fault)	FAULT	FAULT	FAULT	FAULT	HIT	HIT	HIT	FAULT	FAULT	HIT

33. ANSWER:- D

Access pattern is strictly sequential over 100 pages with only 20 frames. Optimal will always evict the page that is farthest in future, which in this case behaves like evicting the most recently used page.

34. ANSWER:- 7

Simulate optimal:

1 Fault
2 Fault
3 Fault
4 Fault (replace 3)
2 hit
1 hit
5 Fault (replace 4)
3 Fault (replace 5)
2 hit
4 Fault (replace 1)
6 Fault (replace 3)

Total faults = 7

35. ANSWER:- 122

On hit: 10 + 80 = 90

On miss: 10 + 80 + 80 = 170

EMAT = 0.6 × 90 + 0.4 × 170
= 54 + 68
= 122

36. ANSWER:- C

Explanation:

- The cache is 1 MB, 16-way set associative, with 64-byte blocks.
- Number of cache sets = (Cache size) / (Block size* associativity) = $\frac{20^{20}}{64\times16} = \frac{2^{20}}{2^{10}} = 2^{10} = 1024$ sets.
- Virtual address = 46 bits, Physical address = 32 bits.
- Page size calculated as 2^{13} bytes = 8 KB (8192 bytes).
- The index bits for cache set = 10 bits (since 1024 sets).
- Out of these indexing bits, 13 bits correspond to page offset, so the remaining bits contributing to cache indexing from the virtual address result in 3 bits (16-13) that can cause synonyms to map to different sets.
- So, we need 2^3 = 8 different page colors to ensure synonyms (different virtual pages mapping to the same physical page) don't map to different cache sets.

37. ANSWER:- B

For FIFO: total no of page faults are 6 (depicted in red)

1	1	1	1	4	4	4	4	4	4
	2	2	2	2	1	1	1	1	1
		3	3	3	3	3	2	2	2

For optimal: total no of page faults are 5 (depicted in red)

1	1	1	1	1	1	1	1	1	1
	2	2	2	4	4	4	4	4	4
		3	3	3	3	3	2	2	2

For LRU: total no of page faults are 9 (depicted in red)

1	1	1	1	4	4	4	2	2	2
	2	2	2	2	2	3	3	3	1
		3	3	3	1	1	1	4	4

The Optimal will be 5, FIFO 6 and LRU 9. so, OPTIMAL < FIFO < LRU option (B) is correct answer.

Check Out Our GATE Courses & Secure Your Dream Rank!

38. ANSWER:- B

The effective memory access time (EAT) is calculated as:

Given:

- Page fault service time S = 10ms = 10000000 ns
- Memory access time ns m = 20
- Page fault rate p = $\frac{1}{10^6}$

Formula:

EAT = (1-p)m + pS

Substitute values:

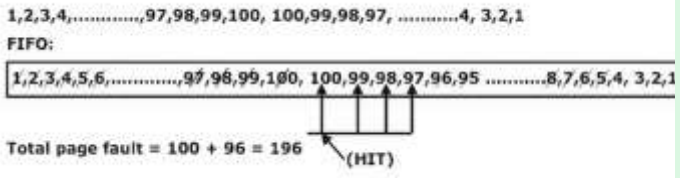
EAT $= \left(1-\frac{1}{10^6}\right)\times 20+\frac{1}{10^6}\times 10,000,000 =$

$20\times\ 0.999999+10=19.99998+10$

Effective access time = 30ns

39. ANSWER:- A

The first 100 accesses cause 100 page faults. Page 1 …….. Page 100 causes 100 faults



Now, when they are accessed in reverse order page 100, 99, 98, 97 are already present. So, we get a total of 196 page faults.

40. ANSWER:- A

Only FIFO and FIFO based algo's suffer from belady's anomaly

41. ANSWER:- B

Virtual Page Number:

The virtual page number (VPN) is not stored inside the page table entry. Instead, the VPN is used as an index to access the page table.

So option (A) is incorrect.

Page Frame Number:

The main purpose of a page table entry (PTE) is to store the physical frame number corresponding to a virtual page.

Thus page frame number is essential.

So option (B) is correct.

Both VPN and PFN:

Since VPN is used as an index and not stored in the entry, this is incorrect.

So option (C) is incorrect.

Access Right Information:

Access bits (read/write/execute, valid bit, dirty bit etc.) are useful but not essential for the basic mapping function.

So option (D) is incorrect.

42. ANSWER:- B

(B) it helps to reduce the size of page table needed to implement the virtual address space of a process.

(A) False → Multilevel paging **increases** access time.

(B) True → Only required page table portions are allocated.

(C) False → TLB works with any page table.

(D) False → Paging structure does not affect page faults.

43. ANSWER:- D

Physical address = 36 bits

Virtual address = 32 bits

Page size = 4 KB = 2^{12}

PTE size = 4 bytes

Number of page frames = $2^{36}/2^{12} = 2^{24}$

Frame number = 24 bits.

Page number = 20 bits.

Virtual address split:

- 2 bits → Level 1
- 9 bits → Level 2
- 9 bits → Level 3
- 12 bits → Offset

The innermost page table refers to the actual page frame. Hence the innermost page table requires 24 bits in it's PTE.

Now to index innermost page table we need 9 bits (given)

So number of such innermost page tables possible in physical memory

$$= 2^{36}/2^{9} = 2^{25}$$

Hence the second level page table requires 25 bits in its PTE to refer to the next page table. Similarly the 3rd level page table requires 24 bits in its PTE to refer to the next page table.

Correct Answer: (D) 25, 25 and 24

Check Out Our GATE Courses & Secure Your Dream Rank!

44. ANSWER:- B

Statement P:

In FIFO page replacement, increasing the number of frames can sometimes increase the page fault rate.

This phenomenon is called **Belady's Anomaly**.

So, P is **true**.

Statement Q:

Some programs do not exhibit locality of reference.

Yes, certain access patterns (like cyclic references larger than frame size) may not show strong locality.

So, Q is **true**.

Is Q the reason for P?

No.

Belady's anomaly occurs due to the nature of FIFO replacement (it does not consider recency or optimality).

It is a property of the algorithm itself, not because programs lack locality.

Hence, Q is not the reason for P.

45. ANSWER:- A

Optimal replacement policy means a page which is "farthest" in the future to be accessed will be replaced next.

Frame 0	1				
Frame 1	~~2~~	~~7~~	~~4~~	~~5~~	6
Frame 2	3				

3 initial page faults for pages 1,2,3 and then for pages 7,4,5,6⟹ 7 page faults occur.

46. ANSWER:- C

LRU: 9 page faults

1	2	1	3	7	4	5	6	3	1
			3	3	3	5	5	5	1
	2	2	2	7	7	7	6	6	6
1	1	1	1	1	4	4	4	3	3
F	F	×	F	F	F	F	F	F	F

9-7 =2

48. ANSWER:- A

1. Multi-user support requires isolation:

- Virtual memory ensures process isolation, preventing one user/process from accessing another's memory
- Without virtual memory, mernory isolation must rely entirely on physical address segmentation, which is less flexible and error-prone.

2. Security Risks:

- No virtual memory = no memory protection mechanisms (like page permissions or protected address spaces)

Other Options Analysis:

- Option B: Cache efficiency doesn't inherently depend on the presence/absence of virtual memory.
- Option C: Hardware support is still required (e.g., for segmentation or direct memory access).
- Option D: CPU scheduling is unrelated to virtual memory

49. ANSWER:- C

With more RAM:

- More pages can stay in memory
- Fewer pages need to be swapped to disk
- Page fault rate decreases

Hence fewer page faults occur

Check Out Our GATE Courses & Secure Your Dream Rank!

5 File System and IO Scheduling

GATE CSE - 2026 NAT SET - 1(2-MARKS)

Q1. Consider a hard disk with a rotational speed of 15000 rpm. The time to move the read/write head from a track to its adjacent track is 1 millisecond. Initially, the head is on track 0. The number of sectors per track is 400, and the sector size is 1024 bytes.

It is necessary to transfer data from 10 randomly located sectors in each of the following tracks in the order: 5, 12, and 7.

The total time for the data transfer (in milliseconds) from the hard disk is _____.

(rounded off to one decimal place)

GATE CSE - 2026 NAT SET - 2(2-MARKS)

Q2. To keep track of free blocks in a file system, one of the two approaches is generally used – using bitmaps (bit vectors) or using linked lists. Consider that the linked list approach is used to keep track of free blocks in a file system.

Assume that:

- Disk size = 16 GB
- Block size = 2 KB
- Block numbers used are 32-bit long
- A single pointer of size 4 bytes is used in each block of the list to point to the next block of the list

The number of blocks required to hold the free disk block numbers is _____.

Note:

$1K = 2^{10}$ and $1G = 2^{30}$.

GATE CSE - 2025 NAT SET - 1(2-MARKS)

Q3. A disk of size 512 M bytes is divided into blocks of 64 K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a NULL pointer (also of 4 bytes). Suppose a file of 1 M bytes needs to be stored in the disk. Assume, 1 K = $2^{10}$and 1 M = 2^{20}. The amount of space in bytes that will be wasted due to internal fragmentation is _____

GATE CSE - 2025 NAT SET - 1(1-MARKS)

Q4. Suppose in a multiprogramming environment, the following C program segment is executed. A process goes into I/O queue whenever an I/O related operation is performed. Assume that there will always be a context switch whenever a process requests for an I/O, and also whenever the process returns from an I/O. The number of times the process will enter the ready queue during its lifetime (not counting the time the process enters the ready queue when it is run initially) is _____

```
Int main()
{
        int x = 0, i = 0;
         scanf("%d", &x);
         for(i = 0; i < 20; i++)
         {
            x = x + 20;
            printf("%d\n", x);
         }
        return 0;
}
```

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2024 NAT SET - 2(2-MARKS)

Q5. Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located randomly on the disk. Assuming average rotational latency, the total time (in seconds, rounded off to 2 decimal places) to read the entire file from the disk is _____

GATE CSE - 2024 NAT SET - 1(2-MARKS)

Q6. Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is _____

GATE CSE - 2022 NAT (2-MARKS)

Q7. Consider two files systems A and B, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B, respectively, then the value of n_A+n_B is_____.

GATE CSE - 2021 MCQ SET - 1(1-MARKS)

Q8. Consider a linear list based directory implementation in a file system. Each directory is a list of nodes, where each node contains the file name along with the file metadata, such as the list of pointers to the data blocks. Consider a given directory foo.

Which of the following operations will necessarily require a full scan of foo for successful completion?

(a) Creation of a new file in foo
(b) Deletion of an existing file from foo
(c) Renaming of an existing file in foo
(d) Opening of an existing file in foo

GATE CSE - 2020 MCQ (2-MARKS)

Q9. Consider the following five disk access requests of the form (request id, cylinder number) that are present in the disk scheduler queue at a given time:

(P, 155), (Q, 85), (R, 110), (S, 30), (T, 115)

Assume the head is positioned at cylinder 100. The scheduler follows Shortest Seek Time First scheduling to service the requests.

Which one of the following statements is FALSE?

(a) T is serviced before P.
(b) Q is serviced after S, but before T.
(c) The head reverses its direction of movement between servicing of Q and P.
(d) R is serviced before P.

GATE CSE - 2019 NAT (2-MARKS)

Q10. The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB.

GATE CSE - 2018 MCQ (1-MARKS)

Q11. The following are some events that occur after a device controller issues an interrupt while process L is under execution:

P. The processor pushes the process status of L onto the control stack
Q. The processor finishes the execution of the current instruction
R. The processor executes the interrupt service routine
S. The processor pops the process status of L from the control stack
T. The processor loads the new PC value based on the interrupt

Which of the following is the correct order in which the events above occur?

(a) QPTRS
(b) PTRSQ
(c) TRPQS
(d) QTPRS

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2017 SET 2 MCQ (1-MARKS)

Q12. In a file allocation system, which of the following allocation scheme(s) can be used if no external fragmentation is allowed?

1. Contiguous
2. Linked
3. Indexed

(a) 1 and 3 only
(b) 2 only
(c) 3 only
(d) 2 and 3 only

GATE CSE - 2018 NAT SET - 1(2-MARKS)

Q13. Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, …, 199), and 256 sectors per track (numbered as 0, 1, … 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2], [180, 134, 1], [60, 20, 0], [212, 86, 3], [56, 116, 2], [118, 16, 1]

Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____ .

GATE CSE - 2016 NAT SET - 1(2-MARKS)

Q14. Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____.

GATE CSE - 2015 NAT SET - 1(2-MARKS)

Q15. Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. The additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution) is _____ tracks.

GATE CSE - 2015 SET 2 NAT (2 MARKS)

Q16. Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of 50×10^6 bytes/sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512-byte sector of the disk is _____

GATE CSE - 2015 SET 1 NAT (2-MARKS)

Q17. Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is _____

GATE CSE - 2014 NAT SET - 2(1-MARKS)

Q18. A FAT (file allocation table) based file system is being used and the total overhead of each entry in the FAT is 4 bytes in size. Given a 100×10^6 bytes disk on which the file system is stored and data block size is 10^3 bytes, the maximum size of a file that can be stored on this disk in units of 10^6 bytes is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2014 NAT SET - 1(2-MARKS)

Q19. Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at cylinder 100, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105, 110, 135 and 145. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 90 is serviced after servicing _____ number of requests.

GATE CSE - 2013 MCQ (2-MARKS)

Q20. Consider a hard disk with 16 recording surfaces (0–15) having 16384 cylinders (0–16383) and each cylinder contains 64 sectors (0–63). Each sector stores 512 bytes. Data are organized cylinder-wise and the addressing format is (cylinder no., surface no., sector no.). A file of size 42797 KB is stored starting at disk location (1200, 9, 40). What is the cylinder number of the last sector of the file, if it is stored contiguously?

GATE CSE - 2012 MCQ (2-MARKS)

Q21. A file system with 300 Gbyte disk uses a file descriptor with 8 direct block addresses, 1 indirect block address and 1 doubly indirect block address. The size of each disk block is 128 Bytes and the size of each disk block address is 8 Bytes. The maximum possible file size in this file system is

(a) 3 KBytes
(b) 35 KBytes
(c) 280 Kbytes
(d) dependent on the size of the disk

GATE CSE - 2011 MCQ (2-MARKS)

Q22. An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 ms. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected.)

(a) 0.50s
(b) 1.50s
(c) 1.25s
(d) 1.00 s

GATE CSE - 2009 MCQ (2-MARKS)

Q23. Consider a disk system with 100 cylinders. The requests to access the cylinders occur in following sequence :
4, 34, 10, 7, 19, 73, 2, 15, 6, 20.
Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and the shortest seek time first policy is used ?

(a) 95 ms
(b) 119 ms
(c) 233 ms
(d) 276 ms

GATE CSE - 2009 MCQ (2-MARKS)

Q24. A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple (c,h,s), where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as ⟨0,0,0⟩, the 1st sector as ⟨0,0,1⟩, and so on.
The address ⟨400,16,29⟩ corresponds to sector number:

(a) 505035
(b) 505036
(c) 505037
(d) 505038

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2009 MCQ (2-MARKS)

Q25. A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple (c,h,s), where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as ⟨0,0,0⟩, the 1st sector as ⟨0,0,1⟩, and so on.
The address of 1039th sector is

(a) ⟨0,15,31⟩
(b) ⟨0,16,30⟩
(c) ⟨0,16,31⟩
(d) ⟨0,17,31⟩

GATE CSE - 2008 MCQ (1-MARKS)

Q26. The data blocks of a very large file in the Unix file system are allocated using

(a) contiguous allocation
(b) linked allocation
(c) indexed allocation
(d) an extension of indexed allocation

GATE CSE - 2008 MCQ (2-MARKS)

Q27. For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to

(a) non-uniform distribution of requests
(b) arm starting and stopping inertia
(c) higher capacity of tracks on the periphery of the platter
(d) use of unfair arm scheduling policies

GATE CSE - 2007 MCQ (1-MARKS)

Q28. Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk pack are respectively:

(a) 256 Mbyte, 19 bits
(b) 256 Mbyte, 28 bits
(c) 512 Mbyte, 20 bits
(d) 64 Gbyte, 28 bits

GATE CSE - 2005 MCQ (1-MARKS)

Q29. What is the swap space in the disk used for?

(a) Saving temporary HTML pages
(b) Saving process data
(c) Storing the super-block
(d) Storing device drivers

GATE CSE - 2005 MCQ (1-MARKS)

Q30. Normally user programs are prevented from handling I/O directly by I/O instructions in them. For CPUs having explicit I/O instructions, such I/O protection is ensured by having the I/O instruction privileged. In a CPU with memory mapped I/O, there is no explicit I/O instruction. Which one of the following is true for a CPU with memory mapped I/O?

(a) I/O protection is ensured by operating system routine(s)
(b) I/O protection is ensured by a hardware trap
(c) I/O protection is ensured during system configuration
(d) I/O protection is not possible

Check Out Our GATE Courses & Secure Your Dream Rank!

Answer key

1. (77.3)	**2.(16417)**	**3.(65468)**	**4. (21)**
5. (30.06)	**6. (4096)**	**7. (153)**	**8. (a,c)**
9.(b)	**10. (4)**	**11. (a)**	**12. (d)**
13. (85)	**14. (346)**	**15. (10)**	**16. (6.1)**
17. (14020)	**18. (99.6)**	**19. (3)**	**20.(1284)**
21. (b)	**22. (b)**	**23. (b)**	**24. (c)**
25. (c)	**26.(d)**	**27. (b)**	**28. (a)**
29. (b)	**30. (a)**		

HINTS & SOLUTIONS

1. **ANSWER:- 77.3**

Rotational speed = 15000 rpm
Tracks movement time = 1 ms per track
Sectors per track = 400
Tracks accessed = 5 → 12 → 7
10 sectors read from each track

Rotation time:

15000 rpm ⇒ 1 rotation = 60/15000 = 0.004 s = 4 ms
Average rotational latency = 4/2 = 2 ms
Total sector accesses = 10 × 3 = 30
Total rotational delay = 30 × 2 = 60 ms

Transfer time:

Time per sector = 4/400 = 0.01 ms
Transfer for 30 sectors = 30 × 0.01 = 0.3 ms

Seek time:

0 → 5 = 5 tracks
5 → 12 = 7 tracks
12 → 7 = 5 tracks
Total movement = 17 tracks
Seek time = 17 × 1 = 17 ms

Total time:

17 + 60 + 0.3 = 77.3 ms

2. **ANSWER:- 16417**

Disk size = 16 GB = 16×2^{30} bytes
Block size = 2 KB = $2 \times 2^{10} = 2^{11}$ bytes
Number of blocks = $(16 \times 2^{30}) / (2^{11})$
$= 16 \times 2^{19}$
$= 2^4 \times 2^{19}$
$= 2^{23}$
= 8,388,608 blocks

Each block number = 32 bits = 4 bytes
Total space needed for storing all block numbers:
= 8,388,608 × 4 bytes
= 33,554,432 bytes
= 32 MB
Block size = 2 KB = 2048 bytes
Each block stores:

- 4 bytes → pointer to next block
- Remaining space → block numbers

Remaining space: 2048 − 4 = 2044 bytes
Number of block numbers per block: 2044 / 4 = 511
Total block numbers = 8,388,608
Blocks needed = ceil(8,388,608 / 511)
= 16,417
Answer: 16,417 blocks

3. **ANSWER:- 65468**

Block size = 2^{16}bytes
Pointer = 4 bytes ⇒ usable data per block = $2^{16} - 4 = 65532$
File size = $2^{20} = 1048576$
Number of blocks = $[1048576/65532] = 17$
Data stored in 16 blocks = $16 \times 65532 = 1048512$
Remaining = $1048576 - 1048512 = 64$
Last block has 64 bytes data, rest wasted:
Wasted = $65532 - 64 = 65468$

4. **ANSWER:- 21**

scanf → 1 I/O
printf inside loop → 20 I/O
Each I/O causes return to ready queue once
Total = 1 + 20 = 21

5. **ANSWER:- 30.06**

Rotation speed = 6000 RPM ⇒ 100 rotations/sec ⇒ 1 rotation = 0.01 s
Average rotational latency = 0.005 s
Time per sector transfer = 0.01 / 500 = 0.00002 s
For each sector (random):
seek + latency + transfer = 0.005 + 0.005 + 0.00002 = 0.01002 s
Total time = 3000 × 0.01002 = 30.06 s

6. **ANSWER:- 4096**

Total size = 512×2^{30}
Data per track = $4096 \times 1024 = 2^{12} \times 2^{10} = 2^{22}$
Tracks per cylinder = 32

Check Out Our GATE Courses & Secure Your Dream Rank!

Data per cylinder = $32 \times 2^{22} = 2^5 \times 2^{22} = 2^{27}$
Number of cylinders = $2^{39}/2^{27} = 2^{12} = 4096$

7. ANSWER:- 153

Fa: Contiguous allocation: 100 blocks (Array)
Fs: Linked allocation: 100 blocks
We have moved total data of 100
1. FA: 50*2 =100+1 (Total disk accesses)
2. Contiguous Allocation can directly traverse to the 50th element as it is already given the file control blocks are already present. Now, we need 50 operations to read the rest of the 50 elements and another 50 operations to write those 50 blocks and 1 operation is needed to write the new block. Therefore, total operations in this case (50+50+1)

- Linked Allocation needs to traverse to the 50th element.

Now, simply change the pointers just like the Linked list where adding an element in the middle of it.
So, here it takes ' 50 operations to read the first 50 elements and 2 operations to change the pointer of the 50 th block and a new block. So, 52 operations in this case.
$n_B = 50+1+1 = 52$
$n_A+n_B = 101+52 = 153$

8. ANSWER:- A,C

Creation requires checking for duplicate name → full scan
Renaming requires checking new name uniqueness → full scan
Deletion and opening can stop once found

9. ANSWER:- B

Consider the following five disk access requests of the form (request id, cylinder number):
(P, 155), (Q, 85), (R, 110), (S, 30), (T, 115)
Head initially at 100. Using SSTF:
Distances from 100:
R(110)=10, T(115)=15, Q(85)=15, P(155)=55, S(30)=70
Order:
R → T → Q → S → P
Check statements:

(a) T before P → true
(b) Q after S but before T → false (Q comes before S and after T)
(c) Direction changes between Q and P → true (85 → 30 → 155)
(d) R before P → true

10. ANSWER:- 4

Given 12 direct, 1 single indirect, 1 double indirect pointers
Size of Disk block =4kB
Disk Block Address =32 bit =4B
Number of addresses/disk block= 4KB/4B = 2^{10}
Maximum possible file size= 12×4 kB + $2^{10} \times 4$kB + $2^{10} \times 2^{10} \times 4$kB
=4.00395 GB≃4 GB
Hence 4GB is the correct answer.

11. ANSWER:- A

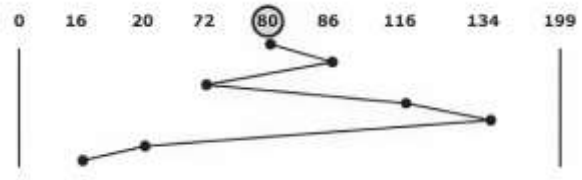
Execution flow:
Finish current instruction → Q
Save state → P
Load ISR address → T
Execute ISR → R
Restore state → S

12. ANSWER:- D

Contiguous allocation suffers from external fragmentation
Linked and indexed do not

13. ANSWER:- 85

Head starts at 80.



In SSTF, Total Head movements = (86-80) + (86-72) + (134-72) + (134-16) = 200
By 200 movements, Power dissipated: P1 = 0.2 * 200 = 40 mW
In reversing head direction once, Power dissipated = 15 mW
The total number of times head changes its direction = 3

Check Out Our GATE Courses & Secure Your Dream Rank!

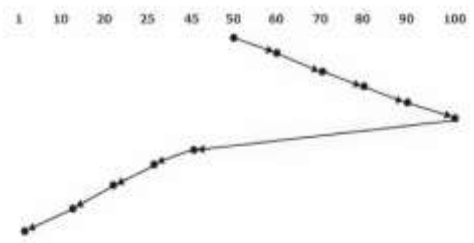
In reversing head direction, power dissipated: P2 = 3 * 15 = 45 mW
Total power consumption (in mW) is -
P1 + P2 = 40 mW + 45 mW = 85 Mw

14. ANSWER:- 346

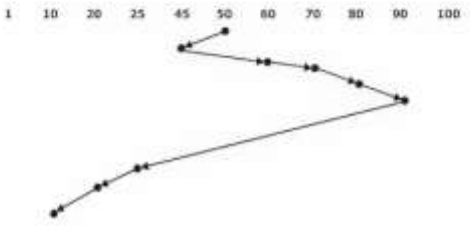
I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10.
The head is initially at cylinder number 63.
C-LOOK scheduling algorithm is used.
Firstly, we start from 63 and move further.
63 → 87 → 92 → 121 → 191 = 24 + 5 + 29 + 70 movements = 128
191 → 10 movements = 181
10 → 11 → 38 → 47 = 1 + 27 + 9 movements = 37
Total = 128 + 181 + 37 = 346

15. ANSWER:- 10

SCAN



∴ Total head movements,
= 10 + 10 + 10 + 10 + 10 + 55 + 20 + 5 + 10
= 140
SSTF:



∴ Total head movements
= 5 + 15 + 10 + 10 + 65 + 5 + 10 + 10
= 130
∴ Additional distance that will be traversed by R/W head is
= 140 - 130
= 10

16. ANSWER:- 6.1

Rotation time = $60/15000 = 0.004$s = 4 ms
Average rotational delay = 2 ms
Average seek time = 2 × 2 = 4 ms
Disk transfer time = $512/(50 \times 10^6) \approx 0.00001024$ s ≈ 0.01024 ms
Controller time = 10 × 0.01024 = 0.1024 ms
Total time = 4 + 2 + 0.01024 + 0.1024 ≈ 6.1 ms

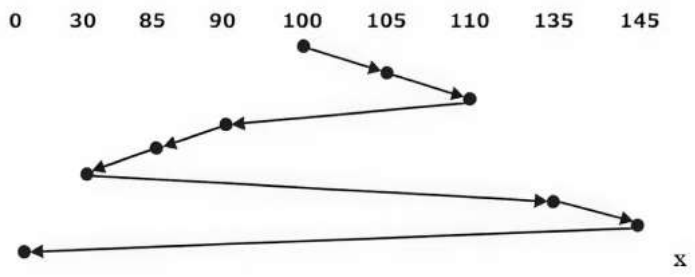
17. ANSWER:- 14020

Rotation time = $60/10000 = 0.006$s = 6 ms
Average rotational latency = 3 ms
Transfer time per sector = $6/600 = 0.01$ms
Time per sector = 4 + 3 + 0.01 = 7.01 ms
Total time = $2000 \times 7.01 = 14020$ms

18. ANSWER:- 99.6

Disk block size is 10^3
Number of entries in FAT
= Total disk size/ disk block size
= $100 \times 10^6 / 10^3$
= 100×10^3
∴ Total overhead = $100 \times 10^3 \times 4$B
= 400×10^3 B
= 0.4×10^6 B
So, maximum size of file that can be stored in disk is,
Total disk size - Total overhead
= $100 \times 10^6 - 0.4 \times 10^6$
= 99.6×10^6 B

19. ANSWER:- 3



After servicing three requests, 90 is serviced, i.e.,
100 → 105 → 110 → 90
So, 3 is the correct answer as 90 is serviced after 100 → 105 → 110 → 90.

20. ANSWER:- 1284

File size = 42797 × 1024bytes
Sector size = 512 bytes ⇒ total sectors = 42797 × 2 = 85594

Check Out Our GATE Courses & Secure Your Dream Rank!

Sectors per cylinder = $16 \times 64 = 1024$
Starting offset within cylinder = $9 \times 64 + 40 = 616$
Total sectors from cylinder start = $616 + 85594 - 1 = 86209$
Cylinders advanced = $86209/1024 \approx 84$
Final cylinder = $1200 + 84 = 1284$

21. ANSWER:- B
It is given data as the disk block is of size 128B.
One direct block addressing will be pointing to 8 disk blocks = 8*128 B = 1 KB
Singly Indirect block addressing will be pointing to 1 disk block which has 128/8 disc block addresses = (128/8)*128 B = 2 KB
Doubly indirect block addressing will be pointing to 1 disk block which has 128/8 addresses to disk blocks which has 128/8 addresses to disk blocks = 16*16*128 B = 32 KB
Maximum file size possible = 1 KB + 2 KB + 32 KB = 35 KB

22. ANSWER:- B
Rotation time = $60/6000 = 0.01s = 10$ ms
Average rotational latency = 5 ms
Time per access = 10 + 5 = 15 ms
Total time = $100 \times 15 = 1500$ms = 1.50 s

23. ANSWER:- B
Head = 50
Requests = 4, 34, 10, 7, 19, 73, 2, 15, 6, 20
Policy = SSTF
1 cylinder movement = 1 ms
Order (SSTF):
$50 \rightarrow 34 \rightarrow 20 \rightarrow 19 \rightarrow 15 \rightarrow 10 \rightarrow 7 \rightarrow 6 \rightarrow 4 \rightarrow 2 \rightarrow 73$
Total movement:
$|50-34| = 16$
$|34-20| = 14$
$|20-19| = 1$
$|19-15| = 4$
$|15-10| = 5$
$|10-7| = 3$
$|7-6| = 1$
$|6-4| = 2$
$|4-2| = 2$
$|2-73| = 71$
Total = 119 ms

24. ANSWER:- C
Surfaces = 10 × 2 = 20
Sectors per track = 63
Sector number formula:
Sector number = c × (surfaces × sectors) + h × sectors + s
= 400 × (20 × 63) + 16 × 63 + 29
= 400 × 1260 + 1008 + 29
= 504000 + 1037
= 505037

25. ANSWER:- C
Surfaces = 10 × 2 = 20
Sectors per track = 63
Since 0th sector = ⟨0,0,0⟩,
Cylinder:
1039 ÷ (20×63) = 1039 ÷ 1260 = 0
h,s position:
1039 ÷ 63 = 16 remainder 31
h = 16
s = 31
option C is correct

26. ANSWER:- D
The Unix file system uses an i-node structure.
An i-node contains:

- Direct block pointers
- Single indirect pointer
- Double indirect pointer
- Triple indirect pointer

For very large files, the system uses multi-level indirect pointers (single, double, triple). This is not simple indexed allocation, but an extension of indexed allocation using multiple levels of indexing.
Hence, the correct answer is (D).

27. ANSWER:- B
Seek time consists of:

- Acceleration phase (arm starts moving)
- Constant velocity phase
- Deceleration phase (arm stops)

Check Out Our GATE Courses & Secure Your Dream Rank!

Because of inertia, the disk arm must accelerate and decelerate.
For short seeks → most time is spent accelerating and decelerating.
For long seeks → arm reaches higher speed and stays longer in constant velocity.
Hence, seek time is non-linear with respect to seek distance.
Options (A), (C), and (D) do not affect the physical seek time mechanism.
Therefore, the correct answer is (B).

28. ANSWER:- A

Total sectors = $16 \times 128 \times 256$

$= 2^4 \times 2^7 \times 2^8 = 2^{19}$ sectors

Total capacity:

Each sector stores 512 bytes

Total capacity = $2^{19} \times 2^9$ bytes = 2^{28} bytes

2^{28} bytes = 256 Mbytes

Bits required to specify a sector:

Total sectors = 2^{19}

So, bits required = 19 bits

Correct option: **(A)**

29. ANSWER:- B

Swap space stores process pages (data, stack, heap) that are temporarily moved out of RAM.

30. ANSWER:- A

In memory-mapped I/O, protection is enforced via memory protection (OS-controlled access to address space)

Check Out Our GATE Courses & Secure Your Dream Rank!

DATABASE MANAGEMENT SYSTEM

“Scan Here for Video Solution”

Link to GfG corresponding Short Notes: GATE

“A well-designed *Database* today, ensures a scalable system tomorrow. Design it right, Normalize it well, and **Optimize** for performance. Because Good Data Design is the foundation of every powerful application.”

LEGEND OF

DATABASE MANAGEMENT SYSTEM

Vijay Kumar Agarwal Sir

An Inspirer. A Mentor. A Guide.

1 Functional Dependency and Normalization

GATE CSE- 2026 SET 2 MCQ (2-MARKS)

Q1. In the context of schema normalization in relational DBMS, consider a set F of functional dependencies. The set of all functional dependencies implied by F is called the closure of F. To compute the closure of F, Armstrong's Axioms can be applied. Consider X, Y, and Z as sets of attributes over a relational schema. The three rules of Armstrong's axioms are described as follows.
Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$
Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any Z
Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$
The additional rule of Union is defined as follows.
Union: If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$
It can be proved that the additional rule of Union is also implied by the three rules of Armstrong's Axioms. Listed below are four combinations of these three rules. Which one of these combinations is both necessary and sufficient for the proof?

(a) Reflexivity, Augmentation, and Transitivity
(b) Reflexivity and Augmentation
(c) Transitivity
(d) Augmentation and Transitivity

GATE CSE-2026 SET 1 MSQ (1-MARKS)

Q2. Let P, Q, R and S be the attributes of a relation in a relational schema. Let $X \rightarrow Y$ indicate functional dependency in the context of a relational database, where $X, Y \subseteq \{P, Q, R, S\}$.
Which of the following options is/are always true?

(a) If $\{P, Q\} \rightarrow \{R\}$ and $\{P\} \rightarrow \{R\}$, then $\{Q\} \rightarrow \{R\}$
(b) If $\{P, Q\} \rightarrow \{R\}$, then ($\{P\} \rightarrow \{R\}$ or $\{Q\} \rightarrow \{R\}$)
(c) If $\{P\} \rightarrow \{R\}$ and $\{Q\} \rightarrow \{S\}$, then $\{P, Q\} \rightarrow \{R, S\}$
(d) If $\{P\} \rightarrow \{R\}$, then $\{P, Q\} \rightarrow \{R\}$

GATE CSE- 2026 SET 1 NAT (2-MARKS)

Q3. Consider a relational database schema with a relation R(A, B, C, D). If {A, B} and {A, C} are the only two candidate keys of the relation R, then the number of super keys of relation R is _____ (answer in integer)

GATE CSE-2026 SET 1 MSQ (1-MARKS)

Q4. In the context of relational database normalization, which of the following statements is/are true?

(a) It is always possible to obtain a dependency-preserving 3NF decomposition of a relation
(b) It is always possible to obtain a dependency-preserving 1NF decomposition of a relation
(c) It is not always possible to obtain a dependency-preserving BCNF decomposition of a relation
(d) It is not always possible to obtain a dependency-preserving 2NF decomposition of a relation

GATE CSE-2025 SET- 2 MSQ (2-MARK)

Q5. Consider the following relational schema along with all the functional dependencies that hold on them.
R1(A,B,C,D,E):{D→E,EA→B,EB→C}
R2(A,B,C,D):{A→D,A→B,C→A}

(a) R1 is in 3NF
(b) R2 is in 3NF
(c) R1 is NOT in 3NF
(d) R2 is NOT in 3NF

GATE CSE-2025 SET- 1 MSQ (2-MARKS)

Q6. Consider a relational schema team(name , city , owner) with functional dependencies {name → city , name → owner} .
The relation team is decomposed into two relations,

t1(name , city) and t2(name , owner). Which of the following statement(s) is/are TRUE?

(a) The relation team is NOT in BCNF.
(b) The relations t1 and t2 are in BCNF.
(c) The decomposition constitutes a lossless join.
(d) The relation team is NOT in 3NF.

GATE CSE-2024 SET- 1 MSQ (2-MARKS)

Q7. Which of the following statements about a relation R in first normal form (1NF) is/are TRUE ?

(a) R can have a multi-attribute key
(b) R cannot have a foreign key
(c) R cannot have a composite attribute
(d) R cannot have more than one candidate key

GATE CSE-2024 SET-1 MSQ (2-MARKS)

Q8. The symbol → indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

(a) (X,Y)→(Z,W) implies X→(Z,W)
(b) (X,Y)→(Z,W) implies (X,Y)→Z
(c) ((X,Y)→Z and W→Y) implies (X,W)→Z
(d) (X→Y and Y→Z) implies X→Z

GATE CSE-2024 SET- 2 NAT (2-MARKS)

Q9. A functional dependency $F: X \rightarrow Y$ is termed as a useful fimctional dependency if and only if it satisfies all the following three conditions:

- X is not the empty set.
- Y is not the empty set.
- Intersection of X and Y is the empty set.

For a relation R with 4 attributes, the total number of possible useful functional dependencies is ___

GATE CSE- 2023 MCQ (1-MARKS)

Q10. Which one of the options given below refers to the degree (or arity) of a relation in relational database systems?

(a) Number of attributes of its relation schema.
(b) Number of tuples stored in the relation.
(c) Number of entries in the relation.
(d) Number of distinct domains of its relation schema.

GATE CSE-2022 NAT (1-MARKS)

Q11. Consider a relation R(A,B,C,D,E) with the following three functional dependencies.
AB→C; BC→D; C→E
The number of super keys in the relation R is...

GATE CSE-2022 MCQ (1-MARKS)

Q12. In a relational data model, which one of the following statements is TRUE?

(a) A relation with only two attributes is always in BCNF.
(b) If all attributes of a relation are prime attributes, then the relation is in BCNF.
(c) Every relation has at least one non-prime attribute.
(d) BCNF decompositions preserve functional dependencies.

GATE CSE-2021 SET-2 MSQ (2-MARKS)

Q13. Suppose the following functional dependencies hold on a relation U with attributes P, Q R, S, and T:

P → QR

RS → T

Which of the following functional dependencies can be inferred from the above functional dependencies?

(a) PS → T
(b) PS → Q
(c) P → R
(d) R → T

GATE CSE-2021 SET-1 MCQ (2-MARKS)

Q14. Consider the relation R(P, Q, S, T, X, Y, Z, W) with the following functional dependencies.
PQ → X; P→YX ; Q→ Y; Y → ZW
Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes.

D1 : R=[(P, Q, S,T); (P,T, X); (Q,Y); (Y, Z,W)]
D2 : R= [(P, Q,S); (T, X); (Q,Y); (Y,Z,W)]
Which one of the following options is correct?

(a) Both D1 and D2 are lossless decompositions.
(b) Both D1 and D2 are lossy decompositions.
(c) D1 is a lossless decomposition, but D2 is a lossy decomposition.
(d) D1 is a lossy decomposition, but D2 is a lossless decomposition.

GATE CSE-2020 MCQ (2-MARKS)

Q15. Consider a relational table R that is in 3NF, but not in BCNF. Which one of the following statements is TRUE?

(a) A cell in R holds a set instead of an atomic value.
(b) R has a nontrivial functional dependency X → A, where x is not a super key and A is a non-prime attribute and X is not a proper subset of any key.
(c) R has a nontrivial functional dependency X → A, where x is not a super key and A is a non-prime attribute and X is a proper subset of some key.
(d) R has a nontrivial functional dependency X → A, where X is not a super key and A is a prime attribute.

GATE CSE-2019 MCQ (2-MARKS)

Q16. Let the set of functional dependencies F = {QR → S, R → P, S → Q} hold on a relation schema X = (PQRS). X is not in BCNF. Suppose X is decomposed into two schemas Y and Z, where Y = (PR) and Z = (QRS). Consider the two statements given below.

I. Both Y and Z are in BCNF

II. Decomposition of X into Y and Z is dependency preserving and lossless.

Which of the statements is/are correct ?

(a) I only
(b) II only
(c) Neither I nor II
(d) Both I and II

GATE CSE – 2018 MCQ (2-MARKS)

Q17. Consider the following four relational schema. For each schema, all non-trivial functional dependencies are listed. The underlined attributes are the respective primary keys.

Schema I: Registration (<u>roll no,</u> courses)
Field 'course' is a set-valued attribute containing the set of courses a student has registered for.
Non-trivial functional dependency
roll no→ courses
Schema II: Registration(<u>roll no</u>, <u>courseid</u>, email)
Non-trivial functional dependencies:
roll no, course id→ email
Email →roll no
Schema III: Registration (<u>roll no</u>, <u>course id</u>, marks, grade)
Non-trivial functional dependencies:
roll no, course id→ marks ,grade
marks→ grade
Schema IV: Registration (<u>roll no</u>, <u>course id</u>, credit)
Non-trivial functional dependencies:
roll no, courseid→ credit
courseid→ credit
Which one of the relational schemas above is in 3NF but not in BCNF?

(a) Schema I
(b) Schema II
(c) Schema III
(d) Schema IV

GATE CSE-2017 SET-2 NAT (1-MARKS)

Q18. Consider the following tables T1 and T2.

T1

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

T2

R	S
2	2
8	3
3	2
9	7
5	7
7	2

In table T1, P is the primary key and Q is the foreign key referencing R in table T2 with on-delete cascade. In table T2, R is the primary key and S is the foreign key referencing P in table T1 with on-delete set NULL and on-update cascade. In order to delete record (3, 8) from table T1, the number of additional records that need to be deleted from table T1 is _____.

GATE CSE-2017 SET- 1 MCQ (1-MARKS)

Q19. The following functional dependencies hold true for the relational schema R{V, W, X, Y, Z}:
V → W
VW → X

Y → VX
Y → Z
Which of the following is irreducible equivalent for this set of functional dependencies?

(a) V → W
V → X
Y → V
Y → Z
(b) V → W
W → X
Y → V
Y → Z
(c) V → W
V → X
Y → V
Y → X
Y → Z
(d) V → W
W → X
Y → V
Y → X
Y → Z

GATE CSE-2016 SET-1 MCQ (1-MARKS)

Q20. A database of research articles in a journal uses the following schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE)

The primary key is (VOLUME, NUMBER, STARTPAGE, ENDPAGE) and the following functional dependencies exist in the schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE) → TITLE

(VOLUME, NUMBER) → YEAR

(VOLUME, NUMBER, STARTPAGE, ENDPAGE) → PRICE

The database is redesigned to use the following schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE)

(VOLUME, NUMBER, YEAR)

Which is the weakest normal form that the new database satisfies, but the old one does not?
(a) 1NF
(b) 2NF
(c) 3NF
(d) BCNF

GATE CSE-2016 SET-1 MCQ (1-MARKS)

Q21. Which of the following is NOT a super key in a relational schema with attributes V, W, X, Y ,Z and primary key VY?

(a) VXYZ
(b) VWXZ
(c) VWXY
(d) VWXYZ

GATE CSE-2015 SET-3 MCQ (1-MARKS)

Q22. Consider the relation X(P, Q, R, S, T, U) with the following set of functional dependencies
F = {{P, R} → {S, T}
{P, S, U} → {Q, R}}
Which of the following is the trivial functional dependency in F^+ is closure of F?

(a) {P, R} → {S, T}
(b) {P, R} → {R, T}
(c) {P, S} → {S}
(d) {P, S, U} → {Q}

GATE CSE-2014 SET-1 MCQ (2-MARKS)

Q23. Given the following two statements:
S1: Every table with two single-valued attributes is in 1NF, 2NF, 3NF and BCNF.
S2: AB → C, D → E, E → C is a minimal cover for the set of functional dependencies AB → C, D → E, AB → E, E → C.
Which one of the following is CORRECT?

(a) S1 is TRUE and S2 is FALSE.
(b) Both S1 and S2 are TRUE.
(c) S1 is FALSE and S2 is TRUE.
(d) Both S1 and S2 are FALSE.

GATE CSE-2014 SET-3 MCQ (1-MARKS)

Q24. A prime attribute of a relation scheme R is an attribute that appears:

(a) in all candidate keys of R,
(b) in some candidate key of R
(c) in a foreign key of R.
(d) only in the primary key of R.

GATE CSE-2014 SET-1 MCQ (1-MARKS)

Q25. Consider the relation scheme R = (E, F, G, H, I, J, K, L, M, N) and the set of functional dependencies {{E,F} → {G}, {F} → {I,J}, {E,H} → {K,L}, {K} → {M}, {L} → {N}} on R. What is the key for R?

(a) {E,F}
(b) {E,F,H}
(c) {E,F,H,K,L}
(d) {E}

GATE CSE-2014 SET-2 NAT (1-MARKS)

Q26. Given an instance of the STUDENTS relation as shown below.
For (StudentName, StudentAge) to be the key for this instance, the value X should NOT be equal to_____

Student ID	Student Name	Student Email	Student Age	CPI
2345	Shankar	shankar@math	X	9.4
1287	Swati	swati@ee	19	9.5
7853	Shankar	shankar@cse	19	9.4
9876	Swati	swati@mech	18	9.3
8765	Ganesh	ganesh@civil	19	8.7

GATE CSE-2013 MCQ (2-MARKS)

Q27. Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values.
F = {CH → G, A → BC, B → CFH, E → A, F → EG} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R.
How many candidate keys does the relation R have?

(a) 3
(b) 4
(c) 5
(d) 6

GATE CSE-2013 MCQ (2-MARKS)

Q28. Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values.
F = {CH → G, A → BC, B → CFH, E → A, F → EG} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R.
The relation R is ?

(a) In 1NF, but not in 2NF.
(b) In 2NF, but not in 3NF.
(c) In 3NF, but not in BCNF.
(d) In BCNF

GATE CSE-2012 MCQ (1-MARKS)

Q29. Which of the following is TRUE?

(a) Every relation in 3NF is also in BCNF
(b) A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
(c) Every relation in BCNF is also in 3NF
(d) No relation can be in both BCNF and 3NF

GATE CSE-2011 MCQ (1-MARKS)

Q30. Consider a relational table with a single record for each registered student with the following attributes.
1. Registration_ Num: Unique registration number of each registered student
2. UID: Unique identity number, unique at the national level for each citizen
3. BankAccount_Num: Unique account number at the bank. A student can have multiple accounts or join accounts. This attribute stores the primary account number.
4. Name: Name of the student
5. Hostel _ Room: Room number of the hostel

Which of the following options is INCORRECT?

(a) Bank Account _ Num is a candidate key.
(b) Registration _ Num can be a primary key.
(c) UID is a candidate key if all students are from the same country.
(d) If S is a super key such that S∩UID is NULL then S ∪ UID is also a super key.

GATE CSE-2009 MCQ (2-MARKS)

Q31. Consider the following relational schema:

- Suppliers(sid:integer, sname:string,city:string ,street:string)
- Parts(pid:integer, pname:string, color:string)
- Catalog(sid:integer, pid:integer, cost:real)

Assume that, in the suppliers relation above, each supplier and each street within a city has unique name, and (sname, city) forms a candidate key. No other functional dependencies are implied other than those implied by primary and candidate keys. Which one of the following is TRUE about the above schema?

(a) The schema is in BCNF
(b) The schema is in 3NF but not in BCNF
(c) The schema is in 2NF but not in 3NF
(d) The schema is not in 2NF

GATE CSE-2008 MCQ (2-MARKS)

Q32. Consider the following relational schemes for a library database:
Book (Title, Author, Catalog_no, Publisher, Year, Price)
Collection (Title, Author, Catalog_no)
with the following functional dependencies:
I. Title Author → Catalog_no
II. Catalog_no → Title Author Publisher Year
III. Publisher Title Year → Price
Assume {Author, Title} is the key for both schemes. Which of the following statements is true?

(a) Both Book and Collection are in BCNF
(b) Both Book and Collection are in 3NF only
(c) Book is in 2NF and Collection in 3NF
(d) Both Book and Collection are in 2NF only

GATE CSE-2008 MCQ (2-MARKS)

Q33. Let R(A, B, C, D, E, P, G) be a relational schema in which the following functional dependencies are known to hold:
AB → CD , DE → P, C → E , P → C , B → G
The relational schema R is

(a) in BCNF
(b) in 3NF, but not in BCNF
(c) in 2NF, but not in 3NF
(d) not in 2NF

GATE CSE-2008 MCQ (2-MARKS)

Q34. Let R(A, B, C, D) be a relational schema with the following functional dependencies:
F={A → B, B → C, C → D, D → B}
The decomposition of R into (A, B), (B, C), (B, D)

(a) gives a lossless join, and is dependency preserving
(b) gives a lossless join, but is not dependency preserving
(c) does not give a lossless join, but is dependency preserving
(d) does not give a lossless join and is not dependency preserving

GATE CSE-2007 MCQ (2-MARKS)

Q35. Which one of the following statements is FALSE?

(a) Any relation with two attributes is in BCNF
(b) A relation in which every key has only one attribute is in 2NF
(c) A prime attribute can be transitively dependent on a key in a 3NF relation
(d) A prime attribute can be transitively dependent on a key in a BCNF relation

GATE CSE-2006 MCQ (2-MARKS)

Q36 The following functional dependencies are given:
AB → CD
AF → D
DE → F
C → G
F → E
G → A
Which one of the following options is false?

(a) $\{CF\}^+ = \{ACDEFG\}$
(b) $\{BG\}^+ = \{ABCDG\}$
(c) $\{AF\}^+ = \{ACDEFG\}$
(d) $\{AB\}^+ = \{ABCDG\}$

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE-2006 MCQ (2-MARKS)

Q37. Consider a relation R with five attributes V, W, X, Y, and Z. The following functional dependencies hold: VY→ W, WX → Z, and ZY → V. Which of the following is a candidate key for R?

(a) VXZ
(b) VXY
(c) VWXY
(d) VWXYZ

GATE CSE-2005 MCQ (2-MARKS)

Q38. Consider a relation schema R = (A, B, C, D, E, H) on which the following functional dependencies hold: { A → B, BC → D, E → C, D → A }. What are the candidate keys of R?

(a) AE, BE
(b) AE, BE, DE
(c) AEH, BEH, BCH
(d) AEH, BEH, DEH

GATE CSE-2005 MCQ (1-MARKS)

Q39. Which one of the following statements about normal forms is FALSE?

(a) BCNF is stricter than 3NF
(b) Lossless, dependency-preserving decomposition into 3NF is always possible
(c) Lossless, dependency-preserving decomposition into BCNF is always possible
(d) Any relation with two attributes is in BCNF

GATE CSE-2005 MCQ (1-MARKS)

Q40. A table has fields Fl, F2, F3, F4, F5 with the following functional dependencies
F1 → F3 F2→ F4 (F1 . F2) → F5
In terms of Normalization, this table is in

(a) 1 NF
(b) 2 NF
(c) 3 NF
(d) none

GATE CSE-2005 MCQ (2-MARKS)

Q41. In a schema with attributes A, B, C, D and E following set of functional dependencies are given
A → B A → C CD → E B → D E → A
Which of the following functional dependencies is NOT implied by the above set?

(a) CD → AC
(b) BD → CD
(c) BC → CD
(d) AC → BC

ANSWER KEY :

1.(d)	**2.(c, d)**	**3.(6)**
4.(a, b, c)	**5.(c, d)**	**6.(b, c)**
7.(a, c)	**8.(b, c, d)**	**9.(50)**
10.(a)	**11.(8)**	**12.(a)**
13.(a, b, c)	**14.(c)**	**15.(d)**
16.(b)	**17.(b)**	**18.(0)**
19.(a)	**20.(b)**	**21.(b)**
22.(c)	**23.(a)**	**24.(b)**
25.(b)	**26.(19)**	**27.(b)**
28.(a)	**29.(c)**	**30.(a)**
31.(a)	**32.(c)**	**33.(d)**
34.(a)	**35.(d)**	**36.(c)**
37.(b)	**38.(d)**	**39.(c)**
40.(a)	**41.(b)**	

HINTS & SOLUTIONS

1. ANSWER:-D
Given
X → Y and X → Z
Need to prove X → YZ
Step 1 (Augmentation)
From X → Y, augment with X:
X → Y
⇒ XX → XY
From X → Z, augment with Y:
X → Z
⇒ XY → YZ
Step 2 (Transitivity)
XX → XY
XY → YZ
⇒ XX → YZ
Since XX = X
⇒ X → YZ
Only Augmentation + Transitivity used.

2. ANSWER:-C & D

1. If {P, Q} → {R} and {P} → {R}, then {Q} → {R}
{P} → {R} means R is functionally dependent on P

- Even though {P, Q} → {R} holds, the dependency is already satisfied by P alone
- Removing P leaves Q alone, which does not necessarily determine R

Not always true

2. If {P, Q} → {R}, then ({P} → {R} or {Q} → {R})

- It is possible that only the combination (P, Q) determines R
- Neither P nor Q individually may determine R

Not always true

3. If {P} → {R} and {Q} → {S}, then {P, Q} → {R, S}

- From {P} → {R}, by augmentation ⇒ {P, Q} → {R}
- From {Q} → {S}, by augmentation ⇒ {P, Q} → {S}
- By union rule ⇒ {P, Q} → {R, S}

Always true

4. If {P} → {R}, then {P, Q} → {R}

- By augmentation rule, adding attributes to LHS preserves dependency

Always true
Final Answer:
C and D

3. ANSWER:- 6

Candidate keys: {A,B} and {A,C}
A superkey is any superset of a candidate key.
Supersets of {A,B} using {C, D}:
{A,B}, {A,B,C}, {A,B,D}, {A,B,C,D} → 4
Supersets of {A,C} using {B, D}:
{A,C}, {A,B,C}, {A,C,D}, {A,B,C,D} → 4
Common sets: {A,B,C}, {A,B,C,D} → 2
Total superkeys = 4 + 4 − 2 = 6

4. ANSWER:-A,B,C

A. Dependency-preserving 3NF decomposition.
The 3NF synthesis algorithm always produces a lossless and dependency-preserving decomposition.
True

B. Dependency-preserving 1NF decomposition. 1NF only requires atomic attributes.
We can always convert a relation to 1NF without decomposing dependencies (even keeping the same relation), so dependency preservation is possible.
True

C. Dependency-preserving BCNF decomposition is not guaranteed to preserve dependencies. Classic result in normalization theory.:
True

D. Not Dependency-preserving
2NF decomposition can be done while preserving dependencies: **False**

5. ANSWER:-C, D

A relation is in 3NF if for every functional dependency X→Y, if X is a super key, or Y is a prime attribute.
R_1(ABCDE){D→E, EA→B, EA→C}
AD is a candidate key.
D→E (Violation: D is not a super key, E is not prime).
EA→B (Violation: EA is not a super key, B is not prime).
EB→C (Violation: EB is not a super key, C is not prime).
R1 is not in 3NF.

R_2(ABCD)
{A→D, A→B, C→A}

C^+ ={A,B,C,D}
C is a candidate key.
A→D (Violation: A is not a super key, D is not prime).
A→B (Violation: A is not a super key, B is not prime).
C→A (No violation: C is a superkey).
R2 is not in 3NF.

6. ANSWER:- B, C

Key of team is name.
LHS of each FD is super key, therefore team is in BCNF.
Every BCNF relation is in 3NF.
t1 (name, city) has FD: name → city is in BCNF. t2 (name, owner) has FD: name → owner is in BCNF.
Both t1 and t2 has common attribute 'name' and is key in t1 and t2 , hence the decomposition is lossless.

7. ANSWER:-A,C

R cannot have a foreign key ,R cannot have more than one candidate key are straight forward false because we can have a foreign key in 1NF and also we can have more than one candidate key.
The correct answer is R can have a multi-attribute key and R cannot have a composite attribute

8. ANSWER:- B,C,D

Option B - Decomposition rule of functional dependencies.
Option C - Apply augmentation rule with X on W -> Y, Then apply Transitivity rule to get X, W -> Z.
Option D - Definition of Transitivity rule for functional dependencies

9. Answer :(50)

(I) for One – three attribute

	No. of FDs	Functional Dependencies
A→ B, C, D	7	A→B, A→C, A→D, A→BC, A→BD, A→CD, A→BCD
B→ A, C, D	7	B→A, B→C, B→D, B→AC, B→AD, B→CD, B→ACD
C→ A, B, D	7	C→A, C→B, C→D, C→AB, C→AD, C→BD, C→ABD
D→A, B, C	7	D→A, D→B, D→C, D→AB, D→AC, D→BC, D→ABC

Total FDs = 28

(ii) for Two - Two Attributes

$AB \rightarrow CD$	$[2^2 - 1]$	$= 3\ \{AB \rightarrow C, AB \rightarrow D, AB \rightarrow CD\}$
$AC \rightarrow BD$	$[2^2 - 1]$	$= 3\ \{AC \rightarrow B, AC \rightarrow D, AC \rightarrow BD\}$
$AD \rightarrow BC$	$[2^2 - 1]$	$= 3\ \{AD \rightarrow B, AD \rightarrow C, AD \rightarrow BC\}$
$BC \rightarrow AD$	$[2^2 - 1]$	$= 3\ \{BC \rightarrow A, BC \rightarrow D, BC \rightarrow AD\}$
$BD \rightarrow AC$	$[2^2 - 1]$	$= 3\ \{BD \rightarrow A, BD \rightarrow C, BC \rightarrow AC\}$
$CD \rightarrow AB$	$[2^2 - 1]$	$= 3\ \{CD \rightarrow A, CD \rightarrow B, CD \rightarrow AB\}$

Total FDs = 18 FD's

(iii) for Three Attributes

$ABC \rightarrow D$
$ABD \rightarrow C$
$BCD \rightarrow A$
$CDA \rightarrow$ B

Total FDs = 4 FD

So Total Useful FDs $= 28 + 18 + 4 = 50$ Ans

OR

$x \rightarrow y$ Useful FD with 4 Attribute

Case I: for a Single-single Attributes

X -> Y $^4C_1$ * $[2^3\text{-}1]$

A->BCD $[2^3\text{-}1]$ L.H.S R.H.S

B->ACD

C->ABD

D->ABC 4 * 7 =28

CASE II : for a Two - Two Attributes

$x \rightarrow y$

2Attribute → 2Attribute $^4C_2 * (2^2 - 1)$

$6 * 3 = (18)$

CASE III:- for Three-Three Attributes

$x \rightarrow y$

3-ttribute attribute

$^4C_3 * (2^1 - 1)$

$4 * 1 = (4)$

Total useful FD $= 28 + 18 + 4 = 50$ Ans

10. ANSWER:-A

Relational Model Definitions:

A relation is a table with columns and rows.

An attribute is a named column of a relation.

A tuple is a row of a relation.

A domain is a set of allowable values for one or more attributes.

The degree of a relation is the number of attributes it contains.

The cardinality of a relation is the number of tuples it contains.

A relational database is a collection of normalized relations with distinct relation names.

The intension of a relation is the structure of the relation including its domains.

The extension of a relation is the set of tuples currently in the relation.

11. ANSWER:- 8

AB → C

BC → D

C → E

are the FDs in the question. If you observe the given FD set, A and B are Independent attributes.

∴ every key must contain A and B.

Let's compute Keys of the relations :

(AB)+=ABCDE

Independent Attributes form the Key, then It is unique and minimal candidate key.

There are 3 more attributes in the given relation, So adding them to Candidate Key results Super key.

Every attribute has two choices (either add to candidate key or left it), and there are 3 such attributes

∴ No . of super keys = 2.2.2 = 8

Note that Every candidate Key is also Super Key.

Alternatively you can think like, set S = {C,D,E} (which are non prime attributes) and it's power set must contains 2^3=8 elements

P(S)={ϕ,{C},{D},{E},{CD},{CE},{DE},{CDE}}

Any Subset combining with 'AB' will form one distinct super key.

12. ANSWER:-A

Option A : A relation with only two attributes is always in BCNF

let relation R(A,B), then possible FD's are

1. A → B, in this case A is minimal Superkey, So this relation should be in BCNF
2. B → A, in this case B is minimal Superkey, So this relation should be in BCNF
3. AB → AB , in this case AB is minimal Superkey, So this relation should be in BCNF

So, given statement is Correct.

Alternatively we can think like, We can't further divide a relation which had only two attributes, then it must be in BCNF.

Option B : If all the attributes of a relation are Prime attributes, then the relation is in BCNF classic counter example : R(A,B,C) with FD set = {AB → C, C → A} in this relation all attributes are prime but this relation is not in BCNF due to " C → A "

Correct statement is :If all the attributes in the relation are Prime attributes, then it is in 3NF.

Option C : Every relation has at least one non-prime attribute counter example : R(A,B,C) with FD set = {AB → C, C → A} in this relation all attributes are prime

Correct statement is :Every relation has atleast one prime attribute.

Option D : BCNF decompositions preserve functional dependencies for any relation, we can guarentee there exist atleast one decomposition which is in 3NF, lossless and Dependency Preserving. But we can notguarentee that atleast one decomposition which is BCNF, lossless and Dependency Preserving.

Option A is correct

13. Answer:A,B,C

Explanation :- A, B, C
In option (D) R → T, we find $(R)^+$.
R^+ = {R} so in R^+ we do not have T so can't be inferred. Hence false.
In option (C) P → R, $(P)^+$ = {PQR} and we find R so it can be inferred. Hence True.
In option (B) PS → Q, $(PS)^+$ = {PSQRT} so it can be inferred. Hence True.
In option (A) PS → T, $(PS)^+$ = {PSQRT} so it can be inferred. Hence True.

14. ANSWER :- C

Explanation :C

Testing Decomposition D_1 (Lossless)
To determine if a multh-relation decomposition is lossless, we can binary-merge the relations step by step using their intersections:

1. Merge (P, Q, S, T) and (P, T, X) :

- Their common attribute intersection is $\{P, T\}$.
  o Since $P \rightarrow YX$, we know that $P \rightarrow X$. Therefore, P (and by extension, $\{P, T\}$) is a superkey for the relation (P, T, X).
- This intersection forms a key for one of the relations, making this merge lossless. The combined relation is (P, Q, S, T, X).

2. Merge (P, Q, S, T, X) and (Q, Y):

- Their intersection is $\{Q\}$.
- The given dependency is $Q \rightarrow Y$, so Q is a superkey for (Q, Y). This merge is lossless. The combined relation becomes (P, Q, S, T, X, Y).

3. Merge (P, Q, S, T, X, Y) and (Y, Z, W) :
   o Their intersection is $\{Y\}$.

- The given dependency is $Y \rightarrow ZW$, so Y is a superkey for (Y, Z, W). This merge is lossless.

Since all relations successfully merge back without losing information, D_1 is a lossless decomposition.

Testing Decomposition D_2 (Lossy)
Let's analyze the relation (T, X) in this scheme:

- To merge (T, X) with any other relation in D_2 (such as (P, Q, S)), we look for a common attribute intersection.
- The intersection between (T, X) and the other relations is empty (0).
- Since the intersection does not contain a candidate key for either relation, we cannot guarantee a lossless join.

Therefore, D_2 is a lossy decomposition.

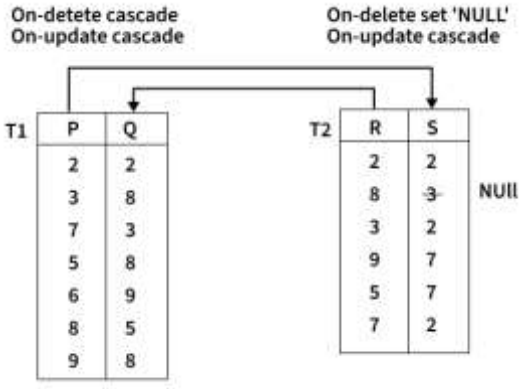
15. **Answer:-D**
Explanation
Consider a relation R(ABCD) and the functional dependencies as
FDs:
AB → C
BC → A
$(BD)^+$ = BD ✗
$(ABD)^+$ = ABDC ✓
$(CBD)^+$ = CBDA ✓
Candidate keys = {ABD, CBD}
• The relation R is in 3NF, as there are no transitive dependencies.
• The relation R is not in BCNF, because the left hand side of both the FD's are not the Super keys.
• In R, BC → A is a non-trivial FD and in which BC is not a Super key and A is a prime attribute.
So, D is the correct answer.

16. **Answer :- B**
Explanation
Y = (PR)
R → P
R^+ = RP
In the dependency R → P, 'R' is a super key. So, Y is in BCNF.
Z = (QRS)
QR → S
S → Q
CK's = QR, RS
In, the dependency S → Q, 'S' is not a super key. So, Z is not in BCNF.
Y is in BCNF and Z is not in BCNF.
'R' is common attribute in the relations Y and Z and R is candidate key for Y. So, the decomposition is lossless.
all dependencies are preserved,
So, the decomposition is dependency preserving.
Hence, Statement II is correct.

17. **Answer:- B**
Explanation
Schema I:
Registration(rollno, courses)
Here: Non-trivial functional dependencies:
rollno → courses
So, here in Registration 'rollno' is a primary key.
So, Schema I is in BCNF and 3NF.
Schema II:
Registrstion (rollno, courseid, email)
Here: Non-trivial functional dependencies:
rollno, courseid → email
email → rollno
From the given schema the candidate key is (rollno + courseid).
There is no part of the key in the left hand of the FD's so, it is in 2NF.
In the FD email→rollno, email is non-prime attribute but rollno is a prime attribute.
So, we don't have a transitive dependency.
As we have no transitive dependencies so, the schema is in 3NF.
But in the second FD: email→rollno, email is not a superkey.
So, it is failing the property of BCNF.
Hence, the schema II is in 3NF but not in BCNF.
Schema III:
Registration(rollno, courseid,marks, grade)
Here: Non-trivial functional dependencies:
rollno, courseid → marks, grade
marks → grade
For the schema the candidate key is (rollno + courseid).
Here, we don't have any partial dependency.
So, the schema is in 2NF.
In the FD:marks → grade, both the attributes marks and grade are non-prime.
So, it is a transitive dependency.
The FD is violating the property of 3NF.
The schema Registration is in 2NF but not in 3NF.
Schema IV:
Registration (rollno, course id, credit)
Here: Non-trivial functional dependencies:
roll no, course id → credit
courseid → credit
The candidate key is (roll no + course id).
Now ,in the FD, course id → credit, course id is part of the key (prime attribute) and credit is non-prime.

The schema is violating the property of 2NF.

18. **Answer:-0**
Explanation



Therefore, number of additional records that needs to be deleted from the table T1 is 0

19. **Answer: A**
Explanation
Step 1:
Remove the redundancy in the RHS.
So, we get: V → W, VW → X, Y → V, Y → X, Y→ Z
Step 2:

Find the closure of each functional dependency with or without that functional dependency to check redundancy in each rule.
V → W, VW → X, Y → V, Y → X, Y→ Z
$(V)^+$ = V ×
$(VW)^+$ = VW ×
$(Y)^+$ = YXZ
$(Y)^+$ = YVW ×
$(Y)^+$ = YVWX
Without Y → X, the closure of Y is deriving 'X' from the remaining attributes.
So, we can remove Y → X as its redundant.
Step 3:
Remove the redundancy in the LHS.
V → W, VW → X, Y → V, Y → Z
$(V)^+$ = VW, the closure of V is deriving W from the remaining FD's.
So, W is redundant. We can remove it.
So, the minimal canonical cover is
V→W, V→X, Y→V, Y→Z

20. **Answer: - B**
Explanation
Journal (V, N, S, E, T, Y, P)
V – VOLUME
N – NUMBER
S – STARTPAGE
E – ENDPAGE
T – TITLE
Y – YEAR
P – PRICE
Primary key: (V, N, S, E)
FD set:
(V, N, S, E) → T
(V, N) → Y
(V, N, S, E) → P
In (V, N) → Y
Now, we check each functional dependency one by one.
In VN→ Y, we have V, N as a part of the candidate key and Y is non-prime attribute.
So, it is a Violation of 2NF
Now, the schema "Journal" is in 1NF but not in 2NF.
So,2NF is the weakest normal form that the new database satisfies, but the old one does not.

21. **Answer:- B**
Explanation
As it is mentioned in the Question that "VY" is a primary key of the relational schema.
So, any superset of "VY" is a super key. So, option B does not contain "Y" and is the correct answer.

22. **Answer:- C**
Explanation
X→Y is trivial if Y⊆ X.
So, PS → S is the trivial FD as S ⊆ PS.

23. **Answer :- A**
Explanation
BCNF (Boyce Codd Normal Form):

- A relation R is in BCNF whenever a non-trivial functional dependency $X \rightarrow A$ holds in R , where X is the super-key of R .
- A binary relation is always in BCNF. A binary relation contains only two attributes.
- Functional dependency that is possible from a binary relation is one.

Example:
Consider R(A, B), in this only one functional dependency is possible either A → B or B → A

In both the cases, left hand side will be the super key. In this way $R(A, B)$ is always in BCNF.

If a relation is in BCNF then it is in 1NF, 2NF and 3 NF
Statement 2: FALSE
Set 1 = {AB → C, D → E, AB → E, E → C}
Set 2 = $\{AB \rightarrow C, D \rightarrow E, E \rightarrow C\}$
Set 2 cannot derive $AB \rightarrow E$ since In set
$Z(AB)^+ = \{A, B, C\}$
The two sets of functional dependencies are not the same and hence one cannot be minimal of other.

24. **Answer :- B**
Explanation
A prime attribute of a relation scheme R is an attribute that appears in some candidate keys of R. It is not necessary to appear in all the candidate keys of Relation R.
Ex: AB, BC, CD are candidate keys of R(ABCD). In the FDs set one attribute may not be part of all the FDs.

25. **Answer:- B**
Explanation
Here , we need to find the candidate key of the Relation R:
So, we take the closure of the attributes and check one by one for each option. $(EF)^+$ = EFGIJ. So, EF is not a key.
$(EFH)^+$ = EFHGIJKLMN, EFH is deriving all the attributes of R. So, EFH is a key.
If EFH is a key, then EFHKL will become the super key.
$(E)^+$ = E

26. **Answer:- 19**
Explanation
As we already have one entry with StudentName as "Shankar" and StudentAge as 19, so, X value should not be 19 to make (StudentName,StudentAge) a key.

27. **Answer :- B**
Explanation
The attribute D is not a part of any FD's,i.e it is an essential attribute. So, D can be a candidate key or it may be part of the candidate key.
Now D+ = {D}.
Hence,we need to add A,B,C,E,F,G,H to D and check which of them can be Candidate keys of size 2.
AD^+ = {ABCDEFGH}
BD^+ = {ABCDEFGH}
ED^+ = {ABCDEFGH}
FD^+= {ABCDEFGH}
But CD^+, GD^+ and HD^+ does not implies all the attributes present in the Relation r and hence CD, GD and HD are not candidate keys.
Here , the total number of Candidate keys are 4: i.e, AD, BD, ED and FD.

28. **Answer :- A**
Explanation
Here, candidate keys are AD, BD, ED and FD.
Partial dependency exists $A \rightarrow BC, B \rightarrow CFH$ and $F \rightarrow EG$ etc. In the following FDs.
For example partial dependency $A \rightarrow C$ exists in $A \rightarrow BC$ and $B \rightarrow C$ and $B \rightarrow H$ in $B \rightarrow CFH$.
etc.
So, given relation is in 1NF ,but not in 2NF.

29. **Answer:- C**
Explanation
As,BCNF is a stronger version of 3NF. So, we can comprehend from the definition of BCNF that: every relation in BCNF will also be in 3NF.

30. **Answer: -A**
Explanation
Here, Bank Account Number cannot be a key, as if two students hold joint account then Bank Account Number will not uniquely determine other attributes.
(a) Bank Account Number is a Candidate key: Incorrect or False Because A Student Can have Multiple Account & joint Account
Example Assume 2 Person Ramesh & Suresh have Joint Account they Both have Same Account Number in a Primary Account Number.
So Bank Account Number Cannot uniquely Determine. and Bank Account Number cannot Determine and Bank Account Number can't be Candidate key
(b) Registration Number can be primary key : True or Correct Because Registration Number is a uniquely for each Registered student so it uniquely determine each tuple of Relation
(c) UID is a candidate if all student are from same Country: Correct (or) True Because UID is Unique at National level
for example in India Aadhar Card is unique for each Person of India.
(d) If S is a super key then SUUID is also super key: correct (or) True Because Any superset of super key is also super key.

31. **Answer :- A**
Explantion:-
Suppliers(sid, sname, city, street)
Given:
- Primary key ⇒sid
- Candidate key ⇒ (sname, city)

So FDs:
- sid → sname, city, street
- (sname, city) → sid, street

Both determinants are candidate keys ⇒ BCNF satisfied
Parts(pid, pname, color)
- Key ⇒pid
- pid → pname, color
- Determinant is key ⇒ BCNF

Catalog(sid, pid, cost)
- Key ⇒ (sid, pid)
- (sid, pid) → cost
- Determinant is key ⇒ BCNF

Conclusion
All relations satisfy BCNF.

1. **32. Answer :- C**
Explanation :
Analyzing the Collection Relation

The relation is Collection(Title, Author, Catalog_no) with the given candidate key {Author, Title}.

- The only applicable functional dependency here is: Title, Author → Catalog no.
- Because the left-hand side (Title, Author) is a candidate key, it perfectly satisfies the requirement for Boyce-Codd Normal Form (BCNF).
- Since any relation in BCNF is automatically in 3NF, Collection is in 3NF.

2. Analyzing the Book Relation

The relation is Book(Title, Author, Catalog_no, Publisher, Year, Price).First, let's find all the candidate keys for this relation:

1. We are given that { Title, Author } is a key.

2. If we compute the attribute closure of Catalog_no using the rule Catalog_no → Title, Author, Publisher, Year, we get those five attributes. Adding the third rule (Publisher, Title, Year → Price), we can also derive Price.
3. Because $(\text{Catalog.no})^+ =$ {Title, Author, Catalog no, Publisher, Year, Price}, Catalog no is also a candlidate key.

   Now we evaluate the normal forms against the rules:

- BCNF / 3NF Violation: Look at the third dependency: Publisher, Title, Year → Price.
- The left-hand side (Publisher, Title, Year) is not a superkey.
- The right-hand side (Price) is a non-prime attribute (it is not part of any candidate key).
- This violates both BCNF and 3 NF .
- 2NF Verification: A relation is in 2NF if there are no partial dependencies where a proper subset of a candidate key points to a non-prime attribute. Since no proper subset of {Title, Author} or {Catalog no} points to a non-prime attribute, it does not violate 2 NF .

  Therefore, Book is in 2NF only. This matches option (c).

33. **Answer :- D**
Explantion :- R(A, B, C, D, E, P, G)
FDs:
AB → CD
DE → P
C → E
P → C
B → G
AB^+ = {A, B, C, D, E, P, G}
So AB is the candidate key.
Prime attributes: A, B
Non-prime: C, D, E, P, G
Since B → G and B is a proper subset of key AB, partial dependency exists.
Therefore, R is not in 2NF.

34. **Answer :- A**
Explanation :-
(A, B), (B, C) — common attribute is B and due to B → C, B is a key for (B, C) and hence ABC can be losslessly decomposed into (A, B) and (B, C).
(A, B, C), (B, D), common attribute is B and B → D is a FD (via B → C, C → D), and hence, B is a key for (B, D). So, decomposition of (A, B, C, D) into (A, B, C) and (B, D) is lossless.
Thus the given decomposition is lossless.
The given decomposition is also dependency preserving as the dependencies A → B is present in (A, B), B → C is present in (B, C), D → B is present in (B, D) and C → D is indirectly present via C → B in (B, C) and B → D in (B, D).

35. **Answer :- D**
Explanation :-A. True — In a two-attribute relation, any non-trivial FD makes one attribute a key, so it is in BCNF.
B. True — If every key is single attribute, no partial dependency is possible ⇒ 2NF satisfied.
C. True — In 3NF, transitive dependency is allowed if RHS is a prime attribute.
D. False — In BCNF, every determinant must be a superkey. So no transitive dependency (even on prime attribute) is allowed.
Final Answer: D

36. **Answer ━C**
Explantion:-
Check closures:
$(CF)^+$
C → G → A
F → E
AF → D
⇒ {A, C, D, E, F, G} (true)
$(BG)^+$
G → A
AB → CD
C → G
⇒ {A, B, C, D, G} (true)
$(AF)^+$
AF → D
F → E
DE → F
⇒ {A, D, E, F} (false) (cannot get C, G)
$(AB)^+$
AB → CD
C → G
G → A
⇒ {A, B, C, D, G} (true)
False option: C

37. **Answer :- B**
Explanation :- FDs:
VY → W
WX → Z
ZY → V
Check $(VXY)^+$
VY → W
WX → Z
Now have V, W, X, Y, Z ⇒ all attributes.

Removing any one of V, X, or Y fails to get all attributes.
So it is minimal.
Answer: b) VXY

38. **Answer :- D**
Explanation :- H is not in RHS of any FD ⇒ must be in every key.
E is not in RHS ⇒ must be in every key.
Check combinations with E and H:
AEH^+ ⇒ all attributes
BEH^+ ⇒ all attributes
DEH^+ ⇒ all attributes
So candidate keys are AEH, BEH, DEH.

39. **Answer :- C**
Explanation :-
A. True — BCNF is stricter than 3NF.
B. True — A lossless and dependency-preserving decomposition into 3NF is always possible.
C. False — BCNF decomposition is always lossless, but may not preserve dependencies.
D. True — Any two-attribute relation is in BCNF.
Correct Answer: C

40. **Answer:-A**
Explanation :-First Normal Form
A relation is in first normal form if every attribute in that relation is singled valued attribute.

Second Normal Form
A relation is in 2NF if it has No Partial Dependency, i.e., no non-prime attribute (attributes which are not part of any candidate key) is dependent on any proper subset of any candidate key of the table.
This table has Partial Dependency f1->f3, f2-> f4
given (F1,F2) is Key
So answer is A

41. **Answer :- B**
Explanation :-FDs:
A → B, C
B → D
CD → E
E → A
Check quickly:
a) CD → AC
CD → E → A → C (true)
b) BD → CD
From BD we get only B, D.
No way to get C (false)
c) BC → CD
B → D ⇒ BC gives C, D (true)
d) AC → BC
A → B ⇒ AC gives B (true)
Not implied: b) BD → CD

2 ER MODEL

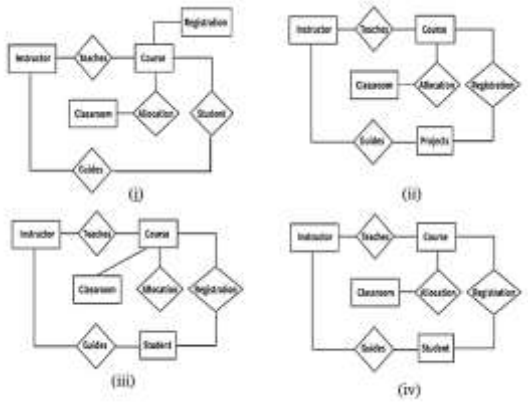
GATE CSE-2024 SET-2 MCQ (1-MARKS)

Q1. In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

(a) The weak entity set MUST have total participation in the identifying relationship
(b) The owner entity set MUST have total participation in the identifying relationship
(c) Both weak and owner entity sets MUST have total participation in the identifying relationship
(d) Neither weak entity set nor owner entity set MUST have total participation in the identifying relationship

GATE CSE-2024 SET-2 MCQ (1-MARKS)

Q2. Let S be the specification: "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?



(a) (i)
(b) (ii)
(c) (iii)
(d) (iv)

GATE CSE-2020 MCQ (1-MARKS)

Q3. Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

(a) Diamonds with double/bold border
(b) Rectangles with double/bold border
(c) Ovals with double/bold border
(d) Ovals that contain underlined identifiers

GATE CSE-2018 MCQ (1-MARKS)

Q4. In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set E1 to entity set E2. Assume that E1 and E2 participate totally in R and that the cardinality of E1 is greater than the cardinality of E2. Which one of the following is true about R?

(a) Every entity in E1 is associated with exactly one entity in E2
(b) Some entity in E1 is associated with more than one entity in E2
(c) Every entity in E2 is associated with exactly one entity in E1
(d) Every entity in E2 is associated with at most one entity in E1

GATE CSE-2017 SET-2 MCQ (1-MARKS)

Q5. An ER model of a database consists of entity types A and B. These are connected by a relationship R which does not have its own attribute. Under which one of the following conditions, can the relational table for R be merged with that of A?

(a) Relationship R is one-to-many and the participation of A in R is total
(b) Relationship R is one-to-many and the participation of A in R is partial
(c) Relationship R is many-to-one and the participation of A in R is total
(d) Relationship R is many-to-one and the participation of A in R is partial

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2015 SET-2 NAT (2-MARKS)

Q6. Consider an Entity-Relationship (ER) model in which entity sets E1 and E2 are connected by an m:n relationship R12. E1 and E3 are connected by a 1:n (1 on the side of E1 and n on the side of E3) relationship R13.

E1 has two-singled attributes a11 and a12 of which a11 is the key attribute. E2 has two singled-valued attributes a21 and a22 of which a21 is the key attribute. E3 has two single-valued attributes a31 and a32 of which a31 is the key attribute. The relationships do not have any attributes.

If a relational model is derived from the above ER model, then the minimum number of relations that would be generated if all relation are in 3NF is_____.

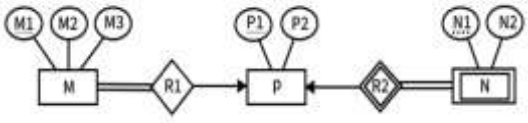
GATE CSE-2012 MCQ (1-MARKS)

Q7. Given the basic ER and relational models, which of the following is INCORRECT?

(a) An attribute of an entity can have more than one value
(b) An attribute of an entity can be composite
(c) In a row of a relational table, an attribute can have more than one value
(d) In a row of a relational table, an attribute can have exactly one value or a NULL value

GATE CSE-2008 MCQ (2-MARKS)

Q8. Consider the following ER diagram

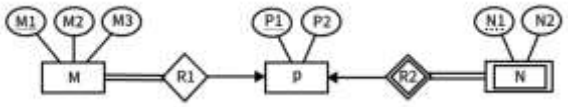


Which of the following is a correct attribute set for one of the tables for the minimum number of tables needed to represent M, N, P, R1, R2 ?

(a) M1, M2, M3, P1
(b) M1, P1, N1, N2
(c) M1, P1, N1
(d) M1, P1

GATE CSE-2008 MCQ (2-MARKS)

Q9. Consider the following ER diagram



The minimum number of tables needed to represent M, N, P, R1, R2 is

(a) 2
(b) 3
(c) 4
(d) 5

GATE CSE-2005 MCQ (2-MARKS)

Q10. Let E_1 and E_2 be two entities in an E/R diagram with simple-valued attributes. R_1 and R_2 are two relationships between E_1 and E_2, where R_1 is one-to-many and R_2 is many-to-many. R_1 and R_2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?

(a) 2
(b) 3
(c) 4
(d) 5

GATE CSE-2005 MCQ (1-MARKS)

Q11. Consider the entities 'hotel room', and 'person' with a many to many relationship 'lodging' as shown below:

If we wish to store information about the rent payment to be made by person (s) occupying different hotel rooms, then this information should appear as an attribute of ?

(a) Person
(b) Hotel Room
(c) Lodging
(d) None of these

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2005 MCQ (2-MARKS)

Q12. The following table has two attributes A and C where A is the primary key and C is the foreign key referencing A with ON DELETE CASCADE.

A	C
2	4
3	4
4	3
5	2
7	2
9	5
6	4

The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple (2, 4) is deleted is:

(a) (3, 4) and (6, 4)
(b) (5, 2) and (7, 2)
(c) (5, 2), (7, 2) and (9, 5)
(d) (3, 4), (4, 3) and (6, 4)

ANSWER KEY

1.(a)	**2.(d)**	**3.(a)**
4.(a)	**5.(c)**	**6.(4)**
7.(c)	**8.(a)**	**9.(b)**
10.(b)	**11(c)**	**12.(c)**

Hints & Solution

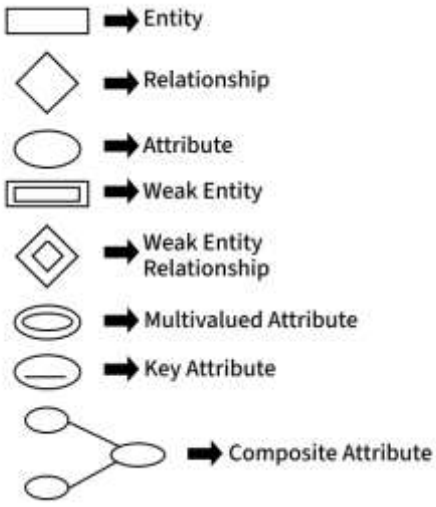
1. **Answer:- A**
Explantion—The weak entity set must participate totally in the identify weak relationship.
Usually, there exist one to many relationship between strong entity and weak entity thus total participation from weak entity set.

2. **Answer:-D**
Explantion :- Option (iv) is correct
There's a relationship b/w :

- Instructors (Teaches) courses
- Students (Registration) courses
- Courses (Allocation) classrooms
- Instructors (Guides) students

Link them and you'll get a figure like (iv) Also obviously : Student, Instructor, Courses and Classroom are all entity.

3. **Answer:-A**
Explantion :- In an ER diagram, the supporting many-one relationship of a weak entity set is called the identifying relationship. It is represented by a diamond with a double (bold) border. Weak entities depend on this relationship for identification, which is why the relationship is shown with a double diamond.



4. **Answer : A**

Explantion:- Since it is a many to one relationship from E1 to E2, therefore:

1. No entity in E1 can be related to more than one entity in E2. (hence B is incorrect)
2. An entity in E2 can be related to more than one entity in E1.(hence C and D are incorrect).

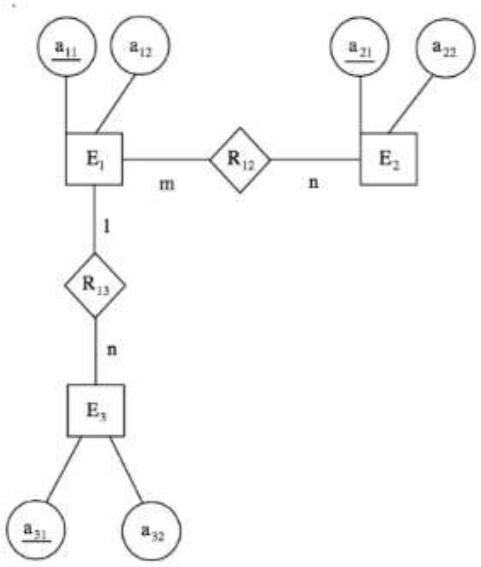
Every entity in E1 is associated with exactly one entity in E2.

5. **Answer:- C**
Explantion :- The relation table for R should always be merged with the entity that has total participation and relationship should be many to one.

Check Out Our GATE Courses & Secure Your Dream Rank!

6. **Answer:-4**

Explantion :-



E_1, E_2 and E_3 are strong entity, one table for each: 3 Tables
R_{12} is m:n hence it requires a separate table: 1 Table
R_{13} is $1:n$ hence it does not require any separate table.
Total: 4 tables

7. **Answer:- C**
Explantion :- In a row of a relational table, an attribute can have more than one value is incorrect as a relational table requires that, in a row, an attribute can have exactly one value or NULL value.

8. **Answer :- A**
Explanation :-
Strong entities → M and P ⇒ separate tables
- R1 between M and P has total participation of M and no attributes
  ⇒ Add key of P (P1) into M as FK
  ⇒ Table M(M1, M2, M3, P1)
- N is a weak entity identified by P via R2
  ⇒ Create table N(P1, N1, N2) with PK = (P1, N1)
- Table P(P1, P2)
  Thus minimum tables = 3 (M, P, N)
  Valid attribute set among options ⇒ M1, M2, M3, P1

9. **Answer : B**
Explanation
3 minimum tables. M, P are strong entities hence they must be represented by separate tables.
Many-to-one and one-to-many relationship sets that are total on the many-side can be represented by adding an extra attribute to the "many" side, containing the primary key of the "one" side. (This way no extra table will be needed for Relationship sets) M table is modified to include primary key of P side(i.e. P1). N is weak entity, and is modified to include primary key of P (i.e. P1). Therefore there would be minimum of 3 tables with schema given below :

M (M1, M2, M3, P1)
P (P1, P2)
N (P1, N1, N2)

10. **Answer:- B**
Solution
• Each entity requires one table ⇒ 2 tables (E_1 and E_2).
• R_1 (one-to-many, no attributes) ⇒ implement using a foreign key in the "many" side ⇒ no extra table needed.
• R_2 (many-to-many) ⇒ requires a separate relation table.
Total tables = 2 (entities) + 1 (for R_2) = 3.
Answer: B. 3

11. **Answer:- C**
Explanation :-
The rent payment depends on both the Person and the Hotel Room. Since it is associated with a specific combination of these two entities, it cannot be an attribute of either entity alone. Therefore, it must be stored as an attribute of the relationship "Lodging".

12. **Answer :- C**
Explanation :-
Given:
A is Primary Key
C is Foreign Key referencing A with ON DELETE CASCADE
Delete (2,4) ⇒ A = 2 is deleted.
Step 1: Delete tuples where C = 2
(5,2)
(7,2)
Step 2: Now A = 5 and A = 7 are deleted.
Check rows with C = 5 or C = 7
(9,5) exists ⇒ delete it.
No row with C = 7.
Step 3: Now A = 9 deleted.
No row with C = 9.
Cascade stops.
Additionally deleted tuples:
(5,2), (7,2), (9,5)
Answer: C

Check Out Our GATE Courses & Secure Your Dream Rank!

3 RELATIONAL ALGEBRA

GATE CSE-2026 SET-1 MCQ (2-MARKS)

Q1. Consider a relational database schema with two relations $R(P, Q)$ and $S(X, Y)$
Let $E = \{\langle u\rangle \mid \exists v \exists w \langle u, v\rangle \in R \wedge \langle v, w\rangle \in S\}$ be a tuple relational calculus expression.
Which one of the following relational algebraic expressions is equivalent to E ?

(a) $\Pi_P(R \bowtie_{R.P=S.X} S)$
(b) $\Pi_P(S \bowtie_{S.X=R.Q} R)$
(c) $\Pi_P(R \bowtie_{R.P=S.Y} S)$
(d) $\Pi_P(S \bowtie_{S.Y=R.Q} R)$

GATE CSE-2025 SET-1 MCQ (2-MARKS)

Q2. Consider two relations describing teams and players in a sports league: teams (tid,$tname$): tid, $tname$ are team-id and team-name, respectively.$players$(pid, $pname$,tid): pid, $pname$, and tid denote player-id, playername and the team-id of the player, respectively. Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having name as 'MI'?

(a) $\{ p.pname \mid p \in players \wedge \exists t \ (t \in teams \wedge p.tid = t.tid \wedge t.tname = 'MI')$
(b) $\{ p.pname \mid p \in teams \wedge \exists t \ (t \in players \wedge p.tid = t.tid \wedge t.tname = 'MI')\}$
(c) $\{ p.pname \mid p \in players \wedge \exists t \ (t \in teams \wedge t.tname = 'MI')\}$
(d) $\{ p.pname \mid p \in teams \wedge \exists t \ (t \in players \wedge t.tname = 'MI')\}$

GATE CSE-2024 SET-1 NAT (1-MARKS)

Q3. Consider the following two relations, R(A,B) and S(A,C):

R	
A	**B**
10	**20**
20	**30**
30	**40**
30	**50**
50	**95**

S	
A	C
10	90
30	45
40	80

The total number of tuples obtained by evaluating the following expression
$\sigma_{B<C}(R \bowtie_{R.A=S.A} S)$ is _____.

GATE CSE-2022 MSQ (1-MARKS)

Q4. Consider the following three relations in a relational database:
Employee(eId, eName)
Brand(bId, bName)
Own(eId, bId)
Which of the following relational algebra expressions returns the set of eIds who own all the brands?

(a) $\pi_{eId}(\pi_{eId,bId}(Own)) / \pi_{bId}(Brand))$
(b) $\pi_{eId}(Own) - \pi_{eId}((\pi_{eId}(Own) \times \pi_{bId}(Brand)) - \pi_{eId,bId} Own))$
(c) $\pi_{eId}(\pi_{eId,bId}(Own) / \pi_{bId}(Own)$
(d) $\pi_{eId}((\pi_{eId}(Own) \times \pi_{bId}(Own)) / \pi_{bId}(Brand))$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2021 SET-1 MCQ (2-MARKS)

Q5. The following relation records the age of 500 employees of a company, where empNo (indicating the employee number) is the key :
empAge(empNo,age)
Consider the following relational algebra expression
$\Pi_{emp \times No}(empAge \bowtie_{(age>age1)} \rho_{empNo1,aye1}(empAge))$

What does the above expression generate?

(a) Employee numbers of all employees whose age is not the minimum.
(b) Employee numbers of all employees whose age is the minimum.
(c) Employee numbers of all employees whose age is more than the age of exactly one other employee.
(d) Employee numbers of only those employees whose age is the maximum.

GATE CSE-2021 SET-1 NAT (1-MARKS)

Q6. A relation r(A, B) in a relational database has 1200 tuples. The attribute A has integer values ranging from 6 to 20, and the attribute B has integer values ranging from 1 to 20. Assume that the attribute A and B are independently distributed. The estimated number of tuples in the output of $\sigma_{(A>10)v(B=18)}(r)$ is _____.

GATE CSE-2021 SET-2 MCQ (2-MARKS)

Q7. Consider the following statements S1 and S2 about the relational data model:
S1: A relation scheme can have at most one foreign key.
S2: A foreign key in a relation scheme R cannot be used to refer to tuples of R.
Which one of the following choices is correct?

(a) S1 is false and S2 is true.
(b) S1 is true and S2 is false.
(c) Both S1 and S2 are false.
(d) Both S1 and S2 are true.

GATE CSE-2019 NAT (2-MARKS)

Q8. Consider the following relations P(X, Y, Z), Q(X, Y, T) and R(Y, V).

Table : P

X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Table : Q

X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

Table : R

Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

How many tuples will be returned by the following relational algebra query?
$\pi_x(\sigma_{(P.Y = R.Y \wedge R.V = V2}(P \times R))) - \pi_x(\sigma_{(Q.Y = R.Y \wedge Q.T > 2}(Q \times R)))$

GATE CSE-2018 MCQ (2-MARKS)

Q9. Consider the relations r(A, B) and s(B, C), where s,B is a primary key and r.B is a foreign key referencing s.B. Consider the query:

Q: $r \bowtie (\sigma_{B<5}(s))$
Let LOJ denote the natural left outer-join operation.
Assume that r and s contain no null values.
Which one of the following queries is NOT equivalent to Q?

(a) $\sigma_{B<5}(r \bowtie s)$
(b) $\sigma_{B<5}(r \text{ LOJ } s)$
(c) $r \text{ LOJ } (\sigma_{B<5}(s))$
(d) $\sigma_{B<5}(r \text{ LOJ } s)$

GATE CSE-2017 SET-1 MCQ (1-MARKS)

Q10. Consider a database that has the relation schemas EMP(Empld, EmpName, Deptld) and DEPT(DeptName, Deptld). Note that the Deptldcan be permitted to a NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.
(I) $\{t|\exists u \in EMP(t[EmpName] = u[EmpName] \wedge \forall \vee \in DEPT(t[Deptld] \neq V[Deptld]))\}$
(II) $\{t|\exists u \in EMP(t[EmpName] = u[EmpName] \wedge \exists \vee \in DEPT(t[Deptld] \neq V[Deptld]))\}$
(III) $\{t|\exists u \in EMP(t[EmpNmae) = u[EmpName] \wedge \exists \vee \in DEPT(t[Deptld] = V[Deptld]))\}$
Which of the above queries are safe?

(a) I and II only
(b) I and III only
(c) II and III only
(d) I, II and III.

GATE CSE-2017 SET-1 NAT (2-MARKS)

Q11. Consider a database that has the relation schema CR(StudentName, CourseName). An instance of the schema CR is as given below:

CR Table

Student Name	Course Name
SA	CA
SA	CB
SA	CC
SB	CB
SB	CC
SC	CA
SC	CB
SC	CC
SD	CA
SD	CB
SD	CC
SD	CD
SE	CD
SE	CA
SE	CB
SF	CA
SF	CB
SF	CC

The following query is run on the database:
T1 ← π CourseName (σ StudentName = 'SA' (CR))
T2 ← CR ÷ T1
The number of rows in T2 is _____.

GATE CSE-2015 SET-2 MCQ (2-MARKS)

Q12. Consider two relations R_1(A, B) with the tuples (1, 5), (3, 7) and R_2(A, C) = (1, 7), (4, 9).
Assume that R(A, B, C) is the full natural outer join of R_1 and R_2. Consider the following tuples of the form (A, B, C): a = (1,5, null), b = (1, null, 7), c = (3, null, 9), d = (4, 7, null), e = (1, 5, 7), f = (3, 7, null), g = (4, null, 9). Which one of the following statements is correct?

(a) R contains a,b,e,f,g but not c, d.
(b) R contains all of a,b,c,d,e,f,g
(c) R contains e,f,g but not a,b
(d) R contains e but not f,g

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2014 SET-3 NAT (2-MARKS)

Q13. Consider the relational schema given below, where eld of the relation dependent is a foreign key referring to empld of the relation employee. Assume that every employee has at least one associated dependent in the dependent relation.
employee(empld,empName,empAge)
dependent(depld, eld,depName, depAge).
Consider the following relational algebra query:
Π_{empId} (employee) − Π_{empId} (employee $\bowtie_{(empId=eID)\wedge(empAge \le depAge)}$ dependent)
The above query evaluates to the set of emplds of employees whose age is greater than that of

(a) Some dependent
(b) All dependents.
(c) Some of his/her dependents
(d) All of his/her dependents

GATE CSE-2014 SET-1 MCQ (1-MARKS)

Q14. Given the following statements:

S1: A foreign key declaration can always be replaced by an equivalent check assertion in SQL.
S2: Given the table R(a,b,c) where a and b together form the primary key, the following is a valid table definition.

CREATE TABLES (
a INTEGER,
d integer,
e integer,
PRIMARY KEY (d),
FOREIGN key (a) references R)
Which one of the following statements is CORRECT?

(a) S1 is TRUE and S2 is FALSE.
(b) Both S1 and S2 are TRUE.
(c) S1 is FALSE and S2 is TRUE
(d) Both S1 and S2 are FALSE.

GATE CSE-2014 SET-2 MCQ (1-MARKS)

Q15. Consider a join (relation algebra) between relations r(R) and s(S) using the nested loop method. There are 3 buffers each of size equal to disk block size, out of which one buffer is reserved for intermediate results.
Assuming size(r(R)) < size(s(S)), the join will have fewer number of disk block accesses if:

(a) relation r(R) is in the outer loop.
(b) relation s(S) is in the outer loop.
(c) join selection factor between r(R) and s(S) is more than 0.5.
(d) join selection factor between r(R) and s(S) is less than 0.5.

GATE CSE-2014 SET-3 MCQ (1-MARKS)

Q16. What is the optimized version of the relation algebra expression $\pi_{A1}(\pi_{A2}(\sigma_{F1}(\sigma_{F2}(r))))$, where A1, A2 are sets of attributes in r with A1 ⊂ A2 and F1, F2 are Boolean expressions based on the attributes in r?

(a) $\pi_{A1}(\sigma_{(F1\wedge F2)}(r))$
(b) $\pi_{A1}(\sigma_{(F1\vee F2)}(r))$
(c) $\pi_{A2}(\sigma_{(F1\wedge F2)}(r))$
(d) $\pi_{A2}(\sigma_{(F1\vee F2)}(r))$

GATE CSE-2013 MCQ (2-MARKS)

Q17. Consider the following relation schema
Students(rollno:integer, sname: string)
Courses(courseno:integer,cname: string)
Registration (rollno: integer, courseno: integer, percent: real)
Which of the following queries are equivalent to this query in English?
"Find the distinct names of all students who score more than 90% in the course numbered 107".
(I) SELECT DISTINCT S.sname
FROM Students as S, Registration as R
WHERE R.rollno = S. rollno AND R.courseno = 107 AND R.percent> 90
(II) $\pi_{sname}(\sigma_{courseno=107\ \wedge\ percent>90}$ (Registration ⋈Students))
(III) {T|∃S ∈ Students, ∃R ∈ Registration (S.rollno = R.rollno ∧ R.courseno = 107 ∧ R.percent> 90 ∧ T.sname = S.sname)}
(IV) $\{<S_N>|\exists S_R \exists R_P(<S_R,S_N>\in Students \wedge <S_R,107,R_P>\in Registration \wedge R_P>90)\}$

(a) I, II, III and IV
(b) I, II and III only
(c) I, II and IV only
(d) II, III and IV only

GATE CSE-2012 MCQ (2-MARKS)

Q18. Suppose R_1(A, B) and R_2(C, D) are two relation schemas. Let r_1 and r_2 be the corresponding relation instances. B is a foreign key that refers to C in R_2. If data in r_1 and r_2 satisfy referential integrity constraints, which of the following is ALWAYS TRUE?

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) $\Pi_B(r_1) - \Pi_c(r_2) = \emptyset$
(b) $\Pi_C(r_2) - \Pi_B(r_1) = \emptyset$
(c) $\Pi_B(r_1) = \Pi_c(r_2)$
(d) $\Pi_B(r_1) - \Pi_c(r_2) \neq \emptyset$

GATE CSE-2011 MCQ (2-MARKS)

Q19. Consider a relational table r with sufficient number of records, having attributes A1, A2,…, An and let 1 <= p <= n. Two queries Q1 and Q2 are given below.
Q1:
$\pi_{A_1, A_2, \ldots, A_p}(\sigma_{A_p = c}(r))$ where c is a constant.
Q2:
$\pi_{A_1, A_2, \ldots, A_p}(\sigma_{c_1 \leq A_p \leq c_2}(r))$ where c_1 and c_2 are constants.
The database can be configured to do ordered indexing on Ap or hashing on Ap. Which of the following statements is TRUE?

(a) Ordered indexing will always outperform hashing for both queries
(b) Hashing will always outperform ordered indexing for both queries
(c) Hashing will outperform ordered indexing on Q1, but not on Q2
(d) Hashing will outperform ordered indexing on Q2, but not on Q1.

GATE CSE-2010 MCQ (2-MARKS)

Q20. The following functional dependencies hold for relations R(A, B, C) and S(B, D, E):
B -> A , A -> C
The relation R contains 200 tuples and the relation S contains 100 tuples. What is the maximum number of tuples possible in the natural join of R and S (R ⋈S) ?

(a) 100
(b) 200
(c) 300
(d) 2000

GATE CSE-2009 MCQ (2-MARKS)

Q21. Let R and S be relational schemes such that R = {a, b, c} and S = {c}. Now consider the following queries on the database:

1 $\pi_{R-S}(r) - \pi_{R-S}\left(\pi_{R-S}(r) \times s - \pi_{R-S,S}(r)\right)$

2. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall u \in s(\exists v \in r(u = v[S] \wedge t = v[R-S]))\}$

3. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall v \in r(\exists u \in s(u = v[S] \wedge t = v[R-S]))\}$

4. Select R.a, R.b

From R, S
Where R.c = S.c
Which of the above queries are equivalent?
(a) 1 and 2
(b) 1 and 3
(c) 2 and 4
(d) 3 and 4

GATE CSE-2008 MCQ (2-MARKS)

Q22. Let R and S be two relations with the following schema:
R(P, Q, R1, R2, R3)
S(P, Q, S1, S2)
where {P, Q} is the key for both schemas. Which of the following queries are equivalent?
1. $\Pi_P(R \bowtie S)$
II. $\Pi_P(R) \bowtie \Pi_P(S)$
III. $\Pi_P\left(\Pi_{P,Q}(R) \cap \Pi_{P,Q}(S)\right)$
IV. $\Pi_P\left(\Pi_{P,Q}(R) - \left(\Pi_{P,Q}(R) - \Pi_{P,Q}(S)\right)\right)$

(a) Only I and II
(b) Only I and III
(c) Only I, II and III
(d) Only I, III and IV

GATE CSE-2008 MCQ (1-MARKS)

Q23. Which of the following tuple relational calculus expression(s) is/are equivalent to ∀t ∈ r (P(t))?
I. ¬∃t ∈ r (P(t))
II. ∃t ∉ r (P(t))
III. ¬∃t ∈ r (¬P(t))
IV. ∃t ∉ r (¬P(t))

(a) I only
(b) II only
(c) III only
(d) III and IV only

GATE CSE-2008 MCQ (2-MARKS)

Q24. Consider the following relational schema:
• Student(school-id, sch-roll-no, sname, saddress)
• School(school-id, sch-name, sch-address, sch-phone)
• Enrolment(school-id, sch-roll-no, erollno, examname)
• ExamResult(erollno, examname, marks)
Consider the following tuple relational calculus query.

Check Out Our GATE Courses & Secure Your Dream Rank!

{ t | ∃E ∈ Enrolment (t = E.school-id ∧
|{ x | x ∈ Enrolment ∧x.school-id = t ∧
(∃B ∈ExamResult (B.erollno =
x.erollno∧B.examname = x.examname∧B.marks> 35))
}| ÷ |{ x | x ∈ Enrolment ∧x.school-id = t }| * 100 > 35
) }
If a student needs to score more than 35 marks to pass an exam, what does the query return?

(a) The empty set
(b) schools with more than 35% of its students enrolled in some exam or the other
(c) Schools with a pass percentage above 35% over all exams taken together
(d) schools with a pass percentage above 35% over each exam

GATE CSE-2007 MCQ (2-MARKS)

Q25. Information about a collection of students is given by the relation
studInfo(studId, name, sex).
The relation enroll(studId, courseId) gives which student has enrolled for (or taken) what course(s).
Assume that every course is taken by at least one male and at least one female student.
What does the following relational algebra expression represent?
$\pi_{courceId}\left((\pi_{studId}(\sigma_{sex = 'female'}(\text{studInfo})) \times \pi_{courseId}(\text{enroll})\right) - \text{enroll})$

(a) Courses in which all the female students are enrolled
(b) Courses in which a proper subset of female students are enrolled.
(c) Courses in which only male students are enrolled.
(d) None of the above.

GATE CSE-2007 MCQ (2-MARKS)

Q26. Consider the relation employee(name, sex, supervisorName) with name as the key.
supervisorName gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?
{e.name | employee(e) ∧ }
(∀x) [¬employee(x) ∨x.supervisorName ≠ e.name
∨x.sex = "male"] }

(a) Names of employees with a male supervisor.
(b) Names of employees with no immediate male subordinates.
(c) Names of employees with no immediate female subordinates.
(d) Names of employees with a female supervisor.

GATE CSE-2007 MCQ (2-MARKS)

Q27. Consider a selection of the form $\sigma_{A \leq 100}(r)$, where r is a relation with 1000 tuples. Assume that the attribute values for A among the tuples are uniformly distributed in the interval [0, 500]. Which one of the following options is the best estimate of the number of tuples returned by the given selection query?

(a) 50
(b) 100
(c) 150
(d) 200

GATE CSE-2007 MCQ (2-MARKS)

Q28. Consider the following relation schemas :

- b -Schema = (b-name, b -city, assets)
- a-Schema = (a-num, b-name, bal)
- d-Schema = (c-name, a-number)

Let branch, account and depositor be respectively instances of the above schemas. Assume that account and depositor relations are much bigger than the branch relation.

Consider the following query:
$\Pi_{c\text{-name}}(\sigma_{b\text{-city}} = \text{"Agra"} \wedge \text{bal} < 0 (\text{branch} \bowtie (\text{account} \bowtie \text{depositor})$
Which one of the following queries is the most efficient version of the above query?
a). $\Pi_{c\text{-name}}(\sigma_{bal<0}(\sigma_{b\text{-city}} = \text{"Agra"} \text{ branch} \bowtie \text{account}) \bowtie \text{depositor})$
b). $\Pi_{c\text{-name}}(\sigma_{b\text{-city}} = \text{"Agra"} \text{ branch} \bowtie (\sigma_{bal<0} \text{ account} \bowtie \text{depositor}))$
c). $\Pi_{c\text{-name}}((\sigma_{b\text{-city}} = \text{"Agra"} \text{ branch} \bowtie \sigma_{b\text{-city}} = \text{"Agra"} \wedge \text{bal} < 0 \text{ account}) \bowtie \text{depositor})$
d). $\Pi_{c\text{-name}}(\sigma_{b\text{-city}} = \text{"Agra"} \text{ branch} \bowtie (\sigma_{b\text{-city}} = \text{"Agra"} \wedge \text{bal} < 0 \text{ account} \bowtie \text{depositor}))$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2006 MCQ (1-MARKS)

Q29. Consider the relations r1(P, Q, R) and r2(R, S, T) with primary keys P and R respectively. The relation r1 contains 2000 tuples and r2 contains 2500 tuples. The maximum size of the join r1⋈ r2 is :

(a) 2000
(b) 2500
(c) 4500
(d) 5000

GATE CSE-2006 MCQ (1-MARKS)

Q30. Which of the following relational query languages have the same expressive power?

1. Relational algebra
2. Tuple relational calculus restricted to safe Expressions
3. Domain relational calculus restricted to safe expressions

(a) II and III only
(b) I and II only
(c) I and III only
(d) I, II and III

GATE CSE-2005 MCQ (1-MARKS)

Q31. Let r be a relation instance with schema R = (A, B, C, D). We define
$r_1 = \pi_{A,B,C}$ (r) and $r_2 = \pi_{A,D}$ (r).
Let s = $r_1 \bowtie r_2$, where ⋈ denotes natural join. Given that the decomposition of r into r_1 and r_2 is lossy, which one of the following is TRUE?

(a) s ⊂ r
(b) r ∪ s = r
(c) r ⊂ s
(d) r ⋈ s = s

ANSWER KEY

1. (b)	**2. (a)**	**3. (2)**	**4. (a,b)**
5.(a)	**6.(820)**	**7. (c)**	**8. (1)**
9.(c)	**10. (d)**	**11. (4)**	**12. (c)**
13.(d)	**14. (d)**	**15. (a)**	**16. (a)**
17.(a)	**18. (a)**	**19. (c)**	**20. (a)**
21.(a)	**22. (d)**	**23. (c)**	**24. (c)**
25.(b)	**26. (c)**	**27. (d)**	**28. (a)**
29.(a)	**30. (d)**	**31. (c)**	

HINTS & SOLUTION

1. Answer :-B
Explantion :-
Given
E = { u | ∃v ∃w ((u, v) ∈ R ∧ (v, w) ∈ S) }
(u, v) ∈ R ⇒ u = P, v = Q
(v, w) ∈ S ⇒ v = X, w = Y
Common variable v ⇒ R.Q = S.X
Result requires only u ⇒ attribute P
So expression becomes:
π_p (R $\bowtie_{R.Q = S.X}$ S)
Option (B) represents the same join condition.
Final Answer: B

2. Answer:- A
Explanation
TRC query to return the name of the players who play for the team having tname as 'M1' is
{p.pname | p ∈ players ∧∃t (t ∈ teams ∧p.tid = t.tid∧t.tname = 'MI')}

- It selects pname from players.
- It ensures that p.tid = t.tid for some team t where tname = 'MI'.

This correctly finds players belonging to the 'MI' team.
This query also works : {p.pname | ∃t∈teams(t.tname='M1′ ∧t.tid=p.tid)}

3. Answer:- 2
Explanation:-
The output of R($\bowtie_{R.A=S.A}$ S) is simply return the number of tuples where R.A value is equal to S.A. So the output is as follow:

A	B	C
10	20	90
30	40	45
30	50	45

from the above table now we have to perform ($\sigma_{B<C}$(R$\bowtie_{R.A=S.A}$ S)). it return the number of rows where B′s value is less than C. so the output is:

A	B	C
10	20	90
30	40	45

So the given expression returns 2 tuples as output.

Check Out Our GATE Courses & Secure Your Dream Rank!

4. **Answer :- A, B**
Explanation :-
π eId (Own) $\div$ π bId (Brand) gives the set of eIds who own **every brand** in the Brand relation.
Option (b) is an expansion of the division operation using basic relational algebra operators.

5. **Answer:-A**
Explanation :-

empAge

empNo	age
1	20
2	40
3	60
× 4	10
5	15

emp No 1	age 1
1	20
2	40
3	60
4	10
5	15

Emp No = 4 will not be selected because the condition age>age$_1$ is not satisfied for this EmpNo.

6. **Answer:- 820**
Explantion :-

The above answer is TRUE for SQL SELECT but not for Relational Algebra as by theory relational algebra operates on a set which means all the elements must be distinct. Since we have 15 distinct possible values for A and 20 distinct possible values for B, in strict relational algebra we'll get

Estimated number of tuples $= \frac{41}{60} \times (15 \times 20) = 205.$
Official Answer: 205 OR 820.

7. **Answer:- C**
Explantion S1: False
Because it is not necessary that we should always have a foreign key(or) more than one FK are also possible in a relation.
S2: False
Foreign Key may refer tuples of the same relation.
Thus, Both S1 and S2 are false. is the correct answer.

8. **Answer :- 1**
Explanation :- $\sigma_{(P.Y=R.Y \wedge R.V=V2)}(P \times R)$

X	Y	Z	V
X2	Y2	Z2	V2

$\Pi_X(\sigma_{(PY=RY \wedge RV=V2)}(P \times R))$

X
X 2

$\sigma_{(QY=RY \wedge QT \times 2)}(Q \times R)$

X	Y	T	V
X 1	Y 1	6	V 1
X 1	Y 2	5	V 3
X 1	Y 2	5	V 2

X
X 1

$\Pi_x(\sigma_{(QY=R,Y \wedge Q,P,2)}(Q \times R))$

$\Pi_x(\sigma_{(PY-RYY_{ARY}=V2)}(P \times R)) - \Pi_x(\sigma_{(QY-RY\ AQTB:2}(Q \times R))$
$= \underset{X2}{X} - \underset{X1}{Y} = 1$

9. **Answer: C**
Explantion :-
Option C is written as:

$$r\text{LOJ}(\sigma_{B<5}(s))$$

A Left Outer Join (LOJ) guarantees that all tuples from the left relation (r) are preserved in the final output, regardless of whether they find a match in the right relation.

- For tuples in r where $B < 5$, they will successfully match with the filtered relation s.
- For tuples in r where $B \geq 5$, they will find no match in the filtered relation $\sigma_{B<5}(s)$.
- Instead of being discarded (like in an inner join), these tuples are still retained in the output, padded with NULL for the attribute C.

Because Option C includes tuples where $B \geq 5$ (filled with NULL s for C), it produces a different result than Q.

10. **Answer:- D**
Explantion:-
An expression is said to be safe, if it yield a finite number of tuples, otherwise unsafe.
(I) Gives EmpNames who do not belong to any Department. So, it is going to be a finite number of tuples as a result.
(II) Gives EmpNames who do not belong to some Department. This is also going to have finite number of tuples.
(III) Gives EmpNames who do not belong to same Department. This one will also give finite number of tuples.
All the expressions I, II and III are giving finite number of tuples. So, all are safe.

Check Out Our GATE Courses & Secure Your Dream Rank!

11. Answer: 4

Explanation:- From SA, the courses are CA, CB, CC.
So, T1 = {CA, CB, CC}.
Now find students who have all these courses:

- SA → has CA, CB, CC
- SB → missing CA
- SC → has CA, CB, CC
- SD → has CA, CB, CC
- SE → missing CC
- SF → has CA, CB, CC

SA ,SC, SD,SF

Total such students = 4

12. Answer:- C

Explantion :-

$*R_1$

A	B
1	5
3	7

$*R_2$

$*R = R_1 \bowtie R_2$

A	B	C
1	5	7
3	7	Null
4	Null	9

A	B
1	7
4	9

So, from the above resultant table, R contains e, f, g only but not a, b.

13. Answer :- D

Explanation :- On evaluating the inner query we get the employees whose age is less than or equal to at least one of his dependents. So, on subtracting it from the set of employees, gives employees whose age is greater than all of his dependents .So, All of his/her dependents is the correct option.

14. Answer:-D

Explantion :-
S1: False
Using a check constraint, we could be having the same effect as foreign key while adding elements to the child table. But on deleting elements from the parent table, the referential integrity constraint is no longer valid. So, a check constraint cannot replace a foreign key.
S2: False:
Foreign key in one table should uniquely identifies a row of other table. In above table definition, table S has a foreign key that refers to field 'a' of R. The field 'a' in table S doesn't uniquely identify a row in table R.

15. Answer:-A

Explantion :- The join relation between r(R) and s(S) which is using nested loop will be as follows:

For each tuple r in R do
For each tuple s in S do
If r and s satisfy the join condition then
output the tuple.
The above algorithm involves $(n_r*b_s+b_r)$ block transfer and (n_r+b_r) seeks.
where,
b_r → no. of blocks in R
b_s → no. of blocks in S
h_r → no. of tuples in R
To have less block accesses, n_r should be less as in question it is given that size(r(R)) < size(s(S)).
So, to have fewer no. of disk block accesses the relation r(R) should be in outer loop.

16. Answer:- A

Explanation :- Note: Nested Queries Logic works with AND operation. Since A1 ⊂ A2 will get only attribute A1 as it is in the outside. So we can remove project A2.
Two Selects with Boolean expression can be combined into one select with AND of two Boolean expressions.

17. Answer:-A

Explanation—The Four queries are given in SQL, RA, TRC and DRC. All four of them are equivalent to the query given in English.

18. Answer:- A

Explantion :- Referential integrity constraints denotes that all the values in foreign key should be present in the primary key.
So we can understand that $r_2(C)$ is the superset of $r_1(B)$.
So (subset - superset) is always empty.
$R_1(AB)$

A	(B)
1	7
2	7
3	8
4	9

C	D
6	1
7	2
8	5
9	6

Ex.

$(a)\quad \pi_B(r_1) - \pi_c(r_2) = \phi$
$(7,8,9] - [6,7,8,9) = \phi$
$(b)\quad \pi_c(r_2) - \pi_B(r_1) = \phi$
$[6,7,8,9) - [7,8,9] = (6) \neq \phi$
$(c)\quad \pi_B(r_1) = \pi_c(r_2)]$
$[7,8,9] \neq [6,7,8,9]$
$(d)\quad \pi_B(r_1) - \pi_c(r_2) \neq \phi$
$[7,8,9] - [6,7,8,9] = \phi$

19. Answer:- C

Explantion :-
Hashing works well on "equal" queries, while ordered indexing works better on range queries.
For example, consider B+ tree, once we have searched a key in B+; we can find range of values via the block

Check Out Our GATE Courses & Secure Your Dream Rank!

pointers pointing to another block of values on the leaf node level.

20. Answer:- A

Explantion:- The maximum number of tuples is 100.
Because B -> A and A -> C in relation R, B is a key for R. This means all 200 B values in R are unique.
The natural join matches on B. Since each of the 100 tuples in S can match at most one unique B value in R, the maximum result size is limited by the number of tuples in S.

21. Answer :- A

Explanation :- Given
R = {a, b, c}, S = {c}
So R − S = {a, b}
Query 4 (SQL)
Select R.a, R.b
From R, S
Where R.c = S.c
⇒ Join on c and project (a,b)
⇒ Return (a,b) tuples of R whose c value exists in S
This is semi join result.

Query 2 (TRC)
t ∈ π_{R−S}(r) AND
∀u ∈ s ∃v ∈ r such that
u = v[S] AND t = v[R−S]
Meaning:
For tuple t = (a,b), for every u in S, there exists a tuple v in R with
- v.c = u
- projection (a,b) = t

⇒ t must appear with all c values in S
⇒ division semantics
Query 3 (TRC)
t ∈ π_{R−S}(r) AND
∀v ∈ r ∃u ∈ s such that
u = v[S] AND t = v[R−S]
Meaning:
For every tuple v in R having projection t, its c must be in S
⇒ All occurrences of (a,b) in R must have c in S
⇒ Equivalent to semijoin result
Query 1 (RA)

$$\pi_{R-S}(r) - \pi_{R-S}\left(\pi_{R-S}(r) \times s - \pi_{R-S,S}(r)\right)$$

This is standard relational division form
⇒ Same as Query 2
Conclusion
- 1 ≡ 2 (division)
- 3 ≡ 4 (semijoin)

Final Answer: A (1 and 2)

22. Answer :- D

Explanation :-
Given
- R(P,Q,…) and S(P,Q,…)
- {P,Q} is key for both

⇒ Natural join R ⋈ S happens on P,Q
I. πP (R ⋈ S)
Join keeps tuples where (P,Q) exist in both relations.
Then project P.
⇒ P values whose (P,Q) pair exists in both.

II. πP(R) ⋈ πP(S)
πP(R) and πP(S) are single-attribute relations.
Natural join on P ⇒ intersection of P values.
But this ignores Q matching.
⇒ Not equivalent to I.

III. πP (πP,Q(R) ∩ πP,Q(S))
- Intersection of (P,Q) pairs
- Then project P

This is exactly what join + projection in I gives.
⇒ Equivalent to I
IV. πP (πP,Q(R) − (πP,Q(R) − πP,Q(S)))
Using identity:
A − (A − B) = A ∩ B
⇒ Expression becomes
πP (πP,Q(R) ∩ πP,Q(S))
⇒ Same as III ⇒ same as I.

23. Answer :-C

Explanation :-
Given
∀t ∈ r (P(t))
Logical equivalence:
∀t P(t) ≡ ¬∃t (¬P(t))
I. ¬∃t ∈ r (P(t))
⇒ No tuple satisfies P
⇒ Opposite meaning
Not equivalent
II. ∃t ∉ r (P(t))
⇒ Tuple outside r satisfies P
⇒ Irrelevant
Not equivalent
III. ¬∃t ∈ r (¬P(t))
⇒ No tuple violates P
⇒ Equivalent to universal statement
✓ Equivalent
IV. ∃t ∉ r (¬P(t))
⇒ Tuple outside r violates P
⇒ Irrelevant
Not equivalent.

24. Answer :- C

Explanation :-
t | ∃E ∈ Enrolment t = E.school-id
Returns school-ids from Enrolment table SUCH THAT

Check Out Our GATE Courses & Secure Your Dream Rank!

• |{x | x ∈ Enrolment ∧x.school-id = t ∧ (∃B ∈ExamResultB.erollno = x.erollno∧B.examname = x.examname∧B.marks> 35)}| ÷
• the number of student enrolments from the school for exams with marks > 35 divides
• |{x | x ∈ Enrolment ∧x.school-id = t}|
• total number of student enrolments from the school
▪ ×100 > 35
▪ percentage of student enrolments with mark > 35 is > 35
Since to pass an exam > 35 mark is needed, this means selecting the school-ids where the pass percentage of students across all the exams taken together is > 35.

25. Answer: B
Explanation:-

STUDENTINFO

1	A	M
2	A	F
3	A	F

ENROLL

1	C1
1	C2
2	C1
2	C2
3	C2

- $\pi_{courceId}(\sigma_{sex="female"}(\text{studInfo})) \times \pi_{courseId}(\text{enroll})$

$$\implies \begin{matrix}2\\3\end{matrix} \times \begin{matrix}C1\\C2\end{matrix} = \begin{matrix}2 & C1\\2 & C2\\3 & C1\\3 & C2\end{matrix}$$

- $(\pi_{studId}(\sigma_{sex="female"}(\text{studInfo})) \times \pi_{courseId}(\text{enroll})) - \text{enroll})$

$$\implies 3 \quad C1$$

- $\pi_{courceId}((\pi_{studId}(\sigma_{sex="female"}(\text{studInfo})) \times \pi_{courseId}(\text{enroll})) - \text{enroll})$

$$\implies C1$$

$C1$ is a course id in which not all girl students enrolled.
i.e. a proper subset of girls students appeared.

Hence **(B)** is the correct answer.

26. Answer :- C
Explantion:-
The condition says:
For all x,
either x is not an employee
or x is not supervised by e
or x is male.
This means:
There does not exist any employee x such that x.supervisorName = e.name AND x.sex ≠ "male".
So, no subordinate of e is female.
Therefore, e has no immediate female subordinates.
Final Answer: C. Names of employees with no immediate female subordinates.

27. Answer:D
Explanation:
Values are uniformly distributed in [0, 500].
Range satisfying A ≤ 100 is [0, 100].
Selectivity = 100 / 500 = 1/5.
Number of tuples = 1000 × (1/5) = 200.
Final Answer: D. 200

28. Answer :- A
Explanation :-
For better record processing, We should always do select before join to avoid unnecessary tuples being considered for join.(For an SQL query this is not strictly required as DBMS will rearrange the query to make it efficient) and processing a smaller table gives more efficiency than the larger one.
option (A)
: Π c-name (σ bal< 0 (σ b-city = "Agra" branch ⋈ account) ⋈ depositor) here we are doing join between relatively smaller table(branch) and larger one(account) and the output that this inner table gives (which is smaller in comparison to joins that we are doing in B) is used for join with depositor table with the selection condition.(same as given query).
option (B)
: Π c-name (σ b-city = "Agra" branch ⋈ (σ bal< 0 account ⋈ depositor)) here we are doing a join between two massive table account and depositor with selection condition(for balance less than zero) and the output that this inner table gives is used for join with branch table(relatively smaller table) with the selection condition (for city Agra). (same as given query) so option A and option B both are same as given query but there is difference between record processing as the record processing of option A will get reduced by some rate. (as filtered conditions are applied after the join between one smaller and one larger table which will give relatively smaller table than the join of two larger tables). So, overall option A is much better than option B.
option (C)
: Π c-name ((σ b-city = "Agra" branch ⋈ σ b-city = "Agra" depositor) Incorrect as there is no b-city column in a-Schema.
option (D)
: Π c-name (σ b-city = "Agra" branch ⋈ (σ b-city = "Agra" depositor)) Incorrect as there is no b-city column in a-Schema. ⋀bal< 0 account) ⋈⋀bal< 0 account ⋈ Note: don't miss this sentence "account and depositor relations are much bigger than the branch relation" because That makes A the best answer.

29. Answer : A
Explanation :-
$r_1 \bowtie r_2$ is a join operation done on the common attribute R. Further R is the primary key of R2
When we take a , the value of common attribute(R2 in this case) should match.The value of R in r2 is matched with corresponding R in r1 . So it will have 2000 tuples. So correct option is (A).

Check Out Our GATE Courses & Secure Your Dream Rank!

30. Answer : D

Explanation : Relational algebra is a procedural query language where we input - relations and it yields relations as output. It provides method to get the result. It is performed recursively on a relation and the in between results are relations(output). Basic set of operations for the relational model. Relational calculus is a non - procedural query language. It provides the query to get result. Higher level declarative language for specifying relational queries. Tuple Relational Calculus operates on each tuple.

Domain Relational Calculus operates on each column or attribute. Safe expression means fixed no. of tuple or column or attribute as a result But all of them has same expressive power. Just different ways to do so.

31. Answer :C

Explanation :- Given:

Here

decomposition of r to r1 and r2 is lossy.

Lossy decomposition :R1 and R2 has a common attribute but the common attribute A is not key in either of R1 or R2.

Due to this when R1 and R2 are joined,we get spurious tuples which are not in Original R.

$s=r1 \bowtie r2$ Therefore, correct answer is $r \subset s$

4 STRUCTURED QUERY LANGUAGE

GATE CSE-2025 SET-2 MSQ (2-MARKS)

Q1. Consider the following relational schema:
Students(rollno: integer, name: string, age: integer, cgpa: real)
Courses(courseno: integer,cname: string, credits: integer)
Enrolled(rollno: integer,courseno: integer,grade: string)
Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e. courseno) 1470?

(a) SELECT S.name FROM Students S WHERE EXISTS (SELECT * FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno)

(b) SELECT S.name FROM Students S WHERE SIZEOF (SELECT * FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno) > 0

(c) SELECT S.name FROM Students S WHERE 0 < (SELECT COUNT(*) FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno)

(d) SELECT S.name FROM Students S NATURAL JOIN Enrolled E WHERE E.courseno = 1470

GATE CSE-2025 SET-1 NAT (2-MARKS)

Q2. Consider the following database tables of a sports league.
1.player (pid, pname, age)team(tid, tname, city, cid)
2. coach (cid, cname) members (pid,tid)

player

pid	pname	age
1	Jasprit	31
2	Atharva	24
3	Ishan	26
4	Axar	30

coach

cid	cname
101	Ricky
102	Mark
103	Trevor

team

tid	tname	city	cid
10	MI	Mumbai	102
20	DC	Delhi	101
30	PK	Mohali	103

members

pid	tid
1	10
2	30
3	10
4	20

SELECT MIN (P.age)
FROM player P
WHERE P.pid IN (SELECT M.pid
FROM team T, coach C, members M
WHERE C.cname = 'Mark'
AND T.cid = C.cid
AND M.tid = T.tid);
The value returned by the given SQL query is _____
(Answer in integer)

GATE CSE-2024 SET-2 NAT (2-MARKS)

Q3. The relation schema Person (pid, city) describes the city of residence for every person uniquely identified by pid. The following relational algebra operators are available: selection, projection, cross product, and rename.
To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of cross product operations that must be used is:

(a) 1
(b) 2
(c) 3
(d) 4

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2023 NAT (2-MARKS)

Q4. Consider the following table named Student in a relational database. The primary key of this table is rollNum.

rollINum	name	gender	marks
1	Naman	M	62
2	Aliya	F	70
3	Aliya	F	80
4	James	M	82
5	Swati	F	65

The SQL query below is executed on this database.

```
SELECT *FROM Student
WHERE gender = 'F' AND  marks  > 65;
```

The number of rows returned by the query is _____.

GATE CSE-2022 NAT (2-MARKS)

Q5. Consider the relational database with the following four schemas and their respective instances.

Student(sNo, sName, dNo)
Dept(dNo, dName)
Course(cNo, cName, dNo)
Register(sNo, cNo)

Student

sNo	sName	dNo
S01	James	D01
S02	Rocky	D02
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Dept

dNo	dName
D01	CSE
D02	EEE

Course

cNo	cName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register

sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

```
SELECT *
FROM Student AS S
WHERE NOT EXISTS (
   (SELECT cNo
    FROM Course
    WHERE dNo = 'D01')
   EXCEPT
   (SELECT cNo
    FROM Register
    WHERE sNo = S.sNo)
);
```

The number of rows returned by the above SQL query is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2021 SET-2 MCQ (2-MARKS)

Q6. The relation scheme given below is used to store information about the employees of a company, where empId is the key and deptId indicates the department to which the employee is assigned. Each employee is assigned to exactly one department.
emp (EMPID, name, gender, salary, deptId)
Consider the following SQL query:
select deptId, count(*)
from emp
where gender = "female" and salary > (select avg(salary) from emp)
group by deptId;
The above query gives, for each department in the company, the number of female employees whose salary is greater than the average salary of:

(a) employees in the company
(b) female employees in the department.
(c) employees in the department.
(d) female employees in the company.

GATE CSE-2020 NAT (2-MARKS)

Q7. Consider a relational database containing the following schemas.

Catalogue

Sno	Pno	cost
S1	P1	150
S1	P2	50
S1	P3	100
S2	P4	200
S2	P5	250
S3	P1	250
S3	P2	150
S3	P5	300
S3	P4	250

Suppliers

Sno	Sname	location
S1	M/s Royal furniture	Delhi
S2	M/s Balaji furniture	Bangalore
S3	M/s Premium furniture	Chennai

Parts

pno	Sname	Location
P1	Table	Wood
P2	Chair	Wood
P3	Table	Steel
P4	Almirah	Steel
P5	Almirah	Wood

The primary key of each table is indicated by underlying the constituent fields.

```
SELECT s.sno, s.sname
FROM Suppliers s, Catalogue c
WHERE s.sno = c.sno AND
      Cost > (SELECT AVG (cost)
          FROM Catalogue
          WHERE pno = 'P4'
          GROUP BY pno);
```

The number of rows returned by the above SQL query is _____.

GATE CSE-2019 NAT (2-MARKS)

Q8. A relational database contains two tables Student and Performance as shown below

Student

Roll_no.	Student_name
1	Amit
2	Priya
3	Vinit
4	Rohan
5	Smita

Performance

Roll_no.	Subject_Code	Marks
1	A	86
1	B	95
1	C	90
2	A	89
2	C	92
3	C	80

The primary key of the Student table is Roll_no. For the Performance table, the columns Roll_no. and Subject_code together from the primary Key. Consider the SQL query given below.

Check Out Our GATE Courses & Secure Your Dream Rank!

```
SELECT S.Student.name, sum (P.Marks)
FROM Student S, Performance P
WHERE P.Marks> 84
GROUP BY S.Student_name;
```

The number of rows returned by the above SQL query is _____?

GATE CSE-2018 MCQ (2-MARKS)

Q9. Consider the following two tables and four queries in SQL .

Book (isbn, bname), Stock (isbn, copies)

```
Query 1:  SELECT B.isbn, S.copies
FROM BOOK B INNER JOIN Stock S
ON B.isbn = S.isbn;
Query 2:  SELECT B.isbn, S.copies
FROM BOOK B LEFT OUTER JOIN Stock S
ON B.isbn=S.isbn;
Query 3: SELECT B.isbn, S.copies
FROM BOOk B RIGHT OUTER JOIN Stock S
ON B.isbn = S.isbn;
Query 4:  SELECT B.isbn, S.copies
FROM BOOk B FULL OUTER JOIN Stock S
ON B.isbn = S.isbn;
```

Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?

(a) Query 1
(b) Query 2
(c) Query 3
(d) Query 4

GATE CSE-2017 SET-2 NAT (2-MARKS)

Q10. Consider the following database table named **top_scorer**:

player	country	goals
Klose	Germany	16
Ronaldo	Brazil	15
G Muller	Germany	14
Fontaine	France	13
Pele	Brazil	12
Klinsmann	Germany	11
Kocsis	Hungary	11
Batistuta	Argentina	10
Cubillas	Peru	10
Lato	Poland	10
Lineker	England	10
T Muller	Germany	10
Rahn	Germany	10

Consider the following SQL query:

```
SELECT ta.player
FROM top_scorer AS ta
WHERE ta.goals > ALL (
   SELECT tb.goals
   FROM top_scorer AS tb
   WHERE tb.country = 'Spain'
)
AND ta.goals > ANY (
   SELECT tc.goals
FROM top_scorer AS tc
 WHERE tc.country = 'Germany');
```

The number of tuples returned by the above SQL query is _____.

GATE CSE-2017 SET-1 NAT (1-MARKS)

Q11. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName). An instance of the schema EMP and a SQL query on it are given below.

EMP		
EmpId	Emp Name	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG (EC.Num)
FROM EC
WHERE (DeptName, Num) IN
(SELECT DeptName, COUNT(EmpId) AS
EC(DeptName, Num)
FROM EMP
GROUP BY DeptName);
```

The output of executing the SQL query is _____?

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2016 SET-2 NAT (2-MARKS)

Q12. Consider the following database table named water_schemes:

water-schemes		
scheme_no	district_name	Capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _____.

```
with total (name, capacity) as
        select district_name, sum(capacity)
        from water_schemes
        group by district_name
with total_avg(capacity) as
        select avg(capacity)
        from total
select name
        from total, total_avg
        where total.capacity>=total_avg.capacity;
```

GATE CSE-2015 SET-3 MCQ (1-MARKS)

Q13. Consider the following relation

Cinema (theater, address, capacity)
Which of the following options will be needed at the end of the SQL query
SELECT P1. address
FROM CINEMA P1
such that it always finds the addresses of theaters with maximum capacity?

(a) WHERE P1.capacity>= All (select P2.capacity from Cinema P2)
(b) WHERE P1.capacity>= Any (select P2.capacity from Cinema P2)
(c) WHERE P1.capacity> All (select max(P2.capacity) from Cinema P2)
(d) WHERE P1.capacity> Any (select max(P2.capacity) from Cinema P2)

GATE CSE-2015 SET-1 MCQ (1-MARKS)

Q14. SELECT operation in SQL is equivalent to

(a) the selection operation in relational algebra
(b) that operations in relations algebra, except that SELECT IN SQL retains duplicates
(c) the projection operation in relational algebra
(d) the projection operation in relational algebra except that SELECT in SQL retains duplicates

GATE CSE-2015 SET-1 NAT (2-MARKS)

Q15. Consider the following relations:

Student

Roll_No	**Student_Name**
1	Raj
2	Rohit
3	Raj

Performances

Roll_No	**Course**	**Marks**
1	Math	80
1	English	70
2	Math	75
3	English	80
2	Physics	65
3	Math	80

```
SELECT S. student_Name, sum(P.Marks)
FROM Student S, Performance P
WHERE S.Roll_No = P.Roll_no
GROUP BY S.Student_Name
```

The number of rows that will be returned by the SQL query is _____?

GATE CSE-2014 SET-3 MCQ (2-MARKS)

Q16. Consider the following relational schema:
employee(empid,empName.empDept)
customer(custid,custName,salesRepId,rating) salesRepId is a foreign key referring to empId of theemployee relation. Assume that each employee makes a sale to at least one customer.
What does the following query return?

Check Out Our GATE Courses & Secure Your Dream Rank!

```
SELECT empName
    FROM employee E
    WHERE NOT EXISTS (SELECT custld
     FROM customer C
    WHERE C.salesRepId= E.empld
    AND(C.rating<>'GOOD');
```

(a) Names of all the employees with at least one of their customers having a 'GOOD' rating.
(b) Names of all the employees with at most one of their customers having a 'GOOD' rating.
(c) Names of all the employees with none of their customers having a 'GOOD' rating.
(d) Names of all the employees with all their customers having a 'GOOD' rating.

GATE CSE-2014 SET-2 MCQ (2-MARKS)

Q17. SQL allows duplicates tuples in relations, and correspondingly defines the multiplicity of tuples in the result of joins. Which one of the following queries always gives the same answer as the nested query shown below:

```
select * from R
where a in (select S.a from S)
```

(a) select R.* from R,S where R.a=S.a
(b) select distinct R.* from R,S where R.a=S.a
(c) select R.* from R,(select distinct a from S) as S1 where R.a = S1.a
(d) select R * from R,S where R.a=S.a and is unique R

GATE CSE-2014 SET-1 MCQ (2-MARKS)

Q18. Given the following schema:
employees(emp-id, first-name, last-name, hire-date, dept-id, salary)
departments(dept-id,dept-name,manager-id, location-id)
You want to display the last names and hire dates of all latest hires in their respective departments in the location ID 1700. You issue the following query:

```
SQL> select last-name, hire-date
FROM employees
WHERE (dept-id, hire-date) IN ( SELECTdept-id,
MAX(hire-date)
FROM employees JOIN departments USING(dept-id)
WHERE location-id = 1700
GROUP BYdept-id);
```

What is the outcome?

(a) It executes but does not give the correct result.
(b) It executes and gives the correct result.
(c) It generates an error because of pairwise comparison.
(d) It generates an error because the GROUP BY clause cannot be used with table joins in a subquery.

GATE CSE-2012 MCQ (2-MARKS)

Q19. Consider the above tables A, B and C. How many tuples does the result of the following SQL query contains?

Table A

Id	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

Table B

Id	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
98	Rohit	11

Table C

Id	Name	Age
10	2200	02
99	2100	01

```
SELECT A.id
FROM   A
WHERE  A.age> ALL (SELECT B.age
FROM   B
WHERE  B. name = "arun");
```

(a) 4
(b) 3
(c) 0
(d) 1

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2012 MCQ (2-MARKS)

Q20. Consider the following relations A, B and C:

A		
ID,	Name,	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B		
ID,	Name,	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C		
ID,	Phone,	Area
10	2200	02
99	2100	01

How many tuples does the result of the following relational algebra expression contain?
Assume that the schema of A ∪ B is the same as that of A.
$(A \cup B) \bowtie_{A.ID > 40 \vee C.ID < 15} C$

(a) 7
(b) 4
(c) 5
(d) 9

GATE CSE-2012 MCQ (2-MARKS)

Q21. Which of the following statements are TRUE about an SQL query?
P: An SQL query can contain a HAVING clause even if it does not have a GROUP BY clause.
Q: An SQL query can contain a HAVING clause only if it has a GROUP BY clause.
R: All attributes used in the GROUP BY clause must appear in the SELECT clause.
S: Not all attributes used in the GROUP BY clause need to appear in the SELECT clause.

(a) P and R
(b) P and S
(c) Q and R
(d) Q and S

GATE CSE-2011 MCQ (2-MARKS)

Q22. Consider a database table T containing two columns X and Y each of type integer. After the creation of the table, one record (X=1, Y=1) is inserted in the table. Let MX and MY denote the respective maximum values of X and Y among all records in the table at any point in time. Using MX and MY, new records are inserted in the table 128 times with X and Y values being MX+1, 2*MY+1 respectively. It may be noted that each time after the insertion, values of MX and MY change.
What will be the output of the following SQL query after the steps mentioned above are carried out?

```
SELECT Y
FROM T
WHERE X=7;
```

(a) 127
(b) 255
(c) 129
(d) 257

GATE CSE-2011 MCQ (2-MARKS)

Q23. Database table by name Loan_Records is given below.

Borrower	Bank_Manager	Loan_Amount
Ramesh	Sunderajan	10000.00
Suresh	Ramgopal	5000.00
Mahesh	Sunderajan	7000.00

What is the output of the following SQL query?

```
SELECT Count(*)
FROM ( ( SELECT Borrower, Bank_Manager
        FROM Loan_Records) AS S
        NATURAL JOIN ( SELECT
        Bank_Manager,
        Loan_Amount
        FROM Loan_Records) AS T );
```

(a) 3
(b) 9
(c) 5
(d) 6

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2010 MCQ (1-MARKS)

Q24. A relational schema for a train reservation database is given below. Passenger (pid , pname, age) Reservation (pid , class, tid)

Table : Passenger

pid	pname	age
0	Sachin	65
1	Rahul	66
2	Sourav	67
3	Anil	69

Table : Reservation

pid	class	tid
0	AC	8200
1	AC	8201
2	SC	8201
5	AC	8203
1	SC	8204
3	AC	8202

What pids are returned by the following SQL query for the above instance of the tables?

```
SELECT pid
FROM Reservation ,
WHERE class='AC' AND
        EXISTS (SELECT *
            FROM Passenger
            WHERE age > 65 AND
            Passenger.pid = Reservation.pid )
```

(a) 1,0
(b) 1,2
(c) 1,3
(d) 1,5

GATE CSE-2009 MCQ (2-MARKS)

Q25. Consider the following relational schema:
Suppliers(sid:integer,sname:string, city:string,street:string)
Parts(pid:integer,pname:string, color:string)
Catalog(sid:integer, pid:integer, cost:real)
Consider the following relational query on the above database:

```
SELECT S.sname
FROM Suppliers S
WHERE S.sid NOT IN (SELECT C.sid
FROM Catalog C
WHERE C.pid NOT IN (SELECT P.pid
FROM Parts P
WHERE P.color<> 'blue'))
```

Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

(a) Find the names of all suppliers who have supplied a non-blue part.
(b) Find the names of all suppliers who have not supplied a non-blue part.
(c) Find the names of all suppliers who have supplied only non-blue part.
(d) Find the names of all suppliers who have not supplied only blue parts.

GATE CSE-2008 MCQ (2-MARKS)

Q26. Consider the following relational schema:
Student (school-id, sch-roll-no, sname, saddress)
School (school-id, sch-name, sch-address, sch-phone)
Enrolment (school-id, sch-rollno, erollno, examname)
ExamResult (erollno, examname, marks)
What does the following SQL query output?

```
SELECT sch-name, COUNT(*)
  FROM School C, Enrolment E, ExamResult R
  WHERE E.school-id = C.school-id
            AND E.examname = R.examname
   AND E.erollno = R.erollno
   AND R.marks = 100
    AND S.school-id IN
    (SELECT school-id
    FROM student
    GROUP BY school-id
     HAVING COUNT(*) > 200)
     GROUP BY school-id;
```

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) for each school with more than 200 students appearing in exams, the name of the school and the number of 100s scored by its students.
(b) for each school with more than 200 students in it, the name of the school and the number of 100s scored by its students.
(c) for each school with more than 200 students in it, the name of the school and the number of its students scoring 100 in at least one exam.
(d) nothing; the query has a syntax error.

GATE CSE-2007 MCQ (2-MARKS)

Q27. Consider the table employee(empId, name, department, salary) and the two queries Q_1, Q_2 below. Assuming that department 5 has more than one employee, and we want to find the employees who get higher salary than anyone in the department 5, which one of the statements is TRUE for any arbitrary employee table?

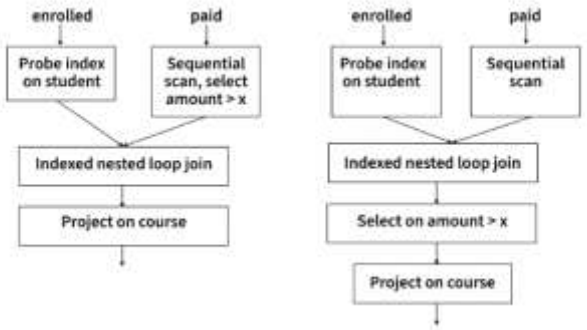
Q_1:
```
Select e.empId
From employee e
Where not exists
(Select * From employee s Where s.department = "5"
and s.salary>= e.salary)
```

Q_2:
```
Select e.empId
From employee e
Where e.salary> Any
(Select distinct salary From employee s Where
s.department = "5")
```

(a) Q_1 is the correct query
(b) Q_2 is the correct query
(c) Both Q_1 and Q_2 produce the same answer
(d) Neither Q_1 nor Q_2 is the correct query

GATE CSE-2006 MCQ (2-MARKS)

Q28. Consider the relation enrolled(student, course) in which (student, course) is the primary key, and the relation paid(student, amount), where student is the primary key. Assume no null values and no foreign keys or integrity constraints. Assume that amounts 6000, 7000, 8000, 9000 and 10000 were each paid by 20% of the students. Consider these query plans (Plan 1 on left, Plan 2 on right) to "list all courses taken by students who have paid more than x".



A disk seek takes 4ms, disk data transfer bandwidth is 300 MB/s and checking a tuple to see if amount is greater than x takes 10 micro-seconds. Which of the following statements is correct?

(a) Plan 1 and Plan 2 will not output identical row sets for all databases.
(b) A course may be listed more than once in the output of Plan 1 for some databases
(c) For x = 5000, Plan 1 executes faster than Plan 2 for all databases.
(d) For x = 9000, Plan I executes slower than Plan 2 for all databases.

GATE CSE-2006 MCQ (2-MARKS)

Q29. Consider the relation "enrolled(student, course)" in which (student, course) is the primary key, and the relation "paid(student, amount)" where student is the primary key. Assume no null values and no foreign keys or integrity constraints. Given the following four queries:

Query1: select student from enrolled where student in (select student from paid)
Query2: select student from paid where student in (select student from enrolled)
Query3: select E.student from enrolled E, paid P where E.student = P.student
Query4: select student from paid where exists (select * from enrolled where enrolled.student= paid.student)

Which one of the following statements is correct?

(a) All queries return identical row sets for any database
(b) Query2 and Query4 return identical row sets for all databases but there exist databases for which Query1 and Query2 return different row sets.
(c) There exist databases for which Query3 returns strictly fewer rows than Query2
(d) There exist databases for which Query4 will encounter an integrity violation at runtime.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2006 MCQ (2-MARKS)

Q30. Consider the relation account (customer, balance) where customer is a primary key and there are no null values. We would like to rank customers according to decreasing balance. The customer with the largest balance gets rank 1. ties are not broke but ranks are skipped: if exactly two customers have the largest balance they each get rank 1 and rank 2 is not assigned
Query1:
select A.customer, count(B.customer)
from account A, account B
where A.balance<=B.balance
group by A.customer
Query2:
select A.customer, 1+count(B.customer)
from account A, account B
where A.balance<B.balance
group by A.customer
Consider these statements about Query1 and Query2.
1. Query1 will produce the same row set as Query2 for some but not all databases.
2. Both Query1 and Query2 are correct implementation of the specification
3. Query1 is a correct implementation of the specification but Query2 is not
4. Neither Query1 nor Query2 is a correct implementation of the specification
5. Assigning rank with a pure relational query takes less time than scanning in decreasing balance order assigning ranks using ODBC.
Which two of the above statements are correct?

(a) 2 and 5
(b) 1 and 3
(c) 1 and 4
(d) 3 and 5

GATE CSE-2006 MCQ (2-MARKS)

Q31. Consider a database with three relation instances shown below. The primary keys for the Drivers and Cars relation are did and cidrespectively and the records are stored in ascending order of these primary keys as given in the tables. No indexing is available in the database.
D: Drivers relation

did	dname	rating	age
22	Karthikeyan	7	25
20	Salman	1	33
31	Boris	8	55
32	Amoldt	8	25
58	Schumacher	10	35
64	Sachin	7	35
71	Senna	10	16
74	Sachin	9	35
83	Rahul	3	25
95	Ralph	3	53

R: Reserves relation

did	cid	day
22	101	10/10/06
22	102	10/10/06
22	103	08/10/06
22	104	07/10/06
31	102	10/11/06
31	103	06/11/06
31	104	12/11/06
64	101	05/09/06
64	102	08/09/06
74	103	08/09/06

C: cars celation

cid	cname	colour
101	Renault	blue
102	Renault	red
103	Ferrari	green
104	Jaguar	red

select D.dname
from Drivers D
where D.didin (

Check Out Our GATE Courses & Secure Your Dream Rank!

```
select R.did
from Cars C, Reserves R
where R.cid = C.cid and C.colour = 'red'
intersect
select R.did
from Cars C, Reserves R
where R.cid  =C.cid and C.colour = 'green'
```

Let n be the number of comparisons performed when the above SQL query is optimally executed. If linear search is used to locate a tuple in a relation using primary key, then n lies in the range.

(a) 36-40
(b) 44-48
(c) 60-64
(d) 100-104

GATE CSE-2006 MCQ (2-MARKS)

Q32. Consider a database with three relation instances shown below. The primary keys for the Drivers and Cars relation are did and cid respectively and the records are stored in ascending order of these primary keys as given in the tables. No indexing is available in the database.

D: Drivers relation

did	dname	rating	age
22	Karthikeyan	7	25
20	Salman	1	33
31	Boris	8	55
32	Amoldt	8	25
58	Schumacher	10	35
64	Sachin	7	35
71	Senna	10	16
74	Sachin	9	35
83	Rahul	3	25
95	Ralph	3	53

R: Reserves relation

did	cid	day
22	101	10/10/06
22	102	10/10/06
22	103	08/10/06
22	104	07/10/06
31	102	10/11/06
31	103	06/11/06
31	104	12/11/06
64	101	05/09/06
64	102	08/09/06
74	103	08/09/06

C: cars celation

cid	cname	colour
101	Renault	blue
102	Renault	red
103	Ferrari	green
104	Jaguar	red

What is the output of the following SQL query?

```
select D.dname
from Drivers D
where D.didin  (
select R.did
from Cars C, Reserves R
where R.cid = C.cid and C.colour = 'green'
```

(a) Karthikeyan, Boris
(b) Sachin, Salman
(c) Karthikeyan, Boris, Sachin
(d) Schumacher, Senna

GATE CSE-2005 MCQ (2-MARKS)

Q33. The relation book(title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

Check Out Our GATE Courses & Secure Your Dream Rank!

```
select title
from book as B
    where (select count(*)
           from book as T
           where T.price>B.price) < 5
```

(a) Titles of the four most expensive books
(b) Title of the fifth most inexpensive book
(c) Title of the fifth most expensive book
(d) Titles of the five most expensive books

GATE CSE-2005 (2-MARKS)

Q34. A table 'student' with schema (roll, name, hostel, marks), and another table 'hobby' with schema (roll, hobbyname) contains records as shown below:

Table: student

Roll	Name	Hostel	Marks
1798	Manoj Rathor	7	95
2154	Soumic Banerjee	5	68
2369	Gumma Reddy	7	86
2581	Pradeep pendse	6	92
2643	Suhas Kulkarni	5	78
2711	Nitin Kadam	8	72
2872	Kiran Vora	5	92
2926	Manoj Kumkalikar	5	94
2959	Hemant Karkhanis	7	88
3125	Rajesh Doshi	5	82

Table: hobby

Roll	Hobby Name
1798	chess
1798	music
2154	music
2369	swimming
2581	cricket
2643	chess
2643	hockey
2711	volleyball
2872	football
2926	cricket
2959	photography
3125	music
3125	chess

The following SQL query is executed on the above tables:
select hostel
from student natural join hobbywhere marks > = 75 and roll between 2000 and 3000;
Relations S and H with the same schema as those of these two tables respectively contain the same information as tuples. A new relation S' is obtained by the following relational algebra operation:
$S' = \prod_{Hostel}((\sigma_{s.roll = H.roll}(\sigma_{marks > 75 \text{ and } roll > 2000 \text{ and } roll < 3000}(S)) X (H))$
The difference between the number of rows output by the SQL statement and the number of tuples in S' is .

(a) 6
(b) 4
(c) 2
(d) 0

GATE CSE-2005 MCQ (2-MARKS)

Q35. In an inventory management system implemented at a trading corporation, there are several tables designed to hold all the information. Amongst these, the following two tables hold information on which items are supplied by which suppliers, and which warehouse keeps which items along with the stock-level of these items. Supply = (supplierid, itemcode) Inventory = (itemcode, warehouse, stocklevel) For a specific information required by the management, following SQL query has been written

```
Select distinct STMP.supplierid
From Supply as STMP
Where not unique (Select ITMP.supplierid
                  From Inventory, Supply as ITMP
                  Where STMP.supplierid = ITMP.supplierid
                  And ITMP.itemcode = Inventory.itemcode
                  And Inventory.warehouse = 'Nagpur');
```

For the warehouse at Nagpur, this query will find all suppliers who

(a) do not supply any item
(b) supply exactly one item
(c) supply one or more items
(d) supply two or more items

GATE CSE-2005 MCQ (2-MARKS)

Q36. A company maintains records of sales made by its salespersons and pays them commission based on each individual's total sales made in a year. This data is maintained in a table with following schema:

salesinfo = (salespersonid, totalsales, commission)

In a certain year, due to better business results, the company decides to further reward its salespersons by enhancing the commission paid to them as per the following formula:

If commission < = 50000, enhance it by 2%
If 50000 < commission < = 100000, enhance it by 4%
If commission > 100000, enhance it by 6%

The IT staff has written three different SQL scripts to calculate enhancement for each slab, each of these scripts is to run as a separate transaction as follows:

Check Out Our GATE Courses & Secure Your Dream Rank!

T1	Update salesinfo Set commission = commission * 1.02 Where commission < = 50000;
T2	Update salesinfo Set commission = commission * 1.04 Where commission > 50000 and commission is < = 100000;
T3	Update salesinfo Set commission = commission * 1.06 Where commission > 100000;

Which of the following options of running these transactions will update the commission of all salespersons correctly

(a) Execute T1 followed by T2 followed by T3
(b) Execute T2, followed by T3; T1 running concurrently throughout
(c) Execute T3 followed by T2; T1 running concurrently throughout
(d) Execute T3 followed by T2 followed by T1

ANSWER KEY

1.(a.c,d)	**2. (26)**	**3. (b)**	**4. (2)**
5.(2)	**6. (a)**	**7. (4)**	**8. (5)**
9.(d)	**10. (7)**	**11. (2. 6)**	**12. (2)**
13.(a)	**14. (d)**	**15. (2)**	**16. (d)**
17.(c)	**18. (b)**	**19. (b)**	**20. (a)**
21.(c)	**22. (a)**	**23. (c)**	**24. (c)**
25.(d)	**26. (d)**	**27. (a)**	**28. (c)**
29.(b)	**30. (c)**	**31. (b)**	**32. (a)**
33.(d)	**34. (b)**	**35.(d)**	**36. (d)**

HINTS & SOLUTION

1. Answer:- A, C D
Explanation
- (A) Using EXISTS: This query correctly checks, for each student, if a corresponding entry exists in the Enrolled table for courseno = 1470. This is a standard and efficient way to solve the problem. This query is correct.
- (B) Using SIZEOF: SIZEOF is not a standard SQL function for checking the number of rows in a result set. This query is syntactically incorrect in virtually all common database systems (like PostgreSQL, MySQL, SQL Server, etc.). This query is incorrect.
- (C) Using COUNT: This query counts the enrollments for each student in course 1470 and checks if that count is greater than zero. This is logically equivalent to the EXISTS clause and is a perfectly valid way to find the required students. This query is correct.
- (D) Using NATURAL JOIN: A NATURAL JOIN between Students and Enrolled will join them on their only common column, rollno. This correctly links students to their enrollments. The WHERE clause then filters for courseno = 1470. This query is also correct.

2. Answer:- 26
Explanation :-
The subquery finds all pids of players who are in a team coached by a coach named 'Mark'.
The outer query finds the minimum age among those players.
2. Subquery Execution
Tables involved in subquery:
- coach C
- team T
- members M

Conditions:
- C.cname = 'Mark' → cid for Mark? From coach table: cid = 102 for Mark.
- T.cid = C.cid → T.cid = 102 → Which team? From team table: tid = 10 (MI, Mumbai, 102).
- M.tid = T.tid → M.tid = 10 → Which players? From members table: pid = 1 (Jasprit), pid = 3 (Ishan) have tid = 10.

So subquery result = {1, 3}.
3. Outer Query
P.pid IN (1, 3) → Players with pid = 1 and pid = 3:
From player table:
- pid 1 → Jasprit → age 31
- pid 3 → Ishan → age 26

4. MIN(P.age)
1. Ages = {31, 26}
   Minimum = 26

3. Answer:- B
Explanation :-
ρ(T1, Person) ρ(T2, Person) ρ(T3, Person)
σ condition (T1 × T2 × T3)
3 tables required.
2 cross product required. Ans

4. Answer:- 2
Explantion:- The output of the given SQL query is simply to return the female students whose marks is >65.

Check Out Our GATE Courses & Secure Your Dream Rank!

we can also solve by first we retrieve all the female students then we can select marks >65.

ROll	Name	Marks	Gender
2	Aliya	70	Female
3	Aliya	80	Female
3	Swati	65	Female

output is as follows:

Roll	Name	Marks	Gender
2	Aliya	70	Female
3	Aliya	80	Female

So the output of the given query is 2.

5. Answer : 2
Explanation :-
The given query is correlated, therefore, for every value of S.sno, we will be running the inner query.
Inner query will have {C11, C12} EXCEPT {C11, C12} = empty relation for sno = S01.
As a consequence, clauses that are not non-existing will return true, and the tuple tih sno = S01 will be added to the resulting relation.
Similarly for sno = S02, Inner query will have {C11, C12} EXCEPT {C11} = {C12}
The non-existing clause will return false, and the resulting relationship will not include it. We'll look for other issues in the same way. This query will return tuples that are enrolled in all of department D01's courses. This query will return tuples that are enrolled in all the courses offered by department D01.
Thus, there will be 2 tuples in the resultant relation:

sno	sname	dno
S01	James	D01
S04	Jane	D01

Hence, the correct answer is 2

6. **Answer: A**
Explanation :- Correct answer is Option A as the sub-query runs on emp table which contains both male and female employee from the company.

7. **Answer :- 4**
Explanation –
(4 to 4)
The inner query "select avg(cost) from catalogue where pno='P4' group by pno;" returns:

AVG(COST)
225

The outer query "select s.sno,s.sname from suppliers s, catalogue c where s.sno=c.sno" returns:

SNO	SNAME
S1	M/s Royal furniture
S1	M/s Royal furniture
S1	M/s Royal furniture
S2	M/s Balaji furniture
S2	M/s Balaji furniture
S3	M/s Premium furniture
S3	M/s Premium furniture
S3	M/s Premium furniture
S3	M/s Premium furniture

So, the final result of the query is:

SN	SNAME
S2	M/s Balaji furniture
S3	M/s Premium furniture
S3	M/s Premium furniture
S3	M/s Premium furniture

Hence, 4 rows will be returned by the query.

8. **Answer:- 5**
Explanation:-

Student	Sum (P.Marks)
Rohan	452
Vinit	452
Priya	452
Amit	452
Smita	452

The number of rows returned is 5.

Check Out Our GATE Courses & Secure Your Dream Rank!

9. Answer:- D

Explanation:-

Given two tables are,
Book (isbn, bname)
Stock (isbn, copies)
isbn is a primary key of Book and isbn is a foreign key of stock referring to Book table.
For example:

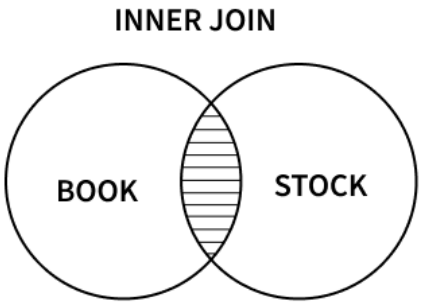
Book:

isbn	Bname
1	c++
2	ds
3	algo
4	co
5	cn

Stock:

isbn	copies
1	10
2	20
3	30
NULL	40
NULL	50

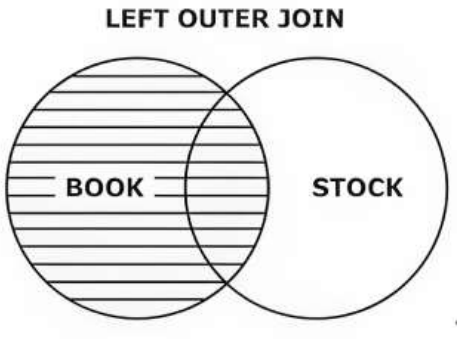
Query 1:
INNER JOIN keyword selects records that have matching values in both tables (Book and Stock).



So, the result of Query 1 is,

isbn	Copies
1	10
2	20
3	30

Query 2:
The LEFT OUTER JOIN keyword will return all records from the left table (Book) and the matched records from the right table (Stock).
The result is NULL from the right side, if there is no match.

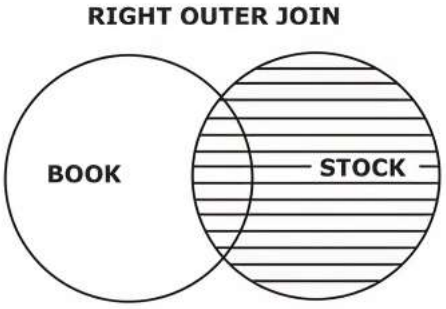


So, the result of Query 2 is,

Isbn	Copies
1	10
2	20
3	30
4	NULL
5	NULL

Query 3:
The RIGHT OUTER JOIN keyword will return records from the right table (Stock), and the matched records from the left table(BOOK).
The result is NULL from the left side, when there is no match.

Isbn	Copies
1	10
2	20
3	30
NULL	40
NULL	50



Query 4:
The FULL OUTER JOIN keyword will return all records when there is a match in either left (Book) or right (Stock) table records.

isbn	Copies
1	10
2	20
3	30
NULL	40
NULL	50
4	NULL
5	NULL

Hence, from the result of above four queries, a superset of the outputs of the Query 1, Query 2 and Query 3 is Query 4.

Note:
If we take isbn as a primary key in both the tables Book and Stock and foreign key, in one of the tables then also will get option D as the answer.
Another Example

Book (B)	
isbn	B name
1	A
2	B
3	C
4	D
5	E
6	F

Stock (S)	
isbn	copies
2	10
4	20
5	30
6	40
7	50

Query I Inner Join

B isbn	S copies
2	10
4	20
5	30
6	40

Query II Left Outer Join

isbn	copies
2	10
4	20
5	30
6	40
1	NULL
3	NULL

Query III B Right outer Join S

isbn	copies
2	10
4	20
5	30
6	40
7	50

Query IV B Full outer Join S

isbn	copies
2	10
4	20
5	30
6	40
1	NULL
3	NULL
7	50

10. Answer 7

Solution : -

```
SELECT ta.player
FROM top_scorer AS ta
WHERE ta.goals > ALL (
  SELECT tb.goals
  FROM top_scorer AS tb
  WHERE tb.country = 'Spain'
)
AND ta.goals > ANY (
  SELECT tc.goals
 FROM top_scorer AS tc
 WHERE tc.country = 'Germany'
);
```

- There are no tuples with country = 'Spain'. Hence, the condition ta.goals > ALL (empty set) is TRUE for all tuples.
- For Germany, the goals are: 16, 14, 11, 10, 10. The condition > ANY means greater than at least one value, i.e., greater than the minimum value (10).

Thus, the query reduces to:
ta.goals>10
Now, counting tuples with goals > 10:
16, 15, 14, 13, 12, 11, 11 → 7 tuples

11. Answer:- 2.6

Explanation:-

(2.6 to 2.6)
The given query is

```
SELECT AVG(EC.Num)FROM EC
WHERE(DeptName,Num)
    IN
(SELECT DeptName, COUNT(EmpId) as
EC(DeptName, Name) From EMP GROUPBY
DeptName; } Inner Query→We start the evaluation from
the inner query.
```

} Outer Query

The inner query forms groups DeptName wise and counts the DeptName wise EmpIds.

Check Out Our GATE Courses & Secure Your Dream Rank!

→ In inner query DeptName, Count(EmpId) is the alias name DeptName, Num.
So, the output of the inner query is,

EC	
DeptName	Num
AA	4
AB	3
AC	3
AD	2
AE	1

The outer query will find the
Avg(Num) as: (4+3+3+2+1)/5 = 2.6

12. Answer:- 2
Explantion:- First we do group by district name and then total capacities are obtained as following:

Name	Capacity
Ajmer	20
Bikaner	40
Churu	30
Dungargarh	10

- Then average capacity is calculated as Average Capacity = (20 + 40 + 30 + 10)/4 = 100/4 = 25
- Finally, the 3rd query will be executed and it's tuples will be considered as output, where name of district and its total capacity should be more than or equal to 25.
- Then average capacity is computed,

Now, Average Capacity = (20 + 40 + 30 + 10)/4 = 100/4 = 25

Name
Bikaner
Churu

Thus, the number of tuples returned by the SQL Query is 2.

13. Answer:-A
Explanation :- We need to find the address of theaters with maximum capacity.
So the theatres which are having maximum capacity will satisfy the condition. Thus, WHERE P1.capacity>= All (select P2.capacity from Cinema P2) is the correct answer.

14. Answer:- D
Explantion :- SELECT operation in SQL is equal to the project operation in Relational Algebra,while SELECT in SQL also retains duplicates. Hence, D is the answer.

15. Answer:- 2
Explanation :- Output table is

Raj	310
Rohit	140

So, the number of tuples returned will be 2

16. Answer:- D
Explantion :-

```
SELECT empName
FROM employee E                    } 1
WHERE NOT EXISTS                   }
```

```
2 { SELECT CustId
    FROM Customer C
    WHERE C.SalesRepRd = E.EmpId.
    AND C.rating<> 'GOOD');
```

The inner query i.e., ② represents all customers having rating other than 'GOOD' while the complete query represents name of all employees with all their customers having a 'good rating'.

17. Answer:- C
Explanation :- Multiplicity of duplicate tuples would be distributed when there is a match between R.a and S.a; and for that match to happen, S.a's value is repeated in each cases except the third case. So, the output of query given in the question matches with the output of select R.* from R,(select distinct a from S) as S1 where R.a = S1.a.

A	B	C
1	4	8
1	4	8
5	6	10
5	6	10

A	D	E
1	4	8
4	5	11
5	8	9
5	8	9

Select * from R where a in (select S. a from S)
a. Select R. * from R, S where R. a = S.a

A	B	C
1	4	8
1	4	8
5	6	10
5	6	10
5	6	10
5	6	10

Check Out Our GATE Courses & Secure Your Dream Rank!

4 × 4 = 16 tuple
3 + 3 = 6 Multiple
B. Select distinct R. * from R, S where R.a = Sa

A	B	C
1	4	8
1	4	8
5	6	10
5	6	10

A	D	E
1	4	8
4	5	11
5	8	9
5	8	9

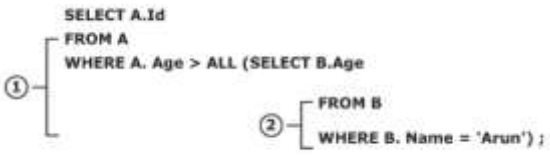
A	B	C
1	4	8
5	6	10

Output of option (b)
C. Select R. * from R, (select distinct a form S) as S1 where R. a = S1.a

A	B	C
1	4	8
1	4	8
5	6	10
5	6	10

18. Answer:- B
Explanation :- The given SQL query will display the last names and hire-dates of all latest hires in their respective departments in the location ID 1700. So, correct option is It executes and gives the correct result.

19. Answer:- B
Explanation :-



On executing the Inner Query or query (2), 0 or no rows will be selected because in relation B there is no Name 'Arun'.
And the outer query (1) results in the following output. (Because inner query returns 0 rows).
12
15
99
Thus, the number of Tuples will be 3.

20. Answer:- A
Explanation :-
Table: A

A_ID	A_Name	A_Age
12	Aarn	60
15	Shreya	24
99	Rohit	11
25	Hadi	40
98	Rohit	20

Table: C

C_ID	C_Phone	C_Area
10	2200	02
99	2100	01

Cartesian Product: A × C

A ID	A. Name	A. Age	C ID	C Phone	C. Area
12	Aam	60	10	2200	02
12	Aam	60	99	2100	01
15	Shreya	24	10	2200	02
15	Shreya	24	99	2100	01
99	Rohit	11	10	2200	02
99	Rohit	11	99	2100	01
25	Hadi	40	10	2200	02
25	Hadi	40	99	2100	01
98	Rohit	20	10	2200	02
98	Rohit	20	99	2100	01

Selection Condition Applied:
$\sigma_{(A.ID > 40 \text{ OR } CID < 15)}$
Since:
- C_ID values = 10 and 99
- C_ID < 15 ⇒ only rows with C_ID = 10
- A_ID > 40 ⇒ rows with A_ID = 99, 98

Final Result

Check Out Our GATE Courses & Secure Your Dream Rank!

A_ID	A_Name	A_Age	C_ID	C_Phone	C_Area
12	Aam	60	10	2200	02
15	Shreya	24	10	2200	02
99	Rohit	11	10	2200	02
99	Rohit	11	99	2100	01
25	Hadi	40	10	2200	02
98	Rohit	20	10	2200	02
98	Rohit	20	99	2100	01

21. Answer:- C
Explantion :- An SQL query can contain a HAVING clause only if it has a GROUP BY clause.
Also, all attributes used in the GROUP BY clause must appear in the SELECT clause.
So, Statement Q and R are correct.

22. Answer:- A
Explantion :-

X	Y
1	1
2	3
3	7
4	15
4	31
6	63
7	127

So, when X =7 , selecting Y will give value 127.

23. Answer:- C
Explantion:-

S

Borrower	Bank-Manager
Ramesh	Sunderajan
Suresh	Ramgopal
Mahesh	Sunderjan

T

Bank-Manager	Loan-Amount
Sunderajan	10000.00
Ramgopal	5000.00
Sunderjan	7000.00

After executing the given query, the output would be

Borrower	Bank Manager	Loan Amount
Ramesh	Sunderajan	10000.00
Ramesh	Sunderajan	7000.00
Suresh	Ramgopal	5000.00
Mahesh	Sunderajan	10000.00
Mahesh	Sunderajan	7000.00

Thus, the output of the SQL query contains 5 tuples.

24. Answer:- C
Explantion :-
Passenger:

Pid	Pname	Age
0	Sachin	65
1	Rahul	66
2	Sourav	67
3	Anil	69

Age > 65

Pid	Class	tid
0	AC	8200
1	AC	8201
2	SC	8201
5	AC	8203
1	SC	8204
3	AC	8202

Class ='AC'

— 1, 3 P_{ids} are returned

25. Answer :- D
Explanation:-
(D) option matched because given query returns suppliers who have not supplied any blue parts. That means it can include other than blue parts.
(A): False, as this may include blue parts and may not include "null" parts.
(B): Obviously false because it returning other than any blue part.
(C): Obviously false because it does not return this.

26. Answer :- D
Explanation:-
If aggregate operator & other attribute are used in the SELECT clause, it is allowed only if the other attribute is in the GROUP BY clause.
In this query (in outer query), Sch_name, COUNT(*) are used, so Sch_name must be in GROUP BY clause. But in the query, School_id is in the GROUP BY clause.
So, the query has a syntax error.

27. Answer :- A
Explanation :-We need employees whose salary is greater than every employee in department 5.
That means:
e.salary> ALL (salaries in department 5)
i.e., greater than the maximum salary in department 5.
Q_1
NOT EXISTS (s.salary ≥ e.salary in dept 5)
This ensures there is no department 5 employee with salary ≥ e.salary.
So e. salary must be strictly greater than all salaries in department 5.
Hence Q_1 is correct.
Q_2
e.salary> ANY (dept 5 salaries)

Check Out Our GATE Courses & Secure Your Dream Rank!

This means salary is greater than at least one employee in department 5.
That is much weaker than required.
Hence Q_2 is incorrect.
Final Answer: A. Q_1 is the correct query.

28. Answer:-C
Explanation :- Answer (C) Assuming that large enough memory is available for all data needed. Both plans need to load both tables courses and enrolled. So disk access time is same for both plans. Plan 2 does lesser number of comparisons compared to plan 1. 1) Join operation will require more comparisons as the second table will have more rows in plan 2 compared to plan 1. 2) The joined table of two tables will have more rows, so more comparisons are needed to find amounts greater than x.

29. Answer :- B
Explanation:- Take an example:
Table enrolled

student	course
abc	c1
xyz	c1
abc	c2
pqr	c1

Table paid

student	amount
abc	20000
xyz	10000
rst	10000

Output of Query 1
abc
abc
xyz

Output of Query 2
abc
xyz

Output of Query 3
abc
xyz

Output of Query 4
abc
xyz
Query 1 and Query 3 may return repetitive student values as "student" is not a key in relation enrolled, however query 2 and query 4 always return same row sets. So, option (B) is correct.

30. Answer :- C
Explanation
Query 1 and Query 2 will give the same result if all the customers have distinct balance. So, for some databases, the result of query 1 and query 2 will be same. Now, let us consider a fact that all the entries in the database have the same value for balance. Ideally, all the customers should have rank 1, but both the queries will give all the customers a rank equal to the number of customers in the database.
So, both the queries do not give us the required output.
Thus, C is the correct choice.
Query 1:
(1) FROM Acc A, Acc B

cid	bal
C1	30
C2	30
C3	25
C4	25
C5	20
C6	15

×

cid	bal
C1	30
C2	30
C3	25
C4	25
C5	20
C6	15

(2) Where A.bal < = B.bal → Group by A.cid

C1 30 C1 30
C1 30 C2 30
C2 30 C1 30
C2 30 C2 30
C3 25 C1 30
C3 25 C2 30
C3 25 C3 25
C3 25 C4 25

(4) select A.cid count (B. cid)

C1	2
C2	2
C3	4
C4	4
C5	5
C6	6

Result Query 1 wrong as per given specifications
Query 2:
(1) From acc A, Acc B
(2) Where A.bal<B.bal
(3) Group by A.cid
(4) Select A.cid, count + 1

cid bal cid bal
C3 25 C1 30
C3 25 C2 30
C4 25 C1 30
C4 25 C2 30

Check Out Our GATE Courses & Secure Your Dream Rank!

C5	20	C1	30
C5	20	C2	30
C5	20	C3	20
C5	20	C4	20
C6	15	C1	30
C6	15	C2	30
C6	15	C3	25
C6	15	C4	25
C6	15	C5	20

C3	3
C4	3
C5	5
C6	6

result

Wrong because 1st rank cust not in result. Both Query 1 Query 2 wrong for given specific between some records of Query 1 Query 2 can same for some database.

31. Answer : B

Explanation:-

Here we have to calculate the number of comparisons performed when the above SQL query is optimally executed.

from the first inner query:

select R.did from Cars C, Reserves R where R.cid = C.cid and C.colour = 'red'

C.color = "Red", comparisons=4 (Cars has four rows)

R.cid=C.cid so there are five rows extracted to this where condition. comparisons=(2 red cars * 10 Reserves rows)=20

from the second inner query: select R.did from Cars C, Reserves R where R.cid = C.cid and C.colour = 'green'

C.color = "Green", comparisons=4 (Cars has four rows)

R.cid=C.cid so there are three rows extracted to this where condition. comparisons=(1 green car*10 Reserves rows)=10

R.did = {22, 22, 31,31, 64} for first inner query R.did = {22, 31, 74} for second inner query

Here unique sets are, R.did={22,31,64} and R.did={22,31,74} respectively for first and second inner queries. so for intersection, 6 comparisons (for 22, we hit on the first try and for 31,we hit on the second try,and for 74,we hit on all three try, so comaprisons=1+2+3) Finally we have to locate the did - 22 and did 31 from the driver table and did is the primary key. As told in the question, we use linear search and for 22, we hit on the first try and for 31 we hit on the third try. So, 1 + 3 = 4 comparisons. so total no of comparisons= 4+20+4+10+6+4=48 therefore B is the answer.

In short: So, first get 2 red cars by scanning 4 tuples of the cars relation. Now, for each of the two 'red' cars, we scan all the 10 tuples of the 'Reserves' relation and thus we get 2*10 + 4 = 24 comparisons.Similarly for the 'green' car we get 4+10 = 14 comparisons.

32. Answer :- A

Explanation:- The sub-query always executes before the execution of the main query. Subqueries are completed first.The result of the subquery is used as input for the outer query. from the first inner query: select R.did from Cars C, Reserves R where R.cid = C.cid and C.colour = 'red'. C.color = "Red", corresponding C.cid={102,104}. R.cid=C.cid so there are five rows extracted to this where condition. R.did = {22, 22, 31,31, 64} from the second inner query: select R.did from Cars C, Reserves R where R.cid = C.cid and C.colour = 'green' C.color = "Green", corresponding C.cid={103} R.cid=C.cid so there are three rows extracted to this where condition. R.did = {22, 31, 74} Finally, the outer query selects driver names whose did = {22,31} which corresponds to Karthikeyan and Boris. so the correct option is A.

33. Answer : D

Explanation :-

For each book B, it counts how many books have price greater than B.price.

Condition:

count(T.price>B.price) < 5

This means: There are fewer than 5 books more expensive than B.

So B is among the top 5 most expensive books. Since prices are unique, exactly the five highest-priced books satisfy this.

Answer: D. Titles of the five most expensive books

Book B	
Title	Price
B_1	10
B_2	20
B_3	30
B_4	40
B_5	5
B_6	15
B_7	60
B_8	70
B_9	80

Check Out Our GATE Courses & Secure Your Dream Rank!

Book T	
Title	Price
B_1	10
B_2	20
B_3	30
B_4	40
B_5	5
B_6	15
B_7	60
B_8	70
B_9	80

(T. price > B. price) < 5

1. T- price > 10 ⇒ 7 < 5 False
2. T. price > 20 ⇒ 5 < 5 False
3. T. price > 30 ⇒ 4 < 5 True
4. T. price > 40 ⇒ 3 < 5 True
5. T. price > 5 ⇒ 8 < 5 False
6. T. Price > 15 ⇒ 6 < 5 False
7. T. price > 60 ⇒ 2 < 5 True
8. T. price > 70 ⇒ 1 < 5 True
9. T. price > 80 ⇒ 0 < 5 True

34. Answer :- B

Explanation :-

Output of above Query

Roll	Hostel
2369	7
2581	6
2643	5
2643	5
2872	5
2926	5
2959	7

Total rows selected by running SQL Query: 7 Total rows by Relation Algebra : 4 i.e 5,6,7 (Unique values only) 7-3=4 Answer is B

35. Answer:- D

Explanation :-

Here [not unique] in nested query ensures that only for those suppliers it return True which supplies more than 1 item in which case supplier *id* in inner query will be repeated for that supplier.
Hence, answer is (D) supply two or more items.

36. Answer : D

Explanation :-

T3 followed by T2 followed by T1
If this sequence is not followed then it may happen that the Officer of one slab gets benefited twice. Say an officer is having commission as 99,999, He must get an updated commission according to T2. But then 99,999*1.04=1,03998 and he again becomes eligible for next slab of commission.
T3 followed by T2 followed by T1 will be check all border cases like above.
So Answer is D

Check Out Our GATE Courses & Secure Your Dream Rank!

5 File Organization & Indexing

GATE CSE-2026 SET-2 MCQ (1-MARKS)

Q1. An index in a DBMS is said to be dense if an index entry appears for every search-key value in the indexed file. Otherwise it is called a sparse index. Consider the following two statements:
S1: A hash index must be a dense index
S2: A B^+ tree index can be a sparse index
Which one of the following options is correct?

(a) Both S1 and S2 are true
(b) Both S1 and S2 are false
(c) S1 is true and S2 is false
(d) S1 is false and S2 is true

GATE CSE-2026 SET-2 MCQ (1-MARKS)

Q2. In the context of DBMS, consider the two sets T and S given below.

T	S
I:Logical Schema Ii:Physical Schema III: External Schema	L: Views M: File organization and indexes N: Relations

Which one of the following is the correct match from T to S?

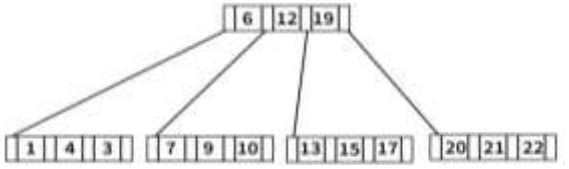
(a) I – L, II – M, III – N
(b) I – M, II – L, III – N
(c) I – N, II – M, III – L
(d) I – N, II – L, III – M

GATE CSE-2025 SET-2 NAT (2-MARKS)

Q3. In a B+ tree where each node can hold at most four key values, a root to leaf path consists of the following nodes:
A = (49, 77, 83,), B = (7, 19, 33, 44), C = (20*, **22***, 25*, **26***)
The *-marked keys signify that these are data entries in a leaf.
Assume that a pointer between keys k_1 and k_2 points to a subtree containing keys in $[k_1, k_2)$, and that when a leaf is created, the smallest key in it is copied up into its parent.
A record with key value 23 is inserted into the B+ tree.
The smallest key value in the parent of the leaf that contains 25* is _____
(Answer in integer)

GATE CSE-2025 SET-1 MSQ (1-MARKS)

Q4. Consider the following *B* + tree with 5 nodes, in which a node can store at most 3 key values. The value 23 is now inserted in the *B* + tree. Which of the following options(s) is/are CORRECT?



(a) None of the nodes will split.
(b) At least one node will split and redistribute
(c) The total number of nodes will remain same
(d) The height of the tree will increase.

GATE CSE- 2024 SET-2 MSQ (1-MARKS)

Q5. Which of the following file organizations is/are I/O efficient for the scan operation in DBMS?

(a) Sorted
(b) Heap
(c) Unclustered tree index
(d) Unclustered hash index

Check Out Our GATE Courses & Secure Your Dream Rank!

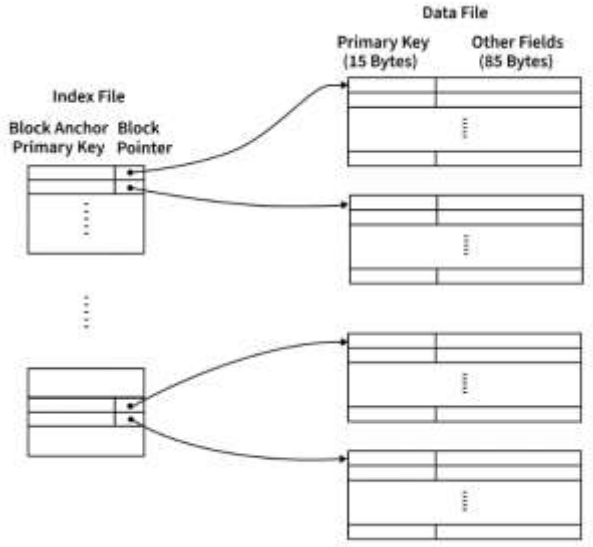
GATE CSE- 2024 SET-1 MCQ (1-MARKS)

Q6. In a B+Tree , the requirement of atleast half-full(50%) node occupancy is relaxed for which one of the following cases?

(a) Only the root node
(b) All leaf nodes
(c) All internal nodes
(d) Only the leftmost leaf node

GATE CSE-2023 NAT (2-MARKS)

Q7. Consider a database of fixed-length records, stored as an ordered file. The database has 25,000 records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. The figure below depicts this indexing scheme.



Suppose the block size of the file system is 1024 bytes, and a pointer to a block occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search on an index file of b blocks takes $\lceil \log_2 b \rceil$ block accesses in the worst case.
Given a key, the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is _____

GATE CSE-2021 SET-2 NAT (1-MARKS)

Q8. A data file consisting of 1,50,000 student-records is stored on a hard disk with block size of 4096 bytes. The data file is sorted on the primary key RollNo. The size of a record pointer for this disk is 7 bytes. Each student-record has a candidate key attribute called ANum of size 12 bytes. Suppose an index file with records consisting of two fields, ANum value and the record pointer to the corresponding student record, is built and stored on the same disk. Assume that the records of data file and index file are not split across disk blocks. The number of blocks in the index file is____.?

GATE CSE- 2020 NAT (2-MARKS)

Q9. Consider a database implemented using B+ tree for file indexing and installed on a disk drive with block size of 4 KB. The size of search key is 12 bytes and the size of tree/disk pointer is 8 bytes. Assume that the database has one million records. Also assume that no node of the B+ tree and no records are present initially in main memory. Consider that each record fits into one disk block. The minimum number of disk accesses required to retrieve any record in the database is _____?

GATE CSE-2019 MCQ (1-MARKS)

Q10. Which one of the following statements is NOT correct about the B+ tree data structure used for creating an index of a relational database table?

(a) Each leaf node has a pointer to the next leaf node
(b) Non-leaf nodes have pointers to data records
(c) B+ tree is a height-balanced tree
(d) Key values in each node are kept in sorted order

GATE CSE-2017 SET-2 NAT (2-MARKS)

Q11. In a B+ tree, if the search-key value is 8 bytes long, the block size is 512 bytes and the block pointer size is 2 bytes, then the maximum order of the B+ tree is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2016 SET-2 MCQ (2-MARKS)

Q12. B+ Trees are considered BALANCED because ...

(a) The lengths of the paths from the root to all leaf nodes are all equal.
(b) The lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
(c) The number of children of any two non-leaf sibling nodes differ by at most 1.
(d) The number of records in any two leaf nodes differ by at most 1.

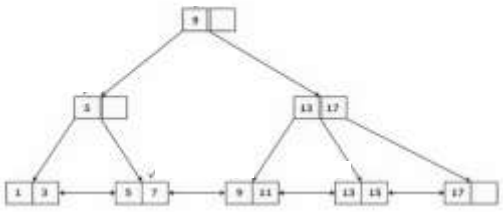
GATE CSE-2015 SET-3 NAT (2-MARKS)

Q13. Consider a B+ tree in which the search key is 12 bytes long, block size is 1024 bytes, recorder pointer is 10 bytes long and the block pointer is 8 bytes long. The maximum number of keys that can be accommodated in each non-leaf node of the tree is _____.

GATE CSE-2015 SET-2 NAT (1-MARKS)

Q14. With reference to the B+ tree index of order 1 shown below, the minimum number of nodes (including the Root node) that must be fetched in order to satisfy the following query:"Get all records with a search key greater than or equal to 7 and less than 15" is

_____



GATE CSE-2015 SET-1 MCQ (1-MARKS)

Q15. A file is organized so that the ordering of data records is the same as or close to the ordering of data entries in some index. Then that index is called

(a) Dense
(b) Sparse
(c) Clustered
(d) Unclustered

GATE CSE-2013 MCQ (1-MARKS)

Q16. An index is clustered, if

(a) It is on a set of fields that form a candidate key.
(b) It is on a set of fields that include the primary key.
(c) The data records of the file are organized in the same order as the data entries of the index.
(d) The data records of the file are organized not in the same order as the data entries of the index.

GATE CSE-2010 MCQ (1-MARKS)

Q17. Consider a B^+-tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?

(a) 1
(b) 2
(c) 3
(d) 4

GATE CSE-2009 MCQ (2-MARKS)

Q18. The following key values are inserted into a B^+-tree in which the order of the internal nodes is 3, and that of the leaf nodes is 2, in the sequence given below. The order of internal nodes is the maximum number of tree pointers in each node, and the order of leaf nodes is the maximum number of data items that can be stored in it. The B^+-tree is initially empty.
10, 3, 6, 8, 4, 2, 1
The maximum number of times leaf nodes would get split up as a result of these insertions is

(a) 2
(b) 3
(c) 4
(d) 5

GATE CSE-2008 MCQ (1-MARKS)

Q19. A clustering index is defined on the fields which are of type

(a) non-key and ordering
(b) non-key and non-ordering
(c) key and ordering
(d) key and non-ordering

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2008 MCQ (2-MARKS)

Q20. A B-tree of order 4 is built from scratch by 10 successive insertions. What is the maximum number of node splitting operations that may take place?

(a) 3
(b) 4
(c) 5
(d) 6

GATE CSE-2008 MCQ (2-MARKS)

Q21. Consider a file of 16384 records. Each record is 32 bytes long and its key field is of size 6 bytes. The file is ordered on a non-key field, and the file organization is unspanned. The file is stored in a file system with block size 1024 bytes, and the size of a block pointer is 10 bytes. If the secondary index is built on the key field of the file, and a multilevel index scheme is used to store the secondary index, then the number of first-level and second-level blocks in the multilevel index are respectively:

(a) 8 and 0
(b) 128 and 6
(c) 256 and 4
(d) 512 and 5

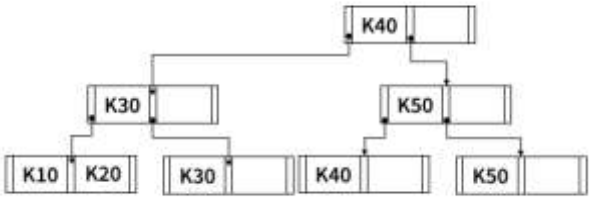
GATE CSE-2007 MCQ (2-MARKS)

Q22. The order of a leaf node in a B^+-tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

(a) 63
(b) 64
(c) 67
(d) 68

GATE CSE-2007 MCQ (2-MARKS)

Q23. Consider the B^+tree in the adjoining figure, where each node has at most two keys and three links.

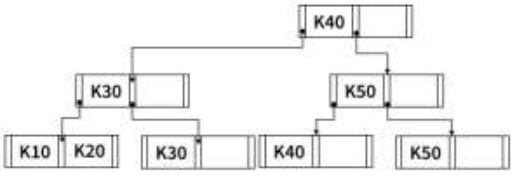


Keys K'15 and then K25 are inserted into this tree in that order. Exactly how many of the following nodes (disregarding the links) will be present in the tree after the two insertions?

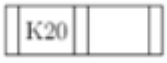
(a) 1
(b) 2
(c) 3
(d) 4

GATE CSE-2006 MCQ (2-MARKS)

Q24. Consider the B^+tree in the adjoining figure, where each node has at most two keys and three links.



Now the key $K50$ is deleted from the B^+tree resulting after the two insertions made earlier. Consider the following statements about the B^+tree resulting after this deletion.
i. The height of the tree remains the same.

ii. The node 

(disregarding the links) is present in the tree.

Check Out Our GATE Courses & Secure Your Dream Rank!

iii. The root node remains unchanged (disregarding the links).
Which one of the following options is true?

(a) Statements (i) and (ii) are true
(b) Statements (ii) and (iii) are true
(c) Statements (iii) and (i) are true
(d) All the statements are false

GATE CSE-2007 MCQ (2-MARKS)

Q25. In a database file structure, the search key field is 9 bytes long, the block size is 512 bytes, a record pointer is 7 bytes and a block pointer is 6 bytes. The largest possible order of a non-leaf node in a B+ tree implementing this file structure is.

(a) 23
(b) 24
(c) 34
(d) 44

GATE CSE-2005 MCQ (1-MARKS)

Q26. Which of the following is a key factor for preferring B^+-trees to binary search trees for indexing database relations?

(a) Database relations have a large number of records
(b) Database relations are sorted on the primary key
(c) B^+-trees require less memory than binary search trees
(d) Data transfer from disks is in blocks

GATE CSE-2005 MCQ (2-MARKS)

Q27. A database table T1 has 2000 records and occupies 80 disk blocks. Another table T2 has 400 records and occupies 20 disk blocks. These two tables have to be joined as per a specified join condition that needs to be evaluated for every pair of records from these two tables. The memory buffer space available can hold exactly one block of records for T1 and one block of records for T2 simultaneously at any point in time. No index is available on either table. If Nested-loop join algorithm is employed to perform the join, with the most appropriate choice of table to be used in outer loop, the number of block accesses required for reading the data are .

(a) 800000
(b) 40080
(c) 32020
(d) 100

GATE CSE- 2005 MCQ (2-MARKS)

Q28. A database table T1 has 2000 records and occupies 80 disk blocks. Another table T2 has 400 records and occupies 20 disk blocks. These two tables have to be joined as per a specified join condition that needs to be evaluated for every pair of records from these two tables. The memory buffer space available can hold exactly one block of records for T1 and one block of records for T2 simultaneously at any point in time. No index is available on either table. If, instead of Nested-loop join, Block nested-loop join is used, again with the most appropriate choice of table in the outer loop, the reduction in number of block accesses required for reading the data will be

(a) 0
(b) 30400
(c) 38400
(d) 798400

GATE CSE - 2005 MCQ (1 – MARKS)

Q29. A B-Tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are:

(a) 5
(b) 4
(c) 3
(d) 2

Check Out Our GATE Courses & Secure Your Dream Rank!

ANSWER KEY

1.(a)	**2.(c)**	**3.(33)**
4.(b, d)	**5.(a,b)**	**6. (a)**
7.(6)	**8.(698)**	**9.(4)**
10.(b)	**11.(52)**	**12.(a)**
13.(50)	**14.(5)**	**15.(c)**
16.(c)	**17.(b)**	**18.(c)**
19.(a)	**20.(c)**	**21.(c)**
22.(a)	**23.(a)**	**24.(a)**
25.(c)	**26.(d)**	**27.(c)**
28.(b)	**29. (a)**	

HINTS & SOLUTION

1. Answer :- A

Explantion :-

S1: A hash index must be a dense index

- Hash indexing maps each search-key value to a bucket
- Direct access requires an entry for each key
- Sparse hashing is not feasible

True

S2: A B^+ tree index can be a sparse index

- B^+ tree can be built on ordered file
- Internal levels act as sparse index over lower levels
- Hence sparse organization is possible

True

2. Answer :- C

Explanation ➖

Logical schema → describes database structure as relations

⇒ I – N

Physical schema → storage details like file organization and indexes

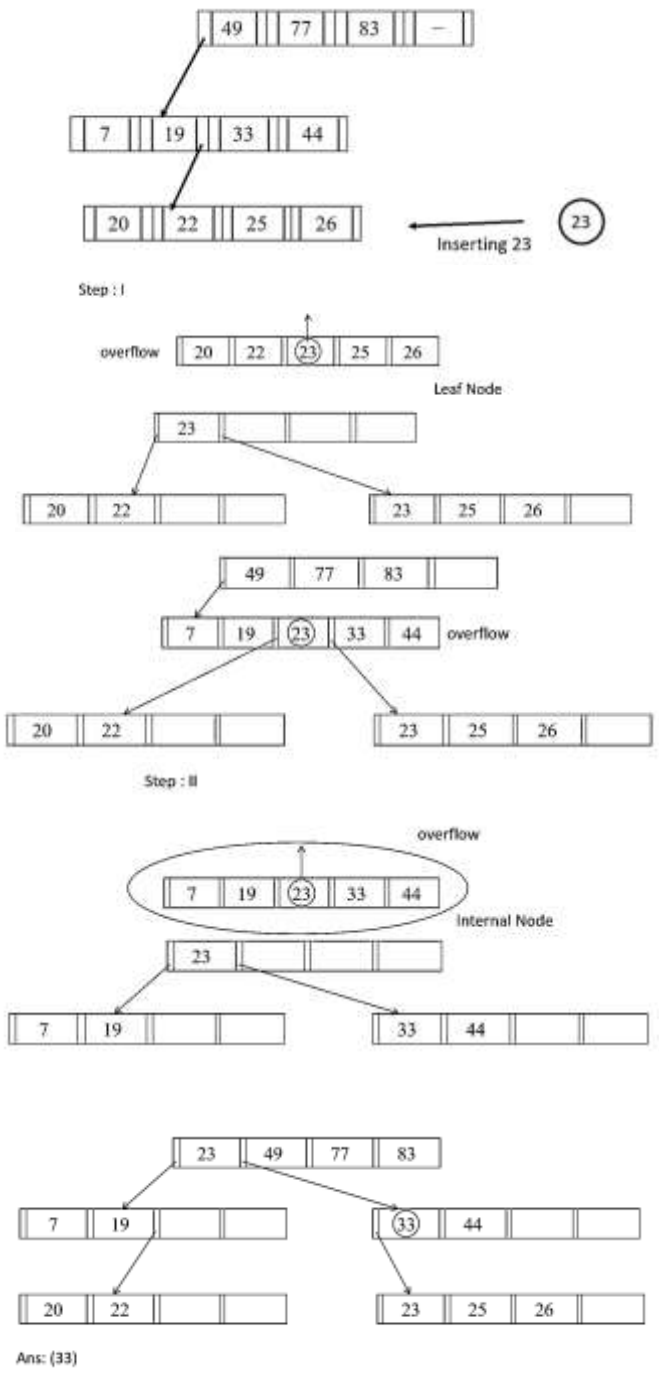
⇒ II – M

External schema → user views of database

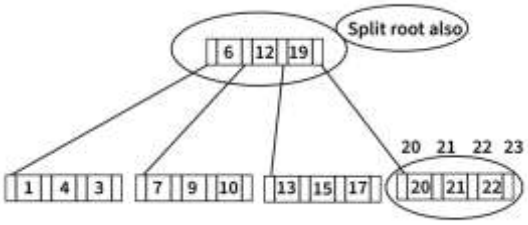
⇒ III – L

3. Answer :- 33

Explanation:-



4. Answer:-B and D
Explanation :-



- Leaf node and root node split required.
- Results increases tree height from 1 to 2.

5. Answer:- A,B
Explanation :- A sorted file organization ensures that records are stored in a sequential order, which is optimal for scan operations since the data can be accessed sequentially with minimal I/O overhead.

- In a heap file organization, records are stored without any specific order. Although not as efficient as sorted files, a heap file still allows sequential scans across all records with moderate I/O efficiency.
- Options C (Unclustered tree index) and D (Unclustered hash index) are not I/O efficient for scans because they involve random access to data blocks, leading to higher I/O costs.
- Hence, the correct answer is A and B.

6. Answer: (A)
Explanation:
In a B-tree of order *m*, every node (except the root) must satisfy the minimum occupancy condition:
Each node must have at least ⌈m/2⌉ children.
This ensures that nodes are at least 50% full.
However, the root node is an exception:
The root is allowed to have fewer than ⌈m/2⌉ children.
It can have as few as 2 children (if it is not a leaf).
If the tree has only one node, the root itself can be a leaf.
Applying the Given Example
(Order = 7)
Internal Nodes:

$$\lceil 7/2 \rceil = \lceil 3.5 \rceil = 4$$

So, every internal node must have **at least 4 children (pointers)**.
This satisfies the **50% occupancy rule**.
Root Node (Exception):
The **root node does NOT need to follow** the above rule.
It can have:
Minimum **2 children (block pointers)**
Which means only **1 key**
This clearly shows that the **50% occupancy condition is relaxed only for the root node**.

Why other options are incorrect:
(B) All leaf nodes: Leaf nodes must still satisfy minimum occupancy (except when the leaf is also the root).
(C) All internal nodes: Internal nodes must follow the minimum occupancy rule.
(D) Only the leftmost leaf node: There is no such special relaxation for any specific leaf.

7. Answer:- 6
Explanation :-
Given that One block should fully contain within block so here it means that they are saying that you should use unspanned file organization.
No of record = 25000
record size = 100B
Key size = 15B
Block pointer size = 5B
1 index record size = 15 + 5 = 20B
Disk Block Size = 1024B
No. of Record per Block=(Bs/Rs) = ⌊1024/100⌋ = ⌊10.24⌋ =10
No of Block Required for blocks
$= \left\lceil \frac{25000}{10} \right\rceil$
= 2500 Block Required to store all records
"The database is indexed by a primary index file" it means that here we have to use sparse indexing
NO of Index per Block =⌊(Bs/Is)⌋ = floorValue(1024/20) = 51
NO of Block Required for indexes =
$\lceil No.\,OfIndex/No.\,OfIndexPerBlock \rceil$
= (2500/51) = 50 Blocks
Given That Binary Search is used to search the desired block and we know that it takes $\log_2(n)$ Time to search in worst case where n= no of comparision Required = No. of Index Block here.

the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is.
=ceil($\log_2(50)$) = 6 blocks

8. Answer :- 698
Explanation :-(698 to 698)
For index file:
records = 150000, block size = 4096 B
key = 12 B, pointer = 7B
No. of index entry/block = 4096 B/19B
4096/19 = 215
[Note: Assume that Records of data file and index file are not split across disc blocks, so it indicates Unspanned Strategy.]

No. of blocks needed for indexing = $\left\lceil \frac{150000}{215} = 698 \right\rceil$

9. Answer: -4
Explanation –
8*p+(p-1)*12<= 4096

Check Out Our GATE Courses & Secure Your Dream Rank!

20p<=4108
therefore p=205

Level	Nodes	Keys	Pointers
1	1	204	205
2	205	204×205	205^2
3	205^2	204×205^2	205^3

Level 3 has approximately 8.5×10^6 records. So we can be sure that a 3-level B+ tree is sufficient to index 10^6 records.

So to access any record (in the worst case), we need 3 block access to search for the record in the index along with 1 more access to actually access the record.

Hence, 4 accesses are required.

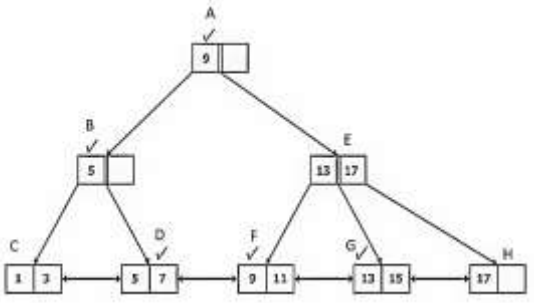
10. Answer:- B
The only incorrect statement about the B^+ tree data structure is that, non-leaf nodes have pointers to data records. So, B is the correct answer.

11. Answer:-52
Explantion:- Given,
Block size = 512 bytes
Block pointer, B_p = 2 bytes
Search key = 8 bytes
Let Maximum order of B^+ tree = p
Using the node structure of B^+ Tree internal node:
So,
$p(B_p) + (p- 1)(k) \leq 512$
$p(2) + (p - 1)(8) \leq 512$
$2p + 8p - 8 \leq 512$
$10p \leq 520$
$p \leq 520/10$
Hence, p = 52

12. Answer :- A
Explanation :- In both B & B^+ trees all the leaf nodes will be present at the same level, so the length of the paths from the root to all leaf nodes are all equal.

13. Answer :-50
Explanation : - (50 to 50)
Using the node structure of B^+ Tree internal node,we have:
$p\times B_p + (p - 1) \times k \leq B$ (for non-leaf node)
B_p= 8 Bytes, k = 12 Bytes and B = 1024 Bytes.
So, $p* 8 + (p-1)12 \leq 1024$
$\Rightarrow 20p - 12 \leq 1024$
$\Rightarrow 20p \leq 1036$
$\Rightarrow p \leq 51.8$
So, p = 51.
Now, maximum number of Keys will be p-1 = 50.

14. Answer:-5
Explanation:-



Searching for key 7 requires **three node accesses**.
In a B+ tree, all leaf nodes are linked using pointers, which allows traversal to adjacent leaves. Hence, to retrieve all records with values greater than or equal to 7, we can continue scanning through linked leaf nodes instead of performing separate searches. This reduces the total node accesses to **15**.
The required nodes to fetch all records ≥ 7 and < 15 are: **A, B, D, F, and G**.

15. Answer :-C
Explanation :-The correct answer is Clustered Index as it is the definition of Clustered Indexing.

16. Answer :- C
Explanation:-
Option (A) and Option (B) are not correct because, index can be created using any column or combination of columns, which need not be unique.
Basically, Indexed column is used to sort the rows of table. Whole data record of file is sorted using index so the correct option is (C).

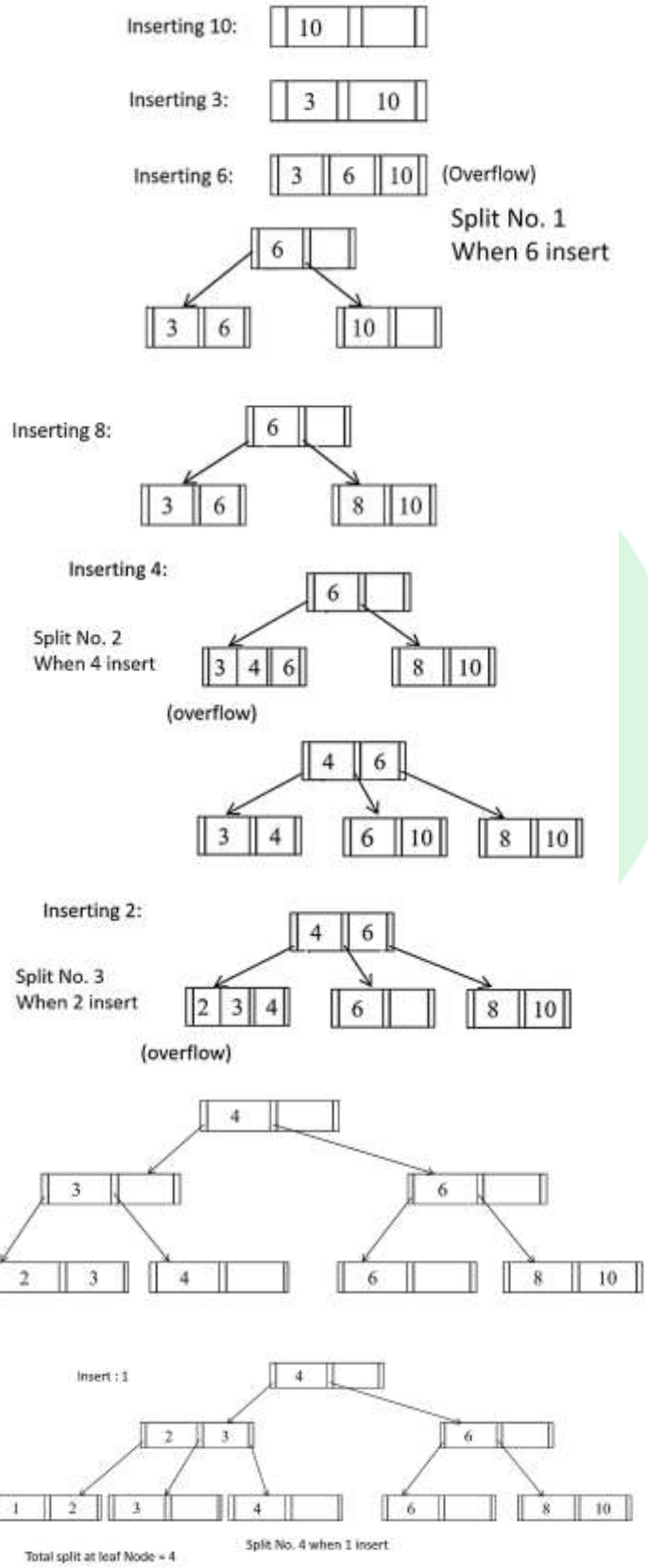
17. Answer :- B
Explanation:-
Maximum number of keys = 5
$\Rightarrow$ Order m=6 (since max keys = m−1)
In a B+-tree,
Minimum number of children = $\lceil m/2 \rceil = \lceil 6/2 \rceil = 3$
Minimum number of keys = children − 1 = 3 − 1 = 2

18. Answer :- C
Explanation :-
Yes — this question depends on leaf split convention.
Given :-

- Leaf order = 2 $\Rightarrow$ max 2 keys per leaf
- On overflow (3 keys), leaf splits

If middle goes to LEFT leaf (common GATE indication)
Leaf splits occur after inserting:

Check Out Our GATE Courses & Secure Your Dream Rank!



19. Answer :- A

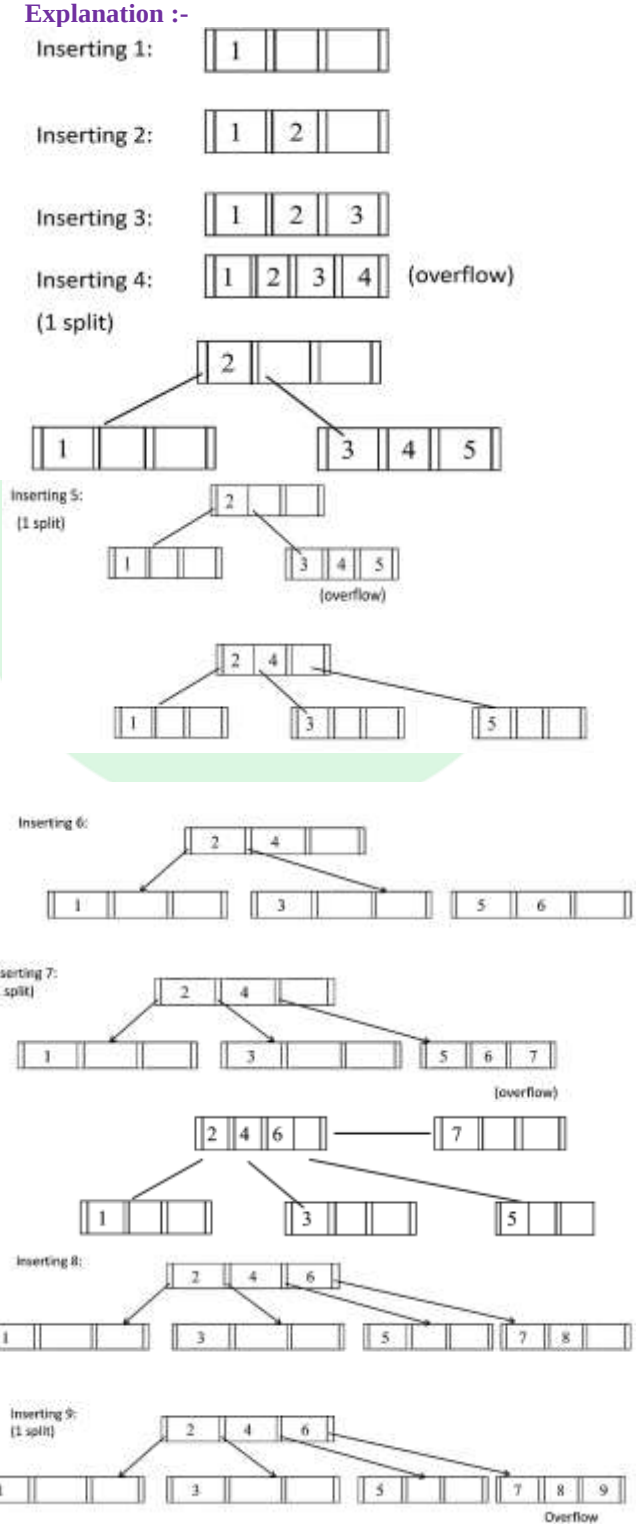
Explanation:-

Clustering index

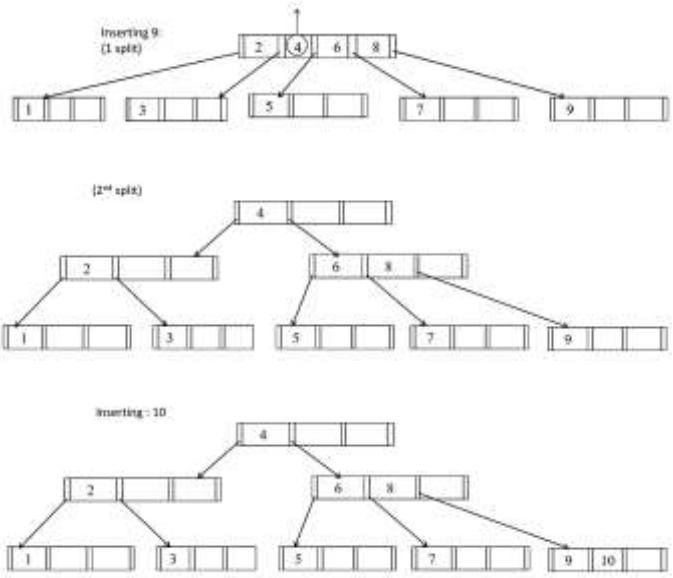
- File is physically ordered on the search key
- Search key is **not unique** (duplicates allowed)
  ⇒ Clustering index is on **non-key ordering field**

20. Answer:-C

Explanation :-



Check Out Our GATE Courses & Secure Your Dream Rank!



Total split = 5

21. Answer:-C

Explanation :-

Record size = 32 bytes, block size = 1024 bytes
⇒ Records per block = 1024 / 32 = 32
Total records = 16384
⇒ Number of data blocks = 16384 / 32 = 512
Secondary index (on key, not ordered file) ⇒ dense index
So, number of index entries = 16384
Each index entry size = key (6 bytes) + pointer (10 bytes) = 16 bytes
⇒ Entries per index block = 1024 / 16 = 64
First-level index blocks:
16384 / 64 = 256
Second-level index:
256 entries (one per first-level block)
⇒ Blocks = 256 / 64 = 4
Answer: (c) 256 and 4

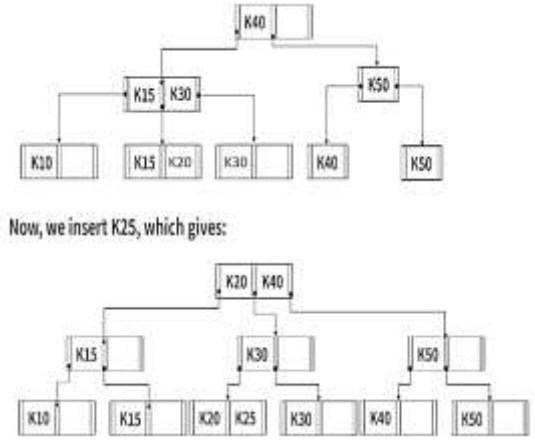
22. Answer :- A

Explanation :-

Given:
Block size = 1K bytes = 1024 bytes
Value field = 9 bytes
Data record pointer = 7 bytes
Block pointer (for next leaf) = 6 bytes
Each (value, record pointer) pair size = 9 + 7 = 16 bytes
Let number of pairs = n
Total size = 16n + 6 ≤ 1024
16n ≤ 1018
n ≤ 1018 / 16
n ≤ 63.625
Maximum integer n = 63

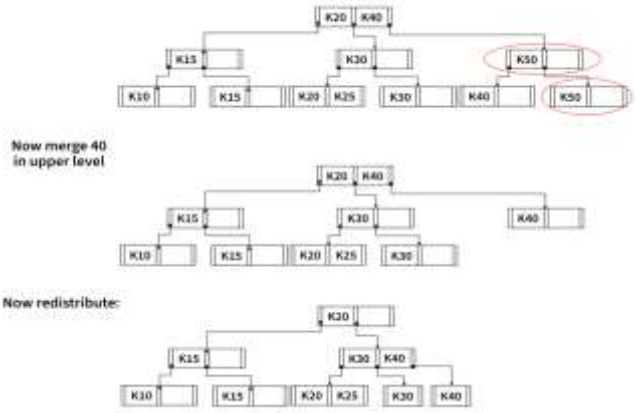
23. Answer : A

Explanation:-



24. Answer:- A

Explanation:-



25. Answer : C

Explanation:

For B+ tree with order n, index pointer p, and block size = B
For a non leaf node it can be given that
n*p + (n-1)*(k) <= B
n*6 + (n-1)*9 <= 512
n <= 34.77

26. Answer :- D

Solution:-

B^+-trees are designed to minimize disk I/O.
Since disks transfer data in blocks (pages), B^+-trees store many keys in one node (block), reducing the number of disk accesses.
Binary search trees are not optimized for block-based disk storage.
Correct Answer: D. Data transfer from disks is in blocks.

27. Answer :- C

Explanation :-

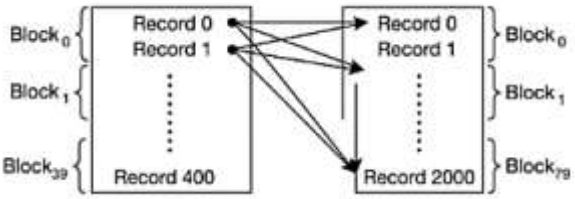
Number of block access = $n_r * b_s + b_r$ where b_r and b_s are number of blocks in relations R and S respectively, and n_r

Check Out Our GATE Courses & Secure Your Dream Rank!

is the number of tuples in relation R.
We select the relation with small number of tuples as outer relation R. So, R is T2.
Number of block access = 400 * 80 + 20 = 32020
Thus, option (C) is correct.

Or

Here condition given only 1 block of T1 and 1 block of T2 can be present simultaneously inside the memory buffer space.



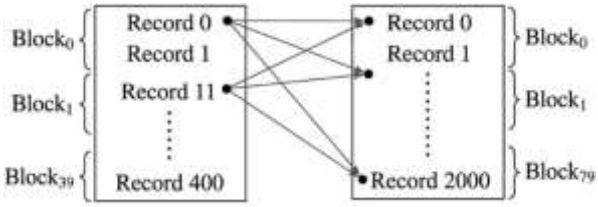
In nested loop join, for every record of one table, every block of the second table is loaded.
So, If T1 is outer = 2000 × 20 + 80 = 40080
If T2 is outer = 400 × 80 + 20 = 32020
So we go with T2 as outer table = 32020

28. Answer : B

Explanation :-Number of block access = $n_r * b_s + b_r$ where b_r and b_s are number of blocks in relations R and S respectively, and n_r is the number of tuples in relation R.
We select the relation with small number of tuples as outer relation R. So, R is T2. Number of block access = 400 * 80 + 20 = 32020
The memory buffer space holds one block of records for T1 and one block of records for T2 simultaneously.
So, number of block accesses is 80 * 20 + 20 = 1620
So, Total number of block access = 32020 - 1620 = 30400
Thus, option (B) is correct.

Or

In a **nested block loop join**, instead of processing each record of table 1 with all blocks of table 2, we process **each block of table 1** with all blocks of table 2.



So,
If **T1 is outer**:
Cost = 80 × 20 + 80 = **1680**
If **T2 is outer**:
Cost = 80 × 20 + 20 = **1620**
Hence, we select **T2 as the outer table** since it gives the lower cost = **1620**

Reduction in block accesses = 32020 − 1620 = **30400**

29. Answer : A

Explanation :- The B-Tree has **4 levels** (including root).
In the worst case, insertion causes **splitting at every level** from leaf to root:

- Leaf split → 1 new node
- Internal splits → 1 new node per level

This gives **4 new nodes**.
Additionally, if the **root splits**, a **new root is created**, adding one more node.
So, total maximum new nodes = 4 + 1 = **5**

Check Out Our GATE Courses & Secure Your Dream Rank!

6 Transaction & Concurrency Control

GATE CSE-2026 SET-2 MCQ (1-MARKS)

Q1. Consider concurrent execution of two transactions T1 and T2 in a DBMS, both of which access a data object A. For these two transactions to not conflict on A, which one of the following statements must be true?

(a) Both T1 and T2 only read A
(b) T1 reads A and T2 writes A
(c) T1 writes A and T2 reads A
(d) Both T1 and T2 write A

GATE CSE-2025 SET-2 MSQ (2-MARKS)

Q2. Consider the database transactions T1and T2, and data items X and Y. Which of the schedule(s) is/are conflict serializable?
Transaction T1 : R1(X) ; W1(Y) ; R1(X) ; W1(X) ; COMMIT(T1)
Transaction T2 : W2(X); W2(Y) ;COMMIT(T2)

(a) R1(X),W2(X),W1(Y),W2(Y),R1(X), W1(X),COM MIT(T2),COMMIT(T1)
(b) W2(X),R1(X),W2(Y),W1(Y),R1(X),COMMIT (T2),W1(X),COMMIT(T1)
(c) R1(X),W1(Y),W2(X),W2(Y),R1(X),W1(X), COMMIT(T1),COMMIT(T2)
(d) W2(X),R1(X),W1(Y),W2(Y),R1(X), COMMIT(T2),W1(X),COMMIT(T1)

GATE CSE-2025 SET-2 MSQ (2-MARKS)

Q3. An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account *A* attempted simultaneous transfers of Rs. 10000 each from account A to account *B*. Both transactions read the same value, Rs. 11000, as the initial balance in *A* and were allowed to go through. *B* was credited Rs. 10000 twice. *A* was debited only once and ended up with a balance of Rs. 1000.
Which of the following properties is/are certain to have been violated by the system?

(a) Atomicity
(b) Consistency
(c) Isolation
(d) Durability

GATE CSE-2025 SET-1 MCQ (1-MARKS)

Q4. A schedule of three database transactions $T1$,$T2$, and$T3$ is shown. $R_i(A)$ and $W_i(A)$ denote read and write of data item A by transaction T_i, i=1,2,3 The transaction T1 aborts at the end. Which other transaction(s) will be required to be rolled back?
$R_1(X)$; $W_1(Y)$; $R_2(X)$; $R_2(Y)$; $R_3(Y)$;ABORT(T1);

(a) Only T2
(b) Only T3
(c) Both T2 and T3
(d) Neither T2 nor T3

GATE CSE-2024 SET-2 MSQ (2-MARKS)

Q5. Which of the following statements about the Two Phase Locking (2PL) protocol is/are TRUE?

(a) 2PL permits only serializable schedules
(b) With 2PL, a transaction always locks the data item being read or written just before every operation and always releases the lock just after the operation
(c) With 2PL, once a lock is released on any data item inside a transaction, no more locks on any data item can be obtained inside that transaction
(d) A deadlock is possible with 2PL

GATE CSE-2024 SET-2 MSQ (2-MARKS)

Q6. Once the DBMS informs the user that a transaction has been successfully completed, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called

(a) Durability
(b) Atomicity
(c) Consistency
(d) Isolation

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2024 SET-2 MSQ (2-MARKS)

Q7. Consider the following read-write schedule *S* over three transactions T1,T2 and T3, where the subscripts in the schedule indicate transaction IDs:

S: r1(z); w1(z);r 2(x); r3(y); w3(y); r2(y); w2(x); w2(y);

Which of the following transaction schedules is/are conflict equivalent to S?

(a) T1T2T3
(b) T1T3T2
(c) T3T2T1
(d) T3T1T2

GATE CSE-2022 MCQ (2-MARKS)

Q8. Let Ri(z) and Wi(z) denote read and write operations on a data element z by a transaction Ti, respectively. Consider the schedule S with four transactions.
S: R4(x) R2(x) R3(x) R1(y) W1(y) W2(x) W3(y) R4(y)
Which one of the following serial schedules is conflict equivalent to S ?

(a) $T1 \rightarrow T3 \rightarrow T4 \rightarrow T2$
(b) $T1 \rightarrow T4 \rightarrow T3 \rightarrow T2$
(c) $T4 \rightarrow T1 \rightarrow T3 \rightarrow T2$
(d) $T3 \rightarrow T1 \rightarrow T4 \rightarrow T2$

GATE CSE-2021 SET-1 MCQ (1-MARKS)

Q9. Suppose a database system crashes again while recovering from a previous crash. Assume checkpointing is not done by the database either during the transactions or during recovery. Which of the following statements is/are correct?

(a) The system cannot recover any further.
(b) The same undo and redo list will be used while recovering again.
(c) The database will become inconsistent.
(d) All the transactions that are already undone and redone will not be recovered again.

GATE CSE 2021 SET-2 MCQ (2-MARKS)

Q10. Let S be the following schedule of operations of three transactions T1, T2 and T3 in a relational database system:
R2(Y), R1(X), R3(Z), R1(Y), W1(X), R2(Z), W2(Y), R3(X), W3(Z)
Consider the statements P and Q below:
P: S is conflict-serializable.
Q: If T3 commits before T1 finishes, then S is recoverable.
Which one of the following choices is correct?

(a) Both P and Q are true.
(b) P is false and Q is true.
(c) Both P and Q are false.
(d) P is true and Q is false.

GATE CSE-2021 SET-1 MCQ (2-MARKS)

Q11. Let $r_i(z)$ and $w_i(z)$ denote read and write operations respectively on a data item z by a transaction Ti. Consider the following two schedules.
S1 : r1(x) r1(y) r2(x) r2(y) w2(y) w1(x)
S2 : r1(x) r2(x) r2(y) w2(y) r1(y) w1(x)
Which one of the following options is correct?

(a) S1 is not conflict serializable, and S2 is conflict serializable.
(b) Both S1 and S2 are conflict serializable.
(c) Neither S1 nor S2 is conflict serializable.
(d) S1 is conflict serializable, and S2 is not conflict serializable.

GATE CSE-2019 MCQ (1-MARKS)

Q12. Consider the following two statements about database transaction schedules:
I. Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable.
II. Timestamp-ordering concurrency control protocol with Thomas write rule can generate view serializable schedules that are not conflict serializable.
Which of the above statements is/are TRUE?

(a) Both I and II
(b) Neither I nor II
(c) II only
(d) I only

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2020 MCQ (2-MARKS)

Q13. Consider a schedule of transactions T_1 and T_2 :

T_1	RA			RC		WD		WD	Commit	
T_2		RB	WB		RD		WC			Commit

Here, RX stands for "Read(X)" and WX stands for "Write(X)". Which one of the following schedules is conflit equivalent to the above schedule?

(a)

T_1					RA	RC	WD	WB	Commit	
T_2	RB	WB	RD	WC						Commit

(b)

T_1				RA	RC	WD	WB		Commit	
T_2	RB	WB	RD					WC		Commit

(c)

T_1	RA	RC	WD	WB					Commit	
T_2					RB	WB	RD	WC		Commit

(d)

T_1	RA	RC	WD				WB		Commit	
T_2				RB	WB	RD		WC		Commit

GATE CSE-2017 SET-1 MCQ (2-MARKS)

Q14. In a database system, unique timestamps are assigned to each transaction using Lamport 's logical clock. Let TS(T1) and TS(T2) be the timestamps of transactions T1 and T2 respectively. Besides, T1 holds a lock on the resource R and T2 has requested a conflicting lock on the same resource R. The following algorithm is used to prevent deadlocks in the database system assuming that a killed transaction is restarted with the same timestamp.
if TS(T2) < TS(T1) then
T1 is killed
else T2 waits.
Assume any transaction that is not killed terminates eventually. Which of the following is TURE about the database system that uses the above algorithm to prevent deadlocks?

(a) The database system is both deadlock-free and starvation-free.
(b) The database system is deadlock-free, but not starvation-free.
(c) The database system is starvation-free, but not deadlock-free.
(d) The database system is neither deadlock-free nor starvation-free.

GATE CSE-2017 SET-2 NAT (2-MARKS)

Q15. Two transactions T1 and T2 are given as:
T1 : r1(X)w1(X)r1(Y)w1(Y)
T2 : r2 (Y)w2(Y)r2(Z)w2(Z)
where r i(V) denotes a read operation by transaction T i on a variable V and w i(V) denotes a write operation by transaction T i on a variable V. The total number of conflict serializable schedules that can be formed by T1 and T2 is _____.

GATE CSE-2016 SET-2 MCQ (1-MARKS)

Q16. Suppose a database schedule S involves transactions T1,…,Tn. Construct the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?

(a) Topological order
(b) Depth-first order
(c) Breadth-first order
(d) Ascending order of the transaction indices.

GATE CSE-2016 SET-1 MCQ (1-MARKS)

Q17. Which one of the following is NOT a part of the ACID properties of database transactions?

(a) Atomicity
(b) Consistency
(c) Isolation
(d) Deadlock-freedom

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2016 SET-1 MCQ (2-MARKS)

Q18. Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects {O1,…..Ok}. This is done in the following manner:
Step 1. T acquires exclusive locks to O1,….Ok in increasing order of their addresses.
Step 2. The required operations are performed.
Step 3. All locks are released.
This protocol will ?

(a) guarantee serializability and deadlock-freedom
(b) guarantee neither serializability nor deadlock-freedom
(c) guarantee serializability but not deadlock-freedom
(d) guarantee deadlock-freedom but not serializability

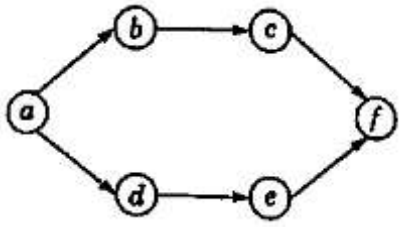
GATE CSE-2016 SET-2 MCQ (2-MARKS)

Q19. Consider the following database schedule with two transactions, T1 and T2.
S – r2(X); r1(X); r2(Y); w1(X); r1(Y); w2(X); a1; a2
Where $r_i(Z)$ denotes a read operation by transaction T_i on a variable Z, $w_i(Z)$ denotes a write operation by T_i on a variable Z and ai denotes an abort by transaction T_i.
Which one of the following statements about the above schedule is TRUE?

(a) S is non-recoverable
(b) S is recoverable, but has a cascading abort
(c) S does not have a cascading abort
(d) S is strict

GATE CSE-2016 SET-1 NAT (1-MARKS)

Q20. Consider the following directed graph:



The number of different topological ordering of the vertices of the graph is ___ .

GATE CSE-2015 SET-3 MCQ (2-MARKS)

Q21. Consider the partial Schedule S involving two transactions T1 and T2. Only the read and the write operations have been shown. The read operation on data item P is denoted by read(P) and write operation on data item P is denoted by write(P).
Schedule S

Time Instance	Transaction ID	
	T1	T2
1	read(*A*)	
2	write (A)	
3		read(C)
4		write (C)
5		read(B)
6		write (B)
7		read(A)
8		commit
9	read(B)	

Suppose that the transaction T1 fails immediately after time instance 9. Which of the following statements is correct?

(a) T2 must be aborted and then both T1 and T2 must be re-started to ensure transaction atomicity
(b) Schedule S is non-recoverable and cannot ensure transaction atomicity
(c) Only T2 must be aborted and then re- started to ensure transaction atomicity
(d) Schedule S is recoverable and can ensure atomicity and nothing else needs to be done.

GATE CSE-2015 SET-2 MCQ (1-MARKS)

Q22. Consider the following transaction involving two bank accounts x and y.
read (x) ;
x : =x - 50;
write (x) ;
read (y);
y : = y + 50;
write (y)
The constraint that the sum of the accounts x and y should remain constant is that of

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) Atomicity
(b) Consistency
(c) Isolation
(d) Durability

GATE CSE-2015 SET-2 MCQ (2-MARKS)

Q23. Consider a simple checkpointing protocol and the following set of operations in the log.
(Start, T4);
(write, T4, y, 2, 3);
(Start, T1);
(commit, T4);
(write, T1, z, 5, 7);
(check point);
(Start, T2);
(write, T2, x, 1, 9);
(commit, T2);
(start, T3);
(write, T3, z, 7, 2);
If a crash happens now and the system tries to recover using both undo and redo operations, what are the contents of the undo list and the redo list?

(a) Undo T3, T1; Redo T2
(b) Undo T3, T1; Redo T2, T4
(c) Undo: none; redo: T2, T4, T3, T1
(d) Undo T3, T1; T4; Redo: T2

GATE CSE-2014 SET-3 MCQ (2-MARKS)

Q24. Consider the transactions T1,T2,andT3 and the schedules S1 and S2 given below.
T1:r1(X);r1(Z);w1(X);w1(Z)
T2:r2(Y);r2(Z);w2(Z)
T3:r3(Y);r3(X);w3(Y)

S1:r1(X);r3(Y);r3(X);r2(Y);r2(Z);w3(Y);w2(Z);r1(Z);
w1(X);w1(Z)
S2:r1(X);r3(Y);r2(Y);r3(X);r1(Z);r2(Z);w3(Y);w1(X);
w2(Z);w1(Z)
Which one of the following statements about the schedules is TRUE?

(a) Only S1 is conflict-serializable.
(b) Only S2 is conflict-serializable.
(c) Both S1 and S2 are conflict-serializable
(d) Neither S1 nor S2 is conflict-serializable.

GATE 2014 SET-1 MCQ (2-MARKS)

Q25. Consider the following four schedules due to three transactions (indicated by the subscript) using read and write on a data item x, denoted by r(x) and w(x) respectively. Which one of them is conflict serializable?

(a) r1 (x); r2 (x);w1 (x); r3 (x); w2 (x)
(b) r2 (x); r1 (x);w2 (x); r3 (x); w1 (x)
(c) r3 (x); r2 (x);r1 (x); w2 (x); w1 (x)
(d) r2 (x); w2 (x);r3 (x); r1 (x); w1 (x)

GATE CSE-2014 SET-2 MCQ (2-MARKS)

Q26. Consider the following schedule S of transactions T1, T2, T3, T4:

T_1	T_3	T3	T.
	Read (X)	Writes (X) Commit	
Writes (X) Commit	Writes (Y) Reads (Z) Commit		Reads (X) Reads (Y) Commit

Which one of the following statements is CORRECT?

(a) S is conflict-serializable but not recoverable
(b) S is not conflict-serializable but is recoverable
(c) S is both conflict-serializable and recoverable
(d) S is neither conflict-serializable nor is it recoverable

GATE CSE-2012 MCQ (2-MARKS)

Q27. Consider the following transactions with data items P and Q initialized to zero:

T1 : read (P);	T2 : read (Q);
read (Q) ;	read (P) ;
if P = 0 then	if Q = 0 then
Q : = Q + 1 ;	P : = P + 1 ;
write (Q) ;	write (P) ;

Any Non-serial interleaving of T1 and T2 for concurrent execution leads to

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) a serializable schedule
(b) a schedule that is not conflict serializable
(c) a conflict serializable schedule
(d) a schedule for which a precedence graph cannot be drawn

GATE CSE-2010 MCQ (1-MARKS)

Q28. Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?
I. 2-phase locking
II. Time-stamp ordering

(a) I only
(b) II only
(c) Both I and II
(d) Neither I nor II

GATE CSE-2010 MCQ (2-MARKS)

Q29. Consider the following schedule for transactions T1, T2 and T3:
Which one of the schedules below is the correct serialization of the above?

T1	T2	T3
Read(X)		
	Read (Y)	
		Read (Y)
	Write (Y)	
Write (X)		
		Write (X)
	Read (X)	
	Write (X)	

(a) T1 → T3 → T2
(b) T2 → T1 → T3
(c) T2 → T3 → T1
(d) T3 → T1 → T2

GATE CSE-2009 MCQ (2-MARKS)

Q30. Consider two transactions T_1 and T_2, and four schedules S_1, S_2, S_3, S_4 of T_1 and T_2 as given below:
$T_1 : R_1[x]\ W_1[x]\ W_1[y]$
$T_2 : R_2[x]\ R_2[y]\ W_2[y]$
$S_1 : R_1[x]\ R_2[x]\ R_2[y]\ W_1[x]\ W_1[y]\ W_2[y]$
$S_2 : R_1[x]\ R_2[x]\ R_2[y]\ W_1[x]\ W_2[y]\ W_1[y]$
$S_3 : R_1[x]\ W_1[x]\ R_2[x]\ W_1[y]\ R_2[y]\ W_2[y]$
$S_4 : R_2[x]\ R_2[y]\ R_1[x]\ W_1[x]\ W_1[y]\ W_2[y]$
Which of the above schedules are conflict-serializable?

(a) S_1 and S_2
(b) S_2 and S_3
(c) S_3 only
(d) S_4 only

GATE CSE-2008 MCQ (2-MARKS)

Q31. Consider the following three schedules of transactions T_1, T_2 and T_3.
(Notations: In the following, NYO represents the action Y (R for read, W for write) performed by transaction N on object O.)
S1:
2RA 2WA 3RC 2WB 3WA 3WC
1RA 1RB 1WA 1WB
S2:
3RC 2RA 2WA 2WB 3WA 1RA
1RB 1WA 1WB 3WC
S3:
2RA 3RC 3WA 2WA 2WB 3WC
1RA 1RB 1WA 1WB
Which of the following statements is TRUE?
(a) S1, S2 and S3 are all conflict equivalent to each other
(b) No two of S1, S2 and S3 are conflict equivalent to each other
(c) S2 is conflict equivalent to S3, but not to S1
(d) S1 is conflict equivalent to S2, but not to S3

GATE CSE-2007 MCQ (2-MARKS)

Q32. Consider the following schedules involving two transactions. Which one of the following statements is TRUE?
$S_1 : r_1(X); r_1(Y); r_2(X); r_2(Y); w_2(Y); w_1(X)$
$S_2 : r_1(X); r_2(X); r_2(Y); w_2(Y); r_1(Y); w_1(X)$
(a) Both S_1 and S_2 are conflict serializable.
(b) S_1 is conflict serializable and S_2 is not conflict serializable.
(c) S_1 is not conflict serializable and S_2 is conflict serializable.
(d) Both S_1 and S_2 are not conflict serializable.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2007 MCQ (2-MARKS)

Q33. Consider the following two transactions: T_1 and T_2.

T_1:	T_2:

read(A);
read(B);
if A = 0 then B ← B + 1;
write(B);
	read(B);
	read(A);
	if B ≠ 0 then A ← A − 1;
	write(A);

Which of the following schedules, using shared and exclusive locks, satisfy the requirements for **strict two-phase locking (Strict 2PL)** for the above transactions?

(a)

S1 :	S2 :
lock S(A);	lock S(B);
read (A);	read (B);
lock S(B);	lock S(A);
read(B);	read(A);
if$A = 0$	if$B \neq 0$
then B ← B + 1;	then A ← A − 1;
write(B);	write(A);
commit;	commit;
unlock(A);	unlock(B);
unlock (B);	unlock (A);

(b)

S1:	S2:
lock X(A);	lock X(B);
read(A);	read(B);
lock X(B);	lock X(A);
read(B);	read(A);
if A = 0 then	if B ≠ 0 then
B ← B + 1;	A ← A − 1;
write(B);	write(A);
unlock(A);	unlock(A);
commit;	commit;
unlock(B);	unlock(A);

(c)

S1:	S2:
lock S(A);	lock S(B);
read(A);	read(B);
lock X(B);	lock X(A);
read(B);	read(A);
if A = 0 then	if B ≠ 0 then
B ← B + 1;	A ← A − 1;
write(B);	write(A);
unlock(A);	unlock(A);
commit;	commit;
unlock (B);	unlock (A);

(d)

S1:	S2:
lock S(A);	lock S(B);
read(A);	read(B);
lock X(B);	lock X(A);
read(B);	read(A);
if A = 0 then	if B ≠ 0 then
B ← B + 1;	A ← A − 1;
write(B);	write(A);
unlock(A);	unlock(A);
unlock(B);	unlock(B);
commit;	commit;

GATE CSE-2006 MCQ (2-MARKS)

Q34. Consider the following log sequence of two transactions on a bank account, with initial balance 12000, that transfer 2000 to a mortgage payment and then apply a 5% interest.

1. T1 start
2. T1 B old=12000 new=10000
3. T1 M old=0 new=2000
4. T1 commit
5. T2 start
6. T2 B old=10000 new=10500
7. T2 commit

Suppose the database system crashes just before log record 7 is written. When the system is restarted, which one statement is true of the recovery procedure?

(a) We must redo log record 6 to set B to 10500
(b) We must undo log record 6 to set B to 10000 and then redo log records 2 and 3.
(c) We need not redo log records 2 and 3 because transaction T1 has committed.
(d) We can apply redo and undo operations in arbitrary order because they are idempotent

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE-2005 MCQ (1-MARKS)

Q35. Amongst the ACID properties of a transaction, the 'Durability' property requires that the changes made to the database by a successful transaction persist

(a) Except in case of an Operating System crash
(b) Except in case of a Disk crash
(c) Except in case of a power failure
(d) Always, even if there is a failure of any kind of failure

ANSWER KEY:

1. (a)	**2.(b)**	**3.(b, c)**
4.(c)	**5.(a,c,d)**	**6.(a)**
7.(b.c,d)	**8.(a)**	**9.(b)**
10.(d)	**11.(a)**	**12.(a)**
13.(b)	**14.(a)**	**15.(54)**
16.(a)	**17.(d)**	**18.(a)**
19.(c)	**20.(6)**	**21.(b)**
22.(b)	**23.(a)**	**24.(a)**
25.(d)	**26.(c)**	**27.(b)**
28.(b)	**29.(a)**	**30.(b)**
31.(d)	**32.(c)**	**33.(c)**
34.(b)	**35.(d)**	

HINTS & SOLUTIONS

1. Answer :- A
Explanation :-
Conflict rules
Two operations conflict if:
- They belong to different transactions
- Access same data item
- At least one is write

Check options:
(A) Both T1 and T2 only read A
Read–Read → no write → no conflict (True)
(B) T1 reads A and T2 writes A → conflict (False)
(C) T1 writes A and T2 reads A → conflict (False)
(D) Both write A → conflict (False)

2. Answer :-B
Explanation :-
While taking precedence graph for each options B is the only option which doesn't contain cycles or deadlocks. I.e T2-> T1 which is conflict serializable.

3. Answer :- B, C
Explanation :-
Given:
Account A has Rs. 11,000.
Two simultaneous transfers of Rs. 10,000 each are attempted.
Account B is credited twice (total Rs. 20,000).
Account A is debited only once (final balance Rs. 1,000).

ACID Properties:

1. Atomicity:
Ensures all parts of a transaction are completed or none are.Each transfer individually was complete (money was debited from A and credited to B),so atomicity is likely not violated
Not necessarily violated.

2. Consistency:
Ensures that the database moves from one valid state to another.
If Rs. 20,000 was credited to B, Rs. 20,000 should be debited from A.
But only Rs. 10,000 was debited. This breaks balance rules.
Violated.
3. Isolation:
Ensures concurrent transactions do not interfere with each other.
Both transactions read the same initial balance (Rs. 11,000)
and proceeded independently, leading to an incorrect final state.
This is a classic lost update / write skew issue.
Violated.
4. Durability:
Ensures that once a transaction is committed, it remains so,
even in case of failure. No crash/rollback problem here.
Not violated.
Conclusion:
Consistency and Isolation are violated.

4. Answer :- C

T_1	T_2	T_3
$R_1(x)$		
$W_1(y)$		
	$R_2(x)$	
	$R_2(y)$	
		$R_3(y)$
Abort		

As $W_1(y)$ is read by both T2 and T3 before it commits aborting T1 required to roll back both T2 and T3.

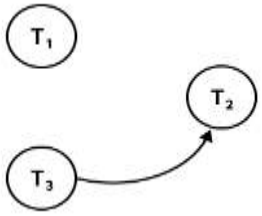
5. Answer :- A,C,D
Explanation :

A. 2PL -> serial schedule
C. Also it's the property of 2pl which states that once a lock is relased no further lock can be taken as the shrinking phase of the transaction starts and you can't acquire any lock in the shrinking phase of a transaction.
D. Deadlock can occur in 2PL as in a schedule 2 transaction can wait for each other to release the locks. This is the disadvantage of normal 2PL

6. Answer :- A
Explanation :

Atomicity: Either execute all operations of a transaction successfully or none of them
Full or None
Consistency: Database remains consistent before and after the transaction Isolation: Transactions execute independently without interference
Durability: Any change in the database must persist for a long period of time

7. Answer :- B,C,D
Explanation:-



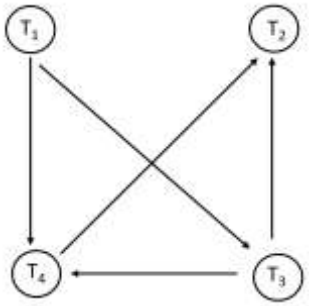
The corresponding polygraph for the given transaction. it has three conflict pairs as T3→T2 as r3(y)→w2(y),w3(y)→r2(y),w3(y)→w2(y).
Because the graph is acyclic it is conflict serializable schedule and its corresponding conflict equal schedule is based on the topological order of the given graph.
Therefore, the following are possible serial schedules:
1. T1,T3,T2
2. T3,T1,T2
3. T3,T2,T1

Option (B,C,D) are correct.

8. Answer :- A
Explanation :- Conflict operations:
1. Transaction Ti reading data item K, after that Transaction Tj writing data item K
2. Transaction Ti writing data item K, after that Transaction Tj reading data item K
3. Transaction Ti writing data item K, after that Transaction Tj writing data item K.

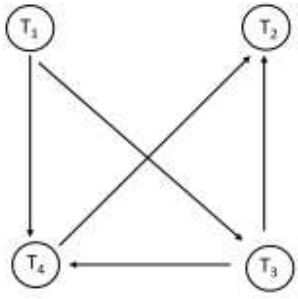
Note that, Transaction Ti writing data item 'K', after that Transaction Tj writing data item 'P' are not conflict operations due to those are different data items.
Tabular representation of given schedule is :
S: R_4 (×) R_2(×)R_3(×)R_1(y)W_1(y)W_2(x)W_3(y)R_4(y)

T_1	T_2	T_3	T_4
			R(x)
	R(x)		
		R(x)	
R(y)			
W(y)			
	W(x)		
		W(y)	
			R(y)



Conflict serializable
< T_1 T_3 T_4 T_2 >
Serializablity order

Alternate approach
S: R_4 (×) R_2(×)R_3(×)R_1(y)W_1(y)W_2(x)W_3(y)R_4(y)
For data item **x**:
$R_4(x) \rightarrow W_2(x) \Rightarrow T_4 \rightarrow T_2$
$R_2(x) \rightarrow W_2(x) \Rightarrow T_2 \rightarrow T_2$ (ignore self-loop)
$R_3(x) \rightarrow W_2(x) \Rightarrow T_3 \rightarrow T_2$
For data item **y**:
$R_1(y) \rightarrow W_3(y) \Rightarrow T_1 \rightarrow T_3$
$W_1(y) \rightarrow W_3(y) \Rightarrow T_1 \rightarrow T_3$
$W_1(y) \rightarrow R_4(y) \Rightarrow T_1 \rightarrow T_4$
$W_3(y) \rightarrow R_4(y) \Rightarrow T_3 \rightarrow T_4$



< T_1 T_3 T_4 T_2 >
Serializablity order

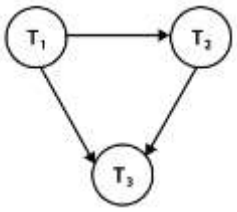
9. Answer :- B

Explanation :-Assuming undo/redo list are persistent, the answer should be The same undo and redo list will be used while recovering again.. As while recovering from the previous crash also a database system will make use of the same undo and redo list to recover the system.

10. Answer :- D

Explanation :-

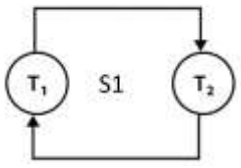
T_1	T_2	T_3
	$R_2(y)$	
$R_1(x)$		
		$R_3(Z)$
$R_1(y)$		
$W_1(x)$		
	$R_2(Z)$	
	$W_2(y)$	
		$R_3(x)$
		$W_3(Z)$ commit



Conflict serialiable s1 : true < T_1 , T_2 , T_3
Irrecoverable or non Recoverable schedule

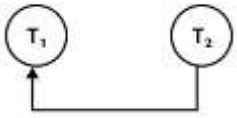
11. Answer :- A

Explanation :- A



S1 is not conflict serializable.

S2 : Cycle does not exist so it is a serializable.



S2 is conflict serializable.

12. Answer :- A

Explanation :- In strict 2PL, we allow only those schedules whose precedence graph is acyclic which means the Schedule is Conflict Serializable .Also here, a transaction T does not release any of its exclusive (write) locks until after it commits or aborts.
Hence, no other transaction can read or write an item that is written by T unless T has committed, leading to a strict schedule for recoverability.
Time stamp ordering schedule with Thomas write rule generates View Serializable Schedule having Blind write. So, due to the presence of Blind write, it would not be Conflict Serial.

13. Answer :-B

T_1	T_2
$r_1(A)$	$r_2(B)$
	$w_2(B)$
$r_1(C)$	
$w_1(D)$	$r_2(D)$
$w_1(B)$	$w_2(c)$
Commit	
	Commit

T_1	T_2
	$r_2(B)$
	$w_2(B)$
	$r_2(D)$
$r_1(A)$	
$r_1(C)$	
$w_1(D)$	
$w_1(B)$	
Commit	$w_2(c)$
	Commit

Both schedules have the same order of conflicting operations, hence they are conflict equivalent.

Check Out Our GATE Courses & Secure Your Dream Rank!

14. Answer :- A

Explanation :- According to the question, it is mentioned that 'unique time stamps' are assigned to each transactions. So, there won't be any equal timestamps.

In case of Lamport's logical clock:

Here, timestamps are assigned in increasing order only.

According to the given algorithm,

$TS(T_2) < TS(T_1)$

then T_1 is killed

else

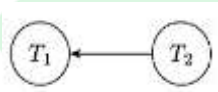
T_2 will wait

So, in both the cases, the database system will be deadlock free and there will be no starvation.

T_1 holds lock on resource R,

T_2 requires conflict lock on same resource R

Wait for graph



if $TS(T_2) < TS(T_1)$

Then T_1 killed [restart with same TS value] else T_2 waits.

Avoids both deadlock and starvation.

15. Answer :-54

Explanation :-

From the given transactions between T_1 and T_2, total number of possible schedules = (4+4)!/4!4!

= 8!/(4×3×2×4×3×2)

= (8×7×6×5×4×3×2×1)/(4×3×2×4×3×2)

= 70

Following two conflict actions are possible :

Number of Permutations = 4 × 3 = 12

Number of permutations = 4 × 1 = 4

Total no. of conflict serial schedules that can be formed by T_1 and T_2 = 70 - 12 - 4 = 54.

Or

T_1: $r_1(x)$ $w_1(x)$ $r_1(y)$ $w_1(y)$

T_2: $r_2(y)$ $w_2(y)$ $r_2(z)$ $w_2(z)$

How to find Conflict Serializable

$T_1 \rightarrow T_2$ OR $T_2 \rightarrow T_1$

$T_1 \rightarrow T_2$

$r_1(y) \rightarrow w_2(y) \Rightarrow$ conflict

$T_1 \rightarrow T_2$: L

Now arrange

T_1: $r_1(x)$ $w_1(x)$ $r_1(y)$ $w_1(y)$

Place T_2 ops: $w_2(y)$, $r_2(z)$, $w_2(z)$

$w_2(y)$ after $r_1(y)$

Case I

$3 \times [3C_1 + 3C_2]$

$= 3 \times (3 + 3)$

Case I = 18 ways

Case II

$2 \times [4C_1 + 4C_2]$

$= 2 \times (4 + 6)$

Case II = 20

Case III

$1 \times [5C_1 + 5C_2]$

$= 5 + 10$

Case III = 15 ways

$T_1 \rightarrow T_2 = 18 + 20 + 15 = 53$ ways

$T_2 \rightarrow T_1$

= 1 way

Final Answer

$T_2 \rightarrow T_1 = 53$ ways

Total Conflict Serializable = 53 + 1 = 54 Ans

16. Answer :- A

Explanation :-When a precedence graph of a Schedule turns out to be Acyclic, then the Topological ordering of the precedence Graph is guaranteed to yield a serial Schedule.

17. Answer :- D

Explanation :-ACID properties of a Database transaction denotes Atomicity, Consistency, Isolation and Durability. So, Deadlock – freedom is not there in the ACID properties.

18. Answer :- A

Explanation :-

In case of Conservative 2PLP:

A transaction has to lock all the items before the transaction can begin execution.

So, the Advantages of conservative 2PLP are :

• Deadlock is not possible.

• Serializability is ensured.

The given scenario (Step1, Step2 and Step3) is following the approach of conservative 2PLP.

19. Answer :- C

Explanation :-

Given schedule is,

T_1	T_2
$r_1(x)$	$r_2(x)$
	$r_2(y)$
$w_1(x)$	
$r_1(y)$	$w_2(x)$
a_1	a_2

Now let's check statements one by one,

A) False, as there is no dirty read. So, it is recoverable.

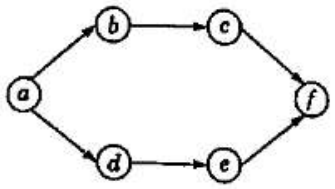
B) False, because there is no dirty read. So, it has no cascading aborts.

C) True.

D) False, because there is Transaction T_2 which writes the value of x which is written by T_1 before T_1 has aborted. So, not strict.

Check Out Our GATE Courses & Secure Your Dream Rank!

20. Answer :- 6
Explanation :-



Number of topological orders: 6

a b c d e f
a b d c e f
a b d e c f
a d b c e f
a d b e c f
a d e b c f

21. Answer :-B
Explanation :-As T1 fails after instance 9, so T1 would rollback, but as T2 has already committed, so it cannot rollback and also T2 has Dirty Read of A from T1, i.e , T2 is dependent on T1.So, to ensure Recoverability; T1 should commit first and then T2.

So, option B satisfies as S is non- recoverable and cannot ensure transaction atomicity.

22. Answer :-B
Explanation :-

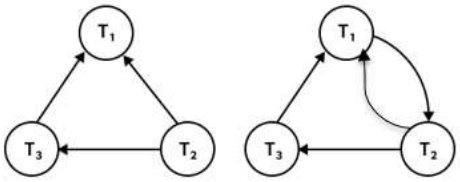
- Consistency: This property ensures that a transaction can only bring the database from one valid state to another, preserving all predefined database invariants, integrity constraints, and business rules. In this specific scenario, the condition that the total money in the system must be conserved (Account x + Account y = Constant) during a transfer is a classic consistency constraint.

Why other options are incorrect:

- (a) Atomicity: Focuses on the "all-or-nothing" execution of the transaction. It ensures that If the system crashes after debiting x but before crediting y, the entire transaction is rolled back so that money isn't lost.
- (c) Isolation: Ensures that concurrent execution of multiple transactions leaves the database in the same state as if they were executed sequentially.
- (d) Durability: Guarantees that once the transaction commits, the updates to accounts x and y survive any subsequent system crashes.

23. Answer :- A
Explanation :- As T_1& T_3 are not yet committed they must be undone, so Undo:T_3 and T_1. We must redo those transactions which are after the latest checkpoint. So T_2 must be redone. We don't need to redo the records which are before the last checkpoint, so T_4 need not be redone. So, Redo: T_2

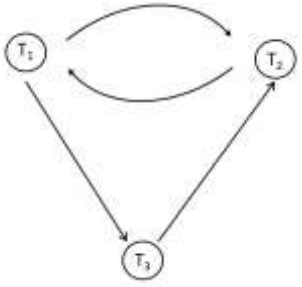
24. Answer :- A
Explanation :- On drawing the Precedence Graph for the Schedule given, we get:



We observe that, S1 has no cycle and is conflict-serializable whereas S2 has cycle and is not conflict serializable.

25. Answer :- D
Explanation :-
A. $r_1(x)$; $r_2(x)$; $w_1(x)$; $r_3(x)$; $w_2(x)$

T_1	T_2	T_3
r(x)		
	r(x)	
w(x)		
		r(x)
	w(x)	

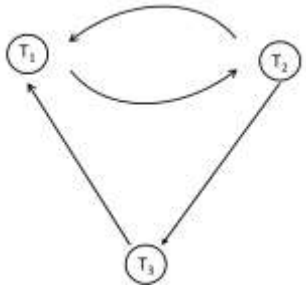


Cycle, Not conflict serializable

B. $r_2(x)$; $r_1(x)$; $w_2(x)$; $r_3(x)$; $w_1(x)$

Check Out Our GATE Courses & Secure Your Dream Rank!

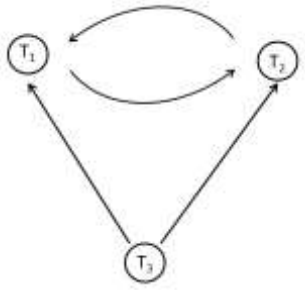
T_1	T_2	T_3
	r(x)	
r(x)		
	w(x)	
		r(x)
w(x)		



Cycle, Not conflict serializable

C. $r_3(x)$; $r_2(x)$; $r_1(x)$; $w_2(x)$; $w_1(x)$

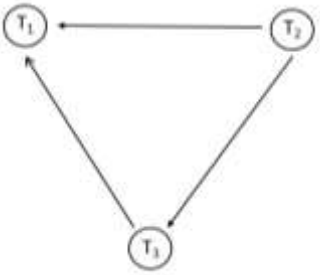
T_1	**T_2**	**T_3**
		r(x)
	r(x)	
r(x)		
	w(x)	
w(x)		



Cycle, Not conflict serializable

D. $r_2(x)$; $w_2(x)$; $r_3(x)$; $r_1(x)$; $w_1(x)$

T_1	T_2	T_3
	r(x)	
	w(x)	
		r(x)
r(x)		
w(x)		

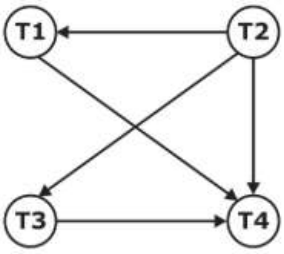


Not Cycle, conflict serializable equivalent serial schedule $< T_2\ T_3\ T_1 >$

26. Answer :- C

Explanation :-

We draw the Precedence Graph for the Schedule of transactions.



As on observing the precedence graph, we have no cycles, so, it is conflict serializable.

Also , the Read(X) in T_4 is of Database and not written by any other transaction, so the schedule has no dirty read

27. Answer :- B

Explanation :-

T_1	**T_2**
	read (Q)
read (P)	read (P)
read (Q)	
write (Q)	write (P)

The above schedule is not conflict serializable due to the presence of cycle in Precedence cycle, i.e $T_1 \rightarrow T_2$ and $T_2 \rightarrow\rightarrow T_1$.

28. Answer :- B

Explanation

— Two-phase locking protocol (2PLP) ensures the schedule to be conflict serializable but it may not be free from deadlock.

— Timestamp ordering protocol ensures both conflict serializability and free from deadlock.

29. Answer :- A

Explanation

Identify conflicting operations:

Check Out Our GATE Courses & Secure Your Dream Rank!

- On **Y**:
  T2 writes Y after T3 reads Y
  ⇒ **T3 → T2**
- On **X**:
  T1 writes X before T3 writes X
  ⇒ **T1 → T3**

  T3 writes X before T2 reads/writes X
  ⇒ **T3 → T2**

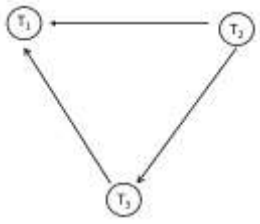
  Combine all orders:
  T1 → T3 → T2
  Graph is acyclic, so schedule is conflict serializable with order:
  T1 → T3 → T2

30. Answer :- B
Explanation:-
• **S1** has a cycle from **T1 → T2** and **T2 → T1**.
• **S2** – It is uni-directional and has only **T2 → T1**.
• **S3** – It is uni-directional and has only **T1 → T2**.
• **S4** – same as **S1**.
A schedule is conflict serializable if there is no cycle in the directed graph made by the schedules.
In the schedules we check for **RW, WR, WW** conflicts between the schedules and only these conflicts contribute in the edges of the graph.

31. Answer :- D
Explanation:-
S1: 2RA 2WA 3RC 2WB 3WA 3WC 1RA 1RB 1WA 1WB

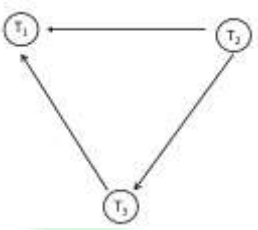
T_1	T_2	T_3
	R(A)	
	W(A)	
		R(C)
	W(R)	
		W(A)
		W(C)
R(A)		
R(B)		
W(A)		
W(B)		

Precedence graph



No cycle conflict serializable
< T_2 T_3 T_1 >
Conflict operations pair
$R_2(A) – W_3(A)$
$R_2(A) – W_1(A)$
$W_2(A) – W_3(A)$
$W_2(A) – R_1(A)$
$W_2(A) – W_1(A)$
$W_2(B) – R_1(B)$
$W_2(B) – W_1(B)$
$W_3(A) – R_1(A)$
$W_3(A) – W_1(A)$
S2: 3RC 2RA 2WA 2WB 3WA 1RA 1RB 1WA 1WB 3WC

T_1	T_2	T_3
		R(C)
	R(A)	
		R(C)
	W(A)	
	W(B)	
		W(A)
R(A)		
R(B)		
W(A)		
W(B)		
		W(C)

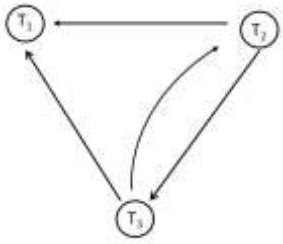


No cycle conflict serializable
< T_2 T_3 T_1 >
Conflict operation pairs
$R_2(A) – W_3(A)$
$R_2(A) – W_1(A)$
$W_2(A) – W_3(A)$
$W_2(A) – R_1(A)$
$W_2(A) – W_1(A)$
$W_2(B) – R_1(B)$
$W_2(B) – W_1(B)$
$W_3(A) – R_1(A)$
$W_3(A) – W_1(A)$

S3: 2RA 3RC 3WA 2WA 2WB 3WC 1RA 1RB 1WA 1WB

T_1	T_2	T_3

	R(A)	
		R(C)
		W(A)
	W(A)	
	W(B)	
		W(C)
R(A)		
R(B)		
W(A)		
W(B)		



S_3 : cycle; so not conflict serializable
$R_3(A) – W_3(A)$
$R_3(A) – W_1(A)$

32. Answer :-C
Explanation :-
Conflicts:
$r_2(X)$ before $w_1(X) \Rightarrow T2 \rightarrow T1$
$r_1(Y)$ before $w_2(Y) \Rightarrow T1 \rightarrow T2$
Cycle exists (T1 ↔ T2).
So S_1 is not conflict serializable.
For S_2
Conflicts:
$r_1(Y)$ after $w_2(Y) \Rightarrow T2 \rightarrow T1$
$r_2(X)$ before $w_1(X) \Rightarrow T2 \rightarrow T1$
Only T2 → T1 edges, no cycle.
So S_2 is conflict serializable (equivalent to T2 → T1).
Final Answer: C. S_1 is not conflict serializable and S_2 is conflict serializable.

33. Answer :-C
Explanation :-
According to the 2-phase locking protocol, no new lock can be acquired after the first unlock operation.
According to strict 2PL, all exclusive locks must be held until the commit operation.
In option (c), no exclusive lock is released before commit, and no new lock is acquired after any unlock.
Hence, it satisfies the strict 2PL protocol.

34. Answer :- B
Explanation :-

When a database system recovers from a crash using a log file, it classifies transactions into two categories based on their status at the time of the crash:

1. Committed Transactions (REDO): If a transaction has both a <start> and a <commit> record in the log, its changes must be preserved.

- Transaction T1 has a T1 commit (record 4) before the crash. Thus, T1 must be redone (log records 2 and 3) to ensure durability.

2. Uncommitted Transactions (UNDO): If a transaction has a <start> record but lacks a <commit> record in the log, its changes must be rolled back to maintain atomicity and consistency.

- Transaction T2 crashes before writing its commit record (record 7). Thus, T2 must be undone (log record 6) to revert any partial modifications.

(a) Incorrect: Log record 6 belongs to T2. Since T2 did not commit, its changes cannot be redone.
(b) Correct: To restore consistency, we roll back the uncommitted change from T2 (undoing log record 6 to reset B to 10000) and re-apply the committed changes from T1 (redoing log records 2 and 3).
(c) Incorrect: Even though T1 committed, standard recovery requires redoing its log records to ensure those changes are safely written to the physical disk (durability).
(d) Incorrect: Redo and undo operations cannot be applied in an arbitrary order; they must follow strict ordering (typically, undoing active transactions backward and redoing committed transactions forward) to avoid corrupting data states.

Check Out Our GATE Courses & Secure Your Dream Rank!

35. Answer :- D

Explanation :-

ACID properties:

Atomicity:

By this, we mean that either the entire transaction takes place at once or doesn't happen at all.

Consistency:

This means that integrity constraints must be maintained so that the database is consistent before and after the transaction.

Isolation:

This property ensures that multiple transactions can occur concurrently without leading to inconsistency of database state.

Durability:

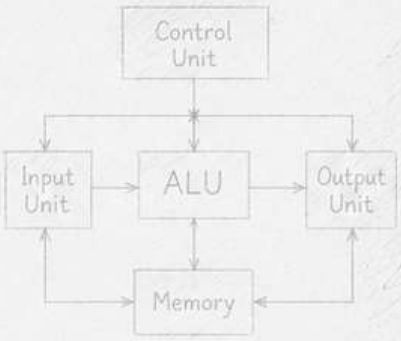
This property ensures that once the transaction has completed execution, the updates and modifications to the database are stored in and written to disk and they persist even if system failure occurs no matter which type of failure.

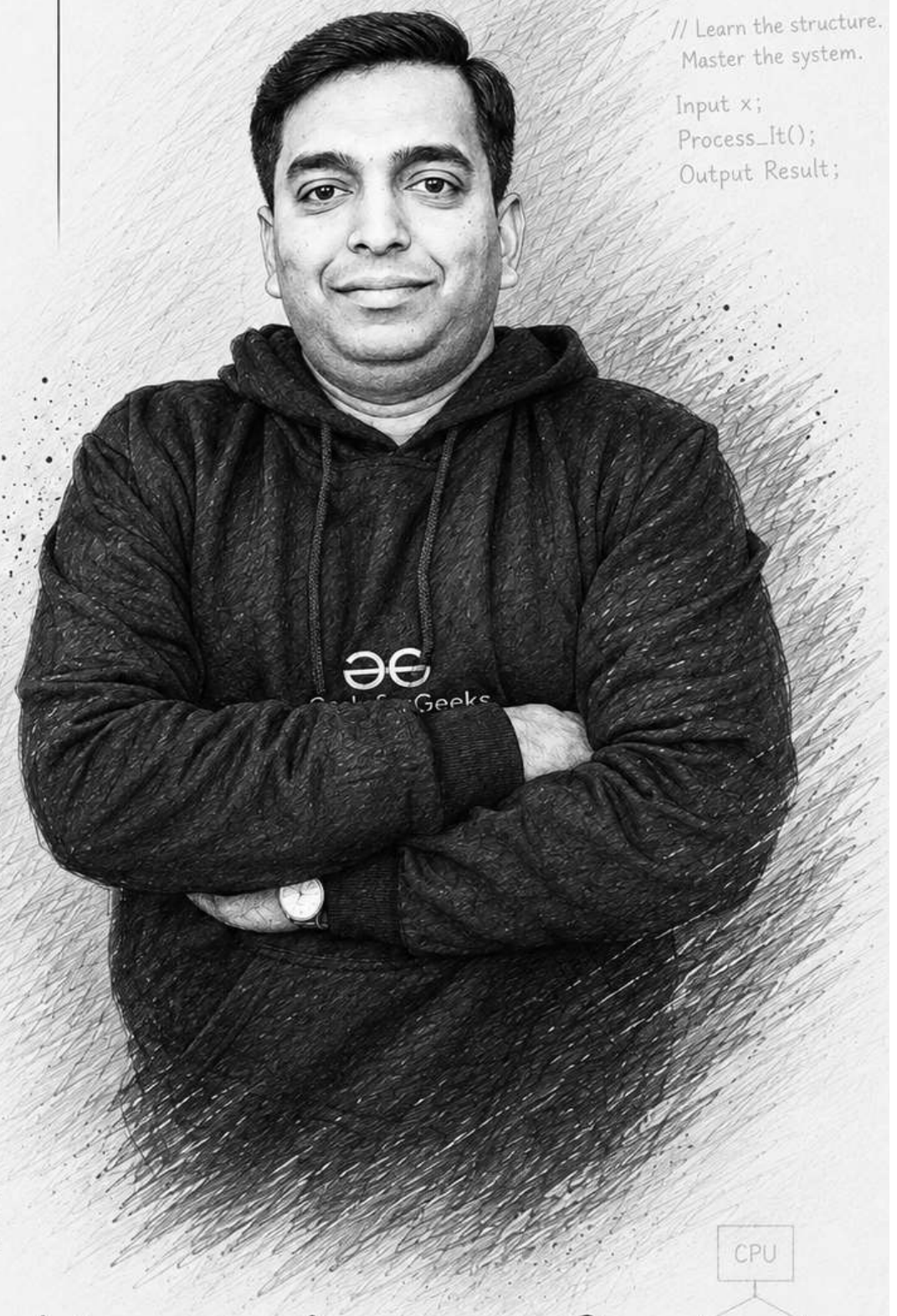
Check Out Our GATE Courses & Secure Your Dream Rank!

COMPUTER ORGANIZATION AND ARCHITECTURE

Link to GfG corresponding Short Notes: GATE FARRAY

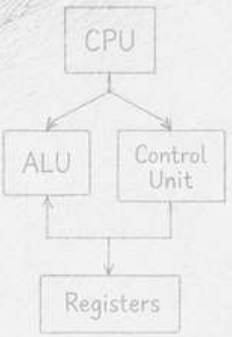
> A well-designed computer system is not magic—it's logic, organized at every level. Understand the **structure, master the performance.**





LEGEND OF

COMPUTER ORGANIZATION AND ARCHITECTURE



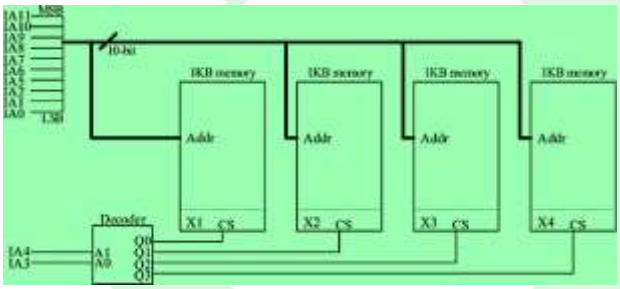
Vijay Kumar Agarwal Sir

An Inspirer. A Mentor. A Guide.

1 INTRODUCTION OF COA

GATE CSE - 2023 MCQ (2-MARKS)

Q1. A 4 kilobyte (KB) byte-addressable memory is realized using four 1 KB memory blocks. Two input address lines (IA4 and IA3) are connected to the chip select (CS) port of these memory blocks through a decoder as shown in the figure. The remaining ten input address lines from IA11-IA0 are connected to the address port of these blocks. The chip select (CS) is active high.



The input memory addresses (IA11-IA0), in decimal, for the starting locations (Addr = 0) of each block (indicated as X1, X2, X3, X4 in the figure) are among the options given below. Which one of the following options is CORRECT?

(a) (0,1,2,3)
(b) (0,1024,2048,3072)
(c) (0,8,16,24)
(d) (0,0,0,0)

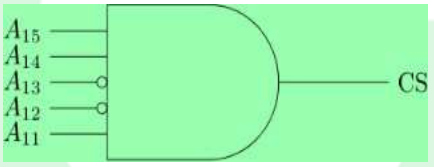
GATE CSE - 2021 MSQ SET - 2 (2-MARKS)

Q2. If the numerical value of a 2-byte unsigned integer on a little endian computer is 255 more than that on a big endian computer, which of the following choices represent(s) the unsigned integer on a little endian computer?

(a) 0x6665
(b) 0x 0001
(c) 0x4243
(d) 0x 0100

GATE CSE - 2019 MCQ (1-MARKS)

Q3. The chip select logic for a certain DRAM chip in a memory system design is shown below. Assume that the memory system has 16 address lines denoted by A15 to A0. What is the range of address (in hexadecimal) of the memory system that can get enabled by the chip select (CS) signal?



(a) C800 to CFFF
(b) CA00 to CAFF
(c) C800 to C8FF
(d) DA00 to DFFF

GATE CSE - 2018 NAT (1-MARKS)

Q4. A 32 −bit wide main memory unit with a capacity of 1 GB is built using $256M \times 4$−bit DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14}. The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closest integer) of the time available for performing the memory read/write operations in the main memory unit is ___.

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE - 2016 NAT (1-MARKS)

Q5. A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of address bus of the processor is at least ___bits.

GATE CSE - 2009 MCQ (1-MARKS)

Q6. How many 32k × 1 RAM chips are needed to provide a memory capacity of 256 K-bytes?

(a) 8
(b) 32
(c) 64
(d) 128

ANSWER KEY

1.(c)	**2. (a, d)**	**3. (a)**
4. (59)	**5. (31)**	**6.(c)**

HINTS & SOLUTIONS

1. **ANSWER: (C)**

Only IA4 and IA3 go to decoder → they select block

Remaining IA11–IA5 and IA2–IA0 go to memory ⇒ total = 10 bits

So internal address bits are NOT contiguous, but still 10 bits → block size = 1 KB

So address format becomes:

[IA11 … IA5][IA4 IA3][IA2 … IA0]

Compute base addresses (put IA2–IA0 = 000 and IA11–IA5 = 0000000):

IA4 IA3 = 00 → 0000000 00 000 → 0

IA4 IA3 = 01 → 0000000 01 000 → 8

IA4 IA3 = 10 → 0000000 10 000 → 16

IA4 IA3 = 11 → 0000000 11 000 → 24

Answer: C

(0, 8, 16, 24)

2. **ANSWER: (A, D)**

1. Understanding Memory Layout

Let the 2-byte unsigned integer be stored in memory at consecutive addresses A and $A + 1$.

Let the byte at the lower address A be B_0 and the byte at the higher address $A + 1$ be B_1.

- Little Endian Computer: Stores the Least Significant Byte (LSB) at the lower address.

LSB $= B_0$

MSB $= B_1$

Value $_{LE} = (B_1 \times 256) + B_0$

- Big Endian Computer: Stores the Most Significant Byte (MSB) at the lower address.

MSB $= B_0$

LSB $= B_1$

Value $_{BE} = (B_0 \times 256) + B_1$

2. Applying the Given Condition

The problem states that the numerical value on the little endian computer is 255 more than on the big endian computer.

$$\text{Value}_{LE} = \text{Value}_{BE} + 255$$

Substitute the equations from step 1:

$$(B_1 \times 256) + B_0 = (B_0 \times 256) + B_1 + 255$$

Now, solve for the relationship between B_1 and B_0 :

$$256B_1 - B_1 - 256B_0 + B_0 = 255$$
$$255B_1 - 255B_0 = 255$$
$$255(B_1 - B_0) = 255$$
$$B_1 - B_0 = 1$$

A. 0 × 6665

$B_1 = 66_{16}$ and $B_0 = 65_{16}$

$66_{16} - 65_{16} = 1$ (Correct)

B. 0 × 0001

$B_1 = 00_{16}$ and $B_0 = 01_{16}$

$00_{16} - 01_{16} = -1 \neq 1$ (Incorrect)

C. 0×4243

$B_1 = 42_{16}$ and $B_0 = 43_{16}$

$42_{16} - 43_{16} = -1 \neq 1$ (Incorrect)

D. 0×0100

$B_1 = 01_{16}$ and $B_0 = 00_{16}$

$01_{16} - 00_{16} = 1$ (Correct)

3. ANSWER: - (A)

- Total address lines = 16 → full range = 0000 to FFFF
- From chip select logic (implied), higher bits are fixed and lower bits vary
- So fixed upper bits → 1100 1xxx xxxx xxxx → corresponds to C800–CFFF

Hence, the correct range is C800 to CFFF.

4. ANSWER: (59)

Memory = 1 GB
Chip = 256M × 4-bit
Rows = 2^{14}
Refresh time per row = 50 ns
Refresh period = 2 ms
Number of rows to refresh:

$$2^{14} = 16384$$

Total refresh time:

$$16384 \times 50 \text{ ns} = 819200 \text{ ns} = 0.8192 \text{ ms}$$

Total time available: 2 ms
Time left for read/write:

$$2 - 0.8192 = 1.1808 \text{ ms}$$

Percentage available:

$$\frac{1.1808}{2} \times 100 \approx 59\%$$

Alternate Approach:

Refresh period = 2 msec
Percentage of time spent in refresh operation
= 0.8192 / 2 = 40.96%
Percentage of time available for read/write operation = 100 − 40.96
= 59.04%

5. ANSWER: (31)

Max memory = 4 GB = 2^{32}bytes
Word size = 2 bytes = 2^{1}bytes
Word-addressable ⇒ addresses count words, not bytes
Number of addressable units =
$2^{32}/2^{1} = 2^{31}$words
Address bits needed:

$$\log_2(2^{31}) = 31$$

6. ANSWER:- (C)

Total number of RAM chips
= Total size / 1 RAM size
= 256 K-bytes / (32k × 1)
Convert to bits:
256 K-bytes = $2^8 \times 2^{10} \times 2^3$ bits
32k × 1 = $2^5 \times 2^{10}$ bits
So, Number of chips
= $(2^8 \times 2^{10} \times 2^3) / (2^5 \times 2^{10})$
= 2^6
= 64

2 PIPELINING

GATE CSE - 2026 MCQ SET - 1(1-MARKS)

Q1. Which one of the following dependencies among the register operands of different instructions can cause a data hazard in a pipelined processor?

(a) Read-after-read
(b) Read-after-write
(c) Write-after-read
(d) Write-after-write

GATE CSE - 2026 NAT SET - 1(2-MARKS)

Q2. The EX stage of a pipelined processor performs the memory read operations for LOAD instructions, and the operations for the arithmetic and logic instructions. Let t_{EX} denote the time taken by the EX stage to perform the operation for an instruction. For each instruction type, the values of t_{EX} and M (the number of instructions of that type in a sequence of 100 instructions for a program P), are given in the table below. The duration of the pipeline clock cycle is 1 nanosecond. Assume that the latch time for the interstage buffers in the pipeline is negligible.

Instruction	t_{EX} (in ns)	M
LOAD	1.8	15
IMUL	1.5	10
IDIV	2.5	5
FADD	1.7	10
FSUB	1.7	5
FMUL	2.8	15
FDIV	3.2	5
All other instructions	Less than 1.0	35

When program P is executed, the number of clock cycles for which the pipeline is stalled due to structural hazards in the EX-stage is _____. (answer in integer).

GATE CSE - 2026 NAT SET - 2(2-MARKS)

Q3. A non-pipelined instruction execution unit that operates at 1.6 GHz clock takes an average of 5 clock cycles to complete the execution of an instruction. To improve the performance, the system was pipelined with a goal of achieving an average throughput of one instruction per clock cycle. However, it could operate only at 1.2 GHz due to pipeline overheads. While executing a program in the pipelined design, 30% of instructions encountered a stall of 2 cycles due to pipeline hazards. The speed-up obtained by the pipelined design over the non-pipelined one for this program is _____. (rounded off to two decimal places)

Note: $1G = 10^9$

GATE CSE - 2025 NAT SET - 2(2-MARKS)

Q4. A 5-stage instruction pipeline has stage delays of 180, 250, 150, 170, and 250, respectively, in nanoseconds. The delay of an inter-stage latch is 10 nanoseconds. Assume that there are no pipeline stalls due to branches and other hazards. The time taken to process 1000 instructions in microseconds is _____. (Rounded off to two decimal places)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2024 MSQ SET - 2(1-MARKS)

Q5. An instruction format has the following structure:
Instruction Number: Opcode destination reg, source reg-1, source reg-2
Consider the following sequence of instructions to be executed in a pipelined processor:
I1 : DIVR3,R1,R2
I2 : SUB R5,R3,R4
I3 : ADDR3,R5,R6
I4: MUL R7,R3,R8
Which of the following statements is/are TRUE?

(a) There is a RAW dependency on R3 between I1 and I2
(b) There is a WAR dependency on R3 between I1 and I3
(c) There is a RAW dependency on R3 between I2 and I3
(d) There is a WAW dependency on R3 between I3 and I4

GATE CSE - 2024 MSQ SET - 1(1-MARKS)

Q6. Consider a 5-stage pipelined processor with Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Register Writeback (WB) stages. Which of the following statements about forwarding is/are CORRECT?

(a) In a pipelined execution, forwarding means the result from a source stage of an earlier instruction is passed on to the destination stage of a later instruction.
(b) In forwarding, data from the output of the MEM stage can be passed on to the input of the EX-stage of the next instruction.
(c) Forwarding cannot prevent all pipeline stalls.
(d) Forwarding does not require any extra hardware to retrieve the data from the pipeline stages.

GATE CSE - 2024 NAT SET - 2(2-MARKS)

Q7. A non-pipelined instruction execution unit operating at 2 GHz takes an average of 6 cycles to execute an instruction of a program P. The unit is then redesigned to operate on a 5-stage pipeline at 2 GHz. Assume that the ideal throughput of the pipelined unit is 1 instruction per cycle. In the execution of program P, 20% instructions incur an average of 2 cycles stall due to data hazards and 20% instructions incur an average of 3 cycles stall due to control hazards. The speedup (rounded off to one decimal place) obtained by the pipelined design over the non-pipelined design is _____

GATE CSE - 2024 NAT SET - 1(2-MARKS)

Q8. The baseline execution time of a program on a 2GHz single core machine is 100 nanoseconds (ns). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is 10 ns when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time.
The number of cores that minimize the execution time of the program is _____.

GATE CSE - 2023 NAT SET - 1(1-MARKS)

Q9. Consider a 3-stage pipelined processor having a delay of 10 ns (nanoseconds), 20 ns, and 14 ns, for the first, second, and the third stages, respectively. Assume that there is no other delay and the processor does not suffer from any pipeline hazards. Also assume that one instruction is fetched every cycle.
The total execution time for executing 100 instructions on this processor is _____ns.

GATE CSE - 2022 NAT (2-MARKS)

Q10. A processor X_1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X_2 operating at the same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X_1 and X_2. If the BPU has a prediction accuracy of 80%, the speed up (rounded off to two decimal places) obtained by X_2 over X_1 in executing P is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2021 NAT SET - 2(2-MARKS)

Q11. Consider a pipelined processor with 5 stages, Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB). Each stage of the pipeline, except the EX-stage, takes one cycle. Assume that the ID stage merely decodes the instruction and the register read is performed in the EX-stage. The EX-stage takes one cycle for ADD instruction and two cycles for MUL instruction. Ignore pipeline register latencies. Consider the following sequence of 8 instructions: ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL

Assume that every MUL instruction is data-dependent on the ADD instruction just before it and every ADD instruction (except the first ADD) is data-dependent on the MUL instruction just before it. The speedup is defined as follows:

$$\text{Speedup} = \frac{\text{Execution time without operand forwarding}}{\text{Execution time with operand forwarding}}$$

The Speedup achieved in executing the given instruction sequence on the pipelined processor (rounded to 2 decimal places) is _____.

GATE CSE - 2021 NAT SET - 1(2-MARKS)

Q12. A five-stage pipeline has stage delays of 150, 120, 150, 160 and 140 nanoseconds. The registers that are used between the pipeline stages have a delay of 5 nanoseconds each.

The total time to execute 100 independent instructions on this pipeline, (assuming there are no pipeline stalls) is _____nanoseconds.

GATE CSE – 2020 NAT (2-MARKS)

Q13. Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5-stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. For this program, the speedup achieved by the pipelined processor over the non-pipelined processor (round off to 2 decimal places) is _____.

GATE CSE - 2018 NAT (2-MARKS)

Q14. The instruction pipeline of a *RISC* processor has the following stages: Instruction Fetch *(IF)*, Instruction Decode *(ID)*, Operand Fetch *(OF)*, Perform Operation *(PO)* and Writeback *(WB)*. The *IF*, *ID*, *OF* and *WB* stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the *PO* stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instructions is _____.

GATE CSE - 2017 NAT SET – 1 (2-MARKS)

Q15. Instruction execution in a processor is divided into 5 stages, Instruction Fetch（IF）, Instruction Decode（ID）, Operand fetch（OF）, Execute（EX）, and Write Back（WB）. These stages take 5, 4, 20, 10 and 3 nanoseconds（ns）respectively. A pipelined implementation of the processor requires buffering between each pair of consecutive stages with a delay of **2 ns**. Two pipelined implementation of the processor are contemplated：

i. A naive pipeline implementation（NP）with 5 stages and

ii. An efficient pipeline（EP）where the OF stage is divided into stages OF1 and OF2 with execution times of **12** ns and **8 ns** respectively.

The speedup（correct to two decimal places）achieved by EP over NP in executing 20 independent instructions with no hazards is ____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2016 NAT SET – 2 (2-MARKS)

Q16. Suppose the functions F and G can be computed in 5 and 3 nanoseconds by functional units U_F and U_G, respectively. Given two instances of U_F and two instances of U_G, it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is _____nanoseconds.

GATE CSE - 2016 NAT SET - 2(2-MARKS)

Q17. Consider a 3 GHz (gigahertz) processor with a three-stage pipeline and stage latencies τ_1, τ_2, and τ_3 such that $\tau_1 = 3\tau_2/4 = 2\tau_3$. If the longest pipeline stage is split into two pipeline stages of equal latency, the new frequency is _____ GHz, ignoring delays in the pipeline registers.

GATE CSE - 2016 NAT SET - 1(2-MARKS)

Q18. The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is percent.

GATE CSE – 2015 NAT SET - 3 (2-MARKS)

Q19. Consider the following reservation table for a pipeline having three stages S1, S2 and S3.

Time →

	1	2	3	4	5
S_1	X				X
S_2		X		X	
S_3			X		

The minimum average latency (MAL) IS

GATE CSE - 2015 NAT SET – 2 (2-MARKS)

Q20. Consider the sequence of machine instructions given below

MUL	R5, R0, R1
DIV	R6, R2, R3
ADD	R7, R5, R6
SUB	R8, R7, R4

In the above sequence, R0 to R8 are general purpose registers. In the instructions shown, the first register stores the result of the operation performed on the second and the third registers. This sequence of instructions is to be executed in a pipelined instruction processor with the following 4 stages:

(1) Instruction Fetch and Decode (IF)
(2) Operand Fetch (OF)
(3) Perform Operation (PO) and
(4) Write back the result (WB).

The IF, OF and WB stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD or SUB instruction, 3 clock cycles for MUL instruction and 5 clock cycles for DIV instruction. The pipelined processor uses operand forwarding from the PO stage to the OF stage. The number of clock cycles taken for the execution of the above sequence of instructions is _____.

GATE CSE - 2015 NAT SET-1 (2-MARKS)

Q21. Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. The speedup achieved in this pipelined processor is ___

,

GATE CSE – 2015 MCQ SET - 3 (2-MARKS)

Q22. Consider the following code sequence having five instructions from I_1 to I_5. Each of these instructions has the following format.

OP Ri, Rj, Rk

Where operation OP is performed on contents of registers Rj and Rk and the result is stored in register

Check Out Our GATE Courses & Secure Your Dream Rank!

Ri.
I_1 : ADD R1, R2, R3
I_2 : MUL R7, R1, R3
I_3 : SUB R4, R1, R5
I_4 : ADD R3, R2, R4
I_5: MUL R7, R8, R9

Consider the following three statements.
S1: There is an anti-dependence between instructions I_2 and I_5
S2: There is an anti-dependence between instructions I_2 and I_4
S3: Within an instruction pipeline an anti-dependence always creates one or more stalls
Which one of the above statements is/are correct?

(a) Only S1 is true
(b) Only S2 is true
(c) Only S1 and S3 are true
(d) Only S2 and S3 are true

GATE CSE - 2014 NAT SET 3 (2-MARKS)

Q23. Consider two processors P_1 and P_2 executing the same instruction set. Assume that under identical conditions, for the same input, a program running on P_2 takes 25% less time but incurs 20% more CPI (clock cycles per instruction) as compared to the program running on P_1. If the clock frequency of P_1 is 1 GHZ , then the clock frequency of P_2 (in GHz) is ____.

GATE CSE - 2014 NAT SET 3 (2-MARKS)

Q24. An instruction pipeline has five stages, namely, instruction fetch (IF), instruction decode and register fetch (ID/RF), instruction execution (EX), memory access (MEM), and register writeback (WB) with stage latencies 1 ns, 2.2 ns, 2 ns, 1 ns, and 0.75 ns, respectively (ns stands for nanoseconds). To gain in terms of performance, the designers have decided to split the ID/RF stage into three stages (ID, RF1, RF2) each of latency 2.2/3 ns. Also, the EX stage is split into two stages (EX1, EX2) each of latency 1 ns. The new design has a total of eight pipeline stages. A program has 20% branch instructions which execute in the EX stage and produce the next instruction pointer at the end of the EX stage in the old design and at the end of the EX2 stage in the new design. The IF stage stalls after fetching a branch until the next instruction pointer is computed. All instructions other than the branch instruction have an average CPI of one in both the designs. The execution times of this program on the old and the new design are P and Q nanoseconds, respectively. The value of P/Q is _____.

GATE CSE - 2014 MCQ SET - 3(1-MARKS)

Q25. Consider the following processors (ns stands for nanoseconds). Assume that the pipeline registers have zero latency.
P1: Four – stage pipeline with stage latencies 1ns, 2 ns, 2 ns, 1 ns.
P2: Four-stage pipeline with stage latencies 1ns, 1.5 ns, 1.5 ns, 1.5 ns.
P3: Five-stage pipeline with stage latencies 0.5 ns, 1 ns, 1 ns, 0.6 ns. 1 ns.
P4: Five-stage pipeline with stage latencies 0.5 ns, 0.5 ns, 1 ns, 1 ns, 1.1 ns.
Which processor has the highest peak clock frequency?

(a) P1
(b) P2
(c) P3
(d) P4

GATE CSE - 2014 NAT SET -1 (2-MARKS)

Q26. Consider a 6-stage instruction pipeline, where all stages are perfectly balanced. Assume that there is no cycle-time overhead of pipelining. When an application is executing on this 6-stage pipeline, the speedup achieved with respect to non-pipelined execution is 25% if the instructions incur 2 pipeline stall cycles is_____.

GATE CSE - 2013 MCQ (2-MARKS)

Q27. Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are

Check Out Our GATE Courses & Secure Your Dream Rank!

intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions I1, I2, I3, …. I12 is executed in this pipelined processor. Instruction I4 is the only branch instruction and its branch target is I9. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

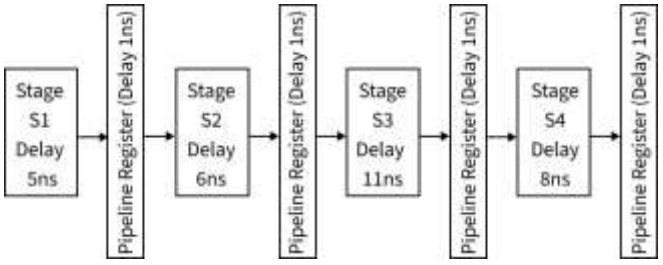
(a) 132
(b) 165
(c) 176
(d) 328

GATE CSE - 2012 MCQ (1-MARKS)

Q28. Register renaming is done in pipelined processors
(a) as an alternative to register allocation at compile time
(b) for efficient access to function parameters and local variables
(c) to handle certain kinds of hazards
(d) as part of address translation

GATE CSE - 2011 MCQ (2-MARKS)

Q29. Consider an instruction pipeline with four stages (S1, S2, S3 and S4) each with a combinational circuit only. The pipeline registers are required between each stage and at the end of the last stage. Delays for the stages and for the pipeline registers are as given in the figure:



What is the approximate speed up of the pipeline in steady state under ideal conditions when compared to the corresponding non-pipeline implementation?

(a) 4.0
(b) 2.5
(c) 1.1
(d) 3.0

GATE CSE - 2010 MCQ (2-MARKS)

Q30. A 5-stage pipelined processor has Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Write Operand (WO) stages. The IF, ID, OF and WO stage take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instructions ,3 clock cycles for MUL instruction, and 6 clock cycles for DIV instruction respectively. Operand forwarding is used in the pipeline. What is the number of clock cycles needed to execute the following sequence of instructions?

Instruction	Meaning of instruction
I_0 : MUL R_2, R_0, R_1	$R_2 \leftarrow R_0 * R_1$
I_1 : DIV R_5, R_3, R_4	$R_5 \leftarrow R_3 / R_4$
I_2 : ADD R_2, R_5, R_2	$R_2 \leftarrow R_5 + R_2$
I_3 : SUB R_5, R_2, R_6	$R_5 \leftarrow R_2 - R_6$

(a) 13
(b) 15
(c) 17
(d) 19

GATE CSE - 2009 MCQ (2-MARKS)

Q31. Consider a 4 stage pipeline processor. The number of cycles needed by the four instructions I1, I2, I3, I4 in stages S1, S2, S3, S4 is shown below.

	S1	S2	S3	S4
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

What is the number of cycles needed to execute the following loop?
For (i = 1 to 2) {I1; I2; I3; I4;}

(a) 16
(b) 23
(c) 28
(d) 30

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2008 MCQ (2-MARKS)

Q32. Delayed branching can help in the handling of control hazards

The following code is to run on a pipelined processor with one branch delay slot:

I_1: ADD $R_2 \leftarrow R_7 + R_8$
I_2: SUB $R_4 \leftarrow R_5 - R_6$
I_3: ADD $R_1 \leftarrow R_2 + R_3$
I_4: STORE Memory $[R_4] \leftarrow R_1$
BRANCH to Label if $R_1 == 0$

Which of the instructions I_1, I_2, I_3 or I_4 can legitimately occupy the delay slot without any other program modification?

(a) I_1
(b) I_2
(c) I_3
(d) I_4

GATE CSE – 2008 MCQ (2-MARKS)

Q33. Which of the following are NOT true in a pipelined processor?

1. Bypassing can handle all RAW hazards.
2. Register renaming can eliminate all register carried WAR hazards.
3. Control hazard penalties can be eliminated by dynamic branch prediction.

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

GATE CSE – 2008 MCQ (2-MARKS)

Q34. Delayed branching can help in the handling of control hazards

For all delayed conditional branch instructions, irrespective of whether the condition evaluates to true or false.

(a) The instruction following the conditional branch instruction in memory is executed
(b). The first instruction in the fall through path is executed
(c). The first instruction in the taken path is executed
(d). The branch takes longer to execute than any other instruction

GATE CSE – 2007 MCQ (2-MARKS)

Q35. A non-pipelined single cycle processor operating at 100MHz is converted into a synchronous pipelined processor with five stages requiring 2.5nsec,1.5nsec,2nsec,1.5nsec and 2.5nsec, respectively. The delay of the latches is 0.5nsec. The speedup of the pipeline processor for a large number of instructions is:

(a) 4.5
(b) 4.0
(c) 3.33
(d) 3.0

GATE CSE – 2007 MCQ (2-MARKS)

Q36. Consider a pipelined processor with the following four stages:

IF: Instruction Fetch
ID: Instruction Decode and Operand Fetch
EX: Execute
WB: Write Back

The IF, ID, and WB stages take one clock cycle each to complete the operation.

The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle, and the MUL instruction needs 3 clock cycles in the EX stage.

Operand forwarding is used in the pipelined processor.

What is the number of clock cycles taken to complete the following sequence of instructions?

ADD	R2, R1, R0	R2 ← R1+R0
MUL	R4, R3, R2	R4 ← R3*R2
SUB	R6, R5, R4	R6 ← R5−R4

(a) 7
(b) 8
(c) 10
(d) 14

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2007 MCQ (2-MARKS)

Q37. A processor takes 12 cycles to complete an instruction 1. The corresponding pipelined processor uses 6 stages with the execution times of 3,2,5,4,6 and 2 cycles respectively. What is the asymptotic speedup assuming that a very large number of instructions are to be executed?

(a) 1.83
(b) 2
(c) 3
(d) 6

GATE CSE - 2006 MCQ (1-MARKS)

Q38. A CPU has a five-stage pipeline and runs at a 1 GHz frequency. Instruction fetch happens in the first stage of the pipeline. A conditional branch instruction computes the target address and evaluates the condition in the third stage of the pipeline.
The processor stops fetching new instructions following a conditional branch until the branch outcome is known.
A program executes 10^9 instructions out of which 20% are conditional branches.
If each instruction takes one cycle to complete on average, then the total execution time of the program is:

(a) 1.0 second
(b) 1.2 Seconds
(c) 1.4 Seconds
(d) 1.6 Seconds

GATE CSE - 2006 MCQ (2-MARKS)

Q39. A pipelined processor uses a 4-stage instruction pipeline with the following stages: Instruction fetch (IF), Instruction decode (ID), Execute (EX) and Writeback (WB). The arithmetic operations as well as the load and store operations are carried out in the EX stage. The sequence of instructions corresponding to the statement $X=(S-R*(P+Q))/T$ is given below. The
values of variables P,Q,R,S and T are available in the registers $R0,R1,R2,R3$ and $R4$ respectively, before the execution of the instruction sequence.
ADD $R5,R0,R1$;$R5$←R0 + R1
MUL $R6,R2,R5$;$R6$←R2 * R5
SUB $R5,R3,R6$;$R5$←R3 -R6
DIV $R6,R5,R4$;$R6$←R5/R4
STORE $R6,X$;X←R6
The IF, ID and WB stages take 1 clock cycle each. The EX stage takes 1 clock cycle each for the ADD, SUB and STORE operations, and 3 clock cycles each for MUL
and DIV operations. Operand forwarding from the EX
stage to the ID stage is used. The number of Read-After-Write (RAW) dependencies, Write-After-Read(WAR) dependencies, and Write After-Write (WAW) dependencies in the sequence of instructions are, respectively,

(a) 2,2,4
(b) 3,2,3
(c) 4,2,2
(d) 3,3,2

GATE CSE - 2006 MCQ (2-MARKS)

Q40. A pipelined processor uses a 4-stage instruction pipeline with the following stages: Instruction fetch (IF), Instruction decode (ID), Execute (EX) and Writeback (WB). The arithmetic operations as well as the load and store operations are carried out in the EX stage. The sequence of instructions corresponding to the statement $X=(S-R*(P+Q))/T$ is given below. The values of variables P,Q,R,S and T are available in the registers $R0,R1,R2,R3$ and $R4$ respectively, before the execution of the instruction sequence.
ADD $R5,R0,R1$;$R5$←R0 + R1
MUL $R6,R2,R5$;$R6$←R2 * R5
SUB $R5,R3,R6$;$R5$←R3 -R6
DIV $R6,R5,R4$;$R6$←R5/R4
STORE $R6,X$;X←R6
The IF, ID and WB stages take 1 clock cycle each. The EX stage takes 1 clock cycle each for the ADD, SUB and STORE operations, and 3 clock cycles each for MUL
and DIV operations. Operand forwarding from the EX
stage to the ID stage is used. The number of clock cycles
required to complete the sequence of instructions i

(a) 10
(b) 12
(c) 14
(d) 16

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2005 MCQ (2-MARKS)

Q41. A 5 stage pipelined CPU has the following sequence of stages:
IF – Instruction fetch from instruction memory
RD – Instruction decode and register read
EX – Execute: ALU operation for data and address computation
MA – Data memory access - for write access, the register read at RD state is used.
WB – Register write back
Consider the following sequence of instructions:
I_1: L R0, Loc1; R0 ← M[Loc1]
I_2: A R0, R0; R0 ← R0 + R0
I_3: A R2, R0; R2 ← R2 − R0
Let each stage takes one clock cycle.
What is the number of clock cycles taken to complete the above sequence of instructions starting from the fetch of I_1?

(a) 8
(b) 10
(c) 12
(d) 15

GATE CSE - 2005 MCQ (2-MARKS)

Q42. We have two designs D1 and D2 for a synchronous pipeline processor. D1 has 5 pipeline stages with execution times of 3 nsec, 2 nsec, 4 nsec, 2 nsec and 3 nsec while the design D2 has 8 pipeline stages each with 2 nsec execution time. How much time can be saved using design D2 over design D1 for executing 100 instructions?

(a) 214 nsec
(b) 202 nsec
(c) 86 nsec
(d) -200 nsec

ANSWER KEY

1. (b)	**2. (95)**	**3. (2.34)**
4. (261.04)	**5.(a)**	**6.(a,b,c)**
7. (2.9 to 3.1)	**8.(3)**	**9. (2040)**
10. (1.42 to 1.43)	**11. (1.87-1.88)**	**12. (17160)**
13. (2.16)	**14. (219)**	**15(1.50 – 1.51)**
16. (28)	**17.(3.9 - 4.1)**	**18. (33-34)**
19.(3)	**20.(13)**	**21. (3.2)**
22.(b)	**23.(1.6)**	**24. (1.54)**
25. (c)	**26.(4)**	**27.(b)**
28.(c)	**29. (b)**	**30.(b)**
31. (b)	**32. (d)**	**33. (b)**
34.(a)	**35. (c)**	**36. (b)**
37.(b)	**38.(c)**	**39.(c)**
40. (b)	**41. (a)**	**42.(b)**

Check Out Our GATE Courses & Secure Your Dream Rank!

HINTS & SOLUTIONS

1. **ANSWER: - (B)**
RR , WR and WW are name dependency or false dependency.
They never create stalls or hazards in the pipeline however result can be messed up if order of execution is changed. RW can create hazard in the pipeline

2. **ANSWER: - (95)**

Pipeline clock cycle = 1 ns When EX-stage takes more than 1 ns then it takes more than 1 pipeline cycle.

Instruction	Execution time	Number of cycle required
LOAD	1.8 ns	2 ⇒ stalls (2 − 1) = 1
IMUL	1.5 ns	2 ⇒ (2 − 1) = 1
IDIV	2.5 ns	3 ⇒ (3 − 1) = 2
FADD	1.7 ns	2 ⇒ (2 − 1) = 1
FSUB	1.7 ns	2 ⇒ (2 − 1) = 1
FMUL	2.8 ns	3 ⇒ (3 − 1) = 2
FDIV	3.2 ns	4 ⇒ (4 − 1) = 3
Other instruction	Less than 1 ns	1 ⇒ (1 − 1) = 0

∴ Number of stalls required in EX-stage = (15 × 1) + (10 × 1) + (5 × 2) + (10 × 1)

(5 × 1) + (15 × 2) + (5 × 3) + (35 × 0)

= 95

3. **ANSWER: - (2.34)**
Non pipeline, ET = CPI × Cycle time
= 5 × 1/1.6 ns = 3.125 ns
Pipeline ET = (1 + Number of stalls / Instructions) × tp
= [1 + (0.3 × 2)] × 1/1.2 ns = 1.333 ns
S = ET non-pipe / ET pipe = 3.125 / 1.33 = 2.34

4. **ANSWER: - (261.04)**
First, determine the clock cycle time of the pipeline. In an ideal pipeline, the clock cycle time is given by the delay of the slowest stage plus the inter-stage latch delay.
Stage delays:
Stage 1 = 180 ns
Stage 2 = 250 ns
Stage 3 = 150 ns
Stage 4 = 170 ns
Stage 5 = 250 ns
Maximum stage delay = 250 ns
Including the latch delay (10 ns):
Clock cycle time (t_cycle) = 250 ns + 10 ns = 260 ns
Next, consider the execution of 1000 instructions. For a pipeline with 5 stages, the first instruction requires 5 cycles to complete, and each subsequent instruction completes in one additional cycle.
Total number of cycles = n + (k − 1)
= 1000 + (5 − 1)
= 1000 + 4
= 1004 cycles
Now, calculate the total execution time:
Total time = 1004 × 260 ns = 261040 ns
Convert nanoseconds to microseconds:
261040 ns = 261040 / 1000 = 261.04 μs
Therefore, the total time required to process 1000 instructions. is 261.04 microseconds.

5. **ANSWER: - (A)**
A RAW dependency exists on R_3 between I_1 and I_2.
A RAW dependency exists on R_5 between I_2 and I_3.
No RAW dependency exists on R_3 between I_2 and I_3.
A WAW dependency exists on R_3 between I_1 and I_3.
Hence, the correct answer is: A RAW dependency exists on R_3 between I_1 and I_2.

Check Out Our GATE Courses & Secure Your Dream Rank!

6. ANSWER: - (A, B, C)

Analysis of each statement about forwarding in a 5-stage pipelined processor (IF, ID, EX, MEM, WB):

1. Statement A:
   In pipelined execution, forwarding means the result from a source stage of an earlier instruction is passed to the destination stage of a later instruction.
   Correct. Forwarding (also called bypassing) sends the result directly to a dependent instruction without waiting for register write back, reducing data hazards.

2. Statement B:
   In forwarding, data from the output of the MEM stage can be passed to the input of the EX stage of the next instruction.
   Correct. A common forwarding path sends the result available at the MEM stage (e.g., from a load instruction) directly to the EX stage of the following instruction.

3. Statement C:
   Forwarding cannot prevent all pipeline stalls.
   Correct. Forwarding reduces some data hazards (like RAW hazards) but cannot eliminate all stalls, such as those caused by control hazards or certain structural hazards.

4. Statement D:
   Forwarding does not require extra hardware to retrieve data from pipeline stages.
   Incorrect. Forwarding requires additional hardware such as multiplexers and control logic to route data between pipeline stages.

Therefore, the correct statements are: A, B, and C

7. ANSWER: - (2.9 to 3.1)

Non-Pipelined Execution Time

Frequency (f) = 2 GHz
Cycle Time (T) = 1 / 2 GHz = 0.5 nsec
Cycles per Instruction (CPI) = 6 cycles
Total Execution Time (ET_Nonpipe)
= CPI * Cycle Time = 6 * 0.5 nsec = 3 nsec

Pipelined Execution Time with Stalls

Cycle time remains 0.5 nsec.
Instruction breakdown based on stalls:
20% of instructions cause 2 stalls (Data Hazards)
20% of instructions cause 3 stalls (Control Hazards)
60% of instructions execute normally with 0 stalls
Average Stalls per Instruction
= (0.20 * 2) + (0.20 * 3) + (0.60 * 0)
Average Stalls = 0.4 + 0.6 = 1 stall per instruction
Average Execution Time (ET_Pipe) =
(1 + Average Stalls) * Cycle Time
ET_Pipe = (1 + 1) * 0.5 nsec = 2 * 0.5 nsec
= 1 nsec

Speedup Calculation

Speedup (S) = ET_Nonpipe / ET_Pipe
S = 3 nsec / 1 nsec = 3

8. ANSWER: - (3)

The portion of code that cannot be parallelized takes 10 ns.
For each additional core, an overhead of 10 ns is added.
Thus, overhead for n cores = $(n - 1) \times 10$ ns.
The parallelizable portion takes 90 ns on one core.
On n cores, it takes 90 / n ns.
Let total execution time be t(n):
$t(n) = 10 + (n - 1) \times 10 + 90 / n$
$= 10n + 90 / n$
Now evaluating:
t(1) = 10(1) + 90/1 = 100 ns
t(2) = 20 + 45 = 65 ns
t(3) = 30 + 30 = 60 ns
t(4) = 40 + 22.5 = 62.5 ns
t(5) = 50 + 18 = 68 ns
Minimum execution time occurs at n = 3.
Answer: 3 cores

Alternate Solution:

Single Core Setup

Total execution time on a single core = 100 nsec
Parallelizable portion (90%) = 90 nsec
Non-parallelizable portion (10%) = 10 nsec

Adding Cores and Overhead

Let the total number of cores = n
Number of newly added cores = (n - 1)
Overhead time per new core = 10 nsec
Total overhead time for new cores = (n - 1) * 10 nsec

Execution Time (ET) Equation

ET = (Parallelizable Time / n) + Total Overhead + Non-parallelizable Time
ET = [90 / n + (n - 1) * 10] + 10
ET = 90 / n + 10n - 10 + 10
ET = 90 / n + 10n
Finding the Optimal Number of Cores (Minimizing ET)

To find the minimum execution time (Maxima/Minima), take the derivative of ET with respect to n and set it to 0:

Check Out Our GATE Courses & Secure Your Dream Rank!

d(ET)/dn = 0
d(ET)/dn = d/dn [90 / n + 10n]
d(ET)/dn = $-90 / n^2 + 10$
Set the derivative to 0:
$-90 / n^2 + 10 = 0$
$10 = 90 / n^2$
$10n^2 = 90$
$n^2 = 9$
n = 3

9. ANSWER: - (2040)

To determine the total execution time for executing 100 instructions on a 3-stage pipelined processor with stage delays of 10 ns, 20 ns, and 14 ns (assuming no pipeline hazards or additional delays), we use the pipeline execution time formula.
First, identify the pipeline cycle time (Tp), which is equal to the delay of the slowest stage.
Maximum of (10 ns, 20 ns, 14 ns) = 20 ns.
Tp = 20 ns
Number of instructions: n = 100
Pipeline depth (number of stages): k = 3
The total execution time for a pipelined processor is given by:
$T = (k + n - 1) \times Tp$
Substituting the values:
$T = (3 + 100 - 1) \times 20$
$T = (3 + 99) \times 20$
$T = 102 \times 20$
T = 2040 ns
Therefore, the total execution time to execute 100 instructions is 2040 ns

10. ANSWER: - (1.42 to 1.43)

Execution Time = No. of Instructions × Clocks per Instruction × Clock cycle time
Speedup = Old system execution time / New system execution time
When no. of Instructions and clock cycle time are same, then
Speedup = Old system CPI / New system CPI
Normal processor CPI = 1, without any pipeline hazards.
Given that, Program P has 30% branch instructions where each instruction will lead to 2 stall cycles.
Processor X1 has NO BPU, therefore
$CPI = 1 + (0.30 \times 2)$
CPI = 1 + 0.60
CPI = 1.60
Processor X2 has BPU (80% correct prediction, 20% wrong prediction).
$CPI = 1 + [0.30 \times ((0.80 \times 0) + (0.20 \times 2))]$
CPI = 1 + [0.30 × (0 + 0.40)]
CPI = 1 + (0.30 × 0.40)
CPI = 1 + 0.12
CPI = 1.12
Speedup = 1.60 / 1.12
Speedup = 1.4285

Alternate Solution:

X1: Processor without Branch Prediction Unit
Clock Frequency = 2 GHz
Cycle Time = 1 / 2 GHz = 0.5 nsec
Instruction Breakdown for X1 (Stalls):
Branch Instructions (30%): 2 Stalls
Non-Branch Instructions (70%): 0 Stalls
Average Stalls per Instruction = (0.30 * 2) + (0.70 * 0) = 0.6
Execution Time for X1 (ET_X1):
Average ET_X1 = (1 + Average Stalls) * Cycle Time
Average ET_X1 = (1 + 0.6) * 0.5 nsec
Average ET_X1 = 1.6 * 0.5 nsec = 0.8 nsec

X2: Processor with Branch Prediction Unit (BPU)
Cycle Time = 0.5 nsec
Instruction Breakdown for X2 (Stalls with BPU):
Non-Branch Instructions (70%): 0 Stalls
Branch Instructions (30%), which are handled by the BPU:
Correct Prediction (80% of branches): 0 Stalls
Incorrect Prediction (20% of branches): 2 Stalls
Average Stalls per Instruction = (0.70 * 0) + (0.30 * 0.80 * 0) + (0.30 * 0.20 * 2)
Average Stalls per Instruction = 0 + 0 + 0.12 = 0.12
Execution Time for X2 (ET_X2):
Average ET_X2 = (1 + Average Stalls) * Cycle Time
Average ET_X2 = (1 + 0.12) * 0.5 nsec
Average ET_X2 = 1.12 * 0.5 nsec = 0.56 nsec
Speedup Calculation (S):
Speedup (S) = ET_X1 / ET_X2
S = 0.8 / 0.56
S = 1.42

11. ANSWER: - (1.87 to 1.88)

Base Pipeline Execution Time
Instruction Sequence: ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL
EX Stage Cycles per instruction: ADD = 1, MUL = 2, ADD = 1, MUL = 2, ADD = 1, MUL = 2, ADD = 1, MUL = 2
Number of Pipeline Stages (k) = 5 Stages
Number of Instructions (n) = 8 Instructions
Ideal Execution Time (ET_PIPE) = [k + (n - 1)] cycles

Check Out Our GATE Courses & Secure Your Dream Rank!

Ideal ET_PIPE = [5 + (8 - 1)] = 12 cycles
Execution Time WITH Operand Forwarding
Each ADD instruction takes 1 cycle in the EX stage.
Each MUL instruction takes 2 cycles in the EX stage.
Out of 8 instructions, 4 are MUL instructions.
These 4 MUL instructions take 1 extra cycle each in the EX stage.
Extra cycles = 4 instructions * 1 extra cycle = 4 cycles
Total ET (With Forwarding) = 12 (Base) + 4 (Extra EX cycles) = 16 cycles
Execution Time WITHOUT Operand Forwarding
Every instruction depends on the result of the previous instruction (Data dependency: ADD -> MUL -> ADD -> MUL ...).
Without forwarding, each data dependency introduces a 2-cycle stall penalty.
Out of 8 instructions, 7 instructions depend on the previous one, thus incurring the 2-cycle stall penalty.
Total Stalls = 7 instructions * 2 stall cycles = 14 cycles
Total ET (Without Forwarding) = Total ET (With Forwarding) + Stall Cycles
Total ET = 16 + 14 = 30 cycles
Speedup Calculation
Speedup = Total ET (Without Forwarding) / Total ET (With Forwarding)
Speedup = 30 / 16 = 1.875

Alternate Solution:

Speedup (defined in question) = $\frac{Time\ without\ Operand\ Forwarding}{Time\ with\ Operand\ Formwarding}$

Without Operand Forwarding:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
ADD	IF	ID	EX	MEM	WB																							
MUL		IF	ID			EX	EX	MEM	WB																			
ADD			IF			ID				EX	MEM	WB																
MUL						IF				ID	EX	EX	EX	EX	MEM	WB												
ADD										IF			ID				EX	MEM	WB									
MUL													IF				ID	EX	EX	EX	EX							
ADD																	IF			ID		EX	MEM	WB				
MUL																				IF		ID	EX	EX	MEM	WB		

Time Taken Without Operand Forwarding = 30

With Operand Forwarding:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
ADD	IF	ID	EX	MEM	WB											
MUL		IF	ID	EX	EX	MEM	WB									
ADD			IF	ID		EX	MEM	WB								
MUL				IF		ID	EX	EX	MEM	WB						
ADD						IF	ID		EX	MEM	WB					
MUL							IF		ID	EX	EX	MEM	WB			
ADD									IF	ID		EX	MEM	WB		
MUL										IF		ID	EX	EX	MEM	WB

Time Taken With Operand Forwarding = 16

Speedup = $\frac{Time\ without\ Operand\ Forwarding}{Time\ with\ Operand\ Forwarding} = \frac{30}{16} = 1.875$

12. ANSWER: - (17160)
For the given pipelined system:
• Total number of stages (k) = 5
• Total number of instructions (n) = 100
• Total delay (t_p) = max(stage delay) + buffer delay
⇒ t_p = max(150, 120, 150, 160, 140) + 5 ns
⇒ t_p = 160 + 5 ns
⇒ t_p = 165 ns
Total execution time of pipeline:
$ET_p = (k + n - 1) \times t_p$
⇒ $ET_p = (5 + 100 - 1) \times 165$ ns
⇒ $ET_p = (5 + 99) \times 165$ ns
⇒ $ET_p = 104 \times 165$ ns
⇒ $ET_p = 17160$ ns
∴ Time required to execute 100 instructions in the given pipeline = 17160 ns.

13. ANSWER: - (2.16)
Given:
Non-pipelined processor:
Frequency = 2.5 GHz

CPI = 5

Pipelined processor:
Number of stages = 5

Frequency = 2 GHz

Instruction mix:
30% memory instructions

60% ALU instructions

10% branch instructions

Additional details:
5% of memory instructions cause 50 cycle stalls

Check Out Our GATE Courses & Secure Your Dream Rank!

50% of branch instructions cause 2 cycle stalls

Non-Pipelined Processor
Clock cycle time = 1 / 2.5 GHz = 0.4 ns
Execution time per instruction = CPI × clock cycle time
= 5 × 0.4 ns= 2 ns
Pipelined Processor
Base CPI = 1
Memory stall contribution: = 0.30 × 0.05 × 50 = 0.75
Branch stall contribution: = 0.10 × 0.50 × 2 = 0.10
Total CPI (pipelined) = 1 + 0.75 + 0.10 = 1.85
Clock cycle time = 1 / 2 GHz = 0.5 ns
Execution time per instruction = 1.85 × 0.5 ns= 0.925 ns
Speedup = Non-pipelined time / Pipelined time
= 2 / 0.925 = 2.16

14. ANSWER: - (219)
Total number of instructions = 100
Number of pipeline stages = 5
For an ideal pipeline (with CPI = 1 and no stalls):
Total cycles = n + (k − 1) = 100 + (5 − 1)= 104 cycles
Now consider the PO (Perform Operation) stage:
40 instructions take 3 cycles → each causes 2 extra stall cycles

35 instructions take 2 cycles → each causes 1 extra stall cycle

25 instructions take 1 cycle → no stall cycles

All other stages (IF, ID, OF, WB) take 1 cycle each and operate ideally.
Extra cycles due to the multi-cycle PO stage:
= (40 × 2) + (35 × 1) + (25 × 0)
= 80 + 35 + 0 = 115 cycles
Therefore,
Total clock cycles
= 104 + 115 = 219 cycles

15. ANSWER: (1.50 to 1.51)
NP (5-stage):
Stage times = 5, 4, 20, 10, 3 ns
Buffer delay = 2 ns
Clock cycle = max stage + buffer = $20 + 2 =$ 22ns
Time for 20 instructions =
$(k + n - 1) \times T = (5 + 20 - 1) \times 22 = 24 \times 22 = 528$ns
EP (6-stage):
Stages = 5, 4, 12, 8, 10, 3
Max = 12 ns
Clock = $12 + 2 = 14$ns
Time = $(6 + 20 - 1) \times 14 = 25 \times 14 = 350$ns
Speedup:

$$\frac{528}{350} \approx 1.508$$

Answer:
1.508

16. ANSWER: - (28)
In this problem, the computation follows a pipelined execution of F(G(Xi)). Function G takes 3 ns and function F takes 5 ns. Since two units of each (UG and UF) are available, the slower stage F becomes the bottleneck.
We need to perform 10 computations. With two units of UF, the total time required for executing F on 10 inputs is 10 × 5 = 50 ns. As two units work in parallel, the effective time becomes 50/2 = 25 ns.
Initially, F must wait for the first output from G, which takes 3 ns. After this, pipelining ensures continuous execution without additional delay.
Therefore, the minimum total time required is:
3 + 25 = 28 ns.

17. ANSWER: - (3.9 to 4.1)

Given: 3-stage pipeline with processor frequency = 3 GHz
Given relation:
$e_1 = (3e_2)/4 = 2e_3$
Assume,
$e_1 = 6x$
Then,
$e_2 = 8x$ and $e_3 = 3x$
Largest stage time = 8x
Processor frequency = 1 / (largest stage time)
⇒ 1 / 8x = 3 GHz
⇒ 1 / x = 24 GHz …… (1)
Now, divide stage e_2 into two equal stages of 4x and 4x.
New pipeline stages become:
6x, 4x, 4x, 3x (4 stages)
Largest stage time = 6x
New processor frequency = 1 / 6x
= (1/x) / 6
= 24 / 6 = 4 GHz [using (1)]

Check Out Our GATE Courses & Secure Your Dream Rank!

Alternate Solution:

Finding the Stage Time Ratios

Given equations: T1 = (3 * T2) / 4 = 2 * T3

From T1 = (3 * T2) / 4, we get the ratio: T1 / T2 = 3 / 4 (which is equivalent to 6 / 8)

From (3 * T2) / 4 = 2 * T3, we get the ratio: T2 / T3 = 8 / 3

Combined ratio for the stages: T1 : T2 : T3 = 6 : 8 : 3

Initial Pipeline Cycle Time

Let 'x' be a constant unit of time.

The stage times are: T1 = 6x, T2 = 8x, T3 = 3x

The pipeline cycle time (tp) is the maximum of the stage times.

tp = max(6x, 8x, 3x)

tp = 8x

Solving for Base Frequency

Frequency = 1 / Cycle Time

The given initial frequency is 3 GHz.

3 GHz = 1 / 8x

Solving for 1/x: 1 / x = 3 GHz * 8 = 24 GHz

New Pipeline Design (Splitting the Bottleneck)

The longest stage (8x) is split into two equal sub-stages.

8x becomes -> 4x and 4x

The new sequence of stage times is: 6x, 4x, 4x, 3x

New pipeline cycle time (tp_new) = max(6x, 4x, 4x, 3x)

tp_new = 6x

Calculating the New Frequency

New Frequency (Freq_new) = 1 / tp_new

Freq_new = 1 / 6x

We can rewrite this as: Freq_new = (1 / 6) * (1 / x)

Substitute the value of (1 / x) we found earlier (24 GHz):

Freq_new = (1 / 6) * 24 GHz

New Frequency = 4 GHz

18. ANSWER: - (33 to 34)

In an ideal pipeline, CPI = 1

So, 1 instruction is completed in 1 cycle.

Throughput = number of instructions completed per unit time.

For Pipeline 1:

Cycle time = maximum stage delay = 800 psec

In 800 psec, 1 instruction is completed

Therefore, throughput of Pipeline 1 = 1/800 instructions per psec

For Pipeline 2:

Cycle time = 600 psec

Throughput of Pipeline 2 = 1/600 instructions per psec

Percentage increase in throughput:

= (New − Old / Old × 100

= (1/600 − 1/800) / (1/800) × 100

= (800 − 600) / 600 × 100

= 200/600 × 100

= 33.33%

19. ANSWER: - (3)

S1 is needed at times 1 and 5, so its forbidden latency is 5 − 1 = 4.

S2 is needed at times 2 and 4, so its forbidden latency is 4 − 2 = 2.

So, forbidden latency = (2, 4, 0) (0 by default is forbidden)

Allowed latency = (1, 3, 5) (any value more than 5 also).

Collision vector (4, 3, 2, 1, 0) = 10101 which is the initial state as well.

From the initial state we can have a transition after "1" or "3" cycles and we reach new states with collision vectors (10101 >> 1 + 10101 = 11111) and (10101 >> 3 + 10101 = 10111) respectively.

These 2 become states 2 and 3 respectively. For "5" cycles we come back to state 1 itself.

From state 2 (11111), the new collision vector is 11111. We can have a transition only when we see the first 0 from the right. So, here it happens on the 5th cycle only which goes to the initial state. (Any transition after 5 or more cycles goes to the initial state as we have 5 time slices).

From state 3 (10111), the new collision vector is 10111. So, we can have a transition on 3, which will give (10111 >> 3 + 10101 = 10111) the third state itself. For 5, we get the initial state. Thus all the transitions are complete.

State/Time	1	3	5
1(10101)	2	3	1
2(11111)	-	-	1
3(10111)	-	3	1

So, the minimum length cycle is of length 3 either from 3-3 or from 1-3,3-1.

Check Out Our GATE Courses & Secure Your Dream Rank!

20. ANSWER: - (13)

	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}	t_{12}	t_{13}	t_{14}	t_{15}
I1	IF	OF	PO	PO	PO	WB									
I2		IF	OF	-	-	PO	PO	PO	PO	PO	WB				
I3			IF	-	-	-	-	-	-	-	OF	PO	WB		
I4				-	-	-	-	-	-	-	IF	-	OF	PO	WB

It is clearly mentioned in the question that operand forwarding occurs from the PO stage to the OF stage (PO → OF) and not within the PO stage itself. Based on this, a straightforward pipeline analysis gives a total of 15 clock cycles.
However, since forwarding happens from PO to OF, the PO stage can produce its output at the rising edge of the clock, and the OF stage can latch (fetch) this forwarded value at the falling edge of the same clock cycle. Hence, the final PO stage and the dependent OF stage effectively overlap within a single clock cycle.
Due to this half-cycle forwarding mechanism, two clock cycles are saved in the overall execution. Therefore, the correct total number of clock cycles is 13.

	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}	t_{12}	t_{13}
I1	IF	OF	PO	PO	PO	WB							
I2		IF	OF	-	-	PO	PO	PO	PO	PO	WB		
I3			IF	-	-	-	-	-	-	OF	PO	WB	
I4				-	-	-	-	-	-	-	OF	PO	WB

Alternate Approach

Instruction	Operation	PO	Stalls (Extra Cycles)
I_1	MUL	3	2
I_2	DIV	5	4
I_3	ADD	1	--
I_4	SUB	1	--

Total: 6 extra cycles (stalls)
Given Variables:

- n = 4
- k = 4

Execution Time (ET) Calculation:
ET = [k + (n-1)]t_p + t_{stalls}
ET = [4 + (4-1)] *1 + 6
ET = 7 + 6
Ans: 13 cycles

21. ANSWER: 3.2
Clock = 2.5 GHz
CPI = 4
Time per instruction = $\frac{4}{2.5}$
Pipelined:
Clock = 2 GHz
CPI ≈ 1 (no stalls)
Time per instruction = $\frac{1}{2}$
Speedup:

$$\frac{(4/2.5)}{(1/2)} = \frac{4 \times 2}{2.5} = \frac{8}{2.5} = 3.2$$

Answer:
3.2

22. ANSWER: (B)
Instructions:
I1: R1 ← R2, R3
I2: R7 ← R1, R3
I3: R4 ← R1, R5
I4: R3 ← R2, R4
I5: R7 ← R8, R9
S1 (I2 and I5):
Both write to R7 ⇒ output dependence (WAW), not anti
S2 (I2 and I4):
I2 reads R3, I4 writes R3 ⇒ anti-dependence (WAR)
S3:
Anti-dependence does not always cause stalls (can be removed via register renaming)
Answer:
Only S2 is true (Option B)

23. ANSWER: (1.6)
Execution time $T = \frac{IC \times CPI}{f}$
Same program ⇒ IC same
P2 takes 25% less time ⇒

$$T_2 = 0.75T_1$$

CPI increases by 20% ⇒

$$CPI_2 = 1.2\ CPI_1$$

Ratio:

$$\frac{T_2}{T_1} = \frac{\text{CPI}_2/f_2}{\text{CPI}_1/f_1} = \frac{1.2 \times f_1}{f_2}$$

Check Out Our GATE Courses & Secure Your Dream Rank!

$$0.75 = \frac{1.2\, f_1}{f_2}$$

$$f_2 = \frac{1.2}{0.75} f_1 = 1.6 f_1$$

Given f_1 = 1GHz

f_2 = 1.6 GHz

Answer:
1.6 GHz

Alternate Solution:

Total Execution Time = Number of Instructions × CPI × Cycle time
Note: In the given question, the Number of Instructions is the same for Processor P_1 and P_2.
Processor P_1:

- Clock Frequency P_1 = 1 GHz
- Cycle time P_1 = 1 / 1 GHz = 10^{-9} sec
- Cycle time P_1 = 1 nsec
- ET_P_1 = CPI × 1 nsec

Processor P_2:

- ET_P_2 = 1.2 × CPI × Cycle time$_2$ (Due to 20% more CPI)
- ET_P_2 = 0.75 × ET_P_1 (Due to 25% less time than that of P_1)

Calculation: Equating the two expressions for ET_P_2:
0.75 × ET_P_1 = 1.2 × CPI × Cycle time$_2$
Substitute the value of ET_P_1: 0.75 × CPI × 1 nsec = 1.2
× CPI × Cycle time$_2$
Cancel out CPI from both sides: 0.75
= 1.2 × Cycle time$_2$
Solve for Cycle time$_2$: Cycle time$_2$ = 0.75 / 1.2 = 0.625 nsec
Calculate the Clock Frequency for P_2: Clock Frequency P_2 = 1 / Cycle time$_2$ Clock Frequency P_2 = 1 / (0.625 × 10^{-9} sec)
Ans: 1.6 GHz

24. ANSWER: - (1.54)
There are five stages:
Instruction Fetch (IF),
instruction decode and register fetch (ID/RF),
Instruction execution (EX),
Memory access (MEM),
and register write back (WB) P
old design:
With stage latencies 1ns, 2.2ns, 2ns, 1ns,
and 0.75ns
Cycle time = MAX (1ns, 2.2ns, 2ns, 1ns, and 0.75ns)
= 2.2nsec
Branch penalty = 3-1 = 2 because the next instruction pointer at the end of the EX stage (which is 3rd stage) in the old design
AVG instruction execution time is
P = Tavg = (1 + no. of stalls * branch penalty) * cycle time
= (1 + 0.20 * 2)2.2
P = 3.08nsec
Q new design:
ID/RF stage is split into three stages (ID, RF1, RF2) each of latency (2.2)/3 ns
= 0.73ns.
The EX stage is split into two stages (EX1, EX2) each of latency 1ns.
The new design has a total of eight pipeline stages. Time of stages in new design = {1ns, 0.73ns, 0.73ns, 0.73ns, 1ns, 1ns, 1ns, and 0.75ns}
Cycle time = MAX (1ns, 0.73ns, 0.73ns, 0.73ns, 1ns, 1ns, 1ns, and 0.75ns) = 1nsec
Branch penalty = 6-1 = 5 because the next instruction pointer at the end of the EX2 stage (which is 6th stage) in the new design.
AVG instruction execution time is
Q = Tavg = (1 + no. of stalls * branch penalty) *cycle time
= (1 + 0.20 * 5)1
Q = 2nsec
Therefore, P/Q = 3.08/2 = 1.54

25. ANSWER: - (C)
P1 = tp = max (1ns, 2ns, 2ns, 1ns) = 2 nsec
P2 = tp = max (1ns, 1.5ns, 1.5ns, 1.5ns) = 1.5 nsec
P3 = tp = max (0.5ns, 1ns, 1ns, 0.6ns, 1ns) = 1 nsec
P4 = tp = max (0.5ns, 0.5ns, 1ns, 1ns, 1.1ns) = 1.1 nsec.
Frequency = 1/tp
Hence p3 has the highest clock frequency.

26. ANSWER: - (4)
Non-Pipelined System: Since the processor has 6 perfectly balanced stages, a nonpipelined design must run through all 6 stages sequentially for each instruction. Thus, $CPI_{non\text{-}pipe} = 6$.

Ideal Pipelined System: Without any hazards or stalls, a pipeline completes 1 instruction every single cycle ($CPI_{ideal} = 1$). This means

Check Out Our GATE Courses & Secure Your Dream Rank!

the absolute maximum ideal speedup possible is $\frac{6}{1} = 6$.

Actual Speedup = Maximum Speedup × 25% = 6 × 0.25 = 1.5

$$\text{Actual Speedup} = \frac{CPI_{non\text{-}pipe}}{CPI_{pipelined}}$$

Substitute our known values into the equation:

$$1.5 = \frac{6}{CPI_{pipelined}}$$

Rearranging to solve for the pipelined execution time per instruction:

$$CPI_{pipelined} = \frac{6}{1.5} = 4$$

27. ANSWER: - (B)

Cycle time = max of all stage delays + buffer delay = max (5ns, 7ns, 10ns, 8ns, 6ns) + 1 = 10 + 1 = 11ns
Out of all the instructions I1, I2, I3. I12 it is given that only I4 is a branch instruction and when I4 takes a branch the control will jump to instruction I9 as I9 is the target instruction.
As can be seen from the timing diagram there is a gap of only 3 stall cycles between I4 and I9 because after I4 enters Decode Instruction (DI) whether there is a branch or not will be known at the end of Execute Instruction (EI) phase. So there are a total 3 phases here namely DI, FO, EI.
After 3 stall cycles I9 will start executing as that is the branch target.
As per the timing diagram total no. of clock cycles to complete the program = 15
Since 1 clock cycle = 11ns, time to complete the program = 15 * 11 = 165ns.

28. ANSWER: - (C)

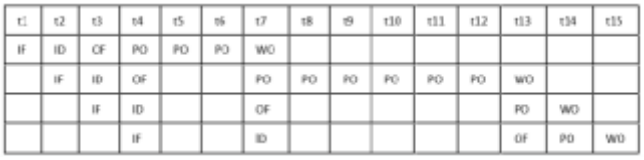
Register renaming is used to eliminate WAR (Write After Read and WAW (Write After Write) dependencies between instructions
Example:
I1: Read A → B
I2: Write C → A
Here, a WAR dependency occurs because the second instruction writes to A before the first instruction completes reading A
To avoid this, register renaming assigns a new register:
Write C → A′ (instead of A)
Thus, the conflict is removed
WAR dependency is also called anti-dependency because there is no true data dependency; the conflict arises only due to the use of the same register name. Register renaming removes such false dependencies. Similarly, it also eliminates WAW dependencies.

29. ANSWER: - (B)

Speedup = Time without pipelining / Time with pipelining
Clock cycle time of pipelining

30. ANSWER: - (B)

It is given that there is operand forwarding. In the case of operand forwarding the update value from previous instruction's PO stage is forwarded to the present instruction's PO stage. Here there's RAW dependency between I1-I2 for R5 and between I2-I3 for R2. These dependencies are resolved by using operand forwarding as shown in the below timeline diagram. The total number of clock cycles needed is 15.



Alternate Approach:

Instruction	Operation	PO Stage	Extra Cycle (Stalls)
I_0	MUL	3	2
I_1	DIV	6	5
I_2	ADD	1	--
I_3	SUB	1	--

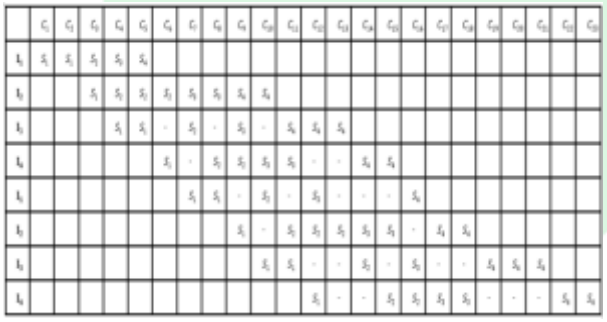
Check Out Our GATE Courses & Secure Your Dream Rank!

Total: 7 Extra Cycles
(Stalls) Given Variables:

- n = 4
- Number of

Stages (k) = 5 Execution
Time (ET) Calculation:
ET = [k + (n - 1)] × tp + tstalls (Extra Cycle Due to Hazards)
ET = [5 + (4 - 1)] × 1 + 7
ET = 8 + 7
Ans: 15 cycles

31. ANSWER: - (B)
Here the bounds of the loop are constants, therefore the compiler will do the loop unrolling (If compiler won't then prefetcher will do) to increase the instruction level parallelism. And after loop unrolling 23 cycles are required for execution. Therefore, the correct answer would be (b).
We assume the buffers between the pipeline stages can store multiple results in the form of a queue.



32. ANSWER: - (D)

An instruction placed in a branch delay slot always executes, regardless of whether the branch is taken or not. To safely fill it with an instruction from before the branch, that instruction must have no data dependencies affecting the branch condition or subsequent logic.
The branch condition depends entirely on register R_1 (if R1 == 0).

α. I_3 cannot be moved: It writes directly to R_1 (ADD R1 +R2 + R3). Moving it below the branch means the branch would evaluate using an old, incorrect value of R_1.

β. I_1 cannot be moved: It computes R_2 (ADD R2 + R7 + R8), which is needed immediately by I_3 to compute the branch register R_1. Moving it would break this critical data chain.

c. I_2 cannot be moved: It computes R_4 (SUB R_4 + R5 − R6), which is needed immediately by the memory write address in I_4.

d. I_4 can be moved: It is a simple store instruction (STORE Memory [R4] & R1). It reads R_4 and R_1 but does not alter any registers that the branch relies on. All of its own source operands are fully calculated by the time the program reaches this block.

33. ANSWER: - (B)
I - False, Bypassing can't handle all RAW hazards, consider when any instruction depends on the result of LOAD instruction, now LOAD updates register value at Memory Access Stage (MA), so data will not be available directly on Execute stage.
II - True, register renaming can eliminate all WAR Hazard.
III- False, it cannot completely eliminate, though it can reduce Control Hazard Penalties

34 ANSWER: (A)
In delayed branching, the instruction immediately after the branch is always executed, regardless of whether branch is taken or not
So:
Execution order =
Branch → next instruction (delay slot) → then branch decision applied
Evaluate options:
A: instruction after branch is executed
B: depends on branch outcome
C: depends on branch outcome
D: unrelated

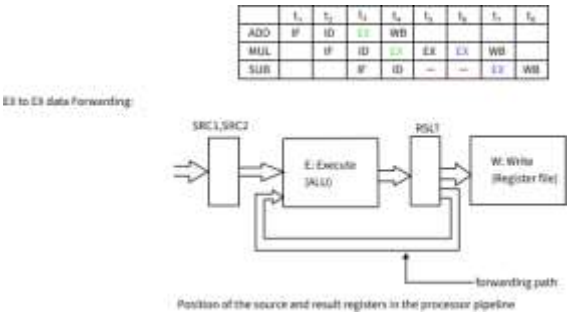
35. Answer: (C)
Non-Pipeline Processor
Frequency = 100 MHz
Cycle time (tn) = 1 / (100 × 10^6 Hz) = 10^{-8} sec
10^{-8} sec = 10 × 10^{-9} sec
Cycle time (tn) = 10 nsec
Part 2: Pipeline Processor Stages (k)
= 5
Stage Delays: 2.5, 1.5, 2, 1.5, 2 nsec Buffer / Latch Delay = 0.5 nsec
Pipeline Cycle Time (tp) = Max (Stage Delay) + Buffer Delay
tp = 2.5 + 0.5
tp = 3 nsec

When a very large number of instructions are executed:
Speedup (S) = tn / tp S =
10 nsec / 3 nsec S = 3.33
Ans: 3.33

Check Out Our GATE Courses & Secure Your Dream Rank!

36. ANSWER: - (B)

Considering EX to EX data forwarding



37. ANSWER: - (B)

For a non-pipelined processor, there are n instructions and each instruction requires 12 cycles, so the total number of cycles is $12n$.

For a pipelined processor, each stage takes 6 ns, and the total time to execute $n$instructions is : $6 \times 6 + (n-1) \times 6$.

$$\lim_{n\to\infty} \frac{12n}{36+(n-1)\times 6} = \frac{12}{6} = 2.$$

38. Answer : (C)

Given Data:

- Cycle time = 1 nsec
- Branch Instruction frequency = 20%
- Target address = 3rd stage
- Program Contains = 10^9 Instructions

Branch Penalty Calculation:

- Branch Penalty = 3 - 1 = 2

Stall Calculation (from tree diagram):

- Total Instructions (100%) split into:
  Normal (80%): 0 Stalls
  Branch (20%): 2 Stalls
- #Stalls / Instruction = (0.80 × 0) + (0.20 × 2)
- #Stalls / Instruction = 0.4

Execution Time (ET) Calculation:

- Average Instruction ET = (1 + #Stalls/Instruction) × Cycle time
- Average Instruction ET = (1 + 0.4) × 1 nsec
- Average Instruction ET = 1.4 nsec

Total Program ET Calculation:

- Program ET = (Total Instructions) × (Average Instruction ET)
- Program ET = $10^9 \times (1.4 \times 10^{-9})$ sec
- Program ET = 1.4 sec

Ans: 1.4 sec

39. Answer : (C)

RAW
1. I1 - I2 (R5)
2. I2 - I3 (R6)
3. I3 - I4 (R5)
4. I4 - I5 (R6)

WAR
1. I2 - I3 (R5)
2. I3 - I4 (R6)

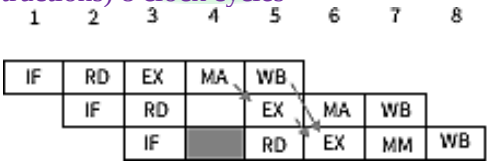
WAW
1. I1 - I3 (R5)
2. I2 - I4 (R6)

40 Answers: (B)

Instruction	CC1	CC2	CC3	CC4	CC5	CC6	CC7	CC8	CC9	CC10	CC11	CC12
1. ADD R5, R0, R1	IF	ID	EX	WB								
2. MUL R6, R2, R5		IF	ID	EX	EX	EX	WB					
3. SUB R5, R3, R6			IF	ID	**Stall**	**Stall**	EX	WB				
4. DIV R6, R5, R4				IF	**Stall**	**Stall**	ID	EX	EX	EX	WB	
5. STORE R6, X							IF	ID	**Stall**	**Stall**	EX	WB

41 ANSWER: - (A)

With data forwarding(Should be the case here as question says no operand forwarding for memory register for STORE instructions) 8 clock cycles



42. Answer: (B)

Execution time for Pipeline= $(K + n - 1) * \text{ET}$ where k = no of stages in pipeline,n = no of instructions
Execution time(ET) = max (all stages execution time)
$D_1 = (5+100-1) * 4 = 416$
$D_2 = (8+100-1) * 2 = 214$
Time saved using D_2
$= 416 - 214 = 202$

3 MACHINE INSTRUCTIONS AND ADDRESSING MODES

GATE CSE - 2026 MCQ SET – 1 (1-MARKS)

Q1. Consider a processor P whose instruction set architecture is the load-store architecture. The instruction format is such that the first operand of any instruction is the destination operand. Which one of the following sequences of instructions corresponds to the high-level language statement Z = X + Y?
Note: X, Y, and Z are memory operands. R0, R1, and R2 are registers.

(a) ADD Z, X, Y
(b) LOAD R0, X
ADD Z, R0, Y
(c) ADD R0, X, Y
STORE Z, R0
(d) LOAD R0, X
LOAD R1, Y
ADD R2, R0, R1
STORE Z, R2

GATE CSE - 2026 MCQ SET – 1 (1-MARKS)

Q2. Match each addressing mode in **List I** with a data element or an element of a data structure (in a high-level language) in **List II**.

List I

- P. Immediate
- Q. Indirect
- R. Base with index
- S. Base with offset/displacement

List II

1. Element of an array
2. Pointer
3. Element of a record
4. Constant

(a) P-4, Q-3, R-1, S-2
(b) P-4, Q-2, R-1, S-3
(c) P-1, Q-4, R-3, S-2
(d) P-2, Q-3, R-1, S-4

GATE CSE - 2026 MCQ SET – 1 (1-MARKS)

Q3. Consider a processor that has 16 general purpose registers and it uses 2-byte instruction format for all its instructions. Variable-sized opcodes are permitted. There are three different types of instructions; M-type, R-type, and C-type. Each M-type instruction has 2 register operands and a 6-bit immediate operand. Each R-type instruction has 3 register operands. Each C-type instruction has a register operand and a 6-bit offset value. If there are 2 unique M-type opcodes and 7 unique R-type opcodes, which one of the following options gives the maximum number of unique opcodes possible for C-type instructions?

(a) 8
(b) 4
(c) 64
(d) 16

GATE CSE - 2025 NAT SET - 2(2-MARKS)

Q4. An application executes 6.4×10^8 number of instructions in 6.3 seconds. There are four types of instructions, the details of which are given in the table. The duration of a clock cycle in nanoseconds is _____(rounded off to one decimal place).

Instruction type	Clock cycles required per instruction (CPI)	Number of instruction executed
Branch	2	2.25×10^8
Load	5	1.20×10^8
Store	4	1.65×10^8
Arithmetic	3	1.30×10^8

GATE CSE - 2025 MCQ SET – 2 (1-MARKS)

Q5. Which of the following is/are part of an Instruction Set Architecture of a processor?

(a) The size of the cache memory
(b) The clock frequency of the processor
(c) The number of cache memory levels
(d) The total number of registers

GATE CSE – 2025 SET – 1 MCQ (2-MARKS)

Q6. A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32-bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction?

$$\text{ADD R1, \#25} \qquad //\text{R1} = \text{R1} + 25$$

(a) 16
(b) 20
(c) 22
(d) 24

GATE CSE – 2024 NAT SET- 2 (2-MARKS)

Q7. A processor with 16 general purpose registers uses a 32-bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, the maximum number of unique opcodes possible for every addressing mode is _____

GATE CSE – 2024 NAT SET- 2 (2- MARKS)

Q8. A processor uses a 32-bit instruction format and supports byte-addressable memory access. The ISA of the processor has 150 distinct instructions. The instructions are equally divided into two types, namely R-type and I-type, whose formats are shown below.
R-type Instruction Format:

I-type Instruction Format:

In the OPCODE, 1 bit is used to distinguish between I-type and R-type instructions, and the remaining bits indicate the operation. The processor has 50 architectural registers, and all register fields in the instructions are of equal size.
Let X be the number of bits used to encode the UNUSED field, Y be the number of bits used to encode the OPCODE field, and Z be the number of bits used to encode the immediate value/address field The value of $X + 2Y + Z$ is _____.

GATE CSE – 2023 MCQ (2-MARKS)

Q9. Consider the given C-code and its corresponding assembly code, with a few operands U1-U4 being unknown. Some useful information as well as the semantics of each unique assembly instruction is annotated as inline comments in the code. The memory is byte-addressable.

	;assembly code (; indicates comments)
	;r1-r5 are 32-bit integer registers
//C-code	;initialize r1=0, r2=10
	;initialize r3, r4 with base address of a, b
	L01: jeq r1, r2, end ;if(r1==r2) goto end
	L02: lw, r5, 0(r4) ;r5 <- Memory[r4+0]
int a[10], b[10], i;	L03: sh1, r5, r5, U1 ;r5 <- r5 << U1
// int is 32 bit	L04: sw, r5, 0(r3) ;Memory[r3+0] <- r5
for(i=0; i<10; i++)	L05: add, r3, r3, U2 ;r3 <- r3+U2
a[i] = b[i] * 8;	L06: add, r4, r4, U3
	L07: add, r1, r1, 1
	L08: jmp U4 ;goto U4
	L09: end

Which one of the following options is a CORRECT replacement for operands in the position (U1, U2, U3, U4) in the above assembly code?

(a) (8, 4, 1, L02)
(b) (3, 4, 4, L01)
(c) (8, 1, 1, L02)
(d) (3, 1, 1, L01)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2021 SET-1 NAT (2-MARKS)

Q10. Consider the following instruction sequence where registers R1, R2 and R3 are general purpose and MEMORY [X] denotes the content at the memory location X .

Instruction	Semantics	Instruction Size (bytes)
MOV R1,(5000)	R1 ← MEMORY[5000]	4
MOV R2,(R3)	R2 ←MEMORY[R3]	4
ADDR2, R1	R2 ← R1 + R2	2
MOV (R3), R2	MEMORY[R3]← R2	4
INC R3	R3 ← R3 + 1	2
DEC R1	R1 ← R1 – 1	2
BNZ 1004	Branch if not zero to the given absolute address	2
HALT	Stop	1

Assume that the content of the memory location 5000 is 10 , and the content of the register R3 is 3000 . The content of each of the memory locations from 3000 to 3010 is 50 . The instruction sequence starts from the memory location 1000. All the numbers are in decimal format. Assume that the memory is byte addressable.

After the execution of the program, the content of memory location 3010 is ___

GATE CSE – 2020 NAT (2-MARKS)

Q11. A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____.

GATE CSE – 2018 NAT (2-MARKS)

Q12. A processor has 16 integer registers (R0, R1, …., R15) and 64 floating point registers (F0, F1, …., F63). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3 and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating-point register operand (1F). The maximum value of N is _____.

GATE CSE – 2018 NAT (1-MARKS)

Q13. The following are some events that occur after a device controller issues an interrupt while the process **L** is under execution.

(P) The processor pushes the process status of **L** onto the control stack.

(Q) The processor finishes the execution of the current instruction.

(R) The processor executes the interrupt service routine.

(S) The processor pops the process status of **L** from the control stack.

(T) The processor loads the new PC value based on the interrupt.

Which one of the following is the correct order in which these events above occur?

(a) QPTRS
(b) PTRSQ
(c) TRPQS
(d) QTPRS

GATE CSE – 2017 NAT SET -1 (2-MARKS)

Q14. Consider a RISC machine where each instruction is exactly 4 bytes long. Conditional and unconditional branch instructions use the PC-relative addressing mode, with the offset specified in bytes to the target location of the branch instruction.
Furthermore, the offset is always with respect to the address of the next instruction in the program sequence.
Consider the following instruction sequence:

Instr. No.	Instruction
i:	add R2, R3, R4
i + 1:	sub R5, R6, R7
i + 2:	cmpR1, R9, R10
i + 3:	beqR1, Offset

If the target of the branch instruction is i, then the decimal value of the Offset is _____.

GATE CSE – 2017 MCQ (1-MARKS)

Q15. Consider the C struct defined below:

```
struct data {
   int marks [100];
   char grade:
   int cnumber;
l:
struct data student:
```

The base address of student is available in register R1. The field student.grade can be accessed efficiently using:

(a) Post-increment addressing mode, (R1)+
(b) Pre-decrement addressing mode, $-(R1)$
(c) Register direct addressing mode, R1
(d) Index addressing mode, $X(R1)$, where X is an offset represented in $2's$ complement 16 -bit representation

GATE CSE – NAT 2016 SET- 2 (1-MARKS)

Q16. A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand The number of bits available for the immediate operand field is _____.

GATE CSE – 2016 NAT SET - 2 (2-MARKS)

Q17. Consider a Processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a twelve-bit immediate value. Each instruction must be stored in memory in a byte aligned fashion. If a program has 100 instructions, the amount of memory (in bytes) consumed by the program text is _____.

GATE CSE – 2015 MCQ SET - 2(2-MARKS)

Q18. Consider a processor with byte-addressable memory. Assume that all registers, including Program Counter (PC) and Program Status Word (PSW), are of size 2 bytes. A stack in the main memory is implemented from memory location $(0100)_{16}$ and it grows upward The stack pointer (SP) points to the top element of the stack. The current value of SP is $(016E)_{16}$. The CALL instruction is of two words, the first word is the op-code and the second word is the starting address of the subroutine (one word = 2 bytes). The CALL instruction is implemented as follows:
• Store the current value of the PC in the stack.
• Store the value of the PSW register in the stack.
• Load the starting address of the subroutine in PC
The content of the PC just before the fetch of a CALL instruction is $(5FA0)_{16}$. After execution of the CALL instruction, the value of the stack pointer is

(a) $(O16A)_{16}$
(b) $(O16C)_{16}$
(c) $(O170)_{16}$
(d) $(O172)_{16}$

GATE CSE – 2015 MCQ SET - 1(1-MARKS)

Q19. For computers based on three-address instruction formats, each address field can be used to specify which of the following:
S1: A memory operand
S2: A processor register
S3: An implied accumulator register

(a) Either S1 or S2
(b) Either S2 or S3
(c) Only S2 and S3
(d) All of S1, S2 and S3

GATE CSE - 2014 NAT SET - 1 (2-MARKS)

Q20. A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, the maximum value of the immediate operand is _____.

GATE MCQ – 2011 MCQ (1-MARKS)

Q21. Consider a hypothetical processor with an instruction of type LW R1, 20(R2), which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of a constant 20 and the contents of register R2. Which of the following best reflects the addressing mode implemented by this instruction for operand in memory?

(a) Immediate Addressing
(b) Register Addressing
(c) Register Indirect Scaled Addressing
(d) Base or Indexed Addressing

GATE MCQ – 2008 MCQ (1-MARKS)

Q22. Which of the following is/are true of the auto-increment addressing mode?
I. It is useful in creating self-relocating code
II. If it is included in an Instruction Set Architecture, then an additional ALU is required for effective address calculation
III. The amount of increment depends on the size of the data item accessed

(a) I only
(b) Il only
(c) III only
(d) II and III only

GATE MCQ – 2008 MCQ (1-MARKS)

Q23. Which of the following must be true for the RFE (Return From Exception) instruction on a general purpose processor?
I.It must be a trap instruction
II. It must be a privileged instruction
III. An exception cannot be allowed to occur during execution of an RFE instruction

(a) I only
(b) II only
(c) I and II only
(d) I, II and III only

GATE MCQ – 2008 MCQ (2-MARKS)

Q24. Assume that EA = (X)+ is the effective address equal to the contents of location X, with X incremented by one word length after the effective address is calculated; EA = −(X) is the effective address equal to the contents of location X, with X decremented by one word length before the effective address is calculated; EA = (X)− is the effective address equal to the contents of location X, with X decremented by one word length after the effective address is calculated. The format of the instruction is (opcode, source, destination), which means (destination ← source op destination). Using X as a stack pointer, which of the following instructions can pop the top two elements from the stack, perform the addition operation and push the result back to the stack.

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) ADD (X)−, (X)
(b) ADD (X), (X)−
(c) ADD −(X), (X)+
(d) ADD −(X), (X)

GATE CSE – 2007 MCQ (2-MARKS)

Q25. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	R1 ← M3000]	2
LOOP:	MOV R2,(R3)	R2←M[R3]	1
	ADD R2,R1	R2 ← R1 + R2	1
	MOV (R3),R2	M[R3] ←R2	1
	INC R3	R3←R3+1	1
	DEC R1	R1 ← R1 – 1	1
	BNZLOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.
Assume that the memory is word addressable. The number of memory references for accessing the data in executing the program completely is

(a) 10
(b) 11
(c) 20
(d) 21

GATE CSE – 2007 MCQ (2-MARKS)

Q26. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	R1 ← M3000]	2
LOOP:	MOV R2,(R3)	R2←M[R3]	1
	ADD R2,R1	R2 ← R1 + R2	1
	MOV (R3),R2	M[R3] ←R2	1
	INC R3	R3←R3+1	1
	DEC R1	R1 ← R1 – 1	1
	BNZLOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal. Assume that the memory is word addressable. After the execution of this program, the content of memory location 2010 is:

(a) 100
(b) 101
(c) 102
(d) 110

GATE CSE – 2007 MCQ (1-MARKS)

Q27. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	R1←M[3000]	2
LOOP:	MOV R2,(R3)	R2 ← [R3]	1
	ADD R2,R1	R2 ← R1+R2	1
	MOV (R3),R2	M[R3] ←R2	1
	INC R3	R3 ← R3+1	1
	DEC R1	R1 ← R1-1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000 . The content of each of the memory locations from 2000 to 2010 is 100 . The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is byte addressable and the word size is 32 bits. If an interrupt occurs during the execution of the instruction "INC R3", what return address will be pushed on to the stack?

(a) 1005
(b) 1020
(c) 1024
(d) 1040

GATE CSE – 2007 MCQ (1-MARKS)

Q28. In a simplified computer the instructions are:

OP R_j, R_i	Perform R_j OP R_i and store the result in register R_j
OP m, R_i	Perform val OP R_i and store the result in register R_i val denotes the content of the memory location m
MOV m, R_i	Moves the content of memory location m to register R_i
MOV R_i, m	Moves the content of register R_i to memory location m

The computer has only two registers, and OP is either ADD or SUB. Consider the following basic block:

- $t_1 = a + b$
- $t_2 = c + d$
- $t_3 = e - t_2$
- $t_4 = t_1 - t_3$

Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instructions in the code generated for this basic block?

(a) 2
(b) 3
(c) 5
(d) 6

GATE CSE – 2007 MCQ (2-MARKS)

Q29. Data forwarding techniques can be used to speed up the operation in the presence of data dependencies. Consider the following replacements of LHS with RHS:

(i) R1 → Loc, Loc → R2 ≡ R1 → R2,R1→Loc
(ii) R1 → Loc, Loc → R2 ≡ R1 → R2
(iii) R1 → Loc, R2 → Loc ≡ R1 → Loc
(iv) R1 → Loc, R2 → Loc ≡ R2 → Loc

In which of the following options will the result of executing the RHS be the same as executing the LHS irrespective of the instruction that follow ?

(a) (i) and (iii)
(b) (i) and (iv)
(c) (ii) and (iii)
(d) (ii) and (iv)

GATE CSE – 2007 NAT (2-MARKS)

Q30. Following table indicates the latencies of operations between the instruction producing the result and instruction using the result.

Instruction producing the result	Instruction using the result	Latency
ALU Operation	ALU Operation	2
ALU Operation	Store	2
Load	ALU Operation	1
Load	Store	0

Consider the following code segment:
Load R1, Loc 1; Load R1 from memory location Loc1
Load R2, Loc 2; Load R2 from memory location Loc 2
Add R1, R2, R1; Add R1 and R2 and save result in R1
Dec R2; Decrement R2
Dec R1; Decrement R1
Mpy R1, R2, R3; Multiply R1 and R2 and save result in R3
Store R3, Loc 3; Store R3 in memory location Loc 3

What is the number of cycles needed to execute the above code segment assuming each instruction takes one cycle to execute?

(a) 7
(b) 10
(c) 13
(d) 14

GATE CSE – 2006 MCQ (1-MARKS)

Q31. A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?

(a) 400
(b) 500
(c) 600
(d) 700

GATE CSE – 2006 MCQ (2-MARKS)

Q32. Which of the following statements about relative addressing mode is FALSE?

(a) It enables reduced instruction size
(b) It allows indexing of array elements with same instruction
(c) It enables easy relocation of data
(d) It enables faster address calculations than absolute addressing

GATE CSE – 2006 MCQ (2-MARKS)

Q33. The memory locations 1000, 1001 and 1020 have data values 18, 1 and 16 respectively before the following program is executed.

MOVI	Rs, 1	Move immediate
LOAD	Rd, 1000(Rs)	Load from memory
ADDI	Rd, 1000	Add immediate
STOREI	0(Rd), 20	Store immediate

Which of the statements below is TRUE after the program is executed?

(a) Memory location 1000 has value 20
(b) Memory location 1020 has value 20
(c) Memory location 1021 has value 20
(d) Memory location 1001 has value 20

GATE CSE - 2005 MCQ (2 -MARKS)

Q34. Consider a three-word machine instruction:
ADD A [R_0], @B
The first operand (destination) "A [R_0]" uses indexed addressing mode with R_0 as the index register.
The second operand (source) "@B" uses indirect addressing mode.
A and B are memory addresses residing at the second and the third words, respectively.
The first word of the instruction specifies the opcode, the index register designation, and the source and destination addressing modes.

Check Out Our GATE Courses & Secure Your Dream Rank!

During execution of the ADD instruction, the two operands are added and stored in the destination (first operand).
The number of memory cycles needed during the execution cycle of the instruction is:

(a) 3
(b) 4
(c) 5
(d) 6

GATE CSE - 2005 MCQ (2 -MARKS)

Q35. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I	List-II
A.[I] = B[J];	1. Indirect addressing
B. while(*A++);	2. Indexed addressing
C. int temp = *x;	3. Autoincrement

	A	B	C
(a)	3	2	1
(b)	1	3	2
(c)	2	3	1
(d)	1	2	3

ANSWER KEY

1.(d)	**2.(b)**	**3.(b)**	**4.(3)**
5.(d)	**6.(b)**	**7.(32)**	**8.(34)**
9.(b)	**10.(50)**	**11.(14)**	**12.(32)**
13.(a)	**14.(-16)**	**15.(d)**	**16.(16)**
17.(500)	**18.(d)**	**19.(a)**	**20.(16383)**
21. (d)	**22.(c)**	**23.(d)**	**24.(a)**
25.(d)	**26.(a)**	**27.(c)**	**28.(b)**
29.(b)	**30.(c)**	**31.(c)**	**32.(d)**
33.(d)	**34.(b)**	**35.(c)**	

HINTS&SOLUTIONS

1. **ANSWER: -** D
In load-store architecture, ALU operations work only on registers, not directly on memory.
So:
First load X and Y into registers
Perform addition in registers
Store result back to memory
Option (d) follows this exactly:
LOAD R0, X
LOAD R1, Y
ADD R2, R0, R1
STORE Z, R2

2. **Answer :- B**

Immediate Am : Constant
Indirect Am : Pointer
Base with Index : Element of Array
Base with Offset/Displacement : Element of a Record

3. **ANSWER: -** B
16 general purpose registers, no of bits required = 4
instruction format = 2 Byte = 16 bit
M inst

Opcode 2bit	Reg 4 bit	Reg 4 bit	*Im 6 bit*

R Inst:

Opcode 4 bit	Reg 4 bit	Reg 4 bit	Reg 4 bit

C Inst:

Opcode 6 bit	Reg 4 bit	Offset 6 bit

Here we use Expand Opcode Technique

Inst. Type	#Total Opcode	#Used Opcode	#Available Opcode
M	$2^2 = 4$	2	$4 - 2 = 2$
R	$2 \times 2^{4-2} = 8$	7	$8 - 7 = 1$
C	$1 \times 2^{6-4} = 4$	-	4

Check Out Our GATE Courses & Secure Your Dream Rank!

4. ANSWER: - 3

Instruction Type	Clock cycles required per instruction (CPI)	Number of instruction executed
Branch	2	2.25×10^8
Load	5	$1.20 \quad 10^8$
Store	4	1.65×10^8
Arithmetic	3	1.30×10^8

Total Execution Time = 6.3 sec
Number of clock cycles = Number of instructions executed × CPI
Clock cycles for each instruction type:
Branch Instructions = $2 \times (2.25 \times 10^8) = 4.5 \times 10^8$
Load Instructions = $5 \times (1.20 \times 10^8) = 6.0 \times 10^8$
Store Instructions = $4 \times (1.65 \times 10^8) = 6.6 \times 10^8$
Arithmetic Instructions = $3 \times (1.30 \times 10^8) = 3.9 \times 10^8$
Total Clock Cycles = $(4.5 \times 10^8) + (6.0 \times 10^8) + (6.6 \times 10^8) + (3.9 \times 10^8) = 21 \times 10^8$ cycles
Execution Time = Total Clock Cycles × Clock Cycle Time
Clock Cycle Time = $6.3 / (21 \times 10^8)$ sec
$= 0.3 \times 10^{-8}$ sec $= 3 \times 10^{-9}$ sec = 3 ns
Answer: Clock Cycle Time = 3 ns

5. ANSWER: - D
ISA (Instruction Set Architecture) defines what is visible to the programmer
It includes:
Instruction formats
Addressing modes
Registers (number, type)
It does NOT include:
Cache size
Cache levels
Clock
frequency

6. ANSWER: - B
we need 6 bits to represent registers
we need 6 bits to represent opcode (we have 50 distinct types of instructions)
we have 32 bits instruction

OPCODE	REGISTER	VALUE
6	6	32-(6+6)

VALUE = 32 - 12 = 20
Maximum 20 bits can be used to store immediate operand.

7. ANSWER: - 32
To determine the maximum number of unique opcodes possible for every addressing mode, we need to break down the 32-bit instruction format and understand how the bits are allocated to each field
Here's the modified solution in direct format:
The 32-bit instruction format consists of the following fields:
Opcode field
Addressing mode field
Two register operand fields
16-bit scalar field
Given details:
The processor has 16 general purpose registers, which means each register operand field needs $\log_2(16) = 4$ bits.
There are 8 addressing modes, so the addressing mode field needs $\log_2(8) = 3$ bits.
The scalar field is fixed at 16 bits.
Let opcode be the number of bits allocated to the Opcode Field The bit allocation is as follows:
Scalar field: 16 bits
Two register operand fields: 4 + 4 = 8 bits
Addressing mode field: 3 bits
Total bits used by these fields:
16 + 8 + 3 = 27 bits
The remaining bits for the opcode field are:
32 − 27 = 5 bits
Therefore, the number of unique opcodes possible for every addressing mode, given 5 bits, is:
$2^5 = 32$
So, the maximum number of unique opcodes possible for every addressing mode is 32.

8. ANSWER: - 34
• Opcode field (Y): There are 150 instructions, equally divided into two types. Therefore, there are 75 I-type instructions and 75 R-type instructions. To encode 75 instructions, we need $\lceil \log_2(75) \rceil = 7$ bits. In addition, 1 bit is required to distinguish between I-type and R-type instructions. Hence, the total number of bits used for the opcode field is Y = 7 + 1 = 8 bits.
• Register field: There are 50 architectural registers. Since all register fields are of equal size, the number of bits required to encode a register is $\lceil \log_2(50) \rceil = 6$ bits.
• Unused field (X): In R-type instructions, the total instruction length is 32 bits. Subtracting the opcode field (8 bits) and three register fields ($3 \times 6 = 18$ bits), the unused field size is
X = 32 − (8 + 18) = 6 bits.
• Immediate value/address field (Z): In I-type instructions, there are two register fields ($2 \times 6 = 12$

bits) and the opcode field (8 bits). Therefore, the immediate/address field size is
$Z = 32 - (8 + 12) = 12$ bits.
Therefore, the values of X, Y, and Z are:
• X = 6 bits (UNUSED field
• Y = 8 bits (OPCODE field
• Z = 12 bits (immediate value/address field
So, the answer is $6 + 2 \times 8 + 12 = 34$.

9. ANSWER: - B
To understand u_1, we must first find out what the register r_5 is storing.
If you check closely the line L_{02}, it's fetching an element from array b and storing it in r_5.
Line L_{04} is storing the value present in r_5 to array a, so we're definitely performing the multiplication by 8 in L_{03}.
L_{03} is a shift-left operator. To multiply by 8, how many bits (in binary) do we need to shift to the left? If you can find out that number, that would be the answer to u_1.
L_{05} and L_{06} are incrementing the values of registers r_3 and r_5 respectively to point to the next element of a and b respectively. But by how much shall we increase it to make it point to the next element?

Suppose at some point in time, r_3 is storing 1000 in decimal and pointing to the 2nd element of array a. Recall that each element is 4Bytes in size (given in question). Now, what is the address of the 3rd element of array a, can you guess? Yes, it would be 1004. Hence, 4 must be added.
Note: Given memory is byte addressable. Each int takes 32bit=4B, so we need to shift by 4 addresses.
u_4 should be the line to make the loop repeat. Note that if we make it L_{02}, the loop termination condition is skipped and would lead to an infinite loop (or unless we got trapped due to accessing illegal memory).
The value of u_1, u_2, u_3, u_4 will be 3, 4, 4, L01.

10. ANSWER: 50
R1 = M[5000] = 10
R3 = 3000
Memory[3000…3020] = 50
Loop runs 10 times (R1 decreases from 10 to 0)
Each iteration:
R2 = M[R3] = 50
R2 = R1 + 50
M[R3] = 50 + R1
R3 = R3 + 1
So updates are:
3000 → 50 + 10 = 60
3001 → 50 + 9 = 59
…
3009 → 50 + 1 = 51
Address 3010 is never touched
Final value at 3010:50

11. ANSWER: - 14
Instruction Length: 16 bits
To distinguish among 64 registers, we need $\log_2(64) = 6$ bits
I-type instruction format:
Opcode | Register | Immediate Value
R-type instruction format:
Opcode | Register | Register
Maximum possible encodings = 2^{16}
It is given that there are 8 I-type instructions. Let the maximum R-type instructions be x.
Therefore,
$(8 \times 2^6 \times 2^4) + (x \times 2^6 \times 2^6) = 2^{16}$
$\Rightarrow x = 16 - 2 = 14$

Alternate Solution:
R-type:
- Opcode = 4 bits → Total combinations = $2^4 = 16$operations
- Given R-type instructions = x
- Free opcodes after allocating R-type = $(16-x)$

I-type:
- Format includes:
  - Opcode = 6 bits
  - Immediate field present
- Total operations in I-type = (Free opcodes) × (variations due to immediate bits)

$$\text{Total I-type operations} = (16 - x) \times 2^{(6-4)}$$
$$= (16 - x) \times 2^2$$

Given:
$$\text{I-type instructions} = 8$$
$$8 = (16 - x) \times 4$$
$$2 = 16 - x$$
$$x = 16 - 2$$
$$x = 14$$

12. ANSWER: - 32

Total Instruction Size: 2 Bytes = 16 bits
Total Encode Combinations Available: $2^{16} = 65,536$
Integer Register (R) : 16 unique options $\Longrightarrow \log_2(16) = 4$ bits
Floating-Point Register (F): 64 unique options $\Longrightarrow \log_2(64) = 6$ bits

Encodings Consumed = (Number of Instructions) $\times 2^{\text{Total bits used by its}}$

- Type-1 (4 instructions, 3Rs): Operands take $4 + 4 + 4 = 12$ bits.

Encodings $_{T1} = 4 \times 2^{12} = 4 \times 4096 = 16,384$

- Type-2 (6 instructions, 2 Fs): Operands take $6 + 6 = 12$ bits.

Encodings $_{T2} = 8 \times 2^{12} = 8 \times 4096 = 32,768$

- Type-3 (14 instructions, 1R + 1F): Operands take $4 + 6 = 10$ bits.

Encodings $_{T3} = 14 \times 2^{10} = 14 \times 1024 = 14,336$

Remaining Combination Pool for Type-4:
Remaining Encodings $= 65,536 - (16,384 + 32,768 + 14,336)$

Remaining Encodings $= 65,536 - 63,488 = 2,048$

- Type-4 Configuration (N instructions, 1F): Operands take 6 bits, which means each unique instruction consumes $2^6 = 64$ binary combinations.

$$N \times 2^6 = 2,048 \Longrightarrow N = \frac{2,048}{64} = 32$$

Alternate Solution:

Type 1
- Opcode = 4 bits
- IR = 4 bits
- IR = 4 bits
- IR = 4 bits

Total no. of operations in Type 1 $= 2^4 = 16$

Given = 4 instructions
Free opcodes after allocating Type 1 $= 16 - 4 = 12$

Type 2
- Opcode = 4 bits
- IR = 4 bits
- IR = 4 bits

Total number of operations in Type 2
$= \text{Free opcodes} \times 2^{\text{increment bits}}$
$= 12 \times 2^{(4-4)} = 12 \times 2^0 = 12$

Given Type 2 = 8 instruction operations Free opcodes after Type 2 $= 12 - 8 = 4$

Type 3
- Opcode = 6 bits
- IC = 4 bits
- FR = 6 bits

Total operations in Type 3
$= \text{Free opcodes} \times 2^{\text{increment bits}}$
$= 4 \times 2^{(6-4)} = 4 \times 4 = 16$

Given Type 3 = 14 instruction operations
Free opcodes after Type 3 $= 16 - 14 = 2$

Type 4
- Opcode = 10 bits
- FR = 6 bits

Total number of operations in Type 4
$= \text{Free opcodes} \times 2^{\text{increment bits}}$
$= 2 \times 2^{(10-6)} = 2 \times 16 = 32$

Final Answer : N = 32

13. Answer :- (A)

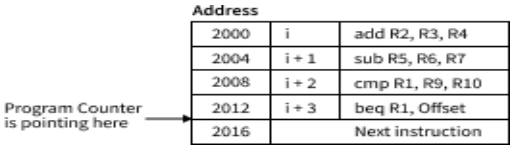
When interrupt occurs, after completion of current instruction, interrupt will be serviced. It push the program counter (PC) value into stack & control transfer to ISR.

Q: Process finish the current instruction execution.
P: Push the PC value into stack.
T: Interrupt → PC
R: Service the interrupt
S: Pop the PC value
Q P T R S:

14. ANSWER: - -16

The Program Counter (PC) is updated with the address of the next instruction before the current instruction completes execution.
Hence, the question states that the PC already contains the address of the next sequential instruction.
Just before executing instruction i+3i + 3i+3, the system state is as follows:
Note: BEQ stands for Branch if Equal.

The question states that the branch target address is instruction 'i', which is located at address 2000 in this example.



Currently, the Program Counter (PC) contains 2016, which is the address of the next sequential instruction.
Therefore, we must branch from 2016 back to 2000.

$$2016 - 2000 = 16$$

Hence, the required offset is –16, meaning 16 must be subtracted from the next instruction address (2016) to reach the target address (2000).

15. **ANSWER**: D

Struct layout:

int marks[100] → 100 × 4 = 400bytes

char grade → next byte ⇒ offset = 400

int number → after padding

So:

student.grade is at R1 + 400

Efficient access:

Need base + offset ⇒ indexed addressing

Answer:

Index addressing mode, X(R1) (Option D)

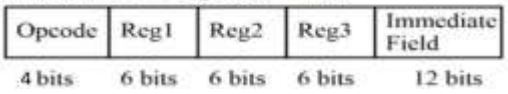
16. **ANSWER: -** 16

Instruction opcode size = $\lceil \log_2 40 \rceil$ = 6 bits
Register operand size = $\lceil \log_2 24 \rceil$ = 5 bits
Total instruction length = 32 bits
Bits required for opcode + two register operands
= 6 + 2 × 5
= 16 bits
Bits available for immediate operand
= 32 − 16
= 16 bits.

17. **ANSWER: -** 500

64 Register,
Instruction set size = 12
5 Fields = Opcode, Source Reg1, S Reg2, D Reg2, 12-bit Immediate field
64 Register → Reg. AF = 6bit

Inst. Set = 12 ⇒ Op code = 4 bit



Instruction size = 4 + 6 + 6 + 6 +12 = 34 bits = 5 Byte
Program having = 100 Inst. Program size =
100 × 5 = 500 byte.

18. **ANSWER: -** D

Given
Current SP value is = $(016E)_{16}$
PC = 2 Byte, PSW = 2 byte
(1) Store the current value of the PC into the stack. Here PC is 2 bytes so stack pointer to increase by 2 bytes
$(016E)_2 + 2 = (0170)_{16}$
(2) Store the value of PSW in the stack Here PSW is also 2-byte long. So SP increased by 2 byte
$(0170)_{16} + 2 = (0172)_{16}$ SP = $(0172)_{16}$
(3) Load the starting address of the subroutine in the PC
PC ← ISR
SP= $(0172)_{16}$

19. **ANSWER: -** A

In a **three-address instruction format**, each address field explicitly specifies an operand location.

Analysis:

- **S1: A memory operand** → Yes, address fields can point to memory locations
- **S2: A processor register** → Yes, address fields can specify registers
- **S3: An implied accumulator register** → No, "implied" means it is **not specified in the instruction field**

Conclusion:

Each address field can specify either memory or register, but not an implied accumulator.

20. **ANSWER: -** 16383

Instruction size = 1 word = 32 bit
45 Operation = 6 bit
64 Register = 6 bit
Immediate field = 32 – (6 + 6 + 6)
= 14 bit
Immediate field Range = 0 to $2^{14} - 1$ = 16383

21. **ANSWER: - D**

Base Index Addressing, as the content of register R2 will serve as the index, and 20 will be the Base address.

22. **ANSWER**: C

I:Auto-increment is for sequential access (arrays), not specifically for self-relocating code

II: No extra ALU is required; increment can be handled within existing hardware

III: Increment depends on data size (byte, word, etc.)

Answer:

III only (Option C)

23. **ANSWER**: D

I:RFE is used to return from exception, not to generate one ⇒ not a trap instruction

II: It restores processor state (PC, flags, mode) ⇒ must be privileged

III: If exception occurs during RFE, state restoration may break ⇒ must be atomic ⇒ no exception allowed

Answer:

II and III only (Option D)

24. **Answer :- A**

Explanation :-

EA = (X)+ → use value at X, then increase X (post-increment)

- EA = −(X) → first decrease X, then use value (pre-decrement)
- EA = (X)− → use value at X, then decrease X (post-decrement)

To add the top two stack elements:

1. First pop the top element → we get a
2. Stack pointer moves down (X decreases)
3. Pop again → we get b
4. Now add them → a + b
5. Store the result back on top of the stack

This is done by:

ADD (X)−, X

25. **ANSWER: - D**

The loop is executed 10 times and there are two memory references in the loop (each MOV is loading 1 word, so 1 memory reference for each MOV inside the loop). So number of memory references inside loop is:

2 (MOV) × 10 (times iteration) × 1 (1-word access / MOV) = 20 memory accesses.

One memory access is outside the loop for the first instruction:

MOV R1, (3000)

So, totally 20 + 1 = 21.

26. **ANSWER**: A

M[3000] = 10 ⇒ R1 = 10 (loop count)

R3 = 2000

Memory 2000–2010 initially = 100

Loop behavior (10 times):

Each iteration:

R2 ← M[R3]

R2 ← R1 + R2

M[R3] ← R2

R3 ← R3 + 1

R1 ← R1 − 1

Track updates:

Iteration 1: addr 2000 → 100 + 10 = 110

Iteration 2: addr 2001 → 100 + 9 = 109

…

Iteration 10: addr 2009 → 100 + 1 = 101

Address 2010 is never modified

Final value at 2010:

100

Answer:

100 (Option A)

27. **ANSWER**: C

Return address = address of next instruction after current instruction

Instruction sizes (in words):

MOV R1,(3000) → 2

MOV R2,(R3) → 1

ADD → 1

MOV → 1

INC → 1

Word size = 4 bytes

Program starts at 1000

Addresses:

MOV R1,(3000) → 1000 to 1007 → next = 1008

MOV R2,(R3) → 1008 to 1011 → next = 1012

ADD → 1012 to 1015 → next = 1016
MOV → 1016 to 1019 → next = 1020
INC → 1020 to 1023 → next = 1024
Interrupt occurs during INC R3 ⇒ return address = next instruction

Answer:
1024 (Option C)

28. **ANSWER**: B
2 registers only ⇒ minimize loads/stores
OP m, Ri allows one operand directly from memory ⇒ reduces MOVs

Compute efficiently:
t1 = a + b
MOV a → R1
ADD b, R1 → t1 in R1
t2 = c + d
MOV c → R2
ADD d, R2 → t2 in R2
t3 = e − t2
SUB e, R2 → R2 = e − t2
t4 = t1 − t3
SUB R2, R1 → R1 = t1 − t3
Final result must be in memory:
MOV R1 → t4

MOV count:
MOV a → R1
MOV c → R2
MOV R1 → memory
Total = 3 MOVs

Answer:
3 (Option B)

29. **Answer :- B**
(i) is correct: both LOC and R2 receive the value from R1 in both LHS and RHS.
(ii) is incorrect: although R2 gets the right value in both cases, LOC is not updated in the RHS.
(iii) is incorrect: in LHS, R2 is overwritten, whereas this does not happen in RHS.
(iv) is correct: the initial write to LOC in LHS has no effect since it is overwritten by the subsequent write.
Therefore, the correct answer is (B).

30. **Answer :- C**
Explanation
Given latencies:
- Load → ALU = 1 cycle
- ALU → ALU = 2 cycles
- ALU → Store = 2 cycles
- Load → Store = 0 cycles

Instruction Sequence:
1. Load R1, Loc1
2. Load R2, Loc2
3. Add R1, R2 → R1
4. Dec R2
5. Dec R1
6. Mpy R1, R2 → R3
7. Store R3, Loc3

Cycle-wise Execution:

Cycle	Instruction	Status	Reason
1	Load R1, Loc 1	Execute	
2	Load R2, Loc 2	Execute	
3	-	Stall	Waiting for Load R2 (Latency 1)
4	Add R1, R2, R1	Execute	
5	Dec R2	Execute	
6	-	Stall	Waiting for Add R1, R2, R1 (Latency 2)
7	Dec R1	Execute	
8	-	Stall	Waiting for Dec R1 (Latency 2)
9	-	Stall	Waiting for Dec R1 (Latency 2)
10	Mpy R1, R2, R3	Execute	
11	-	Stall	Waiting for Mpy R1, R2, R3 (Latency 2)
12	-	Stall	Waiting for Mpy R1, R2, R3 (Latency 2)
13	Store R3, Loc 3	ExecuteTotal number of cycles = 13	

Correct Answer: C

31. ANSWER: - C

Each address is multiple of 3 as the starting address is 300 and each instruction consists of 24 bits, i.e., 3 bytes. Thus, in the given options the valid counter will be the one which is the multiple of 3. Out of the options we can see that only 600 satisfies the condition.

32. Answer:- D

- In absolute addressing, the instruction directly contains the full memory address. In contrast, relative addressing calculates the address by adding an offset to the Program Counter (PC), so an extra computation is required.
- PC-relative addressing is mainly useful for relocating programs during execution.
- It can access nearby memory locations using offsets, which allows limited form of indexing.
- Since it uses a small offset instead of a complete address, the instruction length is reduced.

Hence, the incorrect statement is: (d)

33. Answer :- D

Explanation:-

Given:

Memory[1000] = 18, Memory[1001] = 1, Memory[1020] = 16

Instructions:

1. **MOVI Rs, 1**
   → Rs = 1
2. **LOAD Rd, 1000(Rs)**
   → Effective address = 1000 + Rs = 1001
   → Rd = Memory[1001] = 1
3. **ADDI Rd, 1000**
   → Rd = 1 + 1000 = 1001
4. **STOREI 0(Rd), 20**
   → Effective address = Rd + 0 = 1001
   → Memory[1001] = 20

Final Result:

Memory[1001] = 20

Correct Answer: (d)

34. ANSWER: - B

1 memory read to get the first operand from memory address A + R0 (A is given as part of the instruction)
1 memory read to get the address of the second operand (since second uses indirect addressing)
1 memory read to get the second operand from the address given by the previous memory read
1 memory write to store to first operand (which is the destination)
So, a total of 4 memory cycles once the instruction is fetched.

```
IF      MAR ← PC
        MDR ← mem(MAR)
        PC ← PC + 3
        IR ← MDR
ID      MAR ← MAR + 1
        R1 ← MAR            // R1 is the address of A
        MAR ← MAR + 1
        R2 ← MAR            // R2 is the address of B
OF + EX R3 ← mem(R2)        // R3 is the address of B
        R1' ← mem(R1)       // R3 is the address of B
        R3' ← mem(R3)       // operand fetch A
        R1' ← R1' + R3'     // operand fetch B
        MDR ← mem(R1)
        MDR ← R1'
```

35. Answer :- C

Explanation :-

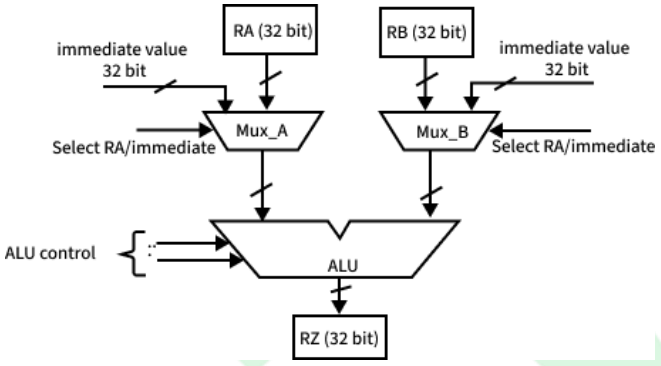
The correct matches are:

1. **A[i] = B[j];** → Indexed Addressing Mode
2. While [*A++]; → Auto-increment Addressing Mode
   int temp = *x; → Indirect Addressing Mode

4 ALU DATA PATH & CONTROL UNIT

GATE CSE – 2025 MSQ SET – 1 (1-MARKS)

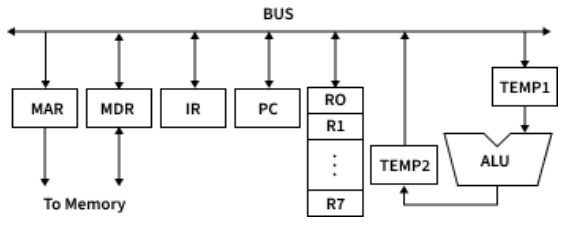
Q1. A partial data path of a processor is given in the figure, where *RA*, *RB*, and *RZ* are 32-bit registers. Which option(s) is/are CORRECT related to arithmetic operations using the data path as shown?



(a) The data path can implement arithmetic operations involving two registers.
(b) The data path can implement arithmetic operations involving one register and one immediate value.
(c) The data path can implement arithmetic operations involving two immediate values.
(d) The data path can only implement arithmetic operations involving one register and one immediate value.

GATE CSE – 2020 MCQ (1-MARKS)

Q2. Consider the following data path diagram



Consider an instruction: R0 ← R1 + R2. The following steps are used to execute it over the given data path. Assume that the PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively.

1. $R2_r$ $TEMP1_r$, ALUadd, $TEMP2_w$
2. $R1_r$ $TEMP1_w$,
3. PC_r, MAR_w, MEM_r
4. $TEMP2_r$, $R0_w$
5. MDR_r, IR_w

Which one of the following is the correct order of execution of the above steps?

(a) 2,1,4,5,3
(b) 1,2,4,3,5
(c) 3,5,2,1,4
(d) 3,5,1,2,4

GATE CSE – 2018 MCQ SET – 2 (1-MARKS)

Q3. Consider the following processor design characteristics:

I. Register-Io-register arithmetic operations only
II. Fixed-length instruction format
III. Hardwired control unit

Which of the characteristics above are used in the design of a RISC processor?

(a) I and II only
(b) II and III only
(c) I and III only
(d) I, II and III

GATE CSE – 2013 MCQ (2-MARKS)

Q4. Consider the following sequence of micro - operations.
MBR ← PC
MAR ← X
PC ← Y
Memory ← MBR
Which one of the following is a possible operation performed by this sequence?

(a) Instruction fetch
(b) Operand fetch
(c) Conditional fetch
(d) Initiation of interrupt service

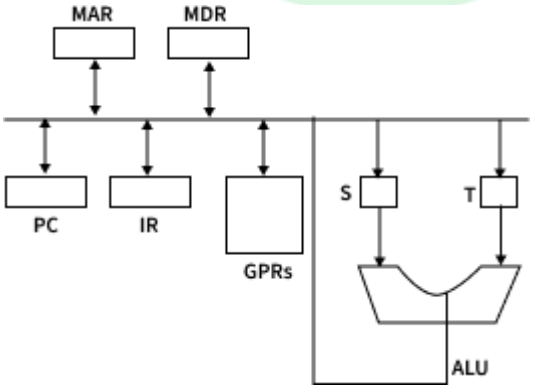
GATE CSE – 2008 MCQ (2-MARKS)

Q5. Consider a CPU where all the instructions require 7 clock cycles to complete execution. There are 140 instructions in the instruction set. It is found that 125 control signals are needed to be generated by the control unit. While designing the horizontal micro-programmed control unit, single address field format is used for branch control logic. What is the minimum size of the control word and control address register?

(a) 125, 7
(b) 125, 10
(c) 135, 9
(d) 135, 10

GATE CSE – 2005 MCQ (2-MARKS)

Q6. The following data path of a CPU

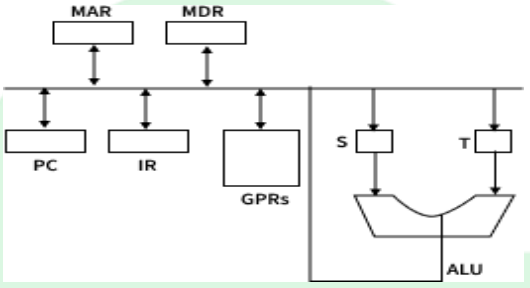


The ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation—the first one for loading addresses in the MAR and the next one for loading data from the memory bus into the MDR.
The instruction "add R0, R1" has the register transfer interpretation
R0 ← R0 + R1.
The minimum number of cycles needed for execution cycle of this instruction is

(a) 2
(b) 3
(c) 4
(d) 5

GATE CSE – 2005 MCQ (2-MARKS)

Q7. Consider the following data path of a CPU.



The ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation - the first one for loading address in the MAR and the next one for loading data from the memory bus into the MDR.
The instruction "call Rn, sub" is a two word instruction. Assuming that PC is incremented during the fetch cycle of the first word of the instruction, its register transfer interpretation is

$$Rn \leftarrow PC + 1$$
$$PC \leftarrow M[PC]$$

The minimum number of CPU clock cycles needed during the execution cycle of this instruction is:

(a) 2
(b) 3
(c) 4
(d) 5

GATE CSE – 2005 MCQ (2-MARKS)

Q8. An instruction set of a processor has 125 signals which can be divided into 5 groups of mutually exclusive signals as follows:

Group 1: 20 signals,
Group 2: 70 signals,
Group 3: 2 signals,
Group 4: 10 signals,
Group 5: 23 signals.

How many bits of the control words can be saved by using vertical microprogramming over horizontal microprogramming?

(a) 0
(b) 103
(c) 22
(d) 55

GATE IT – 2005 MCQ (2-MARKS)

Q9. A hardwired CPU uses 10 control signals S_1 to S_{10}, in various time steps T_1 to T_5, to implement 4 instructions I_1 to I_4 as shown below:

	T_1	T_2	T_3	T_4	T_5
I_1	S_1, S_3, S_5	S_2, S_4, S_6	S_1, S_7	S_{10}	S_3, S_8
I_2	S_1, S_3, S_5	S_8, S_9, S_{10}	$S_5, S_6 S_7$	S_6	S_{10}
I_3	S_1, S_3, S_5	S_7, S_8, S_{10}	S_2, S_6, S_9	S_{10}	S_1, S_3
I_4	S_1, S_3, S_5	S_2, S_6, S_7	S_5, S_{10}	S_6, S_9	S_{10}

Which of the following pairs of expressions represent the circuit for generating control signals S_5 and S_{10} respectively?

($(I_j + I_k)T_n$ indicates that the control signal should be generated in time step T_n if the instruction being executed is I_j or I_k)

(a) $S_5 = T_1 + I_2 \cdot T_3$ and

$S_{10} = (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$

(b) $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and

$S_{10} = (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$

(c) $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and

$S_{10} = (I_2 + I_3 + I_4) \cdot T_2 + (I_1 + I_3) \cdot T_4$

$+(I_2 + I_4) \cdot T_5$

(d) $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and

$S_{10} = (I_2 + I_3) \cdot T_2 + I_4 \cdot T_3 + (I_1$

$+I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$

ANSWER KEY

1. (a, b, c)	**2. (c)**	**3.(d)**
4. (d)	**5. (d)**	**6. (b)**
7. (b)	**8. (b)**	**9. (d)**

HINTS & SOLUTIONS

1. ANSWER: - (A, B, C)
We can choose any input from 1 MUX. and we have different select lines for MUX A & MUX (b) Hence we can have both registers as input, 1 register 1 immediate value as input, or both immediate values as input.
hence, there is no compulsion of 1 input as immediate value and other from register hence option D is false.

2. ANSWER: - (C)
3^{rd} followed by 5^{th} are instruction fetch cycle micro-operations and can be elaborated as follows:

$t_1: MAR_w \leftarrow PC_r \;\; t_2: MDR_w \leftarrow Memory \;\; t_2: IR_w \leftarrow MDR_r$

Now we need to perform Execute cycle micro operations. Just observe the figure and it will be very easy to identify the sequence between

$1^{th}, 2^{nd}, 4^{th}$

2^{nd} is clearly stating that we need to move R1 content to some temporary register named as TEMP1 and it is very clear that before performing ALU operation we need the content in TEMP1. Hence 2^{nd} will be performed next after 5^{th}.

$$TEMP_w \leftarrow R1_r$$

Now we can perform ALU operation and can take second operand directly from R2 and the figure clearly shows us that we need to put the result of ALU back into TEMP2.

All these steps are performed in 1^{st}. So 1^{st} will be next.

$$TEMP2_w \leftarrow TEMP1_r + ALU_w R2_r$$

Lastly we need to put the result present in TEMP2 into R0. This step is performed by 4^{th}.

$RO_w <-$ TEMP2 +

3. ANSWER: (D)

I:

RISC uses register-to-register operations

II:

RISC uses fixed-length instructions

III:

RISC uses hardwired control

Answer:

I, II and III (Option D)

4. ANSWER: - (D)

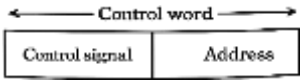
MBR ← PC → current PC is saved
MAR ← X → some memory location X is selected (typically stack or fixed location)
PC ← Y → PC is loaded with a new address (ISR start address)
Memory ← MBR → old PC is stored in memory

This exactly matches interrupt handling

5. Answer : (D)

Control word format consists of:

- Control signals field
- Address field



Step 1: Calculate number of control addresses

Number of control addresses = Number of instructions × Number of clock cycles
= 140 × 7 = 980

Step 2: Calculate size of Control Address Register (CAR)

Number of bits required = $\lceil \log_2(980) \rceil$ = 10 bits

Step 3: Control signals size

Since horizontal microprogramming is used:
Number of control signals = 125 bits

Step 4: Control word size

Control word size = Control signals + Address field
= 125 + 10 = 135 bits

Final Answer:

- Control word size = 135 bits
- Control Address Register (CAR) = 10 bits

Correct Option: (d) 135, 10

6. ANSWER: - (B)

Instruction fetch requires two cycles, but the question asks for the execution part only!
Now for execution:

1.	$R1_{out}$, S_{in}	S ← R0	– 1st cycle
2.	$R2_{out}$, T_{in}	T ← R1	– 2nd cycle
3.	S_{out}, T_{out}, Add $R0_{in}$	R0 ← R0 + R1	– 3rd cycle

So, 3 cycles for execution.
As it is asked for only execution cycles, no of cycles required = 3.
Had it been asked for instruction cycles, then the answer would be 5.
Hence, option B is correct.

7. **ANSWER**: (B)

Rn <= Pc+1

Pc<= M[Pc]

The sequence of instruction take place

I Cycle : PC_{out} , S_{in} , MAR_{in} (MAR can be loaded with PC_{out}

II Cycle: S_{out} , ALU increment , $R_{n\ in}$

III Cycle: MDR_{out} , PC_{in} ,MDR_{out} can be performed once MAR_{in} has been performed) therefore 3 cycle required

8. **Answer: (B)**

In horizontal form, each control signal needs **1 bit**.
So, total bits required:
= 20 + 70 + 2 + 10 + 23
= **125 bits**
In vertical form, each group is encoded using **$\log_2(n)$** bits:

- Group 1 (20 signals) → $\log_2(20) \approx 5$ bits
- Group 2 (70 signals) → $\log_2(70) \approx 7$ bits
- Group 3 (2 signals) → $\log_2(2) = 1$ bit
- Group 4 (10 signals) → $\log_2(10) \approx 4$ bits
- Group 5 (23 signals) → $\log_2(23) \approx 5$ bits

Total bits required:
= 5 + 7 + 1 + 4 + 5
= **22 bits**

Bits saved = Horizontal − Vertical
= 125 − 22
= **103 bits**

9. **Answer: (D)**

First, determine when control signal **S5** is active:

- For instruction **I1**, S5 is present at **T1** → T1·I1
- For instruction **I2**, S5 is present at **T1 and T3** → T1·I2 + T3·I2
- For instruction **I3**, S5 is present at **T1 only** → T1·I3
- For instruction **I4**, S5 is present at **T1 and T3** → T1·I4 + T3·I4

Combining all terms:
S5 = T1·I1 + T1·I2 + T3·I2 + T1·I3 + T1·I4 + T3·I4
= T1(I1 + I2 + I3 + I4) + T3(I2 + I4)
Since (I1 + I2 + I3 + I4 = 1):
S5 = T1 + T3(I2 + I4)
Now, determine when control signal **S10** is active:

- For instruction **I1**, S10 is present at **T4** → T4·I1
- For instruction **I2**, S10 is present at **T2 and T5** → T2·I2 + T5·I2
- For instruction **I3**, S10 is present at **T2 and T4** → T2·I3 + T4·I3
- For instruction **I4**, S10 is present at **T3 and T5** → T3·I4 + T5·I4

Combining all terms:
S10 = T4·I1 + T2·I2 + T5·I2 + T2·I3 + T4·I3 + T3·I4 + T5·I4
= T4(I1 + I3) + T2(I2 + I3) + T5(I2 + I4) + T3·I4

Final Expressions:
S5 = T1 + T3(I2 + I4)
S10 = T4(I1 + I3) + T2(I2 + I3) + T5(I2 + I4) + T3·I4

5 CACHE MEMORY

GATE CSE – 2026 MCQ SET – 1 (2-MARKS)

Q1. The size of the physical address space of a processor is 2^{32} bytes. The capacity of a cache memory unit is 2^{23} bytes. The cache block size is 128 bytes. The cache memory unit can be built as a direct mapped cache or as a K-way set-associative cache, where $K = 2^L$ and $L \in \{1, 2, 3\}$. Let the length of the TAG field be M bits for the direct mapped cache, and N bits for the set-associative cache.

Which one of the following options is true?

(a) $N = M + L$
(b) $N = M - L$
(c) $N = M + K$
(d) $N = M – K$

GATE CSE – 2026 MSQ SET – 2 (2-MARKS)

Q2. Consider a system with a processor and a 4 KB direct mapped cache with block size of 16 bytes. The system has a 16 MB physical memory. Four words P, Q, R, and S are accessed by the processor in the same order 10 times. That is, there are a total of 40 memory references in the sequence P, Q, R, S, P, Q, R, S,... Assume that the cache memory is initially empty. The physical addresses of the words are given below (1 word = 1 byte).

P: 0X845B32, Q: 0X845B26, R: 0X845B36, S: 0X846B32
Which of the following statements is/are true?
Note: $1\text{ K} = 2^{10}$ and $1\text{M} = 2^{20}$

(a) Every access to P results in a cache miss
(b) Every access to R results in a cache hit
(c) Every access to Q results in a cache miss
(d) Except the first access to S, all subsequent accesses to S result in cache hits

GATE CSE – 2026 NAT SET – 2 (2-MARKS)

Q3. Consider a system with 1 MB physical memory and a word length of 1 byte. The system uses a direct mapped cache, with block numbers starting from 0. The word with physical address 0XA2C28 is mapped to the cache block number $(176)_{10}$. The maximum possible size of the cache (in KB) for this configuration is _____. (answer in integer)
Note: $1\text{K} = 2^{10}$ and $1\text{M} = 2^{20}$

GATE CSE – 2025 NAT SET – 2 (2-MARKS)

Q4. Given a computing system with two levels of cache (L1 and L2) and a main memory. The first level (L1) cache access time is 1 nanosecond (ns) and the "hit rate" for L1 cache is 90% while the processor is accessing the data from L1 cache. Whereas, for the second level (L2) cache, the "hit rate" is 80% and the "miss penalty" for transferring data from L2 cache to L1 cache is 10 ns. The "miss penalty" for the data to be transferred from main memory to L2 cache is 100 ns.
Then the average memory access time in this system in nanoseconds is _____. (rounded off to one decimal place)

GATE CSE – 2025 MCQ SET – 2 (2-MARKS)

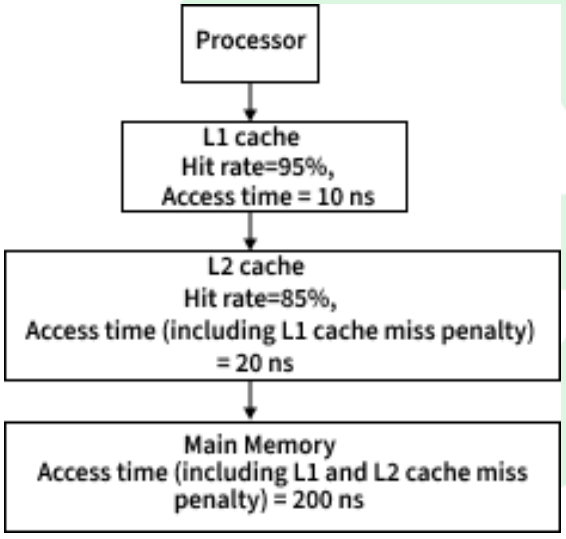
Q5. For a direct-mapped cache, 4 bits are used for the tag field and 12 bits are used to index into a cache block. The size of each cache block is one byte. Assume that there is no other information stored for each cache block.
Which ONE of the following is the CORRECT option for the sizes of the main memory and the cache memory in this system (byte addressable), respectively?

(a) 64 KB and 4 KB
(b) 128 KB and 16 KB
(c) 64 KB and 8 KB
(d) 128 KB and 6 KB

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2025 NAT SET – 1 (2-MARKS)

Q6. A computer has a memory hierarchy consisting of two-level cache (L1 and L2) and a main memory. If the processor needs to access data from memory, it first looks into L1 cache. If the data is not found in L1 cache, it goes to L2 cache. If it fails to get the data from L2 cache, it goes to main memory, where the data is definitely available. Hit rates and access times of various memory units are shown in the figure. The average memory access time in nanoseconds (ns) is _____. (rounded off to two decimal places)



GATE CSE – 2025 MCQ SET – 1 (2-MARKS)

Q7. Consider a memory system with 1M bytes of main memory and 16K bytes of cache memory. Assume that the processor generates a 20-bit memory address, and the cache block size is 16 bytes. If the cache uses direct mapping, how many bits will be required to store all the tag values? [Assume memory is byte addressable, $1K = 2^{10}$, $1M = 2^{20}$.]

(a) 6×2^{10}
(b) 8×2^{10}
(c) 2^{12}
(d) 2^{14}

GATE CSE – 2024 NAT SET-1 (2-MARKS)

Q8. A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (rounded off to two decimal places) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

GATE CSE – 2024 MSQ SET – 1 (2-MARKS)

Q9. Consider two set-associative cache memory architectures: WBC, which uses the write back policy, and WTC, which uses the write through policy. Both of them use the LRU (Least Recently Used) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are TRUE?

(a) A read miss in WBC never evicts a dirty block
(b) A read miss in WTC never triggers a write back operation of a cache block to main memory
(c) A write hit in WBC can modify the value of the dirty bit of a cache block
(d) A write miss in WTC always writes the victim cache block to main memory before loading the missed block to the cache

GATE CSE – 2023 NAT (2-MARKS)

Q10. An 8-way set associative cache of size 64 KB (1 KB = 1024 bytes) is used in a system with a 32-bit address. The address is sub-divided into TAG, INDEX, and BLOCK OFFSET.
The number of bits in the TAG is ____.

GATE CSE – 2022 MSQ (2-MARKS)

Q11. Consider a system with a 2 KB direct-mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRS…). Hence, there are 40 accesses to

the data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0XA248, 0XC28A, 0XCA8A, and 0XA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

(a) Every access to S is a hit.
(b) Once P is brought to the cache it is never evicted.
(c) At the end of the execution-only R and S reside in the cache.
(d) Every access to R evicts Q from the cache.

GATE CSE – 2022 NAT (1-MARKS)

Q12. A cache memory that has a hit rate of 0.8 has an access latency 10 ns and miss penalty 100 ns. Optimization is done on the cache to reduce the miss rate. However, the optimization results in an increase of cache access latency to 15 ns, whereas the miss penalty is not affected The minimum hit rate (rounded off to two decimal places) needed after the optimization such that it should not increase the average memory access time is_____.

GATE CSE – 2022 MSQ (1-MARKS)

Q13. Let WB and WT be two set associative cache organizations that use LRU algorithm for cache block replacement. WB is a write back cache and WT is a write through cache. Which of the following statements is/are FALSE?

(a) Each cache block in WB and WT has a dirty bit.
(b) Every write hit in WB leads to a data transfer from cache to main memory.
(c) Eviction of a block from WT will not lead to data transfer from cache to main memory.
(d) A read miss in WB will never lead to eviction of a dirty block from WB.

GATE CSE – 2021 MCQ SET – 2 (2-MARKS)

Q14. Assume a two-level inclusive cache hierarchy, L1 and L2, where L2 is the larger of the two. Consider the following statements.
S1: Read misses in a write through L1 cache do not result in writebacks of dirty lines to the L2
S2: Write allocate policy must be used in conjunction with write through caches and no-write allocate policy is used with writeback caches.
Which of the following statements is correct?

(a) S1 is true and S2 is false
(b) S1 is false and S2 is true
(c) S1 is true and S2 is true
(d) S1 is false and S2 is false

GATE CSE – 2021 NAT SET- 2 (1-MARKS)

Q15. Consider a set-associative cache of size 2KB (1KB=2^{10} bytes) with cache block size of 64 bytes. Assume that the cache is byte-addressable and a 32 - bit address is used for accessing the cache. If the width of the tag field is 22 bits, the associativity of the cache is _____.

GATE CSE – 2021 NAT SET-1 (1-MARKS)

Q16. Consider a computer system with a byte-addressable primary memory of size 2^{32} bytes. Assume the computer system has a direct-mapped cache of size 32 KB (1 KB = 2^{10} bytes), and each cache block is of size 64 bytes.
The size of the tag field is _____bits.

GATE CSE – 2020 MCQ (2-MARKS)

Q17. A computer system with a word length of 32 bits has a 16 MB byte- addressable main memory and a 64 KB, 4-way set associative cache memory with a block size of 256 bytes. Consider the following four physical addresses represented in hexadecimal notation.
A1=0x42C8A4,
A2=0x546888,
A3=0x6A289C,
A4=0x5E4880
Which one of the following is TRUE?

(a) A1 and A4 are mapped to different cache sets
(b) A2 and A3 are mapped to the same cache set.
(c) A3 and A4 are mapped to the same cache set.
(d) A1 and A3 are mapped to the same cache set.

GATE CSE – 2020 NAT (1-MARKS)

Q18. A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. The average memory access time in ns (round off to 1 decimal place) is_____.

GATE CSE – 2019 NAT (2-MARKS)

Q19. A certain processor deploys a single-level cache. The cache block size is 8 words and the word size is 4 bytes. The memory system uses a 60-MHz clock. To service a cache-miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is _____$\times 10^6$ bytes/sec.

GATE CSE – 2019 MCQ (1-MARKS)

Q20. A certain processor uses a fully associative cache of size 16 kB. The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the Tag and the Index fields respectively in the addresses generated by the processor?

(a) 24 bits and 0 bits
(b) 28 bits and 4 bits
(c) 24 bits and 4 bits
(d) 28 bits and 0 bits

GATE CSE – 2018 MCQ (2-MARKS)

Q21. The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block is 2^M words. For a K-way set-associative cache memory, the length (in number of bits) of the tag field is

(a) $P - N - \log_2 K$
(b) $P - N + \log_2 K$
(c) $P - N - M - W - \log_2 K$
(d) $P - N - M - W + \log_2 K$

GATE CSE – 2017 NAT SET-2 (2-MARKS)

Q22. Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____.

GATE CSE – 2017 NAT SET-2 (2-MARKS)

Q23. The read access times and the hit ratios for different caches in a memory hierarchy are as given below.

Cache	Read access time (in nanoseconds)	Ht ratio
I-cache	2	0.8
D-cache	2	0.9
L2-cache	8	0.9

The read access time of main memory is 90 nanoseconds. Assume that the caches use the referred word-first read policy and the write back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is_____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2017 MCQ SET-2 (2 MARKS)

Q24. In a two-level cache system, the access times of L_1 and L_2 1 and 8 clock cycles, respectively. The miss penalty from the L_2 cache to main memory is 18 clock cycles. The miss rate of L_1 cache is twice that of L_2. The average memory access time(AMAT) of this cache system is 2 cycles. The miss rates of L_1 and L_2 respectively are:

(a) 0.111 and 0.056
(b) 0.056 and 0.111
(c) 0.0892 and 0.1784
(d) 0.1784 and 0.0892

GATE CSE – 2017 NAT SET-1 (2-MARKS)

Q25. A cache memory unit with capacity of N words and block size of B words is to be designed. If it is designed as a direct mapped cache, the length of the TAG field is 10 bits. If the cache unit is now designed as a 16-way set-associative cache, the length of the TAG field is _____bits.

GATE CSE – 2017 NAT SET-1 (2-MARKS)

Q26. Consider a 2-way set associative cache with 256 blocks and use LRU replacement, initially the cache is empty. Conflict misses are those misses which occur due to the contention of multiple blocks for the same cache set. Compulsory misses occur due to first time access to the block. The following sequence of accesses to memory blocks.

{0,128,256,128,0,128,256,128,1,129,257,129,1,129,257,129}

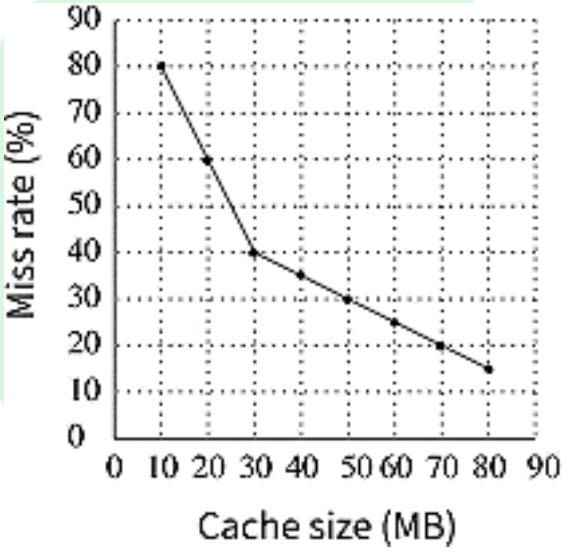
is repeated 10 times. The number of conflict misses experienced by the cache is _____.

GATE CSE – 2017 NAT SET-1 (1-MARKS)

Q27. Consider a two-level cache hierarchy with L1 and L2 caches. An application incurs 1.4 memory accesses per instruction on average. For this application, the miss rate of L1 cache 0.1, the L2 cache experiences, on average, 7 misses per 1000 instructions. The miss rate of L2 expressed correct to two decimal places is _____.

GATE CSE – 2016 NAT SET-2 (2-MARKS)

Q28. A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is _____ MB.

GATE CSE – 2016 NAT SET-2 (2-MARKS)

Q29. The width of the physical address on a machine is 40 bits. The width of the tag field in a 512KB 8-way set associative cache is _____bits.

GATE CSE – 2015 MCQ SET-3 (1-MARKS)

Q30. Consider a machine with a byte addressable main memory of 2^{20} bytes, block size of 16 bytes and a direct mapped cache having 2^{12} cache lines. Let the addresses of two consecutive bytes in main memory be $(E201F)_{16}$ and $(E2020)_{16}$. What are the tag and

cache line address (in hex) for main memory address $(E201F)_{16}$?

(a) E, 201
(b) F, 201
(c) E, E20
(d) 2, 01F

GATE CSE – 2015 NAT SET-2 (1-MARKS)

Q31. Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

GATE CSE – 2014 NAT SET 3 (2-MARKS)

Q32. The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is _____

GATE CSE – 2014 MCQ SET – 2 (2-MARKS)

Q33. If the associativity of a processor cache is doubled while keeping the capacity and block size unchanged, which one of the following is guaranteed to be NOT affected?

(a) Width of tag comparator
(b) Width of set index decoder
(c) Width of way selection multiplexor
(d) Width of processor to main memory data bus

GATE CSE – 2014 MCQ SET – 2 (MARKS)

Q34. In designing a computer's cache system, the cache block (or cache line) size is an important parameter. Which one of the following statements is correct in this context?

(a) A smaller block size implies better spatial locality
(b) A smaller block size implies a smaller cache tag and hence lower cache tag overhead
(c) A smaller block size implies a larger cache tag and hence lower cache hit time
(d) A smaller block size incurs a lower cache miss penalty

GATE CSE – 2014 NAT SET 2 (1-MARKS)

Q35. A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is _____.

GATE CSE – 2014 NAT SET 2 (2-MARKS)

Q36. Consider a main memory system that consists of 8 memory modules attached to the system bus, which is one word wide. When a write request is made, the bus is occupied for 100 nanoseconds (ns) by the data, address, and control signals. During the same 100 ns, and for 500 ns thereafter, the addressed memory module executes one cycle accepting and storing the data. The (internal) operation of different memory modules may overlap in time, but only one request can be on the bus at any time. The maximum number of stores (of one word each) that can be initiated in 1 millisecond is _____.

GATE CSE – 2014 MCQ SET – 1 (2-MARKS)

Q37. An access sequence of cache block addresses is of length N and contains n unique block addresses. The number of unique block addresses between two consecutive accesses to the same block address is bounded above K. What is the miss ratio if the access sequence is passed through a cache of associativity A≥k exercising least-recently-used

replacement policy?

a) n/N
b)1/N
c)1/A
d)k/n

GATE CSE – 2013 MCQ (1-MARKS)

Q38. In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from

(a) (j mod v) *k to (j mod v) *k + (k-1)
(b) (j mod v) to (j mod v) +(k-1)
(c) (j mod k) to (j mod k) +(v-1)
(d) (j mod k) *v to (j mod k) *v + (v-1)

GATE CSE – 2012 MCQ (2-MARKS)

Q39. A computer has a 256 -KByte, 4 -way set associative, write back data cache with block size of 32 Bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.
The size of the cache tag directory is:

(a) 160 Kbits
(b) 136 Kbits
(c) 40 Kbits
(d) 32 Kbits

GATE CSE – 2012 MCQ (2-MARKS)

Q40. A computer has a 256-KByte, 4-way set associative, write back data cache with block size of 32-Bytes. The processor sends 32-bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.
The number of bits in the tag field of an address is

a) 11
b) 14
c) 16
d) 27

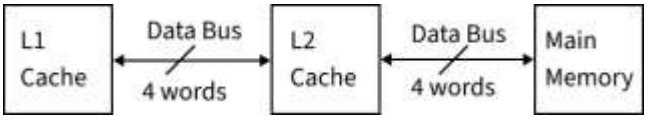
GATE CSE – 2011 MCQ (2-MARKS)

Q41. An 8KB direct mapped write-back cache is organized as multiple blocks, each of size 32-bytes. The processor generates 32-bit addresses. The cache controller maintains the tag information for each cache block consisting of the following.
1 Valid bit, 1 Modified bit as many bits as the minimum needed to identify the memory block mapped in the cache.
What is the total size of memory needed at the cache controller to store metadata (tags) for the cache?

(a) 4864 bits
(b) 6144 bits
(c) 6656 bits
(d) 5376 bits

GATE CSE – 2010 MCQ (2-MARKS)

Q42. A computer system has an L1 cache, an L2 cache, and a main memory unit connected as shown below. The block size in L1 cache is 4 words. The block size in L2 cache is 16 words. The memory access times are 2 nanoseconds. 20 nanoseconds and 200 nanoseconds for L1 cache, L2 cache and main memory unit respectively.

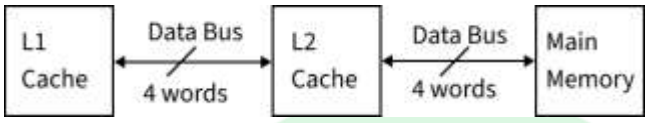


When there is a miss in both $L1$ cache and $L2$ cache, first a block is transferred from main memory to $L2$ cache, and then a block is transferred from $L2$ cache to $L1$ cache. What is the total time taken for these transfers?

(a) 222 nanoseconds
(b) 888 nanoseconds
(c) 902 nanoseconds
(d) 968 nanoseconds

GATE CSE – 2010 MCQ (2-MARKS)

Q43. A computer system has an L1 cache, an L2 cache, and a main memory unit connected as shown below. The block size in L1 cache is 4 words. The block size in L2 cache is 16 words. The memory access times are 2 nanoseconds. 20 nanoseconds and 200 nanoseconds for L1 cache, L2 cache and main memory unit respectively.



When there is a miss in L1 cache and a hit in L2 cache, a block is transferred from L2 cache to L1 cache. What is the time taken for this transfer?

(a) 2 nanoseconds
(b) 20 nanoseconds
(c) 22 nanoseconds
(d) 88 nanoseconds

GATE CSE – 2009 MCQ (2-MARKS)

Q44 Consider a 4-way set associative cache (initially empty) with total 16 cache blocks. The main memory consists of 256 blocks and the request for memory blocks are in the following order:

0,255,1,4,3,8,133,159,216,129,63,8,48,32,73,92,155.

Which one of the following memory block will NOT be in cache if LRU replacement policy is used?

(a) 3
(b) 8
(c) 129
(d) 216

GATE CSE – 2008 MCQ (2- MARKS)

Q45. For inclusion to hold between two cache levels $\mathbf{L_1}$ and $\mathbf{L_2}$ in a multilevel cache hierarchy, which of the following are necessary?

1. L_1 must be a write-through cache

2. L_2 must be a write-through cache

3. The associativity of L_2 must be greater than that of L_1

4. The L_2 cache must be at least as large as the L_1 cache

(a) 4 only
(b) 1 and 2 only
(c) 1, 2 and 4 only
(d) 1, 2, 3 and 4

GATE CSE – 2008 MCQ (2-MARKS)

Q46 Consider a machine with a 2 -way set associative data cache of size 64 Kbytes and block size 16 bytes. The cache is managed using 32 bit virtual addresses and the page size is 4 Kbytes . A program to be run on this machine begins as follows:

```
double ARR[1024][1024];
int i, j;

for(i = 0; i < 1024; i++)
        for(j= 0; j < 1024; j++)
        ARR[i][j] = 0.0;
```

The size of double is 8Bytes. Array ARR is located in memory starting at the beginning of virtual page 0 xFF000 and stored in row major order. The cache is initially empty and no pre-fetching is done. The only data memory references made by the program are those to array ARR.
The total size of the tags in the cache directory is:

(a) 32 Kbits
(b) 34 Kbits
(c) 64 Kbits
(d) 68 Kbits

GATE CSE – 2008 MCQ (2-MARKS)

Q47. Consider a machine with a 2 -way set associative data cache of size 64 Kbytes and block size 16 bytes. The cache is managed using 32 bit virtual addresses and the page size is 4 Kbytes . A program to be run on this machine begins as follows:

```
double ARR[1024][1024];
int i, j;

for(i = 0; i < 1024; i++)
      for(j= 0; j < 1024; j++)
    ARR[i][j] = 0.0;
```

The size of double is 8Bytes. Array ARR is located in memory starting at the beginning of virtual page 0 xFF000 and stored in row major order. The cache is initially empty and no pre-fetching is done. The only data memory references made by the program are those to array ARR.
Which of the following array elements have the same cache index as ARR[0][0] ?

(a) ARR[0][4]
(b) ARR[4][0]
(c) ARR[0][5]
(d) ARR[5][0]

GATE CSE – 2008 MCQ (2-MARKS)

Q48. Consider a machine with a 2 -way set associative data cache of size 64 Kbytes and block size 16 bytes. The cache is managed using 32 bit virtual addresses and the page size is 4 Kbytes . A program to be run on this machine begins as follows:

```
double ARR[1024][1024];
int i, j;
for(i = 0; i < 1024; i++)
   for( = 0; j < 1024; j++)
       ARR[i][j] = 0.0;
```

The size of double is 8 bytes. Array ARR is located in memory starting at the beginning of virtual page 0 xFF000 and stored in row major order. The cache is initially empty and no pre-fetching is done. The only data memory references made by the program are those to array ARR.
The cache hit ratio for this initialization loop is:

(a) 0%
(b) 25%
(c) 50%
(d) 75%

GATE CSE – 2008 MCQ (2-MARKS)

Q49. Consider a computer with a 4-ways set-associative mapped cache of the following characteristics: a total of 1 MB of main memory, a word size of 1 byte, a block size of 128 words and a cache size of 8 KB.

The number of bits in the TAG, SET and WORD fields, respectively are:

(a) 7, 6, 7
(b) 8, 5, 7
(c) 8, 6, 6
(d) 9, 4, 7

GATE CSE – 2008 MCQ (2-MARKS)

Q50. Consider a computer with a 4-ways set-associative mapped cache of the following characteristics: a total of 1 MB of main memory, a word size of 1 byte, a block size of 128 words and a cache size of 8 KB. While accessing the memory location 0C795H by the CPU, the contents of the TAG field of the corresponding cache line is:

(a) 000011000
(b) 110001111
(c) 00011000
(d) 110010101

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2007 MCQ (2-MARKS)

Q51. Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50×50 two-dimensional array of bytes is stored in the main memory starting from memory location 1100 H . Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.
Which of the following lines of the data cache will be replaced by new blocks in accessing the array for the second time?

(a) line 4 to line 11
(b) line 4 to line 12
(c) line 0 to line 7
(d) line 0 to line 8

GATE CSE – 2007 MCQ (1-MARKS)

Q52. Consider a 4-way set associative cache consisting of 128 lines with a line size of 64 words. The CPU generates a 20-bit address of a word in main memory. The numbers of bits in the TAG, SET, and WORD fields are respectively:

(a) 9, 6, 5
(b) 7, 7, 6
(c) 7, 5, 8
(d) 9, 5, 6

GATE CSE – 2007 MCQ (2-MARKS)

Q53. Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50×50 two-dimensional array of 1 byte is stored in the main memory starting from memory location 1100H. The data cache is initially empty.
The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.
How many data cache misses will occur in total?

(a) 48
(b) 50
(c) 56
(d) 59

GATE CSE – 2006 MCQ (2-MARKS)

Q54. A CPU has a cache with block size 64 bytes. The main memory has k banks, each bank being c bytes wide. Consecutive c-byte chunks are mapped on consecutive banks, with wrap-around All the k banks can be accessed in parallel, but two accesses to the same bank must be serialized. A cache block access may involve multiple iterations of parallel bank accesses depending on the amount of data obtained by accessing all the k banks in parallel. Each iteration requires decoding the bank numbers to be accessed in parallel and this takes k/2 ns. The latency of one bank access is 80 ns.
If c = 2 and k = 24, then the latency of retrieving a cache block starting at address zero from main memory is:

(a) 92 ns
(b) 104 ns
(c) 172 ns
(d) 184 ns

GATE CSE – 2006 MCQ (2-MARKS)

Q55. Consider two cache organizations: The first one is a 32 KB 2-way set associative with 32-byte block size. The second one is of the same size but directly mapped. The size of an address is 32 bits in both cases. A 2-to-1 multiplexer has latency of 0.6 ns while a k-bit comparator has a latency of k/10 ns. The hit latency of the set associative organization is h_1 while that of the direct mapped one is h_2.
The value of h_1 is :

(a) 2.4 ns
(b) 2.3 ns
(c) 1.8 ns
(d) 1.7 ns

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2006 MCQ (2-MARKS)

Q56. Consider two cache organizations: The first one is a 32 KB 2-way set associative with 32-byte block size. The second one is of the same size but directly mapped. The size of an address is 32 bits in both cases. A 2-to-1 multiplexer has latency of 0.6 ns while a k-bit comparator has a latency of k/10 ns. The hit latency of the set associative organization is h_1 while that of the direct mapped one is h_2.
The value of h_2 is :

(a) 2.4 ns
(b) 2.3 ns
(c) 1.8 ns
(d) 1.7 ns

GATE CSE – 2006 MCQ (2-MARKS)

Q57. A CPU has a $32KB$ direct mapped cache with 128 byte - block size . Suppose A is two dimensional array of size 512×512 with elements that occupy 8 - bytes each . Consider the following two C code segments , $P1$ and $P2$.

P1 :

```
for(i=0; i<512  ; i++)

{
   for (j=0;j<512; j++)
   }
      x +=A[i][j]
   }
```

P2:

```
for(i=0; i<512  ; i++)

{

for ( j=0:j<512:j++ )
   {
      x +=A[j][i]
   1
```

$P1$ and $P2$ are executed independently with the same initial state , namely , the array A is not in the cache and i, j , x are in registers . Let the number of cache misses experienced by $P1$ be $M1$ and that for $P2$ be $M2$.
The value of the ratio $\frac{M1}{M2}$:

(a) 0
(b) $\frac{1}{16}$
(c) $\frac{1}{8}$
(d) 16

GATE CSE – 2006 MCQ (2-MARKS)

Q58. A CPU has a $32KB$ direct mapped cache with 128 byte - block size . Suppose A is two dimensional array of size 512×512 with elements that occupy 8 - bytes each . Consider the following two C code segments , $P1$ and $P2$.

P1 :

```
for(i=0; i<512  ; i++)

{
   for (j=0;j<512; j++)
 }
      x +=A[i][j]
 }
```

P2 :

```
for(i=0; i<512  ; i++)

{

for ( j=0:j<512:j++ )
   {
      x +=A[j][i]
   1
```

$P1$ and $P2$ are executed independently with the same initial state , namely , the array A is not in the cache and i, j , x are in registers . Let the number of cache misses experienced by $P1$ be $M1$ and that for $P2$ be $M2$.
The value of M_1 :

(a) 0
(b) 2048
(c) 16384
(d) 262144

GATE CSE – 2005 MCQ (2-MARKS)

Q59. Consider a direct mapped cache of size 32 KB with block size 32 bytes. The CPU generates 32-bit addresses. The number of bits needed for cache indexing and the number of tag bits are respectively:

(a) 10, 17
(b) 10, 22
(c) 15, 17
(d) 5, 17

GATE IT – 2005 MCQ (2-MARKS)

Q60. Consider a 2-way set associative cache memory with 4 sets and total 8 cache blocks (0–7) and main memory with 128 blocks (0–127). Which memory blocks will be present in the cache after the following sequence of memory block references if LRU policy is used for cache block replacement? If initially the cache did not have any memory block from the current job?

0 5 3 9 7 0 16 55

(a) 0 3 5 7 16 55
(b) 0 3 5 7 9 16 55
(c) 0 5 7 9 16 55
(d) 3 5 7 9 16 55

GATE IT – 2005 MCQ (2-MARKS)

Q61. A computer system has a level-1 instruction cache (I-cache), a level-1 data cache (D-cache) and a level-2 cache (L2-cache) with the following specifications:

Cache	Capacity	Mapping method	Block size
I-cache	4K words	Direct mapping	4 Words
D-cache	4K words	2-way set-associative mapping	4 Words
L2-cache	64K words	4-way set-associative mapping	16 Words

Capacity Mapping method Block size I-cache 4K words Direct mapping 4 Words D-cache 4K words 2-way set-associative mapping 4 Words L2-cache 64K words 4-way set-associative mapping 16 Words

The length of the physical address of a word in the main memory is 30 bits. The capacity of the tag memory in the I-cache, D-cache and L2-cache is, respectively,

(a) 1K × 18-bit, 1K × 19-bit, 4K × 16-bit
(b) 1K × 16-bit, 1K × 19-bit, 4K × 18-bit
(c) 1K × 16-bit, 512 × 18-bit, 1K × 16-bit
(d) 1K × 18-bit, 512 × 18-bit, 1K × 18-bit

Check Out Our GATE Courses & Secure Your Dream Rank!

ANSWER KEY

1. (a)	**2. (a, b)**	**3. (128)**
4. (4 to 4)	**5. (a)**	**6. (11.85)**
7. (a)	**8. (3)**	**9. (b, c)**
10. (19)	**11. (a, b, d)**	**12. (0.85)**
13. (a, b, d)	**14. (a)**	**15.(2)**
16. (17)	**17. (b)**	**18. (13.3-13.5)**
19. (160)	**20. (d)**	**21. (b)**
22. (18)	**23. (4.72)**	**24. (a)**
25. (14)	**26. (76)**	**27. (0.05)**
28. (30)	**29. (24)**	**30. (a)**
31. (14)	**32. (1.68)**	**33. (d)**
34. (d)	**35. (20)**	**36. (10000)**
37. (a)	**38. (a)**	**39. (a)**
40. (c)	**41. (d)**	**42. (c)**
43. (c)	**44. (d)**	**45. (a)**
46.(d)	**47. (b)**	**48. (c)**
49. (d)	**50. (a)**	**51. (a)**
52. (d)	**53. (c)**	**54. (d)**
55. (a)	**56. (d)**	**57. (b)**
58. (c)	**59. (a)**	**60. (c)**
61. (a)		

HINTS & SOLUTIONS

1. ANSWER:- A

Direct cache address format:

| tag | LO | WO |

m-bit

Set associative cache address format:

| tag | SO | WO |

n-bit

Here, N-value increases depend on the associativity of a cache.

i.e., S = Number of lines / P-way = Number of lines / $2^L = 2^{(LO-L)}$

So, index value decreases w.r.t. P-value. ∴ Tag space increases w.r.t. associativity. Hence, N = (M + L).

2. ANSWER:- (A, B)

CACHE SIZE = 4 KB = 2^{12} B.

CACHE BLOCK SIZE = 16 B = 2^4 B.

NUMBER OF CACHE LINES SINCE IT IS DIRECT MAPPED = CACHE SIZE / CACHE BLOCK SIZE = $2^{12} / 2^4 = 2^8$ = 256 CACHE LINES.

CACHE LINE BITS = log (base 2) 2^8 = 8 bits.

BLOCK OFFSET = log (base 2) 2^4 = 4 bits.

P: 1000 0100 0101 1011 0011 0010

Q: 1000 0100 0101 1011 0010 0110

R: 1000 0100 0101 1011 0011 0110

S: 1000 0100 0110 1011 0011 0010

as you can see . when you access P . since it is initially empty . the entire block is placed in the cache . R is also present in the block .

when you access R eventually it will be hit every time because its block offset is different from P and for Q it is an entirely different line of cache . so it will not be a problem for R .

since P and S have the same cache line bits and offset bits . it will result in tag mismatch every time you access S. It will replace P. So S and P are always a miss .

SO , the answer is : A , B .

Check Out Our GATE Courses & Secure Your Dream Rank!

3. ANSWER:- 128

(A2C28)H : 1010 0010 1100 00 101000

$(176)_{10}$ WO

Here $(176)_{10}$ = 00010110000 - 11 bit

Word offset = 6 bit

∴ Cache address = LO + WO = 11 + 6 = 17 bit

So, Cache size = 2^{17} B = 128 KB

4. ANSWER:- 4 TO 4

To calculate the average memory access time for a computing system with two levels of cache (L1 and L2) and a main memory, we consider the cache hit and miss rates along with their respective penalties. Here's the breakdown:

L1 Cache:

Access time: 1 nanosecond (ns)

Hit rate: 90% or 0.9

L2 Cache:

Hit rate: 80% or 0.8

Miss penalty (L2 to L1): 10 ns

Main Memory:

Miss penalty (Memory to L2): 100 ns

The formula to compute average memory access time (T_avg) is:

$T_avg = H1 \cdot T1 + (1 - H1) \cdot H2 \cdot (T2 + T1) + (1 - H1) \cdot (1 - H2) \cdot (T3 + T2 + T1)$

Where:

H1 is the hit rate for L1 cache.

T1 is the access time for L1 cache.

H2 is the hit rate for L2 cache.

T2 includes the miss penalty for L2 cache.

T3 includes the miss penalty for main memory.

Substituting the given values:

$T_avg = (0.9 \cdot 1) + (1 - 0.9) \cdot 0.8 \cdot (10 + 1) + (1 - 0.9) \cdot (1 - 0.8) \cdot (100 + 10 + 1)$

$= 0.9 + 0.08 \cdot 11 + 0.02 \cdot 111$

$= 0.9 + 0.88 + 2.22$

$= 4$ ns

Thus, the average memory access time is 4 nanoseconds.

Alternate Solution:

$T_1 = 1$ µsec, $H_1 = 90\%$

$T_2 = 10$ µsec, $H_2 = 80\%$

$T_m = 100$ µsec

$(1 - H_1) = 0.1$

$(1 - H_2) = 0.2$

Formula:

$T_avg = T_1 + (1 - H_1)T_2 + (1 - H_1)(1 - H_2)T_m$

$= T_1 + (1 - H_1) [T_2 + (1 - H_2)T_m]$

Substitution:

$= 1 + 0.1 [10 + 0.2(100)]$

$= 1 + 0.1 (10 + 20)$

$= 1 + 0.1 (30)$

$= 1 + 3$

$= 4$ µsec

5. ANSWER:- A

For the direct-mapped cache, 4 bits are used for the tag field, and 12 bits are used to index into a cache block. The block size is 1 byte.

To find the size of the main memory and the cache, we use:

The total number of blocks in the cache is determined by the 12 bits for indexing, which means there are 2^{12} = 4096 blocks.

The size of each block is 1 byte.

So, the cache size is 4096 bytes, or 4 Kb

The address field consists of 4 bits for the tag and 12 bits for the index, making the total address size 16 bits. This implies the main memory is 2^{16} = 65536 bytes, or 64 Kb

Alternate Solution :

Tag bits = 4 bits

Cache has 16 blocks

Block size = 1 Byte = 8 bits

Mapping type = Direct Mapped Cache

1. Address Format:

Address is divided into three parts:

TAG | Line (Index) | Word Offset

TAG = 4 bits

Remaining bits (Line + Offset) = 12 bits

Total address size = 16 bits

2. Cache Size (CM):

Number of blocks = 16 = 2^4

Block size = 1 Byte

Cache size = 2^4 × 1 Byte = 16 Bytes

3. Main Memory Size (MM):
Total address bits = 16
Main memory size = 2^{16} Bytes = 64 KB

4. Direct Mapping Formula:
Number of tags = MM size / CM size
= (64 KB) / (16 B)
= (2^{16}) / (2^4)
= 2^{12}

5. Tag Bits Calculation:
Tag bits = log2(Number of tags)
= log2(2^{12})
= 12 bits

6. General Formula:
Tag bits = log2(MM size / CM size)

7. Additional Examples:
Case 1:
MM size = 64 KB
CM size = 4 KB
Number of tags = 64 KB / 4 KB = 16 = 2^4
Tag bits = 4 bits

Case 2:
MM size = 128 KB
CM size = 1 KB
Number of tags = 128 KB / 1 KB = 128 = 2^7
Tag bits = 7 bits

Final Answer:
Tag bits = 4 bits

6. ANSWER:- 11.85

Given Information:
L1 cache hit rate (H1) = 95% = 0.95
L1 cache access time (T1) = 10 ns
L2 cache hit rate (H2) = 85% = 0.85
L2 cache access time (T2) (including the L1 cache miss penalty) = 20 ns
Main memory access time (T3) (including L1 and L2 cache miss penalties) = 200 ns
Formula for Average Memory Access Time (AMAT):
The formula for the AMAT for this hierarchical memory system is:
AMAT = H1 × T1 + (1 − H1) × H2 × T2 + (1 − H1) × (1 − H2) × T3
Where:
H1 is the hit rate for L1 cache.
T1 is the access time for L1 cache.
H2 is the hit rate for L2 cache.
T2 is the access time for L2 cache (including L1 miss penalty).
T3 is the access time for main memory (including L1 and L2 miss penalties).
Calculation:
Using the given values:
AMAT=0.95 × 10 + (1 − 0.95) × 0.85 × 20 + (1 − 0.95) × (1 − 0.85) × 200
Breaking this down:
1. L1 hit case: 0.95 × 10 = 9.5 ns
2. L1 miss and L2 hit case: (1 − 0.95) × 0.85 × 20 = 0.05 × 0.85 × 20 = 0.85 ns
3. L1 miss and L2 miss case: (1 − 0.95) × (1 − 0.85) × 200 = 0.05 × 0.15 × 200 = 1.5 ns

Now, adding them together:
AMAT = 9.5 + 0.85 + 1.5 = 11.85 ns

7. ANSWER:- A

TAG | line | offset
6 10 4
(20 bits total)
Cache block size = 16B = 2^4 B
lines = Cache Mem Size / block size
= 16KB / 2^4 B
= 2^{14} B / 2^4 B
= 2^{10} lines
TAG = 20 − (10 + 4) = 6
bits required to store all tag values
= TAG bits × # lines
= 6 × 2^{10}

8. ANSWER:- 3

CPI with a Perfect Cache: 2

CPI with the Actual Cache: 2 (ideally) +0.02 × 100 (stall cycles for instruction cache miss) +0.25 × 0.08 × 100 (stall cycles for data cache miss) = 6
Speedup with a perfect cache = 6/2 = 3.

Speedup with a perfect cache = 5.4 / 1.8 = 3.

9. ANSWER:- B,C

In Write-Through Cache WTC, data is immediately written to the main memory on a write, so there are no dirty blocks. A read miss only fetches data without requiring a write-back.
(b) A read miss in WTC never triggers a write back operation of a cache block to main memory is TRUE. In WBC, a write hit updates the cache block and sets

the dirty bit if it wasn't already set, indicating the block has been modified.
Hence, (c) A write hit in WBC can modify the value of the dirty bit of a cache block is also TRUE.

10. ANSWER:- 19

S_1 Given that for set associative mapping technique:
Cache memory(CM's)=64KB=2^{16} bytes
Main memory size=32 bits
P=8 way SAM
Tag bit size (y) =?
Since block size is not given we assume it is 2^x byte.
Number of cache block(N)=Cache memory size/Block size=2^{16-x}
Number of sets(S)=N/P=2^{13-x}
as we know for set associative mapping technique:
TAG+SLO+B'S=MM
y+13-x+x=32 bits
y=19 bits

11. ANSWER:- A, B, D

Physical memory $= 64\ KB$ ===> 16 bits required to represent Physical memory
Cache memory = 2KB ===> 11 bits for cache memory
Block size $= 64\ B = 32$ words ==> 6 bits because the system is Byte addressable.

Tag $= 16 - 11 = 5$ bits
Cache index = 11-6 = 5 bits
Block offset = 6 bits

P = 0 × A248 = 1010001001001000
=1010001001001000(Tag- cache index - Block offset)

Q = 0 × C28A = 1100001010001010 = 110000101000101010

R = 0 × CA8A = 1100101010001010 = 1100101010001010

S = 0 × A262 = 1010001001100010 = 10100010011000010

Given that, Direct mapped cache.
If we observe, P and S belong to the same Block (Tag and cache bits are the same). Therefore every access of S should result in a hit due to neither Q nor R competing for the same cache block and once P brought to the cache, it is never evicted.

If we observe Q and R, those are competing for the same cache block. So at the end R only present in the cache due to R is accessed at last. (Comparing to Q) and every access to R evicts Q from the Cache.

Therefore at the end, P,R and S in the Cache.

Options A,B and D are true.

12. ANSWER:- 0.85

Given:
Cache Hit Rate (h) = 0.8
Cache Access Time (t_n or t_c) = 10 ns
Miss Penalty (MP) = 100 ns
1. Average Access Time Formula:
$T_{avg} = h \times t_c + (1 - h) \times (MP + t_c)$
2. Substitution:
$T_{avg} = 0.8 \times 10 + (1 - 0.8)(100 + 10)$
$= 0.8 \times 10 + 0.2 \times 110$
$= 8 + 22$
$= 30$ ns
Alternative Simplified Formula:
$T_{avg} = t_c + (1 - h) \times MP$
$= 10 + (1 - 0.8) \times 100$
$= 10 + 0.2 \times 100$
$= 10 + 20$
$= 30$ ns
Final Answer:
Average Access Time = 30 ns
3. Optimization Case
Goal: Reduce miss rate (i.e., increase hit rate)
Given:
New cache access time (t_{new}) = 15 ns
Miss penalty (MP) = 100 ns (unchanged)
Target average time (T_{avg}) = 30 ns

Formula:
$T_{avg} = t_{new} + (1 - h_{new}) \times MP$
Substitution:
$30 = 15 + (1 - h_{new}) \times 100$
$30 = 15 + 100 - 100h_new$
$30 = 115 - 100h_new$
Solving:
$100h_new = 115 - 30$
$100h_new = 85$
$h_{new} = 85 / 100$

h_{new} = 0.85
Final Answer:
Required new hit rate = 0.85 (or 85%)

13. ANSWER:- A, B, D

WB Cache: Updates to the cache block don't result in Updates to the Main Memory immediately. Burst Writes are preferred for higher throughput and those needing higher write performance.
WT Cache: Updates to the Cache block are reflected on Main Memory before carrying out other processes. Consistency is preferred

A. WB must necessarily have a dirty bit to avoid redundant writes to Main Memory. Whereas a WT cache needn't as the change is reflected after a write. False

B. WB cache's primary use is to increase throughput or useful work. Multiple writes to the same cache block will not be reflected immediately, avoiding unnecessary data transfer time. False

C. WT cache's main goal is to prefer consistency over write performance, and a cache block is made to reflect the current main memory. True

D. LRU doesn't have specific replacement strategies for dirty and regular blocks. Hence a read miss might evict a dirty block. False

14. ANSWER:- A

S_1 : A Read Miss in a write-through L1 cache results in read allocate. No write back is required here, because in a write-through L1 cache, both L1 and L2 are updated on every write (no dirty blocks are maintained as in a write-back cache). Therefore, during a Read Miss, the missed block is fetched from L2 into L1, possibly replacing an existing block in L1. The replaced block does not require a write back since it is already consistent with L2. Hence, S_1 is TRUE.
S_2 : This statement is FALSE. Both write-through and write-back cache policies can technically use either write-allocate or no-write-allocate on a write miss, although they are commonly paired in a typical manner.
• In write-through policy, no-write-allocate is generally preferred since every write updates both L1 and L2. Allocating the block on a write miss offers little benefit because each write still accesses L2.
• In write-back policy, write-allocate is usually used so that after the first write miss, the block is brought into L1. Subsequent writes can then hit in L1, reducing costly L2 accesses.

15. ANSWER:- 2

Number of lines in Cache
Cache size / Block Size
So number of lines
$2^{11}/2^6$
2^5
No of sets = Number of lines/ Number of lines in one set
Total bits = Tag bits + Set bits + Offset bits
Set bits = 32-22-6 = 4
$2^4 = 2^5$ / Number of lines in one set
On solving we get associativity = 2.

16. ANSWER:- 17

Tag bits = PAS_bits − $\log_2$(Cache Size) + $\log_2$(K) (where K is associativity)
= 32 − 15 + 0 = 17 bits.

17. ANSWER:- B

Total Physical Address Size = 24 bits Main Memory is 16MB = 2^{24} bytes. Therefore, any direct pointer or address is 24 bits long (represented by 6 hexadecimal digits).
Block Offset Field = 8 bits The block size is 256 bytes = 2^8 bytes. This means the lower 8 bits are reserved to locate a specific byte within a cache block.

Set Index Field = 6 bits First, calculate the total number of blocks in the cache:

$$\text{Total Cache Blocks} = \frac{64 \text{ KB Cache Size}}{256 \text{ bytes Block Size}} = \frac{2^{16}}{2^8} = 256 \text{ blocks}$$

Since it is a 4-way set associative cache, we divide these blocks into sets:

$$\text{Total Sets} = \frac{256 \text{ blocks}}{4 \text{ blocks per set}} = 64 \text{ sets} = 2^6 \text{ sets}$$

This requires 6 bits to index into the correct set.

2. Identifying the Target Set for A2 and A3

An address maps to a cache set based entirely on the 6 bits of the Set Index field (bits 8 to 13).

Let's inspect the lower 16 bits (the last 4 hex digits) of both addresses in binary:

- For A2 (0 × 546888): Focus on the lower hex digits 6888 → 0110100010001000
- Block Offset (lowest 8 bits): 10001000
- Set Index (next 6 bits): 011010 (Decimal 26)
- For A3 (0x6A289C): Focus on the lower hex digits 289C → 0010100010011100
- Block Offset (lowest 8 bits): 10011100
- Set Index (next 6 bits): 011010 (Decimal 26)

Because both A2 and A3 share the exact same binary pattern (011010) in their set index fields, they are routed to the exact same cache set (Set 26). This makes statement (b) definitively true.

18. ANSWER:- 13.3 – 13.5

Block size is 256 Bytes, word size is 64 bits or 8 bytes. So block size in words is 32 words.
Number of words per block = 32
Time to fetch a block from main memory to cache:
The first word takes 20 ns and the remaining 31 words take 5 ns each.
Total time = 20 + (31 × 5)
= 20 + 155
= 175 ns
So average memory access time (AMAT) is:
AMAT = 0.94(3) + 0.06(3 + 175)
= 2.82 + 10.68
= 13.5 ns

19. ANSWER:- 160

Time to transfer a cache block = 1 + 3 + 8 = 12 cycles.
i.e., 4 bytes × 8 = 32 bytes in 12 cycles.
So, memory bandwidth = 32 / (12 cycle time)
= 32 / (12 / (60 × 10^6))
= 160 × 10^6 bytes/s

20. ANSWER:- D

Given that cache is Fully Associative.

Tag Bits	Block Offset
28	4

There are no index bits in fully associative cache because every main memory block can go to any location in the cache
⇒ Index bits = 0
Given that memory is byte addressable and uses a 32-bit address.
Cache Block size is 16 Bytes ⇒ Number of bits required for Block Offset = $\lfloor \log_2 16 \rfloor$ = 4 bits
∴ Number of Tag bits = 32 − 4 = 28.

21. ANSWER:- B

Physical Address Space = 2^P Bytes. Word Length is 2^W bytes, which means each word is of size 2^W bytes.
Cache memory size = 2^N Bytes and Tag Size = 2^X Bytes.
Physical address is P - W bits
Number of blocks in cache = $2^{(N-W-M)}$

It is a K-way set associative cache memory, each set in cache will have K-blocks.
So, Number of sets = $2^{(N-W-M)}$/ K
SET bits will be N-W-M-logk
Offset bits will be M

We know,
TAG bits = Main memory bits - SET bits - offset bits
So, TAG bits(x) = P - W - (N-M-W-logk)- M
= P - W - N + M + W + logk - M
x = P - N + logk

22. ANSWER:- 18

No. of blocks of main Memory = 2^{32} / 2^5 = 2^{27}
And there are 512 = 2^9 lines in Cache Memory.
Tag bits tell us how many blocks does 1 line in Cache memory points to
1 cache line points to 2^{27} / 2^9 = 2^{18} lines
So, 18 bits are required as TAG bits.

23. ANSWER:- 4.72

L2 cache is shared between Instruction and Data (is it always? see below)
So, average read time
= Fraction of Instruction Fetch * Average Instruction fetch time
Fraction of Data Fetch * Average Data Fetch Time

Average Instruction fetch Time
= L1 access time + L1 miss rate * L2 access time
L1 miss rate * L2 miss rate * Memory access time

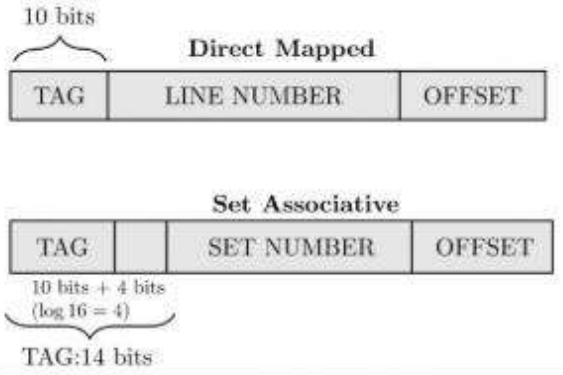
= 2 + 0.2 × 8 + 0.2 × 0.1 × 90
= 5.4 ns
Average Data fetch Time

= L1 access time + L1 miss rate * L2 access time
L1 miss rate * L2 miss rate * Memory access time
= 2 + 0.1 × 8 + 0.1 × 0.1 × 90
= 3.7 ns
So, average memory access time
= 0.6 × 5.4 + 0.4 × 3.7 = 4.72 ns

24. ANSWER:- A

Access time of L1 = 1
Access Time of L2 = 8
Let miss rate of L_2 cache =a
miss rate of L_1 cache ($2*L_{2)}$ = 2*a
AMAT (average memory access time) =2
Miss penalty of L2 cache = 18
AMAT = Access time of L1 + (MissRate L1 * miss penalty L
where miss penalty L1 = Access time of L2 + (MissRate L2 * miss penalty L2)
2 = 1+ 2*a *(8 + a* 18)
Solving the equation, a=0.111
Hence option A is correct

25. ANSWER:- 14



In set-associative 1 set = 16 lines. So the number of index bits will be 4 less than the direct mapped case.
So, Tag bits increased to 14 bits.

26. ANSWER:- 76

Total Sets: The cache has 256 blocks and is 2 -way set-associative, giving $\frac{256}{2}$ = 128 sets.

Mapping Rule: A memory block maps to a set using Set = Block Number (mod 128).
king at the sequence:

Set 0 handles blocks 0,128 , and 256 (since they all equal 0(mod128)).
Set 1 handles blocks 1,129 , and 257 (since they all equal 1(mod128)).

Iteration 1

- 0, 128, 256 are accessed for the first time → 3 Compulsory Misses.
- Accessing 256 evicts 0 from the cache.
- When the sequence loops back to access 0 and 256 again later in the string, they have already been kicked out → 2 Conflict Misses.
- The iteration ends with the cache holding [256, 128] .

Iterations 2 to 10 (Next 9 loops)
Because the cache starts full but contains the wrong combination of blocks, every time the sequence asks for 0 or 256 , it results in a miss that evicts the other.

- This causes exactly **4** Conflict Misses per iteration for Set **0**.

3. Total Calculation

Summing the conflict misses across both independent sets:

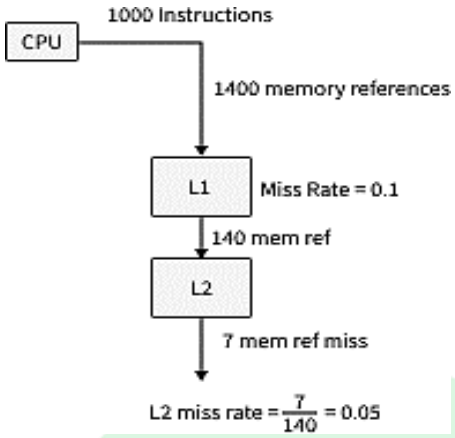
- Iteration 1: 2 misses (Set 0) + 2 misses (Set 1) = 4 conflict misses
- Iterations 2-10 (9 loops):9 loops × (4 misses from Set 0 + 4 misses from Set 1) = 9 × 8 = 72 conflict misses

Total Conflict Misses = 4 + 72 = 76

27. ANSWER:- 0.05

Assuming there are 1000 instructions for the ease of calculation, which means there are 7 memory reference misses from the L2 cache.
A cache experiences misses with memory references. Thus, it is essential to determine the counts of incoming memory references and the counts of memory references hitting or missing in order to calculate the cache hit rate or miss rate.
Upon reading the question, it becomes apparent that

the incoming memory references to the L2 cache are unknown, and we must derive this information using the provided L1 cache information.



28. ANSWER:- 30

When the CPU needs to search for data, and finds it in cache, it's called a HIT, else wise MISS. If data is not found in the cache, then the CPU searches it in main memory. Consider x to be MISS ratio, then (1-x) would be HIT ratio. Whenever there is a hit, latency is 1ms and 10ms upon miss. Time to read from main memory(disk) for all misses = x * 10 ms Time to read for all hits from cache = (1-x)*1 ms Average time: 10x + 1 -x = 9x + 1 As asked in the question, average read latency should be less than 6 ms.

$9x + 1 < 6$

$9x < 5$

$x < 0.5556$

For 20 MB, miss rate is 60% and for 30 MB, it is 40%. Thus, the smallest cache size required to ensure an average read latency of less than 6 ms is 30 MB

29. ANSWER:- 24

An easy approach would be we know that physical address is 40 bytes

We know cache-size = no.of.sets + lines-per-set + block-size

Let us assume no of sets = 2^x

And block size= 2^y

So applying it in formulas.

$2^{19} = 2^x + 8 + 2^y$;

So $x + y = 16$

Now we know that to address block size and set number we need 16 bits so remaining bits must be for tag

i.e., 40 - 16 = 24

The answer is 24 bits

30. ANSWER:- A

Block Size = 16 bytes

Block Offset = 4

No. of sets or cache lines = 2^{12}

Number of index bits = 12

Size of main memory = 2^{20}

Number of tag bits = 20 - 12 - 4 = 4

Let us consider the hex address E201F

Tag lines = First 4 bits = E (in hex)

Cache lines = Next 12 bits = 201 (In Hex)

31. ANSWER:- 14

The average read access time in nanoseconds = 0.8 * 5 + 0.2*50 = 14

32. ANSWER:- 1.68

To calculate the Average Memory Access Time (AMAT), we evaluate the cost per operation type based on the 90% hit rate (0.9) and 10% miss rate (0.1).

Step A: Find Cost per Operation Type

1. Instruction Fetch & Operand Read Cost (both are read operations):

$$\text{Cost}_{\text{read}} = (0.9 \times 1\ \text{ns}) + (0.1 \times 5\ \text{ns}) = 0.9 + 0.5 = 1.4\ \text{ns}$$

2. Operand Write Cost:

$$\text{Cost}_{\text{write}} = (0.9 \times 2\ \text{ns}) + (0.1 \times 10\ \text{ns}) = 1.8 + 1.0 = 2.8\ \text{ns}$$

Step B: Compute Total System Time

Multiply the cost of each operation by the frequency it occurs in the sequence:

- Instruction Fetches(100 operations): $100 \times 1.4\ \text{ns} = 140\ \text{ns}$
- Operand Reads (60 operations): $60 \times 1.4\ \text{ns} = 84\ \text{ns}$
- Operand Writes(40 operations): $40 \times 2.8\ \text{ns} = 112\ \text{ns}$

Total Time $= 140 + 84 + 112 = 336\ \text{ns}$

Step C: Calculate AMAT
Divide the accumulated time by the 200 total memory access operations:

$$\text{AMAT} = \frac{336 \text{ ns}}{200 \text{ operations}} = 1.68 \text{ ns}$$

33. ANSWER:- D

If associativity is doubled, keeping the capacity and block size constant, then the number of sets gets halved So, the width of the set index decoder can surely decrease -
(b) is false. The width of the way-selection multiplexer must be increased as we have to double the ways to choose from-
(c) is false As the number of sets gets decreased, the number of possible cache block entries that a set maps to gets increased So, we need more tag bits to identify the correct entry. So, (a) is also false.
(d) is the correct answer- main memory data bus has nothing to do with cache associativity- this can be answered without even looking at other options.

34. ANSWER:- D

Block : The memory is divided into equal size segments. Each segment is called a block. Data in cache is retrieved in the form of blocks. The idea is to use Spatial Locality (Once a location is retrieved, it is highly probable that the nearby locations would be retrieved in near future).
TAG bits : Each cache block is given a set of TAG bits to identify which main memory block is present in that cache block.
Option A : If the block size is small, there would be less number of near-by addresses for future references by CPU to be present in that block. Hence this is not a better spatial locality.
Option B : If the block size is smaller, no of blocks would be more in cache, hence more cache tag bits would be needed, not less.
Option C : Cache tag bits are more (because more no of blocks due to smaller block size), but more cache tag bits can't lower the hit time (even if it will increase).
Option D : If there is a miss at cache memory (i.e. the needed block by the CPU is not present in the cache memory), then that block has to be moved from next lower level of memory (lets say main memory) in the memory hierarchy, and if the block size is lower, then it takes less time to be placed into cache memory, hence less miss penalty.
Hence option D is correct.

35. ANSWER:- 20

Number of sets = cache size / size of a set
Size of a set = block size × number of blocks in a set
= 8 words × 4 (4-way set-associative)
= 8 × 4 × 4 (since a word is 32 bits = 4 bytes)
= 128 bytes
So, number of sets = 16 KB / 128 B = 128
Now, we can divide the physical address space equally between these 128 sets.
So, the number of bytes each set can access = 4 GB / 128
= 32 MB
= 32 / 4 = 8 M words = 1 M blocks (2^{20} blocks)
So, we need 20 tag bits to identify these 2^{20} blocks.

36. Answer : (10000)

0 to 100 → Module 0 ⇒ 100 ns + 500 = 600 ns
100 to 200 → Module 1 ⇒ 200 + 500 = 700 ns
200 to 300 → Module 2 ⇒ 300 + 500 = 800 ns
300 to 400 → Module 3 ⇒ 400 + 500 = 900 ns
400 to 500 → Module 4 ⇒ 500 + 500 = 1000 ns
500 to 600 → Module 5 ⇒ 600 + 500 = 1100 ns
600 to 700 → Module 6 ⇒ 700 + 500 = 1200 ns
700 to 800 → Module 7 ⇒ 800 + 500 = 1300 ns
bus → initiate request (1 word request)
In every 100 ns, initiate 1 word data to the bus.
for 1 milli second, #words we can initiate
⇒ 1 milli sec / 100 nsec = 10^{-3} / (100 × 10^{-9})
= 10^4 = 10,000 words

37. ANSWER:- A

There are N accesses to cache.
Out of these n are unique block addresses.
Now, we need to find the number of misses.
(Minimum n misses are guaranteed whatever be the access sequence due to n unique block addresses).
We are given that between two consecutive accesses to the same block, there can be only k unique block addresses. So, for a block to get replaced we can assume that all the next k block addresses go to the same set (given cache is set-associative) which will be the worst case scenario (they may also go to a different set but then there is less chance of a replacement). Now, if associativity size ≥ k, and if we use LRU (Least Recently Used) replacement policy, we can guarantee that these k accesses won't throw out our previously accessed cache entry (for

that we need at least k accesses). So, this means we are at the best-cache scenario for cache replacement — out of N accesses we miss only n (which are unique and cannot be helped from getting missed and there is no block replacement in cache). So, the miss ratio is n/N.

38. ANSWER:- A

Number of sets in cache = v. So, main memory block j will be mapped to set (j mod v), which will be any one of the cache lines from (j mod v) * k to (j mod v) * k + (k-1).
(Associativity plays no role in mapping- k-way associativity means there are k spaces for a block and hence reduces the chances of replacement.)

Alternate Solution:

- Number of sets = V
- Each set contains K lines
- Mapping type = Set Associative Mapping

1. Mapping Function:
Set number = (Main Memory Block Number) mod V

2. Example: (V = 4 sets)
Set assignment:
Set 0 → Blocks: 0, 4, 8, 12, ...
Set 1 → Blocks: 1, 5, 9, 13, ...
Set 2 → Blocks: 2, 6, 10, 14, ...
Set 3 → Blocks: 3, 7, 11, 15, ...

3. General Representation:
For any memory block **j**:
Set number = j mod V
Blocks in a set follow the pattern:
j, j + V, j + 2V, j + 3V, ...

4. Cache Organization:
Each set contains K lines
Total cache lines = V × K

5. Cache Line Range for a Set:
Each set occupies K consecutive cache lines
For set number **s**:
Start line = s × K
End line = (s × K) + (K − 1)

6. Example: (K = 8 lines per set)
Set 0 → Lines 0 to 7
Set 1 → Lines 8 to 15
Set 2 → Lines 16 to 23
Set 3 → Lines 24 to 31

7. General Formula:
For block **j**:
Set number = j mod V
Cache line range = (j mod V) × K to (j mod V) × K + (K − 1)

39. ANSWER: A

Cache = 256 KB = 2^{18}bytes
Block size = 32 B = 2^5
4-way set associative
Address = 32 bits

Number of blocks:

$$2^{18}/2^5 = 2^{13}$$

Number of sets:

$$2^{13}/4 = 2^{11}$$

Address split:
Block offset = 5 bits
Index = 11 bits
Tag = $32 - (11 + 5) = 16$bits

Bits per entry:
Tag = 16
2 valid bits
1 modified bit
1 replacement bit
= 20 bits

Entries:
Each set has 4 entries ⇒ total entries = $2^{11} \times 4 = 2^{13}$

Total size:
$2^{13} \times 20 = 163840$bits = 160 Kbits

Answer:
160 Kbits

40. ANSWER:- C

Total cache size = 256 KB
Cache block size = 32 Bytes
So, number of cache entries = 256K / 32 = 8K
Number of sets in cache = 8K / 4 = 2K as cache is 4-way associative.
So, $\log_2(2048) = 11$ bits are needed for accessing a set. Inside a set we need to identify the cache entry.
No. of memory block possible = Memory size / Cache block size
= $2^{32} / 32 = 2^{27}$
So, no. of memory block that can go to a single

cache set
$= 2^{27} / 2^{11}$
$= 2^{16}$
So, we need 16 tag bits along with each cache entry to identify which of the possible 2^{16} blocks is being mapped there.

41. ANSWER:- D
Number of cache blocks = cache size / size of a block
= 8 KB / 32 B
= 256
So, we need 8-bits for indexing the 256 blocks of the cache. And since a block is 32 bytes we need 5 WORD bits to address each byte. So, out of the remaining 19-bits (32 - 8 - 5) should be tag bits.
So, a tag entry size = 19 + 1 (valid bit) + 1 (modified bit) = 21 bits.
Total size of metadata = 21 × Number of cache blocks
= 21 × 256
= 5376 bits

42. ANSWER:- C
Since the block size of L2 cache is 16 words and the bandwidth of mainmem->L2 cache is 4 words, it requires a transfer of 4 words 4 times and then a transfer of required 4 words from L2 cache to L1 cache.
So total time is 4*(200 + 20) + 1*(20 + 2) = 902 nanoseconds.

43. ANSWER:- C
A block to access in L2 cache requires 20 nanoseconds, and 2 nanoseconds to place in L1-cache.

The block size in L1 cache is 4 words, so total time is = time to access L2 + time to place in L1 = 20+2 = 22 ns..

44. ANSWER: D
Total blocks in cache = 16 4-way ⇒ sets = $16/4 = 4$
Set index = block number mod 4

Group requests by set:
Set 0 → (block % 4 = 0):
0, 4, 8, 216, 8, 48, 32, 92
Set 1 → 1, 133, 129, 73
Set 2 → 2-type none
Set 3 → 255, 3, 159, 63, 155
Apply LRU (only final content matters):
Set 0 (capacity 4):
Final → 8, 48, 32, 92
Set 1:
Final → 1, 133, 129, 73
Set 3 (capacity 4):
Sequence → 255, 3, 159, 63 → full
Next 155 → evicts LRU (255)
Final → 3, 159, 63, 155
Check options:
3 → present
8 → present
129 → present
216 → NOT present
Answer:
216

45. ANSWER:- A
1st is not correct as data need not be exactly the same at the same point of time and so write-back policy can be used in this.
2nd is not needed when talking only about L1 and L2.
For 3rd, associativity can be equal.
So, only the 4th statement is necessarily true

46. ANSWER: D
Total Cache Size: $64\ \text{KB} = 64 \times 1024\ \text{Bytes} = 2^{16}$ Bytes
Block Size: 16 Bytes $= 2^4$ Bytes
Total Blocks (Lines) in Cache:

$$\text{Total Blocks} = \frac{\text{Cache Size}}{\text{Block Size}} = \frac{2^{16}}{2^4} = 2^{12}$$
$$= 4096 \text{ blocks}$$

Since the cache configuration is 2-way set-associative, every set contains exactly 2 parallel blocks;

1. Block Offset Field: The number of bits needed to address individual bytes inside a single 16-byte block.

Block Offset bits $= \log_2(16) = 4$ bits

2. Set Index Field: The number of bits needed to uniquely select one of the 2048 sets.

Set Index bits $= \log_2(2048) = 11$ bits

3. Tag Field: The remaining bits of the address structure used to distinguish different memory blocks mapping to the same set. Given the 32 -bit address space width:

Tag bits = Total Address Width - (Set Index bits + Block Offset bits)
Tag bits = $32 - (11 + 4) = 32 - 15 = 17$ bits
Total Tag Size = Total Blocks × Tag bits per block

Total Tag Size = $4096 \times 17 = 69{,}632$ bits

To convert the raw bit capacity into kilobits (Kbits), divide the value by 1024:

Total Tag Size in Kbits = $\frac{69{,}632 \text{ bits}}{1024} =$ 68Kbits

47. **ANSWER**: B
Block size = 16 B ⇒ offset = 4 bits
Cache = 64 KB, 2-way ⇒ sets = 2^{11}⇒ index = 11 bits
Key idea:
Same cache index ⇒ same (index bits) ⇒ addresses differ only in tag or offset, not index
Array: row-major
Address of ARR[i][j] = base + $(i \times 1024 + j) \times 8$
Check differences from ARR[0][0]:
ARR[0][4]:
offset = $4 \times 8 = 32$bytes ⇒ crosses 2 blocks ⇒ index changes
ARR[4][0]:
offset = $4 \times 1024 \times 8 = 32768 = 2^{15}$
This changes tag, not index
ARR[0][5]:
offset = 40 bytes ⇒ different block ⇒ index changes
ARR[5][0]:
offset = $5 \times 1024 \times 8 = 40960$⇒ affects index
Answer:
ARR[4][0] (Option B)

48. **ANSWER**: C
Block size = 16 B
double = 8 B ⇒ 1 block holds 2 elements
Access pattern:
Row-major ⇒ sequential access
So access sequence:
Element 0 → miss (block loaded)
Element 1 → hit (same block)
Element 2 → miss
Element 3 → hit
… continues
Thus:
For every 2 accesses → 1 miss + 1 hit
Hit ratio:
$\frac{1}{2} = 50\%$
Cache size/associativity doesn't matter here because no reuse beyond block
Answer:
50% (Option C)

49. Answer :(D)
Main memory = **1 MB = 2^{20} bytes**
Word size = **1 byte ⇒ byte-addressable**
Block size = **128 words = 128 bytes = 2^7 bytes**
Cache size = **8 KB = 2^{13} bytes**
Cache type = **4-way set associative**

Step 1: Word (Block Offset) bits
Block size = 2^7 ⇒
Word bits = 7
Step 2: Number of cache lines
Cache size / Block size =
$2^{13} / 2^7 = 2^6 =$ **64 lines**
Step 3: Number of sets
4-way set associative ⇒
Number of sets = 64 / 4 = **16 = 2^4**
⇒ Set bits = 4
Step 4: Tag bits
Total address bits = 20
Tag = Total − (Set + Word)
= 20 − (4 + 7)
= **9 bits**

50. Answer : (A)

Given address
= **0C795H**

Hex → Binary:
0 = 0000
C = 1100
7 = 0111
9 = 1001
5 = 0101

So,
0C795H = 0000 1100 0111 1001 0101 (20 bits)
Step 2: Split into TAG | SET | WORD
Structure:
TAG = first 9 bits
SET = next 4 bits
WORD = last 7 bits

Binary:
000011000 | 1111 | 0010101
Step 3: Extract TAG
TAG = 000011000

51. ANSWER: A

Line index = $(\text{address}/64) \bmod 32$
Starting address = 1100H = 4352
Starting line:
$4352/64 = 68$
$68 \bmod 32 = 4 \Rightarrow$ first block maps to line 4
Total blocks = 40
So blocks map to lines:
4,5,6,...,31,0,1,2,3,4,5,6,7,8,9,10,11
After first pass:
Lines 4 to 11 contain the last 8 blocks
Second pass:
When we restart, first 8 blocks again map to lines 4 to 11, replacing those blocks
Answer:
line 4 to line 11

52. ANSWER:- D

Number of sets = cache size / (block size × number of blocks per set)
= (128 × 64) / (64 × 4)
= 32 sets
Number of index (line) bits = $\log_2(32) = 5$
Number of word (block offset) bits = $\log_2(64) = 6$
Total address bits = 20
Tag bits = 20 − 5 − 6
= 9

53. ANSWER:- C

Bits to represent the address = $\log_2(2^{16}) = 16$Each cache line size = 64 bytes → offset = 6 bits Total number of cache lines = 32 → index = 5 bits Tag bits = 16 − 6 − 5 = 5 bits
We have a 2D array, each element size = 1 byte
Total array size = 50 × 50 × 1 = 2500 bytes
Number of cache lines required = 2500 / 64 ≈ 39.0625 ≈ 40 lines
Starting address = 1100H = 0001 0001 0000 0000
Index bits indicate cache line number 4 → first array block maps to line 4
Since 40 memory blocks map into 32 cache lines, some blocks map to the same cache line.
Mapping pattern:
0 → A28
1 → A29
2 → A30
3 → A31
4 → A0, A32
5 → A1, A33
6 → A2, A34
7 → A3, A35
8 → A4, A36
9 → A5, A37
10 → A6, A38
11 → A7, A39
12 → A8
...
30 → A26
31 → A27
Accessing the complete array twice:
First pass: 40 compulsory misses
Second pass: 16 conflict misses (because 8 cache lines from index 4 to 11 are replaced
Total misses = 40 + 16 = 56 misses
Hence, the answer is option (c).

54. ANSWER:- D

According to the original question, there are k banks with k = 24, and each bank has c bytes where c = 2.
So in one iteration, the total data transferred is:
2 × 24 = 48 bytes
Now we calculate the latency for one iteration.
Decoding time per iteration = k / 2 = 24 / 2 = 12 ns
Each bank latency = 80 ns
Total time for one iteration = decoding latency + bank latency
= 12 ns + 80 ns
= 92 ns
Since one iteration transfers 48 bytes and the cache block size is 64 bytes, we need 2 iterations.
Total transfer time = 2 × 92 ns
= 184 ns

Check Out Our GATE Courses & Secure Your Dream Rank!

55. ANSWER:- A

Cache size is 32 KB and cache block size is 32 B

So,

Number of sets = $\frac{cache\ \ size}{no.of\ blocks\ in\ a\ set \times\ block\ size}$ = $32\ \frac{KB}{2 \times 32\ B} - 512$

So, the number of index bits needed = 9 (since 2^9 = 512).

Number of offset bits = 5 (since 2^5 = 32; B is the block size and assuming byte addressing).

So, the number of tag bits = 32 − 9 − 5 = 18 (as the memory address is 32 bits).

So, time for comparing the data = Time to compare the data + Time to select the block in set

= 0.6 + 18/10 ns

= 2.4 ns.

56. ANSWER: D

Number of blocks:

= 2^{15} / 2^5 = 2^{10} blocks

Index bits:

= $\log_2(2^{10})$ = 10

Block offset bits:

= 5

Tag bits:

= 32 − (10 + 5) = 17 bits

Hit latency components (Direct Mapped):

Tag comparison latency = k/10 = 17/10 = 1.7 ns

No multiplexer needed (only one block per index)

So,

h_2 = 1.7 ns

57. ANSWER: B

Cache = 32 KB, block = 128 B ⇒ elements per block = $128/8 = 16$

Array = 512×512, row-major

P1 (row-wise access):

A[i][j] → contiguous access

Every 16 elements → 1 miss

Total elements = 512×512

$$M1 = \frac{512 \times 512}{16} = 16384$$

P2 (column-wise access):

A[j][i] → stride = $512 \times 8 = 4096$bytes

Each access goes to a different block ⇒ no reuse

Every access = miss

$$M2 = 512 \times 512 = 262144$$

Ratio:

$$\frac{M1}{M2} = \frac{16384}{262144} = \frac{1}{16}$$

Answer:

1/16 (Option B)

58. ANSWER: C

Block size = 128 B ⇒ elements per block = $128/8 = 16$

Array = 512×512, row-major

P1 (row-wise):

Sequential access ⇒ every 16 elements → 1 miss

Total elements = 512×512

$$M1 = \frac{512 \times 512}{16} = 16384$$

Cache size doesn't cause extra misses because no conflict within a row

Answer:

16384 (Option C)

59. ANSWER:- A

Number of blocks = cache size / block size = 32 KB / 32 B = 1024

So, indexing requires 10 bits.

Number of OFFSET bits required to access 32-byte block = 5.

So, the number of TAG bits = 32 − 10 − 5 = 17.

60. ANSWER : (C)

- 2-way set associative cache
- Total cache blocks = 8 ⇒ **Number of sets = 8 / 2 = 4 sets**
- Main memory blocks = 128
- Mapping:
  Set number = (Block number) mod 4

Reference string:

0, 5, 3, 9, 7, 0, 16, 55

Step-by-step simulation (LRU):

Set mapping:

- Set = Block mod 4

Block	Set
0	0
5	1
3	3
9	1
7	3
0	0
16	0
55	3

Now simulate each set (2-way, LRU):
Set 0: (blocks → 0, 0, 16)
- Insert 0 → [0]
- Access 0 → hit
- Insert 16 → [0,16] (no eviction yet)

Final Set 0 → **0, 16**

Set 1: (blocks → 5, 9)
- Insert 5 → [5]
- Insert 9 → [5,9]

Final Set 1 → **5, 9**

Set 3: (blocks → 3, 7, 55)
- Insert 3 → [3]
- Insert 7 → [3,7]
- Insert 55 → LRU = 3 ⇒ replace → [7,55]

Final Set 3 → **7, 55**

Set 2:
- No blocks mapped → empty

Final cache contents:
All blocks present:
0, 16, 5, 9, 7, 55
Sorted:
0, 5, 7, 9, 16, 55

61. Answer : (A)

- Physical address size = 30 bits
- I-cache: 4K words, Direct mapping, Block size = 4 words
- D-cache: 4K words, 2-way set associative, Block size = 4 words
- L2-cache: 64K words, 4-way set associative, Block size = 16 words

1. I-Cache (Direct Mapping)

Capacity = 4K words = 2^{12} words
Block size = 4 words = 2^2
Number of cache lines = $2^{12} / 2^2 = 2^{10} = 1K$
Word offset = $\log_2(4) = 2$ bits
Index = $\log_2(1K) = 10$ bits
Tag bits = $30 - (10 + 2) = 18$ bits
Tag memory = 1K × 18 bits

2. D-Cache (2-way Set Associative)
Capacity = 4K words = 2^{12} words
Block size = 4 words = 2^2
Total blocks = $2^{12} / 2^2 = 2^{10}$
Number of sets = $2^{10} / 2 = 2^9 = 512$
Word offset = 2 bits
Set index = $\log_2(512) = 9$ bits
Tag bits = $30 - (9 + 2) = 19$ bits
Total entries = total blocks = $2^{10} = 1K$
Tag memory = 1K × 19 bits

3. L2-Cache (4-way Set Associative)
Capacity = 64K words = 2^{16} words
Block size = 16 words = 2^4
Total blocks = $2^{16} / 2^4 = 2^{12}$
Number of sets = $2^{12} / 4 = 2^{10} = 1K$
Word offset = 4 bits
Set index = 10 bits
Tag bits = $30 - (10 + 4) = 16$ bits
Total entries = total blocks = $2^{12} = 4K$
Tag memory = 4K × 16 bits

Check Out Our GATE Courses & Secure Your Dream Rank!

6 SECONDARY MEMORY & I/O ORGANIZATION

GATE CSE – 2026 MCQ SET - 2(1-MARKS)

Q1. Consider the following two statements about interrupt handling mechanisms in a CPU,
S1: In non-vectored interrupt mechanism, it usually takes more time to start the Interrupt Service Routine (ISR) when compared to that in a vectored interrupt mechanism.
S2: In daisy-chain interrupt mechanism, the CPU polls all the input devices individually to determine the source of the interrupt.
Which one of the following options is correct with respect to SI and S2?

(a) Both S1 and S2 are true
(b) Both S1 and S2 are false
(c) S1 is true and S2 is false
(d) S1 is false and S2 is true

GATE CSE – 2025 MCQ SET - 1(1-MARKS)

Q2. Suppose a program is running on a non-pipelined single processor computer system. The computer is connected to an external device that can interrupt the processor asynchronously. The processor needs to execute the interrupt service routine (ISR) to serve this interrupt. The following steps (not necessarily in order) are taken by the processor when the interrupt arrives:
(i) The processor saves the content of the program counter.
(ii) The program counter is loaded with the start address of the ISR.
(iii) The processor finishes the present instruction.
Which ONE of the following is the CORRECT sequence of steps?

(a) (iii), (i), (ii)
(b) (i), (iii), (ii)
(c) (i), (ii), (iii)
(d) (iii), (ii), (i)

GATE CSE – 2024 MCQ SET - 2(1-MARKS)

Q3. Consider a computer with a 4 MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate *(in bits per second)* of the DMA controller if 1% of the processor cycles are used for DMA?

(a) 2,56,000
(b) 3,200
(c) 25,60,000
(d 32,000

GATE CSE – 2024 MCQ SET - 1(1-MARKS)

Q4. Which one of the following statements is FALSE?

(a) In the cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle
(b) For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode
(c) Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism
(d) The CPU can start executing an interrupt service routine faster with vectored interrupts than with non-vectored interrupts

GATE CSE – 2023 NAT (1-MARKS)

Q5. A keyboard connected to a computer is used at a rate of 1 keystroke per second The computer system polls the keyboard every 10 ms to check for a keystroke and consumes 100 μs (micro seconds) for each poll. If it is determined after polling that a key has been pressed, the system consumes an additional

Check Out Our GATE Courses & Secure Your Dream Rank!

200 μs to process the keystroke. Let T_1 denote the fraction of a second spent in polling and processing a keystroke.
In an alternative implementation, the system uses interrupts instead of polling. An interrupt is raised for every keystroke. It takes a total of 1 ms for servicing an interrupt and processing a keystroke. Let T_2 denote the fraction of a second spent in servicing the interrupt and processing a keystroke. The ratio T_1 / T_2 is _____. (Rounded off to one decimal place)

GATE CSE – 2022 MCQ (1-MARKS)

Q6. Which one of the following facilitates transfer of bulk data from hard disk to main memory with the highest throughput?

(a) DMA based I/O transfer
(b) Interrupt driven I/O transfer
(c) Polling based I/O transfer
(d) Programmed I/O transfer

GATE CSE – 2021 SET2 NAT (1-MARKS)

Q7. Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through cycle stealing at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is _____bits per second.

GATE CSE – 2020 MCQ (1-MARKS)

Q8. Consider the following statements.
I. Daisy chaining is used to assign priorities in attending interrupts.
II. When a device raises a vectored interrupt, the CPU does polling to identify the source of the interrupt.
III. In polling, the CPU periodically checks the status bits to know if any device needs its attention.
IV. During DMA, both the CPU and DMA controller can be bus masters at the same time.
Which of the above statements is/are TRUE?

(a) I and IV only
(b) I and II only
(c) III only
(d) I and III only

GATE CSE – 2016 Set 1 MCQ (2-MARKS)

Q9. The size of the data count register of a DMA controller is 16 bits . The processor needs to transfer a file of 29,154 kilobytes from disk to main memory . The memory is byte addressable
The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is ___

GATE CSE – 2015 SET-1 MCQ (2-MARKS)

Q10. Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of 50×10^6 bytes /sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512-byte sector of the disk is .

GATE CSE – 2015 SET-1 MCQ (2-MARKS)

Q11. Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors Assume that every sector access necessitates a seek, And the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is

GATE CSE – 2013 MCQ (2-MARKS)

Q12. Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each

cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and addressing format is <cylinder no., surface no., sector no>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is < 1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

(a) 1281
(b) 1282
(c) 1283
(d) 1284

GATE CSE – 2011 MCQ (2-MARKS)

Q13. An application loads 100 libraries at start-up. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 ms . Rotational speed of disk is 6000 rpm . If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected.)

(a) 0.50 s
(b) 1.50 s
(c) 1.25 s
(d) 1.00 s

GATE CSE – 2011 MCQ (1-MARKS)

Q14. On a non-pipelined sequential processor, a program segment, which is the part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

Intialize the address register
Intilatize the count to 500
LOOP: Load a byte from device
Store in memory at address given by address register
Increment the address register
Decrement the count
If count != 0 go to LOOP

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.
The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.
What is the approximate speed up when the DMA controller based design is used in a place of the interrupt driven program based input-output?

(a) 3.4
(b) 4.4
(c) 5.1
(d) 6.7

GATE CSE – 2009 MCQ (1-MARKS)

Q15. A CPU generally handles an interrupt by executing an interrupt service routine

(a) as soon as an interrupt is raised
(b) by checking the interrupt register at the end of fetch cycle
(c) by checking the interrupt register after finishing the execution of the current instruction
(d) by checking the interrupt register at fixed time intervals

GATE CSE – 2005 MCQ (2-MARKS)

Q16. A device with data transfer rate 10 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 4 μ sec The byte transfer time between the device interface register and CPU or memory is negligible.
What is the minimum performance gain of operating the device under interrupt mode over operating it under program controlled mode?

(a) 15
(b) 25
(c) 35
(d) 45

GATE CSE – 2005 MCQ (2-MARKS)

Q17. Consider the disk drive with the following specifications: 16 surfaces, 512 tracks/surface, 512 sectors/track, 1 KB/sector, rotation speed 3000 rpm. The disk is operated in cycle stealing mode whereby whenever one byte is ready it is sent to memory, similarly, for writing, the disk interface reads a 4-byte word from the memory in each DMA cycle. Memory cycle time is 40 nsec The maximum percentage of time that the CPU gets blocked during DMA operation is

(a) 10
(b) 25
(c) 40
(d) 50

ANSWER KEY

1. (c)	**2. (a)**	**3. (c)**
4. (c)	**5. (10.2)**	**6. (a)**
7. (80000)	**8. (d)**	**9. (456)**
10. (6.1)	**11. (14020)**	**12. (d)**
13. (b)	**14. (a)**	**15.(c)**
16. (b)	**17. (b)**	

HINTS & SOLUTIONS

1. ANSWER: (C)
S1 is true and S2 is false

- **S1:** True
  In a **non-vectored interrupt**, the CPU has to identify the interrupt source (often by polling or additional steps) before jumping to the ISR. This increases the response time.
  In a **vectored interrupt**, the ISR address is directly provided, so execution starts faster.
- **S2:** False
  In a **daisy-chain interrupt mechanism**, devices are connected in a chain and priority is resolved through hardware propagation of the interrupt acknowledge signal. The CPU does **not** poll each device individually; instead, the device with the highest priority in the chain responds first.

So, **S1 is correct, S2 is incorrect → Option (c).**

2. ANSWER:- A
Let's analyze the steps that a processor must take when an interrupt occurs on a non-pipelined single-processor system:
The processor must first finish executing the current instruction. This ensures that the instruction, which might be partially completed, is fully executed before transferring control to the interrupt service routine (ISR).
Once the instruction finishes, the processor saves the content of the program counter (PC). This step is crucial because it stores the address of the next instruction to execute, which allows the processor to resume normal operations after the ISR is complete.
Finally, the processor loads the PC with the start address of the ISR, thereby transferring control to the interrupt handler.
In summary, the correct sequence is:
(iii) Finish the present instruction.
(i) Save the content of the program counter.
(ii) Load the PC with the start address of the ISR.
This corresponds to Option (a)

3. ANSWER:- C
1 sec → 4×10^6 cycles
1 cycle → 8 B transferred
1% (Processor cycle) = $(1/100) \times 4 \times 10^6 = 40{,}000$ cycles
Data transfer rate = $40{,}000 \times 8 \times 8$ Bps
$= 2{,}560{,}000$

4. ANSWER:- C
(a) In the cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle
True (as definition)
(b) For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode
True (as define, through as compared to bulk data, for burst mode throughput will be equal to 1 as bulk of data will be transferred at once but with cycle stealing mode transfer will take more time and throughput will be less)
(c) Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism

False (Programmed i/o is like polling(can remember to recall the concept) so to check whether there is i/o who needs cpu ,cpu has to do a lot of work, but in interrupt driven i/o ,i/o has given privileged to tell cpu that i need you by sending the interrupt so cpu in is more utilized in interrupt driven i/o)

(d) The CPU can start executing an interrupt service routine faster with vectored interrupts than with non-vectored interrupts
true (vectored interrupt means address of program which handles that interrupt is send with the interrupt itself where in non-vectored interrupt it is not send so cpu has to run default service routine to identify the address of that routine and then run that routing)

5. ANSWER:- 10.2

1. Polling System (T1)
CPU polls (asks) keyboard every 10 ms and asks did you get any keystrokes? The keyboard answers "Yes" only once per second
In one second, the CPU polls the keyboard 100 times (1 sec = 1000 ms, and the CPU polls every 10 ms).
Time Spent in Polling = 100 × 100 microsec = 10000 microseconds or 10 ms.
Time Spent in Processing the Keystroke (Only once per second = 200 microsec or 0.2 ms.
CPU's Time wasted in Polling System = Time Spent in Polling + Time Spent in Processing = 10.2 ms
2. Interrupt System (T2)

The CPU doesn't ask the keyboard periodically about the keystroke, but the keyboard tells the CPU via interrupt about a keystroke.
Whenever there is a keystroke, the keyboard interrupts the CPU. CPU executes corresponding Interrupt Service Routine (ISR) which takes 1 ms. That's all.
Speed Up of Interrupt over Polling System (S) = T1 / T2 = 10.2 ms / 1 ms = 10.2

6. ANSWER:- A

In DMA, a bulk amount of data will be transferred from the secondary memory to main memory without the involvement of a CPU.

7. ANSWER:- 80000

To complete one cycle at 2 MHz it will take 1 / (2 × 10^6) seconds. So the total number of CPU cycles in one second will be 2 × 10^6.
Now 0.5% of these cycles are taken by DMA to transfer the data.
So total number of cycles taken to transfer the data will be (0.5 / 100) × 2 × 10^6 = 10,000 and in each cycle 8 bits are transferred
So, data transfer rate in bits per second = 8 × 10000 = 80,000.

8. ANSWER: - D

Statement I: True – Daisy chaining is used to assign priorities in attending interrupts.
Statement II: False – Vectored interrupts do not involve polling; it is the non-vectored interrupts that involve polling.
Statement III: True – Polling means that the CPU periodically checks the status bits to know if any device needs attention.
Statement IV: False – During DMA, only one of the CPU or the DMA controller can be the bus master at a time.
Correct Statements: I and III

9 ANSWER: 456

Data count register = 16 bits ⇒ max count = 2^{16} = 65536bytes per transfer
File size = 29,154 KB
= 29154 × 1024 = 29,853,696bytes
Number of DMA transfers:

$$\left\lceil \frac{29853696}{65536} \right\rceil = \lceil 455.5 \rceil = 456$$

Answer:
456

10. Answer :- (6.1)

$$Average\ Disk\ access\ time\ Or\ Avg\ read/write\ time = Avg\ Seek\ Time(ST) + Avg\ Rotation\ Time(R) + Disk\ Tranfer\ Time(DTT) + Overhead\ (if\ any\ given)$$

Given,
Disk RPM = 15000
i.e. 15000 Rotation in 60 sec.
In 1 Rotation = 60 / 15000 sec × 1000 / 1000
1 Rotation = 4 msec
Avg ST = 2 × Avg RT
⇒ 2 × 2 = 4 msec

Avg ST = 4 msec
Avg Rotational Latency = 1/2 Rotation time
⇒ 1/2 × 4 msec
Avg Rotational Latency = 2 msec
Transfer Rate = 50×10^6 Bytes/sec
50×10^6 Byte — 1 sec
1 Byte transfer time = $1 / (50 \times 10^6)$ sec
Given 1 sector = 512 Bytes
512 Byte transfer time = 512 / (50 × 10^6)
$= 10.24 \times 10^{-6}$
= 0.01024 msec
D.T.T = 0.01024 msec
Controller time = 10 × 0.01024 msec
= 0.1024 msec
ST + RL + DTT + overhead
= 4 + 2 + 0.01024 + 0.1024
= 6.11264 msec ≈ 6.1 msec (Ans)

11. Answer : (14020)
Time Required to Read One Sector = Seek Time +Average Rotational Latency + Data transfer time
Seek Time $(ST) = 4$msec

Each sector = 512 Byte

600 sector/track X 512 Byte /sector $= 600 *$ 512 Byte

10000 Rotation in 60 sec .
1 Rotation = $\frac{60}{1000} \Rightarrow 6 \times 10^{-3}$sec.
1 Rotation takes = 6 msec
Average Rotational Latency $= 1/2 \times 6$
$R = 3$ msec

1 Track Capacity $= 600 \times 512$ Byte
In 1 Rotation 1 Complete Track is transferred
600 × 512 Byte takes→ 6 msec

1Byte→ $\frac{6}{60\times512}$ msse.
512 Byte (1 Sector) → $\frac{512*6}{600*512}$

DTT=0.01 msec.
Total Time Required for 1 sector = $4 + 3 + 0.01$

$= 7.01$ msec.

Total Time Required to 2000 sector $= 2000 *$ 7.01ms

=14020 msec

12. Answer :- (D)
Surfaces = 16
Sector/ Track = 64
16384 cylinder.
Sector per Cylinder=$16 \times 64 = 1024$ sec/cylinders.
Sector Starting Address: $< 1200,9,40 >$
To traverse 1200 cylinders : $1200 \times 16 \times 64$
To traverse 9 surface $= 9 \times 64$
To traverse 40 sector $= 40$

Starting Address of the File: 1229416

File size = 42797 kB Each sector = 512 Byte
Sector required to Store this file $= \left[\frac{42797kB}{512B}\right] = 85594$ sector
First Sector (Starting) address of the File = 1229416
Lost Sector address of the $File$ is

$$= 1229416 + (85594 - 1)$$

This file Log Secter = 1315009
Cylinder No = $\frac{1315009}{16\times44} = 1284$ Ans

13. Answer : (B)
Total 100 libraries
Seek time = 10 msec
DTT (negligible)
Rotational Speed = 6000 rpm
Average Disk access time Or Avg read/ write time = Avg Seek Time(ST) + Avg Rotation Time(R) + Disk Tranfer Time(DTT) + Overhead (if any given)
Avg. RT = 1/2 Rotation Time
6000 Rotation = 60 sec
1 Rotation = 60 / 6000 sec × 1000 / 1000 = 10 msec
Avg. RT = 1/2 × 10 msec = 5 msec
Total time to access 1 library = ST + RL + DTT
⇒ 10 + 5 = 15 msec
Total time taken for 100 libraries

= 100 × 15 msec
= 100 × 15 × 10^{-3} sec
= 1.5 sec

14. ANSWER: (A)

Interrupt-driven program:
Per iteration (500 times):
Load = 2 cycles
Store = 2 cycles
Increment address = 1
Decrement count = 1
Check & branch = 1
Total per iteration = 7 cycles
Total = $500 \times 7 = 3500$cycles
Initialization (2 instructions):
= 2 cycles
Total ≈ 3502 cycles
DMA approach:
Initialization = 20 cycles
Transfer = $500 \times 2 = 1000$cycles
Total = 1020 cycles
Speedup:

$$\frac{3502}{1020} \approx 3.43 \approx 3.4$$

15. ANSWER: (C)
After finishing the execution of each instruction the CPU reads the interrupt pins to recognize the interrupts.
INTR = 1 = Interrupt is present. (Service the Interrupt)
INTR= 0 = Interrupt is not present. (Go to next Instruction fetch from user program)

16 ANSWER: (B)
In Programmed I/O, the CPU issues a command and waits for I/O operations to complete.
So here, the CPU will wait for 1 sec to transfer 10 KB of data.
The minimum performance gain for interrupt mode happens for the smallest unit of data transfer – which here is 1 byte.
Time to transfer 1 byte of data in programmed I/O mode = 1 / (10 KBps) = 100 μs
In Interrupt mode, to transfer 1 byte of data, overhead is
4×10^{-6} s = 4 μs
Performance gain = 100 / 4 = 25.

17. ANSWER: (B)
Track Capacity: 512 sectors × 1 KB = 512 KB (using 1000 bytes for streaming data rate).

Rotation Time: $\frac{60 \text{ seconds}}{3000 \text{ RPM}} = 0.02$ seconds. Data Transfer Rate: $\frac{512 \text{ KB}}{0.02a} = 25{,}600{,}000$ bytes/ sec.

The DMA controller transfers a 4-byte word per cycle. The time required for the disk to ready or consume 4 bytes is:

$$\text{I ime Interval} = \frac{4 \text{ bytes}}{25{,}600{,}000 \text{ bytes /sec}} = 156.25\text{nsec}$$

Every 156.25 osech, the DMA controller steals one memory cycle (40 oseck), during which the CPU is blocked from accessing memory:

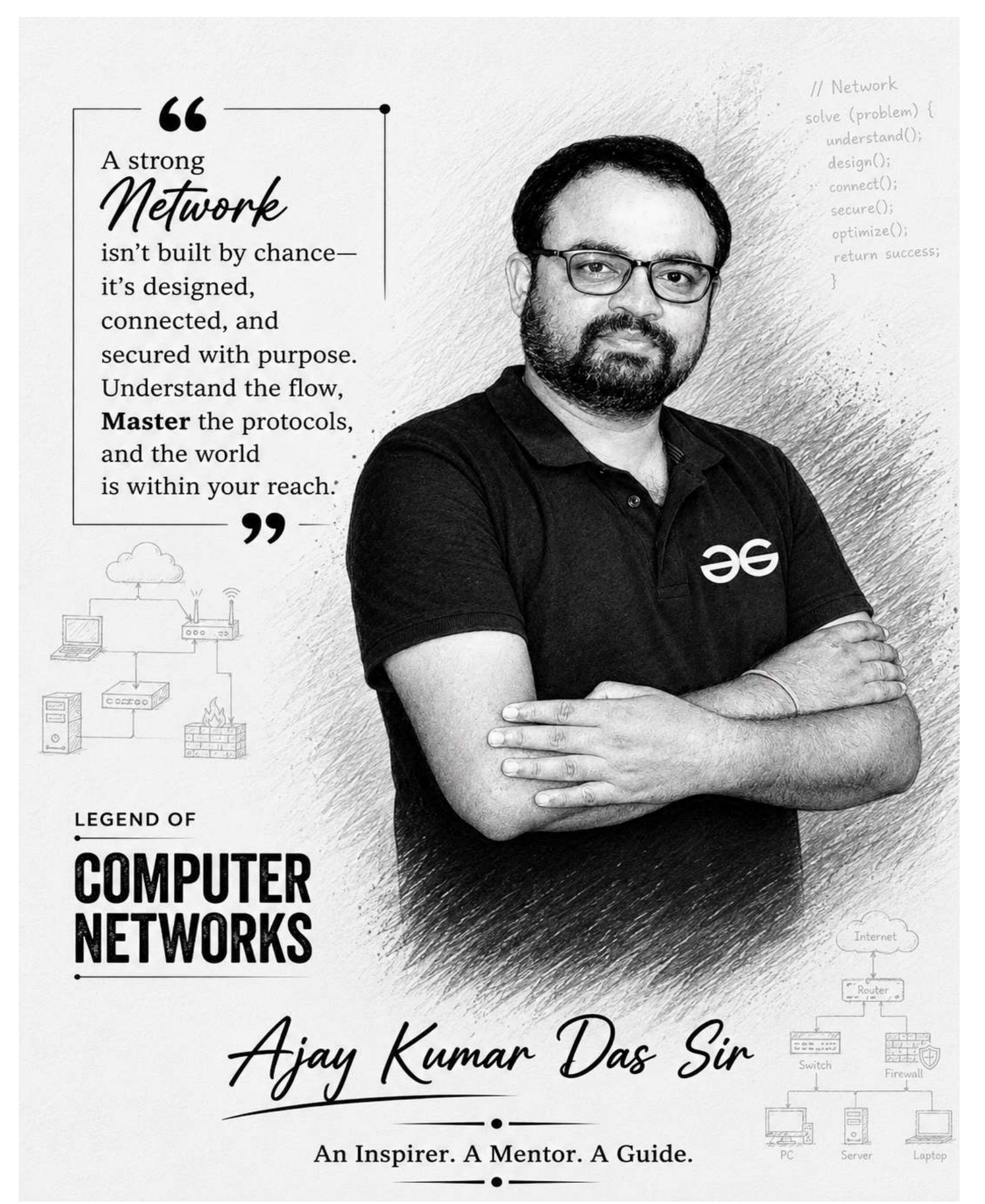
$$\text{Blocked \%} = \frac{\text{Memory Cycle Time}}{\text{DMA Interval}} \times 100 - \frac{40\text{nsec}}{156.25\text{nscc}} \times 100 - 40$$

The closest value among the options is 25%.

COMPUTER NETWORK

“Scan Here for Video Solution”

Link to GfG corresponding Short Notes: GATE FARRAY


// Network
solve (problem) {
understand();
design();
connect();
secure();
optimize();
return success;
}
A strong
Network
isn't built by chance—
it's designed,
connected, and
secured with purpose.
Understand the flow,
Master the protocols,
and the world
is within your reach.
LEGEND OF
COMPUTER
NETWORKS
Ajay Kumar Das Sir
An Inspirer. A Mentor. A Guide.
Internet
Router
Switch
Firewall
PC
Server
Laptop

1 OSI STACK AND PHYSICAL LAYER

GATE CSE - 2025 MCQ SET - 1(1-MARKS)

Q1. Identify the ONE CORRECT matching between the OSI layers and their corresponding functionalities as shown.

OSI Layers	Functionalities
(a) Network layer	(I) Packet routing
(b) Transport layer	(II) Framing and error handling
(c) Datalink layer	(III) Host to host communication

(a) (a)-(I), (b)-(II), (c)-(III)
(b) (a)-(I), (b)-(III), (c)-(II)
(c) (a)-(II), (b)-(III), (c)-(I)
(d) (a)-(III), (b)-(II), (c)-(I)

GATE CSE - 2014 MCQ SET - 3(1-MARKS)

Q2. In the following pairs of OSI protocol layer/sub-layer and its functionality, the INCORRECT pair is

(a) Network layer and Routing
(b) Data Link Layer and Bit synchronization
(c) Transport layer and End-to-end process communication
(d) Medium Access Control sub-layer and Channel sharing

GATE CSE – 2013 MCQ (1-MARKS)

Q3. Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet must visit the network layer and the data link layer during a transmission from S to D.

(a) Network layer - 4 times and Data link layer - 4 times
(b) Network layer - 4 times and Data link layer - 3 times
(c) Network layer - 4 times and Data link layer - 6 times
(d) Network layer - 2 times and Data link layer - 6 times

GATE CSE - 2007 MCQ (1-MARKS)

Q4. In Ethernet, when Manchester encoding is used, the bit rate is:

(a) Half the baud rate
(b) Twice the baud rate
(c) Same as the baud rate
(d) None of the above

ANSWER KEY

1. (b)	**2. (b)**	**3. (c)**	**4. (a)**

HINTS & SOLUTIONS

1. ANSWER: - B
Explanation:
- Packet Routing is the responsibility of the Network Layer (a) (I).
- Framing and error (bit level) handling is the responsibility of the Data Link Layer. Hence, (c) (II).
- Host-to-Host communication (end-to-end Communication) is the responsibility of the Transport Layer (Practically). Hence, (b) (III).

Therefore, option B is correct.

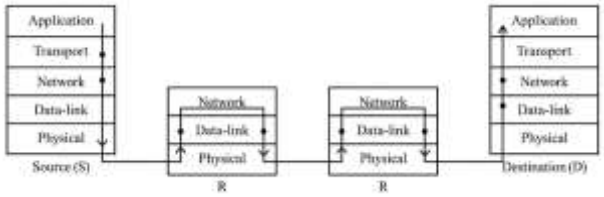
2. ANSWER: - B
Explanation:
Option (A) is CORRECT: Routing is a primary responsibility of the Network Layer.

Option (B) is INCORRECT: Bit synchronization is strictly a function of the Physical Layer, not the Data Link Layer.

Option (C) is CORRECT: The Transport Layer is responsible for managing end-to-end, process-to-process communication.

Option (D) is CORRECT: It utilizes three main categories of protocols: Random Access, Controlled Access, and Channelized Access.

3. ANSWER: - C
Explanation:



4. ANSWER: - A
Explanation:
In Manchester encoding, each bit is represented by two signal level changes (one transition per bit period).
So,
baud rate = 2 × bit rate
⇒ bit rate = baud rate / 2

2 DATA LINK LAYER

GATE CSE – 2026 MCQ (1-MARKS)

Q1. Consider a file of size 4 million bytes being transferred between two hosts connected via a path consisting of three consecutive links of bandwidth 2 Mbps, 500 kbps, and 1 Mbps, respectively.
All processing delays and propagation delays are negligible. Assume there is no other background traffic over the path and no additional overhead to transfer the file.
Which one of the following is the total time (in seconds) to transfer the file?
Note: $1M = 10^6$, $1k = 10^3$

(a) 731
(b) 64
(c) 8
(d) 16

GATE CSE - 2026 MCQ SET - 1(2-MARKS)

Q2. Consider the implementation of a sliding window protocol over a lossless link, with a window size of W frames, where each frame is of size 1000 bits (including header).
The bandwidth of the link is 100 kbps ($1k = 10^3$) and the one-way propagation delay is 100 milliseconds. Assume that processing times at the sender and receiver are zero and the transmission time of acknowledgements is also zero.
Which one of the following options gives the minimum size of W (in number of frames) required to achieve 100% link utilization?

(a) 10
(b) 21
(c) 20
(d) 11

GATE CSE - 2025 MCQ SET - 2(2-MARKS)

Q3. Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits per second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgment packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?

(a) 88.23
(b) 93.75
(c) 85.44
(d) 66.67

GATE CSE – 2024 NAT SET - 2(2-MARKS)

Q4. Consider an Ethernet segment with a transmission speed of 10^8 bits/sec and a maximum segment length of 500 meters. If the maximum speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (in bits) required for collision detection is _____.

GATE CSE – 2023 MCQ (1-MARKS)

Q5. Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.

(a) Longer link length and lower transmission rate
(b) Longer link length and higher transmission rate
(c) Shorter link length and lower transmission rate
(d) Shorter link length and higher transmission rate

GATE CSE – 2022 NAT (2-MARKS)

Q6. Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of

Check Out Our GATE Courses & Secure Your Dream Rank!

2100 km. The signal propagates at a speed of 3×10^8 m/s. The time taken (in milliseconds, rounded off to two decimal places) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is: _____

GATE CSE – 2021 MCQ (2-MARKS)

Q7. Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial x^3+x+1. Suppose the message $m_4m_3m_2m_1m_0=11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. The value of the check bit sequence $c_2c_1c_0$ is:

(a) 110
(b) 111
(c) 100
(d) 101

GATE CSE – 2021 MCQ (2-MARKS)

Q8. Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables:

Data Bits	d8	d7	d6	d5	d4	d3	d2	d1
	1	1	0	x	0	1	0	1

Check Bits	c8	c4	c2	c1
	y	0	1	0

Which one of the following choices gives the correct values of x and y?

(a) x is 0 and y is 0
(b) x is 0 and y is 1
(c) x is 1 and y is 0
(d) x is 1 and y is 1

GATE CSE – 2021 NAT (2-MARKS)

Q9. Consider the sliding window flow-control protocol operating between a sender and a receiver over a full-duplex error-free link.
Assume the following:

- The time taken for processing the data frame by the receiver is negligible.
- The time taken for processing the acknowledgement frame by the sender is negligible.
- The sender has an infinite number of frames available for transmission.
- The size of the data frame is 2000 bits, and the size of the acknowledgement frame is 10 bits.
- The link data rate in each direction is 1 Mbps (= 10^6 bits per second).
- One-way propagation delay of the link is 100 milliseconds.

The minimum value of the sender's window size in terms of the number of frames, (rounded to the nearest integer) needed to achieve a link utilization of 50% is _____.

GATE CSE – 2021 NAT SET-2 (2-MARKS)

Q10. Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1Mbps($=10^6$ bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is _____

GATE CSE – 2018 NAT (2-MARKS)

Q11. Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless).
The nodes in the system use the following carrier-sense based medium access protocol.
A node that receives a packet to transmit will carrier-sense the medium for 5 units of time.

Check Out Our GATE Courses & Secure Your Dream Rank!

If the node does not detect any other transmission in this duration, it starts transmitting its packet in the next time unit.
If the node detects another transmission, it waits until this other transmission finishes and then begins to carrier-sense for 5 time units again.
Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs.
All transmissions last for 20 units of time.
Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium.
Assume that the system has two nodes P and Q, located at a distance d meter from each other.
P starts transmitting a packet at time t=0 after successfully completing its carrier-sense phase. Node Q has a packet to transmit at time t=0 and begins to carrier-sense the medium.
The maximum distance d (in meters, rounded to the closest integer) that allows Q to avoid a collision between its proposed transmission and P's ongoing transmission is _____.

GATE CSE – 2017 MCQ SET - 2(2-MARKS)

Q12. Consider a binary code that consists of only four valid codewords as given below: 00000, 01011, 10101, 11110.
Let the minimum Hamming distance of the code be p and the maximum number of erroneous bits that can be corrected by the code be q. Then the values of p and q are:

(a) p=3 and q=1
(b) p=3 and q=2
(c) p=4 and q=1
(d) p=4 and q=2

GATE CSE – 2017 MCQ SET - 1(2-MARKS)

Q13. A computer network uses polynomials over GF(2) for error checking with 8 bits as information bits and uses x^3+x+1 as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as:

(a) 01011011010
(b) 01011011011
(c) 01011011101
(d) 01011011100

GATE CSE – 2017 NAT SET – 1 (2-MARKS)

Q14. The values of parameters for the Stop-and-Wait ARQ protocol are as given below:

- Bit rate of the transmission channel = 1 Mbps.
- Propagation delay from sender to receiver = 0.75 ms.
- Time to process a frame = 0.25 ms.
- Number of bytes in the information frame = 1980.
- Number of bytes in the acknowledge frame = 20.
- Number of overhead bytes in the information frame = 20.

Assume that there are no transmission errors. Then, the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _____ (correct to 2 decimal places).

GATE CSE – 2016 MCQ SET - 2(2-MARKS)

Q15. Consider two hosts X and Y , connected by a single direct link of rate 10^6 bits /sec . The distance between the two hosts is 10,000 km and the propagation speed along the link is 2×10^8 m/sec . Host X sends a file of 50,000 bytes as one large message to host Y continuously . Let the transmission and propagation delays be p milliseconds and q milliseconds respectively. Then the value of p and q are:

(a) p = 50 and q = 100
(b) p = 50 and q = 400
(c) p = 100 and q = 50
(d) p = 400 and q = 50

GATE CSE – 2016 NAT SET - 2(2-MARKS)

Q16. Consider a 128×10^3 bits/second satellite communication link with one-way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgment. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2016 NAT SET - 1(2-MARKS)

Q17. A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1 Kbps = 1000 bits/second). Size of an acknowledgment is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, the sender throughput is _____ bytes/second.

GATE CSE – 2016 NAT SET - 2(2-MARKS)

Q18. A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.

GATE CSE – 2016 MCQ SET - 2(2-MARKS)

Q19. In an Ethernet local area network, which one of the following statements is TRUE?

(a) A station stops to sense the channel once it starts transmitting a frame.
(b) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.
(c) A station continues to transmit the packet even after the collision is detected.
(d) The exponential backoff mechanism reduces the probability of collision on retransmissions

GATE CSE – 2015 MCQ SET - 3(1-MARKS)

Q20. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

(a) 8000
(b) 10000
(c) 16000
(d) 20000

GATE CSE – 2015 NAT SET - 1(2-MARKS)

Q21. Consider a LAN with four nodes S1,S2,S3,S4. Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmits in the same slot. The probabilities of generation of a frame in a time slot by S1,S2,S3 and S4 are 0.1, 0.2, 0.3, and 0.4, respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is _____.

GATE CSE – 2015 NAT SET - 2(1-MARKS)

Q22. A link has a transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgment has negligible transmission delay, and that its propagation delay is the same as the data propagation delay. Also assume that the processing delays at nodes are negligible. The efficiency of the stop-and-wait protocol in this setup is exactly 25%. The value of the one-way propagation delay (in milliseconds) is:

GATE CSE – 2015 NAT SET - 3(2-MARKS)

Q23. Two hosts are connected via a packet switch with 10^7 bits per second links. Each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits, the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds is:

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2015 NAT SET - 3(2-MARKS)

Q24. Consider a network connecting two systems located 8000 kilometers apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It is needed to design a Go-Back-N sliding window protocol for this network. The average packet size is 10^7 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then, the minimum size in bits of the sequence number field has to be

GATE CSE – 2015 NAT SET - 1(2-MARKS)

Q25. Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgment and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% is: _____

GATE CSE - 2014 MCQ SET - 1(1-MARKS)

Q26. Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec . To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is ___ .

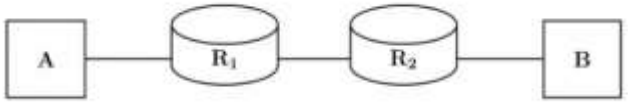
GATE CSE - 2014 MCQ SET - 3(1-MARKS)

Q27. A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 011111100101, then the input bit-string is:

(a) 0111110100
(b) 0111110101
(c) 0111111101
(d) 0111111111

GATE CSE – 2014 MCQ SET - 2 (2-MARKS)

Q28. Consider the store and forward packet switched network given below. Assume that the bandwidth of each link is 10^6 bytes/sec. A user on host A sends a file of size 10^3 bytes to host B through routers R1 and R2 in three different ways. In the first case, a single packet containing the complete file is transmitted from A to B. In the second case, the file is transmitted from A to B. In the third case, the file is split into 20 equal parts and these packets are sent from A to B. Each packet contains 100 bytes of header information along with the user data. Consider only transmission time and ignore processing, queuing and propagation delays. Also assume that there are no errors during transmission. Let T1, T2, and T3 be the times taken to transmit the file in the first, second, and third case respectively. Which one of the following is CORRECT?



(a) T1 < T2 < T3
(b) T1 > T2 > T3
(c) T2 = T3, T3 < T1
(d) T1 = T3, T3 > T2

GATE CSE – 2013 MCQ (2-MARKS)

Q29. Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

(a) 1
(b) 2
(c) 2.5
(d) 5

GATE CSE – 2012 MCQ (2-MARKS)

Q30. Consider a source computer (S) transmitting a file of size 10^6 bits to a destination computer (D) over a network of two routers (R1 and R2) and three links (L1, L2, and L3).

Check Out Our GATE Courses & Secure Your Dream Rank!

L1 connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D. Let each link be of length 100 km.
Assume signals travel over each link at a speed of 10^8meters per second.
Assume that the link bandwidth on each link is 1 Mbps. Let the file be broken down into 1000 packets, each of size 1000 bits.
Find the total sum of transmission and propagation delays in transmitting the file from S to D?

(a) 1005 ms
(b) 1010 ms
(c) 3000 ms
(d) 3003 ms

GATE CSE – 2009 MCQ (2-MARKS)

Q31. Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25 ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).
What is the minimum number of bits (l) required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

(a) $I = 2$
(b) $I = 3$
(c) $I = 4$
(d) $I = 5$

GATE CSE – 2009 MCQ (2-MARKS)

Q32. Suppose that the sliding window protocol is used with the sender window size of 2^l, where l is the number of bits identified in the earlier part and acknowledgements are always piggybacked. After sending 2^l frames, what is the minimum time the sender will have to wait before starting transmission of the next frame? (Ignore frame processing time.)

(a) 16 ms
(b) 18 ms
(c) 20 ms
(d) 22 ms

GATE CSE – 2009 MCQ (2-MARKS)

Q33. Let $G(x)$ be the generator polynomial used for CRC checking. What is the condition that should be satisfied by $G(x)$ to detect odd number of bits in error?

(a) $G(x)$ contains more than two terms
(b) $G(x)$ does not divide $1 + x^k$, for any k not exceeding the frame length
(c) $1 + x$ is a factor of $G(x)$
(d) $G(x)$ has an odd number of terms.

GATE CSE – 2007 MCQ (2-MARKS)

Q34. The distance between two stations M and N is L kilometers. All frames are K bits long. The propagation delay per kilometer is t seconds. Let R bits/second be the channel capacity. Assuming that processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the sliding window protocol is used, is:

(a) $\lceil log_2 \frac{(2LtR+2K)}{K} \rceil$
(b) $\lceil log_2 \left(\frac{2LtR}{K}\right) \rceil$
(c) $\lceil log_2 (\frac{2LtR + K)}{K}) \rceil$
(d) $\lceil log_2 (\frac{(2LtR + K)}{(2K)}) \rceil$

GATE CSE – 2007 MCQ (2-MARKS)

Q35. The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is:

(a) 11001001000
(b) 11001001011
(c) 11001010
(d) 110010010011

GATE CSE – 2007 MCQ (2-MARKS)

Q36. In a token ring network, the transmission speed is 10^7 bps and the propagation speed is 200 metres/μs. The 1-bit delay in this network is equivalent to:

(a) 500 metres of cable
(b) 200 metres of cable
(c) 20 metres of cable
(d) 50 metres of cable

GATE CSE – 2007 MCQ (2-MARKS)

Q37. There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot?

(a) $np(1-p)^{n-1}$
(b) $(1-p)^{n-1}$
(c) $p(1-p)^{n-1}$
(d) $1-(1-p)^{n-1}$

GATE CSE – 2006 MCQ (2-MARKS)

Q38. Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?

(a) 20
(b) 40
(c) 160
(d) 320

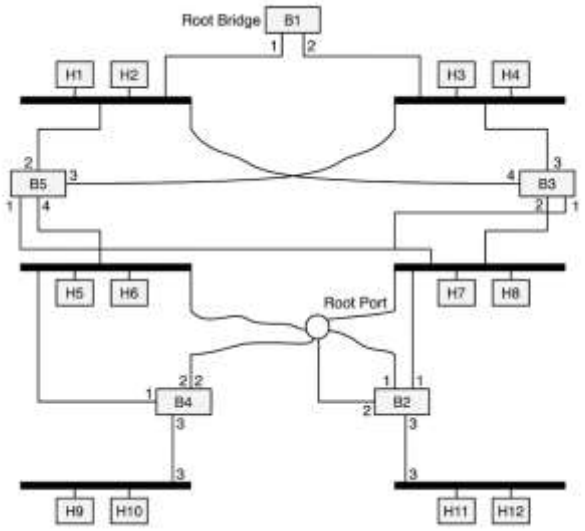
GATE CSE – 2006 MCQ (2-MARKS)

Q39. Station A needs to send a message consisting of 9 packets of Station B using sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

(a) 12
(b) 14
(c) 16
(d) 18

GATE CSE – 2006 MCQ (2-MARKS)

Q40. Consider the diagram shown below where a number of LANs are connected by (transparent) bridges. In order to avoid packets looping through circuits in the graph, the bridges organize themselves in a spanning tree. First, the root bridge is identified as the bridge with the least serial number. Next, the root sends out (one or more) data units to enable the setting up of the spanning tree of shortest paths from the root bridge to each bridge.
Each bridge identifies a port (the root port) through which it will forward frames to the root bridge. Port conflicts are always resolved in favour of the port with the lower index value. When there is a possibility of multiple bridges forwarding to the same LAN (but not through the root port), ties are broken as follows: bridges closest to the root get preference and between such bridges, the one with the lowest serial number is preferred.



For the given connection of LANs by bridges, which one of the following choices represents the depth first traversal of the spanning tree of bridges?

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) B1, B5, B3, B4, B2
(b) B1, B3, B5, B2, B4
(c) B1, B5, B2, B3, B4
(d) B1, B3, B4, B5, B2

GATE CSE – 2006 MCQ (2-MARKS)

Q41. Consider the correct spanning tree for the previous question. Let host H1 send out a broadcast ping packet. Which of the following options represents the correct forwarding table on B3?

(a)

Hosts	Port
H1, H2, H3, H4	3
H5, H6, H9, H10	1
H7, H8, H11, H12	2

(b)

Hosts	Port
H1, H2	4
H3, H4	3
H5, H6	1
H7, H8, H9, H10, H11, H12	2

(c)

Hosts	Port
H3, H4	3
H5, H6, H9, H10	1
H1, H2	4
H7, H8, H11, H12	2

(d)

Hosts	Port
H1, H2, H3, H4	3
H5, H7, H9, H10	1
H7, H8, H11, H12	4

GATE CSE – 2005 MCQ (1-MARKS)

Q42. The maximum window size for data transmission using the selective reject protocol with n-bit frame sequence numbers is:

(a) 2^n
(b) 2^{n-1}
(c) $2^n - 1$
(d) 2^{n-2}

GATE CSE – 2005 MCQ (1-MARKS)

Q43. In a network of LANs connected by bridges, packets are sent from one LAN to another through intermediate bridges. Since more than one path may exist between two LANs, packets may have to be routed through multiple bridges. Why is the spanning tree algorithm used for bridge-routing?

(a) For shortest path routing between LANs
(b) For avoiding loops in the routing paths
(c) For fault tolerance
(d) For minimizing collisions

GATE CSE – 2005 MCQ (2-MARKS)

Q44. Suppose the round trip propagation delay for a 10 Mbps Ethernet having 48-bit jamming signal is 46.4 μs. The minimum frame size is:

(a) 94
(b) 416
(c) 464
(d) 512

Answer key

1. (b)	**2. (b)**	**3. (a)**
4. (500)	**5. (b)**	**6 . (7.08)**
7. (c)	**8. (a)**	**9. (51)**
10.(135)	**11.(50)**	**12.(a)**
13. (c)	**14.(86.5-89.5)**	**15.(d)**
16. (4)	**17. (2500)**	**18. (200)**
19. (d)	**20. (d)**	**21. (0.4404)**
22. (12)	**23. (1575)**	**24. (8)**
25. (320)	**26.(5)**	**27.(b)**
28.(d)	**29. (b)**	**30. (a)**
31.(d)	**32. (c)**	**33.(c)**
34. (c)	**35. (b)**	**36. (c)**
37. (a)	**38. (b)**	**39. (c)**
40. (a)	**41. (a)**	**42. (c)**
43. (b)	**44. (d)**	

HINTS & SOLUTIONS

1. **ANSWER:- B**
File size in bits:
4 million bytes = 4×10^6 bytes

Check Out Our GATE Courses & Secure Your Dream Rank!

$= 4 \times 10^6 \times 8$ bits
$= 32 \times 10^6$ bits
Bottleneck bandwidth:
Link rates:
2 Mbps = 2×10^6 bps
500 kbps = 5×10^5 bps
1 Mbps = 1×10^6 bps
Minimum bandwidth = 500 kbps = 5×10^5 bps
Total transfer time:
For a large continuous transfer across multiple links, throughput is limited by the **bottleneck link**.
Time = File size / Bottleneck rate
time = $(32 \times 10^6) / (5 \times 10^5)$
time = 64 seconds

2. ANSWER:- B
For 100% utilization in sliding window:
$W \geq 1 + 2a$
where
a = propagation delay / transmission delay
Transmission delay:
Frame size / bandwidth
$= 1000 / (100 \times 10^3)$
= 0.01 s
= 10 ms
Propagation delay:
100 ms
a = 100 / 10 = 10
Window size:
$W \geq 1 + 2a$
$W \geq 1 + 20$
$W \geq 21$
Minimum W = 21

3. ANSWER:- A
- Frame size 3000 bits
- Transmission rate = 2000 bps
- Propagation delay (T) = 100 milliseconds

The formula for channel utilization is:
Channel utilization $\frac{T_{data}}{T_{data}+2T_p} \times 100T$
Where:
$T_{data} = \frac{Frame\ size}{Tranmission\ rate} = \frac{300bits}{2000\ bps} = 1.5\ sec.$
So, the channel utilization becomes:
Channel utilization $= \frac{1.5}{\frac{1.5}{+2}\times 01} \times 100 = \frac{1.5}{1.7} \times 100$
= 88.23

4. ANSWER:- 500
Bw = 10^8 bits/sec
D = 500 m
V = 2×10^8 m/sec
For ethernet,
$T_t >= 2 . T_p$
where T_t is transmission time and T_p is propagation time
$\Rightarrow \frac{f_s}{Bw} \geq 2 \times \frac{d}{v}$
$\Rightarrow f_s \geq 2 \times \frac{d}{v} \times Bw$
$\Rightarrow f_s \geq 2 \times \frac{500\ m}{2\times10^8\ m/sec} \times 10^8 bits/sec$

$\Rightarrow f_s \geq 500$ bits
$\therefore f_{s\,min} = 500\ bit$

5. ANSWER:- B
Utilisation of Link = Efficiency
Efficiency of Stop and wait protocol is
$$\eta = \left(\frac{1}{1+\left(\frac{2T_{\{p\}}}{T_{\{t\}}}\right)}\right) = \left(\frac{1}{1+\left(\frac{2*dB}{vL}\right)}\right)$$
where d = link length (distance), v = speed, B = bandwidth (Transmission Rate) and L = packet Length
As Efficiency η is inversely proportional to distance d, So Longer Distance means lower efficiency.
As Efficiency η is also inversely proportional to Bandwidth B, So Higher Bandwidth means lower efficiency.
Hence option B is correct.

6. ANSWER:- 7.08
Bandwidth (Bw) = 100 Mbps
distance (d) = 2100Km = 2100×10^3 m
velocity (v) = 3×10^8 m / s
packet size (L) = 1000 bytes = 8000 bits
Here, sender is station and receiver is satellite
So, Total time = Transmission time + Propagation time
Transmission time = $\frac{L}{Bw} = \frac{800\ b}{100\times10^5 b/sec}$ =0.08 ms
Propagation time ms = $\frac{d}{v} = \frac{2100\times10^3}{3\times10^8\ m/sec} = 7$
$\therefore$ Total time = 0.08 + 7 = 7.08 ms

7. ANSWER:- C
The given polynomial $z^3 + z + 1$ is written as 1011, which consists of 4 bits so, append 3 bits of Os to the message.
M = 11000000

Check Out Our GATE Courses & Secure Your Dream Rank!

```
1011)11000000
     1011
      1110
      1011
       1010
       1011
          100
```

There the check bits sequence is 100.

8. ANSWER:- A

$$c_1 = d_{11} \oplus d_9 \oplus d_7 \oplus d_5 \oplus d_3$$

For even parity, the total modulo-2 sum of the check bit and its associated data bits must equal 0 :

$$c_1 \oplus d_{11} \oplus d_9 \oplus d_7 \oplus d_5 \oplus d_3 = 0 \ (\text{mod}2)$$

Substitute the known values from the tables:

$$0 \oplus 1 \oplus x \oplus 0 \oplus 0 \oplus 1 = 0 \ (\text{mod}2)$$

Simplify the numbers (remembering that $1 \oplus 1 = 0$):

$$0 \oplus x = 0 \ (\text{mod}2)$$

$$c_8 = d_{12} \oplus d_{11} \oplus d_{10} \oplus d_9$$

For even parity, the total modulo-2 sum of c_8 and these data bits must equal 0 :

$$c_8 \oplus d_{12} \oplus d_{11} \oplus d_{10} \oplus d_9 = 0 \ (\text{mod}2)$$

Substitute the known values, including $c_8 = y$ and the value of $x = 0$ we just calculated:

$$y \oplus 1 \oplus 1 \oplus 0 \oplus 0 = 0 \ (\text{mod}2)$$

Simplify the expression:

$$y \oplus 0 = 0 \ (\text{mod}2)$$

$$\mathbf{y = 0}$$

9. ANSWER:- 51

The time taken for a packet (Tt(packet)):

Tt(packet)=L/B.W=2000 bits/10^6 bps=2×10^{-3} sec= 2 msec

The time taken for an acknowledgement (Tt(Ack)):

Tt(Ack) = 10 bits/10^6 bps = 10^{-5} sec=10^{-2} msec=0.01 msec

Tp=100 msec

The total time (Total time):

Totaltime=Tt(packet)+2*Tp+Tt(Ack)=2 msec+200 msec+0.01 msec=202.01

Efficiency Calculation:

Efficiency (e) = useful time/total time

e = n* Tt(packet)/(Totaltime)

0.5 <= n*2/202.01

n >= 0.5*202.01/2 >= 101.005/2 >=50.5025

hence the window size is 51.

10. Answer: 135

Frame length = 1000 bits

Transmission rate = 1 Mbps = 10^6 bits/sec

Arrival rate λ = 1000 frames/sec

Step 1: Frame transmission time (T):

T = frame length / transmission rate

T = 1000 / 10^6 = 0.001 sec

Step 2: Offered load (G):

G = λ × T

G = 1000 × 0.001 = 1

Step 3: Throughput formula for Pure ALOHA:

S = G e^(−2G)

S = 1 × e^(−2) ≈ 0.1353

(This is in frames per frame-time)

Step 4: Convert to frames per second:

Throughput = S / T

Throughput = 0.1353 / 0.001 ≈ 135.3

11. ANSWER:- 50

At time t=0, station P starts transmitting packet (since it already senses the channel is free). At the same time, station Q is sensing the channel to check if the channel is free or not.

To avoid collision:

Sensing of node Q ≥ Transmitting a packet i.e., if the channel is found free at 5 unit time or more than 5 unit time, station Q data will be collied with station P anywhere in the middle.

So, the distance between P and Q = sensing of node Q ≥ transmitting a packet.

d=5 time unit

d=5 unit × 10 meter/unit

d=50 meter

Note: The idea is simple, the 1st bit of node P data reaches node Q within 5 time units or at exact 5 time unit, then node Q understands that someone is transmitting the data and waits for the next time unit to sense the channel is free or not.

12. ANSWER:- A

code1: 00000, code2: 01011, code3: 10101, code4: 411110

Hamming distance = min of all Hamming distances.

Which is 3 between code1 and code2, so p=3.

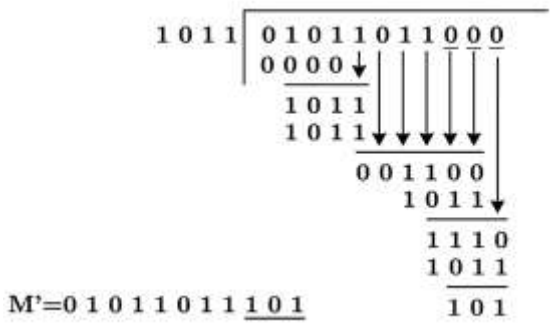
Now to correct d bit error we need Hamming distance = 2d+1.

So, 2d+1=3 will give d=1.

13. ANSWER:- C

The generator polynomial has degree 3. So, we append 3 zeroes to the original message.

Check Out Our GATE Courses & Secure Your Dream Rank!



14. ANSWER:- 86.5-89.5

Processing time of frame = 0.25 msec

Transmission time of data = $\frac{Data\ size}{Bandwidth}$

$\frac{(1980+20)bytes}{106\ bits/sec}$ =16 msec

Propagation time = 0.75 msec

Transmission time of ACK

$\frac{20\times8\ bits}{10^6\ bit/sec}$ = 0.16 msec

Transmission efficiency of Stop-and-Wait ARQ

$\frac{16}{0.25+16+1.5+0.16} = \frac{16}{17.91} \times 100 = 89.33\%$

15. ANSWER:- D

• Transmission time = $\frac{Data\ size}{Brand\ width}$

$= \frac{50000 \times 8\ bits}{10^6\ bits/sec} = 400\ msec$

• Propagation time = $\frac{Distance}{Velocity}$

$= \frac{10000 \times 10^3\ m}{2\times10^8\ m/sec} = 50 \times 10^{-8}$ sec $= 50\ msec$

16. ANSWER:- 4

BW= 128×10^3 bits/sec

L=1KB=$1 \times 10^3 \times 8$ bits

T_p = 150 m sec

η = 100%

$T_t = \frac{L}{BW} = \frac{1\times10^3\times8}{128\times10}$ = 0.0625 sec = 62.5 msec

$a = \frac{T_p}{T_t} = \frac{150}{62.5} = 2.4$

$\eta = \frac{N}{1+2a} = 1 \Rightarrow \frac{N}{1+2\times2.4} = 1 \Rightarrow N = 5.8$

$\therefore W_s = W_r = 6$

Available Sequence Number $\geq$ 12

Min No. of bits required 4 bits

17. ANSWER:- 2500

Sender Throughput = Data/Total Time

= 1000 bytes/(0.1+0.1+0.1+0.1)

= 2500 bytes/sec

18. ANSWER:- 200

because bandwidth is $20*10^6$ bits

L=2* propagation delay * bw

L = 2* $(40/10^6) * 20*10^6$

L = 1600 bits(because of bandwidth in bits)

L=1600/8

L=200 Bytes

19. ANSWER:- D

exponential backoff of algorithm reduces the probability of collision in next iteration.

20. ANSWER:- D

bandwidth ,b=100 Mbps

distance , d= 1km

no repeaters

I = 1250 * 8 bits

tf >= 2tp

I/b >= 2* d/v

v = 2 * 1000 * 100,000000/(1250 *8)

v = 20000 km/sec

21. ANSWER:- 0.4404

The probability of sending a frame in the first slot without any collision by any of these four stations is sum of following 4 probabilities

Probability that S1 sends a frame and no one else does + Probability thatS2 sends a frame and no one else does + Probability thatS3 sends a frame and no one else does +

Probability thatS4 sends a frame and no one else does

= 0.1 * (1 - 0.2) * (1 - 0.3) *(1 - 0.4) + (1 -0.1) * 0.2 * (1 - 0.3) *(1 - 0.4) + (1 -0.1) * (1 - 0.2) * 0.3 *(1 - 0.4) + (1 -0.1) * (1 - 0.2) * (1 - 0.3) * 0.4

= 0.4404

22. ANSWER:- 12

Efficiency of Stop & Wait = 1/1+2a

0.25=1/1+2a

$a=P_d/t_d$, t_d=8 ms

=> Pd=12 ms

23. ANSWER:- 1575

Extra delay at switch = 35 microseconds for each packet.

Data = 10000 bits

Number of packets = 10000/5000=2 packets

Transmission delay for one packet = 500 microseconds.

Check Out Our GATE Courses & Secure Your Dream Rank!

- At t=500 μsec, last bit of packet 1 is placed on link 1 by A and transmission of packet 1 begins.
- At t=520 μsec, last bit of packet 1 reaches switch.
- At t=555 μsec, first bit of packet 1 is placed on link 2 by the switch.
- At t=1000 μsec, last bit of packet 2 is placed on link 1 by A.
- At t=1020 μsec, last bit of packet 2 reaches switch.
- Note: Packet 2 needs to wait until 1055 μsec before the switch transfers it.
- Last bit of packet 1 will be placed on link 2 by switch at 1055 μsec.
- Hence, no additional delay for packet 2.
- At t=1055 μsec, packet 2's first bit is placed on link 2.
- At t=1575 μsec, last bit of packet 2 reaches B.
- ∴1575 μsec is required.

24. ANSWER:- 8

d=8000km
Bandwidth = 500×10^{6} bps
Propagation delay:
$Pd=8000\times10^{3}/4\times10^{6}=2$ sec
Packet size = 10^{7} bits
Transmission delay:
$t_d=10^{7}/500\times10^{6}=1/50$ sec
n=100%
100/100=N/(1+2a)
$a=P_d/t_d=2/(1/50)$ sec=100
100/100=N/201
N=201
Number of packets = 201
8 bits are required for the sequence number.

25. ANSWER:- 320

efficiency = $t_d / (t_d + 2\times p_d)$
For 50% efficiency:
$1/2 = t_d / (t_d + 2\times p_d)$
$t_d = (1/2) * (t_d + 2\times p_d)$
$t_d = (1/2) * (t_d) + p_d$
$t_d / 2 = p_d = 20$ ms
$t_d = 40$ ms
Minimum frame size / 64Kbps = 40 ms
Minimum frame size = 64Kbps * 40 ms =2560 bits = 320 bytes

26. Answer: 5

Frame size = 1 KB = 1024 bytes = 8192 bits
Bandwidth = 1.5 Mbps = 1.5×10^{6} bits/sec
One-way delay = 50 ms ⇒ RTT = 100 ms
Target utilization = 0.6
Transmission time (Tt):
$Tt = 8192 / (1.5 \times 10^{6}) \approx 0.00546$ sec
Using utilization formula for sliding window:
$U = \min(1, W / (1 + 2a))$
where $a = Tp / Tt$
$a = 0.05 / 0.00546 \approx 9.16$
So, $1 + 2a \approx 1 + 18.32 = 19.32$
To achieve U = 0.6:
$W / 19.32 = 0.6$
$W \approx 0.6 \times 19.32 \approx 11.59$
Minimum window size W = 12
Selective Repeat condition:
$W \leq 2^{(k-1)}$
So,
$2^{(k-1)} \geq 12$
$k-1 = 4 \Rightarrow k = 5$

27. ANSWER:- B

Input string: 0111110101
Output string: 01111100101
After five consecutive 1's in the input, a bit 0 is inserted.

28. ANSWER:- D

Bandwidth: 10^6 bps
Packet size: 1000 bytes
1st transmission: 1000 + 100 = 1100 bytes transmitted at a time from A
T_1:
1000+100(overhead)/106=1100×8/106=8800/106=8.8 msec
$T_1 = 8.8 \times 3 = 26.4$ msec
2nd transmission: 100 + 100 = 200 bytes transmitted 10 times from A
T_2:
100+100(overhead)/106=200×8/106=1600/106=1.6 msec
$T_2 = (3 \times 1.6) + (9 \times 1.6) = 19.2$ msec
3rd transmission: 50 + 100 = 150 bytes transmitted 20 times from A
T_3:
50+100(overhead)/106=150×8/106=1200/106=1.2 m sec
$T_3 = (3 \times 1.2) + (19 \times 1.2) = 26.4$ msec
$T_1 = T_3 = 26.4$ msec
$T2 < T1 = T_3$

29. ANSWER:- B

transmission time > round trip time of 1 bit
transmission time ≥ 2 × propagation time

Check Out Our GATE Courses & Secure Your Dream Rank!

$\frac{10{,}000\ bits}{500\ Mbps} \geq 2 \times \frac{d}{2\times10^5\ km\ per\ sec}$

2 km ≥ d

30. ANSWER:- A

Routers are store and forward devices.

- **Propagation time** = 100 km/10^8 m/s = 1 ms
- **Transmission time for a packet** = 1000 bits/10^6 bits/sec = 1 ms

Packets will be forwarded in a pipelined manner after the first packet reaches the receiver, with every 1 ms a new one arriving.

Now, the time taken by packet number 1 to reach the destination is:

1 ms (Tx at sender) + 1 ms (Tp from sender to R1) + 1 ms (Tx at R1)

- 1 ms (Tp from R1 to R2) + 1 ms (Tx at R2) + 1 ms (Tp from R2 to destination) = 6 ms

So, for packet 1000:

6 ms + 999 ms = 1005 ms

31. ANSWER:- D

Propagation time is given as 25 ms .

Bandwidth = 10^6bps.

So, to fully utilize the channel, we must send 10^6 bits into the channel in a second, which will be 1000 frames per second as each frame is 1000 bits.

Now, since the propagation time is 25 ms , to fully pack the link we need to send at least $1000 \times 25 \times 10^{-3} = 25$ frames.

So, we need $\lceil \log_2 25 \rceil = 5$ bits

32. ANSWER:- C

1. Timing Parameters

- Transmission Delay of Data Frame (T_t) :

$$T_t = \frac{1000\text{bits}}{10^6\text{bps}} = 1\text{ ms}$$

- Round Trip Time (RTT): Since acknowledgments are piggybacked onto full-sized reverse data frames, the acknowledgment transmission delay is also 1 ms .

$$\text{RTT} = T_t(\text{ Data }) + T_p + T_t(\text{ACK}) + T_p$$

$$\text{RTT} = 1\text{ ms} + 25\text{ ms} + 1\text{ ms} + 25\text{ ms} = 52\text{ ms}$$

2. Sender Activity and Idle Window

- Window Size (W): Given as $2^5 = 32$ frames.
- Sender Busy Time: The time taken to transmit the entire window continuously without stopping:

Busy Time = 32 frames × 1 ms/ frame = 32 ms

- Sender Waiting Time: The idle time spent waiting for the first acknowledgment to arrive after the transmission of the window finishes:

Waiting Time = RTT - Busy Time

Waiting Time = 52 ms − 32 ms = 20 ms

33. Answer: C

To detect any odd number of bit errors, the generator polynomial must ensure that any error pattern with odd weight is not divisible by $G(x)$.

An error pattern with an odd number of bit errors always has $(1+x)$as a factor.

So, to detect all such errors, $G(x)$must also contain $(1+x)$as a factor.

34. ANSWER:- C

Propagation delay = Lt

RTT = 2Lt

Transmission time per frame = K / R

Maximum frames in transit (for full utilization):

= RTT / transmission time

= (2Lt) / (K/R)

= 2LtR / K

For continuous transmission, window size W must satisfy:

W ≥ 1 + 2LtR/K

So,

W ≥ (2LtR + K) / K

Sequence number space ≥ W ⇒ 2^l ≥ W

l ≥ $\log_2((2LtR + K)/K)$

Minimum integer value:

l = $\lceil \log_2((2LtR + K)/K) \rceil$

35. ANSWER:- B

Generator polynomial: $x^3 + 1$ ⇒ binary = 1001

Step 1: Append 3 zeros (degree = 3)

11001001 → 11001001000

Step 2: Perform modulo-2 division of 11001001000 by 1001

After division, remainder = 011

Step 3: Append remainder to original message

Final transmitted message = 11001001 + 011

= 11001001011

36. ANSWER:- C

Bit rate = 10^7 bps ⇒ time for 1 bit = 1 / 10^7 s = 0.1 μs

Propagation speed = 200 m/μs

Distance travelled in 0.1 μs = 200 × 0.1 = 20 metres

Check Out Our GATE Courses & Secure Your Dream Rank!

37. ANSWER:- A

For exactly one station to transmit:
One station transmits ⇒ probability = p
Remaining (n − 1) stations do not transmit ⇒ probability = $(1-p)^{n-1}$
Choose which station transmits: n ways
Total probability = $n \times p \times (1-p)^{n-1}$

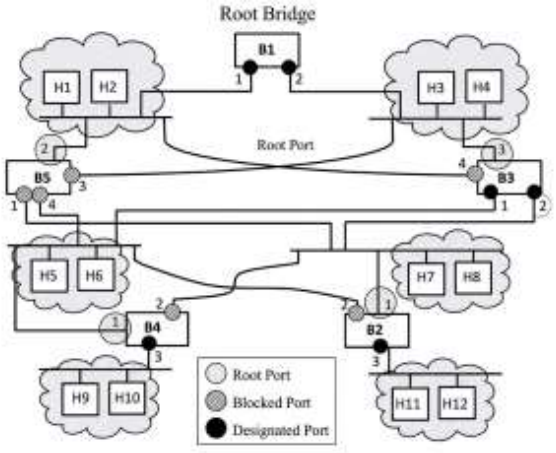
38. ANSWER:- B

Packet size = 32 bytes = 32 × 8 = 256 bits
Bandwidth = 128 kbps = 128000 bits/s
Transmission time per packet = 256 / 128000 = 0.002 s = 2 ms
RTT = 80 ms
Optimal window size = RTT / transmission time = 80 / 2 = 40

39. ANSWER:- C

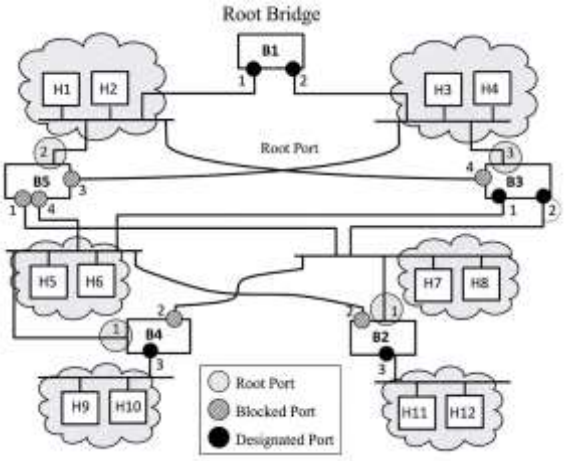
Window size = 3, Go-Back-N
Packets to send = 9
Every 5th transmission is lost (count transmissions, not packet number)
Transmit sequence:
1, 2, 3,4 → success
5, 6, 7 → here 5th transmission (packet 5) is lost
Due to Go-Back-N, packets after lost one are discarded → resend from packet 5
Now retransmit:
5, 6 → all success
Next:
7,8,9
again 7th packet is lost
Now retransmit:
7,8 → all success
and 9 lost
Resend 9
Total no of transmissions = 16

40. ANSWER:- A



The correct DFS traversal will be (a) B1, B5, B3, B4, B2

41. ANSWER:- A



Correct Answer: a

42. ANSWER:- B

In Selective Repeat (Selective Reject), to avoid ambiguity between old and new frames:
Window size ≤ 2^{n-1}
So maximum window size = 2^{n-1}

43. ANSWER:- B

Since, in a spanning tree, there is a unique path from a source to the destination, which avoids loops, since it is a tree, and contains all the nodes, since it is a spanning tree.

44. ANSWER:- D

Tt >=2td + time for sending jamming signal

$RTT = 2^{*}Td = 46.6\mu s$
time for sending jamming signal = 48 bits /10Mbps = $4.8\mu s$

Now,

Tt >=2*td + time for sending jamming signal
$Tt >= 46.6\mu s + 4.8\mu s$
$Tt >= 51.2\mu s$
$L >= 51.2\mu s^{*}10Mbps$
$L >= 512$

Check Out Our GATE Courses & Secure Your Dream Rank!

3 TCP UDP AND IP

GATE CSE – 2026 NAT SET - 2(1-MARKS)

Q1. If an IP network uses a subnet mask of 255.255.240.0, the maximum number of IP addresses that can be assigned to network interfaces is _____. (answer in integer)

GATE CSE – 2026 MSQ (2-MARKS)

Q2. An ISP having an address block 202.16.0.0/15 assigns a block of 6000 IP addresses to a client using the Classless Internet Domain Routing (CIDR) supernetting approach.
Which of the following address blocks can be assigned by the ISP?

(a) 202.16.0.0/19
(b) 202.17.64.0/19
(c) 202.16.32.0/19
(d) 202.17.24.0/19

GATE CSE – 2026 MCQ SET - 1(1-MARKS)

Q3. A TCP sender successfully establishes a connection with a TCP receiver and starts the transmission of segments. The TCP congestion control mechanism's slow-start threshold is set to 10000 segments. Assume that the round-trip time is fixed at 1 millisecond. Assume that the sender always has data to send, the segments are numbered from 1, and no segment is lost.
Let t denote the time (in milliseconds) at which the transmission of segment number 2000 starts.
Which one of the following options is correct?

(a) $9 \leq t < 10$
(b) $10 \leq t < 11$
(c) $11 \leq t < 12$
(d) $12 \leq t < 13$

GATE CSE – 2026 MCQ SET - 1(1-MARKS)

Q4. With respect to a TCP connection between a client and a server, which one of the following statements is true?

(a) The client and server use a two-way handshake mechanism before the start of data transmission
(b) The server cannot initiate closing of the connection before the client initiates closing of the connection
(c) The TCP connection is half-duplex
(d) The client and server can initiate closing of the connection at the same time

GATE CSE - 2026 NAT SET - 2(2-MARKS)

Q5. It is necessary to design a link-layer protocol between two hosts that are directly connected over a lossless link of length 3000 km. The link bandwidth is 10^8 bits per second, and the propagation delay is 5 ns per meter. Every transmitted data byte is assigned a unique sequence number.
Let N be the minimum number of bits needed for the sequence number field such that:
i. the sequence numbers do not wrap around before 60 seconds, and
ii. the maximum utilization of the link is achieved.
Find **N**. (answer in integer)

GATE CSE – 2026 NAT SET - 2(2-MARKS)

Q6. Consider a new TCP connection between a sender and a receiver. The receiver advertised window is constant at 48 KB, the maximum segment size (MSS) is 2 KB, and the slow start threshold (ssthresh) is 16 KB.
Assume there are no timeouts or duplicate acknowledgements. The number of rounds of transmission required for the congestion control

Check Out Our GATE Courses & Secure Your Dream Rank!

algorithm to reach the congestion avoidance phase is _____.
Note: $1K = 2^{10}$

GATE CSE – 2025 MCQ SET – 2 (1-MARKS)

Q7. Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes. Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?

(a) 5 fragments, 1488 bytes
(b) 6 fragments, 88 bytes
(c) 6 fragments, 108 bytes
(d) 6 fragments, 116 bytes

GATE CSE – 2025 MCQ SET – 2 (1-MARKS)

Q8. Consider the following statements:

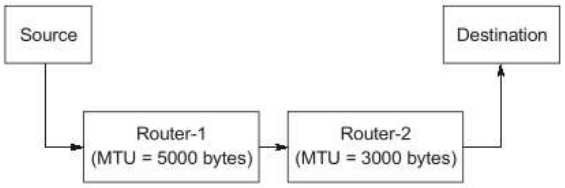
- (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address.
- (ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgment for a segment from R to S.

Which ONE of the following is CORRECT?

(a) Both (i) and (ii) are TRUE
(b) (i) is TRUE and (ii) is FALSE
(c) (i) is FALSE and (ii) is TRUE
(d) Both (i) and (ii) are FALSE

GATE CSE – 2025 NAT SET – 1 (2-MARKS)

Q9. Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including the IP header. The number of fragments that will be delivered to the destination is _____.



GATE CSE – 2025 MCQ SET – 1 (2-MARKS)

Q10. A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

Subnet Address	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	E1
145.36.128.0	/17	E2
145.36.64.0	/18	E3
145.36.255.0	/24	E4
Default	--	E5

(a) E3
(b) E1
(c) E2
(d) E5

GATE CSE – 2025 MCQ SET – 2 (1-MARKS)

Q11. Consider the routing protocols given in List-I and the names given in List-II:

List-I	List-II
(i) Distance vector routing	(a) Bellman-Ford
(ii) Link state routing	(b) Dijkstra

For matching of items in List-I with those in List-II, which ONE of the following options is CORRECT?

(a) (i) – (a) and (ii) – (b)
(b) (i) – (a) and (ii) – (a)
(c) (i) – (b) and (ii) – (a)
(d) (i) – (b) and (ii) – (b)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2025 MCQ SET – 2 (1-MARKS)

Q12. A machine receives an IPv4 datagram. The protocol field of the IPv4 header has the protocol number of a protocol X.
Which ONE of the following is NOT a possible candidate for X?

(a) Internet Control Message Protocol (ICMP)
(b) Internet Group Management Protocol (IGMP)
(c) Open Shortest Path First (OSPF)
(d) Routing Information Protocol (RIP)

GATE CSE – 2025 MSQ SET – 1 (1-MARKS)

Q13. Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

(a) P3: SYN = 1, ACK = 1
(b) P2: SYN = 1, ACK = 1
(c) P2: SYN = 0, ACK = 1
(d) P1: SYN = 1

GATE CSE – 2024 MCQ SET – 1 (2-MARKS)

Q14. Consider a network path P → Q → R between nodes P and R via router Q. Node P sends a file of size 10^6 bytes to R via this path, splitting the file into chunks one after the other, each of size 10^3 bytes. Node P sends these chunks without any wait time between the successive chunk transmissions. Assume that the size of the extra headers added to these chunks is negligible, and that the chunk size is less than the MTU. Each of the links P → Q and Q → R has a bandwidth of 10^6 bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queuing delays.
Assume P starts transmitting the chunks at time t = 0. Which one of the following options gives the time (in seconds, rounded off to 3 decimal places) at which R receives all the chunks of the file?

(a) 16.000
(b) 8.000
(c) 8.008
(d) 15.992

GATE CSE – 2024 MCQ SET – 2 (2-MARKS)

Q15. Which one of the following CIDR prefixes exactly represents the range of IP address 10.12.2.0 to 10.12.3.255?

(a) 10.12.2.0/24
(b) 10.12.2.0/22
(c) 10.12.0.0/22
(d) 10.12.2.0/23

GATE CSE – 2024 NAT SET – 2 (2-MARKS)

Q16. Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _____.

GATE CSE – 2024 NAT SET – 1 (2-MARKS)

Q17. Consider the entries shown below in the forwarding table of an IP router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation "/N" in a prefix indicates a subnet mask with the most significant N bits set to 1.

Prefix	Next hop router
10.1.1.0/24	R1
10.1.1.128/25	R2
10.1.1.64/26	R3
10.1.1.192/26	R4

This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are 10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191, and 10.1.1.205. The number of packets forwarded via the next hop router R2 is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2024 NAT SET – 1 (2-MARKS)

Q18. Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____.

GATE CSE – 2024 MSQ SET – 2 (1-MARKS)

Q19. Which of the following statements about IPv4 fragmentation is/are TRUE?

(a) The fragmentation of an IP datagram is performed only at the source of the datagram.
(b) The reassembly of fragments is performed at all intermediate routers along the path from the source to the destination.
(c) The reassembly of fragments is performed only at the destination of the datagram.
(d) The fragmentation of an IP datagram is performed at any IP router which finds that the size of the datagram to be transmitted exceeds the MTU.

GATE CSE – 2024 MSQ SET – 2 (1-MARKS)

Q20. Which of the following fields of an IP header is/are always modified by any router before it forwards the IP packet?

(a) Time to Live (TTL)
(b) Header Checksum
(c) Protocol
(d) Source IP Address

GATE CSE – 2024 MSQ SET – 2 (1-MARKS)

Q21. Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

(a) Total Length
(b) Source IP
(c) Destination IP
(d) Header Checksum

GATE CSE – 2024 MSQ SET – 1 (1-MARKS)

Q22. TCP client P successfully establishes a connection to TCP server Q. Let N_P denote the sequence number in the SYN sent from P to Q. Let N_Q denote the acknowledgment number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

(a) The acknowledgment number N_Q is equal to N_P
(b) The sequence number N_P is always 0 for a new connection
(c) The acknowledgment number N_Q is equal to $N_P + 1$
(d) The sequence number N_P is chosen randomly by P

GATE CSE – 2024 MCQ SET - 2(1-MARKS)

Q23. Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection. Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

(a) The destination IP address is the IP address of Y
(b) The destination MAC address is the MAC address of Y
(c) The destination MAC address is the MAC address of S
(d) The destination IP address is the IP address of R

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2023 MSQ (1-MARKS)

Q24. Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?

(a) OSPF implements Bellman-Ford algorithm to find shortest paths.
(b) OSPF uses Dijkstra's shortest path algorithm to implement least-cost path routing.
(c) OSPF is used as an inter-domain routing protocol.
(d) OSPF implements hierarchical routing.

GATE CSE – 2023 NAT (2-MARKS)

Q25. The forwarding table of a router is shown below:

Subnet Number	Subnet mask	Interface ID
200.150.0.0	255.250.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3
200.150.68.64	255.255.255.224	4
Default		0

A packet addressed to a destination address 200.150.68.118
arrives at the router. It will be forwarded to the interface
with ID _____.

GATE CSE – 2023 MSQ (2-MARKS)

Q26. Suppose You are asked to design a new reliable byte-stream transport protocol like TCP, named myTCP, which runs over a 100 Mbps network with a Round Trip Time (RTT) of 150 milliseconds and a maximum segment lifetime of 2 minutes.
Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

(a) 30 bits
(b) 32 bits
(c) 34 bits
(d) 36 bits

GATE CSE – 2022 NAT (2-MARKS)

Q27. Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity. The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q-R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (rounded off to one decimal place) between P and Q, leading to the count-to-infinity problem, is _____.

GATE CSE – 2022 MSQ (2-MARKS)

Q28. Consider the routing table of an organization's router shown below:

Subnet number	Subnet mask	Next hop
12.20.164.0	255.255.252.0	R1
12.20.170.0	255.255.254.0	R2
12.20.168.0	255.255.254.0	Interface 0
12.20.166.0	255.255.254.0	Interface 1
default		R3

Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

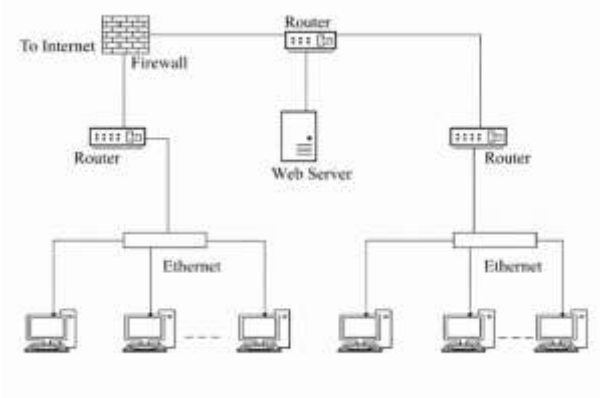
(a) 12.20.164.0/20
(b) 12.20.164.0/22
(c) 12.20.164.0/21
(d) 12.20.168.0/22

GATE CSE – 2022 MCQ (1-MARKS)

Q29. Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via

Check Out Our GATE Courses & Secure Your Dream Rank!

three routers as shown below:



What is the number of subnets inside the enterprise network?

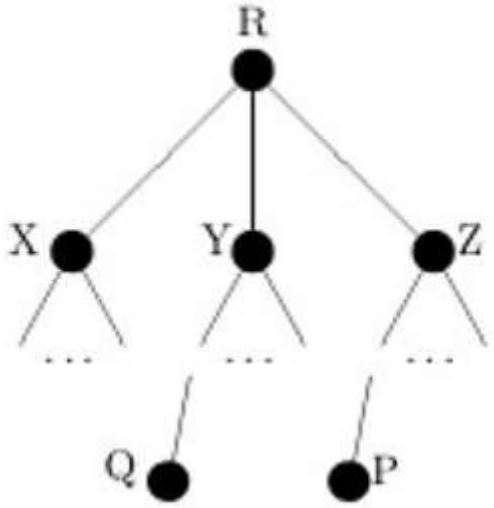
(a) 3
(b) 12
(c) 6
(d) 8

GATE CSE – 2022 NAT (2-MARKS)

Q30. Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is _____.

GATE CSE – 2021 MSQ SET – 2 (2-MARKS)

Q31. Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is as shown below.
The objective is to find the shortest-cost path from the router R to routers P and Q.
Assume that R does not initially know the shortest routes to P and Q.
Assume that R has three neighbouring routers denoted as X, Y, and Z.
During one iteration, R measures its distance to its neighbours X, Y, and Z as 3, 2, and 5, respectively.
Router R gets routing vectors from its neighbours that indicate that the distance to router P from routers X, Y, and Z are 7, 6, and 5, respectively.
The routing vector also indicates that the distance to router Q from routers X, Y, and Z are 4, 6, and 8, respectively. Which of the following statement(s) is/are correct with respect to the new routing table of R, after update during this iteration?



(a) The distance from R to Q will be stored as 7.
(b) The distance from R to P will be stored as 10.
(c) The next hop router for a packet from R to Q is Z.
(d) The next hop router for a packet from R to P is Y.

GATE CSE – 2021 MSQ (2-MARKS)

Q32. Consider two hosts P and Q connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes.
A TCP segment of size 1400 bytes was transferred from P to Q through R, with IP identification value as 0x1234. Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented, i.e., Don't Fragment (DF) flag in the IP header is not set by P.
Which of the following statements is/are correct?

(a) Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes.
(b) If the second fragment is lost, P is required to resend the whole TCP segment.
(c) TCP destination port can be determined by analyzing only the second fragment.
(d) If the second fragment is lost, R will resend the fragment with the IP identification value 0x1234.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2021 MSQ SET – 1 (2-MARKS)

Q33. A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer.
Which of the following behaviors is/are possible?

(a) If the client sends a packet after the server reboot, it will receive a FIN segment.
(b) If the client was waiting to receive a packet, it may wait indefinitely.
(c) The TCP server application on S can listen on P after reboot.
(d) If the client sends a packet after the server reboot, it will receive a RST segment.

GATE CSE – 2021 MCQ SET – 2 (1-MARKS)

Q34. Consider the three-way handshake mechanism followed during TCP connection establishment between hosts P and Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a TCP connection request message to Q with a TCP segment having SYN bit = 1, SEQ number = X, and ACK bit = 0. Suppose Q accepts the connection request. Which one of the following choices represents the information present in the TCP segment header that is sent by Q to P?

(a) SYN bit = 1, SEQ number = Y, ACK bit = 1, ACK number = X + 1, FIN bit = 0
(b) SYN bit = 1, SEQ number = Y, ACK bit = 1, ACK number = X, FIN bit = 0
(c) SYN bit = 0, SEQ number = X + 1, ACK bit = 0, ACK number = Y, FIN bit = 1
(d) SYN bit = 1, SEQ number = X + 1, ACK bit = 1, ACK number = Y, FIN bit = 0

GATE CSE – 2021 MCQ (1-MARKS)

Q35. Consider the following two statements:
- S1: Destination MAC address of an ARP reply is a broadcast address.
- S2: Destination MAC address of an ARP request is a broadcast address.

Which one of the following choices is correct?

(a) S_1 is false and S_2 is true.
(b) Both S_1 and S_2 are false.
(c) Both S_1 and S_2 are true.
(d) S_1 is true and S_2 is false.

GATE CSE – 2020 MSQ (1-MARKS)

Q36. Consider the following statements about the functionality of an IP-based router.
I. A router does not modify the IP packets during forwarding.
II. It is not necessary for a router to implement any routing protocol.
III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet.
Which of the above statements is/are TRUE?

(a) I only
(b) I and II only
(c) II and III only
(d) II only

GATE CSE – 2020 NAT (2-MARKS)

Q37. Consider a TCP connection between a client and a server with the following specifications: the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time t = 0. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time t + 60 ms after all acknowledgments are processed is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2020 MCQ (2-MARKS)

Q38. An organization requires a range of IP addresses to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space 202.61.0.0/17. The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one to the organization?
I. 202.61.84.0/21
II. 202.61.104.0/21
III. 202.61.64.0/21
IV. 202.61.144.0/21

(a) I and IV only
(b) III and IV only
(c) I and II only
(d) II and III only

GATE CSE – 2019 MCQ (2-MARKS)

Q39. Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

(a) X sends an ARP request packet with broadcast IP address in its local subnet
(b) X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X
(c) X sends an ARP request packet with broadcast MAC address in its local subnet
(d) X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X

GATE CSE – 2019 NAT (2-MARKS)

Q40. Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _____.

GATE CSE – 2019 MCQ (2-MARKS)

Q41. Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true?

(a) M, N and P all belong to the same subnet
(b) Only N and P belong to the same subnet
(c) M, N, and P belong to three different subnets
(d) Only M and N belong to the same subnet

GATE CSE – 2018 NAT (2-MARKS)

Q42. Consider an IP packet with a length of 4500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.
The fragmentation offset value stored in the third fragment is _____.

GATE CSE – 2018 NAT (1-MARKS)

Q43. Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps = (10^9 bits per second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is

GATE CSE – 2018 MCQ (1-MARKS)

Q44. Match the following:

Field	Length (bits)
P. UDP Header's Port Number	i. 48
Q. Ethernet MAC Address	ii. 8
R. IPv6 Next Header	iii. 32
S. TCP Header's Sequence Number	iv. 16

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) P-III, Q-IV, R-II, S-I
(b) P-II, Q-I, R-IV, S-III
(c) P-IV, Q-I, R-II, S-III
(d) P-IV, Q-I, R-III, S-II

GATE CSE – 2018 MCQ (1-MARKS)

Q45. Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note the cwnd stands for the TCP congestion window and MSS denotes the Maximum Segment Size.
(i) The cwnd increases by 2 MSS on every successful acknowledgment.
(ii) The cwnd approximately doubles on every successful acknowledgment.
(iii) The cwnd increases by 1 MSS every round trip time.
(iv) The cwnd approximately doubles every round trip time.
Which one of the following is correct?

(a) Only (ii) and (iii) are true
(b) Only (i) and (iii) are true
(c) Only (iv) is true
(d) Only (i) and (iv) are true

GATE CSE – 2017 MCQ (1-MARKS)

Q46. Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK, which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server-side TCP?

(a) LAST-ACK
(b) TIME-WAIT
(c) FIN-WAIT-1
(d) FIN-WAIT-2

GATE CSE – 2017 MCQ SET - 2(1-MARKS)

Q47. Consider the following statements about the routing protocols, Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.
I. RIP uses distance vector routing
II. RIP packets are sent using UDP
III. OSPF packets are sent using TCP
IV. OSPF operation is based on link-state routing protocol
Which of the statements above are CORRECT?

(a) I and IV only
(b) I, II and III only
(c) I, II and IV only
(d) II, III and IV only

GATE CSE – 2017 MCQ SET - 2(1-MARKS)

Q48. Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which connect function has already been called. Which of the following statements is/are CORRECT?
I. A connected UDP socket can be used to communicate with multiple peers simultaneously.
II. A process can successfully call connect function again for an already connected UDP socket.

(a) I only
(b) II only
(c) Both I and II
(d) Neither I nor II

GATE CSE – 2017 NAT SET - 2 (1-MARKS)

Q49. The maximum number of IPv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is _____.

GATE CSE – 2016 NAT SET - 1 (2-MARKS)

Q50. For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The

Check Out Our GATE Courses & Secure Your Dream Rank!

minimum time required to transmit the data is _____ seconds.

GATE CSE – 2016 NAT SET - 1 (2-MARKS)

Q51. An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. The number of fragments that the IP datagram will be divided into for transmission is _____.

GATE CSE - 2016 MCQ SET - 1(2-MARKS)

Q52. Which one of the following protocols is NOT used to resolve one form of address to another one?

(a) DNS
(b) ARP
(c) DHCP
(d) RARP

GATE CSE – 2015 NAT SET - 3 (2-MARKS)

Q53. In the network 200.10.11.144/27, the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is _____.

GATE CSE – 2015 MCQ SET - 2 (2-MARKS)

Q54. Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e., MTU = 1500 bytes). Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in the IP header. How many total number of IP fragments will be transmitted and what will be the contents of the offset field in the last fragment?

(a) 6 and 925
(b) 6 and 7400
(c) 7 and 1110
(d) 7 and 8880

GATE CSE – 2015 MCQ SET - 2 (2-MARKS)

Q55. Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are:

(a) 63 milliseconds 65535×2^{14}
(b) 63 milliseconds 65535×2^{16}
(c) 500 milliseconds 65535×2^{14}
(d) 500 milliseconds 65535×2^{16}

GATE CSE – 2015 MCQ SET - 3 (1-MARKS)

Q56. Consider the following statements:
I. TCP connections are full duplex.
II. TCP has no option for selective acknowledgment.
III. TCP connections are message streams.

(a) Only I is correct
(b) Only I and II are correct
(c) Only II and III are correct
(d) All of I, II, and III are correct

GATE CSE – 2015 MCQ SET - 1 (1-MARKS)

Q57. Which one of the following fields of an IP header is NOT modified by a typical IP router?

(a) Checksum
(b) Source address
(c) Time to Live(TTL)
(d) Length

GATE CSE – 2015 MCQ SET - 2 (1-MARKS)

Q58. Identify the correct order in which a server process must invoke the function calls accept, bind, listen, and recv according to UNIX socket API.

(a) listen, accept, bind, recv
(b) bind, listen, accept, recv
(c) bind, accept, listen, recv
(d) accept, listen, bind, recv

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2015 MCQ SET - 1 (1-MARKS)

Q59. Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are FALSE with respect to the TCP connection?

1. If the sequence number of a segment is m, then the sequence number of the subsequent segment is always m+1.
2. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec.
3. The size of the advertised window never changes during the course of the TCP connection.
4. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window.

(a) 3 only
(b) 1 and 3 only
(c) 1 and 4 only
(d) 2 and 4 only

GATE CSE – 2015 MCQ SET - 2 (2-MARKS)

Q60. Consider the following routing table at an IP router:

Network No	Net Mask	Next Hop
128.96 .170 .0	255.255 .254 .0	Interface 0
128.96 .168 .0	255.255 .254 .0	Interface 1
128.96 .166 .0	255.255 .254 .0	R2
128.96 .164 .0	255.255 .252 .0	R3
0.0 .0 .0	Default	R4

For each IP address in Group I Identify the correct choice of the next hop from Group II using the entries from the routing table above.

Group I	Group II
i) 128.96 .171 .92	a) Interface 0
ii) 128.96 .167 .151	b) Interface 1
iii) 128.96 .163 .151	c) R2
iv) 128.96 .164 .121	d) R3
	e) R4

A. i-a, ii-c, iii-e, iv-d
B. i-a, ii-d, iii-b, iv-e
c, j-b, ii-c, iii-d, iv-e
D. j-b, ij-c, iii-e. iv-d

GATE CSE – 2014 MCQ SET - 2 (1-MARKS)

Q61. Which one of the following is TRUE about the interior gateway routing protocols - Routing Information Protocol (RIP) and Open Shortest Path First (OSPF)?

(a) RIP uses distance vector routing and OSPF uses link state routing.
(b) OSPF uses distance vector routing and RIP uses link state routing.
(c) Both RIP and OSPF use link state routing.
(d) Both RIP and OSPF use distance vector routing.

GATE CSE – 2014 MCQ SET - 1 (1-MARKS)

Q62. Consider the following three statements about link state and distance vector routing protocols for a large network with 500 network nodes and 4000 links.
S1: The computational overhead in link state protocols is higher than in distance vector protocols.
S2: A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol.
S3: After a topology change, a link state protocol will converge faster than a distance vector protocol.
Which one of the following is correct about S1, S2, and S3?

(a) S1, S2, and S3 are all true.
(b) S1, S2, and S3 are all false.
(c) S1 and S2 are true, but S3 is false.
(d) S1 and S3 are true, but S2 is false

Check Out Our GATE Courses & Secure Your Dream Rank!

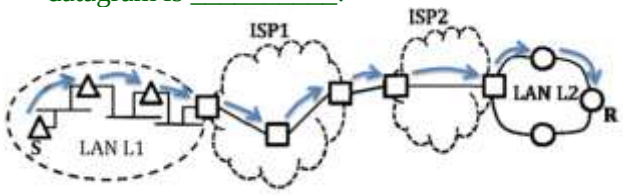
GATE CSE – 2014 MCQ SET - 3 (2-MARKS)

Q63. An IP router with a Maximum Transmission Unit (MTU) of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. The values of the relevant fields in the header of the third IP fragment generated by the router for this packet are:

(a) MF bit: 0, Datagram Length: 1444; Offset: 370
(b) MF bit: 1, Datagram Length: 1424; Offset: 185
(c) MF bit: 1, Datagram Length: 1500; Offset: 370
(d) MF bit: 0, Datagram Length: 1424; Offset: 2960

GATE CSE – 2014 NAT SET - 2 (1-MARKS)

Q64. In the diagram shown below, L1 is an Ethernet LAN and L2 is a Token-Ring LAN. An IP packet originates from sender S and traverses to R, as shown. The links within each ISP and across the two ISPs, are all point-to-point optical links. The initial value of the TTL field is 32. The maximum possible value of the TTL field when R receives the datagram is _____.



GATE CSE – 2014 MCQ SET - 3 (1-MARKS)

Q65. Host A (on TCP/IP v4 network A) sends an IP datagram D to host B (also on TCP/IP v4 network B). Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D?
(i) TTL
(ii) Checksum
(iii) Fragment Offset

(a) (i) only
(b) (i) and (ii) only
(c) (ii) and (iii) only
(d) (i),(ii) and (iii)

GATE CSE – 2014 MCQ SET - 2 (1-MARKS)

Q66. Which one of the following socket API functions converts an unconnected active TCP socket into a passive socket?
(a) connect
(b) bind
(c) listen
(d) accept

GATE CSE – 2014 MCQ SET - 3 (2-MARKS)

Q67. Every host in an IPv4 network has a 1-second resolution real-time clock with battery backup. Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique IPv4 address. Design a 50-bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around?

GATE CSE – 2014 NAT SET - 3 (2-MARKS)

Q68. An IP router implementing Classless Interdomain Routing (CIDR) receives a packet with address 131.23.151.76. The router's routing table has the following entries:

Prefix	Output Interface Identifier
131.16.0/12	3
131.28.0.0/4	5
131.19.0.0/16	2
131.22.0.0/15	1

The identifier of the output interface on which this packet will be forwarded is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2014 MCQ SET - 1 (2-MARKS)

Q69. Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is _____.

GATE CSE – 2014 NAT SET - 1 (2-MARKS)

Q70. Let the size of the congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is _____.

GATE CSE – 2013 MCQ (2-MARKS)

Q71. In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the first fragment in the datagram, the sequence numbers of the payload, and the last bytes of the payload, respectively, are

(a) Last fragment, 2400 and 2789
(b) First fragment, 2400 and 2759
(c) Last fragment, 2400 and 2759
(d) Middle fragment, 300 and 689

GATE CSE – 2012 MCQ (1-MARKS)

Q72. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the windows size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a time-out occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

(a) 8 MSS
(b) 14 MSS
(c) 7 MSS
(d) 12 MSS

GATE CSE – 2012 MCQ (2-MARKS)

Q73. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

(a) 245.248.136.0/21 and 245.248.128.0/22
(b) 245.248.128.0/21 and 245.248.128.0/22
(c) 245.248.136.0/22 and 245.248.132.0/21
(d) 245.248.136.0/24 and 245.248.132.0/21

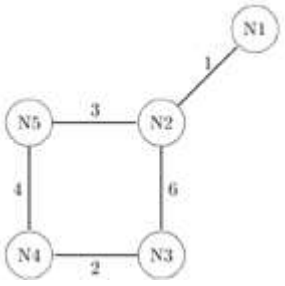
GATE CSE – 2012 MCQ (1-MARKS)

Q74. In the IPv4 addressing format, the number of networks allowed under Class C addresses is

(a) 2^{14}
(b) 2^{7}
(c) 2^{21}
(d) 2^{24}

GATE CSE – 2011 MCQ (2-MARKS)

Q75. Consider a network with five nodes, $N1$ to $N5$, as shown as below.



The network uses a Distance Vector Routing protocol. Once the routes have been stabilized, the distance vectors at different nodes are as follows.
N1: (0,1,7,8,4)
N2: (1,0,6,7,3)
N3: (7,6,0,2,6)
N4: (8, 7, 2, 0,4)
N5: (4, 3, 6, 4, 0)
Each distance vector is the distance of the best known path at that instance to nodes, $N1$ to $N5$,

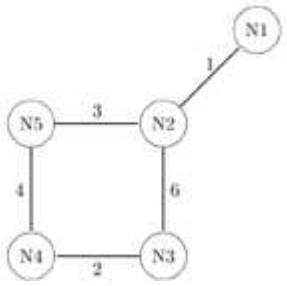
Check Out Our GATE Courses & Secure Your Dream Rank!

where the distance to itself is 0 . Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.
The cost of link $N2 - N3$ reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, $N3$?

(a) (3,2,0,2,5)
(b) (3,2,0,2,6)
(c) (7,2,0,2,5)
(d) (7,2,0,2,6)

GATE CSE – 2011 MCQ (2-MARKS)

Q76 Consider a network with five nodes, $N1$ to $N5$, as shown as below.



The network uses a Distance Vector Routing protocol. Once the routes have been stabilized, the distance vectors at different nodes are as follows.

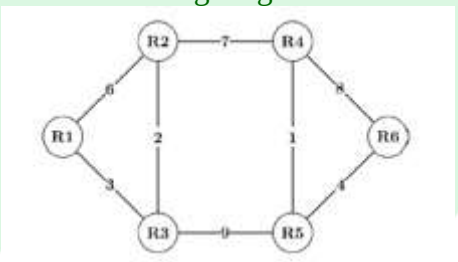
- N1: (0,1,7,8,4)
- N2: (1,0,6,7,3)
- N3: (7,6,0,2,6)
- N4: (8,7,2,0,4)
- N5: (4, 3, 6, 4, 0)

Each distance vector is the distance of the best known path at that instance to nodes, N1 to N5, where the distance to itself is 0 . Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.
The cost of link N2 – N3 reduces to 2 (in both directions). After the next round of updates, the link N1 – N2 goes down. $N2$ will reflect this change immediately in its distance vector as cost, ∞. After the NEXT ROUND of update, what will be the cost to $N1$ in the distance vector of $N3$?

(a) 3
(b) 9
(c) 10
(d) ∞

GATE CSE – 2010 MCQ (2-MARKS)

Q77. Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram.



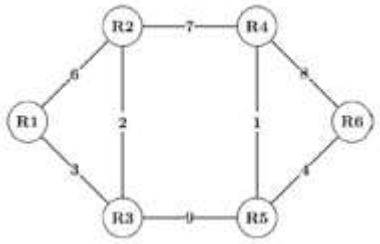
All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

(a) 4
(b) 3
(c) 2
(d) 1

GATE CSE – 2010 MCQ (2-MARKS)

Q78 Consider a network with 6 routers $R1$ to $R6$ connected with links having weights as shown in the following diagram.

Check Out Our GATE Courses & Secure Your Dream Rank!



Suppose the weights of all unused links are changed to 2 and the distance vector algorithm is used again until all routing tables stabilize. How many links will now remain unused?

(a) 0
(b) 1
(c) 2
(d) 3

GATE CSE – 2010 MCQ (2-MARKS)

Q79. Suppose computers A and B have IP addresses 10.105.1.113 and 10.105.1.91 respectively. They both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network?

(a) 255.255.255.0
(b) 255.255.255.128
(c) 255.255.255.192
(d) 255.255.255.224

GATE CSE – 2010 MCQ (1-MARKS)

Q80. One of the header fields in an IP datagram is the Time-to-Live (TTL) field. Which of the following statements best explains the need for this field?

(a) It can be used to prioritize packets
(b) It can be used to reduce delays
(c) It can be used to optimize throughput
(d) It can be used to prevent packet looping

GATE CSE – 2009 MCQ (2-MARKS)

Q81. While opening a TCP connection, the initial sequence number is to be derived using a time-of-day (ToD) clock that keeps running even when the host is down. The low order 32 bits of the counter of the ToD clock is to be used for the initial sequence numbers. The clock counter increments once per millisecond. The maximum packet lifetime is given to be 64 s.
Which one of the choices given below is closest to the minimum permissible rate at which sequence numbers used for packets of a connection can increase?

(a) 0.015/s
(b) 0.064/s
(c) 0.135/s
(d) 0.327/s

GATE CSE – 2008 MCQ (2-MARKS)

Q82. In the slow start phase of the TCP congestion control algorithm, the size of the congestion window

(a) does not increase
(b) increases linearly
(c) increases quadratically
(d) increases exponentially

GATE CSE – 2008 MCQ (2-MARKS)

Q83. If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

(a) 1022
(b) 1023
(c) 2046
(d) 2047

GATE CSE – 2008 MCQ (2-MARKS)

Q84. A computer on a 10 Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2 Mbps. It is initially filled to capacity with 16 Megabits. What is the maximum duration for which the computer can transmit at the full 10 Mbps?

(a) 1.6 seconds
(b) 2 seconds
(c) 5 seconds
(d) 8 seconds

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2008 MCQ (2-MARKS)

Q85. A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a socket(), a bind() and a listen() system call in that order, following which it is preempted. Subsequently, the client process P executes a socket() system call followed by connect() system call to connect to the server process S. The server process has not executed any accept() system call. Which one of the following events could take place?

(a) connect() system call returns successfully
(b) connect() system call blocks
(c) connect() system call returns an error
(d) connect() system call results in a core dump

GATE CSE – 2008 MCQ (1-MARKS)

Q86. Which of the following system calls results in the sending of SYN packets?

(a) socket
(b) bind
(c) listen
(d) connect

GATE CSE – 2007 MCQ (2-MARKS)

Q87. The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

(a) 62 subnets and 262142 hosts
(b) 64 subnets and 262142 hosts
(c) 62 subnets and 1022 hosts
(d) 64 subnets and 1024 hosts

GATE CSE – 2006 MCQ (1-MARKS)

Q88. For which one of the following reasons does Internet Protocol (IP) use the time-to-live (TTL) field in the IP datagram header?

(a) Ensure packets reach destination within that time
(b) Discard packets that reach later than that time
(c) Prevent packets from looping indefinitely
(d) Limit the time for which a packet gets queued in intermediate routers

GATE CSE – 2006 MCQ (2-MARKS)

Q89. Two computers C1 and C2 are configured as follows:
C1 → IP: 203.197.2.53, Mask: 255.255.128.0
C2 → IP: 203.197.75.201, Mask: 255.255.192.0
Which statement is true?

(a) C1 and C2 both assume they are on the same network
(b) C2 assumes C1 is on same network, but C1 assumes C2 is on a different network
(c) C1 assumes C2 is on same network, but C2 assumes C1 is on a different network
(d) C1 and C2 both assume they are on different networks

GATE CSE – 2005 MCQ (1-MARKS)

Q90. Packets of the same session may be routed through different paths in:

(a) TCP, but not UDP
(b) TCP and UDP
(c) UDP, but not TCP
(d) Neither TCP nor UDP

GATE CSE – 2005 MCQ (1-MARKS)

Q91. An organization has a class B network and wishes to form subnets for 64 departments. The subnet mask would be:

(a) 255.255.0.0
(b) 255.255.64.0
(c) 255.255.128.0
(d) 255.255.252.0

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2005 MCQ (2-MARKS)

Q92. In a packet switching network, packets are routed from source to destination along a single path having two intermediate nodes. If the message size is 24 bytes and each packet contains a header of 3 bytes, then the optimum packet size is:

(a) 4
(b) 6
(c) 7
(d) 9

GATE CSE – 2005 MCQ (1-MARKS)

Q93. The address resolution protocol (ARP) is used for:

(a) Finding the IP address from the DNS
(b) Finding the IP address of the default gateway
(c) Finding the IP address that corresponds to a MAC address
(d) Finding the MAC address that corresponds to an IP address

Answer key

1. (4094)	2. (a, b, c)	3. (b)
4. (d)	5. (30)	6. (4)
7. (d)	8. (b)	9. (7)
10. (a)	11. (a)	12. (d)
13. (b; d)	14. (c)	15. (d)
16.(4)	17. (40)	18. (6)
19. (c, d)	20. (a, b)	21. (b, d)
22. (c, d)	23.(a)	24. (a, c)
25. (3)	26. (b, c, d)	27. (0.5)
28. (b, d)	29. (c)	30. (33)
31. (a.d)	32. (a.b)	33. (b, c, d)
34. (a)	35.(a)	36. (d)
37. (44)	38. (d)	39.(c)
40.(3)	41. (b)	42. (144)
43. (34-35)	44. (c)	45. (c)
46. (d)	47. (c)	48. (b)
49. (9)	50. (1.1)	51. (13)
52.(c)	53. (158)	54. (c)
55. (c)	56. (a)	57. (b)
58. (b)	59 (b)	60.(a)
61.(a)	62. (d)	63. (a)
64. (26)	65. (d)	66. (c)
67. (256)	68. (1)	69. (5)
70. (1100)	71. (c)	72. (c)
73. (a)	74. (c)	75.(a)
76.(c)	77.(c)	78.(c)
79. (d)	80. (d)	81. (a)
82. (d)	83. (c)	84. (b)
85. (c)	86.(d)	87.(c)
88. (c)	89. (c)	90.(b)
91. (d)	92. (d)	93. (d)

Check Out Our GATE Courses & Secure Your Dream Rank!

HINTS & SOLUTIONS

1. ANSWER:- 4094

Subnet mask in binary:
255.255.240.0
255 → 11111111
255 → 11111111
240 → 11110000
0 → 00000000
Total **network bits = 20**
Host bits:
32 − 20 = 12
Total host addresses:
$2^{12} = 4096$
Assignable addresses:
4096 − 2
= 4094

2. ANSWER:- A; B; C

Number of addresses required = 6000
CIDR blocks must be power of 2.
$2^{12} = 4096$ (not enough)
$2^{13} = 8192$ (sufficient)
So the ISP must allocate a block of 8192 addresses.
Prefix length:
$8192 = 2^{13}$
32 − 13 = /19
Thus the client must receive a /19 block.
All bits in HID must be zero.
(A) 202.16.0.0/19
Valid boundary and inside ISP range
(B) 202.17.64.0/19
Valid boundary and inside ISP range
(C) 202.16.32.0/19
Valid boundary and inside ISP range
(D) 202.17.24.0/19
Not valid /19 boundary

3. ANSWER:- B

Segments sent per RTT:
RTT 0: 1
RTT 1: 2
RTT 2: 4
RTT 3: 8
RTT 4: 16
RTT 5: 32
RTT 6: 64
RTT 7: 128
RTT 8: 256
RTT 9: 512
RTT 10: 1024

Cumulative segments sent
After RTT 0: 1
After RTT 1: 3
After RTT 2: 7
After RTT 3: 15
After RTT 4: 31
After RTT 5: 63
After RTT 6: 127
After RTT 7: 255
After RTT 8: 511
After RTT 9: 1023
After RTT 10: 2047
Segment 2000 lies between 1024 and 2047, which are sent during RTT 10.
Since each RTT = 1 ms
$10 \le t < 11$

4. ANSWER:- D

(A): False
TCP uses a three-way handshake, not a two-way handshake.
(B): False
Either the client or the server can initiate connection termination.
(C): False
TCP is full-duplex, not half-duplex.
(D): True
Both sides can initiate closing simultaneously, called simultaneous TCP close.

5. ANSWER:- 30

Bandwidth = 10^8 bits/sec
= $(10^8 / 8)$ bytes/sec
= 1.25×10^7 bytes/sec
Bytes sent in 60 sec:
$1.25 \times 10^7 \times 60$
= 7.5×10^8 bytes
Sequence space must be $\ge 7.5 \times 10^8$
$2^N \ge 7.5 \times 10^8$
$\log_2(7.5 \times 10^8) \approx 29.5$
So,
$N \ge 30$
Condition 2: Full link utilization
Distance = 3000 km = 3×10^6 m
Propagation delay:
$3 \times 10^6 \times 5$ ns
= 15×10^6 ns
= 0.015 sec
RTT = 2 × 0.015 = 0.03 sec
Bandwidth–delay product:
$10^8 \times 0.03$
= 3×10^6 bits
= 3.75×10^5 bytes

Check Out Our GATE Courses & Secure Your Dream Rank!

Sequence numbers needed ≥ 375000

$2^N \geq 375000 \rightarrow N \geq 19$

Final N:

Must satisfy both conditions:

max(30, 19) = 30

6. ANSWER:- 4

Initial congestion window:

In TCP, initial cwnd = **1 MSS = 2 KB**

Slow start phase:

cwnd doubles every round until it reaches **ssthresh = 16 KB**

Round 1:
cwnd = 2 KB
Round 2:
cwnd = 4 KB
Round 3:
cwnd = 8 KB
Round 4:
cwnd = 16 KB

At 16 KB, cwnd reaches ssthresh, so TCP enters congestion avoidance.

Number of rounds: 4

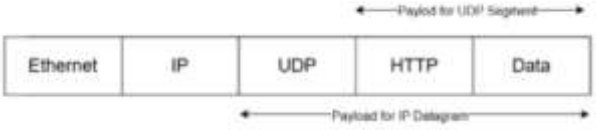
7. ANSWER:- D

MTU = 1500 B

It is given that IPv4 headers do not use any options field, that means IPv4 Header Size is 20 B.

IP Header Size = 20 B

So, MTU without IP Header = (1500 – 20) Bytes = 1480 Bytes



Now, Payload for UDP Segment is 7488 Bytes.

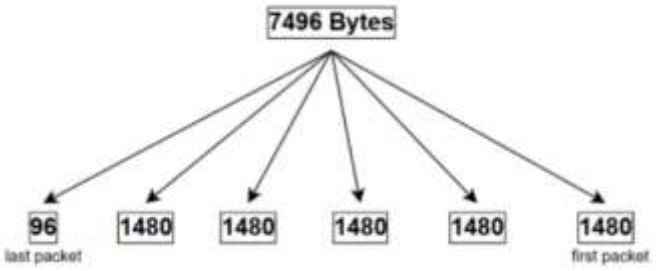
UDP Header is of 8 Bytes.

So,

Payload for IP datagram = Payload for UDP segment + UDP Header.

$= 7488 + 8 = 7496$

Now, fragmenting this Payload for IP Datagram using MTU without IP Header



Total Fragments = 6

Last packet data without header = 96 Bytes

Last packet size with IP Header = 96 + 20 = 116 Bytes.

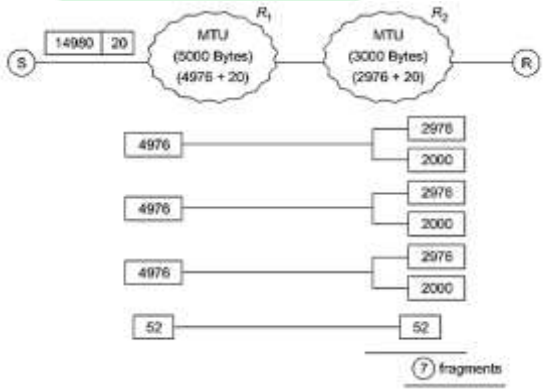
So, Total no of fragments are 6 and size of last packet including IP Header is 116 Bytes.

8. ANSWER:- B

ARP is a network protocol used to find the MAC (Media Access Control) address corresponding to a given IP address. So, (i) is true.

TCP uses piggybacking to send an acknowledgment (ACK) along with data in the same segment. So, (ii) is false.

9. ANSWER:- 7



10. ANSWER:- A

Since the first two octets (145.36) match everywhere, we evaluate the third octet (109 , which is 01101101 in binary):

- Interface E1(/16 mask): * Mask leaves the first two octets unchanged → 145.36.0.0 .
- Result: MATCH (Prefix length = 16)
- Interface E2 (/17 mask): *3rd octet mask is 10000000 .
- 01101101(109) AND 16000000 = 06000000(0) → 145.36.0.0.
- Result: MISMATCH (Table expects 145.36.128.0)

Check Out Our GATE Courses & Secure Your Dream Rank!

- Interface E3 (/ 18 mask): *3rd octet mask is 11000000 .
  o 01101101(109) AND 11000000 = 01000000(64) → 145.36.64.0.
- Result: MATCH (Prefix length = 18)
- Interface E4 (/ 24 mask): * Mask leaves the first three octets unchanged → 145.36.109.0.
- Result: MISMATCH (Table expects 145.36.255.0)

Final Decision
Both E1 (16) and E3 (18) are valid matches.
Under the Longest Prefix Match rule, the router selects the more specific route (18 > 16).

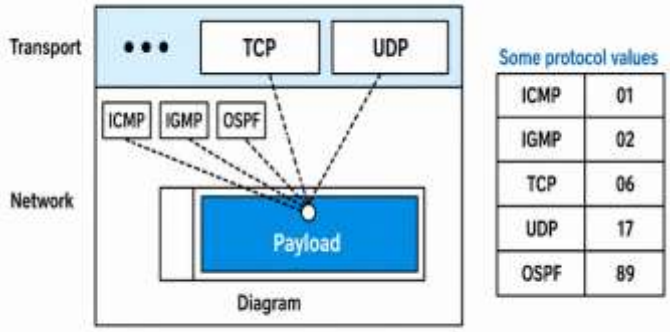
The packet is forwarded to interface E3.

11. ANSWER:- A

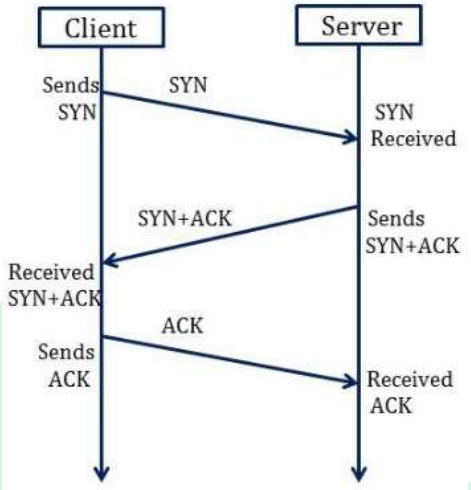
- DVR algorithm uses Bellman-Ford Algorithm to find out the optimal path.
- Link State Routing algorithm uses Dijkstra's Algorithm to construct the SPT.

12. ANSWER:- D

Multiplexing and multiplexing using the value of the protocol field



13. ANSWER:- B, D



P1: SYN = 1, ACK = 0
P2: SYN = 1, ACK = 1
P3: SYN = 0, ACK = 1

14. ANSWER:- C

The network has three nodes P,Q,R connected as P-Q-R. P sends 1000 packets each of size 1000 bytes.
The transmission time of one packet between two nodes is packet size / Bandwidth.
we are given bandwitdth is 1 mega bits per second.
hence we have 8000 / 10^6 = 8 milliseconds.
The total transmission time = (Number of packets + n) * T_t | here n is the number of nodes on the path in between source and destination.
Hence total time taken = (1000+1)* 8ms = 8008ms = 8.008s

15. ANSWER:- D

As per the given data,
Range of IP Address = 10 · 12 · 2 · 0 to 10.12.3.255
Expanding the last two octets of these IP Addresses:

10.12.00000010.00000000
10.12.00000011.11111111

As you can see above, the 23 -bit common prefix and the last 9 bits are all zeros to all ones.
10.12.2.0/23

Check Out Our GATE Courses & Secure Your Dream Rank!

16. ANSWER:- 4

Given cwnd = 12 MSS
Now timeout occurs which makes,
cwnd=1 and ssthreshold = 12/2= 6 MSS
Again slow start phase will begin,
After 1 RTT, cwnd = 2MSS
After 2 RTT, cwnd = 4MSS
During 3rd RTT cwnd will be 4 MSS

17. ANSWER:- 40

Forwarding Table

IPs	For Understanding
10.1.1.16	10.1.1.0001 0000 (maximum matches with R1)
10.1.1.72	10.1.1.0100 1000 (maximum matches with R3)
10.1.1.132	10.1.1.1000 0100 (maximum matches with R2)
10.1.1.191	10.1.1.1011 1111 (maximum matches with R2)
10.1.1.205	10.1.1.1100 1101 (maximum matches with R4)

Given 5 IPs represented in the table below

PREFIX	Next Hop	For understanding IPs (BOLD FIXED FOR NET ID)
10.1.1.0/24	R1	10.1.1.0000 0000
10.1.1.128/25	R2	10.1.1.1000 0000
10.1.1.64/26	R3	10.1.1.0100 0000
10.1.1.192/26	R4	10.1.1.1100 0000

From the Longest Prefix matching, we can get 10.1.1.132 and 10.1.1.191 are the IPs visiting Router R2.
Router forwarded packets to each host is 20 so 20* 2= 40

18. ANSWER:- 6

Given intial ip datagram with 1400bytes data+20 bytes header
Sender-------MTU(542 bytes)-------router----------
At sender 1400 bytes of data is fragmented as
520 bytes(D)+20bytes(H)
360bytes(D)+20bytes(H)

At router there are 3 fragments to send to receiver the MTU size is 360 Bytes so, again fragmentation is required

The 1st fragment of 520 Bytes of data is fragmented to
336 bytes(D) + 20 bytes (H)
184 bytes(D) +20 bytes(H)

The 2nd fragment of 520 Bytes of data is fragmented to
336 bytes(D) + 20 bytes (H)
184 bytes(D) +20 bytes(H)

The 3rd fragment of 360 Bytes of data is fragmented to
336 bytes(D) + 20 bytes (H)
24 bytes(D) +20 bytes(H)

so, the final total fragments at the receiver is 6 fragment

19. ANSWER:- C,D

Option C : True

Reassembly of fragmented Datagrams is done only after reaching the destination . It cannot take place at any intermediary place because :

1. All fragments may not meet at a router .
2. Further fragmentation of fragmented packets can take place .
3. fragmented datagrams may reach destination via an independent path .

Option D: True

4. Fragmentation is done by intermediary devices such as Routers .
5. When a packet needs to be fragmented because it exceeds the Maximum Transmission Unit (MTU) of a network segment, the original packet is divided into smaller fragments that fit within the MTU limits. The original packet's data is split into pieces, each typically including an IP header with appropriate fragmentation information.

20. ANSWER:- A; B

Source IP address remains unchanged as it identifies the original sender and is used for sending acknowledgement by destination or an ICMP message by router/destination.
Protocol indicates the type of data carried (e.g., TCP, UDP) and isn't altered by routers.
Time to Live(TTL) value is decremented by 1 by each router that forwards the packet.
As TTL value of IP header is changed by every router, corresponding Header Checksum is recalculated based on the new value of TTL.

Check Out Our GATE Courses & Secure Your Dream Rank!

21. ANSWER:- B; D

NAT device takes the outgoing packet and replaces the source IP address from private address pool to a IP address from its public address pool.

Since the source IP address is changed, the header checksum will also be recalculated so that the router does not get error while checking the header checksum.

Answer: B,D

22. ANSWER:- C, D

TCP always starts connection with an initial sequence number that is randomly generated in the range of available sequence numbers, to avoid conflicts with delayed packets from previous sessions. This is a safety measure to reduce the probability of accepting delayed packets from a past closed connection to be accepted by a new connected made from same host and same port. TCP ACK always sends the next expected byte sequence number hence N_p+1 , will be sent as ack number

23. ANSWER:- A

Source IP address (public IP) and Destination IP address always remain unchanged throughout the journey of Packet from Node X to Node Y.

The destination IP address is always that of Node Y. (At destination, public destination IP address will be changed to private IP of receiving node due to NAT)

When the receiving node is outside the source node's local network, the source node sends packets to the default gateway (usually a router).

In this case, the destination MAC address will be that of the router.

The switch in the network simply forwards the packets along the path without altering their content.

24. ANSWER:- A, C

OSPF uses Dijkstra's shortest path algorithm to implement least-cost path routing

OSPF implements hierarchical routing

25. ANSWER:- 3

Interface	Subnet Mask	SubnetRange Calculation	Does it Match?
1	255.250.0.0	Mask is non-contiguous. Broad match possible, but not the most specific.	Likely
2	255.255.224.0	118 AND 224 = 96 → Subnet becomes 200.150.64.0	Yes
3	255.255.255.0	118 AND 255 = 118 → Subnet becomes 200.150.68.0	Yes
4	255.255.255.224	118 AND 224 = 96 → Expected subnet 200.150.68.96 ≠ 200.150.68.64	No

Due to **longest prefix matching rule**, the packet is forwarded to the interface with the most specific matching subnet.

Interface ID = 3

26. ANSWER:- B, C, D

Given, Bandwidth = 100 Mbps

i.e. in 1 second = 100×2^{20} bits can be transferred.

Since MSL is 2 minutes than, number of bits that can be transferred in 2 minutes

= $120 \times 100 \times 2^{20}$ bits

Note: Sequence number is assigned to each byte in TCP.

Total bytes = 1500×2^{20} or 31 bits at least

27. ANSWER:- 0.5

It is given that P and Q are sending routing updates at the same average rate, so in a particular time period of sending updates, there are two possibilities, one is P sends before Q and other is Q sends before P. Now, We have 2 conditions:

1. Q sends update to P first, then P finds the value of R as ∞ and count-to-infinity will not occur.
2. P sends update to Q first, then Q found that P has a way to reach R, so it is count-to-infinity problem. As Q - R is broken, So R cannot send update to anyone.

   From above two conditions (total possibilities), only one condition leads to count-to-infinity problem and each possibility has equal probability of occurrence, so probability is 1/2 = 0.5 for count-to-infinity.

28. ANSWER:- B; D

12.20.164.0/22 → 12.20.10100100.00000000
12.20.170.0/23 → 12.20.10101010.00000000
12.20.168.0/23 → 12.20.10101000.00000000
12.20.166.0/23 → 12.20.10100110.00000000

12.20.170.0 and 12.20.168.0 can be combined to 12.20.10101000.0 (12.20.168.0/22).

12.20.166.0/23 comes inside 12.20.164.0/22. So, 12.20.164.0/22 can be used for route aggregation of both.

12.20.168.0/22 and 12.20.164.0/22 can't be combined further.

So, answer is option B and D.

29. ANSWER:- C

top most router have 3 different interfaces having 3 different NIDs (goes to firewall, webserver and right most router respectively)

left most router's 2 diff interfaces goes to firewall

Check Out Our GATE Courses & Secure Your Dream Rank!

and left most ethernet respectively
right most router's 1 interface goes to right most ethernet
There are total 6 subnets

30. ANSWER:- 33
let suppose SEQUNCE NO = X BITS
WINDOW SIZE=2^X B
WRAP AROUND TIME $(2^X \times 2^3)/2^{30}$ SEC
WRAP AROUND TIME > MSL
$(2^x \times 2^3)/2^{30} > 60$ (why taking greater bcz WAT should be greater than MSL bcz atleast after these many seconds it can repeat the sequence number)= 33

31. ANSWER:- A, D
Given R gets the distance vector (3, 2, 5). After the one iteration, the distance vector from X to P, Y to P, and Z to P is (7, 6, 5) respectively.
The distance vector from R to P via XYZ is (3 + 7, 2 + 6, 5 + 5) = (10, 8, 10).
So, take the minimum distance from R to P which is 8 via Y.
After the iteration, the distance vector from X to Q, Z to Q is (4, 6, 8) respectively.
The distance vector from R to Q via XYZ is (3 + 4, 2 + 6, 5 + 8) = (7, 8, 13).
So, take the minimum distance from R to Q which is 7 via X.

32. ANSWER:- A, B
First Fragement = 800(Data) + 20(Header) = 820 B
Second Fragement = 1400-800 = 620 B

33. ANSWER:-B, C, D
A. Option A is false as the FIN is used to close connection.
B. Option B is correct
C. Option C is correct as once the current session will be terminated(due to reboot) and new connection can take place on the same port.
D. Option D is also correct

34. ANSWER:- A
Q will send the SYN bit = 1 to the connection establishment.
Q seq number will be Y different from X. ACK bit = 1 because sending the ACK. ACK number = X + 1 (Next seq number id) FIN bit = 0 (Because establishing the connection)

35. ANSWER:- A
ARP request is broadcasting.
ARP reply is unicasting.

36. ANSWER:- D
I. A router modifies the IP packets during forwarding because TTL is changing.
II. A router can also be used in a LAN network; it does not require a routing protocol at that time.
III. Packet fragmentation is done if packet size is more than MTU.

37. ANSWER:- 44
The state of congestion window changes as below
Note: As specified in question, there are no errors and timeouts
- At t: 1 MSS
- At t+6:2 MSS
- At t+12:4 MSS
- At t+18:8 MSS
- At t+24:16 MSS

Now here since threshold value is reached, it is no longer in slow start phase and enters congestion avoidance
phase
- At t+30:17 MSS
- At t+36:18 MSS
- At t+42:19 MSS
- At t+48: 20 MSS
- At t+54:21 MSS
- At t+60:22 MSS

So at time t+60, the congestion window size is 22 MSS i.e., 44 KB.

38. ANSWER:- D
There is a rule in supernetting, first ip address must be divisible by total size of the supernet,ie,the host bits must be 0.
II and III follow it

39. ANSWER:- C
1. The sender (X) knows its own IP address and MAC address. X also knows the IP address of the target (Y) . It needs to find MAC address of Y
2. IP asks ARP to create an ARP request message, filling in X's IP and MAC address and Y's IP address. The destination MAC address is set to al l 0s.
3. The message is passed to Data Link layer where it is encapsulated in a frame using MAC address of X as the source address

Check Out Our GATE Courses & Secure Your Dream Rank!

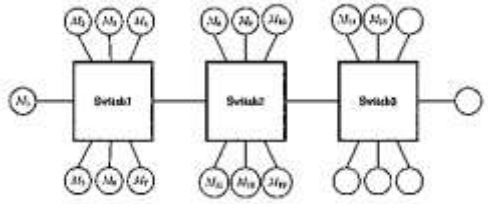
and physical broadcast address (all 1s) as the destination address.

4. Every host in subnet receives the request message because we have used broadcast MAC address in the destination address. All machines except Y (we have specified Y's IP address in our ARP request message) drop the ARP request message.
5. Y replies with an ARP reply message that contains its MAC address.
6. When X receives this message, it gets to know the MAC address of Y.

note that, a host uses its own IP address and network mask to decide if target IP address is in its own network or not.
If it is in its network, it uses ARP to resolve the MAC address.
If target IP address is not in its network, it takes the help of default gateway to resolve MAC address using ARP.
Since in question it is clearly mentioned that X (source) and Y (destination) both are in the same subnet, we need not send ARP request message targeted at gateway and then expect it to give the MAC address of Y to X.

40. ANSWER:- 3

3 switches of ethernet are required to connect 15 computers.



41. ANSWER:- B

We can avoid BITWISE AND here. The concept is, in subnet mask we need to take a note of positions of set bits. For those positions, in IP addresses we should've matching bits.

here, 1st, 2nd and 3rd octets of M,N,P are same.
Their 4th octet is given as
M: 0000 0010; N: 0000 0101; P: 0000 0110
4th octet of Subnet Mask: 1111 1100
It means, we should've matching bits for starting 6 bits in given IP addresses.
Only N and P satisfies this. B is the correct option here.

42. ANSWER:- 144

IP length = 4500B. Payload = 4480B.
MTU = 600B.
So, max packet we can make here is: Header(20) + Payload (580). But the payload isn't a multiple of 8 here.
Hence legally the max packet here is: Header(20) + Payload(576).
Seven such packets make up a total payload of 4032B.
So, last packet would be Header(20) + Payload (448).
Offset of first packet = 0.
Offset of second packet = 576B = 72
Offset of third packet = 72 + 72 = 144.

43. ANSWER:- 34-35

remember tcp is byte(not bit) stream protocol
sequence number field of tcp header is 32 bit, so only (2^{32}=4G) sequence numbers are available so wrap arount time is after how much time we use same sequence number
bandwidth = 1Gbps = 10^9 / 8 Byte per second
10^9/8 sequence no. consume - 1 sec (since bandwidth=10^9/8)
1 seq no. consume - 8 / 10^9 sec
2^{32} seq no. consume - 2 ^ 32 * (8 / 10^9) sec = 34.35 sec (approx 34 sec)

44. ANSWER:- C

UDP Header's Port Number → 16 bit
Ethernet MAC Address → 48 bit
IPv6 Next Header → 8 bit
TCP Header's Sequence Number → 32 bit

45. ANSWER:- C

cwnd approximately doubles every round trip time

46. ANSWER:- D

FIN WAIT-1 --> if client sent FIN segment to the server and waiting for the ACK of its own FIN segment
if client receives ACK for its own FIN then client move to FIN-wait-2 and waiting for FIN from server side

47. ANSWER:- C

RIP uses distance vector routing. OSPF uses link-state routing protocols.
RIP uses UDP as transport protocol. OSPF neither uses TCP nor UDP

Check Out Our GATE Courses & Secure Your Dream Rank!

48. ANSWER:- B

Bind() function creates local address. Connect() function is specifying remote address. An unconnected UDP socket is just a Bind() function.

A connected UDP socket is one more step above i.e. connect() function [Just behaves like TCP].

49. ANSWER:- 9

Record route is an optional field used to record address.

In IPv4 header, 40 bytes are reserved for OPTIONS.

For Record Route to stores, 1 byte is used to store type of option, 1 byte for length and 1 byte for pointer. Out of 40, 37 bytes are left. An IP4 address takes 32 bits or 4 bytes. We can store at most floor(37/4) = 9 router addresses.

50. ANSWER:- 1.1 sec

arrival rate is 10 MB/sec
outgoing is 20 MB/sec
relative rate at which bucket is being empty = 20-10 = 10 MB/sec
to empty bucket , 1/10 sec = 0.1 sec required
date sent in 0.1 second = 0.1 * 20 = 2 MB
remaining data = 10 MB
now output rate will become input rate i.e. 10 MB/sec
to send 10 MB , 10/10 = 1 sec required
total time = 1.1 sec

51. ANSWER:- 13

1. Extract Payload & Max Capacity

- Total Datagram Data: 1000 bytes total - 20 bytes header = 980 bytes payload.
- Max Fragment Payload: 100 bytes MTU - 20 bytes header = 80 bytes.

2. Check 8-Byte Alignment

The payload size for all fragments (except the last) must be a multiple of 8 due to the Fragment Offset field. Since 80 is divisible by 8 $\left(\frac{80}{8} = 10\right)$, the maximum payload per fragment remains exactly 80 bytes.

3. Calculate Fragments

$$\text{Number of Fragments} = \left\lceil \frac{980 \text{ bytes total payload}}{80 \text{ bytes per fragment}} \right\rceil = \lceil 12.25 \rceil = 13$$

- Fragments 1-12: 80 bytes payload + 20 bytes header = 100 bytes each.
- Fragment 13:20 bytes remaining payload + 20 bytes header = 40 bytes.

The datagram is divided into 13 fragments.

52. ANSWER:- C

- **DNS** (Domain Name System) is used for mapping a hostname to an IP address.
- **ARP** (Address Resolution Protocol) is used for mapping an IP address to a MAC address.
- **RARP** (Reverse Address Resolution Protocol) is used for mapping a MAC address to an IP address.
- **DHCP** is used to assign ip address not to map one address form to other form.

53. ANSWER:- 158

144 in binary = 10010000
out of this 3 bits in left are subnet bits. (27 bits are used for subnet, which means top 3 bytes and leftmost 3 bits from the last byte)
So, the 4th octet in the last IP address of the network which can be assigned to a host is 10011110. (its not 10011111 because its network broadcast address)
So, 10011110 is 158 in decimal.

54. ANSWER:- C

UDP data = 8880 bytes
UDP header = 8 bytes
IP Header = 20 bytes
Total Size excluding IP Header = 8888 bytes.
Number of fragments = ⌈ 8888 / 1480 ⌉ = 7
Offset of last segment = (1480 * 6) / 8 = 1110

55. ANSWER:- C

In TCP when the bandwidth-delay product increases beyond 64 K receiver window scaling is needed.

The bandwidth-delay product is the maximum amount of data on the network circuit at any time and is measured as RTT Bandwidth. This is not the time for sending data rather just the time for sending data without acknowledgement.

So, here, we have bandwidth delay product (1048560/8) Bα= 64 K α = (64 K*8)/10485600.5s 500 milliseconds.

When window scaling happens, a 14 bit shift count is used in TCP header. So, the maximum possible window size gets increased from 2161 to $(2^{16} - 1) * 2^{14}$ or from 65535 to $65535 * 2^{14}$

56. ANSWER:- A

TCP is a byte stream protocol. Hence lll is false
TCP can use both selective ACK and Cumulative Acknowledgement. Hence ll is false
TCP Connection are full duplex. Hence i is true

Check Out Our GATE Courses & Secure Your Dream Rank!

57. ANSWER:- B
Only Source address field of IP header is not modified by a router

58. ANSWER:- B
order: bind, listen, accept, recv

59. ANSWER:- B
Statement 1 is false because, the sequence number of the subsequent segment depends on the number of byte characters in the current segment. Statement 2 is true, because RTO timer is always ≥ Estimated RTT.Statement 3 is false. Receiver window is used to give the sender an idea of how much free buffer space is available at the receiver. Statement 4 is true, TCP is not permitted to overflow the allocated receiver buffer.
Hence, when the sender cannot send any more data, receiver window would be 0 and hence, the buffer would have unacknowledged data.

60. ANSWER:A
Using longest prefix matching (most specific Subnet mask wins)

Routing Table Interpretation:
128.96.170.0 /23 → covers 170–171 → Interface 0
128.96.168.0 /23 → covers 168–169 → Interface 1
128.96.166.0 /23 → covers 166–167 → R2
128.96.164.0 /22 → covers 164–167 → R3
Default → R4

Matching:
i) 128.96.171.92 → falls in 170–171 → Interface 0 →a
ii) 128.96.167.151 → matches both /23 (166–167 → R2) and /22 (164–167 → R3)
Longest prefix = /23 → R2 → c
iii) 128.96.163.151 → does not match any → Default → R4 → e
iv) 128.96.164.121 → matches only /22 → R3 → d

61. ANSWER:- A
Both Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) are Interior Gateway Protocol, i.e., they both are used within an autonomous system. RIP is an old protocol (not used anymore) based on distance vector routing. OSPF is based Link State Routing.

62. ANSWER:- D
The computational overhead in link state protocols is higher than in distance vector protocols because LSR is based upon global knowledge whereas DVR is based upon local knowledge.
Persistent looping can be avoided with the help of split horizon in DVR. There is no concept of persistent looping in LSR, in LSR only temporary loop exists and can automatically be solved by system or router. **S2** is false.

And, after a topology change, a link state protocol will converge faster than a distance vector protocol. **S3** is true.
Answer is option **D.**

63. ANSWER:- A
MTU = 1500 bytes
Header length = 20 bytes
Packet size = 4404 bytes
Fragment 1: 1480 + 20
Fragment 2: 1480 + 20
Fragment 3: 1424 + 20
Datagram length for third IP fragment = 1444
MF bit = 0
Offset=1480/4=370

64. ANSWER:- 26
There are 6 routers in the above diagram.
Initially TTL value was 32, so at the Receiver it will become 32 - 6 = 26.

65. ANSWER:- D
TTL changes from one hope to the next.
checksum changes on each hop due to TTL change.
Fragement offset may be compurted again.
TTL, check sum and fragement offset may be changed in the transmission.

66. ANSWER:- C
Listen() converts an unconnected active TCP socket into a passive socket.

67. ANSWER:- 256
Each host needs to generate 1000 unique identifiers per second which requires
⌈log2 1000⌉=10 bits
Now, these 10 bits along with 32-bit globally unique IP address will give a globally unique 42-bit ID's which stays constant. Since we are allowed 50 bits, we can use the next 8 bits using the clock which

Check Out Our GATE Courses & Secure Your Dream Rank!

changes every second. Thus, our IDs will wrap around once in 28=256 seconds.

68. ANSWER:- 1

since first octet is same for all and given address , hence only second octet matters as all sid bits are greater than 8 and less than 17
second octet : 00010111
second octet of first address : 00010000 (given mask bits 12) -> matched
second octet of second address : 00011100 (given mask bits 14)
second octet of third address : 00010011 (given mask bits 16)
second octet of fourth address : 00010110 (given mask bits 15) -> matched
1st and 4th entry is being matched with given address
due to longest subnet mask match rule , this packet will be forwarded to interface 1.

69. ANSWER:- 5

Bandwidth delay product = Bandwidth × RTT

$\rightarrow 1.5 \times 10^6 \times 2 \times 50 \times 10^{-3} = 1.5 \times 10^5$ bits.

$$\text{Utilization} = \frac{N \times \text{packet size}}{\text{Bandwidth delay product}}$$

$$\rightarrow 0.6 = \frac{N \times 8 \times 1000}{1.5 \times 10^6} \rightarrow 6 \times 15 = N \times 8$$

$$\rightarrow N^T = \frac{45}{4} = 11.25 \approx 12$$

In selective repeat, $N \leq \frac{2^n}{2} \rightarrow 12 \leq \frac{2^n}{2} \rightarrow 2^n \geq 24 \rightarrow n \geq \log_2(24)$ or $n \geq 5$.

70. ANSWER:- 1100

Given that at the time of Time Out, Congestion Window Size is 32 KB and **RTT** = 100 ms,
When Time Out occurs, for the next round of Slow Start,

$$\text{Threshold} = \frac{\text{size of congestion window}}{2},$$

$$\text{Threshold} = 16 \text{ KB}$$

Suppose we have a slow start ⇒ 2 KB|4 KB|8 KB | 16 KB (As the threshold is reached, Additive increase starts)

|18 KB|20 KB|22 KB|24 KB|26 KB|28 KB|30 KB|32 KB
Here | (vertical line) is representing RTT so the total number of vertical lines is 11×100 ms = 1100msec and so this is the answer.

71. ANSWER:- C

Since M=0; It is indication of last fragment.
Data length=Total Length-header length
Data length=400-10*4=360
Fragments offset=300
Number of data byte ahead from this fragments is=300*8=2400
Last byte of last frame is=2399+360=2759

72. ANSWER:- C

Explanation:
Since Slow Start is used, window size is increased by the number of segments successfully sent. This happens until either threshold value is reached or time out occurs.
In both of the above situations AIMD is used to avoid congestion. If threshold is reached, window size will be increased linearly. If there is timeout, window size will be reduced to half.
Window size for 1st transmission = 2 MSS
Window size for 2nd transmission = 4 MSS
Window size for 3rd transmission = 8 MSS
threshold reached, increase linearly (according to AIMD)
Window size for 4th transmission = 9 MSS
Window size for 5th transmission = 10 MSS
time out occurs, resend 5th with window size starts with as slow start.
Window size for 6th transmission = 1 MSS
Window size for 7th transmission = 2 MSS
Window size for 8th transmission = 4 MSS
threshold reached, now increase linearly (according to AIMD)
Additive Increase: 5 MSS (since 8 MSS isn't permissible anymore)
Window size for 9th transmission = 5 MSS
Window size for 10th transmission = 6 MSS
Window size at the end of 10th transmission = 6+1 =7 MSS

73. ANSWER:- A

The prefix is given is 20. So, the first 20 bits denote the network id bits, and the next 12 bits indicate host bits. To divide the network into two halves, we borrow bits from the host bits. Hence, for half (2 partitions), the prefix will be 20 + 1 and for 4 partitions, the prefix will be 20 + 2 = 22. So prefix for A will be 21 and prefix for B will be 22. This eliminates options C and D.
For A, we need to consider the 21st bit. It can have values 0 or 1. Let's take its value as 1. So, the network id becomes 245.248.136.0/21.

Check Out Our GATE Courses & Secure Your Dream Rank!

For B, we need to consider the 21st and 22nd bits. We can have four possibilities: 00, 01, 10, 11. Out of these, 10, and 11 belong to network A. So, we take 00 ie 245.248.128.0/22 (Network B) and 01 as 245.248.132.0/22.

74. ANSWER:- C

Number of Networks in class 'C' are 2^{21} as in class 'C' there
24 bits Net id
8 bits host id
Out of 24 bits, 3 bits are used for representation class 'C' i.e. 110
Thus, 21 bits with 2^{11} we can make 2^{21} networks.

75. Answer: A

Initial DV at N3 = (7, 6, 0, 2, 6)
Link cost N2–N3 reduces from 6 → 2
So N3 updates only that entry immediately:
New DV at N3 (before exchange) = (7, 2, 0, 2, 6)

Neighbors of N3:
From distances → neighbors are N2 and N4

Distance vectors received:
From N2: (1, 0, 6, 7, 3)
From N4: (8, 7, 2, 0, 4)

Update using Bellman-Ford:
Formula:
$D_3(x) = \min [\text{cost}(3\to2) + D_2(x), \text{cost}(3\to4) + D_4(x)]$
Cost(3→2) = 2
Cost(3→4) = 2

Compute each entry:
To N1:
via N2 → 2 + 1 = 3
via N4 → 2 + 8 = 10 → min = 3
To N2:
via N2 → 2 + 0 = 2
via N4 → 2 + 7 = 9 → min = 2
To N3: 0
To N4:
via N2 → 2 + 7 = 9
via N4 → 2 + 0 = 2 → min = 2
To N5:
via N2 → 2 + 3 = 5
via N4 → 2 + 4 = 6 → min = 5

Final Distance Vector at N3: (3, 2, 0, 2, 5)

76. Answer: C

Step 1: After first update (given from previous part)
Distance vector at N3 becomes:
(3, 2, 0, 2, 5)

Step 2: Link N1–N2 goes down
At N2: distance to N1 becomes ∞
So updated DV at N2:
(∞, 0, 2, 7, 3)

Step 3: Next round update at N3
Neighbors of N3: N2 and N4
Costs:
C(3→2) = 2
C(3→4) = 2

Compute distance to N1 at N3:
Via N2:
2 + D2(N1) = 2 + ∞ = ∞
Via N4:
2 + D4(N1) = 2 + 8 = 10

Take minimum:
min(∞, 10) = 10

77. Answer: C

After distance vector stabilizes, only shortest path links are used.
So we need to find links that are never part of any shortest path between any pair of nodes.

Edges from diagram:
R1–R2 = 6
R1–R3 = 3
R2–R3 = 2
R2–R4 = 7
R3–R5 = 9
R4–R5 = 1
R4–R6 = 8
R5–R6 = 4

Find useful shortest paths:
R1 → R2: via R3 → cost = 3 + 2 = 5 (better than 6)
→ edge R1–R2 NOT used
R3 → R5: direct = 9
via R4: 2 + 7 + 1 = 10 (worse)
via R1? not better → so R3–R5 IS used
R2 → R4: direct = 7
via R3–R5–R4 = 2 + 9 + 1 = 12 (worse) → used
R4 → R6:
via R5 = 1 + 4 = 5 (better than 8) → edge R4–R6 NOT used

Check Out Our GATE Courses & Secure Your Dream Rank!

Unused links:
R1–R2
R4–R6

78. Answer: C

Initially unused links:
R1–R2 and R4–R6
After changing their weights to 2:
R1–R2 = 2 → becomes useful (clearly shortest between R1 and R2 and used in multiple paths)
R4–R6 = 2 → check carefully:
R5→R6: direct = 4
via R4 = 1 + 2 = 3 → better
So traffic from R5 to R6 will go via R4
Now check if R5–R6 (cost 4) is used anywhere:
R5→R6 prefers via R4 (cost 3)
R3→R6 prefers via R2–R4 (2+7+2=11) instead of R3–R5–R6 (9+4=13)
R1→R6 prefers via R2–R4 (2+7+2=11) instead of R1–R3–R5–R6 (3+9+4=16)
So R5–R6 is never on any shortest path
All other links are used somewhere.

79. ANSWER:- D

When we perform bitwise AND operation between IP address 10.105.1.113 and 255.255.255.224, the result is 10.105.1.96.
When we perform bitwise AND operation between IP address 10.105.1.91 and 255.255.255.224, the result is 10.105.1.64.
10.105.1.96 and 10.105.1.64 are different networks, so D is the correct answer.

80. ANSWER:- D

In the IP datagram there is time to live (TTL) field. It is mainly used to prevent packet looping. Means every packet is associated with certain time stamp (TTL), then this packet is discarded so it can be used to prevent packet looping.

81. ANSWER:- A

Here Minimum Permissible rate may be confusing but think like that if life time of packet is given 64s so this is max . Means if we increment seq.no per 64 sec is allowed but more than 64s should not be allowed.
If we keep more than 64s for 1 sequence number we need to wait for next sequence without any reason.

so ans should be=1/64s = 0.015/s

82. ANSWER:- D

(D) increases exponentially

83. ANSWER:- C

Subnet mask = 255.255.248.0
Binary:
255.255.248.0 =
11111111.11111111.11111000.00000000
Number of network bits = 21
So host bits = 32 − 21 = 11
Maximum hosts per subnet = $2^{11} - 2$
= 2048 − 2
= 2046

84. ANSWER:- B

Transmission rate (M)= 10 Mbps
Token generation rate (r) = 2 Mbps
Net token consumption rate (M-r) = 10 − 2 = 8 Mbps
Initial tokens (b) = 16 Megabits
Time = b/(M-r)=16/(10-2) = 2

85. ANSWER:- C

Suppose connect() is in default blocking mode then calling connect() sends SYN packet to the server. Since server has not executed any accept() call it can not acknowledge the SYN packet. Connect() in blocking mode keep sending SYN packets at fixed intervals(first after 6 sec, second after 24 sec typically until 75 sec latest). This is done until an error ETIMEDOUT is returned by the TCP.(in this case,else there are several other type of errors returned in case No port exists for that connection or server id not listening etc.)
Here, option (B) saying that connect() blocks is not entirely wrong but since we know that accept() call is not made by server, connect() WILL NOT WAIT FOREVER and SO IT CAN NOT BLOCK. It will ultimately return with an ERROR message.
So, option (C) is CORRECT.

86. Answer: D

SYN packet is sent when a TCP connection is initiated (3-way handshake)
Check options:
socket → just creates endpoint
bind → assigns local address
listen → marks socket as passive (server side)
connect → initiates connection → sends SYN

Check Out Our GATE Courses & Secure Your Dream Rank!

87. ANSWER:- C
Class B has 16 host bits.
6 bits used for subnet → remaining host bits = 16 − 6 = 10
Subnets = $2^6 - 2 = 62$
Hosts per subnet = $2^{10} - 2 = 1022$

88. ANSWER:- C
TTL is decremented at each router hop. When it becomes 0 → packet is discarded.
Purpose: avoid infinite routing loops in the network.

89. ANSWER:- C
C1:
Mask = /17 → block size = 128
Network of C1 = 203.197.0.0 – 203.197.127.255
C2 IP = 203.197.75.201 → lies in this range
So, C1 thinks C2 is in same network
C2:
Mask = /18 → block size = 64
Network of C2 = 203.197.64.0 – 203.197.127.255
C1 IP = 203.197.2.53 → NOT in this range
So, C2 thinks C1 is in different network
Answer: (c) C1 assumes C2 is on same network, but C2 assumes C1 is on a different network

90. ANSWER:- B
Routing happens at the IP layer (network layer), not at TCP/UDP.
IP is connectionless → each packet is routed independently.
So even in TCP (though it is connection-oriented), packets can take different paths. Same applies to UDP.
Answer: (b) TCP and UDP

91. ANSWER:- D
Class B default = /16
Need ≥ 64 subnets → $2^6 = 64$ → borrow 6 bits
New mask = 16 + 6 = /22
/22 in decimal = 255.255.252.0
Answer: (d) 255.255.252.0

92. ANSWER:- D
Number of links = 2 intermediate nodes → 3 links
Let payload per packet = x
Total packet size = x + 3
Number of packets = 24 / x
Total transmission time ∝ (number of packets + number of links − 1) × packet size
$= (24/x + 3 - 1)(x + 3)$
$= (24/x + 2)(x + 3)$
Minimize:
$= (24/x)(x+3) + 2(x+3)$
$= 24 + 72/x + 2x + 6$
$= 30 + 72/x + 2x$
Differentiate:
$d/dx = -72/x^2 + 2 = 0$
$\Rightarrow 2 = 72/x^2$
$\Rightarrow x^2 = 36$
$\Rightarrow x = 6$
Packet size = x + header = 6 + 3 = 9
Answer: (d) 9

93. ANSWER:- D
(d) Finding the MAC address that corresponds to an IP address

Check Out Our GATE Courses & Secure Your Dream Rank!

4 APPLICATION LAYER

GATE CSE – 2026 MSQ SET – 1 (1-MARKS)

Q1. Which of the following statements is/are true with respect to the interaction of a web browser with a web server using HTTP 1.1?

(a) HTTP 1.1 facilitates downloading multiple objects of the same webpage over the same TCP connection, if the objects are stored in the same server
(b) HTTP 1.1 facilitates downloading multiple objects of the same webpage over the same TCP connection, even if they are stored in different servers
(c) HTTP 1.1 facilitates sending a request for downloading one object without waiting for a previously requested object to be downloaded completely
(d) HTTP 1.1 facilitates downloading multiple webpages on the same server to be downloaded over a single TCP connection

GATE CSE – 2026 MCQ SET – 2 (2-MARKS)

Q2. Which one of the following protocols may need to **broadcast some of its messages?**

(a) SMTP
(b) FTP
(c) DHCP
(d) HTTP

GATE CSE – 2024 MCQ SET – 2 (1-MARKS)

Q3. A user starts browsing a webpage hosted at a remote server. The browser opens a single connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.
(i) HTTP GET request for the index page.
(ii) DNS request to resolve the web server's name to its IP address.
(iii) HTTP GET request for an image object.
(iv) TCP SYN to open a connection to the web server.
Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer?

(a) (iv), (ii), (iii), (i)
(b) (iv), (ii), (i), (iii)
(c) (ii), (iv), (i), (iii)
(d) (ii), (iv), (iii), (i)

GATE CSE – 2023 MSQ (2-MARKS)

Q4. Suppose in a web browser, you click on the www . gate-2023 . in URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers. Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting www.gate-2023.in is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server.
Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

(a) 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
(b) 5 RTTs, in case of persistent HTTP with pipelining.
(c) 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
(d) 6 RTTs, in case of persistent HTTP with pipelining.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2022 NAT (1-MARKS)

Q5. Consider the resolution of the domain name www.gate.org.in by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.

GATE CSE – 2020 NAT (1-MARKS)

Q6. Assume that you have made a request for a web page through your web browser to a web server. Initially, the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is _____.

GATE CSE – 2019 MCQ (1-MARKS)

Q7. Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

(a) IMAP, SMTP
(b) IMAP, POP3
(c) SMTP, MIME
(d) SMTP, POP3

GATE CSE – 2016 MCQ SET - 1(2-MARKS)

Q8. Which of the following is/are examples of stateful application layer protocols?
(i) HTTP
(ii) FTP
(iii) TCP
(iv) POP3

(a) (i) and (ii) only
(b) (ii) and (iii) only
(c) (ii) and (iv) only
(d) (iv) only

GATE CSE – 2016 MCQ SET - 1(1-MARKS)

Q9. Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

(a) HTTP GET request, DNS query, TCP SYN
(b) DNS query, HTTP GET request, TCP SYN
(c) DNS query, TCP SYN, HTTP GET request
(d) TCP SYN, DNS query, HTTP GET request

GATE CSE – 2015 MCQ SET - 1(1-MARKS)

Q10. In one of the pairs of protocols given below, both the protocols can use multiple TCP connections between the same client and the server. Which one is that?

(a) HTTP, FTP
(b) HTTP, TELNET
(c) FTP, SMTP
(d) HTTP, SMTP

GATE CSE – 2014 MCQ SET - 2(2-MARKS)

Q11. An IP machine Q has a path to another IP machine H via three IP routers R1, R2, and R3.
Q — R1 — R2 — R3 — H
H acts as an HTTP server, and Q connects to H via HTTP and downloads a file. Session layer encryption is used, with DES as the shared key encryption protocol. Consider the following four pieces of information:
[**I1**] The URL of the file downloaded by Q
[**I2**] The TCP port numbers at Q and H
[**I3**] The IP addresses of Q and H
[**I4**] The link layer addresses of Q and H
Which of I1, I2, I3, and I4 can an intruder learn through sniffing at R2 alone?

(a) Only I1 and I2
(b) Only I1
(c) Only I2 and I3
(d) Only I3 and I4

GATE CSE – 2013 MCQ (1-MARKS)

Q12. The transport layer protocols used for real-time multimedia, file transfer, DNS, and email respectively are:

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) TCP, UDP, TCP, and TCP
(b) UDP, TCP, TCP, and UDP
(c) UDP, TCP, UDP, and TCP
(d) TCP, UDP, TCP, and UDP

GATE CSE – 2012 MCQ (1-MARKS)

Q13. The Protocol Data Unit (PDU) for the application layer in the Internet stack is

(a) Segment
(b) Datagram
(c) Message
(d) Frame

GATE CSE – 2012 MCQ (1-MARKS)

Q14. Which of the following transport layer protocols is used to support electronic mail?

(a) SMTP
(b) IP
(c) TCP
(d) UDP

GATE CSE – 2011 MCQ(1-MARKS)

Q15 A layer-4 firewall (a device that can look at all protocol headers up to the transport layer) CANNOT

(a) block entire HTTP traffic during 9:00 pm and 5:00 am
(b) block all ICMP traffic
(c) stop incoming traffic from a specific IP address but allow outgoing traffic to the same IP address
(d) block TCP traffic from a specific user on a multi-user system during 9:00 pm and 5:00 am

GATE CSE – 2011 MCQ(1-MARKS)

Q16 Consider the different activities related to email.

- ml : Send an email from mail client to mail server
- m 2 : Download an email from mailbox server to a mail client
- m3 ; Checking email in a web browser

Which is the application level protocol used in each activity?

(a) m1: HTTP m2: SMTP m3: POP
(b) m1 : SMTP m2 : FTP m3 : HTTP
(c) m1: SMTP m2: POP m3: HTTP
(d) m1 : POP m2 : SMTP m 3 : IMAP

GATE CSE – 2010 MCQ (1-MARKS)

Q17. Which one of the following is not a client-server application?

(a) Internet chat
(b) Web browsing
(c) E-mail
(d) Ping

GATE CSE – 2008 MCQ (1-MARKS)

Q18. What is the maximum size of data that the application layer can pass on to the TCP layer below?

(a) Any size
(b) 2^{16} bytes − size of TCP header
(c) 2^{16} bytes
(d) 1500 bytes

GATE CSE – 2007 MCQ (1-MARKS)

Q19. Which one of the following uses UDP as the transport protocol?

(a) HTTP
(b) Telnet
(c) DNS
(d) SMTP

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2007 MCQ (1-MARKS)

Q20. Match the following:

Protocol	OSI Layer
SMTP	Application layer
BGP	Network layer
TCP	Transport layer
PPP	Data link layer
	Physical Layer

(a) P-2, Q-1, R-3, S-5
(b) P-1, Q-4, R-2, S-3
(c) P-1, Q-4, R-2, S-5
(d) P-2, Q-4, R-1, S-3

Answer key

1. (a, c, d)	**2. (c)**	**3. (c)**
4. (c; d)	**5. (4)**	**6. (6)**
7. (d)	**8. (c)**	**9. (c)**
10. (a)	**11. (c)**	**12. (c)**
13. (c)	**14. (c)**	**15. (d)**
16. (c)	**17. (d)**	**18. (a)**
19. (c)	**20. (b)**	

HINTS & SOLUTIONS

1. ANSWER:- A, C, D

(A): True
Persistent connections allow multiple objects (HTML, images, CSS, etc.) of the same webpage from the same server to be downloaded over a single TCP connection.
(B): False
A TCP connection is established with a specific server.
Objects stored on different servers require different TCP connections.
(C): True
HTTP/1.1 supports pipelining, where the client can send another request without waiting for the previous response to complete.
(D): True
Persistent connections allow the browser to request multiple webpages from the same server over the same TCP connection.Bottom of Form

2. ANSWER:- C

DHCP (Dynamic Host Configuration Protocol): When a new device joins a network, it does not have an IP address and does not know the IP address of the DHCP server. To discover an available server and request network configurations, it must send its initial DHCPDISCOVER message as a broadcast (sent to the destination IP 255.255.255.255) so that every device on the local network segment receives it, including the DHCP server.

3. ANSWER:- C

The standard sequence of operations when trying to reach a server are as follows.

1. Device checks with local DNS server to identify the destination IP address.
2. Since HTTP uses TCP network protocol, it has to setup a link with the webserver, hence it sends a syn packet to the server's IP that was resolved using DNS query.
3. After a TCP connection has been established, The host would request for an index of the webpage using the HTTP GET command.
4. Based on the Index recieved, Host would make further requests to get the other information that it made the connection for.

Option C would be the best fit for the discussed sequence

4. ANSWER:- C, D

persistent http with pipelining will take 3 RTTs (one for connection + one for opening web page + one for 10 objects)
And in iterative mode for DNS is 3 RTTs (Root + Top domain + Authoritative).
So total 6 RTTs in case of persistent HTTP with pipeline.
For non-persistent with 5 parallel TCP:
DNS lookup will take 3 RTTs.
TCP connection: 2 RTTs (for opening webpage)
HTTP request/response: Since we are using non-persistent HTTP, each TCP connection can be used to request one object at a time. Since there are 10 objects in total, and we have 5 parallel TCP connections, it will take 2 rounds of HTTP requests/responses to fetch all the objects. Therefore, this step will take a minimum of 4 RTTs (2 RTTs for

Check Out Our GATE Courses & Secure Your Dream Rank!

the first round and 2 RTTs for the second round).
∴ 3 + 2 + 4 RTTs i.e. 9 RTTs

5. ANSWER:- 4

1. Count the Hierarchy Labels (Right-to-Left)

The domain name www. gate. org. in consists of 4 labels across the DNS hierarchy:

1. . in (Country Code Top-Level Domain)
2. .org. in (Sub-domain zone)
3. gate.org. in (Authoritative domain)
4. *www* (Specific host record)
5. Follow the Iterative Queries

Because nothing is cached, the local resolver must send an independent request-response pair to each consecutive layer of the infrastructure:

- Pair 1: Queries a Root Server → Points to the .in TLD server.
- Pair 2: Queries the .in TLD Server → Points to the .org. in server.
- Pair 3: Queries the .org. in Server → Points to the gate.org. in Authoritative server.
- Pair 4: Queries the gate. org. in Authoritative Server → Returns the final IP address for wnw .

Conclusion The complete resolution requires exactly 4 DNS query-response pairs.

6. ANSWER:- 6

In non-persistent HTTP, for every object, there is a TCP connection required.
Hence, 1 TCP connection for text and 5 TCP connections for images are required.
So, a total of 6 connections are required.

7. ANSWER:- D

SMTP is push protocol and to send e-mail and POP3 is pull protocol i.e. to retrieve e-mail.

8. ANSWER:- C

SMTP is push protocol and to send e-mail and

HTTP	Application	**Stateless**
FTP	Application	**Stateful**
TCP	Transport	**Stateful**
POP3	Application	**Stateful**

9. ANSWER:- C

DNS query, TCP SYN, HTTP GET request

10. ANSWER:- A

HTTP and FTP protocols can use multiple TCP Connection between the same same client and server.

11. ANSWER:- C

I1 is encrypted by DES at the Session layer so intruder can't see that.

I2 is a part of TCP header(below Session layer), so it is not encrypted. Obv., the intruder can see that.

I3 is a part of the IP header(below TCP layer), so again, it is not encrypted and the intruder can see that as well.

I4 is not a part of the DLL header(below the IP layer), since the DLL always contains the Mac Addresses of the immediate sender(previous hop-R1) and the immediate destination(next hop-R3), so the intruder cannot see I4.

So, answer - (C)

12. ANSWER:- C

UDP, TCP, UDP and TCP

Suppose u are looking at the video on you tube and quality of video is not HD then u are just using UDP. When u want to see video in 720HD, 1080HD or 2160HD in that case TCP is used.

For file transfer we use TCP but for file downloading we use FTP or for small file we use TFTP.

DNS,RIP,OSPF,BOOTP,DHCP,NNP,NAT,TIMES TAMP and quote of day all uses UDP.

SMTP is application layer protocol but it uses TCP for sending mail.

13. ANSWER:- C

Message is the answer.
For Application, Presentation and Session layers, the PDU is message
For Transport layer, PDU is segment for TCP and datagram for UDP
For Network layer, PDU is packet
For Datalink layer, PDU is frames
For physical layer, PDU is stream of bits

Check Out Our GATE Courses & Secure Your Dream Rank!

14. ANSWER:- C

TCP is used in transport layer to carry out mail which is iniated by application layer protocol SMTP.
So correct option will be -TCP

15. ANSWER:- D

A is wrong
Because we're asked if we can block "entire" HTTP traffic. This is doable by a L4 firewall by disabling port no. 80.
B is wrong
ICMP is a NL protocol, which comes under TL, so we're good.
C is wrong
IP addresses are the characteristics of NL, which comes under TL.
D is right
Because we're asked to manipulate a specific user. We can't do that with a firewall that can see upto TL.

16. ANSWER:- C

Sending mail → SMTP
Downloading mail → POP (or IMAP, but POP in options)
Checking via browser → HTTP
Mapping:
m1 → SMTP
m2 → POP
m3 → HTTP
Final Answer:
C. m1: SMTP, m2: POP, m3: HTTP

17. ANSWER:- D

(A) Typical internet chat is a client/server app, since a client makes a connection to a central server and receives update. There are some chat systems that work on a P2P model.

(B) Web browsing is client/server, where a browser is the client and the web page is served up by a server.
(C) E-mail is typically thought of as a client/server app, although hops will be made between POP servers.

(D) Ping uses ICMP protocol, and a ping client sends a request to a destination and gets back a response. It is a connection testing tool.
So, D is the answer

18. ANSWER:- A

TCP is a stream-oriented protocol.
Application layer can pass any amount of data to TCP.
TCP itself segments the data into suitable sizes.
Answer: (A) Any size

19. ANSWER:- C

HTTP → TCP

Telnet → TCP

SMTP → TCP

DNS → uses UDP (for normal queries)

20. ANSWER:- B

SMTP → Application layer (1)

BGP → Network layer (4)

TCP → Transport layer (2)

PPP → Data link layer (3)

Answer: (B) P-1, Q-4, R-2, S-3

Check Out Our GATE Courses & Secure Your Dream Rank!

DISCRETE MATHEMATICS

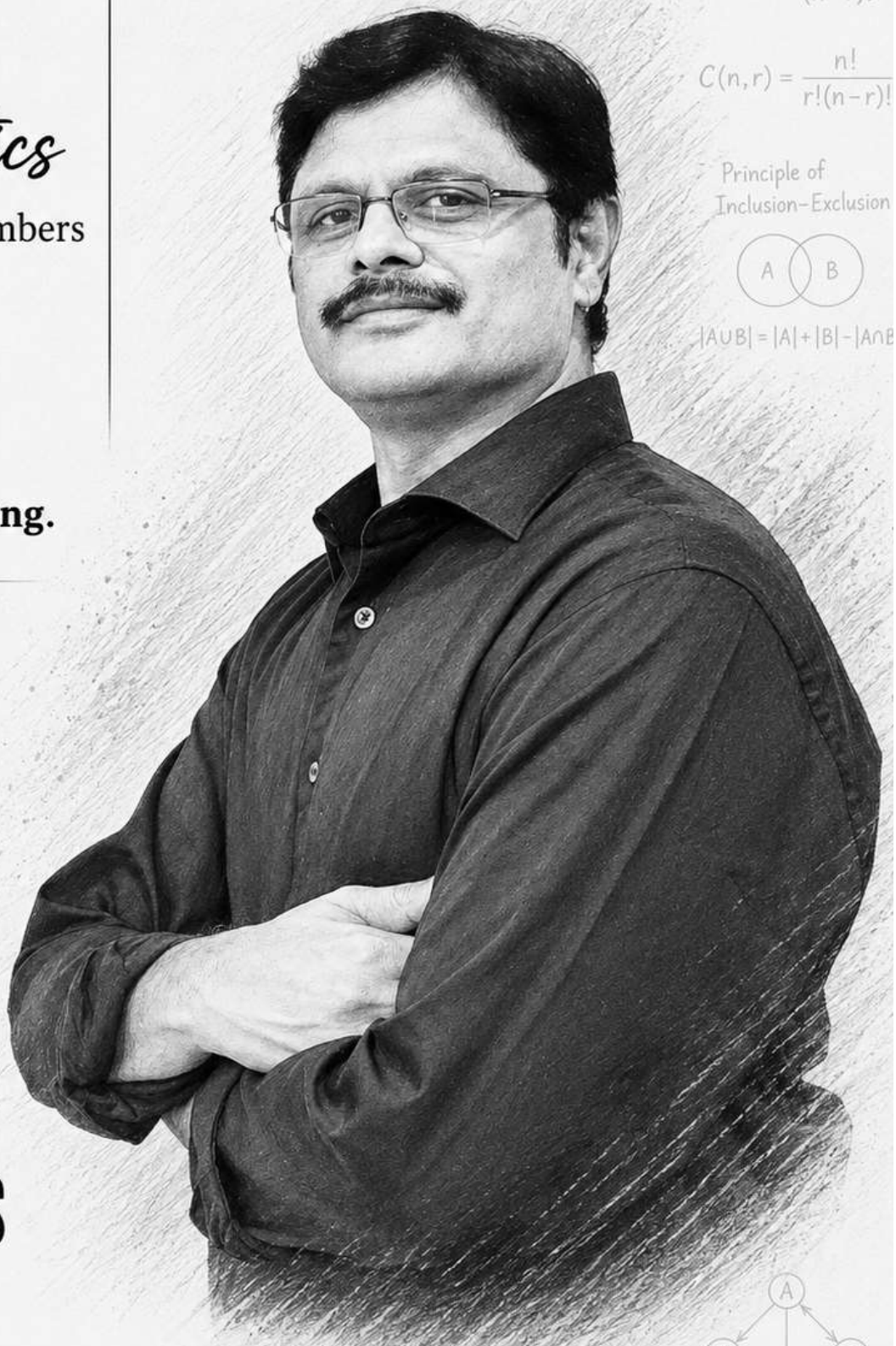
"Scan Here for Video Solution"

Link to GfG corresponding Short Notes: GATE

“

Discrete
Mathematics
is not just about numbers
and structures—
it’s about building
the foundation of
logical thinking
and **problem solving.**

”



LEGEND OF

DISCRETE MATHEMATICS

Sridhar Dhulipala Sir

An Inspirer. A Mentor. A Guide.

1 COMBINATORS

GATE CSE - 2023 MCQ (2-MARKS)

Q1. Let U = {1, 2,...,n}, where n is a large positive integer greater than 1000. Let k be a positive integer less than n. Let A, B be subsets of U with |A| = |B| = k and A ∩ B = ∅. We say that a permutation of U separates A from B if one of the following is true.
- All members of A appear in the permutation before any of the members of B.
- All members of B appear in the permutation before any of the members of A.
How many permutations of U separate A from B?

a) n!
b) $\binom{n}{2k}$ (n-2k)!
c) $\binom{n}{2k}$ (n-2k)!(k!)^2
d) $2\binom{n}{2k}$ (n-2k)!(k!)^2

GATE CSE - 2020 NAT (2-MARKS)

Q2. The number of permutations of the characters in LILAC so that no character appears in its original position, if the two L's are indistinguishable, is _____.

GATE CSE – 2018 MCQ (2-MARKS)

Q3. Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n+3$ for all n = 0, 1, 2, …?

a) $3/(1 – x)^2$
b) $3x/(1–x)^2$
c) $(2–x)/(1–x)^2$
d) $(3–x)/(1–x)^2$

GATE CSE – 2017 NAT SET - 2(2-MARKS)

Q4. If the ordinary generating function of a sequence $\{a_n\}^{\infty}_{n=0}$ is $(1+z)/(1-z)^3$ then $a_3 – a_0$ is equal to _____.

GATE CSE - 2016 NAT SET –1 (2-MARKS)

Q5. Consider the recurrence relation $a_1 = 8$, $a_n = 6n^2 + 2n + a_{n-1}$. Let $a_{99} = K \times 10^4$. The value of K is _____.

GATE CSE – 2016 NAT SET - 1(2-MARKS)

Q6. The coefficient of x^{12} in $(x^3 + x^4 + x^5 + x^6 +...)^3$ is

GATE CSE – 2016 MCQ SET - 1(1-MARKS)

Q7. Let a_n be the number of n-bit strings that do NOT contain two consecutive 1s. Which one of the following is the recurrence relation for a_n?

a) $a_n = a_{n–1} + 2a_{n–2}$
b) $a_n = a_{n–1} + a_{n–2}$
c) $a_n = 2a_{n–1} + a_{n–2}$
d) $a_n = 2a_{n–1} + 2a_{n–2}$

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE – 2015 MCQ SET - 1(2-MARKS)

Q8. Let a_n represent the number of bit strings of length n containing two consecutive 1s. What is the recurrence relation for a_n?

a) $a_{n-2} + a_{n-1} + 2^{n-2}$
b) $a_{n-2} + 2a_{n-1} + 2^{n-2}$
c) $2a_{n-2} + a_{n-1} + 2^{n-2}$
d) $2a_{n-2} + 2a_{n-1} + 2^{n-2}$

GATE CSE – 2015 NAT SET - 1(2-MARKS)

Q9. $\sum_1^{99} 1/x(x+1) =$ _____.

GATE CSE – 2015 NAT SET - 2 (1-MARKS)

Q10. The number of divisors of 2100 is _____.

GATE CSE - 2015 NAT SET-3(1-MARKS)

Q11. The number of 4-digit numbers having their digits in non-decreasing order (from left to right) constructed by using the digits belonging to the set {1, 2, 3} is _____.

GATE CSE - 2014 NAT SET-1(2-MARKS)

Q12. A pennant is a sequence of numbers, each number being 1 or 2. An n-pennant is a sequence of numbers with sum equal to n.
For example, (1,1,2) is a 4-pennant. The set of all possible 1-pennants is {(1)}, the set of all possible 2-pennants is {(2), (1,1)}and the set of all 3-pennants is {(2,1), (1,1,1), (1,2)}.
Note that the pennant (1,2) is not the same as the pennant (2,1). The number of 10-pennants is _____.

GATE CSE - 2014 NAT SET-2(2-MARKS)

Q13. The number of distinct positive integral factors of 2014 is _____

GATE CSE - 2014 NAT SET-2(1-MARKS)

Q14. Each of the nine words in the sentence
"The quick brown fox jumps over the lazy dog" is written on a separate piece of paper. These nine pieces of paper are kept in a box. One of the pieces is drawn at random from the box. The expected length of the word drawn is _____.
(The answer should be rounded to one decimal place.)

GATE IT - 2008 MCQ (2-MARKS)

Q15. In how many ways can b blue balls and r red balls be distributed in n distinct boxes?

a) $\frac{(n+b-1)!(n+r-1)!}{(n-1)!b!(n-1)!r!}$
b) $\frac{(n+(b+r)-1)!}{(n-1)!(n-1)!(b+r)!}$
c) $\frac{n!}{b!r!}$
d) $\frac{(n+(b+r)-1)!}{n!(b+r+1)}$

ANSWER KEY

1. (d)	**2. (12)**	**3. (d)**	**4. (15)**
5. (198)	**6. (10)**	**7. (b)**	**8. (a)**
9. (0.99)	**10.(36)**	**11.(15)**	**12.(89)**
13.(8)	**14.(3.9)**	**15.(a)**	

HINTS&SOLUTIONS

1. ANSWER:- D

Instead of taking variables lets take a small example to first understand the question, once that's done, the solutions would seem pretty intuitive!
Let n = 6
U = {1,2,3,4,5,6}
A = {1,2} and B = {4,5}
Here k = 2 and A ∩ B = Φ
Now, coming to the solution,
Step 1 : We first select 2k elements from n elements for creating A and B i.e ${}^{n}C_{2k}$
Step 2 : Elements in A and B can permute amongst themselves respectively so k!×k!(i.e., ${}^{n}C_{2k}$×k!×k!)
Step 3 : Now there can be two cases – AB and BA i.e ${}^{n}C_{2k}$×k!×k!×2
Step 4 : The remaining elements (n-2k) can arrange in any way so (n−2k)!
Thus, the expression becomes:
${}^{n}C_{2k}$×k!×k!×2×(n−2k)!
Thus, our solution becomes Option D.

2. ANSWER:- 12

There are 5 places
— — — — —
Given: L I L A C
There are three cases possible-
Case-1: _L _ L _
In this case, we put C in first and third place
So, C L A L I and C L I L A
A L C L I and I L C L A
There are 4 ways possible in this case.
Case-2: _L _ _ L
In this case, we put A in first and third place
So, A L C I L and A L I C L
C L A I L and I L A C L
There are 4 ways possible in this case.
Case-3: _ _ _ LL
In this case, we put I in first and third place
So, I A C L L and I C A L L
C A I L L and A C I L L
There are 4 ways possible in this case.
Thus, We get a total of 12 ways by adding all the above cases.

3. ANSWER:- D

Given sequence $a_n = 2n+3$
Generating function "G(x)" for the sequence a_n is
$G(x) = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 2n(x^n) + 3(x)^n$
$= 2(0 + x + 2x^2 + 3x^3 + \ldots) + 3(1 + x + x^2 + \ldots)$
We know,
$(1 - x)^{-1} = 1 + x + x^2 + \ldots$ (1)
$x + 2x^2 + 3x^3 + \ldots = x(1 + 2x + 3x^2 + 4x^3 + \ldots.)$
$= x(1 - x)^{-2}$
$= \frac{x}{(1-x)^2}$ ... (2)
Substitute (1) and (2) in G(x)
$G(x) = 2\left(\frac{x}{(1-x^2)}\right) + 3\left(\frac{1}{1-x}\right)$
$= \frac{2x + 3(1-x)}{(1-x)^2} = \frac{2x + 3 - 3x}{(1-x)^2}$
$= \frac{3 - x}{(1-x)^2}$

4. ANSWER:- 15

$\frac{1+z}{(1-z)^3} = (1+z)(1-z)^{-3}$
$(1 - z)^{-3} = 1 + \binom{3}{1}z + \binom{4}{2}z^2 + \binom{5}{3}z^3 + \cdots \infty$
$(1+z)(1-z)^{-3} = (1+z) * (1 + \binom{3}{1}z$
$+ \binom{4}{1}z^2 + \binom{5}{3}z^3 + \cdots \infty)$
α_0 is the first term in the expansion of above series and α_3 is the fourth term (or) coefficient of z^3
α_0 = coefficient of $z^0 = 1$
α_3 = coefficient of $z^3 = \binom{5}{3} + \binom{4}{2} = 10 + 6$
$\Rightarrow \alpha_3 - \alpha_0 = 16 - 1 = 15$

5. ANSWER:- 198

$a_n = 6n^2 + 2n + a_{n-1}$
Replace $a_{(n-1)}$
$\Rightarrow a_n = 6n^2 + 2n + 6(n-1)^2 + 2(n-1)$
$\Rightarrow a_n = 6n^2 + 2n + 6(n-1)^2 + 2(n-1) + 6(n-2)^2 + 2(n-2) + \cdots a_1$
Given that $a_1 = 8$, replace it
$\Rightarrow a_n = 6n^2 + 2n + 6(n-1)^2 + 2(n-1) + 6(n-2)^2 + 2(n-2) + \cdots 8$
$= 6n^2 + 2n + 6(n-1)^2 + 2(n-1) + 6(n-2)^2 + 2(n-2) + \cdots + 6(1)^2 + 2(1)$
Observe that $6n^2 + 2n$ (replace n with n-1) is repeated with each 'n'.
$= 6(n^2 + (n-1)^2 + (n-2)^2 + \cdots + 2^2 + 1^2) + 2(n + (n-1) + \cdots 1)$
Sum of $n^2 = (n(n+1)(2n+1))/6$
Sum of $n = (n(n+1))/2$
$= 6 \times (n(n+1)(2n+1))/6 + 2 \times (n(n+1))/2$
$= n(n+1)[1+2n+1]$
$= n(n+1)[2n+2]$
$= 2n(n+1)^2$
Given $a_{99} = k \times 10^4$
$a_{99} = 2(99)(100)^2 = 198 \times 10^4$
$\therefore k = 198$

6. ANSWER:- 10

Coefficient of x^{12} in $(x^3 + x^4 + x^5 + x^6 + ...)^3$ is
$\Rightarrow [x^3 (1 + x + x^2 + x^3 + ...)]^3$
$= x^9 (1 + x + x^2 + x^3 + ...)^3$
First Reduction:
As x^9 is out of the series, we need to find the co-efficient of x^3 in $(1 + x + x^2 + \cdots)^3$

$$(1 + x + x^3 + \cdots)^m = \sum_{k=0}^{\infty} \binom{m + k - 1}{k} x^k$$

Here, m=3, k=3, the coefficient
$= {}^5C_3 = 5!/2!3! = 10$

7. ANSWER:- B

Let: a_n = number of n-bit strings that do not have two consecutive 1's.
If n=1, possible strings {0,1}
$a_1 = 2$
If n=2, possible strings {00,01,10}
$a_2 = 3$
If n=3,
possible strings {000,001,010,100,101}
$a_3 = 5$
It is evident that $a_3 = a_1 + a_2$
Similarly, $a_n = a_{n-1} + a_{n-2}$

8. ANSWER:- A

For a string of length 1, there is '0' consecutive 1's.
So, $a_1 = 0$
For string of length 2,
$a_2 = 1$
Similarly, $a_3 = 3$
$a_4 = 8$
Only option A will satisfy the above values.

9. ANSWER:- 0.99

$$\sum_{x=1}^{99} \frac{1}{x(x+1)}$$
$$= \sum_{x=1}^{99} \left(\frac{1}{x} - \frac{1}{x+1}\right)$$
$$= \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{99} - \frac{1}{100}$$
$$= 1 - \frac{1}{100}$$
$$= 0.99$$

10. ANSWER:- 36

Let N = 2100
$= 2^2 \times 3 \times 5^2 \times 7$ (i.e., the product of primes)
Then, the number of divisors of 2100 is:
$(2+1)\cdot(1+1)\cdot(2+1)\cdot(1+1)$
i.e., (3)(2)(3)(2)
i.e., 36.

11. ANSWER:- 15

To form a four-digit number with non-decreasing digits from the set {1, 2, 3}, we can look at the possible combinations:
1. All digits are 1s: There's only one such number, 1111.
2. One digit is 2 and the rest are 1s: There are four such numbers, 1112, 1122, 1222, and 2222.
3. One digit is 3 and the rest are 1s: There are four such numbers, 1113, 1133, 1333, and 3333.
4. One digit is 2, one digit is 3, and the rest are 1s: There are six such numbers, 1123, 1133, 1223, 1233, 1333, and 2233.
Adding these up, we have a total of 1+4+4+6=15 four-digit numbers with non-decreasing digits from the set {1, 2, 3}.

Alternative approach

Since the digits must be non-decreasing, once we decide how many 1's, 2's, and 3's are used, the number is automatically fixed. So we only need to count how many ways we can choose these digits.

Let x1 ,x2 ,x3 be the number of 1's, 2's, and 3's respectively. Then: x1+x2+ x3=4, x1,x2,x3≥0.

This is a standard stars and bars problem. The number of non-negative integer solutions is:$({}^{4+3-1}C_{3-1}) = ({}^6C_2)=15$ So, the answer is 15.

12. ANSWER:- 89

No two's: 1111111111⇒ 1 pennant
Single two: 211111111
⇒ 9!/8!1! = 9 pennants
Two two's: 22111111 ⇒ 8!/6!2! = 28
Three two's: 2221111 ⇒ 7!/3!4! = 35
Four two's: 222211 ⇒ 6!/4!2! = 15
Five two's: 22222 ⇒ 1
Total = 89 pennants.

13. ANSWER:- 8

Let's find prime factorization of $2014 = 2^1 \times 19^1 \times 53^1$
Then number of distinct positive integral factors of 2014 will be,
$(1+1)\times(1+1)\times(1+1) = 2\times2\times2 = 8$

14. ANSWER:- 3.9

Given sentence is

"The quick brown fox jumps over the lazy dog"

Number of 3 letter words=4

(The, fox, the, dog)

Number of 4 letter words=2

(over, lazy)

Number of 5 letter words=3

(quick, brown, jumps)

Let X: Length of the word

Hence the probability distribution for X is

X	3	4	5
$p(x)$	$4/9$	$2/9$	$3/9$

Expected value of x is

$$E(X) = \sum x\, p(x)$$

$$= 3 \times \frac{4}{9} + 4 \times \frac{2}{9} + 5 \times \frac{3}{9}$$

$$= \frac{35}{9} = 3.88 = 3.9$$

15. ANSWER:- a

r red balls can be distributed into n – distinct boxes in

C (n + r −1, r) = $\frac{(n+r-1)!}{(n-1)!r!}$

b blue balls can be distributed in C (n + b – 1, b) = $\frac{(n+b-1)!}{(n-1)!b!}$

By product rule total ways are $\frac{(n+b-1)!(n+r-1)!}{(n-1)!b!(n-1)!r!}$

So the answer is (A)

2 SET THEORY

GATE CSE - 2026 MSQ SET - 2(1-MARKS)

Q1. Let R be a binary relation on the set $\{1, 2, \ldots, 10\}$, where $(x, y) \in R$ if the product of x and y is square of an integer. Which of the following properties is/are satisfied by R?

a) Reflexive
b) Symmetric
c) Transitive
d) Antisymmetric

GATE CSE - 2025 MSQ SET - 2(2-MARKS)

Q2. Let F be the set of all functions from $\{1,\ldots,n\}$ to $\{0,1\}$.
Define the binary relation $\preceq$ on F as follows:
$\forall f,g \in F, f \preceq g$ if and only if $\forall x \in \{1,\ldots,n\}$, $f(x) \leq g(x)$
Which of the following statement(s) is/are TRUE?

a) $\preceq$ is a symmetric relation.
b) $(F, \preceq)$ is a partial order.
c) $(F, \preceq)$ is a lattice.
d) $\preceq$ is an equivalence relation.

GATE CSE - 2025 MSQ SET - 1(2-MARKS)

Q3. $A=\{0,1,2,3,\ldots\}$ is the set of non-negative integers.
Let F be the set of functions from A to itself.
For any two functions $f_1, f_2 \in F$, define
$(f_1 \Theta f_2)(n) = f_1(n) + f_2(n)$
for every number $n \in A$
Which of the following is/are CORRECT about the mathematical structure (F, Θ) ?

a) (F, Θ) is an Abelian group.
b) (F, Θ) is an Abelian monoid.
c) (F, Θ) is a non-Abelian group.
d) (F, Θ) is a non-Abelian monoid.

GATE CSE - 2025 MCQ SET - 1(1-MARKS)

Q4. g(.) is a function from A to B, f(.) is a function from B to C, and their composition defined as f(g(.)) is a mapping from A to C.
If f(.) and f(g(.)) are onto (surjective) functions, which ONE of the following is TRUE about the function g(.)?

a) g(.) must be an onto (surjective) function.
b) g(.) must be a one-to-one (injective) function.
c) g(.) must be a bijective function, that is, both one-to-one and onto.
d) g(.) is not required to be a one-to-one or onto function.

GATE CSE - 2024 NAT SET - 1(1-MARKS)

Q5. Let A and B be non-empty finite sets such that there exist one-to-one and onto functions:
1. From A to B.
2. From $A \times A$ to $A \cup B$
The number of possible values of $|A|$ is _____.

GATE CSE - 2024 MSQ SET - 1(2-MARKS)

Q6. Consider the operators $\Diamond$ and $\square$ defined by:

- $a \Diamond b = a + 2b$
- $a \square b = ab$

for positive integers. Which of the following statements is/are TRUE?

a) Operator $\Diamond$ obeys the associative law
b) Operator $\square$ obeys the associative law
c) Operator $\Diamond$ over the operator $\square$ obeys the distributive law
d) Operator $\square$ over the operator $\Diamond$ obeys the distributive law

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2024 NAT SET - 2(2-MARKS)

Q7. Let Z_n be the group of integers $\{0, 1, 2, \ldots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is _____

GATE CSE - 2023 MSQ (2-MARKS)

Q8. Let f : A → B be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation ∼ on the set A as
$a_1 \sim a_2$ if $f(a_1) = f(a_2)$,
where $a_1, a_2 \in A$. Let $\varepsilon = \{[x] : x \in A\}$ be the set of all the equivalence classes under ∼. Define a new mapping F : ε→ B as
F([x]) = f(x), for all the equivalence classes [x] in ε.
Which of the following statements is/are TRUE?

a) F is NOT well-defined.
b) F is an onto (or surjective) function.
c) F is a one-to-one (or injective) function.
d) F is a bijective function.

GATE CSE - 2023 MSQ (2-MARKS)

Q9. Which of the following statements is/are TRUE for a group G?

a) If for all $x, y \in G$, $(xy)^2 = x^2 y^2$, then G is commutative.
b) If for all $x \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the identity element of G.
c) If the order of G is 2, then G is commutative.
d) If G is commutative, then a subgroup of G need
not be commutative.

GATE CSE - 2021 NAT SET - 2(2-MARKS)

Q10 Let S be a set of consisting of 10 elements.
The number of tuples of the form (A, B) such that A and B are subsets of S, and A ⊆ B is _____

GATE CSE – 2021 MSQ SET- 2 (1-MARKS)

Q11. Consider the following sets, where $n \geq 2$:
S_1 : Set of all n × n. matrices with entries from the set {a, b, c}
S_2 : Set of all functions from the set $\{0, 1, 2, \ldots, n^2 - 1\}$ to the set {0,1,2}
Which of the following choice(s) is/are correct?

a) There exists a bijection from S_1 to S_2.
b) There exists a surjection from S_1 to S_2.
c) There does not exist an injection from S_1 to S_2.
d) There does not exist a bijection from S_1 to S_2

GATE CSE – 2021 MSQ SET- 1 (2-MARKS)

Q12. A relation R is said to be circular if aRb and bRc together imply cRa.
Which of the following options is/are correct?

a) If a relation S is reflexive and symmetric, then S is an equivalence relation.
b) If a relation S is reflexive and circular, then S is an equivalence relation.
c) If a relation S is circular and symmetric, then S is an equivalence relation.
d) If a relation S is transitive and circular, then S is an equivalence relation.

GATE CSE - 2021 MCQ SET-1 (2-MARKS)

Q13. Let G be a group of order 6, and H be subgroup of G such that $1 < |H| < 6$.
Which one of the following options is correct?

Check Out Our GATE Courses & Secure Your Dream Rank!

a) G is always cyclic, but H may not be cyclic.
b) G may not be cyclic, but H is always cyclic.
c) Both G and H are always cyclic.
d) Both G and H may not be cyclic.

GATE CSE - 2020 NAT (1-MARKS)

Q14. Let R be the set of all binary relations on the set {1,2,3}. Suppose a relation is chosen from R at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is _____.

GATE CSE - 2020 NAT (1-MARKS)

Q15. Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is _____.

GATE CSE – 2019 MCQ (1-MARKS)

Q16. Let G be an arbitrary group. Consider the following relations on G:
R_1 : ∀a,b ∈ G, aR_1 b if and only if ∃g ∈ G such that $a = g^{-1} bg$
R_2 : ∀a,b ∈ G, aR_2 b if and only if $a = b^{-1}$
Which of the above is/are equivalence relation/relations?

a) R_2 only
b) R_1 and R_2
c) Neither R_1 nor R_2
d) R_1 only

GATE CSE - 2018 MCQ (2-MARKS)

Q17. Let N be the set of natural numbers.
Consider the following sets,
P: Set of Rational numbers (positive and negative)
Q: Set of functions from {0, 1} to N
R: Set of functions from N to {0, 1}
S: Set of finite subsets of N.

Which of the above sets are countable?

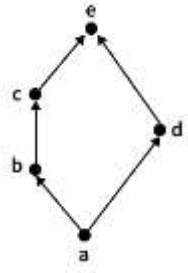
a) Q and S only
b) P and S only
c) P and R only
d) P, Q and S only

GATE CSE - 2018 NAT (1-MARKS)

Q18. Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____.

GATE CSE - 2017 NAT SET - 2(1-MARKS)

Q19. Consider the set X = {a,b,c,d e} under the partial ordering
R = {(a,a),(a,b),(a,c),(a,d),(a,e),(b,b),(b,c),(b,e),(c,c),(c,e),(d,d),(d,e),(e,e)}.
The Hasse diagram of the partial order (X,R) is shown below.



The minimum number of ordered pairs that need to be added to R to make (X, R) a lattice is _____.

GATE CSE - 2017 NAT SET -1 (1-MARKS)

Q20. The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is _____.

GATE CSE - 2016 NAT SET -1 (2-MARKS)

Q21. A function f: $N^+ \rightarrow N^+$, defined on the set of positive integers N^+ , satisfies the following properties:

Check Out Our GATE Courses & Secure Your Dream Rank!

f(n) = f(n/2) if n is even
f(n) = f(n+5) if n is odd
Let R = {i|∃j: f(j)=i} be the set of distinct values that f takes. The maximum possible size of R is _____.

GATE CSE - 2016 MCQ SET - 2(2-MARKS)

Q22. Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U. Consider the following statements:

I. Each compound in U\S reacts with an odd number of compounds.
II. At least one compound in U\S reacts with an odd number of compounds.
III. Each compound in U\S reacts with an even number of compounds.

Which one of the above statements is ALWAYS TRUE?

a) Only I
b) Only II
c) Only III
d) None

GATE CSE - 2016 MCQ SET - 2(2-MARKS)

Q23. A binary relation R on $\mathbb{N}\times\mathbb{N}$ is defined as follows: (a,b)R(c,d) if $a \leq c$ or $b \leq d$.
Consider the following propositions:
P: R is reflexive
Q: R is transitive
Which one of the following statements is TRUE?

a) Both P and Q are true.
b) P is true, and Q is false.
c) P is false, and Q is true.
d) Both P and Q are false.

GATE CSE - 2015 NAT SET - 2(2-MARKS)

Q24. The number of onto functions (surjective function) from set X = {1, 2, 3, 4} to set Y = {a, b, c} is _____.

GATE CSE – 2015 NAT SET - 2 (2-MARKS)

Q25. The cardinality of the power set of {0, 1, 2, … 10} is _____

GATE CSE – 2015 MCQ SET - 1 (2-MARKS)

Q26. The binary operator ≠ is defined by the following truth table

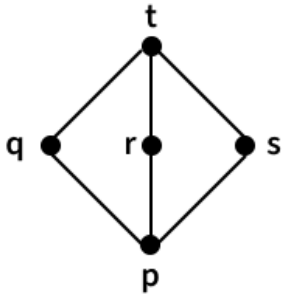
P	q	p ≠ q
0	0	0
0	1	1
1	0	1
1	1	0

Which one of the following is true about the binary operator ≠?

a) Both commutative and associative
b) Commutative but not associative
c) Not commutative but associative
d) Neither commutative nor associative

GATE CSE – 2015 MCQ SET - 1 (2-MARKS)

Q27. Suppose L = {p, q, r, s, t} is a lattice represented by the following Hasse diagram:



For any x, y ∈ L, not necessarily distinct, x ∨ y and x ∧ y are join and meet of x, y respectively. Let L^3 = {(x,y,z): x, y, z ∈ L} be the set of all ordered triplets of the elements of L. Let p_r be the probability that an element (x,y,z) ∈ L^3 chosen equiprobably satisfies x ∨ (y ∧ z) = (x ∨ y) ∧ (x ∨ z). Then

Check Out Our GATE Courses & Secure Your Dream Rank!

a) $p_r = 0$
b) $p_r = 1$
c) $0 < p_r \leq 1/5$
d) $1/5 < p_r < 1$

GATE CSE - 2015 MCQ SET – 3 (2-MARKS)

Q28. Let R be a relation on the set of ordered pairs of positive integers such that ((p,q),(r,s)) ∈ R if and only if p - s = q - r.
Which one of the following is true about R?

a) Both reflexive and symmetric
b) Reflexive but not symmetric
c) Not reflexive but symmetric
d) Neither reflexive nor symmetric

GATE CSE – 2015 MCQ SET - 2 (1-MARKS)

Q29. Let R be the relation on the set of positive integers such that aRb if and only if a and b are distinct and have a common divisor other than 1. Which one of the following statements about R is true?

a) R is symmetric and reflexive but not transitive
b) R is reflexive but not symmetric and not transitive
c) R is transitive but not reflexive and not symmetric
d) R is symmetric but not reflexive and not transitive

GATE CSE – 2015 MCQ SET - 3 (1-MARKS)

Q30. Suppose U is the power set of the set S = {1, 2, 3, 4, 5, 6}. For any T ∈ U, let |T| denote the number of elements in T and T' denote the complement of T. For any T, R ∈ U, let T \ R be the set of all elements in T which are not in R. Which one of the following is true?

a) $\forall X \in U (|X| = |X'|)$
b) $\exists X \in U \exists Y \in U (|X| = 5, |Y| = 5 \text{ and } X \cap Y = \emptyset)$
c) $\forall X \in U \forall Y \in U (|X| = 2, |Y| = 3 \text{ and } X \setminus Y = \emptyset)$
d) $\forall X \in U \forall Y \in U (X \setminus Y = Y' \setminus X')$

GATE CSE - 2015 MCQ SET - 1(1-MARKS)

Q31. For a set A, the power set of A is denoted by 2^A . If A = {5,{6},{7}}, which of the following options are TRUE?
I. $\emptyset \in 2^A$
II. $\emptyset \subseteq 2^A$
III. $\{5,\{6\}\} \in 2^A$
IV. $\{5,\{6\}\} \subseteq 2^A$

a) I and III only
b) II and III only
c) I, II and III only
d) I, II and IV only

GATE CSE - 2014 NAT SET – 1 (2-MARKS)

Q32. Let S denote the set of all functions f: $\{0,1\}^4 \rightarrow \{0,1\}$. Denote by N the number of functions from S to the set {0,1}. The value of $\log_2\log_2 N$ is _____.

GATE CSE – 2014 NAT SET - 3 (2-MARKS)

Q33. There are two elements x,y in a group (G, ∗) such that every element in the group can be written as a product of some number of x's and y's in some order. It is known that
x*x = y*y =x*y*x*y = y*x*y*x = e where e is the identity element. The maximum number of elements in such a group is____.

GATE CSE – 2014 NAT SET- 3 (1-MARKS)

Q34. Let G be a group with 15 elements. Let L be a subgroup of G. It is known that L ≠ G and that the size of L is at least 4. The size of L is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2014 MCQ SET - 2 (2-MARKS)

Q35. Consider the following relation on subsets of the set S of integers between 1 and 2014. For two distinct subsets U and V of S we say U < V if the minimum element in
the symmetric difference of the two sets is in U.
Consider the following two statements:
S1: There is a subset of S that is larger than every other subset.
S2: There is a subset of S that is smaller than every other subset.
Which one of the following is CORRECT?

a) Both S1 and S2 are true
b) S1 is true and S2 is false
c) S2 is true and S1 is false
d) Neither S1 nor S2 is true

GATE CSE – 2014 MCQ SET - 3 (2-MARKS)

Q36. Consider the set of all functions f: {0, 1,....., 2014} → {0, 1, ..., 2014} such that f(f(i)) = i, for all $0 \leq i \leq 2014$. Consider the following statements:
P. For each such function it must be the case that for every i, f(i) = i.
Q. For each function it must be the case that for some i, f(i) = i.
R. Each such function must be onto.
Which one of the following is CORRECT?

a) P, Q and R are true
b) Only Q and R are true
c) Only P and Q are true
d) Only R is true

GATE CSE – 2014 MCQ SET – 3 (1-MARKS)

Q37. Let X and Y be finite sets and f: X → Y be a function. Which one of the following statements is TRUE?

a) For any subsets A and B of X, $| f(A \cup B) | = |f(A)| + |f(B)|$
b) For any subsets A and B of X, $f(A \cap B) = f(A) \cap f(B)$
c) For any subsets A and B of X, $|f(A \cap B)| = \min\{ |f(A)|, |f|(B)|\}$
d) For any subsets S and T of Y, $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

GATE CSE – 2013 MCQ (1-MARKS)

Q38. A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is TRUE about $\oplus$?

a) Commutative but not associative
b) Both commutative and associative
c) Associative but not commutative
d) Neither commutative nor associative

GATE CSE – 2012 MCQ (2-MARKS)

Q39. How many onto (or surjective) functions are there from an n-element ($n \geq 2$) set to a 2-element set?

a) 2^n
b) $2^n - 1$
c) $2^n - 2$
d) $2(2^n - 2)$

GATE CSE – 2010 MCQ (1-MARKS)

Q40. Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 are cube roots of unity. If * denotes the multiplication operation, the structure (S, *) forms.

a) A group
b) A ring
c) An integral domain
d) A field

GATE CSE – 2010 MCQ (1-MARKS)

Q41. What is the possible number of reflexive relations on a set of 5 elements?

a) 2^{10}
b) 2^{15}
c) 2^{20}
d) 2^{25}

Check Out Our GATE Courses & Secure Your Dream Rank!

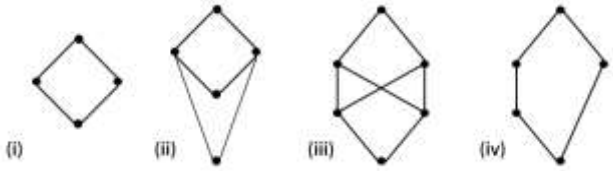
GATE CSE – 2009 MCQ (1-MARKS)

Q42. Consider the binary relation R = {(x, y), (x, z), (z, x), (z, y)} on the set {x, y, z}. Which one of the following is TRUE?

a) R is symmetric but NOT antisymmetric.
b) R is NOT symmetric but antisymmetric.
c) R is both symmetric and antisymmetric.
d) R is neither symmetric nor antisymmetric.

GATE IT - 2008 MCQ (2- MARKS)

Q43. Consider the following Hasse diagrams



Which all of the above represent a lattice?

a) (i) and (iv) only
b) (ii) and (iii) only
c) (iii) only
d) (i), (ii) and (iv) only

GATE IT – 2008 MCQ (2-MARKS)

Q44. Consider the field C of complex numbers with addition and multiplication. Which of the following form(s) a subfield of C with addition and multiplication?

- S1: the set of real numbers
- S2: {(a + ib) | a and b are rational numbers}
- S3: {a+ib | $(a^2 + b^2) \leq 1$}
- S4: {ia | a is real}

a) only S1
b) S1 and S3
c) S2 and S3
d) S1 and S2

GATE IT - 2008 MCQ (2 - MARKS)

Q45. The exponent of 11 in the prime factorization of 300! is

a) 27
b) 28
c) 29
d) 30

GATE CSE – 2007 MCQ (1-MARKS)

Q46. Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are

a) n and n
b) n^2 and n
c) n^2 and 0
d) n and 1

GATE IT – 2007 MCQ (2-MARKS)

Q47. A partial order P is defined on the set of natural numbers as follows. Here x/y denotes integer division.
i. (0,0) ∈ P.
ii. (a, b) ∈ P if and only if (a%10) ≤ (b%10) and (a%10, b%10) ∈ P.
Consider the following ordered pairs:
i. (101, 22)
ii. (22, 101)
iii. (145, 265)
iv. (0, 153)
Which of these ordered pairs of natural numbers are contained in P?

a) (i) and (iii)
b) (ii) and (iv)
c) (i) and (iv)
d) (iii) and (iv)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE IT – 2007 MCQ (2-MARKS)

Q48. How many different non-isomorphic Abelian groups of order 4 are there?

a) 2
b) 3
c) 4
d) 5

GATE CSE – 2006 MCQ (1-MARKS)

Q49. A relation is defined on ordered pairs of integers as follows:
(x,y)R(u,v) **if** x<u and y>v
Then R is:

a) Neither a Partial Order nor an Equivalence Relation
b) A Partial Order but not a Total Order
c) A total Order
d) An Equivalence Relation

GATE CSE – 2006 MCQ (2-MARKS)

Q50. Let E, F and G be finite sets. Let
$X=(E\cap F)-(E\cap G)$ and $Y=(E-(E\cap G))-(E-F)$
Which one of the following is true?

a) $X \subset Y$
b) $X \supset Y$
c) $X = Y$
d) $X - Y \neq \emptyset$ and $Y - X \neq \emptyset$

GATE IT - 2006 MCQ (2 -MARKS)

Q51. What is the cardinality of the set of integers X defined below?
$X = \{n \mid 1 \leq n \leq 123$, n is not divisible by either 2, 3 or 5$\}$

a) 28
b) 33
c) 37
d) 44

GATE IT - 2006 MCQ (2 -MARKS)

Q52. Let P, Q and R be sets let Δ denote the symmetric difference operator defined as $P\Delta Q = (PUQ)-(P\cap Q)$. Using Venn diagrams, determine which of the following is/are TRUE?
I. $P\Delta(Q\cap R) = (P\Delta Q) \cap (P\Delta R)$
II. $P\cap (Q\cap R) = (P\cap Q) \Delta (P\Delta R)$

a) I only
b) II only
c) Neither I nor II
d) Both I and II

GATE IT - 2006 MCQ (1 -MARKS)

Q53. For the set N of natural numbers and a binary operation $f : N \times N \rightarrow N$, an element $z \in N$ is called an identity for f, if $f(a, z) = a = f(z, a)$, for all $a \in N$. Which of the following binary operations have an identity?
I. $f(x, y) = x + y - 3$
II. $f(x, y) = \max(x, y)$
III. $f(x, y) = x^y$

a) I and II only
b) II and III only
c) I and III only
d) None of these

GATE CSE – 2005 MCQ (1-MARKS)

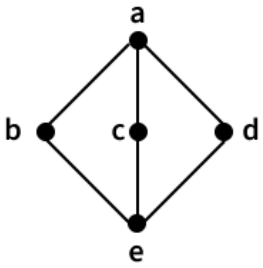
Q54. Let A, B and C be non-empty sets and let
$X = (A - B) - C$ and $Y = (A - C) - (B - C)$.
Which one of the following is TRUE?

a) X = Y
b) $X \subset Y$
c) $Y \subset X$
d) None of these

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2005 MCQ (1-MARKS)

Q55. The following is the Hasse diagram of the poset [{a,b,c,d,e}, $\prec$]



The poset is :

a) not a lattice
b) a lattice but not a distributive lattice
c) a distributive lattice but not a Boolean algebra
d) a Boolean algebra

GATE CSE – 2005 MCQ (2-MARKS)

Q56. Let R and S be any two equivalence relations on a non-empty set A. Which one of the following statements is TRUE?

a) R ∪ S, R ∩ S are both equivalence relations
b) R ∪ S is an equivalence relation
c) R ∩ S is an equivalence relation
d) Neither R ∪ S nor R ∩ S are equivalence relations

GATE IT – 2005 MCQ (2-MARKS)

Q57. Let $n = p^2q$, where p and q are distinct prime numbers. How many numbers m satisfy $1 \leq m \leq n$ and gcd (m, n) = 1? Note that gcd (m, n) is the greatest common divisor of m and n.

a) p(q-1)
b) pq
c) $(p^2 - 1)(q-1)$
d) p(p-1)(q-1)

GATE IT – 2005 MCQ (2-MARKS)

Q58. Let A be a set with n elements. Let C be a collection of distinct subsets of A such that for any two subsets S1 and S2 in C, either $S1 \subset S2$ or $S2 \subset S1$. What is the maximum cardinality of C?

a) n
b) n+1
c) $2^{n-1} + 1$
d) n!

GATE IT – 2005 MCQ (2-MARKS)

Q59. Let f be a function from a set A to a set B, g a function from B to C, and h a function from A to C, such that h(a) = g(f(a)) for all a ∈ A. Which of the following statements is always true for all such functions *f* and g?

a) g is onto ⟹ h is onto
b) h is onto ⟹ f is onto
c) h is onto ⟹ g is onto
d) h is onto ⟹ *f* and g are onto

ANSWER KEY

1. (a,b,c)	**2. (b.c)**	**3.(b)**	**4. (d)**
5. (2)	**6. (b, d)**	**7. (4)**	**8. (b,c,d)**
9. (a,b,c)	**10. (59049)**	**11.(a,b)**	**12. (b)**
13.(b)	**14. (0.125)**	**15. (7)**	**16. (d)**
17. (d)	**18. (42)**	**19. (0)**	**20. (271)**
21. (2)	**22. (b)**	**23. (b)**	**24. (36)**
25. (2048)	**26. (a)**	**27. (d)**	**28. (c)**
29. (d)	**30. (d)**	**31. (c)**	**32. (16)**
33. (4)	**34. (5)**	**35. (a)**	**36. (b)**
37. (d)	**38. (a)**	**39. (c)**	**40. (a)**
41. (c)	**42. (d)**	**43. (a)**	**44. (d)**
45. (c)	**46.(b)**	**47.(d)**	**48. (a)**
49. (a)	**50. (c)**	**51. (b)**	**52. (c)**
53. (a)	**54. (a)**	**55. (b)**	**56. (c)**
57. (d)	**58. (b)**	**59. (c)**	

Check Out Our GATE Courses & Secure Your Dream Rank!

HINTS&SOLUTIONS

1. ANSWER:- A, B, C

Given : $(x, y) \in R$ iff xy is a perfect square, where the set is $\{1, 2, \dots, 10\}$

Reflexive

For every x, $x.x = x^2$ which is always a perfect square.
So, Reflexive.

Symmetric

If xy is a perfect square,then xy = yx which is always a perfect square.
So, Symmetric.

Transitive

If xy and yz are perfect squares, then in prime factorization form, the exponents of primes in both products are even.
This implies the parity of exponents in x and z matches, hence xz is also a perfect square.
So, Transitive.

Antisymmetric

Example:
2·8=16, which is a perfect square.
So (2,8)∈R and (8,2)∈R, but 2≠8.
So, Not antisymmetric.
So, the final Answers are a, b, c.

2. ANSWER:- B, C

(A) $f \preceq g$, and $f \neq g$, $g \preceq f$ does not hold because $f(x) \leq g(x) \not\Rightarrow g(x) \leq f(x)$ So $\preceq$ is not symmetric relation.

(B) Reflexive: $f(x) \leq f(x)$ for all x, so $f \preceq f$.
Antisymmetric: If $f \preceq g$ and $g \preceq f$, then $f(x) = g(x)$ for all x, so $f = g$.
$\forall$ a, b$(a \leq b \wedge b \leq a \rightarrow a = b)$
$\forall$ a, b $(a \leq b \wedge a \neq b \rightarrow b \not\succ a)$
Transitive: If $f \preceq g$ and $g \preceq h$, then $f(x) \leq g(x)$, $g(x) \leq h(x) \Rightarrow f(x) \leq h(x)$, so $f \preceq h$.
RAT property satisfy, so partial order / Correct

(C) $(f \vee g)(x) = \max(f(x), g(x)) \rightarrow$ LUB
$(f \wedge g)(x) = \min(f(x), g(x)) \rightarrow$ GLB
Meet exists for every pair: $f \wedge g = \min(f(x), g(x))$
Join exists for every pair: $f \vee g = \max(f(x), g(x))$
So Lattice , Hence Option-C is correct.

(D) Not symmetric, option A ⇒ So not equivalence relation.
So, B & C are correct.

3. ANSWER:- B

Closure: The sum of two functions from F is still in F. True
Associativity: Addition is associative, so ⊙ is associative. True
Identity: e=0 acts as an identity element.
$(f \odot e)(n)=f(n)+0=f(n)$. True
A valid inverse does not exist for nonzero functions.
Since, A contains only non-negative numbers
Hence, it is an abelian monoid.
option – b

4. ANSWER:- D

g: A→B
f: B → C are two functions
Hence fog: A→C is a composite function and also given that fog: A→C is an onto function
Hence f: B→C is necessary to be an onto function, where as g: A→B need not be onto (or) one-one.

5. ANSWER:- 2

1-1 and onto function exists from A to B
$|A| = |B| = n$
1-1 and onto function exists from $A \times A$ to $A \cup B$
$|A \times A| = |A \cup B|$
$|A \times A| = |A| + |B| - |A \cap B|$
$n^2 = n + n - k$
$n^2 - 2n + k = 0$
$n = \frac{2 \pm \sqrt{4-4k}}{2}$
But n must integer
$4 - 4k \geq 0$
$k \leq 1$
When k=0 , $n^2 - 2n = 0 \rightarrow n = 0,2$
When k=1,
$n^2 - 2n + 1 = 0 \rightarrow n = 1$
Hence, n=1,2
$|A| = n$
Hence Number of possible values for A are 1,2

6. ANSWER:- B, D

An operation # is associative if (a#b)#c=a#(b#c).

Option A: Operation 1 follows Associativity?
(a◊b) ◊c = (a+2b) ◊) c = a + 2b + 2c
a ◊(b◊c) a◊ (b + 2c) = a + 2b + 4c
Here, (a ◊ b) ◊ c ≠ a◊ (b◊ c).
So, Option A is rejected.
Option B: Operation 2 follows Associativity?
(a□ b) □ c = (ab) c = abc.
= a □ (b □ c) = a (bc) = abc
Here, (a □ b) □ c = a □ (b□ c).
So, Option B is Accepted.
NOTE : An operator % is distributive over @ if a % (b @ c) = (a % b) @ (a % c).

Check Out Our GATE Courses & Secure Your Dream Rank!

Option C: Operator ◇ is distributive over□?
Is a◊ (b□c) = (a ◊ b) (a◊ c)?
a◊ (b□c) a◊ (bc) = a + 2bc
(a ◊ b) □ (a ◊ c) = (a + 2b) □ (a+2c) = a^2+2ac+2ab+ 4bc.
Here, a ◊ (b□c) ≠ (a ◊ b) □ (a ◊ c).
So, Option C is rejected.
Option D: Operator is distributive over ◊?
Is a □(b◊c) = (a□b) ◊ (a□c)?
a□ (b◊c) = a □ (b+2c) = ab + 2ac
(a□b)◊(a□c) = (ab) ◊ (ac) = ab + 2ac.
Here, a□(b◊c) = (a□b) ◊ (a□c).
So, Option D is Accepted.

7. ANSWER:- 4

To solve this problem using a table, we need to list all elements of $Z_2 \times Z_3 \times Z_4$ and determine which elements are their own inverses. An element (a,b,c) is its own inverse if:
(a,b,c)+(a,b,c)=(0,0,0)(modulo 2,3,4 respectively)
This condition simplifies to:
(2a mod 2, 2bmod 3, 2c mod 4)=(0,0,0)
Let's construct a table for all elements of $Z_2 \times Z_3 \times Z_4$ and check which ones satisfy the condition.

a	b	c	2a mod 2, 2b mod 3, 2c mod 4
0	0	0	(0,0,0)
0	0	1	(0,0,2)
0	0	2	(0,0,0)
0	0	3	(0,0,2)
0	1	0	(0,2,0)
0	1	1	(0,2,2)
0	1	2	(0,2,0)
0	1	3	(0,2,2)
0	2	0	(0,1,0)
0	2	1	(0,1,2)
0	2	2	(0,1,0)
0	2	3	(0,1,2)
1	0	0	(0,0,0)
1	0	1	(0,0,2)
1	0	2	(0,0,0)
1	0	3	(0,0,2)
1	1	0	(0,2,0)
1	1	1	(0,2,2)
1	1	2	(0,2,0)
1	1	3	(0,2,2)
1	2	0	(0,1,0)
1	2	1	(0,1,2)
1	2	2	(0,1,0)
1	2	3	(0,1,2)

To find elements that are their own inverses, we need to look for rows where the result is (0,0,0)(0, 0, 0)(0,0,0).
These elements are:

- (0,0,0)
- (0,0,2)
- (1,0,0)
- (1,0,2)

Thus, the number of elements in $Z_2 \times Z_3 \times Z_4$ that are their own inverses is 4.

8. ANSWER:- B, C, D

Here, equivalence relation ~ on the set *A* is defined as:
$a_1 \sim a_2$ if $f(a_1)=f(a_2)$
where $a_1, a_2 \in A$
So, consider $a_i,b_i,c_i,\cdots \in A$ and the mapping as:
$a_1 \mapsto \alpha$, $a_2 \mapsto \alpha$, $a_3 \mapsto \alpha$, …,$a_m \mapsto \alpha$
Similarly,
$b1 \mapsto \beta, b_2 \mapsto \beta, b_3 \mapsto \beta, \ldots, b_n \mapsto \beta$
$c_1 \mapsto \gamma, c_2 \mapsto \gamma, c_3 \mapsto \gamma, \ldots, c_p \mapsto \gamma$
and so on for the non-empty sets A and B, and it is given that f:A→B is an onto function.
According to the definition of Equivalence relation, Equivalence class is:
$[a_1]=[a_2]=[a_3]=\cdots=[a_m]$
$[b_1]=[b_2]=[b_3]=\cdots=[b_n]$
$[c_1]=[c_2]=[c_3]=\cdots=[c_p]$
and so on.

Check Out Our GATE Courses & Secure Your Dream Rank!

Now, set of equivalence classes under relation ~ is defined as:
$E=\{[a_1],[b_1],[c_1],\dots\}$
Now, given the new mapping $F:\varepsilon \to B$
$F([x])=f(x)$ for all $[x] \in \varepsilon$
It means the mapping would be:
$a_1 \mapsto \alpha$
$b_1 \mapsto \beta$
$c_1 \mapsto \gamma$
and so on.
Since all distinct $a_1,b_1,c_1,\dots$ maps to different elements of set B and so, F is an injective function. Here, we have taken a_1,b_1,c_1 as leaders for their own equivalence classes, you can take $a_2,b_2,c_2,....$ so on.
It is cleared from the mapping of F that all the elements of set B are covered, so F is a surjective function because each element from their own equivalence class maps according to the mapping of f. We have taken one element from the group of a′s and mapped it to a′s and similarly for other groups of b′s, c′s and so on.
Since F is both injective and surjective, hence F is a bijective function.
Here, we say, function F is well-defined or single-valued when:
1. $F \subseteq [x] \times f(x)$
2. The domain of F is [x]
3. If ([x],y),([x],z) then y=z

Since from the mapping of F, it is clear that all the above three conditions are true because we have taken one element from the group of a′s,b′s,c′s, and so on and mapped to α,β,γ,… and so f is a well-defined function.
Therefore, (B), (C), (D)

9. ANSWER:- A,B,C

Option A : If for all $x, y \in G$, $(xy)^2 = x^2y^2$, then G is commutative.
True,
$(xy)^2 = x^2y^2$
xy xy = xx yy
Take x^{-1} and y^{-1} on both sides.
x^{-1} xy xy $y^{-1} = x^{-1}$ xx yy y^{-1}
yx = xy
$\forall x, y \in G$, yx = xy, Here G is commutative.
Option B : If for all $x \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the identity element of G.
True, $\forall x \in G$, $x^2 = 1$
If every element has its own inverse in a graph, then the graph is commutative.
Option C : If the order of G is 2, then G is commutative.
True, Every group of prime order is commutative, so if O(G) = 2, the group 'G' is commutative. If the Group order is a prime number, then it is a cyclic group. And we know that every Cyclic group is an abelian group.
Alternatively, we can think like this: Given that the Order of Group G is 2, one element is the Identity element, so another element x is inverse to itself. By Option 3, G is commutative.
Option D : If G is commutative, then a subgroup of G need not be commutative.
False, If group G is commutative, then a subgroup of G need not be commutative.
Theorem: Every subgroup of an abelian group has to be abelian.
Proof: Let G be an abelian group, and suppose that $H \subseteq G$.
Now we have to check that, for any $a, b \in H$, do we have ab = ba?
Since we have: $a, b \in H \subseteq G$, so a, b are elements of G and since G is abelian, so ab = ba.
Therefore, H is abelian as well.

10. ANSWER:- 59049

A set having n elements will have 2^n subsets. Now, from the given set we can have:
${}^nC_n$ subsets of size n (which will have 2^n subsets)
${}^nC_{n-1}$ subsets of size n – 1 (which will have 2^{n-1} subset which are also subsets of original set)
${}^nC_{n-2}$ subsets of size n – 2 (which will have 2^{n-2} subsets which are also subsets of original set)
${}^nC_1$ subsets of size 1
${}^nC_0$ subsets of size 0.
So, total number of ordered pairs satisfying the subset order will be:
$2^n\,{}^nC_n + 2^{n-1}\,{}^nC_{n-1} + 2^{n-2}\,{}^nC_{n-2} + \dots + 2^0\,{}^nC_0$
$= (x + 1)^n$ where x = 2,
$= 3^n$
$= 3^{10}$
= 59049

11. ANSWER:- A, B

S1: There are n^2 elements in the matrix, we have 3 choices for each elements, so number of such matrices = 3^{n^2}
S2: We are having n^2 total elements with 3 choices for each element, so total number of functions possible = 3^{n^2}
Since, the cardinality of both the sets are the same, we can establish a bijection from one set to another, as bijection is possible,

Check Out Our GATE Courses & Secure Your Dream Rank!

so surjection is also possible.
So, option A and B are correct.

12. ANSWER:- B

It a relation S is reflexive and circular
⇒ Symmetry let (a, b) ∈ S (a, b ∈ A)
Since S is reflexive (b, b) ∈ S
Since S is circular and (a, b) ∈ S and (b, b) ∈ S
(b, a) ∈ S
Thus, S is a symmetric relation (since (b, a) ∈ S whenever (a, b) ∈ S)
⇒ Transitivity Let (a, b) ∈ S and (b, c) ∈S
Since S is circular: (c, a) ∈ S
we have already shown that S is symmetric, thus let use the fact that S is symmetric: (a, c) ∈ S
Thus, S is a transitive relation (since (a, c) ∈ s whenever (a, c) ∈ S and (b, c) ∈ S)
∴ S is an equivalence relation, between S is reflexive, symmetric, and transitive.

13. ANSWER:- B

As order of group is 6 (given) ,so according to Lagrange's: order of subgroup is a divisor of order of group.
i.e., O(G) / O(H) divides O(G)
O(G) = 6 (not a prime no)
So, G may not be a cyclic group.
Now, 1 < |H| < 6 so order of H may be 2,3, 4, 5 but only 2, 3 divides 6 so order of H is either 2 or 3 and both are prime number so H is cyclic group.

14. ANSWER:- 0.125

For a set with n elements,
The number of reflexive relations is $2^{n(n-1)}$
The total number of possible relations on a set with n elements is 2^{n*n}.
Thus, the probability of choosing the reflexive relation out of the set of relations is:
$= 2^{n(n-1)} / 2^{n*n}$
$= 2^{(n*n - n*n - n)} = 2^{(-n)}$
Given n=3, the probability will be
$2^{-3} = 0.125$

15. ANSWER:- 7

Lagrange's Theorem:
If 'H" is a subgroup of finite group (G, *) then O(H) is the divisor of O(G).
Now, we are given that the order of group is 35. It's divisors are 1,5,7,35.
Now, we need to find the size of largest possible subgroup other than G, so it will be 7.

16. ANSWER:- D

Given relations are

R_1: ∀ a, b ∈ G, a R_1 b if and only if ∃ g ∈G such that $a=g^{-1}bg$
R_2: ∀ a, b ∈ G, a R_2 b if and only $a= b^{-1}$
A relation is said to be equivalence relation if it is

I. Reflexive
II. Symmetric
III. Transitive

Relation R_1:

Reflexive property:
For any a ∈ G,
$a=e^{-1}ae$, where 'e' is identity
⇒ (a, a)∈ R_1
So R_1 is reflexive

Symmetric property:
Let (a, b)∈ R_1
⇒∃ g ∈G such that $a=g^{-1}bg$
$\Rightarrow b=(g^{-1})^{-1}a(g^{-1})$
⇒(b, a)∈ R_1
So R_1 is symmetric

Transitive property:
Let (a, b)∈ R_1, (b, c) ∈R_1
⇒∃ g_1∈G, g_2∈G such that $a={g_1}^{-1}bg_1$ and $b={g_2}^{-1}cg_2$
$\Rightarrow a = {g_1}^{-1}{g_2}^{-1}cg_2g_1 = (g_2g_1)^{-1}cg_2g_1$
⇒(a, c)∈ R_1
So R_1 is transitive
Hence R_1 is an equivalence relation.

Relation R_2:
Reflexive property:
For any a ∈ G,
$a=a^{-1}$ is not true always
⇒ (a, a) ∉ R_2 for all a in G
So R_2 is not reflexive and hence R_2 is not an equivalence relation
Therefore only R_1 is an equivalence relation

17. ANSWER:- D

P: Set of Rational numbers are ***countable***.
Rational numbers are of the form $\frac{p}{q}$ =(p,q) where p, q are integers. Proof using cantors diagonal argument.

Check Out Our GATE Courses & Secure Your Dream Rank!

Q: Set of functions from {0,1} to N. There are N^2 such functions. Hence ***countable***.
R: Set of functions from N to {0,1}. There are 2^N such functions.
Theorem: If a set S is countable, then P(S) i.e 2^S is uncountable.
Hence, statement R is ***uncountable***.
S: Set of finite subsets of N. They are ***countable***.
Theorem: Every subset of a countable set is either countable or finite.

18. ANSWER:- 42
Lagrange's Theorem:
Order of subgroup is a divisor of order of group.
i.e., O(H)/O(G)
Given, order n = 84.
Then the order of subgroups = {1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84}
As per the proper subgroup definition, the answer should be 42.

19. ANSWER:- 0
Given: R = {(a,a), (a,b), (a,c), (a,d),(a,e), (b,b), (b,c), (b,e), (c,c), (c,e), (d,d),(d,e), (e,e)}
As per the definition of lattice, each pair should have GLB, LUB.
The given partial ordering R has GLB, LUB for each and every pair.
So,we don't need to add any extra pair.
Thus, the number of required pairs such that the Hasse diagram becomes lattice is 0.

20. ANSWER:- 271
Let we have:
A → the number of integers between 1 and 500 which are divisible by 3.
B → the number of integers between 1 and 500 which are divisible by 5.
C → the number of integers between 1 and 500 which are divisible by 7.
Thus, the required answer is :
n (A ∪ B ∪ C) = n(A) + n(B) + n(C) – n (A∩ B) – n (A ∩ C) – n(B ∩ C) + n (A ∩ B ∩ C)
= ⌊500/3 ⌋ + ⌊500/5 ⌋ + ⌊500/7 ⌋ - ⌊500/15 ⌋ - ⌊500/21⌋ - ⌊500/35 ⌋ + ⌊500/105 ⌋
= 166 + 100 + 71 – 33 – 23 – 14 + 4
= 271

21. ANSWER:- C
Given:
f(n)= f($\frac{n}{2}$) , if n is even
f(n)= f(n+5) , if n is odd

N	1	2	3	4	5
f(n))	f(9))	f(1))	f(8))	f(2))	f(10))

N	6	7	8	9	10
f(n))	f(3))	f(12))	f(4))	f(1)4)	f(5))

We can observe that:
f(8) = f(4) = f(2) = f(1)
f(1) = f(0) = f(3)
and f(5) = f(10) = f(15) = f(20)
We observe that f(11) = f(8).
f(12) = f(6) = f(3)
f(13) = f(9) = f(14) = f(7) = f(12) = f(6) = f(3)
f(14) = f(9) = f(12) = f(6) = f(3)
f(16) = f(8) = f(4) = f(2) = f(1)
[repeating]
So, we can conclude that 'R' can have size only of two. [one: multiple of 5's, other: other than 5's multiple]

22. ANSWER:- B
The sum of the degrees of all the vertices in a graph is equal to twice the number of edges.
Here, Set S has 9 elements and each element has a degree of 3 as each element in S reacts to exactly 3 elements. So 9*3 + Sum. of degrees of vertices in U-S = 2*e
Since 2*e is even no, Sum of degrees of vertices in U-S is odd .
Thus, some vertex in U-S must have odd degree. At least one compound in U-S reacts with an odd number of compounds
Option B is the correct answer.

23. ANSWER:- B
For every a,b∈ N,
a≤c ∨ b≤d
Let a≤a ∨ b≤b is true for all a,b∈ N.
So, there exists (a,b)R(a,b) for (a,b)
Thus, it is a Reflexive relation.
Consider an example:
c = (a,b)R(c,d) and (c,d)R(e,f) then (a,b)R(e,f)
This does not hold for every (a > e) or (b > f)
e.g.:

Check Out Our GATE Courses & Secure Your Dream Rank!

(2,2)R(1,2) as 2≤2
(1,2)R(1,1) as 1≤1
but (2,2) R (1,1) is False
So, not transitive.

24. ANSWER:- 36
Number of onto functions from set X to set Y with |X| = m, |Y| = n is
$\sum_{k=0}^{n-1}(-1)^k . n_{C_k} . (n-k)^m$
m = 4. n = 3 ⇒ number of onto function is
$\sum_{k=0}^{2}(-1)^k . 3_{C_k} . (3-k)^4$
$= 3^4 - (2)^4 + (3)(1)$
= 81 – 48 + 3
= 36

25. ANSWER:- 2048
Let S be the given set {0, 1, 2, …, 10}, then cardinality of S, |S| = 11.
So, cardinality of power set |P(S)| = 2^{11} i.e., 2048.

26. ANSWER:- A
It can be clearly seen from the truth table, that the binary operation ≠ is equivalent to XOR i.e., ⊕, which satisfies both commutative and associative i.e., p≠q, q≠p and p≠(q≠r), (p≠q)≠r.

27. ANSWER:- D
Total number of ordered triplets (x,y,z) in L^3 are 5×5×5 i.e., 125. So,n(S) = 125.
Let F be the event that an element (x,y,z)∈ L^3 satisfies x ∨ (y∧z) = (x ∨ y) ∧ (x ∨ z).Since q ∨ (r ∧ s) = q ∨ p = q.
And (q∨r) ∧ (q∨s) = t ∧ t=t,
q ∨ (r∧s) ≠ (q∨r)∧(q∨s)
Therefore,
(x,y,z) = (q,r,s), (q,s,r), (r,q,s), (r,s,q), (s,r,q), (s,q,r)
i.e., 3! = 6 elements will not satisfy distributive law and all other (x,y,z) choices satisfy distributive law.
n(F) = 125-6 = 119
∴ required probability (p_r) = n(F)/n(S) = 119/125
⇒ i.e. 1/5< p_r <1

28. ANSWER:- C
Since p-q ≠ q-p
∴(p,q) R (p,q) is not true.
⇒ R is not reflexive.
Let (p,q) R (r,s) then p-s = q-r
⇒ r-q = s-p
⇒ (r,s) R (p,q)
⇒ R is symmetric.
So, R is not Reflexive but Symmetric

29. ANSWER:- D
For Reflexive:
In a given relation, aRb iff 'a' and 'b' are distinct. So, it can never be reflexive.
For Symmetric:
In a given relation, aRb, if 'a' and 'b' have a common divisor other than 1, then bRa, i.e., 'b' and 'a' also will have a common divisor other than 1. So, it is symmetric.
So, R is symmetric but not reflexive and not transitive
For Transitive:
Take (3, 6) and (6, 2) elements of R. For transitivity (3, 2) must be the element of R, but 3 and 2 don't have a common divisor. So, not transitive.

30. ANSWER:- D
As X and Y are elements of U, so X and Y are subsets of S.
A. False. Consider X = {1,2}. Therefore, X' = {3,4,5,6}, |X| = 2 and |X'| = 4.
B. False. Because for any two possible subsets of S with 5 elements, we should have at least 4 elements in common.
Hence X∩Y cannot be null.
C. False. Consider X = {1,4}, Y= {1,2,3} then X\Y = {4} which is not null.
D. True. Take any possible cases.

31. ANSWER:- C
A = {5, {6}, {7}}
I. ∅ ∈ 2^A is true. Any power set contains an empty set.
II. ∅ ⊆ 2^A is true. Empty set is subset of any set.
III. {5, {6}} ∈ 2^A is true. power set of A contain {5, {6}} as a 2-element subset of A.
IV. {5, {6}} ⊆ 2^A is false. Power set of A not contain 5 and {6} as elements.
So, {5, {6}} cannot be subset of 2^A .

32. ANSWER:- 16
The number of functions from A to B where size of A=|A| and size of B=|B| is $|B|^{|A|}$
Now, $\{0,1\}^4$ = {0,1}×{0,1}×{0,1}×{0,1} =16
|S|=2^{16}
N=$2^{|S|}$
log(logN)=log(log$2^{|S|}$) =log|S| =log2^{16} =16

33. ANSWER:- 4
Given that
$x * x = y * y = x * y * x * y = y * x * y * x = e$
$x * x = e \Rightarrow x = x^{-1}$

Check Out Our GATE Courses & Secure Your Dream Rank!

$y * y = e \Rightarrow y = y^{-1}$
$(x * y) * (x * y) = e \Rightarrow x * y = (x * y)^{-1}$
$(y * x) * (y * x) = e \Rightarrow y * x = (y * x)^{-1}$
Consider

$$(x * y) * (y * x) = x * (y * y) * x$$
$$= x * e * x$$
$$= x * x = e$$

So $(x * y)^{-1} = (y * x)$ but $(x * y)^{-1} = x * y$
Hence $x * y = y * x$
So given group is commutative group
Also given that every element in the group can be written as a product of some number of x' s and y's in some order. Any such combination is equal to one of the elements e, x, y, xy
[For example:

$$xxx = xe = x$$
$$xyy = xe = x$$
$$xxy = ey = y$$
$$xyx = xxy = y$$
$$yx = xy$$

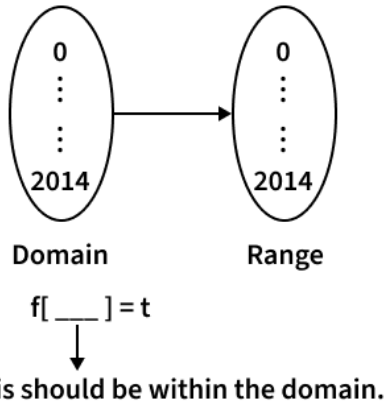
34. ANSWER:- 5
According to Lagrange's theorem, in the mathematics of group theory, for any finite group G, the order (number of elements) of every subgroup H of G divides the order of group G.
So, 15 is divided by {1, 3, 5, 15}.
As minimum order is 4 and not 15, so we eliminate 1,3,15.
So, the answer is 5.

35. ANSWER:- A

1. $A \Delta U = A^c$, Hence the minimum element in the symmetric difference of the two sets is in U. U is smallest then every set.

2. Since $A \Delta \phi = A$, Hence the minimum element in the symmetric difference of the two sets is in A. A is smaller than ϕ
Both statements are correct

36 ANSWER:- B
f: {0,1,…,2014}→{0,1,…,2014} and f(f(i)) = i

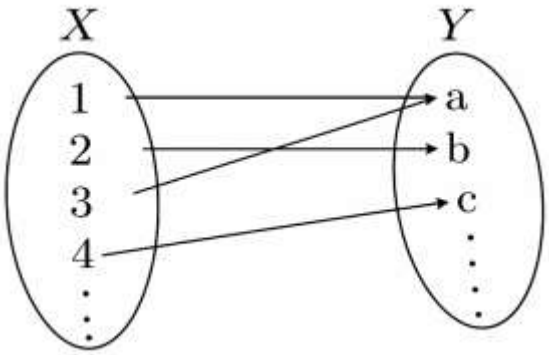


So, f(i) should result only in {0, 1,…2014}.
So, every element in the range has a result value to the domain. This is onto. (Option R is correct)
We have '2015' elements in the domain. So at least one element can have f(i) = i, So, option 'Q' is also True.
∴ Q and R are correct.

37. ANSWER:- D
Given X and Y are finite sets and $f: X \to Y$ is a function.

Let us verify the options

Options (A), (B), (C) are false. Let us use the following example



Let $A = \{1, 2\}, B = \{3, 4\}$

Option (A): |f (A∪ B)| = |f(A)| + |f(B)|
A∪ B={1, 2,3,4} ⇒f (A∪ B) = {a, b, c}
⇒|f (A∪ B)|=3
f (A)={a, b} ⇒|f(A)|=2
f(B)={a, c}⇒|f(B)|=2
⇒|f(A)| + |f(B)|=2+2=4
Therefore |f (A∪ B)| ≠ |f(A)| + |f(B)|
So option (A) is false
Option (B): f (A ∩ B) = f (A) ∩ f(B)

$A \cap B = \phi \Rightarrow f(A \cap B) = \phi$
$f(A)=\{a, b\}$ and $f(B)=\{a, c\}$
$f(A) \cap f(B)=\{a\}$
Therefore $f(A \cap B) \neq f(A) \cap f(B)$
So option (B) is false

Option (C):

$|f(A \cap B)| = \min\{|f(A)|, |f(B)|\}$
$A \cap B = \phi \Rightarrow f(A \cap B) = \phi$
$\Rightarrow |f(A \cap B)|=0$
$f(A)=\{a, b\} \Rightarrow |f(A)|=2$
$f(B)=\{a, c\} \Rightarrow |f(B)|=2$
$\min\{|f(A)|, |f(B)|\}=\min\{2, 2\}=2$
Therefore
$|f(A \cap B)| \neq \min\{|f(A)|, |f(B)|\}$
So option (C) is false

Option (D):

$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$
$x \in f^{-1}(S \cap T)$
$\Leftrightarrow f(x) \in S \cap T$
$\Leftrightarrow f(x) \in S$ and $f(x) \in T$
$\Leftrightarrow x \in f^{-1}(S)$ and $x \in f^{-1}(T)$
$\Leftrightarrow x \in f^{-1}(S) \cap f^{-1}(T)$
Therefore
$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$
Or Option D
$f(A \cap B) = f(A) \cap f(B)$ iff f is 1-1,
since $f^{-1} is\ 1-1\ and\ onto$
$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

38. ANSWER:- A

Commutative property:
A binary operation on a set S is called commutative if $a*b = b*a\ \forall x, y \in S$.
Associative property:
A binary operation on set is called associative if $(a*b)*c = a*(b*c)\ \forall x,y \in S$.
Given $x \oplus y = x^2 + y^2$ --------(1)
Replace x, y in (1):
$y \oplus x = y^2 + x^2$ which is same as (1), so this is commutative.
$(x \oplus y) \oplus z = (x^2 + y^2) \oplus z$
$= (x^2 + y^2)^2 + z^2$
$= x^2 + y^2 + z^2 + 2x^2y^2$ ----------(2)
$x \oplus (y \oplus z) = x \oplus (y^2 + z^2)$
$= x^2 + (y^2 + z^2)^2$
$= x^2 + y^2 + z^2 + 2y^2z^2$ ---------- (3)
(2) & (3) are not same, so this is not associative.

39. ANSWER:- C

Onto function is possible if $m \geq n$. So, number of onto functions possible is:
$n^m - {}^nC_1 (n-1)^m + {}^nC_2 (n-2)^m + ...$
According to the Question:
$m = n, n = 2$
So, number of onto functions:
$= 2^n - {}^2C_1 (2-1)^n + {}^2C_2 (2-2)^n$
$= 2^n - 2 \times 1 + 0$
$= 2^n - 2$

40. ANSWER:- A

A group is a set of elements together with an operation that combines any two of its elements to form a third element also in the set while satisfying four conditions called the group axioms, namely closure, associativity, identity and invertibility.
(S, *) is a group with identity as 1

41. ANSWER:- C

Total number of reflexive relations = $2^{n(n-1)}$
For the given question, n = 5.
The number of reflexive relations $=2^{(25-5)} =2^{20}$

42. ANSWER:- D

A binary relation R over a set X is symmetric if it holds for all a and b in X that if a is related to a.
$\forall a,b \in X, aRb \Rightarrow bRa$.
Here (x, y) is there in R but (y, x) is not there.
Not Symmetric.
For Antisymmetric Relations: $\forall a,b \in X$, R(a, b) & R(b, a) $\Rightarrow a = b$.
Here (x, z) is there in R also (z, x) is there violating the antisymmetric rule.
Not AntiSymmetric.

43. ANSWER:- A

In a lattice L, every pair of elements has both LUB and GLB

Let us verify the given Hasse diagrams

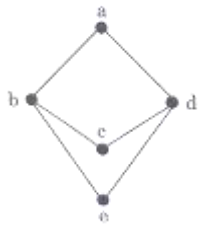
Hasse diagram1:

Every pair has LUB and GLB

So it is a lattice

Check Out Our GATE Courses & Secure Your Dream Rank!

Hasse diagram2:

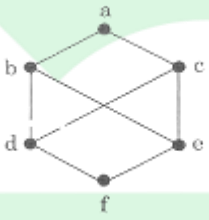


Lower bounds of {b, d} are 'c', 'e' and these two are not comparable

So GLB({b, d}) not exists

So it is not a lattice

Hasse diagram3:



Lower bounds of {b, c} are 'd', 'e' and 'f'. But there is greatest element among lower bounds 'd', 'e' and 'f'

So GLB({b, c}) not exists

So it is not a lattice

Hasse diagram4:

Every pair has LUB and GLB

So it is a lattice

Therefore only Hasse diagrams 1 and 4 are lattices

44. ANSWER:- D

A field is an algebraic structure over two operations + and * if:

1. It is closed under both + and *.
2. + and * are both commutative and associative. (+ and in this question are already commutative and associative, so no need to check)
3. Existence of additive identity(0) and multiplicative identity (1)
4. Existence of additive and multiplicative inverses for each non-zero element.
5. Distributive property of * over + (This is also satisfied in question)

So, for each option, we have to check only properties 1, 3 and 4.

(S1): set of all real numbers

1. Closed: Yes, real + real = real, real * real = real
3. Additive and multiplicative identity: Yes, 0 and 1 are real numbers
4. Additive and multiplicative inverse for each non-zero element. Yes, for any real number a, additive inverse is -a, which is also a real number, and multiplicative inverse is 1/a, which is also a real number. (S2): {(a+ib) | a and b are rational numbers}

1. Closed: Yes, rational + rational = rational, rational * rational = rational
2. Additive and multiplicative identity: Yes, 0+0i (additive identity) and 1+0i (multiplicative identity) belong to given set.
3. Additive and multiplicative inverse for each non-zero element: Additive inverse is -a - ib, which belongs to given set. Multiplicative identity is $1/(a+ib) = (a - ib) / (a^2 + b^2) = a/(a^2+b^2) + i(-b)/(a^2+b^2)$, which also belongs to given set.

(S3): $\{a+ib \mid (a^2 + b^2) \leq 1\}$

1. Closed: No, for example: (0.3 +0.4i) + (0.7+0.6%)=1+i. Here, both complex numbers which were added were in the given set, but resultant complex number is not. (S4): {ia | a is real}

Here, this set does not contain 1 (multiplicative identity)

So, (S1) and (S2) are subfields of C. Option D is correct.

45. ANSWER:- C

300! is 1*2*3*...*300.

Now there are 27 multiples of 11 from 1 to 300, so they will include 11 as a prime factor atleast once. 121 and 242 will contain an extra 11, all other will contain 11 as a factor only once.

So, total number of 11's = 27+2 = 29.

So, exponent of 11 is 29 i.e. option C.

Check Out Our GATE Courses & Secure Your Dream Rank!

46. ANSWER:- B

Equivalence relation means it is reflexive, symmetric and transitive
If a relation is reflexive then it must have all the pairs of diagonal elements and relation with only diagonal elements is also symmetric and transitive. Therefore the smallest such relation is of size n.
With diagonal elements, we can include all the elements as well. Therefore the largest equivalence relation is of size n^2.
So, Correct Option is b

47. ANSWER:- D

For ordered pair (a, b), to be in P, each digit in a starting from unit place must not be larger than the corresponding digit in b.
This condition is satisfied by options
(iii) (145, 265) → 5<=5, 4 < 6 and 1 < 2.
(iv) (0, 153) → 0 < 3 and no need to examine further.

48. ANSWER:- A

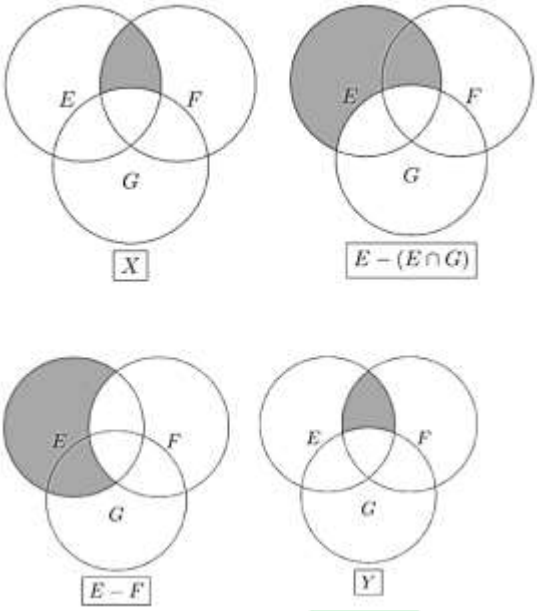
The number of Abelian groups of order P^k, where P is prime, is the number of partitions of k.
Here, order is $4 = 2^2$
{1,1},
{2}
Partition of 2 are 2.

Total 2 partition so number of different abelian groups are 2 .

49. ANSWER:- A

For reflexivity, (x,y) R (x,y), which here requires x<x and y<y and this is not possible in this relation.
This relations is Not Symmetric as
(x,y)R(u,v) iff x<u and y>v
(u,v)R(x,y) iff u<x and v>y in completely opposite conditions!!
Since, reflexive and symmetric pairs are not allowed, the matrix of this relation would contain only either lower triangular or upper triangular elements as 1.
So this relation is Anti-Symmetric.
This relation is Irreflexive as not even a single case of reflexivity is possible.
Since the relation is Irreflexive and Anti-Symmetric, it is Asymmetric.
This relation is also transitive as
(x,y)R(u,v) iff x<u and y>v
(u,v)R(a,b) iff u<a and v>b
both imply
x<u<a and y>v>b
implies
x<a and y>b which makes (x,y)R(a,b)
Hence, transitive.
So, Neither a Partial Order nor an Equivalence Relation

50. ANSWER:- 2



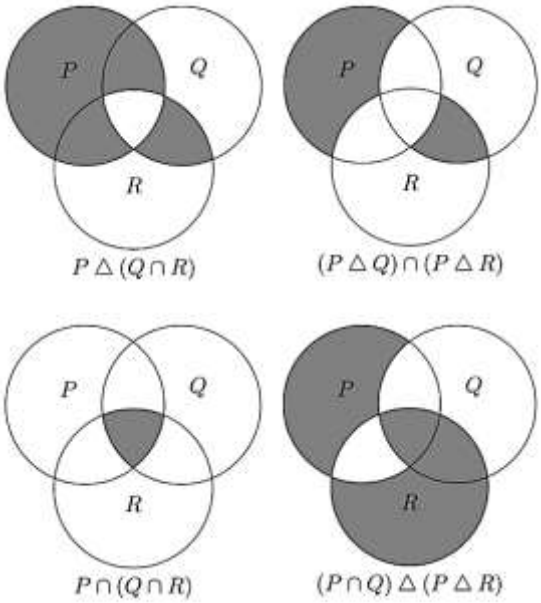
X = Y
Option C is the answer.

51. ANSWER:- B

Number's divisible by 2 in X = 61
[= integer(123/2)]
Number's divisible by 3 in X = 41
Number's divisible by 5 in X = 24
Number's divisible by 2 and 3 i.e. by 6 = 20
Number's divisible by 2 and 5 i.e by 10 = 12
Number's divisible by 3 and 5 i.e by 15 = 8
Number's divisible by 2 and 3 and 5 i.e by 30 = 4
Number's divisible by either 2 or 3 or 5 = N(AUBUC) =
N(A) + N(B) + N(C) – N(A∩B) – N(B∩C) – N(A∩C) + N(A∩B∩C)
= 61 +41 + 24-20-12-8+ 4 = 90
X={n, 1 ≤ n ≤ 123, n is not divisible by either 2, 3 or 5 }
Cardinality = 123 - 90=33
So, the correct Answer is B.

Check Out Our GATE Courses & Secure Your Dream Rank!

52. ANSWER:- C



So the correct answer is C.

53. ANSWER:- A

An Element z ∈ N is called an Identity element for f if f(a, z) = f(z, a) = a, ∀a ∈ N

I f(x,y) = x+y-3

We will say that z is an identity element if f(x, z) = x = f(z, x), Hence, x+z-3=x=z+x-3

So, We have z = 3.

II: f(x,y) = max(x, y)

We will say that z is an identity element if f(x, z) = x = f(z, x), hence, max(x, z) = x = max(z, x), ∀x ∈N.

So, We have z = 0 (if N consists of 0)

Or We have z = 1 (if N does not consists of 0

NOTE: Whether 0 belongs to Set of natural numbers or not, is an ambiguous thing. In Number theory, generally we don't consider 0 as a Natural number. But in Computer Science, 0 is considered as a Natural number by many authors.

In this question, It doesn't matter whether or not 0 belong to N or not. We can say that the Set N will have an Identity element in both the cases. (Also See below NOTE 3)

III: $f(x, y) = x^y$

We will say that z is an identity element if $f(x, z) = x = f(z, x)$, hence $x^z = x = z^x$, $\forall x \in \mathbb{N}$.

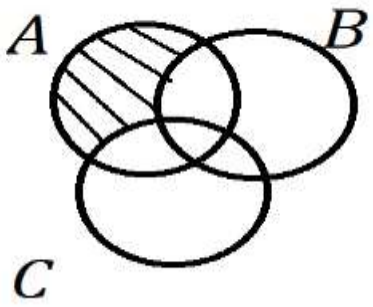
- For the first part, $x^z = x$ gives $z = 1$.
- For the second part, substituting $z = 1$ gives $1^x = x$. However, $1^x = 1$, which is not equal to x for all $x > 1$.

Since no single value of z satisfies both equations simultaneously, there is no identity element for this binary operation.

54. ANSWER:- A

Method 1:

If we draw venn diagram, both X and Y is equal to the shaded region as follows



Method 2:

$$X = (A - B) - C$$
$$= (A \cap B') - C$$
$$= (A \cap B') \cap C'$$
$$Y = (A - C) - (B - C)$$
$$= (A \cap C') - (B \cap C')$$
$$= (A \cap C') \cap (B \cap C')'$$
$$= (A \cap C') \cap (B' + C)$$
$$= (A \cap C') \cap B' + (A \cap C') \cap C$$
$$= A \cap (C' \cap B') + A \cap (C' \cap C)$$
$$= A \cap (B' \cap C') + A \cap \phi$$
$$= (A \cap B') \cap C' + \phi$$
$$= (A \cap B') \cap C'$$

Hence $X = Y$

55. ANSWER:- B

A lattice has a least upper bound (lub) and a greatest upper bound (glb), but to be distributive every element of the lattice should have at most one complement. Here, elements b,c,d are complements of each other and hence the given lattice is not distributive.

56. ANSWER:- C

If R and S are equivalence relation then R ∩ S is always an equivalence relation.

But R ∪ S may or may not be an equivalence relation because it might not be transitive always.

So, the correct answer is (C) R ∩ S is an equivalence relation.

Check Out Our GATE Courses & Secure Your Dream Rank!

57. ANSWER:- D

$n = p^2q$, where p and q are prime.
So, number of multiples of p in n = pq
Number of multiples of q in $n = p^2$
Number of multiples of pq in n = p
Since, prime factorization of n consists of only p and q, gcd(m, n) will be a multiple of these or 1.
So, number of possible m such that gcd(m, n) is 1 will be
n-number of multiples of either p or q
$= n-p^2 - pq+p$
$= p^2q-p^2 - pq+p$
$= p(pq-p-q+1)$
$= p(p-1)(q - 1)$
So, the correct answer is d.

58. ANSWER:- B

Let's take an example set {a, b, c}.
Now let's try to create the required set of subsets, say S.
Let's start by adding sets of size 1 to S. We can only add one of the sets {a}, {b}, {c}
Lets say we add {a}, so S now becomes {{a}}.
Now lets add sets of size 2 to S. Again we see that we can only add one of {a, b}, {a, c} or {b, c}, and we cannot add {b, c} since we have already added {a}.
Continuing this way we see we can add only one set for a all size till n.
So, the answer should be (B) n + 1 (include the empty set).

59. ANSWER:- C

Let h be onto (onto means co-domain = range).
So, h maps to every element in C from A. Since h(a) = g(f(a)), g should also map to all elements in C. So, g is also onto.

Check Out Our GATE Courses & Secure Your Dream Rank!

3 MATHEMATICAL LOGICS

GATE CSE – 2026 MCQ SET – 2 (1-MARKS)

Q1. For two different persons x and y, the predicate $M(x, y)$ denotes that x knows y.
Consider the following statement.
There is a person who does not know anyone else, but that person is known by everyone else.
Which one of the following expressions represents the above statement?

a) $(\exists y)(\forall x)\ ((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
b) $(\forall y)(\exists x)\ ((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
c) $(\exists y)(\exists x)\ ((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$
d) $(\forall y)(\forall x)\ ((x \neq y) \rightarrow (M(x, y) \wedge \neg M(y, x)))$

GATE CSE – 2025 MCQ SET – 2 (1-MARKS)

Q2. Let P(x) be an arbitrary predicate over the domain of natural numbers.
Which ONE of the following statements is TRUE?

a) $(P(0) \wedge \forall x[P(x) \Rightarrow P(x+1)]) \Rightarrow (\forall x P(x))$
b) $(P(0) \wedge \forall x[P(x) \Rightarrow P(x-1)]) \Rightarrow (\forall x P(x))$
c) $(P(1000) \wedge \forall x[P(x) \Rightarrow P(x-1)]) \Rightarrow (\forall x P(x))$
d) $(P(1000) \wedge \forall x[P(x) \Rightarrow P(x+1)]) \Rightarrow (\forall x P(x))$

GATE CSE – 2025 MSQ SET – 1 (2-MARKS)

Q3. Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement:
"Everyone has exactly one mother"?
The meanings of the predicates used are:
- mother(y,x): y is the mother of x
- noteq(x,y): x and y are not equal

a) $\forall x \exists y \exists z(\text{mother}(y,x) \wedge \neg \text{mother}(z,x))$
b) $\forall x \exists y[\text{mother}(y,x) \wedge \forall z (\text{noteq}(z,y) \rightarrow \neg \text{mother}(z,x))]$
c) $\forall x \forall y[\text{mother}(y,x) \rightarrow \exists z(\text{mother}(z,x) \wedge \neg \text{noteq}(z,y))]$
d) $\forall x \exists y[\text{mother}(y,x) \wedge \neg \exists z(\text{noteq}(z,y) \wedge \text{mother}(z,x))]$

GATE CSE – 2024 MCQ SET – 2 (1-MARKS)

Q4. Let p and q be the following propositions:
- p: Fail grade can be given.
- q: Student scores more than 50% marks.

Consider the statement: "Fail grade cannot be given when student scores more than 50% marks."
Which one of the following is the CORRECT representation of the above statement in propositional logic?

a) $q \rightarrow \neg p$
b) $q \rightarrow p$
c) $p \rightarrow q$
d) $\neg p \rightarrow q$

GATE CSE – 2023 MSQ (1-MARKS)

Q5. Geetha has a conjecture about integers, which is of the form
$\forall x(P(x) \Rightarrow \exists y Q(x, y))$,
where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would imply Geetha's conjecture?

a) $\exists x(P(x) \wedge \forall y Q(x, y))$
b) $\forall x \forall y Q(x, y)$
c) $\exists y \forall x(P(x) \Rightarrow Q(x, y))$
d) $\exists x(P(x) \wedge \exists y Q(x, y))$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2021 MSQ SET – 2 (1-MARKS)

Q6. Choose the correct choice(s) regarding the following propositional logic assertion
S: ((P ∧ Q) → R) → ((P ∧ Q) → (Q → R))
Which of the following is true?

a) S is neither a tautology nor a contradiction.
b) S is a tautology.
c) S is a contradiction.
d) The antecedent of S is logically equivalent to the consequent of S.

GATE CSE – 2021 MCQ SET – 1 (1-MARKS)

Q7. Let p and q be two propositions. Consider the following two formulae in propositional logic.
S_1:~p ∧ (p ∨ q)→q
S_2: q → (~p ∧ (p ∨ q))
Which one of the following choices is correct?

a) Neither S_1 nor S_2 is a tautology.
b) S_1 is not a tautology but S_2 is a tautology.
c) S_1 is a tautology but S_2 is not a tautology.
d) Both S_1 and S_2 are tautologies.

GATE CSE – 2020 MCQ (2-MARKS)

Q8. Which one of the following predicate formulae is NOT logically valid?
Note that W is a predicate formula without any free occurrence of x.

a) ∃x(p(x) → W) ≡ ∀xp(x) → W
b) ∀x(p(x) → W) ≡ ∀x p(x) → W
c) ∃x(P(x) ∧ W) ≡ ∃x P(x) ∧ W
d) ∀x(p(x) ∨ W) ≡ ∀ xp(x) ∨ W

GATE CSE – 2019 MCQ (2-MARKS)

Q9. Consider the first order predicate formula φ: ∀x[(∀z z|x⇒ ((z = x) ∨ (z = 1))) ⇒∃w (w >x) ∧ (∀z z|w⇒ ((w = z) ∨ (z = 1)))]
Here 'a|b' denotes that 'a divides b' where a and b are integers. Consider the following sets:
S1. {1, 2, 3, ..., 100}
S2. Set of all positive integers
S3. Set of all integers

Which of the above sets satisfy φ?

a) S1 and S3
b) S1, S2 and S3
c) S2 and S3
d) S1 and S2

GATE CSE – 2018 MCQ (2-MARKS)

Q10. Consider the first-order logic sentence: φ ≡ ∃s∃t∃u∀v∀w∀x∀yψ(s,t,u,v,w,x,y) where ψ(s,t,u,v,w,x,y) is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements.
Which one of the following statements is necessarily true?

a) There exists at least one model of φ with universe of size less than or equal to 3.
b) There exists no model of φ with universe of size less than or equal to 3.
c) There exists no model of φ with universe of size greater than 7.
d) Every model of φ has a universe of size equal to 7

GATE CSE – 2017 MCQ SET – 1 (2-MARKS)

Q11. Let p, q and r be prepositions and the expression (p → q) → r be a contradiction.
Then, the expression (r → p) → q is

Check Out Our GATE Courses & Secure Your Dream Rank!

a) a tautology
b) a contradiction
c) always TRUE when p is FALSE
d) always TRUE when q is TRUE

GATE CSE – 2017 MCQ SET- 2 (1-MARKS)

Q12. Consider the first-order logic sentence F: $\forall x(\exists y R(x,y))$. Assuming non-empty logical domains, which of the sentences below are implied by F?
I. $\exists y(\exists x R(x,y))$
II. $\exists y(\forall x R(x,y))$
III. $\forall y(\exists x R(x,y))$
IV. $\neg\exists x(\forall y \neg R(x,y))$

a) IV only
b) I and IV only
c) II only
d) II and III only

GATE CSE – 2017 MCQ SET-1 (1-MARKS)

Q13. The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?
I. $p \Rightarrow q$
II. $q \Rightarrow p$
III. $(\neg q) \vee p$
IV. $(\neg p) \vee q$

a) I only
b) I and IV only
c) II only
d) II and III only

GATE CSE – 2017 MCQ SET-2 (1-MARKS)

Q14. Let p, q, r denote the statements "It is raining", "It is cold", and "It is pleasant", respectively. Then the statement "It is not raining, and it is pleasant, and it is not pleasant only if it is raining and it is cold" is represented by

a) $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$
b) $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$
c) $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$
d) $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

GATE CSE – 2016 MCQ SET-2 (2-MARKS)

Q15. Which one of the following well-formed formulae in predicate calculus is NOT valid?

a) $(\forall x\, p(x) \Rightarrow \forall x\, q(x)) \Rightarrow (\exists x \neg p(x) \vee \forall x\, q(x))$
b) $(\exists x\, p(x) \vee \exists x\, q(x)) \Rightarrow \exists x\, (p(x) \vee q(x))$
c) $\exists x\, (p(x) \wedge q(x)) \Rightarrow (\exists x\, p(x) \wedge \exists x\, q(x))$
d) $\forall x\, (p(x) \vee q(x)) \Rightarrow (\forall x\, p(x) \vee \forall x\, q(x))$

GATE CSE – 2016 NAT SET-1 (1-MARKS)

Q16. Let p,q,r,s represent the following propositions.
p: $x \in \{8,9,10,11,12\}$
q: x is a composite number
r: x is a perfect square
s: x is a prime number
The integer $x \geq 2$ which satisfies $\neg ((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is _____.

GATE CSE – 2016 NAT SET-2 (1-MARKS)

Q17. Consider the following expressions:
(i) false
(ii) Q
(iii) true
(iv) $P \vee Q$
(v) $\neg Q \vee P$
The number of expressions given above that are logically implied by $P \wedge (P \rightarrow Q)$ is

GATE CSE – 2015 MCQ SET-2 (2-MARKS)

Q18. Which one of the following well-formed formulae is a tautology?

Check Out Our GATE Courses & Secure Your Dream Rank!

a) $\forall x\ \exists y\ R(x,y) \leftrightarrow \exists y\ \forall x\ R(x,y)$
b) $(\forall x\ [\exists y\ R(x,y) \rightarrow S(x,y)]) \rightarrow \forall x \exists y\ S(x,y)$
c) $[\forall x\ \exists y\ (P(x,y) \rightarrow R(x,y)] \leftrightarrow [\forall x\ \exists y(\neg P(x,y) \vee R(x,y))]$
d) $\forall x\ \forall y\ P(x,y) \rightarrow \forall x\ \forall y\ P(y,x)$

GATE CSE – 2015 MCQ SET-3 (1-MARKS)

Q19. In a room there are only two types of people, namely Type 1 and Type 2. Type 1 people always tell the truth and Type 2 people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking, the person replies to the following: "The result of the toss is head if and only if I am telling the truth."
Which of the following options is correct?

a) The result is head
b) The result is tail
c) If the person is of Type 2, then the result is tail
d) If the person is of Type 1, then the result is tail

GATE CSE – 2015 MCQ SET-2 (1-MARKS)

Q20. Consider the following two statements.
S1: If a candidate is known to be corrupt, then he will not be elected
S2: If a candidate is kind, he will be elected
Which one of the following statements follows from S1 and S2 as per sound inference rules of logic?

a) If a person is known to be corrupt, he is kind
b) If a person is not known to be corrupt, he is not kind
c) If a person is kind, he is not known to be corrupt
d) If a person is not kind, he is not known to be corrupt.

GATE CSE – 2015 MCQ SET-1 (1-MARKS)

Q21. Which one of the following is NOT equivalent to $p \leftrightarrow q$?

a) $(\neg p \vee q) \wedge (p \vee \neg q)$
b) $(\neg p \vee q) \wedge (q \rightarrow p)$
c) $(\neg p \wedge q) \vee (p \wedge \neg q)$
d) $(\neg p \wedge \neg q) \vee (p \wedge q)$

GATE CSE – 2014 MCQ SET-1 (2-MARKS)

Q22. Which one of the following propositional logic formulas is TRUE when exactly two of p, q, and r are TRUE?

a) $((p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
b) $(\sim (p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
c) $((p \rightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
d) $(\sim (p \leftrightarrow q) \wedge r) \wedge (p \wedge q \wedge \sim r)$

GATE CSE – 2014 MCQ SET-2 (2-MARKS)

Q23. Which one of the following Boolean expressions is NOT a tautology?

a) $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$
b) $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$
c) $(a \wedge b \wedge c) \rightarrow (c \vee a)$
d) $a \rightarrow (b \rightarrow a)$

GATE CSE – 2014 MCQ SET-3 (2-MARKS)

Q24. The CORRECT formula for the sentence, "not all rainy days are cold" is

a) $\forall d\ (Rainy(d) \wedge \sim Cold(d))$
b) $\forall d\ (\sim Rainy(d) \rightarrow Cold(d))$
c) $\exists d\ (\sim Rainy(d) \rightarrow Cold(d))$
d) $\exists d\ (Rainy(d) \wedge \sim Cold(d))$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2014 MCQ SET-1 (1-MARKS)

Q25. Consider the statement:
"Not all that glitters is gold"
Predicate glitters(x) is true if x glitters and predicate gold(x) is true if x is gold. Which one of the following logical formulae represents the above statement?

a) $\forall x$ glitters(x) $\Rightarrow \neg$gold(x)
b) $\forall x$ gold(x) $\Rightarrow$glitters(x)
c) $\exists x$ gold(x)$\wedge\neg$glitters(x)
d) $\exists x$ glitters(x)$\wedge\neg$gold(x)

GATE CSE – 2014 MCQ SET-3 (1-MARKS)

Q26. Consider the following statements:
P: Good mobile phones are not cheap
Q: Cheap mobile phones are not good
L :P implies Q
M: Q implies P
N: P is equivalent to Q
Which one of the following about L, M, and N is CORRECT?

a) Only L is TRUE.
b) Only M is TRUE.
c) Only N is TRUE.
d) L, M and N are TRUE.

GATE CSE – 2013 MSQ (2-MARKS)

Q27. Which one of the following is NOT logically equivalent to $\neg\exists x(\forall y(\alpha) \wedge \forall z(\beta))$?

a) $\forall x(\exists z(\neg\beta)\rightarrow\forall y(\alpha))$
b) $\forall x(\forall z(\beta)\rightarrow\exists y(\neg\alpha))$
c) $\forall x(\forall y(\alpha)\rightarrow\exists z(\neg\beta))$
d) $\forall x(\exists y(\neg\alpha) \rightarrow \exists z(\neg\beta))$

GATE CSE – 2013 MCQ (2-MARKS)

Q28. What is the logical translation of the following statement?
"None of my friends are perfect"

a) $\exists x(F(x) \wedge \neg P(x))$
b) $\exists x(\neg F(x) \wedge P(x))$
c) $\exists x(\neg F(x) \wedge \neg P(x))$
d) $\neg\exists x(F(x) \wedge P(x))$

GATE CSE – 2012 MCQ (1-MARKS)

Q29. What is the correct translation of the following statement into mathematical logic?
"Some real numbers are rational"

a) $\exists x$ (real(x) $\vee$ rational(x))
b) $\forall x$ (real(x) $\rightarrow$ rational(x))
c) $\exists x$ (real(x) $\wedge$ rational(x))
d) $\exists x$ (rational(x) $\rightarrow$ real(x))

GATE CSE – 2012 MCQ (1-MARKS)

Q30. Consider the following logical inferences.
I_1 : If it rains then the cricket match will not be played. The cricket match was played.
Inference: There was no rain.
I_2 : If it rains then the cricket match will not be played. It did not rain.
Inference: The cricket match was played.
Which of the following is TRUE?

a) Both I_1 and I_2 are correct inferences
b) I_1 is correct but I_2 is not a correct inference
c) I_1 is not correct but I_2 is a correct inference
d) Both I_1 and I_2 are not correct inferences.

GATE CSE – 2011 MCQ (2-MARKS)

Q31. Which one of the following options is CORRECT given three positive integers x,y and z, and a predicate

Check Out Our GATE Courses & Secure Your Dream Rank!

$P(x) = \neg(x=1) \wedge \forall y(\exists z(x=y*z) \Rightarrow (y=x) \vee (y=1)$

a) P(x) being true means that x is a prime number
b) P(x) being true means that x is a number other than 1
c) P(x) is always true irrespective of the value of x
d) P(x) being true means that x has exactly two factors other than 1 and x

GATE CSE – 2010 MCQ (2-MARKS)

Q32. Suppose the predicate F(x, y, t) is used to represent the statement that person x can fool person y at time t. Which one of the statements below expresses best the meaning of the formula $\forall x \exists y \exists t(\neg F(x, y, t))$?

a) Everyone can fool some person at some time
b) No one can fool everyone all the time
c) Everyone cannot fool some person all the time
d) No one can fool some person at some time

GATE CSE – 2009 MCQ (2-MARKS)

Q33. Which one of the following is the most appropriate logical formula to represent the statement:
"Gold and silver ornaments are precious"
The following notations are used :
G(x): x is a gold ornament.
S(x): x is a silver ornament.
P(x): x is precious.

a) $\forall x(P(x) \rightarrow (G(x) \wedge S(x)))$
b) $\forall x((G(x) \wedge S(x)) \rightarrow P(x))$
c) $\exists x((G(x) \wedge S(x)) \rightarrow P(x))$
d) $\forall x((G(x) \vee S(x)) \rightarrow P(x))$

GATE CSE – 2009 MCQ (2-MARKS)

Q34. The binary operation is defined as follows:

P	Q	P □ Q
T	T	T
T	F	T
F	T	F
F	F	T

Which one of the following is equivalent to P ∨ Q?

a) ¬Q □ ¬P
b) P □ ¬Q
c) ¬P □ Q
d) ¬P □ ¬Q

GATE CSE – 2009 MCQ (2-MARKS)

Q35. Consider the following well-formed formulae :
I. $\neg\forall x(P(x))$
II. $\neg\exists x(P(x))$
III. $\neg\exists x(\neg P(x))$
IV. $\exists x(\neg P(x))$
Which of the above are equivalent?

a) I and III
b) I and IV
c) II and III
d) II and IV

GATE CSE – 2009 MCQ (1-MARKS)

Q36. Which one of the following is NOT necessarily a property of a Group?

a) Commutativity
b) Associativity
c) Existence of inverse for every element
d) Existence of identity

GATE CSE – 2008 MCQ (2-MARKS)

Q37. Let fsa and pda be two predicates such that fsa(x) means x is a finite state automaton, and pda(y)

Check Out Our GATE Courses & Secure Your Dream Rank!

means that y is a pushdown automaton. Let equivalent be another predicate such that equivalent(a,b) means a and b are equivalent. Which of the following first order logic statements represents the following:
Each finite state automaton has an equivalent pushdown automaton.

a) $(\forall x\ fsa(x)) \Rightarrow (\exists y\ pda(y) \wedge equivalent(x, y))$
b) $\sim\forall y\ (\exists x\ fsa(x) \Rightarrow pda(y) \wedge equivalent(x, y))$
c) $\forall x \exists y\ (fsa(x) \wedge pda(y) \wedge equivalent(x, y))$
d) $\forall x \exists y\ (fsa(y) \wedge pda(x) \wedge equivalent(x, y))$

GATE CSE – 2008 MCQ (2-MARKS)

Q38. P and Q are two propositions. Which of the following logical expressions are equivalent?
I. $P \vee \neg Q$
II. $\neg (\neg P \wedge Q)$
III. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$
IV. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$

a) Only I and II
b) Only I, II and III
c) Only I, II and IV
d) All of I, II, III and IV

GATE IT – 2008 MCQ (2-MARKS)

Q39. Which of the following is the negation of $[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, y))]$

a) $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, y))]$
b) $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \neg y))]$
c) $[\forall x, \neg\alpha \rightarrow (\exists y, \neg\beta \rightarrow (\forall u, \exists v, \neg y))]$
d) $[\exists x, \alpha \wedge (\forall y, \beta \wedge (\exists u, \forall v, \neg y))]$

GATE IT – 2008 MCQ (2-MARKS)

Q40. Which of the following first-order formulae is logically valid? Here $\alpha(x)$ is a first-order formula with x as a free variable, and β is a first-order formula with no free variable.

a) $[\beta \rightarrow (\exists x \alpha(x))] \rightarrow [\forall x (\beta \rightarrow \alpha(x))]$
b) $[\exists x\ (\beta \rightarrow \alpha(x))\ ] \rightarrow [\ \beta \rightarrow (\forall x\ \alpha(x))\ ]$
c) $[(\exists x \alpha(x)) \rightarrow \beta] \rightarrow [\forall x (\alpha(x) \rightarrow \beta)]$
d) $[(\forall x \alpha(x)) \rightarrow \beta] \rightarrow [\forall x (\alpha(x) \rightarrow \beta)]$

GATE CSE – 2007 MCQ (2-MARKS)

Q41. Let Graph(x) be a predicate which denotes that x is a graph. Let Connected(x) be a predicate which denotes that x is connected. Which of the following first order logic sentences DOES NOT represent the statement: "Not every graph is connected"?

a) $\neg\forall x\ (Graph(x) \Rightarrow Connected(x))$
b) $\exists x\ (Graph(x) \wedge \neg\ Connected(x))$
c) $\neg\ \forall x (\neg\ Graph(x)\ V\ Connected(x))$
d) $\forall x\ (Graph(x) \Rightarrow \neg\ Connected(x))$

GATE IT – 2007 MCQ (2-MARKS)

Q42. Which one of these first-order logic formulae is valid?

a) $(\forall x(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x P(x)) \Rightarrow (\forall x Q(x)))$
b) $(\exists x(P(x) \vee Q(x))) \Rightarrow ((\exists x P(x)) \Rightarrow (\exists x Q(x)))$
c) $(\exists x(P(x) \wedge Q(x))) \Leftrightarrow ((\exists x P(x)) \wedge (\exists x Q(x)))$
d) $(\forall x \exists y P(x,y)) \Rightarrow (\exists y \forall x P(x,y))$

GATE CSE – 2006 MCQ (2-MARKS)

Q43. Which one of the first order predicate calculus statements given below correctly expresses the following English statement?
Tigers and lions attack if they are hungry or threatened.

a) $\forall x[(tiger(x) \wedge lion(x)) \rightarrow (hungry(x)\ V\ threatened(x)) \rightarrow attacks(x)]$
b) $\forall x[(tiger(x)\ V\ lion(x)) \rightarrow (hungry(x)\ V\ threatened(x)) \wedge attacks(x)]$
c) $\forall x[(tiger(x)\ V\ lion(x)) \rightarrow attacks(x) \rightarrow (hungry(x)\ V\ threatened(x))]$
d) $\forall x[(tiger(x)\ V\ lion(x)) \rightarrow (hungry(x)\ V\ threatened(x)) \rightarrow attacks(x)]$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2006 MCQ (2-MARKS)

Q44. Consider the following propositional statements:
P1: $((A \wedge B) \rightarrow C) \equiv ((A \rightarrow C) \wedge (B \rightarrow C))$
P2: $((A \vee B) \rightarrow C) \equiv ((A \rightarrow C) \vee (B \rightarrow C))$
Which one of the following is true?

a) P1 is a tautology, but not P2
b) P2 is a tautology, but not P1
c) P1 and P2 are both tautologies
d) Both P1 and P2 are not tautologies

GATE IT – 2006 MCQ (2-MARKS)

Q45. Consider the following first order logic formula in which R is a binary relation symbol.
$\forall x \forall y\ (R(x,y) \Rightarrow R(y,x))$
The formula is

a) satisfiable and valid
b) satisfiable and so is its negation
c) unsatisfiable but its negation is valid
d) satisfiable but its negation is unsatisfiable

GATE IT – 2006 MCQ (1-MARKS)

Q46. The set {1,2,3,5,7,8,9} under multiplication modulo 10 is not a group. Given below are four possible reasons. Which one of them is false?

a) It is not closed
b) 2 does not have an inverse
c) 3 does not have an inverse
d) 8 does not have an inverse

GATE IT – 2006 MCQ (2-MARKS)

Q47. A logical binary relation $\odot$ is defined as follows:

A	B	A⊙B
True	True	True
True	False	True
False	True	False
False	False	True

Let ~ be the unary negation (NOT) operator, with higher precedence than $\odot$.
Which one of the following is equivalent to $A \wedge B$?

a) $(\sim A \odot B)$
b) $\sim (A \odot \sim B)$
c) $\sim (\sim A \odot \sim B)$
d) $\sim (\sim A \odot B)$

GATE CSE – 2005 MCQ (2-MARKS)

Q48. Let P, Q, and R be three atomic propositional assertions.
Let X denote $(P \vee Q) \rightarrow R$ and Y denote $(P \rightarrow R) \vee (Q \rightarrow R)$.
Which one of the following is a tautology?

a) $X \equiv Y$
b) $X \rightarrow Y$
c) $Y \rightarrow X$
d) $\neg Y \rightarrow X$

GATE CSE – 2005 MCQ (2-MARKS)

Q49. What is the first order predicate calculus statement equivalent to the following?
"Every teacher is liked by some student"

a) $\forall x\ [teacher(x) \rightarrow \exists y\ (student(y) \rightarrow likes(y, x))]$
b) $\forall x\ [teacher(x) \rightarrow \exists y\ (student(y) \wedge likes(y, x))]$
c) $\exists y\ \forall x\ [teacher(x) \rightarrow (student(y) \wedge likes(y, x))]$
d) $\forall x\ [teacher(x) \wedge \exists y\ (student(y) \rightarrow likes(y, x))]$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE IT – 2005 MCQ (2-MARKS)

Q50. Let P(x) and Q(x) be arbitrary predicates. Which of the following statements is always TRUE?

a) $(\forall x(P(x) \vee Q(x))) \Rightarrow ((\forall x P(x)) \vee (\forall x Q(x)))$
b) $(\forall x(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x P(x)) \Rightarrow (\forall x Q(x)))$
c) $(\forall x(P(x)) \Rightarrow \forall x(Q(x)))) \Rightarrow (\forall x(P(x) \Rightarrow Q(x)))$
d) $(\forall x(P(x)) \Leftrightarrow (\forall x Q(x))) \Rightarrow (\forall x(P(x) \Leftrightarrow Q(x)))$

Check Out Our GATE Courses & Secure Your Dream Rank!

ANSWER KEY

1. (a)	**2. (a)**	**3. (b,d)**	**4. (a)**
5. (b,c)	**6. (b,d)**	**7. (c)**	**8. (b)**
9. (c)	**10. (a)**	**11. (d)**	**12.(b)**
13. (d)	**14.(a)**	**15. (d)**	**16. (11)**
17. (4)	**18. (c)**	**19. (a)**	**20. (c)**
21. (c)	**22. (b)**	**23. (b)**	**24. (d)**
25. (d)	**26. (d)**	**27. (a,d)**	**28. (d)**
29. (c)	**30. (b)**	**31. (a)**	**32. (b)**
33. (d)	**34. (b)**	**35. (b)**	**36. (a)**
37. (a)	**38. (b)**	**39. (d)**	**40. (c)**
41. (d)	**42. (a)**	**43. (d)**	**44. (d)**
45. (b)	**46. (c)**	**47. (d)**	**48. (b)**
49. (b)	**50. (b)**		

HINTS&SOLUTIONS

1. ANSWER:- A

Statement meaning:
There exists a person y such that
- everyone else knows y
- y knows no one else

Formally:
- "There is a person" → ∃y
- "For every other person x≠y" → ∀x
- "x knows y" → M(x,y)
- "y does not know x" → ¬M(y,x)

So we get:
(∃y)(∀x)((x≠y)→(M(x,y)∧¬M(y,x)))
That matches option (a).

2. ANSWER:- A

Option (A) represents mathematical induction, which is a proof technique often used for predicates over natural numbers:
Base Case: P(0) is true (the starting point).
Inductive Step: ∀x[P(x)⇒P(x+1)], meaning if the predicate P(x) holds for an arbitrary natural number x, it will also hold for x+1.
Conclusion: ∀xP(x), meaning the predicate P(x) holds for all natural numbers x.

3. ANSWER:- B, D

The logical representation is:
∀x∃y(mother(y,x)∧∀z(mother(z,x)→¬Noteq(z,y)))
which is same as
∀x∃y[mother(y,x)∧¬∃z(noteq(z,y)∧mother(z,x))]
Answer: B & D match.

4. ANSWER:- A

Q when P means P→Q.
Given statement,
P: Fail grade can be given.
Q: Student score more than 50% marks.
¬P : Fail grade cannot be given
So,
Fail grade cannot be given when student score more than 50% marks.
is written as Q→¬P
Correct answer is option A.

5. ANSWER:- B,C

L: ∀x[P(x) → ∃y Q(x, y)]
"For every x, if P(x) is true, then there exists some y such that Q(x, y) will be true."
Option:
(i) ∃x(P(x) ∧ ∀yQ(x, y))
"For some x, P(x) is true, and for all y, Q(x, y) is true, which does not imply L."
(ii) ∀x ∀y Q(x, y)
"For every x and every y, Q(x, y) is true, which implies L."
(iii) ∃y ∀x[P(x) → Q(x, y)]
"There exists some y such that for every x, if P(x) is true, then Q(x, y) is also true, which implies L."
(iv) ∃xP(x) ∧ ∃yQ(x, y)
"There exists some x for which P(x) is true, and also for some y, Q(x, y) is true, which cannot imply L."

6. ANSWER:- B,D

Antecedent of S : (P ∧ Q) → R
≡ ~(P∧Q) v R
≡ ~P v ~Q v R
Consequent of S : (P ∧ Q) → (Q → R)
≡ (P ∧ Q) → (~Q v R)
≡ ~(P∧Q) v (~Q v R)
≡ ~P v ~Q v (~Q v R)
≡ ~P v ~Q v R
Antecedent of S is equivalent to Consequent of S. Hence Option (D) is right. A → A is always a Tautology. Option B is right.

Check Out Our GATE Courses & Secure Your Dream Rank!

7. ANSWER:- C

S_1: ¬p∧(p∨q)→q
If consequence is false and hypothesis is true, then we will get False in the truth table.
Lets assume q is false. So consequence is FALSE.
Can it make Hypothesis TRUE?
Hypothesis:
¬p∧(p∨q)≡¬p∧(p∨FALSE)≡¬p∧(p)≡FALSE.
Hypothesis can't be true, So we can't get False in the Truth Table.
∴ S1 is Tautology
S_2 : q → (∼p ∧ (p ∨ q))
∼q ∨ (∼p ∧ (p ∨ q))
∼ q ∨ (() ∨ (∼ p ∧ q))
∼ q ∨ (F ∨ (∼p ∧ q))
∼ q ∨ (∼ p ∧ q)
(∼q ∨∼ p) ∧ T
it can be false when q=false and p=false
S_2 is not a tautology

8. ANSWER:- B

∀x (p(x) → W) ≡ ∀x p(x) → W is wrong. Since
∀x [p(x) → W]
≡ ∀x [¬p(x) ∨ W]
≡ ∀x (¬p(x) ∨ W
≡ ¬(∃x p(x)) ∨ W
≡ ∃x p(x) → W
Option ∀x(p(x) → W) ≡ ∀xp(x) → W is correct.

9 ANSWER:- C

The first-order logic means that if z is a prime number, then there exists another prime number in the set, which is larger than it.
One of the cases: If -7 is a prime number (i.e, it is either divided by -7 or 1 only), then there exists some number like -3, which is larger than -7 and it satisfies the property (either divided by -3 or 1 only).
So, S3 is correct
It's true for all integers too.

10. ANSWER:- A

φ = ∃s ∃t ∃u ∀v ∀w ∀x ∀y ψ(s, t, u, v, w, x,y)
Let's interpret the problem this way: ∀ are always True and ∃ are always False for empty sets.
So, there exists at least one model with universe of size 3 (or less than) or equal to 3.
Therefore, option A is necessarily TRUE.

11. ANSWER:- D

Given that (p→q)→r is a contradiction.
So, r = F and (p→q) = T.
So, we have to evaluate the expression (r→p)→q
Since r = F, (r→p) = T (As F→p, is always true)
The final expression is T→q, which is true when q is true.

12. ANSWER:- B

Let's convert the given first-order logic sentence into some English sentence.
Given:- F: ∀x(∃yR(x, y))
: For all girls there exist a boyfriend (x for girls and y for boys)
I: ∃y(∃xR(x, y))
: There exist some boys who have girlfriends. (Subset of statement F, so True)
II: ∃y(∀xR(x, y))
: There exist some boys for which all the girls are girlfriend. (False)
III: ∀y(∃xR(x, y))
: For all boys exists a girlfriend. (False)
IV: ∼∃x(∀y∼R(x, y))
= ∀x(∼∀y∼∼R(x, y))
= ∀x(∃yR(x, y)) (∵ ∼∀y=∃y, ∼∃x=∀x) (True)

13. ANSWER:- D

(I) ¬p⇒¬q ≡ q ⇒ p (contrapositive)
And q ⇒ p ≡ ¬q ∨ p
So, statements (II) and (III) are equivalent to the statement given in the question.

14. ANSWER:- A

p: It is raining
q: It is cold
r: It is pleasant
"If it is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold."
We can divide the statement into two parts with "Conjunction".

$\left(\frac{If\ it\ is\ not\ raining}{\neg p}\right)$ and $\left(\frac{It\ is\ pleasant}{r}\right) \rightarrow (1)$
$\left(\frac{It\ is\ not\ pleasant}{\neg p}\right)$ only if $\left(\frac{It\ is\ raining\ and\ \ it\ is\ cold}{p \wedge q}\right)$
i.e., $\neg r \rightarrow (p \wedge q) \rightarrow (2)$
From (1) & (2), the given statement can be represented as $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$

15. ANSWER:- D

d) $\forall x(p(x) \vee q(x)) \Rightarrow (\forall x p(x) \wedge \forall x q(x))$

- This states that if every clement in the domain satisfies either p or q, then all elements must satisfy p AND all elements must satisfy q. This is a logical fallacy.
- Counterexample: *Let the domain be the set of natural numbers: {1,2}.
- Let $p(x)$ mean " x is odd" and $q(x)$ mean " x is even".

Check Out Our GATE Courses & Secure Your Dream Rank!

- The premise $\forall x(p(x) \vee q(x))$ is True because every number is either odd or even.
- The conclusion $(\forall x p(x) \wedge \forall x q(x))$ is False because it is not true that all numbers are odd, nor is it true that all numbers are even.
- Since True ⇒ False results in False, this formula is NOT valid.

16. ANSWER:- 11

Given,
~((p→q) ∧ (~r ∨ ~s))
⇒ first simplify the given statement by converting ∧ to ∨ and ∨ to ∧.
⇒ [~(p→q) ∨ (~(~r ∨ ~s)]
Demorgan's law:
⇒ [~(~p ∨ q) ∨ (r ∧ s)]
∵p→q≡ ~p ∨ q
⇒ [(p ∧ ~q) ∨ (r ∧ s)]
p ∧ ~q is {8,9,10,11,12} ∧ {not a composite number} i.e. {11}

17. ANSWER:- 4

(P ∧ (P → Q)) = p ∧ (~p ∨ Q) = (p ∧ ~p) ∨(p ∧ Q) = p ∧ Q
(P ∧ Q) → True is always a tautology.
(P ∧ Q) → False is not a tautology
(P ∧ Q) → Q is a tautology
(P ∧ Q) →¬Q ∨ P is a tautology
(P ∧ Q) → P ∨ Q is a tautology
So, there are 4 expressions logically implied by (P ∧ (P → Q)).

18. ANSWER:- C

Since, we know that: P→R = ¬P ∨ R
[∀x ∃y (P(x,y)→R(x,y)] ↔ [∀x ∃y (¬ P(x,y)∨R(x,y)] is a tautology.

19. ANSWER:- A

As we do not know the type of person from whom those words are coming, so we can have two possible cases.
Case 1:
The person who speaks truth. This implies that the result of the toss is Head.
Case 2:
The person who lies. In this case , the statement itself must be false.
The negation of (x⇔y) is : (~x and y) or (x and ~y)
so there are two possibilities for this statement to be false.
→ It is head and lies spoken.
→ It is not head and truth spoken.
Clearly, the second one cannot speak the truth. So finally, it is head.
Hence, option (A) is the correct answer.

20. ANSWER:- C

S1: If a candidate is known to be corrupt, then he will not be elected.
S2: If a candidate is kind, he will be elected.
If p → q, then ¬q → ¬p
So from S1, elected → not corrupt
and S2 is, kind → elected
Therefore, kind → not corrupt

21. ANSWER:- C

Here, option (a) and (b) can be reduced to (p → q) ^ (q → p) and hence it is equivalent to p ↔ q.
Option (d) is p'q' + pq ≡ p ↔ q.
Option (c) is p'q + pq' = p ⊕ q which is not equivalent to p ↔ q.

22. ANSWER:- B

We directly check with one of the {TTF, TFT, FTT} options.
As there are two T's in each option, replace them and check with the third value.
E.g.: Let p=q= T
(~(p↔q) ∧r) ∨ (p∧q∧~r)
= (~(T↔T) ∧ r) ∨ (T∧T∧~r)
= (~(T)∧r) ∨ (T∧~r)
= (F∧ r) ∨ (T∧~r)
= (F)∨(~r)
= ~r
This is true for r=F.
Similarly, we can have p=r=T and q=F.
q=r=T and p=F
So, Option (~ (p↔ q)∧r)∨(p ∧ q∧~r) is the correct answer.

23. ANSWER:- B

A: ((a → b) ∧ (b → c)) → (a → c)
it can be false only when (a → c) is false i.e. a=T , c=F and left is true
then left part: ((T → b) ∧ (b → F))
to make it true T-> b must be true i.e. b must be true and simultaneosly b-> F must be true i.e. b must be false.
which is not possible .
This is a tautology.
B: (a ↔ c) ⟶ (~b ⟶ (a ∧ c))
Let a = c = F
(F ↔ F) → (~b⟶ (F ∩ F))
T → (~b → F)

Check Out Our GATE Courses & Secure Your Dream Rank!

This is False for b = F
So, this is not True.
C: (a ∧ b ∧ c) ⟶ (c ∨ a)
(c ∨ a) is False only for a = c = F
if (c ∨ a) is False
(F ∧ b ∧ F) ⟶ F
F ⟶ F which is a Tautology
True always.
D: a ⟶ (b ⟶ a)
a ⟶ (~b ∨ a)
(~a ∨ a) ∨ ~b = T ∨ ~b = T which is a tautology

24. ANSWER:- D
Not all rainy days are cold
= ~ [∀rainy days are cold]
= ~ [∀ days (rainy days ⇒ cold days]
= ∃ days [~ (cold days ∨~rainy days)]
= ∃ days [rainy days ∧~cold days]

25. ANSWER:- D
Not (all that glitters is gold)
~ (∀x glitters(x) ⇒ gold(x))
⇒ (~∀x) (~ (glitters(x) ⇒ gold(x))
⇒∃x (~ (~glitters(x) ∨ gold(x))
⇒∃x (glitters(x) ∧ ~gold(x))

26. ANSWER:- D
Observe that in the given statements, "not cheap & cheap", "good & not good" are used.
So, we can divide the given statements to utilize the negation of these atomic statements.
Suppose, X is a good mobile and Y is a cheap one, then
P: (Good(x) ⟶ ~cheap(x)) ↔ (~good(x) ∨ ~cheap(x))
Q: cheap(x) ⟶ ¬good(x) ⟺ ((¬cheap(x) ∨ good(x)) ⟺ ¬good(x) ∨¬cheap(x))
All these are contrapositive.
All L, M, N are true.

27. ANSWER:- A,D
Option A:
∀x(∃z(¬β) ⟶∀y(α))
⇒∀x(¬∃z(¬β) ∨ ∀y(α))
⇒∀x(∀z(β) ∨ ∀y(α))
⇒¬∃x¬(∀z(β) ∨ ∀y(α))
⇒¬∃x(¬∀z(β)∧¬∀y(α))
A is not equivalent to the given.
Option B:
∀x(∀z(β)⟶∃y(¬α))
⇒∀x(¬∀z(β)∨∃y(¬α))
⇒¬∃x¬(¬∀z(β)∨∃y(¬α))
⇒¬∃x(∀z(β)∨∀y(α))
B is Matching and equivalent to given.
Option C:
∀x(∀y(α)⟶∃z(¬β))
⇒∀x(¬∀y(α) ∨ ∃z(¬β))
⇒¬∃x¬(¬∀y(α) ∨ ∃z(¬β))
⇒¬∃x(∀y(α)∧ ∀z(β))
C is equivalent to the given.
Option D:
∀x(∃y(¬α)⟶∃z(¬β))
⇒∀x(¬∃y(¬α) ∨ ∃z(¬β))
⇒∀x(∀y(α) ∨ ∃z(¬β))
⇒¬∃x¬(∀y(α)∨∃z(¬β))
⇒¬∃x(¬∀y(α)∧¬∃z(¬β))
⇒¬∃x(¬∀y(α)∧∀z(β))
So D is Not equivalent to the given.
Observe that in the given statements,

28. ANSWER:- D
A. some of my friends are not perfect
B. some of those who are not my friends are perfect
C. some of those who are not my friends are not perfect
D. NOT (some of my friends are perfect) / none of my friends are perfect

29. ANSWER:- C
"Some real numbers are rational"
∃x (real(x) ∧ rational(x))
(A) ∃x(real(x) ∨ rational(x))
means "Some number, which are either real or rational".
(B) ∀x (real(x)⟶rational(x))
Means "If a number is real, then it is rational".
(D) ∃x (rational(x)⟶real(x))
means "There exists a number such that if it is rational then it is real".
it will be true even if all real numbers are real but not rational. which is not correct.
only C is correct.

30. ANSWER:- B
Given:
I_1: If it rains then the cricket match will not be played. The cricket match was played.
Let's break the above statement into two as-
p = it rains
q = playing cricket/ match played
If (it rains) then (the match will not be played)
p ⇒ (~q)
q ⇒(~p) (contrapositive)
Inference: There was no rain. (i.e., p = F)
this inference is valid.

Check Out Our GATE Courses & Secure Your Dream Rank!

I_2: If it rains then the cricket match will not be played.
It did not rain.
$p \Rightarrow (\sim q)$
but $\sim p \Rightarrow \sim(\sim q) = q$ need not be true
Inference: The cricket match was played.

31. ANSWER:- A
Statement: x is not equal to 1, and if there exists some z for all y such that the product of y and z is x, then y is either the number itself or 1.
This is the definition of prime numbers.
So, P(x) being true means that x is a prime number is the correct answer.

32. ANSWER:- B
F(x,y,t) ⇒ Person 'x' can fool person 'y' at time 't'.
Now we pull the negation sign by applying Demorgan's law.
$\forall x \exists y \exists t(\neg F(x, y, t)) \equiv \neg \exists x \forall y \forall t(F(x,y,t))$
$\neg \exists x \forall y \forall t(F(x,y,t)) \Rightarrow$ No person can fool all the person all the time.
Which is same as "No one can fool everyone all the time"

33. ANSWER:- D
The statement could be translated as, if x is either Gold or Silver, then it would be precious. Rather than,
If x is both Gold and Silver, as an item cannot both Gold and silver at the same time.
Hence Ans is (D).

34. ANSWER:- B
Answer is B because the truth values for option B is the same as that of P "or" Q.
The given truth table is for $Q \rightarrow P$ which is $\neg Q \vee P$.
Now, with B option we get $(\neg(\neg Q)) \vee P = P \vee Q$

35. ANSWER:- B
Let P(x) ⇒ Student x is pass
- I ⇒ Not all students are pass. (which means "Some students are fail")
- II ⇒ There does not exist a student who is pass. (which means "Every student is fail")
- III ⇒There does not exist a student who is not pass (which means "Every student is pass")
- IV ⇒Some students are not pass. (which means "Some students are fail")

So, I and IV are equivalent.

36. ANSWER:- A
Group properties are
- Closure
- Associative
- Identity Element
- Inverse Element

Commutativity is not required for a mathematical structure to become a group.
So, Answer should be Option A

37. ANSWER:- A
We are given:
- fsa(x): "x is a finite state automaton"
- pda(y): "y is a pushdown automaton"
- equivalent(a, b): "a and b are equivalent"

We must formalize:
Each finite state automaton has an equivalent pushdown automaton
"Each finite state automaton …"
$\forall x \ (fsa(x) \Rightarrow \dots)$
"… has an equivalent pushdown automaton"
$\exists y \ (pda(y) \wedge equivalent(x,y))$
$\forall x(fsa(x) \Rightarrow \exists y(pda(y) \wedge equivalent(x,y)))$
This corresponds exactly to:
$(\forall x fsa(x)) \Rightarrow (\exists y pda(y) \wedge equivalent(x,y))$
Interpreting properly with scope:
$\forall x(fsa(x) \Rightarrow \exists y(pda(y) \wedge equivalent(x,y)))$
Why others are wrong:
(B) Wrong quantifier order (starts with ∀y).
(C) Says: for every x there exists y such that x is FSA AND y is PDA…
This incorrectly forces every x in the universe to be considered.
(D) Swaps arguments incorrectly (pda(x) instead of pda(y)).

38. ANSWER:- B
I and II are present in all options so need to check.
For III and IV
$(P \wedge Q) \vee (P \wedge \sim Q) = P \wedge (Q \vee \sim Q)$ (By distributive law)
$= P \wedge T$
$= P$
For III.
$P \vee (\sim P \wedge \sim Q) \equiv P \vee \sim Q$ (By Absorption Law)
For IV.
$P \vee (\sim P \wedge Q) \equiv P \vee Q$ (By Absorption Law)
So Option B is correct.
I, II, III are logically equivalent.

39. ANSWER:- D
Given : $[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, y))]$
Now,

Check Out Our GATE Courses & Secure Your Dream Rank!

= ∃x ¬[α → (∃y [β → (∀u ∃v y)])]
= ∃x [α ∧ ¬(∃y [β → (∀u ∃v y)])]
= ∃x [α ∧ ∀y ¬(β → (∀u ∃v y))]
= ∃x [α ∧ ∀y (β ∧ ¬(∀u ∃v y))]
= ∃x [α ∧ ∀y (β ∧ (∃u ∀v ¬y))]
So, the correct answer is d.

40. ANSWER:- C

A. [β→(∃x α(x))]→[∀x (β→α(x))]
LHS: If β (some condition) is true, then there exists an x for which α(x) is true.
RHS: For all x, if β is true then α(x) is true. This is same as saying if β is true then for all x, α(x) is true.
(β ⇒ ∀x α(x)).
So, RHS ⇒ LHS and LHS ⇏ RHS.
B. [∃x (β→α(x))]→[β→(∀x α(x))]
LHS: There exists an x such that if β is true then α(x) is true.
RHS: If β is true then for all x, α(x) is true.
So, RHS ⇒ LHS and LHS ⇏ RHS.
C.[(∃x α(x))→β]→[∀x (α(x)→β)]
LHS: If there is an x such that α(x) is true, then β is true.
RHS: For all x, if α(x) is true, then β is true.
Here, both LHS and RHS are in fact same as β is a formula which is independent of x.(If β is true for one x, it is true for every x and vice versa.)
So,RHS ⇔ LHS.
D.[(∀x α(x))→β]→[∀x (α(x)→β)]
LHS: If α(x) is true for every x, then β is true.
RHS: For every x, if α(x) is true then β is true.
So, RHS ⇒ LHS and LHS ⇏ RHS.
So, answer here is option C.
Any of options A, B, or D could be valid if their implication is reversed. For option C, LHS and RHS being equivalent, even if the implication is reversed (or changed to double implies) it remains valid.

41. ANSWER:- D

Just writing to simplify each options
A)¬∀x(Graph(x)⟹ Connected(x))
It is not the case that every graph then will be connected, which implies that some graph may be disconnected.
B)∃x(Graph(x)∧¬ Connected(x))
There exists some graphs which are disconnected which implies not every graph is connected.
C)¬∀x(¬ Graph(x)∨ Connected(x))
Here no need to express it in english just solve first using
de morgen's law we will get it as option B)
D)∀x(Graph(x)⟹¬ Connected(x))
For all graph it is always disconnected...makes it FALSE

42. ANSWER:- A

A. LHS: For every x, if P holds then Q holds
RHS: If P(x) holds for all x, then Q(x) holds for all x.
LHS implies RHS but RHS does not imply LHS.

B. LHS: An x exist for which either P(x) is true or Q(x) is true.
RHS: If an x exist for which P(x) is true then another x exist for which Q(x) is true.
LHS does not imply RHS, but on RHS if we change ∃xP(x) to ~∃xP(x), implication becomes TRUE.

C. LHS: There exist an x for which both P(x) and Q(x) are true.
RHS: There exist an x for which P(x) is true and there exist an x for which Q(x) is true.
LHS implies RHS but RHS does not imply LHS as the 'x' for P and Q can be different on the RHS

D. LHS: For every x, there exist a y such that P(x, y) holds.
RHS: There exist a y such that for all x P(x, y) holds.
Here RHS implies LHS but LHS does not imply RHS as the y on LHS can be different for each x.
So, the correct answer is A.

43. ANSWER:- D

∀x [(tiger(x) ∨ lion(x)) ∧ (hungry(x) ∨ threatened(x)) → attack(x)]
We can represent also in this way,because of Exportation law : (P→ (Q→ R)) ≡ ((P ∧ Q) → R)
So, D is correct ans.
∀x[(tiger(x) ∨ lion(x)) → {(hungry(x) ∨ threatened(x)) → attacks(x)}]

44. ANSWER:- D

P1:
Let A = True, B = False
LHS of P1 becomes True, but RHS becomes False
Hence, P1 is not a tautology
P2:
The forward implication is true, but the reverse is not always true
Let A = False, B = True, C = False
RHS becomes True, but LHS becomes False
Hence, P2 is not a tautology

Check Out Our GATE Courses & Secure Your Dream Rank!

Simplification of LHS of P2:
((A ∨ B) → C)
≡ ¬(A ∨ B) ∨ C
≡ (¬A ∧ ¬B) ∨ C
≡ (¬A ∨ C) ∧ (¬B ∨ C)
≡ (A → C) ∧ (B → C)
Final Conclusion:
Both P1 and P2 are not tautologies

45. ANSWER:- B
The given relation represent the symmetry property. We have both symmetric relations possible as well as anti-symmetric but neither always holds for all sets. So they both are not valid but are satisfiable.
So, the correct answer is b.

46. ANSWER:- C

$\otimes_{10}$	1235789
1	1235789
2	2460468
3	3695147
5	5055505
7	7415963
8	8640642
9	9875321

From the above composition table, it is clear that 1 is the identity element. Hence, inverses exist only for some elements.

1. it is not closed: true, because elements like 0,4,6 ∉ the set {1,2,3,5,7,8,9}
2. 2 doesn't have an inverse: true
3. 3 does not have an inverse: false, inverse (3) = 7
4. 8 does not have an inverse: true.

So Option C is correct.

47. ANSWER:- D
A ⊙ B ≡ B → A, i.e. (¬B ∨ A).
Now, let us look at Option D
¬(¬A ⊙ B)
≡ ¬(B → ¬A)
≡ ¬(¬B ∨ ¬A)
≡ B ∧ A
So, the answer is D.

48. ANSWER:- B
X ≡ (P ∨ Q) → R
≡ ¬(P ∨ Q) ∨ R
≡ (¬P ∧ ¬Q) ∨ R
≡ (¬P ∨ R) ∧ (¬Q ∨ R)
≡ (P → R) ∧ (Q → R)
So, X → Y is true as (A ∧ B) → (A ∨ B) is always TRUE but reverse implication is not always true.
Hence, B is the correct answer.

49. ANSWER:- B
Answer is Option B. In a simpler way we can say "If x is a teacher then there exists some y who is a student and likes x".
Option A choice: If x is a teacher, then there exists a y such that if y is a student, then y likes x.
Option C choice: There exists a student who likes all teachers.
Option D choice: Everyone is a teacher and there exists a y such that if y is student then y likes x.
Assuming one cannot be both student and teacher at same time, this just means, everyone is a teacher.

50. ANSWER:- B
By digital logic method : Assume LHS as true based on some condition then try to make RHS false under some conditions and vice versa.
(A) $(\forall x(P(x)\vee Q(x)))\Rightarrow((\forall xP(x))\vee(\forall xQ(x)))$
$(P_1\vee Q_1)\wedge(P_2\vee Q_2)\Rightarrow(P_1\wedge P_2)\vee(Q_1\wedge Q_2)$
$(T\vee F)\wedge(F\vee T) \Rightarrow (T\wedge F)\vee(F\wedge T)$
T ⇒ F [false]
(C) $((\forall x(P(x))\Rightarrow\forall x(Q(x))))\Rightarrow(\forall x(P(x)\Rightarrow Q(x)))$
$[P_1\Rightarrow(Q_1\wedge Q_2)\wedge P_2\Rightarrow(Q_3\wedge Q_4)]\Rightarrow[(P_1\Rightarrow Q_1)\wedge(P_2\Rightarrow Q_2)]$
F ⇒ Anything∧T ⇒ T∧T ⇒ F ⇒ Anything∧T ⇒ F
T ⇒ F [false]
(D) $(\forall x(P(x))\Leftrightarrow(\forall xQ(x)))\Rightarrow(\forall x(P(x)\Leftrightarrow Q(x)))$
$[(P1\wedge P2)\Leftrightarrow(Q1\wedge Q2)]\Rightarrow(P1\Leftrightarrow Q1)\wedge(P2\Leftrightarrow Q2)$
$T\wedge F\Leftrightarrow F\wedge F\Rightarrow T\Leftrightarrow F\wedge F\Leftrightarrow F$
T ⇒ F [false]
So, B is the correct answer as we can't make it false.

Check Out Our GATE Courses & Secure Your Dream Rank!

4 GRAPH THEORY

GATE CSE – 2026 MSQ SET - 1(2-MARKS)

Q1. Let $G(V, E)$ be a simple, undirected graph. A vertex cover of G is a subset $V' \subseteq V$ such that for every $(u, v) \in E$, $u \in V'$ or $v \in V'$. Let the size of the smallest vertex cover in G be k. Let S be any vertex cover of size k.
For a vertex $v \in V$, which of the following constraints will always ensure that $v \in S$?

a) The degree of v is at least $k + 1$
b) The vertex v is on a path of length $k + 1$
c) The vertex v is on a cycle of length $k + 1$
d) The vertex v is a part of a clique of size k

GATE CSE – 2026 NAT SET - 1(2-MARKS)

Q2. Let G be an undirected graph, which is a path on 8 vertices. The number of matchings in G is _____. (answer in integer)

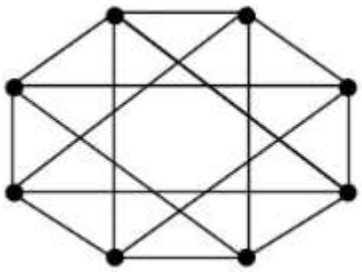
GATE CSE – 2026 MCQ SET - 2(2-MARKS)

Q3. Consider a complete graph K_n with n vertices ($n > 4$). Note that multiple spanning trees can be constructed over K_n. Each of these spanning trees is represented as a set of edges. The Jaccard coefficient between any two sets is defined as the ratio of the size of the intersection of the two sets to the size of the union of the two sets.
Which one of the following options gives the lowest possible value for the Jaccard coefficient between any two spanning trees of K_n?

a) 1/n
b) 1/(2n-3)
c) 0
d) 1/(n-1)

GATE CSE – 2024 NAT SET - 2(2-MARKS)

Q4. The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. The chromatic number of the following graph is _____.



GATE CSE – 2024 MSQ SET - 1(2-MARKS)

Q5. The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. Let G be any graph with n vertices and chromatic number k. Which of the following statements is/are always TRUE?

a) G contains a complete subgraph with k vertices.
b) G contains an independent set of size at least n/k.
c) G contains at least k(k−1)/2 edges.
d) G contains a vertex of degree at least k.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2023 MSQ (2-MARKS)

Q6. Let G be a simple, finite, undirected graph with vertex set $\{v_1,...,v_n\}$.
Let $\Delta(G)$ denote the maximum degree of G and let N = {1, 2,...} denote the set of all possible colors.
Color the vertices of G using the following greedy strategy:
for i = 1,...,n
color(v_i) ← min {j ∈ N : no neighbour of v_i is colored j}
Which of the following statements is/are TRUE?

a) This procedure results in a proper vertex coloring of G.
b) The number of colors used in at most Δ (G) + 1.
c) The number of colors used is at most Δ (G).
d) The number of colors used is equal to the chromatic number of G.

GATE CSE – 2022 NAT (2-MARKS)

Q7. Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is ____ .

GATE CSE – 2022 NAT (2-MARKS)

Q8. Let G(V, E) be a directed graph, where V = {1, 2, 3, 4, 5} is the set of vertices and E is the set of directed edges, as defined by the following adjacency matrix A.
A[i][j] = 1, $1 \le j \le i \le 5$
A[i][j] = 0, otherwise
A[i][j] = 1 indicates a directed edge from node i to node j. A directed spanning tree of G, rooted at r ∈ V, is defined as a subgraph T of G such that the undirected version of T is a tree, and T contains a directed path from r to every other vertex in V. The number of such directed spanning trees rooted at vertex 5 is _____.

GATE CSE – 2022 MCQ (2-MARKS)

Q9. Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of ___.

a) A^3
b) A^3 divided by 2
c) A^3 divided by 3
d) A^3 divided by 6

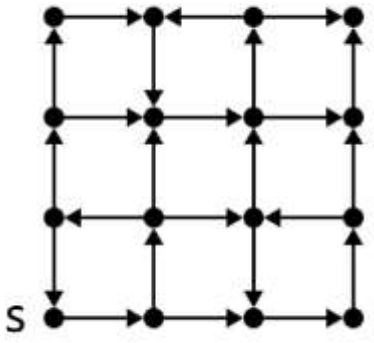
GATE CSE – 2022 MCQ (2-MARKS)

Q10. Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?

a) The diagonal entries of A^2 are the degrees of the vertices of the graph.
b) If the graph is connected, then none of the entries of $A^{n-1} + I_n$ can be zero.
c) If the sum of all the elements of A is at most 2(n-1), then the graph must be acyclic.
d) If there is at least a 1 in each of A's rows and columns, then the graph must be connected.

GATE CSE – 2021 MSQ SET-2(2-MARKS)

Q11. Consider the following directed graph:



Which of the following is/are correct about the graph?

Check Out Our GATE Courses & Secure Your Dream Rank!

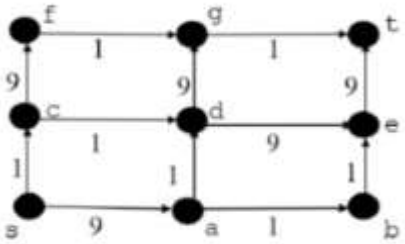
a) The graph does not have a strongly connected component.
b) The graph does not have a topological order.
c) A depth-first traversal starting at vertex S classifies three directed edges as back edges.
d) For each pair of vertices u and v, there is a directed path from u to v.

GATE CSE – 2021 NAT SET-1(1-MARKS)

Q12. In an undirected connected planar graph G, there are eight vertices and five faces. The number of edges in G is_____

GATE CSE – 2021 NAT SET-2(1-MARKS)

Q13. In a directed acyclic graph with a source vertex s, the quality-score of a directed path is defined to be the product of the weights of the edges on the path. Further,
for a vertex v other than s, the quality-score of v is defined to be the maximum among the quality-scores of all the paths from s to v. The quality-score of s is assumed to be 1.



The sum of the quality-scores of all the vertices in the graph shown above is _____.

GATE CSE – 2020 NAT (2-MARKS)

Q14. Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is _____.

GATE CSE – 2019 MCQ (1-MARKS)

Q15. Let G be an undirected complete graph on n vertices, where n > 2. Then, the number of different Hamiltonian cycles in G is equal to ____.

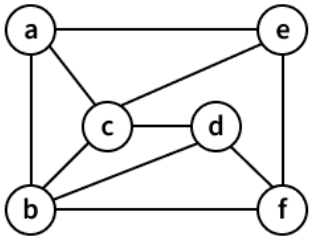
a) n!
b) 1
c) (n-1)!
d) (n-1)!/2

GATE CSE – 2018 NAT (2-MARKS)

Q16. Let G be a graph with 100! vertices, with each vertex labelled by a distinct permutation of the numbers 1, 2, …, 100.
There is an edge between vertices u and v if and only if the label of u can be obtained by swapping two adjacent numbers in the label of v. Let y denote the degree of a vertex in G and z denote the number of connected components in G. Then, y + 10z = _____.

GATE CSE – 2018 NAT (1-MARKS)

Q17. The chromatic number of the following graph is _____.



Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2017 NAT SET - 1(1-MARKS)

Q18. Let T be a tree with 10 vertices. The sum of the degrees of all the vertices in T is_____.

GATE CSE – 2017 NAT SET - 2(1-MARKS)

Q19. G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

GATE CSE – 2016 NAT SET - 2(1-MARKS)

Q20. The minimum number of colours that is sufficient to vertex-colour any planar graph is _____.

GATE CSE – 2015 NAT SET - 1(2-MARKS)

Q21. Let G be a connected planar graph with 10 vertices. If the number of edges on each face is three, then the number of edges in G is _____.

GATE CSE – 2015 MCQ (2-MARKS)

Q22. In a connected graph, a bridge is an edge whose removal disconnects a graph. Which one of the following statements is true?

a) A tree has no bridges
b) A bridge cannot be part of a simple cycle
c) Every edge of a clique with size > = 3 is a bridge (A clique is any complete sub graph of a graph)
d) A graph with bridges cannot have a cycle

GATE CSE – 2015 MCQ SET - 2(2-MARKS)

Q23. A graph is self-complementary if it is isomorphic to its complement. For all self-complementary graphs on n vertices, n is ____.

a) A multiple of 4
b) Even
c) Odd
d) Congruent to 0 mod 4, or, 1 mod 4

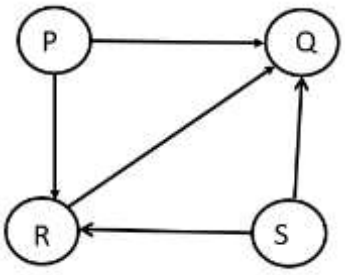
GATE CSE – 2014 MCQ SET - 1(1-MARKS)

Q24. Let G = (V,E) be a directed graph where V is the set of vertices and E the set of edges. Then which one of the following graphs has the same strongly connected components as G?

a) $G_1 = (V, E_1)$ where $E_1 = \{(u,v) \mid (u,v) \notin E\}$
b) $G_2 = (V,E_2)$ where $E_2 = \{(u,v) \mid (v,u) \in E\}$
c) $G_3 = (V,E_3)$ where $E_3 = \{(u,v) \mid$ there is a path of length ≤ 2 from u to v in $E\}$
d) $G_4 = (V_4, E)$ where V_4 is the set of vertices in G which are not isolated

GATE CSE – 2014 MCQ SET - 1(1-MARKS)

Q25. Consider the directed graph given below.



Which one of the following is TRUE?

a) The graph does not have any topological ordering.
b) Both PQRS and SRQP are topological.
c) Both PSRQ and SPRQ are topological orderings.
d) PSRQ is the only topological ordering.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2014 NAT (2-MARKS)

Q26. A cycle on n vertices is isomorphic to its complement. The value of n is _____.

GATE CSE – 2013 MCQ (1-MARKS)

Q27. Consider an undirected random graph of eight vertices. The probability that there is an edge between a pair of vertices is 1/2. What is the expected number of unordered cycles of length three?

a) 1/8
b) 1
c) 7
d) 8

GATE CSE – 2013 MCQ (2-MARKS)

Q28. The line graph L(G) of a simple graph G is defined as follows:
There is exactly one vertex v(e) in L(G) for each edge e in G.
For any two edges e and e' in G, L(G) has an edge between v(e) and v(e'), if and only if e and e' are incident with the same vertex in G.
Which of the following statements is/are TRUE?
(P) The line graph of a cycle is a cycle.
(Q) The line graph of a clique is a clique.
(R) The line graph of a planar graph is planar.
(S) The line graph of a tree is a tree.

a) P only
b) P and R only
c) R only
d) P, Q and S only

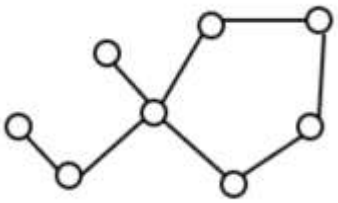
GATE CSE – 2013 MCQ (1-MARKS)

Q29. Which of the following statements is/are TRUE for undirected graphs?
P: Number of odd degree vertices is even.
Q: Sum of degrees of all vertices is even.

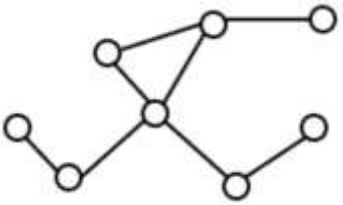
a) P only
b) Q only
c) Both P and Q
d) Neither P nor Q

GATE CSE – 2012 MCQ (2-MARKS)

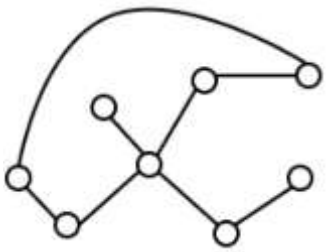
Q30. Which of the following graphs is isomorphic to



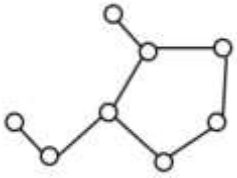
a)



b)

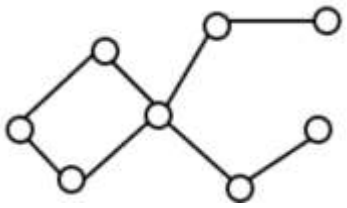


c)



Check Out Our GATE Courses & Secure Your Dream Rank!

d)



GATE CSE – 2012 MCQ (2-MARKS)

Q31. Let G be a complete undirected graph on 6 vertices. If vertices of G are labelled, then the number of distinct cycles of length 4 in G is equal to ____.

a) 15
b) 30
c) 45
d) 360

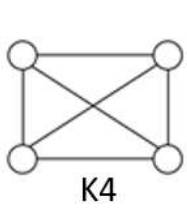
GATE CSE – 2012 MCQ (1-MARKS)

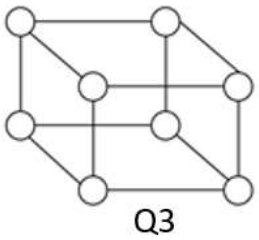
Q32. Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of bounded faces in any embedding of G on the plane is equal to ____.

a) 3
b) 4
c) 5
d) 6

GATE CSE – 2011 MCQ (1-MARKS)

Q33. K4 and Q3 are graphs with the following structures.





Which one of the following statements is TRUE in relation to these graphs?

a) K4 is planar while Q3 is not
b) Both K4 and Q3 are planar
c) Q3 is planar while K4 is not
d) Neither K4 nor Q3 is planar

GATE CSE – 2010 MCQ (1-MARKS)

Q34. Let G = (V, E) be a graph. Define $\xi(G) = \Sigma\ i_d * d$, where i_d is the number of vertices of degree d in G. If S and T are two different trees with $\xi(S) = \xi(T)$, then

a) $|S| = 2|T|$
b) $|S| = |T| - 1$
c) $|S| = |T|$
d) $|S| = |T| + 1$

GATE CSE – 2010 MCQ (2-MARKS)

Q35. The degree sequence of a simple graph is the sequence of the degrees of the nodes in the graph in decreasing order. Which of the following sequences can not be the degree sequence of any graph?

I) 7,6,5,4,4,3,2,1
II) 6,6,6,6,3,3,2,2
III) 7,6,6,4,4,3,2,2
IV) 8,7,7,6,4,2,1,1

a) I and II
b) III and IV
c) IV only
d) II and IV

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2009 MCQ (1-MARKS)

Q36. What is the chromatic number of an n-vertex simple connected graph which does not contain any odd length cycle ? Assume n ≥ 2.

a) 2
b) 3
c) n-1
d) n

GATE CSE – 2009 MCQ (1-MARKS)

Q37. Which one of the following is TRUE for any simple connected undirected graph with more than 2 vertices?

a) No two vertices have the same degree.
b) At least two vertices have the same degree.
c) At least three vertices have the same degree.
d) All vertices have the same degree.

GATE CSE – 2008 MCQ (2-MARKS)

Q38. Which of the following statements is true for every planar graph on n vertices?

a) The graph is connected
b) The graph is Eulerian
c) The graph has a vertex-cover of size at most 3n/4
d) The graph has an independent set of size at least n/3

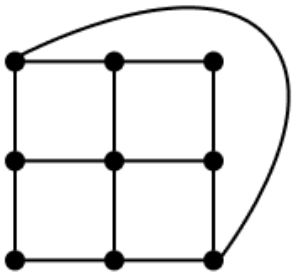
GATE IT – 2008 MCQ (2-MARKS)

Q39. G is a simple undirected graph. Some vertices of G are of odd degree. Add a node v to G and make it adjacent to each odd degree vertex of G. The resultant graph is sure to be

a) Regular
b) Complete
c) Hamiltonian
d) Euler

GATE IT – 2008 MCQ (1-MARKS)

Q40. What is the chromatic number of the following graph?



a) 2
b) 3
c) 4
d) 5

GATE CSE – 2007 MCQ (1-MARKS)

Q41. Let G be the non-planar graph with the minimum possible number of edges. Then G has

a) 9 edges and 5 vertices
b) 9 edges and 6 vertices
c) 10 edges and 5 vertices
d) 10 edges and 6 vertices

GATE CSE – 2007 MCQ (2-MARKS)

Q42. Which of the following graphs has an Eulerian circuit?

a) Any k-regular graph where k is an even number.
b) A complete graph on 90 vertices.
c) The complement of a cycle on 25 vertices.
d) None of the above.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE IT – 2006 MCQ (2-MARKS)

Q43. Consider the undirected graph G defined as follows. The vertices of G are bit strings of length n. We have an edge between vertex u and vertex v if and only if u and v differ in exactly one bit position (in other words, v can be obtained from u by flipping a single bit). The ratio of the chromatic number of G to the diameter of G is

a) $1/2^{n-1}$
b) 1/n
c) 2/n
d) 3/n

GATE IT – 2006 MCQ (1-MARKS)

Q44. If all the edge weights of an undirected graph are positive, then any subset of edges that connects all the vertices and has minimum total weight is a

a) Hamiltonian cycle
b) grid
c) hypercube
d) tree

GATE CSE – 2005 MCQ (2-MARKS)

Q45. Let G be a simple connected planar graph with 13 vertices and 19 edges. Then, the number of faces in the planar embedding of the graph is:

a) 6
b) 8
c) 9
d) 13

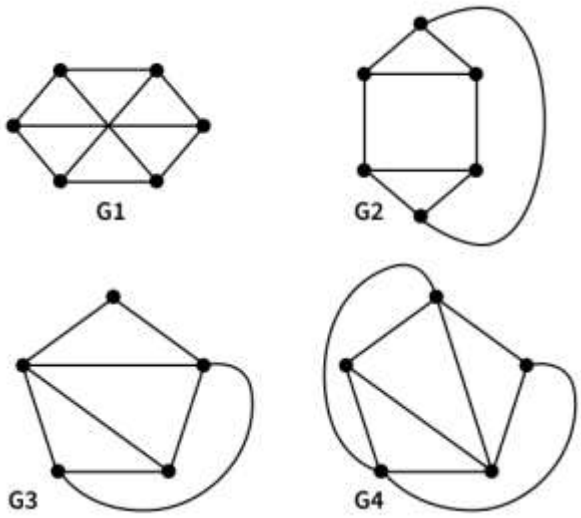
GATE CSE – 2005 MCQ (1-MARKS)

Q46. Let G be a simple graph with 20 vertices and 100 edges. The size of the minimum vertex cover of G is 8. Then, the size of the maximum independent set of G is:

a) 12
b) 8
c) less than 8
d) more than 12

GATE CSE – 2005 MCQ (2-MARKS)

Q47. Which one of the following graphs is NOT planar?



a) G1
b) G2
c) G3
d) G4

ANSWER KEY

1. (a)	**2. (34)**	**3. (c)**	**4. (2)**
5. (b.c)	**6. (a,b)**	**7. (36)**	**8. (24)**
9. (d)	**10. (a)**	**11. (b.c)**	**12.(11)**
13. (929)	**14.(7)**	**15. (d)**	**16.(109)**
17. (3)	**18. (18)**	**19. (16)**	**20. (4)**
21. (24)	**22. (b)**	**23. (d)**	**24. (b)**
25. (c)	**26. (5)**	**27. (c)**	**28. (a)**
29. (c)	**30. (b)**	**31. (c)**	**32. (d)**
33. (b)	**34. (c)**	**35. (d)**	**36. (a)**
37. (b)	**38. (c)**	**39. (d)**	**40. (b)**
41. (b)	**42. (c)**	**43. (c)**	**44. (d)**
45. (b)	**46. (a)**	**47. (a)**	

Check Out Our GATE Courses & Secure Your Dream Rank!

HINTS & SOLUTIONS

1. ANSWER:- A

We analyze each option given that:

- G(V,E) is a simple undirected graph.
- The minimum vertex cover size is k.
- S is a vertex cover of size k (so S is a minimum vertex cover)

We want a condition that forces a vertex v to belong to *every* minimum vertex cover of size k.

Option (A) Degree(v) $\geq k+1$

If $v \notin S$, then all its neighbors must be in S. But $\deg(v) \geq k+1$ means at least k+1 neighbors are needed in S, which is impossible since $|S|=k$. So v must be included.

Option (B) On a path of length k+1

A path of length k+1 needs only about (k+1)/2 vertices in a cover. Not forced.

Option (C) On a cycle of length k+1

A cycle needs about (k+1)/2 vertices. Not forced.

Option (D) In a clique of size k

A clique of size k needs only $k-1$ vertices in a cover. One vertex can be excluded.

2. ANSWER:- 34

For a path on n vertices, the number of matchings follows:

$M(n)=M(n-1)+M(n-2)$

Reason:

- Either the first vertex is not matched → reduce to path of $n-1$
- Or it is matched with its neighbor → reduce to path of $n-2$

Base cases:

$M(0)=1, M(1)=1$

Compute sequentially:

$M(2)=2$
$M(3)=3$
$M(4)=5$
$M(5)=8$
$M(6)=13$
$M(7)=21$
$M(8)=34$

So, the final answer is 34.

3. ANSWER:- C

- Each spanning tree of K_n has $n-1$ edges.
- The Jaccard coefficient is

$J=|T1 \cap T2|/|T1 \cup T2|$

- To minimize J, we minimize the intersection.

In a complete graph K_n (with $n>4$), it is possible to construct two edge-disjoint spanning trees (they share no common edges).

So:

$|T1 \cap T2|=0$

Hence,

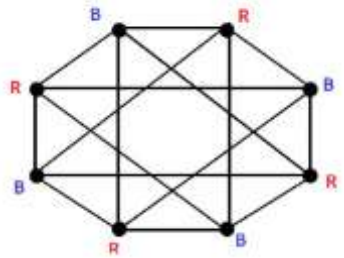
$J=0/(|T1 \cup T2|)=0$

So, the answer is c.

4. ANSWER:- 2

Theorem - graph is bipartite ↔ graph has no odd length cycles notice that graph doesn't have any odd length cycles so graph is bipartite.

A bipartite graph with atleast 1 edge always has a chromatic number of 2.



5. ANSWER:- B,C

First Option is false because chromatic number 'k' not necessarily means that graph contains a clique on 'k' vertices. For example, Chromatic Number of a Cycle Graph on odd vertices is 3 but it doesn't contain any complete subgraph on 3 vertices.

Second Option is True by virtue of Pigeon-Hole Principle. If there are 'k' color classes and n vertices, then one color class size is atleast n/k. Note that, a color class is nothing but an independent set.

Third Option is True because minimum number of edges between k color classes is C(K,2) or 'k choose 2'

Fourth Option is False because take a cycle graph on odd vertices, chromatic number is 3 but degree is not 3 for any vertex. Similarly, for any complete graph, degree for every vertex is n-1 but chromatic number is n

6. ANSWER:- A,B

Greedy means Assign the colour only if required otherwise use the existing assigned colours.

So here colour is getting assigned only if none of the neighbours have that colour and that assigning happening in greedy mode.

Check Out Our GATE Courses & Secure Your Dream Rank!

So it'll result in proper vertex colouring of G and also at max Chromatic Number will be <= 1+ Max Degree (There is a theorem on this.)

7. ANSWER:- 36

To make a disconnected graph with maximum edges, we can make a fully connected graph with (n-1) edges and isolate 1 vertex.
Now, no. of edges in fully connected graph with (n-1) edges will be: ${}^{(n-1)}C_2$.
Therefore, the maximum possible number of edges is:
${}^{9}C_2 = 9*8/2 = 36$

8. ANSWER:- 24

In a directed graph where edges only go from higher-numbered nodes to lower-numbered nodes, every node (except the root) must choose exactly one parent from the nodes higher than itself. Because edges only go downwards, choosing any higher node as a parent is guaranteed to form a valid tree without any cycles.

Counting the Choices

- Node 4 can only choose: 5 (1 choice)
- Node 3 can choose: 4 or 5 (2 choices)
- Node 2 can choose: 3, 4, or 5 (3 choices)
- Node 1 can choose: 2, 3, 4, or 5 (4 choices)

Multiply the independent choices together:

$$\text{Total Trees} = 1 \times 2 \times 3 \times 4 = 24$$

9. ANSWER:- D

Take a cycle graph on 3 vertices i.e. C_3.
Write the adjacency matrix A of C_3, find A^3. The trace of A^3 is 6, but we have only one 3-cycle (3 length cycle) in C_3.
So, we have to divide the trace of A^3 by 6 to find out the number of 3-cycles. That's our answer.
Option A^3 divided by 6 .

10. ANSWER:- A

In an adjacency matrix,
the $(i,j)^{th}$ entry in A^K denotes the number of walks of length k from i to j
so in A^2 , the $(i,i)^{th}$ entry denotes the number of walks of length 2 from i to i i.e. the degree of the vertex 'i'
hence The diagonal entries of A^2 are the degrees of the vertices of the graph.

11. ANSWER:- B,C

- False, The graph does have a strongly connected component.
- True, because of cycle in the bottom left corner of the graph.
- True
- False, not a strongly connected graph.

12. ANSWER:- 11

$|V| - |E| + |R| = 2$ [According to Euler's Formula]
Here, $|V| = 8$, $|E| = ?$, $|R| = 5$
$8 - |E| + 5 = 2$
$|E| = 11$

13. ANSWER:- 929

Let Q(V) denote the quality-score of vertex V.
Q(S) = 1 (Given)
Q(C) = 1 (S → C)
Q(F) = 1 * 9 (S → C → F)
Q(A) = 9 (S → A)
Q(D) = 9*1 (S → A → D)
Q(G) = 9 * 1 * 9 (S → A → D → G)
Q(B) = 9 * 1 (S → A → B)
Q(E) = 9 * 1 * 9 (S → A → D → E)
Q(T) = 9*1*9*9 (S → A → D → E → T)
Sum of quality-score of all vertices,
= 1 + 1 + 9 + 9 + 9 + 81 + 9 + 81 + 729
= 929

14. ANSWER:- 7

In $k_{3,4}$ we have two sets with sizes as 3,4 (its a complete bipartite graph).
The vertex in the set having size 3 has 4 edges and is connected to the set having 4 vertices. So, the edge color of G is max(3,4) i.e. 4.
When a vertex is added to the graph having 7 vertices (K 3,4 has 7 vertices), we would be having 7 edges associated to that new vertex.Now, as per the edge coloring
"no two adjacent edges have same color". So,as the new vertex with 7 edges need to be colored with 7 colors, the edge color of graph G is 7.

15. ANSWER:- D

A Hamiltonian cycle is a cycle having a closed loop on a graph in which every node(vertex) is visited exactly once.
The Total number of Hamiltonian cycles in a Complete graph is given by: (n-1)!/2, where n is number of vertices in the graph.

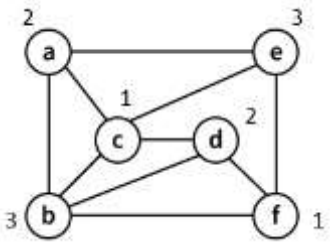
Check Out Our GATE Courses & Secure Your Dream Rank!

16. ANSWER:- 109

G is a graph with 100! vertices. Label of each vertex is obtained from a distinct permutation of numbers "1, 2, … 100".
There exists an edge between two vertices iff the label of 'u' is obtained by swapping two adjacent numbers in the label of 'v'.
The sets of the swapping numbers be (1,2) (2, 3) (3, 4) … (99,100).
The no. of such sets are 99 i.e., no. of edges = 99.
As this is regular graph, each vertex has '99' edges correspond to it.
So, degree of each vertex = 99 = y.
As the vertices are connected, the number of components formed = 1 = z
y + 10z = 99 + 10(1) = 109

17. ANSWER:- 3

Chromatic number of the graph is "3"



18. ANSWER:- 18

We know that a Tree T with n vertices has n-1 number of edges. So, Number of edges in T is : 10-1 = 9.
Now, by Handshaking Lemma, Sum of Degree of vertices is equal to twice the number of edges , so 2E = Sum of Degree.
Thus, Sum of degree = 2*9 = 18.

19. ANSWER:- 16

An undirected graph 'G' is given with 'n' vertices & 25 edges.
Degree of each vertex ≥ 3
Sum of degree vertices = 2 (no. of edges)
So, |v| = 2|E|
The relation Between max and min degree of a graph is given by:
$m \leq 2|E| / |v| \leq M$
Given minimum degree = 3.
So, $3 \leq 2|E| / |v|$
$3|v| \leq 2|E|$
$3(n) \leq 2(25)$
$n \leq 50/3$
$n \leq 16.6$
So, n = 16

20. ANSWER:- 4

The 4-colour theorem states that we require at most 4 colors to color any planar graph.
Here it asks about the sufficient number of colors.
So, we can color any planar graph with the 4 colors in the worst case.

21. ANSWER:- 24

By Euler formula for connected planar graph,

$$n - e + f = 2$$

$10 - 3f/2 + f = 2$ (An edge is part of two faces)

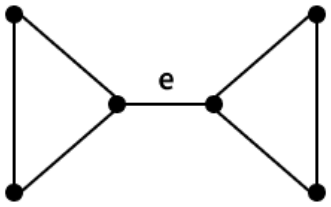
$$f/2 = 8$$

$$f = 16$$

$$e = 3f/2 = 24$$

22. ANSWER:- B

Since, every edge in a tree is a bridge
∴ Option (A) is not True.
As, every edge in a complete graph k_n ($n \geq 3$) is not a bridge ⇒
∴ (C) is not True.
Let us consider the graph G:



The given graph has a bridge i.e., edge 'e' and a cycle of length '3'.
So, (D) is not True.
Since, in a cycle every edge is not a bridge
∴ (B) is true and is the correct answer.

23. ANSWER:- D

We have been given an n vertex self-complementary graph.
Now, an n vertex graph which is self complementary has exactly half the number of edges of the complete graph, i.e.,it has n(n-1)/4 number of edges.
Since n(n – 1) must be divisible by 4, n must be congruent to 0 or 1 modulus 4.

Check Out Our GATE Courses & Secure Your Dream Rank!

24. ANSWER:- B

G(V, E) is a directed graph.

A) $G_1 = (V, E_1)$ where $E_1 = \{(u,v) | (u,v) \notin E\}$

If (u, v) does not belong to the edge set 'E', then it indicates there are no edges present. So, it is not connected.

B) $G_2 = (V, E_2)$ where $E_2 = \{(u,v) \mid (v,u) \in E\}$

Given that 'G' is a directed graph, i.e., it has path from each vertex to every other vertex in the graph.

Though here direction is changed from (u,v) to (v, u), it is still a connected component and is same as 'G'.

C) $G_3 = (V, E_3)$ where $E_3 = \{(u,v) \mid$ there is a path of length≤ 2 from u to v in E$\}$

This is not always true.

it can add extra edges.

D) $G_4 = (V_4, E)$ where V_4 is the set of vertices in G which are not isolated.

If 'G' has same 'x' number of isolated vertices, one strongly connected component then no. of SCC = x + 1

G_4 contain only '1' component, which is not same as G.

25. ANSWER:- C

No cycles are there in the graph, so topological orderings do exist.

We can consider P & S as starting vertex, followed by R & Q.

Hence, PSRQ & SPRQ are the topological orderings.

26. ANSWER:- 5

According to Complement of a graph, we need to remove the existing edges from the graph and place the new edges which were not present in the actual graph.

In a cycle of n vertices, each vertex is connected to the other two vertices. So each vertex's degree is 2.

When we complement it, each vertex will be connected to the remaining n-3 vertices (one is self and two other vertices in the actual graph).

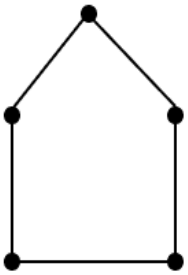
As given in the question,

n-3 =2

n = 3 + 2

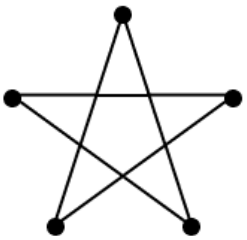
n = 5

Cycle of 5 vertices is



Graph – 1

Complement of the above graph 1 is



Graph - 2

Graph 1 and Graph 2 are complements of each other.

So, the value of n is 5.

OR

Given the graph is cyclic what we can deduce is that the number of vertices= number of edges=n

Now, it is isomorphic to its complement so no. of edges in its complement graph will also be n .

No. of edges in graph G + No. Of edges in graph $G' =$ no. Of edges in the complete graph $\left(\frac{n(n-1)}{2}\right)$.

$$n + n = \frac{n(n-1)}{2}$$
$$2n = \frac{n(n-1)}{2}$$
$$4n = n(n-1)$$
$$4n = n^2 - n$$
$$n^2 - 5n = 0$$
$$n(n-5) = 0$$
$$n = 5$$

27. ANSWER:- C

1. Find the total number of possible 3-cycles

A cycle of length three is formed by choosing any 3 distinct vertices out of the total 8 vertices available in the graph.

The number of ways to choose 3 vertices from 8 is given by the combination formula $\binom{n}{r}$:

$$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

Check Out Our GATE Courses & Secure Your Dream Rank!

So, there are 56 potential 3 -cycles in the graph.

2. Find the probability that a specific 3-cycle exists

For any specific set of 3 vertices to form a cycle, all 3 edges connecting them must exist simultaneously.

- The probability of any single edge existing is $P(e) = \frac{1}{2}$.
- Since edge existences in a standard random graph (Erdös-Rényi model) are independent events, we multiply the individual probabilities:

$$P(\text{ all 3 edges exist }) = \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) = \frac{1}{8}$$

3. Calculate the Expected Value

By the linearity of expectation, the total expected number of 3 -cycles is the total number of potential cycles multiplied by the probability that any single one exists:

$$E[X] = \text{ Total potential cycles } \times P \text{ (cycle exists)}$$

$$E[X] = 56 \times \frac{1}{8} = 7$$

28. ANSWER:- A

P) True.
Because every edge in the cycle graph will become a vertex in the new graph L(G) and every vertex of cycle graph will become an edge in the new graph.
R) False.
We can give an counter- example. Let a graph G has 5 vertices and 9 edges and it is planar. Assume degree of one vertex is 2 and of all others are 4.
Now, L(G) has 9 vertices (because G has 9 edges) and 25 edges. But for a graph to be planar,
$|E| \leq 3|v| - 6$
For 9 vertices $|E| \leq 3 \times 9 - 6$
$\Rightarrow |E| \leq 27 - 6$
$\Rightarrow |E| \leq 21$.
But L(G) has 25 edges and so is not planar.
As option (R) is false, option (B) & (C) are eliminated.

29. ANSWER:- C

P is true: For undirected graph as adding an edge always increases degree of two vertices by 1.
Q is true: If we consider sum of degrees and subtract all even degrees, we get an even number because every edge increases the sum of degrees by 2. So total number of odd degree vertices must be even.

30. ANSWER:- B

(A) 3-cycle graph not in original one.
(B) is the correct graph as we have 5-cycles & max degree is 4.
(C) As the Original graph doesn't have a degree of 3, so it is incorrect.
(D) 4-cycles not in original one.

31. ANSWER:- C

There can be total ${}^6C_4$ ways to pick 4 vertices from 6. The value of ${}^6C_4$ is 15.

Note that the given graph is complete so any 4 vertices can form a cycle.

There can be 6 different cycle with 4 vertices. For example, consider 4 vertices as a, b, c and d. The three distinct cycles are

cycles should be like this
(a, b, c, d,a)
(a, b, d, c,a)
(a, c, b, d,a)
(a, c, d, b,a)
(a, d, b, c,a)
(a, d, c, b,a)

and

(a, b, c, d,a) and (a, d, c, b,a)
(a, b, d, c,a) and (a, c, d, b,a)
(a, c, b, d,a) and (a, d, b, c,a)
are same cycles.

So total number of distinct cycles is (15*3) = 45.

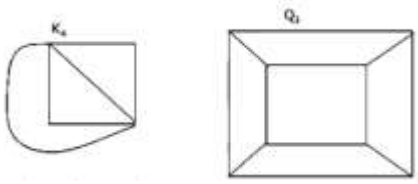
32. ANSWER:- D

As the given graph is planar, then we have to consider Euler's formula which is:
v-e+f=2
Given 10 vertices & 15 edges, so
10-15+f=2
f=2+15-10
f=7
There will be an unbounded face always.
So, the number of bounded faces = 6.

33. ANSWER:- B

A Graph is said to be planar if it can be drawn in a plane without any edges crossing each other.

Check Out Our GATE Courses & Secure Your Dream Rank!



34. ANSWER:- C

Sum of degrees in a graph = 2|E|, as each edge contributes two to the sum of degrees. So, when the sum of degrees is the same, the number of edges must also be the same.

Trees with equal no of edges will have equal no of vertices as :
No of Edges = No of vertices - 1 (in a tree.)

So, should be |S| = |T|

35. ANSWER:- D

By Havel–Hakimi Theorem we can check the graphical sequence :

I. 7, 6, 5, 4, 4, 3, 2, 1
→5, 4, 3, 3, 2, 1, 0
→ 3, 2, 2, 1, 0, 0
→ 1, 1, 0, 0, 0
→0, 0, 0, 0 so it is graphical.

II. 6, 6, 6, 3, 3, 2, 2
→ 5, 5, 5, 2, 2, 1, 2 (arrange in descending order)
→ 5, 5, 5, 2, 2, 2, 1
→ 4, 4, 1, 1, 1, 1
→ 3, 0, 0, 0, 1
we cannot continue to get all 0's, so it is not graphical.

III. 7, 6, 6, 4, 4, 3, 2, 2
→ 5, 5, 3, 3, 2, 1, 1
→ 4, 2, 2, 1, 1, 0
→ 1, 1, 0, 0, 0
→ 0, 0, 0, 0
so it is graphical.

IV. 8, 7, 7, 6, 4, 2, 1, 1, here degree of a vertex is 8 and total number of vertices are 8, which is impossible, hence it is not graphical.

Hence, only option (I) and (III) are graphic sequence and answer is option-(D)

36. ANSWER:- A

A simple connected graph with no odd-length cycle is, by definition, a bipartite graph.

A graph contains no odd cycle if and only if it is bipartite.

In a bipartite graph:

- The vertex set can be divided into two disjoint sets
- No two vertices within the same set are adjacent
- Therefore, it can be properly colored using only 2 colors

Since the graph is:

- Simple
- Connected
- Has $n \geq 2$ vertices
- Contains no odd cycle

It must be a connected bipartite graph, and any connected bipartite graph with at least one edge requires exactly 2 colors.

37. ANSWER:- B

There are n vertices and at least (n-1) edges. So, for each vertex, the degree should range from 1 (since the graph is connected) to (n-1) (since the graph is simple).

But we have n such vertices- filling n things with (n-1) numbers.

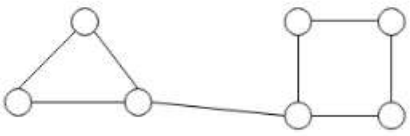
$$\left\lceil \frac{n}{n-1} \right\rceil = \lceil 1.\sim \rceil = 2$$

So, at least 2 of them must be equal (pigeonhole principle)

38. ANSWER:- C

A) **Incorrect**. Consider the following disconnected planar graph.

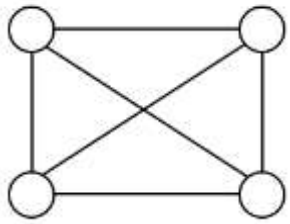
B) **Incorrect**. A graph is Eulerian if all its vertices have even degree, but a planar graph can have vertices with odd degree.



C) **Correct**. Any planar graph can be vertex-colored with at most 4 colors (by the 4-color theorem) such that no two adjacent vertices have the same color.

Check Out Our GATE Courses & Secure Your Dream Rank!

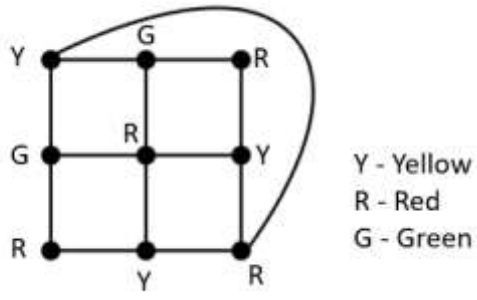
Consider a planar graph with nnn vertices having chromatic number 4. Then we have at most 4 sets of vertices, one for each color, each having size n/4 (assuming they are evenly distributed). If we remove one set, the remaining vertices still include at least one endpoint of every edge. Hence, we obtain a vertex cover of size
$n-(n/4)=3n/4$.
D) **Incorrect**. Consider the K_4 graph. It has an independent set of size 1, which is less than 4/3.



39. ANSWER:- D

In any simple undirected graph, total degree of all vertices is even (since each edge contributes 2 degrees). So number of vertices having odd degrees must be even, otherwise their sum would have been odd, making total degree also odd.
Now Single vertex v is connected to all these even number of vertices (which have odd degrees). So degree of v is also even. Moreover, now degree of all vertices which are connected to v is increased by 1, hence vertices which had odd degree earlier now have even degree.
So now, all vertices in graph have even degree, which is necessary and sufficient condition for euler graph. So (D) is correct.

40. ANSWER:- B



The chromatic number of a graph is the smallest number of colors needed to color the vertices so that no two adjacent vertices share the same color. Hence minimum number of colors needed to color given graph is equal to 3

For odd length cycles we need minimum 3 colors for vertex coloring and for even length cycles we need just 2.
So, the correct answer is b.

41. ANSWER:- B

Non planar graph with smallest number of edges is $K_{3,3}$; it has 9 edges and 6 vertices
K_5 is also non planar but it has 10 edges and 5 vertices.

42. ANSWER:- C

A connected graph G has an Eulerian circuit if and only if every vertex in the graph has an even degree.
Option A:In a k-regular graph:
- Every vertex has degree k.
- If k is even → all vertices have even degree.

If the graph is connected, it is Eulerian.
However, the option does not guarantee connectivity.
A disconnected k-regular graph (with multiple components) would not have a single Eulerian circuit.
So this is not always true.
Option B:
For a complete graph K_n:
Each vertex has degree $n-1$.
Here:
$n=90 \Rightarrow \text{degree}=89$
Since 89 is odd, all vertices have an odd degree.
So no Eulerian circuit.
Option C:
Cycle C_{25}:
- Each vertex has degree 2.

In complement:
Total possible degree = 24
So complement degree:
$24-2=22$
Every vertex has degree 22 (even).
Also, the complement of C_{25} (for $n \geq 5$) is connected.
Thus:
- Connected
- All degrees even

Eulerian circuit exists.
Option D:
Incorrect since (c) works.
So, the correct answer is C

Check Out Our GATE Courses & Secure Your Dream Rank!

43. ANSWER:- C

Bipartite graph:-
A bipartite graph is a graph G(V,E) where vertices can be decomposed into two disjoint sets such that no two vertices within the same set are adjacent.
Diameter of a graph:-
The longest shortest path in between any two vertices of a graph The given graph is a bipartite graph => chromatic number is equals to 2 The diameter of graph is equals to n because at most we need to traverse n-1 edges. The ratio = 2/n.
So, the correct answer is c.

44. ANSWER:- D

Hamiltonian cycle : This is a cycle. A cycle will not only connect all vertices, it will have 1 extra edge than necessary. So I can just remove that edge & get better cost "subset of edges" which connect all vertices. So, this is FALSE.
grid : A grid graph has cycles and so this is FALSE for same reason as option A.
Hypercube : A hypercube graph also has cycles. So, this also is FALSE.
Tree : This is answer. We need to have Minimum spanning tree to be exact.
"If all the edge weights of an undirected graph are positive, then any subset of edges that connects all the vertices and has minimum total weight is a Minimum Spanning Tree".
So, the correct answer is tree

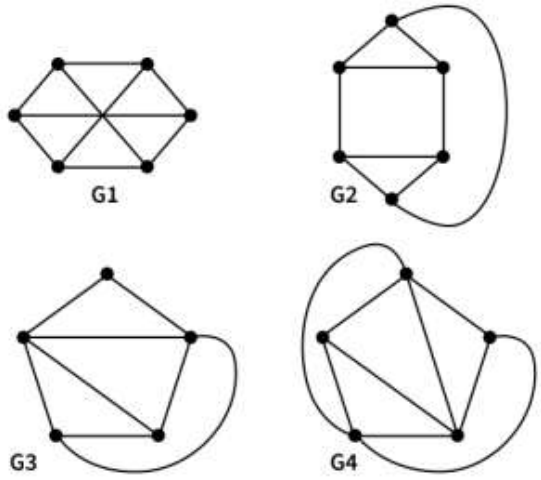
45. ANSWER:- B

The number of faces in the planar embedding of the graph is
f=e-n+(k+1)
f=number of faces,
e=number of edges,
n=number of vertices,
k=number of connected components,
for connected graph k=1, so
f=e-n+(1+1)
f=e-n+2
f=19-13+2
f=8

46. ANSWER:- A

Vertex cover: A set of vertices such that each edge of the graph is incident to at least one vertex of the set.
Therefore, removing all the vertices of the vertex cover from the graph results in an isolated graph and the same set of nodes would be the independent set in the original graph.
Size of minimum vertex cover = 8
Size of maximum independent set = 20-8 =12
Therefore, the correct answer would be (A).

47. ANSWER:- A



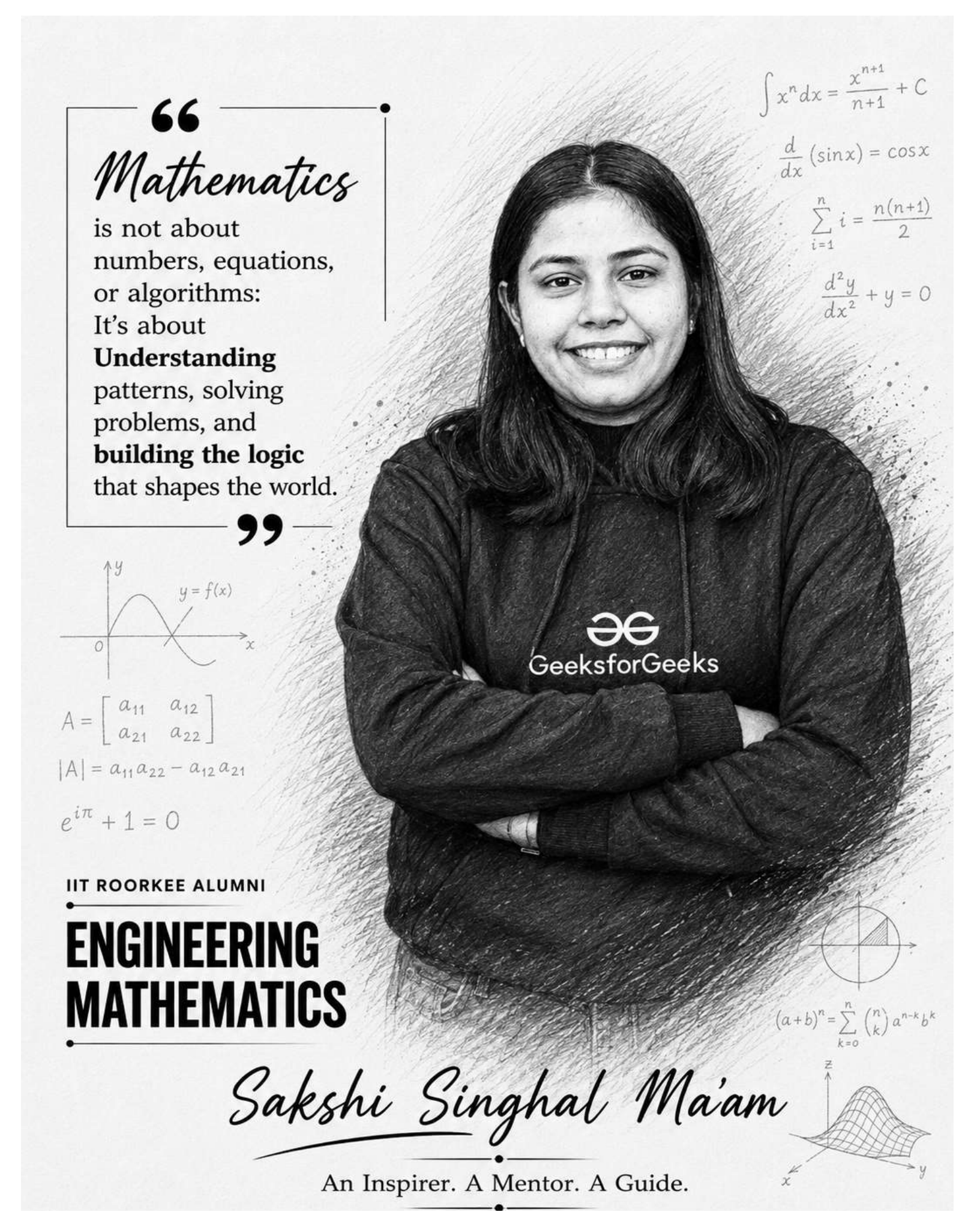
We can form a planar graph for all except G1. Hence G1 is not a planar graph.

Check Out Our GATE Courses & Secure Your Dream Rank!

ENGINEERING MATHEMATICS

"Scan Here for Video Solution"

Link to GfG corresponding Short Notes: GATE


"Mathematics is not about numbers, equations, or algorithms: It's about **Understanding** patterns, solving problems, and **building the logic** that shapes the world."
$\int x^n dx = \frac{x^{n+1}}{n+1} + C$
$\frac{d}{dx}(\sin x) = \cos x$
$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$
$\frac{d^2y}{dx^2} + y = 0$
$y = f(x)$
$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$
$|A| = a_{11}a_{22} - a_{12}a_{21}$
$e^{i\pi} + 1 = 0$
GeeksforGeeks
IIT ROORKEE ALUMNI
ENGINEERING MATHEMATICS
$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$
Sakshi Singhal Ma'am
An Inspirer. A Mentor. A Guide.

1 PROBABILITY & STATISTICS

GATE CSE - 2026 NAT SET - 1(2-MARKS)

Q1. Let X be a random variable which takes values in the set {1, 2, 3, 4, 5, 6, 7, 8}. Further,
Pr(X = 1) = Pr(X = 2) = Pr(X = 5) = Pr(X = 7) = 1/6
Pr(X = 3) = Pr(X = 4) = Pr(X = 6) = Pr(X = 8) = 1/12
The expected value of X, denoted by E[X], is equal to ____. (Round off to two decimal places)

GATE CSE - 2026 MCQ SET - 1(2-MARKS)

Q2. An urn contains one red ball and one blue ball. At each step, a ball is picked uniformly at random from the urn, and this ball together with another ball of the same color is put back in the urn. The probability that there are equal number of red and blue balls after two steps is

(a) 1/4
(b) 1/3
(c) 1/2
(d) 2/3

GATE CSE - 2026 MCQ SET - 2(1-MARKS)

Q3. The probability density function $f(x)$ of a random variable X which takes real values is

$$f(x) = \frac{1}{3\sqrt{2\pi}} \exp\left(-\frac{x^2}{18}\right), x \in (-\infty, +\infty)$$

Which one of the following statements is correct about the random variable X ?

(a) X is an exponential random variable
(b) X is a normal random variable
(c) X is a Poisson random variable
(d) X is a uniform random variable

GATE CSE - 2026 NAT SET - 2(2-MARKS)

Q4. Suppose an unbiased coin is tossed 6 times. Each coin toss is independent of all previous coin tosses. Let E1 be the event that among the second, fourth, and sixth coin tosses, there are at least two heads. Let E2 be the event that among the first, second, third, and fifth coin tosses, there are equal number of heads and tails.
The conditional probability P(E1 | E2) is equal to the _____.(rounded off to one decimal place)

GATE CSE - 2025 NAT SET - 1(1-MARKS)

Q5. A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, P(head) = 0.5 and for a fake coin P(head) = 1.
You pick a coin at random and toss it twice, and get two heads.
The probability that the coin you have chosen is the fake coin is _____ (rounded off to two decimal places).

GATE CSE - 2025 NAT SET - 2(2-MARKS)

Q6. The unit interval (0,1) is divided at a point chosen uniformly distributed over (0,1) into two disjoint subintervals.
The expected length of the subinterval that contains 0.4 is _____ (rounded off to two decimal places).

GATE CSE - 2024 NAT SET - 1(2-MARKS)

Q7. A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement.
Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is _____.

GATE CSE - 2024 MSQ SET - 1(1-MARKS)

Q8. Let A and B be two events in a probability space with P(A) = 0.3, P(B) = 0.5, and P(A∩B) = 0.1 Which of the following statements is/are TRUE?

(a) The two events A and B are independent
(b) $P(A \cup B) = 0.7$
(c) $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B
(d) $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B, respectively

GATE CSE - 2024 MCQ SET - 2(1-MARKS)

Q9. When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (i.e., 1, 2, 3, 4, 5, and 6) is:

(a) 1/324
(b) 5/324
(c) 7/324
(d) 11/324

GATE CSE – 2024 MCQ SET- 2 (2-MARKS)

Q10. Let x and y be random variables, not necessarily independent, that take real values in the interval [0,1]. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}$, $\bar{y}$, $\bar{z}$, respectively. Which one of the following statements is TRUE?

(a) $\bar{z} = \bar{x}\,\bar{y}$
(b) $\bar{z} <= \bar{x}\,\bar{y}$
(c) $\bar{z} >= \bar{x}\,\bar{y}$
(d) $\bar{z} <= \bar{x}$

GATE CSE - 2023 MSQ (2-MARKS)

Q11. Two fair coins are tossed. Let A = HEAD on both throws, B = HEAD on the first throw, C = HEAD on the second throw.
Which of the following statements is/are TRUE?

(a) A and B are independent
(b) A and C are independent
(c) B and C are independent
(d) P(B|C) = P(B)

GATE CSE - 20221 MCQ (2-MARKS)

Q12. In an examination, a student can choose the order in which two questions (QuesA and QuesB) must be attempted.

- If the first question is answered wrong, the student gets zero marks.
- If the first question is answered correctly and the second question is not answered correctly, the student gets the marks only for the first question.
- If both the questions are answered correctly, the student gets the sum of the marks of the two questions.

The following table shows the probability of correctly answering a question and the marks of the question respectively.

question	probabilty of answering correctly	marks
QuesA	0.8	10
QuesB	0.5	20

Assuming that the student always wants to maximize her expected marks in the examination, in which order should she attempt the questions and what is the expected marks for that order (assume that the questions are independent)?

(a) First QuesA and then QuesB. Expected marks 14.
(b) First QuesB and then QuesA. Expected marks 14.
(c) First QuesB and then QuesA. Expected marks 22.
(d) First QuesA and then QuesB. Expected marks 16.

GATE CSE - 2021 MCQ SET - 2(2-MARKS)

Q13. A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls

in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?

(a) r/(r+b)
(b) r/(r+b+3)
(c) (r+3)/(r+b+3)
(d) $\left(\frac{r}{r+b}\right)\left(\frac{r+1}{r+b+1}\right)\left(\frac{r+2}{r+b+2}\right)\left(\frac{r+3}{r+b+3}\right)$

GATE CSE - 2021 MCQ SET - 1(2-MARKS)

Q14. Consider the following two statements:
S1: There exist random variables X and Y such that $(\mathbb{E}[(X–E(X))(Y–E(Y))])^2 > Var[X]\cdot Var[Y]$
S2: For all random variables X and Y, $Cov[X,Y] = \mathbb{E}[|X– \mathbb{E}[X]|\cdot|Y– \mathbb{E}[Y]|]$
Which one of the following choices is correct?

(a) Both S_1 and S_2 are true
(b) S_1 is true, but S_2 is false
(c) S_1 is false, but S_2 is true
(d) Both S_1 and S_2 are false

GATE CSE - 2021 NAT SET - 1(1-MARKS)

Q15. The lifetime of a component of a certain type is a random variable whose probability density function is exponentially distributed with parameter 2 . For a randomly picked component of this type, the probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is ___ .

GATE CSE - 2021 NAT SET - 2(1-MARKS)

Q16. For a given biased coin, the probability that the outcome of a toss is a head is 0.4 . This coin is tossed 1,000 times. Let X denote the random variable whose value is the number of times that head appeared in these 1,000 tosses. The standard deviation of X (rounded to 2 decimal place) is ___

GATE CSE - 2018 NAT (1-MARKS)

Q17. Two people, P and Q, decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6 . The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q. Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is ___

GATE CSE - 2017 MCQ SET - 2(2-MARKS)

Q18. P and Q are considering to apply for a job. The probability that P applies for the job is $\frac{1}{4}$, the probability that P applies for the job given that Q applies for the job is $\frac{1}{2}$, and the probability that Q applies for the job given that P applies for the job is $\frac{1}{3}$. Then the probability that P does not apply for the job given that Q does not apply for this job is

(a) $\left(\frac{4}{5}\right)$
(b) $\left(\frac{5}{6}\right)$
(c) $\left(\frac{7}{8}\right)$
(d) $\left(\frac{11}{12}\right)$

GATE CSE - 2017 MCQ SET - 2(2-MARKS)

Q19. In hypothesis testing, N independent tests are conducted. Each test independently has a probability β of a Type II error (false negative).
What is the expected number of tests that result in a Type II error?

(a) Nβ(1−β)
(b) Nβ
(c) N(1−β)
(d) Not expressible in terms of N and β alone

GATE CSE - 2017 NAT SET - 2(2-MARKS)

Q20. If a random variable X has a Poisson distribution with mean 5, then E[(X+2)²] equals _____.

GATE CSE - 2017 NAT SET - 1(1-MARKS)

Q21. Let X be a Gaussian random variable with mean 0 and variance σ^2. Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b. The median of Y is ___ .

GATE CSE - 2016 NAT SET - 1(2-MARKS)

Q22. Consider the following experiment.
Step 1. Flip a fair coin twice.
Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.
Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.
The probability that the output of the experiment is Y is (up to two decimal places)

GATE CSE - 2016 NAT SET - 1(2-MARKS)

Q23. A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is ___ .

GATE CSE - 2016 NAT SET - 2(1-MARKS)

Q24. Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7 , and given that it is of Type 2 is 0.4 . The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is ___ .

GATE CSE - 2015 NAT SET - 1(2-MARKS)

Q25. Suppose X_i for $i = 1,2,3$ are independent and identically distributed random variables whose probability mass functions are $\Pr[X_i = 0] = \Pr[X_i = 1] = \frac{1}{2}$ for $i = 1,2,3$. Define another random variable $Y = X_1 X_2 \oplus X_3$, where $\oplus$ denotes XOR. Then $\Pr[Y = 0 \mid X_3 = 0] =$ ___ .

GATE CSE - 2014 NAT SET - 3(2-MARKS)

Q26. Let S be a sample space and two mutually exclusive events A and B be such that $A \cup B = S$. If P(.) denotes the probability of the event, the maximum value of $P(A)P(B)$ is ___ .

GATE CSE - 2014 NAT SET - 2(2-MARKS)

Q27. The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2,3 or 5 is ____.

GATE CSE - 2014 NAT SET - 1(2-MARKS)

Q28. Four fair six-sided dice are rolled. The probability that the sum of the results is 22 is X/1296.
The value of X is _____.

GATE CSE - 2014 NAT SET - 2(1-MARKS)

Q29. The security system at an IT office is composed of 10 computers of which exactly four are working. To check whether the system is functional, the officials inspect four of the computers picked at random (without replacement). The system is deemed functional if at least three of the four computers inspected are working. Let the probability that the system is deemed functional be denoted by p. Then $100p =$ ___ .

GATE CSE - 2014 MCQ SET - 2(1-MARKS)

Q30. The number of distinct positive integral factors of 2014 is _____.

(a) 7
(b) 8
(c) 9
(d) 10

GATE CSE - 2013 MCQ (1-MARKS)

Q31. Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM , and p has a Poisson distribution with mean 3 . What is the probability of observing fewer than 3 cars during any given minute in this interval?

(a) $\frac{8}{(2e^3)}$
(b) $\frac{9}{(2e^3)}$
(c) $\frac{17}{(2e^3)}$
(d) $\frac{26}{(2e^3)}$

GATE CSE - 2012 MCQ (2-MARKS)

Q32. Suppose a fair six-sided die is rolled once. If the value on the die is 1,2 , or 3 , the die is rolled a second time. What is the probability that the sum total of values that turn up is at least 6 ?

(a) $\frac{10}{21}$
(b) $\frac{5}{12}$
(c) $\frac{2}{3}$
(d) $\frac{1}{6}$

GATE CSE - 2012 MCQ (1-MARKS)

Q33. Consider a random variable X that takes values +1 and -1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and +1 are

(a) 0 and 0.5
(b) 0 and 1
(c) 0.5 and 1
(d) 0.25 and 0.75

GATE CSE – 2011 MCQ (2-MARKS)

Q34. If Two fair coins are flipped and at least one of the outcome is known to be a head.
What is the probability that both outcomes are heads?

(a) 1/3
(b) 1/4
(c) 1/2
(d) 2/3

GATE CSE – 2011 MCQ (2-MARKS)

Q35. Consider a finite sequence of random values X={x_1,x_2…..x_n}. Let μ_x be the mean and σ_x be the standard deviation of X. Let another finite sequence Y of equal length be derived from this as $y_i = a * x_i + b$, where a and b are positive constants. Let μ_y be the mean and σ_y be the standard deviation of this sequence.

Which one of the following statements is INCORRECT?

(a) Index position of mode of X in X is the same as the index position of mode of Y in Y
(b) Index position of median of X in X is the same as the index position of median of Y in Y
(c) $\mu_y = a\mu_x + b$
(d) $\sigma_y = a\sigma_x + b$

GATE CSE – 2011 MCQ (2-MARKS)

Q36. A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

(a) $\left(\frac{1}{5}\right)$
(b) $\left(\frac{4}{25}\right)$
(c) $\left(\frac{1}{4}\right)$
(d) $\left(\frac{2}{5}\right)$

GATE CSE – 2011 MCQ (2-MARKS)

Q37. If the difference between the expectation of the square of a random variable ($E[X^2]$) and the square of the expectation of the random variable $(E[X])^2$ is denoted by R, then

(a) $R = 0$
(b) $R < 0$
(c) $R \geq 0$
(d) $R > 0$

GATE CSE – 2010 MCQ (2-MARKS)

Q38. What is the probability that a divisor of 10^{99} is a multiple of 10^{96}?

(a) $\left(\frac{1}{625}\right)$
(b) $\left(\frac{4}{625}\right)$
(c) $\left(\frac{12}{625}\right)$
(d) $\left(\frac{16}{625}\right)$

GATE CSE – 2010 MCQ (2-MARKS)

Q39. Consider a company that assembles computers. The probability of a faulty assembly of any computer is p. The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q. What is the probability of a computer being declared faulty?

(a) $pq + (1-p)(1-q)$
(b) $(1-q)p$
(c) $(1-p)q$
(d) pq

GATE CSE – 2009 MCQ (2-MARKS)

Q40. An unbalanced dice (with 66 faces, numbered from 11 to 66) is thrown. The probability that the face value is odd is 90% of the probability that the face value is even. The probability of getting any even numbered face is the same. If the probability that the face is even given that it is greater than 33 is 0.75, which one of the following options is closest to the probability that the face value exceeds 33?

(a) 0.453
(b) 0.468
(c) 0.485
(d) 0.492

GATE CSE – 2008 MCQ (2-MARKS)

Q41. Let X be a random variable following normal distribution with mean +1 and variance 4 . Let Y be another normal variable with mean -1 and variance unknown. If $P(X \leq -1) = P(Y \geq 2)$, the standard deviation of Y is

(a) 3
(b) 2
(c) $\sqrt{2}$
(d) 1

GATE CSE – 2008 MCQ (2-MARKS)

Q42. Aishwarya studies either computer science or mathematics everyday. If she studies computer science on a day, then the probability that she studies mathematics the next day is 0.6 . If she studies mathematics on a day, then the probability that she studies computer science the next day is 0.4 . Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday?

(a) 0.24
(b) 0.36
(c) 0.4
(d) 0.6

GATE CSE – 2008 MCQ (2-MARKS)

Q43. What is the probability that in a randomly chosen group of r people at least three people have the same birthday?

(a) $1-\frac{365-364\ldots(365-r+1)}{365^r}$

(b) $\frac{365\cdot364\ldots(365-r+1)}{365^r}+{}^{r}C_1\cdot 365.\frac{364.363\ldots(364-(r-2)+1)}{364^{r+2}}$

(c) $1-\frac{365\cdot364\ldots(365-r+1)}{365^r}-{}^{r}C_2\cdot 365.\frac{364\cdot363\ldots(364-(r-2)+1)}{364^{r-2}}$

(d) $\frac{365\cdot364\ldots(365-r+1)}{365^r}$

GATE CSE – 2007 MCQ (1-MARKS)

Q44. Box P has 2 red balls and 3 blue balls and box Q has 3 red balls and 1 blue ball. A ball is selected as follows: (i) select a box (ii) choose a ball from the selected box such that each ball in the box is equally likely to be chosen. The probabilities of selecting boxes P and Q are $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Given that a ball selected in the above process is a red ball, the probability that it came from the box P is:

(a) $\frac{4}{19}$
(b) $\frac{5}{19}$
(c) $\frac{2}{9}$
(d) $\frac{19}{30}$

GATE CSE – 2007 MCQ (1-MARKS)

Q45. Suppose we uniformly and randomly select a permutation from the 20 ! permutations of $1,2,3\ldots,20$. What is the probability that 2 appears at an earlier position than any other even number in the selected permutation?

(a) $\left(\frac{1}{2}\right)$
(b) $\left(\frac{1}{10}\right)$
(c) $\left(\frac{9!}{20!}\right)$
(d) None of these

GATE CSE – 2006 MCQ (1-MARKS)

Q46. In a certain town, the probability that it will rain TODAY is 0.5.
The probability that it will rain tomorrow is 0.6.
The probability that it will rain on BOTH today and tomorrow is 0.3.
The probability that it will rain today OR tomorrow is:

(a) 0.2
(b) 0.5
(c) 0.8
(d) 1.0

GATE CSE – 2006 MCQ (2-MARKS)

Q47. When a coin is tossed, the probability of getting a Head is $p, 0<p<1$. Let N be the random variable denoting the number of tosses till the first Head appears, including the toss where the Head appears. Assuming that successive tosses are independent, the

expected value of N is

(a) $\frac{1}{p}$
(b) $\frac{1}{(1-p)}$
(c) $\frac{1}{p^2}$
(d) $\frac{1}{(1-p^2)}$

GATE CSE – 2005 MCQ (2-MARKS)

Q48. Let $f(x)$ be the continuous probability density function of a random variable x, the probability that $a < x \leq b$, is:

(a) $f(b-a)$
(b) $f(b)-f(a)$
(c) $\int_a^b f(x)dx$
(d) $\int_a^b xf(x)dx$

GATE CSE – 2005 MCQ (2-MARKS)

Q49. A random bit string of length n is constructed by tossing a fair coin n times and
setting a bit to 0 or 1 depending on outcomes HEAD and TAIL, respectively.
The probability that two such randomly generated strings are NOT identical is:

(a) $1/2^n$
(b) $1 - 1/n$
(c) $1/n!$
(d) $1 - 1/2^n$

GATE CSE – 2005 MCQ (2-MARKS)

Q50. A bag contains 10 blue marbles, 20 green marbles and 30 red marbles. A marble is drawn from the bag, its colour recorded and it is put back in the bag. This process is repeated 3 times. The probability that no two of the marbles drawn have the same colour is

(a) 1/36
(b) 1/6
(c) 1/4
(d) 1/3

Answer key

1. (4.25)	**2. (b)**	**3. (b)**	**4. (0.5)**
5. (0.50)	**6. (0.74)**	**7. (0.375)**	**8. (b,c)**
9. (b)	**10.(d)**	**11. (c,d)**	**12. (d)**
13. (a)	**14. (d)**	**15. (0.35 - 0.39)**	
16. (15.00 – 16.00)		**17. (0.0230 - 0.0232)**	
18. (a)	**19.(b)**	**20. (54)**	**21. (0)**
22. (0.33 - 0.34)		**23. (0.5)**	**24.(0.55)**
25. (0.75)	**26. (0.25)**	**27. (0.259 - 0.261)**	
28. (10)	**29. (11.85 - 11.95)**		**30. (c)**
31. (c)	**32. (b)**	**33. (c)**	**34. (a)**
35. (d)	**36. (a)**	**37. (c)**	**38. (a)**
39. (a)	**40. (b)**	**41. (a)**	**42. (c)**
43. (c)	**44. (a)**	**45. (b)**	**46. (c)**
47. (a)	**48. (c)**	**49. (d)**	**50. (b)**

HINTS & SOLUTIONS

1. ANSWER:- 4.25

First, interpret the probabilities correctly (there's a small typo in the question). The intended distribution is:

- $P(X=1)=P(X=2)=P(X=5)=P(X=7)=\frac{1}{6}$
- $P(X=3)=P(X=4)=P(X=6)=P(X=8)=\frac{1}{12}$

Check total probability:

$$4\cdot\frac{1}{6}+4\cdot\frac{1}{12}=\frac{4}{6}+\frac{4}{12}=\frac{2}{3}+\frac{1}{3}=1$$

Now compute the expected value:

$$E[X]=\sum x\cdot P(X=x)$$

First group:

$$\frac{1+2+5+7}{6}=\frac{15}{6}=2.5$$

Second group:

$$\frac{3+4+6+8}{12}=\frac{21}{12}=1.75$$

Total:

$$E[X]=2.5+1.75=4.25$$

2. ANSWER:- B

After 2 draws we will have total of 4 balls in the urn. In our case we want equal number of Red and Blue balls i.e. $2R\&2B$

So one draw should be Blue and another should be Red, the 2 sequence can be RB & BR

$$\rightarrow P(RB)=P(\text{" first is " } R)\times P(\text{" second is " } B \mid \text{" first is " } R)$$

$$=\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)=\frac{1}{6}$$

$$\rightarrow P(BR)=\text{" same case as " } P(RB)=\frac{1}{6}$$

Total Probability $P(\text{ Equal } R \text{ \& } B)=\frac{1}{6}+\frac{1}{6}=\frac{1}{3}$

3. ANSWER:- B

Poisson is Discrete, eliminates Poisson immediately.

Now check Domain:

- Exponential PDF: $f(x)=\lambda e\text{^}(-\lambda x)$ for $\mathbf{x}\geq 0$ → eliminates Exponential (question says $\mathbf{x}\in(-\infty,+\infty)$)
- Uniform: defined on $(\mathbf{a},\mathbf{b})$ → finite interval → eliminates Uniform
- Normal: only distribution that lives on $(-\infty,+\infty)$

Answer: Option B

4. ANSWER:- 0.5

Let

- E1: At least 2 heads on tosses 2, 4, and 6
- E2: Number of heads = number of tails on tosses 1, 2, 3, and 5

We need to find P(E1|E2). So the sample space is restricted to E2.

Sample Space under E2

(Tosses 1,2,3,5 → equal heads and tails)

	1	2	3	4	5	6
(A)	H	H	T	_	T	_
(B)	H	T	H	_	T	_
(C)	H	T	T	_	H	_
(D)	T	T	H	_	H	_
(E)	T	H	T	_	H	_
(F)	T	H	H	_	T	_

Tosses 4 and 6 are free → 2^2=4 possibilities each.

Total cases = 6×4=24

Favorable Cases (E1):

For tosses 2, 4, 6 → at least 2 heads.

Possible combinations for (4,6): HH, HT, TH, TT

- For (A), (E), (F): 3 valid combinations each → 3×3=9
- For (B), (C), (D): 1 valid combination each → 3×1=3

Total favorable cases = 9+3=12

P(E1/E2) = 12/24 = 0.5

5. ANSWER:- 0.50

Using Bayes' Theorem:

P(HH | fake) = 1 × 1 = 1

P(HH | regular) = 0.5 × 0.5 = 0.25

P(fake) = 1/5, P(regular) = 4/5

P(HH) = 1×(1/5) + 0.25×(4/5) = 0.2 + 0.2 = 0.4

P(fake | HH) = P(HH|fake)×P(fake) / P(HH) = (1 × 1/5) / 0.4 = 0.2/0.4 = 0.50

6. ANSWER:- 0.74

Let U ~ Uniform(0,1) be the random division point.

Case 1: U < 0.4 → the subinterval containing 0.4 is [U, 1], length = 1 − U.

Case 2: U > 0.4 → the subinterval containing 0.4 is [0, U], length = U.

E[length] = P(U<0.4)·E[1−U | U<0.4] + P(U>0.4)·E[U | U>0.4]

= 0.4 × 0.8 + 0.6 × 0.7 = 0.32 + 0.42 = 0.74

7. ANSWER:- 0.375
Given the first ball is red, the bag now has 9 red and 15 blue balls (24 total).
P(2nd ball is red | 1st is red) = 9/24 = 0.375.

8. ANSWER:- B and C
(A) INCORRECT: For independence, $P(A \cap B)$ must equal $P(A) \times P(B) = 0.15$, but $P(A \cap B) = 0.1 \neq 0.15$.
(B) CORRECT: $P(A \cup B) = 0.3 + 0.5 - 0.1 = 0.7$.
(C) CORRECT: $P(A \cap B^c) = P(A) - P(A \cap B) = 0.3 - 0.1 = 0.2$.
(D) INCORRECT: $P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - 0.7 = 0.3 \neq 0.4$.

9. ANSWER:- B
Total outcomes = $6^6 = 46656$.
Favourable outcomes = 6! = 720 (all permutations of {1,2,3,4,5,6}).
Probability = 720 / 46656 = 5/324.

10. ANSWER:- D
As given in the question, X and Y can take values in the interval of [0,1]. From this, we can write 0 <= Y <= 1.
This can also be rewritten as 0 <= XY <= X.
E(X–c) = E(X)–c where c is a constant. Also called: mathematical expectation.
Taking the expectation of the above inequality, we will get this:
E(0) <= E(XY) <= E(X)
This is also equal to 0 <= E(XY) <= E(X)
This can only be true in the case when the range is specified between 0 and 1.
Therefore, the correct option is D.

11. ANSWER:- C, D
Sample space: {HH, HT, TH, TT}, each with prob 1/4.
P(A)=1/4, P(B)=1/2, P(C)=1/2.
(i) $P(A \cap B)=1/4 \neq P(A) \times P(B)=1/8$. NOT independent.
(ii) $P(A \cap C)=1/4 \neq P(A) \times P(C)=1/8$. NOT independent.
(iii) $P(B \cap C)=P(HH)=1/4 = P(B) \times P(C)=1/4$. INDEPENDENT.
(iv) Since B and C are independent, P(B|C) = P(B). TRUE.

12. ANSWER:- D
Case 1: Attempt Qa first, then Qb

- Probability of correct Qa $= 0.8$, incorrect $= 0.2$
- Probability of correct Qb $= 0.5$, incorrect $= 0.5$

Outcomes:

- Both correct $\rightarrow 0.8 \times 0.5 = 0.4$, marks $= 10 + 20 = 30$
- Only Qa correct $\rightarrow 0.8 \times 0.5 = 0.4$, marks $= 10$
- If Qa is incorrect (0.2), no marks (as second attempt doesn't help)

Expected marks:

$$E_1 = (0.4 \times 30) + (0.4 \times 10) = 12 + 4 = 16$$

Case 2: Attempt Qb first, then Qa

- Probability of correct Qb $= 0.5$, incorrect $= 0.5$
- Probability of correct Qa $= 0.8$, incorrect $= 0.2$

Outcomes:

- Both correct $\rightarrow 0.5 \times 0.8 = 0.4$, marks $= 30$
- Only Qb correct $\rightarrow 0.5 \times 0.2 = 0.1$, marks $= 20$

Expected marks:

$$E_2 = (0.4 \times 30) + (0.1 \times 20) = 12 + 2 = 14$$

Final Comparison

$$E_1 = 16 > E_2 = 14$$

13. ANSWER:- A
This is Pólya's Urn model.
By symmetry (exchangeability), the probability of drawing a red ball on any trial is always r/(r+b), regardless of the trial number.
Verification for 2nd trial:
P(red 2nd) = (r/(r+b))×(r+1)/(r+b+1) + (b/(r+b))×r/(r+b+1)
= r(r+1+b)/[(r+b)(r+b+1)] = r/(r+b). ✓

14. ANSWER:- D
S_1 is FALSE: By Cauchy-Schwarz, $(Cov(X,Y))^2 \leq Var(X) \cdot Var(Y)$. The strict reverse is impossible.
S_2 is FALSE: Cov[X,Y] = E[(X–E[X])(Y–E[Y])], which can be negative.
E[|X–E[X]|·|Y–E[Y]|] is always non-negative, so they are not equal in general.

15. ANSWER:- 0.35 - 0.39
Expected lifetime $E[X] = 1/\lambda = 1/2$.
$P(X > 1/2) = e^{(-\lambda \times 1/2)} = e^{(-2 \times 0.5)} = e^{(-1)} \approx 0.3679 \approx 0.37$.

16. ANSWER:- 15.00 - 16.00
$X \sim$ Binomial($n=1000$, $p=0.4$).
Variance $= n \times p \times (1-p) = 1000 \times 0.4 \times 0.6 = 240$.
Standard Deviation $= \sqrt{240} \approx 15.49$.

17. ANSWER:- 0.023
P(tie) = 6/36 = 1/6
{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)}
P(no tie) = 5/6.
Win on 3rd trial $\Longrightarrow$ trials 1 and 2 are ties, trial 3 is not a tie:
$P = (1/6) \times (1/6) \times (5/6) = 5/216 \approx 0.023$.

18. ANSWER:- A
Given: P(P)=1/4, P(P|Q)=1/2, P(Q|P)=1/3.
By Bayes: P(P|Q) = P(Q|P)×P(P)/P(Q)
$\Longrightarrow$ 1/2 = (1/3 × 1/4)/P(Q) $\Longrightarrow$ P(Q) = 1/6, P(Q') = 5/6.
P(Q'|P) = 1 − P(Q|P) = 2/3.
P(P|Q') = P(Q'|P)×P(P)/P(Q') = (2/3 × 1/4)/(5/6) = (1/6)/(5/6) = 1/5.
P(P'|Q') = 1 − 1/5 = 4/5.

19. ANSWER:- B
Each of the N tests independently has probability β of resulting in a Type II error.
Let X = number of tests with Type II error. Then $X \sim$ Binomial(N, β).
$E[X] = N \times \beta = N\beta$.
Note: $N\beta(1-\beta)$ is the variance of X, not the expectation.

20. ANSWER:- 54
For Poisson: $E[X] = Var[X] = \lambda = 5$.
$E[X^2] = Var[X] + (E[X])^2 = 5 + 25 = 30$.
$E[(X+2)^2] = E[X^2 + 4X + 4] = E[X^2] + 4E[X] + 4$
$= 30 + 20 + 4 = 54$.

21. ANSWER:- 0
$X \sim N(0, \sigma^2)$: $P(X \le 0) = 0.5$ and $P(X > 0) = 0.5$.
When $X \le 0$, $Y = 0$; when $X > 0$, $Y = X > 0$.
So $P(Y = 0) = 0.5$ and $P(Y > 0) = 0.5$.
Median m satisfies $P(Y \le m) \ge 0.5$ and $P(Y \ge m) \ge 0.5$.
At m = 0: $P(Y \le 0) = 0.5$ ✓ and $P(Y \ge 0) = 1$ ✓.
Hence, median = 0.

22. ANSWER:- 0.33
Each round: TH→Y (prob 1/4), HH or HT→N (prob 2/4), TT→restart (prob 1/4).
Let p = P(output Y).
$p = 1/4 + (1/4) \times p \Longrightarrow (3/4)p = 1/4 \Longrightarrow p = 1/3 \approx 0.33$.

23. ANSWER:- 0.5

A probability density function is a function of continuous random variable, whose Integral across
an interval gives the probability that the value of the variable lies within the same interval.

Calculation:
Integral value must be equal to 1 .

$$\int_a^1 \frac{1}{a^1} = \left[\frac{-1}{a}\right]_a^1 = \left[-1 + \frac{1}{a}\right]$$

Equate this to 1.

$$-1 + \frac{1}{a} = 1$$
$$a = 0.5$$

24. ANSWER:- 0.55
P(Type 1) = P(Type 2) = 0.5.
By total probability: P(lasts > 100) = 0.7×0.5 + 0.4×0.5 = 0.35 + 0.20 = 0.55.

25. ANSWER:- 0.75
Given $X_3 = 0$: $Y = X_1 \cdot X_2 \oplus 0 = X_1 \cdot X_2$.
$P[Y=0 \mid X_3=0] = P[X_1 \cdot X_2 = 0]$.
$X_1 \cdot X_2 = 1$ only when $X_1=1$ AND $X_2=1$.
$P(X_1 \cdot X_2=1) = 1/2 \times 1/2 = 1/4$.
$P[Y=0 \mid X_3=0] = 1 - 1/4 = 3/4 = 0.75$.

26. ANSWER:- 0.25
Since $A \cap B = \emptyset$ and $A \cup B = S$: $P(A) + P(B) = 1$.
By AM-GM inequality: $(P(A)+P(B))/2 \ge \sqrt{(P(A) \cdot P(B))}$
$\Longrightarrow 1/2 \ge \sqrt{(P(A) \cdot P(B))} \Longrightarrow P(A) \cdot P(B) \le 1/4$.
Equality when P(A) = P(B) = 1/2. Maximum = 0.25.

27. ANSWER:- 0.26
n(divisible by 2)=50, n(by 3)=33, n(by 5)=20.
n(by 2∩3)=16, n(by 3∩5)=6, n(by 2∩5)=10, n(by 2∩3∩5)=3.
$|A \cup B \cup C| = 50+33+20-16-6-10+3 = 74$.
P(NOT divisible by 2, 3, or 5) = 1 − 74/100 = 0.26.

28. ANSWER:- 10

Need $x_1+x_2+x_3+x_4 = 22$, each $x_i \in \{1,\ldots,6\}$.
Maximum = 24.
Combination (6,6,6,4): permutations = 4!/3! = 4.
Combination (6,6,5,5): permutations = 4!/(2!2!) = 6.
Total = 4 + 6 = 10. ∴ X = 10.

29. ANSWER:- 11.90

Initially P (working computer) $= \frac{4}{10}$, P (non-working computer) $= \frac{6}{10}$.

Case 1 : three computers are functional : There are 4 sub-cases
$WWWN, WWNW, WNWW, NWWW$, where W means working, N means non-working, but $P(WWWN) = P(WWNW) = P(WNWW) = P(NWWW)$, because for example $P(WWWN) = \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{6}{7} = \frac{144}{5040}$

In all other 3 sub-cases, we get same numerators and denominators (in different order), so total prob in this case is

$$\frac{4 \times 144}{5040} = \frac{576}{5040}.$$

Case 2 : all 4 are working

$$P(WWWW) = \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} = \frac{24}{5040}$$

$$P\left(\text{ atleast 3 are working}\right) = \frac{600}{5040}$$

So $100 \times p = 11.90$

30. ANSWER:- C

Prime factorisation of 2014 = $2^1 \times 19^1 \times 53^1$.
Number of factors = (1+1)×(1+1)×(1+1) = 2×2×2 = 8.
The 8 factors are: 1, 2, 19, 38, 53, 106, 1007, 2014.

31. ANSWER:- C

Poisson Probability Mass Function (with mean λ) $= \frac{\lambda^k}{(e^{\lambda} k!)}$,

We have to sum the probability mass function for $k = 0,1$ and 2 and $\lambda = 3$ (thus finding the cumulative mass function)

$$= \left(\frac{1}{e^3}\right) + \left(\frac{3}{e^3}\right) + \left(\frac{9}{2e^3}\right)$$
$$= \frac{17}{(2e^3)}$$

32. ANSWER:- B

Favourable outcomes: 6 alone, (1,5),(1,6),(2,4),(2,5),(2,6),(3,3),(3,4),(3,5),(3,6).
P(6 on first roll) = 1/6.
P(first=1, second≥5) = (1/6)(2/6); P(first=2, second≥4) = (1/6)(3/6); P(first=3, second≥3) = (1/6)(4/6).
P = 1/6 + 2/36 + 3/36 + 4/36 = 6/36 + 9/36 = 15/36 = 5/12.

33. ANSWER:- C

F(x) = P(X ≤ x).
F(−1) = P(X ≤ −1) = P(X = −1) = 0.5.
F(+1) = P(X ≤ +1) = P(X=−1) + P(X=+1) = 0.5 + 0.5 = 1

34. ANSWER:- A

Given at least one head, the reduced sample space = {HH, HT, TH} (3 equally likely outcomes).
Only {HH} satisfies 'both heads'.
P(both heads | at least one head) = 1/3.

35. ANSWER:- D

When $y_i = a \cdot x_i + b$: $\text{Var}(Y) = a^2 \cdot \text{Var}(X) \implies \sigma y = a \cdot \sigma x$.
The constant b shifts the distribution but does NOT affect the spread.
So σy = aσx (not aσx + b). All other options (A), (B), (C) are correct

36. ANSWER:- A

Favourable ordered pairs (first, second) where first = second + 1:
(2,1), (3,2), (4,3), (5,4) → 4 pairs.
Total ordered pairs = 5 × 4 = 20.
Probability = 4/20 = 1/5.

37. ANSWER:- C

$R = E[X^2] - (E[X])^2 = \text{Var}(X)$, which is always ≥ 0.
R = 0 only when X is a constant, so R > 0 does not always hold.
∴ Correct answer is R ≥ 0.

38. ANSWER:- A

$10^{99} = 2^{99} \times 5^{99}$. Total divisors = 100 × 100 = 10000.
A multiple of 10^{96} that divides 10^{99} = 10^{96} × (divisor of 10^3).
$10^3 = 2^3 \times 5^3$. Number of divisors of 10^3 = (3+1)(3+1) = 16.
Probability = 16/10000 = 1/625.

39. ANSWER:- A

Declared faulty in two cases:
1. IS faulty AND test correctly identifies it: p × q.
2. NOT faulty AND test incorrectly declares it faulty: (1−p) × (1−q).
P(declared faulty) = pq + (1−p)(1−q).

40. ANSWER:- B

$P(\{1,3,5\}) = 0.9P(\{2,4,6\})$ and their sum must be 1 . So,

$$P(\{1,3,5\}) = \frac{0.9}{1.9} = 0.4736 \text{ and}$$

$$P(\{2,4,6\}) = \frac{1}{1.9} = 0.5263$$

Given that probability of getting 2 or 4 or 6 is same.

So, $P(2) = P(4) = P(6) = \frac{0.5263}{3} = 0.1754$

$$P(\{4,6\}) \mid x > 3) = 0.75$$
$$\Rightarrow P(5 \mid x > 3) = 0.25$$
$$\Rightarrow P(5) = \frac{1}{3}(P(4) + P(6)) \because x > 3$$
$$\Rightarrow x \in \{4,5,6\}$$
$$\Rightarrow P(5) = \frac{2.P(4)}{3} \because P(4) = P(6).$$

So, $P(x > 3) = P(4) + P(5) + P(6) = \frac{8}{3} \times 0.1754 = 0.468$

41. ANSWER:- A

Step-by-Step Explanation:
$X \sim N(\mu=+1, \sigma^2=4)$, so $\sigma_X = 2$.

Standardise $P(X \le -1)$:
$z = (-1 - 1)/2 = -2/2 = -1$
$\therefore P(X \le -1) = \Phi(-1)$

Standardise $P(Y \ge 2)$: $Y \sim N(-1, \sigma_Y^2)$
$z = (2 - (-1)) / \sigma_Y = 3/\sigma_Y$
$\therefore P(Y \ge 2) = P(Z \ge 3/\sigma_Y) = 1 - \Phi(3/\sigma_Y)$

Set equal: $\Phi(-1) = 1 - \Phi(3/\sigma_Y)$
By symmetry of standard normal: $\Phi(-a) = 1 - \Phi(a)$

$\therefore 1 - \Phi(1) = 1 - \Phi(3/\sigma_Y) \rightarrow 3/\sigma_Y = 1 \rightarrow \sigma_Y = 3$.

42. ANSWER:- C

Transition matrix (row = today, column = next day):
P(CS→CS) = 1 − 0.6 = 0.4 P(CS→M) = 0.6
P(M→CS) = 0.4 P(M→M) = 1 − 0.4 = 0.6

P(Wed = CS | Mon = CS)
= P(Wed=CS | Tue=CS) × P(Tue=CS | Mon=CS)
+ P(Wed=CS | Tue=M) × P(Tue=M | Mon=CS)
= (0.4)(0.4) + (0.4)(0.6)
= 0.16 + 0.24 = 0.40

43. ANSWER:- C

Strategy: complement method.
P(at least 3 share) = 1 − P(all birthdays different) − P(exactly one pair shares)

Term 1 — P(all r birthdays different):
= 365 × 364 × 363 × … × (365 − r + 1) / 365^r
= 365! / [(365 − r)! × 365^r] (falling factorial form)

Term 2 — P(exactly one pair shares, rest all different from each other & from pair):
Step a: Choose which 2 people share: C(r, 2) ways.
Step b: Choose their common birthday: 365 ways.
Step c: Assign distinct birthdays ≠ pair's to remaining (r−2) people:
364 × 363 × … × (367 − r) → (r−2) terms, starting at 364.
Step d: Divide by total outcomes 365^r.

P(exactly one pair) = C(r,2) × 365 × [364 × 363 × … × (367−r)] / 365^r
= C(r,2) × [364 × 363 × … × (367−r)] / 365^(r−1)

∴ P(at least 3 share) = 1 − [365×364×…×(366−r)]/365^r − C(r,2)×[364×363×…×(367−r)]/365^(r−1)
This matches Option (C). ✓

44. ANSWER:- A

The probability of selecting a red ball,

$$= \left(\frac{1}{3}\right) * \left(\frac{2}{5}\right) + \left(\frac{2}{3}\right) * \left(\frac{3}{4}\right)$$
$$= \frac{2}{15} + \frac{1}{2} = \frac{19}{30}$$

Probability of selecting a red ball from box,

$$P = \left(\frac{1}{3}\right) * \left(\frac{2}{5}\right) = \frac{2}{15}$$

Given that a ball selected in the above process is a red ball, the probability that it came from the

$$\text{box } P \text{ is } = \left(\frac{2}{15} \div \frac{19}{30}\right) = \frac{4}{19}$$

45. ANSWER:- B

The even numbers in {1, 2, …, 20} are: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20.
There are exactly 10 even numbers.

Key Insight: The odd numbers are irrelevant. We only need to consider the RELATIVE ORDER of the 10 even numbers among themselves within the permutation.

Since the permutation is chosen uniformly at random, every one of the 10! orderings of {2, 4, 6, 8, 10, 12, 14, 16, 18, 20} is equally likely.

P(2 is first among all even numbers) = 1/10.
Intuition: By symmetry, each of the 10 even numbers is equally likely to appear first (leftmost) among all even numbers. So the probability = 1/10.

46. ANSWER:- C

Using the Addition Rule of Probability:
P(A ∪ B) = P(A) + P(B) − P(A ∩ B)
= 0.5 + 0.6 − 0.3
= 0.8

Verification: since P(A ∩ B) = 0.3 > 0 the events are NOT mutually exclusive, so we must subtract the intersection to avoid double-counting.

47. ANSWER:- A

$E = 1 \times p + 2 \times (1-p)p + 3 \times (1-p)(1-p)p + \cdots$

multiply both side with $(1-p)$ and subtract:

$E - (1-p)E = 1 \times p + (1-p)p + (1-p)(1-p)p + \cdots$
$= p/(1-(1-p)) = 1$ (because it is now forming a GP)
$\Longrightarrow (1-1+p)E = 1$
$\Longrightarrow E = 1/p$

So, option (A).

48. ANSWER:- C

A. This gives the probability at the point of $b-a$ which is not having any significant w.r.t a and b.
B. This gives the difference of the probabilities at b and a. Note: This is different from cumulative distribution function $F(b) - F(a)$.
C. This is Probability Density Function.
D. This is expected value of continuous random variable.

49. ANSWER:- D

Each bit string of length n is produced by n independent fair coin tosses.
Total distinct strings = 2^n, each generated with equal probability.

Fix string S_1 (the first generated string — any specific string).
The second string S_2 is generated independently.

P(S_2 identical to S_1)
= P(all n bits of S_2 match S_1)
= $(1/2)^1 \times (1/2)^1 \times \ldots \times (1/2)^1$ [n times, independent]
= $(1/2)^n = 1/2^n$

P(S_2 NOT identical to S_1)
= $1 - 1/2^n$ ✓

50. ANSWER:- B

Total marbles = 10 + 20 + 30 = 60.

Individual probabilities (with replacement, so constant each draw):
P(blue) = 10/60 = 1/6
P(green) = 20/60 = 1/3
P(red) = 30/60 = 1/2

"No two same colour" means all THREE draws are DIFFERENT colours.
There are 3! = 6 orderings of (blue, green, red).

P(one blue, one green, one red in any order)
= 3! × P(blue) × P(green) × P(red)
= 6 × (1/6) × (1/3) × (1/2)
= 6 × 1/36
= 6/36 = 1/6

2 CALCULUS

GATE CSE – 2026 NAT SET- 1 (1-MARKS)

Q1. Consider the function $f:\mathbb{R}\to\mathbb{R}$ defined as follows:

$$f(x)=\begin{cases}c_1e^x-c_2\log_e\left(\frac{1}{x}\right), & \text{if } x>0\\ 3 & \text{otherwise}\end{cases}$$

where $c1$,$c2 \in\mathbb{R}$.
If f is continuous at $x = 0$, then $c1+c2$ = _____. (answer in integer)

GATE CSE – 2026 MSQ SET- 1 (2-MARKS)

Q2. Let $f:\mathbb{R}\to\mathbb{R}$ be defined as follows:

$$f(x)=((|x|/2)-x)(x-(|x|/2))$$

Which of the following statements is/are true?

(a) f has a local maximum
(b) f has a local minimum
(c) f' is continuous over $\mathbb{R}$
(d) f' is not differentiable over $\mathbb{R}$

GATE CSE – 2026 MSQ SET- 2 (1-MARKS)

Q3. For a real number a, let

$$I(a)=\int_{-1}^{1}(3x^2-ax+1)dx$$

Which of the following statements is/are true?

(a) The value of $I(a)$is independent of the value of a
(b) The value of $I(a)$can vary with the value of a
(c) There exists $a\in(-\infty,+\infty)$such that $I(a)$is a positive real number.
(d) There exists $a\in(-\infty,+\infty)$such that $I(a)$is a negative real number.

GATE CSE – 2025 MCQ SET - 2 (1-MARKS)

Q4. The value of x such that $x > 1$, satisfying the equation
$\int_1^x t\ln t\,dt=\frac{1}{4}$ is

(a) √e
(b) e
(c) e^2
(d) e-1

GATE CSE – 2025 NAT SET- 1 (1- MARKS)

Q5. Consider the given function $f(x)$.

$$f(x)=\begin{cases}ax+b & \text{for } x<1\\ x^3+x^2+1 & \text{for } x\geq 1\end{cases}$$

If the function is differentiable everywhere, the value of b must be _____. (rounded off to one decimal place)

GATE CSE – 2024 MCQ SET - 2 (1-MARKS)

Q6. Let f(x) be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that
f(x) = 1 - f(2-x)
Which one of the following options is the CORRECT value of of $\int_0^2 f(x)dx$?

(a) 0
(b) 1
(c) 2
(d) -1

GATE CSE – 2023 NAT SET - 1 (1-MARKS)

Q7. The value of the definite integral

$$\int_{-3}^{3}\int_{-2}^{2}\int_{-1}^{1}(4x^2y-z^3)\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2023 MSQ (1-MARKS)

Q8. Let f (x) = x^3+ $15x^2$ − 33x − 36 be a real-valued function.
Which of the following statements is/are TRUE?

(a) f(x) has a local maximum.
(b) f(x) does not have a local maximum
(c) f(x) has a local minimum.
(d) f(x) does not have a local minimum.

GATE CSE - 2020 MCQ (1-MARKS)

Q9. Consider the functions
I. e^{-x}
II. x^2 - sin x
III. $\sqrt{(x^3+1)}$
Which of the above functions is/are increasing everywhere in [0, 1]?

(a) III only
(b) II only
(c) II and III only
(d) I and III only

GATE CSE – 2015 MCQ (2-MARKS)

Q10. If $g(x) = 1 - x$ and $h(x) = \frac{x}{x-1}$, then $\frac{g(h(x))}{h(g(x))}$ is

(a) h(x)/g(x)
(b) $-1/x$
(c) g(x)/h(x)
(d) $x/(1-x)^2$

GATE CSE – 2015 NAT (2-MARKS)

Q11. Solve:
$\int_{1/\pi}^{2/\pi} \frac{\cos(1/x)}{x^2} dx =$

GATE CSE – 2015 MCQ (2-MARKS)

Q12. Let f(x) = $x^{-(1/3)}$ and A denote the area of the region bounded by f(x) and the X-axis, when x varies from –1 to 1. Which of the following statements is/are TRUE?
(I) f is continuous in [–1, 1]
(II) f is not bounded in [–1, 1]
(III) A is nonzero and finite

(a) II only
(b) III only
(c) II and III only
(d) I, II and III

GATE CSE – 2015 NAT (1-MARKS)

Q13. Solve
$\sum_{x=1}^{99} \frac{1}{x(x+1)} =$ ___ .

(a) 0.99
(b) 1.00
(c) 2.00
(d) 3.00

GATE CSE – 2015 MCQ SET – 3 (2-MARKS)

Q14. The value of $\lim_{x\to\infty} (1+x^2)^{e^{-x}}$ is

(a) 0
(b) 1/2
(c) 1
(d) ∞

GATE CSE – 2014 NAT (1-MARKS)

Q15. The function f(x) = x sinx satisfies the following equation f''(x) + f(x) + t cosx = 0. The value of t is ____.

GATE CSE – 2014 MCQ (2-MARKS)

Q16. A function f(x) is continuous in the interval [0,2]. It is known that f(0) = f(2) = -1 and f(1) = 1. Which one of the following statements must be true?

(a) There exists a y in the interval (0,1) such that f(y) = f(y + 1)
(b) For every y in the interval (0,1),f(y) = f(2 − y)
(c) The maximum value of the function in the interval (0,2) is 1.
(d) There exists a y in the interval (0, 1) such that f(y) = −f(2 − y)

GATE CSE – 2014 MCQ (2-MARKS)

Q17. The value of the integral given below is

$$\int_0^{\pi} x^2 \cos x dx$$

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) -2π
(b) π
(c) $-\pi$
(d) 2π

GATE CSE – 2014 MCQ (1-MARKS)

Q18. A non-zero polynomial f(x) of degree 3 has roots at x = 1, x = 2 and x = 3. Which one of the following must be TRUE?

(a) f(0)f(4) < 0
(b) f(0)f(4) > 0
(c) f(0) + f(4) > 0
(d) f(0) + f(4) < 0

GATE CSE – 2014 MCQ (1-MARKS)

Q19. If $\int_0^{2\pi} |x \sin x| dx = k\pi$, then the value of k is equal to ___ .

(a) 4
(b) 5
(c) 6
(d) 7

GATE CSE – 2013 MCQ (1-MARKS)

Q20. Which one of the following functions is continuous at x = 3?

(a) $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x-1, & \text{if } x > 3 \\ \frac{x+3}{3}, & \text{if } x < 3 \end{cases}$

(b) $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8-x & \text{if } x \neq 3 \end{cases}$

(c) $f(x) = \begin{cases} x+3, & \text{if } x \leq 3 \\ x-4, & \text{if } x > 3 \end{cases}$

(d) $f(x) = \frac{1}{x^3-27}$, if x ≠ 3

GATE CSE – 2011 MCQ (2-MARKS)

Q21. Given i = $\sqrt{-1}$, what will be the evaluation of the definite integral

$$\int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} dx$$

(a) 0
(b) 2
(c) -i
(d) i

GATE CSE – 2010 MCQ (1-MARKS)

Q22. What is the value of $\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^{2n}$?

(a) 0
(b) e^{-2}
(c) $e^{-1/2}$
(d) 1

GATE CSE – 2010 MCQ (1-MARKS)

Q23. The value of x at which y is minimum for $y = x^2 - 3x + 1$ is

(a) -3/2
(b) 3/2
(c) 0
(d) -5/4

GATE CSE – 2009 MCQ (2-MARKS)

Q24. $\int_0^{\pi/4} (1 - \tan x)/(1 + \tan x) dx$ evaluates to

(a) 0
(b) 1
(c) ln 2
(d) 1/2ln 2

GATE CSE – 2008 MCQ (2-MARKS)

Q25. If $f(x)$ is defined as follows, what is the minimum value of $f(x)$ for $x \in (0,2]$?

$$f(x) = \begin{cases} \frac{25}{8x} \text{ when } x \leq \frac{3}{2} \\ x + \frac{1}{x} \text{ otherwise} \end{cases}$$

(a) 2
(b) $2\frac{1}{12}$
(c) $2\frac{1}{6}$
(d) $2\frac{1}{2}$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2008 MCQ (2-MARKS)

Q26. $\lim_{x\to\infty} \frac{x-\sin x}{x+\cos x}$ equals

(a) 1
(b) -1
(c) ∞
(d) -∞

GATE CSE – 2007 MCQ (1-MARKS)

Q27. Consider the following two statements about the function f(x)=|x|:
P. f(x) is continuous for all real values of x
Q. f(x) is differentiable for all real values of x
Which of the following is TRUE?

(a) P is true and Q is false.
(b) P is false and Q is true.
(c) Both P and Q are true.
(d) Both P and Q are false.

GATE CSE – 2005 MCQ (1-MARKS)

Q28. What is the value of $\int_0^{2\pi} (x-\pi)^2 (\sin x)\,dx$

(a) -1
(b) 0
(c) 1
(d) π

ANSWER KEY

1.(3)	**2.(a,c,d)**	**3.(a,c)**	**4.(a)**
5.(-2.0)	**6.(b)**	**7.(0)**	**8.(a,c)**
9. (a)	**10.(a)**	**11.(-1)**	**12.(c)**
13.(a)	**14.(c)**	**15.(-2)**	**16.(a,d)**
17.(a)	**18.(a)**	**19.(a)**	**20.(a)**
21.(d)	**22.(b)**	**23.(b)**	**24.(d)**
25.(b)	**26.(a)**	**27.(a)**	**28.(b)**

HINTS & SOLUTIONS

1. ANSWER:- 3
$e^x \to 1, \log(1/x) \to \infty$
So, $f(x) \approx c_1 - c_2 \cdot \infty$
For limit to be finite ⇒ must have $c_2 = 0$
Then limit becomes c_1
So, $c_1 = 3$
$c_1 + c_2 = 3 + 0 = 3$

2. ANSWER:- A,C, D
f(x) = (|x|/2 − x)(x − |x|/2)
= −(|x|/2 − x)²
Casewise:
- For x ≥ 0: f(x) = −x²/4
- For x < 0: f(x) = −9x²/4

f′(x):
- For x > 0: f′(x) = −x/2
- For x < 0: f′(x) = −9x/2
- At x = 0: f′(0) = 0 ⇒ continuous

f″(x):
- For x > 0: −1/2
- For x < 0: −9/2 ⇒ not defined at x = 0

Conclusions:
(a) True (local maximum at x = 0)
(b) False
(c) True (f′ continuous)
(d) True (f′ not differentiable at x = 0)

3. ANSWER:- A,C
$\int_{-1}^{1} x\,dx = 0$(odd function over symmetric interval)
So, $I(a)$becomes independent of a
Now,
$$I(a) = \int_{-1}^{1} (3x^2 + 1)\,dx = 3 \cdot \frac{2}{3} + 2 = 2 + 2 = 4$$

4. ANSWER:- A
x = √e
Work:
∫ t ln t dt = (t²/2) ln t − t²/4 + C
So,
$\int_1^x$ t ln t dt = [(x²/2) ln x − x²/4] − [0 − 1/4]
= (x²/2) ln x − x²/4 + 1/4
Set equal to 1/4 and solve:
=(x²/2) ln x − x²/4 = 0
⇒ ln x = 1/2
⇒ x = e (1/2) = √e

5. ANSWER:- (-2.0)
Problem Analysis:
The function f(x) is defined as:
- For x < 1, f(x) = ax + b
- For x ≥ 1, f(x) = $x^3 + x^2 + 1$

We need to find the value of b so that the function is differentiable everywhere.

Check Out Our GATE Courses & Secure Your Dream Rank!

Step 1: Ensure Continuity at x = 1
For the function to be continuous at x = 1, the left-hand limit must equal the right-hand limit.

- For x < 1, the left-hand limit at x = 1 is: $f(1^-)$ = a + b
- For x ≥ 1, the right-hand limit at x = 1 is: $f(1^+)$ = 3

For continuity, we require: a + b = 3. This gives us Equation 1.
Step 2: Ensure Differentiability at x = 1
For differentiability at x = 1, the derivatives from both sides must match at x = 1.

- For x < 1, the derivative of f(x) = ax + b is: f'(x) = a
- For x ≥ 1, the derivative of $f(x) = x^3 + x^2 + 1$ is: $f'(x) = 3x^2 + 2x$. At x = 1, $f'(1^+) = 5$.

For differentiability, we require: a = 5.
Step 3: Solve for b
From Equation 1, we know that a + b = 3.
Substituting a = 5 into this equation, we get: 5 + b = 3 Solving for b, we get b = -2.
Final Answer:
The value of b is -2.0 (rounded to one decimal place).

6. ANSWER:- B

Let $u = a - x$, we have $du = -dx$, then

$$\int_0^a f(a-x)\,dx = \int_{u=a}^{u=0} -f(u)\,du$$

$$= -\int_a^0 f(u)\,du = \int_0^a f(u)\,du$$

So,

$$\int_0^a f(a-x)\,dx = \int_0^a f(u)\,du = \int_0^a f(x)\,dx$$

(∵ variable of integration is a dummy variable)
Now,

$$\int_0^2 f(x)\,dx = \int_0^2 f(2-x)\,dx = \int_0^2 [1-f(x)]\,dx$$

$$= 2 - \int_0^2 f(x)\,dx$$

$$2\int_0^2 f(x)\,dx = 2$$

$$\int_0^2 f(x)\,dx = 1$$

7. ANSWER:- 0

$$\int_{-3}^{3}\int_{-2}^{2}\int_{-1}^{1} (4x^2y - z^3)\,dz\,dy\,dx$$

$$= \int_{-3}^{3}\int_{-2}^{2} \left[4x^2yz - \frac{z^4}{4}\right]_{-1}^{1} dy\,dx$$

$$= \int_{-3}^{3}\int_{-2}^{2} (8x^2y)\,dy\,dx$$

$$= \int_{-3}^{3} [4x^2y^2]_{-2}^{2}\,dx$$

$$= 0$$

8. ANSWER:- A, C

the real valued function $f(x) = x^3 + 15x^2 - 33x - 36 = 0$

1. find $f'(x) = 0$

$$\Rightarrow 3x^2 + 30x - 33 = 0$$
$$\Rightarrow x^2 + 10x - 11 = 0$$
$$\Rightarrow x^2 + 11x - x - 11 = 0$$
$$\Rightarrow (x+11)(x-1) = 0$$
$$\Rightarrow x = -11, 1$$

2. find $f''(x)$ we get: $f''(x) = 6x + 30$
3. $f''(1) = 6 + 30 = 36 > 0$ it is gives local minima
   $4f''(-11) = -66 + 30 = -36 < 0$ it is gives local maxima.
   so given function $f(x)$ will give local maxima at $x = -11$ and local minima at $x = 1$
   ∴ Option A, C is correct.

9. ANSWER:- A

Concept: Increasing Function
A function $f(x)$ is increasing on an interval if:

$$x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)$$

That is, as x increases, $f(x)$ should not decrease.

Checking the Options
Option 1
Given:

$$f(0) = 1, f(1) = 0.368$$

Since:

$$f(1) < f(0)$$

Check Out Our GATE Courses & Secure Your Dream Rank!

So, as x increases from 0 to 1, $f(x)$ decreases.
Not an increasing function

Option II
Given:

$$f(x) = x^2 - \sin x$$

Check values:

- $f(0) = 0$
- $f(0.5) = 0.25 - \sin(0.5)$

Now, note:
The angle must be in radians, so:

$$\sin(0.5) \approx 0.479$$

Thus:

$$f(0.5) \approx 0.25 - 0.479 = -0.229$$

Since:

$$f(0.5) < f(0)$$

Again, $f(x)$ decreases when x increases.
Not an increasing function

Option III
Here, $f(x)$ behaves such that:

- When x increases → $f(x)$ increases
- When x decreases → $f(x)$ decreases

This satisfies the definition of an increasing function on [0,1].

10. ANSWER:- A

g(h(x)) = g(x/(x-1)) = 1 - x/(x-1) = -1/(x-1)
h(g(x)) = h(1-x) = (1-x)/((1-x) - 1) = -(1-x)/x
g(h(x)) / h(g(x)) = [-1/(x-1)] / [-(1-x)/x]
= $-x/(x-1)^2$
Therefore: $-x/(x-1)^2$ is same as h(x) / g(x).

11. ANSWER:- (−1)

Let $u = \frac{1}{x} = x^{-1}$.
Now, find the differential du by differentiating with respect to x :

$$du = -x^{-2}dx = -\frac{1}{x^2}dx$$

$$-du = \frac{1}{x^2}dx$$

Lower limit $(x = 1/\pi)$:

$$u = \frac{1}{1/\pi} = \pi$$

o Upper limit $(x = 2/\pi)$:

$$u = \frac{1}{2/\pi} = \frac{\pi}{2}$$

$$\int_{\pi}^{\pi/2} \cos(u) \cdot (-du)$$

We can eliminate the negative sign by flipping the integration limits:

$$\int_{\pi/2}^{\pi} \cos(u)du$$

The artiderivative of con(u) is sin(u):

$$[\sin(u)]_{\pi/2}^{\pi} = \sin(\pi) - \sin\left(\frac{\pi}{2}\right)$$

Substitute the standard trigonometric values ($\sin(\pi) = 0$ and $\sin(\pi/2) = 1$):

$$0 - 1 = -1$$

12. ANSWER:- C

Since $f(0) \to \infty$
$\therefore f$ is not bounced in $[-1,1]$ and hence f is not continuous in $[-1,1]$.

$$A = \int_{-1}^{1} f(x)dx = -\int_{-1}^{0} x^{-1/3}dx + \int_{0}^{1} x^{-1/3}dx$$

$$= -\frac{3}{2}\left(x^{2/3}\right)_{-1}^{0} + \frac{3}{2}\left(x^{2/3}\right)_{0}^{1} = \frac{3}{2} + \frac{3}{2} = 3$$

which is non-zero and finite
$\therefore$ Statement II & III are true.

13. ANSWER:- A

S = 1/1*2 + 1/2*3 + 1/3*4 + ... + 1/99*100
= (1 - 1/2) + (1/2 - 1/3) + (1/3 - 1/4) + ... + (1/99 -

Check Out Our GATE Courses & Secure Your Dream Rank!

1/100)
= (1 + 1/2 + 1/3 .... 1/99) - (1/2 + 1/3 + 1/4 ... 1/100)
= 1 - 1/100
= 99/100
= 0.99

14 ANSWER:- C

Use Logarithms

Let $y = (1 + x^2)^{e^{-x}}$. Taking the natural logarithm gives $\ln(y) = e^{-x}\ln(1 + x^2) = \frac{\ln(1+x^2)}{e^x}$.

Evaluate the Limit of $\ln(y)$

Apply L'Hôpital's Rule to the $\frac{\infty}{\infty}$ form:

$$\lim_{x\to\infty} \frac{\frac{d}{dx}[\ln(1 + x^2)]}{\frac{d}{dx}[e^x]} = \lim_{x\to\infty} \frac{\frac{2x}{1+x^2}}{e^x}$$

$$= \lim_{x\to\infty} \frac{2x}{e^x(1 + x^2)}$$

Apply L'Hôpital's Rule a second time as it remains in $\frac{\infty}{\infty}$ form:

$$\lim_{x\to\infty} \frac{\frac{d}{dx}[2x]}{\frac{d}{dx}[e^x(1 + x^2)]}$$

$$= \lim_{x\to\infty} \frac{2}{e^x(1 + x^2) + e^x(2x)}$$

$$= \lim_{x\to\infty} \frac{2}{e^x(x + 1)^2}$$

Since the denominator is now infinitely large, $\lim_{x\to\infty} \ln(y) = 0$.

Solve for y

Since $\ln(y) \to 0$, then $y \to e^0 = 1$.

Final Answer: The value of the limit is 1.

15. ANSWER:- -2

We have f(x) = x sin x

⇒ f'(x) = x cos x + sin x

⇒ f''(x) = x (− sin x) + cos x + cos x = (−x sin x) + 2 cos x

Now, it is given that f(x) = x sin x satisfies the equation f''(x) + f(x) + t cos x = 0

⇒ (−x sin x) + 2 cos x + x sin x + t cos x = 0

⇒ 2 cos x + t cos x = 0

⇒ cos x (t + 2) = 0

⇒ t + 2 = 0

⇒ t = −2

16. ANSWER:- A, D

For Option A: comparing $f(x)$ and $f(x + 1)$

$f(0) = -1, f(1) = 1$

- $f(1) = 1, f(2) = -1$

So for $x \in [0,1]$:

- $f(x)$ goes from $-1 - 1$
- $f(x + 1)$ goes from $1 \to -1$

Now define:

$$g(x) = f(x) - f(x + 1)$$

Then:

- $g(0) = -1 - 1 = -2$
- $g(1) = 1 - (-1) = 2$

Since $g(x)$ is continuous, it must cross 0 somewhere in $(0,1)$.

So there exists $c \in (0,1)$ such that:

$$f(c) = f(c + 1)$$

So Option A is TRUE,

For Option D: comparing $f(y)$ and $-f(2 - y)$

- Reflect → shift → reflect gives $-f(2 - y)$

Now check values on $[0,1]$:

- $f(0) = -1, f(1) = 1$
- $-f(2 - 0) = -f(2) = -(-1) = 1 \downarrow$
- $-f(2 - 1) = -f(1) = -1$

So:

- $f(y)$: goes $-1 \to 1$
- $-f(2 - y)$: goes $1 \to -1$

Define:

$$h(y) = f(y) + f(2 - y)$$

Check Out Our GATE Courses & Secure Your Dream Rank!

Then:

- $h(0) = -1 + (-1) = -2$
- $h(1) = 1 + 1 = 2$

Again, by continuity, $h(y) = 0$ somewhere in $(0,1)$, i.e.:

$$f(y) = -f(2 - y)$$

So Option D is also TRUE.

17. ANSWER:- A

$$\int_0^{\pi} x^2 \cos x dx = x^2[\sin x]_0^{\pi} - \int_0^{\pi} 2x \cdot \sin x$$
$$= (\pi^2(0) - 0) - \left[[-2x\cos x]_0^{\pi} - \int_0^{\pi} 2\cos x dx\right]$$
$$= 0 - [2\pi - 0 + 2[\sin x]_0^{\pi}]$$
$$= -[2\pi + 2(0 - 0)]$$
$$= -2\pi$$

18. ANSWER:- A

Roots of polynomial of degree '3' are 1, 2, 3
Polynomial will be
F(x) = (x-1)(x-2)(x-3)
f(0) = -1 × -2 × -3 = -6
f(4) = 3 × 2 × 1 = 6
f(0)·f(4) = - 36
f(0) + f(4) = 6 - 6 = 0
Option (A) is correct.

19. ANSWER:- A

Given:

$$\int_0^{2\pi} |x\sin x| dx = k\pi$$

Break integral:

$$= \int_0^{\pi} x\sin x dx - \int_{\pi}^{2\pi} x\sin x dx$$

Using integration by parts:

$$\int x\sin x dx = -x\cos x + \sin x + C$$

Evaluate:

$$\int_0^{\pi} x\sin x dx = \pi$$
$$\int_{\pi}^{2\pi} x\sin x dx = -3\pi$$

So,

$$\int_0^{2\pi} |x\sin x| dx = \pi - (-3\pi) = 4\pi$$

Therefore,

$$k = 4$$

20. ANSWER:- A

A function is continuous at some point c,
Value of f(x) defined for x > c = Value of f(x) defined for x < c = Value of f(x) defined for x = c
All values are 2 in option A, thus A is continuous at x = 3.

21. ANSWER:- D

We know that,

$$e^{ix} = \cos x + i\sin x$$

So, $I = \int_0^{\pi/2} \frac{e^{ix}}{e^{-ix}} dx$

$$= \int_0^{\pi/2} e^{i2x} dx$$
$$= \left[\frac{e^{i2x}}{2i}\right]_0^{\pi/2}$$
$$= \frac{1}{2i}[\cos 2x + i\sin 2x]_0^{\pi/2}$$
$$= \frac{1}{2i} \times (-2)$$

= -1/i
= i

22. ANSWER:- B

Let $L = \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^{2n}$
Take the natural log of both sides: $\ln(L) = \lim_{n\to\infty} 2n\ln\left(1 - \frac{1}{n}\right)$
Rewrite as a fraction to apply L'Hôpital's Rule:
$\ln(L) = \lim_{n\to\infty} \frac{2\ln(1-1/n)}{1/n}$
Applying L'Hôpital's Rule Differentiate the numerator and the denominator: $\ln(L) =$

$$\lim_{n\to\infty} \frac{2 \cdot \frac{1}{1-1/n} \cdot \frac{1}{n^2}}{-1/n^2}$$
$$\ln(L) = \lim_{n\to\infty} \frac{-2}{1-1/n}$$

Evaluate the limit as $n \to \infty$: $\ln(L) = -2$
$L = e^{-2}$

23. ANSWER:- B

$y=x^2-3x+1$
$y'=2x-3$
$y''=2$ // So, y has minimum value
the value at point $(2x-3)=0$
or, $x=3/2$

Check Out Our GATE Courses & Secure Your Dream Rank!

24. ANSWER:- D

Let $I = \int_0^{\frac{\pi}{4}} \frac{1-\tan x}{1+\tan x} dx = \int_0^{\frac{\pi}{4}} \frac{\cos x - \sin x}{\cos x + \sin x} dx$

Now put $\cos x + \sin x = t$, Then $(-\sin x + \cos x)dx = dt$ and changing limit So we get $I = \int_1^{\sqrt{2}} \frac{1}{t} dt = [\ln t] = \ln(\sqrt{2}) = \frac{\ln 2}{2}$

25. ANSWER:- B

For $x \leq \frac{3}{2}$:

$f(x) = \frac{25}{8x} \longrightarrow$ decreasing $\Rightarrow$ min at $x = \frac{3}{2}$

$$f = \frac{25}{12}$$

For $x > \frac{3}{2}$:

$f(x) = x + \frac{1}{x} \longrightarrow$ increasing in this interval $\Rightarrow$ min near $\frac{3}{2}$

$$f = \frac{13}{6}$$

Compare:

$$\frac{25}{12} < \frac{13}{6}$$

Minimum = $2\frac{1}{12}$(Option B)

26. ANSWER:- A

$$\lim_{x \to \infty} \frac{x - \sin x}{x + \cos x}$$

Divide numerator and denominator by x :

$$= \frac{1 - \frac{\sin x}{x}}{1 + \frac{\cos x}{x}}$$

As $x \to \infty$:

$$\frac{\sin x}{x} \to 0, \frac{\cos x}{x} \to 0$$

So,

$$\lim = \frac{1-0}{1+0} = 1$$

Final Answer: 1

27. ANSWER:- A

$f(x) = |x|$. Here, for all values of x, $f(x)$ exists. Therefore, it is continuous for all real values of x. At $x = 0, f(x)$ is not differentiable. Because if we take the left hand derivative here, it is negative while the right hand derivative is positive making LHD $\neq$ RHD.

28. ANSWER:- B

The definite integral evaluates to **0** due to symmetry.

Using the substitution $u = x - \pi$, the integral shifts from the interval $[0, 2\pi]$ to the symmetric interval $[-\pi, \pi]$:

$$\int_{-\pi}^{\pi} u^2 \sin(u + \pi) du = \int_{-\pi}^{\pi} -u^2 \sin u du$$

The function $f(u) = -u^2 \sin u$ is an odd function because plugging in $-u$ negates the entire expression:

$$f(-u) = -(-u)^2 \sin(-u) = u^2 \sin u = -f(u)$$

The integral of any odd function over a symmetric interval $[-a, a]$ is always 0 because the areas on the left and right sides of the y-axis completely cancel each other out.

Check Out Our GATE Courses & Secure Your Dream Rank!

3 LINEAR ALGEBRA

GATE CSE – 2026 MCQ SET- 1 (1-MARKS)

Q1. Consider 4 × 4 matrices with their elements from {**0**, **1**}. What is the number of such matrices with even number of **1**s in every row and every column is

(a) 512
(b) 1025
(c) 1023
(d) 255

GATE CSE – 2026 MCQ SET- 1 (1-MARKS)

Q2. For $n > 1$, the maximum multiplicity of any eigenvalue of an $n \times n$ matrix with elements from $\mathbb{R}$ is

(a) n
(b) $n-1$
(c) 1
(d) $n+1$

GATE CSE – 2026 MSQ SET- 1 (1-MARKS)

Q3. Let $n > 1$. Consider an $n \times n$ matrix M with its elements from $\mathbb{R}$. Let the vector $(0, 1, 0, 0, \ldots, 0) \in \mathbb{R}^n$ be in the null space of M. Which of the following options is/are always correct?

(a) Determinant of M is 1
(b) Determinant of M is 0
(c) Rank of M is 1
(d) There are at least two non-zero vectors in the null space of M

GATE CSE – 2026 MCQ SET- 2 (MARKS)

Q4. Consider the system of linear equations given below.

$$ax + y = b$$
$$16x + ay = 24$$

Suppose the values of a and b are chosen such that the system of linear equations produce multiple solutions. Then the product of a and b is ___ . (answer in integer)

GATE CSE – 2025 MCQ SET- 1 (2-MARKS)

Q5. Let A be a 2 × 2 matrix as given.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

What are the eigenvalues of the matrix A^{13}?

(a) $64\sqrt{2}, -64\sqrt{2}$
(b) $2\sqrt{2}, -2\sqrt{2}$
(c) $4\sqrt{2}, -4\sqrt{2}$
(d) $1, -1$

GATE CSE – 2025 MCQ SET- 1 (2-MARKS)

Q6. The product of all eigen value the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(a) -1
(b) 0
(c) 1
(d) 2

GATE CSE – 2025 MCQ SET- 1 (2-MARKS)

Q7. Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

(a) There exist at least $m - n$ linearly independent solutions to this system
(b) There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors

Check Out Our GATE Courses & Secure Your Dream Rank!

(c) There exists a non-zero solution in which at least $m - n$ variables are 0
(d) There exists a solution in which at least n variables are non-zero

GATE CSE – 2025 MCQ SET- 2 (1-MARKS)

Q8. Let L, M, and N be non-singular matrices of order 3 satisfying the equations $L^2 = L^{-1}$, $M = L^8$ and $N = L^2$. Which ONE of the following is the value of the determinant of $(M - N)$?

(a) 0
(b) 1
(c) 2
(d) 3

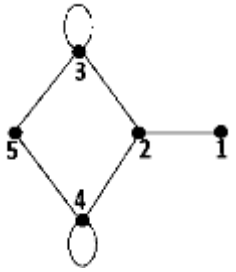
GATE CSE – 2024 MCQ SET- 2 (2-MARKS)

Q9. Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

(a) The determinant of B is the negative of the determinant of A
(b) If A is invertible, then B is also invertible
(c) If A is symmetric, then B is also symmetric
(d) If the trace of A is zero, then the trace of B is also zero

GATE CSE – 2024 MCQ SET- 2 (2-MARKS)

Q10. Let A be the adjacency matrix of the graph with vertices {1, 2, 3, 4, 5}.



Let λ1, λ2, λ3, λ4, and λ5 be the five eigenvalues of A. Note that these eigenvalues need not be distinct.
The value of λ1 + λ2 + λ3 + λ4 + λ5 = _____.

GATE CSE – 2023 MCQ (1-MARKS)

Q11. Let

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}.$$

Let det(A) and det(B) denote the determinants of the matrices A and B, respectively.
Which one of the options given below is TRUE?

(a) det(A) = det(B)
(b) det(B) = −det(A)
(c) det(A) = 0
(d) det(AB) = det(A) + det(B)

GATE CSE – 2022 MCQ (1-MARKS)

Q12. Consider the following two statements with respect to the matrices $A_{m\times n}$, $B_{n\times m}$, $C_{n\times n}$ and $D_{n\times n}$
Statement 1: tr(AB) = tr(BA)
Statement 2: tr(CD) = tr(DC)
where tr() represents the trace of a matrix. Which one of the following holds?

(a) Statement 1 is correct and Statement 2 is wrong
(b) Statement 1 is wrong and Statement 2 is correct.
(c) Both Statement 1 and Statement 2 are correct.
(d) Both Statement 1 and Statement 2 are wrong.

GATE CSE – 2022 MCQ (2-MARKS)

Q13. Consider solving the following system of simultaneous equations using LU decomposition.
$x_1 + x_2 - 2x_3 = 4$
$x_1 + 3x_2 - x_3 = 7$
$2x_1 + x_2 - 5x_3 = 7$
where L and U are denoted as

$$L = \begin{pmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{pmatrix}, U = \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{pmatrix}$$

Which one of the following is the correct combination of values for L_{32}, U_{33}, and x_1?

(a) $L_{32} = 2$, $U_{33} = -1/2$, $x_1 = -1$
(b) $L_{32} = 2$, $U_{33} = 2$, $x_1 = -1$
(c) $L_{32} = -1/2$, $U_{33} = 2$, $x_1 = 0$
(d) $L_{32} = -1/2$, $U_{33} = -1/2$, $x_1 = 0$

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2022 MCQ (2-MARKS)

Q14. Which of the following is/are the eigenvector(s) for the matrix given below?

$$\begin{pmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{pmatrix}$$

(a) $\begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$

(c) $\begin{pmatrix} -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}$

(d) $\begin{pmatrix} 0 \\ 1 \\ -3 \\ 0 \end{pmatrix}$

GATE CSE – 2021 NAT SET – 2 (1-MARKS)

Q15. Suppose that P is a 4 × 5 matrix such that every solution of the equation Px = 0 is a scalar multiple of $[2\ 5\ 4\ 3\ 1]^T$. The rank of P is _____

GATE CSE – 2021 NAT SET - 1(2-MARKS)

Q16. Consider the following matrix.

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

The largest eigenvalue of the above matrix is _____.

GATE CSE – 2020 MCQ (2-MARKS)

Q17. Let A and B be two n×n matrices over real numbers. Let rank(M) and det(M) denote the rank and determinant of a matrix M, respectively.
Consider the following statements.
I. rank(AB) = rank(A) × rank (B)
II. det(AB) = det(A) × det(B)
III. rank(A+B) ≤ rank(A) + rank(B)
IV. det(A+B) ≤ det(A) + det(B)
Which of the above statements are TRUE?

(a) I and II only
(b) I and IV only
(c) III and IV only
(d) II and III only

GATE CSE – 2019 MCQ (1-MARKS)

Q18. Let X be a square matrix. Consider the following two statements on X.
I. X is invertible.
II. Determinant of X is non-zero.
Which one of the following is TRUE?

(a) I implies II; II does not imply I.
(b) II implies I; I does not imply II.
(c) I and II are equivalent statements.
(d) I does not imply II; II does not imply I.

GATE CSE – 2019 MCQ (1-MARKS)

Q19. Let U = {1, 2, …,n} and A = {(x, X)| x∈X, X⊆U }.
Consider the following two statements on |A|.
I. $|A| = n \times 2^{n-1}$
II. $|A| = \sum_{k=1}^{n} k \binom{n}{k}$
Which of the above statements is/are TRUE?

(a) Only II
(b) Only I
(c) Both I nor II
(d) Neither I nor II

GATE CSE – 2019 NAT (1-MARKS)

Q20. Consider the following matrix

$$R = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}$$

The absolute value of the product of Eigen value of R is _____.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2018 MCQ (2-MARKS)

Q21. Assume that multiplying a matrix G_1 of dimension p × q with another matrix G_2 of dimension q × r requires pqr scalar multiplications. Computing the product of *n* matrices $G_1G_2G_3…Gn$ can be done by parenthesizing in different ways. Define G_iG_{i+1} as an **explicitly computed pair** for a given paranthesization if they are directly multiplied. For example, in the matrix multiplication chain $G_1G_2G_3G_4G_5G_6$ using parenthesization $(G_1(G_2G_3))(G_4(G_5G_6))$,$G_2G_3$ and G_5G_6, are only explicitly computed pairs. Consider a matrix multiplication chain $F_1F_2F_3F_4F_5$, where matrices F_1,F_2,F_3,F_4 and F_5 are of dimensions 2×25,25 ×3,3 ×16,16 ×1 and 1 ×1000, respectively.
In the parenthesization of $F_1F_2F_3F_4F_5$ that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are

(a) F_1F_2 and F_3F_4 only
(b) F_2F_3 only
(c) F_3F_4 only
(d) F_1F_2 and F_4F_5 only

GATE CSE – 2018 MCQ (2-MARKS)

Q22. Consider a matrix P whose only eigenvectors are the multiples of $\begin{bmatrix}1\\4\end{bmatrix}$.
Consider the following statements.
(I) P does not have an inverse
(II) P has a repeated eigenvalue
(III) P cannot be diagonalized
Which one of the following options is correct?

(a) Only I and III are necessarily true
(b) Only II is necessarily true
(c) Only I and II are necessarily true
(d) Only II and III are necessarily true

GATE CSE – 2018 MCQ (1-MARKS)

Q23. Consider a matrix A = uv^T where u = $\begin{pmatrix}1\\2\end{pmatrix}$, $v = \begin{pmatrix}1\\1\end{pmatrix}$. Note that V^T denotes the transpose of v. The largest eigenvalue of A is _____.

GATE CSE – 2017 MCQ SET - 2(1-MARKS)

Q24. Let P $\begin{bmatrix}1 & 1 & -1\\2 & -3 & 4\\3 & -2 & 3\end{bmatrix}$ and Q = $\begin{bmatrix}-1 & -2 & -1\\6 & 12 & 6\\5 & 10 & 5\end{bmatrix}$ be two matrices. Then the rank of P + Q is _____.

GATE CSE – 2017 NAT SET - 2(1-MARKS)

Q25. If the characteristic polynomial of a 3 × 3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is _____.

GATE CSE – 2017 MCQ SET - 1(2-MARKS)

Q26. Let $c_1 \ldots . c_n$ be scalars, not all zero, such that $\sum_{i=1}^{n} c_i a_i = 0$ where a_i are column vectors in R^n. Consider the set of linear equations

$$Ax = b$$

where $A = [a_1 \ldots . a_n]$ and $b = \sum_{i=1}^{n} a_i$. The set of equations has

(a) a unique solution at $x = J_n$ where J_n denotes a n-dimensional vector of all 1.
(b) no solution
(c) infinitely many solutions
(d) finitely many solutions

GATE CSE – 2017 MCQ SET - 1(2-MARKS)

Q27. Let A be m×n real valued square symmetric matrix of rank 2 with expression given below.
$\sum_{i=j}^{n}\sum_{j=1}^{n} A_{ij}^2 = 50$
Consider the following statements
(i) One eigenvalue must be in [-5, 5].
(ii) The eigenvalue with the largest magnitude must be strictly greater than 5.
Which of the above statements about eigenvalues of A is/are necessarily CORRECT?

(a) Both (I) and (II)
(b) (I) only
(c) (II) only
(d) Neither (I) nor (II)

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2016 MCQ SET - 2(1-MARKS)

Q28. Consider the systems, each consisting of m linear equations in n variables.
I. If m < n, then all such systems have a solution
II. If m > n, then none of these systems has a solution
III. If m = n, then there exists a system which has a solution
Which one of the following is CORRECT?
(a) I, II and III are true
(b) Only II and III are true
(c) Only III is true
(d) None of them is true

GATE CSE – 2016 NAT SET - 2(1-MARKS)

Q29. Suppose that the eigenvalues of matrix A are 1, 2, 4. The determinant of $(A^{-1})^T$ is _____.

GATE CSE – 2016 NAT SET - 2(1-MARKS)

Q30. Two eigenvalues of a 3 × 3 real matrix P are (2 + √-1) and 3. The determinant of P is _____.

GATE CSE – 2015 MCQ SET - 3(1-MARKS)

Q31. In the given matrix $\begin{bmatrix}1 & -1 & 2\\0 & 1 & 0\\1 & 2 & 1\end{bmatrix}$, one of the eigenvalues is 1 . The eigenvectors corresponding to the eigenvalue 1 are

(a) $\{a(4,2,1) \mid a \neq 0, a \in \mathbb{R}\}$
(b) $\{a(-4,2,1) \mid a \neq 0, a \in \mathbb{R}\}$
(c) $\{a(\sqrt{2},0,1) \mid a \neq 0, a \in \mathbb{R}\}$
(d) $\{a(-\sqrt{2},0,1) \mid a \neq 0, a \in \mathbb{R}\}$

GATE CSE – 2015 MCQ SET - 2(1-MARKS)

Q32. The larger of the two eigenvalues of the matrix $\begin{bmatrix}4 & 5\\2 & 1\end{bmatrix}$ is

(a) 6
(b) 7
(c) 8
(d) 9

GATE CSE – 2015 MCQ SET - 3(2-MARKS)

Q33. If the following system has non-trivial solution,
px + qy + rz = 0 qx + ry + pz = 0 rx + py + qz = 0
then which one of the following options is True?

(a) p-q+r = 0 or p = q = -r
(b) p+q-r = 0 or p = -q = r
(c) p+q+r = 0 or p = q = r
(d) p-q+r = 0 or p = -q = -r

GATE CSE – 2015 MCQ SET - 1(2-MARKS)

Q34. Consider the following 2 × 2 matrix A where two elements are unknown and are marked by a and b. The eigenvalues of this matrix are –1 and 7. What are the values of a and b?
$A = \begin{pmatrix}1 & 4\\b & a\end{pmatrix}$

(a) a = 6, b = 4
(b) a = 4, b = 6
(c) a = 3, b = 5
(d) a = 5, b = 3

GATE CSE – 2015 NAT SET - 1(2-MARKS)

Q35. In the LU decomposition of the matrix $\begin{bmatrix}2 & 2\\4 & 9\end{bmatrix}$, if the diagonal elements of U are both 1, then the lower diagonal entry l_{22} of L is _____

GATE CSE – 2015 NAT SET - 2(2-MARKS)

Q36. Perform the following operations on the matrix

$$\begin{bmatrix}3 & 4 & 45\\7 & 9 & 105\\13 & 2 & 195\end{bmatrix}$$

(i) Add the third row to the second row.
(ii) Subtract the third column from the first column.
The determinant of the resultant matrix is _____

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2015 NAT SET - 2(2-MARKS)

Q37. Which one of the following statements is TRUE about every n × n matrix with only real eigenvalues?

(a) If the trace of the matrix is positive and the determinant of the matrix is negative, at least one of its eigenvalues is negative.
(b) If the trace of the matrix is positive, all its eigenvalues are positive.
(c) If the determinant of the matrix is positive, all its eigenvalues are positive.
(d) If the product of the trace and determinant of the matrix is positive, all its eigenvalues are positive.

GATE CSE – 2014 NAT SET - 2(2-MARKS)

Q38. If the matrix A is such that

$$A = \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} [1 \quad 9 \quad 5]$$

then the determinant of A is equal to _____.

GATE CSE – 2014 NAT SET - 1(1-MARKS)

Q39. The value of the dot product of the eigenvectors corresponding to any pair of different eigenvalues of a 4-by-4 symmetric positive definite matrix is _____.

GATE CSE – 2014 NAT SET - 2(1-MARKS)

Q40. The product of the non-zero eigenvalues of the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

is _____

s

GATE CSE – 2013 MCQ (1-MARKS)

Q41. Which one of the following does NOT equal to

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} ?$$

(a) $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$

(b) $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

(c) $\begin{vmatrix} 1 & x-y & x^2-y^2 \\ 1 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$

(d) $\begin{vmatrix} 1 & x+y & x^2+y^2 \\ 1 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

GATE CSE – 2012 MCQ (1-MARKS)

Q42. Let A be a 2 × 2 matrix with elements $a_{11} = a_{12} = a_{21} = +1$ and $a_{22} = -1$. Then the eigenvalues of the matrix A^{19} are

(a) 1024 and –1024
(b) $1024\sqrt{2}$ and $-1024\sqrt{2}$
(c) $4\sqrt{2}$ and $-4\sqrt{2}$
(d) $512\sqrt{2}$ and $-512\sqrt{2}$

GATE CSE – 2011 MCQ (2-MARKS)

Q43. Consider the matrix as given below.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 7 \\ 0 & 0 & 3 \end{bmatrix}$$

Which one of the following provides the CORRECT values of eigenvalues of the matrix?

(a) 1, 4, 3
(b) 3, 7, 3
(c) 7, 3, 2
(d) 1, 2, 3

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2010 MCQ SET - 1(2-MARKS)

Q44. Consider the following matrix

$$A = \begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$$

If the eigenvalues of A are 4 and 8, then

(a) x=4, y=10
(b) x=5, y=8
(c) x=-3, y=9
(d) x=-4, y=10

GATE CSE – 2008 MCQ (2-MARKS)

Q45. The following system of equations

- $x_1 + x_2 + 2x_3 = 1$
- $x_1 + 2x_2 + 3x_3 = 2$
- $x_1 + 4x_2 + \alpha x_3 = 4$

has a unique solution. The only possible value(s) for α is/are

(a) 0
(b) either 0 or 1
(c) one of 0,1 , or -1
(d) any real number except 5.

GATE CSE – 2007 MCQ (2-MARKS)

Q46. Let A be a 4×4 matrix with eigen values $-5, -2, 1, 4$. Which of the following is an eigen value of the matrix $\begin{bmatrix} A & I \\ I & A \end{bmatrix}$, where I is the 4×4 identity matrix?

(a) -5
(b) $\mathbf{-7}$
(c) 2
(d) 1

GATE CSE – 2007 MCQ (2-MARKS)

Q47. Consider the set of (column) vectors defined by $X = \{x \in R^3 \mid x_1 + x_2 + x_3 = 0, \text{ where } x^T = [x_1, x_2, x_3]^T\}$

Which of the following is TRUE?

(a) $\{[1,-1,0]^T, [1,0,-1]^T\}$ is a basis for the subspace X.
(b) $\{[1,-1,0]^T, [1,0,-1]^T\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X.
(c) X is not a subspace of R^3.
(d) None of the above

GATE CSE – 2006 MCQ (2-MARKS)

Q48. F is an $n \times n$ real matrix. b is an $n \times 1$ real vector. Suppose there are two $n \times 1$ vectors, u and v such that, $u \neq v$ and $Fu = b, Fv = b$. Which one of the following statements is false?

(a) Determinant of F is zero.
(b) There are an infinite number of solutions to $Fx = b$
(c) There is an $x \neq 0$ such that $Fx = 0$
(d) F must have two identical rows

GATE CSE – 2005 MCQ (2-MARKS)

Q49. Consider the set H of all $3 * 3$ matrices of the type

$$\begin{pmatrix} a & f & e \\ 0 & b & d \\ 0 & 0 & c \end{pmatrix}$$

where a, b, c, d, e and f are real numbers and $abc \neq 0$. Under the matrix multiplication operation, the set H is:

(a) a group
(b) a monoid but not a group
(c) a semi group but not a monoid
(d) neither a group nor a semi group

GATE CSE – 2005 MCQ (2-MARKS)

Q50. Consider the following system of linear equations :

$$2x_1 - x_2 + 3x_3 = 1$$
$$3x_1 + 2x_2 + 5x_3 = 2$$
$$-x_1 + 4x_2 + x_3 = 3$$

The system of equations has

(a) no solution
(b) a unique solution
(c) more than one but a finite number of
(d) Solutions an infinite number of solutions

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE – 2005 MCQ (2-MARKS)

Q51. What are the eigenvalues of the following 2 × 2 matrix?

$$\begin{pmatrix} 2 & -1 \\ -4 & 5 \end{pmatrix}$$

(a) -1 and 1
(b) 1 and 6
(c) 2 and 5
(d) 4 and -1

ANSWER KEY

1.(a)	**2.(a)**	**3.(b,d)**	**4.(24)**
5.(a)	**6.(b)**	**7. (a)**	**8. (a)**
9.(a, b)	**10.(2)**	**11.(b)**	**12.(c)**
13.(d)	**14.(a, c, d)**	**15.(4:4)**	**16. (3)**
17.(d)	**18.(c)**	**19.(c)**	**20.(12)**
21.(c)	**22.(d)**	**23.(3)**	**24.(2)**
25.(5)	**26.(c)**	**27.(b)**	**28.(c)**
29.(0.125)	**30.(15)**	**31.(b)**	**32.(a)**
33.(c)	**34.(d)**	**35. (5)**	**36.(0)**
37.(a)	**38.(0)**	**39.(0)**	**40.(6)**
41.(d)	**42.(d)**	**43.(a)**	**44. (d)**
45.(d)	**46.(c)**	**47.(a)**	**48.(d)**
49.(a)	**50.(b)**	**51.(b)**	

HINTS & SOLUTIONS

1. ANSWER:- A

Explanation : We need 4×4 matrices with entries {0,1} such that **each row and each column has even number of 1s**.
Key idea: Think in mod 2 (even = sum = 0).
Step 1: First 3 rows and 3 columns can be filled freely.
So we can choose a **3×3 block arbitrarily**.
Number of ways:
2^9
Step 2: Now fix the last column.
For each of the first 3 rows, choose the 4th element so that row sum becomes even.
So last column entries are uniquely determined for first 3 rows.
Step 3: Fix the last row.
For each of first 3 columns, choose entry so column sum becomes even.
Again uniquely determined.
Step 4: The bottom-right element is automatically fixed (both row and column condition match).
So total number of valid matrices:
2^9=512

Answer : 512

2. ANSWER:- A

Multiplicity of an eigenvalue means how many times it appears (algebraic multiplicity).
For an n×n matrix, the characteristic polynomial has degree n, so total multiplicities of all eigenvalues add up to n.
Maximum possible multiplicity of a single eigenvalue occurs when **all eigenvalues are same** (e.g., identity matrix or scalar matrix).
So maximum multiplicity = n.

3. ANSWER:- B, D

Given: $M\mathbf{v} = 0$ where $\mathbf{v} = (0,1,0,\dots,0) \neq 0$
So null space contains a non-zero vector ⇒ matrix is singular.
Hence:
- Determinant must be 0 ⇒ (B) true
- (A) false

Rank:
We only know rank < n, not necessarily 1 ⇒ (C) false
Null space:
If one non-zero vector is in null space, then all its scalar multiples are also in null space ⇒ infinitely many non-zero vectors ⇒ at least two exist ⇒ (D) true
Answer: B and D

4. ANSWER:- 24

System:

$$ax + y = b$$
$$16x + ay = 24$$

For multiple (infinite) solutions, the two equations must be dependent.
So ratios must be equal:

$$\frac{a}{16} = \frac{1}{a} = \frac{b}{24}$$

From:

$$\frac{a}{16} = \frac{1}{a} \Rightarrow a^2 = 16 \Rightarrow a = \pm 4$$

Now:

$$\frac{a}{16} = \frac{b}{24} \Rightarrow b = \frac{24a}{16} = \frac{3a}{2}$$

If $a = 4, b = 6 \Rightarrow ab = 24$
If $a = -4, b = -6 \Rightarrow ab = 24$
Final Answer: 24

Check Out Our GATE Courses & Secure Your Dream Rank!

5. ANSWER:- A

To find the eigenvalues of A, we need to solve the characteristic equation:
$\det(A - \lambda I) = 0$
Where:

- A is the given matrix,
- λ is the eigenvalue,
- I is the identity matrix.

The characteristic equation becomes:
The determinant of this 2x2 matrix is:
$(1 - \lambda)(-1 - \lambda) - 1$
Simplify this:
$\lambda^2 - 2 = 0$
So:
$\lambda^2 = 2$
Taking the square root:
$\lambda = \pm\sqrt{2}$
So, the eigenvalues of A are $\sqrt{2}$ and $-\sqrt{2}$.
Step 2: Eigenvalues of A^{13}
The eigenvalues of powers of a matrix are simply the eigenvalues raised to the power of 13. For A^{13}:

- The eigenvalue $\sqrt{2}$ becomes $(\sqrt{2})^{13} = 2^{(13/2)}$.
- The eigenvalue $-\sqrt{2}$ becomes $(-\sqrt{2})^{13} = -(2^{13/2})$.

Simplifying $2^{(13/2)}$:
$2^{(13/2)} = 64\sqrt{2}$
Thus, the eigenvalues of A^{13} are:
$64\sqrt{2}$ and $-64\sqrt{2}$.
Final Answer:
The eigenvalues of A^{13} are $64\sqrt{2}$ and $-64\sqrt{2}$.
So, the correct answer is $64\sqrt{2}, -64\sqrt{2}$.

6. ANSWER:- B

We know that the product of any eigen value is equal to it's determinant.
So, determinant of the given matrix is:
$|A|=1(5*9-8*6)-2(9*4-7*6)+3(4*8-7*5)|A|=1(45-48)-2(36-42)+3(32-35)|A|=1(-3)-2(-6)+3(-3)|A|=-3+12-9=0$Since the determinant of the given matrix is zero, one of the eigenvalues must also be zero. Therefore, we can conclude that the product of the eigenvalues is zero.

7. ANSWER:- A

In a homogeneous system $Ax = 0$ with m variables and n equations, the dimension of the solution space (nullity) is $m - \text{rank}(A)$.
Since $\text{rank}(A) \leq n$, then nullity $= m - \text{rank}(A) \geq m - n$. This ensures at least $m - n$ linearly independent vectors in the solution set.

8. ANSWER:- A

The given equations $L^2 = L^{-1}$, $M = L^8$, and $N = L^2$ lead to $\det(L)^3 = 1$, so $\det(L)$ is either 1.
The determinant of $M-N = L^8 - L^2 = L^2 (L^6-I)$ is 0 because $L^6 = I$, making $\det(L^6-I) = 0$.
Therefore, $\det(M-N)=0$.

9. ANSWER:- A, B

Option A.

Swapping two rows of a matrix multiplies its determinant by -1.
Correct statement → TRUE

Option B.

Row swapping is an elementary row operation that does not affect invertibility. It only changes the determinant by a nonzero factor (-1), so:

- Matrix is invertible iff determinant $\neq 0$
- This condition is preserved

Correct statement → TRUE
(But the given justification is incomplete—should refer to preservation of invertibility under elementary operations.)

Option C.

A matrix is symmetric if $A=A^T$
Swapping only rows does not preserve this condition in general, because symmetry requires coordinated changes in both rows and columns.
So the statement "this is not always true" is logically correct.
Correct statement → TRUE

Option D.

Trace is the sum of diagonal elements.
Row swapping does not have a fixed effect on the diagonal entries—it may change or preserve them depending on the swap.

So the claim "trace will never remain same" is not universally true.

Incorrect statement → FALSE

10. ANSWER:- 2

Firstly we will convert this graph into an adjacency matrix and then we will look further

Check Out Our GATE Courses & Secure Your Dream Rank!

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Now we know the standard definition of the graph the summation of the eigenvalue is equal to the trace of the matrix

- $\lambda 1=0$
- $\lambda 2=0$
- $\lambda 3=1$
- $\lambda 4=1$
- $\lambda 5=0$

$\therefore \lambda 1 +\lambda 2 +\lambda 3 +\lambda 4+ \lambda 5=0+0+1+1+0=2$

So, the summation of the diagonal is 2.

11. ANSWER:- B

The elements of both the determinant A and B are the same just two rows are interchanged R1, R3

According to the property of any determinant if two rows (or 2 columns) of a determinant are interchanged the sign of the value of the determinant is changed.

Here only one time rows are changed so the determinant should be multiplied by (-1).

12. ANSWER:- C

To compute the trace of AB, we focus on the sum of the diagonal elements of AB:

$$\text{tr}(AB) = \sum_{i=1}^{m} (AB)_{ii}$$

Similarly, for BA, the trace is the sum of the diagonal elements:

$\text{tr}(BA) = \sum_{i=1}^{m} (BA)_{ii}$

Now, there is a key property in matrix theory known as the cyclic property of the trace, which states that for matrices A and B where the products AB and BA are defined (i.e., the dimensions match), the trace is invariant under cyclic permutations of the matrices.

So, option C is the correct answer.

13. ANSWER:- D

Initial matrix

R_2 - (1) $R_1 \rightarrow R_2$

Subtract 1 row 1 from row 2

U[2][1] = 1 -(1*1)/1 = 0

U[2]2] = 3-(1*1)/1 = 2

U[2][3] =-1-(1*-2)/1 = 1

L[2][1] = 1

R_2 - (2)$R_1 \rightarrow R_3$

Subtract 1 row 1 from row 3

U[3][1] = 2 -(2*1)/1 = 0

U[3]2]=1 -(2*1)/1 = -1

U[3][3] = -5 -(2*-2)/1 = -1

L[3][1] = 2

R_3 - (-1/2) $R_2 \rightarrow R_3$

Subtract 2 row 2 from row 2

U[3][2] = 1 - (-1*2)/1 = 0

U[3]3] = -1 - (-1*1)/2 = -1/2

L[3][2] = -1/2

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -2 \\ 1 & 3 & -1 \\ 2 & 1 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & -2 \\ 0 & 2 & 1 \\ 2 & 1 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & -2 \\ 0 & 2 & 1 \\ 0 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & \frac{-1}{2} & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{-1}{2} \end{bmatrix}$$

14. ANSWER:- A, C, D

Step 1: Matrix

$$A = \begin{pmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{pmatrix}$$

We want to determine which of the following vectors are eigenvectors of A :

$$\begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -3 \\ 0 \end{pmatrix}$$

Step 2: Definition

A vector v is an eigenvector of A if there exists a scalar λ such that

$$Av = \lambda v$$

where λ is the eigenvalue corresponding to v.

Step 3: Check each vector

Vector 1:

$$v = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

$$Av = \begin{pmatrix} -9(-1) + (-6)(1) + (-2)(0) + (-4)(1) \\ -8(-1) + (-6)(1) + (-3)(0) + (-1)(1) \\ 20(-1) + 15(1) + 8(0) + 5(1) \\ 32(-1) + 21(1) + 7(0) + 12(1) \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

$$Av = 1 \cdot v$$

Therefore, this vector is an eigenvector with eigenvalue $\lambda = 1$.

Vector 2:

Check Out Our GATE Courses & Secure Your Dream Rank!

$$v = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$$

$$Av = \begin{pmatrix} -9(1) + (-6)(0) + (-2)(-1) + (-4)(0) \\ -8(1) + (-6)(0) + (-3)(-1) + (-1)(0) \\ 20(1) + 15(0) + 8(-1) + 5(0) \\ 32(1) + 21(0) + 7(-1) + 12(0) \end{pmatrix} = \begin{pmatrix} -7 \\ -5 \\ 12 \\ 25 \end{pmatrix}$$

since Av is not a scalar multiple of v, the vector is not an eigenvector of the matrix.

Vector 3:

$$v = \begin{pmatrix} -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}$$

$$Av = \begin{pmatrix} -9(-1) + (-6)(0) + (-2)(2) + (-4)(2) \\ -8(-1) + (-6)(0) + (-3)(2) + (-1)(2) \\ 20(-1) + 15(0) + 8(2) + 5(2) \\ 32(-1) + 21(0) + 7(2) + 12(2) \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 6 \\ 6 \end{pmatrix}$$

$$Av = 3 \cdot v$$

Therefore, this vector is an eigenvector with eigenvalue $\lambda = 3$.

Vector 4:

$$v = \begin{pmatrix} 0 \\ 1 \\ -3 \\ 0 \end{pmatrix}$$

$$Av = \begin{pmatrix} -9(0) + (-6)(1) + (-2)(-3) + (-4)(0) \\ -8(0) + (-6)(1) + (-3)(-3) + (-1)(0) \\ 20(0) + 15(1) + 8(-3) + 5(0) \\ 32(0) + 21(1) + 7(-3) + 12(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ -9 \\ 0 \end{pmatrix}$$

$$Av = 3 \cdot v$$

Therefore, this vector is an eigenvector with eigenvalue $\lambda = 3$.

15. ANSWER:- 4

Since every solution of $Px = 0$ is a scalar multiple of the given vector, the null space is one-dimensional, so nullity(P) = 1. Using the Rank–Nullity Theorem,
rank(P) + nullity(P) = 5, we get
rank(P) = 4. Therefore, the rank of $P$is 4.

16. ANSWER:- 3

Let $B = A + I$

So, $B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$

All rows are identical
Only 1 independent row
So, rank of $(B) = 1$
Also, rank = Number of nonzero eigen values $= 1$

So, eigen values of $B \Rightarrow \lambda, 0, 0, 0$
And $\lambda + 0 + 0 + 0 =$ trace $= 1 + 1 + 1 + 1$
So, $\lambda = 4$
$B \rightarrow 4, 0, 0, 0$
$A \rightarrow B - I \rightarrow 3, -1, -1, -1$
Maximum eigen value $= 3$

17. ANSWER:- D

Option 1 justification: There is no result like $P(AB) = P(A) + P(B) - n$ Option 3 Justification is not required as it is irrelevant of the given inequality. Option 4, better to give counter example.

Take $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
Here $Det(A) = Det(B) = 0$ but $Det(A + B) = 1$.

18. ANSWER:- C

For a square matrix X:
If X is invertible ⇒ there exists X^{-1}
⇒ det(X) ≠ 0
⇒ I ⇒ II
If det(X) ≠ 0 ⇒ matrix is non-singular
⇒ inverse exists
⇒ X is invertible
⇒ II ⇒ I
Therefore, both imply each other.

19. ANSWER:- C

Given the set $U = \{1, 2, \ldots, n\}$ and the set

$$A = \{(x, X) \mid x \in X, X \subseteq U\},$$

the cardinality $|A|$ equals the total number of pairs (x, X) such that x is an element of subset X.
For each fixed element $x \in U$, the number of subsets X containing x is 2^{n-1} because X can be any subset of the remaining $n - 1$ elements.
Since there are n such elements x,

$$|A| = n \times 2^{n-1}$$

This matches the expression

$$|A| = \sum_{k=1}^{n} k \binom{n}{k},$$

because

$$\sum_{k=1}^{n} k \binom{n}{k} = n2^{n-1}.$$

Therefore, both statements I and II are true.

Check Out Our GATE Courses & Secure Your Dream Rank!

20. ANSWER:- 12

$$R = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}$$

Apply $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1,$

$$R_4 \to R_4 - R_1$$
$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 1 & 5 & 19 \\ 0 & 2 & 12 & 56 \\ 0 & 2 & 16 & 98 \end{bmatrix}$$

Apply $R_3 \to R_3 - 2R_2, R_4 \to R_4 - R_3$

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 1 & 5 & 19 \\ 0 & 0 & 2 & 18 \\ 0 & 0 & 4 & 42 \end{bmatrix}$$

Apply $R_4 \to R_4 - 2R_3$

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 1 & 5 & 19 \\ 0 & 0 & 2 & 18 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Product of Eigen values $= 1 * 1 * 2 * 6 = 12$

21. ANSWER:- C

Matrix F_5 is of dimension 1 × 1000, which is going to cause very much multiplication cost. So evaluating F_5 at last is optimal.
Total number of scalar multiplications are 48 + 75 + 50 + 2000 = 2173 and optimal parenthesis is $((F_1(F_2(F_3\ F_4)))F_5)$.

As concluded, F_3, F_4 are explicitly computed pairs.

Option (C) is Correct.

22. ANSWER:- D

Since all eigenvectors of *P*are multiples of a single vector, the eigen space is one dimensional. For a 2∗2 matrix, this implies that there is only one linearly independent eigenvector, so the matrix must have a repeated eigenvalue. Hence, statement (II) is true. Also, a matrix is diagonalizable only if it has two linearly independent eigenvectors. Since here there is only one, the matrix is not diagonalizable. Hence, statement (III) is true.

However, having only one eigenvector does not imply that the matrix is non-invertible. A matrix can still be invertible if its eigenvalue is non-zero. Hence, statement (I) is not necessarily true.

23. ANSWER:- 3

$$\mathrm{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathrm{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

→ Correction in Explanation:

$$\mathrm{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathrm{v}^{\mathrm{T}} = [1 \quad 1]$$
$$\mathrm{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}[1 \quad 1] = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$
$$\det(\mathrm{A} - \lambda \mathrm{I}) = 0$$
$$\Rightarrow \begin{bmatrix} 1-\lambda & 1 \\ 2 & 2-\lambda \end{bmatrix} = 0$$
$$\Rightarrow (1-\lambda)(2-\lambda) - 2 = 0$$
$$\Rightarrow \lambda^2 - 3\lambda = 0$$
$$\lambda = 0,3$$

So maximum is 3 .

24. ANSWER:- 2

Determinant of matrix P + Q = 0.
so, rank ≠ 3
now, the 1st minor of P+Q matrix = 8 ≠ 0
so, rank of P + Q is 2.

25. ANSWER:- 5

For a 3 × 3 matrix 'M', the characteristic equation |A – λI| is
$\lambda^3 - 4\lambda^2 + a\lambda + 30 = 0 \to (1)$
One eigen value is '2', so substitute it
$2^3 - 4(2)^2 + a(2) + 30 = 0$
$8 - 16 + 2a + 30 = 0$
$2a = -22$
$a = -11$
Substitute in (1),
$\lambda^3 - 4\lambda^2 - 11 + 30 = 0$

$$\lambda = 2 \Big| \begin{matrix} 1 & -4 & -11 & 30 \\ & 2 & -4 & -30 \end{matrix}$$

So, (1) can be written as

$$(\lambda - 2)(\lambda^2 - 2\lambda - 15) = 0$$
$$(\lambda - 2)(\lambda^2 - 5\lambda + 3\lambda - 15) = 0$$
$$(\lambda - 2)(\lambda + 3)(\lambda - 5) = 0$$
$$\lambda = 2, -3, 5$$
$$\mathrm{Max}\lambda = 5$$

26. ANSWER:- C

Let $c^1 \cdot c^n$ be scalars not all zero. Such that the following expression holds:
$\sum_{i=1}^{n} c_i a_i = 0$ where a_i is column vectors in R^n.
Consider the set of linear equations.

Check Out Our GATE Courses & Secure Your Dream Rank!

$Ax = B$. where $A = [a_1 \ldots . a_n]$ and $b = \sum_{i=1}^{n} a_i$
Then, set of equations has
The equation $\sum_i c_i a_i = 0$ with $\exists i$: $c_i \neq 0$ indicates that the column vectors of A are linearly dependent. Consequently, the determinant of matrix A is zero. This implies that the equation Ax=b either has no solution or infinitely many solutions. Given $\sum_i a_i = b$, it is evident that an n-dimensional vector of all ones satisfies the equation Ax=b. Therefore, Ax=b must have infinitely many solutions. The correct answer is (C)

27. ANSWER:- B

Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
be a real valued, rank = 2 matrix.
$\sum_{i=1}^{n} \sum_{i=1}^{n} A_{ij}^2 = 50$
$a^2 + b^2 + c^2 + d^2 = 50$
Square values are of order 0, 1, 4, 9, 16, 25, 36, …
So consider (0, 0, 5, 5) then sum of this square = 0 + 0 + 25 + 25 = 0
To get rank 2 , the 2 × 2 matrix can be

$$A = \begin{bmatrix} 0 & 5 \\ 5 & 0 \end{bmatrix}$$

The eigen values are, $|A - \lambda I| = 0$ (The characteristic equation)

$$\begin{vmatrix} 0-\lambda & 5 \\ 5 & 0-\lambda \end{vmatrix} = 0$$
$$-\lambda(-\lambda) - 25 = 0$$
$$\lambda^2 - 25 = 0$$
$$\lambda = \pm 5$$

So, the eigen values are within $[-5,5]$, Statement I is correct.
The eigen values with largest magnitude must be strictly greater than 5: False.
So, only Statement I is correct.

28. ANSWER:- C

i) If m < n
In AX = B,
If R(A) ≠ R(A|B)
then there will be no solution.
ii) False, because if R(A) = R(A|B),
then there will be solution possible.
iii) True, if R(A) = R(A|B),
then there exists a solution.

29. ANSWER:- 0.125

Determinant of a matrix is product of the eigen values.
Given that eigen values are 1, 2, 4.
So, its determinant is 1*2*4 = 8
The determinant of $(A^{-1})^T = 1/ A^T = 1/|A| = 1/8 = 0.125$

30. ANSWER:- 15

If an eigen value of a matrix is a complex number, then there will be other eigen value, which is conjugate of the complex eigen value.
So, For the given 3×3 matrix there would be 3 eigen values.
Given eigen values are : 2+i and 3.
So the third eigen value should be 2-i.
As per the theorems, the determinant of the matrix is the product of the eigen values.
So the determinant is (2+i)*(2-i)*3 = 15.

31. ANSWER:- B

Given matrix:

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Eigenvalue $\lambda = 1$.
To find eigenvectors, solve:

$$(A - I)\mathbf{x} = 0$$

where I is the identity matrix

$$A - I = \begin{bmatrix} 0 & -1 & 2 \\ 0 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix}$$

Let $\mathbf{x} = (x, y, z)^T$. Then:

$$\begin{cases} 0 \cdot x - 1 \cdot y + 2 \cdot z = 0 \\ 0 = 0 \\ 1 \cdot x + 2 \cdot y + 0 \cdot z = 0 \end{cases}$$

From the first and third equations:
1. $y + 2z = 0 \Longrightarrow y = 2z$
2. $x + 2y = 0 \Longrightarrow x = -2y = -2(2z) = -4z$

Eigenvector in terms of z :

$$\mathbf{x} = (x, y, z) = (-4z, 2z, z) = z(-4,2,1)$$

with $z \neq 0$.
Answer: (b) $\{a(-4,2,1) \mid a \neq 0, a \in \mathbb{R}\}$

32. ANSWER:- A

The character equation for the given matrix is

$$\begin{vmatrix} 4-\lambda & 5 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

⇒ (4 - λ) × (1 - λ) - 10 = 0
⇒ λ^2 - 5λ - 6 = 0
⇒ (λ + 1)(λ - 6) = 0
⇒ λ = -1, 6
Larger eigen value is 6

Check Out Our GATE Courses & Secure Your Dream Rank!

33. ANSWER:- C

For non-trivial solution the value of determinant should be zero.

$$\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$$

$R_1 \leftarrow R_1 + R_2 + R_3$

$$\begin{vmatrix} p+q+r & p+q+r & p+q+r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$$

$$(p+q+r)\begin{bmatrix} 1 & 1 & 1 \\ q & r & p \\ r & p & q \end{bmatrix} = 0$$

$C_1 \leftarrow C_1 - C_2$

$$(p+q+r)\begin{vmatrix} 0 & 1 & 1 \\ q-r & r & p \\ r-p & p & q \end{vmatrix} = 0$$

$C_2 \leftarrow C_2 - C_3$

$$(p+q+r)\begin{vmatrix} 0 & 0 & 1 \\ q-r & r-p & p \\ r-p & p-q & q \end{vmatrix} = 0$$

$$(p+q+r)(pq - rp + qr - q^2 - r^2 - p^2 + 2pr = 0$$

and for other part to be zero,

$p = q = r$

34. ANSWER:- D

The sum of the diagonal elements equals the sum of the eigervalues:

$$1 + a = -1 + 7$$

$$1 + a = 6 \Longrightarrow a = 5$$

The determinant of the matrix equals the product of the eigenvalues:

$$(1 \cdot a) - (4 \cdot b) = -1 \cdot 7$$
$$a - 4b = -7$$

Substitute $a = 5$:

$$5 - 4b = -7$$
$$= -12 \Longrightarrow b = 3$$

35. ANSWER:- 5

$A = LU$

$$\begin{bmatrix} 2 & 2 \\ 4 & 9 \end{bmatrix} = \begin{bmatrix} I_{11} & 0 \\ I_{21} & I_{22} \end{bmatrix}\begin{bmatrix} 1 & u_{12} \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 \\ 4 & 9 \end{bmatrix} = \begin{bmatrix} I_{11} & I_{11}u_{12} \\ I_{21} & I_{21}u_{12} + I_{22} \end{bmatrix}$$

$I_{11} = 2 \cdots - (1)$

$I_{11}u_{12} = 2$

$u_{12} = 2/2$

$u_{12} = 1 \cdots - (2)$

$I_{21} = 4 \cdots - (3)$

$I_{21}u_{12} + I_{22} = 9$

$I_{22} = 9 - I_{21}u_{12} = 9 - 4 \times 1 = 5$

36. ANSWER:- 0

Let $A = \begin{bmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{bmatrix}$

$R_2 \rightarrow R_2 + R_3$

$$\sim \begin{bmatrix} 3 & 4 & 45 \\ 20 & 11 & 300 \\ 13 & 2 & 195 \end{bmatrix}$$

$C_1 \rightarrow C_2 - C_3$

$$\sim \begin{bmatrix} -42 & 4 & 45 \\ -280 & 11 & 300 \\ -182 & 2 & 195 \end{bmatrix} = B \text{ (resultant matrix)}$$

Now $|B| = \begin{bmatrix} -42 & 4 & 45 \\ 280 & 11 & 300 \\ -182 & 2 & 195 \end{bmatrix}$

$$= (-14)(15)\begin{vmatrix} 3 & 4 & 3 \\ 20 & 11 & 20 \\ 13 & 2 & 13 \end{vmatrix}$$

$= 0$

Method 2: Determinant is unaltered by the operations (i) and (ii)

$\therefore$ Determinant of the resultant matrix = Determinant of the given matrix

$$= \begin{vmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{vmatrix}$$

$= 0$

(Since C_1, C_3 are proportional i.e., $C_3 = 15C_1$)

37. ANSWER:- A

The trace of a matrix is the sum of the elements of the principal diagonal of the matrix.

Fact - The sum of Eigen values of a matrix is equal to it's trace.

Fact - The product of Eigen values of a matrix is equal to its determinant value.

Since it's given that the trace is positive and the determinant is negative, there must be atleast one negative Eigen value. In general there can be an odd number of negative Eigen values in this case since the determinant value is negative.

Check Out Our GATE Courses & Secure Your Dream Rank!

38. ANSWER:- 0

$$A = \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} [1 \quad 9 \quad 5]$$

$$A = \begin{bmatrix} 2 & 18 & 10 \\ -4 & -36 & -20 \\ 7 & 63 & 35 \end{bmatrix}$$

$$\text{Determinant } (A) = \begin{bmatrix} 2 & 18 & 10 \\ -4 & -36 & -20 \\ 7 & 63 & 35 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 9 & 5 \\ -4 & -36 & -20 \\ 7 & 63 & 35 \end{bmatrix}$$

$$= 2 \times (-4) \begin{bmatrix} 1 & 9 & 5 \\ 1 & 9 & 5 \\ 7 & 63 & 35 \end{bmatrix}$$

$$= 2 \times (-4) \begin{bmatrix} 1 & 9 & 5 \\ 1 & 9 & 5 \\ 7 & 9 & 35 \end{bmatrix} R_3 \to R_3 - R_1$$

$$= 2 \times (-4) \times 7(0)$$

$$= 0$$

39. ANSWER:- 0

The eigen vectors corresponding to different eigen values of a real symmetric matrix are orthogonal to each other. And dot product of orthogonal vectors is 0.

40. ANSWER:- 6

Characteristic equation for matrix ' A ' is

$$AX = \lambda X$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

$x_1 + x_5 = \lambda x_1$

$x_1 + x_5 = \lambda x_5$

$(1) + (2) \Rightarrow 2(x_1 + x_5) = \lambda(x_1 + x_5)$

$\Rightarrow \lambda_1 = 2$

$x_2 + x_3 + x_4 = \lambda \cdot x_2 - (2)$

$x_2 + x_3 + x_4 = \lambda \cdot x_3 - \cdots - (5)$

$x_2 + x_3 + x_4 = \lambda \cdot x_4 - \cdots - \cdots (6)$

$(4) + (5) + (6) = 3(x_2 + x_3 + x_4)$

$= \lambda(x_2 + x_3 + x_4) \Rightarrow \lambda_2 = 3$

Product $= \lambda_1 \times \lambda_2 = 2 \times 3 = 6$

41. ANSWER:- D

This is a Vandermonde determinant, and its value is:Option D is not reducing to the determinant provided in question

(x−y)(y−z)(z−x)(x - y)(y - z)(z - x)(x−y)(y−z)(z−x)

So any expression not algebraically equivalent to this value is the wrong one.

Now, check each option to see which matches or doesn't.

This mixes two variables in each row — violates the original form.

Cannot be transformed into Vandermonde form.

42. ANSWER:- D

$a_{11} = a_{12} = a_{21} = 1, a_{22} = -1$

The 2×2 matrix $= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Cayley Hamilton theorem:

If matrix A has ' λ ' as eigen value, A^n has eigen value as λ^n.

Eigen value of $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & -(1+\lambda) \end{vmatrix} = 0$$

$$-(1-\lambda)(1+\lambda) - 1 = 0$$

$$-(1-\lambda^2) - 1 = 0$$

$$-1 = 1 - \lambda^2$$

$$\lambda^2 = 2$$

$$\lambda = \pm\sqrt{2}$$

Eigenvalues of $A^{19} = (\lambda 1)^{19}, (\lambda 2)^{19}$

So,Eigenvalues of A^{19} are $\mathbf{512}\sqrt{2}$ and $-512\sqrt{2}$

43. ANSWER:- A

Given matrix is upper triangular matrix and its diagonal elements are its eigen values = 1, 4, 3

44. ANSWER:- D

The trace is the sum of the diagonal elements of the matrix, and it is also the sum of the eigenvalues.

Thus, 2 + y = 4 + 8

⇒ 2 + y = 12

⇒ y = 10

The determinant is the product of the eigenvalues.

Thus, 2y − 3x = 8 × 4

⇒ 2(10) - 3x = 32

⇒ 20 - 3x = 32

⇒ 3x = 20 - 32 = -12

⇒ x = -4

Thus, x = -4 and y = 10

45. ANSWER:- D

System:

$x_1 + x_2 + 2x_3 = 1$

$x_1 + 2x_2 + 3x_3 = 2$

$x_1 + 4x_2 + \alpha x_3 = 4$

For a unique solution, determinant of coefficient matrix $\neq 0$.

Check Out Our GATE Courses & Secure Your Dream Rank!

Matrix:

$$\begin{bmatrix}1 & 1 & 2\\ 1 & 2 & 3\\ 1 & 4 & \alpha\end{bmatrix}$$

Compute determinant:

$$= 1\begin{vmatrix}2 & 3\\ 4 & \alpha\end{vmatrix} - 1\begin{vmatrix}1 & 3\\ 1 & \alpha\end{vmatrix} + 2\begin{vmatrix}1 & 2\\ 1 & 4\end{vmatrix}$$

$$= 1(2\alpha - 12) - 1(\alpha - 3) + 2(4 - 2)$$

$$= 2\alpha - 12 - \alpha + 3 + 4 = \alpha - 5$$

For unique solution:

$$\alpha - 5 \neq 0 \Rightarrow \alpha \neq 5$$

So solution is unique for all real values except 5.

46. ANSWER:- C

Given eigenvalues of A: $-5, -2, 1, 4$

Matrix:

$$\begin{pmatrix}A & I\\ I & A\end{pmatrix}$$

Key idea: If λ is an eigenvalue of A, then eigenvalues of this block matrix are:

$\lambda + 1$ and $\lambda - 1$

Now compute:

From $\lambda = -5$: $-4, -6$

From $\lambda = -2$: $-1, -3$

From $\lambda = 1$: $2, 0$

From $\lambda = 4$: $5, 3$

All eigenvalues:

$$-6, -4, -3, -1, 0, 2, 3, 5$$

From options, only 2 is present.

Answer: C. 2

47. ANSWER:- A

Given:

$$X = \{x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}$$

This is a plane through origin ⇒ it is a subspace of $\mathbb{R}^3$.

So option C is false.

Dimension of X :

One linear constraint ⇒ dimension $= 2$

Now check vectors:

$$v_1 = [1, -1, 0]^T, v_2 = [1, 0, -1]^T$$

Check if they belong to X :

For v_1: $1 - 1 + 0 = 0$✓

For v_2: $1 + 0 - 1 = 0$✓

So both are in X.

Check linear independence:

They are not scalar multiples ⇒ independent.

Since:

- 2 independent vectors
- subspace dimension $= 2$

They form a basis of X.

Answer: A

48. ANSWER:- D

Given: $Fu = b$ and $Fv = b$ with $u \neq v$

Subtract:

$$F(u - v) = 0$$

Let $x = u - v \neq 0$, so:

$$Fx = 0$$

This means a non-zero solution exists for homogeneous system, so columns of F are linearly dependent.

Now check options:

A. Determinant $= 0 \rightarrow$ True (matrix is singular)

B. Infinite solutions for $Fx = b \rightarrow$ True (since at least two solutions exist, so infinite)

C. There exists $x \neq 0$ such that $Fx = 0 \rightarrow$ True (proved above)

D. Must have two identical rows → False (linear dependence does not require identical rows)

Answer: D.

49. ANSWER:- A

We are given upper triangular matrices with $a, b, c \neq 0$.

Check properties:

Closure: Product of two upper triangular matrices is again upper triangular, and diagonal entries multiply, so still nonzero. Hence closed.

Associativity: Matrix multiplication is associative.

Identity: Identity matrix (with 1,1,1 on diagonal) is in the set since all diagonal entries are nonzero.

Inverse: Any upper triangular matrix with nonzero diagonal entries has an inverse, and that inverse is also upper triangular with nonzero diagonal.

All group properties are satisfied.

Answer: A. a group

50. ANSWER:- B

We check how many independent equations are there.

Write coefficient matrix:

$$\begin{bmatrix}2 & -1 & 3\\ 3 & 2 & 5\\ -1 & 4 & 1\end{bmatrix}$$

Now check determinant:

$$= 2\begin{vmatrix}2 & 5\\ 4 & 1\end{vmatrix} + 1\begin{vmatrix}3 & 5\\ -1 & 1\end{vmatrix} + 3\begin{vmatrix}3 & 2\\ -1 & 4\end{vmatrix}$$

$$= 2(2 \cdot 1 - 5 \cdot 4) + 1(3 \cdot 1 - 5(-1)) + 3(3 \cdot 4 - 2(-1))$$

$$= 2(2 - 20) + 1(3 + 5) + 3(12 + 2)$$

$$= 2(-18) + 8 + 3(14) = -36 + 8 + 42 = 14 \neq 0$$

Since determinant $\neq 0$, matrix is invertible.

So system has exactly one solution.

Answer: B. a unique solution

51. ANSWER:- B

Matrix:

$$\det\begin{pmatrix}2 - \lambda & -1\\ -4 & 5 - \lambda\end{pmatrix} = 0$$

2. Compute the Determinant

$$(2 - \lambda)(5 - \lambda) - (-1)(-4) = 0$$

$$(10 - 2\lambda - 5\lambda + \lambda^2) - 4 = 0$$

$$\lambda^2 - 7\lambda + 6 = 0$$

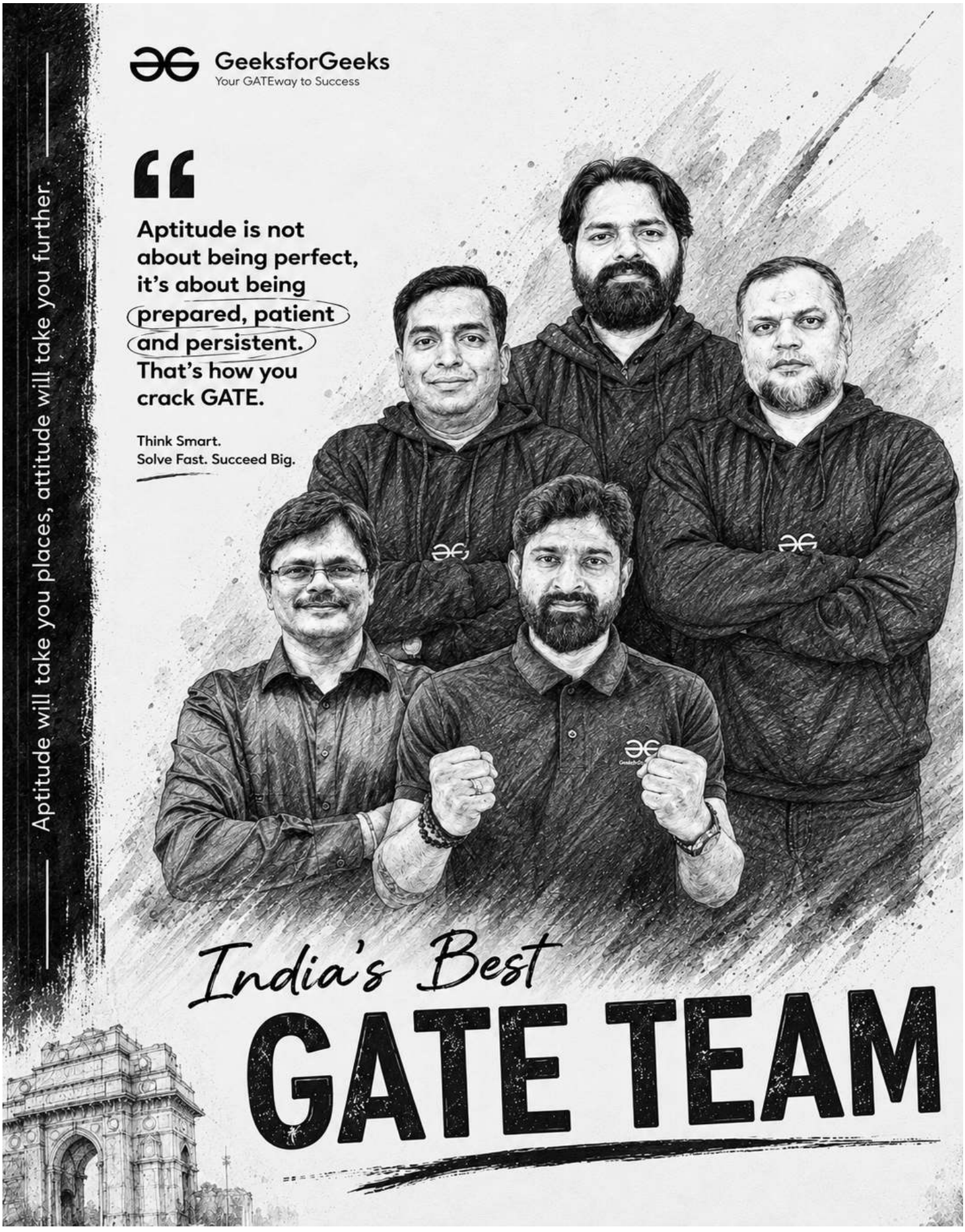
$$\lambda = 1 \text{ and } \lambda = 6$$

Check Out Our GATE Courses & Secure Your Dream Rank!

GENERAL APTITUDE

"Scan Here for Video Solution"

Link to GfG corresponding Short Notes: GATE


GeeksforGeeks
Your GATEway to Success
Aptitude will take you places, attitude will take you further.
Aptitude is not about being perfect, it's about being prepared, patient and persistent. That's how you crack GATE.
Think Smart.
Solve Fast. Succeed Big.
India's Best
GATE TEAM

1 QUANTITATIVE APTITUDE

GATE CSE - 2026 MCQ SET – 1 (1-MARKS)

Q1. Consider a knock-out women's badminton singles tournament where there are no ties. The loser in each game is eliminated from the tournament. Every player plays until she is defeated or remains the last undefeated player. The last undefeated player is declared the winner of the tournament. If there are 64 players in the beginning of the tournament, how many games should be played in total to declare the winner of the tournament?

(a) 127
(b) 64
(c) 63
(d) 32

GATE CSE - 2026 MCQ SET – 1 (1-MARKS)

Q2. A student needs to enroll for a minimum of 60 credits. A student cannot enroll for more than 70 credits. The credits are divided amongst project and three distinct sets of courses namely, core courses, specialization courses, and elective courses. It is compulsory for a student to enroll for exactly 15 credits of core courses and exactly 20 credits of project. In addition, a student has to enroll for a minimum of 10 credits of specialization courses. The maximum credits of elective courses that a student can enroll for is _____

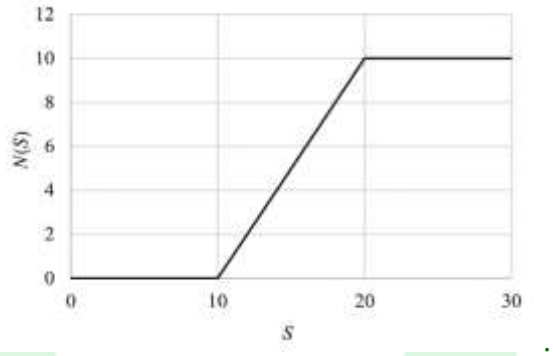
(a) 10
(b) 15
(c) 20
(d) 25

GATE CSE - 2026 MCQ SET – 1 (2-MARKS)

Q3. For positive real numbers S and K, the function $H_K(S)$ is defined as: $H_K(S) = \max(S - K, 0)$. The max function is defined as:

$$\max(a, b) = \begin{cases} a, & \text{when } a > b \\ b, & \text{when } a \leq b \end{cases}$$

The graph below shows the plot of a function $N(S)$ versus S.
$N(S)$ can be expressed as ___



(a) $H_{10}(S) - H_{20}(S)$
(b) $H_{10}(S) - 2H_{20}(S)$
(c) $-H_{10}(S) + H_{20}(S)$
(d) $H_{15}(S) - H_{20}(S)$

GATE CSE - 2026 MCQ SET – 1 (2-MARKS)

Q4. An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the number appearing in the second roll is an integer multiple of the number appearing in the first roll is _____

(a) 1/6
(b) 5/18
(c) 7/18
(d) 5/6

GATE CSE - 2026 MCQ SET – 2 (1-MARKS)

Q5. A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day?

(a) 1/4
(b) 3/4
(c) 2/3
(d) 3/8

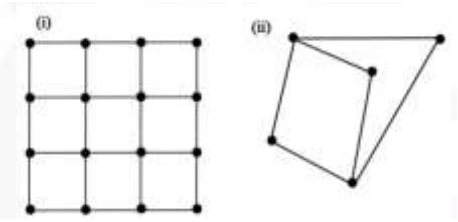
GATE CSE - 2026 MCQ SET – 2 (1-MARKS)

Q6. The values of Stock A and Stock B on a particular day are Rs. 50 and Rs. 80, respectively. An investor invests Rs. 100 in Stock A and Rs. 80 in Stock B. He sells all the stocks the next day when the value of Stock A is Rs. 55 and Stock B is Rs. 70. The profit made by the investor is Rs. _____

(a) 0
(b) 5
(c) 10
(d) 20

GATE CSE - 2026 MCQ SET – 2 (2-MARKS)

Q7. Figures (i) and (ii) represent intercity highway systems. The black dots represent cities and the line segments between them represent intercity highways. A salesperson needs to make a trip. She needs to start from a city, visit each of the remaining cities exactly once, and finally return to the same city from which she started.



Which one of the following options is then true?

(a) Such a trip is possible for (i), but not for (ii).
(b) Such a trip is possible for (ii), but not for (i).
(c) Such a trip is possible for both (i) and (ii).
(d) Such a trip is possible neither for (i) nor for (ii)

GATE CSE - 2026 MCQ SET – 2 (2-MARKS)

Q8. An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the sum of the two recorded numbers is a prime number is _____

(a) 3/36
(b) 13/36
(c) 15/36
(d) 19/36

GATE CSE - 2025 MCQ SET – 1 (1-MARKS)

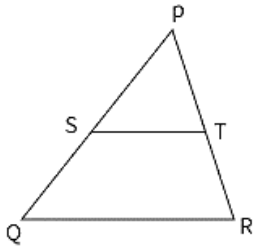
Q9. The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?

(a) 25
(b) 28
(c) 30
(d) 32

GATE CSE - 2025 MCQ SET – 1 (2-MARKS)

Q10. In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT?

Note: The figure shown is representative.



(a) 1/3
(b) 1/4
(c) 2/5
(d) 1/2

GATE CSE - 2025 MCQ SET – 1 (2-MARKS)

Q11. A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once?

(a) 75/216
(b) 1/6
(c) 1/18
(d) 25/216

GATE CSE - 2025 MCQ SET – 1 (2-MARKS)

Q12. A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. The order in which the scoops are purchased is inconsequential. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?

(a) 4
(b) 20
(c) 24
(d) 48

GATE CSE - 2025 MCQ SET – 2 (1-MARKS)

Q13. If $Pe^{x} = Qe^{-x}$ for all real values of x, which one of the following statements is true?

(a) $P = Q = 0$
(b) $P = Q = 1$
(c) $P = 1; Q = -1$
(d) $P/Q = 0$

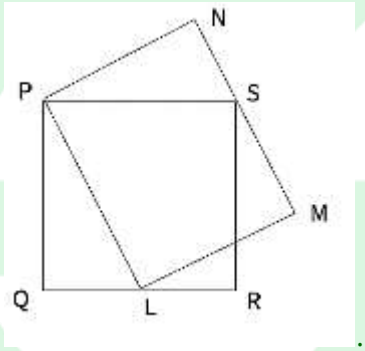
GATE CSE - 2025 MCQ SET – 2 (1-MARKS)

Q14. Let $p1$ and $p2$ denote two arbitrary prime numbers. Which one of the following statements is correct for all values of $p1$ and $p2$?

(a) $p1 + p2$ is not a prime number
(b) $p1p2$ is not a prime number
(c) $p1 + p2 + 1$ is a prime number
(d) $p1p2 + 1$ is a prime number

GATE CSE - 2025 MCQ SET – 2 (2-MARKS)

Q15. In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square. What is the area (in cm^2) of the rectangle PLMN? Note: The figure shown is representative.



(a) $2\sqrt{2}$
(b) 2
(c) 8
(d) 4

GATE CSE - 2024 MCQ SET – 1 (1-MARKS)

Q16. If two distinct non-zero real variables x and y such that x + y is proportional to (x-y) then the value of x/y ?

(a) depends on xy
(b) depends only on x and not on y
(c) depends only on y and not on x
(d) is a constant

GATE CSE - 2024 MCQ SET – 1 (1-MARKS)

Q17. Consider the following sample of numbers

9, 18, 11, 14, 15, 17, 10, 69, 11, 13

The median of the sample is

(a) 13.5
(b) 14
(c) 11
(d) 18.7

GATE CSE - 2024 MCQ SET – 1 (1-MARKS)

Q18. The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in ₹5 coins is

(a) 21%
(b) $14\frac{2}{7}\%$
(c) 10%
(d) 30%

GATE CSE - 2024 MCQ SET – 1 (1-MARKS)

Q19. For positive non-zero real variables p and q, if

$\log(p^2+q^2)=\log p + \log q + 2\log 3$

then, the value of $(p^4+q^4)/p^2q^2$ is

(a) 9
(b) 79
(c) 81
(d) 83

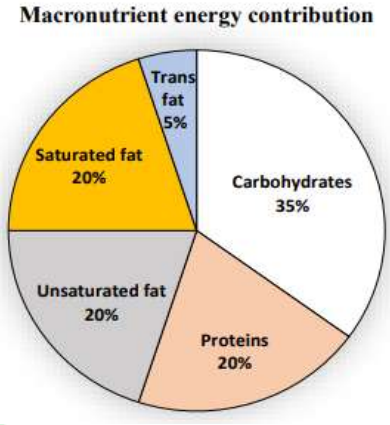
GATE CSE - 2024 MCQ SET – 1 (2-MARKS)

Q20. A rectangular paper sheet of dimensions 54 cm × 4 cm is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken. Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

(a) $1/\pi$
(b) $2/\pi$
(c) $3/\pi$
(d) $4/\pi$

GATE CSE - 2024 MCQ SET – 1 (2-MARKS)

Q21. The pie chart presents the percentage contribution of different macronutrients to a typical 2,000 kcal diet of a person.



The typical energy density (kcal/g) of these macronutrients is given in the table.

Macronutrient Energy	density (kcal/g)
Carbohydrates	4
Proteins	4
Unsaturated fat	9
Saturated fat	9
Trans fat	9

The total fat (all three types), in grams, this person consumes is

(a) 44.9
(b) 77.8
(c) 100
(d) 3600

GATE CSE - 2024 MCQ SET – 2 (1-MARKS)

Q22. Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work. How many attempts does

each wizard make before coming to this conclusion, independently?

(a) 24
(b) 48
(c) 16
(d) 12

GATE CSE - 2024 MCQ SET – 2 (1-MARKS)

Q23. In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is 1/4th of the number of students who like other branches. The number of students who like both their core and other branches is 500. The number of students who like their core branches is

(a) 1800
(b) 3500
(c) 1600
(d) 1500

GATE CSE - 2024 MCQ SET – 2 (1-MARKS)

Q24. For positive non-zero real variables x and y, if

$\ln((x+y)/2) = 1/2[\ln(x) + \ln(y)]$

then, the value of $x/y + y/x$

(a) 1
(b) 1/2
(c) 2
(d) 4

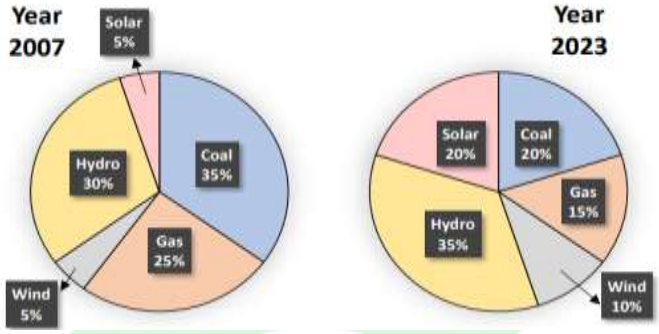
GATE CSE - 2024 MCQ SET – 2 (2-MARKS)

Q25. A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

(a) 1% profit
(b) 2% profit
(c) 1% loss
(d) 2% loss

GATE CSE - 2024 MCQ SET – 2 (2-MARKS)

Q26. The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023.



The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generation of a country for the years 2007 and 2023. what is the percentage increase in the share of the renewable sources of electricity generation over this period?

(a) 25%
(b) 50%
(c) 77.5%
(d) 62.5%

GATE CSE - 2023 MCQ (2-MARKS)

Q27. Consider the following

$f(t)=0.01t^2$

$g(t)=4t$

where 0<t<INFINITY

Now, consider the following two statements

(i)For some t>0,g(t)>f(t)

(ii)There exists a T, such that f(t)>g(t) for all t>T?

Which one of the following options is TRUE?

(a) Only (i) is correct
(b) Only (ii) is correct
(c) Both (i) and (ii) are correct
(d) Neither (i) nor (ii) is correct

GATE CSE - 2023 MCQ (2-MARKS)

Q28. f(x) and g(y) are functions of x and y, respectively, and f (x) = g (y) for all real values of x and y. Which of the following option is necessarily TRUE for all conditions?

(a) f(x)=0 and g(y)=0
(b) f(x)=g(y)=constant
(c) f(x)=constant and g(y) not constant
(d) f(x)+g(y)=f(x)-g(y)

GATE CSE - 2022 MCQ (1-MARKS)

Q29. A function y(x) is defined in the interval [0, 1] on the x-axis as :

$$y(x) = \begin{cases} 2 & \text{if} \quad 0 \le x < \frac{1}{3} \\ 3 & \text{if} \quad \frac{1}{3} \le x < \frac{3}{4} \\ 1 & \text{if} \quad \frac{3}{4} \le x \le 1 \end{cases}$$

Which one of the following is the area under the curve for the interval [0, 1] on the x-axis?

(a) 5/6
(b) 6/5
(c) 13/6
(d) 6/13

GATE CSE - 2022 MCQ (1-MARKS)

Q30. Let r be a root of the equation $x^2 + 2x + 6 = 0$. Then the value of the expression (r+2) (r+3)(r+4)(r+5) is

(a) 51
(b) -51
(c) 126
(d) -126

GATE CSE - 2021 MCQ SET – 1 (1-MARKS)

Q31. The ratio of boys to girls in a class is 7 to 3. Among the options below, an acceptable value for the total number of students in the class is:

(a) 21
(b) 37
(c) 50
(d) 73

GATE CSE - 2021 MCQ SET – 1 (2-MARKS)

Q32. We have 2 rectangular sheets of paper, M and N, of dimensions 6 cm × 1 cm each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____.

(a) π/2
(b) 3/π
(c) 9/π
(d) 3π

GATE CSE - 2021 MCQ SET – 1 (2-MARKS)

Q33.

Items	Cost	Profit %	Marked price
P	5,400	….	5,860
Q	…..	25	10000

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is 3:4. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost

$$\left(\text{Profit } \% = \frac{\text{Selling price-Cost}}{\text{Cost}} \times 100\right).$$

The discount on item Q, as a percentage of its marked price, is ___

(a) 25
(b) 12.5
(c) 10
(d) 5

GATE CSE - 2021 MCQ SET – 1 (2-MARKS)

Q34. There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag. The probability that at least two chocolates are identical is _____ .

(a) 0.3024
(b) 0.4235
(c) 0.6976
(d) 0.8125

GATE CSE - 2021 MCQ SET – 2 (1-MARKS)

Q35. If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, cosθ =

(a) 1/2
(b) 1/√3
(c) 1/√2
(d) √3/2

GATE CSE - 2021 MCQ SET – 2 (1-MARKS)

Q36. If $\left(x-\frac{1}{2}\right)^2-\left(x-\frac{3}{2}\right)^2=x+2$, then the value of x is

(a) 2
(b) 4
(c) 6
(d) 8

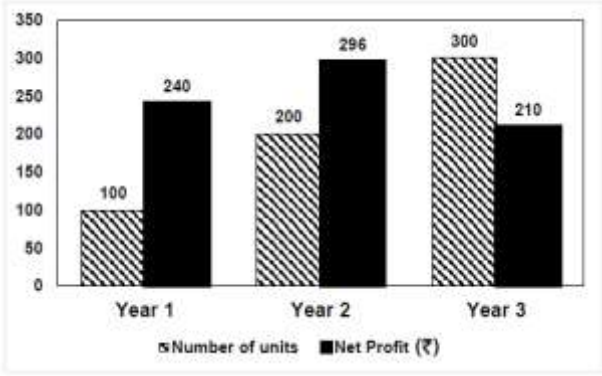
GATE CSE - 2021 MCQ SET – 2 (2-MARKS)

Q37. The number of students in three classes is in the ratio 3:13:6. If 18 students are added to each class, the ratio changes to 15:35:21. The total number of students in all the three classes in the beginning was:

(a) 22
(b) 66
(c) 88
(d) 110

GATE CSE - 2021 MCQ SET – 2 (2-MARKS)

Q38.

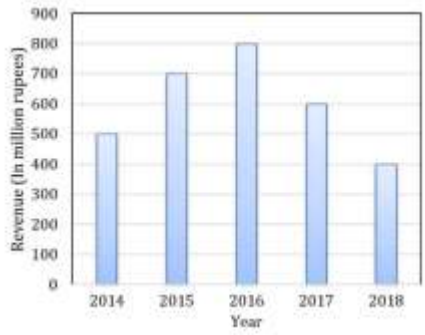


The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was Re. 1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13%, and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year. The ratio of the selling price in Year 2 to the selling price in Year 3 is _____.

(a) 4:3
(b) 1:1
(c) 3:4
(d) 1:2

GATE CSE - 2020 MCQ (-MARKS)

Q39. The total revenue of a company during 2014–2018 is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during 2014–2018 is



(a) 16.67% profit
(b) 16.67% loss
(c) 20% profit
(d) 20% loss

GATE CSE - 2019 MCQ (1-MARKS)

Q40. Two cars start at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is _____.

(a) 1
(b) 2
(c) 3
(d) 6

GATE CSE - 2019 MCQ (1-MARKS)

Q41. Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs 150 more. The cost of the gift was Rs

(a) 666
(b) 3000
(c) 6000
(d) 12000

GATE CSE - 2019 MCQ (2-MARKS)

Q42. In a college, there are three student clubs. Sixty students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in the Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all the clubs. If 75% of the students in the college are not in any of these clubs, the total number of students in the college is _____.

(a) 1000
(b) 975
(c) 900
(d) 225

GATE CSE - 2018 MCQ (1-MARKS)

Q43. The area of a square is d. What is the area of the circle which has the diagonal of the square as its diameter?

(a) πd
(b) πd^2
(c) $(1/4)\pi d^2$
(d) $(1/2)\pi d$

GATE CSE - 2018 MCQ (1-MARKS)

Q44. What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

(a) 3047
(b) 6047
(c) 7987
(d) 63847

GATE CSE - 2018 MCQ (1-MARKS)

Q45. What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, __, 0.
(a) 2880
(b) 1440
(c) 720
(d) 0

GATE CSE - 2018 MCQ (2-MARKS)

Q46. A six-sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

(a) Three green faces and four red faces
(b) Four green faces and three red faces
(c) Five green faces and two red faces
(d) Six green faces and one red face

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE - 2017 MCQ SET -1 (1-MARKS)

Q47. Find the smallest number y such that y×162 is a perfect cube.

(a) 24
(b) 27
(c) 32
(d) 36

GATE CSE - 2017 MCQ SET -1 (1-MARKS)

Q48. The probability that a k-digit number does NOT contain the digits 0, 5, or 9 is

(a) 0.3^k
(b) 0.6^k
(c) 0.7^k
(d) 0.9^k

GATE CSE - 2017 MCQ SET -2 (1-MARKS)

Q49. A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

(a) 12
(b) 15
(c) 18
(d) 19

GATE CSE - 2017 MCQ SET 2 (1-MARKS)

Q50. There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is

(a) 1/5
(b) 7/30
(c) 1/4
(d) 4/15

GATE CSE - 2016 MCQ SET -1 (1-MARKS)

Q51. A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____

(a) 56
(b) 64
(c) 72
(d) 96

GATE CSE - 2016 MCQ SET – 1 (2-MARKS)

Q52. A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78, and Executive at Rs. 173 per piece. The numbers of each razor sold in each quarter of a year are given.

Quarter \Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

(a) Elegance
(b) Executive
(c) Smooth
(d) Soft

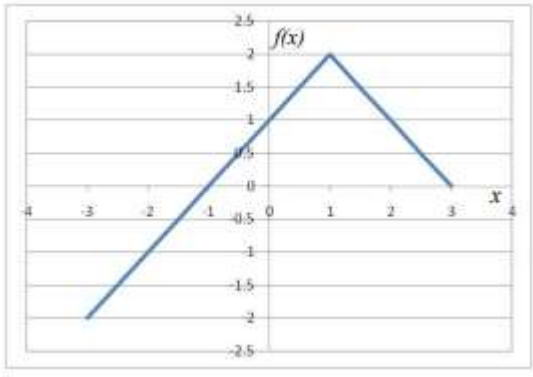
GATE CSE - 2016 MCQ SET – 1 (2-MARKS)

Q53. If $f(x) = 2x^2 + 3x - 5$, which of the following is a factor?

(a) $(x^2 + 8)$
(b) $(x - 1)$
(c) $(2x - 5)$
(d) $(x + 1)$

GATE CSE - 2016 MCQ SET – 2 (2-MARKS)

Q54. Choose the correct expression for f(x) given in the graph.



(a) $f(x) = 1 - |x - 1|$
(b) $f(x) = 1 + |x - 1|$
(c) $f(x) = 2 - |x - 1|$
(d) $f(x) = 2 + |x - 1|$

GATE CSE – 2017 MCQ SET – 1 (1-MARKS)

Q55. Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

(a) 21
(b) 18
(c) 16
(d) 14

GATE CSE - 2017 MCQ SET – 2 (2-MARKS)

Q56. X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

(a) 90 digits
(b) 91 digits
(c) 92 digits
(d) 93 digits

GATE CSE - 2017 MCQ SET – 2 (2-MARKS)

Q57. The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

(a) 0
(b) 1
(c) 2
(d) 3

GATE CSE - 2016 MCQ SET – 1 (2-MARKS)

Q58. In a process, the number of cycles to failure decreases exponentially with increase in load. At load 80 units, it takes 100 cycles for failure. When load is halved, it takes 10000 cycles.
Find load for failure at 5000 cycles.

(a) 40.00
(b) 46.02
(c) 60.01
(d) 92.02

GATE CSE - 2016 MCQ SET – 2 (2-MARKS)

Q59. In a quadratic function, the value of the product of the roots (α, β) is 4 . Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

(a) n^4
(b) 4^n
(c) 2^{2n-1}
(d) 4^{n-1}

GATE CSE - 2016 MCQ SET – 2 (1-MARKS)

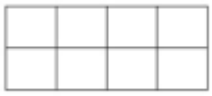
Q60. Among 150 faculty members in an institute, 55 are connected with each other through Facebook and 85 are connected through WhatsApp. 30 faculty members do not have Facebook or WhatsApp accounts. The number of faculty members connected only through Facebook accounts is _____.

(a) 35
(b) 45
(c) 65
(d) 90

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE - 2016 MCQ SET – 2 (1-MARKS)

Q61. In a 2 × 4 rectangle grid shown, each cell is a rectangle. How many rectangles can be observed in the grid?



(a) 21
(b) 27
(c) 30
(d) 36

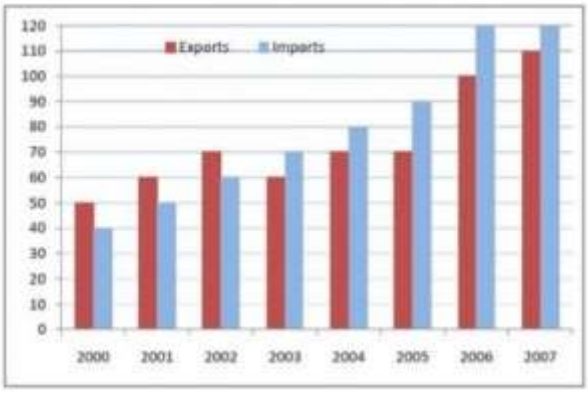
GATE CSE - 2015 MCQ SET – 3 (1-MARKS)

Q62. A function f(x) is linear and has a value of 29 at x= -2 and 39 at x=3. Find its value at x=5.

(a) 59
(b) 45
(c) 43
(d) 35

GATE CSE - 2015 NAT SET – 3 (2-MARKS)

Q63. The exports and imports (in crores of Rs.) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest?



GATE CSE - 2015 MCQ SET – 2 (2-MARKS)

Q64. If the list of letters P, R, S, T, U is an arithmetic sequence, which of the following are also in arithmetic sequence?
I. 2P, 2R, 2S, 2T, 2U
II. $P-3, R-3, S-3, T-3, U-3$
III. P^2, R^2, S^2, T^2, U^2

(a) I only
(b) I and II
(c) II and III
(d) I and III

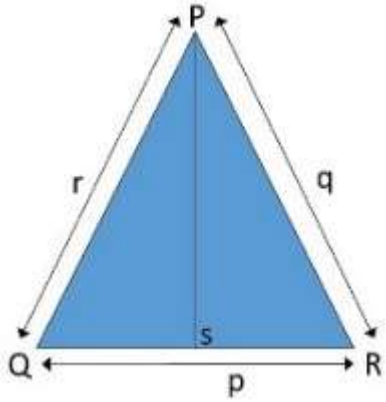
GATE CSE - 2015 MCQ SET – 2 (2-MARKS)

Q65. Four branches of a company are located at M, N, O and P. M is north of N at a distance of 4km; P is south of O at a distance of 2 km; N is southeast of O by 1km. What is the distance between M and P in km?

(a) 5.34
(b) 6.74
(c) 28.5
(d) 45.49

GATE CSE - 2015 MCQ SET – 2 (2-MARKS)

Q66. In a triangle PQR,PS PQR,PS is the angle bisector of ∠QPR and ∠QPS=60∘ ∠QPR and ∠QPS=60 What is the length of PS ?



(a) $(q + r)/qr$
(b) $qr/(q + r)$
(c) $\sqrt{(q^2 + r^2)}$
(d) $(q + r)^2/qr$

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE - 2015 NAT SET – 2 (2-MARKS)

Q67. If p, q, r, s are distinct integers such that: $f(p, q, r, s) = \max(p, q, r, s)$

$g(p, q, r, s) = \min(p, q, r, s)$

$h(p, q, r, s) =$ remainder of $\frac{(p \times q)}{(r \times s)}$ if $(p \times q) > (r \times s)$

or remainder of $\frac{(r \times s)}{(p \times q)}$ if $(r \times s) > (p \times q)$

Also a function $fgh(p, q, r, s) = f(p, q, r, s) \times g(p, q, r, s) \times h(p, q, r, s)$

Also the same operations are valid with two variable functions of the form

$$f(p, q)$$

What is the value of $fg(h(2,5,7,3),4,6,8)$?

GATE CSE - 2015 MCQ SET – 2 (1-MARKS)

Q68. Consider a function $f(x) = 1 - |x|$ on $-1 \leq x \leq 1$. The value of x at which the function attains a maximum, and the maximum value of the function are:

(a) 0, -1
(b) -1, 0
(c) 0, 1
(d) -1, 2

GATE CSE - 2015 MCQ SET – 2 (2-MARKS)

Q69. Based on the given statements, select the most appropriate option to solve the given question.What will be the total weight of 10 poles each of the same weight?

Statements: I. One fourth of the weight of the pole is 5 Kg.

Statement II. The total weight of these poles is 160 Kg more than the total weight of two poles.

(a) Statement I alone is not sufficient.
(b) Statement II alone is not sufficient.
(c) Either I or II alone is sufficient.
(d) Both statements I and II together are not sufficient.

GATE CSE - 2015 MCQ SET – 1 (2-MARKS)

Q70. The number of students who answered correctly, wrongly or not attempted each question in an exam:

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

(a) 2.29
(b) 2.97
(c) 6.795
(d) 8.795

GATE CSE - 2015 MCQ SET – 1 (2-MARKS)

Q71. The probabilities that a student passes in mathematics, physics and chemistry are m, p and c respectively. The student has 75% chance of passing in at least one, 50% chance of passing in at least two and 40% chance of passing in exactly two. Following relations are drawn in m, p, c:

I. p+m+c = 27/20

II. p+m+c = 13/20

III. (p)(m)(c) = 1/10

(a) Only relation I is true.
(b) Only relation II is true.
(c) Relations II and III are true.
(d) Relations I and III are true.

GATE CSE - 2015 MCQ SET – 1 (1-MARKS)

Q72. Given Set A = {2,3,4,5} and Set B = {11,12,13,14,15}, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?

(a) 0.20
(b) 0.25
(c) 0.3
(d) 0.33

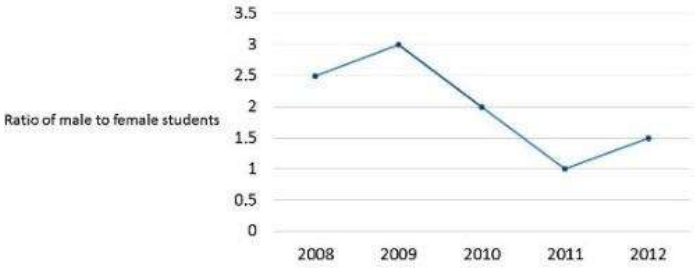
GATE CSE - 2014 MCQ SET – 3 (2-MARKS)

Q73. The Gross Domestic Product (GDP) in Rupees grew at 7% during 2012-2013. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period 2012-2013 the exchange rate for the USD increased from Rs. 50/ USD to Rs. 60/ USD. India's GDP in USD during the period 2012-2013

(a) increased by 5%
(b) decreased by 13%
(c) decreased by 20%
(d) decreased by 11%

GATE CSE - 2014 MCQ SET – 3 (2-MARKS)

Q74. The ratio of male to female students in a college for five years is plotted in a line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011?



(a) 1:1
(b) 2:1
(c) 1.5:1
(d) 2.5:1

GATE CSE - 2014 MCQ SET – 3 (2-MARKS)

Q75. Consider the equation: $(7526)_8 - (Y)_8 = (4364)_8$, where $(X)_n$ stands for X to the base N. Find Y.

(a) 1634
(b) 1737
(c) 3142
(d) 3162

GATE CSE - 2014 MCQ SET – 3 (1-MARKS)

Q76. Which number does not belong in the series below?
2, 5, 10, 17, 26, 37, 50, 64

(a) 17
(b) 37
(c) 64
(d) 26

GATE CSE - 2014 MCQ SET – 3 (1-MARKS)

Q77. The table below has question-wise data on the performance of students in an examination. There is no negative or partial marking.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

(a) 1.34
(b) 1.74
(c) 3.02
(d) 3.91

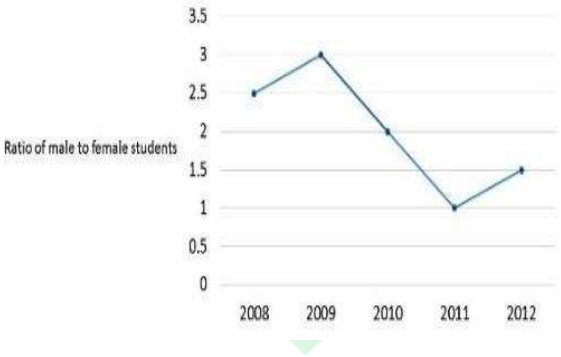
GATE CSE - 2014 MCQ SET – 2 (2-MARKS)

Q78. If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include

(a) 2, 4
(b) 2, 14
(c) 4, 52
(d) 14, 52

GATE CSE - 2014 NAT SET – 2 (2-MARKS)

Q79. The ratio of male to female students in a college for five years is plotted in a line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?



GATE CSE - 2014 MCQ SET – 2 (2-MARKS)

Q80. At what time between 6 a.m. and 7 a.m. will the minute hand and hour hand of a clock make an angle closest to 60°?

(a) 6:22 a.m.
(b) 6:27 a.m.
(c) 6:38 a.m.
(d) 6:45 a.m.

GATE CSE - 2014 MCQ SET – 2 (1-MARKS)

Q81. What is the average of all multiples of 10 from 2 to 198?

(a) 90
(b) 100
(c) 110
(d) 120

GATE CSE - 2014 MCQ SET – 2 (1-MARKS)

Q82. The value of $\sqrt{12+\sqrt{12+\sqrt{12+\cdots}}}$ is

(a) 3.464
(b) 3.932
(c) 4.000
(d) 4.444

GATE CSE - 2014 NAT SET – 1 (2-MARKS)

Q83. When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines?

GATE CSE - 2014 MCQ SET – 1 (2-MARKS)

Q84. Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is Rs 100, a group of 5 tourists purchasing round-trip tickets will be charged Rs _____.

GATE CSE - 2014 NAT SET – 1 (1-MARKS)

Q85. If $(z + 1/z)^2 = 98$, compute $(z^2 + 1/z^2)$

GATE CSE - 2014 MCQ SET – 1 (1-MARKS)

Q86. The roots of $ax^2 + bx + c = 0$ are real and positive. a, b and c are real. Then $ax^2 + b|x| + c = 0$ has

(a) no roots
(b) 2 real roots
(c) 3 real roots
(d) 4 real roots

GATE CSE - 2013 MCQ (2-MARKS)

Q87. Find the sum of the expression $1/(\sqrt{1}+\sqrt{2}) + 1/(\sqrt{2}+\sqrt{3}) + 1/(\sqrt{3}+\sqrt{4}) + \ldots\ldots\ldots + 1/(\sqrt{80}+\sqrt{81})$

(a) 7
(b) 8
(c) 9
(d) 10

Link to GfG corresponding Short Notes: GATE FARRAY

GATE CSE - 2013 MCQ (2-MARKS)

Q88. Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

(a) 13/90
(b) 12/90
(c) 78/90
(d) 77/90

GATE CSE - 2013 MCQ (2-MARKS)

Q89. A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

(a) 36
(b) 30
(c) 24
(d) 18

GATE CSE - 2013 MCQ (2-MARKS)

Q90. The current erection cost of a structure is Rs. 13,200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in Rs. is

(a) 16500
(b) 15180
(c) 11000
(d) 10120

GATE CSE - 2013 MCQ (1-MARKS)

Q91. What will be the maximum sum of 44, 42, 40, ... ?

(a) 502
(b) 504
(c) 506
(d) 500

GATE CSE - 2012 MCQ (2-MARKS)

Q92. A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is :

(a) 8 meters
(b) 10 meters
(c) 12 meters
(d) 14 meters

GATE CSE - 2012 MCQ (2-MARKS)

Q93. An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable. The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

(a) 0.288
(b) 0.334
(c) 0.667
(d) 0.72

GATE CSE - 2012 MCQ (1-MARKS)

Q94. The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of Rs. 50 per unit. The number of units to be produced by the firm such that the profit is maximized is:

(a) 5
(b) 10
(c) 15
(d) 25

GATE CSE - 2011 MCQ (2-MARKS)

Q95. If log(P)=(1/2)log(Q)=(1/3)log(R) log(P)=(1/2)log(Q)=(1/3)log(R), then which of the following options is TRUE?

(a) $P^2 = Q^3R^2$
(b) $Q^2 = PR$

(c) $Q^2 = R^3P$
(d) $R = P^2Q^2$

GATE CSE - 2011 MCQ (2-MARKS)

Q96. The variable cost (V) of manufacturing a product varies according to the equation V=4q, where q is the quantity produced. The fixed cost (F) of production of the same product reduces with q according to the equation F=100/q. How many units should be produced to minimize the total cost (V+F)

(a) 4
(b) 5
(c) 7
(d) 6

GATE CSE - 2011 MCQ (2-MARKS)

Q97. A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of 5th day?

(a) 4
(b) 5
(c) 6
(d) 7

GATE CSE - 2011 MCQ (2-MARKS)

Q98. A container originally contains 10 litres of pure spirit. From this container, 1 litre of spirit is replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

(a) 7.58 litres
(b) 7.84 litres
(c) 7 litres
(d) 7.29 litres

GATE CSE - 2010 MCQ (2-MARKS)

Q99. If 137+276=435 how much is 731+672?

(a) 534
(b) 1403
(c) 1623
(d) 1513

GATE CSE - 2010 MCQ (1-MARKS)

Q100. 25 people are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

(a) 2
(b) 17
(c) 13
(d) 3

GATE CSE - 2010 MCQ (2-MARKS)

Q101. 5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

(a) 20 days
(b) 18 days
(c) 16 days
(d) 15 days

GATE CSE - 2010 MCQ (2-MARKS)

Q102. Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?

(a) 50
(b) 51
(c) 52
(d) 54

ANSWER KEY

1. (c)	2. (d)	3. (a)
4. (c)	5. (a)	6. (a)
7. (a)	8. (c)	9. (c)
10. (a)	11. (a)	12. (b)
13. (a)	14. (b)	15. (d)
16. (d)	17. (a)	18. (c)
19. (b)	20. (a)	21. (c)
22. (a)	23. (a)	24. (c)
25. (c)	26. (d)	27. (c)
28. (b)	29. (c)	30. (d)
31. (c)	32. (c)	33. (c)
34. (c)	35. (b)	36. (b)
37. (c)	38. (a)	39. (c)
40. (b)	41. (c)	42. (c)
43. (d)	44. (c)	45. (b)
46. (c)	47. (d)	48. (c)
49. (b)	50. (d)	51. (d)
52. (b)	53. (b)	54.(c)
55.(d)	56. (a)	57. (d)
58. (b)	59. (b)	60. (a)
61. (c)	62.(c)	63.(2006)
64. (b)	65.(a)	66. (b)
67. (8)	68. (c)	69. (c)
70. (c)	71. (d)	72. (a)
73. (d)	74. (c)	75. (c)
76. (c)	77. (c)	78. (d)
79. (140%)	80. (a)	81. (b)
82. (c)	83. (6)	84.(850)
85. (96)	86. (d)	87. (b)
88. (d)	89. (c)	90. (b)
91. (c)	92. (b)	93. (b)
94. (a)	95. (b)	96. (b)
97. (c)	98. (d)	99.(c)
100. (d)	101. (d)	102. (b)

HINTS & SOLUTIONS

1. ANSWER: - (C)

Number of players left after each match: 64 → 32 → 16→8→4→2→1

Number of matches being played: 32 +16+8+4+2+1=63

So, the correct answer is C.

2. ANSWER: - (D)

60 <= Credits <= 70

Credits = Project + Core Course + Special. Course + Elective Course, since we want maximum credits for Elective course,assign minimum values to each project, core course and Special Course.

45+ Elective Course = 70

Elective Course = 25

3. ANSWER: - (A)

$H_K(S)=\max(S-K,0)$ → starts from S=K, slope = 1

From Graph:

- N(S)=0 for S<10
- Linear increase from 10 to 20
- Constant (=10) after S=20

So it behaves like:

- Starts at 10 → use $H_{10}(S)$
- Stops increasing after 20 → subtract $H_{20}(S)$

Final Expression:$N(S)=H_{10}(S)-H_{20}(S)$

4. ANSWER: - (C)

We need Probability that second number is a multiple of the first.

Total Outcomes: Dice rolled twice → 6×6=36.

Favorable Cases:Fix first roll and count valid second rolls:

First Roll	Multiples in (1-6)	Count
1	1,2,3,4,5,6	6
2	2,4,6	3
3	3,6	2
4	4	1
5	5	1
6	6	1

Total favorable = 6+3+2+1+1+1=14

Probability:14/36=7/18.

So, the correct answer is C.

Link to GfG corresponding Short Notes: GATE FARRAY

5. ANSWER: - (A)
A day can either be cloudy or sunny only.

If probability of a day being cloudy is 0.5 then it being sunny is also 0.5.

Use binomial ditribution :

Out of 4 days, choose 3 days for cloudy.
$\binom{4}{3} * \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right) = 4 * \left(\frac{1}{16}\right) = \frac{1}{4}$

6. ANSWER: - (A)
Investor investing = Rs 100 in A means 2 stocks of A of 50rs each and Rs 80 in B means 1 stock of 80rs.

The next day the values of the stocks change to A = Rs 55 and B = Rs 70.

He sold all the stocks he had, meaning two stock of A and a stock of B, thus total sold at = 55x2 + 70 = 180.

Invested = 100 + 80 = 180.

Net profit = sale - buy = 180 - 180 = 0.

So, the correct answer is A.

7. ANSWER: - (A)

We check for a Hamiltonian cycle.

(i) Grid graph (4×4)

This graph does have a Hamiltonian cycle.

You can trace a loop around the grid in a "zig-zag"/snake pattern and return to the start.

Possible

(ii) Irregular graph
There is a vertex (the inner one) that effectively creates a bottleneck.

To include it, you must enter and leave through the same limited connections, which forces revisiting or skipping another vertex.
Not possible

Possible for (i), but not for (ii).

So, the correct answer is A.

8. ANSWER: - (C)
When a six-faced die is rolled twice, the total number of possible outcomes in the sample space S is:

$$n(S) = 6 \times 6 = 36$$

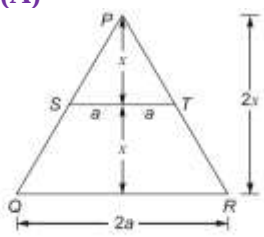
here we use six-sided dice as a pair, then the minimum possible sum will be $1 + 1 = 2$, and the maximum will be $6 + 6 = 12$. so the required prime numbers between 2 and 12 are: {2,3,5,7,11}.

Now, we have to find the pairs (x, y) that result in these sums:

- Sum = 2: (1,1) → **1** possibilities
- Sum = 3: (1,2), (2,1) → 2 possibilities
- Sum = 5: (1,4), (4,1), (2,3), (3,2) → 4 possibilities
- Sum = 7: (1,6), (6,1), (2,5), (5,2), (3,4), (4,3) → 6 possibilities
- Sum = 11: (5,6), (6,5) → **2** possibilities
  therefore total favorable outcome is: $n(E) = 1 + 2 + 4 + 6 + 2 = 15$
  $P(\text{ Sum is Prime }) = \frac{n(E)}{n(S)} = \frac{15}{36}$
  Option (C) is correct.

9. ANSWER: - (C)
The difference in average is $31.4 - 30.8 = 0.6$
The total correction was $42 - 24 = 18$
So, number of students = $18 \div 0.6 = 30$
So, the correct answer is C.

10. ANSWER: - (A)



$\frac{\Delta PST}{T,\ QRST} = \frac{\frac{1}{2} \times 2a \times x}{\frac{1}{2} \times (4a+2a)} = \frac{2ax}{6ax} = \frac{1}{3}$

11. ANSWER: - (A)

$$\text{Required probability} = A \times \bar{B} \times \bar{C} + \bar{A} \times \bar{B} \times C + \bar{A} \times B \times \bar{C}$$
$$= \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6}$$
$$= \frac{75}{216}$$

12. ANSWER: - (B)

Given:

- 4 flavors
- 3 scoops
- Order doesn't matter
- Repetition allowed

Formula:

Number of ways to choose rrr items from nnn types with repetition:$({}^{n+r-1}C_r)$

Apply: $({}^{4+3-1}C_3)=({}^{6}C_3)=20$

13. ANSWER: - (A)

The given equation is: $Pe^{x} = Qe^{-x}$

$=> Pe^{2x} = Q$

For this to be true for all x, Pe^{2x} must always be equal to a constant Q. This is only possible if both sides are constant. The only way this holds is if both P=0 and Q=0, so that both sides remain 0.

Hence, the correct answer is P=Q=0

14. ANSWER: - (B)

Let's consider p1=2 and p2=3 as two prime numbers

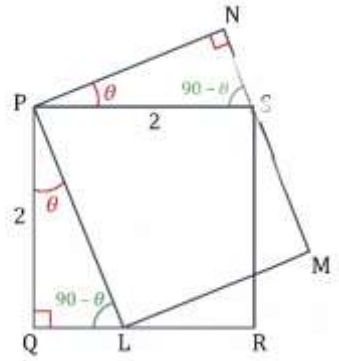
p1+p2=5 is a prime number. Hence, Option (A) is incorrect.

p1p2=6 is not a prime number. Hence, Option (B) is correct.

p1+p2+1=6 is not a prime number. Hence, Option (C) is incorrect.

p1p2+1=7 is a prime number in this case but it not valid for all prime numbers. Take, p1=5 and p2=7 then, p1p2+1=36 which is not prime. Hence, Option (D) is incorrect.

15. ANSWER: - (D)



$\triangle PQL$ similica to $\triangle PNS$

So,

$$\frac{PQ}{PN} = \frac{PL}{PS} \Rightarrow \frac{2}{PN} = \frac{PL}{2}$$
$$\Rightarrow PL \times PN = 4 \Rightarrow \text{Area of PLMN}$$

16. ANSWER: - (D)

(x + y) ∝ (x - y)
(x + y) =K (x - y)
x/y = (K+1)/(K-1) = K' (constant)

17. ANSWER: - (A)

Arrange the data in ascending order:

9,10,11,11,13,14,15,17,18,69

Total numbers=n=10(even).Hence,

Median=[(n/2)th term + (n/2 +1)th term] /2

=(13+14)/2 = 13.5

18. ANSWER: - (C)

Given the ratio of coins: ₹1 : ₹5 : ₹10 = 5x : 3x : 13x

Total number of coins = 5x + 3x + 13x = 21x

Total amount of money = ₹(5x + 15x + 130x) = ₹150x

Percentage of money in ₹5 coins = (Amount contributed by ₹5 coins / Total amount) * 100%
= (15x / 150x) * 100%
= 10%

Link to GfG corresponding Short Notes: GATE FARRAY

19. ANSWER: - (B)

By applying Product rule and power rule for logarithms, we get log (p^2+q^2)= log (9pq)

On comparing and squaring both sides, $(p^2+q^2)^2=(9pq)^2$. This expression evaluates to 79.

20. ANSWER: - (A)

Area of rectangular sheet = $54\times4=216$ cm^2

Surface area of cube =216= 6 $(side)^3$
Side=6

Volume of cube = $6^3 = 216$

Radius of cylinder = $4/2\pi$

Volume of cylinder = $\pi r^2h=216/\pi$

Volume of cylinder/Volume of cube=$(216/\pi)/216$ = $1/\pi$

21. ANSWER: - (C)
The total percentage contribution of fat macronutrients = 45% of 2000 = 900

The total fat (all three types), in grams = 900/9 = 100 (since energy density (kcal/g) of all fat macronutrients is 9)

22. ANSWER: - (A)

As per the given data in the question, there are four elements. Hence, we will use a combination of four elements.
Therefore, the result is 4 * 3 * 2 * 1 = 24.

Hence, the correct option is A.

23. ANSWER: - (A)

From the given data, the Total number of students = 10000.
Number of students who are not interested in both = 1500.
Remaining Students = 10000-1500 = 8500.

Let x represent core students, therefore, 4x will represent not core students.
=> x+4x = 8500+500.
5x = 9000.
X = 1800.

Hence, number of core students is 1800.
Hence, the correct option is A.

24. ANSWER: - (C)

As given in the question,

$\ln((x+y)/2) = 1/2[\ln x + \ln y]$

$\ln((x+y)/2) = 1/2 [\ln xy]$

$\ln((x+y)/2) = \ln\sqrt{(xy)}$

Removing ln from both sides, we get

$(x+y)/2 = \sqrt{(xy)}$

$x^2 + y^2 + 2xy = 4xy$

$x/y + y/x + 2 = 4$

$x/y + y/x = 2$

Hence, the correct option is C.

25. ANSWER: - (C)

As per the given data in the question, the selling price is the same for both items. Profit and loss are also the same for both, then overall it will give loss.
Hence, Loss% = $x^2/100 = 10^2/100 = 1\%$.

Therefore, the correct option is C.

26. ANSWER: - (D)

As per the information given in the question,
Considering year 2007, Electricity generated through renewable sources = Hydro + Solar + Wind = 30% + 5% + 5% = 40 %

Similarly, if we consider for the year 2023,
Electricity generated through renewable sources = 35% + 20% + 10% = 65%.

Therefore, Percentage increase in the electricity generation = (65-40)/40 % = 62.5%
Hence, the correct option is D.

27. ANSWER: - (C)

$f(t) = 0.01t^2$

$g(t) = 4t$, where $t > 0$

To compare f(t) and g(t), solve:

$0.01t^2 = 4t$

$0.01t^2 - 4t = 0$

$t(0.01t - 4) = 0$

So, $t = 0$ or $t = 400$

Since $t > 0$, we consider $t = 400$

Now check intervals:

For $0 < t < 400$:

Take $t = 100$

$f(100) = 0.01 \times 100^2 = 100$

$g(100) = 4 \times 100 = 400$

So, $g(t) > f(t)$

Hence, statement (i) is true

For $t > 400$:

Take $t = 500$

$f(500) = 0.01 \times 500^2 = 2500$

$g(500) = 4 \times 500 = 2000$

So, $f(t) > g(t)$

Also, for all $t > 400$, f(t) will always be greater than g(t)

So, statement (ii) is true (T = 400)

Final Answer:

(c) Both (i) and (ii) are correct

28. ANSWER: - (B)

Let's analyze from the given information:
We know that f(x) and g(y) are functions of x and y, respectively, and f(x)
= g(y) for all real values of x and y.
Since f(x) and g(y) are equal for all real values of x and y, this means that
both functions are constant.
Now let's consider the options:
(A) f(x) = 0 and g(y) = 0: This is not necessarily true because the
functions can be any constant value, not necessarily zero.
(B) f(x) = g(y) = constant: This is correct. We already established that
both functions are constant, so they are equal to some constant value.
(C) f(x) ≠ constant and g(y) ≠ constant: This is not true, as both functions are
constant.

29. ANSWER: - (C)

Area $= \int_0^1 y(x)dx$. Add the areas of the three constant pieces:

- On $\left[0, \frac{1}{3}\right)$: height 2 , width $\frac{1}{3} \Rightarrow$ area $2 \cdot \frac{1}{3} = \frac{2}{3}$.
- On $\left[\frac{1}{3}, \frac{3}{4}\right)$: height 3 , width $\frac{3}{4} - \frac{1}{3} = \frac{5}{12} \Rightarrow$ area $3 \cdot \frac{5}{12} = \frac{5}{4}$.
- On $\left[\frac{3}{4}, 1\right]$: height 1 , width $\frac{1}{4} \Rightarrow$ area $1 \cdot \frac{1}{4} = \frac{1}{4}$.

Total area $= \frac{2}{3} + \frac{5}{4} + \frac{1}{4} = \frac{2}{3} + \frac{6}{4} = \frac{2}{3} + \frac{3}{2} = \frac{13}{6}$.

30. ANSWER: - (D)

$$= (r^2 + 7r + 10)(r^2 + 7r + 12)$$

$$= [(\mathbf{r}^2 + 2\mathbf{r}) + 5r + 10][(\mathbf{r}^2 + 2\mathbf{r}) + 5r + 12]$$
$$= (-6 + 5r + 10)(-6 + 5r + 12)$$

$$= (5r + 4)(5r + 6)$$

Expend the terms:

$$= 25r^2 + 30r + 20r + 24$$
$$= 25r^2 + 50r + 21$$

Factor out 25

$$= 25(r^2 + 2r) + 24$$

Substitute $r^2 + 2r = -6$ one last time:

$$= 25(-6) + 24$$

=-150 +24 =-126

31. ANSWER: - (C)

To solve this problem, the key is understanding that the total number of students in the class must be divisible by the sum of the ratio values (7 for boys and 3 for girls).

Ratio of boys to girls = 7 : 3, The total number of students in the class must be a multiple of 10 (since the distribution follows a ratio of 7:3).

Hence, from given option, only 50 satisfies = 35 : 15 = 7 : 3 (50: Divisible by 10, so acceptable.)

32. ANSWER: - (C)

- Sheet M (Cylinder): > *Circumference of the base $= 6$ cm $\Longrightarrow r = \frac{6}{2\pi} = \frac{3}{\pi}$ cm
- Height $(h) = 1$ cm
- $V_{\text{cylinder}} = \pi r^2 h = \pi \left(\frac{3}{\pi}\right)^2 (1) = \frac{9}{\pi}$ cm^3
- Sheet N (Cube): > * Total area $= 6$ cm $\times$ 1 cm $= 6$ cm^2
- Cut into 6 equal square faces, the area of each face $= \frac{6}{6} = 1$ cm$^2 \Longrightarrow$ side $(a) = 1$ cm
- $V_{\text{cube}} = a^3 = 1^3 = 1$ cm^3
- Ratio: $>* \frac{V_{\text{cylinder}}}{V_{\text{cube}}} = \frac{\frac{9}{\pi}}{1} = \frac{9}{\pi}$

33. ANSWER: - (C)

Given, ratio of cost of item P to cost of item Q is 3:4

So, $\text{Cost}_P/\text{Cost}_Q = 3/4$

$5400/\text{Cost}_Q = 3/4 = 7200$

Acc. to Question: Profit % = 25 %

Now, Selling Price (SP) = 7200+ 720 (10% profit) + 720 (10% profit) + 360 (5% profit) = 9,000
[Here we break down the 25% profit into 3 parts]

So the Marked Price is = 10,000 & Selling price = 9000, which clearly shows the difference of 10%.

That means Discount is of 10%.

34. ANSWER: - (C)

- Bags are identical but not chocolates. Chocolates are distinct.
- We picked one chocolates from each bag. Order does not matter here.
- The probability that at least two chocolates are identical = 1 - (all are distinct chocolates) = 1 - (10/10*9/10*8/10*7/10*6/10) = 1 - 30240 / 100000 = 69760 / 100000 = 0.6976

35. ANSWER: - (B)

Take a cube of side a.

- Longest diagonal (space diagonal) vector: (a,a,a)
- Edge vector (say along x-axis): (a,0,0)

Using dot product:

$\cos\theta = (a,a,a)\cdot(a,0,0)/|(a,a,a)|\cdot|(a,0,0)|$

$= a^2/(\sqrt{3}a)(a) = 1/\sqrt{3}$

So, the correct answer is B.

36. ANSWER: - (B)

Given: $(x-1/2)^2-(x-3/2)^2=x+2$

Use identity: $a^2-b^2=(a-b)(a+b)$

Let $a=x-1/2$, $b=x-3/2$

Then: $a-b=1, a+b=2x-2$

So LHS: $1\cdot(2x-2)=2x-2$

Now: $2x-2=x+2 \Rightarrow x=4$

So, the correct answer is 4.

37. ANSWER: - (C)
According to given conditions, 3x+18 : 13x+18 : 6x+18 = 15 : 35 : 21.

On solving, (3x+18) / (13x+18) = 15/35 105x + 630 = 195x + 270 195x - 105x = 630 - 270 90x = 360 x = 360/90 = 4

So total number of students in these three classes, 3*4 + 13*4 + 6*4 = 88 (In total 88 students are there).

38. ANSWER: - (A)
From the above graph we obtained following information-

Year	Cost/Unit	Cost Price	No. of Units
1	3	300	100
2	2	400	200
3	1	300	300

Let selling price of year 2 = SP2 and Selling price of year3 = SP3

Given that-
Taxes for year 2 = 13% of SP2 = 0.13SP2
Taxes for year 3 = 15% of SP3 = 0.15SP3
Cost per unit of year 3 = 1
(Cost per unit of year 2)/2 = 1
So now Cost / unit of year 2 = 2×1 = 2

Net profit formula now = Selling Price - (Cost Price + (Tax% × selling price)

For year 2
=> 296 = SP2- (200×2 + 0.13×SP2)

=> SP2 = 696×100/87 SP2 = 800

For year 3
=> 210 = SP3 - (300×1+0.15×SP3)

=> SP3 = 510×100/85 SP3 = 600

Required Ratio 800/600= 4/3 (Correct Option A)

39. ANSWER: - (C)

Total expenditure during each year = 500 million rupees
Total expenditure for 5 years = 500 × 5 = 2500 million rupees

- Revenue in 2014 = 500 million rupees
- Revenue in 2015 = 700 million rupees
- Revenue in 2016 = 800 million rupees
- Revenue in 2017 = 600 million rupees
- Revenue in 2018 = 400 million rupees

Total revenue = 500 + 700 + 800 + 600 + 400 = 3000 million rupees
Profit = 3000 − 2500 = 500 million rupees
Profit percentage = $\left(\frac{500}{2500}\right) \times 100 = 20\%$.

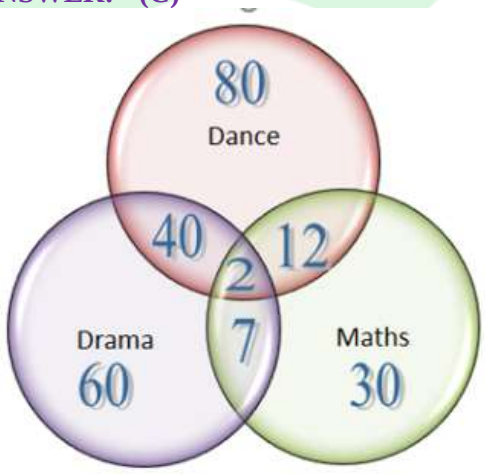
40. ANSWER: - (B)

Solution:
Relative speed = 60 − 50 = 10 km/h
Time = Distance / Relative speed = 20 / 10 = 2 hours

41. ANSWER: - (C)

Let total cost be x
Original share = x/10
New share = x/8
x/8 − x/10 = 150
(5x − 4x)/40 = 150
x/40 = 150
x = 6000

42. ANSWER: - (C)



Read question carefully. We cannot directly use the formula

$$P(A \cup B \cup C)=P(A)+P(B)+P(C)-P(A \cap B)-P(B \cap C)-P(C \cap A)+P(A \cap B \cap C)$$

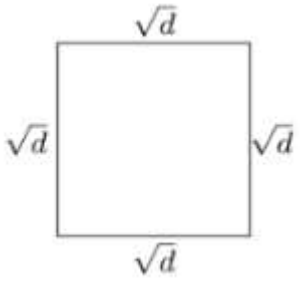
because of the only keyword which means we do not directly know what is P(A), P(B) or P(C)
Here, according to question statement, we can calculate total number of students belonging to at least one club :

$$P(A \cup B \cup C)=60+80+30+(40-2)+(12-2)+(7-2)+2=225$$

This is just 25% of the total no of students since 75% do not belong to any club.
So, total number of students = 225 ÷ 25% = 225 × 4 = 900.

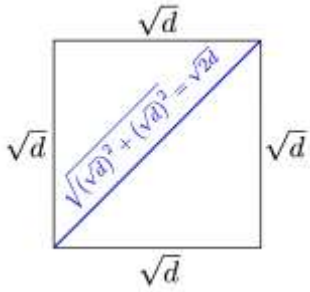
43. ANSWER: - (D)

The area of a square is d
We know that, if a square has a units as the length of its side, then its area will be a^2.

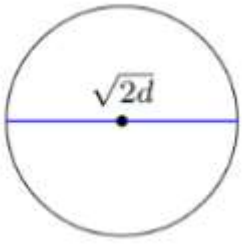


⇒ every side of the square will be $\sqrt{d}$
Now, the diagonal of the square will be

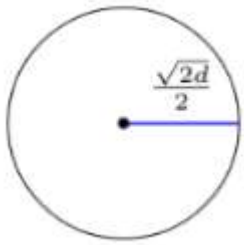
$$\sqrt{(\sqrt{d})^2 + (\sqrt{d})^2} = \sqrt{2d}$$



Now, $\sqrt{2d}$ is the diameter of a circle.



∴ Radius of the circle will be $\frac{\sqrt{2d}}{2}$



We know Area of the circle is πr^2, where, r = radius of the circle

$$= \frac{22}{7} \times \frac{\sqrt{2d}}{2} \times \frac{\sqrt{2d}}{2}$$
$$= \frac{22}{7} \times \frac{2d}{2 \times 2}$$
$$= \frac{22}{7} \times \frac{d}{2}$$

We know Area of the circle is πr^2, where, r = radius of the circle

Option (D)

44. ANSWER: - (C)

Required number = LCM(20, 42, 76) + 7
LCM:
$20 = 2^2 \times 5$
$42 = 2 \times 3 \times 7$
$76 = 2^2 \times 19$

LCM = $2^2 \times 3 \times 5 \times 7 \times 19 = 7980$
Number = 7980 + 7 = 7987

45. ANSWER: - (B)

$2 \times 6 = 12$
$12 \times 5 = 60$
$60 \times 4 = 240$
$240 \times 3 = 720$
$720 \times 2 = 1440$
$1440 \times 1 = 1440$

So, option B, 1440 is the next term.

46. ANSWER: - (C)

Probability of green = 4/6 = 2/3

Expected value = n × p = 7 × (2/3) ≈ 4.67

Closest integer = 5

Hence most likely outcome is 5 green and 2 red.

47. ANSWER: - (D)

$162 = 2 \times 3^4$
Make powers multiple of 3 → multiply by $2^2 * 3^2$
= 36

48. ANSWER: - (C)

Digits allowed = 1–9 except {5,9} → also exclude 0
Allowed digits = 7 out of 10
Probability per digit = 0.7
So total = 0.7^k

49. ANSWER: - (B)

Let MCQs = x, essays = y

x + y = 20

3x + 11y = 100

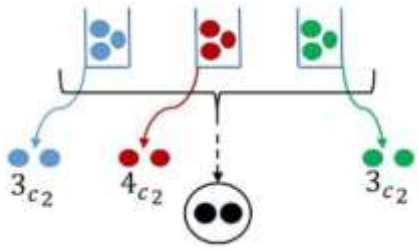
Substitute y = 20 − x:

3x + 11(20 − x) = 100

3x + 220 − 11x = 100

−8x = −120 → x = 15

50. ANSWER: - (D)



$$\frac{{}^3C_2 + {}^4C_2 + {}^3C_2}{{}^{10}C_2} = \frac{4}{15}.$$

assuming ball are not identical
Correct Answer: D

51. ANSWER: - (D)

64 cubes → cube of side 4

Original surface area = 6 × 4² = 96

Removing 8 corner cubes:
Each removal removes 3 faces but exposes 3 new faces → no net change per corner

Total surface area remains 96.

52. ANSWER: - (B)

To calculate the revenue generated by each product in a year we need to first calculate total units of each category sold, and then multiply it with the cost per unit.

Revenue generated by products

	Total units sold	Price per unit	Total revenue
Elegance	102,510	48	4,920,480
Smooth	80,059	63	5,043,717
Soft	72,186	78	5,630,508
Executive	39,284	173	6,796,132

From the table above we can see that the max revenue is generated by Executive.
Hence answer is B.

53. ANSWER: - (B)

Check x = 1:
f(1) = 2 + 3 − 5 = 0

Hence (x − 1) is a factor.

54. ANSWER :- (C)

Graph is inverted V-shape with peak at (1,2) → form:
f(x) = 2 − |x − 1|

55. ANSWER :- (D)

Total possibilities = 4! = 24.
No. of ways 'Arun chooses red' or 'Shwetha chooses white'
= (no.of ways 'Arun chooses red')
+(no.of ways 'Shwetha chooses white')
-(no.of ways 'Arun chooses red' and 'Shwetha chooses white')
= 6 + 6 − 2 = 10
required = 24 − 10 = 14.

56. ANSWER :- (A)

X = 4777 ... (729 times)

It can be written as $X = 4.7777 \cdots * 10^{29}$

$$X^3 = (4.777 \cdots * 10^{29})^3 = (4.777 \ldots)^3 * 10^{87}$$

Now, even if we round up 4.777 ... to 5 , we could represent $5^3 = 125$ in 3 digits. So, We can say $(4.77 \ldots)^3$ also has 3 digits before decimal point.

So, X^3 requires 3 + 87 = 90 digits.
Correct Answer: A

57. ANSWER :- (D)

Let $f(x) = e^x + 0.5x^2 - 2$
Check values:
f(0) = 1 − 2 = −1
f(1) = e + 0.5 − 2 > 0 → one root between 0 and 1

$f(-2) = e^{(-2)} + 2 - 2 \approx$ positive
$f(-1) = e^{(-1)} + 0.5 - 2 < 0$ → another root between −2 and −1

Hence total roots = 2.

58. ANSWER: - (B)
Exponential relation:
$N = A\, e^{(-kL)}$
Using given points:
100 → 80
10000 → 40
Solve ratio:
10000/100 = e^(k×40)
$100 = e^{(40k)}$
Take ln:
ln100 = 40k → k = ln100/40

Now for N = 5000:
5000/100 = e^(k(80 − L))
$50 = e^{(k(80 - L))}$

Solve:
ln50 = k(80 − L)

Substitute k:
80 − L = (ln50)/(ln100/40)
≈ 33.98

L ≈ 46.02

59. ANSWER: - (B)

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$
$$= \frac{\alpha^n + \beta^n}{\left(\frac{1}{\alpha^n} + \frac{1}{\beta^n}\right)}$$
$$= \frac{\alpha^n + \beta^n}{\left(\frac{\alpha^n + \beta^n}{\alpha^n \beta^n}\right)}$$
$$= (\alpha\beta)^n$$
$$= 4^n$$

as product of roots, αβ = 4
Correct Answer: B

60. ANSWER: - (A)

Total = 150
Neither = 30 → at least one = 120

FB = 55, WA = 85

Intersection = 55 + 85 − 120 = 20

Only FB = 55 − 20 = 35

61. ANSWER: - (C)
Number of rectangles = (m(m+1)/2) × (n(n+1)/2)

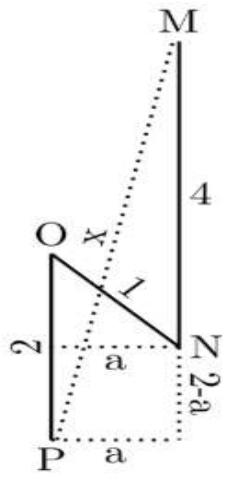
= (2×3/2) × (4×5/2)
= 3 × 10 = 30

62. ANSWER: - (C)
f(x) is linear means it is of the form ax + b
given f(-2) and f(3)
solve the equation and find out value for a and b
then find f(5). it will be 43
Correct Answer: C

63. ANSWER: - (2006)
Answer is 2006
No need to do any calculations, just observe the bar chart.
Here % increase is asked, so just check for previous year & next year pair, where you can find maximum positive change.
Consider year 2006.
Red Bar is above 3 steps than 2005 AND Blue Bar is 3 steps more than 2005, total 3+36, it's maximum.

64. ANSWER: - (B)
Because If we multiply the terms of an AP with a constant, the common difference will get multiplied by the constant and if we subtract the terms of an AP with a constant, the common difference will be same but if we make square of AP element then difference will not be same.

65. ANSWER: - (A)



$$a = \frac{1}{\sqrt{2}}$$

$$x = \sqrt{[4 + (2 - a)]^2 + a^2}$$

Solving we get x = 5.34.
Correct Answer: A

66. ANSWER: - (B)
Area of a $\triangle = \frac{1}{2} \times a\sin B = \frac{1}{2} \times b\sin A = \frac{1}{2}ab\sin C$
so, Here area ($\triangle$ PQR) = area ($\triangle$ PQS) + area ($\triangle$ PRS)

$$\frac{1}{2}rq\sin 120 = \frac{1}{2}PS \times r\sin 60 + \frac{1}{2}PS \times q\sin 60$$

$$\Rightarrow PS = \frac{rq}{(r+q)}$$

so, choice (B) is correct..

67. ANSWER: - (8)
To find the value of f(g(h(2, 5, 7, 3), 4, 6, 8)), we solve it from the inside out.

1. Evaluate the inner function: h(2, 5, 7, 3)

The function h calculates the remainder of the larger product divided by the smaller product of the pairs (p, q) and (r, s).

- **First pair (p * q):** 2 X 5 = 10
- **Second pair (r * s):** 7 X 3 = 21
- **Calculate Remainder:** 21 / 10 = 2 with a remainder of **1**.
- **Result:** h(2, 5, 7, 3) = 1.

2. Evaluate the outer function: fg(1, 4, 6, 8)

Now, substitute the result back into the expression: fg(1, 4, 6, 8).

The function fg is defined as f(set) X g(set).

- **Find f (Maximum):** max(1, 4, 6, 8) = 8
- **Find g (Minimum):** min(1, 4, 6, 8) = 1
- **Final Calculation:** 8 X 1 = 8

Final Answer:

8

68. ANSWER: - (C)

f(x) = 1 − |x| is maximum when |x| is minimum. |x| is minimum at x = 0,

giving f(0) = 1 − 0 = 1. So the function attains a maximum value of 1 at x = 0.

69. ANSWER: - (C)

From Statement I: one pole = 20 Kg,

so 10 poles = 200 Kg.

From Statement II: 10x - 2x = 160,

so 8x = 160, x = 20 Kg, 10 poles = 200 Kg.

Both statements independently give the answer, so either alone is sufficient.

70. ANSWER: - (C)

Total students = 21+17+6 = 44.

Total marks = (21x2)+(15x3)+(11x1)+(23x2)+(31x5) = 42+45+11+46+155 = 299.

Average = 299/44 = 6.795.

71. ANSWER: - (D)

P(at least one) = p+m+c - P(exactly two) - 2P(all three) = 0.75.

P(exactly two) = 0.40,

P(at least two) = 0.50,

so P(all three) = 0.10.

Thus p+m+c = 0.75 + 0.40 + 0.10 = 1.35 = 27/20 and (p)(m)(c) = 1/10. So I and III are true.

72. ANSWER: - (A)

Total pairs = 4 x 5 = 20. Pairs that sum to 16: (2,14), (3,13), (4,12), (5,11) = 4 pairs. Probability = 4/20 = 0.2.

73. ANSWER: - (D)

Let India's GDP $= \text{Rs.}\, x$

$\text{Rs. } 50 = 1 \text{ USD}$

$\text{Rs. } x = \frac{1}{50} \times x\text{USD}$

$\text{New GDP} = \text{GDP} + 0.07 \times \text{GDP}$

$= \text{Rs}1.07x$

$\text{Rs}60 = 1\text{USD}$

$\text{Rs } 1.07x = \frac{1}{60} \times 1.07x$

$\text{Change in GDP} = \frac{\text{new old}}{\text{old}} \times 100$

$$= \frac{\left(\frac{1.07x}{60}\right) - \left(\frac{x}{50}\right)}{\frac{x}{50}} \times 100$$

$$= \frac{\left(\frac{-13x}{6000}\right)}{\left(\frac{x}{50}\right)} \times 100$$

$$= \left(\frac{-13}{120}\right) \times 100$$

$$= -0.10833 \times 100$$

$$= -10.833$$

$$\approx -11$$

So, there is an 11% decrease

Answer = option D

74. ANSWER: - (C)

From graph, ratio in 2011 = 1 and in 2012 = 1.5. Since female students are equal in both years, male students in 2012/male students in 2011 = 1.5/1 = 1.5:1.

75. ANSWER: - (C)

Subtracting in octal: $7526_8 - 4364_8 = 3142_8$. Borrowing is done in base 8 instead of base 10. So $Y = 3142_8$.

76. ANSWER: - (C)

The pattern is $n^2 + 1$: 1+1=2, 4+1=5, 9+1=10, 16+1=17, 25+1=26, 36+1=37, 49+1=50, 64+1=65. So 64 should be 65, making 64 the wrong number.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

77. ANSWER: - (C)

The question paper has only 3 questions. There are 3 ways a student can attempt a question.

1. Attempt correctly
2. Attempt wrongly
3. Not attempted

Each row lists number of students who attempted the question correctly, wrongly and who did not attempt.
Sum of these, of any row gives the total strength of class, $21 + 17 + 6 = 44$.
Average marks obtained by the class in the examination,

$$= \frac{(\text{Total marks(Q1)} + \text{Total marks(Q2)} + \text{Total marks (Q3)})}{\text{Total strength of class}}$$
$$= \frac{((2 \times 21) + (3 \times 15) + (2 \times 23))}{44}$$
$$= \frac{133}{44} = 3.02$$

Note : There is no negative or partial marking for the wrongly attempted and the non attempted questions.

Correct Answer: C

78. ANSWER: - (D)

$|x^2 - 2x + 3| = 11$ gives $x^2 - 2x + 3 = 11$ or $x^2 - 2x + 3 = -11$.

First gives $x^2 - 2x - 8 = 0$,

so $x = 4$ or $x = -2$.

For $x = 4$: $|-64 + 16 - 4| = 52$.

For $x = -2$: $|8 + 4 + 2| = 14$.

So values are 14 and 52.

79. ANSWER: - (140%)

From the graph, ratio in 2008 = 2.5 and in 2009 = 3. If female students doubled, let female = f in 2008, so female = 2f in 2009. Male in 2008 = 2.5f, male in 2009 = 3 x 2f = 6f. Increase = (6f - 2.5f)/2.5f x 100 = 140%.

80. ANSWER: - (A)

At 6:00, the hour hand is at 180°. The minute hand gains 5.5° per minute over the hour hand. For a 60° angle, we need 180° - 60° = 120°.

So 120/5.5 = 21.8 ≈ 22 minutes. Hence 6:22 a.m.

81. ANSWER: - (B)

Multiples of 10 from 2 to 198 are 10, 20, 30, ... 190. This is an arithmetic series with first term 10, last term 190. Average = (first + last)/2 = (10 + 190)/2 = 100.

82. ANSWER: - (C)

Let $x = \sqrt{(12 + \sqrt{(12 + \sqrt{(12 + ...)})})}$.
So $x = \sqrt{(12 + x)}$.
Squaring both sides, $x^2 = 12 + x$.
So, $x^2 - x - 12 = 0$, giving $(x-4)(x+3) = 0$.

Since x is positive, $x = 4$.

83. ANSWER: - (6)

Tetrahedron has 4 corner points. So, it forms 4 planes. Now, we add an internal point making a total of 5 points. Any three combination of points leads to a plane and thus we can get $^5C_3$ planes.
So, newly created planes = 10-4 = 6

84. ANSWER: - (850)

1. **Calculate the Base Fare:**
   - One-way fare per person = Rs. 100
   - Round-trip fare per person = Rs. 200
   - Total fare for 5 people = **Rs. 1,000**
2. **Apply the Discounts:**
   - **Round-trip discount (10%):** 10% of 1000 = Rs. 100
   - **Group discount (5%):** 5% of 1000 = Rs. 50
   - **Total Discount:** 100+50 = Rs. 150
3. **Final Amount:**
   - Total Fare - Total Discount = **Rs. 850**

85. ANSWER: - (96)

$$\left(Z+\frac{1}{Z}\right)^2$$

$$=\left(z^2+2(z)\left(\frac{1}{z}\right)+\left(\frac{1}{z}\right)^2\right)=\left(z^2+\frac{1}{z^2}\right)+2=98$$

$\Rightarrow 98-2=96$ is answer..

86. ANSWER: - (D)

Since $ax^2 + bx + c = 0$ has real and positive roots, say p and q,

then for $ax^2 + b|x| + c = 0$,

substituting |x| means both +p, +q and -p, -q satisfy the equation, giving 4 real roots.

87. ANSWER: - (B)

Rationalizing each term by multiplying numerator and denominator by $(\sqrt{(n+1)} - \sqrt{n})$, each term becomes $\sqrt{(n+1)} - \sqrt{n}$. The series is telescoping, so the sum

$= \sqrt{81} - \sqrt{1}$

$= 9 - 1$

$= 8.$

88. ANSWER: - (D)

Total 2-digit numbers = 90 (from 10 to 99). Numbers divisible by 7 between 10 and 99 are: 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, 91, 98 = 13 numbers. So probability of not divisible by 7 = (90-13)/90 = 77/90.

89. ANSWER: - (C)

Let total distance = 4d.

Time by train = 2d/60,

Time by bus = d/30,

Time by cycle = d/10.

Total time = d/30 + d/30 + d/10

= d/30 + d/30 + 3d/30 = 5d/30.

Average speed = 4d / (5d/30)

= 4d x 30/5d

= 24 km/h.

90. ANSWER: - (B)

Given Data

- **Current Cost (C):** Rs. 13,200
- **Wage Increase:** 1/5 of current wages.
- **Working Hours Decrease:** 1/24 of the current period.

1. Define the Relationship

The total cost of erection is generally the product of the wage rate and the time (duration) spent on the project:
Cost = Wage X Time

2. Determine the New Scaling Factors

Let the original wage be W and the original time period be T.

- **New Wage (W_new):**
  W_new = W + 1/5W = 6/5W
- **New Time (T_new):**
  T_new = T - 1/24T = 23/24T

3. Calculate the New Cost (C_2)

We can express the new cost as a ratio of the old cost:

$$C_2 = W_{new} \times T_{new}$$
$$C_2 = \left(\frac{6}{5}W\right) \times \left(\frac{23}{24}T\right)$$
$$C_2 = \left(\frac{6}{5} \times \frac{23}{24}\right) \times (W \times T)$$

Since the original cost (W X T) = 13,200

C_2 = 6/5 X 23/24 X 13,200

4. Simplify and Solve

First, simplify the fraction $\frac{6}{24}$ to $\frac{1}{4}$:

$$C_2 = \frac{1}{5} \times \frac{23}{4} \times 13{,}200$$
$$C_2 = \frac{23}{20} \times 13{,}200$$

Divide 13, 200 by 20 :

$$C_2 = 23 \times 660$$
$$C_2 = 15{,}180$$

Final Result

The new cost of erection is **Rs. 15,180**.

91. ANSWER: - (C)

This is a decreasing arithmetic series ending at 2.

Sum = (n/2)(first + last) = (22/2)(44 + 2) = 11 x 46 = 506.

Ans. C) 506.

92. ANSWER: - (B)

To find maximum height, differentiate

$y = 2x - 0.1x^2$ and set $dy/dx = 0$.

So $2 - 0.2x = 0$, giving $x = 10$.

Substituting back, $y = 2(10) - 0.1(100) = 20 - 10 = 10$ meters.

93. ANSWER: - (B)

Using Bayes' theorem,

P(Y|Reliable) = P(Reliable|Y) x P(Y) / P(Reliable). P(Reliable)

= (0.96 x 0.60) + (0.72 x 0.40)

= 0.576 + 0.288 = 0.864.

So P(Y|Reliable) = 0.288 / 0.864 = 0.334.

94. ANSWER: - (A)

Profit = Revenue - Cost = $50q - 5q^2$. To maximize profit, differentiate with respect to q and set equal to zero

: dP/dq = 50 - 10q = 0

which gives q = 5.

Since the second derivative is negative, this confirms it is a maximum.

95. ANSWER: - (B)

$$\log(P) = \frac{1}{2}\log(Q) = \frac{1}{3}\log(R) = k$$
$$\log(P) = k, \frac{1}{2}\log(Q) = k, \frac{1}{3}\log(R) = k$$
$$\log(P) = k, \log(Q) = 2k, \log(R) = 3k$$
$$P = 10^k, Q = 10^{2k}, R = 10^{3k}$$

by looking at the values of P, Q, R we can easily guess that option B is True.

$$Q^2 = PR$$
$$(10^{2t})^2 = 10^t \cdot 10^{3t}$$
$$10^{2\times 2t} = 10^{t+3t}$$
$$10^{4t} = 10^{4t}$$
L.H.S. = R.H.S

96. ANSWER: - (B)

Given Data

- **Variable Cost (**V**):** 4q
- **Fixed Cost (**F**):** 100/q
- **Objective Function (Total Cost** C**):** C(q) = V + F = 4q + 100/q

Method 1: Calculus Approach (Differentiation)

To find the minimum value of a function, we look for the point where its rate of change (slope) is zero.

Step 1: Find the First Derivative

We differentiate the cost function C(q) with respect to q:

$$\frac{dC}{dq} = \frac{d}{dq}(4q + 100q^{-1})$$

Using the power rule $\left(\frac{d}{dx}x^n = nx^{n-1}\right)$:

$$\frac{dC}{dq} = 4 - 100q^{-2} = 4 - \frac{100}{q^2}$$

Step 2: Set the Derivative to Zero

For a local minimum or maximum, the first derivative must be zero:

$$4 - \frac{100}{q^2} = 0$$
$$4 = \frac{100}{q^2}$$
$$4q^2 = 100$$
$$q^2 = 25$$
$$q = \sqrt{25} = 5$$

(Note: We ignore q = -5 because production quantity cannot be negative.)

Step 3: Verify the Minimum (Second Derivative Test)

To confirm q=5 is a minimum, we find the second derivative:

$$\frac{d^2C}{dq^2} = \frac{d}{dq}(4 - 100q^{-2}) = 0 - (-2)\cdot 100q^{-3}$$
$$= \frac{200}{q^3}$$

At q = 5:

$$\frac{200}{5^3} = \frac{200}{125} = 1.6$$

Since 1.6 > 0 (positive), the function is concave up at q=5 , confirming it is a **minimum**.

Method 2: Alternative Approach (AM-GM Inequality)

The Arithmetic Mean-Geometric Mean (AM-GM) inequality states that for non-negative real numbers a and b:

$$\frac{a+b}{2} \geq \sqrt{ab}$$

The minimum value occurs when a = b.

Step 1: Apply to the Cost Function

Let a = 4q and b = 100/q . The total cost is a + b. According to the inequality:

$$\frac{4q + \frac{100}{q}}{2} \geq \sqrt{4q \cdot \frac{100}{q}}$$

$$4q + \frac{100}{q} \geq 2\sqrt{4 \cdot 100}$$

$$C(q) \geq 2\sqrt{400}$$

$$C(q) \geq 2 \cdot 20 = 40$$

Step 2: Find the value of q

The minimum value (40) is reached only when the two terms being added are equal:

$$4q = \frac{100}{q}$$

$$4q^2 = 100$$

$$q^2 = 25$$

$$q = 5$$

Both methods yield the same result. To minimize the total cost, **5 units** should be produced.

Correct Option: (b) 5

97. ANSWER: - (C)

Let the amount of orders received per day be x, the amount of pending orders be y and the amount of orders carried by a truck each day be z.

$$7z \times 4 = 4x + y \rightarrow (1)$$
$$3z \times 10 = 10x + y \rightarrow (2)$$

$$(2) - (1) \Longrightarrow 2z = 6x, z = 3x, y = 80x$$

We want to find the number of trucks to finish the orders in 5 days. Let it be A .

$$Az \times 5 = 5x + y$$
$$15Ax = 5x + 80x$$

$$A = [85/15] = [17/3] = 6$$

So, minimum 6 trucks must be used.
Correct Answer: C

98. ANSWER: - (D)

Explanation: Each time you replace 1/10th of the volume, you are keeping **9/10ths** of the spirit. After three rounds, you simply have:

10 * 0.9 * 0.9 * 0.9 = 7.29

99. ANSWER: - (C)

Assume base is x then convert numbers in given equation to decimal then find the value of x

$x^2+3x+7+2x^2+7x+6=4x^2+3x+5$

x = 8,-1

base cannot be -1 so x = 1623

$(137)_8 + (276)_8 = (435)_8$

So basically the numbers are given in Octal base. Similarly, addition of 731, 672 gives 1623 in octal.

100. ANSWER: - (D)

No. of persons who play either football or hockey = 15 + 17 -10 = 22

No. of persons playing neither hockey nor football = 25-22 = 3

101. ANSWER: - (D)

5 skilled → 1 wall in 20 days
⇒ 1 skilled in 1 day = 1 / (5 × 20) = 1/100

8 semi-skilled → 1 wall in 25 days
⇒ 1 semi-skilled in 1 day = 1 / (8 × 25) = 1/200

10 unskilled → 1 wall in 30 days
⇒ 1 unskilled in 1 day = 1 / (10 × 30) = 1/300

2 skilled = 2/100 = 1/50
6 semi-skilled = 6/200 = 3/100
5 unskilled = 5/300 = 1/60
Total work per day = 1/50 + 3/100 + 1/60
LCM = 300
= 6/300 + 9/300 + 5/300 = 20/300 = 1/15
Time = 15 days

Answer: **15 days**

102. ANSWER: - (B)

first place should be occupied by either **3** or **4**

Case 1: First digit is 3

The number looks like 3.

- **Remaining digits in set:**{2, 2, 3, 4, 4, 4, 4}
- **Total theoretical ways:** If we had an unlimited supply of 2s, 3s, and 4s, there would be 3^3 = 27 ways to fill the remaining three spots.
- **The Constraint:** We only have **two** 2s and **one** 3 left.
- **Impossible Case:** The only number we cannot form is **3333**, because we only have three 3s total (one used in the first spot, two left in the set).
  - Note: We have enough 2s to fill the remaining spots? No, we only have two 2s. However, since there are 3 spots left, the only "impossible" 2-heavy number would be 2222, which is impossible anyway since we only have two 2s total.
- **Calculation:** 27 - 1 (for 3333) - 1 {for 2222) = 25

Case 2: First digit is 4

The number looks like 4 .

- **Remaining digits in set:** {2, 2, 3, 3, 4, 4, 4}
- **Total theoretical ways:** 3^3 = 27 ways.
- **The Constraint:** We only have **two** 2s and **two** 3s left.
- **Impossible Cases:** We cannot form numbers that require three of a kind for 2 or 3.

1. **2222** (Requires three 2s in the blanks; we only have two).
2. **3333** (Requires three 3s in the blanks; we only have two).

- **Calculation:** 27 - 2 = 25

Final Summation

By adding the valid permutations from both independent cases:

26 + 25 = 51 ways

2 ANALYTICAL APTITUDE

GATE CSE - 2026 MCQ SET – 1 (1-MARKS)

Q1. 'When the teacher is in the room, all students stand silently.'

If the above statement is true, which one of the following statements is not necessarily true?

(a) If any student is not standing silently, then the teacher is not in the room.
(b) When the teacher is in the room, all students are silent.
(c) If all students are standing, then the teacher is in the room.
(d) When the teacher is in the room, all students are standing.

GATE CSE - 2026 MCQ SET – 1 (2-MARKS)

Q2. Combinatorics deals with problems involving counting. For example, "How many distinct arrangements of N distinct objects in M spaces on a circle are possible?" is a typical problem in combinatorics. This kind of counting is sometimes used in the modeling of several physical phenomena. Often, in such models, the different combinatorial possibilities are assigned probability values. Assigning probabilities enables the computation of the average values of physical quantities.

Consider the following statements:

P: Combinatorics is always invoked in the modeling of physical phenomena.

Q: Modeling some physical phenomena involves assigning probabilities to combinatorial possibilities in order to compute average values of physical quantities.

Based on the passage above, what can be inferred about statements P and Q?

(a) P is False and Q is False
(b) P is False and Q is True
(c) P is True and Q is False
(d) P is True and Q is True

GATE CSE - 2026 MCQ SET – 1 (2-MARKS)

Q3. In the 2020 summer Olympics' Javelin throw finals, Neeraj Chopra exhibited a spectacular performance to win the gold medal. The silver medal was won by Jakub Vadlejch and the bronze medal was won by Vitezlav Vesely. There were six rounds of throws with each athlete having one throw per round. The best of all the throws of each athlete is considered for the medal. Following were the observations about the throws:

i. The first and second rounds were dominated by Neeraj Chopra with a gold medal performance in his second throw, while the other two athletes did not have any medal winning throws in these rounds.

ii. The throws in the last round by both Jakub Vadlejch and Vitezlav Vesely were fouls and were not considered for scoring.

iii. After four rounds, Vitezlav Vesely was in the second position and could not improve upon his best throw in the succeeding rounds.

iv. In the fourth round, the throw by Jakub Vadlejch was the best in that round.

In which round did Vitezlav Vesely have his best throw?

(a) Third
(b) Fourth
(c) Fifth
(d) Sixth

GATE CSE - 2026 MCQ SET – 2 (1-MARKS)

Q4. 'When it is raining, peacocks dance.'

Based only on this sentence, which one of the following options is necessarily true?

(a) Peacocks dance only when it is raining.
(b) When peacocks dance, it is raining.
(c) When peacocks are not dancing, it is not raining.
(d) When it is not raining, peacocks do not dance.

Check Out Our GATE Courses & Secure Your Dream Rank!

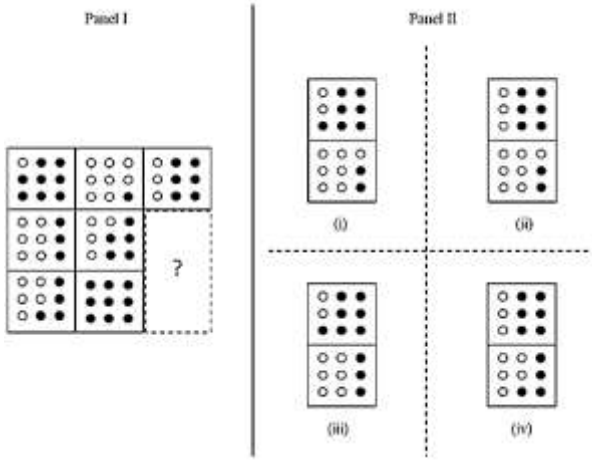
GATE CSE - 2026 MCQ SET – 2 (2-MARKS)

Q5. Water : P :: Food : Q
Choose the P and Q combination from the options below to form a meaningful analogy.

(a) P = Thirst; Q = Hunger
(b) P = Drink; Q = Hunger
(c) P = Thirst; Q = Satiated
(d) P = Wet; Q = Critic

GATE CSE - 2026 MCQ SET – 2 (2-MARKS)

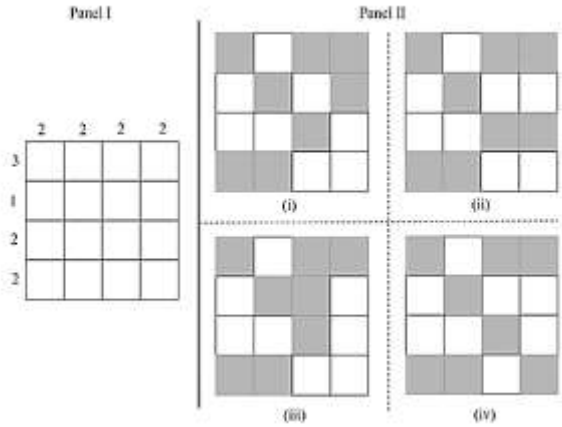
Q6. Two tiles are missing in Panel I. Which one of the options in Panel II is the appropriate choice for the missing tiles?



(a) (i)
(b) (ii)
(c) (iii)
(d) (iv)

GATE CSE - 2026 MCQ SET – 2 (2-MARKS)

Q7. The figure in Panel I below is a grid of cells with four rows and four columns. The numbers on the top and on the left represent the number of cells that are to be shaded in that column and row, respectively. Which one of the options shown in Panel II below represents the grid shaded correctly?



(a) (i)
(b) (ii)
(c) (iii)
(d) (iv)

GATE CSE - 2025 MCQ SET – 1 (1-MARKS)

Q8. Consider the relationships among P, Q, R, S, and T:

- P is the brother of Q.
- S is the daughter of Q.
- T is the sister of S.
- R is the mother of Q.

The following statements are made based on the relationships given above.

(1) R is the grandmother of S.
(2) P is the uncle of S and T.
(3) R has only one son.
(4) Q has only one daughter.

Which one of the following options is correct?

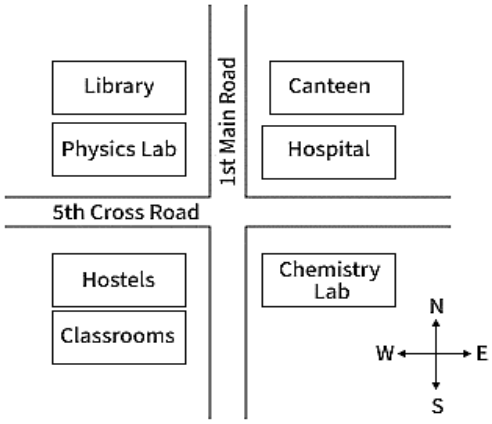
(a) Both (1) and (2) are true.
(b) Both (1) and (3) are true.
(c) Only (3) is true.
(d) Only (4) is true.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2025 MCQ SET – 1 (1-MARKS)

Q9. According to the map shown in the figure, which one of the following statements is correct?

Note: The figure shown is representative.



(a) The library is located to the northwest of the canteen.
(b) The hospital is located to the east of the chemistry lab.
(c) The chemistry lab is to the southeast of physics lab.
(d) The classrooms and canteen are next to each other.

GATE CSE - 2025 MCQ SET – 2 (2-MARKS)

Q10. Which one of the following options is correct for the given data in the table?

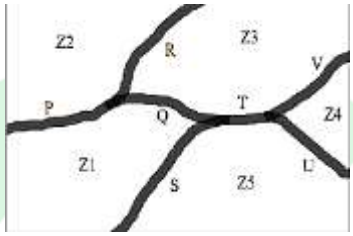
Iteration (i)	0	1	2	3
Input (1)	20	-4	10	15
Output (x)	20	16	26	41
Output (Y)	20	-80	-800	-12000

(a) $X(i) = X(i-1) + I(i);\ Y(i) = Y(i-1)I(i);\ i > 0$
(b) $X(i) = X(i-1)I(i);\ Y(i) = Y(i-1) + I(i);\ i > 0$
(c) $X(i) = X(i-1)I(i);\ Y(i) = Y(i-1)I(i);\ i > 0$
(d) $X(i) = X(i-1) + I(i);\ Y(i) = Y(i-1)I(i-1);\ i > 0$

GATE CSE - 2025 MCQ SET – 2 (1-MARKS)

Q11. The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V. It splits the land into 5 zones, marked Z1, Z2, Z3, Z4, and Z5. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct?

Note: The figure shown is representative.



(a) Bridges on P, Q, and T
(b) Bridges on P, Q, S, and T
(c) Bridges on Q, R, T, and V
(d) Bridges on P, Q, S, U, and V

GATE CSE - 2025 MCQ SET – 2 (2-MARKS)

Q12. If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH ?

(a) CEADP
(b) IDBFC
(c) JGIBC
(d) IBCEC

GATE CSE - 2024 MCQ SET – 2 (1-MARKS)

Q13. In the sequence 6, 9, 14, x, 30, 41, a possible value of

(a) 25
(b) 21
(c) 18
(d) 20

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2024 MCQ SET – 2 (2-MARKS)

Q14. In the 4 × 4 array shown below, each cell of the first three rows has either a cross (X) or a number.

1	X	4	3
X	5	5	4
3	X	6	X

The number in a cell represents the count of the immediate neighboring cells (left, right, top, bottom, diagonals) NOT having a cross (X). Given that the last row has no crosses (X), the sum of the four numbers to be filled in the last row is

(a) 11
(b) 10
(c) 12
(d) 9

GATE CSE - 2023 MCQ (1-MARKS)

Q15. A series of natural numbers $F_1,F_2,F_3,F_4,F_5,F_6,F_7$ obeys $F_{n+1}=F_n+F_{n-1}$ for all integers n>=2.

If $F_6 = 37$ and $F_7 = 60$ Then what is the value of F_1?

(a) 4
(b) 5
(c) 6
(d) 7

GATE CSE - 2023 MCQ (1-MARKS)

Q16. A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers. Given this information, which one of the following statements can be inferred with certainty?

(a) More than half of the pregnant women received medical care at least once from a doctor.
(b) Less than half of the pregnant women received medical care at least once from a doctor.
(c) More than half of pregnant women received medical care at most once from a doctor.
(d) Less than half of the pregnant women received medical care at most once from a doctor.

GATE CSE - 2022 MCQ (1-MARKS)

Q17. Given below are four statements.

Statement 1: All students are inquisitive.

Statement 2: Some students are inquisitive.

Statement 3: No student is inquisitive.

Statement 4: Some students are not inquisitive.

From the given four statements, find the two statements that **CANNOT BE TRUE** simultaneously, assuming that there is at least one student in the class.

(a) Statement 1 and Statement 3
(b) Statement 1 and Statement 2
(c) Statement 2 and Statement 4
(d) Statement 3 and Statement 4

GATE CSE - 2022 MCQ (2-MARKS)

Q18. In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates.

Which one of the following is the correct logical inference based on the information provided in the above passage?

(a) Equal number of boys and girls qualified
(b) Equal number of boys and girls appeared for the test
(c) The number of boys who appeared for the test is less than the number of girls who appeared
(d) The number of boys who qualified for the test is less than the number of girls who qualified

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2022 MCQ (2-MARKS)

Q19. A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball.

First ball is drawn. What is the probability of getting an orange ball in the next draw?

(a) 1/2
(b) 8/25
(c) 19/50
(d) 23/50

GATE CSE - 2022 MCQ (2-MARKS)

Q20. The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T, and U, but not necessarily in the same order. Consider the following statements:

- The line joining P and R is parallel to the line joining Q and S.
- P is placed on the side opposite to corner T.
- S and U cannot be placed on the same side.

Which one of the following statements is correct based on the above information?

(a) P cannot be placed at a corner
(b) S cannot be placed at a corner
(c) U cannot be placed at a mid-point
(d) R cannot be placed at a corner

GATE CSE - 2021 MCQ SET - 1(2-MARKS)

Q21. Given below are two statements 1 and 2, and two conclusions I and II

- Statement 1: All bacteria are microorganisms.
- Statement 2: All pathogens are microorganisms.
- Conclusion I: Some pathogens are bacteria.
- Conclusion II: All pathogens are not bacteria.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

(a) Only conclusion I is correct
(b) Only conclusion II is correct
(c) Either conclusion I or II is correct
(d) Neither conclusion I nor II is correct

GATE CSE - 2021 MCQ SET - 2(2-MARKS)

Q22. Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations.

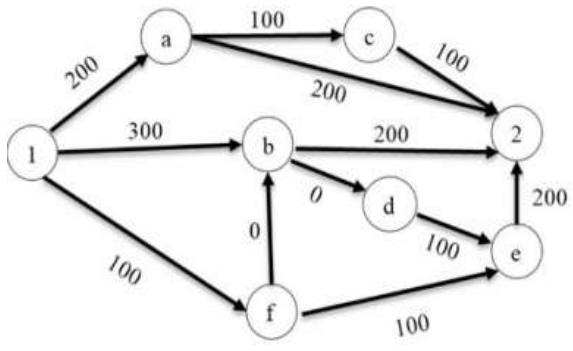
- Observation I: S is taller than R.
- Observation II: Q is the shortest of all.
- Observation III: U is taller than only one student.
- Observation IV: T is taller than S but is not the tallest.

The number of students that are taller than R is the same as the number of students shorter than _____.

(a) T
(b) R
(c) S
(d) P

GATE CSE - 2020 MCQ (1 -MARKS)

Q23. There are multiple routes to reach from node 1 to node 2, as shown in the network. The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200, and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2



Check Out Our GATE Courses & Secure Your Dream Rank!

(a) 1-a-c-2
(b) 1-f-b-2
(c) 1-b-2
(d) 1-f-e-2

GATE CSE - 2020 MCQ (2-MARKS)

Q24. If P = 3, R = 27, T = 243. What is Q + S?

(a) 84
(b) 72
(c) 90
(d) 108

GATE CSE - 2019 MCQ (2-MARKS)

Q25. The police arrested four criminals — P, Q, R and S. The criminals knew each other. They made the following statements:

P says "Q committed the crime."

Q says "S committed the crime."

R says "I did not do it."

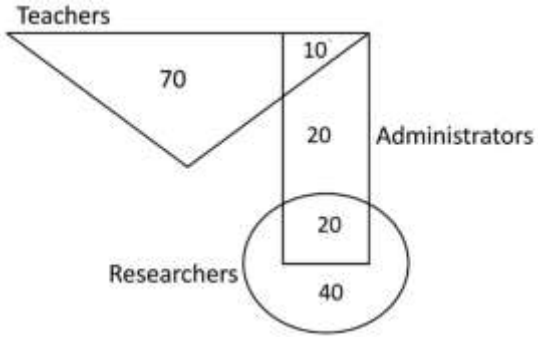
S says "What Q said about me is false."

Assume only one of the arrested four committed the crime and only one of the statements made above is true. Who committed the crime?

(a) P
(b) R
(c) S
(d) Q

GATE CSE - 2019 MCQ (2-MARKS)

Q26. In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage administrators shall be in the range of _____.



(a) 0 to 15
(b) 16 to 30
(c) 31 to 45
(d) 46 to 60

GATE CSE - 2018 MCQ (2-MARKS)

Q27. In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs 750 to each male senior citizen and Rs 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only 8/9 of the eligible men and 2/3 of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away?

(a) 1,50,000
(b) 2,00,000
(c) 1,75,000
(d) 1,51,000

GATE CSE - 2018 MCQ (2-MARKS)

Q28. If $pqr \neq 0$ and $p^{-x} = \frac{1}{q}, q^{-y} = \frac{1}{r}, r^{-z} = \frac{1}{p}$, what is the value of the product xyz?

(a) -1
(b) $\frac{1}{pqr}$
(c) 1
(d) pqr

Check Out Our GATE Courses & Secure Your Dream Rank!

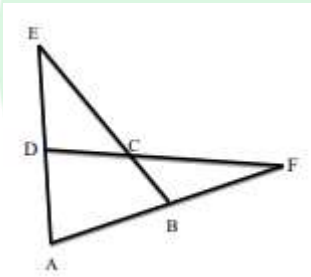
GATE CSE - 2018 MCQ (2-MARKS)

Q29. In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees?

(a) 2:3
(b) 1:1
(c) 3:2
(d) 2:1

GATE CSE - 2018 MCQ (2-MARKS)

Q30. In the figure below, ∠DEC + ∠BFC is equal to _____.



(a) ∠BCD − ∠BAD
(b) ∠BAD + ∠BCF
(c) ∠BAD + ∠BCD
(d) ∠CBA + ∠ADC

GATE CSE - 2017 MCQ SET - 1(2-MARKS)

Q31. Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which pair are seated opposite?

(a) Rahul and Murali
(b) Srinivas and Arul
(c) Srinivas and Murali
(d) Srinivas and Rahul

GATE CSE - 2017 MCQ SET - 1(1-MARKS)

Q32. Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is

(a) 2
(b) 3
(c) 4
(d) Cannot be determined

GATE CSE - 2017 MCQ SET - 1(2-MARKS)

Q33. The expression $\frac{(x+y)-|x-y|}{2}$

is equal to

(a) maximum of x and y
(b) minimum of x and y
(c) 1
(d) none

GATE CSE - 2017 MCQ SET - 2(1-MARKS)

Q34. There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?.

(a) V
(b) W
(c) X
(d) Y

GATE CSE - 2017 MCQ SET – 2 (1-MARKS)

Q35. There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labelled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box

Check Out Our GATE Courses & Secure Your Dream Rank!

would you open to determine the contents of all three boxes?

(a) The box labelled 'Apples'
(b) The box labelled 'Apples and Oranges'
(c) The box labelled 'Oranges'
(d) Cannot be determined

GATE CSE - 2016 MCQ SET – 1 (2-MARKS)

Q36. Indian currency notes show the denomination in at least seventeen languages. This is not an indication of the nation's diversity, nothing else is. Which inference is correct?

(a) India has exactly seventeen languages
(b) Linguistic pluralism is the only indicator of diversity
(c) Notes have space for all languages
(d) Linguistic pluralism strongly indicates diversity

GATE CSE - 2016 MCQ SET – 1 (2-MARKS)

Q37. Consider the following statements relating to the level of poker play of four players P, Q, R and S.

1. P always beats Q
II. R always beats S
III. S loses to P only sometimes.
IV. R always loses to Q

Which of the following can be logically inferred from the above statements?
i. P is likely to beat all the three other players
ii. S is the absolute worst player in the set
(a) (i). only
(b) (ii) only
(c) (i) and (ii) only'
(d) neither (i) nor (ii)

GATE CSE - 2016 MCQ SET – 2 (2-MARKS)

Q38. Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.
Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?
(i) The author believes that computers are not good for us.
(ii) Mobile computers and the internet are both intended inventions.
(a) (i) only
(b) (ii) only
(c) Both (i) and (ii)
(d) Neither (i) nor (ii)

GATE CSE - 2016 MCQ SET – 2 (2-MARKS)

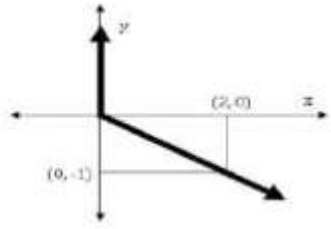
Q39. All hill-stations have a lake. Ooty has two lakes. Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

(i) Ooty is not a hill-station.
(ii) No hill-station can have more than one lake.

(a) (i) only
(b) (ii) only
(c) both (i) and (ii)
(d) neither (i) nor (ii)

GATE CSE - 2015 MCQ SET – 3 (2-MARKS)

Q40. Choose the most appropriate equation for the function drawn as a thick line, in the plot below.



Check Out Our GATE Courses & Secure Your Dream Rank!

(a) x=y - |y|
(b) x= -(y -|y|)
(c) x=y + |y|
(d) x= -(y + |y|)

GATE CSE - 2015 MCQ SET – 3 (2-MARKS)

Q41. The head of the newly formed government desires to appoint five of the six selected members P, Q, R, S, T and U to portfolios of Home, Power, Defense, Telecom, and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one.

Which is the valid distribution of portfolios?

(a) P-Home, Q-Power, R-Defense, S-Telecom, T-Finance
(b) R-Home, S-Power, P-Defense, Q-Telecom, T-Finance
(c) P-Home, Q-Power, T-Defense, S-Telecom, U-Finance
(d) Q-Home, U-Power, T-Defense, R-Telecom, P-Finance

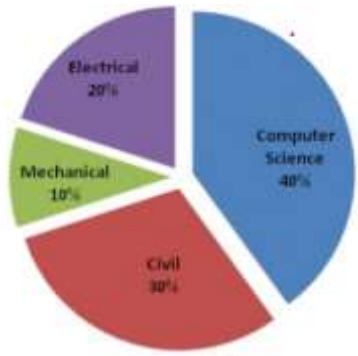
GATE CSE - 2015 MCQ SET – 3 (1-MARKS)

Q42. If ROAD is written as URDG, then SWAN should be written as:

(a) VXDQ
(b) VZDQ
(c) VZDP
(d) UXDQ

GATE CSE - 2015 MCQ SET – 1 (2-MARKS)

Q43. The pie chart shows the breakup of students from different departments. Electrical = 20%, Computer Science = 40%, Civil = 30%, Mechanical = 10%. Male to female ratio is 5:4 in each department. There are 40 males in Electrical Engineering. What is the difference between female students in Civil and female students in Mechanical?



GATE CSE - 2015 MCQ SET – 1 (1-MARKS)

Q44. Based on the given statements, select the most appropriate option to solve the given question.

If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor? Statements: I. Each step is 3/4-foot high. II. Each step is 1 foot wide.

(a) Statement I alone is sufficient, but statement II alone is not sufficient.
(b) Statement II alone is sufficient, but statement I alone is not sufficient.
(c) Both statements together are sufficient, but neither statement alone is sufficient.
(d) Statements I and II together are not sufficient.

GATE CSE - 2015 MCQ SET – 1 (2-MARKS)

Q45. The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option.
Statement: There has been a significant drop in the water level in the lakes supplying water to the city.
Course of action: I.
The water supply authority should impose a partial cut in supply to tackle the situation.
II. The government should appeal to all the residents through mass media for
minimal use of water.
III. The government should ban the water supply in lower areas.

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) Statements I and II follow
(b) Statements II and III follow
(c) Statements I and III follow
(d) All statements follow

GATE CSE - 2014 MCQ SET – 2 (1-MARKS)

Q46. Match the columns:

Column I	Column 2
1. Eradicate	P. misrepresent
2. distort	Q. soak completely
3. Saturate	R. use
4. utilize	S. desroy utterly

(a) 1:S, 2:P, 3:Q, 4:R
(b) 1:P, 2:Q, 3:R, 4:S
(c) 1:Q, 2:R, 3:S, 4:P
(d) 1:S, 2:P, 3:R, 4:Q

GATE CSE - 2014 NAT SET – 1 (2-MARKS)

Q47. In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter?

	Men	**Women**
Own vehicle - Car	40	34
Own vehicle - Scooter	30	20
Own vehicle - Both	60	46
Do not own vehicle	20	50

GATE CSE - 2012 NAT SET – 1 (2-MARKS)

Q48. Given the sequence of terms, AD CG FK JP, the next term is

(a) OV
(b) OW
(c) PV
(d) PW

GATE CSE - 2012 MCQ (2-MARKS)

Q49. Which of the following assertions are CORRECT?

P: Adding 7 to each entry in a list adds 7 to the mean of the list

Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list R: Doubling each entry in a list doubles the mean of the list

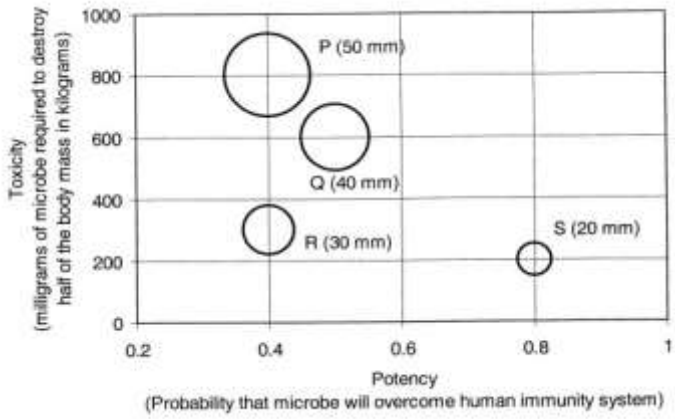
S: Doubling each entry in a list leaves the standard deviation of the list unchanged

(a) P, Q
(b) Q, R
(c) P, R
(d) R, S

GATE CSE - 2012 MCQ (2-MARKS)

Q50. P,Q,R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below :

Check Out Our GATE Courses & Secure Your Dream Rank!



A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

(a) P
(b) Q
(c) R
(d) S

GATE CSE - 2010 MCQ (2-MARKS)

Q51. Hari(H), Gita(G), Irfan(I) and Saira(S) are siblings (i.e., brothers and sisters). All were born on 1st January. The age difference between any two successive siblings (that is born one after another) is less than three years. Given the following facts:

i. Hari's age + Gita's age > Irfan's age + Saira's age

ii. The age difference between Gita and Saira is one year. However Gita is not the oldest and Saira is not the youngest.

iii. There are no twins.

In what order they were born (oldest first)?

(a) HSIG
(b) SGHI
(c) IGSH
(d) IHSG

Check Out Our GATE Courses & Secure Your Dream Rank!

ANSWER KEY

1. (c)	**2. (b)**	**3. (a)**
4. (c)	**5. (a)**	**6. (a)**
7. (b)	**8. (a)**	**9. (c)**
10. (a)	**11. (c)**	**12. (b)**
13. (b)	**14. (a)**	**15. (a)**
16. (a)	**17. (a)**	**18. (d)**
19. (d)	**20. (b)**	**21.(c,d)**
22. (c)	**23. (b)**	**24. (c)**
25.(b)	**26. (c)**	**27. (b)**
28. (c)	**29. (b)**	**30. (a)**
31. (c)	**32. (a)**	**33. (b)**
34. (a)	**35. (b)**	**36. (d)**
37. (d)	**38. (d)**	**39. (d)**
40. (b)	**41. (b)**	**42. (b)**
43. (32)	**44. (a)**	**45. (a)**
46. (a)	**47. (48%)**	**48. (a)**
49. (c)	**50. (d)**	**51. (b)**

HINTS & SOLUTIONS

1. ANSWER: - (C)

'When the teacher is in the room, all students stand silently.'
Negation of this statement:
A. 'If any student is not standing silently, then the teacher is not in the room'.
So, A is true.
B. 'When the teacher is in the room, all students are silent.'
This is true as per given statement. So, B true.
C. 'If all students are standing, then the teacher is in the room.'
This is not necessarily true as per the given statement when teacher is in the room then only all students are standing silently. Also when any one student is not standing silently then it implies that the teacher is not present in the room. So, C' doesn't directly imply from the given statement and not necessarily true.
D. 'When the teacher is in the room, all students are standing'
This is true as it directly implies from given statement. So, D is true.

2. ANSWER: - (B)

P: the passage in 3rd line mentions that Combinatorics can be used to sometimes model physical phenomena.
using it to model all physical phenomena is a strong statement which is not implied by the passsge.

Q: can be directly inferred form the given passage.

So, the correct answer is B.

3. ANSWER: - (A)

Neeraj Chopra
Got gold with 2^{nd} throw
No one else had medal-level throws in first 2 rounds
So best throws of others are after Round 2
Vesely
After 4 rounds → 2nd position

Could NOT improve after that
Best throw is within first 4 rounds
Last Round (6^{th})
Vesely = foul
Not possible
Jakub
Best in 4^{th} round
So in Round 4, Jakub throws better than Vesely
Vesely's best ≠ Round 4
Narrow Down:
Not Round 6
Not after Round 4
Not Round 4
Left: Round 3
So, the correct answer is A.

4. ANSWER: - (C)

Let P and Q be two statements such that:
P: It is raining.
Q: Peacocks Dance.
Converting given statement in terms of P and Q provides:
$P = Q$
$\because P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$
$\neg Q \Longrightarrow \neg P \equiv$ "When peacocks are not dancing, it is not raining".

Check Out Our GATE Courses & Secure Your Dream Rank!

5. **ANSWER: - (A)**
This preposition is a relation of needed : when :: needed : when .
Here water is needed when thirsty, and food when hunger.

So, the correct answer is A.

6. **ANSWER: - (A)**

Given a 3×3 arrangement of tiles in Panel I, one tile is missing. Each tile has two halves (top and bottom) containing dot patterns (filled and hollow).

Step 1: Observe row pattern

Moving from left to right in each row, the arrangement of filled (black) dots increases or shifts in a consistent manner.

Step 2: Observe column pattern

Moving from top to bottom in each column, the dot pattern transforms systematically. The top and bottom halves follow a fixed progression.

Step 3: Analyze the missing tile (bottom-right position)

• Its top half must follow the pattern of the third column.

• Its bottom half must continue the pattern of the third row.

Step 4: Match with given options

Only option (i) satisfies both conditions:

• The top half matches the required column progression.

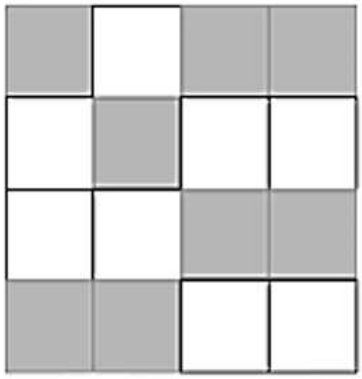
• The bottom half correctly continues the row pattern.

• The overall arrangement remains consistent with the grid logic.

Final Answer:

(i)

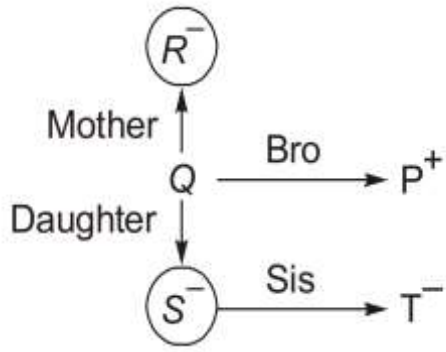
7. **ANSWER: - (B)**



(ii)

.

8. **ANSWER: - (A)**



So, the correct answer is A.

9. **ANSWER: - (C)**

- Library → top-left
- Canteen → top-right
- Physics Lab → left, slightly below library
- Hospital → right, slightly below canteen
- Chemistry Lab → bottom-right
- Classrooms → bottom-left

Check options:

(a) Library northwest of canteen
Wrong — Library is directly west, not northwest.

(b) Hospital east of chemistry lab
Wrong — Hospital is north of chemistry lab.

(c) Chemistry lab southeast of physics lab
Correct — from Physics Lab (left-middle) to Chemistry Lab (bottom-right) is down + right = southeast.

Check Out Our GATE Courses & Secure Your Dream Rank!

(d) Classrooms and canteen next to each other
Wrong — they are diagonally far apart.

10. ANSWER: - (A)

From the given values:

- X(0)=20
- X(1)=X(0)+I(1)=20+(−4)=16
- X(2)=X(1)+I(2)=16+10=26
- X(3)=X(2)+I(3)=26+15=41

This follows: X(i)=X(i−1)+I(i)

Now, let's verify Y(i):

From the given values:

- Y(0)=20
- Y(1)=Y(0)×I(1)=20×(−4)=−80
- Y(2)=Y(1)×I(2)=−80×10=−800
- Y(3)=Y(2)×I(3)=−800×15=−12000

This follows: Y(i)=Y(i−1)×I(i)

11. ANSWER: - (C)

In Option 1 (Bridges on P, Q, and T) and Option 2 (Bridges on P, Q, S, and T): Z4 may not be reachable.

Option 3 (Bridges on Q, R, T, and V) and Option 4 (Bridges on P, Q, S, U, and V) ensures all zones are connected but option 3 contains least number of bridges.

Hence, option C is correct.

12. ANSWER: - (B)

For IMAGE:

- I → J (next letter of I)
- M → N (next letter of M)
- A → B (next letter of A)
- G → H (next letter of G)
- E → F (next letter of E)

Then, the letters are placed in reverse order: FHBNJ

For FIELD:

- F → G
- I → J
- E → F
- L → M
- D → E

Then, the letters are placed in reverse order: EMFJG

Now, for beach

- B → C
- E → F
- A → B
- C → D
- H → I

Then, placing the letters in reverse order: IDBFC

13. ANSWER: - (B)

9-6 = 3
14-9 = 5 (3+2)
X-14 = y (3+2+2)
30-y = y+2.

As per the given sequence, y = 3+2+2 = 7.
Therefore, x-14=7.
And, x = 14+7=21.
Hence, the correct option is B.

14. ANSWER: - (A)

As per the given information in the question, we can draw the following.

1	X	4	3
X	5	5	4
3	X	6	X
2	4	3	2

Hence, the sum of the fourth row = 2 +4 +3 +2 = 11.
Therefore, the correct option is A.

15. ANSWER: - (A)

According to the given data we have to find the value of F_1 using the recurrence relation

Check Out Our GATE Courses & Secure Your Dream Rank!

$F_{n+1}=F_n+F_{n-1}$

While putting the value of F_6 and F_7 we will get the value of F_5 which is 23

$F_{n+1}=F_n+F_{n-1}$ so after this we will look for F_4 which will 14

And by following F_4 and F_5 we will calculate the value of F_3 which is 9

and so on value of F_1 is 4 .

16. ANSWER: - (A)

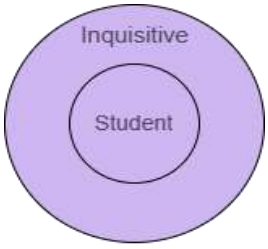
Total women = 100
Pregnant women received medical care at least.
Once before giving birth = 90% of 100 = 90
Out of 90 women 60% received medical care from doctors = 60% of 90 = 54

While 40% received from other health care providers = 40% of 90 =36
Here, more than half of pregnant women (54) received medical care at least once from a doctor.

17. ANSWER: - (A)

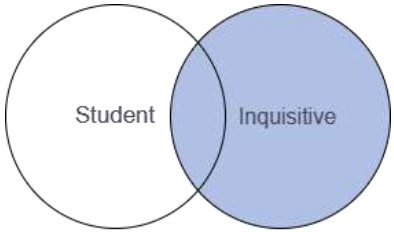
The following statements can be easily understood with the Venn diagram

Statement 1: All students are inquisitive.



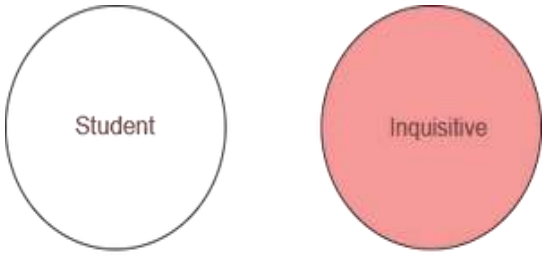
Statement 1

Statement 2: Some students are inquisitive.



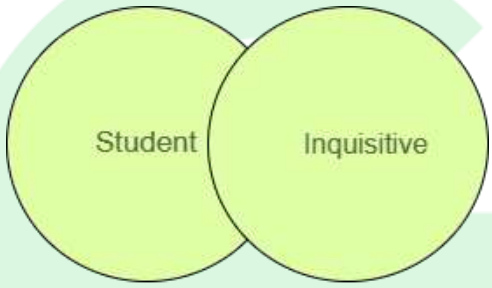
Statement 1

Statement 3: No student is inquisitive.



Statement 3

Statement 4: Some students are not inquisitive.



Statement 4

From the above diagram, it is clear that Statement 1 and Statement 3 are conflicting statements and both cannot be true at a time.

18. ANSWER: - (D)

Since the pool of qualified candidates consists entirely of boys and girls:

- Girls Qualified: 60% of the qualified pool
- Boys Qualified: $100\% - 60\% = 40\%$ of the qualified pool

Comparing the two percentages within this same population:

$$\mathbf{40}\% \text{ (Boys)} < \mathbf{60}\% \text{ (Girls)}$$

Therefore, the number of boys who qualified is guaranteed to be less than the number of girls who qualified, regardless of how many total students appeared or passed.

Why other options are incorrect:

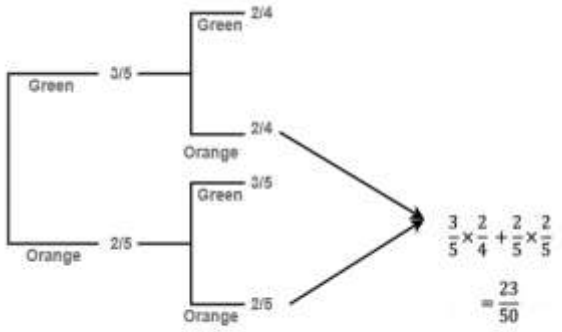
- (a) Incorrect because the qualified percentages are unequal (40% vs 60%).

Check Out Our GATE Courses & Secure Your Dream Rank!

- (b) & (c) Incorrect because boys made up 65% of the appeared candidates, meaning significantly more boys appeared than girls (35%).

19. ANSWER: - (D)

We have a box with 5 balls in total, three of which are green and two of which are orange. The chance of obtaining an orange ball in the second draw must be calculated. So, for the first ball drawn, there will be two scenarios. Green or orange might be used for the first ball.



Case 1: If the first ball drawn is green, we will only have four balls in the box instead of five if we get the orange ball in the second draw.

3/5 * 2/4 = 3/10

Case 2: If the first ball drawn is orange, we will be left with just 5 balls in the second draw since the orange ball will be replaced with a new orange ball.

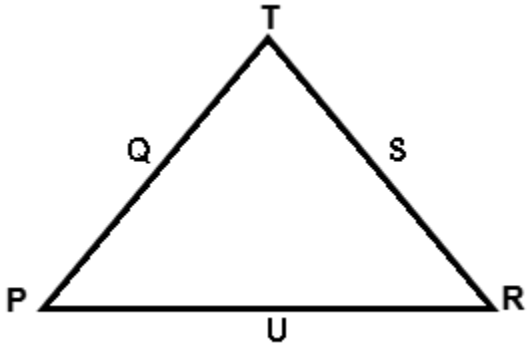
2/5 * 25 = 4/25

Now, the probability of getting the orange ball in the 2nd draw will be

3/10 + 4/25 = (15+8)/50 = 23/50

20. ANSWER: - (B)

We need to draw the diagram based on the given conditions in the question.



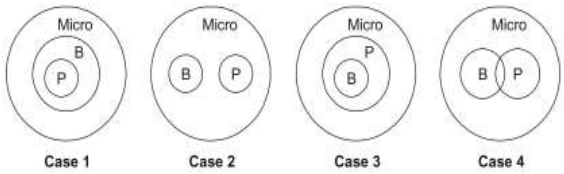
Now we can observe the second stated condition, which states that P is on the side opposite to corner T. This means that if we consider T at any corner point, P will be on the other side.

Similarly, if we examine the first condition, which states that P and R are parallel to Q and S, we can see that S cannot be on any of the corners when analyzed with the use of a diagram.

Also, we need to check the third condition before confirming option B.

21. ANSWER: - (D)

We have to find counter example for both given Conclusion, if possible.



- Case-1 is counter example of Conclusion II, because of all pathogens are bacteria in this diagram. Conclusion I would be true in this case, because all so also some. (So, Conclusion I is true and II is false.).
- Case-2 is counter example of Conclusion I, because no pathogens are bacteria. Conclusion II would be true in this case, because all so also some. (So, Conclusion II is false and II is true.).
- Case-3: Conclusion I is true, and Conclusion II is false as some pathogens are *already* bacteria.
- Case-4: Conclusion I is true, and Conclusion II is false as some pathogens are *already* bacteria.

So, we found counter example for both Conclusion.

Check Out Our GATE Courses & Secure Your Dream Rank!

22. ANSWER: - (C)

From the given observations, the relative order of heights can be deduced as follows:

1. Q is the shortest (Q=1st).
2. U>Q and U is taller than only one student (U=2nd).
3. R must come next (R=3rd), as S>R and S is not shorter than U.
4. S>R (S=4th).
5. T>S, but T is not the tallest (T=5th).
6. P must be the tallest (P=6th).

Thus, the order of heights is: Q<U<R<S<T<P

So,

- Students taller than R: S, T, P (3 students).
- Students shorter than S: Q, U, R (3 students).

Therefore, the number of students taller than R is the same as the number of students shorter than S.

23. ANSWER: - (B)

Option (A): 1 → a → c → 2

Edge cost = 200 + 100 + 100 = 400

Toll cost = 200 (a) + 100 (c) = 300

Total = 700

Option (B): 1 → f → b → 2

Edge cost = 100 + 0 + 200 = 300

Toll cost = 100 (f) + 100 (b) = 200

Total = 500

Option (C): 1 → b → 2

Edge cost = 300 + 200 = 500

Toll cost = 100 (b)

Total = 600

Option (D): 1 → f → e → 2

Edge cost = 100 + 100 + 200 = 400

Toll cost = 100 (f) + 200 (e) = 300

Total = 700

Final Answer: (B) 1-f-b-2 (Minimum cost = 500)

24. ANSWER: - (C)

P = 3

R = 27

T = 243

Step 1: Identify the pattern

$3 = 3^1$

$27 = 3^3$

$243 = 3^5$

So letters are mapped as powers of 3.

Step 2: Check positions

P → 16th letter → exponent = 1

R → 18th letter → exponent = 3

T → 20th letter → exponent = 5

So pattern:

Exponent = (Position − 15)

Step 3: Find Q and S

Q → position 17
Exponent = 17 − 15 = 2
$Q = 3^2 = 9$
S → position 19
Exponent = 19 − 15 = 4
$S = 3^4 = 81$
Step 4: Final answer
Q + S = 9 + 81 = 90
Final Answer: (C) 90

Check Out Our GATE Courses & Secure Your Dream Rank!

25. ANSWER: - (B)

Option B. R

Assuming S is saying the truth.

P says Q committed the crime (FALSE) means Q has not Q says S committed the crime(FALSE) means S has not R says I did not (FALSE) means R has committed the crime.

26. ANSWER: - (C)

- No of administrator $= 10 + 20 + 20 = 50$
- Total no of person $= 80 + 20 + 20 + 40 = 160$
- Required $\% = (50/160) * 100 = 31.25\%$

$\therefore$ Option C. 31 to 45 is the correct answer.

27. ANSWER: - (B)

Let men = x, women = y

x + y = 300

Money = 750 × (8/9)x + 1000 × (2/3)y

= (2000/3)x + (2000/3)y

= (2000/3)(300) = 200000

28. ANSWER: - (C)

From relations:

$p^{-x} = q^{-1} \rightarrow x = r$

$q^{-y} = r^{-1} \rightarrow y = r$

$r^{-z} = p^{-1} \rightarrow z = r$

Thus xyz = 1

29. ANSWER: - (B)

Assume total = 100

Males = 60, females = 40

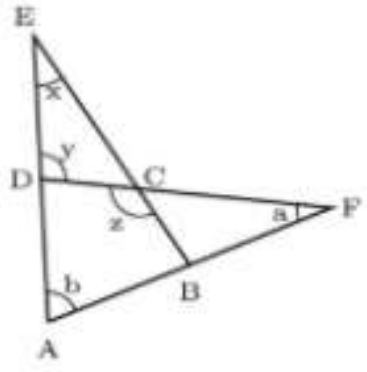
All females attended = 40

Total attended = 80

Males attended = 40

Ratio = 40 : 40 = 1:1

30. ANSWER: - (A)



$$\angle z = \angle x + \angle y$$

This is a triangle property that tells us that "the exterior angle of a triangle is the sum of the two opposite interior angle."

Similarly.

$$\angle y = \angle a + \angle b$$

By substituting, we get,

$$\angle Z = \angle x + \angle a + \angle b.$$

Now $\angle Z = \angle BCD$

$$\angle b = \angle BAD$$
$$\angle BCD - \angle BAD = \angle Z - \angle b$$
$$= \angle x + \angle a + \angle b - \angle b$$
$$= \angle x + \angle a$$
$$= \angle DEC + \angle BFC$$

So, option A is the answer.

31. ANSWER: - (C)

Arul and Rahul will sit opposite to each other.

Murali and Srinivas will sit opposite to each other in the square table.

32. ANSWER: - (A)

If we take 2 and try to assign all the conditions given it is successfull but for 3 and 4 we can not impose all conditions given so 2 is the answer

A is correct answer here

Check Out Our GATE Courses & Secure Your Dream Rank!

33. ANSWER: - (B)

Case 1: When $x \geq y$
If x is greater than or equal to y, the absolute value term opens positively $(|x-y| = x-y)$:

$$\frac{(x+y)-(x-y)}{2} = \frac{x+y-x+y}{2} = \frac{2y}{2} = y$$

Case 2: When $x < y$
If x is less than y, the absolute value term flips signs to stay positive $(|x-y| = -(x-y) = y-x)$:

$$\frac{(x+y)-(y-x)}{2} = \frac{x+y-y+x}{2} = \frac{2x}{2} = x$$

In both scenarios, the expression simplifies directly to the smaller of the two variables.

34. ANSWER: - (A)

Given order: (West on left)

$$\text{VW}, \text{XZ}, \text{ZV}, \text{WY}$$

Considering them together:

$$\text{XZ}, \text{ZV} \Longrightarrow \text{XZV}$$
$$\text{VW}, \text{WY} \Longrightarrow \text{VWY}$$
$$\text{XZV}, \text{VWY} \Longrightarrow \text{XZVWY}$$

So, V is the building in the middle.

35. ANSWER: - (B)

All labels are incorrect. So the box labelled "Apples and Oranges" cannot contain both.

Case:
- If you pick from this box and get an apple → it must be "Apples"
- If you get an orange → it must be "Oranges"
Then remaining boxes can be correctly assigned using elimination.

Hence, open the box labelled "Apples and Oranges".

36. ANSWER: - (D)

Statement emphasizes language diversity reflects national diversity. Hence D.

37. ANSWER: - (D)

This option is the most logical inference because the presence of multiple languages on the currency notes reflects the linguistic diversity of the nation. The other options are either too specific, not supported by the statement, or not a logical inference based on the information provided. Remember, the statement highlights the diversity of languages as an indication of the nation's diversity, not as the only indicator or that it is related to the physical space on the currency notes.

38. ANSWER: - (D)

"Many believe that the internet itself is an unintended consequence of the original invention. So (ii) statement does not follow from the passage."
The author has no where said that computers are bad, authoring is talking about the way computers are being used today and the author questions this way. So, (i) statement does not follow.
Hence, option (D) is the answer.

39. ANSWER: - (D)

All hill stations have a lake $\Rightarrow \forall x(h(x) \rightarrow (\exists y, l(y) \wedge has(x, y)))$
Ooty has two lakes $\Rightarrow \exists x(o(x) \wedge \exists y, z(has(x, y, z) \wedge l(y) \wedge l(z) \wedge (z \neq y))$
Here.

- $h(x) \rightarrow$ x is hill station
- $l(x) \rightarrow$ x is lake
- $has(x, y) \rightarrow$ x has y
- $has\ (x, y, z) \rightarrow$ x has y, z
- $o(x) \rightarrow$ x is Ooty
  i. Ooty is not a hill station: We can not derive this above arguments, Ooty has two lakes already, if Ooty had 0 lakes only then this can become true.
  ii. No hill station can have more than one lake.

All arguments here are saying are if we have hill station, it can have lake. It is nowhere told that how

Check Out Our GATE Courses & Secure Your Dream Rank!

many lakes it has. So, this is false.
Answer: (D) neither (i) nor (ii)

40. ANSWER: - (B)
When y is -1, x is 2 .

When y is positive x is 0 .
So, $x = -(y - |y|)$
Correct Answer: B

41. ANSWER: - (B)
" U does not want any portfolio if S gets one of the five"
So, S and U cannot come together. Option C eliminated.
" R wants either Home or Finance or no portfolio"
So, options A and D eliminated.
So, answer is B.
Just to confirm:
Q says that if S gets Power or Telecom, then she must get the other one
In B, S gets Power and Q gets Telecom
"T insists on a portfolio if P gets one"
In B, T is getting a portfolio.

42. ANSWER: - (B)
R + 3 = U
O + 3 = R
A + 3 = D
D + 3 = G
Applying same logic to SWAN:
S + 3 = V
W + 3 = Z
A + 3 = D
N + 3 = Q

43. ANSWER: - (32)
Total in Electrical = 40 x 9/5 = 72. Total students = 72/0.20 = 360. Civil = 360 x 0.30 = 108, females = 108 x 4/9 = 48. Mechanical = 360 x 0.10 = 36, females = 36 x 4/9 = 16. Difference = 48 - 16 = 32.

44. ANSWER: - (A)
To find the number of steps, we need the height of each step and total height. Statement I gives step height = 3/4 foot, so steps = 9 ÷ 3/4 = 12.
Statement II gives width which is irrelevant for counting steps. So Statement I alone is sufficient.

45. ANSWER: - (A)
Statement talks about a **water shortage**, so actions should aim at **conservation and fair distribution**.

I. Imposing a partial cut in supply is a reasonable administrative step to manage limited water → follows.

II. Appealing to residents to use water minimally helps reduce demand → follows.

III. Banning water supply in lower areas is arbitrary and unfair, with no logical link to solving the shortage → does not follow.

Therefore, only I and II follow.

46. ANSWER: - (A)

"eradicate" means "destroy utterly" "distort" matches with "misrepresent" "saturate" matches with "soak completely" "utilize" matches with "use"

47. ANSWER: - (48)

Not having scooter:
From Men = 40 (car owners) + 20 (nothing) = 60
From Women = 34 (car owners) + 50 (nothing) = 84
Total not having scooter = 60 + 84 = 144
Percentage = 144 / 300 = 0.48 = 48%

48. ANSWER: - (A)

The gap between first letters: A, C, F, J, O increases by 2, 3, 4, 5. The gap between second letters: D, G, K, P, V increases by 3, 4, 5, 6. So the next term is OV.

49. ANSWER: - (C)

Suppose we double each entry of a list

Initial Mean $(M_I) = \frac{\sum_{i=1}^{n} z_i}{n}$

New Mean $(M_N) = \frac{\sum_{i=1}^{n} 2 \times x_i}{n}$

$$= \frac{2}{n} \sum_{i=1}^{n} x_i$$

Check Out Our GATE Courses & Secure Your Dream Rank!

So, when each entry in the list is doubled, mean also gets doubled.

Standard Deviation $\sigma_I = \sqrt{\frac{1}{N}\sum_{i=1}^{n} (M_I - x_i)^2}$

New Standard Deviation $\sigma_N = \sqrt{\frac{1}{N}\sum_{i=1}^{n} (M_N - 2 \times x_i)^2}$

$$= \sqrt{\frac{1}{N}\sum_{i=1}^{n} \left(2 \times (M_I - x_i)\right)^2}$$

$$= 2\sigma_I$$

So, when each entry is doubled, standard deviation also gets doubled.

When we add a constant to each element of the list, it gets added to the mean as well. This can be seen from the formula of mean.

When we add a constant to each element of the list, the standard deviation (or variance) remains unchanged. This is because, the mean also gets added by the same constant and hence the deviation from the mean remains the same for each element.

So, here P and R are correct.
Correct Answer: C

50. ANSWER: - (D)

As per the question, it is quite clear that the danger of a microbe to human being will be directly proportional to potency and growth. At the same time it is inversely proportional to toxicity, defined as(more dangerous will a microbe be if lesser of its milligram is required).

So, Level Of Danger (D) $\propto$ Growth (G) $\propto$ Potency (P) $\frac{1}{\propto}$ Toxicity (T)

$$D = \frac{KGP}{T}$$

where K is constant of proportionality.
So level of danger will be maximum for S.
Given by,

$$D_S = \frac{0.8 \times \pi(10)^2}{200}$$
$$= 1.256$$

Similar calculations for D_P, D_Q, D_R can be done. Which will consequently lead to D_S being the most dangerous and hence will be targeted first.

51. ANSWER: - (B)

Given that,

•H+G> S+I

• No twins,

• And either G = S +1 or S = G+1

• G is not the oldest and S is not the youngest.

Lets analyze each case,

Case 1:

G=S+1,

It implies, H > I – 1, Since twins are not possible H > I

Case 2:

G=S-1,

It implies H >I+1

Therefore, both cases imply H is older than I, also G and S must be placed successively.

Only option B satisfies these implications.

Check Out Our GATE Courses & Secure Your Dream Rank!

3 VERBAL APTITUDE

GATE CSE - 2026 MCQ SET – 1 (1-MARKS)

Q1. The antonym of the word protagonist is _____.

(a) agnostic
(b) antagonist
(c) arsonist
(d) anarchist

GATE CSE - 2026 MCQ SET – 2 (1-MARKS)

Q2. Expedite, Hasten, Hurry, _____
Fill the blank by choosing a word with a meaning similar to that of the words given above.

(a) Accelerate
(b) Retard
(c) Provide
(d) Disable

GATE CSE - 2025 MCQ SET – 1 (1-MARKS)

Q3. Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select the option with the correct sequence of articles to fill in the blanks.

(a) a; a; an
(b) the; an; a
(c) a; an; a
(d) an; an; a

GATE CSE - 2025 MCQ SET – 1 (1-MARKS)

Q4. The CEO's decision to downsize the workforce was considered myopic because it sacrificed long-term stability to accommodate short-term gains. Select the most appropriate option that can replace the word "myopic" without changing the meaning of the sentence.

(a) visionary
(b) shortsighted
(c) progressive
(d) innovative

GATE CSE - 2025 MCQ SET – 1 (2-MARKS)

Q5. "I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past."

(From G.K. Chesterton's "A Piece of Chalk")

Based only on the information provided in the above passage, which one of the following statements is true?

(a) The author of the passage carries a mirror in his pocket to reflect upon things.
(b) The author of the passage had decided to write a poem on epics.
(c) The pocket-knife is described as the infant of the sword.
(d) Epics are described as too inconvenient to write.

GATE CSE - 2025 MCQ SET – 2 (1-MARKS)

Q6. Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered.

Select the most appropriate option to complete the above sentence.

Check Out Our GATE Courses & Secure Your Dream Rank!

(a) ambivalence
(b) satisfaction
(c) resolve
(d) revolve

GATE CSE - 2025 MCQ SET – 2 (1-MARKS)

Q7. Bird : Nest :: Bee : _____

Select the correct option to complete the analogy.

(a) Kennel
(b) Hammock
(c) Hive
(d) Lair

GATE CSE - 2025 MCQ SET – 2 (2-MARKS)

Q8. Based only on the conversation below, identify the logically correct inference:

"Even if I had known that you were in the hospital, I would not have gone there to see you", Ramya told Josephine.

(a) Ramya knew that Josephine was in the hospital.
(b) Ramya did not know that Josephine was in the hospital.
(c) Ramya and Josephine were once close friends; but now, they are not.
(d) Josephine was in the hospital due to an injury to her leg.

GATE CSE - 2024 MCQ SET – 1 (1-MARKS)

Q9. If '→' denotes increasing order of intensity, then the meaning of the words [dry → arid → parched] is analogous to [diet → fast → _____]. Which one of the given options is appropriate to fill the blank?

(a) starve
(b) reject
(c) feast
(d) deny

GATE CSE - 2024 MCQ SET – 1 (2-MARKS)

Q10. In the given text, the blanks are numbered (i)–(iv). Select the best match for all the blanks.

Steve was advised to keep his head (i) before heading (ii) to bat; for, while he had a head (iii) batting, he could only do so with a cool head (iv) his shoulders.

(a) (i) down (ii) down (iii) on (iv) for
(b) (i) on (ii) down (iii) for (iv) on
(c) (i) down (ii) out (iii) for (iv) on
(d) (i) on (ii) out (iii) on (iv) for

GATE CSE - 2024 MCQ SET – 2 (1-MARKS)

Q11. If '→' denotes increasing order of intensity, then the meaning of the words [walk → jog → sprint] is analogous to [bothered → _____ → daunted]. Which one of the given options is appropriate to fill the blank?

(a) phased
(b) phrased
(c) fazed
(d) fused

GATE CSE - 2024 MCQ SET – 2 (2-MARKS)

Q12. Sequence the following sentences in a coherent passage.

P: This fortuitous geological event generated a colossal amount of energy and heat that resulted in the rocks rising to an average height of 4 km across the contact zone.

Q: Thus, the geophysicists tend to think of the Himalayas as an active geological event rather than as a static geological feature.

R: The natural process of the cooling of this massive edifice absorbed large quantities of atmospheric carbon dioxide, altering the earth's atmosphere and making it better suited for life.

Check Out Our GATE Courses & Secure Your Dream Rank!

S: Many millennia ago, a breakaway chunk of bedrock from the Antarctic Plate collided with the massive Eurasian Plate.

(a) QPSR
(b) QSPR
(c) SPRQ
(d) SRPQ

GATE CSE - 2023 MCQ (1-MARKS)

Q13. We reached the station late, and _____ missed the train.

(a) Near
(b) Nearly
(c) Utterly
(d) Mostly

GATE CSE - 2023 MCQ (1-MARKS)

Q14. Kind : _____ : : Often : Frequently

(By word meaning)

(a) Mean
(b) Type
(c) Cruel
(d) Kindly

GATE CSE - 2023 MCQ (2-MARKS)

Q15. The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work.

Based only on the information provided above, which one of the following statements can be logically inferred with certainty?

(a) All the working population of Zombieland will henceforth ride bicycles to work.
(b) Riding bicycles will ensure that all of the working population of Zombieland is free of health issues.
(c) The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory.
(d) The Government of Zombieland believes that riding bicycles is a form of physical exercise.

GATE CSE - 2023 MCQ (1-MARKS)

Q16. Which of the following sentence sequence create a coherent narrative?

(i) Once on the terrace, on her way to her small room in the corner, she notices the man right away.

(ii) She begins to pant by the time she has climbed all the stairs.

(iii) Mina has bought vegetables and rice at the market, so her bags are heavy.

(iv) He was leaning against the parapet, watching the traffic below.

(a) (i), (ii), (iv), (iii)
(b) (ii), (iii), (i), (iv)
(c) (iv), (ii), (i), (iii)
(d) (iii), (ii), (i), (iv)

GATE CSE - 2022 MCQ (1-MARKS)

Q17. The _____ is too high for it to be considered _____.

(a) fair / fare
(b) faer / fair
(c) fare / fare
(d) fare / fair

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2022 MCQ (2-MARKS)

Q18. Some people believe that "what gets measured, improves". Some others believe that "what gets measured, gets gamed". One possible reason for the difference in beliefs is the work culture in organizations. In organizations with good work culture, metrics help improve outcomes. However, the same metrics are counterproductive in organizations with the poor work cultures.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

(a) Metrics are useful in organizations with poor work culture
(b) Metrics are useful in organizations with good work culture
(c) Metrics are always counterproductive in organizations with good work culture
(d) Metrics are never useful in organizations with good work culture

GATE CSE - 2021 MCQ SET – 1 (1-MARKS)

Q19. Consider the following sentences:

- **(i)** Everybody in the class is prepared for the exam.
- **(ii)** Babu invited Danish to his home because he enjoys playing chess.

Which of the following is the CORRECT observation about the above two sentences?

(a) (i) is grammatically correct and (ii) is unambiguous
(b) (i) is grammatically incorrect and (ii) is unambiguous
(c) (i) is grammatically correct and (ii) is ambiguous
(d) (i) is grammatically incorrect and (ii) is ambiguous

GATE CSE - 2021 MCQ SET – 1 (1-MARKS)

Q20. _____ is to surgery as writer is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

(a) Plan, outline
(b) Hospital, library
(c) Doctor, book
(d) Medicine, grammar

GATE CSE - 2021 MCQ SET – 1 (2-MARKS)

Q21. Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality. Which one of the following statements summarizes the passage?

(a) The proposed AOM addresses the core problems that cause obesity
(b) If obesity reduces, poverty will naturally reduce, since obesity causes poverty
(c) AOM are addressing the core problems and are likely to succeed
(d) AOM are addressing the problem superficially

GATE CSE - 2021 MCQ SET – 2 (1-MARKS)

Q22. Gauri said that she can play the keyboard _____ her sister.

(a) as well as
(b) as better as
(c) as nicest as
(d) as worse as

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2021 MCQ SET – 2 (1-MARKS)

Q23. Pen : Write :: Knife : _____

Which one of the following options maintains a similar logical relation in the above?

(a) Vegetables
(b) Sharp
(c) Cut
(d) Blunt

GATE CSE - 2021 MCQ SET – 2 (2-MARKS)

Q24. Listening to music during exercise improves performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive.

Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music.

Which one of the following statements is the CORRECT inference of the above passage?

(a) Listening to music has no effect on learning and a positive effect on physical exercise
(b) Listening to music has a clear positive effect both in physical exercise and on learning
(c) Listening to music has a clear positive effect on physical exercise. Music has a positive effect on learning only in some students
(d) Listening to music has a clear positive effect on learning in all students. Music has a positive effect only in some students who exercise

GATE CSE - 2020 MCQ (1-MARKS)

Q25. Raman is confident of speaking English ____ six months as he has been practising regularly ____ the last three weeks.

(a) during, for
(b) for, since
(c) for, in
(d) within, for

GATE CSE - 2020 MCQ (1-MARKS)

Q26. His knowledge of the subject was excellent but his classroom performance was _____.

(a) extremely poor
(b) good
(c) desirable
(d) praiseworthy

GATE CSE - 2020 MCQ (1-MARKS)

Q27. Select the correct that fits the analogy:

Cook : Cooking :: Fly : _____

(a) Flyer
(b) Flying
(c) Flew
(d) Flight

GATE CSE - 2020 MCQ (1-MARKS)

Q28. The dawn of the 21st century witnessed the melting glaciers oscillating between giving too much and too little to billions of people who depend on them for fresh water. The UN climate report estimates that without deep cuts to man-made emissions, at least 30% of the northern hemisphere's surface permafrost could melt by the end of the century. Given this situation of imminent global exodus of billions of people displaced by rising seas, nation-states need to rethink their carbon footprint for political concerns, if not for environmental ones.Which one of the following statements can be inferred from the given passage?

(a) Nation-states do not have environmental concerns.
(b) Nation-states are responsible for providing fresh water to billions of people.
(c) Billions of people are responsible for man-made emissions.
(d) Billions of people are affected by melting glaciers.

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2020 MCQ (2-MARKS)

Q29. Goods and Services Tax (GST) is an indirect tax introduced in India in 2017 that is imposed on the supply of goods and services, and it subsumes all indirect taxes except few. It is a destination-based tax imposed on goods and services used, and it is not imposed at the point of origin from where goods come. GST also has a few components specific to state governments, central government and Union Territories (UTs).Which one of the following statements can be inferred from the given passage?

(a) GST is imposed on the production of goods and services.
(b) GST does not have a component specific to UT.
(c) GST is imposed at the point of usage of goods and services.
(d) GST includes all indirect taxes.

GATE CSE - 2019 MCQ (1-MARKS)

Q30. The expenditure on the project _____ as follows: equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

(a) break down
(b) break
(c) breaks down
(d) breaks

GATE CSE - 2019 MCQ (1-MARKS)

Q31. The search engine's business model _____ around the fulcrum of trust.

(a) revolves
(b) plays
(c) sinks
(d) bursts

GATE CSE - 2019 MCQ (1-MARKS)

Q32. A court is to a judge as _____ is to a teacher.

(a) a student
(b) a punishment
(c) a syllabus
(d) a school

GATE CSE - 2019 MCQ (1-MARKS)

Q33. A recent High Court judgement has sought to dispel the idea of begging as a disease — which leads to its stigmatization and criminalization — and to regard it as a symptom. The underlying disease is the failure of the state to protect citizens who fall through the social security net.

Which one of the following statements can be inferred from the given passage?

(a) Beggars are lazy people who beg because they are unwilling to work
(b) Beggars are created because of lack of social welfare schemes
(c) Begging is an offence that has to be dealt with firmly
(d) Begging has to be banned because it adversely affects the welfare of the state

GATE CSE - 2018 MCQ (1-MARKS)

Q34. "From where are they bringing their books? _____ bringing _____ books from _____."

The words that best fill the blanks in the above sentence are

(a) Their, they're, there
(b) They're, their, there
(c) There, their, they're
(d) They're, there, there

GATE CSE - 2018 MCQ (1-MARKS)

Q35. "A _____ investigation can sometimes yield new facts, but typically organized ones are more successful."

(a) meandering
(b) timely
(c) consistent
(d) systematic

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2017 MCQ SET – 1 (1-MARKS)

Q36. After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.

(a) was wishing
(b) is wishing
(c) wished
(d) had wished

GATE CSE - 2017 MCQ SET – 1 (1-MARKS)

Q37. Research in the workplace reveals that people work for many reasons _____.

(a) money beside
(b) beside money
(c) money besides
(d) besides money

GATE CSE - 2017 MCQ SET – 1 (2-MARKS)

Q38. "The hold of the nationalist imagination on our colonial past is such that anything inadequately or improperly nationalist is just not history."Which of the following statements best reflects the author's opinion?

(a) Nationalists are highly imaginative
(b) History is viewed through the filter of nationalism
(c) Our colonial past never happened
(d) Nationalism has to be both adequately and properly imagined

GATE CSE - 2017 MCQ SET – 2 (1-MARKS)

Q39. Choose the option with words that are not synonyms.

(a) aversion, dislike
(b) luminous, radiant
(c) plunder, loot
(d) yielding, resistant

GATE CSE - 2017 MCQ SET -21(1-MARKS)

\

Q40. Saturn is _____ to be seen on a clear night with the naked eye. .

(a) enough bright
(b) bright enough
(c) as enough bright
(d) bright as enough

GATE CSE - 2016 MCQ SET – 1 (1-MARKS)

Q41. Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world." The sentence above is an example of a _____ statement.

(a) figurative
(b) collateral
(c) literal
(d) figurine

GATE CSE - 2016 MCQ SET - 1(1-MARKS)

Q42. Out of the following four sentences, select the most suitable sentence with respect to grammar and usage

(a) I will not leave the place until the minister does not meet me.
(b) I will not leave the place until the minister doesn't meet me.
(c) I will not leave the place until the minister meet me.
(d) I will not leave the place until the minister meets me.

GATE CSE - 2016 MCQ SET - 1(1-MARKS)

Q43. A rewording of something written or spoken is a _____.

(a) paraphrase
(b) paradox
(c) paradigm
(d) paraffin

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2016 MCQ SET - 1(1-MARKS)

Q44. If 'reltaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean aftercare?

(a) zentaga
(b) tagafer
(c) tagazen
(d) reltfer

GATE CSE - 2016 MCQ SET - 2(1-MARKS)

Q45. The man who is now Municipal Commissioner worked as _____.

(a) the security guard at a university
(b) a security guard at the university
(c) a security guard at university
(d) the security guard at the university

GATE CSE - 2016 MCQ SET - 2(1-MARKS)

Q46. Nobody knows how the Indian cricket team is going to **cope with** the difficult and seamer-friendly wickets in Australia. Choose the option which is closest in meaning to the underlined phrase.

(a) put up with
(b) put in with
(c) put down to
(d) put up against

GATE CSE - 2016 MCQ SET - 2(1-MARKS)

Q47. Find the odd one in the following group of words.

mock, deride, praise, jeer

(a) mock
(b) deride
(c) praise
(d) jeer

GATE CSE - 2016 MCQ SET - 2(1-MARKS)

Q48. Pick the odd one from the following options.

(a) CADBE
(b) IHKJL
(c) XVYWZ
(d) ONPMQ

GATE CSE - 2015 MCQ SET – 3 (2-MARKS)

Q49. Most experts feel that in spite of possessing all the technical skills required to be a batsman of the highest order, he is unlikely to be so due to lack of requisite temperament. He was guilty of throwing away his wicket several time after working hard to lay a strong foundation. His critics pointed out that until he addressed his problem, success at the highest level will continue to elude him.

Which of the statement(s) below is/are logically valid and can be inferred from the above passage?

i. He was already a successful batsman at the highest level.
ii. He was to improve his temperament in order to become a great batsman.
iii. He failed to make many of his good starts count.
iv. Improving his technical skills will guarantee success.

(a) iii and iv
(b) ii and iii
(c) i, ii and iii
(d) ii only

GATE CSE - 2015 MCQ SET – 3 (1-MARKS)

Q50. Extreme focus on syllabus and studying for tests has become such a dominant concern of Indian students that they close their minds to anything _____ to the requirements of the exam.

(a) related
(b) extraneous
(c) outside
(d) useful

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2015 MCQ SET - 3(1-MARKS)

Q51. Select the pair of best expresses a relationship similar to that expressed in the pair:

Children : Pediatrician

(a) Adult : Orthopaedist
(b) Females : Gynaecologist
(c) Kidney : Nephrologist
(d) Skin : Dermatologist

GATE CSE - 2015 MCQ SET – 3 (1-MARKS)

Q52. The Tamil version of _____ John Abraham-starrer Madras Cafe _____ cleared by the Censor Board with no cuts last week, but the film's distributor _____ no takers among the exhibitors for a release in Tamilnadu _____ this Friday.

(a) Mr., was, found, on
(b) a, was, found, at
(c) the, was, found, on
(d) a, being, find at

GATE CSE - 2015 MCQ SET – 2 (2-MARKS)

Q53. Out of the following 4 sentences, select the most suitable sentence with respect to grammar and usage:

(a) Since the report lacked needed information, it was of no use to them.
(b) The report was useless to them because there were no needed information in it.
(c) Since the report did not contain the needed information, it was not real useful to them.
(d) Since the report lacked needed information, it would not had been useful to them.

GATE CSE - 2015 MCQ SET – 2 (1-MARKS)

Q54. We _____ our friend's birthday and we _____ how to make it up to him.

(a) completely forgot --- don't just know
(b) forgot completely --- don't just know
(c) completely forgot --- just don't know
(d) forgot completely --- just don't know

GATE CSE - 2015 MCQ SET – 2 (1-MARKS)

Q55. Choose the statement where the underlined word is used correctly.

(a) The industrialist had a personnel jet.
(b) I write my experience in my personnel diary.
(c) All personnel are being given the day off.
(d) Being religious is a personnel aspect.

GATE CSE - 2015 MCQ SET – 2 (1-MARKS)

Q56. A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt is

(a) fabric
(b) textile
(c) fiber
(d) apparel

GATE CSE - 2015 MCQ SET – 1 (2-MARKS)

Q57. Select the alternative meaning of the underlined part of the sentence. The chain snatchers took to their heels when the police party arrived.

(a) Took shelter in a thick jungle
(b) Open indiscriminate fire
(c) Took to flight
(d) Unconditionally surrendered

GATE CSE - 2015 MCQ SET – 1 (1-MARKS)

Q58. Which of the following options is the closest in meaning of the sentence below?

She enjoyed herself immensely at the party.

(a) She had a terrible time at the party
(b) She had a horrible time at the party
(c) She had a terrific time at the party
(d) She had a terrifying time at the party

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2015 MCQ SET – 1 (1-MARKS)

Q59. Didn't you buy _____ when you went shopping?

(a) any paper
(b) much paper
(c) no paper
(d) a few paper

GATE CSE - 2015 MCQ SET – 1 (2-MARKS)

Q60. Statement: There has been a significant drop in the water level in the lakes supplying water to the city. Course of action: I. The water supply authority should impose a partial cut in supply. II. The government should appeal to residents through mass media for minimal use of water. III. The government should ban the water supply in lower areas.

(a) Statements I and II follow.
(b) Statements I and III follow.
(c) Statements II and III follow.
(d) All the statements follow.

GATE CSE - 2015 MCQ SET – 1 (1-MARKS)

Q61. Which one of the following combinations is incorrect?

(a) Acquiescence - Submission
(b) Wheedle - Roundabout
(c) Flippancy - Lightness
(d) Profligate – Extravagant

GATE CSE - 2014 MCQ SET – 3 (2-MARKS)

Q62. A dance programme is scheduled for 10.00 a.m. Some students are participating in the programme and they need to come an hour earlier than the start of the event. These students should be accompanied by a parent. Other students and parents should come in time for the programme. The instruction you think that is appropriate for this is

(a) Students should come at 9.00 a.m. and parents should come at 10.00 a.m.
(b) Participating students should come at 9.00 a.m. accompanied by a parent, and other parents and students should come by 10.00 a.m.
(c) Students who are not participating should come by 10.00 a.m. and they should not bring their parents. Participating students should come at 9.00 a.m.
(d) Participating students should come before 9.00 a.m. Parents who accompany them should come at 9.00 a.m. All others should come at 10.00 a.m.

GATE CSE - 2014 MCQ SET – 3 (2-MARKS)

Q63. By the beginning of the 20th century, several hypotheses were being proposed, suggesting a paradigm shift in our understanding of the universe. The clinching evidence was provided by experimental measurements of the position of a star directly behind our sun. Which of the following inference(s) may be drawn?

i. Our understanding of the universe changes based on the positions of stars ii. Paradigm shifts usually occur at the beginning of centuries iii. Stars are important objects in the universe iv. Experimental evidence was important in confirming this paradigm shift

(a) i, ii and iv
(b) iii only
(c) i and iv
(d) iv only

GATE CSE - 2014 MCQ SET – 3 (1-MARKS)

Q64. While trying to collect an envelope from under the table, Mr. X fell down and was losing consciousness.

The sentence above is divided into four parts as shown below:

- **I**: While trying to collect an envelope from under the table
- **II**: Mr. X

Check Out Our GATE Courses & Secure Your Dream Rank!

- **III**: fell down
- **IV**: and was losing consciousness

Which one of the above underlined parts of the sentence is **NOT appropriate**?

(a) I
(b) II
(c) III
(d) IV

GATE CSE - 2014 MCQ SET – 3 (2-MARKS)

Q65. If she _____ how to calibrate the instrument, she _____ done the experiment.

(a) knows, will have
(b) knew, had
(c) had known, could have
(d) should have known, would have

GATE CSE - 2014 MCQ SET – 3 (2-MARKS)

Q66. Choose the word that is opposite in meaning to the word "coherent".

(a) sticky
(b) well-connected
(c) rambling
(d) friendly

GATE CSE - 2014 MCQ SET – 2 (2-MARKS)

Q67. The old city of Koenigsberg, which had a German majority population before World War 2, is now called Kaliningrad. After the events of the war, Kaliningrad is now a Russian territory and has a predominantly Russian population. It is bordered by the Baltic Sea on the north and the countries of Poland to the south and west and Lithuania to the east respectively. Which of the statements below can be inferred from this passage?

(a) Kaliningrad was historically Russian in its ethnic make up

(b) Kaliningrad is a part of Russia despite it not being contiguous with the rest of Russia

(C) Koenigsberg was renamed Kaliningrad, as that was its original Russian name

(d) Poland and Lithuania are on the route from Kaliningrad to the rest of Russia

GATE CSE - 2014 MCQ SET – 2 (2-MARKS)

Q68. The number of people diagnosed with dengue fever in North India is twice the number diagnosed last year. Municipal authorities concluded that measures to control the mosquito population have failed. Which one of the following statements, if true, does not contradict this conclusion?

(a) A high proportion of the affected population has returned from neighbouring countries where dengue is prevalent

(b) More cases of dengue are now reported because of an increase in the Municipal Office's administrative efficiency

(c) Many more cases of dengue are being diagnosed this year since the introduction of a new and effective diagnostic test

(d) The number of people with malarial fever (also contracted from mosquito bites) has increased this year

GATE CSE - 2014 MCQ SET – 2 (1-MARKS)

Q69. Choose the most appropriate phrase from the options given below to complete the following sentence.

India is a post-colonial country because

(a) it was a former British colony
(b) Indian Information Technology professionals have colonized the world
(c) India does not follow any colonial practices
(d) India has helped other countries gain freedom

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2014 MCQ SET – 2 (1-MARKS)

Q70. Who _____ was coming to see us this evening?

(a) you said
(b) did you say
(c) did you say that
(d) had you said

GATE CSE - 2014 MCQ SET – 1 (2-MARKS)

Q71. The Paighat Gap (or Palakkad Gap), a region about 30 km wide in the southern part of the Western Ghats in India, is lower than the hilly terrain to its north and south. It results in the neighbouring regions of Tamil Nadu getting more rainfall from the South West monsoon and the neighbouring regions of Kerala having higher summer temperatures. What can be inferred from this passage?

(a) The Paighat gap is caused by high rainfall and high temperatures in southern Tamil Nadu and Kerala
(b) The regions in Tamil Nadu and Kerala that are near the Paighat Gap are low-lying
(c) The low terrain of the Paighat Gap has a significant impact on weather patterns in neighbouring parts of Tamil Nadu and Kerala
(d) Higher summer temperatures result in higher rainfall near the Paighat Gap area.

GATE CSE - 2014 MCQ SET – 1 (2-MARKS)

Q72. Geneticists say that they are very close to confirming the genetic roots of psychiatric illnesses such as depression and schizophrenia, and consequently, that doctors will be able to eradicate these diseases through early identification and gene therapy. On which of the following assumptions does the statement above rely?

(a) Strategies are now available for eliminating psychiatric illnesses
(b) Certain psychiatric illnesses have a genetic basis
(c) All human diseases can be traced back to genes and how they are expressed
(d) In the future, genetics will become the only relevant field for identifying psychiatric illnesses

GATE CSE - 2014 MCQ SET – 1 (1-MARKS)

Q73. Which of the following options is the closest in meaning to the phrase in bold in the sentence below?
It is fascinating to see life forms **cope with** varied environmental conditions.

(a) Adapt to
(b) Adopt to
(c) Adept in
(d) Accept with

GATE CSE - 2014 MCQ SET – 1 (1-MARKS)

Q74. He could not understand the judges awarding her the first prize, because he thought that her performance was quite _____.

(a) superb
(b) medium
(c) mediocre
(d) exhilarating

GATE CSE - 2014 MCQ SET – 1 (1-MARKS)

Q75. In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement?

(a) He wants all the money
(b) He will return the money
(c) He will assume final responsibility
(d) He will resist all enquiries

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2013 MCQ (2-MARKS)

Q76. After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

(a) Failure is the pillar of success.
(b) Honesty is the best policy.
(c) Life begins and ends with adventures.
(d) No adversity justifies giving up hope.

GATE CSE - 2013 MCQ (1-MARKS)

Q77. Which one of the following options is the closest in meaning to the word given below?

Nadir

(a) Highest
(b) Lowest
(c) Medium
(d) Integration

GATE CSE - 2013 MCQ (1-MARKS)

Q78. Complete the sentence:

Universalism is to particularism as diffuseness is to _____.

(a) specificity
(b) neutrality
(c) generality
(d) adaptation

GATE CSE - 2013 MCQ (1-MARKS)

Q79. Were you a bird, you _____ in the sky.

(a) would fly
(b) shall fly
(c) should fly
(d) shall have flown

GATE CSE - 2013 MCQ (1-MARKS)

Q80. Choose the grammatically INCORRECT sentence:

(a) He is of Asian origin.
(b) They belonged to Africa.
(c) She is an European.
(d) They migrated from India to Australia.

GATE CSE - 2012 MCQ (2-MARKS)

Q81. Wanted Temporary, Part-time persons for the post of Field Interviewer to conduct personal interviews to collect and collate economic data. Requirements: High School-pass, must be available for Day, Evening and Saturday work. Transportation paid, expenses reimbursed. Which one of the following is the best inference from the above advertisement?

(a) Gender-discriminatory
(b) Xenophobic
(c) Not designed to make the post attractive
(d) Not gender-discriminatory

GATE CSE - 2012 MCQ (1-MARKS)

Q82. Choose the most appropriate alternative from the options given below to complete the following sentence:
Despite several _____ the mission succeeded in its attempt to resolve the conflict.

(a) attempts
(b) setbacks
(c) meetings
(d) delegations

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2012 MCQ (1-MARKS)

Q83. Which one of the following options is the closest in meaning to the word: **MITIGATE**

(a) Diminish
(b) Divulge
(c) Dedicate
(d) Denote

GATE CSE - 2012 MCQ (1-MARKS)

Q84. Choose the grammatically INCORRECT sentence:

(a) They gave us the money back less the service charges of Three Hundred rupees.
(b) This country's expenditure is not less than that of Bangladesh.
(c) The committee initially asked for a funding of Fifty Lakh rupees, but later settled for a lesser sum.
(d) This country's expenditure on educational reforms is very less.

GATE CSE - 2012 MCQ (1-MARKS)

Q85. Choose the most appropriate alternative from the options given below to complete the following sentence:

Suresh's dog is the one _____ was hurt in the stampede.

(a) that
(b) which
(c) who
(d) whom

GATE CSE - 2011 MCQ (1-MARKS)

Q86. Which of the following options is the closest in the meaning to the word below:
Inexplicable

(a) Incomprehensible
(b) Indelible
(c) Inextricable
(d) Infallible

GATE CSE - 2011 MCQ (2-MARKS)

Q87. Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.

Based on the above passage, which topic would not be included in a unit on bereavement?

(a) how to write a letter of condolence
(b) what emotional stages are passed through in the healing process
(c) what the leading causes of death are
(d) how to give support to a grieving friend

GATE CSE - 2011 MCQ (1-MARKS)

Q88. Choose the most appropriate word(s) from the options given below to complete the following sentence.

I contemplated _____ Singapore for my vacation but decided against it..

(a) To visit
(b) Having to visit
(c) Visiting
(d) For a visit

GATE CSE - 2011 MCQ (1-MARKS)

Q89. Choose the most appropriate word from the options given below to complete the following

If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or

(a) Hyperbolic
(b) Argumentative
(c) Restrained
(d) Indifferent

Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2011 MCQ (1-MARKS)

Q90. Choose the word from the options given below that is most nearly opposite in the meaning to the given word

Amalgamate

(a) Merge
(b) Split
(c) Collect
(d) Separate

GATE CSE - 2010 MCQ (1-MARKS)

Q91. Choose the most appropriate word from the options given below to complete the following sentence:

His rather casual remarks on politics _____ his lack of seriousness about the subject.

(a) Masked
(b) Belied
(c) Betrayed
(d) Suppressed

GATE CSE - 2010 MCQ (1-MARKS)

Q92. Which of the following options is the closest in meaning to the word given below:
Circuitous

(a) Cyclic
(b) Indirect
(c) Confusing
(d) Crooked

GATE CSE - 2010 MCQ (1-MARKS)

Q93. Choose the most appropriate word from the options given below to complete the following sentence:
If we manage to _____ our natural resources, we would leave a better planet for our children.

(a) Uphold
(b) Restrain'
(c) cherish
(d) Conserve

GATE CSE - 2010 MCQ (1-MARKS)

Q94. The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

Unemployed : Worker

(a) fallow : land
(b) unaware : sleeper
(c) Wit : jester
(d) Renovated : house

GATE CSE - 2010 MCQ (2-MARKS)

Q95. Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare; and regretfully, there exist people in military establishments who think that chemical agents are useful tools for their cause.

Which of the following statements best sums up the meaning of the above passage ?

(a) Modern warfare has resulted in civil strife.
(b) Chemical agents are useful in modern warfare.
(c) Use of chemical agents in warfare would be undesirable.
(d) People in military establishments like to use chemical agents in war.

Check Out Our GATE Courses & Secure Your Dream Rank!

ANSWER KEY

1. (b)	2. (a)	3. (a)
4. (b)	5. (c)	6. (c)
7. (c)	8. (b)	9. (a)
10. (c)	11. (c)	12. (c)
13. (b)	14. (b)	15. (d)
16. (d)	17. (d)	18. (b)
19. (c)	20. (c)	21. (d)
22. (a)	23. (c)	24. (c)
25. (d)	26. (a)	27. (a)
28. (d)	29. (a)	30. (c)
31. (a)	32. (d)	33. (b)
34. (b)	35. (a)	36. (c)
37. (d)	38. (b)	39. (d)
40. (b)	41. (a)	42. (d)
43. (a)	44. (c)	45. (b)
46. (a)	47. (c)	48. (d)
49. (b)	50. (b)	51. (b)
52. (c)	53. (a)	54. (c)
55. (c)	56. (d)	57. (c)
58. (c)	59. (a)	60.(a)
61. (b)	62. (b)	63.(d)
64. (d)	65. (c)	66.(c)
67. (b)	68. (d)	69.(a)
70. (b)	71. (c)	72.(b)
73. (a)	74. (c)	75.(c)
76. (d)	77. (b)	78.(a)
79. (a)	80. (c)	81.(d)
82. (b)	83. (a)	84.(d)
85. (a)	86. (a)	87.(c)
88. (c)	89. (c)	90.(d)
91. (c)	92. (b)	93.(d)
94. (a)	95. (d)	

HINTS & SOLUTIONS

1. ANSWER: - (B)

The protagonist is the main character or central figure in a story, often the one the audience follows most closely and empathizes with.

Examples : In J.K. Rowling's Harry Potter series, Harry Potter is the protagonist

The antagonist is the direct antonym of the protagonist. This character or force acts in opposition to the protagonist, creating the conflict that drives the story.

Example :In the Harry potter series, Lord Voldemort is the antagonist

2. ANSWER: - (A)

Expedite - to make something happen more quickly

Hasten - to be quick to do or say something.

Hurry - to do something quickly

Retard - to delay something

Out of the given options, Accelerate is the correct option. Hence, Answer is A.

3. ANSWER: - (A)

The word younger is a comparative adjective, and we generally use "a" before singular, non-specific countable nouns.

"university" → The word "university" begins with the letter 'u', which is a vowel, but it has a "yu" (consonant sound) as in you-niversity.

"honorable man" → The word "honorable" begins with 'h', but the 'h' is silent. It is pronounced "onorable", which starts with a vowel sound.

We use "an" before words that start with a vowel sound.

4. ANSWER: - (B)

"Myopic" literally means nearsighted (unable to see far).

Figuratively, it means shortsighted, lacking foresight, or focusing only on immediate results rather than considering long-term consequences.

(A) Visionary

Check Out Our GATE Courses & Secure Your Dream Rank!

Incorrect. Opposite of myopic. A visionary person has long-term foresight.

(B) Shortsighted

Correct. It directly means lacking long-term perspective, which fits the meaning of "myopic."

(C) Progressive

Incorrect. Progressive means forward-thinking and future-oriented, which is the opposite of myopic.

(D) Innovative

Incorrect. Innovative means creative and introducing new ideas, which does not match the meaning of myopic.

5. ANSWER: - (C)

The correct answer is:

The pocket-knife is described as the infant of the sword.

In the passage, the author mentions the pocket-knife as "the infant of the sword," comparing it to a basic, early form of a more sophisticated tool (the sword).

6. ANSWER: - (C)

"Resolve" means strong determination. The phrase "never wavered" suggests unwavering commitment, making "resolve" the best fit.

Ambivalence → Means uncertainty which contradicts the sentence.

Satisfaction → Refers to contentment not determination.

Revolve → Means to rotate which doesn't fit the context.

7. ANSWER: - (C)

This is a relationship analogy where the first word is an animal and the second word is its dwelling.

- A bird lives in a nest.
- A bee lives in a hive.

8. ANSWER: - (B)

Ramya says, "Even if I had known you were in the hospital, I would not have gone there to see you."

This implies that Ramya did not know Josephine was in the hospital. If she had known, she would have visited, but the statement suggests that she was unaware of it at the time.

9. ANSWER: - (A)

Arid meaning - having little or no rain; too dry

Parched meaning - dried out with heat

Thus the words has to be in increasing order of intensity.

So for diet -> fast -> starve

10. ANSWER: - (C)

Only "down" fits in first blank, meaning "not to engage" and stay focused. In second, Both "down" and "out" work, with similar meanings in cricket. In third, both "on" and "for" are valid while in fourth only "on" fits contextually.

11. ANSWER: - (C)

The appropriate word that fits in this sequence is Fazed. The meaning of this word is to make someone nervous.

Hence, the correct option is C.

12. ANSWER: - (C)

The correct sequence for this question is S -> P -> R -> Q.

Hence the correct answer is C.

13. ANSWER: - (B)

The most suitable word is nearly, Hence option (b) is correct.

14. ANSWER: - (B)

Frequently is synonym of often, similarly, type is synonym of kind. Hence, the correct answer is (b).

Check Out Our GATE Courses & Secure Your Dream Rank!

15. ANSWER: - (D)

The statement "All the working population of Zombieland will henceforth ride bicycles to work" is illogical, because the information provided does not imply that everyone will begin riding bicycles to work.
The statement "Riding bicycles will ensure that all of Zombieland's working population is free of health issues" cannot be logically inferred because other health issues may be affecting the population.
The statement "The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory" cannot be logically inferred because no information on what the health experts recommended to the government is provided.

Bicycling is considered a form of physical exercise by the Zombieland government. It can be logically inferred with certainty.

16. ANSWER: - (D)

The narrative flows in the logically in this order ,with mina carry heavy bags after shop, then panting as she climbs the stairs and finally ,on the terrace, she notices the man is leaning against he parapet.

17. ANSWER: - (D)

Fare means the amount or charge you have to pay for any service, and Fair meaning is appropriate or acceptable in a particular situation.

So, in the above sentence fare/ fair is the correct option. The sentence will become, The fare is too high to be considered fair, means The charge/amount is too high to be considered appropriate.

18. ANSWER: - (B)

The second statement of the passage states that 'In organizations with good work culture, metrics help improve outcomes.' As a result, we can confidently assert that metrics are beneficial in firms with the positive work culture. As a result, option B is right.

19. ANSWER: - (C)

Sentence (i): This sentence is grammatically correct because it follows proper subject-verb agreement where "Everybody" (a singular subject) matches the singular verb "is." and there is no ambiguity in this sentence.

Sentence (ii): This sentence is ambiguous because it is unclear who "he" refers to. It could mean: Babu enjoys playing chess, Danish enjoys playing chess. This ambiguity arises from the unclear pronoun reference ("he").

20. ANSWER: - (C)

"Doctor is to surgery" makes sense, as doctors perform surgeries.

"Writer is to book" also makes sense, as writers write books.

This option maintains the logical relation.

21. ANSWER: - (D)

The problems mentioned are poverty and inequality. But AOMs are not addressing the main problems directly.

But it is a kind of site step taken to address obesity.

So AOMs are only addressing the obesity problem superficially but not addressing the real problems**.**

22. ANSWER: - (A)

'worse', 'better' are comparative words which need the word 'than' for comparison.

Out of the remaining two options "nicest" is superlative degree which expresses the highest form of an adjective so it is not suitable here.

But 'as well as' expresses equality in doing some action or skill so it fits perfectly & is the correct answer.

Correct Option (A).

Check Out Our GATE Courses & Secure Your Dream Rank!

23. ANSWER: - (C)

A pen is used to write. Similarly, a knife is used to cut. Correct Option (C)

24. ANSWER: - (C)

From the first statement “Listening to music during exercise improves exercise performance and reduces discomfort. “ It is clear that listening to music has a positive effect on physical exercise.

From the statement “Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. “ it is clear that music's effect on studying depends on the student (some benefit, some do not), making the results inconclusive.

Only on some students music has a positive effect. Which Passage 3 totally aligning with the message.

25. ANSWER: - (D)

The correct answer is **"within, for"** because "within six months" indicates a time frame in the future, and "for the last three weeks" refers to the duration of time he's been practicing.

26. ANSWER: - (A)

The word “but” shows contrast, so the second part must be opposite of “excellent”. Hence, “extremely poor” is correct.

27. ANSWER: - (B)

The analogy represents the relationship between a person or thing (Cook, Fly) and the action they perform (Cooking, Flying).

Here is the breakdown:

- Cook (noun/verb) → Cooking (action/gerund)
- Fly (noun/verb) → Flying (action/gerund)

28. ANSWER: - (D)

The passage clearly states that billions depend on glaciers for fresh water and are impacted by their melting.

29. ANSWER: - (C)

(c) is correct because the text defines it as a **"destination-based tax"** on goods and services **"used"**, rather than at the point of origin.

30. ANSWER: - (C)

The subject “expenditure” is singular, so the verb must also be singular. Hence, “breaks down” is correct.

31. ANSWER: - (A)

The correct phrase is “revolves around,” which means centered on something.

32. ANSWER: - (D)

A judge works in a court, and a teacher works in a school. This is a place-based analogy.

33. ANSWER: - (B)

The passage states that begging is a symptom caused by failure of the state’s social security system. This implies lack of welfare schemes leads to begging.

34. ANSWER: - (B)

“They’re” = they are

“Their” = possession

“There” = place

Correct sequence: They’re, their, there.

35. ANSWER: - (A)

“Meandering” means unstructured or wandering, opposite of “organized”, hence correct.

36. ANSWER: - (C)

Past action → simple past required → **wished**

37. ANSWER: - (D)

“Besides” = in addition to → correct phrase = **besides money**

Check Out Our GATE Courses & Secure Your Dream Rank!

38. ANSWER: - (B)

Statement says history is interpreted via nationalism → **filter of nationalism**

39. ANSWER: - (D)

"Yielding" means flexible or giving way, while "resistant" is opposite → not synonyms.

40. ANSWER: - (B)

Correct structure: adjective + enough → "bright enough".

41. ANSWER: - (A)

The statement is symbolic/exaggerated, not literal. Hence figurative.

42. ANSWER: - (D)

Correct structure: "until + subject + verb (positive form)"

Double negative is incorrect. Hence "meets" is correct.

43. ANSWER: - (A)

Paraphrase means expressing the same idea in different words.

44. ANSWER: - (C)

Based on the common components of the words provided:

1. Identify the root: The cormon ending in all three given words is taga, which corresponds to the common word care.
2. Determine the order: The modifier comes first, followed by the root word (Modfifier + Care). Therefore, aftercare must end with taga (i.e. "after" + taga).
3. Select the prefici The prefa for "after" must be a new, unused root. zen fits this criterion.

The correct option is (a) zentaga.

45. ANSWER: - (B)

Correct article usage: "a security guard at the university" is grammatically appropriate.

46. ANSWER: - (A)

"Cope with" means to tolerate or manage → closest is "put up with".

47. ANSWER: - (C)

Mock, deride, jeer all mean to insult. Praise is opposite.

48. ANSWER: - (D)

A. $\underbrace{CADBE}_{3\,1\,4\,2\,5}$
B. $\underbrace{JHKIL}_{3\,1\,4\,2\,5}$
C. $\underbrace{XVYWZ}_{3\,1\,4\,4\,2\,5}$

They are all making sequence of $1-2-3-4-5$ and as in alphabetical order

49. ANSWER: - (B)

Has technical skills but lacks temperament

50. ANSWER: - (B)

To use outside it must be in the form "outside of".

51. ANSWER: - (B)

Orthopaedist: An orthopaedic surgeon, a doctor who corrects congenital or functional abnormalities of the bones with surgery, casting, and bracing.

Gynaecologist a physician or surgeon qualified to practice in functions and diseases specific to women and girls, especially those affecting the reproductive system.

Nephrologist is a medical doctor who specializes in kidney care and treating diseases of the kidneys.

Dermatologist is a medical practitioner qualified to diagnose and treat skin disorders.

Kidneys and skin are parts of the body. Females is the only option which represents **group of people** like children which is the correct option

Check Out Our GATE Courses & Secure Your Dream Rank!

52. ANSWER: - (C)
Here, we are talking about the movie and not about John Abraham. Article "the" is used to define a particular object here(The Movie).

53. ANSWER: - (A)
Correct versions of other 3:

B. The report was useless to them because there **was** no needed information in it.

C. Since the report did not contain the needed information, it was not **really** useful to them.

D. Since the report lacked needed information, it would not **have** been useful to them

54. ANSWER: - (C)
"Completely forgot" is correct as the adverb "completely" should come before the verb it modifies. "Just don't know" is correct as "just" is used as an adverb to emphasize uncertainty. So option C has both parts in the correct word order.

55. ANSWER: - (A)

"Personnel" means the people employed in an organization. In option C, it correctly refers to employees being given a day off. In other options, the word "personal" should be used instead of "personnel."

56. ANSWER: - (D)
"Apparel" is the generic term for all types of clothing items. Fabric, textile and fiber refer to the material used to make clothes, not the clothing items themselves. So apparel is the correct answer.

57. ANSWER: - (C)

"Took to their heels" is an idiom meaning to run away quickly. "Took to flight" means to flee or escape, which is the closest alternative meaning. The chain snatchers ran away as soon as the police arrived.

58. ANSWER: - (C)
means, she had a good time at the party.

A. She had a terrible time at the party - means her time was bad at the party.
B. She had a horrible time at the party - same as above; her time was bad at the party.
C. She had a terrific time at the party - means she had a good time at the party.
D. She had a terrifying time at the party - means she had a frightening (not good) time at the party.

So, correct option: C.

59. ANSWER: - (A)
In negative and interrogative sentences, "any" is used instead of "some." "Much" is used in statements, "no paper" would make a double negative, and "a few" is used for countable nouns. So "any paper" is the correct choice.

60. ANSWER: - (A)
Both I and II are practical and proportionate responses to a drop in water level. Statement III is too extreme as a complete ban is not justified by a drop in water level alone. So only I and II are appropriate courses of action.

61. ANSWER: - (B)
Acquiescence means submission, Flippancy means lightness, and ProfliGATE means extravagant — all correct pairs. Wheedle means to persuade through flattery or coaxing, not roundabout. So B is the incorrect combination.

62. ANSWER: - (B)
Participating students should come at 9.00 a.m. accompanied by a parent, and other parents and students should come by 10.00 a.m

63. ANSWER: - (D)
Only inference iv is directly supported by the passage, which explicitly states that experimental measurements provided clinching evidence. Inference i is too broad, ii is not mentioned, and iii is not implied by the passage.

64. ANSWER: - (D)
Part IV "was losing consciousness" is incorrect because it implies a gradual ongoing process, whereas losing consciousness after a fall is a sudden

Check Out Our GATE Courses & Secure Your Dream Rank!

event. The correct phrase should be "lost consciousness" in simple past tense

65. ANSWER: - (C)
This is a third conditional sentence expressing an unreal past condition. The correct structure is "If + past perfect, subject + could/would have + past participle." So "had known, could have" is the grammatically correct choice.

66. ANSWER: - (C)
"Coherent" means logical, clear and well-structured. "Rambling" means disorganized, unclear and wandering without a clear point, which is the direct opposite of coherent.

67. ANSWER: - (B)
The passage states Kaliningrad is a Russian territory bordered by Poland and Lithuania, meaning it has no direct land connection to the rest of Russia. This clearly supports option B that Kaliningrad is part of Russia despite not being contiguous with it.

68. ANSWER: - (D)
Options A, B, and C all provide alternative explanations for the rise in dengue cases that contradict the conclusion of failed mosquito control. Option D about malarial fever increasing actually supports the conclusion that mosquito control has failed, so it does not contradict it.

69. ANSWER: - (A)
"Post-colonial" refers to a country that was formerly under colonial rule and has since gained independence. India was under British colonial rule until 1947, making option A the only factually and logically correct reason for India being called a post-colonial country.

70. ANSWER: - (B)
The sentence is an indirect question embedded in a regular question structure. "Who did you say was coming to see us this evening?" is the grammatically correct form, using the auxiliary verb "did" for past tense question formation.

71. ANSWER: - (C)
The passage directly states that the low terrain of the Paighat Gap results in more rainfall in Tamil Nadu and higher temperatures in Kerala. This clearly supports option C that the low terrain significantly impacts weather patterns in neighbouring regions.

72. ANSWER: - (B)
The statement assumes that psychiatric illnesses like depression and schizophrenia have genetic roots. Without this assumption, the claim about early identification and gene therapy would have no basis. Option B directly supports this underlying assumption.

73. ANSWER: - (A)
"Cope with" means to deal with or adapt to difficult conditions. "Adapt to" is the closest in meaning as it conveys adjusting or modifying oneself to suit different environmental conditions.

74. ANSWER: - (C)
Since he could not understand why she won the first prize, he clearly thought her performance was poor. "Mediocre" meaning average or below average fits perfectly, as it contrasts with winning first prize.

75. ANSWER: - (C)
"The buck stops here" is an idiomatic expression meaning the person accepts final responsibility and will not pass blame to anyone else. It conveys accountability and ownership of the situation.

76. ANSWER: - (D)
The passage clearly shows that despite extreme adversity and hopelessness, Bruce was inspired by the spider's persistence to never give up. The central message is that no matter how difficult the situation, one should never abandon hope.

77. ANSWER: - (B)
"Nadir" refers to the lowest point of something, either literally or figuratively. It is the opposite of "zenith" which means the highest point. So "lowest" is the closest in meaning.

78. ANSWER: - (A)

Check Out Our GATE Courses & Secure Your Dream Rank!

Universalism and particularism are opposites, so the answer must be the opposite of diffuseness. Diffuseness means being broad and spread out, while specificity means being precise and focused, making them direct opposites.

79. ANSWER: - (A)
This is a second conditional sentence expressing an imaginary or hypothetical situation. The correct structure is "If + past tense, subject + would + base verb." So "would fly" is the correct choice.

80. ANSWER: - (C)
"An" is used before words starting with a vowel sound. "European" starts with a "Y" sound (consonant sound), so "a European" is correct, not "an European."

81. ANSWER: - (D)
The advertisement uses the word "persons" and does not mention any gender preference or restriction anywhere. It is open to all genders, making it not gender-discriminatory. The ad also offers transport and expense reimbursement, making it reasonably attractive.

82. ANSWER: - (B)
The word "despite" indicates a contrast, suggesting that something negative was overcome. "Setbacks" fits perfectly as it means obstacles or difficulties, making the success of the mission more meaningful. Other options like "attempts" or "meetings" do not create a logical contrast.

83. ANSWER: - (A)
"MITIGATE" means to make something less severe or intense.
"Diminish" means to reduce or lessen, which is the closest in meaning.
"Divulge" means to reveal,
"Dedicate" means to commit, and
"Denote" means to indicate.

84. ANSWER: - (D)
"Very less" is grammatically incorrect in English. "Less" cannot be modified by "very"; instead, "very little" should be used. All other sentences use "less" and "lesser" correctly in their respective contexts.

85. ANSWER: - (A)
"That" is used for animals and things when referring to them in a defining relative clause. "Who" and "whom" are used for people, and "which" is typically used in non-defining clauses. Since the subject is a dog, "that" is the most appropriate choice.

86. ANSWER: - (A)
Inexplicable - difficult or impossible to explain
Incomprehensible - difficult or impossible to understand or comprehend - Most appropriate
Indelible - impossible to remove, erase or wash away - Not appropriate
Inextricable - unavoidable , inescapable - Not appropriate
Infallible - completely dependable or trustworthy - irrelevant

87. ANSWER: - (C)
Based on the passage, the unit on bereavement must include topics which can help one overcome losses from death and parting. So, options (A), (B) and (D) are suitable topics but (C) is not helping here.

88. ANSWER: - (C)
Some verb can be followed by either an infinitive or a gerund, but there will be a difference in meaning.

89. ANSWER: - (C)
Hyperbolic: Exaggerated or overstated.

1. **Restrained** : Controlled or moderate in behavior or expression.
2. **Argumentative** : Prone to arguing or disputing.
3. **Indifferent** : Having no particular interest or concern.

The most appropriate word to complete the sentence is **"restrained"** because being restrained means holding back or being moderate, which contrasts with making a strong impression

90. ANSWER: - (D)
- **Amalgamate:** Means to combine or unite two or more separate things to form one organization or structure.
- **Separate:** This is the direct antonym, as it means to move or keep apart.
- **Why (b) Split is less accurate:** A "split" usually refers to a forceful separation of something that was originally a single unit (like splitting wood).

Check Out Our GATE Courses & Secure Your Dream Rank!

Since "amalgamate" refers to bringing different things together, "separate" is the logically superior opposite

Amalgamate= combine or unite (two separate or different things)to form one organization or structure (like alloy)

Split = a forceful separation of something that was one thing(eg. split of wood)

Separate= take two things that were already different,further apart.(eg. separation of husband wife or two joint companies)

91. ANSWER: - (C)

(a) Masked : Hide under a false appearance **- opposite**

(b) Belied : Be in contradiction with **- not appropriate**

(c) Betrayed : Reveal unintentionally **- most appropriate**

(d) Suppressed: To put down by force or authority **- irrelevant**

92. ANSWER: - (B)

Circuitous means roundabout, not direct, taking a longer path.

Among the options:

- Cyclic = repetitive
- Indirect = not direct (closest meaning)
- Confusing = unclear
- Crooked = bent/dishonest

93. ANSWER: - (D)

Uphold : cause to remain **- Not appropriate**

Restrain : keep under control **- Not appropriate**

Cherish : be fond of **- Not related**

Conserve : keep in safety and protect from harm , decay, loss or destruction **- most appropriate**

94. ANSWER: - (A)

"Unemployed : Worker" → A **worker who currently has no work** (i.e., a worker in an inactive state).

Similarly,

"fallow : land" → **Land that is currently not being used** (inactive land).

Relation:

Both describe something that **has the capacity/role but is presently inactive or unused**.

Why others are wrong:

- unaware : sleeper → not the same inactive-role relation
- wit : jester → different (quality vs person)
- renovated : house → opposite (improved, not inactive)

Final: fallow : land

95. ANSWER: - (D)

Statement 1:

The phrase *"civil strife"* refers to conflict within a country, such as between groups or regions. This idea is not mentioned in the passage, so this statement does not fit the context and is incorrect.

Statement 2:

Although the passage says that chemical agents seem suitable for modern warfare, it does not provide any evidence or justification proving their usefulness. Therefore, this statement is incorrect.

Statement 3:

This statement claims that chemical agents are undesirable. However, the passage suggests that such agents work silently and are considered useful by some people in military establishments. This implies a level of acceptance or desirability, making this statement incorrect.

Statement 4:

The passage mentions that certain people in military establishments believe chemical agents are useful tools. This aligns directly with the idea that such individuals favor their use in warfare. Hence, this statement correctly represents the passage.

Check Out Our GATE Courses & Secure Your Dream Rank!

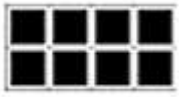
4 SPATIAL APTITUDE

GATE CSE - 2026 MCQ SET-1 (1-MARKS)

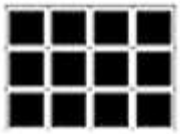
Q1. The figure shows two 4-tile patterns.

Either one or both of the patterns can be used any number of times and in any orientation to construct a new pattern. Which one of the options below cannot be constructed by using only these two 4-tile patterns assuming there are no overlaps among them?

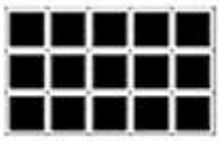
(a)



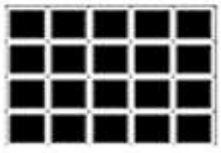
(b)



(c)

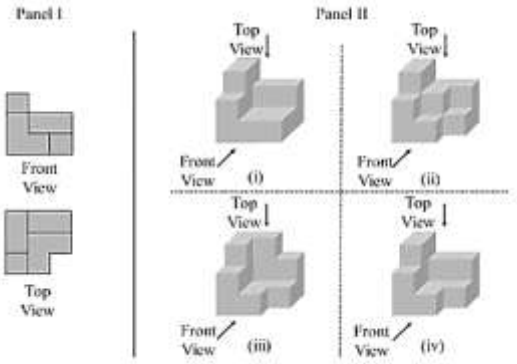


(d)



GATE CSE - 2026 MCQ SET – 1 (2-MARKS)

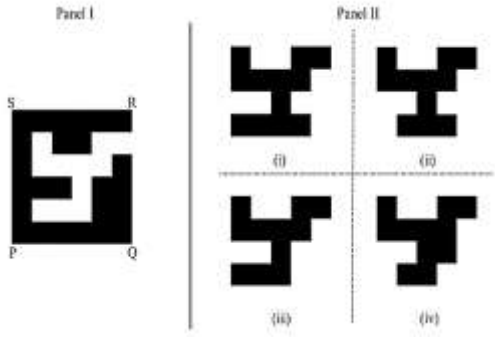
Q2. In Panel I of the figure below, the front view and top view of a structure are shown. Which one of the 3D structures shown in Panel II possesses the views shown in Panel I?



(a) (i)
(b) (ii)
(c) (iii)
(d) (iv)

GATE CSE - 2026 MCQ SET – 2 (1-MARKS)

Q3. A black square PQRS has been cut into two parts. One part of it is shown in Panel I. Which one of the shapes in Panel II is the other part?



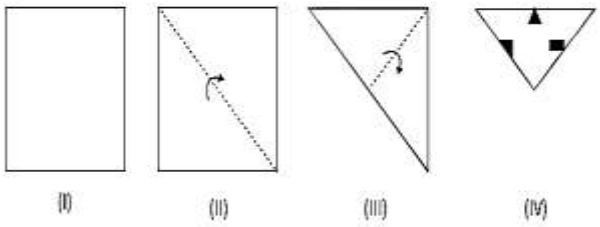
(a) (i)
(b) (ii)
(c) (iii)
(d) (iv)

Check Out Our GATE Courses & Secure Your Dream Rank!

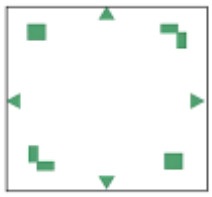
GATE CSE - 2026 MCQ SET – 1 (1-MARKS)

Q4. A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded?

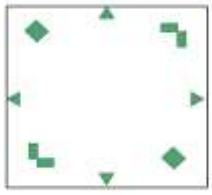
Note: The figures shown are representative.



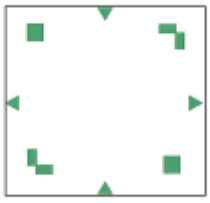
(a)



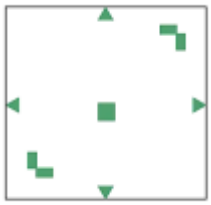
(b)



(c)



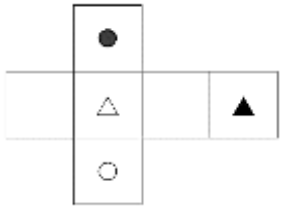
(d)



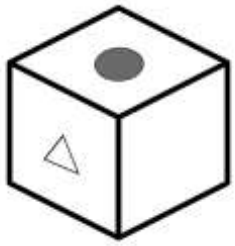
GATE CSE - 2025 MCQ SET – 2 (1-MARKS)

Q5. The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube?

Note: The figures shown are representative

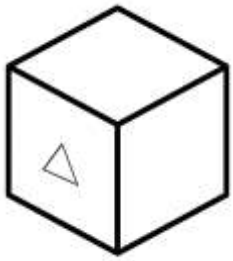


(a)



(b)

(c)



(d)

Check Out Our GATE Courses & Secure Your Dream Rank!

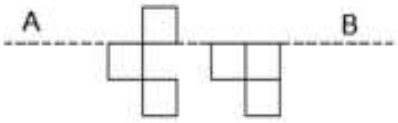
GATE CSE - 2024 MCQ SET – 1 (2-MARKS)

Q6. A rectangular paper of 20 cm × 8 cm is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

(a) 18
(b) 24
(c) 20
(d) 21

GATE CSE - 2024 MCQ SET – 1 (2-MARKS)

Q7. The least number of squares to be added in the figure to make AB a line of symmetry is



(a) 6
(b) 4
(c) 5
(d) 7

GATE CSE - 2024 MCQ SET – 2 (2-MARKS)

Q8. A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube.

The minimum number of such cuts required is

(a) 3
(b) 4
(c) 7
(d) 8

GATE CSE - 2023 MCQ (1-MARKS)

Q9. Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE?

(a) The surface of the object must be concave everywhere.
(b) The surface of the object must be convex everywhere.
(c) The surface of the object may be concave in some places and convex in other places.
(d) The object can have edges, but no corners.

GATE CSE - 2023 MCQ (2-MARKS)

Q10. Which one of the options best describes the transformation of the 2-dimensional figure **P** to **Q**, and then to **R**, as shown?

(a) Operation 1: A clockwise rotation by 90° about an axis perpendicular to the plane of the figure Operation 2: A reflection along a horizontal line
(b) Operation 1: A counter clockwise rotation by 90° about an axis perpendicular to the plane of the figure Operation 2: A reflection along a horizontal line
(c) Operation 1: A clockwise rotation by 90° about an axis perpendicular to the plane of the figure Operation 2: A reflection along a vertical line
(d) Operation 1: A counter clockwise rotation by 180° about an axis perpendicular to the plane of the figure Operation 2: A reflection along a vertical line

GATE CSE - 2022 MCQ (1-MARKS)

Q11. A palindrome is a word that reads the same forwards and backward. In a game of words, a player has the following two plates painted with letters.

From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome. The player should use all the five plates exactly once. The plates can be rotated in their plane.
(a)

(b)

Check Out Our GATE Courses & Secure Your Dream Rank!

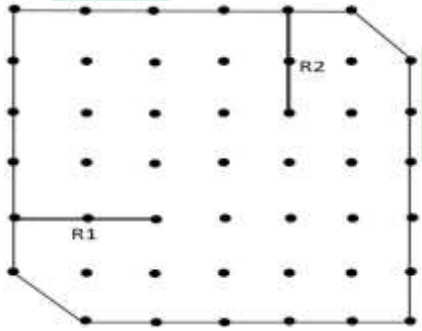
(c)

(d)

GATE CSE - 2022 MCQ (2-MARKS)

Q12. A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved.

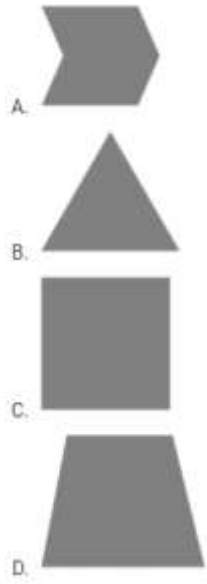
What is the least number of additional straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line.



(a) 2
(b) 4
(c) 5
(d) 3

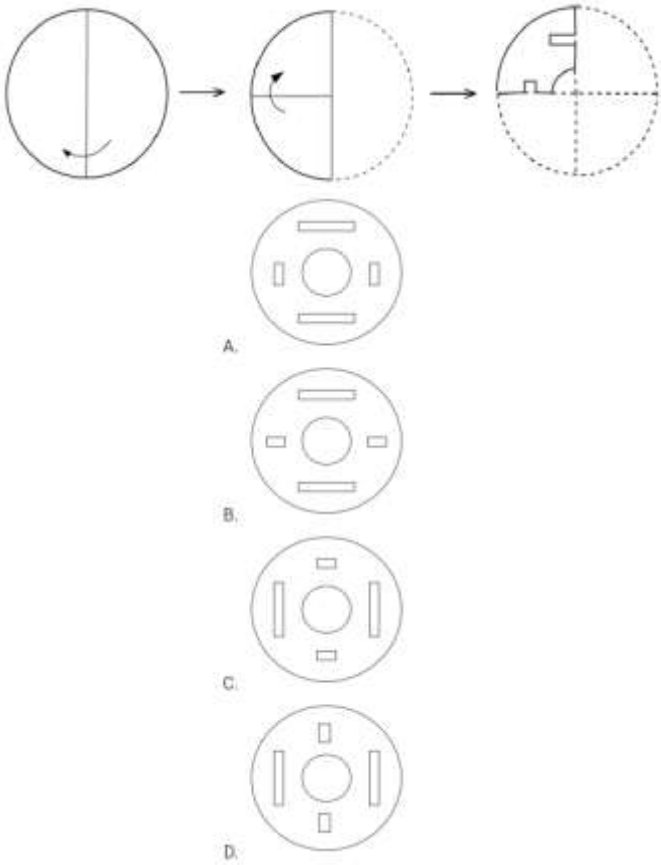
GATE CSE - 2021 MCQ SET – 1 (1-MARKS)

Q13. A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon.
Which one of the following is NOT a convex polygon?



GATE CSE - 2021 MCQ SET – 1 (1-MARKS)

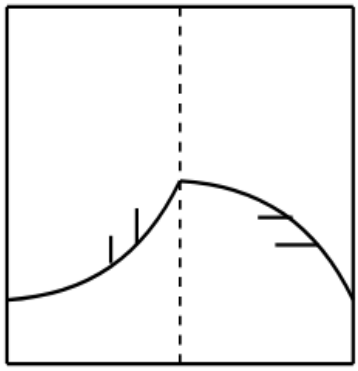
Q14. A circular sheet of paper is folded along the lines in the direction shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.



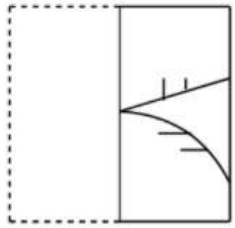
Check Out Our GATE Courses & Secure Your Dream Rank!

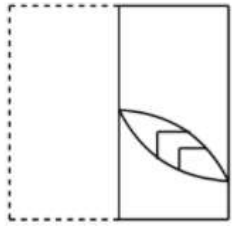
GATE CSE - 2021 MCQ SET – 2 (1-MARKS)

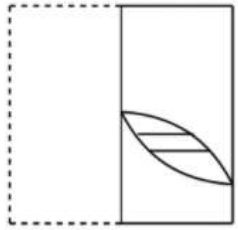
Q15.

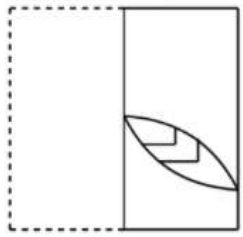


A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

A.


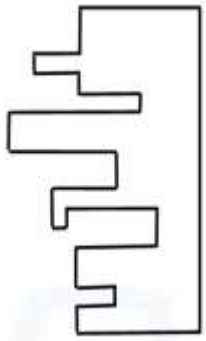
B.


C.


D.


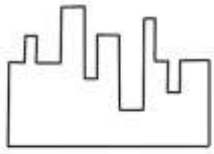
GATE CSE - 2021 MCQ SET – 2 (2-MARKS)

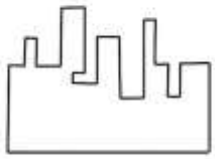
Q16.

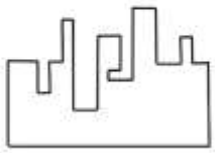


A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

(A)

(B)


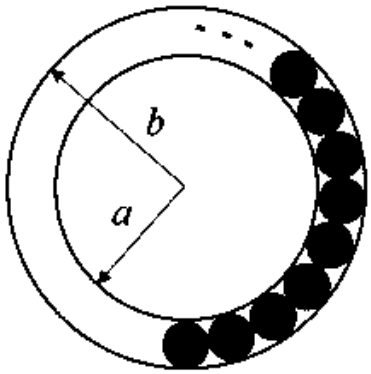
(C)


(D)


Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2020 MCQ (2-MARKS)

Q17. The figure below shows an annular ring with outer and inner radii as b and a, respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is ____.



(a). $\pi\left[(b^2-a^2)-\frac{n}{4}(b-a)^2\right]$
(b). $\pi[(b^2-a^2)-n(b-a)^2]$
(c). $\pi\left[(b^2-a^2)+\frac{n}{4}(b-a)^2\right]$
(d). $\pi[(b^2-a^2)+n(b-a)^2]$

GATE CSE - 2020 MCQ (2-MARKS)

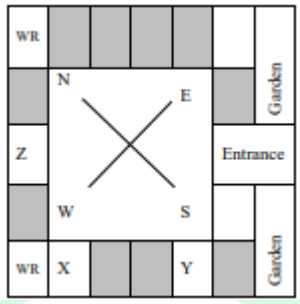
Q18. Two straight lines are drawn perpendicular to each other in X-Y plane. If α and β are the acute angles the straight lines make with the X-axis, then α + β is _

(a) 180°
(b) 120°
(c) 90°
(d) 60°

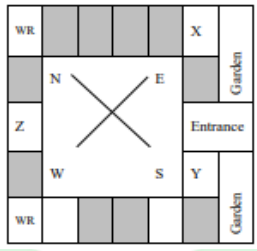
GATE CSE - 2019 MCQ (2-MARKS)

Q19. Three of the five students allocated to a hostel put in special requests to the warden. Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.Request by X: Due to pollen allergy, I want to avoid a wing next to the garden.Request by Y: I want to live as far from the washrooms as possible, since I am very sensitive to smell.
Request by Z: I believe in Vaastu and so I want to stay in the South-west wing.The shaded rooms are already occupied. WR is washroom.

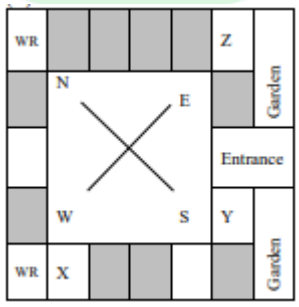
(a)



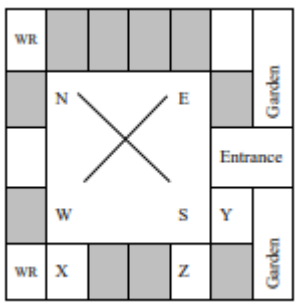
(b)



(c)



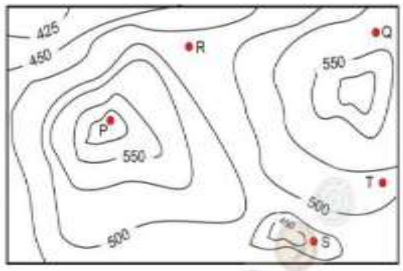
(d)



Check Out Our GATE Courses & Secure Your Dream Rank!

GATE CSE - 2017 MCQ SET – 1 (2-MARKS)

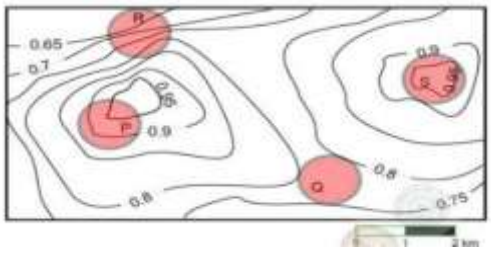
Q20. A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to 525 m, which of the villages P, Q, R, S, T get submerged?



(a) P, Q
(b) P, Q, T
(c) R, S, T
(d) Q, R, S

GATE CSE - 2017 MCQ SET – 2 (2-MARKS)

Q21. An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.



a

If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?

(a) P
(b) Q
(c) R
(d) S

ANSWER KEY

1. (c)	**2. (d)**	**3. (c)**
4. (a)	**5. (a)**	**6. (a)**
7. (a)	**8. (a)**	**9. (c)**
10. (a)	**11. (b)**	**12. (d)**
13. (a)	**14. (a)**	**15. (b)**
16. (a)	**17. (a)**	**18. (c)**
19. (d)	**20. (c)**	**21. (c)**

HINT & SOLUTION

1. ANSWER :- (C)

Two types of tiles are given:

(1) A 2 × 2 square tile

(2) A 1 × 4 rectangular tile (can be placed horizontally or vertically)

Each tile covers exactly 4 unit squares.

Necessary Condition:

The total number of unit squares in the pattern must be divisible by 4.

Checking the options:

(A)Total squares = 4

This can be formed using one 1 × 4 tile.

Hence, possible.

(B)Total squares = 12

This can be formed using three 1 × 4 tiles.

Hence, possible.

(C) Total squares = 20

Although 20 is divisible by 4, the width of the pattern is 5 (odd).

A 2 × 2 tile requires even dimensions, and a 1 × 4 tile cannot exactly cover a row of length 5.

Hence, complete tiling is not possible.

(D)Total squares = 16

This can be formed using four 2 × 2 tiles or four 1 × 4 tiles.

Hence, possible.

Final Answer: (C)

Check Out Our GATE Courses & Secure Your Dream Rank!

2. ANSWER :- (D)

- Top View → footprint (block positions)
- Front View → heights

Step 1: Match Top View

- Required shape: specific L-type footprint
- Only (iv) matches exactly
  (i), (ii), (iii) eliminated

Step 2: Match Front View

- L-shape with correct height variation.
- Only (iv) satisfies it.

So, the correct answer is D.

3. ANSWER :- (C)

The given shape is one part of a square, so the missing piece must be its exact complement—filling all gaps perfectly without overlap.

Observe the given figure:

- It has specific inward cuts (notches) and step-like boundaries
- The missing piece must have matching outward projections at exactly those positions
- Orientation is critical—only rotation (not reflection mismatch) that fits inside the square is valid
- On comparing options:
- (i), (ii), and (iv) have mismatched edges or incorrect step alignment
- Only (iii) has:
  - Projections exactly matching the gaps
  - Correct orientation of all steps
  - Perfect boundary completion of the square

Hence, (iii) is the exact complementary piece that completes the square.

4. ANSWER :- (A)

- Two folds ⇒ paper has 4 layers ⇒ every cut appears 4 times.
- Cuts reflect across both diagonals ⇒ final pattern must have perfect diagonal symmetry.

From Figure (IV):

- ▲ triangle → appears 4 times, all pointing correctly after reflection
- ■ square → appears 4 times, symmetric positions
- corner notch → shows up at all 4 corresponding corners

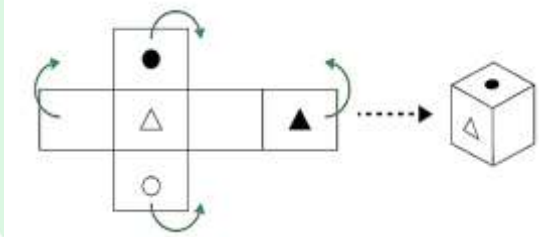
Option (A) matches:

- correct count (×4)
- correct symmetry
- correct orientation

Others break symmetry or misplace shapes.

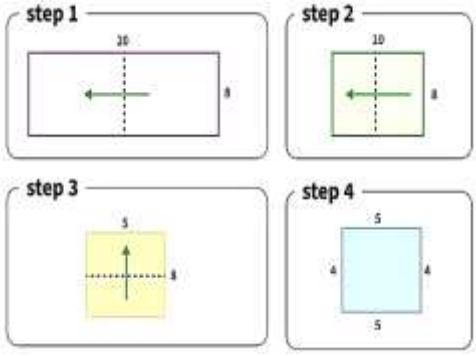
5. ANSWER :- (A)

Two folds ⇒ paper



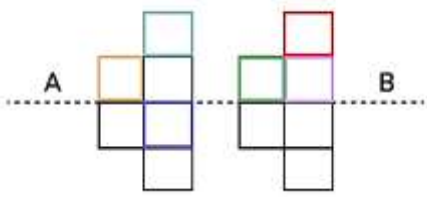
6. ANSWER :- (A)

The sheet would be folded as shown and resulting perimeter= (5+4)*2 = 18



Check Out Our GATE Courses & Secure Your Dream Rank!

7. ANSWER :- (A)



8. ANSWER :- (A)

Let us suppose x cuts happen on each edge.

For x cut on each edge, number of small cubes = $(x+1)^3 = 8$ [as per the data given]

Therefore, x = 1 cuts will happen on each edge.

Hence, a total of 3 cuts have to take place.

So, the correct option is A.

9. ANSWER :- (C)

A smooth 3-dimensional object can have following surfaces:

1. Convex in some regions.
2. Concave in some regions.
3. Flat in other regions.

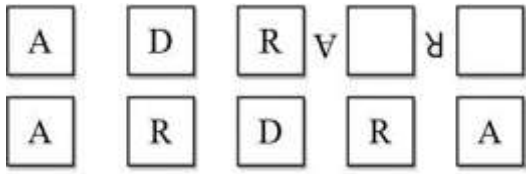
for example:

Torus shape (Doughnut like).

10. ANSWER :- (A)

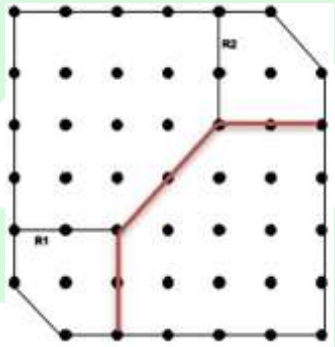
To determine the transformation of the 2-dimensional figure P to Q and then to R, let's analyze the changes in the figure step by step:
From P to Q:
The figure is rotated clockwise by 90º about an axis perpendicular to the plane of the figure. The triangular portion with the dot moves to the right.
Based on the change, option (A) "Operation 1: A clockwise rotation by 90º about an axis perpendicular to the plane of the figure" seems correct.
From Q to R:
The figure is reflected along a horizontal line. The triangular portion with the dot now appears below the line.
Based on the change, option (A) "Operation 2: A reflection along a horizontal line" seems correct.
Therefore, the correct answer is (A)

11. ANSWER :- (B)



12. ANSWER :- (D)

It is clear that 3 lines can divide the plot into similar shapes.



13. ANSWER :- (A)

This shape A polygon "curves inward," meaning it has a concave region where a line segment connecting two points might lie outside the polygon. Thus, Shape A is NOT convex.

These shapes B, C, and D are regular polygons or convex shapes with no inward curvature and all angles $\leq 180\circ$.

So, the correct answer is A is the shape that is NOT a convex polygon.

14. ANSWER :- (A)

After careful observations you will see that:

Since the paper was folded into a quarter-circle, the punching pattern will be symmetrically replicated in all four quadrants of the original circular sheet when fully unfolded.

The exact pattern of punches depends on the shapes and positions shown in the folded quarter-circle. Upon unfolding the punched pattern will be reflected symmetrically across the folds due to the way the paper was folded and the punched pattern will

Check Out Our GATE Courses & Secure Your Dream Rank!

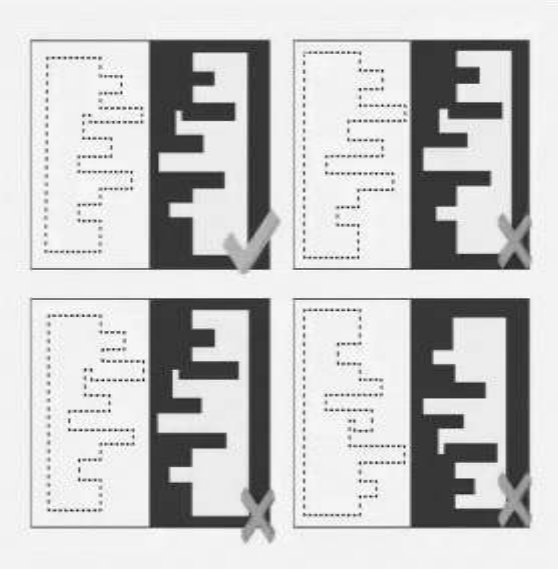
correspond to Option A.

15. ANSWER :- (B)

When the sheet is folded, the vertical lines of the sheet will be above than the horizontal lines. So option (B) is true.

16. ANSWER :- (A)

1. When we join the missing piece, A rectangle should be formed. If we carefully notice the L shape part of the shape, we can easily eliminate option B and option D.
2. Now if we notice option A and option C, option A is perfectly matching to complete the shape.



17. ANSWER :- (A)

Area of ring = Area of outer circle − Area of inner circle
$= \pi b^2 - \pi a^2$
$= \pi(b^2 - a^2)$
Diameter of each small circle = (b − a)
Radius = (b − a)/2
Area of one small circle = $\pi[(b - a)/2]^2 = \pi(b - a)^2/4$
Area of n such circles = $n \times \pi(b - a)^2/4$
Required unpainted area = Area of ring − Area of n small circles
$= \pi(b^2 - a^2) - n\pi(b - a)^2/4$
$= \pi[(b^2 - a^2) - n/4\ (b - a)^2]$

18. ANSWER :- (C)

Perpendicular lines → sum = 90°.

19. ANSWER :- (D)

X want not live nearby garden due to pollen allergy

Y want not live nearby wash room(WR
Z want to live in the South West wing

Clear it is option D

Because x is far from garden Y is far from wash room and Z is staying in South West wing.

20. ANSWER :- (C)

Villages located below 525 m will be submerged.
From the contour map:
- R is below 525 m
- S is below 525 m
- T is below 525 m
- P and Q are above 525 m

Hence, submerged villages are R, S, T.

21. ANSWER :- (C)

Region with closest contour lines → highest pressure gradient → strongest change → R

Check Out Our GATE Courses & Secure Your Dream Rank!